MATRIX SPENCER BOUNDS FROM FINITE-ENTROPY LIFTS

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Abstract. The Matrix Spencer conjecture asks whether every collection of self-adjoint $n \times n$ contractions $A_1, \dots, A_n$ admits signs $\varepsilon_i \in \{-1, 1\}$ for which $\| \sum_i \varepsilon_i A_i \|_{\mathrm{op}} = O(\sqrt{n})$. We formulate a reusable finite-lift principle for structured instances. Let

$$\mathcal{B} = \bigoplus_{\alpha=1}^{m} \mathrm{M}_{r_\alpha}(\mathbb{C}), \qquad h(\mathcal{B}) = \max_\alpha r_\alpha, \qquad R(\mathcal{B}) = \sum_\alpha r_\alpha,$$

and set $\mathsf{E}(\mathcal{B}) = h(\mathcal{B})R(\mathcal{B})$. If a tuple of matrices has a K-bounded lift to $\mathcal{B}$ through a contraction on the lifted span, then

$$\mathrm{disc}_{\mathrm{op}}(A_1, \dots, A_n) \le CK \sqrt{n \log\left(e + \frac{\mathsf{E}(\mathcal{B})}{n}\right)}.$$

This is a direct consequence of the block-diagonal Matrix Spencer theorem of Dadush, Jiang and Reis. The same argument also covers matrix-valued multiplier models after sampling on a finite norming set.

The framework gives deterministic Matrix Spencer bounds for scalar Toeplitz and Hankel contractions, fixed block-size multilevel Toeplitz and Hankel classes at any fixed level (full bandwidth, with level-dependent constants) together with dimension-free constants and a matching lower bound in the banded case, matrices in a common finite-dimensional C*-algebra, finite-group convolution algebras with controlled irreducible dimensions, and the usual compression and bounded-image operations. The paper also records the correct entropy accounting for finite sums: for direct sums one has $\mathsf{E}(\bigoplus_\ell \mathcal{B}_\ell) = (\max_\ell h(\mathcal{B}_\ell)) \sum_\ell R(\mathcal{B}_\ell)$, not generally $\sum_\ell \mathsf{E}(\mathcal{B}_\ell)$. The same direct-sum bookkeeping gives a finite-menu common-sign corollary: many finite-lift tuples have one signing whenever their combined lift entropy is $O(n)$, and this supplies deterministic common core signings for the later active-prefix random-amplitude perturbation menus. A packaged corollary therefore covers finite-lift core libraries plus localized random-amplitude matrix noise under the same union-count and energy schedules. Finally, an elementary dimension argument shows that any exact, or even fixed-error approximate faithful, finite block lift whose image covers the full unit ball of $\mathrm{M}_D(\mathbb{C})$ has entropy at least D^2; the associated lift-entropy invariant records the exact complexity seen by this finite-lift engine and explains why the full-unit-ball finite-lift route cannot by itself settle the square case of the full Matrix Spencer conjecture. We also record the June 2026 algebraic theorem of Akbas–Sra, which improves the exact finite-dimensional C*-algebra input from the elementary hR parameter to intrinsic dimension $\dim_{\mathbb{C}} \mathcal{A}$; this gives a sharper AF-stage corollary when the approximating stages have $\dim_{\mathbb{C}} A_k = O(n)$. Their low-Frobenius perturbation theorem also gives a matrix-representation AF corollary: an AF-stage algebraic core of intrinsic dimension $O(n)$, perturbed by self-adjoint tails with $\|L_i\|_{\mathrm{F}}^2 \le n/\log^3(en)$ – in particular by operator-norm-one tails of rank at most $n/\log^3(en)$ – still has discrepancy $O(\sqrt{n})$ up to the AF approximation error. The same perturbative input applies to the square group-convolution regime: if $|G| = O(n)$, arbitrary self-adjoint convolution contractions remain at discrepancy $O(\sqrt{n})$ after adding self-adjoint low-Frobenius, hence low-rank, perturbations. The same deterministic perturbative input is packaged as a distribution-free random-matrix corollary: with the same probability as the structural event, arbitrary random algebraic cores of intrinsic dimension $O(n)$, including full $D \times D$ random self-adjoint models for $D \lesssim \sqrt{n}$, remain Spencer-scale after arbitrary dependent random low-Frobenius, low-rank, spiked, compressed, or coordinate-supported self-adjoint tails. A separate average-case corollary covers full centered, possibly dependent, random matrix ensembles in every ambient dimension. First, a distribution-free all-plus gauge says that arbitrary self-adjoint or rectangular random matrices become a Matrix-Spencer-scale instance after pathwise scaling by the larger of the largest individual operator norm and the all-plus norm divided

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1. Introduction

Spencer's theorem that "six standard deviations suffice" is a central result of discrepancy theory [24, 20, 8]. In one standard form, n vectors in the unit cube of $\mathbb{R}^n$ can be signed so that their signed sum has ℓ_∞ norm $O(\sqrt{n})$. The Matrix Spencer conjecture asks for the analogous statement with vectors replaced by self-adjoint matrices and ℓ_∞ norm replaced by operator norm.

Definition 1.1 (Matrix discrepancy)**.** For self-adjoint matrices $A_1, \dots, A_n$, set

$$\operatorname{disc}(A_1, \dots, A_n) = \min_{\varepsilon \in \{-1,1\}^n} \left\| \sum_{i=1}^n \varepsilon_i A_i \right\|_{\mathrm{op}}.$$

For arbitrary complex matrices we use the same formula and write

$$\operatorname{disc}_{\mathrm{op}}(A_1, \dots, A_n) = \min_{\varepsilon \in \{-1,1\}^n} \left\| \sum_{i=1}^n \varepsilon_i A_i \right\|_{\mathrm{op}}.$$

Question 1.2 (Matrix Spencer)**.** *Is there a universal constant C such that every n self-adjoint matrices $A_1, \dots, A_n \in \mathrm{M}_n(\mathbb{C})$ satisfying $\|A_i\|_{\mathrm{op}} \le 1$ obey*

$$\operatorname{disc}(A_1, \dots, A_n) \le C\sqrt{n}?$$

More generally, one asks for the same bound for $d \times d$ matrices with $d \le n$, with a constant independent of d.

The general problem remains open. Noncommutative Khintchine inequalities give logarithmic losses [18, 23]. Important recent advances include the intrinsic-freeness concentration theory of Bandeira, Boedihardjo and van Handel [1]; the quantum-communication route of Hopkins, Raghavendra and Shetty [16]; the polylogarithmic-rank theorem of Bansal, Jiang and Meka [4]; and the mirror-descent matrix-discrepancy framework of Dadush, Jiang and Reis [11]. The engine we use is the block-diagonal case: a bound of $O(\sqrt{n}\,\sqrt{\log h})$ for block size h was first proved by Levy, Ramadas and Rothvoss [14] via the multiplicative weight update method, and Dadush, Jiang and Reis [11] later removed the restriction $h \le n$ through their mirror-descent partial-coloring framework. These results do not directly cover many full-rank structured families, such as normalized Toeplitz, Hankel, or convolution matrices in the square regime.

The point of this paper is to isolate a simple sufficient condition that does cover these structured classes. The condition is the existence of a bounded finite block lift. If each input matrix is a compression, quotient, or bounded linear image of a block-diagonal element whose total block complexity is small, one signs the lifted block-diagonal matrices and then maps back.

1.1. The finite-lift principle.

Let

$$\mathcal{B} = \bigoplus_{\alpha=1}^m \mathrm{M}_{r_\alpha}(\mathbb{C}), \qquad \|Y\|_{\mathcal{B}} = \max_\alpha \|Y_\alpha\|_{\mathrm{op}}.$$

We write

$$h(\mathcal{B}) = \max_\alpha r_\alpha, \qquad R(\mathcal{B}) = \sum_{\alpha=1}^m r_\alpha, \qquad \mathsf{E}(\mathcal{B}) = h(\mathcal{B})R(\mathcal{B}).$$

The number $\mathsf{E}(\mathcal{B})$ is the entropy dimension seen by block-diagonal Matrix Spencer. It is D for a diagonal algebra of dimension D, D^2 for the full matrix algebra M_D, and bD for a block-diagonal algebra on $\mathbb{C}^D$ with blocks of size at most b.

Theorem 1.3 (Finite block-lift Matrix Spencer)**.** *There is a universal constant C with the following property. Let $A_1, \dots, A_n \in \mathrm{M}_D(\mathbb{C})$ be arbitrary complex matrices with $\|A_i\|_{\mathrm{op}} \le 1$. Suppose there are elements $Y_i \in \mathcal{B}$ and a fixed linear map*

$$\Pi : \operatorname{span}\{Y_1, \dots, Y_n\} \longrightarrow \mathrm{M}_D(\mathbb{C})$$

such that

$$\Pi(Y_i) = A_i, \qquad \|\Pi(Y)\|_{\mathrm{op}} \le \Lambda \|Y\|_{\mathcal{B}} \quad (Y \in \operatorname{span}\{Y_1, \dots, Y_n\}),$$

and

$$\|Y_i\|_{\mathcal{B}} \le K \qquad (1 \le i \le n).$$

Then

$$\operatorname{disc}_{\mathrm{op}}(A_1, \dots, A_n) \le C\Lambda K \sqrt{n \log\left(e + \frac{\mathsf{E}(\mathcal{B})}{n}\right)}.$$

If the A_i are self-adjoint and the Y_i may be chosen self-adjoint, the same estimate holds for $\operatorname{disc}(A_1, \dots, A_n)$.

Thus every class with uniformly bounded lift loss and $\mathsf{E}(\mathcal{B}) = O(n)$ satisfies the conjectured $O(\sqrt{n})$ Matrix Spencer bound. Scalar Toeplitz matrices are the commutative case: after a bounded finite Toeplitz extension and Fourier sampling, the lift algebra is diagonal of dimension $O(D)$. Finite group convolution and block Toeplitz matrices are noncommutative variants in which the finite lift has bounded block size.

The theorem has a built-in limitation. If the full unit ball of $\mathrm{M}_D(\mathbb{C})$ is lifted exactly through a finite block algebra, or even approximated to fixed operator-norm error < 1 by such a lift, dimension forces $\sum_\alpha r_\alpha^2 \ge D^2$, hence $\mathsf{E}(\mathcal{B}) \ge D^2$. Therefore this full-unit-ball route cannot, by itself, turn arbitrary $D \times D$ matrices in the square regime $D = n$ into an $O(\sqrt{n})$ theorem. It yields instead the usual $\sqrt{n \log(e + D^2/n)}$ scale. The point is not that finite lifts are the only possible method, but that this reusable class-level sufficient condition applies when the structured class has small finite-lift entropy.

Since this note is organized around finite lifts, it keeps the elementary hR theorem and the corresponding obstruction bookkeeping explicit. The current strongest general input for exact finite-dimensional C*-algebra containment is stronger: Akbas and Sra [2] prove that contractions contained in a finite-dimensional C*-algebra of intrinsic dimension $O(n)$ already satisfy the $O(\sqrt{n})$ Matrix Spencer bound. We use that result below as an external upgrade for finite stages and for group convolution algebras. We also use their perturbative theorem for algebraic families plus low-Frobenius self-adjoint tails to record a matrix-representation AF corollary and a distribution-free random algebraic-core ensemble theorem. The finite-lift theorem remains the mechanism for the Toeplitz, Hankel, multiplier, and obstruction statements proved here.

1.2. **Main deterministic consequences.** The abstract theorem gives the following consequences. Constants are universal unless a dependence is displayed in the subscript.

(1) **Scalar Toeplitz and Hankel.** Every collection of arbitrary complex $d \times d$ Toeplitz or Hankel contractions satisfies

$$\operatorname{disc}_{\mathrm{op}}(A_1, \dots, A_n) \le C\sqrt{n \log\left(e + \frac{d}{n}\right)}.$$

In particular, self-adjoint Toeplitz and Hankel matrices satisfy the Matrix Spencer bound when $d \le n$.

(2) **Fixed-block and fixed-level multilevel Toeplitz/Hankel.** For fixed matrix block size b, all one-level $b \times b$ block Toeplitz and block Hankel contractions of total dimension D satisfy

$$\operatorname{disc}_{\mathrm{op}}(A_1, \dots, A_n) \le C_b\sqrt{n \log\left(e + \frac{D}{n}\right)}.$$

More generally, for any fixed multilevel order $r \ge 1$ and block size b, all r-level $b \times b$ block Toeplitz and block Hankel contractions of full bandwidth and ambient dimension $D = b\prod_j d_j$ satisfy

$$\operatorname{disc}_{\mathrm{op}}(A_1, \dots, A_n) \le C_{r,b}\sqrt{n \log\left(e + \frac{D}{n}\right)} \qquad \text{(Theorem 5.22)},$$

so this is $O_{r,b}(\sqrt{n})$ whenever $D \le n$. The constant $C_{r,b}$ here is non-constructive and grows with r, since it passes through a multilevel Toeplitz symbol extension (Theorem 5.10) whose constant cannot be taken uniform in r. A complementary, fully explicit form is available for banded symbols: if the matrix symbols have per-coordinate degree at most w (with each box dimension at least $4w$), then

$$\operatorname{disc}_{\mathrm{op}}(A_1, \dots, A_n) \le C_{r,b}\sqrt{n \log\left(e + \frac{w^r}{n}\right)}$$

with a dimension-free, explicit constant and no appeal to the symbol extension. This banded bound is sample-optimal: the factor w^r is forced by a norming-set dimension count, and a matching lower bound of the same form is realized by multilevel circulants, which are exactly complex vector Spencer in disguise. The formal anisotropic version replaces w^r by $\prod_j(2w_j + 1)$, and the periodic circulant lower bound has the same product form. In the self-adjoint product-bandwidth regime $\prod_j(2w_j + 1) \le n$ it is $O_{r,b}(\sqrt{n})$ with D arbitrary, so the class escapes the D^2 entropy barrier.

(3) **Common finite-dimensional C*-algebras.** Suppose the A_i lie in a common finite-dimensional C*-algebra represented as

$$\mathcal{A} \simeq \bigoplus_{\alpha=1}^{m} (\mathrm{M}_{r_\alpha}(\mathbb{C}) \otimes I_{m_\alpha}) \subseteq \mathrm{M}_D(\mathbb{C}).$$

Then

$$\operatorname{disc}_{\mathrm{op}}(A_1,\dots,A_n) \le C\sqrt{n\log\left(e+\frac{(\max_\alpha r_\alpha)\sum_\alpha r_\alpha}{n}\right)}.$$

This includes simultaneous diagonalization, common bounded-block diagonalization, and commutative association-scheme algebras.

(4) **Finite group convolution.** For left convolution operators on $\ell^2(G)$, where G is finite and $q(G)=\max_{\pi\in\widehat{G}}\dim\pi$,

$$\operatorname{disc}_{\mathrm{op}}(L_{f_1},\dots,L_{f_n}) \le C\sqrt{n\log\left(e+\frac{q(G)\sum_{\pi\in\widehat{G}}\dim\pi}{n}\right)}.$$

For finite abelian groups this becomes $C\sqrt{n\log(e+|G|/n)}$. Using the Akbas–Sra intrinsic-dimension theorem, the square regime $|G|=O(n)$ has the sharper $O(\sqrt{n})$ bound for arbitrary finite groups, with no bounded-irrep hypothesis; the low-Frobenius perturbative version also allows self-adjoint rank-$n/\log^3(en)$ noise on top of those convolution cores.

(5) **Closure operations.** The framework is stable under compression, bounded linear images, completely bounded fixed-size tensor amplification, and finite sums when the combined direct-sum entropy remains controlled. The last condition is important: the entropy of a direct sum is not the sum of the individual entropies in general.

(6) **Structured random inputs.** All of the preceding statements are pathwise deterministic. They therefore apply to random Toeplitz, Hankel, fixed-block multilevel Toeplitz/Hankel, circulant, and group-convolution samples after normalization by their realized operator norms; a complementary parity obstruction to the stability of random banded Toeplitz matrices is established in the companion paper [27]. The Akbas–Sra perturbative input also gives a distribution-free random theorem for arbitrary random algebraic cores of intrinsic dimension $O(n)$, including $D\times D$ full random self-adjoint models for $D^2=O(n)$, after adding arbitrary dependent low-Frobenius, low-rank, spiked, compressed, or coordinate-supported tails. At the completely distribution-free level, the all-plus gauge

$$\max\left\{\max_i \|X_i\|_{\mathrm{op}}, n^{-1/2}\left\|\sum_i X_i\right\|_{\mathrm{op}}\right\}$$

normalizes arbitrary self-adjoint or rectangular random matrices into a Spencer-scale instance, and L^p moment or quantile budgets for this gauge give checkable deterministic almost-sure schedules, with explicit all-integer and sparse-dimension thresholds for polynomial active menus. Direct tail-profile bounds, including sub-Weibull gauge tails, give the corresponding logarithmic deterministic normalizations without independence. The finite-confidence version gives one-row or finite-library

certificates with explicit dependence on the requested failure probability δ, and its conditional form allows the normalizing scales to be chosen from side information through conditional moment or tail envelopes, and a random-active finite-confidence form charges only the selected menu. The corresponding conditional Borel–Cantelli form gives almost-sure adaptive schedules when the conditional failure budgets are summable, including random active submenus chosen by the environment. For full centered random matrix ensembles in arbitrary dimension, even with dependence among the matrices, an operator-tail hypothesis stable under scalar ℓ_2 sums gives another average-case route: after scaling by $K(\sqrt{D} + \sqrt{\log(en)})$, all matrices are contractions and the all-plus signing has norm at most $\sqrt{n}$ with high probability. The corresponding active-menu version is simultaneous over finite or countable libraries under the operator-tail union cost $J_m(n_m + 1)e^{-\tau_m^2}$, with no coordinate-wise sign-symmetry requirement. If the active operator-tail streams are selected randomly by the environment, the same proof pays only the selected conditional tail cost. The rectangular version has the same selected-cost random-active form and uses the same all-plus certificate and the left/right dimension sum $d_1 + d_2$. Bilinear scalar sub-Gaussian tails for $\langle u, \sum_i a_i X_i v\rangle$ verify that rectangular operator-tail hypothesis by a two-net argument, and their random-active form pays only the selected bilinear tail cost; rectangular Gaussian and feature-map families only need a uniform bilinear proxy q. The Bernstein bilinear analogue also has a random-active selected-cost form, with the additional jump term $b(\sqrt{d_1 + d_2} + \tau)^2$, and its independent-entry specialization is checkable from selected variance and jump proxies; a further entrywise Bernstein-mgf corollary verifies those proxies from conditional independence of the individual matrix entries, and its finite-confidence form gives explicit one-row failure budgets. For bounded conditionally independent entries the same finite-confidence certificate needs only conditional entry variances and uniform entry caps. The bounded matrix-Bernstein all-plus route also has a finite-confidence selected-library version, using only matrix norm caps and all-plus variance budgets, with a rectangular form obtained by self-adjoint dilation. The adapted bounded matrix-Freedman route has the same finite-confidence version with predictable row and column variance budgets. The tail-capped Bernstein/Freedman route likewise has a finite-confidence selected-library version for unbounded streams after paying the selected cap-failure budget. Combining the core-signing and operator-tail inputs, the paper also allows any random core class with an environment-measurable $K\sqrt{n}$ signing, including dimension-controlled algebraic cores, finite-lift cores, Toeplitz cores, and Hankel cores, to absorb full random operator-tail noise whose scalar linear combinations obey the same Gaussian tail conditionally on the core, in arbitrary ambient dimension after the same natural spectral scaling of the noise. For active-prefix finite-lift core libraries,

the same mechanism is simultaneous after paying $J_m(n_m+1)e^{-\tau_m^2}$ and scaling the full noise by $\sqrt{D_m}+\tau_m$; the concrete schedules are arbitrary-growth $\tau_m^2 \gtrsim \log(J_m(n_m+1)) + \log m$, all-integer $\tau_D^2 = R\log(eD)$ with $R > \Gamma+1$, and sparse $R > \Gamma$ when $J_m(n_m+1) \le (eD_m)^\Gamma$. For Gaussian, Rademacher, or scalar sub-Gaussian matrix-series noise with predictable self-adjoint coefficient directions, a matrix-Laplace tail improves the finite-lift perturbation scale to $q_{m,h}(\sqrt{\log(e\kappa_{m,h})}+\tau_m)$, assuming the realized individual and signed-sum coefficient variances are bounded by $q_{m,h}^2$ and $n_m q_{m,h}^2$, with coefficient square-function ranks at most $\kappa_{m,h}$. Taking $\kappa_{m,h} = D_m$ recovers the full-rank fallback. A common-projection specialization packages this as low-rank matrix-series noise: if all coefficient directions live in a rank-$\kappa_{m,h}$ subspace, the rank hypotheses are automatic. A norm-budget specialization then replaces the square-function hypotheses by the scalar checks $\max_i \sum_r b_{m,h,i,r}^2 \le q_{m,h}^2$ and $\sum_{i,r} b_{m,h,i,r}^2 \le n_m q_{m,h}^2$, whenever $\|G_{m,h,i,r}\|_{\mathrm{op}} \le b_{m,h,i,r}$ inside the common support. A block-load specialization lets these atoms range over an orthogonal block family, replacing total atom energy by the maximum incident block energy. A bounded weighted block-design specialization covers regular, matching, permutation, and balanced without-replacement designs with no load-tail cost once their weighted block degrees are deterministic or fail only summably. The bounded-overlap block-hypergraph specialization makes this a direct $q/\sqrt{d}$ rule for block designs, regular block hypergraphs, and balanced block dictionaries with overlap degree d. The heterogeneous bounded-overlap version replaces d by local row-block and total-block degree profiles, so irregular block hypergraphs need not be normalized by their worst active block. The row-wise block-permutation specialization makes this explicit for d-round permutation, matching, and balanced without-replacement assignments with atom amplitudes at scale $q/\sqrt{d}$. A random block-load specialization verifies the incident-energy event by conditional scalar Bernstein, with union cost over active blocks and atoms. A uniform random block-support specialization makes the same test explicit in terms of each atom's expected support size divided by the number of blocks. A profile schedule version requires only the resulting Bernstein exponents to dominate the active block-count logarithm. A homogeneous uniform version further reduces those exponents to $N_{\mathrm{ind}}s/B$ and $N_{\mathrm{tot}}s/B$, yielding all-integer threshold $8(\Gamma+1)/3$ and sparse threshold $8\Gamma/3$ for the incidence counts. A weighted version applies the same argument directly to random incident weights, covering random amplitudes, bounded compound-thinned atom fields, and random weighted dictionaries. A predictable block-support profile specialization removes independence for sequential support exposure by using one-step inclusion probabilities and predictable amplitude envelopes. Its profile schedule has the same active block-count logarithm as the weighted Freedman

schedule, with the predictable-support exponents in place of the weighted incident-load exponents. An inhomogeneous incidence-mass specialization replaces the raw exponents by row-block and total-block predictable probability masses, retaining the same 8/3 threshold constants for biased adaptive block samplers. Its homogeneous predictable uniform schedule has the same $8(\Gamma+1)/3$ all-integer and $8\Gamma/3$ sparse incidence thresholds as the independent homogeneous block-support schedule. A predictable weighted version replaces independence by scalar Freedman control of the incident weights, allowing adaptive atom sampling and without-replacement block exposure. A profile schedule form covers both weighted variants once the corresponding Bernstein or Freedman exponents dominate the active block-count logarithm. A bounded weighted incidence specialization needs only the normalized row-block and total-block mean loads ρ/ω, with the same 8/3 threshold constants. A quadratic-form-to-operator-tail lemma then verifies the same hypothesis from conditional scalar sub-Gaussian bounds on $\langle u, \sum_i a_i Y_i u\rangle$, so Gaussian, correlated Gaussian, and Wigner-type active-prefix noise libraries only need their one-dimensional quadratic-form proxy $q_{m,h}$, and the natural scale becomes $q_{m,h}(\sqrt{D_m}+\tau_m)$. For conditionally independent upper-triangular sub-Gaussian entries, that proxy is controlled by the entry square-function envelope $Q_{m,h}$; uniform entry scale $\sigma_{m,h}$ gives $Q_{m,h} \le \sqrt{2}\,\sigma_{m,h}$, hence the usual $\sigma_{m,h}(\sqrt{D_m}+\tau_m)$ Wigner/GOE/GUE normalization. A row-variance specialization replaces the vector supremum by $\max_p(\alpha_p^2+2\sum_{q\neq p}\beta_{pq}^2)^{1/2}$, giving sparse and banded entry-profile libraries the explicit scale $(\alpha^2+2\Delta\beta^2)^{1/2}(\sqrt{D_m}+\tau_m)$. At normalized graph scale $\beta \lesssim \Delta^{-1/2}$, the same all-integer and sparse-dimension active-prefix schedules apply with no degree loss. The same row-variance reduction now covers independent random graph masks after adding the scalar Bernstein max-degree failure probability; for inhomogeneous Erdos–Renyi masks this uses expected-degree normalization whenever the expected degree is larger than the menu union logarithm. Random regular, matching, and other dependent bounded-degree masks skip this support exception. A weighted random-mask refinement controls the actual row-variance profile $2\sum_q \eta_{pq}c_{pq}^2$ by scalar Bernstein, so highly inhomogeneous variance-profile masks need not be reduced to their maximum degree; the same arbitrary-growth, all-integer, and sparse-dimension schedules follow when the support Bernstein exponent dominates the active-prefix union logarithm. A bounded-incidence specialization reduces this support condition to the normalized mean row variance ρ/ω, keeping the same 8/3 threshold constants for bounded inhomogeneous entry masks. A predictable-profile variant lets the whole intensity and proxy field be chosen from the core environment, paying only the probability that the conditional Bernstein envelopes exceed their deterministic schedule budgets. A dependent weighted-budget variant

removes support independence entirely once the realized row budgets are controlled outside a summable event. A bounded weighted-degree specialization recovers the normalized $d^{-1/2}$ edge scale for arbitrary dependent weighted-degree-bounded masks with no support exception. For coordinatewise sign-symmetric random tuples, the paper also records a direct transfer from all-plus operator-norm tails to discrepancy tails for the same realization-normalized ensemble, and a Borel–Cantelli active-menu version turns summable all-plus tails into simultaneous eventual signings for finite or countably exhausted sign-symmetric libraries. A matrix-Bernstein specialization turns independent or conditionally independent bounded sign-symmetric arrays with summable variance-envelope failures into simultaneous eventual normalized signings, with explicit all-integer and sparse schedules once $n_m \gtrsim \log^2(eD_m)$, including closed-form constants for the admissible discrepancy margin. Symmetric scalar-multiplier menus therefore have the same closed-form margins under the conditional second-moment profile $\|\sum_i q_i B_i^2\|_{\mathrm{op}} = O(n_m/\log(eD_m))$. Unbounded symmetric sub-Gaussian multipliers are also covered after deterministic cap normalization; the matrix-Laplace tail gives all-integer and sparse margins $K > \sqrt{2\beta(\Gamma+1)}$ and $K > \sqrt{2\beta\Gamma}$, without the bounded-jump correction. A tail-capped companion covers arbitrary symmetric multiplier tails after summable cap failures; in the sub-Weibull case the cap cost is $2\sum_i \exp\{-(L/a_i)^\alpha\}$, while the truncated second-moment budget inherits the same bounded-Bernstein margins $\Theta(\Gamma+1, a, \beta)$ and $\Theta(\Gamma, a, \beta)$. A polynomial moment-proxy specialization covers every tail index $p > 0$: writing $r = p \wedge 2$, the variance budget is controlled by $L^{-r}\|\sum_i a_i^r B_i^2\|_{\mathrm{op}}$ and the cap cost by $\sum_i (a_i/L)^p$. The same conclusions hold when the caps and moment proxies are chosen from the conditioning environment, provided the conditional cap-failure series is summable. A noncommutative-Khintchine square-function certificate gives a variance-profile random route as well: any realization with $\|\sum_i A_i^2\|_{\mathrm{op}} = O(n/\log(eD))$ is Spencer-scale; more precisely, the logarithm can be taken at the realized rank of $\sum_i A_i^2$. Hermitian dilation gives the same certificate for rectangular or non-self-adjoint operators from the two natural profiles $\|\sum_i X_i X_i^*\|_{\mathrm{op}}$ and $\|\sum_i X_i^* X_i\|_{\mathrm{op}}$, with the logarithm at $d_1 + d_2$ or at the realized left-plus-right profile rank. Independent random sign search finds such a rank-sensitive signing after m trials except with an additional 2^{-m} failure probability. A common-sign self-adjoint menu version direct-sums the active square-profile streams and gives one signing for all active models at the scale $\log(eJ_m D_m)$, with a shared sampled sign list losing only 2^{-M_m}. A rank-sensitive common-sign version replaces $J_m D_m$ by the certified total active square-function rank, so low-rank active libraries pay $\log(eR_m)$ instead of the ambient menu dimension. The rectangular common-sign version has the same rank-sensitive upgrade with the total active left-plus-right square-profile

rank in place of the active ambient rectangular dimension. Matrix Bernstein turns this into a probabilistic theorem for independent random contractions with $\|\sum_i \mathbb{E}A_i^2\|_{\mathrm{op}} = O(n/\log(eD))$; Haar-random rotations of fixed or conditionally random spectra are included whenever the average Frobenius-square is $O(D/\log(eD))$. More generally, conditional square-twirl rotations and exact finite-design rotations, including finite irreducible unitary-group rotations and signed-permutation/monomial twirls, obey the same theorem with the same average Frobenius-square profile. A block-design twirl version covers partially randomized bases and fixed orthogonal decompositions by replacing the scalar Frobenius energy with the corresponding block-energy operator norm; a dominated approximate-twirl version also covers approximate scalar or block designs with the aggregate PSD error added to the same profile. A matrix-Freedman variant covers adapted dependent random contractions with predictable square profile $\|\sum_i \mathbb{E}(A_i^2 \mid \mathcal{F}_{i-1})\|_{\mathrm{op}} = O(n/\log(eD))$, with the same randomized sign-search certificate. For Hermitian matrices with conditionally independent centered upper-triangular entries, this profile is checked by the maximum cumulative row variance, and whole-matrix cap tails give sub-Weibull or finite-moment Wigner/GOE/GUE-type variants with one sign per whole matrix. The rectangular variant only asks for the corresponding predictable left and right square profiles to be $O(n/\log(e(d_1+d_2)))$ and has the same active-menu Borel–Cantelli schedule. For centered rectangular matrices with conditionally independent entries, this reduces to the maximum cumulative row and column variance, giving one discrepancy sign per whole random matrix after deterministic operator-norm normalization. A tail-capped version replaces the almost-sure operator-norm bound by summable whole-matrix cap tails, with explicit sub-Weibull and finite-moment cap schedules. Its low-overlap specialization reduces sparse rectangular and bipartite support models to the two weighted occupancy profiles on rows and columns. A realized-profile common-sign menu version direct-sums the active rectangular library, pays only the scale $\log(eJ_m(d_{m,1}+d_{m,2}))$, and removes the active-menu factor from a sampled shared sign search. The dependent overlap-budget active-menu version needs only summable certification of the realized left and right overlap budgets, and a tail-capped dependent version adds sub-Weibull or finite-moment coefficients by paying cap-tail summability plus the capped realized overlap certificate. Scalar Bernstein controls inhomogeneous coordinate supports and matrix Bernstein controls independent random row/column subspaces. Another tail-capped version allows unbounded random low-rank or block coefficients by paying their operator-norm cap exceptions. A cap-selection schedule makes this explicit for sub-Weibull and finite-moment operator tails, including the all-integer $R > \Gamma + 1$ and sparse $R > \Gamma$ cap-tail thresholds. The rank-one specialization covers sampled rectangular dyads from arbitrary frame laws by the two expected weighted

frame operators, including Haar-isotropic dyads, randomized features, and nonuniform sampled atom dictionaries, with the same active-menu schedule for countable dyad libraries. A tail-capped version adds Gaussian, sub-Gaussian, sub-exponential, and heavy-tailed scalar weights by paying only their cap-exception probabilities plus the capped frame-Bernstein term. Its cap-selection schedule gives explicit sub-Weibull and finite-moment scalar-weight criteria for dyad/frame menus with the same all-integer and sparse cap-tail thresholds; in the homogeneous isotropic case this reduces to the scalar frame cap $q^2 \lesssim d_*/\log(eD)$ and the explicit growth thresholds $R > \Gamma + 1,\ c_0 a > \Gamma + 2$ on all integer dimensions, or $R > \Gamma,\ c_0 a > \Gamma + 1$ on sparse stretched-exponential schedules. A coordinate-sampled specialization replaces the Haar frame budget by the largest row or column marginal load, giving the same schedules for iid sampled-entry, random bipartite-edge, and sparse observation matrices. Its predictable version removes independence between samples: conditional marginal-load envelopes cover adaptive sampling, without-replacement exposure, and permutation or matching windows after the same cap-tail accounting. A profile-capped form keeps the full row and column load arrays, so nonuniform coordinate laws and varying amplitude caps pay only their maximum cumulative capped marginal mass. Its finite-menu schedule gives general, all-integer, sparse-dimension, and sampled-certificate thresholds for active nonuniform coordinate libraries. An entry-intensity specialization lets adaptive inhomogeneous sparse matrices be checked directly from their predictable weighted row and column degrees. A closed-form cap schedule then reduces sub-Weibull or finite-moment entry weights to this maximum predictable degree budget. A position-dependent cap refinement charges rare heavy coordinates by $\sum q_{i,r,s}^2 p_{i,r,s}$, rather than by a worst-case scalar cap times the whole expected degree. If the stream is supported in a common random κ-dimensional corner, the same theorem uses $\log(e\kappa)$ and a κ-dimensional Freedman tail, with active menus paying $\sum_h \kappa_{m,h}$. A two-scale square-function certificate keeps one signing for a rank-κ core plus a full ambient residual, and its predictable menu schedule pays the low-rank structural cost $\sum_h \kappa_{m,h}$ together with the residual structural cost $J_m D_m$. A finite-depth multiscale extension handles several low-rank layers plus one ambient residual, with active structural cost $\sum_{h,q} \kappa_{m,h,q} + J_m D_m$. A summable-depth extension permits countably many low-rank layers under a summable $\sum_q \sqrt{a_q}$ budget and uses the corresponding $\sum_{m,h,q}$ structural tail in active menus. A flag-compressible specialization turns any nested initial projection flag into summable compression shells, so multiresolution, wavelet, spectral-window, and hierarchical basis models only need summable shell square profiles. A compression-error version bounds those shell profiles by predictable approximation errors, so common-flag spectral truncation or low-rank approximation tails can be checked without computing

the shell squares. An explicit tail-envelope version covers polynomial approximation tails against stretched-exponential flag ranks and power tails against geometrically growing rank windows, with active menus paying only summable envelope exceptions and sampled-search losses. The corresponding dominated square-twirl corollary makes Haar, finite-design, signed-permutation, and approximate random-basis rotations inside a common random low-rank corner pay only the κ-dimensional profile $O(n/\log(e\kappa))$, and the matching finite-menu dominated-twirl schedule gives simultaneous eventual signings for active libraries at structural cost $\sum_h \kappa_{m,h}$. The predictable dominated-twirl theorem also has a rank-sensitive common-corner version, so online adaptive low-rank random-basis streams and their active-menu schedules pay the same $\sum_h \kappa_{m,h}$ structural count. A finite-menu predictable square-profile schedule gives simultaneous almost-sure signings for active libraries of arbitrary adapted random-matrix streams after replacing D_m by $J_m D_m$, with common sampled-sign batches paying $J_m 2^{-M_m}$. Its predictable dominated-twirl specialization lets the spectrum, block decomposition, and approximate random-basis sampler be chosen from the past at each step; only the one-step PSD twirl-error profile enters the same square-function budget. The corresponding Borel–Cantelli array version yields almost-sure eventual Matrix-Spencer signings, and sampled sign certificates, when the row twirl-budget tails and 2^{-M_m} search losses are summable. Explicit schedule consequences include all integer dimensions once $n(D) \gtrsim \log^2(eD)$, and sparse stretched-exponential dimension schedules under the one-row logarithmic-square threshold. A finite-menu predictable twirl schedule gives simultaneous eventual signings for active families of online random-basis samplers after replacing the one-row structural count D_m by $J_m D_m$; common sampled-sign batches add only $J_m 2^{-M_m}$. Its subspace-localized specialization covers adaptive coordinate supports, online random subspaces, and conditionally sampled localized Wigner/GOE/GUE blocks under the same predictable overlap scale. A finite-menu subspace-localized schedule makes these adaptive support/subspace models simultaneous on active menus, with the same $J_m D_m$ structural count and common sampled-sign penalty $J_m 2^{-M_m}$. A conditional scalar-multiplier specialization also covers arbitrary adversarial matrix directions after independent bounded thinning or amplitudes with conditional second moments $O(1/\log(eD))$, while the predictable specialization allows the directions and amplitudes to be generated adaptively. Its triangular-array form gives the same all-integer and sparse-dimension almost-sure schedules under the half-budget conditional second-moment bound. Finite menus of such adaptive scalar-multiplier streams are simultaneous under the same schedules after replacing D_m by $|\mathcal{H}_m| D_m$; common sampled-sign batches add only $|\mathcal{H}_m| 2^{-M_m}$. An exhaustive countable-menu version controls the active

prefix $h \le J_m$ after replacing D_m by $J_m D_m$, hence every fixed adaptive scalar-multiplier stream is eventually signed. The same centered scalar-multiplier noise can be placed on top of any core class whose signs are already controlled, including algebraic or finite-lift cores, at the cost of an explicit matrix-Bernstein tail under the core signing. A matrix-valued martingale-difference version replaces conditional independence by one-step centering; a scalar variance-domination corollary only assumes $\mathbb{E}(Y_i^2 \mid \mathcal{F}_{i-1}) \preceq \theta_i \rho_i^2 I$ and gives a scalar-Bernstein tail for the random variance budget, while a matrix-profile version allows anisotropic random PSD variance profiles and pays a matrix-Bernstein budget tail. A trace-budget variant gives a dimension-free scalar budget exception when only the total trace of the random variance profile is controlled; the predictable matrix martingale-noise branch now has an almost-sure triangular-array schedule form for all integer dimensions and sparse stretched-exponential dimension subsequences, with the scalar martingale-noise case included. Its active-prefix countable-menu version pays $J_m D_m$ and therefore eventually covers every fixed already-signed core-plus-noise model in a countable library. The sampled localized specialization gives the same Borel–Cantelli schedules for adaptive coordinate-support and conditional Haar-subspace matrix-noise arrays when the predictable weighted occupancy, or the weighted average-rank budget, is $O(n/\log(eD))$. The same argument accepts an active-prefix countable localized-noise library, replacing the variance budget by the corresponding weighted-overlap or Haar average-rank budget for each active model, and also accepts independent trace envelopes, giving explicit low-rank and effective-rank variance-profile tails from $\operatorname{rank}(V_i)\|V_i\|_{\mathrm{op}}$, and a Schatten-envelope corollary interpolates between trace, Frobenius-plus-rank, and operator-plus-rank control. A quantile trace-envelope variant only needs a high-probability bound, or a finite moment, for the total weighted trace budget; it therefore covers heavy-tailed trace-class random covariance and spike profiles outside the Bernstein regime. Its summable version gives almost-sure eventual Spencer-scale signings for triangular arrays whose trace-budget quantiles are at the $n/\log(eD)$ scale; a normalized-moment specialization makes this checkable by summing moments of $S_m/(n_m/\log(eD_m))$, and a termwise moment version reduces this to individual heavy-tail atom moments without dependence assumptions. A termwise-tail version only asks for summable individual truncation failures, and a scale-mixture specialization converts this into an aggregate tail check for deterministic atom weights. A common-scale version removes the atom-count union factor when all atoms share one heavy-tailed multiplier, and finite or countable grouped-scale versions pay one tail per volatility group. An optimized tail-profile form allows exact survival functions and log-Weibull/lognormal tails, while the heterogeneous polynomial and sub-Weibull specializations have optimized

share-cost criteria. Profile-menu aggregation groups volatility components by tail class and optimizes only over class-level trace shares. A finite heterogeneous polynomial-index menu is controlled by its worst tail index once the budget dominates the largest class-count/trace product. A countable weighted heterogeneous version gives the same worst-index budget rule for power-decaying coefficient and trace tails whenever a summable class-share distribution dominates the trace tail. For a finite heterogeneous stretched-exponential profile menu, the corresponding logarithmic budget rule has exponent $\delta = c_0 a^{\beta_0}$ and pays the active class-count exponent w, giving $\delta > p + w + 1$ with the same log-log endpoint convention. A countable weighted stretched-exponential version removes this count penalty under class-index trace decay faster than the summable share weights, giving $\delta > p + 1$. The analogous finite log-Weibull menu has the same $\delta > p + w + 1$ rule with a subpolynomial budget margin $\exp\{a(\Lambda_m + bM_m)^{1/\beta_0}\}$, while a countable weighted log-Weibull version removes the count penalty once the residual class-index log-Weibull series is summable. For a common polynomial tail index this infimum has a closed-form effective class mass, and the resulting polynomial/logarithmic schedule has the rule $\alpha b > (\alpha + 1)p + 1$. If the number of active classes grows like m^w, the finite-menu rule becomes $\alpha b > p + \alpha u + (\alpha + 1)w + 1$. For countably many classes with power-decaying coefficients and trace masses, the rule becomes $\alpha b > p + \alpha u + 1$ once the class-index decay is summable. When all groups share one tail profile, proportional mass shares give a single aggregate criterion. A stretched-exponential schedule gives the concrete rule $ca^\beta > p + 1$, with a log-log endpoint $q - ca^\beta b < -1$, while a log-Weibull schedule gives the subpolynomial rule $ca^\beta > p + 1$. A regularly varying schedule with log corrections gives the matching exponent rule $\alpha a > p + 1$, with endpoint condition $q + \gamma - \alpha b < -1$; the common-index polynomial-tail form also has an explicit effective group-tail mass. A sub-gamma trace-envelope variant removes the bounded-envelope hypothesis and only asks for a Bernstein-type tail for the total weighted envelope; a termwise version verifies this from independent sub-gamma envelopes with summable variance proxies. A gamma-dominated specialization gives explicit chi-square and Wishart-type trace-envelope tails. Gamma-atom specializations allow heterogeneous finite or countable gamma and exponential atom scales in the same weighted trace envelope. A compound-Poisson dominated specialization covers random counts of bounded trace-envelope atoms and bounded spike-count variance profiles. Atomized bounded-envelope specializations give finite or countable Bernoulli/binomial spike and low-rank atom sums an atom-level jump parameter. A mixed-atom calculus combines bounded, gamma, and compound-Poisson trace atoms in the same envelope with one Bernstein parameter. A weighted chi-square specialization covers unequal Gaussian directions and real

Wishart trace envelopes. A weighted exponential specialization covers complex Gaussian directions and complex Wishart trace envelopes. The same calculation with deterministic shifts and residual constants covers noncentral and correlated real or complex Gaussian/Wishart trace envelopes. A scalar specialization allows predictable noise directions with the same matrix-Freedman tail. Localized versions control online coordinate-supported, subspace-localized, or frame-localized martingale noise by realized predictable overlap, conditional predictable overlap, or weighted-overlap operators. Thus random localized, projection, frame, and sparse-support ensembles are covered whenever their weighted empirical support overlap, or weighted frame operator, has norm $O(n/\log(eD))$. For independent inhomogeneous random coordinate supports, scalar Bernstein controls the weighted coordinate occupancies; in the homogeneous case support density $O(1/\log(eD))$ is enough for a high-probability $O(\sqrt{n})$ theorem. The same variance route also has a noncommutative concentration form: independent random subspace projections with $\|\sum_i \rho_i^2 \mathbb{E}P_i\|_{\mathrm{op}} = O(n/\log(eD))$ have the required weighted empirical frame bound with matrix-Bernstein probability; Haar ranks enter through $D^{-1}\sum_i \rho_i^2 r_i$. The same weighted-support theorem allows independent random amplitudes: scalar Bernstein controls $\max_j \sum_i \rho_i^2 \mathbf{1}_{\{j\in S_i\}}$ from $\max_j \sum_i p_{ij}\mathbb{E}\rho_i^2$, while matrix Bernstein controls $\|\sum_i \rho_i^2 P_i\|_{\mathrm{op}}$ from $\|\sum_i \mathbb{E}(\rho_i^2 P_i)\|_{\mathrm{op}}$. The coordinate and subspace Bernstein tails also give an almost-sure triangular-array version whenever the row failure probabilities are summable. Under the half-budget condition these tails are summable along all integer dimensions once $n_D \geq a\log^2(eD)$ with $a > 32/(3C_0)$, and along sparse stretched-exponential dimension schedules once $a > 16/(3C_0)$; the explicit random-amplitude schedule form also covers arbitrary growth by paying the $\log(e+m)$ summability margin. A fixed orthogonal block decomposition has the same random-amplitude support theorem with D in the union bound replaced by the block count B, and its schedules depend on $\log(eB_m)$ in the summability margin. A random-profile version permits conditional inclusion profiles Q_{ij} or $Q_{i\alpha}$, replacing deterministic budgets by their ρ_i^2-weighted expectations and justifying random support-size budgets even when sizes and amplitudes are correlated. Its triangular-array form is variance-sensitive: summability of

$$N_m \exp\{-t_m^2/(2(\sigma_m^2 + t_m/3))\}, \qquad N_m = D_m \text{ or } B_m,$$

gives eventual almost-sure profile signings, and the half-budget case is summable under

$$n_m \geq \frac{16}{3C_0}\log(eD_m)\big(\log(eN_m) + (1+\gamma)\log(e+m)\big).$$

A predictable profile/Freedman version further permits sequentially sampled coordinate or block supports whose conditional inclusion profiles are $\mathcal{F}_{i-1}$-measurable; the same conclusion holds outside the realized

predictable-overlap budget exception, and sampled signs still add only 2^{-M}. Its explicit schedule form uses $N_m = D_m$ or B_m: arbitrary growth needs

$$n_m \geq \frac{16}{3C_0} \log(eD_m)\big(\log(eN_m) + (1+\gamma)\log(e+m)\big),$$

while $N_m \leq (eD_m)^\Gamma$ gives the all-integer threshold $a > 16(\Gamma+1)/(3C_0)$ and the sparse stretched-exponential threshold $a > 16\Gamma/(3C_0)$, up to the summable predictable-budget exceptions. A finite-menu predictable-profile schedule gives simultaneous eventual signings for many online coordinate/block support models after replacing N_m by $L_m = \sum_h N_{m,h}$ and summing the predictable-budget exceptions. An exhaustive countable-menu schedule controls the active prefix $h \leq J_m$ after replacing N_m by $L_m^{(J)} = \sum_{h \leq J_m} N_{m,h}$, hence every fixed member of a countable online support library is eventually signed. A deterministic-envelope specialization has no such exception: if the adaptive conditional inclusion probabilities are bounded by fixed p_{ij} or $q_{i\alpha}$, and the predictable amplitudes are bounded by deterministic caps a_i, the budget is simply $\max_j \sum_i a_i^2 p_{ij}$ or $\max_\alpha \sum_i a_i^2 q_{i\alpha}$. The deterministic-envelope schedule form imposes only this fixed half-budget, so the arbitrary-growth, all-integer, sparse, and sampled-certificate thresholds hold with no budget-exception summability term. For finite menus of deterministic-envelope coordinate/block models, the same simultaneous conclusion holds after replacing the structural count by $L_m = \sum_{h \in \mathcal{H}_m} N_{m,h}$, and common sampled-sign batches pay only $|\mathcal{H}_m| 2^{-M_m}$. For a countable deterministic-envelope library, the exhaustive-prefix version controls $h \leq J_m$ at cost $L_m^{(J)} = \sum_{h \leq J_m} N_{m,h}$, so every fixed envelope model is eventually signed without adding any budget-exception series. Also, M independent sign trials add only 2^{-M} to the finite failure probability, so logarithmically many trials give eventual sampled certificates: $(1+\delta)\log_2(eD)$ trials along all integer dimensions and $(1+\delta)\log_2(e+m)$ trials along sparse schedules. For arbitrary triangular arrays, the same proof gives a direct growth rule:

$$n_m \geq \frac{16}{3C_0} \log(eD_m)\big(\log(eD_m) + (1+\gamma)\log(e+m)\big)$$

makes both the matrix failure and, with $M_m \geq (1+\delta)\log_2(e+m)$, the sampled-search failure summable. The signed-core martingale-noise version has the analogous conditional forms: coordinate-dependent adaptive inclusion probabilities and conditional Haar ranks control the same core signing through a random weighted budget W, with failure term $\mathbb{P}\{W > v\}$, and through the overlap $\|\sum_i \rho_i^2 \mathbb{E}(P_i \mid \mathcal{F}_{i-1})\|_{\mathrm{op}}$, recovering the unweighted deterministic form when $\rho_i = 1$ and W is bounded. Its triangular-array form gives all-integer and sparse stretched-exponential almost-sure schedules under the same $n/\log(eD)$ weighted-overlap or

average-rank scale. For coordinate supports with deterministic, or independent random, inclusion profiles and independent random amplitudes, a scalar Bernstein union bound makes this budget tail explicit; in the homogeneous support-density case the effective overlap is pA, where $A = \sum_i \mathbb{E}\rho_i^2$. The same coordinate argument works for fixed orthogonal block decompositions, replacing the coordinate union by a union over the block count. For fixed-rank Haar subspaces with independent random amplitudes the analogous scalar Bernstein bound has mean $\sum_i (r_i/D)\mathbb{E}\rho_i^2$, with no coordinate-union loss in the budget exception. The same budget-tail statement holds for independent random ranks R_i, replacing r_i/D by R_i/D inside the scalar Bernstein mean and variance proxy. More generally, any conditionally isotropic projection law has the same bound after replacing R_i/D by the random intensity α_i. The equality can be weakened to the spectral domination $\mathbb{E}(P_i \mid \mathcal{F}_{i-1}) \preceq \alpha_i I$, which covers leverage-bounded and approximate-design projection laws. The resulting random-amplitude localized menu has a Borel–Cantelli array theorem: if the scalar Bernstein budget exceptions and the matrix-Freedman row tails are summable, the same core signing is eventually $O(\sqrt{n})$, with explicit all-integer and sparse dimension schedules under half-budget $n/\log(eD)$ profiles. Its countable-menu form controls active prefixes by replacing N_m with $\sum_{h\le J_m} N_{m,h}$, hence every fixed localized random-amplitude model is eventually covered. In the homogeneous subcases this becomes an effective-energy criterion: the budget scale is βH and the scalar budget exception is summable once $H = \sum_i \mathbb{E}\rho_i^2$ dominates $\log(eD)$ by the appropriate union-count margin. If the signed cores share one active-prefix signing, a common-sign countable form preserves that same signing through all random-amplitude localized perturbations with the same $\sum_{h\le J_m} N_{m,h}$ payment. A finite-lift common-menu corollary supplies this shared core signing deterministically whenever the active-prefix direct-sum lift entropy is $O(n)$. Thus finite-lift core libraries absorb localized random-amplitude noise under the same active-prefix union-count and homogeneous energy schedules. The homogeneous criterion also has a countable active-prefix version with $\Lambda_m^{(J)} = \sum_{h\le J_m} N_{m,h}$, giving all-integer coverage for $R > 8(\Gamma+1)/3$ when $\Lambda_D^{(J)} \le (eD)^\Gamma$, and sparse coverage for $R > 8\Gamma/3$. The same energy rule has a common-sign version, so one active-prefix core signing remains valid for every homogeneous random-amplitude perturbation in the prefix. The coordinate and Haar branches can also be mixed sample-by-sample: if $M_i \in \{0,1\}$ selects the coordinate model and $\beta_{ij} = M_i Q_{ij} + (1-M_i)R_i/D$, then the same tail holds with $\max_j \sum_i \mathbb{E}(\beta_{ij}\rho_i^2)$ and $\max_j \sum_i \mathbb{E}(\beta_{ij}^2\rho_i^2)$.

2. Discrepancy inputs

We use two discrepancy inputs. The first is the rectangular vector Spencer theorem. All logarithms are natural.

Theorem 2.1 (Vector Spencer)**.** *There is a universal constant C_S such that, for every $x_1, \dots, x_n \in \mathbb{C}^m$ with $\|x_i\|_\infty \le 1$, there are signs $\varepsilon \in \{-1,1\}^n$ satisfying*

$$\left\| \sum_{i=1}^n \varepsilon_i x_i \right\|_\infty \le C_S \sqrt{n \log\left(e + \frac{m}{n}\right)}.$$

The complex case follows from the real theorem by applying it to real and imaginary parts and absorbing the harmless factor of two into the constant.

The second input is the block-diagonal Matrix Spencer theorem. For block size h it was first established by Levy, Ramadas and Rothvoss [14], with the restriction $h \le n$, and the restriction was later removed by Dadush, Jiang and Reis [11]. We state it in the standard form with total matrix dimension R and maximum block size h.

Theorem 2.2 (Block-diagonal Matrix Spencer)**.** *There is a universal constant C_{BD} such that the following holds. Let $\mathbb{C}^R = \bigoplus_\alpha \mathbb{C}^{r_\alpha}$ be a fixed block decomposition with $h = \max_\alpha r_\alpha$. Let $Y_1, \dots, Y_n$ be self-adjoint matrices that are block diagonal with respect to this decomposition and satisfy $\|Y_i\|_{\mathrm{op}} \le 1$. Then there are signs $\varepsilon \in \{-1,1\}^n$ such that*

$$\left\| \sum_{i=1}^n \varepsilon_i Y_i \right\|_{\mathrm{op}} \le C_{BD} \sqrt{n \log\left(e + \frac{hR}{n}\right)}.$$

Equivalently, for $\mathcal{B} = \bigoplus_\alpha \mathrm{M}_{r_\alpha}(\mathbb{C})$,

$$\left\| \sum_{i=1}^n \varepsilon_i Y_i \right\|_{\mathcal{B}} \le C_{BD} \sqrt{n \log\left(e + \frac{\mathsf{E}(\mathcal{B})}{n}\right)}.$$

Remark 2.3. Dadush, Jiang and Reis prove this as an application of their partial-coloring framework; in their notation the logarithmic factor is of the form $\log(hm/n)$ for $m \times m$ block-diagonal matrices with block size at most h [11]. Their m is the total matrix dimension, which is our R, so $\log(hm/n) = \log(hR/n)$. The displayed version inserts $e+$ to cover all regimes.

Remark 2.4 (Realification). Theorem 2.2 is stated for real symmetric matrices. One passes to $\mathbb{C}$ through the realification

$$\iota(A) = \begin{pmatrix} \operatorname{Re} A & -\operatorname{Im} A \\ \operatorname{Im} A & \operatorname{Re} A \end{pmatrix},$$

which is $\mathbb{R}$-linear, sends self-adjoint matrices to real symmetric ones, preserves the operator norm, and respects the block decomposition. It doubles the block size and the total dimension, so the entropy is multiplied by four; this is absorbed into the universal constant.

For non-self-adjoint block-diagonal matrices we use Hermitian dilation. For a matrix Y, set

$$\mathscr{H}(Y) = \begin{pmatrix} 0 & Y \\ Y^* & 0 \end{pmatrix}.$$

Then $\mathscr{H}(Y)$ is self-adjoint and $\|\mathscr{H}(Y)\|_{\mathrm{op}} = \|Y\|_{\mathrm{op}}$. Applying Theorem 2.2 to the dilations doubles all block sizes and changes only the universal constant.

Corollary 2.5 (Non-self-adjoint block diagonal signing)**.** *Let $Y_1, \dots, Y_n \in \mathcal{B}$ be arbitrary complex block-diagonal matrices with $\|Y_i\|_{\mathcal{B}} \le 1$. Then*

$$\min_{\varepsilon \in \{-1,1\}^n} \left\| \sum_{i=1}^n \varepsilon_i Y_i \right\|_{\mathcal{B}} \le C \sqrt{n \log\left(e + \frac{\mathsf{E}(\mathcal{B})}{n}\right)}.$$

3. Finite block and sampled multiplier lifts

The lift formulation is tuple-level: for a given input collection it is enough to choose bounded lifts of the individual matrices and a contraction on their span.

Definition 3.1 (Finite block lift)**.** Let $A_1, \dots, A_n \in \mathrm{M}_D(\mathbb{C})$ and $\mathcal{B} = \bigoplus_{\alpha=1}^m \mathrm{M}_{r_\alpha}(\mathbb{C})$. We say that the tuple admits a (K, Λ)-lift to $\mathcal{B}$ if there exist elements $Y_i \in \mathcal{B}$ and a fixed linear map

$$\Pi : \operatorname{span}\{Y_1, \dots, Y_n\} \to \mathrm{M}_D(\mathbb{C})$$

such that

$$\Pi(Y_i) = A_i, \qquad \|\Pi(Y)\|_{\mathrm{op}} \le \Lambda \|Y\|_{\mathcal{B}} \quad (Y \in \operatorname{span}\{Y_1, \dots, Y_n\}),$$

and $\|Y_i\|_{\mathcal{B}} \le K$ for all i.

The map Π is usually a compression, a quotient map, or a fixed bounded linear image. Contractive lifts correspond to $\Lambda = 1$.

Proof of Theorem 1.3. If $K = 0$ then all $Y_i = 0$, so all $A_i = 0$ and the bound is trivial; assume $K > 0$. First assume that the Y_i are self-adjoint. Apply Theorem 2.2 to Y_i/K. There are signs for which

$$\left\| \sum_i \varepsilon_i Y_i \right\|_{\mathcal{B}} \le CK \sqrt{n \log\left(e + \frac{\mathsf{E}(\mathcal{B})}{n}\right)}.$$

Since Π is linear and has norm at most Λ on the relevant span,

$$\left\| \sum_i \varepsilon_i A_i \right\|_{\mathrm{op}} = \left\| \Pi\left(\sum_i \varepsilon_i Y_i \right) \right\|_{\mathrm{op}} \le \Lambda \left\| \sum_i \varepsilon_i Y_i \right\|_{\mathcal{B}}.$$

This proves the self-adjoint case. In the arbitrary complex case, replace each block $Y_{i,\alpha}$ by its Hermitian dilation. The block sizes and total reduced dimension are both doubled, so the entropy changes by a factor of four, and the identity $\|\sum_i \varepsilon_i \mathscr{H}(Y_i)\| = \|\sum_i \varepsilon_i Y_i\|$ transfers the bound. Since Theorem 2.2 is over $\mathbb{R}$, one further applies the realification ι of Remark 2.4 before invoking it, which again doubles the block size and the total dimension. The general complex case thus uses both Hermitian dilation and realification; each doubles the dimension, so the entropy grows by a factor of at most 16, absorbed into the universal constant C. □

Corollary 3.2 (Common signs for finite menus of finite-lift tuples)**.** *Let $\mathcal{H}$ be finite and nonempty. For each $h \in \mathcal{H}$, let $A_1^{(h)}, \dots, A_n^{(h)} \in \mathrm{M}_{D_h}(\mathbb{C})$ be contractions which admit a (K_h, Λ_h)-lift to*

$$\mathcal{B}_h = \bigoplus_{\alpha=1}^{m_h} \mathrm{M}_{r_{h,\alpha}}(\mathbb{C}).$$

Put

$$M = \max_{h\in\mathcal{H}} \Lambda_h K_h, \qquad h_* = \max_{h,\alpha} r_{h,\alpha},$$

$$R_* = \sum_{h\in\mathcal{H}} \sum_{\alpha=1}^{m_h} r_{h,\alpha}, \quad E_* = h_* R_*.$$

Then there is one sign vector $\varepsilon \in \{-1,1\}^n$ such that

$$\max_{h\in\mathcal{H}} \left\| \sum_{i=1}^n \varepsilon_i A_i^{(h)} \right\|_{\mathrm{op}} \le CM \sqrt{n \log\left(e + \frac{E_*}{n}\right)}.$$

If all tuples and their chosen lifts are self-adjoint, the same bound holds for self-adjoint discrepancy. In particular, if $E_* \le C_E n$ and $M \le M_0$, then the whole finite menu has one $O_{C_E, M_0}(\sqrt{n})$ signing.

Consequently, in a triangular-array countable active-prefix library, if each row m and active prefix $h \le J_m$ has contraction cores $B_{m,1}^{(h)}, \dots, B_{m,n_m}^{(h)}$ admitting such finite lifts with $M_m^{(J)} \le M_0$ and $E_m^{(J)} \le C_E n_m$, then the common-core signing hypotheses in Corollaries 7.304 and 7.305 hold deterministically, with no exceptional core event.

Proof. For $K_h > 0$, choose lifts $Y_i^{(h)} \in \mathcal{B}_h$ and set $Z_i^{(h)} = Y_i^{(h)}/K_h$. If $K_h = 0$, then $A_i^{(h)} = 0$ for all i, and we set $Z_i^{(h)} = 0$. Form the direct-sum algebra

$$\mathcal{B}_* = \bigoplus_{h\in\mathcal{H}} \mathcal{B}_h$$

and $Z_i = (Z_i^{(h)})_{h\in\mathcal{H}} \in \mathcal{B}_*$. Then $\|Z_i\|_{\mathcal{B}_*} \le 1$, and $\mathsf{E}(\mathcal{B}_*) = h_* R_* = E_*$. Define, on the span of the Z_i, the map whose h-component is

$$\Theta(W)^{(h)} = \begin{cases} K_h \Pi_h(W^{(h)}), & K_h > 0, \\ 0, & K_h = 0, \end{cases} \qquad \Theta(W) \in \bigoplus_{h\in\mathcal{H}} \mathrm{M}_{D_h}(\mathbb{C}).$$

For $W \in \operatorname{span}\{Z_1, \dots, Z_n\}$,

$$\|\Theta(W)\|_{\mathrm{op}} \le \max_{h\in\mathcal{H}} \Lambda_h K_h \, \|W\|_{\mathcal{B}_*} = M \, \|W\|_{\mathcal{B}_*},$$

and $\Theta(Z_i) = \bigoplus_{h\in\mathcal{H}} A_i^{(h)}$. Applying Theorem 1.3 to the block-diagonal target tuple $\bigoplus_h A_i^{(h)}$ gives the displayed estimate, because the operator norm in the target direct sum is the maximum over h. The self-adjoint case uses self-adjoint direct sums and the self-adjoint part of Theorem 1.3. The final assertion applies the finite-menu statement row by row to the deterministic active prefix. □

3.1. **Sampled multiplier lifts.** Toeplitz arguments naturally produce continuous multiplier lifts rather than finite block lifts at the first step. The following sampling principle converts them into finite block lifts.

Definition 3.3 (Norming set)**.** Let Ω be a compact space and let $\mathcal{F} \subset C(\Omega; \mathrm{M}_b(\mathbb{C}))$ be a finite-dimensional linear space. A finite set $Q \subset \Omega$ is L_Q-norming for $\mathcal{F}$ if

$$\|F\|_{L^\infty(\Omega;\mathrm{M}_b)} \le L_Q \max_{\omega\in Q} \|F(\omega)\|_{\mathrm{op}} \qquad (F \in \mathcal{F}).$$

Definition 3.4 (Sampled multiplier lift)**.** A tuple $A_1, \dots, A_n \in \mathrm{M}_D(\mathbb{C})$ admits a (K, Λ, L_Q, Q, b) sampled multiplier lift if there are functions $F_i \in \mathcal{F} \subset C(\Omega; \mathrm{M}_b(\mathbb{C}))$ and a fixed linear map

$$\Pi : \operatorname{span}\{F_1, \dots, F_n\} \to \mathrm{M}_D(\mathbb{C})$$

such that

$$\Pi(F_i) = A_i, \qquad \|\Pi(F)\|_{\mathrm{op}} \le \Lambda \|F\|_{L^\infty(\Omega;\mathrm{M}_b)},$$

$\|F_i\|_{L^\infty} \le K$, and Q is L_Q-norming for $\operatorname{span}\{F_1, \dots, F_n\}$.

Theorem 3.5 (Sampled multiplier Matrix Spencer)**.** *If $A_1, \dots, A_n$ admit a (K, Λ, L_Q, Q, b) sampled multiplier lift, then*

$$\operatorname{disc}_{\mathrm{op}}(A_1, \dots, A_n) \le CK\Lambda L_Q \sqrt{n \log\left(e + \frac{b^2|Q|}{n}\right)}.$$

For $b = 1$ this is the vector-Spencer scale $CK\Lambda L_Q\sqrt{n\log(e + |Q|/n)}$.

Proof. Evaluate each F_i on Q and form

$$Y_i = \bigoplus_{\omega\in Q} F_i(\omega) \in \bigoplus_{\omega\in Q} \mathrm{M}_b(\mathbb{C}).$$

This finite block algebra has $h = b$, $R = b|Q|$, and entropy $b^2|Q|$. By Corollary 2.5, there are signs such that

$$\max_{\omega\in Q} \left\| \sum_i \varepsilon_i F_i(\omega) \right\|_{\mathrm{op}} \le CK \sqrt{n \log\left(e + \frac{b^2|Q|}{n}\right)}.$$

Since Q is L_Q-norming,

$$\left\| \sum_i \varepsilon_i F_i \right\|_{L^\infty} \le CKL_Q \sqrt{n \log\left(e + \frac{b^2|Q|}{n}\right)}.$$

Applying Π gives the result. □

3.2. **A grid lemma for trigonometric polynomial lifts.** We shall use the following elementary norming lemma. It is stated for matrix-valued trigonometric polynomials because this is the form needed for block Toeplitz matrices.

Lemma 3.6 (Grid norming)**.** *Fix $r \geq 1$. Let $P : \mathbb{T}^r \to \mathrm{M}_b(\mathbb{C})$ be a matrix-valued trigonometric polynomial with coordinate degrees at most $N_1, \dots, N_r$. For each j, let G_j be a uniform grid in $\mathbb{T}$ of size at least $8rN_j$ when $N_j \geq 1$, and of size 1 when $N_j = 0$. Put $G = G_1 \times \cdots \times G_r$. Then*

$$\|P\|_{L^\infty(\mathbb{T}^r;\mathrm{M}_b)} \leq 2 \max_{\theta \in G} \|P(\theta)\|_{\mathrm{op}}.$$

In particular, $|G| = O_r(\prod_{j=1}^r (1 + N_j))$.

Proof. The Banach-valued Bernstein inequality gives $\|\partial_j P\|_\infty \leq N_j \|P\|_\infty$ for each coordinate. One way to see this is to apply the scalar Bernstein inequality to $\varphi(P(\cdot))$ for norm-one linear functionals φ on M_b. Choose θ^* with $\|P(\theta^*)\| \geq \|P\|_\infty - \delta$, and choose $\theta \in G$ with $|\theta_j - \theta_j^*| \leq \pi/|G_j|$ for every j. Then

$$\|P(\theta^*) - P(\theta)\| \leq \sum_{j=1}^r \frac{\pi}{|G_j|} N_j \|P\|_\infty \leq \frac{\pi}{8} \|P\|_\infty.$$

Letting $\delta \downarrow 0$ gives $\|P\|_\infty \leq (1 - \pi/8)^{-1} \max_G \|P\| \leq 2 \max_G \|P\|$. □

4. Finite-dimensional C*-algebras and group convolution

The cleanest applications are matrices that already lie in a common finite-dimensional C*-algebra.

Theorem 4.1 (Common finite-dimensional C*-algebra)**.** *Let $\mathcal{A} \subset \mathrm{M}_D(\mathbb{C})$ be a finite-dimensional C*-algebra. On its support, and omitting any zero summand, write its representation after unitary conjugacy as*

$$\mathcal{A} \simeq \bigoplus_{\alpha=1}^m (\mathrm{M}_{r_\alpha}(\mathbb{C}) \otimes I_{m_\alpha}).$$

Let $h = \max_\alpha r_\alpha$ and $R = \sum_\alpha r_\alpha$. If $A_1, \dots, A_n \in \mathcal{A}$ satisfy $\|A_i\|_{\mathrm{op}} \leq 1$, then

$$\mathrm{disc}_{\mathrm{op}}(A_1, \dots, A_n) \leq C \sqrt{n \log\left(e + \frac{hR}{n}\right)}.$$

If the matrices are self-adjoint, this is a self-adjoint discrepancy bound.

Proof. An element of $\mathcal{A}$ is unitarily equivalent to $0 \oplus \bigoplus_\alpha (B_\alpha \otimes I_{m_\alpha})$, with the initial zero summand absent in the unital-on-$\mathbb{C}^D$ case. Delete the zero summand and the multiplicity identities to obtain the reduced algebra $\mathcal{B} = \bigoplus_\alpha \mathrm{M}_{r_\alpha}(\mathbb{C})$. The deletion map and the inverse map $\bigoplus B_\alpha \mapsto \bigoplus (B_\alpha \otimes I_{m_\alpha})$ are isometric for the operator norm. Theorem 1.3 applies with $K = \Lambda = 1$. □

Theorem 4.2 (Akbas–Sra algebraic Matrix Spencer)**.** *For every $C_0 \geq 1$ there is a constant $C(C_0)$ with the following property. Let $\mathcal{A}$ be a finite-dimensional C*-algebra with $\dim_{\mathbb{C}} \mathcal{A} \leq C_0 n$. If $A_1, \dots, A_n \in \mathcal{A}$ are contractions, then*

$$\operatorname{disc}_{\mathrm{op}}(A_1, \dots, A_n) \leq C(C_0)\sqrt{n}.$$

The self-adjoint version holds for self-adjoint tuples.

Proof. This is the algebraic Matrix Spencer theorem of Akbas and Sra [2, Thm. 1.3]. Their proof reduces a finite-dimensional C*-algebra by Wedderburn decomposition to common block-diagonal matrices and uses a multi-scale partial-coloring argument whose complexity is the intrinsic square-sum $\sum_\alpha r_\alpha^2 = \dim_{\mathbb{C}} \mathcal{A}$, rather than the cruder $(\max_\alpha r_\alpha) \sum_\alpha r_\alpha$ used in Theorem 4.1. □

Theorem 4.3 (Akbas–Sra low-Frobenius perturbation)**.** *For every $C_0 \geq 1$ there is $C(C_0)$ with the following property. Let*

$$X_i = B_i + L_i, \qquad i = 1, \dots, n,$$

be self-adjoint matrices. Suppose $B_i \in \mathcal{A}$, where $\mathcal{A}$ is a finite-dimensional C-algebra with $\dim_{\mathbb{C}} \mathcal{A} \leq C_0 n$, and suppose*

$$\|B_i\|_{\mathrm{op}} \leq 1, \qquad \|L_i\|_{\mathrm{op}} \leq 1, \qquad \|L_i\|_{\mathrm{F}}^2 \leq \frac{n}{\log^3(en)}$$

for all i. Then there are signs $\varepsilon_i \in \{-1, 1\}$ such that

$$\left\| \sum_{i=1}^n \varepsilon_i X_i \right\|_{\mathrm{op}} \leq C(C_0)\sqrt{n}.$$

Proof. This is [2, Thm. 1.5]. It combines their algebraic partial-coloring body with the Bansal–Jiang–Meka low-Frobenius/rank-constrained Matrix Spencer input. □

Corollary 4.4 (Low-rank perturbations of algebraic cores)**.** *For every $C_0 \geq 1$ there is $C(C_0)$ with the following property. Let*

$$X_i = B_i + L_i, \qquad i = 1, \dots, n,$$

be self-adjoint matrices. Suppose $B_i \in \mathcal{A}$, where $\mathcal{A}$ is a finite-dimensional C-algebra with $\dim_{\mathbb{C}} \mathcal{A} \leq C_0 n$, and suppose*

$$\|B_i\|_{\mathrm{op}} \leq 1, \qquad \|L_i\|_{\mathrm{op}} \leq 1, \qquad \operatorname{rank}(L_i) \leq \frac{n}{\log^3(en)}$$

for all i. Then there are signs $\varepsilon_i \in \{-1, 1\}$ such that

$$\left\| \sum_{i=1}^n \varepsilon_i X_i \right\|_{\mathrm{op}} \leq C(C_0)\sqrt{n}.$$

Proof. Since L_i is self-adjoint,

$$\|L_i\|_{\mathrm{F}}^2 = \sum_j s_j(L_i)^2 \leq \operatorname{rank}(L_i)\, \|L_i\|_{\mathrm{op}}^2 \leq \frac{n}{\log^3(en)}.$$

The result is therefore Theorem 4.3. □

Corollary 4.5 (Simultaneous diagonalization)**.** *If $A_1, \dots, A_n$ are normal matrices diagonalized by a common unitary and $\|A_i\|_{\mathrm{op}} \le 1$, then*

$$\mathrm{disc}_{\mathrm{op}}(A_1, \dots, A_n) \le C\sqrt{n \log\left(e + \frac{D}{n}\right)}.$$

Thus the conjectured $O(\sqrt{n})$ bound holds when $D \le n$.

Corollary 4.6 (Common bounded-block diagonalization)**.** *Suppose all A_i are simultaneously unitarily block diagonal with common block sizes at most b. Let D be the ambient dimension. Then*

$$\mathrm{disc}_{\mathrm{op}}(A_1, \dots, A_n) \le C\sqrt{n \log\left(e + \frac{bD}{n}\right)}.$$

For fixed b and $D \le n$, this is $O_b(\sqrt{n})$.

Corollary 4.7 (Commutative association schemes)**.** *Let $\mathcal{A} \subset \mathrm{M}_D(\mathbb{C})$ be a commutative self-adjoint matrix algebra with s minimal spectral projections. If $A_1, \dots, A_n \in \mathcal{A}$ are contractions, then*

$$\mathrm{disc}_{\mathrm{op}}(A_1, \dots, A_n) \le C\sqrt{n \log\left(e + \frac{s}{n}\right)}.$$

In particular, any family contained in the Bose–Mesner algebra of a symmetric association scheme with s primitive idempotents satisfies the Matrix Spencer bound whenever $s = O(n)$, independently of the number of vertices; a d-class symmetric scheme contributes $s = d + 1$ primitive idempotents.

4.1. **Finite group convolution.** Let G be a finite group. A left convolution operator on $\ell^2(G)$ has the form

$$L_f \xi(x) = \sum_{y \in G} f(xy^{-1})\xi(y).$$

By the finite Fourier transform,

$$L_f \simeq \bigoplus_{\pi \in \widehat{G}} \left(\widehat{f}(\pi) \otimes I_{d_\pi}\right), \qquad d_\pi = \dim \pi,$$

and $\|L_f\|_{\mathrm{op}} = \max_{\pi \in \widehat{G}} \left\|\widehat{f}(\pi)\right\|_{\mathrm{op}}$. Thus the reduced convolution algebra is $\bigoplus_{\pi \in \widehat{G}} \mathrm{M}_{d_\pi}(\mathbb{C})$.

Theorem 4.8 (Finite group convolution)**.** *Let G be finite, let $q(G) = \max_{\pi \in \widehat{G}} d_\pi$, and put $S(G) = \sum_{\pi \in \widehat{G}} d_\pi$. If $L_{f_1}, \dots, L_{f_n}$ are left convolution operators on $\ell^2(G)$ with $\|L_{f_i}\|_{\mathrm{op}} \le 1$, then*

$$\mathrm{disc}_{\mathrm{op}}(L_{f_1}, \dots, L_{f_n}) \le C\sqrt{n \log\left(e + \frac{q(G)S(G)}{n}\right)}.$$

Consequently, for finite abelian groups,

$$\mathrm{disc}_{\mathrm{op}}(L_{f_1},\dots,L_{f_n}) \le C\sqrt{n\log\left(e+\frac{|G|}{n}\right)}.$$

For any family of groups with $q(G)\le q$ fixed, arbitrary convolution kernels satisfy $O_q(\sqrt{n})$ whenever $|G|\le n$.

Proof. Apply Theorem 4.1 to the group convolution algebra. In the abelian case all irreducible representations have dimension one and $S(G)=|G|$. In general $S(G)\le|G|$, and therefore $q(G)S(G)\le q|G|$ on bounded-irrep families. □

Corollary 4.9 (Group convolution with no irrep bound)**.** *For every $C_0\ge 1$ there is $C(C_0)$ such that the following holds. Let G be a finite group with $|G|\le C_0 n$. If $L_{f_1},\dots,L_{f_n}$ are arbitrary left convolution contractions on $\ell^2(G)$, then*

$$\mathrm{disc}_{\mathrm{op}}(L_{f_1},\dots,L_{f_n}) \le C(C_0)\sqrt{n}.$$

Proof. The left convolution algebra is finite-dimensional and, by the Fourier decomposition above, is isomorphic to $\bigoplus_{\pi\in\widehat{G}}\mathrm{M}_{d_\pi}(\mathbb{C})$. Its intrinsic dimension is

$$\sum_{\pi\in\widehat{G}} d_\pi^2 = |G|.$$

Apply Theorem 4.2. □

Corollary 4.10 (Group convolution with low-rank noise)**.** *For every $C_0\ge 1$ there is $C(C_0)$ such that the following holds. Let G be a finite group with $|G|\le C_0 n$, and let*

$$X_i = L_{f_i} + L_i, \qquad i=1,\dots,n,$$

be self-adjoint operators on $\ell^2(G)$. Suppose that each L_{f_i} is a self-adjoint left convolution contraction and that

$$\|L_i\|_{\mathrm{op}}\le 1, \qquad \|L_i\|_{\mathrm{F}}^2 \le \frac{n}{\log^3(en)} \qquad (1\le i\le n).$$

Then

$$\mathrm{disc}_{\mathrm{op}}(X_1,\dots,X_n) \le C(C_0)\sqrt{n}.$$

In particular the same conclusion holds if $\mathrm{rank}(L_i)\le n/\log^3(en)$ and $\|L_i\|_{\mathrm{op}}\le 1$.

Proof. The self-adjoint left convolution operators form the self-adjoint part of the finite-dimensional convolution C*-algebra $\bigoplus_{\pi\in\widehat{G}}\mathrm{M}_{d_\pi}(\mathbb{C})$, whose intrinsic dimension is $\sum_{\pi\in\widehat{G}} d_\pi^2 = |G|\le C_0 n$. Apply Theorem 4.3 with $B_i = L_{f_i}$. The rank specialization follows from

$$\|L_i\|_{\mathrm{F}}^2 \le \mathrm{rank}(L_i)\|L_i\|_{\mathrm{op}}^2 \le \frac{n}{\log^3(en)}.$$

□

Remark 4.11. Theorem 4.8 signs arbitrary convolution kernels. This is different from the finite-group regular-representation theorem of Bandeira and Bölcskei [3], which signs the specific list of left-regular representation matrices indexed by group elements with a universal $C\sqrt{|G|}$ bound for every finite group. The two results are complementary: Theorem 4.8 is the elementary finite-lift estimate and handles arbitrary kernels with an explicit $q(G)S(G)$ entropy parameter; the Akbas–Sra corollary above is stronger in the square group-algebra regime $|G| = O(n)$, because it uses the intrinsic dimension $|G|$ and does not require the irreducible dimensions to be controlled. Corollary 4.10 adds the perturbative version in the same square regime: arbitrary self-adjoint low-Frobenius, in particular low-rank, noise can be placed on top of the convolution cores.

5. Toeplitz, Hankel, and multilevel Toeplitz classes

We now recover and extend the Toeplitz and Hankel endpoint. The full-bandwidth results, at one level and at any fixed level, use bounded finite Toeplitz extension; at higher level the extension constant depends on the level. The banded result uses an elementary finite-section estimate, is dimension-free in its constant, and is included with full details.

5.1. **Scalar Toeplitz matrices.** For a finite sequence $(a_k)_{-(d-1)\le k\le d-1}$, write

$$T_d(a) = (a_{r-s})_{0\le r,s<d}.$$

If $\varphi \in L^\infty(\mathbb{T})$ has Fourier coefficients $\widehat{\varphi}(k) = a_k$ for $|k| < d$, then φ induces $T_d(a)$.

Theorem 5.1 (Finite Toeplitz extension)**.** *There is an absolute constant K_T such that every complex $d \times d$ Toeplitz matrix $A = T_d(a)$ has an inducer $\varphi_A \in L^\infty(\mathbb{T})$ satisfying*

$$\widehat{\varphi}_A(k) = a_k \quad (|k| < d), \qquad \|\varphi_A\|_{L^\infty} \le K_T \|A\|_{\mathrm{op}}.$$

One may take $K_T \le 3$.

This is the finite Toeplitz extension theorem of Nikolskaya and Farforovskaya [21]; Courtney and Sarason record the bound and note that the same reasoning applies in the non-self-adjoint case [9].

We also use a standard de la Vallée Poussin kernel V_N satisfying

$$\widehat{V}_N(k) = 1 \quad (|k| \le N), \qquad \widehat{V}_N(k) = 0 \quad (|k| > 2N), \qquad \|V_N\|_{L^1} \le C_V.$$

Theorem 5.2 (Arbitrary scalar Toeplitz signing)**.** *There is a universal constant C such that, for every collection of arbitrary complex Toeplitz matrices $A_1, \dots, A_n \in \mathrm{M}_d(\mathbb{C})$ with $\|A_i\|_{\mathrm{op}} \le 1$,*

$$\mathrm{disc}_{\mathrm{op}}(A_1, \dots, A_n) \le C\sqrt{n \log\left(e + \frac{d}{n}\right)}.$$

In particular, self-adjoint Toeplitz matrices satisfy Matrix Spencer in the regime $d \le n$.

Proof. For each i, choose an inducer φ_i from Theorem 5.1. Put $p_i = \varphi_i * V_{d-1}$. Then p_i is a trigonometric polynomial of degree at most $2d-2$, its Fourier coefficients agree with those of φ_i for $|k| < d$, and

$$\|p_i\|_\infty \le C_V K_T.$$

Hence $T_d(p_i) = A_i$.

Choose a uniform grid $Q \subset \mathbb{T}$ of size $O(d)$ that is 2-norming for trigonometric polynomials of degree at most $2d$ by Lemma 3.6. Apply Theorem 3.5 with $b = 1$, $K = C_V K_T$, and $\Lambda = 1$. There are signs for which the signed polynomial $P = \sum_i \varepsilon_i p_i$ satisfies

$$\|P\|_\infty \le C\sqrt{n \log\left(e + \frac{d}{n}\right)}.$$

Toeplitz compression is contractive from $L^\infty(\mathbb{T})$ to operator norm, so

$$\left\|\sum_i \varepsilon_i A_i\right\|_{\mathrm{op}} = \|T_d(P)\|_{\mathrm{op}} \le \|P\|_\infty .$$

□

Remark 5.3 (Circulant-lift interpretation). If the grid size is larger than twice the degree of the p_i, then the vector $(p_i(\theta))_{\theta \in Q}$ defines a circulant matrix whose upper-left $d \times d$ block is A_i. Thus the preceding proof is a bounded circulant-lift proof.

5.2. **Scalar Hankel matrices.** Let J_d be the reversal matrix, $(J_d x)_r = x_{d-1-r}$. If H is Hankel, then HJ_d is Toeplitz and $\|HJ_d\| = \|H\|$.

Corollary 5.4 (Arbitrary scalar Hankel signing)**.** *For arbitrary complex Hankel matrices $H_1, \dots, H_n \in \mathrm{M}_d(\mathbb{C})$ with $\|H_i\|_{\mathrm{op}} \le 1$,*

$$\mathrm{disc}_{\mathrm{op}}(H_1, \dots, H_n) \le C\sqrt{n \log\left(e + \frac{d}{n}\right)}.$$

Thus self-adjoint Hankel matrices satisfy Matrix Spencer when $d \le n$.

Proof. Apply Theorem 5.2 to the Toeplitz matrices $H_i J_d$ and observe that

$$\left\|\sum_i \varepsilon_i H_i\right\| = \left\|\left(\sum_i \varepsilon_i H_i J_d\right) J_d\right\| = \left\|\sum_i \varepsilon_i H_i J_d\right\|.$$

□

5.3. **Banded Toeplitz matrices.** For bandwidth $w \ll d$, one needs only $O(w)$ Fourier samples. The following finite-section lemma replaces the bounded extension theorem in this case; for the spectral theory of banded Toeplitz matrices in general see Böttcher and Grudsky [5].

Lemma 5.5 (Finite-section lower bound for banded symbols)**.** *Let $p(e^{it}) = \sum_{|k|\le w} a_k e^{ikt}$ and let $T_d(p) = (a_{r-s})_{0\le r,s<d}$. If $d > 2w$, then*

$$\|p\|_\infty \le \left(1 - \frac{2w}{d}\right)^{-1/2} \|T_d(p)\|_{\mathrm{op}}\,.$$

In particular, if $d \ge 4w$, then $\|p\|_\infty \le \sqrt{2}\,\|T_d(p)\|$.

Proof. Choose t_0 such that $|p(e^{-it_0})| = \|p\|_\infty$, and set $x_s = d^{-1/2}e^{ist_0}$ for $0 \le s < d$. Then $\|x\|_2 = 1$. For every interior row $w \le r \le d-1-w$,

$$(T_d(p)x)_r = \frac{1}{\sqrt{d}} \sum_{|k|\le w} a_k e^{i(r-k)t_0} = \frac{e^{irt_0}}{\sqrt{d}} p(e^{-it_0}).$$

Therefore

$$\|T_d(p)\|_{\mathrm{op}}^2 \ge \|T_d(p)x\|_2^2 \ge \frac{d-2w}{d}\,\|p\|_\infty^2\,.$$

□

Theorem 5.6 (Banded scalar Toeplitz signing)**.** *There is a universal constant C such that the following holds. Let $A_i = T_d(p_i)$ be arbitrary complex Toeplitz matrices whose symbols have degree at most w, and assume $\|A_i\|_{\mathrm{op}} \le 1$. If $d \ge 4w$ and $w \ge 1$, then*

$$\mathrm{disc}_{\mathrm{op}}(A_1,\dots,A_n) \le C\sqrt{n \log\left(e + \frac{w}{n}\right)}.$$

For $w = 0$ the same statement holds with w replaced by 1 inside the logarithm.

Proof. By Lemma 5.5, $\|p_i\|_\infty \le \sqrt{2}$ for all i. Choose an $O(w)$-point uniform grid that is 2-norming for degree-w trigonometric polynomials by Lemma 3.6. Apply Theorem 3.5 with $b = 1$ to obtain signs with

$$\left\|\sum_i \varepsilon_i p_i\right\|_\infty \le C\sqrt{n \log\left(e + \frac{w}{n}\right)}.$$

Contractivity of Toeplitz compression gives the same bound for $\sum_i \varepsilon_i A_i$. □

5.4. **Block and multilevel Toeplitz matrices.** An r-level $b \times b$ block Toeplitz matrix is indexed by a box

$$\Lambda = \prod_{j=1}^{r} \{0,\dots,d_j - 1\}$$

and has $b \times b$ blocks depending only on coordinate differences. Its total dimension is

$$D = b \prod_{j=1}^{r} d_j.$$

At one level the bounded finite Toeplitz extension of Theorem 5.1 supplies a symbol of comparable norm, and this drives the sampling argument. We record one consequence of this fact for emphasis.

Theorem 5.7 (Finite Toeplitz extension; one level)**.** *There is an absolute constant K_T (one may take $K_T \le 3$) such that every $d \times d$ scalar Toeplitz matrix $A = T_d(a)$ has an inducer $\varphi_A \in L^\infty(\mathbb{T})$ with $\widehat{\varphi}_A(k) = a_k$ for $|k| < d$ and $\|\varphi_A\|_{L^\infty} \le K_T \|A\|_{\mathrm{op}}$.*

This is Theorem 5.1, restated for reference; we use it below at one level. At higher level the extension still holds for each fixed r, by Theorem 5.10 below, but the constant is no longer absolute: it depends on r and cannot be taken uniform in r.

Remark 5.8. The one-level constant K_T has no analogue that is uniform in the number of levels. For each fixed r a bounded symbol $\varphi \in L^\infty(\mathbb{T}^r)$ of comparable norm does exist (Theorem 5.10), but the best constant K_r grows with r: the underlying polydisc Nehari constant is at least $(\pi^2/8)^{r/4}$ [22], so there is no level-free bound and no infinite-dimensional analogue. For the purposes of the present method this means the symbol route is available at every fixed level, but with an r-dependent loss in the constant rather than the absolute constant of the one-level case.

Theorem 5.9 (Fixed-block, one-level block Toeplitz/Hankel)**.** *Fix $b \ge 1$. There is a constant C_b such that every collection of $b \times b$ (one-level) block Toeplitz, or block Hankel, contractions $A_1, \dots, A_n \in \mathrm{M}_D(\mathbb{C})$ satisfies*

$$\mathrm{disc}_{\mathrm{op}}(A_1, \dots, A_n) \le C_b \sqrt{n \log\left(e + \frac{D}{n}\right)}.$$

Proof. First consider block Toeplitz matrices, so $r = 1$. For a fixed pair of block coordinates $p, q \in \{1, \dots, b\}$, the scalar Toeplitz matrix formed by the (p, q) entries of all blocks is a compression of A_i between coordinate subspaces, hence has norm at most $\|A_i\|_{\mathrm{op}}$. By Theorem 5.1, each scalar entry has an $L^\infty(\mathbb{T})$ inducer with norm at most $K_T \|A_i\|_{\mathrm{op}}$. Combining the b^2 scalar inducers gives a matrix-valued inducer $\Phi_i \in L^\infty(\mathbb{T}; \mathrm{M}_b)$ with

$$\|\Phi_i\|_{L^\infty(\mathbb{T};\mathrm{M}_b)} \le b K_T \|A_i\|_{\mathrm{op}}.$$

Convolve Φ_i with the de la Vallée Poussin kernel $V_{D/b-1}$ to obtain a matrix trigonometric polynomial P_i of degree $O(D/b)$ with the same finite Toeplitz coefficients as A_i and $\|P_i\|_{L^\infty(\mathbb{T};\mathrm{M}_b)} \le C_b \|A_i\|_{\mathrm{op}}$. By Lemma 3.6 with $r = 1$ there is a 2-norming grid $Q \subset \mathbb{T}$ of size $O(D/b)$. Theorem 3.5 with block

size b gives

$$\operatorname{disc}_{\mathrm{op}}(A_1,\dots,A_n) \le C_b\sqrt{n\log\left(e+\frac{b^2|Q|}{n}\right)} \le C_b\sqrt{n\log\left(e+\frac{D}{n}\right)},$$

since b is fixed and $b^2|Q| = O_b(D)$, using contractivity of Toeplitz compression to transfer the bound to $\sum_i \varepsilon_i A_i$. The block Hankel statement follows by multiplying by the reversal matrix, exactly as in Corollary 5.4. □

5.5. **The multilevel symbol extension theorem.** The finite Toeplitz extension of Theorem 5.1 is multilevel. For each fixed number of levels there is a bounded $L^\infty(\mathbb{T}^r)$ symbol carrying the matrix entries, with a constant that depends on r but not on the box.

Theorem 5.10 (Finite multilevel Toeplitz symbol extension)**.** *For each fixed $r \ge 1$ there is a finite constant K_r such that every finite scalar r-level Toeplitz matrix A on a box $\prod_j\{0,\dots,d_j-1\}$ admits a symbol $\varphi \in L^\infty(\mathbb{T}^r)$ with*

$$\widehat{\varphi}(k) = a_k \quad \text{on the difference indices } k \in \prod_j\{-(d_j-1),\dots,d_j-1\}, \qquad \|\varphi\|_{L^\infty(\mathbb{T}^r)} \le K_r\,\|A\|_{\mathrm{op}}\,.$$

This is the multivariable finite Toeplitz extension theorem of Carlsson and Perfekt [7, Thm. 7.1 and Cor. 5.4], whose proof realizes the matrix as a bounded convolution operator with an L^∞ symbol. For $r = 1$ it is Theorem 5.1. For $r \ge 2$ it rests on the polydisc Nehari theorem, proved for $r = 2$ by Ferguson and Lacey [12] and for $r \ge 3$ by Lacey and Terwilleger [13].

Remark 5.11. The constant K_r cannot be taken uniform in r. The corresponding polydisc Nehari constant is at least $(\pi^2/8)^{r/4}$ by the lower bound of Ortega-Cerdà and Seip [22], so it grows at least exponentially in r, and there is no infinite-dimensional analogue of the extension with a level-free constant.

For higher level the symbol of Theorem 5.10 is bounded only by the r-dependent constant K_r of Remark 5.11. If the symbols are in addition banded, an elementary finite-section estimate replaces the extension theorem with no loss of constant and no dependence on r. The next lemma is the multilevel analogue of Lemma 5.5.

Lemma 5.12 (Multilevel banded finite-section bound)**.** *Let $\mathbf{w} = (w_1,\dots,w_r)$, and let*

$$p(\theta_1,\dots,\theta_r) = \sum_{|k_j|\le w_j\ \forall j} a_k e^{i\langle k,\theta\rangle}$$

be a trigonometric polynomial of per-coordinate half-bandwidth at most w_j, and let $T_{\mathbf{d}}(p)$ be its r-level Toeplitz matrix on $\prod_j\{0,\dots,d_j-1\}$. If $d_j > 2w_j$ for all j, then

$$\|p\|_{L^\infty(\mathbb{T}^r)} \le \left(\prod_{j=1}^{r}\left(1-\frac{2w_j}{d_j}\right)\right)^{-1/2} \|T_{\mathbf{d}}(p)\|_{\mathrm{op}}\,.$$

In particular, if $d_j \geq 4w_j$ for all j, then $\|p\|_\infty \leq 2^{r/2}\,\|T_{\mathbf{d}}(p)\|_{\mathrm{op}}$.

Proof. Pick $(t_1^0,\dots,t_r^0)$ with $|p(e^{-it_1^0},\dots,e^{-it_r^0})| = \|p\|_\infty$, and take the tensor unit vector $x_{\mathbf{s}} = \prod_j d_j^{-1/2} e^{is_j t_j^0}$, so $\|x\|_2 = 1$. For every multi-index $\mathbf{m}$ interior in every coordinate ($w_j \leq m_j \leq d_j - 1 - w_j$) the multilevel-Toeplitz action factors coordinatewise, giving

$$(T_{\mathbf{d}}(p)x)_{\mathbf{m}} = \Big(\prod_j d_j^{-1/2}\Big) e^{i\langle \mathbf{m},\mathbf{t}^0\rangle} p(e^{-it_1^0},\dots,e^{-it_r^0}).$$

The number of fully-interior indices is $\prod_j (d_j - 2w_j)$, so

$$\|T_{\mathbf{d}}(p)\|_{\mathrm{op}}^2 \geq \|T_{\mathbf{d}}(p)x\|_2^2 \geq \prod_{j=1}^r \frac{d_j - 2w_j}{d_j}\;\|p\|_\infty^2 .$$

This is Lemma 5.5 run on a product test vector; no extension theorem is used. □

Theorem 5.13 (Banded multilevel block Toeplitz/Hankel)**.** *Fix $r, b \geq 1$ and bandwidth $w \geq 1$. There is a constant $C_{r,b}$ such that for arbitrary complex r-level $b \times b$ block Toeplitz contractions $A_1,\dots,A_n \in \mathrm{M}_D(\mathbb{C})$ whose matrix symbols have per-coordinate degree at most w, with $d_j \geq 4w$ for all j,*

$$\mathrm{disc}_{\mathrm{op}}(A_1,\dots,A_n) \leq C_{r,b}\sqrt{n \log\left(e + \frac{w^r}{n}\right)}.$$

The same holds for block Hankel matrices for which $H_i J_{\mathbf{d}}$ is a block Toeplitz matrix satisfying the same bandwidth hypothesis, where $J_{\mathbf{d}}$ is the tensor-product reversal. For fixed r, b, w with the box large ($d_j \geq 4w$) this is $O_{r,b,w}(\sqrt{n})$.

Proof. The matrix symbols $P_i \in L^\infty(\mathbb{T}^r; \mathrm{M}_b)$ of the A_i are already trigonometric polynomials of per-axis degree at most w, so no extension step is needed. For each block-coordinate pair, the corresponding scalar finite section is a compression of A_i between coordinate subspaces, hence has operator norm at most $\|A_i\|_{\mathrm{op}}$. Applying Lemma 5.12 entrywise to the b^2 scalar symbols gives

$$\|P_i\|_{L^\infty(\mathbb{T}^r;\mathrm{M}_b)} \leq 2^{r/2} b\;\|A_i\|_{\mathrm{op}}\,.$$

By Lemma 3.6 there is a 2-norming grid $Q \subset \mathbb{T}^r$ of size $|Q| = O_r(w^r)$. Theorem 3.5 with block size b gives

$$\mathrm{disc}_{\mathrm{op}}(A_1,\dots,A_n) \leq C_{r,b}\sqrt{n \log\left(e + \frac{b^2|Q|}{n}\right)} \leq C_{r,b}\sqrt{n \log\left(e + \frac{w^r}{n}\right)},$$

and multilevel Toeplitz-compression contractivity transfers the bound to $\sum_i \varepsilon_i A_i$. For the Hankel case, apply the Toeplitz statement to $H_i J_{\mathbf{d}}$ and then multiply on the right by the unitary $J_{\mathbf{d}}$, exactly as in Corollary 5.4. □

5.6. **Banded multilevel block Hankel matrices.** The Hankel half of Theorem 5.13 was phrased as a hypothesis on $H_i J_{\mathbf{d}}$. We now make the reduction unconditional by recording the exact properties of the tensor-product reversal $J_{\mathbf{d}}$, so that an arbitrary multilevel block Hankel matrix is turned into a multilevel block Toeplitz matrix of the same operator norm, bandwidth, and box.

Lemma 5.14 (Tensor reversal)**.** *For $d \geq 1$ let J_d be the reversal matrix $(J_d)_{rs} = \delta_{s,\,d-1-r}$ on $\mathbb{C}^d$, and set*

$$\mathbf{J}_\Lambda = J_{d_1} \otimes \cdots \otimes J_{d_r} \otimes I_b$$

on the box $\Lambda = \prod_{j=1}^{r} \{0, \dots, d_j - 1\}$ with $b \times b$ blocks. Then $\mathbf{J}_\Lambda$ is real, symmetric, orthogonal, and an involution. If H is an r-level $b \times b$ block Hankel matrix, that is, its block at $(\mathbf{m}, \mathbf{m}')$ depends only on $\mathbf{m} + \mathbf{m}'$, then $H\mathbf{J}_\Lambda$ is r-level $b \times b$ block Toeplitz on the same box,

$$\|H\mathbf{J}_\Lambda\|_{\mathrm{op}} = \|H\|_{\mathrm{op}}\,,$$

and per-coordinate bandedness with half-bandwidth w is preserved.

Proof. Each J_{d_j} is a permutation matrix, hence orthogonal, and it is symmetric with $J_{d_j}^2 = I$, so it is a real symmetric orthogonal involution; the same is true of I_b. A tensor product of real symmetric orthogonal involutions is again one, so $\mathbf{J}_\Lambda$ has all four properties. Write the block of H at $(\mathbf{m}, \mathbf{m}')$ as $C_{\mathbf{m}+\mathbf{m}'}$. Since $\mathbf{J}_\Lambda$ acts on the index $\mathbf{m}'$ by the coordinatewise reflection $\rho(\mathbf{m}')_j = d_j - 1 - m'_j$ and is the identity on the block index,

$$(H\mathbf{J}_\Lambda)_{\mathbf{m},\mathbf{m}'} = C_{\mathbf{m}+\rho(\mathbf{m}')}, \qquad \big(\mathbf{m} + \rho(\mathbf{m}')\big)_j = (d_j - 1) + (m_j - m'_j).$$

The right-hand side depends only on the differences $m_j - m'_j$, so $H\mathbf{J}_\Lambda$ is r-level block Toeplitz; its symbol coefficient at lag $\mathbf{k}$ is $C_{(\mathbf{d}-\mathbf{1})+\mathbf{k}}$, which is nonzero only when $|k_j| \leq w$ for every j exactly when H has per-coordinate half-bandwidth w. Finally $\mathbf{J}_\Lambda$ is orthogonal, so right multiplication by it is an isometry for the operator norm. □

Theorem 5.15 (Banded multilevel block Hankel signing)**.** *Fix $r, b \geq 1$ and bandwidth $w \geq 1$. There is a constant $C_{r,b}$ such that every collection of arbitrary complex r-level $b \times b$ block Hankel contractions $H_1, \dots, H_n \in \mathrm{M}_D(\mathbb{C})$ with per-coordinate half-bandwidth at most w, on a box with $d_j \geq 4w$ for all j, satisfies*

$$\mathrm{disc}_{\mathrm{op}}(H_1, \dots, H_n) \leq C_{r,b} \sqrt{n \log\left(e + \frac{w^r}{n}\right)}.$$

Proof. Put $A_i = H_i \mathbf{J}_\Lambda$. By Lemma 5.14 each A_i is an r-level $b \times b$ block Toeplitz contraction with the same per-coordinate half-bandwidth w on the same box, so the hypotheses of Theorem 5.13 hold. Because $\mathbf{J}_\Lambda$ is one fixed orthogonal matrix, independent of i,

$$\sum_i \varepsilon_i H_i = \Big(\sum_i \varepsilon_i A_i\Big)\mathbf{J}_\Lambda, \qquad \Big\|\sum_i \varepsilon_i H_i\Big\|_{\mathrm{op}} = \Big\|\sum_i \varepsilon_i A_i\Big\|_{\mathrm{op}},$$

and the signs furnished by Theorem 5.13 give the bound. □

5.7. **Sample-optimality of the grid.** The grids used above have size $O_r(w^r)$ in the isotropic case, and $O_r \prod_j (2w_j+1)$ in the anisotropic case. The following dimension count shows that no norming set can be smaller, so the bandwidth factor inside the logarithm is intrinsic to the sampling method.

Proposition 5.16 (Norming-set lower bound)**.** *Let $V \subset C(\mathbb{T}^r; \mathrm{M}_b(\mathbb{C}))$ be the space of M_b-valued trigonometric polynomials of degree at most w_j in the j-th coordinate, so that*

$$\dim_{\mathbb{C}} V = b^2 \prod_{j=1}^{r} (2w_j + 1).$$

If $Q \subset \mathbb{T}^r$ is an L_Q-norming set for V in the sense of the norming definition of §3, then the evaluation map $V \to \bigoplus_{\omega \in Q} \mathrm{M}_b(\mathbb{C})$, $F \mapsto (F(\omega))_{\omega \in Q}$, is injective, and therefore

$$b^2\,|Q| \geq \dim_{\mathbb{C}} V = b^2 \prod_{j=1}^{r} (2w_j + 1), \qquad \text{that is} \qquad |Q| \geq \prod_{j=1}^{r} (2w_j + 1).$$

In particular, when $w_1 = \cdots = w_r = w$, the grid size $O_r(w^r)$ of Lemma 3.6 is optimal up to the constant, and the appearance of w^r inside the logarithm in Theorems 5.13 and 5.15 is inherent to any norming-set argument for this class.

Proof. If $F \in V$ vanishes on Q then $\max_{\omega \in Q} \|F(\omega)\|_{\mathrm{op}} = 0$, so the norming inequality forces $\|F\|_{L^\infty} = 0$ and $F = 0$; thus evaluation is injective. An injective linear map cannot increase dimension, and $\dim_{\mathbb{C}} \bigoplus_{\omega \in Q} \mathrm{M}_b(\mathbb{C}) = b^2\,|Q|$, which gives the stated inequality. □

Corollary 5.17 (Anisotropic banded multilevel block Toeplitz/Hankel)**.** *With per-axis half-bandwidths $w_1, \ldots, w_r$ the same arguments, using the anisotropic grid of Lemma 3.6 and the count $\dim_{\mathbb{C}} V = b^2 \prod_j (2w_j + 1)$, give the sharp form*

$$\mathrm{disc}_{\mathrm{op}}(A_1, \ldots, A_n) \leq C_{r,b} \sqrt{n \log \left(e + \frac{\prod_{j=1}^{r} (2w_j + 1)}{n} \right)},$$

for arbitrary complex r-level $b \times b$ block Toeplitz contractions whose matrix symbols have per-coordinate degree at most w_j, on boxes with $d_j \geq 4w_j$. The same bound holds for arbitrary complex r-level $b \times b$ block Hankel contractions with the same anisotropic half-bandwidths. Moreover any norming-set proof for this anisotropic symbol class must use at least $\prod_j (2w_j + 1)$ sample points.

Proof. For the Toeplitz case, repeat the proof of Theorem 5.13 with the anisotropic grid from Lemma 3.6. Lemma 5.12 gives the same entrywise symbol norm control, while the grid has $|Q| = O_r \prod_j (2w_j + 1)$. Theorem 3.5 then gives the displayed bound. The Hankel case follows by applying the Toeplitz statement to $H_i \mathbf{J}_\Lambda$ and using Lemma 5.14. The lower bound on

the number of sample points is Proposition 5.16 with $\dim_{\mathbb{C}} V = b^2 \prod_j (2w_j + 1)$. □

5.8. **The self-adjoint regime and the entropy obstruction.** For fixed r, b and total anisotropic bandwidth small relative to n, the banded multilevel results give the conjectured square-root bound with no restriction on the ambient dimension $D = b \prod_j d_j$.

Corollary 5.18 (Self-adjoint anisotropic banded multilevel regime)**.** *Fix $r, b \geq 1$ and half-bandwidths $w_1, \ldots, w_r$ with*

$$\prod_{j=1}^{r} (2w_j + 1) \leq n.$$

Let $A_1, \ldots, A_n$ be self-adjoint banded r-level $b \times b$ block Toeplitz or block Hankel contractions with per-coordinate half-bandwidth at most w_j, on any box with $d_j \geq 4w_j$ for all j. Then

$$\operatorname{disc}(A_1, \ldots, A_n) = O_{r,b}(\sqrt{n}),$$

with the ambient dimension $D = b \prod_{j=1}^{r} d_j$ unrestricted.

Proof. The logarithm in Corollary 5.17 is at most $\log(e + 1)$, a constant, so the bound reads $O_{r,b}(\sqrt{n})$. The hypotheses constrain only the symbol and the bandwidths, never $\prod_j d_j$, so D may be arbitrarily large. □

This avoids the full-unit-ball obstruction of Proposition 8.1: the banded symbol family has only $b^2 \prod_j (2w_j + 1) = O_{r,b}(n)$ effective scalar degrees of freedom, whereas faithful finite-block reductions of the whole unit ball/full algebra $\mathrm{M}_D(\mathbb{C})$ must pay D^2 entropy.

5.9. **A matching lower bound from circulants.** The upper bounds above are sharp in the sampled regime. We show that for the periodic model the discrepancy problem is, on the nose, complex vector Spencer, and we read off a matching lower bound from the tightness of Spencer's theorem.

Fix $N_1, \ldots, N_r \geq 1$ and work on the product torus $\Lambda = \prod_{j=1}^{r} \mathbb{Z}_{N_j}$, with $M = \prod_j N_j$ points. For a lag function $a : \Lambda \to \mathbb{C}$ the r-level circulant

$$A = \sum_{\mathbf{k} \in \Lambda} a_{\mathbf{k}} S^{\mathbf{k}}, \qquad S^{\mathbf{k}} = S_1^{k_1} \cdots S_r^{k_r},$$

is built from the coordinate cyclic shifts S_j on $\mathbb{Z}_{N_j}$. All r-level circulants are simultaneously diagonalized by the product discrete Fourier transform, and the eigenvalue at the character indexed by $\xi \in \Lambda$ is

$$v(A)_\xi = \sum_{\mathbf{k} \in \Lambda} a_{\mathbf{k}} \prod_{j=1}^{r} \omega_j^{\xi_j k_j}, \qquad \omega_j = e^{2\pi i / N_j}.$$

Thus

$$\|A\|_{\mathrm{op}} = \max_{\xi \in \Lambda} |v(A)_\xi| = \|v(A)\|_\infty,$$

and $a \mapsto v(A)$ is a (scaled) Fourier bijection $\mathbb{C}^M \to \mathbb{C}^M$.

Theorem 5.19 (Anisotropic circulant lower bound)**.** *Let $w_1, \dots, w_r \geq 0$ be integers, let $N_j = 2w_j+1$, let $\Lambda = \prod_j \mathbb{Z}_{N_j}$, and let $M = \prod_j N_j = \prod_j (2w_j+1)$. Since every residue in $\mathbb{Z}_{N_j}$ has a representative in $[-w_j, w_j]$, these are the periodic analogues of anisotropic half-bandwidths w_j. Signing r-level circulants on Λ is exactly complex vector Spencer on M coordinates: for any circulants $A_1, \dots, A_n$,*

$$\left\| \sum_{i=1}^{n} \varepsilon_i A_i \right\|_{\mathrm{op}} = \left\| \sum_{i=1}^{n} \varepsilon_i v(A_i) \right\|_{\infty} \qquad (\varepsilon \in \{-1,1\}^n),$$

where $v(A_i) \in \mathbb{C}^M$ are the Fourier symbols. Consequently, for $n \leq M$ there exist circulant contractions $A_1, \dots, A_n$ with

$$\operatorname{disc}(A_1, \dots, A_n) \geq c \sqrt{n \log\left(e + \frac{\prod_{j=1}^{r} (2w_j+1)}{n} \right)}$$

for an absolute constant $c > 0$. The same lower bound already holds among self-adjoint scalar circulants, and hence among self-adjoint $b \times b$ block circulants for every $b \geq 1$. In particular this matches the upper bound of Corollary 5.17 up to constants.

Proof. The displayed identity is immediate from $\|A\|_{\mathrm{op}} = \|v(A)\|_{\infty}$ and the linearity of $a \mapsto v(A)$ over the common signs. Because $a \mapsto v(A)$ is a bijection $\mathbb{C}^M \to \mathbb{C}^M$, choosing the symbols $v(A_i)$ freely realizes any tuple of vectors in $\mathbb{C}^M$ as circulant symbols, the contraction normalization $\|A_i\|_{\mathrm{op}} \leq 1$ being exactly $\|v(A_i)\|_{\infty} \leq 1$. The lower bound is then the tightness of vector Spencer: for $n \leq M$ there are vectors $x_1, \dots, x_n \in \mathbb{C}^M$ with $\|x_i\|_{\infty} \leq 1$ for which every signing has $\|\sum_i \varepsilon_i x_i\|_{\infty} \geq c\sqrt{n \log(e + M/n)}$. One may take the x_i to be independent uniform signs on a suitable subset of coordinates; the matching $\Omega(\sqrt{n \log(e + M/n)})$ lower bound for the rectangular ℓ_∞ problem is classical [20, Ch. 4], and is also a case of Gluskin's lower bounds for ℓ_∞-balls [15]. Substituting $M = \prod_j (2w_j + 1)$ and transporting through the bijection v gives the claim.

For the self-adjoint version, choose the lower-bound vectors x_i real. The Fourier transform maps self-adjoint scalar circulants bijectively onto real functions on the dual product torus: if $v(A)$ is real, then the inverse Fourier coefficients satisfy $a_{-\mathbf{k}} = \overline{a_{\mathbf{k}}}$, and the circulant is self-adjoint. Thus the same real ℓ_∞^M lower bound is realized by self-adjoint scalar circulants. Finally, for block size b, fix a rank-one projection $P \in \mathrm{M}_b(\mathbb{C})$ and replace A_i by $A_i \otimes P$. This is a self-adjoint $b \times b$ block circulant contraction and

$$\left\| \sum_i \varepsilon_i (A_i \otimes P) \right\|_{\mathrm{op}} = \left\| \sum_i \varepsilon_i A_i \right\|_{\mathrm{op}},$$

so the scalar lower bound is inherited unchanged. □

Corollary 5.20 (The clean one-level case)**.** *For $r = 1$, $N = 2w + 1$, and $n \le N$, there are scalar circulant contractions $A_1, \dots, A_n \in \mathrm{M}_N(\mathbb{C})$ with*

$$\operatorname{disc}(A_1, \dots, A_n) \ge c\sqrt{n \log\left(e + \frac{2w+1}{n}\right)}.$$

Since scalar circulants are the periodic analogue of banded scalar Toeplitz matrices, this gives the exact lower-bound companion for the sampled periodic model behind Theorem 5.6.

Proof. This is Theorem 5.19 at $r = 1$, where $M = N = 2w+1$ and a circulant is a Toeplitz matrix with cyclically wrapped, hence w-banded, diagonals. □

Question 5.21 (Non-periodic matching bound)**.** *The lower bound of Theorem 5.19 is for the periodic (circulant) model, where the symbol-to-operator passage is an exact isometry. Does the same*

$$\Omega\left(\sqrt{n \log\left(e + \frac{\prod_j (2w_j + 1)}{n}\right)}\right)$$

lower bound hold for genuine finite-section banded Toeplitz matrices $T_{\mathbf{d}}(p)$ on a box with d_j comparable to w_j? The finite-section symbol map is only a near-isometry: Lemma 5.12 controls it from below by a constant, but transporting a worst-case symbol to the matrix incurs the Lebesgue-type loss of the grid, which for the natural isotropic construction degrades the bound by a factor $(\log w)^{-r}$. Closing this residual gap, so that the finite-section banded class matches the anisotropic circulant lower bound exactly, is open.

5.10. **The full-bandwidth multilevel case.** For fixed level the full-bandwidth class is also covered, at the Spencer rate, once the multilevel symbol extension of Theorem 5.10 is fed into the sampling argument. No bandwidth restriction is needed; only r and b are fixed.

Theorem 5.22 (Fixed-level multilevel block Toeplitz/Hankel)**.** *Fix $r, b \ge 1$. There is a constant $C_{r,b}$ such that every collection of r-level $b \times b$ block Toeplitz, or block Hankel, contractions $A_1, \dots, A_n \in \mathrm{M}_D(\mathbb{C})$ of full bandwidth and ambient dimension $D = b \prod_j d_j$ satisfies*

$$\operatorname{disc}_{\mathrm{op}}(A_1, \dots, A_n) \le C_{r,b}\sqrt{n \log\left(e + \frac{D}{n}\right)}.$$

In particular, for $D \le n$ this is $O_{r,b}(\sqrt{n})$, with the box $\prod_j d_j$ otherwise unrestricted.

Proof. Treat the block Toeplitz case first. For each ordered pair of block coordinates $p, q \in \{1, \dots, b\}$, the scalar r-level Toeplitz matrix formed by the (p, q) entries of the blocks of A_i is a compression of A_i between coordinate subspaces, hence has operator norm at most $\|A_i\|_{\mathrm{op}}$. By Theorem 5.10 each of these b^2 scalar entries has an $L^\infty(\mathbb{T}^r)$ symbol agreeing with the matrix

on the difference indices and of norm at most $K_r \|A_i\|_{\mathrm{op}}$. Assembling the b^2 scalar symbols gives a matrix-valued inducer $\Phi_i \in L^\infty(\mathbb{T}^r; \mathrm{M}_b)$ with

$$\|\Phi_i\|_{L^\infty(\mathbb{T}^r;\mathrm{M}_b)} \le bK_r \|A_i\|_{\mathrm{op}}.$$

Convolve Φ_i with the product de la Vallée Poussin kernel $\prod_j V_{d_j-1}(\theta_j)$ to obtain a matrix trigonometric polynomial P_i of per-axis degree $O(d_j)$ that retains the finite Toeplitz coefficients of A_i and satisfies $\|P_i\|_{L^\infty(\mathbb{T}^r;\mathrm{M}_b)} \le C_{r,b}\|A_i\|_{\mathrm{op}}$. By Lemma 3.6 there is a 2-norming grid $Q \subset \mathbb{T}^r$ of size $|Q| = O_r(\prod_j d_j) = O(D/b)$. Theorem 3.5 with block size b then has entropy $b^2|Q| = O_{r,b}(D)$ and gives

$$\mathrm{disc}_{\mathrm{op}}(A_1,\dots,A_n) \le C_{r,b}\sqrt{n\log\left(e+\frac{b^2|Q|}{n}\right)} \le C_{r,b}\sqrt{n\log\left(e+\frac{D}{n}\right)},$$

where multilevel Toeplitz-compression contractivity transfers the bound to $\sum_i \varepsilon_i A_i$. The block Hankel case follows by applying the Toeplitz statement to $H_i\mathbf{J}_\Lambda$ and multiplying on the right by the unitary tensor reversal $\mathbf{J}_\Lambda$ of Lemma 5.14, exactly as in Corollary 5.4. □

Remark 5.23 (An alternative, weaker operator route). There is a second, purely operator-theoretic route to a bound that does not pass through a symbol. Carlsson and Perfekt [7] also extend a finite-section multilevel Toeplitz matrix to a bounded ℓ^2 multilevel Toeplitz operator with a level-only constant; taking that operator as a single block of size D and applying Theorem 1.3 gives the unconditional but weaker rate $\sqrt{n\log(e+D^2/n)}$, with the generic D^2 entropy of Proposition 8.1. The symbol route of Theorem 5.22 feeds entropy $O(D)$ rather than D^2, so it gives the better dimension dependence and upgrades the fixed-level full-bandwidth case from this trivial operator rate to a theorem at the Spencer rate.

Remark 5.24 (Comparison with the banded results). Theorem 5.22 is non-constructive and its constant K_r depends on r through the polydisc Nehari theorem (Remark 5.11). The banded results of Theorem 5.13 onward are complementary: there the symbol is the matrix symbol itself, so the finite section bound of Lemma 5.12 gives a dimension-free, explicit constant with no appeal to Nehari's theorem, and Theorem 5.19 supplies a matching lower bound. The full-bandwidth theorem trades these explicit, matched constants for coverage of the entire fixed-level class.

6. Closure operations

The finite-lift framework is stable under the operations normally used to build structured matrices. The entropy accounting in this section is explicit because this is a common source of mistakes.

Proposition 6.1 (Compressions and bounded images)**.** *Suppose B_i satisfy a discrepancy bound*

$$\mathrm{disc}_{\mathrm{op}}(B_1,\dots,B_n) \le \Delta.$$

If $A_i = V^* B_i V$ for a fixed isometry V, then $\operatorname{disc}_{\mathrm{op}}(A_1, \dots, A_n) \le \Delta$. *More generally, if $A_i = \Phi(B_i)$ for a fixed linear map Φ with $\|\Phi\| \le L$ on the relevant span, then*

$$\operatorname{disc}_{\mathrm{op}}(A_1, \dots, A_n) \le L\Delta.$$

Proof. Use the same signs and the inequality $\|\sum_i \varepsilon_i \Phi(B_i)\| \le L \, \|\sum_i \varepsilon_i B_i\|$. □

Proposition 6.2 (Finite sums and direct-sum entropy)**.** *Suppose*

$$A_i = A_i^{(1)} + \cdots + A_i^{(s)}.$$

For each ℓ, assume the tuple $(A_i^{(\ell)})_{i=1}^n$ has a (K_ℓ, Λ_ℓ)-lift to

$$\mathcal{B}_\ell = \bigoplus_\alpha \mathrm{M}_{r_{\ell,\alpha}}(\mathbb{C}).$$

Put

$$h = \max_{\ell,\alpha} r_{\ell,\alpha}, \qquad R = \sum_{\ell=1}^{s} \sum_{\alpha} r_{\ell,\alpha}.$$

Then $(A_i)_{i=1}^n$ has a lift to $\bigoplus_{\ell=1}^s \mathcal{B}_\ell$ and

$$\operatorname{disc}_{\mathrm{op}}(A_1, \dots, A_n) \le C \left(\sum_{\ell=1}^{s} \Lambda_\ell K_\ell \right) \sqrt{n \log \left(e + \frac{hR}{n} \right)}.$$

Proof. For each component choose lifts $Y_i^{(\ell)}$ in $\mathcal{B}_\ell$. If $K_\ell > 0$, set $Z_i^{(\ell)} = Y_i^{(\ell)} / K_\ell$; zero components may be omitted. Then $Z_i = (Z_i^{(1)}, \dots, Z_i^{(s)})$ has norm at most one in $\bigoplus_\ell \mathcal{B}_\ell$. Define

$$\Theta(Z^{(1)}, \dots, Z^{(s)}) = \sum_{\ell=1}^{s} K_\ell \Pi_\ell(Z^{(\ell)}).$$

Then $\Theta(Z_i) = A_i$ and $\|\Theta\| \le \sum_\ell \Lambda_\ell K_\ell$ on the relevant span. Apply Theorem 1.3 to the direct-sum algebra, whose maximum block size is h and whose reduced dimension is R. □

Remark 6.3 (Direct sums are not additive in entropy). For $\mathcal{B} = \bigoplus_{\ell=1}^s \mathcal{B}_\ell$ one has

$$\mathsf{E}(\mathcal{B}) = \left(\max_\ell h(\mathcal{B}_\ell) \right) \sum_{\ell=1}^{s} R(\mathcal{B}_\ell),$$

not generally $\sum_\ell \mathsf{E}(\mathcal{B}_\ell)$. Consequently, a finite sum of classes that are individually in the Spencer regime remains in the Spencer regime only when this combined entropy is $O(n)$. This condition is automatic, for example, for a fixed number of scalar sampled multiplier classes of total sample size $O(n)$, and for a fixed number of uniformly bounded-block classes with total reduced dimension $O(n)$.

Corollary 6.4 (Toeplitz plus Hankel with bounded decomposition)**.** *If $A_i = T_i + H_i$, where T_i is Toeplitz, H_i is Hankel, and*

$$\|T_i\|_{\mathrm{op}} + \|H_i\|_{\mathrm{op}} \le K \qquad (1 \le i \le n),$$

then

$$\mathrm{disc}_{\mathrm{op}}(A_1, \dots, A_n) \le CK\sqrt{n \log\left(e + \frac{d}{n}\right)}.$$

Proof. Use the scalar Toeplitz and Hankel multiplier lifts from Theorem 5.2 and Corollary 5.4. Their sampled lift algebras are diagonal with $O(d)$ total samples. The direct sum of the two sampled algebras is still diagonal and has $O(d)$ total samples. Proposition 6.2 gives the result. □

Proposition 6.5 (Completely bounded tensor amplification)**.** *Let $S \subset \mathrm{M}_D(\mathbb{C})$ be an operator space and suppose there are linear maps*

$$\sigma : S \to \mathcal{B} = \bigoplus_{\alpha} \mathrm{M}_{r_\alpha}(\mathbb{C}), \qquad \Pi : \sigma(S) \to S,$$

with $\Pi\sigma = \mathrm{id}_S$, $\|\sigma\|_{cb} \le K$, and $\|\Pi\|_{cb} \le \Lambda$. Then, for each $c \ge 1$, the amplified space $S \otimes \mathrm{M}_c$ has a (K, Λ) lift to

$$\mathcal{B} \otimes \mathrm{M}_c \simeq \bigoplus_{\alpha} \mathrm{M}_{cr_\alpha}(\mathbb{C}).$$

The entropy is multiplied by c^2. Hence fixed-size completely bounded tensor amplification preserves the Spencer-regime conclusion.

Proof. Use the maps $\sigma \otimes \mathrm{id}_{\mathrm{M}_c}$ and $\Pi \otimes \mathrm{id}_{\mathrm{M}_c}$. The cb-norm assumptions give the same lift constants at matrix level c. The new block sizes are cr_α, so both h and R are multiplied by c. □

Remark 6.6. The completely bounded hypothesis in Proposition 6.5 is necessary for this general formulation. For concrete lifts coming from $*$-homomorphisms, compressions, or C*-quotient maps it is automatically satisfied with the expected constants.

7. Random structured inputs

The preceding results are deterministic. They therefore apply pathwise to random samples from the covered structured classes after normalization. For random banded Toeplitz matrices a complementary structural obstruction, a parity obstruction to stability, is established in the companion paper [27].

Corollary 7.1 (Pathwise random structured matrices)**.** *Let $A_1, \dots, A_n$ be random matrices which, almost surely, lie in one of the classes covered above: scalar Toeplitz, scalar Hankel, one-level block Toeplitz/Hankel or its banded multilevel form, a fixed finite-dimensional C*-algebra of entropy E, or a finite group convolution algebra. On each realization define*

$$\widetilde{A}_i = \begin{cases} A_i / \|A_i\|_{\mathrm{op}}, & A_i \neq 0, \\ 0, & A_i = 0. \end{cases}$$

Then the corresponding deterministic bound holds for $\widetilde{A}_1, \dots, \widetilde{A}_n$ on that realization.

Corollary 7.2 (Distribution-free random algebraic-core ensembles)**.** *For every $C_0 \geq 1$ there is $C(C_0)$ with the following property. Let $X_1, \dots, X_n$ be random self-adjoint matrices in a common ambient $\mathrm{M}_D(\mathbb{C})$. Suppose that, outside an exceptional event E, there is a finite-dimensional C^*-subalgebra $\mathcal{A}_\omega \subset \mathrm{M}_D(\mathbb{C})$ and a decomposition*

$$X_i(\omega) = B_i(\omega) + L_i(\omega), \qquad i = 1, \dots, n,$$

such that

$$\dim_{\mathbb{C}} \mathcal{A}_\omega \leq C_0 n, \qquad B_i(\omega) \in \mathcal{A}_\omega, \qquad \|B_i(\omega)\|_{\mathrm{op}} \leq 1,$$

and

$$L_i(\omega) = L_i(\omega)^*, \qquad \|L_i(\omega)\|_{\mathrm{op}} \leq 1, \qquad \|L_i(\omega)\|_{\mathrm{F}}^2 \leq \frac{n}{\log^3(en)}$$

for every i. Then

$$\mathbb{P}\left\{\mathrm{disc}(X_1, \dots, X_n) > C(C_0)\sqrt{n}\right\} \leq \mathbb{P}(E).$$

No independence, identical distribution, or deterministic choice of $\mathcal{A}_\omega$ is required.

Consequently the Matrix Spencer bound holds almost surely, or with the same high probability as the structural event above, for all of the following random ensembles:

(i) *arbitrary random self-adjoint contractions in a full matrix algebra $\mathrm{M}_D(\mathbb{C})$ whenever $D^2 \leq C_0 n$; this includes normalized* GOE, GUE, *Wigner, sample-covariance, and Wishart-type models in the $D \lesssim \sqrt{n}$ square-block range;*

(ii) *arbitrary random self-adjoint contractions in a random common block algebra $U_\omega(\bigoplus_\alpha \mathrm{M}_{r_\alpha}(\mathbb{C}))U_\omega^*$ whenever $\sum_\alpha r_\alpha^2 \leq C_0 n$;*

(iii) *any of the preceding algebraic-core ensembles after adding arbitrary dependent random self-adjoint tails of rank at most $n/\log^3(en)$ and operator norm at most 1;*

(iv) *in particular, random spiked tails $U_i D_i U_i^*$, random compressions $P_i Y_i P_i$, and random coordinate-supported tails $P_{S_i} Y_i P_{S_i}$, provided the projections or supports have rank/cardinality at most $n/\log^3(en)$ and the inserted self-adjoint operators have operator norm at most 1.*

Proof. On each realization outside E, apply the Akbas–Sra low-Frobenius perturbation theorem, Theorem 4.3, to the finite-dimensional C*-algebra $\mathcal{A}_\omega$ and to the decomposition $X_i = B_i + L_i$. This gives the displayed discrepancy bound on that realization. The probability bound is therefore immediate.

The listed examples are just specializations of the structural event. In $\mathrm{M}_D(\mathbb{C})$ the intrinsic algebra dimension is D^2. A common block algebra has

intrinsic dimension $\sum_\alpha r_\alpha^2$, unitary conjugation does not change it, and no distributional assumption enters the deterministic theorem. Finally,

$$\|L_i\|_{\mathrm{F}}^2 \le \operatorname{rank}(L_i)\|L_i\|_{\mathrm{op}}^2 \le \frac{n}{\log^3(en)}$$

for every rank-controlled tail with $\|L_i\|_{\mathrm{op}} \le 1$. Spiked, compressed, and coordinate-supported tails have rank bounded by the displayed projection rank or support size. □

Corollary 7.3 (Distribution-free all-plus gauge for arbitrary random matrices)**.** *Let $X_1, \dots, X_n$ be arbitrary random self-adjoint matrices in a common $\mathrm{M}_D(\mathbb{C})$. No independence, centering, identical distribution, tail, or moment assumption is imposed. Define the pathwise gauge*

$$G(X) := \max\left\{ \max_{i\le n} \|X_i\|_{\mathrm{op}}, \frac{1}{\sqrt{n}} \left\| \sum_{i=1}^n X_i \right\|_{\mathrm{op}} \right\}.$$

Set $A_i = 0$ on $\{G(X) = 0\}$, and $A_i = X_i/G(X)$ otherwise. Then, pathwise,

$$\max_{i\le n} \|A_i\|_{\mathrm{op}} \le 1, \qquad \operatorname{disc}(A_1, \dots, A_n) \le \sqrt{n}.$$

The same assertion holds for arbitrary rectangular matrices $X_i \in \mathrm{M}_{d_1\times d_2}(\mathbb{C})$, with $\operatorname{disc}_{\mathrm{op}}$ in place of disc*.*

More generally, for an active triangular menu $(X_{m,h,i})_{i\le n_m}$, $1 \le h \le J_m$, of arbitrary self-adjoint or rectangular random matrices, let $G_{m,h}$ be the corresponding all-plus gauge. If $s_{m,h} > 0$ are deterministic or environment-measurable scales and

$$\sum_m \mathbb{P}\left\{ \max_{h\le J_m} \frac{G_{m,h}}{s_{m,h}} > 1 \right\} < \infty,$$

then almost surely, for all sufficiently large m, every active scaled tuple

$$X_{m,h,1}/s_{m,h}, \dots, X_{m,h,n_m}/s_{m,h}$$

is contraction-normalized and has the all-plus certificate

$$\operatorname{disc}\left(X_{m,h,1}/s_{m,h}, \dots, X_{m,h,n_m}/s_{m,h}\right) \le \sqrt{n_m}$$

in the self-adjoint case, and the analogous $\operatorname{disc}_{\mathrm{op}}$ bound in the rectangular case.

Proof. The pathwise statement is immediate from the definition of $G(X)$. If $G(X) = 0$, then all normalized matrices are zero. Otherwise,

$$\|A_i\|_{\mathrm{op}} \le 1 \qquad \text{and} \qquad \left\| \sum_{i=1}^n A_i \right\|_{\mathrm{op}} \le \sqrt{n}.$$

The all-plus sign vector is an admissible signing, so it gives the displayed discrepancy bound. The rectangular assertion is the same argument with $\operatorname{disc}_{\mathrm{op}}$, since the all-plus sum is still a valid signed sum. For the active triangular statement, Borel–Cantelli gives $G_{m,h} \le s_{m,h}$ for every active

$h \le J_m$ and all large m, and the preceding pathwise argument applies to each active scaled tuple. □

Corollary 7.4 (Moment schedules for the distribution-free gauge)**.** *In the active triangular menu of Corollary 7.3, let $G_{m,h}$ be the all-plus gauge of the h-th active tuple. Fix $p > 0$, and let $s_{m,h} > 0$ be deterministic scales. If*

$$\sum_m \sum_{h=1}^{J_m} \mathbb{E}\left(\frac{G_{m,h}}{s_{m,h}}\right)^p < \infty,$$

then almost surely, for all sufficiently large m, every active scaled tuple

$$X_{m,h,1}/s_{m,h}, \dots, X_{m,h,n_m}/s_{m,h}$$

is contraction-normalized and has the all-plus Matrix-Spencer certificate

$$\operatorname{disc}\left(X_{m,h,1}/s_{m,h}, \dots, X_{m,h,n_m}/s_{m,h}\right) \le \sqrt{n_m}$$

in the self-adjoint case, with the analogous $\operatorname{disc}_{\mathrm{op}}$ *bound in the rectangular case.*

Equivalently, if $\eta_{m,h} > 0$ satisfy $\sum_m \sum_{h \le J_m} \eta_{m,h} < \infty$, then it is enough to choose

$$s_{m,h} \ge \eta_{m,h}^{-1/p} \left(\mathbb{E} G_{m,h}^p\right)^{1/p}.$$

In particular, a common row scale s_m works whenever

$$s_m \ge \eta_m^{-1/p} J_m^{1/p} \max_{h \le J_m} \left(\mathbb{E} G_{m,h}^p\right)^{1/p} \qquad \text{and} \qquad \sum_m \eta_m < \infty.$$

No independence between the matrices, between menu elements, or across dimensions is required.

Proof. Markov's inequality gives

$$\mathbb{P}\left\{\max_{h \le J_m} \frac{G_{m,h}}{s_{m,h}} > 1\right\} \le \sum_{h=1}^{J_m} \mathbb{E}\left(\frac{G_{m,h}}{s_{m,h}}\right)^p.$$

The first displayed summability therefore implies the tail summability hypothesis of Corollary 7.3, and that corollary gives the almost-sure conclusion. The quantile-budget and common row-scale forms are the same estimate with

$$\mathbb{E}(G_{m,h}/s_{m,h})^p \le \eta_{m,h} \quad \text{or} \quad \sum_{h \le J_m} \mathbb{E}(G_{m,h}/s_m)^p \le \eta_m.$$

□

Corollary 7.5 (Closed-form distribution-free gauge schedules)**.** *In the active triangular menu of Corollary 7.3, let $G_{m,h}$ be the all-plus gauge of the h-th active tuple. Fix $p > 0$, and suppose that deterministic envelopes $r_{m,h} > 0$ satisfy*

$$\mathbb{E} G_{m,h}^p \le r_{m,h}^p \qquad (m \ge 1,\ 1 \le h \le J_m).$$

Then the following deterministic normalizations are valid for arbitrary self-adjoint or rectangular random matrices, with no independence assumptions.

(i) *For arbitrary growth of J_m, if $\gamma > 0$ and*

$$s_{m,h} \geq r_{m,h}\big(J_m(e+m)^{1+\gamma}\big)^{1/p},$$

then almost surely all active scaled tuples are eventually contraction-normalized and have the all-plus $\sqrt{n_m}$ discrepancy certificate.

(ii) *If $D_m \geq 2$, $J_m \leq (eD_m)^{\Gamma}$, and*

$$\max_{h \leq J_m} r_{m,h} \leq r_m,$$

then the common row scale

$$s_m = r_m(eD_m)^B$$

is valid along all integer dimensions $D_m = m$ whenever $pB > \Gamma + 1$. Along every sparse stretched-exponential dimension schedule

$$D_m \geq \exp(cm^{\kappa}) \qquad (c, \kappa > 0),$$

the same scale is valid whenever $pB > \Gamma$.

(iii) *More generally, if $L_m := \sum_{h=1}^{J_m} r_{m,h}^p$, then any common row scale s_m satisfying*

$$\sum_m \frac{L_m}{s_m^p} < \infty$$

is valid.

Proof. Corollary 7.4 shows that it is enough to verify

$$\sum_m \sum_{h=1}^{J_m} \mathbb{E}(G_{m,h}/s_{m,h})^p < \infty.$$

For (i), the m-th row contribution is at most

$$\sum_{h=1}^{J_m} \frac{r_{m,h}^p}{r_{m,h}^p J_m (e+m)^{1+\gamma}} = (e+m)^{-1-\gamma},$$

which is summable. For (ii), the row contribution is bounded by

$$J_m(eD_m)^{-pB} \leq (eD_m)^{\Gamma - pB}.$$

This is summable over all integer dimensions when $pB > \Gamma + 1$, and is stretched-exponentially summable under $D_m \geq \exp(cm^{\kappa})$ when $pB > \Gamma$. The last assertion is exactly the same moment-summability criterion with $s_{m,h} = s_m$. □

Corollary 7.6 (Tail-profile distribution-free gauge schedules)**.** *In the active triangular menu of Corollary 7.3, let $G_{m,h}$ be the all-plus gauge of the h-th active tuple. Suppose that there are deterministic scales $r_{m,h} > 0$ and tail controls $\Phi_{m,h} : [1, \infty) \to [0, 1]$ such that*

$$\mathbb{P}\{G_{m,h} > r_{m,h}u\} \leq \Phi_{m,h}(u) \qquad (u \geq 1).$$

If $u_{m,h} \geq 1$ satisfy

$$\sum_{m} \sum_{h=1}^{J_m} \Phi_{m,h}(u_{m,h}) < \infty,$$

then the deterministic scales $s_{m,h} = r_{m,h}u_{m,h}$ are valid: almost surely, for all sufficiently large m, every active scaled self-adjoint or rectangular tuple is contraction-normalized and has the all-plus $\sqrt{n_m}$ discrepancy certificate.

In particular, assume the uniform sub-Weibull gauge tail

$$\mathbb{P}\{G_{m,h} > r_{m,h}u\} \leq C_0 \exp(-c_0 u^q) \qquad (u \geq 1)$$

with $C_0, c_0, q > 0$. Then, for every $\gamma > 0$, it is enough to take

$$u_m = \max\left\{1, \left(\frac{\max\{0, \log(C_0 J_m) + (1+\gamma)\log(e+m)\}}{c_0}\right)^{1/q}\right\}$$

and $s_{m,h} = r_{m,h}u_m$. If $D_m \geq 2$, $J_m \leq (eD_m)^\Gamma$, and $\max_{h \leq J_m} r_{m,h} \leq r_m$, then the common scale

$$s_m = r_m A(\log(eD_m))^{1/q}$$

is valid along all integer dimensions $D_m = m$ whenever $c_0 A^q > \Gamma + 1$. Along every sparse stretched-exponential dimension schedule

$$D_m \geq \exp(cm^\kappa) \qquad (c, \kappa > 0),$$

the same common scale is valid whenever $c_0 A^q > \Gamma$.

Proof. The first assertion is exactly the active-menu hypothesis in Corollary 7.3, since

$$\mathbb{P}\left\{\max_{h \leq J_m} \frac{G_{m,h}}{r_{m,h}u_{m,h}} > 1\right\} \leq \sum_{h=1}^{J_m} \Phi_{m,h}(u_{m,h}).$$

For the sub-Weibull choice of u_m, the m-th row contribution is at most

$$J_m C_0 \exp(-c_0 u_m^q) \leq (e+m)^{-1-\gamma},$$

which is summable. For the polynomial-menu common scale, the row contribution is bounded by

$$C_0 J_m (eD_m)^{-c_0 A^q} \leq C_0 (eD_m)^{\Gamma - c_0 A^q}.$$

This is summable over all integer dimensions if $c_0 A^q > \Gamma + 1$, and is stretched-exponentially summable under $D_m \geq \exp(cm^\kappa)$ if $c_0 A^q > \Gamma$. □

Corollary 7.7 (Finite-confidence distribution-free gauge certificates)**.** *Let $(X_{h,i})_{i \leq n_h}$, $1 \leq h \leq J$, be a finite menu of arbitrary self-adjoint or rectangular random matrix tuples, and let G_h be the all-plus gauge of the h-th tuple. Fix $0 < \delta < 1$.*

(i) *If $p > 0$ and deterministic envelopes $r_h > 0$ satisfy*

$$\mathbb{E}G_h^p \le r_h^p,$$

then the deterministic scales

$$s_h = r_h \left(\frac{J}{\delta}\right)^{1/p}$$

make all J scaled tuples contraction-normalized and all-plus certified with probability at least $1 - \delta$.

(ii) *More generally, if*

$$\mathbb{P}\{G_h > r_h u\} \le \Phi_h(u) \qquad (u \ge 1)$$

and $u_h \ge 1$ are chosen so that

$$\sum_{h=1}^{J} \Phi_h(u_h) \le \delta,$$

then the scales $s_h = r_h u_h$ have the same $1 - \delta$ conclusion.

(iii) *In particular, under the uniform sub-Weibull gauge tail*

$$\mathbb{P}\{G_h > r_h u\} \le C_0 \exp(-c_0 u^q) \qquad (u \ge 1),$$

the common multiplier

$$u = \max\left\{1, \left(\frac{\max\{0, \log(C_0 J/\delta)\}}{c_0}\right)^{1/q}\right\}$$

and scales $s_h = r_h u$ have the same $1 - \delta$ conclusion.

The conclusion means, simultaneously for all $h \le J$, that the scaled self-adjoint tuples obey

$$\operatorname{disc}(X_{h,1}/s_h, \dots, X_{h,n_h}/s_h) \le \sqrt{n_h},$$

with the analogous $\operatorname{disc}_{\mathrm{op}}$ *bound in the rectangular case. No independence, centering, or identical-distribution assumption is imposed.*

Proof. For the moment form, Markov's inequality and the union bound give

$$\mathbb{P}\left\{\max_{h \le J} \frac{G_h}{s_h} > 1\right\} \le \sum_{h=1}^{J} \left(\frac{r_h}{s_h}\right)^p \le \delta.$$

The tail-profile form is the same union bound:

$$\mathbb{P}\left\{\max_{h \le J} \frac{G_h}{r_h u_h} > 1\right\} \le \sum_{h=1}^{J} \Phi_h(u_h) \le \delta.$$

The sub-Weibull scale makes this last sum at most $JC_0 \exp(-c_0 u^q) \le \delta$. On the complementary event, each scaled tuple has gauge at most one, and Corollary 7.3 gives the pathwise all-plus discrepancy certificate for every menu element. □

Corollary 7.8 (Conditional gauge certificates)**.** *Let $(X_{h,i})_{i\le n_h}$, $1 \le h \le J$, be a finite menu of arbitrary self-adjoint or rectangular random matrix tuples, let G_h be the all-plus gauge of the h-th tuple, and let $\mathcal{F}$ be a sub-σ-algebra. Fix $0 < \delta < 1$.*

(i) *If $p > 0$ and positive $\mathcal{F}$-measurable random variables R_h satisfy*

$$\mathbb{E}(G_h^p \mid \mathcal{F}) \le R_h^p \qquad \text{a.s.},$$

then the $\mathcal{F}$-measurable scales

$$S_h := R_h \left(\frac{J}{\delta}\right)^{1/p}$$

make all J scaled tuples contraction-normalized and all-plus certified with probability at least $1 - \delta$.

(ii) *More generally, suppose positive $\mathcal{F}$-measurable R_h, $\mathcal{F}$-measurable $U_h \ge 1$, and $\mathcal{F}$-measurable error budgets $\eta_h \ge 0$ satisfy the chosen-threshold conditional tail bounds*

$$\mathbb{P}\{G_h > R_h U_h \mid \mathcal{F}\} \le \eta_h$$

and

$$\sum_{h=1}^{J} \eta_h \le \delta \qquad \text{a.s.}$$

Then $S_h := R_h U_h$ has the same $1 - \delta$ conclusion.

(iii) *In particular, if the conditional sub-Weibull gauge tails*

$$\mathbb{P}\{G_h > R_h u \mid \mathcal{F}\} \le C_0 \exp(-c_0 u^q) \qquad (u \ge 1)$$

hold a.s. for positive $\mathcal{F}$-measurable R_h, then

$$u = \max\left\{1, \left(\frac{\max\{0, \log(C_0 J/\delta)\}}{c_0}\right)^{1/q}\right\}, \qquad S_h := R_h u,$$

has the same $1 - \delta$ conclusion.

The conclusion is simultaneous over all $h \le J$: after division by the random scales S_h, the self-adjoint tuples have discrepancy at most $\sqrt{n_h}$, with the analogous $\mathrm{disc}_{\mathrm{op}}$ *bound for rectangular tuples. The side information $\mathcal{F}$ may contain arbitrary dependence between menu elements; no conditional independence, centering, or identical-distribution assumption is imposed.*

Proof. For the moment form, conditional Markov and the conditional union bound give

$$\mathbb{P}\left\{\max_{h\le J} \frac{G_h}{S_h} > 1 \mid \mathcal{F}\right\} \le \sum_{h=1}^{J} \frac{\mathbb{E}(G_h^p \mid \mathcal{F})}{S_h^p} \le \delta \qquad \text{a.s.}$$

Taking expectations gives the unconditional failure probability bound. The tail-profile form is identical:

$$\mathbb{P}\left\{\max_{h\le J} \frac{G_h}{R_h U_h} > 1 \mid \mathcal{F}\right\} \le \sum_{h=1}^{J} \eta_h \le \delta \qquad \text{a.s.}$$

The sub-Weibull choice makes this last sum at most $JC_0\exp(-c_0u^q)\le\delta$. On the complementary event, every scaled tuple has all-plus gauge at most one, so Corollary 7.3 gives the pathwise discrepancy certificate. □

Corollary 7.9 (Random active finite-confidence gauge certificates)**.** *Let $\mathcal{F}$ be side information and let $\mathcal{H}\subset\mathbb{N}$ be an almost surely finite $\mathcal{F}$-measurable active set. For each $h\in\mathbb{N}$, let $(X_{h,i})_{i\le n_h}$ be an arbitrary self-adjoint or rectangular random matrix tuple, and let G_h be its all-plus gauge. Fix $0<\delta<1$.*

(i) *Fix $p>0$. Suppose positive $\mathcal{F}$-measurable random variables R_h and positive $\mathcal{F}$-measurable budgets η_h satisfy*

$$\mathbb{E}(G_h^p\mid\mathcal{F})\le R_h^p\qquad a.s.$$

and

$$\mathbb{E}\sum_{h\in\mathcal{H}}\eta_h\le\delta.$$

Then the random scales

$$S_h:=R_h\eta_h^{-1/p}$$

make every active scaled tuple contraction-normalized and all-plus certified with probability at least $1-\delta$.

(ii) *More generally, suppose positive $\mathcal{F}$-measurable R_h, $\mathcal{F}$-measurable $U_h\ge1$, and nonnegative $\mathcal{F}$-measurable budgets η_h satisfy*

$$\mathbb{P}\{G_h>R_hU_h\mid\mathcal{F}\}\le\eta_h\qquad a.s.$$

and

$$\mathbb{E}\sum_{h\in\mathcal{H}}\eta_h\le\delta.$$

Then $S_h:=R_hU_h$ has the same $1-\delta$ conclusion on the random active menu.

(iii) *In particular, suppose the conditional sub-Weibull gauge tails*

$$\mathbb{P}\{G_h>R_hu\mid\mathcal{F}\}\le C_0\exp(-c_0u^q)\qquad(u\ge1)$$

hold a.s. for positive $\mathcal{F}$-measurable R_h, with fixed $C_0,c_0,q>0$. If $M\ge1$ is $\mathcal{F}$-measurable, $|\mathcal{H}|\le M$ a.s., and $u\ge1$ is $\mathcal{F}$-measurable with

$$\mathbb{E}\left[MC_0e^{-c_0u^q}\right]\le\delta,$$

then $S_h:=R_hu$ has the same $1-\delta$ conclusion. In particular, this holds for the random multiplier

$$u=\max\left\{1,\left(\frac{\max\{0,\log(C_0M/\delta)\}}{c_0}\right)^{1/q}\right\}.$$

Thus a finite-confidence certificate may be requested only for the active random catalogue selected by the environment. The confidence cost is the selected conditional budget, not the size of a deterministic ambient library. No independence, conditional independence, centering, or identical-distribution assumption is imposed.

Proof. For the moment form, conditional Markov gives

$$\mathbb{P}\{G_h > S_h \mid \mathcal{F}\} \le \eta_h \qquad \text{a.s.}$$

Since $\mathcal{H}$ is $\mathcal{F}$-measurable and finite,

$$\mathbb{P}\left\{\exists h \in \mathcal{H} : G_h > S_h\right\} \le \mathbb{E} \sum_{h\in\mathcal{H}} \eta_h \le \delta.$$

On the complementary event, every selected scaled tuple has all-plus gauge at most one, so Corollary 7.3 gives the simultaneous all-plus discrepancy certificate.

The chosen-threshold form is identical with

$$\mathbb{P}\{G_h > R_h U_h \mid \mathcal{F}\} \le \eta_h.$$

For the conditional sub-Weibull form, the selected failure probability is at most

$$\mathbb{E} \sum_{h\in\mathcal{H}} C_0 e^{-c_0 u^q} \le \mathbb{E}\left[M C_0 e^{-c_0 u^q}\right] \le \delta.$$

The displayed random multiplier makes $MC_0e^{-c_0u^q} \le \delta$ almost surely. □

Corollary 7.10 (Conditional gauge schedules)**.** *In the active triangular menu of Corollary 7.3, let $\mathcal{F}_m$ be side information for the m-th row and let $G_{m,h}$ be the all-plus gauge of the h-th active tuple.*

(i) *Fix $p > 0$. Suppose positive $\mathcal{F}_m$-measurable random variables $R_{m,h}$ and positive deterministic budgets $\eta_{m,h}$ satisfy*

$$\mathbb{E}(G_{m,h}^p \mid \mathcal{F}_m) \le R_{m,h}^p \qquad \text{a.s.}$$

and

$$\sum_m \sum_{h=1}^{J_m} \eta_{m,h} < \infty.$$

Then the random scales

$$S_{m,h} := R_{m,h}\eta_{m,h}^{-1/p}$$

are valid: almost surely, for all sufficiently large m, every active scaled self-adjoint or rectangular tuple is contraction-normalized and has the all-plus $\sqrt{n_m}$ discrepancy certificate.

(ii) *More generally, suppose positive $\mathcal{F}_m$-measurable $R_{m,h}$, $\mathcal{F}_m$-measurable $U_{m,h} \ge 1$, and nonnegative $\mathcal{F}_m$-measurable budgets $\eta_{m,h}$ satisfy*

$$\mathbb{P}\{G_{m,h} > R_{m,h}U_{m,h} \mid \mathcal{F}_m\} \le \eta_{m,h} \qquad \text{a.s.}$$

and

$$\sum_m \sum_{h=1}^{J_m} \mathbb{E}\eta_{m,h} < \infty.$$

Then $S_{m,h} := R_{m,h}U_{m,h}$ is valid in the same almost-sure sense.

(iii) *In particular, if the conditional sub-Weibull gauge tails*

$$\mathbb{P}\{G_{m,h} > R_{m,h}u \mid \mathcal{F}_m\} \le C_0 \exp(-c_0 u^q) \qquad (u \ge 1)$$

hold a.s. for positive $\mathcal{F}_m$-measurable $R_{m,h}$, with fixed $C_0, c_0, q > 0$, then for every $\gamma > 0$ the row multiplier

$$u_m = \max\left\{1, \left(\frac{\max\{0, \log(C_0 J_m) + (1+\gamma)\log(e+m)\}}{c_0}\right)^{1/q}\right\}$$

and scales $S_{m,h} := R_{m,h}u_m$ are valid.

The scales may depend on the side information and on the active environment. No independence, conditional independence, centering, or identical-distribution assumption is imposed.

Proof. For the moment form, conditional Markov gives

$$\mathbb{P}\{G_{m,h} > S_{m,h} \mid \mathcal{F}_m\} \le \frac{\mathbb{E}(G_{m,h}^p \mid \mathcal{F}_m)}{S_{m,h}^p} \le \eta_{m,h} \qquad \text{a.s.}$$

After taking expectations and summing over m, h, the bad events $\{G_{m,h} > S_{m,h}\}$ have summable probabilities. Borel–Cantelli therefore implies that almost surely only finitely many active scale failures occur. On the complementary eventual event, the pathwise all-plus gauge certificate of Corollary 7.3 applies to each scaled tuple.

The chosen-threshold tail form is identical, since

$$\mathbb{P}\{G_{m,h} > R_{m,h}U_{m,h}\} = \mathbb{E}\,\mathbb{P}\{G_{m,h} > R_{m,h}U_{m,h} \mid \mathcal{F}_m\} \le \mathbb{E}\eta_{m,h}.$$

For the conditional sub-Weibull choice, the m-th row failure probability is bounded by

$$\sum_{h=1}^{J_m} C_0 \exp(-c_0 u_m^q) \le (e+m)^{-1-\gamma},$$

which is summable. The preceding Borel–Cantelli argument completes the proof. □

Corollary 7.11 (Random active gauge schedules)**.** *For each m, let $\mathcal{F}_m$ be side information and let $\mathcal{H}_m \subset \mathbb{N}$ be an almost surely finite $\mathcal{F}_m$-measurable active set. For each $h \in \mathbb{N}$, let $(X_{m,h,i})_{i \le n_m}$ be an arbitrary self-adjoint or rectangular random matrix tuple, and let $G_{m,h}$ be its all-plus gauge.*

(i) *Fix $p > 0$. Suppose positive $\mathcal{F}_m$-measurable random variables $R_{m,h}$ and positive $\mathcal{F}_m$-measurable budgets $\eta_{m,h}$ satisfy*

$$\mathbb{E}(G_{m,h}^p \mid \mathcal{F}_m) \le R_{m,h}^p \qquad \text{a.s.}$$

and

$$\sum_m \mathbb{E} \sum_{h \in \mathcal{H}_m} \eta_{m,h} < \infty.$$

Then the random scales

$$S_{m,h} := R_{m,h}\eta_{m,h}^{-1/p}$$

are valid on the random active menu: almost surely, for all sufficiently large m and every $h \in \mathcal{H}_m$, the scaled tuple is contraction-normalized and has the all-plus $\sqrt{n_m}$ discrepancy certificate.

(ii) *More generally, suppose positive $\mathcal{F}_m$-measurable $R_{m,h}$, $\mathcal{F}_m$-measurable $U_{m,h} \geq 1$, and nonnegative $\mathcal{F}_m$-measurable budgets $\eta_{m,h}$ satisfy*

$$\mathbb{P}\{G_{m,h} > R_{m,h}U_{m,h} \mid \mathcal{F}_m\} \leq \eta_{m,h} \qquad a.s.$$

and

$$\sum_m \mathbb{E} \sum_{h \in \mathcal{H}_m} \eta_{m,h} < \infty.$$

Then $S_{m,h} := R_{m,h}U_{m,h}$ is valid on the random active menu in the same almost-sure sense.

(iii) *In particular, suppose the conditional sub-Weibull gauge tails*

$$\mathbb{P}\{G_{m,h} > R_{m,h}u \mid \mathcal{F}_m\} \leq C_0 \exp(-c_0 u^q) \qquad (u \geq 1)$$

hold a.s. for positive $\mathcal{F}_m$-measurable $R_{m,h}$, with fixed $C_0, c_0, q > 0$. If $M_m \geq 1$ is an $\mathcal{F}_m$-measurable random variable such that $|\mathcal{H}_m| \leq M_m$ a.s. and

$$\sum_m \mathbb{E}\left[M_m C_0 e^{-c_0 u_m^q}\right] < \infty,$$

then $S_{m,h} := R_{m,h}u_m$ is valid on the random active menu. In particular, this holds for

$$u_m = \max\left\{1, \left(\frac{\max\{0, \log(C_0 M_m) + (1+\gamma)\log(e+m)\}}{c_0}\right)^{1/q}\right\}$$

whenever M_m is deterministic, or more generally whenever the displayed series is summable.

Thus the active catalogue itself may be random and environment-selected. The cost is only the selected conditional budget, not a deterministic ambient library size. No independence, conditional independence, centering, or identical-distribution assumption is imposed.

Proof. For the moment form, conditional Markov gives

$$\mathbb{P}\{G_{m,h} > S_{m,h} \mid \mathcal{F}_m\} \leq \eta_{m,h} \qquad \text{a.s.}$$

Since $\mathcal{H}_m$ is $\mathcal{F}_m$-measurable and finite,

$$\mathbb{P}\left\{\exists h \in \mathcal{H}_m : G_{m,h} > S_{m,h}\right\} \leq \mathbb{E} \sum_{h \in \mathcal{H}_m} \eta_{m,h}.$$

The assumed summability and Borel–Cantelli give eventual simultaneous control on every selected active set. On that event, Corollary 7.3 gives the pathwise all-plus discrepancy certificate for each selected scaled tuple.

The chosen-threshold tail form is the same calculation with

$$\mathbb{P}\{G_{m,h} > R_{m,h}U_{m,h} \mid \mathcal{F}_m\} \leq \eta_{m,h}.$$

For the conditional sub-Weibull form, the selected-row failure probability is at most

$$\mathbb{E}\sum_{h\in\mathcal{H}_m} C_0 e^{-c_0 u_m^q} \le \mathbb{E}\left[M_m C_0 e^{-c_0 u_m^q}\right],$$

which is summable by assumption. If M_m is deterministic, the displayed choice gives $M_m C_0 e^{-c_0 u_m^q} \le (e+m)^{-1-\gamma}$, and hence summability. □

Corollary 7.12 (Full random matrices with Gaussian operator tails)**.** *Let $Y_1,\dots,Y_n$ be centered self-adjoint random matrices in $\mathrm{M}_D(\mathbb{C})$. Suppose that there is $K_0 \ge 1$ such that for every $a=(a_1,\dots,a_n)\in\mathbb{R}^n$ and every $t\ge 0$,*

$$\mathbb{P}\left\{\left\|\sum_{i=1}^n a_i Y_i\right\|_{\mathrm{op}} > K_0\|a\|_2(\sqrt{D}+t)\right\} \le 2e^{-t^2}.$$

Put

$$s_{D,n} := \sqrt{D}+\sqrt{3\log(en)}, \qquad A_i := \frac{Y_i}{K_0 s_{D,n}}.$$

Then, with probability at least $1-2(n+1)(en)^{-3}$, all A_i are contractions and

$$\mathrm{disc}(A_1,\dots,A_n) \le \sqrt{n}.$$

Thus, after the deterministic spectral scaling $K_0(\sqrt{D}+\sqrt{\log(en)})$, full random matrix ensembles satisfying the displayed operator-tail stability obey the Matrix Spencer bound in every ambient dimension D, with no independence among the Y_i needed beyond the displayed tail. This includes the usual independent GOE, GUE*, and centered sub-Gaussian Wigner-type ensembles, as well as correlated centered ensembles for which the same linear-combination operator-norm tail holds.*

Proof. Take $t=\sqrt{3\log(en)}$. Applying the assumed tail with $a=e_i$ gives

$$\mathbb{P}\{\|Y_i\|_{\mathrm{op}} > K_0 s_{D,n}\} \le 2(en)^{-3}.$$

A union bound over $i=1,\dots,n$ shows that all A_i are contractions except with probability at most $2n(en)^{-3}$. Applying the same tail with $a=(1,\dots,1)$ gives

$$\mathbb{P}\left\{\left\|\sum_{i=1}^n Y_i\right\|_{\mathrm{op}} > K_0\sqrt{n}\, s_{D,n}\right\} \le 2(en)^{-3}.$$

On the intersection of these two events, the all-plus signing certifies

$$\mathrm{disc}(A_1,\dots,A_n) \le \left\|\sum_{i=1}^n A_i\right\|_{\mathrm{op}} \le \sqrt{n}.$$

□

Corollary 7.13 (Active menus of full operator-tail random matrices)**.** *For each $m\ge 1$, let $D_m\ge 2$, $n_m\ge 1$, and $J_m\ge 1$. Let $\mathcal{H}_m$ be an environment, and for every active $1\le h\le J_m$ let*

$$Y_{m,h,1},\dots,Y_{m,h,n_m}\in \mathrm{M}_{D_m}(\mathbb{C})$$

be self-adjoint random matrices. Suppose that $K_{m,h} > 0$ is $\mathcal{H}_m$-measurable and that, conditionally on $\mathcal{H}_m$, for every $a = (a_1, \dots, a_{n_m}) \in \mathbb{R}^{n_m}$ and every $t \geq 0$,

$$\mathbb{P}\left\{ \left\| \sum_{i=1}^{n_m} a_i Y_{m,h,i} \right\|_{\mathrm{op}} > K_{m,h} \|a\|_2 (\sqrt{D_m} + t) \,\middle|\, \mathcal{H}_m \right\} \leq 2e^{-t^2}.$$

Let $\tau_m > 0$ be deterministic and define the normalized tuple

$$A_{m,h,i} := \frac{Y_{m,h,i}}{K_{m,h}(\sqrt{D_m} + \tau_m)}.$$

If

$$\sum_m J_m (n_m + 1) e^{-\tau_m^2} < \infty,$$

then, almost surely, for all sufficiently large m, every active full operator-tail stream is contraction-normalized and, for $1 \leq h \leq J_m$, has the all-plus certificate

$$\max_{i \leq n_m} \|A_{m,h,i}\|_{\mathrm{op}} \leq 1, \qquad \mathrm{disc}(A_{m,h,1}, \dots, A_{m,h,n_m}) \leq \sqrt{n_m}.$$

No coordinatewise sign symmetry is required. If $J_m \to \infty$ and the active prefixes exhaust a countable library, then every fixed library element is eventually covered.

The summability condition has the following explicit schedules. If, for some $\gamma > 0$,

$$\tau_m^2 \geq \log\bigl(2J_m(n_m + 1)\bigr) + (1 + \gamma)\log(e + m)$$

eventually, then the active-menu conclusion holds. If

$$J_m(n_m + 1) \leq (eD_m)^{\Gamma}$$

eventually, then along all integer dimensions $D_m = m$ it is enough to take

$$\tau_m^2 = R \log(eD_m), \qquad R > \Gamma + 1,$$

while along every stretched-exponential sparse dimension schedule $D_m \geq \exp(cm^{\kappa})$ it is enough that $R > \Gamma$.

Consequently, Gaussian, Wigner/GOE/GUE, correlated Gaussian, and other full random matrix libraries whose scalar linear combinations satisfy the displayed operator-tail bound are simultaneously Matrix-Spencer scale after their natural spectral normalization $K_{m,h}(\sqrt{D_m} + \tau_m)$, even when the normalizing tail scale is chosen from the environment.

Proof. Fix m and condition on $\mathcal{H}_m$. For a fixed active stream h, apply the conditional operator-tail hypothesis with $a = e_i$, $1 \leq i \leq n_m$, and with $a = (1, \dots, 1)$, all at $t = \tau_m$. A union bound gives conditional failure probability at most

$$2(n_m + 1)e^{-\tau_m^2}$$

for the event that either some $\|A_{m,h,i}\|_{\mathrm{op}} > 1$, or

$$\left\|\sum_{i=1}^{n_m} A_{m,h,i}\right\|_{\mathrm{op}} > \sqrt{n_m}.$$

Union over $h \le J_m$ and then average over $\mathcal{H}_m$. The displayed summability and Borel–Cantelli imply that, almost surely, all these events fail for every active h and all sufficiently large m. On that event the all-plus sign vector gives the discrepancy bound.

The arbitrary-growth schedule gives

$$2J_m(n_m+1)e^{-\tau_m^2} \le (e+m)^{-1-\gamma},$$

which is summable. Under $J_m(n_m+1) \le (eD_m)^\Gamma$ and $\tau_m^2 = R\log(eD_m)$, the failure term is at most a constant multiple of $(eD_m)^{\Gamma-R}$. This is summable over integer dimensions when $R > \Gamma + 1$, and is a stretched-exponential summable sequence along the sparse dimension schedules when $R > \Gamma$. □

Corollary 7.14 (Random active full operator-tail random matrices)**.** *For each $m \ge 1$, let $D_m \ge 2$, $n_m \ge 1$, let $\mathcal{F}_m$ be side information, and let $\mathcal{A}_m \subset \mathbb{N}$ be an almost surely finite $\mathcal{F}_m$-measurable active set. For every $h \in \mathbb{N}$, let*

$$Y_{m,h,1}, \dots, Y_{m,h,n_m} \in \mathrm{M}_{D_m}(\mathbb{C})$$

be self-adjoint random matrices. Suppose that $K_{m,h} > 0$ and $\tau_{m,h} > 0$ are $\mathcal{F}_m$-measurable and that, conditionally on $\mathcal{F}_m$, for every h, every $a \in \mathbb{R}^{n_m}$, and every $t \ge 0$,

$$\mathbb{P}\left\{\left\|\sum_{i=1}^{n_m} a_i Y_{m,h,i}\right\|_{\mathrm{op}} > K_{m,h}\|a\|_2(\sqrt{D_m}+t) \,\middle|\, \mathcal{F}_m\right\} \le 2e^{-t^2}.$$

Define

$$A_{m,h,i} := \frac{Y_{m,h,i}}{K_{m,h}(\sqrt{D_m}+\tau_{m,h})}.$$

If

$$\sum_m \mathbb{E}\sum_{h\in\mathcal{A}_m} 2(n_m+1)e^{-\tau_{m,h}^2} < \infty,$$

then, almost surely, for all sufficiently large m, every selected active full operator-tail stream is contraction-normalized and has the all-plus certificate

$$\max_{i\le n_m}\|A_{m,h,i}\|_{\mathrm{op}} \le 1, \qquad \mathrm{disc}(A_{m,h,1},\dots,A_{m,h,n_m}) \le \sqrt{n_m} \qquad (h\in\mathcal{A}_m).$$

Thus random active operator-tail libraries pay only their selected tail cost, not a deterministic ambient library size, and no coordinatewise sign symmetry is required.

In particular, if $M_m \ge 1$ is $\mathcal{F}_m$-measurable, $|\mathcal{A}_m| \le M_m$ a.s., and a common $\tau_m > 0$ satisfies

$$\sum_m \mathbb{E}\left[2M_m(n_m+1)e^{-\tau_m^2}\right] < \infty,$$

then the conclusion holds with $\tau_{m,h} = \tau_m$. If M_m is deterministic, it is enough to take, for any $\gamma > 0$,

$$\tau_m^2 \geq \log(2M_m(n_m+1)) + (1+\gamma)\log(e+m)$$

eventually. More generally, if

$$\mathbb{E}[M_m](n_m+1) \leq (eD_m)^{\Gamma}$$

eventually, then $\tau_m^2 = R\log(eD_m)$ is enough along all integer dimensions $D_m = m$ whenever $R > \Gamma + 1$, and along every stretched-exponential sparse dimension schedule $D_m \geq \exp(cm^{\kappa})$ whenever $R > \Gamma$.

Proof. Fix m and condition on $\mathcal{F}_m$. For a selected stream h, apply the conditional operator-tail hypothesis with $a = e_i$, $1 \leq i \leq n_m$, and with $a = (1, \dots, 1)$, all at $t = \tau_{m,h}$. The conditional probability that either some $\|A_{m,h,i}\|_{\mathrm{op}} > 1$ or

$$\left\| \sum_{i=1}^{n_m} A_{m,h,i} \right\|_{\mathrm{op}} > \sqrt{n_m}$$

is at most $2(n_m+1)e^{-\tau_{m,h}^2}$. Since $\mathcal{A}_m$ is $\mathcal{F}_m$-measurable and finite, the conditional selected-row failure probability is at most

$$\sum_{h \in \mathcal{A}_m} 2(n_m+1)e^{-\tau_{m,h}^2}.$$

After taking expectations, the assumed summability and Borel–Cantelli give eventual simultaneous control on every selected active set. On that event the all-plus sign vector gives the discrepancy bound.

The common-τ_m assertion follows by the bound

$$\sum_{h \in \mathcal{A}_m} 2(n_m+1)e^{-\tau_m^2} \leq 2M_m(n_m+1)e^{-\tau_m^2}.$$

The deterministic-size schedule makes this upper bound at most $(e+m)^{-1-\gamma}$. Under $\mathbb{E}[M_m](n_m+1) \leq (eD_m)^{\Gamma}$ and $\tau_m^2 = R\log(eD_m)$, the expected failure contribution is at most a constant multiple of $(eD_m)^{\Gamma - R}$, which is summable in the two stated regimes. □

Corollary 7.15 (Active menus from quadratic-form tails)**.** *In the setting of Corollary 7.13, replace the operator-tail hypothesis by the following scalar quadratic-form hypothesis. For every active $1 \leq h \leq J_m$, suppose that $q_{m,h} > 0$ is $\mathcal{H}_m$-measurable and that, conditionally on $\mathcal{H}_m$, the self-adjoint matrices $Y_{m,h,i}$ are centered and, for every $a \in \mathbb{R}^{n_m}$, every deterministic unit vector $u \in \mathbb{C}^{D_m}$, and every $s \geq 0$,*

$$\mathbb{P}\left\{ \left| \left\langle u, \sum_{i=1}^{n_m} a_i Y_{m,h,i} u \right\rangle \right| > q_{m,h}\|a\|_2 s \,\middle|\, \mathcal{H}_m \right\} \leq 2e^{-s^2}.$$

Set

$$\widehat{A}_{m,h,i} := \frac{Y_{m,h,i}}{5q_{m,h}(\sqrt{D_m} + \tau_m)}.$$

If

$$\sum_m J_m(n_m+1)e^{-\tau_m^2} < \infty,$$

then, almost surely, for all sufficiently large m, all active $\widehat{A}_{m,h,i}$ are contractions and the all-plus signing gives

$$\operatorname{disc}(\widehat{A}_{m,h,1},\dots,\widehat{A}_{m,h,n_m}) \le \sqrt{n_m} \qquad (1 \le h \le J_m).$$

The arbitrary-growth, all-integer, and sparse schedules of Corollary 7.13 apply with quadratic-form scale $5q_{m,h}(\sqrt{D_m}+\tau_m)$.

Consequently, Gaussian, correlated Gaussian, Wigner-type, and other self-adjoint random matrix libraries with uniform conditional scalar quadratic-form sub-Gaussian proxies are simultaneously Matrix-Spencer scale without requiring sign symmetry or entry independence.

Proof. Apply Lemma 7.111 conditionally on $\mathcal{H}_m$, for each active h. This gives the operator-tail hypothesis of Corollary 7.13 with $K_{m,h} = 5q_{m,h}$. The conclusions and schedules are therefore those of that corollary. □

Corollary 7.16 (Hermitian coefficient-series quadratic-form proxies)**.** *For each $m \ge 1$, let $D_m \ge 2$, $n_m \ge 1$, and $J_m \ge 1$. Let $\mathcal{H}_m$ be an environment. For each active $1 \le h \le J_m$, suppose that*

$$Y_{m,h,i} = \sum_{\alpha \in \mathcal{A}_{m,h,i}} \xi_{m,h,i,\alpha} B_{m,h,i,\alpha} \in \mathrm{M}_{D_m}(\mathbb{C}), \qquad 1 \le i \le n_m,$$

where the atom sets $\mathcal{A}_{m,h,i}$ are finite, the atoms $B_{m,h,i,\alpha}$ are self-adjoint, and the atoms and coefficient proxies $\sigma_{m,h,i,\alpha} \ge 0$ are $\mathcal{H}_m$-measurable. Assume that, conditionally on $\mathcal{H}_m$, the real scalar coefficients are centered and have the sub-Gaussian linear-combination estimate

$$\mathbb{P}\left\{ \left| \sum_{i,\alpha} c_{i,\alpha}\xi_{m,h,i,\alpha} \right| > C_{\mathrm{hser}} \left(\sum_{i,\alpha} c_{i,\alpha}^2 \sigma_{m,h,i,\alpha}^2 \right)^{1/2} t \,\middle|\, \mathcal{H}_m \right\} \le 2e^{-t^2}$$

for every real coefficient array $c = (c_{i,\alpha})$ and every $t \ge 0$. This is implied, after changing C_{hser}, by conditional independence and the usual real scalar sub-Gaussian mgf bounds. Define

$$Q_{m,h}^{\mathrm{h}} := \max_{1 \le i \le n_m} \sup_{\|u\|_2=1} \left(\sum_{\alpha \in \mathcal{A}_{m,h,i}} \sigma_{m,h,i,\alpha}^2 |\langle u, B_{m,h,i,\alpha} u\rangle|^2 \right)^{1/2}.$$

Assume $Q_{m,h}^{\mathrm{h}} > 0$ for the nonzero active streams, and set

$$H_{m,h,i} := \frac{Y_{m,h,i}}{5C_{\mathrm{hser}} Q_{m,h}^{\mathrm{h}} (\sqrt{D_m}+\tau_m)}.$$

If

$$\sum_m J_m(n_m+1)e^{-\tau_m^2} < \infty,$$

then, almost surely, for all sufficiently large m, all active $H_{m,h,i}$ are contractions and the all-plus signing gives

$$\operatorname{disc}(H_{m,h,1},\dots,H_{m,h,n_m}) \le \sqrt{n_m} \qquad (1 \le h \le J_m).$$

The arbitrary-growth, all-integer, and sparse schedules of Corollary 7.13 apply with scale

$$5C_{\mathrm{hser}}Q^{\mathrm{h}}_{m,h}(\sqrt{D_m}+\tau_m).$$

Moreover

$$(Q^{\mathrm{h}}_{m,h})^2 \le \max_{1\le i\le n_m} \left\| \sum_{\alpha\in\mathcal{A}_{m,h,i}} \sigma^2_{m,h,i,\alpha} B^2_{m,h,i,\alpha} \right\|_{\mathrm{op}}.$$

Thus Hermitian sub-Gaussian atom expansions, correlated Wigner-type patterns, random Hermitian feature blocks, and structured self-adjoint measurement libraries are covered once this atom square function is controlled.

Proof. Fix m,h and condition on $\mathcal{H}_m$. For a deterministic unit vector $u\in\mathbb{C}^{D_m}$ and $a\in\mathbb{R}^{n_m}$, apply the scalar coefficient tail with

$$c_{i,\alpha} = a_i\langle u, B_{m,h,i,\alpha}u\rangle.$$

Since the atoms are self-adjoint, these coefficients are real. The variance proxy is bounded by

$$\begin{aligned}\sum_{i,\alpha} c^2_{i,\alpha}\sigma^2_{m,h,i,\alpha} &= \sum_{i=1}^{n_m} a_i^2 \sum_{\alpha\in\mathcal{A}_{m,h,i}} \sigma^2_{m,h,i,\alpha}|\langle u, B_{m,h,i,\alpha}u\rangle|^2 \\ &\le (Q^{\mathrm{h}}_{m,h})^2\|a\|_2^2.\end{aligned}$$

Therefore the hypothesis of Corollary 7.15 holds with $q_{m,h}=C_{\mathrm{hser}}Q^{\mathrm{h}}_{m,h}$, and that corollary gives the simultaneous all-plus conclusion and schedules.

Finally, for each self-adjoint atom B and unit vector u, $|\langle u,Bu\rangle|^2 \le \langle u,B^2u\rangle$. Summing this inequality over α, taking the supremum over u, and then the maximum over i gives the displayed square-function bound. □

Corollary 7.17 (Independent-entry Hermitian quadratic-form proxies)**.** *For each $m\ge 1$, let $D_m\ge 2$, $n_m\ge 1$, and $J_m\ge 1$. Let $\mathcal{H}_m$ be an environment. For each active $1\le h\le J_m$, let*

$$\begin{aligned}Y_{m,h,i} = &\sum_{p=1}^{D_m} \xi_{m,h,i,p}E_{pp} \\ &+ \sum_{1\le p<q\le D_m} \left(\zeta_{m,h,i,pq}E_{pq} + \overline{\zeta_{m,h,i,pq}}E_{qp}\right), \qquad 1\le i\le n_m,\end{aligned}$$

be centered self-adjoint random matrices. Suppose that, conditionally on $\mathcal{H}_m$, all upper-triangular entries in each active stream are independent. Assume that $\xi_{m,h,i,p}$ is real sub-Gaussian with proxy $\alpha_{m,h,i,p}$, and that every real

projection $\mathrm{Re}(e^{\sqrt{-1}\theta}\zeta_{m,h,i,pq})$ *is sub-Gaussian with proxy* $\beta_{m,h,i,pq}$*, uniformly in* $\theta \in \mathbb{R}$*. Define*

$$Q^{\mathrm{ent}}_{m,h} := \max_{1\le i\le n_m} \sup_{\|u\|_2=1} \left[\sum_p \alpha^2_{m,h,i,p}|u_p|^4 + 4\sum_{p<q} \beta^2_{m,h,i,pq}|u_p|^2|u_q|^2\right]^{1/2}.$$

Assume $Q^{\mathrm{ent}}_{m,h} > 0$ *for the nonzero active streams, and set*

$$G_{m,h,i} := \frac{Y_{m,h,i}}{5C_{\mathrm{ent}}Q^{\mathrm{ent}}_{m,h}(\sqrt{D_m}+\tau_m)}.$$

If

$$\sum_m J_m(n_m+1)e^{-\tau_m^2} < \infty,$$

then, almost surely, for all sufficiently large m*, all active* $G_{m,h,i}$ *are contractions and the all-plus signing gives*

$$\mathrm{disc}(G_{m,h,1},\dots,G_{m,h,n_m}) \le \sqrt{n_m} \qquad (1\le h\le J_m).$$

The arbitrary-growth, all-integer, and sparse schedules of Corollary 7.13 apply with scale

$$5C_{\mathrm{ent}}Q^{\mathrm{ent}}_{m,h}(\sqrt{D_m}+\tau_m).$$

If all displayed entry proxies are bounded by $\sigma_{m,h}$*, then*

$$Q^{\mathrm{ent}}_{m,h} \le \sqrt{2}\,\sigma_{m,h}.$$

More generally, if

$$R^2_{m,h} \ge \max_{i,p}\left(\alpha^2_{m,h,i,p} + 2\sum_{q\ne p}\beta^2_{m,h,i,\{p,q\}}\right),$$

where $\beta_{m,h,i,\{p,q\}}$ *denotes the corresponding off-diagonal proxy, then* $Q^{\mathrm{ent}}_{m,h} \le R_{m,h}$*. Thus homogeneous Wigner/GOE/GUE libraries, variance-profile Hermitian ensembles, and sparse or banded upper-triangular models are covered at the natural* $Q(\sqrt{D}+\tau)$ *spectral scale by the direct all-plus route.*

Proof. Fix m, h and condition on $\mathcal{H}_m$. For a deterministic unit vector $u \in \mathbb{C}^{D_m}$, the quadratic form is

$$\langle u, Y_{m,h,i}u\rangle = \sum_p \xi_{m,h,i,p}|u_p|^2 + 2\sum_{p<q}\mathrm{Re}\left(\zeta_{m,h,i,pq}\overline{u_p}u_q\right).$$

By the standard linear-combination rule for independent sub-Gaussian variables, this scalar is conditionally sub-Gaussian with proxy at most

$$C_{\mathrm{ent}}\left[\sum_p \alpha^2_{m,h,i,p}|u_p|^4 + 4\sum_{p<q}\beta^2_{m,h,i,pq}|u_p|^2|u_q|^2\right]^{1/2}.$$

Since the entries are independent across the whole active stream, for every $a \in \mathbb{R}^{n_m}$ the quadratic form $\langle u, \sum_i a_i Y_{m,h,i}u\rangle$ has proxy at most $C_{\mathrm{ent}}Q^{\mathrm{ent}}_{m,h}\|a\|_2$, after increasing the universal constant to match the $2e^{-s^2}$ tail normalization.

Therefore Corollary 7.15 applies with $q_{m,h} = C_{\mathrm{ent}} Q^{\mathrm{ent}}_{m,h}$, giving the active all-plus conclusion and schedules.

If all proxies are at most $\sigma_{m,h}$, then, with $x_p = |u_p|^2$,

$$\sum_p x_p^2 + 4\sum_{p<q} x_p x_q = 2 - \sum_p x_p^2 \le 2,$$

which gives $Q^{\mathrm{ent}}_{m,h} \le \sqrt{2}\,\sigma_{m,h}$. For the row-variance bound, use $2x_p x_q \le x_p^2 + x_q^2$ to get

$$4\sum_{p<q} \beta^2_{m,h,i,pq} x_p x_q \le 2\sum_p x_p^2 \sum_{q\neq p} \beta^2_{m,h,i,\{p,q\}}.$$

The displayed row-budget hypothesis then bounds the whole square function by $R^2_{m,h}\sum_p x_p^2 \le R^2_{m,h}$. □

Corollary 7.18 (Bernstein quadratic-form active menus)**.** *For each $m \ge 1$, let $D_m \ge 2$, $n_m \ge 1$, and $J_m \ge 1$. Let $\mathcal{H}_m$ be an environment. For each active $1 \le h \le J_m$, let*

$$Y_{m,h,1}, \dots, Y_{m,h,n_m} \in \mathrm{M}_{D_m}(\mathbb{C})$$

be conditionally centered and self-adjoint. Suppose that $q_{m,h} \ge 0$ and $b_{m,h} \ge 0$ are $\mathcal{H}_m$-measurable. Assume that, conditionally on $\mathcal{H}_m$, for every $a \in \mathbb{R}^{n_m}$, every deterministic unit vector $u \in \mathbb{C}^{D_m}$, and every $s \ge 0$,

$$\mathbb{P}\left\{ \left| \left\langle u, \sum_{i=1}^{n_m} a_i Y_{m,h,i} u \right\rangle \right| > q_{m,h}\|a\|_2 s + b_{m,h}\|a\|_\infty s^2 \,\middle|\, \mathcal{H}_m \right\} \le 2e^{-s^2}.$$

Put

$$S_{m,h} := C_{\mathrm{Bq}} \left[q_{m,h}(\sqrt{D_m} + \tau_m) + b_{m,h}(\sqrt{D_m} + \tau_m)^2 \right],$$

where C_{Bq} is a sufficiently large numerical constant, and assume $S_{m,h} > 0$ for nonzero active streams. Define

$$B_{m,h,i} := \frac{Y_{m,h,i}}{S_{m,h}}.$$

If

$$\sum_m J_m(n_m + 1)e^{-\tau_m^2} < \infty,$$

then, almost surely, for all sufficiently large m, all active $B_{m,h,i}$ are contractions and the all-plus signing gives

$$\mathrm{disc}(B_{m,h,1}, \dots, B_{m,h,n_m}) \le \sqrt{n_m} \qquad (1 \le h \le J_m).$$

The arbitrary-growth, all-integer, and sparse schedules of Corollary 7.13 apply with the Bernstein scale $S_{m,h}$. Thus self-adjoint random matrix libraries whose fixed quadratic forms have a Bernstein rather than sub-Gaussian linear-tail are still covered by the direct all-plus route.

Proof. Fix m, h and condition on $\mathcal{H}_m$. The usual 1/4-net argument from Lemma 7.111, applied at scalar level $c_0(\sqrt{D_m} + \tau_m)$ with c_0 large enough, gives the following bound. Put $A_m = \sqrt{D_m} + \tau_m$ and

$$R_{m,h}(a) := C_{\mathrm{Bq}} \left[q_{m,h} \|a\|_2 A_m + b_{m,h} \|a\|_\infty A_m^2 \right].$$

Then for every deterministic $a \in \mathbb{R}^{n_m}$,

$$\mathbb{P}\left\{ \left\| \sum_i a_i Y_{m,h,i} \right\|_{\mathrm{op}} > R_{m,h}(a) \,\middle|\, \mathcal{H}_m \right\} \le 2e^{-\tau_m^2}.$$

Indeed, the net has cardinality at most 9^{2D_m}, and the constant c_0 absorbs this entropy exactly as in Lemma 7.111. Apply the displayed bound to $a = e_i$, $1 \le i \le n_m$, and to $a = (1, \dots, 1)$. The first choices give $\|B_{m,h,i}\|_{\mathrm{op}} \le 1$ outside conditional failure probability $2n_m e^{-\tau_m^2}$. The all-plus choice gives

$$\begin{aligned} \left\| \sum_i Y_{m,h,i} \right\|_{\mathrm{op}} &\le C_{\mathrm{Bq}} \left[q_{m,h} \sqrt{n_m} A_m + b_{m,h} A_m^2 \right] \\ &\le \sqrt{n_m}\, S_{m,h}, \end{aligned}$$

because $n_m \ge 1$. Thus the all-plus normalized sum has norm at most $\sqrt{n_m}$. Union over $h \le J_m$, average over the environment, and use Borel–Cantelli. The schedule statements are the same summability estimates as in Corollary 7.13. □

Corollary 7.19 (Centered covariance and Wishart active menus)**.** *For each $m \ge 1$, let $D_m \ge 2$, $n_m \ge 1$, and $J_m \ge 1$. Let $\mathcal{H}_m$ be an environment. For each active $1 \le h \le J_m$, let $r_{m,h,i} \ge 1$ and let*

$$x_{m,h,i,\ell} \in \mathbb{C}^{D_m}, \qquad 1 \le i \le n_m, \quad 1 \le \ell \le r_{m,h,i},$$

be conditionally independent random vectors. Assume that, conditionally on $\mathcal{H}_m$, every one-dimensional real projection is centered sub-Gaussian with proxy $\kappa_{m,h,i,\ell}$: for every deterministic unit vector $u \in \mathbb{C}^{D_m}$, every $\theta \in \mathbb{R}$, and every $t \ge 0$,

$$\mathbb{P}\left\{ \left| \mathrm{Re}\left(e^{\sqrt{-1}\theta} \langle x_{m,h,i,\ell}, u \rangle \right) \right| > \kappa_{m,h,i,\ell} t \,\middle|\, \mathcal{H}_m \right\} \le 2e^{-t^2}.$$

Set

$$Y_{m,h,i} := \frac{1}{r_{m,h,i}} \sum_{\ell=1}^{r_{m,h,i}} \left(x_{m,h,i,\ell} x_{m,h,i,\ell}^* - \mathbb{E}(x_{m,h,i,\ell} x_{m,h,i,\ell}^* \mid \mathcal{H}_m) \right),$$

and define

$$q_{m,h}^{\mathrm{cov}} := C_{\mathrm{cov}} \max_{1 \le i \le n_m} \left(\frac{1}{r_{m,h,i}^2} \sum_{\ell=1}^{r_{m,h,i}} \kappa_{m,h,i,\ell}^4 \right)^{1/2},$$

$$b_{m,h}^{\mathrm{cov}} := C_{\mathrm{cov}} \max_{i,\ell} \frac{\kappa_{m,h,i,\ell}^2}{r_{m,h,i}}.$$

With

$$S^{\mathrm{cov}}_{m,h} := C_{\mathrm{Bq}}\left[q^{\mathrm{cov}}_{m,h}(\sqrt{D_m}+\tau_m)+b^{\mathrm{cov}}_{m,h}(\sqrt{D_m}+\tau_m)^2\right],$$

the normalized matrices $Y_{m,h,i}/S^{\mathrm{cov}}_{m,h}$ *satisfy the conclusion of Corollary 7.18. In particular, if all* $\kappa_{m,h,i,\ell} \le \kappa_{m,h}$ *and all* $r_{m,h,i} \ge r_{m,h}$*, then the active covariance/Wishart scale is*

$$C\kappa^2_{m,h}\left[\frac{\sqrt{D_m}+\tau_m}{\sqrt{r_{m,h}}}+\frac{(\sqrt{D_m}+\tau_m)^2}{r_{m,h}}\right].$$

Rank-one centered covariance matrices correspond to $r_{m,h}=1$*, while averaged real or complex Wishart increments and random feature covariance blocks gain the displayed* $r^{-1/2}$ *and* r^{-1} *improvements.*

Proof. Fix m,h and condition on $\mathcal{H}_m$. For a unit vector u, put

$$Z_{i,\ell}(u) := |\langle x_{m,h,i,\ell},u\rangle|^2 - \mathbb{E}\left(|\langle x_{m,h,i,\ell},u\rangle|^2 \,\middle|\, \mathcal{H}_m\right).$$

The one-dimensional sub-Gaussian hypothesis implies the standard scalar Bernstein estimate for centered squares: for every real coefficient array $c_{i,\ell}$ and every $s \ge 0$,

$$V_c := \left(\sum_{i,\ell} c^2_{i,\ell}\kappa^4_{m,h,i,\ell}\right)^{1/2}, \qquad M_c := \max_{i,\ell}|c_{i,\ell}|\kappa^2_{m,h,i,\ell},$$

$$\mathbb{P}\left\{\left|\sum_{i,\ell} c_{i,\ell}Z_{i,\ell}(u)\right| > C_{\mathrm{cov}}\left(V_c s + M_c s^2\right)\,\middle|\,\mathcal{H}_m\right\} \le 2e^{-s^2}.$$

Apply this with $c_{i,\ell}=a_i/r_{m,h,i}$. Since

$$\left\langle u, \sum_i a_i Y_{m,h,i}u\right\rangle = \sum_{i,\ell}\frac{a_i}{r_{m,h,i}}Z_{i,\ell}(u),$$

the variance and jump terms are bounded by

$$q^{\mathrm{cov}}_{m,h}\|a\|_2, \qquad b^{\mathrm{cov}}_{m,h}\|a\|_\infty,$$

respectively. Hence the hypothesis of Corollary 7.18 holds with $q_{m,h}=q^{\mathrm{cov}}_{m,h}$ and $b_{m,h}=b^{\mathrm{cov}}_{m,h}$, and that corollary gives the active-menu conclusion and schedules. The homogeneous display follows by the definitions of $q^{\mathrm{cov}}_{m,h}$ and $b^{\mathrm{cov}}_{m,h}$. □

Corollary 7.20 (Independent-entry Hermitian Bernstein-form proxies)**.** *For each* $m \ge 1$*, let* $D_m \ge 2$*,* $n_m \ge 1$*, and* $J_m \ge 1$*. Let* $\mathcal{H}_m$ *be an environment. For each active* $1 \le h \le J_m$*, let*

$$Y_{m,h,i} = \sum_{p=1}^{D_m}\xi_{m,h,i,p}E_{pp} + \sum_{1\le p<q\le D_m}\left(\zeta_{m,h,i,pq}E_{pq}+\overline{\zeta_{m,h,i,pq}}E_{qp}\right), \qquad 1\le i\le n_m,$$

be centered self-adjoint random matrices. Suppose that, conditionally on $\mathcal{H}_m$, the upper-triangular variables in each active stream satisfy the following scalar Bernstein linear-combination estimate. There are $\mathcal{H}_m$-measurable proxies $\alpha_{m,h,i,p}, \lambda_{m,h,i,p} \geq 0$ and $\beta_{m,h,i,pq}, \mu_{m,h,i,pq} \geq 0$ such that, for every real coefficient arrays $c_{i,p}$, $d_{i,pq}$, every phases $\theta_{i,pq} \in \mathbb{R}$, and every $s \geq 0$,

$$Z(c,d,\theta) := \sum_{i,p} c_{i,p}\xi_{m,h,i,p} + \sum_{i,p<q} d_{i,pq}\,\mathrm{Re}\left(e^{\sqrt{-1}\theta_{i,pq}}\zeta_{m,h,i,pq}\right).$$

Then

$$\mathbb{P}\left\{|Z(c,d,\theta)| > C_{\mathrm{eB}}\left(Vs + Ms^2\right) \,\middle|\, \mathcal{H}_m\right\} \leq 2e^{-s^2},$$

where

$$V^2 := \sum_{i,p} c_{i,p}^2\alpha_{m,h,i,p}^2 + \sum_{i,p<q} d_{i,pq}^2\beta_{m,h,i,pq}^2,$$

$$M := \max\left\{\max_{i,p}|c_{i,p}|\lambda_{m,h,i,p}, \max_{i,p<q}|d_{i,pq}|\mu_{m,h,i,pq}\right\}.$$

This estimate is implied, after changing C_{eB}, by conditional independence and the usual scalar Bernstein mgf bounds for the displayed real variables. Define

$$Q_{m,h}^{\mathrm{eB}} := \max_{1\leq i\leq n_m}\sup_{\|u\|_2=1}\left[\sum_p \alpha_{m,h,i,p}^2|u_p|^4 + 4\sum_{p<q}\beta_{m,h,i,pq}^2|u_p|^2|u_q|^2\right]^{1/2}$$

and

$$B_{m,h}^{\mathrm{eB}} := \max_{1\leq i\leq n_m}\max\left\{\max_p \lambda_{m,h,i,p}, \max_{p<q}\mu_{m,h,i,pq}\right\}.$$

Set

$$S_{m,h}^{\mathrm{eB}} := C_{\mathrm{Bq}}C_{\mathrm{eB}}\left[Q_{m,h}^{\mathrm{eB}}(\sqrt{D_m}+\tau_m) + B_{m,h}^{\mathrm{eB}}(\sqrt{D_m}+\tau_m)^2\right],$$

and assume this scale is positive for nonzero active streams. If

$$\sum_m J_m(n_m+1)e^{-\tau_m^2} < \infty,$$

then, almost surely, for all sufficiently large m, all active matrices $Y_{m,h,i}/S_{m,h}^{\mathrm{eB}}$ are contractions and the all-plus signing gives

$$\mathrm{disc}\left(Y_{m,h,1}/S_{m,h}^{\mathrm{eB}}, \ldots, Y_{m,h,n_m}/S_{m,h}^{\mathrm{eB}}\right) \leq \sqrt{n_m}.$$

The same arbitrary-growth, all-integer, and sparse schedules apply. If $\alpha, \beta \leq \sigma_{m,h}$ and $\lambda, \mu \leq L_{m,h}$ throughout the active stream, then

$$Q_{m,h}^{\mathrm{eB}} \leq \sqrt{2}\,\sigma_{m,h}, \qquad B_{m,h}^{\mathrm{eB}} \leq L_{m,h},$$

so centered bounded, Poissonized, exponential-entry, gamma-entry, and other Bernstein-entry Hermitian libraries enter at scale

$$C\left[\sigma_{m,h}(\sqrt{D_m}+\tau_m) + L_{m,h}(\sqrt{D_m}+\tau_m)^2\right].$$

Proof. Fix m, h and condition on $\mathcal{H}_m$. For a deterministic unit vector u, write $x_p = |u_p|^2$. The quadratic form is

$$\langle u, Y_{m,h,i} u\rangle = \sum_p \xi_{m,h,i,p} x_p + 2\sum_{p<q} \mathrm{Re}\left(\zeta_{m,h,i,pq}\overline{u_p} u_q\right).$$

For $\sum_i a_i \langle u, Y_{m,h,i}u\rangle$, choose

$$c_{i,p} = a_i x_p, \qquad d_{i,pq} = 2a_i |u_p||u_q|,$$

and choose the phases so that the second term agrees with $Z(c,d,\theta)$'s rotated off-diagonal real parts. The variance term is bounded by $(Q^{\mathrm{eB}}_{m,h})^2 \|a\|_2^2$. Also $|c_{i,p}| \le |a_i|$ and

$$|d_{i,pq}| = 2|a_i||u_p||u_q| \le |a_i|,$$

so the jump term is bounded by $B^{\mathrm{eB}}_{m,h}\|a\|_\infty$. Therefore the hypothesis of Corollary 7.18 holds with $q_{m,h} = C_{\mathrm{eB}} Q^{\mathrm{eB}}_{m,h}$ and $b_{m,h} = C_{\mathrm{eB}} B^{\mathrm{eB}}_{m,h}$, giving the active all-plus conclusion. The homogeneous bound for Q^{eB} is the same computation as in Corollary 7.17; the bound for B^{eB} is immediate. □

Corollary 7.21 (Active menus of rectangular full operator-tail matrices)**.** *For each $m \ge 1$, let $d_{m,1}, d_{m,2} \ge 1$,*

$$D^{\mathrm{rec}}_m := d_{m,1} + d_{m,2},$$

let $n_m \ge 1$, and let $J_m \ge 1$. Let $\mathcal{H}_m$ be an environment, and for every active $1 \le h \le J_m$ let

$$X_{m,h,1}, \dots, X_{m,h,n_m} \in \mathrm{M}_{d_{m,1}\times d_{m,2}}(\mathbb{C})$$

be random rectangular matrices. Suppose that $K_{m,h} > 0$ is $\mathcal{H}_m$-measurable and that, conditionally on $\mathcal{H}_m$, for every $a = (a_1, \dots, a_{n_m}) \in \mathbb{R}^{n_m}$ and every $t \ge 0$,

$$\mathbb{P}\left\{ \left\| \sum_{i=1}^{n_m} a_i X_{m,h,i} \right\|_{\mathrm{op}} > K_{m,h}\|a\|_2 (\sqrt{D^{\mathrm{rec}}_m} + t) \,\middle|\, \mathcal{H}_m \right\} \le 2e^{-t^2}.$$

Let $\tau_m > 0$ be deterministic and define

$$Z_{m,h,i} := \frac{X_{m,h,i}}{K_{m,h}(\sqrt{D^{\mathrm{rec}}_m} + \tau_m)}.$$

If

$$\sum_m J_m (n_m + 1) e^{-\tau_m^2} < \infty,$$

then, almost surely, for all sufficiently large m, every active rectangular full operator-tail stream is contraction-normalized and, for $1 \le h \le J_m$, has the all-plus certificate

$$\max_{i \le n_m} \|Z_{m,h,i}\|_{\mathrm{op}} \le 1,$$

$$\begin{aligned} \mathrm{disc}(Z_{m,h,1}, \dots, Z_{m,h,n_m}) &\le \left\| \sum_{i=1}^{n_m} Z_{m,h,i} \right\|_{\mathrm{op}} \\ &\le \sqrt{n_m}. \end{aligned}$$

No coordinatewise sign symmetry, entry independence, or self-adjointness is required beyond the displayed full operator-tail hypothesis. If $J_m \to \infty$ and the active prefixes exhaust a countable library, then every fixed library element is eventually covered.

The same schedules as in Corollary 7.13 hold with D_m replaced by D_m^{rec}. In particular, if, for some $\gamma > 0$,

$$\tau_m^2 \geq \log(2J_m(n_m+1)) + (1+\gamma)\log(e+m)$$

eventually, then the active-menu conclusion holds. If

$$J_m(n_m+1) \leq (eD_m^{\mathrm{rec}})^{\Gamma}$$

eventually, then along all integer left-plus-right dimensions $D_m^{\mathrm{rec}} = m$ it is enough to take

$$\tau_m^2 = R\log(eD_m^{\mathrm{rec}}), \qquad R > \Gamma + 1,$$

while along every stretched-exponential sparse schedule $D_m^{\mathrm{rec}} \geq \exp(cm^{\kappa})$ it is enough that $R > \Gamma$.

Consequently, rectangular Gaussian, rectangular sub-Gaussian, correlated rectangular matrix-series, randomized feature-block, measurement-matrix, and other non-self-adjoint full operator-tail libraries are simultaneously Matrix-Spencer scale after their natural spectral normalization $K_{m,h}(\sqrt{D_m^{\mathrm{rec}}} + \tau_m)$.

Proof. Fix m and condition on $\mathcal{H}_m$. For a fixed active stream h, apply the conditional operator-tail hypothesis with $a = e_i$, $1 \leq i \leq n_m$, and with $a = (1, \dots, 1)$, all at $t = \tau_m$. The conditional probability that some $\|Z_{m,h,i}\|_{\mathrm{op}} > 1$, or that

$$\left\| \sum_{i=1}^{n_m} Z_{m,h,i} \right\|_{\mathrm{op}} > \sqrt{n_m},$$

is at most $2(n_m+1)e^{-\tau_m^2}$. Union over $h \leq J_m$, average over $\mathcal{H}_m$, and apply Borel–Cantelli to the displayed summability assumption. On the resulting eventual event, the all-plus sign vector is an admissible rectangular signing and gives the discrepancy bound.

The schedule estimates are identical to those in Corollary 7.13: the arbitrary-growth hypothesis gives

$$2J_m(n_m+1)e^{-\tau_m^2} \leq (e+m)^{-1-\gamma},$$

and the polynomial active-menu condition with $\tau_m^2 = R\log(eD_m^{\mathrm{rec}})$ gives a failure term bounded by a constant multiple of $(eD_m^{\mathrm{rec}})^{\Gamma - R}$. This is summable over all integer left-plus-right dimensions for $R > \Gamma + 1$, and along the sparse schedules for $R > \Gamma$. □

Corollary 7.22 (Random active rectangular full operator-tail matrices)**.** *For each $m \geq 1$, let $d_{m,1}, d_{m,2} \geq 1$,*

$$D_m^{\mathrm{rec}} := d_{m,1} + d_{m,2},$$

let $n_m \geq 1$, let $\mathcal{F}_m$ be side information, and let $\mathcal{A}_m \subset \mathbb{N}$ be an almost surely finite $\mathcal{F}_m$-measurable active set. For every $h \in \mathbb{N}$, let

$$X_{m,h,1}, \dots, X_{m,h,n_m} \in \mathrm{M}_{d_{m,1} \times d_{m,2}}(\mathbb{C})$$

be random rectangular matrices. Suppose that $K_{m,h} > 0$ and $\tau_{m,h} > 0$ are $\mathcal{F}_m$-measurable and that, conditionally on $\mathcal{F}_m$, for every h, every $a \in \mathbb{R}^{n_m}$, and every $t \geq 0$,

$$\mathbb{P}\left\{ \left\| \sum_{i=1}^{n_m} a_i X_{m,h,i} \right\|_{\mathrm{op}} > K_{m,h} \|a\|_2 (\sqrt{D_m^{\mathrm{rec}}} + t) \,\middle|\, \mathcal{F}_m \right\} \leq 2e^{-t^2}.$$

Define

$$Z_{m,h,i} := \frac{X_{m,h,i}}{K_{m,h}(\sqrt{D_m^{\mathrm{rec}}} + \tau_{m,h})}.$$

If

$$\sum_m \mathbb{E} \sum_{h \in \mathcal{A}_m} 2(n_m + 1) e^{-\tau_{m,h}^2} < \infty,$$

then, almost surely, for all sufficiently large m, every selected active rectangular full operator-tail stream is contraction-normalized and has the all-plus certificate

$$\max_{i \leq n_m} \|Z_{m,h,i}\|_{\mathrm{op}} \leq 1,$$

$$\mathrm{disc}(Z_{m,h,1}, \dots, Z_{m,h,n_m}) \leq \left\| \sum_{i=1}^{n_m} Z_{m,h,i} \right\|_{\mathrm{op}} \leq \sqrt{n_m} \qquad (h \in \mathcal{A}_m).$$

Thus random active rectangular or non-self-adjoint operator-tail libraries pay only their selected tail cost, not a deterministic ambient library size, and no coordinatewise sign symmetry, entry independence, or self-adjointness is required beyond the displayed full operator-tail hypothesis.

In particular, if $M_m \geq 1$ is $\mathcal{F}_m$-measurable, $|\mathcal{A}_m| \leq M_m$ a.s., and a common $\tau_m > 0$ satisfies

$$\sum_m \mathbb{E}\left[2M_m(n_m + 1) e^{-\tau_m^2} \right] < \infty,$$

then the conclusion holds with $\tau_{m,h} = \tau_m$. If M_m is deterministic, it is enough to take, for any $\gamma > 0$,

$$\tau_m^2 \geq \log(2M_m(n_m + 1)) + (1 + \gamma)\log(e + m)$$

eventually. More generally, if

$$\mathbb{E}[M_m](n_m + 1) \leq (eD_m^{\mathrm{rec}})^{\Gamma}$$

eventually, then $\tau_m^2 = R\log(eD_m^{\mathrm{rec}})$ is enough along all integer left-plus-right dimensions $D_m^{\mathrm{rec}} = m$ whenever $R > \Gamma + 1$, and along every stretched-exponential sparse schedule $D_m^{\mathrm{rec}} \geq \exp(cm^{\kappa})$ whenever $R > \Gamma$.

Proof. Fix m and condition on $\mathcal{F}_m$. For a selected stream h, apply the conditional operator-tail hypothesis with $a = e_i$, $1 \le i \le n_m$, and with $a = (1, \dots, 1)$, all at $t = \tau_{m,h}$. The conditional probability that either some $\|Z_{m,h,i}\|_{\mathrm{op}} > 1$ or

$$\left\| \sum_{i=1}^{n_m} Z_{m,h,i} \right\|_{\mathrm{op}} > \sqrt{n_m}$$

is at most $2(n_m + 1)e^{-\tau_{m,h}^2}$. Since $\mathcal{A}_m$ is $\mathcal{F}_m$-measurable and finite, the conditional selected-row failure probability is at most

$$\sum_{h \in \mathcal{A}_m} 2(n_m + 1)e^{-\tau_{m,h}^2}.$$

Taking expectations and using Borel–Cantelli gives eventual simultaneous control on every selected active set. On that event the all-plus sign vector is an admissible rectangular signing and gives the discrepancy bound.

The common-τ_m assertion follows from

$$\sum_{h \in \mathcal{A}_m} 2(n_m + 1)e^{-\tau_m^2} \le 2M_m(n_m + 1)e^{-\tau_m^2}.$$

The deterministic-size and polynomial expected-size schedules are the same calculations as in Corollary 7.14, with D_m replaced by D_m^{rec}. □

Lemma 7.23 (Bilinear-form tails imply rectangular operator tails)**.** *Let $X_1, \dots, X_n \in \mathrm{M}_{d_1 \times d_2}(\mathbb{C})$ be random rectangular matrices relative to a sigma-field $\mathcal{H}$, and put $D_{\mathrm{rec}} = d_1 + d_2$. Suppose that there is $q > 0$ such that, conditionally on $\mathcal{H}$, for every $a \in \mathbb{R}^n$, every pair of deterministic unit vectors $u \in \mathbb{C}^{d_1}$, $v \in \mathbb{C}^{d_2}$, and every $s \ge 0$,*

$$\mathbb{P}\left\{ \left| \left\langle u, \sum_{i=1}^{n} a_i X_i v \right\rangle \right| > q\|a\|_2 s \,\middle|\, \mathcal{H} \right\} \le 2e^{-s^2}.$$

Then, conditionally on $\mathcal{H}$, for every $a \in \mathbb{R}^n$ and every $t \ge 0$,

$$\mathbb{P}\left\{ \left\| \sum_{i=1}^{n} a_i X_i \right\|_{\mathrm{op}} > 5q\|a\|_2(\sqrt{D_{\mathrm{rec}}} + t) \,\middle|\, \mathcal{H} \right\} \le 2e^{-t^2}.$$

Proof. Fix a and t, and put $S = \sum_i a_i X_i$. Let $\mathcal{N}_1$ and $\mathcal{N}_2$ be deterministic $1/4$-nets of the unit spheres of $\mathbb{C}^{d_1}$ and $\mathbb{C}^{d_2}$, viewed as real spheres, with $|\mathcal{N}_j| \le 9^{2d_j}$. The standard two-net estimate gives

$$\|S\|_{\mathrm{op}} \le 2 \max_{u \in \mathcal{N}_1,\ v \in \mathcal{N}_2} |\langle u, Sv \rangle|.$$

Therefore the conditional failure probability is at most

$$\begin{aligned} &2|\mathcal{N}_1||\mathcal{N}_2| \exp\left(-\frac{25}{4}(\sqrt{D_{\mathrm{rec}}} + t)^2 \right) \\ &\qquad \le 2 \exp\left(2D_{\mathrm{rec}} \log 9 - \frac{25}{4} D_{\mathrm{rec}} - \frac{25}{4} t^2 \right) \le 2e^{-t^2}, \end{aligned}$$

since $25/4 > 2 \log 9$. □

Corollary 7.24 (Active menus of bilinear-form rectangular matrices)**.** *In the setting of Corollary 7.21, replace the rectangular operator-tail hypothesis by the following scalar bilinear-form hypothesis. For each active $1 \le h \le J_m$, suppose that $q_{m,h} > 0$ is $\mathcal{H}_m$-measurable and that, conditionally on $\mathcal{H}_m$, for every $a \in \mathbb{R}^{n_m}$, every deterministic unit $u \in \mathbb{C}^{d_{m,1}}$, every deterministic unit $v \in \mathbb{C}^{d_{m,2}}$, and every $s \ge 0$,*

$$\mathbb{P}\left\{\left|\left\langle u, \sum_{i=1}^{n_m} a_i X_{m,h,i} v \right\rangle\right| > q_{m,h}\|a\|_2 s \,\middle|\, \mathcal{H}_m\right\} \le 2e^{-s^2}.$$

Let $\tau_m > 0$ be deterministic and set

$$\widetilde{Z}_{m,h,i} := \frac{X_{m,h,i}}{5q_{m,h}(\sqrt{D_m^{\mathrm{rec}}} + \tau_m)}.$$

If

$$\sum_m J_m(n_m + 1)e^{-\tau_m^2} < \infty,$$

then, almost surely, for all sufficiently large m, all active $\widetilde{Z}_{m,h,i}$ are contractions and the all-plus signing gives

$$\operatorname{disc}(\widetilde{Z}_{m,h,1}, \dots, \widetilde{Z}_{m,h,n_m}) \le \sqrt{n_m} \qquad (1 \le h \le J_m).$$

The arbitrary-growth, all-integer, and sparse schedules of Corollary 7.21 apply with the bilinear scale $5q_{m,h}(\sqrt{D_m^{\mathrm{rec}}} + \tau_m)$.

Consequently, conditionally Gaussian rectangular matrix series, correlated rectangular Gaussian blocks, independent-entry rectangular sub-Gaussian matrices, randomized feature blocks, and measurement-matrix libraries are covered whenever all fixed bilinear forms have conditional sub-Gaussian proxy at most $q_{m,h}\|a\|_2$.

Proof. Apply Lemma 7.23 conditionally on $\mathcal{H}_m$, for each active h. This gives the operator-tail hypothesis of Corollary 7.21 with $K_{m,h} = 5q_{m,h}$, and all conclusions follow from that corollary. The listed examples are exactly the standard one-dimensional Gaussian or sub-Gaussian fixed-bilinear-form estimates plugged into the displayed hypothesis. □

Corollary 7.25 (Random active bilinear-form rectangular matrices)**.** *In the setting of Corollary 7.22, replace the rectangular operator-tail hypothesis by the following scalar bilinear-form hypothesis. Suppose that $q_{m,h} > 0$ and $\tau_{m,h} > 0$ are $\mathcal{F}_m$-measurable and that, conditionally on $\mathcal{F}_m$, for every h, every $a \in \mathbb{R}^{n_m}$, every deterministic unit $u \in \mathbb{C}^{d_{m,1}}$, every deterministic unit $v \in \mathbb{C}^{d_{m,2}}$, and every $s \ge 0$,*

$$\mathbb{P}\left\{\left|\left\langle u, \sum_{i=1}^{n_m} a_i X_{m,h,i} v \right\rangle\right| > q_{m,h}\|a\|_2 s \,\middle|\, \mathcal{F}_m\right\} \le 2e^{-s^2}.$$

Define

$$\widetilde{Z}_{m,h,i} := \frac{X_{m,h,i}}{5q_{m,h}(\sqrt{D_m^{\mathrm{rec}}} + \tau_{m,h})}.$$

If

$$\sum_m \mathbb{E} \sum_{h\in\mathcal{A}_m} 2(n_m+1)e^{-\tau_{m,h}^2} < \infty,$$

then, almost surely, for all sufficiently large m, every selected active bilinear-form rectangular stream is contraction-normalized and has the all-plus certificate

$$\operatorname{disc}(\widetilde{Z}_{m,h,1},\dots,\widetilde{Z}_{m,h,n_m}) \le \sqrt{n_m} \qquad (h\in\mathcal{A}_m).$$

The common-τ_m, deterministic-size, and dimension-schedule conclusions from the random-active rectangular operator-tail corollary apply with the bilinear scale

$$5q_{m,h}(\sqrt{D_m^{\mathrm{rec}}}+\tau_{m,h}).$$

Thus random active conditionally Gaussian rectangular series, correlated rectangular Gaussian blocks, independent-entry rectangular sub-Gaussian matrices, randomized feature blocks, and measurement-matrix libraries pay only their selected expected bilinear tail count.

Proof. Apply Lemma 7.23 conditionally on $\mathcal{F}_m$, for each selected h. This gives the operator-tail hypothesis of Corollary 7.22 with $K_{m,h}=5q_{m,h}$. The conclusion and schedules are those of that corollary. □

Corollary 7.26 (Haar unitary and orthogonal active menus)**.** *For each $m\ge 1$, let $D_m\ge 2$, $n_m\ge 1$, and $J_m\ge 1$. Let $\mathcal{H}_m$ be an environment. For each active $1\le h\le J_m$, let*

$$U_{m,h,1},\dots,U_{m,h,n_m}\in \mathrm{M}_{D_m}(\mathbb{C})$$

be conditionally independent random matrices, each of which is either Haar unitary on $\mathbb{C}^{D_m}$ or Haar orthogonal on $\mathbb{R}^{D_m}$, embedded in $\mathrm{M}_{D_m}(\mathbb{C})$. The choices of unitary versus orthogonal may depend on $\mathcal{H}_m$. Put

$$q_m^{\mathrm{Haar}} := C_{\mathrm{Haar}} D_m^{-1/2}$$

and define

$$\widehat{U}_{m,h,i} := \frac{U_{m,h,i}}{5q_m^{\mathrm{Haar}}(\sqrt{2D_m}+\tau_m)}.$$

If

$$\sum_m J_m(n_m+1)e^{-\tau_m^2} < \infty,$$

then, almost surely, for all sufficiently large m, all active $\widehat{U}_{m,h,i}$ are contractions and the all-plus signing gives

$$\operatorname{disc}(\widehat{U}_{m,h,1},\dots,\widehat{U}_{m,h,n_m}) \le \sqrt{n_m} \qquad (1\le h\le J_m).$$

The arbitrary-growth, all-integer, and sparse schedules of Corollary 7.21 apply with $D_m^{\mathrm{rec}}=2D_m$. In particular, if $\tau_m\le c\sqrt{D_m}$ eventually, then the normalizing denominator is bounded above and below by positive constants depending only on c. Thus active libraries of independent Haar unitary, Haar orthogonal, and mixed Haar unitary/orthogonal matrices are Matrix-Spencer scale after only a universal constant normalization.

Proof. Fix m, h and condition on $\mathcal{H}_m$. For deterministic unit vectors $u, v \in \mathbb{C}^{D_m}$, if U is Haar unitary, then Uv is uniform on the complex unit sphere, and the standard spherical-coordinate tail gives

$$\mathbb{P}\left\{\left|\mathrm{Re}\left(e^{\sqrt{-1}\theta}\langle u, Uv\rangle\right)\right| > CD_m^{-1/2}s \,\middle|\, \mathcal{H}_m\right\} \le 2e^{-s^2}$$

for every $\theta \in \mathbb{R}$. For Haar orthogonal matrices embedded in $\mathrm{M}_{D_m}(\mathbb{C})$, decompose u and v into real and imaginary parts and apply the same real spherical-coordinate tail to the resulting four real bilinear forms, increasing the numerical constant. Conditional independence therefore gives, for every $a \in \mathbb{R}^{n_m}$,

$$\mathbb{P}\left\{\left|\left\langle u, \sum_i a_i U_{m,h,i} v\right\rangle\right| > q_m^{\mathrm{Haar}}\|a\|_2 s \,\middle|\, \mathcal{H}_m\right\} \le 2e^{-s^2}.$$

Apply Corollary 7.24 with $d_{m,1} = d_{m,2} = D_m$ and $q_{m,h} = q_m^{\mathrm{Haar}}$. The final constant-scale claim follows from

$$5C_{\mathrm{Haar}}D_m^{-1/2}(\sqrt{2D_m} + \tau_m) = 5C_{\mathrm{Haar}}(\sqrt{2} + \tau_m/\sqrt{D_m}).$$

□

Corollary 7.27 (Independent-entry rectangular bilinear proxies)**.** *For each $m \ge 1$, let $d_{m,1}, d_{m,2} \ge 1$, $D_m^{\mathrm{rec}} = d_{m,1} + d_{m,2}$, $n_m \ge 1$, and $J_m \ge 1$. Let $\mathcal{H}_m$ be an environment. For each active $1 \le h \le J_m$, let*

$$X_{m,h,i} = (x_{m,h,i,r,s})_{1\le r\le d_{m,1},\, 1\le s\le d_{m,2}} \in \mathrm{M}_{d_{m,1}\times d_{m,2}}(\mathbb{C}), \qquad 1 \le i \le n_m,$$

be centered rectangular random matrices. Suppose that, conditionally on $\mathcal{H}_m$, the entries in the active stream are independent and have $\mathcal{H}_m$-measurable sub-Gaussian proxies $\sigma_{m,h,i,r,s} \ge 0$ in the following linear-combination sense: for every coefficient array $c = (c_{i,r,s})$ and every $t \ge 0$,

$$\mathbb{P}\left\{\left|\sum_{i,r,s} c_{i,r,s}x_{m,h,i,r,s}\right| > C_{\mathrm{ent}}\left(\sum_{i,r,s}|c_{i,r,s}|^2\sigma_{m,h,i,r,s}^2\right)^{1/2} t \,\middle|\, \mathcal{H}_m\right\} \le 2e^{-t^2},$$

where C_{ent} is an absolute constant. This displayed hypothesis is implied, after changing C_{ent}, by the usual conditional real or complex sub-Gaussian entry mgf bounds. Define

$$Q_{m,h} := \max_{1\le i\le n_m} \sup_{\substack{\|u\|_2=1\\ \|v\|_2=1}} \left(\sum_{r=1}^{d_{m,1}}\sum_{s=1}^{d_{m,2}} |u_r|^2|v_s|^2\sigma_{m,h,i,r,s}^2\right)^{1/2}.$$

Assume $Q_{m,h} > 0$ for the nonzero active streams, and define

$$W_{m,h,i} := \frac{X_{m,h,i}}{5C_{\mathrm{ent}}Q_{m,h}(\sqrt{D_m^{\mathrm{rec}}} + \tau_m)}.$$

If

$$\sum_m J_m(n_m+1)e^{-\tau_m^2} < \infty,$$

then, almost surely, for all sufficiently large m, all active $W_{m,h,i}$ are contractions and the all-plus signing gives

$$\operatorname{disc}(W_{m,h,1},\dots,W_{m,h,n_m}) \le \sqrt{n_m} \qquad (1 \le h \le J_m).$$

The arbitrary-growth, all-integer, and sparse schedules of Corollary 7.21 apply with scale

$$5C_{\mathrm{ent}}Q_{m,h}(\sqrt{D_m^{\mathrm{rec}}} + \tau_m).$$

In particular,

$$Q_{m,h} \le \max_{i,r,s} \sigma_{m,h,i,r,s},$$

so homogeneous independent-entry rectangular sub-Gaussian matrices with entry proxy $\sigma_{m,h}$ are covered at the natural spectral scale $\sigma_{m,h}(\sqrt{D_m^{\mathrm{rec}}} + \tau_m)$, up to the universal constant.

Proof. Fix an active stream and condition on $\mathcal{H}_m$. For deterministic unit vectors $u \in \mathbb{C}^{d_{m,1}}$, $v \in \mathbb{C}^{d_{m,2}}$, and $a \in \mathbb{R}^{n_m}$, apply the displayed scalar tail with

$$c_{i,r,s} = a_i \overline{u_r} v_s.$$

The variance proxy is bounded by

$$\begin{aligned}\sum_{i,r,s} |c_{i,r,s}|^2 \sigma_{m,h,i,r,s}^2 &= \sum_{i=1}^{n_m} a_i^2 \sum_{r,s} |u_r|^2 |v_s|^2 \sigma_{m,h,i,r,s}^2 \\ &\le Q_{m,h}^2 \|a\|_2^2.\end{aligned}$$

Thus the bilinear-form hypothesis of Corollary 7.24 holds with $q_{m,h} = C_{\mathrm{ent}}Q_{m,h}$. That corollary gives the simultaneous all-plus conclusion and schedules. Finally, $Q_{m,h} \le \max_{i,r,s} \sigma_{m,h,i,r,s}$, since $\sum_{r,s} |u_r|^2 |v_s|^2 = 1$. □

Corollary 7.28 (Sub-Gaussian coefficient-series rectangular bilinear proxies)**.** *For each $m \ge 1$, let $d_{m,1}, d_{m,2} \ge 1$, $D_m^{\mathrm{rec}} = d_{m,1} + d_{m,2}$, $n_m \ge 1$, and $J_m \ge 1$. Let $\mathcal{H}_m$ be an environment. For each active $1 \le h \le J_m$, suppose that*

$$X_{m,h,i} = \sum_{\alpha \in \mathcal{A}_{m,h,i}} \xi_{m,h,i,\alpha} B_{m,h,i,\alpha} \in \mathrm{M}_{d_{m,1} \times d_{m,2}}(\mathbb{C}), \qquad 1 \le i \le n_m,$$

where the atom sets $\mathcal{A}_{m,h,i}$ are finite and the atom matrices $B_{m,h,i,\alpha}$ and coefficient proxies $\sigma_{m,h,i,\alpha} \ge 0$ are $\mathcal{H}_m$-measurable. Assume that, conditionally on $\mathcal{H}_m$, the scalar coefficients are centered and have the following sub-Gaussian linear-combination estimate: for every coefficient array $c = (c_{i,\alpha})$ and every $t \ge 0$,

$$V_c := \left(\sum_{i=1}^{n_m} \sum_{\alpha \in \mathcal{A}_{m,h,i}} |c_{i,\alpha}|^2 \sigma_{m,h,i,\alpha}^2 \right)^{1/2} t$$

satisfies

$$\mathbb{P}\left\{ \left| \sum_{i=1}^{n_m} \sum_{\alpha \in \mathcal{A}_{m,h,i}} c_{i,\alpha} \xi_{m,h,i,\alpha} \right| > C_{\mathrm{ser}} V_c \,\middle|\, \mathcal{H}_m \right\} \le 2e^{-t^2}.$$

This is implied, after changing the absolute constant C_{ser}, by conditional independence and the usual real or complex scalar sub-Gaussian mgf bounds. Define the bilinear atom-energy proxy

$$R_{m,h} := \max_{1\le i\le n_m} \sup_{\substack{\|u\|_2=1\\ \|v\|_2=1}} \left(\sum_{\alpha\in\mathcal{A}_{m,h,i}} \sigma_{m,h,i,\alpha}^2 |\langle u, B_{m,h,i,\alpha} v\rangle|^2 \right)^{1/2}.$$

Assume $R_{m,h} > 0$ for the nonzero active streams, and set

$$U_{m,h,i} := \frac{X_{m,h,i}}{5C_{\mathrm{ser}} R_{m,h}(\sqrt{D_m^{\mathrm{rec}}} + \tau_m)}.$$

If

$$\sum_m J_m(n_m+1)e^{-\tau_m^2} < \infty,$$

then, almost surely, for all sufficiently large m, all active $U_{m,h,i}$ are contractions and the all-plus signing gives

$$\mathrm{disc}(U_{m,h,1},\dots,U_{m,h,n_m}) \le \sqrt{n_m} \qquad (1\le h\le J_m).$$

The arbitrary-growth, all-integer, and sparse schedules of Corollary 7.21 apply with scale

$$5C_{\mathrm{ser}} R_{m,h}(\sqrt{D_m^{\mathrm{rec}}} + \tau_m).$$

Moreover

$$R_{m,h}^2 \le \max_{1\le i\le n_m} \min\{L_{m,h,i}^{\mathrm{at}}, R_{m,h,i}^{\mathrm{at}}\},$$

where

$$L_{m,h,i}^{\mathrm{at}} := \left\| \sum_{\alpha\in\mathcal{A}_{m,h,i}} \sigma_{m,h,i,\alpha}^2 B_{m,h,i,\alpha} B_{m,h,i,\alpha}^* \right\|_{\mathrm{op}},$$

$$R_{m,h,i}^{\mathrm{at}} := \left\| \sum_{\alpha\in\mathcal{A}_{m,h,i}} \sigma_{m,h,i,\alpha}^2 B_{m,h,i,\alpha}^* B_{m,h,i,\alpha} \right\|_{\mathrm{op}}.$$

Consequently this covers atomized rectangular sub-Gaussian series, common-feature correlated entries, random feature blocks, and structured measurement libraries once the fixed-bilinear atom energy is controlled.

Proof. Fix m, h and condition on $\mathcal{H}_m$. For deterministic unit vectors $u \in \mathbb{C}^{d_{m,1}}$, $v \in \mathbb{C}^{d_{m,2}}$, and $a \in \mathbb{R}^{n_m}$, apply the scalar coefficient tail with

$$c_{i,\alpha} = a_i \langle u, B_{m,h,i,\alpha} v\rangle.$$

The variance proxy is bounded by

$$\begin{aligned}\sum_{i,\alpha} |c_{i,\alpha}|^2 \sigma_{m,h,i,\alpha}^2 &= \sum_{i=1}^{n_m} a_i^2 \sum_{\alpha\in\mathcal{A}_{m,h,i}} \sigma_{m,h,i,\alpha}^2 |\langle u, B_{m,h,i,\alpha} v\rangle|^2 \\ &\le R_{m,h}^2 \|a\|_2^2.\end{aligned}$$

Therefore the hypothesis of Corollary 7.24 holds with $q_{m,h} = C_{\mathrm{ser}} R_{m,h}$, and that corollary gives the simultaneous all-plus conclusion and schedules.

For the displayed computable bound, for each i, u, v,

$$\sum_{\alpha} \sigma^2_{m,h,i,\alpha} |\langle u, B_{m,h,i,\alpha} v\rangle|^2 \le \left\| \sum_{\alpha} \sigma^2_{m,h,i,\alpha} B_{m,h,i,\alpha} B^*_{m,h,i,\alpha} \right\|_{\mathrm{op}},$$

because $vv^* \le I$. The same argument with $uu^* \le I$ gives the right-variance bound with $B^*_{m,h,i,\alpha} B_{m,h,i,\alpha}$. Taking the minimum of the two bounds and then the maximum over i proves the claim. □

Corollary 7.29 (Bernstein bilinear-form rectangular active menus)**.** *In the setting of Corollary 7.21, replace the rectangular operator-tail hypothesis by the following scalar Bernstein bilinear-form hypothesis. For each active $1 \le h \le J_m$, suppose that $q_{m,h} \ge 0$ and $b_{m,h} \ge 0$ are $\mathcal{H}_m$-measurable and that, conditionally on $\mathcal{H}_m$, the rectangular matrices $X_{m,h,i}$ are centered and, for every $a \in \mathbb{R}^{n_m}$, every deterministic unit $u \in \mathbb{C}^{d_{m,1}}$, every deterministic unit $v \in \mathbb{C}^{d_{m,2}}$, and every $s \ge 0$,*

$$\mathbb{P}\left\{ \left| \left\langle u, \sum_{i=1}^{n_m} a_i X_{m,h,i} v \right\rangle \right| > q_{m,h}\|a\|_2 s + b_{m,h}\|a\|_\infty s^2 \,\middle|\, \mathcal{H}_m \right\} \le 2e^{-s^2}.$$

Put

$$T_{m,h} := C_{\mathrm{Bb}} \left[q_{m,h}(\sqrt{D^{\mathrm{rec}}_m} + \tau_m) + b_{m,h}(\sqrt{D^{\mathrm{rec}}_m} + \tau_m)^2 \right],$$

where C_{Bb} is a sufficiently large numerical constant, and assume $T_{m,h} > 0$ for nonzero active streams. Define

$$V_{m,h,i} := \frac{X_{m,h,i}}{T_{m,h}}.$$

If

$$\sum_m J_m (n_m + 1) e^{-\tau_m^2} < \infty,$$

then, almost surely, for all sufficiently large m, all active $V_{m,h,i}$ are contractions and the all-plus signing gives

$$\mathrm{disc}(V_{m,h,1}, \dots, V_{m,h,n_m}) \le \sqrt{n_m} \qquad (1 \le h \le J_m).$$

The arbitrary-growth, all-integer, and sparse schedules of Corollary 7.21 apply with the Bernstein bilinear scale $T_{m,h}$.

Proof. Fix m, h and condition on $\mathcal{H}_m$. Let $A_m = \sqrt{D^{\mathrm{rec}}_m} + \tau_m$. The same two-net argument as in Lemma 7.23, using the scalar Bernstein hypothesis at level cA_m with c large enough, gives for every deterministic $a \in \mathbb{R}^{n_m}$

$$\mathbb{P}\left\{ \left\| \sum_i a_i X_{m,h,i} \right\|_{\mathrm{op}} > C_{\mathrm{Bb}} \left[q_{m,h}\|a\|_2 A_m + b_{m,h}\|a\|_\infty A_m^2 \right] \,\middle|\, \mathcal{H}_m \right\} \le 2e^{-\tau_m^2}.$$

Apply this bound to $a = e_i$, $1 \le i \le n_m$, and to $a = (1, \dots, 1)$. The e_i choices give $\|V_{m,h,i}\|_{\mathrm{op}} \le 1$ outside conditional failure probability $2n_m e^{-\tau_m^2}$. The all-plus choice gives

$$\left\| \sum_i X_{m,h,i} \right\|_{\mathrm{op}} \le C_{\mathrm{Bb}} \left[q_{m,h} \sqrt{n_m} A_m + b_{m,h} A_m^2 \right] \le \sqrt{n_m}\, T_{m,h},$$

since $n_m \ge 1$. Union over $h \le J_m$, average over the environment, and apply Borel–Cantelli. The schedule statements are the same summability estimates as in the rectangular full operator-tail corollary. □

Corollary 7.30 (Random active Bernstein bilinear-form rectangular matrices)**.** *In the setting of Corollary 7.22, replace the rectangular operator-tail hypothesis by the following scalar Bernstein bilinear-form hypothesis. Suppose that $q_{m,h} \ge 0$, $b_{m,h} \ge 0$, and $\tau_{m,h} > 0$ are $\mathcal{F}_m$-measurable and that, conditionally on $\mathcal{F}_m$, the rectangular matrices $X_{m,h,i}$ are centered and, for every h, every $a \in \mathbb{R}^{n_m}$, every deterministic unit $u \in \mathbb{C}^{d_{m,1}}$, every deterministic unit $v \in \mathbb{C}^{d_{m,2}}$, and every $s \ge 0$,*

$$\mathbb{P}\left\{ \left| \left\langle u, \sum_{i=1}^{n_m} a_i X_{m,h,i} v \right\rangle \right| > q_{m,h} \|a\|_2 s + b_{m,h} \|a\|_\infty s^2 \,\middle|\, \mathcal{F}_m \right\} \le 2e^{-s^2}.$$

Put

$$A_{m,h} := \sqrt{D_m^{\mathrm{rec}}} + \tau_{m,h}, \qquad T_{m,h} := C_{\mathrm{Bb}}(q_{m,h} A_{m,h} + b_{m,h} A_{m,h}^2),$$

where C_{Bb} is a sufficiently large numerical constant, and define

$$\widehat{Z}_{m,h,i} := \begin{cases} X_{m,h,i}/T_{m,h}, & T_{m,h} > 0, \\ 0, & T_{m,h} = 0. \end{cases}$$

If

$$\sum_m \mathbb{E} \sum_{h \in \mathcal{A}_m} 2(n_m + 1) e^{-\tau_{m,h}^2} < \infty,$$

then, almost surely, for all sufficiently large m, every selected active Bernstein bilinear-form rectangular stream is contraction-normalized and has the all-plus certificate

$$\mathrm{disc}(\widehat{Z}_{m,h,1}, \dots, \widehat{Z}_{m,h,n_m}) \le \sqrt{n_m} \qquad (h \in \mathcal{A}_m).$$

The common-τ_m, deterministic-size, and dimension-schedule conclusions from the random-active rectangular operator-tail corollary apply with the Bernstein bilinear scale $T_{m,h}$. Thus selected bounded-entry, Poissonized, exponential-entry, gamma-entry, and balanced random-dyad rectangular libraries pay only their selected expected Bernstein bilinear tail count.

Proof. Condition on $\mathcal{F}_m$ and fix a selected h. Let $A_{m,h} = \sqrt{D_m^{\mathrm{rec}}} + \tau_{m,h}$. The two-net argument used in Corollary 7.29 gives, for every deterministic

$a \in \mathbb{R}^{n_m}$,

$$\mathbb{P}\left\{ \left\| \sum_i a_i X_{m,h,i} \right\|_{\mathrm{op}} > C_{\mathrm{Bb}} \left[q_{m,h} \|a\|_2 A_{m,h} + b_{m,h} \|a\|_\infty A_{m,h}^2 \right] \middle| \mathcal{F}_m \right\} \leq 2e^{-\tau_{m,h}^2}.$$

Apply this estimate to $a = e_i$, $1 \leq i \leq n_m$, and to $a = (1, \dots, 1)$. These n_m+1 estimates show that, outside conditional failure probability $2(n_m+1)e^{-\tau_{m,h}^2}$, every $\widehat{Z}_{m,h,i}$ is a contraction and

$$\left\| \sum_i \widehat{Z}_{m,h,i} \right\|_{\mathrm{op}} \leq \sqrt{n_m}.$$

Union over $h \in \mathcal{A}_m$, average over the side information, and apply Borel–Cantelli. The common-τ_m and dimension-schedule statements are the same summability estimates as in Corollary 7.22. □

Corollary 7.31 (Independent-entry rectangular Bernstein-form proxies)**.** *For each $m \geq 1$, let $d_{m,1}, d_{m,2} \geq 1$, $D_m^{\mathrm{rec}} = d_{m,1} + d_{m,2}$, $n_m \geq 1$, and $J_m \geq 1$. Let $\mathcal{H}_m$ be an environment. For each active $1 \leq h \leq J_m$, let*

$$X_{m,h,i} = (x_{m,h,i,r,s})_{1 \leq r \leq d_{m,1},\, 1 \leq s \leq d_{m,2}} \in \mathrm{M}_{d_{m,1} \times d_{m,2}}(\mathbb{C}), \qquad 1 \leq i \leq n_m,$$

be centered rectangular random matrices. Suppose that, conditionally on $\mathcal{H}_m$, the entries in the active stream satisfy the following complex scalar Bernstein linear-combination estimate. There are $\mathcal{H}_m$-measurable proxies $\sigma_{m,h,i,r,s}, L_{m,h,i,r,s} \geq 0$ such that, for every coefficient array $c = (c_{i,r,s})$ and every $t \geq 0$,

$$\mathbb{P}\left\{ \left| \sum_{i,r,s} c_{i,r,s} x_{m,h,i,r,s} \right| > C_{\mathrm{rB}} \left(V_c t + M_c t^2 \right) \middle| \mathcal{H}_m \right\} \leq 2e^{-t^2},$$

where

$$V_c^2 := \sum_{i,r,s} |c_{i,r,s}|^2 \sigma_{m,h,i,r,s}^2, \qquad M_c := \max_{i,r,s} |c_{i,r,s}| L_{m,h,i,r,s}.$$

This estimate is implied, after changing C_{rB}, by conditional independence and the usual real and imaginary scalar Bernstein mgf bounds. Define

$$Q_{m,h}^{\mathrm{rB}} := \max_{1 \leq i \leq n_m} \sup_{\substack{\|u\|_2 = 1 \\ \|v\|_2 = 1}} \left(\sum_{r=1}^{d_{m,1}} \sum_{s=1}^{d_{m,2}} |u_r|^2 |v_s|^2 \sigma_{m,h,i,r,s}^2 \right)^{1/2}$$

and

$$B_{m,h}^{\mathrm{rB}} := \max_{i,r,s} L_{m,h,i,r,s}.$$

Set

$$T_{m,h}^{\mathrm{rB}} := C_{\mathrm{Bb}} C_{\mathrm{rB}} \left[Q_{m,h}^{\mathrm{rB}} (\sqrt{D_m^{\mathrm{rec}}} + \tau_m) + B_{m,h}^{\mathrm{rB}} (\sqrt{D_m^{\mathrm{rec}}} + \tau_m)^2 \right],$$

and assume this scale is positive for nonzero active streams. Then, under

$$\sum_m J_m(n_m+1)e^{-\tau_m^2} < \infty,$$

the normalized matrices $X_{m,h,i}/T^{\mathrm{rB}}_{m,h}$ are eventually contractions and the all-plus signing gives discrepancy at most $\sqrt{n_m}$ simultaneously for every active h. If $\sigma_{m,h,i,r,s} \le \sigma_{m,h}$ and $L_{m,h,i,r,s} \le L_{m,h}$, then

$$Q^{\mathrm{rB}}_{m,h} \le \sigma_{m,h}, \qquad B^{\mathrm{rB}}_{m,h} \le L_{m,h},$$

so bounded, Poissonized, exponential-entry, gamma-entry, and other light-tailed non-sub-Gaussian rectangular libraries are covered at scale

$$C\left[\sigma_{m,h}(\sqrt{D^{\mathrm{rec}}_m}+\tau_m) + L_{m,h}(\sqrt{D^{\mathrm{rec}}_m}+\tau_m)^2\right].$$

Proof. Fix m, h and condition on $\mathcal{H}_m$. For deterministic unit vectors $u \in \mathbb{C}^{d_{m,1}}$, $v \in \mathbb{C}^{d_{m,2}}$, and $a \in \mathbb{R}^{n_m}$, apply the displayed scalar Bernstein estimate with

$$c_{i,r,s} = a_i \overline{u_r} v_s.$$

Then $V_c \le Q^{\mathrm{rB}}_{m,h}\|a\|_2$, while

$$M_c \le B^{\mathrm{rB}}_{m,h}\|a\|_\infty$$

because $|u_r| \le 1$ and $|v_s| \le 1$. Hence the hypothesis of Corollary 7.29 holds with $q_{m,h} = C_{\mathrm{rB}} Q^{\mathrm{rB}}_{m,h}$ and $b_{m,h} = C_{\mathrm{rB}} B^{\mathrm{rB}}_{m,h}$. The homogeneous bound follows from $\sum_{r,s} |u_r|^2|v_s|^2 = 1$. □

Corollary 7.32 (Random active independent-entry rectangular Bernstein proxies)**.** *For each $m \ge 1$, let $d_{m,1}, d_{m,2} \ge 1$, $D^{\mathrm{rec}}_m = d_{m,1} + d_{m,2}$, $n_m \ge 1$, let $\mathcal{F}_m$ be side information, and let $\mathcal{A}_m \subset \mathbb{N}$ be an almost surely finite $\mathcal{F}_m$-measurable active set. For every $h \in \mathbb{N}$, let*

$$X_{m,h,i} = (x_{m,h,i,r,s})_{1\le r\le d_{m,1},\, 1\le s\le d_{m,2}} \in \mathrm{M}_{d_{m,1}\times d_{m,2}}(\mathbb{C}), \qquad 1 \le i \le n_m,$$

be centered rectangular random matrices. Suppose that, conditionally on $\mathcal{F}_m$, the entries in every selected stream satisfy the following complex scalar Bernstein linear-combination estimate. There are $\mathcal{F}_m$-measurable proxies $\sigma_{m,h,i,r,s}, L_{m,h,i,r,s} \ge 0$ such that, for every h, every coefficient array $c = (c_{i,r,s})$, and every $t \ge 0$,

$$\mathbb{P}\left\{\left|\sum_{i,r,s} c_{i,r,s} x_{m,h,i,r,s}\right| > C_{\mathrm{rB}}\left(V_{m,h,c}\,t + M_{m,h,c}\,t^2\right) \,\middle|\, \mathcal{F}_m\right\} \le 2e^{-t^2},$$

where

$$V^2_{m,h,c} := \sum_{i,r,s} |c_{i,r,s}|^2 \sigma^2_{m,h,i,r,s}, \qquad M_{m,h,c} := \max_{i,r,s} |c_{i,r,s}| L_{m,h,i,r,s}.$$

Define

$$Q^{\mathrm{rB}}_{m,h} := \max_{1\le i\le n_m} \sup_{\substack{\|u\|_2=1\\ \|v\|_2=1}} \left(\sum_{r=1}^{d_{m,1}} \sum_{s=1}^{d_{m,2}} |u_r|^2 |v_s|^2 \sigma^2_{m,h,i,r,s} \right)^{1/2},$$

and

$$B^{\mathrm{rB}}_{m,h} := \max_{i,r,s} L_{m,h,i,r,s}.$$

Let $\tau_{m,h} > 0$ be $\mathcal{F}_m$-measurable and put

$$T^{\mathrm{rB}}_{m,h} := C_{\mathrm{Bb}} C_{\mathrm{rB}} \left[Q^{\mathrm{rB}}_{m,h} (\sqrt{D^{\mathrm{rec}}_m} + \tau_{m,h}) + B^{\mathrm{rB}}_{m,h} (\sqrt{D^{\mathrm{rec}}_m} + \tau_{m,h})^2 \right].$$

Define

$$Z^{\mathrm{rB}}_{m,h,i} := \begin{cases} X_{m,h,i}/T^{\mathrm{rB}}_{m,h}, & T^{\mathrm{rB}}_{m,h} > 0, \\ 0, & T^{\mathrm{rB}}_{m,h} = 0. \end{cases}$$

If

$$\sum_m \mathbb{E} \sum_{h\in\mathcal{A}_m} 2(n_m+1) e^{-\tau^2_{m,h}} < \infty,$$

then, almost surely, for all sufficiently large m, every selected independent-entry Bernstein stream is contraction-normalized after the displayed normalization and has the all-plus certificate

$$\operatorname{disc}\left(Z^{\mathrm{rB}}_{m,h,1}, \dots, Z^{\mathrm{rB}}_{m,h,n_m} \right) \le \sqrt{n_m} \qquad (h \in \mathcal{A}_m).$$

If $\sigma_{m,h,i,r,s} \le \sigma_{m,h}$ and $L_{m,h,i,r,s} \le L_{m,h}$, then one may use

$$C \left[\sigma_{m,h} (\sqrt{D^{\mathrm{rec}}_m} + \tau_{m,h}) + L_{m,h} (\sqrt{D^{\mathrm{rec}}_m} + \tau_{m,h})^2 \right]$$

instead. Thus random active bounded-entry, Poissonized, exponential-entry, gamma-entry, and other light-tailed independent-entry rectangular libraries pay only their selected expected Bernstein entry-tail count.

Proof. For deterministic unit vectors $u \in \mathbb{C}^{d_{m,1}}$, $v \in \mathbb{C}^{d_{m,2}}$, and $a \in \mathbb{R}^{n_m}$, use the scalar Bernstein estimate with

$$c_{i,r,s} = a_i \overline{u_r} v_s.$$

Then $V_{m,h,c} \le Q^{\mathrm{rB}}_{m,h} \|a\|_2$ and $M_{m,h,c} \le B^{\mathrm{rB}}_{m,h} \|a\|_\infty$. Hence the hypothesis of Corollary 7.30 holds with $q_{m,h} = C_{\mathrm{rB}} Q^{\mathrm{rB}}_{m,h}$ and $b_{m,h} = C_{\mathrm{rB}} B^{\mathrm{rB}}_{m,h}$. That corollary gives the selected active conclusion. The homogeneous scale follows from $\sum_{r,s} |u_r|^2 |v_s|^2 = 1$. □

Corollary 7.33 (Random active conditionally independent Bernstein entries)**.** *In Corollary 7.32, replace the scalar linear-combination hypothesis by the following entrywise condition. Conditionally on $\mathcal{F}_m$, for every selected h the variables $(x_{m,h,i,r,s})_{i,r,s}$ are independent, centered, and satisfy the uniform*

rotated Bernstein mgf bound: for every phase $\theta \in \mathbb{R}$ and every real λ with $|\lambda| L_{m,h,i,r,s} < 1$,

$$\mathbb{E}\left[\exp\left\{\lambda \operatorname{Re}(e^{-i\theta} x_{m,h,i,r,s})\right\} \,\middle|\, \mathcal{F}_m\right] \le \exp\left\{\frac{\lambda^2 \sigma^2_{m,h,i,r,s}}{2(1-|\lambda| L_{m,h,i,r,s})}\right\}.$$

Then the conclusions of Corollary 7.32 hold, after changing only the absolute numerical constant in the normalization. In particular, the same selected summability condition

$$\sum_m \mathbb{E} \sum_{h \in \mathcal{A}_m} 2(n_m+1) e^{-\tau^2_{m,h}} < \infty$$

gives eventual all-plus Matrix-Spencer certificates for bounded, Poissonized, centered exponential or gamma, and other conditionally independent rectangular entry models with the displayed variance and jump proxies.

Proof. Fix m, h and condition on $\mathcal{F}_m$. For a deterministic coefficient array $c = (c_{i,r,s})$, put

$$\begin{aligned} Z_c &:= \sum_{i,r,s} c_{i,r,s} x_{m,h,i,r,s}, \\ V_c^2 &:= \sum_{i,r,s} |c_{i,r,s}|^2 \sigma^2_{m,h,i,r,s}, \\ M_c &:= \max_{i,r,s} |c_{i,r,s}| L_{m,h,i,r,s}. \end{aligned}$$

For every fixed phase ϕ, independence and the entrywise mgf bound give

$$\mathbb{E}\left[\exp\{\lambda \operatorname{Re}(e^{-i\phi} Z_c)\} \,\middle|\, \mathcal{F}_m\right] \le \exp\left\{\frac{\lambda^2 V_c^2}{2(1-|\lambda| M_c)}\right\}, \qquad |\lambda| M_c < 1.$$

The scalar Bernstein inequality therefore gives

$$\mathbb{P}\left\{|\operatorname{Re}(e^{-i\phi} Z_c)| > C(V_c t + M_c t^2) \,\middle|\, \mathcal{F}_m\right\} \le 2e^{-t^2}.$$

Taking $\phi = 0$ and $\phi = \pi/2$, and using $|Z_c| \le \sqrt{2} \max\{|\operatorname{Re} Z_c|, |\operatorname{Im} Z_c|\}$, gives the scalar complex Bernstein linear-combination estimate required in Corollary 7.32, with a changed absolute constant. Applying that corollary proves the claim. □

Corollary 7.34 (Finite-confidence entrywise Bernstein random matrices)**.** *Fix one dimension pair $d_1, d_2 \ge 1$, put $D^{\mathrm{rec}} = d_1 + d_2$, and let $\mathcal{F}$ be side information. Let $\mathcal{A} \subset \mathbb{N}$ be an almost surely finite $\mathcal{F}$-measurable active set. For every h, let*

$$X_{h,i} = (x_{h,i,r,s})_{1 \le r \le d_1,\, 1 \le s \le d_2}, \qquad 1 \le i \le n,$$

be centered rectangular matrices whose selected entries satisfy the conditional independence and rotated Bernstein mgf hypothesis of Corollary 7.33, with

proxies $\sigma_{h,i,r,s}$ and $L_{h,i,r,s}$. Define Q_h^{rB}, B_h^{rB}, T_h^{rB}, and $Z_{h,i}^{\mathrm{rB}}$ as in Corollary 7.32, with an $\mathcal{F}$-measurable choice $\tau_h > 0$. If

$$\mathbb{E} \sum_{h \in \mathcal{A}} 2(n+1)e^{-\tau_h^2} \le \delta,$$

then, with probability at least $1-\delta$, every selected entrywise Bernstein stream is contraction-normalized and has the all-plus certificate

$$\operatorname{disc}\left(Z_{h,1}^{\mathrm{rB}}, \dots, Z_{h,n}^{\mathrm{rB}}\right) \le \sqrt{n} \qquad (h \in \mathcal{A}).$$

In particular, if $|\mathcal{A}| \le M$ almost surely, the common choice

$$\tau^2 \ge \log \frac{2M(n+1)}{\delta}$$

gives a simultaneous $1-\delta$ certificate after the homogeneous scale

$$C\left[\sigma_h(\sqrt{D^{\mathrm{rec}}} + \tau) + L_h(\sqrt{D^{\mathrm{rec}}} + \tau)^2\right]$$

whenever $\sigma_{h,i,r,s} \le \sigma_h$ and $L_{h,i,r,s} \le L_h$.

Proof. Condition on $\mathcal{F}$. By the proof of Corollary 7.33, each selected stream satisfies the scalar Bernstein linear-combination hypothesis of Corollary 7.32. The conditional failure probability for a fixed selected h is therefore at most $2(n+1)e^{-\tau_h^2}$. Summing over the $\mathcal{F}$-measurable active set and then taking expectation gives total failure at most the displayed budget δ. The common-τ and homogeneous scale statements are the corresponding specializations. □

Corollary 7.35 (Finite-confidence bounded-entry random matrices)**.** *Fix one dimension pair $d_1, d_2 \ge 1$, put $D^{\mathrm{rec}} = d_1 + d_2$, and let $\mathcal{F}$ be side information. Let $\mathcal{A} \subset \mathbb{N}$ be an almost surely finite $\mathcal{F}$-measurable active set. For every h, let*

$$X_{h,i} = (x_{h,i,r,s})_{1 \le r \le d_1,\, 1 \le s \le d_2}, \qquad 1 \le i \le n,$$

be centered rectangular matrices. Assume that, conditionally on $\mathcal{F}$, for every selected h the entries $(x_{h,i,r,s})_{i,r,s}$ are independent and there are $\mathcal{F}$-measurable proxies $\nu_{h,i,r,s}, R_{h,i,r,s} \ge 0$ such that

$$\mathbb{E}\left(|x_{h,i,r,s}|^2 \mid \mathcal{F}\right) \le \nu_{h,i,r,s}^2, \qquad |x_{h,i,r,s}| \le R_{h,i,r,s} \quad \text{a.s.}$$

Define

$$Q_h^{\mathrm{bd}} := \max_{1 \le i \le n} \sup_{\substack{\|u\|_2 = 1 \\ \|v\|_2 = 1}} \left(\sum_{r=1}^{d_1} \sum_{s=1}^{d_2} |u_r|^2 |v_s|^2 \nu_{h,i,r,s}^2 \right)^{1/2},$$

and

$$R_h^{\mathrm{bd}} := \max_{i,r,s} R_{h,i,r,s}.$$

Let $\tau_h > 0$ be $\mathcal{F}$-measurable and put

$$T_h^{\mathrm{bd}} := C_{\mathrm{bd}}\left[Q_h^{\mathrm{bd}}(\sqrt{D^{\mathrm{rec}}} + \tau_h) + R_h^{\mathrm{bd}}(\sqrt{D^{\mathrm{rec}}} + \tau_h)^2\right],$$

with

$$Z_{h,i}^{\mathrm{bd}} := \begin{cases} X_{h,i}/T_h^{\mathrm{bd}}, & T_h^{\mathrm{bd}} > 0, \\ 0, & T_h^{\mathrm{bd}} = 0. \end{cases}$$

If

$$\mathbb{E} \sum_{h \in \mathcal{A}} 2(n+1) e^{-\tau_h^2} \le \delta,$$

then, with probability at least $1-\delta$*, every selected bounded-entry stream is contraction-normalized and has the all-plus certificate*

$$\operatorname{disc}\left(Z_{h,1}^{\mathrm{bd}}, \dots, Z_{h,n}^{\mathrm{bd}}\right) \le \sqrt{n} \qquad (h \in \mathcal{A}).$$

In particular, if $|\mathcal{A}| \le M$ *almost surely, the common choice*

$$\tau^2 \ge \log \frac{2M(n+1)}{\delta}$$

gives a simultaneous $1-\delta$ *certificate after the homogeneous scale*

$$C\left[\nu_h(\sqrt{D^{\mathrm{rec}}} + \tau) + R_h(\sqrt{D^{\mathrm{rec}}} + \tau)^2\right]$$

whenever $\nu_{h,i,r,s} \le \nu_h$ *and* $R_{h,i,r,s} \le R_h$*.*

Proof. Condition on $\mathcal{F}$ and fix a selected stream. For every phase $\theta \in \mathbb{R}$, the scalar variable

$$Y_{h,i,r,s,\theta} := \mathrm{Re}(e^{-i\theta} x_{h,i,r,s})$$

is centered, has conditional variance at most $\nu_{h,i,r,s}^2$, and satisfies $|Y_{h,i,r,s,\theta}| \le R_{h,i,r,s}$. The standard bounded-variable Bernstein mgf bound gives, for $|\lambda| R_{h,i,r,s} < 1$,

$$\mathbb{E}\left[\exp\{\lambda Y_{h,i,r,s,\theta}\} \mid \mathcal{F}\right] \le \exp\left\{\frac{C\lambda^2 \nu_{h,i,r,s}^2}{2(1 - |\lambda| R_{h,i,r,s})}\right\}.$$

Absorbing the absolute constant into the final normalization, this is precisely the rotated Bernstein mgf hypothesis of Corollary 7.34, with $\sigma_{h,i,r,s} = C\nu_{h,i,r,s}$ and $L_{h,i,r,s} = R_{h,i,r,s}$. That corollary gives the finite-confidence certificate, and the homogeneous bound follows from the displayed definitions of Q_h^{bd} and R_h^{bd}. □

Corollary 7.36 (Haar rank-one dyad active menus)**.** *For each* $m \ge 1$*, let* $d_{m,1}, d_{m,2} \ge 2$*,* $D_m^{\mathrm{rec}} = d_{m,1} + d_{m,2}$*,* $n_m \ge 1$*, and* $J_m \ge 1$*. Let* $\mathcal{H}_m$ *be an environment. For each active* $1 \le h \le J_m$*, let*

$$x_{m,h,i} \in \mathbb{C}^{d_{m,1}}, \qquad y_{m,h,i} \in \mathbb{C}^{d_{m,2}}, \qquad 1 \le i \le n_m,$$

be conditionally independent unit vectors, each Haar-distributed on the corresponding complex unit sphere, and let $\rho_{m,h,i}$ *be* $\mathcal{H}_m$*-measurable amplitudes with* $|\rho_{m,h,i}| \le \rho_{m,h}$*. Real Haar unit vectors, embedded in the complex spaces, are also allowed after changing the absolute constant. Put*

$$X_{m,h,i} := \rho_{m,h,i}\, x_{m,h,i} y_{m,h,i}^*$$

and

$$T^{\mathrm{dyad}}_{m,h} := C_{\mathrm{dyad}}\rho_{m,h}\left[\frac{\sqrt{D^{\mathrm{rec}}_m}+\tau_m}{\sqrt{d_{m,1}d_{m,2}}}+\frac{(\sqrt{D^{\mathrm{rec}}_m}+\tau_m)^2}{\sqrt{d_{m,1}d_{m,2}}}\right].$$

Set $Z^{\mathrm{dyad}}_{m,h,i} := X_{m,h,i}/T^{\mathrm{dyad}}_{m,h}$. *If*

$$\sum_m J_m(n_m+1)e^{-\tau_m^2}<\infty,$$

then, almost surely, for all sufficiently large m, *all active* $Z^{\mathrm{dyad}}_{m,h,i}$ *are contractions and the all-plus signing gives*

$$\mathrm{disc}\left(Z^{\mathrm{dyad}}_{m,h,1},\dots,Z^{\mathrm{dyad}}_{m,h,n_m}\right)\le\sqrt{n_m}.$$

In particular, if $d_{m,1}$ *and* $d_{m,2}$ *are comparable and* $\tau_m=O(\sqrt{D^{\mathrm{rec}}_m})$, *then independent Haar rank-one dyads, random feature outer products, and mixed real/complex Haar dyad libraries are Matrix-Spencer scale after only a universal constant multiple of the amplitude normalization.*

Proof. Fix m,h and condition on $\mathcal{H}_m$. For deterministic unit vectors $u\in\mathbb{C}^{d_{m,1}}$ and $v\in\mathbb{C}^{d_{m,2}}$,

$$\langle u,X_{m,h,i}v\rangle=\rho_{m,h,i}\langle u,x_{m,h,i}\rangle\langle y_{m,h,i},v\rangle.$$

The spherical-coordinate tail gives sub-Gaussian proxies $Cd_{m,1}^{-1/2}$ and $Cd_{m,2}^{-1/2}$ for the two scalar factors, uniformly over real projections. The product of two independent sub-Gaussian variables is sub-exponential with proxy the product of their proxies. Therefore the standard scalar Bernstein estimate for independent sub-exponential variables gives, for every $a\in\mathbb{R}^{n_m}$,

$$\mathbb{P}\left\{\left|\left\langle u,\sum_i a_iX_{m,h,i}v\right\rangle\right|>C\rho_{m,h}\left(\frac{\|a\|_2 s}{\sqrt{d_{m,1}d_{m,2}}}+\frac{\|a\|_\infty s^2}{\sqrt{d_{m,1}d_{m,2}}}\right)\,\middle|\,\mathcal{H}_m\right\}\le 2e^{-s^2}.$$

The same proof applies to real Haar vectors after decomposing complex bilinear forms into real and imaginary parts and increasing the numerical constant. Thus Corollary 7.29 applies with

$$q_{m,h}=C\rho_{m,h}(d_{m,1}d_{m,2})^{-1/2},\qquad b_{m,h}=C\rho_{m,h}(d_{m,1}d_{m,2})^{-1/2}.$$

The displayed dyad scale is the resulting Bernstein bilinear scale. If $d_{m,1}\asymp d_{m,2}$ and $\tau_m=O(\sqrt{D^{\mathrm{rec}}_m})$, then $(D^{\mathrm{rec}}_m)/\sqrt{d_{m,1}d_{m,2}}$ is bounded, giving the final constant-normalization claim. □

Corollary 7.37 (Bounded matrix-Bernstein all-plus active menus)**.** *For each* $m\ge 1$, *let* $D_m\ge 2$, $n_m\ge 1$, *and* $J_m\ge 1$. *Let* $\mathcal{H}_m$ *be an environment. For each active* $1\le h\le J_m$, *let*

$$X_{m,h,1},\dots,X_{m,h,n_m}\in\mathrm{M}_{D_m}(\mathbb{C})$$

be conditionally independent centered self-adjoint random matrices. Let $E_{m,h} \in \mathcal{H}_m$ be an exceptional event and let $s_{m,h}, v_{m,h}, u_{m,h} > 0$ be deterministic. Assume that, on $E_{m,h}^c$,

$$\|X_{m,h,i}\|_{\mathrm{op}} \le s_{m,h} \qquad (1 \le i \le n_m)$$

almost surely and

$$\left\| \sum_{i=1}^{n_m} \mathbb{E}(X_{m,h,i}^2 \mid \mathcal{H}_m) \right\|_{\mathrm{op}} \le v_{m,h}.$$

Set

$$S_{m,h} := \max\{s_{m,h},\, u_{m,h}/\sqrt{n_m}\}.$$

If

$$\sum_m \sum_{h=1}^{J_m} \left[\mathbb{P}(E_{m,h}) + 2D_m \exp\left\{ -\frac{u_{m,h}^2}{2(v_{m,h} + s_{m,h}u_{m,h}/3)} \right\} \right] < \infty,$$

then, almost surely, for all sufficiently large m, all active matrices $X_{m,h,i}/S_{m,h}$ are contractions and the all-plus signing gives

$$\mathrm{disc}(X_{m,h,1}/S_{m,h}, \dots, X_{m,h,n_m}/S_{m,h}) \le \sqrt{n_m} \qquad (1 \le h \le J_m).$$

No coordinatewise sign symmetry is required. In particular, if $v_{m,h} \le \beta s_{m,h}^2 n_m / \log(eD_m)$ and $u_{m,h} = K s_{m,h}\sqrt{n_m}$, this gives a constant multiple of the $s_{m,h}$-normalization whenever the displayed Bernstein failure series is summable. This covers centered bounded random projections, bounded sparse adjacency increments, bounded random block matrices, and centered bounded self-adjoint measurement libraries under their all-plus variance budgets.

Proof. Fix m, h. On $E_{m,h}^c$, the two-sided matrix Bernstein inequality [25] gives

$$\mathbb{P}\left\{ \left\| \sum_i X_{m,h,i} \right\|_{\mathrm{op}} > u_{m,h} \,\middle|\, \mathcal{H}_m \right\} \le 2D_m \exp\left\{ -\frac{u_{m,h}^2}{2(v_{m,h} + s_{m,h}u_{m,h}/3)} \right\}.$$

Therefore the unconditional probability that either $E_{m,h}$ occurs or the all-plus sum exceeds $u_{m,h}$ is bounded by the displayed summand. The first Borel–Cantelli lemma gives, almost surely, eventual simultaneous control for all active h. On that event, $\|X_{m,h,i}/S_{m,h}\|_{\mathrm{op}} \le 1$ and

$$\left\| \sum_i X_{m,h,i}/S_{m,h} \right\|_{\mathrm{op}} \le u_{m,h}/S_{m,h} \le \sqrt{n_m},$$

so the all-plus signing is an admissible Matrix-Spencer certificate. □

Corollary 7.38 (Rectangular bounded matrix-Bernstein all-plus active menus)**.** *For each $m \ge 1$, let $d_{m,1}, d_{m,2} \ge 1$, $D_m^{\mathrm{rec}} = d_{m,1} + d_{m,2}$, $n_m \ge 1$, and $J_m \ge 1$. Let $\mathcal{H}_m$ be an environment. For each active $1 \le h \le J_m$, let*

$$X_{m,h,1}, \dots, X_{m,h,n_m} \in \mathrm{M}_{d_{m,1} \times d_{m,2}}(\mathbb{C})$$

be conditionally independent centered rectangular random matrices. Let $E_{m,h} \in \mathcal{H}_m$ *and let* $s_{m,h}, v_{m,h}, u_{m,h} > 0$ *be deterministic. Assume that, on* $E_{m,h}^c$,

$$\|X_{m,h,i}\|_{\mathrm{op}} \le s_{m,h} \qquad (1 \le i \le n_m)$$

almost surely and

$$\left\| \sum_i \mathbb{E}(X_{m,h,i} X_{m,h,i}^* \mid \mathcal{H}_m) \right\|_{\mathrm{op}} \le v_{m,h},$$

$$\left\| \sum_i \mathbb{E}(X_{m,h,i}^* X_{m,h,i} \mid \mathcal{H}_m) \right\|_{\mathrm{op}} \le v_{m,h}.$$

Set $S_{m,h} := \max\{s_{m,h}, u_{m,h}/\sqrt{n_m}\}$. *If*

$$\sum_m \sum_{h=1}^{J_m} \left[\mathbb{P}(E_{m,h}) + 2D_m^{\mathrm{rec}} \exp\left\{ -\frac{u_{m,h}^2}{2(v_{m,h} + s_{m,h}u_{m,h}/3)} \right\} \right] < \infty,$$

then, almost surely, for all sufficiently large m, *all active rectangular matrices* $X_{m,h,i}/S_{m,h}$ *are contractions and the all-plus signing gives*

$$\mathrm{disc}(X_{m,h,1}/S_{m,h}, \dots, X_{m,h,n_m}/S_{m,h}) \le \sqrt{n_m} \qquad (1 \le h \le J_m).$$

Thus centered bounded rectangular blocks, sparse rectangular increments, centered bounded random feature blocks, and bounded measurement matrices are Matrix-Spencer scale whenever their row and column all-plus variance budgets make the displayed Bernstein series summable.

Proof. Apply Corollary 7.37 to the self-adjoint dilations

$$\mathscr{D}(X_{m,h,i}) = \begin{pmatrix} 0 & X_{m,h,i} \\ X_{m,h,i}^* & 0 \end{pmatrix} \in \mathrm{M}_{D_m^{\mathrm{rec}}}(\mathbb{C}).$$

The dilation has operator norm $\|X_{m,h,i}\|_{\mathrm{op}}$, and its variance is block diagonal with diagonal blocks $\sum_i \mathbb{E}(X_{m,h,i} X_{m,h,i}^* \mid \mathcal{H}_m)$ and $\sum_i \mathbb{E}(X_{m,h,i}^* X_{m,h,i} \mid \mathcal{H}_m)$. Hence the self-adjoint corollary applies with dimension D_m^{rec}, jump $s_{m,h}$, and variance bound $v_{m,h}$. Finally, $\|\sum_i \mathscr{D}(X_{m,h,i})\|_{\mathrm{op}} = \|\sum_i X_{m,h,i}\|_{\mathrm{op}}$, so the dilation certificate is exactly the rectangular all-plus certificate. □

Corollary 7.39 (Finite-confidence bounded matrix-Bernstein all-plus certificates)**.** *Let* $\mathcal{F}$ *be side information, let* $\mathcal{A} \subset \mathbb{N}$ *be an almost surely finite* $\mathcal{F}$*-measurable active set, and fix* $n \ge 1$ *and* $0 < \delta < 1$. *For each* h, *let*

$$X_{h,1}, \dots, X_{h,n} \in \mathrm{M}_D(\mathbb{C})$$

be centered self-adjoint random matrices. Assume that, conditionally on $\mathcal{F}$, *every selected stream is independent and has* $\mathcal{F}$*-measurable proxies* $s_h, v_h, u_h > 0$ *satisfying*

$$\|X_{h,i}\|_{\mathrm{op}} \le s_h \qquad (1 \le i \le n)$$

almost surely and

$$\left\| \sum_{i=1}^{n} \mathbb{E}(X_{h,i}^2 \mid \mathcal{F}) \right\|_{\mathrm{op}} \le v_h.$$

Set

$$S_h := \max\{s_h, u_h/\sqrt{n}\}.$$

If

$$\mathbb{E} \sum_{h \in \mathcal{A}} 2D \exp\left\{ -\frac{u_h^2}{2(v_h + s_h u_h/3)} \right\} \le \delta,$$

then, with probability at least $1-\delta$*, every selected stream is contraction-normalized and all-plus certified:*

$$\mathrm{disc}(X_{h,1}/S_h, \dots, X_{h,n}/S_h) \le \sqrt{n} \qquad (h \in \mathcal{A}).$$

The same statement holds for rectangular $X_{h,i} \in \mathrm{M}_{d_1 \times d_2}(\mathbb{C})$*, with* $D^{\mathrm{rec}} = d_1 + d_2$ *replacing* D*, provided*

$$\|X_{h,i}\|_{\mathrm{op}} \le s_h$$

and

$$\left\| \sum_i \mathbb{E}(X_{h,i} X_{h,i}^* \mid \mathcal{F}) \right\|_{\mathrm{op}} \le v_h,$$

$$\left\| \sum_i \mathbb{E}(X_{h,i}^* X_{h,i} \mid \mathcal{F}) \right\|_{\mathrm{op}} \le v_h.$$

In particular, if $|\mathcal{A}| \le M$ *almost surely and*

$$L = \log \frac{2M\Delta}{\delta}, \qquad \Delta = D \text{ or } D^{\mathrm{rec}},$$

then the homogeneous choice

$$u = C(\sqrt{vL} + sL)$$

gives a simultaneous $1-\delta$ *certificate after normalizing by*

$$C \max\left\{ s, \sqrt{\frac{vL}{n}} + \frac{sL}{\sqrt{n}} \right\},$$

whenever $s_h \le s$ *and* $v_h \le v$ *for all selected* h*.*

Proof. Condition on $\mathcal{F}$. For a fixed selected self-adjoint stream, the two-sided matrix Bernstein inequality [25] gives

$$\mathbb{P}\left\{ \left\| \sum_i X_{h,i} \right\|_{\mathrm{op}} > u_h \,\middle|\, \mathcal{F} \right\} \le 2D \exp\left\{ -\frac{u_h^2}{2(v_h + s_h u_h/3)} \right\}.$$

Summing over the $\mathcal{F}$-measurable active set and taking expectation gives total failure at most δ. On the complementary event, $\|X_{h,i}/S_h\|_{\mathrm{op}} \le 1$ and

$$\left\|\sum_i X_{h,i}/S_h\right\|_{\mathrm{op}} \le u_h/S_h \le \sqrt{n},$$

so the all-plus signing is a Matrix-Spencer certificate.

The rectangular assertion follows by applying the self-adjoint statement to the dilations $\mathscr{D}(X_{h,i})$. The dilation has dimension D^{rec}, operator norm $\|X_{h,i}\|_{\mathrm{op}}$, and variance bounded by the two displayed row and column variance budgets. Finally, $\|\sum_i \mathscr{D}(X_{h,i})\|_{\mathrm{op}} = \|\sum_i X_{h,i}\|_{\mathrm{op}}$.

For the homogeneous finite-menu form, the usual Bernstein algebra gives

$$\frac{u^2}{2(v + su/3)} \ge L$$

after increasing the numerical constant in $u = C(\sqrt{vL} + sL)$. The displayed scale is $\max\{s, u/\sqrt{n}\}$, up to the same absolute constant. □

Corollary 7.40 (Finite-confidence quantum effect fluctuations)**.** *Let $\mathcal{F}$ be side information, let $\mathcal{A} \subset \mathbb{N}$ be an almost surely finite $\mathcal{F}$-measurable active set, and fix $n, D \ge 1$ and $0 < \delta < 1$. For each h, let*

$$E_{h,1}, \dots, E_{h,n} \in \mathrm{M}_D(\mathbb{C})$$

be conditionally independent quantum effects:

$$0 \preceq E_{h,i} \preceq I_D \qquad (1 \le i \le n).$$

Put

$$\mu_{h,i} := \mathbb{E}(E_{h,i} \mid \mathcal{F}), \qquad X_{h,i} := E_{h,i} - \mu_{h,i},$$

and suppose that $v_h, u_h > 0$ are $\mathcal{F}$-measurable proxies satisfying

$$\left\|\sum_{i=1}^n \mu_{h,i}\right\|_{\mathrm{op}} \le v_h \qquad (h \in \mathcal{A}).$$

Set

$$S_h := \max\{1, u_h/\sqrt{n}\}.$$

If

$$\mathbb{E} \sum_{h \in \mathcal{A}} 2D \exp\left\{-\frac{u_h^2}{2(v_h + u_h/3)}\right\} \le \delta,$$

then, with probability at least $1 - \delta$, every selected centered quantum-effect stream is contraction-normalized and all-plus certified:

$$\mathrm{disc}(X_{h,1}/S_h, \dots, X_{h,n}/S_h) \le \sqrt{n} \qquad (h \in \mathcal{A}).$$

In particular, if $|\mathcal{A}| \le M$ almost surely and

$$L = \log \frac{2MD}{\delta}, \qquad v_h \le v \quad (h \in \mathcal{A}),$$

then the homogeneous normalization

$$S = C \max\left\{1, \sqrt{\frac{vL}{n}} + \frac{L}{\sqrt{n}}\right\}$$

gives the same simultaneous $1-\delta$ *certificate. If* $C(\sqrt{vL}+L) \le \sqrt{n}$*, the unscaled centered effects* $X_{h,i}$ *are already all-plus Matrix-Spencer certified.*

For random density matrices with conditionally maximally mixed means $\mu_{h,i} = I_D/D$*, one may take* $v = n/D$*. Hence, for a finite active menu, the unscaled centered density-matrix fluctuations are all-plus certified as soon as* $D \gtrsim L$ *and* $n \gtrsim L^2$*.*

Proof. Condition on $\mathcal{F}$. Since $0 \preceq E_{h,i} \preceq I_D$ and $0 \preceq \mu_{h,i} \preceq I_D$, we have

$$-I_D \preceq -\mu_{h,i} \preceq X_{h,i} \preceq E_{h,i} \preceq I_D,$$

so $\|X_{h,i}\|_{\mathrm{op}} \le 1$. Also,

$$\mathbb{E}(X_{h,i}^2 \mid \mathcal{F}) = \mathbb{E}(E_{h,i}^2 \mid \mathcal{F}) - \mu_{h,i}^2 \preceq \mathbb{E}(E_{h,i}^2 \mid \mathcal{F}) \preceq \mu_{h,i},$$

because $E_{h,i}^2 \preceq E_{h,i}$. Thus

$$\left\| \sum_i \mathbb{E}(X_{h,i}^2 \mid \mathcal{F}) \right\|_{\mathrm{op}} \le \left\| \sum_i \mu_{h,i} \right\|_{\mathrm{op}} \le v_h.$$

Corollary 7.39 applies with $s_h = 1$. The homogeneous scale and the unscaled criterion are the displayed specializations of $u = C(\sqrt{vL}+L)$. If $\mu_{h,i} = I_D/D$, then $\| \sum_i \mu_{h,i} \|_{\mathrm{op}} = n/D$. □

Corollary 7.41 (Bounded matrix-Freedman all-plus active menus)**.** *For each* $m \ge 1$*, let* $D_m, n_m, J_m \ge 1$*. For each active* $1 \le h \le J_m$*, let*

$$(X_{m,h,i}, \mathcal{F}_{m,h,i})_{i=1}^{n_m}$$

be an adapted self-adjoint matrix martingale-difference sequence in $\mathrm{M}_{D_m}(\mathbb{C})$*. Let* $E_{m,h}$ *be an event, and let* $s_{m,h}, v_{m,h}, u_{m,h} > 0$ *be deterministic. Assume that*

$$\|X_{m,h,i}\|_{\mathrm{op}} \le s_{m,h} \qquad (1 \le i \le n_m)$$

almost surely and that, on $E_{m,h}^c$*,*

$$\left\| \sum_{i=1}^{n_m} \mathbb{E}(X_{m,h,i}^2 \mid \mathcal{F}_{m,h,i-1}) \right\|_{\mathrm{op}} \le v_{m,h}.$$

Set $S_{m,h} := \max\{s_{m,h}, u_{m,h}/\sqrt{n_m}\}$*. If*

$$\sum_m \sum_{h=1}^{J_m} \left[\mathbb{P}(E_{m,h}) + 2D_m \exp\left\{ -\frac{u_{m,h}^2}{2(v_{m,h} + s_{m,h}u_{m,h}/3)} \right\} \right] < \infty,$$

then, almost surely, for all sufficiently large m*, all active matrices* $X_{m,h,i}/S_{m,h}$ *are contractions and the all-plus signing gives*

$$\mathrm{disc}(X_{m,h,1}/S_{m,h}, \dots, X_{m,h,n_m}/S_{m,h}) \le \sqrt{n_m} \qquad (1 \le h \le J_m).$$

Thus the bounded Bernstein all-plus route remains valid for online or adapted centered self-adjoint streams after replacing the ordinary variance by the predictable quadratic variation.

Proof. Fix m, h and put

$$\nu_{m,h} := \left\| \sum_i \mathbb{E}(X_{m,h,i}^2 \mid \mathcal{F}_{m,h,i-1}) \right\|_{\mathrm{op}}.$$

The two-sided matrix Freedman inequality [25], applied to the martingale differences $X_{m,h,i}$ and $-X_{m,h,i}$, gives

$$\mathbb{P}\left\{ \left\| \sum_i X_{m,h,i} \right\|_{\mathrm{op}} > u_{m,h},\ \nu_{m,h} \le v_{m,h} \right\} \le 2D_m \exp\left\{ -\frac{u_{m,h}^2}{2(v_{m,h} + s_{m,h}u_{m,h}/3)} \right\}.$$

On $E_{m,h}^c$ the event $\nu_{m,h} \le v_{m,h}$ holds, so the displayed series controls all active all-plus failures together with the exceptional events. Borel–Cantelli and the same normalization calculation as in Corollary 7.37 finish the proof. □

Corollary 7.42 (Rectangular bounded matrix-Freedman all-plus active menus)**.** *For each $m \ge 1$, let $d_{m,1}, d_{m,2} \ge 1$, $D_m^{\mathrm{rec}} = d_{m,1} + d_{m,2}$, $n_m \ge 1$, and $J_m \ge 1$. For each active $1 \le h \le J_m$, let*

$$(X_{m,h,i}, \mathcal{F}_{m,h,i})_{i=1}^{n_m}$$

be adapted rectangular martingale differences with values in

$$\mathrm{M}_{d_{m,1} \times d_{m,2}}(\mathbb{C}).$$

Let $E_{m,h}$ be an event, and let $s_{m,h}, v_{m,h}, u_{m,h} > 0$ be deterministic. Assume that

$$\|X_{m,h,i}\|_{\mathrm{op}} \le s_{m,h} \qquad (1 \le i \le n_m)$$

almost surely and that, on $E_{m,h}^c$,

$$\left\| \sum_i \mathbb{E}(X_{m,h,i} X_{m,h,i}^* \mid \mathcal{F}_{m,h,i-1}) \right\|_{\mathrm{op}} \le v_{m,h},$$

$$\left\| \sum_i \mathbb{E}(X_{m,h,i}^* X_{m,h,i} \mid \mathcal{F}_{m,h,i-1}) \right\|_{\mathrm{op}} \le v_{m,h}.$$

Set $S_{m,h} := \max\{s_{m,h}, u_{m,h}/\sqrt{n_m}\}$. If

$$\sum_m \sum_{h=1}^{J_m} \left[\mathbb{P}(E_{m,h}) + 2D_m^{\mathrm{rec}} \exp\left\{ -\frac{u_{m,h}^2}{2(v_{m,h} + s_{m,h}u_{m,h}/3)} \right\} \right] < \infty,$$

then, almost surely, for all sufficiently large m, all active rectangular matrices $X_{m,h,i}/S_{m,h}$ are contractions and the all-plus signing gives

$$\mathrm{disc}(X_{m,h,1}/S_{m,h}, \dots, X_{m,h,n_m}/S_{m,h}) \le \sqrt{n_m} \qquad (1 \le h \le J_m).$$

This covers online rectangular feature blocks, adaptive measurement increments, and other non-self-adjoint martingale-difference streams under row and column predictable variance budgets.

Proof. Apply Corollary 7.41 to the self-adjoint dilations $\mathscr{D}(X_{m,h,i})$. The dilation is a self-adjoint martingale difference, has norm $\|X_{m,h,i}\|_{\mathrm{op}}$, and has predictable quadratic variation block diagonal with the two displayed row and column variance sums. The operator norm of the dilated all-plus sum is the operator norm of the rectangular all-plus sum, so the self-adjoint conclusion is exactly the desired rectangular certificate. □

Corollary 7.43 (Finite-confidence bounded matrix-Freedman all-plus certificates)**.** *Let $\mathcal{G}$ be side information, let $\mathcal{A} \subset \mathbb{N}$ be an almost surely finite $\mathcal{G}$-measurable active set, and fix $n \geq 1$ and $0 < \delta < 1$. For each h, let*

$$(\mathcal{F}_{h,i})_{i=0}^{n}$$

be a filtration with $\mathcal{G} \subset \mathcal{F}_{h,0}$, and let $X_{h,1}, \dots, X_{h,n} \in \mathrm{M}_D(\mathbb{C})$ be a self-adjoint matrix martingale-difference sequence adapted to it. Assume that every selected stream has $\mathcal{G}$-measurable proxies $s_h, v_h, u_h > 0$ satisfying

$$\|X_{h,i}\|_{\mathrm{op}} \leq s_h \qquad (1 \leq i \leq n)$$

almost surely and

$$\left\| \sum_{i=1}^{n} \mathbb{E}(X_{h,i}^2 \mid \mathcal{F}_{h,i-1}) \right\|_{\mathrm{op}} \leq v_h$$

almost surely. Set

$$S_h := \max\{s_h, u_h/\sqrt{n}\}.$$

If

$$\mathbb{E} \sum_{h \in \mathcal{A}} 2D \exp\left\{ -\frac{u_h^2}{2(v_h + s_h u_h/3)} \right\} \leq \delta,$$

then, with probability at least $1 - \delta$, every selected stream is contraction-normalized and all-plus certified:

$$\mathrm{disc}(X_{h,1}/S_h, \dots, X_{h,n}/S_h) \leq \sqrt{n} \qquad (h \in \mathcal{A}).$$

The same statement holds for rectangular martingale differences $X_{h,i} \in \mathrm{M}_{d_1 \times d_2}(\mathbb{C})$, with $D^{\mathrm{rec}} = d_1 + d_2$ replacing D, provided

$$\|X_{h,i}\|_{\mathrm{op}} \leq s_h$$

and

$$\left\| \sum_{i} \mathbb{E}(X_{h,i} X_{h,i}^* \mid \mathcal{F}_{h,i-1}) \right\|_{\mathrm{op}} \leq v_h,$$

$$\left\| \sum_{i} \mathbb{E}(X_{h,i}^* X_{h,i} \mid \mathcal{F}_{h,i-1}) \right\|_{\mathrm{op}} \leq v_h$$

almost surely. In particular, if $|\mathcal{A}| \le M$ *almost surely and*

$$L = \log \frac{2M\Delta}{\delta}, \qquad \Delta = D \text{ or } D^{\mathrm{rec}},$$

then the homogeneous choice

$$u = C(\sqrt{vL} + sL)$$

gives a simultaneous $1-\delta$ *certificate after normalizing by*

$$C \max\left\{ s, \sqrt{\frac{vL}{n}} + \frac{sL}{\sqrt{n}} \right\},$$

whenever $s_h \le s$ *and* $v_h \le v$ *for all selected* h.

Proof. Condition on $\mathcal{G}$. For a fixed selected self-adjoint stream, put

$$\nu_h := \left\| \sum_i \mathbb{E}(X_{h,i}^2 \mid \mathcal{F}_{h,i-1}) \right\|_{\mathrm{op}}.$$

The two-sided matrix Freedman inequality [25], applied conditionally on the initial side information, gives

$$\mathbb{P}\left\{ \left\| \sum_i X_{h,i} \right\|_{\mathrm{op}} > u_h,\ \nu_h \le v_h \,\middle|\, \mathcal{G} \right\} \le 2D \exp\left\{ -\frac{u_h^2}{2(v_h + s_h u_h/3)} \right\}.$$

The predictable variance bound removes the event $\nu_h \le v_h$. Summing over the $\mathcal{G}$-measurable active set and taking expectation gives total failure at most δ. On the complementary event, $\|X_{h,i}/S_h\|_{\mathrm{op}} \le 1$ and

$$\left\| \sum_i X_{h,i}/S_h \right\|_{\mathrm{op}} \le u_h/S_h \le \sqrt{n},$$

so the all-plus signing is a Matrix-Spencer certificate.

The rectangular assertion follows by applying the self-adjoint statement to the dilations $\mathscr{D}(X_{h,i})$. The dilation is a self-adjoint martingale difference, has dimension D^{rec}, has operator norm $\|X_{h,i}\|_{\mathrm{op}}$, and has predictable quadratic variation bounded by the two displayed row and column variance budgets. The all-plus norm is preserved by dilation.

The homogeneous finite-menu scale is the same Bernstein algebra used in Corollary 7.39. □

Corollary 7.44 (Explicit bounded Bernstein/Freedman all-plus schedules)**.** *Consider any one of the four bounded all-plus settings in Corollaries 7.37, 7.38, 7.41, and 7.42. Let* $\Delta_m = D_m$ *in the self-adjoint case and* $\Delta_m = D_m^{\mathrm{rec}}$

in the rectangular case, and put $\ell_m = \log(e\Delta_m)$. *Fix* $K \geq 1$, $\beta \geq 0$, *and* $B > 0$. *Assume that, outside the exceptional events* $E_{m,h}$,

$$v_{m,h} \leq \beta s_{m,h}^2 \frac{n_m}{\ell_m} \qquad (1 \leq h \leq J_m),$$

that

$$\sum_m \sum_{h=1}^{J_m} \mathbb{P}(E_{m,h}) < \infty,$$

and that $\ell_m \leq B\sqrt{n_m}$ *eventually. Define*

$$\gamma(K, \beta, B) := \frac{K^2}{2(\beta + KB/3)}.$$

If

$$\sum_m J_m \Delta_m (e\Delta_m)^{-\gamma(K,\beta,B)} < \infty,$$

then, almost surely, for all sufficiently large m, *every active self-adjoint or rectangular bounded Bernstein/Freedman stream is contraction-normalized by* $Ks_{m,h}$, *and the all-plus signing gives*

$$\operatorname{disc}(X_{m,h,1}/(Ks_{m,h}), \dots, X_{m,h,n_m}/(Ks_{m,h})) \leq \sqrt{n_m} \qquad (1 \leq h \leq J_m).$$

In particular, if $J_m\Delta_m \leq (e\Delta_m)^\Gamma$ *eventually and* $n_m \geq a\ell_m^2$, *then the conclusion holds along all integer dimensions* $\Delta_m = m$ *whenever*

$$\gamma(K, \beta, a^{-1/2}) > \Gamma + 1,$$

and along every stretched-exponential sparse dimension schedule whenever

$$\gamma(K, \beta, a^{-1/2}) > \Gamma.$$

Proof. Use the appropriate preceding self-adjoint or rectangular bounded all-plus corollary with $u_{m,h} = Ks_{m,h}\sqrt{n_m}$. Since $K \geq 1$, its normalizing scale is

$$S_{m,h} = \max\{s_{m,h}, u_{m,h}/\sqrt{n_m}\} = Ks_{m,h}.$$

On $E_{m,h}^c$, the common Bernstein/Freedman exponent is at least

$$\begin{aligned} \frac{u_{m,h}^2}{2(v_{m,h} + s_{m,h}u_{m,h}/3)} &\geq \frac{K^2 n_m}{2(\beta n_m/\ell_m + K\sqrt{n_m}/3)} \\ &\geq \gamma(K, \beta, B)\ell_m, \end{aligned}$$

because $\ell_m \leq B\sqrt{n_m}$. Hence the active-row failure probability is bounded by a constant multiple of

$$J_m\Delta_m(e\Delta_m)^{-\gamma(K,\beta,B)}$$

plus the exceptional-event probabilities. The displayed summability and Borel–Cantelli give the general schedule. If $J_m\Delta_m \leq (e\Delta_m)^\Gamma$, the remaining summand is bounded by a constant multiple of $(e\Delta_m)^{\Gamma-\gamma}$. This is summable over all integer dimensions for $\gamma > \Gamma + 1$, and is summable along every stretched-exponential sparse schedule for $\gamma > \Gamma$. □

Corollary 7.45 (Proxy-scale bounded all-plus schedules)**.** *Consider any one of the four bounded all-plus settings in Corollaries 7.37, 7.38, 7.41, and 7.42. Let $\Delta_m = D_m$ in the self-adjoint case and $\Delta_m = D_m^{\mathrm{rec}}$ in the rectangular case, and put $\ell_m = \log(e\Delta_m)$. Fix $K > 0$, $R > 0$, $\beta \geq 0$, and $B > 0$. For each active stream let $q_{m,h} > 0$ be a deterministic proxy scale and let $b_{m,h}$ be a deterministic jump bound, so that all increments have operator norm at most $b_{m,h}$. Assume that, outside $E_{m,h}$,*

$$b_{m,h} \leq R q_{m,h}, \qquad v_{m,h} \leq \beta q_{m,h}^2 \frac{n_m}{\ell_m} \qquad (1 \leq h \leq J_m),$$

that

$$\sum_m \sum_{h=1}^{J_m} \mathbb{P}(E_{m,h}) < \infty,$$

and that $\ell_m \leq B\sqrt{n_m}$ eventually. Put

$$\gamma_{\mathrm{pr}}(K, R, \beta, B) := \frac{K^2}{2(\beta + KRB/3)}.$$

If

$$\sum_m J_m \Delta_m (e\Delta_m)^{-\gamma_{\mathrm{pr}}(K,R,\beta,B)} < \infty,$$

then, almost surely, for all sufficiently large m, every active self-adjoint or rectangular bounded Bernstein/Freedman stream is contraction-normalized by

$$M q_{m,h}, \qquad M := \max\{R, K\},$$

and the all-plus signing gives

$$\mathrm{disc}(X_{m,h,1}/(Mq_{m,h}), \ldots, X_{m,h,n_m}/(Mq_{m,h})) \leq \sqrt{n_m} \qquad (1 \leq h \leq J_m).$$

In particular, if $J_m \Delta_m \leq (e\Delta_m)^{\Gamma}$ eventually and $n_m \geq a\ell_m^2$, then the conclusion holds along all integer dimensions whenever

$$\gamma_{\mathrm{pr}}(K, R, \beta, a^{-1/2}) > \Gamma + 1,$$

and along every stretched-exponential sparse dimension schedule whenever

$$\gamma_{\mathrm{pr}}(K, R, \beta, a^{-1/2}) > \Gamma.$$

This lets heterogeneous bounded libraries use a typical variance scale $q_{m,h}$ rather than the worst increment scale, provided the worst increment is within a fixed factor R of $q_{m,h}$.

Proof. Apply the appropriate bounded Bernstein or bounded Freedman all-plus corollary with common jump $s_{m,h} = b_{m,h}$ and

$$u_{m,h} = K q_{m,h} \sqrt{n_m}.$$

The normalizing scale in that corollary is at most

$$\max\{b_{m,h}, u_{m,h}/\sqrt{n_m}\} \leq \max\{R, K\} q_{m,h} = M q_{m,h}.$$

Using the larger scale $Mq_{m,h}$ preserves contraction and only reduces the all-plus norm. On $E_{m,h}^c$, the common Bernstein/Freedman exponent is at least

$$\begin{aligned}\frac{u_{m,h}^2}{2(v_{m,h}+b_{m,h}u_{m,h}/3)} &\geq \frac{K^2 n_m}{2(\beta n_m/\ell_m + KR\sqrt{n_m}/3)}\\ &\geq \gamma_{\mathrm{pr}}(K,R,\beta,B)\ell_m.\end{aligned}$$

The same Borel–Cantelli and summability argument as in Corollary 7.44 gives the active-menu, all-integer, and sparse conclusions. □

Corollary 7.46 (Tail-capped centered Bernstein all-plus schedules)**.** *For each $m \geq 1$, consider either self-adjoint dimensions $D_m \geq 2$ or rectangular dimensions $d_{m,1}, d_{m,2} \geq 1$. Put $\Delta_m = D_m$ in the self-adjoint case, put $\Delta_m = d_{m,1} + d_{m,2}$ in the rectangular case, and let $\ell_m = \log(e\Delta_m)$. Let $\mathcal{H}_m$ be an environment. For each active $1 \leq h \leq J_m$, let*

$$X_{m,h,1}, \dots, X_{m,h,n_m}$$

be conditionally independent random matrices, self-adjoint in $\mathrm{M}_{D_m}(\mathbb{C})$ in the first case and rectangular in $\mathrm{M}_{d_{m,1}\times d_{m,2}}(\mathbb{C})$ in the second. Let $E_{m,h} \in \mathcal{H}_m$ be exceptional events. Let $q_{m,h} > 0$ and $b_{m,h} > 0$ be deterministic scales, and define the capped variables

$$T_{m,h,i} := X_{m,h,i}\mathbf{1}_{\{\|X_{m,h,i}\|_{\mathrm{op}} \leq b_{m,h}\}}, \qquad \mu_{m,h,i} := \mathbb{E}(T_{m,h,i} \mid \mathcal{H}_m),$$

$$Y_{m,h,i} := T_{m,h,i} - \mu_{m,h,i}, \qquad F_{m,h}^{\mathrm{cap}} := \left\{ \max_{1\leq i\leq n_m} \|X_{m,h,i}\|_{\mathrm{op}} > b_{m,h} \right\}.$$

Fix constants $K > 0$, $R > 0$, $\beta \geq 0$, $\eta \geq 0$, and $B > 0$. Assume that $b_{m,h} \leq Rq_{m,h}$, that

$$\sum_m \sum_{h=1}^{J_m} [\mathbb{P}(E_{m,h}) + \mathbb{P}(F_{m,h}^{\mathrm{cap}})] < \infty,$$

and that, outside $E_{m,h}$,

$$\left\| \sum_{i=1}^{n_m} \mu_{m,h,i} \right\|_{\mathrm{op}} \leq \eta q_{m,h}\sqrt{n_m}.$$

In the self-adjoint case assume also, outside $E_{m,h}$, that

$$\left\| \sum_{i=1}^{n_m} \mathbb{E}(Y_{m,h,i}^2 \mid \mathcal{H}_m) \right\|_{\mathrm{op}} \leq \beta q_{m,h}^2 \frac{n_m}{\ell_m}.$$

In the rectangular case replace this by the two row and column bounds

$$\left\| \sum_i \mathbb{E}(Y_{m,h,i}Y_{m,h,i}^* \mid \mathcal{H}_m) \right\|_{\mathrm{op}} \leq \beta q_{m,h}^2 \frac{n_m}{\ell_m},$$

$$\left\| \sum_i \mathbb{E}(Y_{m,h,i}^*Y_{m,h,i} \mid \mathcal{H}_m) \right\|_{\mathrm{op}} \leq \beta q_{m,h}^2 \frac{n_m}{\ell_m}.$$

Suppose that $\ell_m \le B\sqrt{n_m}$ eventually and put

$$\gamma_{\mathrm{tc}}(K,R,\beta,B) := \frac{K^2}{2(\beta + 2KRB/3)}.$$

If

$$\sum_m J_m \Delta_m (e\Delta_m)^{-\gamma_{\mathrm{tc}}(K,R,\beta,B)} < \infty,$$

then, almost surely, for all sufficiently large m, every active original stream is contraction-normalized by

$$Mq_{m,h}, \qquad M := \max\{R, K+\eta\},$$

and the all-plus signing gives

$$\operatorname{disc}(X_{m,h,1}/(Mq_{m,h}), \dots, X_{m,h,n_m}/(Mq_{m,h})) \le \sqrt{n_m} \qquad (1 \le h \le J_m).$$

For $A \ge 0$ define the closed-form margin

$$\Theta_{\mathrm{tc}}(A,a,\beta,R) := \frac{2AR}{3\sqrt{a}} + \sqrt{\frac{4A^2R^2}{9a} + 2A\beta}.$$

If $J_m\Delta_m \le (e\Delta_m)^{\Gamma}$ eventually and $n_m \ge a\ell_m^2$, then the conclusion holds along all integer dimensions $\Delta_m = m$ whenever

$$K > \Theta_{\mathrm{tc}}(\Gamma+1, a, \beta, R),$$

and along every stretched-exponential sparse dimension schedule whenever

$$K > \Theta_{\mathrm{tc}}(\Gamma, a, \beta, R).$$

When the original variables are conditionally centered, the drift condition is only a tail-mean budget, since $\mu_{m,h,i} = -\mathbb{E}(X_{m,h,i}\mathbf{1}_{\{\|X_{m,h,i}\|_{\mathrm{op}} > b_{m,h}\}} \mid \mathcal{H}_m)$.

Proof. Fix m, h and condition on $\mathcal{H}_m$. Outside $E_{m,h}$, the matrices $Y_{m,h,i}$ are conditionally independent and centered. Moreover,

$$\|Y_{m,h,i}\|_{\mathrm{op}} \le \|T_{m,h,i}\|_{\mathrm{op}} + \|\mathbb{E}(T_{m,h,i} \mid \mathcal{H}_m)\|_{\mathrm{op}} \le 2b_{m,h} \le 2Rq_{m,h}.$$

In the self-adjoint case, matrix Bernstein gives

$$\mathbb{P}\left\{ \left\| \sum_i Y_{m,h,i} \right\|_{\mathrm{op}} > Kq_{m,h}\sqrt{n_m} \,\middle|\, \mathcal{H}_m \right\} \le 2\Delta_m \exp\left\{ -\frac{K^2 n_m}{2(\beta n_m/\ell_m + 2RK\sqrt{n_m}/3)} \right\}.$$

The rectangular case is identical after applying the same inequality to the self-adjoint dilations; the two displayed row and column variance bounds are exactly the dilation variance bound. Since $\ell_m \le B\sqrt{n_m}$, the exponent is at least $\gamma_{\mathrm{tc}}(K,R,\beta,B)\ell_m$. Hence the active centered-sum failure probabilities are bounded by a constant multiple of

$$J_m\Delta_m(e\Delta_m)^{-\gamma_{\mathrm{tc}}(K,R,\beta,B)}.$$

Together with the summable $E_{m,h}$ and cap failures, Borel–Cantelli gives, almost surely, eventual simultaneous control over all active h.

On that eventual event, $F_{m,h}^{\mathrm{cap}}$ does not occur, so $T_{m,h,i} = X_{m,h,i}$ for every i. Therefore

$$\left\|\sum_i X_{m,h,i}\right\|_{\mathrm{op}} \le \left\|\sum_i Y_{m,h,i}\right\|_{\mathrm{op}} + \left\|\sum_i \mu_{m,h,i}\right\|_{\mathrm{op}} \le (K+\eta) q_{m,h}\sqrt{n_m},$$

while $\max_i \|X_{m,h,i}\|_{\mathrm{op}} \le b_{m,h} \le R q_{m,h}$. Dividing by $M q_{m,h}$, where $M = \max\{R, K+\eta\}$, gives contractions and an all-plus sum of norm at most $\sqrt{n_m}$.

For the schedule statements, $n_m \ge a\ell_m^2$ lets us run the preceding argument with $B = a^{-1/2}$. If $J_m \Delta_m \le (e\Delta_m)^{\Gamma}$, the remaining row summand is $O((e\Delta_m)^{\Gamma-\gamma_{\mathrm{tc}}})$, summable over all integer dimensions when $\gamma_{\mathrm{tc}} > \Gamma + 1$ and along stretched-exponential schedules when $\gamma_{\mathrm{tc}} > \Gamma$. Finally,

$$\gamma_{\mathrm{tc}}(K, R, \beta, a^{-1/2}) > A$$

is equivalent to

$$K^2 - \frac{4AR}{3\sqrt{a}} K - 2A\beta > 0,$$

whose positive root is $\Theta_{\mathrm{tc}}(A, a, \beta, R)$. □

Corollary 7.47 (Tail-capped matrix-Freedman all-plus schedules)**.** *For each $m \ge 1$, consider either self-adjoint dimensions $D_m \ge 2$ or rectangular dimensions $d_{m,1}, d_{m,2} \ge 1$. Put $\Delta_m = D_m$ in the self-adjoint case, put $\Delta_m = d_{m,1} + d_{m,2}$ in the rectangular case, and let $\ell_m = \log(e\Delta_m)$. For each active $1 \le h \le J_m$, let*

$$(X_{m,h,i}, \mathcal{F}_{m,h,i})_{i=1}^{n_m}$$

be an adapted martingale-difference sequence, self-adjoint in $\mathrm{M}_{D_m}(\mathbb{C})$ in the first case and rectangular in $\mathrm{M}_{d_{m,1}\times d_{m,2}}(\mathbb{C})$ in the second. Let $E_{m,h}$ be an event, let $q_{m,h} > 0$ and $b_{m,h} > 0$ be deterministic scales, and define

$$T_{m,h,i} := X_{m,h,i}\mathbf{1}_{\{\|X_{m,h,i}\|_{\mathrm{op}} \le b_{m,h}\}}, \qquad \mu_{m,h,i} := \mathbb{E}(T_{m,h,i} \mid \mathcal{F}_{m,h,i-1}),$$

$$Y_{m,h,i} := T_{m,h,i} - \mu_{m,h,i}, \qquad F_{m,h}^{\mathrm{cap}} := \left\{\max_{1\le i\le n_m} \|X_{m,h,i}\|_{\mathrm{op}} > b_{m,h}\right\}.$$

Fix constants $K > 0$, $R > 0$, $\beta \ge 0$, $\eta \ge 0$, and $B > 0$. Assume that $b_{m,h} \le R q_{m,h}$, that

$$\sum_m \sum_{h=1}^{J_m} [\mathbb{P}(E_{m,h}) + \mathbb{P}(F_{m,h}^{\mathrm{cap}})] < \infty,$$

and that, outside $E_{m,h}$,

$$\left\|\sum_{i=1}^{n_m} \mu_{m,h,i}\right\|_{\mathrm{op}} \le \eta q_{m,h}\sqrt{n_m}.$$

In the self-adjoint case assume also, outside $E_{m,h}$, that

$$\left\|\sum_{i=1}^{n_m}\mathbb{E}(Y_{m,h,i}^2 \mid \mathcal{F}_{m,h,i-1})\right\|_{\mathrm{op}} \le \beta q_{m,h}^2\,\frac{n_m}{\ell_m}.$$

In the rectangular case replace this by the two predictable row and column bounds

$$\left\|\sum_i \mathbb{E}(Y_{m,h,i}Y_{m,h,i}^* \mid \mathcal{F}_{m,h,i-1})\right\|_{\mathrm{op}} \le \beta q_{m,h}^2\,\frac{n_m}{\ell_m},$$

$$\left\|\sum_i \mathbb{E}(Y_{m,h,i}^*Y_{m,h,i} \mid \mathcal{F}_{m,h,i-1})\right\|_{\mathrm{op}} \le \beta q_{m,h}^2\,\frac{n_m}{\ell_m}.$$

Suppose that $\ell_m \le B\sqrt{n_m}$ eventually. If

$$\sum_m J_m\Delta_m(e\Delta_m)^{-\gamma_{\mathrm{tc}}(K,R,\beta,B)} < \infty,$$

where γ_{tc} is the exponent from Corollary 7.46, then, almost surely, for all sufficiently large m, every active original stream is contraction-normalized by

$$Mq_{m,h}, \qquad M := \max\{R, K+\eta\},$$

and the all-plus signing gives

$$\mathrm{disc}(X_{m,h,1}/(Mq_{m,h}),\dots,X_{m,h,n_m}/(Mq_{m,h})) \le \sqrt{n_m} \qquad (1 \le h \le J_m).$$

If $J_m\Delta_m \le (e\Delta_m)^{\Gamma}$ eventually and $n_m \ge a\ell_m^2$, then the same closed-form margin Θ_{tc} gives the all-integer threshold

$$K > \Theta_{\mathrm{tc}}(\Gamma+1, a, \beta, R)$$

and the stretched-exponential sparse threshold

$$K > \Theta_{\mathrm{tc}}(\Gamma, a, \beta, R).$$

Since the original stream is a martingale difference,

$$\mu_{m,h,i} = -\mathbb{E}\left(X_{m,h,i}\mathbf{1}_{\{\|X_{m,h,i}\|_{\mathrm{op}} > b_{m,h}\}} \mid \mathcal{F}_{m,h,i-1}\right),$$

so the drift hypothesis is again only a predictable tail-mean budget.

Proof. Fix m, h. The variables $Y_{m,h,i}$ are adapted martingale differences, and

$$\|Y_{m,h,i}\|_{\mathrm{op}} \le 2b_{m,h} \le 2Rq_{m,h}.$$

In the self-adjoint case, matrix Freedman applied to $Y_{m,h,i}$ and $-Y_{m,h,i}$ gives

$$\mathbb{P}\left\{\left\|\sum_i Y_{m,h,i}\right\|_{\mathrm{op}} > Kq_{m,h}\sqrt{n_m},\ \nu_{m,h} \le \beta q_{m,h}^2 n_m/\ell_m\right\}$$
$$\le 2\Delta_m \exp\left\{-\frac{K^2 n_m}{2(\beta n_m/\ell_m + 2RK\sqrt{n_m}/3)}\right\},$$

where

$$\nu_{m,h} := \left\| \sum_i \mathbb{E}(Y_{m,h,i}^2 \mid \mathcal{F}_{m,h,i-1}) \right\|_{\mathrm{op}}.$$

In the rectangular case, apply the same argument to the self-adjoint dilations; the two displayed predictable row and column bounds are exactly the dilation variance bound. The exponent is at least $\gamma_{\mathrm{tc}}(K, R, \beta, B)\ell_m$ whenever $\ell_m \le B\sqrt{n_m}$. The same Borel–Cantelli argument as in Corollary 7.46 gives eventual simultaneous control of the centered capped sums, together with the summable exceptional and cap-failure events.

On the eventual no-cap event,

$$X_{m,h,i} = T_{m,h,i} = Y_{m,h,i} + \mu_{m,h,i},$$

so the drift estimate gives

$$\left\| \sum_i X_{m,h,i} \right\|_{\mathrm{op}} \le (K + \eta) q_{m,h} \sqrt{n_m}.$$

The same event gives $\max_i \|X_{m,h,i}\|_{\mathrm{op}} \le b_{m,h} \le R q_{m,h}$. Dividing by $M q_{m,h}$ proves the all-plus certificate. The all-integer and sparse thresholds are the identical summability calculation and quadratic-margin calculation from Corollary 7.46. □

Corollary 7.48 (Finite-confidence tail-capped all-plus certificates)**.** *Let $\mathcal{G}$ be side information, let $\mathcal{A} \subset \mathbb{N}$ be an almost surely finite $\mathcal{G}$-measurable active set, and fix $n \ge 1$ and $0 < \delta < 1$. Work in one of the following two settings.*

Independent setting. *Conditionally on $\mathcal{G}$, the variables $X_{h,1}, \dots, X_{h,n}$ in each selected stream are independent. They are either self-adjoint in $\mathrm{M}_D(\mathbb{C})$ or rectangular in $\mathrm{M}_{d_1 \times d_2}(\mathbb{C})$. Put $\mathcal{P}_{h,i} = \mathcal{G}$.*

Adapted setting. *For each selected h, let $(\mathcal{F}_{h,i})_{i=0}^n$ be a filtration with $\mathcal{G} \subset \mathcal{F}_{h,0}$, and let $X_{h,1}, \dots, X_{h,n}$ be a self-adjoint or rectangular martingale-difference sequence adapted to it. Put $\mathcal{P}_{h,i} = \mathcal{F}_{h,i-1}$.*

Let $\Delta = D$ in the self-adjoint case and $\Delta = d_1 + d_2$ in the rectangular case. Let $b_h, v_h, u_h, r_h > 0$ be $\mathcal{G}$-measurable proxies. Define

$$T_{h,i} := X_{h,i} \mathbf{1}_{\{\|X_{h,i}\|_{\mathrm{op}} \le b_h\}}, \qquad \mu_{h,i} := \mathbb{E}(T_{h,i} \mid \mathcal{P}_{h,i}),$$

$$Y_{h,i} := T_{h,i} - \mu_{h,i}, \qquad F_h^{\mathrm{cap}} := \left\{ \max_{1 \le i \le n} \|X_{h,i}\|_{\mathrm{op}} > b_h \right\}.$$

Assume that the selected streams satisfy the drift bound

$$\left\| \sum_{i=1}^n \mu_{h,i} \right\|_{\mathrm{op}} \le r_h$$

almost surely. In the self-adjoint case assume

$$\left\| \sum_{i=1}^n \mathbb{E}(Y_{h,i}^2 \mid \mathcal{P}_{h,i}) \right\|_{\mathrm{op}} \le v_h$$

almost surely. In the rectangular case replace this by

$$\left\|\sum_i \mathbb{E}(Y_{h,i}Y_{h,i}^* \mid \mathcal{P}_{h,i})\right\|_{\mathrm{op}} \le v_h,$$
$$\left\|\sum_i \mathbb{E}(Y_{h,i}^*Y_{h,i} \mid \mathcal{P}_{h,i})\right\|_{\mathrm{op}} \le v_h$$

almost surely. If

$$\mathbb{E}\sum_{h\in\mathcal{A}}\left[\mathbb{P}(F_h^{\mathrm{cap}} \mid \mathcal{G}) + 2\Delta\exp\left\{-\frac{u_h^2}{2(v_h+2b_hu_h/3)}\right\}\right] \le \delta,$$

then, with probability at least $1-\delta$*, every selected original stream is all-plus certified after the contraction-normalizing scale*

$$S_h := \max\left\{b_h, \frac{u_h+r_h}{\sqrt{n}}\right\}:$$

$$\mathrm{disc}(X_{h,1}/S_h,\dots,X_{h,n}/S_h) \le \sqrt{n} \qquad (h\in\mathcal{A}).$$

In particular, if $|\mathcal{A}| \le M$ *almost surely,* $\mathbb{E}\sum_{h\in\mathcal{A}}\mathbb{P}(F_h^{\mathrm{cap}} \mid \mathcal{G}) \le \delta/2$*, and* $b_h \le b$*,* $v_h \le v$*,* $r_h \le r$ *on the selected set, then with*

$$L = \log\frac{4M\Delta}{\delta}, \qquad u = C(\sqrt{vL}+bL),$$

the same conclusion holds with probability at least $1-\delta$ *after the homogeneous normalization*

$$C\max\left\{b, \frac{r+\sqrt{vL}+bL}{\sqrt{n}}\right\}.$$

Proof. Condition on $\mathcal{G}$ and fix a selected stream. In the independent setting the matrices $Y_{h,i}$ are conditionally independent and centered. In the adapted setting they are martingale differences. In either case,

$$\|Y_{h,i}\|_{\mathrm{op}} \le \|T_{h,i}\|_{\mathrm{op}} + \|\mathbb{E}(T_{h,i} \mid \mathcal{P}_{h,i})\|_{\mathrm{op}} \le 2b_h.$$

The two-sided matrix Bernstein inequality in the independent self-adjoint case, and the two-sided matrix Freedman inequality in the adapted self-adjoint case, give

$$\mathbb{P}\left\{\left\|\sum_i Y_{h,i}\right\|_{\mathrm{op}} > u_h \,\middle|\, \mathcal{G}\right\} \le 2D\exp\left\{-\frac{u_h^2}{2(v_h+2b_hu_h/3)}\right\}.$$

For rectangular streams apply the same argument to the self-adjoint dilations; the row and column variance hypotheses are exactly the dilation variance bound, and the dimension is $\Delta = d_1+d_2$.

Taking the union bound over the $\mathcal{G}$-measurable active set and then expectation gives total failure at most δ. On the complementary event no selected

cap failure occurs, so $X_{h,i} = T_{h,i}$ and $\|X_{h,i}\|_{\mathrm{op}} \le b_h$. Also,

$$\left\|\sum_i X_{h,i}\right\|_{\mathrm{op}} \le \left\|\sum_i Y_{h,i}\right\|_{\mathrm{op}} + \left\|\sum_i \mu_{h,i}\right\|_{\mathrm{op}} \le u_h + r_h.$$

Dividing by S_h gives contractions and all-plus sum norm at most $\sqrt{n}$.

For the homogeneous form, the cap failures cost at most $\delta/2$. The usual Bernstein algebra gives

$$\frac{u^2}{2(v + 2bu/3)} \ge L$$

after increasing the numerical constant in $u = C(\sqrt{vL} + bL)$, so the centered capped-sum failures also cost at most $\delta/2$. □

Corollary 7.49 (Finite-confidence sub-Weibull norm-tail certificates)**.** *Work in the independent or adapted setting of Corollary 7.48. Fix $\alpha > 0$ and $A_* \ge 1$. For each selected stream let $s_h > 0$, $L_h \ge 1$, $v_h > 0$, $u_h > 0$, and $r_h \ge 0$ be $\mathcal{G}$-measurable, and put*

$$b_h := s_h L_h^{1/\alpha}.$$

Assume the conditional operator-norm cap tails

$$\mathbb{P}\{\|X_{h,i}\|_{\mathrm{op}} > b_h \mid \mathcal{P}_{h,i}\} \le A_* e^{-L_h} \qquad (1 \le i \le n).$$

Define

$$T_{h,i} := X_{h,i}\mathbf{1}_{\{\|X_{h,i}\|_{\mathrm{op}} \le b_h\}}, \qquad \mu_{h,i} := \mathbb{E}(T_{h,i} \mid \mathcal{P}_{h,i}).$$

Let

$$Y_{h,i} := T_{h,i} - \mu_{h,i}, \qquad F_h^{\mathrm{cap}} := \left\{\max_{1\le i\le n} \|X_{h,i}\|_{\mathrm{op}} > b_h\right\}.$$

Assume the truncated drift bound

$$\left\|\sum_i \mu_{h,i}\right\|_{\mathrm{op}} \le r_h.$$

In the self-adjoint case assume the uncentered capped variance bound

$$\left\|\sum_i \mathbb{E}(T_{h,i}^2 \mid \mathcal{P}_{h,i})\right\|_{\mathrm{op}} \le v_h.$$

In the rectangular case assume

$$\left\|\sum_i \mathbb{E}(T_{h,i}T_{h,i}^* \mid \mathcal{P}_{h,i})\right\|_{\mathrm{op}} \le v_h,$$

$$\left\|\sum_i \mathbb{E}(T_{h,i}^*T_{h,i} \mid \mathcal{P}_{h,i})\right\|_{\mathrm{op}} \le v_h.$$

If

$$\mathbb{E}\sum_{h\in\mathcal{A}}\left[A_* n e^{-L_h} + 2\Delta \exp\left\{-\frac{u_h^2}{2(v_h + 2b_h u_h/3)}\right\}\right] \le \delta,$$

then, with probability at least $1-\delta$, every selected original stream is all-plus certified after the contraction-normalizing scale

$$S_h := \max\left\{b_h, \frac{u_h + r_h}{\sqrt{n}}\right\}.$$

Equivalently,

$$\operatorname{disc}(X_{h,1}/S_h, \dots, X_{h,n}/S_h) \le \sqrt{n} \qquad (h \in \mathcal{A}).$$

In particular, suppose that $|\mathcal{A}| \le M$ almost surely, with $M \ge 1$ deterministic, and that on the selected set $s_h \le s$, $v_h \le v$, and $r_h \le r$. Suppose also that the tails hold at all levels,

$$\mathbb{P}\{\|X_{h,i}\|_{\mathrm{op}} > s_h t \mid \mathcal{P}_{h,i}\} \le A_* e^{-t^\alpha} \qquad (t \ge 1).$$

Set

$$L_{\mathrm{cap}} := \max\left\{1, \log\frac{2A_* Mn}{\delta}\right\}, \qquad L_{\mathrm{mat}} := \log\frac{4M\Delta}{\delta}, \qquad b := sL_{\mathrm{cap}}^{1/\alpha}.$$

Then the same conclusion holds with probability at least $1-\delta$ after the homogeneous normalization

$$C\max\left\{b, \frac{r + \sqrt{vL_{\mathrm{mat}}} + bL_{\mathrm{mat}}}{\sqrt{n}}\right\}.$$

Proof. For each selected h, the conditional union bound and the tower property give

$$\begin{aligned}\mathbb{P}(F_h^{\mathrm{cap}} \mid \mathcal{G}) &\le \sum_{i=1}^{n} \mathbb{E}\left[\mathbb{P}\{\|X_{h,i}\|_{\mathrm{op}} > b_h \mid \mathcal{P}_{h,i}\} \,\middle|\, \mathcal{G}\right] \\ &\le A_* n e^{-L_h}.\end{aligned}$$

It remains to compare the variance hypothesis with the centered capped variance required in Corollary 7.48. In the self-adjoint case,

$$\mathbb{E}(Y_{h,i}^2 \mid \mathcal{P}_{h,i}) = \mathbb{E}(T_{h,i}^2 \mid \mathcal{P}_{h,i}) - \mu_{h,i}^2 \preceq \mathbb{E}(T_{h,i}^2 \mid \mathcal{P}_{h,i}).$$

In the rectangular case,

$$\mathbb{E}(Y_{h,i}Y_{h,i}^* \mid \mathcal{P}_{h,i}) = \mathbb{E}(T_{h,i}T_{h,i}^* \mid \mathcal{P}_{h,i}) - \mu_{h,i}\mu_{h,i}^*$$

and the analogous identity holds for $Y_{h,i}^* Y_{h,i}$. Thus the displayed uncentered capped variance bounds imply the centered variance bounds of Corollary 7.48. Applying that corollary proves the first assertion.

For the homogeneous assertion take $L_h = L_{\mathrm{cap}}$ and $b_h = s_h L_{\mathrm{cap}}^{1/\alpha}$. The selected cap cost is at most $\delta/2$. With

$$u = C(\sqrt{vL_{\mathrm{mat}}} + bL_{\mathrm{mat}}),$$

the usual Bernstein algebra gives

$$\frac{u^2}{2(v + 2bu/3)} \ge L_{\mathrm{mat}}$$

after increasing the numerical constant C. Hence the selected centered matrix-tail cost is at most $\delta/2$, and the first assertion gives the homogeneous scale. □

Corollary 7.50 (Finite-confidence finite-moment norm-tail certificates)**.** *Work in the independent or adapted setting of Corollary 7.48. Fix $p > 0$ and $A_* \geq 1$. For each selected stream let $s_h > 0$, $U_h \geq 1$, $v_h > 0$, $u_h > 0$, and $r_h \geq 0$ be $\mathcal{G}$-measurable, and put*

$$b_h := s_h U_h^{1/p}.$$

Assume the conditional operator-norm moment bound

$$\mathbb{E}(\|X_{h,i}\|_{\mathrm{op}}^p \mid \mathcal{P}_{h,i}) \leq A_* s_h^p \qquad (1 \leq i \leq n).$$

Define

$$T_{h,i} := X_{h,i}\mathbf{1}_{\{\|X_{h,i}\|_{\mathrm{op}} \leq b_h\}}, \qquad \mu_{h,i} := \mathbb{E}(T_{h,i} \mid \mathcal{P}_{h,i}).$$

Let

$$F_h^{\mathrm{cap}} := \left\{ \max_{1 \leq i \leq n} \|X_{h,i}\|_{\mathrm{op}} > b_h \right\}.$$

Assume the truncated drift bound

$$\left\| \sum_i \mu_{h,i} \right\|_{\mathrm{op}} \leq r_h.$$

In the self-adjoint case assume the uncentered capped variance bound

$$\left\| \sum_i \mathbb{E}(T_{h,i}^2 \mid \mathcal{P}_{h,i}) \right\|_{\mathrm{op}} \leq v_h.$$

In the rectangular case assume

$$\left\| \sum_i \mathbb{E}(T_{h,i} T_{h,i}^* \mid \mathcal{P}_{h,i}) \right\|_{\mathrm{op}} \leq v_h,$$

$$\left\| \sum_i \mathbb{E}(T_{h,i}^* T_{h,i} \mid \mathcal{P}_{h,i}) \right\|_{\mathrm{op}} \leq v_h.$$

If

$$\mathbb{E} \sum_{h \in \mathcal{A}} \left[A_* n U_h^{-1} + 2\Delta \exp\left\{ -\frac{u_h^2}{2(v_h + 2b_h u_h/3)} \right\} \right] \leq \delta,$$

then, with probability at least $1 - \delta$, every selected original stream satisfies

$$\mathrm{disc}(X_{h,1}/S_h, \ldots, X_{h,n}/S_h) \leq \sqrt{n}, \qquad S_h := \max\left\{ b_h, \frac{u_h + r_h}{\sqrt{n}} \right\}.$$

The uncentered capped variance hypotheses are automatic with

$$v_h := A_* s_h^2 n\, U_h^{(2-p)_+/p}.$$

If, in addition, the independent streams are conditionally centered given $\mathcal{G}$, or the adapted streams are martingale differences, and $p \geq 1$, then the truncated drift bound is automatic with

$$r_h := A_* n s_h \, U_h^{(1-p)/p}.$$

In particular, suppose that $|\mathcal{A}| \leq M$ almost surely, with $M \geq 1$ deterministic, that $s_h \leq s$ on the selected set, and that the displayed moment bound holds. Set

$$U := \max\left\{1, \frac{2A_* M n}{\delta}\right\}, \qquad L := \log \frac{4M\Delta}{\delta}, \qquad b := sU^{1/p}.$$

If $v_h \leq v$ and $r_h \leq r$ on the selected set, then the same conclusion holds with probability at least $1 - \delta$ after the homogeneous normalization

$$C \max\left\{b, \frac{r + \sqrt{vL} + bL}{\sqrt{n}}\right\}.$$

One may take

$$v = A_* s^2 n \, U^{(2-p)_+/p},$$

and, in the centered or martingale-difference case with $p \geq 1$,

$$r = A_* n s \, U^{(1-p)/p}.$$

Proof. Markov's inequality gives, for every selected h and i,

$$\mathbb{P}\{\|X_{h,i}\|_{\mathrm{op}} > b_h \mid \mathcal{P}_{h,i}\} \leq A_* U_h^{-1}.$$

Thus the same conditional union-bound calculation as in Corollary 7.49 gives

$$\mathbb{P}(F_h^{\mathrm{cap}} \mid \mathcal{G}) \leq A_* n U_h^{-1}.$$

The uncentered capped variance bounds imply the centered capped variance bounds by the same subtraction-of-squares argument used in Corollary 7.49. Hence Corollary 7.48 proves the first assertion.

It remains to justify the automatic proxies. In the self-adjoint case, $T_{h,i}^2 \preceq \|T_{h,i}\|_{\mathrm{op}}^2 I$; in the rectangular case, $T_{h,i} T_{h,i}^*$ and $T_{h,i}^* T_{h,i}$ are bounded by the same scalar multiple of the identity. Therefore

$$\left\|\mathbb{E}(T_{h,i}^2 \mid \mathcal{P}_{h,i})\right\|_{\mathrm{op}} \leq \mathbb{E}(\|T_{h,i}\|_{\mathrm{op}}^2 \mid \mathcal{P}_{h,i})$$

and similarly for the rectangular row and column terms. If $p \geq 2$, Jensen's inequality and $A_* \geq 1$ give

$$\mathbb{E}(\|T_{h,i}\|_{\mathrm{op}}^2 \mid \mathcal{P}_{h,i}) \leq A_* s_h^2.$$

If $0 < p < 2$, then on the cap event $\|T_{h,i}\|_{\mathrm{op}}^2 \leq b_h^{2-p} \|X_{h,i}\|_{\mathrm{op}}^p$, so the same conditional moment bound gives

$$\mathbb{E}(\|T_{h,i}\|_{\mathrm{op}}^2 \mid \mathcal{P}_{h,i}) \leq A_* s_h^2 U_h^{(2-p)/p}.$$

Summing over i gives the displayed v_h.

If the original stream is conditionally centered or a martingale difference, then

$$\mu_{h,i} = -\mathbb{E}\left(X_{h,i}\mathbf{1}_{\{\|X_{h,i}\|_{\mathrm{op}}>b_h\}} \,\middle|\, \mathcal{P}_{h,i}\right).$$

For $p \geq 1$,

$$\|\mu_{h,i}\|_{\mathrm{op}} \leq b_h^{1-p}\mathbb{E}(\|X_{h,i}\|_{\mathrm{op}}^p \mid \mathcal{P}_{h,i}) \leq A_* s_h U_h^{(1-p)/p}.$$

Summing over i gives the displayed r_h.

For the homogeneous form take $U_h = U$. The selected cap cost is at most $\delta/2$. With $u = C(\sqrt{vL} + bL)$, the usual Bernstein algebra gives

$$\frac{u^2}{2(v + 2bu/3)} \geq L$$

after increasing C, so the selected matrix-tail cost is also at most $\delta/2$. □

Corollary 7.51 (Finite-confidence heterogeneous cap-profile certificates)**.** *Work in the independent or adapted setting of Corollary 7.48. For each selected stream let $b_{h,i} > 0$ and $\eta_{h,i} \geq 0$ be $\mathcal{G}$-measurable, and assume*

$$\mathbb{P}\{\|X_{h,i}\|_{\mathrm{op}} > b_{h,i} \mid \mathcal{P}_{h,i}\} \leq \eta_{h,i} \qquad (1 \leq i \leq n).$$

Let $v_h > 0$, $u_h > 0$, and $r_h \geq 0$ be $\mathcal{G}$-measurable. Put

$$b_h^* := \max_{1\leq i\leq n} b_{h,i}.$$

Define the heterogeneously capped variables

$$T_{h,i} := X_{h,i}\mathbf{1}_{\{\|X_{h,i}\|_{\mathrm{op}}\leq b_{h,i}\}}, \qquad \mu_{h,i} := \mathbb{E}(T_{h,i} \mid \mathcal{P}_{h,i}),$$

and assume the drift bound

$$\left\|\sum_i \mu_{h,i}\right\|_{\mathrm{op}} \leq r_h.$$

In the self-adjoint case assume the uncentered capped variance bound

$$\left\|\sum_i \mathbb{E}(T_{h,i}^2 \mid \mathcal{P}_{h,i})\right\|_{\mathrm{op}} \leq v_h.$$

In the rectangular case assume

$$\left\|\sum_i \mathbb{E}(T_{h,i}T_{h,i}^* \mid \mathcal{P}_{h,i})\right\|_{\mathrm{op}} \leq v_h,$$

$$\left\|\sum_i \mathbb{E}(T_{h,i}^*T_{h,i} \mid \mathcal{P}_{h,i})\right\|_{\mathrm{op}} \leq v_h.$$

If

$$\mathbb{E}\sum_{h\in\mathcal{A}}\left[\sum_{i=1}^n \eta_{h,i} + 2\Delta\exp\left\{-\frac{u_h^2}{2(v_h + 2b_h^*u_h/3)}\right\}\right] \leq \delta,$$

then, with probability at least $1-\delta$*, every selected original stream satisfies*

$$\operatorname{disc}(X_{h,1}/S_h,\dots,X_{h,n}/S_h)\le\sqrt{n},\qquad S_h:=\max\left\{b_h^*,\frac{u_h+r_h}{\sqrt{n}}\right\}.$$

In particular, if $|\mathcal{A}|\le M$ *almost surely,*

$$\mathbb{E}\sum_{h\in\mathcal{A}}\sum_i\eta_{h,i}\le\delta/2,$$

and $b_h^*\le b$*,* $v_h\le v$*,* $r_h\le r$ *on the selected set, then with*

$$L=\log\frac{4M\Delta}{\delta},\qquad u=C(\sqrt{vL}+bL),$$

the same conclusion holds with probability at least $1-\delta$ *after the homogeneous normalization*

$$C\max\left\{b,\frac{r+\sqrt{vL}+bL}{\sqrt{n}}\right\}.$$

Proof. Let

$$F_h^{\mathrm{cap}}:=\left\{\max_i\|X_{h,i}\|_{\mathrm{op}}>b_{h,i}\right\}.$$

The conditional union bound and the tower property give

$$\mathbb{P}(F_h^{\mathrm{cap}}\mid\mathcal{G})\le\sum_i\eta_{h,i}.$$

Set $Y_{h,i}:=T_{h,i}-\mu_{h,i}$. In both the independent and adapted settings these are centered independent variables, conditionally on $\mathcal{G}$, or martingale differences. Moreover

$$\|Y_{h,i}\|_{\mathrm{op}}\le 2b_h^*.$$

The uncentered capped variance bounds imply the centered capped variance bounds by the same subtraction-of-squares argument used in Corollary 7.49.

Matrix Bernstein in the independent case and matrix Freedman in the adapted case, with self-adjoint dilation for rectangular streams, give

$$\mathbb{P}\left\{\left\|\sum_i Y_{h,i}\right\|_{\mathrm{op}}>u_h\,\middle|\,\mathcal{G}\right\}\le 2\Delta\exp\left\{-\frac{u_h^2}{2(v_h+2b_h^*u_h/3)}\right\}.$$

Taking the union bound over the active set and then expectation gives total failure at most δ. On the complementary event $X_{h,i}=T_{h,i}$, $\|X_{h,i}\|_{\mathrm{op}}\le b_{h,i}\le b_h^*$, and

$$\left\|\sum_i X_{h,i}\right\|_{\mathrm{op}}\le u_h+r_h.$$

Dividing by S_h proves the first assertion. The homogeneous form is the same Bernstein algebra used in Corollary 7.48. □

Corollary 7.52 (Finite-confidence heterogeneous finite-moment certificates)**.** *Work in the independent or adapted setting of Corollary 7.48. Fix $A_* \geq 1$. For each selected stream and each $1 \leq i \leq n$, let $p_{h,i} > 0$, $s_{h,i} > 0$, and $U_{h,i} \geq 1$ be $\mathcal{G}$-measurable, and assume the conditional moment envelope*

$$\mathbb{E}(\|X_{h,i}\|_{\mathrm{op}}^{p_{h,i}} \mid \mathcal{P}_{h,i}) \leq A_* s_{h,i}^{p_{h,i}}.$$

Put

$$b_{h,i} := s_{h,i} U_{h,i}^{1/p_{h,i}}, \qquad b_h^* := \max_{1 \leq i \leq n} b_{h,i},$$

and

$$V_h := A_* \sum_{i=1}^{n} s_{h,i}^2 U_{h,i}^{(2-p_{h,i})_+/p_{h,i}}.$$

For $\mathcal{G}$-measurable $u_h > 0$, define the heterogeneously capped variables

$$T_{h,i} := X_{h,i} \mathbf{1}_{\{\|X_{h,i}\|_{\mathrm{op}} \leq b_{h,i}\}}, \qquad \mu_{h,i} := \mathbb{E}(T_{h,i} \mid \mathcal{P}_{h,i}).$$

Assume the truncated drift bound

$$\left\| \sum_i \mu_{h,i} \right\|_{\mathrm{op}} \leq r_h$$

for some $\mathcal{G}$-measurable $r_h \geq 0$. If

$$\mathbb{E} \sum_{h \in \mathcal{A}} \left[A_* \sum_{i=1}^{n} U_{h,i}^{-1} + 2\Delta \exp\left\{ -\frac{u_h^2}{2(V_h + 2b_h^* u_h/3)} \right\} \right] \leq \delta,$$

then, with probability at least $1 - \delta$, every selected original stream satisfies

$$\mathrm{disc}(X_{h,1}/S_h, \ldots, X_{h,n}/S_h) \leq \sqrt{n}, \qquad S_h := \max\left\{ b_h^*, \frac{u_h + r_h}{\sqrt{n}} \right\}.$$

The same conclusion holds in the rectangular case, with the same Δ-convention as in Corollary 7.51.

If the independent streams are conditionally centered given $\mathcal{G}$, or the adapted streams are martingale differences, and $p_{h,i} \geq 1$ for every selected h, i, then the drift hypothesis is automatic with

$$r_h := A_* \sum_{i=1}^{n} s_{h,i} U_{h,i}^{(1-p_{h,i})/p_{h,i}}.$$

In particular, if $|\mathcal{A}| \leq M$ almost surely,

$$\mathbb{E} \sum_{h \in \mathcal{A}} A_* \sum_{i=1}^{n} U_{h,i}^{-1} \leq \delta/2,$$

and $b_h^* \leq b$, $V_h \leq v$, $r_h \leq r$ on the selected set, then with

$$L = \log \frac{4M\Delta}{\delta}, \qquad u = C(\sqrt{vL} + bL),$$

the same conclusion holds with probability at least $1-\delta$ *after the homogeneous normalization*

$$C \max\left\{ b, \frac{r+\sqrt{vL}+bL}{\sqrt{n}} \right\}.$$

Proof. Markov's inequality gives

$$\mathbb{P}\{\|X_{h,i}\|_{\mathrm{op}} > b_{h,i} \mid \mathcal{P}_{h,i}\} \le A_* U_{h,i}^{-1}.$$

It remains only to verify the capped second-moment input in Corollary 7.51. In the self-adjoint case, $T_{h,i}^2 \preceq \|T_{h,i}\|_{\mathrm{op}}^2 I$; in the rectangular case, the row and column squares have the same scalar upper bound. If $p_{h,i} \ge 2$, Jensen's inequality and $A_* \ge 1$ give

$$\mathbb{E}(\|T_{h,i}\|_{\mathrm{op}}^2 \mid \mathcal{P}_{h,i}) \le A_* s_{h,i}^2.$$

If $0 < p_{h,i} < 2$, then on the cap event

$$\|T_{h,i}\|_{\mathrm{op}}^2 \le b_{h,i}^{2-p_{h,i}} \|X_{h,i}\|_{\mathrm{op}}^{p_{h,i}},$$

so

$$\mathbb{E}(\|T_{h,i}\|_{\mathrm{op}}^2 \mid \mathcal{P}_{h,i}) \le A_* s_{h,i}^2 U_{h,i}^{(2-p_{h,i})/p_{h,i}}.$$

Summing these bounds gives the self-adjoint, and rectangular row and column, uncentered capped variance bound with V_h. Applying Corollary 7.51 proves the first assertion.

In the centered or martingale-difference case,

$$\mu_{h,i} = -\mathbb{E}\left(X_{h,i} \mathbf{1}_{\{\|X_{h,i}\|_{\mathrm{op}} > b_{h,i}\}} \,\middle|\, \mathcal{P}_{h,i} \right).$$

For $p_{h,i} \ge 1$,

$$\|\mu_{h,i}\|_{\mathrm{op}} \le b_{h,i}^{1-p_{h,i}} \mathbb{E}(\|X_{h,i}\|_{\mathrm{op}}^{p_{h,i}} \mid \mathcal{P}_{h,i}) \le A_* s_{h,i} U_{h,i}^{(1-p_{h,i})/p_{h,i}}.$$

Summing over i gives the displayed automatic drift proxy.

For the homogeneous form, the selected cap cost is at most $\delta/2$. With $u = C(\sqrt{vL} + bL)$, the usual Bernstein algebra gives

$$\frac{u^2}{2(v + 2bu/3)} \ge L$$

after increasing C, so the selected centered matrix-tail cost is also at most $\delta/2$. □

Corollary 7.53 (Finite-confidence tail-profile certificates)**.** *Work in the independent or adapted setting of Corollary 7.48. For each selected stream and each* $1 \le i \le n$*, let* $s_{h,i} > 0$*,* $\lambda_{h,i} > 0$*, and* $u_h > 0$ *be* $\mathcal{G}$*-measurable, and let* $\pi_{h,i} : [0,\infty) \to [0,1]$ *be deterministic measurable profiles satisfying*

$$\mathbb{P}\{\|X_{h,i}\|_{\mathrm{op}} > s_{h,i} t \mid \mathcal{P}_{h,i}\} \le \pi_{h,i}(t) \qquad (t \ge 0).$$

Put

$$b_{h,i} := s_{h,i}\lambda_{h,i}, \qquad b_h^* := \max_{1 \le i \le n} b_{h,i},$$

and

$$V_h := 2\sum_{i=1}^{n} s_{h,i}^2 \int_0^{\lambda_{h,i}} t\,\pi_{h,i}(t)\,dt.$$

Define

$$T_{h,i} := X_{h,i}\mathbf{1}_{\{\|X_{h,i}\|_{\mathrm{op}} \le b_{h,i}\}}, \qquad \mu_{h,i} := \mathbb{E}(T_{h,i} \mid \mathcal{P}_{h,i}).$$

Assume

$$\left\|\sum_i \mu_{h,i}\right\|_{\mathrm{op}} \le r_h$$

for some $\mathcal{G}$-measurable $r_h \ge 0$. If

$$\mathbb{E}\sum_{h\in\mathcal{A}}\left[\sum_{i=1}^{n}\pi_{h,i}(\lambda_{h,i}) + 2\Delta\exp\left\{-\frac{u_h^2}{2(V_h + 2b_h^* u_h/3)}\right\}\right] \le \delta,$$

then, with probability at least $1-\delta$, every selected original stream satisfies

$$\operatorname{disc}(X_{h,1}/S_h, \dots, X_{h,n}/S_h) \le \sqrt{n}, \qquad S_h := \max\left\{b_h^*, \frac{u_h + r_h}{\sqrt{n}}\right\}.$$

The same conclusion holds in the rectangular case, with the usual $\Delta = d_1 + d_2$ dilation convention.

If the independent streams are conditionally centered given $\mathcal{G}$, or the adapted streams are martingale differences, then the drift hypothesis is automatic whenever

$$\sum_i s_{h,i}\left[\lambda_{h,i}\pi_{h,i}(\lambda_{h,i}) + \int_{\lambda_{h,i}}^{\infty}\pi_{h,i}(t)\,dt\right] \le r_h.$$

In particular, if $|\mathcal{A}| \le M$ almost surely,

$$\mathbb{E}\sum_{h\in\mathcal{A}}\sum_i \pi_{h,i}(\lambda_{h,i}) \le \delta/2,$$

and $b_h^* \le b$, $V_h \le v$, $r_h \le r$ on the selected set, then with

$$L = \log\frac{4M\Delta}{\delta}, \qquad u = C(\sqrt{vL} + bL),$$

the same conclusion holds with probability at least $1-\delta$ after the homogeneous normalization

$$C\max\left\{b, \frac{r + \sqrt{vL} + bL}{\sqrt{n}}\right\}.$$

Proof. The profile tail bound gives

$$\mathbb{P}\{\|X_{h,i}\|_{\mathrm{op}} > b_{h,i} \mid \mathcal{P}_{h,i}\} \le \pi_{h,i}(\lambda_{h,i}).$$

For the capped second moment, in the self-adjoint case $T_{h,i}^2 \preceq \|T_{h,i}\|_{\mathrm{op}}^2 I$, and similarly for rectangular row and column squares. By layer cake and the

profile tail bound,

$$\mathbb{E}(\|T_{h,i}\|_{\mathrm{op}}^2 \mid \mathcal{P}_{h,i}) \le s_{h,i}^2 \int_0^{\lambda_{h,i}^2} \mathbb{P}\{\|X_{h,i}\|_{\mathrm{op}}^2 > s_{h,i}^2 r \mid \mathcal{P}_{h,i}\}\, dr$$
$$\le 2s_{h,i}^2 \int_0^{\lambda_{h,i}} t\, \pi_{h,i}(t)\, dt.$$

Summing gives the self-adjoint and rectangular uncentered capped variance bound with V_h. Corollary 7.51 therefore gives the first assertion.

For centered or martingale-difference streams,

$$\mu_{h,i} = -\mathbb{E}\left(X_{h,i}\mathbf{1}_{\{\|X_{h,i}\|_{\mathrm{op}} > b_{h,i}\}} \,\middle|\, \mathcal{P}_{h,i}\right),$$

and another layer-cake estimate gives

$$\|\mu_{h,i}\|_{\mathrm{op}} \le s_{h,i}\left[\lambda_{h,i}\pi_{h,i}(\lambda_{h,i}) + \int_{\lambda_{h,i}}^{\infty} \pi_{h,i}(t)\, dt\right].$$

Summing proves the drift claim. The homogeneous form is the same Bernstein algebra used in Corollary 7.51. □

Corollary 7.54 (Sub-Weibull norm-tail cap selection)**.** *Work in either the independent setting of Corollary 7.46 or the adapted martingale-difference setting of Corollary 7.47. Let $\mathcal{P}_{m,h,i}$ denote the conditioning field $\mathcal{H}_m$ in the independent case and $\mathcal{F}_{m,h,i-1}$ in the adapted case. Fix $\alpha > 0$, $A_* \ge 1$, $K > 0$, $\beta \ge 0$, $\eta \ge 0$, and $B > 0$. For each active stream let $s_{m,h} > 0$ be a deterministic norm-tail scale, and let $L_m \ge 1$ be deterministic. Put*

$$b_{m,h} := s_{m,h}L_m^{1/\alpha}, \qquad q_{m,h} := s_{m,h}\max\{L_m^{1/\alpha}, \sqrt{\ell_m}\}.$$

Assume the conditional operator-norm tails

$$\mathbb{P}\{\|X_{m,h,i}\|_{\mathrm{op}} > b_{m,h} \mid \mathcal{P}_{m,h,i}\} \le A_* e^{-L_m} \qquad (1 \le i \le n_m)$$

and let

$$T_{m,h,i} := X_{m,h,i}\mathbf{1}_{\{\|X_{m,h,i}\|_{\mathrm{op}} \le b_{m,h}\}}, \qquad \mu_{m,h,i} := \mathbb{E}(T_{m,h,i} \mid \mathcal{P}_{m,h,i}).$$

Outside events $E_{m,h}$, assume the truncated-tail drift bound

$$\left\|\sum_{i=1}^{n_m} \mu_{m,h,i}\right\|_{\mathrm{op}} \le \eta q_{m,h}\sqrt{n_m}.$$

In the self-adjoint case assume also, outside $E_{m,h}$, that

$$\left\|\sum_{i=1}^{n_m} \mathbb{E}(T_{m,h,i}^2 \mid \mathcal{P}_{m,h,i})\right\|_{\mathrm{op}} \le \beta s_{m,h}^2 n_m.$$

In the rectangular case replace this by the two truncated row and column profile bounds

$$\left\|\sum_i \mathbb{E}(T_{m,h,i}T_{m,h,i}^* \mid \mathcal{P}_{m,h,i})\right\|_{\mathrm{op}} \le \beta s_{m,h}^2 n_m,$$
$$\left\|\sum_i \mathbb{E}(T_{m,h,i}^*T_{m,h,i} \mid \mathcal{P}_{m,h,i})\right\|_{\mathrm{op}} \le \beta s_{m,h}^2 n_m.$$

Suppose that $\ell_m \le B\sqrt{n_m}$ *eventually and that*

$$\sum_m \sum_{h=1}^{J_m} \left[\mathbb{P}(E_{m,h}) + A_* n_m e^{-L_m}\right] < \infty$$

and

$$\sum_m J_m \Delta_m (e\Delta_m)^{-\gamma_{\mathrm{tc}}(K,1,\beta,B)} < \infty.$$

Then, almost surely, for all sufficiently large m*, every active original stream is contraction-normalized by*

$$Mq_{m,h}, \qquad M := \max\{1, K+\eta\},$$

and the all-plus signing gives

$$\mathrm{disc}(X_{m,h,1}/(Mq_{m,h}), \dots, X_{m,h,n_m}/(Mq_{m,h})) \le \sqrt{n_m} \qquad (1 \le h \le J_m).$$

Consequently, if

$$J_m\Delta_m \le (e\Delta_m)^{\Gamma}, \qquad J_m n_m \le (e\Delta_m)^{\Lambda}, \qquad n_m \ge a\ell_m^2$$

eventually and $L_m = c\ell_m$*, then the conclusion holds along all integer dimensions whenever*

$$c > \Lambda + 1, \qquad K > \Theta_{\mathrm{tc}}(\Gamma + 1, a, \beta, 1),$$

and along every stretched-exponential sparse dimension schedule whenever

$$c > \Lambda, \qquad K > \Theta_{\mathrm{tc}}(\Gamma, a, \beta, 1).$$

In these closed-form schedules the normalizing scale is

$$q_{m,h} = s_{m,h} \max\{(c\ell_m)^{1/\alpha}, \sqrt{\ell_m}\}.$$

For conditionally centered independent streams, and for the martingale streams above, the displayed sub-Weibull tail bound makes the drift hypothesis automatic for every fixed $\eta > 0$ *under the same closed-form cap choices.*

Proof. The conditional tail bound and a union bound give

$$\mathbb{P}(F_{m,h}^{\mathrm{cap}}) \le A_* n_m e^{-L_m}.$$

Moreover $b_{m,h} \le q_{m,h}$, so the tail-capped corollaries apply with $R = 1$ once their capped variance hypotheses are verified. In the self-adjoint case,

$$\mathbb{E}((T_{m,h,i}-\mu_{m,h,i})^2 \mid \mathcal{P}_{m,h,i}) = \mathbb{E}(T_{m,h,i}^2 \mid \mathcal{P}_{m,h,i}) - \mu_{m,h,i}^2 \preceq \mathbb{E}(T_{m,h,i}^2 \mid \mathcal{P}_{m,h,i}).$$

The rectangular row and column identities are the same with TT^* and T^*T. Since

$$q_{m,h}^2/\ell_m = s_{m,h}^2 \max\{L_m^{2/\alpha}/\ell_m, 1\} \geq s_{m,h}^2,$$

the displayed truncated profiles imply the required ordinary or predictable variance bounds with the same β. Corollaries 7.46 and 7.47, with $R = 1$, now give the general conclusion.

For the closed-form schedules, the cap-tail summand is at most $(e\Delta_m)^{\Lambda-c}$, up to the fixed factor A_*. This is summable over all integer dimensions when $c > \Lambda + 1$, and along every stretched-exponential sparse schedule when $c > \Lambda$. The all-plus matrix-tail summability is exactly the corresponding Θ_{tc} calculation from the two tail-capped corollaries, again with $R = 1$.

Finally suppose the original variables are centered relative to $\mathcal{P}_{m,h,i}$. For

$$\psi_\alpha(u) := ue^{-u^\alpha} + \int_u^\infty e^{-t^\alpha}\, dt$$

the conditional tail bound gives

$$\|\mu_{m,h,i}\|_{\mathrm{op}} \leq A_* s_{m,h} \psi_\alpha(L_m^{1/\alpha}).$$

With $L_m = c\ell_m$ and $J_m n_m \leq (e\Delta_m)^\Lambda$,

$$\frac{\sum_i \|\mu_{m,h,i}\|_{\mathrm{op}}}{q_{m,h}\sqrt{n_m}} \leq A_* \frac{\sqrt{n_m}\, \psi_\alpha((c\ell_m)^{1/\alpha})}{\max\{(c\ell_m)^{1/\alpha}, \sqrt{\ell_m}\}} \longrightarrow 0$$

under the displayed closed-form cap choices: indeed $\sqrt{n_m} \leq (e\Delta_m)^{\Lambda/2}$, while $\psi_\alpha((c\ell_m)^{1/\alpha})$ is a polynomial factor in ℓ_m times $(e\Delta_m)^{-c}$. Hence the drift bound holds eventually for every fixed $\eta > 0$. □

Corollary 7.55 (Finite-moment norm-tail cap selection)**.** *Work in the same independent or adapted setting, with the same conditioning fields $\mathcal{P}_{m,h,i}$, as in Corollary 7.54. Fix $p > 0$, $A_* \geq 1$, $K > 0$, $\eta \geq 0$, and $B > 0$. For each active stream let $s_{m,h} > 0$ be deterministic and assume the conditional moment envelope*

$$\mathbb{E}(\|X_{m,h,i}\|_{\mathrm{op}}^p \mid \mathcal{P}_{m,h,i}) \leq A_* s_{m,h}^p \qquad (1 \leq i \leq n_m).$$

Let $U_m \geq 1$ be deterministic and put

$$\theta_p := \frac{(2-p)_+}{2p}, \qquad b_{m,h} := s_{m,h} U_m^{1/p},$$

$$q_{m,h} := s_{m,h} \max\{U_m^{1/p}, \sqrt{\ell_m}\, U_m^{\theta_p}\}.$$

Define $T_{m,h,i}$ and $\mu_{m,h,i}$ as in Corollary 7.54. Outside events $E_{m,h}$, assume the truncated-tail drift bound

$$\left\| \sum_{i=1}^{n_m} \mu_{m,h,i} \right\|_{\mathrm{op}} \leq \eta q_{m,h} \sqrt{n_m}.$$

Suppose that $\ell_m \le B\sqrt{n_m}$ *eventually and that*

$$\sum_m \sum_{h=1}^{J_m} \left[\mathbb{P}(E_{m,h}) + A_* n_m U_m^{-1}\right] < \infty$$

and

$$\sum_m J_m \Delta_m (e\Delta_m)^{-\gamma_{\mathrm{tc}}(K,1,A_*,B)} < \infty.$$

Then, almost surely, for all sufficiently large m*, every active original stream is contraction-normalized by*

$$Mq_{m,h}, \qquad M := \max\{1, K+\eta\},$$

and the all-plus signing gives

$$\operatorname{disc}(X_{m,h,1}/(Mq_{m,h}), \dots, X_{m,h,n_m}/(Mq_{m,h})) \le \sqrt{n_m} \qquad (1 \le h \le J_m).$$

Consequently, if

$$J_m \Delta_m \le (e\Delta_m)^{\Gamma}, \qquad J_m n_m \le (e\Delta_m)^{\Lambda}, \qquad n_m \ge a\ell_m^2$$

eventually and $U_m = (e\Delta_m)^c$*, then the conclusion holds along all integer dimensions whenever*

$$c > \Lambda + 1, \qquad K > \Theta_{\mathrm{tc}}(\Gamma + 1, a, A_*, 1),$$

and along every stretched-exponential sparse dimension schedule whenever

$$c > \Lambda, \qquad K > \Theta_{\mathrm{tc}}(\Gamma, a, A_*, 1).$$

In these closed-form schedules the normalizing scale is

$$q_{m,h} = s_{m,h} \max\{(e\Delta_m)^{c/p}, \sqrt{\ell_m}\,(e\Delta_m)^{c\theta_p}\}.$$

If $p \ge 1$*, and if the original streams are conditionally centered in the independent case or martingale differences in the adapted case, then the moment envelope makes the drift hypothesis automatic for every fixed* $\eta > 0$ *under the same closed-form cap choices.*

Proof. Markov's inequality gives

$$\mathbb{P}(F_{m,h}^{\mathrm{cap}}) \le A_* n_m U_m^{-1}.$$

Also $b_{m,h} \le q_{m,h}$, so the tail-capped corollaries may be used with $R = 1$ once the capped variance budget is verified. In the self-adjoint case,

$$T_{m,h,i}^2 \preceq \|T_{m,h,i}\|_{\mathrm{op}}^2 I.$$

If $0 < p < 2$, then on the cap event

$$\|T_{m,h,i}\|_{\mathrm{op}}^2 \le b_{m,h}^{2-p} \|X_{m,h,i}\|_{\mathrm{op}}^p,$$

while if $p \ge 2$, conditional Lyapunov monotonicity gives

$$\mathbb{E}(\|T_{m,h,i}\|_{\mathrm{op}}^2 \mid \mathcal{P}_{m,h,i}) \le A_* s_{m,h}^2.$$

Thus, in all cases,

$$\mathbb{E}(T_{m,h,i}^2 \mid \mathcal{P}_{m,h,i}) \preceq A_* s_{m,h}^2 U_m^{(2-p)_+/p} I.$$

The same argument with TT^* and T^*T gives the rectangular row and column bounds. Since

$$q_{m,h}^2/\ell_m \geq s_{m,h}^2 U_m^{(2-p)_+/p},$$

the centered capped ordinary or predictable variance profile is at most $A_* q_{m,h}^2 n_m/\ell_m$. Corollaries 7.46 and 7.47, with $R = 1$ and $\beta = A_*$, prove the general conclusion.

For the closed-form schedules, the cap-tail summand is bounded by $(e\Delta_m)^{\Lambda-c}$, up to A_*, so it is summable over all integer dimensions when $c > \Lambda + 1$ and along every stretched-exponential sparse schedule when $c > \Lambda$. The matrix-tail summability is exactly the same Θ_{tc} calculation with $\beta = A_*$.

If $p \geq 1$ and the original variables are centered relative to $\mathcal{P}_{m,h,i}$, then

$$\|\mu_{m,h,i}\|_{\mathrm{op}} \leq A_* s_{m,h} U_m^{-(p-1)/p}.$$

Because $q_{m,h} \geq s_{m,h} U_m^{1/p}$, we get

$$\frac{\sum_i \|\mu_{m,h,i}\|_{\mathrm{op}}}{q_{m,h}\sqrt{n_m}} \leq A_* \sqrt{n_m}\, U_m^{-1}.$$

With $U_m = (e\Delta_m)^c$ and $J_m n_m \leq (e\Delta_m)^\Lambda$, this tends to 0 under the displayed closed-form cap choices. Hence the drift hypothesis holds eventually for every fixed $\eta > 0$. □

Corollary 7.56 (Heterogeneous norm-tail cap profiles)**.** *Work in the same independent or adapted setting, with the same conditioning fields $\mathcal{P}_{m,h,i}$, as in Corollary 7.54. For each active stream let $q_{m,h} > 0$ be deterministic, and let $b_{m,h,i} > 0$ be deterministic caps satisfying*

$$b_{m,h,i} \leq R q_{m,h} \qquad (1 \leq i \leq n_m)$$

for a fixed $R > 0$. Define

$$T_{m,h,i} := X_{m,h,i} \mathbf{1}_{\{\|X_{m,h,i}\|_{\mathrm{op}} \leq b_{m,h,i}\}}, \qquad \mu_{m,h,i} := \mathbb{E}(T_{m,h,i} \mid \mathcal{P}_{m,h,i}),$$

and

$$F_{m,h}^{\mathrm{cap}} := \{\exists i : \|X_{m,h,i}\|_{\mathrm{op}} > b_{m,h,i}\}.$$

Fix $K > 0$, $\beta \geq 0$, $\eta \geq 0$, and $B > 0$. Assume

$$\sum_m \sum_{h=1}^{J_m} [\mathbb{P}(E_{m,h}) + \mathbb{P}(F_{m,h}^{\mathrm{cap}})] < \infty$$

and, outside $E_{m,h}$,

$$\left\| \sum_{i=1}^{n_m} \mu_{m,h,i} \right\|_{\mathrm{op}} \leq \eta q_{m,h} \sqrt{n_m}.$$

In the self-adjoint case assume also, outside $E_{m,h}$, that

$$\left\| \sum_i \mathbb{E}(T_{m,h,i}^2 \mid \mathcal{P}_{m,h,i}) \right\|_{\mathrm{op}} \leq \beta q_{m,h}^2 \frac{n_m}{\ell_m}.$$

In the rectangular case replace this by the two row and column bounds

$$\left\|\sum_i \mathbb{E}(T_{m,h,i}T_{m,h,i}^* \mid \mathcal{P}_{m,h,i})\right\|_{\mathrm{op}} \le \beta q_{m,h}^2 \frac{n_m}{\ell_m},$$

$$\left\|\sum_i \mathbb{E}(T_{m,h,i}^* T_{m,h,i} \mid \mathcal{P}_{m,h,i})\right\|_{\mathrm{op}} \le \beta q_{m,h}^2 \frac{n_m}{\ell_m}.$$

If $\ell_m \le B\sqrt{n_m}$ eventually and

$$\sum_m J_m \Delta_m (e\Delta_m)^{-\gamma_{\mathrm{tc}}(K,R,\beta,B)} < \infty,$$

then, almost surely, for all sufficiently large m, every active original stream is contraction-normalized by

$$Mq_{m,h}, \qquad M := \max\{R, K+\eta\},$$

and the all-plus signing gives

$$\mathrm{disc}(X_{m,h,1}/(Mq_{m,h}), \dots, X_{m,h,n_m}/(Mq_{m,h})) \le \sqrt{n_m} \qquad (1 \le h \le J_m).$$

If $J_m\Delta_m \le (e\Delta_m)^{\Gamma}$ eventually and $n_m \ge a\ell_m^2$, then the same closed-form margins as in the tail-capped corollaries give the all-integer threshold

$$K > \Theta_{\mathrm{tc}}(\Gamma+1, a, \beta, R)$$

and the stretched-exponential sparse threshold

$$K > \Theta_{\mathrm{tc}}(\Gamma, a, \beta, R).$$

In particular, suppose the conditional sub-Weibull norm tails

$$\mathbb{P}\{\|X_{m,h,i}\|_{\mathrm{op}} > s_{m,h,i}u \mid \mathcal{P}_{m,h,i}\} \le A_* e^{-u^\alpha} \qquad (u > 0)$$

hold. With $L_m \ge 1$, choose

$$b_{m,h,i} := s_{m,h,i} L_m^{1/\alpha}$$

and

$$q_{m,h} := \max\left\{\max_i b_{m,h,i}, \left(\frac{\ell_m}{n_m}\sum_i s_{m,h,i}^2\right)^{1/2}\right\}.$$

Then the preceding theorem applies with $\beta = C_\alpha A_*$, where C_α is an absolute constant depending only on α, whenever the displayed drift bound holds and

$$\sum_m \sum_{h=1}^{J_m} \left[\mathbb{P}(E_{m,h}) + A_* n_m e^{-L_m}\right] < \infty.$$

Proof. Let $Y_{m,h,i} = T_{m,h,i} - \mu_{m,h,i}$. In both the independent and adapted settings these are centered, respectively martingale-difference, capped increments relative to $\mathcal{P}_{m,h,i}$, and

$$\|Y_{m,h,i}\|_{\mathrm{op}} \le 2b_{m,h,i} \le 2Rq_{m,h}.$$

The displayed uncentered second-moment profiles dominate the corresponding centered variance profiles:

$$\mathbb{E}(Y_{m,h,i}^2 \mid \mathcal{P}_{m,h,i}) \preceq \mathbb{E}(T_{m,h,i}^2 \mid \mathcal{P}_{m,h,i})$$

in the self-adjoint case, and similarly with YY^*, TT^* and Y^*Y, T^*T in the rectangular case. Matrix Bernstein in the independent case, and matrix Freedman in the adapted case, therefore give the same all-plus tail used in Corollaries 7.46 and 7.47, with the common jump $2Rq_{m,h}$ and variance parameter $\beta q_{m,h}^2 n_m/\ell_m$. Borel–Cantelli, the drift estimate, and the cap event then give the stated contraction normalization and all-plus discrepancy bound. The all-integer and sparse thresholds are the same summability calculation with $B = a^{-1/2}$.

For the sub-Weibull specialization, the cap-failure probability is bounded by

$$\mathbb{P}(F_{m,h}^{\mathrm{cap}}) \le A_* n_m e^{-L_m}.$$

The same tail integration used in Corollary 7.54 gives

$$\mathbb{E}(\|T_{m,h,i}\|_{\mathrm{op}}^2 \mid \mathcal{P}_{m,h,i}) \le C_\alpha A_* s_{m,h,i}^2.$$

Hence the self-adjoint and rectangular uncentered row/column variance profiles are at most $C_\alpha A_* \sum_i s_{m,h,i}^2$, which is bounded by $C_\alpha A_* q_{m,h}^2 n_m/\ell_m$ by the definition of $q_{m,h}$. □

Corollary 7.57 (Heterogeneous finite-moment norm-tail selection)**.** *Work in the same independent or adapted setting, with the same conditioning fields $\mathcal{P}_{m,h,i}$, as in Corollary 7.56. Fix $A_* \ge 1$, $K > 0$, $\eta \ge 0$, and $B > 0$. For each active stream let $p_{m,h,i} > 0$, $s_{m,h,i} > 0$, and $U_{m,h,i} \ge 1$ be deterministic, and assume the conditional moment envelopes*

$$\mathbb{E}(\|X_{m,h,i}\|_{\mathrm{op}}^{p_{m,h,i}} \mid \mathcal{P}_{m,h,i}) \le A_* s_{m,h,i}^{p_{m,h,i}} \qquad (1 \le i \le n_m).$$

Put

$$b_{m,h,i} := s_{m,h,i} U_{m,h,i}^{1/p_{m,h,i}}, \qquad \rho_{m,h,i} := s_{m,h,i}^2 U_{m,h,i}^{(2-p_{m,h,i})_+/p_{m,h,i}},$$

and choose

$$q_{m,h} := \max\left\{ \max_i b_{m,h,i}, \left(\frac{\ell_m}{n_m} \sum_i \rho_{m,h,i} \right)^{1/2} \right\}.$$

Define $T_{m,h,i}$ and $\mu_{m,h,i}$ using these individual caps as in Corollary 7.56. Outside events $E_{m,h}$, assume the truncated-tail drift bound

$$\left\| \sum_{i=1}^{n_m} \mu_{m,h,i} \right\|_{\mathrm{op}} \le \eta q_{m,h} \sqrt{n_m}.$$

Suppose that $\ell_m \le B\sqrt{n_m}$ eventually and that

$$\sum_m \sum_{h=1}^{J_m} \left[\mathbb{P}(E_{m,h}) + A_* \sum_{i=1}^{n_m} U_{m,h,i}^{-1} \right] < \infty$$

and

$$\sum_m J_m \Delta_m (e\Delta_m)^{-\gamma_{\mathrm{tc}}(K,1,A_*,B)} < \infty.$$

Then, almost surely, for all sufficiently large m, every active original stream is contraction-normalized by

$$Mq_{m,h}, \qquad M := \max\{1, K+\eta\},$$

and the all-plus signing gives

$$\operatorname{disc}(X_{m,h,1}/(Mq_{m,h}), \dots, X_{m,h,n_m}/(Mq_{m,h})) \le \sqrt{n_m} \qquad (1 \le h \le J_m).$$

In particular, assume $U_{m,h,i} = U_m = (e\Delta_m)^c$ for every active (h,i), and suppose

$$J_m \Delta_m \le (e\Delta_m)^{\Gamma}, \qquad J_m n_m \le (e\Delta_m)^{\Lambda}, \qquad n_m \ge a\ell_m^2$$

eventually. Then the conclusion holds along all integer dimensions whenever

$$c > \Lambda + 1, \qquad K > \Theta_{\mathrm{tc}}(\Gamma + 1, a, A_*, 1),$$

and along every stretched-exponential sparse dimension schedule whenever

$$c > \Lambda, \qquad K > \Theta_{\mathrm{tc}}(\Gamma, a, A_*, 1).$$

Under these common-U_m schedules, write

$$Q_{m,h}^{\mathrm{cap}} := \max_i s_{m,h,i} (e\Delta_m)^{c/p_{m,h,i}},$$

$$Q_{m,h}^{\mathrm{var}} := \left(\frac{\ell_m}{n_m} \sum_i s_{m,h,i}^2 (e\Delta_m)^{c(2-p_{m,h,i})_+/p_{m,h,i}} \right)^{1/2}.$$

Then $q_{m,h} = \max\{Q_{m,h}^{\mathrm{cap}}, Q_{m,h}^{\mathrm{var}}\}$. If $p_{m,h,i} \ge 1$ for all i, and if the original streams are conditionally centered in the independent case or martingale differences in the adapted case, then the displayed drift hypothesis is automatic for every fixed $\eta > 0$ under the same common-U_m closed-form cap choices.

Proof. Markov's inequality gives the cap-failure bound

$$\mathbb{P}(F_{m,h}^{\mathrm{cap}}) \le A_* \sum_{i=1}^{n_m} U_{m,h,i}^{-1}.$$

Also $b_{m,h,i} \le q_{m,h}$, so Corollary 7.56 can be used with $R = 1$ once the uncentered second-moment profiles are checked. In the self-adjoint case,

$$T_{m,h,i}^2 \preceq \|T_{m,h,i}\|_{\mathrm{op}}^2 I.$$

If $p_{m,h,i} < 2$, then on the cap event

$$\|T_{m,h,i}\|_{\mathrm{op}}^2 \le b_{m,h,i}^{2-p_{m,h,i}} \|X_{m,h,i}\|_{\mathrm{op}}^{p_{m,h,i}}.$$

If $p_{m,h,i} \ge 2$, conditional Lyapunov monotonicity and $A_* \ge 1$ give

$$\mathbb{E}(\|T_{m,h,i}\|_{\mathrm{op}}^2 \mid \mathcal{P}_{m,h,i}) \le A_* s_{m,h,i}^2.$$

Thus in all cases

$$\mathbb{E}(T_{m,h,i}^2 \mid \mathcal{P}_{m,h,i}) \preceq A_* \rho_{m,h,i} I.$$

The same scalar domination with TT^* and T^*T gives the rectangular row and column profiles. Hence each uncentered profile is bounded by

$$A_* \sum_i \rho_{m,h,i} \le A_* q_{m,h}^2 \frac{n_m}{\ell_m},$$

so the preceding heterogeneous cap-profile corollary applies with $\beta = A_*$.

For the common-U_m schedules, the cap-tail summand is bounded by $(e\Delta_m)^{\Lambda - c}$, up to the fixed factor A_*. This is summable over all integer dimensions when $c > \Lambda + 1$, and along every stretched-exponential sparse schedule when $c > \Lambda$. The matrix-tail summability is the same Θ_{tc} calculation with $\beta = A_*$ and $R = 1$.

Finally suppose $p_{m,h,i} \ge 1$ and the original variables are centered relative to $\mathcal{P}_{m,h,i}$. Then, under the common-U_m cap choice,

$$\|\mu_{m,h,i}\|_{\mathrm{op}} \le A_* s_{m,h,i} U_m^{-1+1/p_{m,h,i}} = A_* U_m^{-1} b_{m,h,i}.$$

Therefore

$$\frac{\sum_i \|\mu_{m,h,i}\|_{\mathrm{op}}}{q_{m,h}\sqrt{n_m}} \le A_* \sqrt{n_m}\, U_m^{-1}.$$

Since $n_m \le (e\Delta_m)^{\Lambda}$ and $U_m = (e\Delta_m)^c$, this tends to zero under either displayed cap-tail margin. Hence the drift bound holds eventually for every fixed $\eta > 0$. □

Corollary 7.58 (Heterogeneous tail-profile norm-tail selection)**.** *Work in the same independent or adapted setting, with the same conditioning fields $\mathcal{P}_{m,h,i}$, as in Corollary 7.56. Fix $K > 0$, $\eta \ge 0$, and $B > 0$. For each active stream let $s_{m,h,i} > 0$, $u_{m,h,i} > 0$, and deterministic measurable profiles $\pi_{m,h,i} : [0,\infty) \to [0,1]$ satisfy the conditional norm-tail bounds*

$$\mathbb{P}\{\|X_{m,h,i}\|_{\mathrm{op}} > s_{m,h,i} t \mid \mathcal{P}_{m,h,i}\} \le \pi_{m,h,i}(t) \qquad (t \ge 0).$$

Set

$$b_{m,h,i} := s_{m,h,i} u_{m,h,i}, \qquad \rho_{m,h,i} := 2 s_{m,h,i}^2 \int_0^{u_{m,h,i}} t\, \pi_{m,h,i}(t)\, dt,$$

and choose

$$q_{m,h} := \max\left\{ \max_i b_{m,h,i}, \left(\frac{\ell_m}{n_m} \sum_i \rho_{m,h,i} \right)^{1/2} \right\}.$$

Define $T_{m,h,i}$ and $\mu_{m,h,i}$ using these individual caps as in Corollary 7.56. Outside events $E_{m,h}$, assume the truncated-tail drift bound

$$\left\| \sum_{i=1}^{n_m} \mu_{m,h,i} \right\|_{\mathrm{op}} \le \eta q_{m,h} \sqrt{n_m}.$$

Suppose that $\ell_m \le B\sqrt{n_m}$ eventually and that

$$\sum_m \sum_{h=1}^{J_m} \left[\mathbb{P}(E_{m,h}) + \sum_{i=1}^{n_m} \pi_{m,h,i}(u_{m,h,i}) \right] < \infty$$

and

$$\sum_m J_m \Delta_m (e\Delta_m)^{-\gamma_{\mathrm{tc}}(K,1,1,B)} < \infty.$$

Then, almost surely, for all sufficiently large m, every active original stream is contraction-normalized by

$$Mq_{m,h}, \qquad M := \max\{1, K+\eta\},$$

and the all-plus signing gives

$$\operatorname{disc}(X_{m,h,1}/(Mq_{m,h}), \dots, X_{m,h,n_m}/(Mq_{m,h})) \le \sqrt{n_m} \qquad (1 \le h \le J_m).$$

If $J_m \Delta_m \le (e\Delta_m)^{\Gamma}$ eventually and $n_m \ge a\ell_m^2$, then the matrix-tail part is summable along all integer dimensions whenever

$$K > \Theta_{\mathrm{tc}}(\Gamma + 1, a, 1, 1),$$

and along every stretched-exponential sparse dimension schedule whenever

$$K > \Theta_{\mathrm{tc}}(\Gamma, a, 1, 1),$$

provided the displayed profile cap-tail series is summable.

Moreover, if the original streams are conditionally centered in the independent case or martingale differences in the adapted case, then the drift hypothesis is implied by the deterministic tail-drift profile bound

$$\sum_i s_{m,h,i} \left[u_{m,h,i} \pi_{m,h,i}(u_{m,h,i}) + \int_{u_{m,h,i}}^{\infty} \pi_{m,h,i}(t)\, dt \right] \le \eta q_{m,h} \sqrt{n_m}$$

outside $E_{m,h}$.

Proof. The cap-failure event satisfies

$$\mathbb{P}(F_{m,h}^{\mathrm{cap}}) \le \sum_{i=1}^{n_m} \pi_{m,h,i}(u_{m,h,i}).$$

Also $b_{m,h,i} \le q_{m,h}$, so it remains to verify the uncentered second-moment profiles in Corollary 7.56 with $R = 1$ and $\beta = 1$. In the self-adjoint case,

$$T_{m,h,i}^2 \preceq \|T_{m,h,i}\|_{\mathrm{op}}^2 I.$$

The conditional layer-cake formula and the displayed tail profile give

$$\begin{aligned} \mathbb{E}(\|T_{m,h,i}\|_{\mathrm{op}}^2 \mid \mathcal{P}_{m,h,i}) &\le s_{m,h,i}^2 \int_0^{u_{m,h,i}^2} \mathbb{P}\{\|X_{m,h,i}\|_{\mathrm{op}}^2 > s_{m,h,i}^2 r \mid \mathcal{P}_{m,h,i}\}\, dr \\ &\le 2 s_{m,h,i}^2 \int_0^{u_{m,h,i}} t\, \pi_{m,h,i}(t)\, dt = \rho_{m,h,i}. \end{aligned}$$

The same scalar domination gives the rectangular TT^* and T^*T profiles. Therefore every uncentered profile is bounded by

$$\sum_i \rho_{m,h,i} \le q_{m,h}^2 \frac{n_m}{\ell_m},$$

and the heterogeneous cap-profile corollary proves the general conclusion. The two closed-form margins are exactly its all-integer and sparse Θ_{tc} calculation with $\beta = R = 1$.

For the final assertion, conditional centering gives

$$\mu_{m,h,i} = -\mathbb{E}\Big(X_{m,h,i}\mathbf{1}_{\{\|X_{m,h,i}\|_{\mathrm{op}}>b_{m,h,i}\}} \mid \mathcal{P}_{m,h,i}\Big).$$

Another layer-cake estimate bounds

$$\|\mu_{m,h,i}\|_{\mathrm{op}} \le s_{m,h,i}\left[u_{m,h,i}\pi_{m,h,i}(u_{m,h,i}) + \int_{u_{m,h,i}}^{\infty} \pi_{m,h,i}(t)\,dt\right].$$

Summing in i gives the stated sufficient drift condition. □

Corollary 7.59 (Quantile-budget norm-tail profile selection)**.** *Work in the setting of Corollary 7.58, with the same deterministic profiles $\pi_{m,h,i}$ and scales $s_{m,h,i}$. For $\delta \in (0,1]$, let $Q_{m,h,i}(\delta)$ be any deterministic upper δ-tail level for the profile, meaning*

$$\pi_{m,h,i}(Q_{m,h,i}(\delta)) \le \delta.$$

For each active stream choose a deterministic cap-failure budget $\varepsilon_{m,h} \in (0,1]$ and deterministic shares $\omega_{m,h,i} > 0$ with $\sum_i \omega_{m,h,i} \le 1$. Put

$$u_{m,h,i} := Q_{m,h,i}(\varepsilon_{m,h}\omega_{m,h,i})$$

and define

$$b_{m,h,i} := s_{m,h,i}u_{m,h,i},$$
$$\rho_{m,h,i} := 2s_{m,h,i}^2 \int_0^{u_{m,h,i}} t\,\pi_{m,h,i}(t)\,dt,$$
$$q_{m,h} := \max\left\{\max_i b_{m,h,i}, \left(\frac{\ell_m}{n_m}\sum_i \rho_{m,h,i}\right)^{1/2}\right\}.$$

Assume the truncated-tail drift bound from Corollary 7.58, with these caps, holds outside $E_{m,h}$. If $\ell_m \le B\sqrt{n_m}$ eventually,

$$\sum_m \sum_{h=1}^{J_m} \big[\mathbb{P}(E_{m,h}) + \varepsilon_{m,h}\big] < \infty,$$

and

$$\sum_m J_m\Delta_m(e\Delta_m)^{-\gamma_{\mathrm{tc}}(K,1,1,B)} < \infty,$$

then the original streams are eventually contraction-normalized by $\max\{1, K+\eta\}q_{m,h}$, and the all-plus signing gives discrepancy at most $\sqrt{n_m}$, simultaneously for all active h.

Consequently, if $J_m\Delta_m \le (e\Delta_m)^{\Gamma}$, $J_m \le (e\Delta_m)^{\Lambda}$, $n_m \ge a\ell_m^2$ eventually, and $\varepsilon_{m,h} = (e\Delta_m)^{-c}$, then the cap-budget part is summable for all integer dimensions when $c > \Lambda + 1$, and along every stretched-exponential sparse dimension schedule when $c > \Lambda$. The corresponding matrix-tail margins are

$$K > \Theta_{\mathrm{tc}}(\Gamma + 1, a, 1, 1)$$

in the all-integer case and

$$K > \Theta_{\mathrm{tc}}(\Gamma, a, 1, 1)$$

in the sparse case. The same conclusion holds for any deterministic choice of shares, including a near-minimizer of the displayed scale $q_{m,h}$ over the simplex $\sum_i \omega_{m,h,i} \le 1$.

Proof. The selected upper quantiles give

$$\sum_i \pi_{m,h,i}(u_{m,h,i}) \le \varepsilon_{m,h} \sum_i \omega_{m,h,i} \le \varepsilon_{m,h}.$$

Thus all hypotheses of Corollary 7.58 are exactly the displayed assumptions here, and that corollary gives the general conclusion. If $\varepsilon_{m,h} = (e\Delta_m)^{-c}$ and $J_m \le (e\Delta_m)^{\Lambda}$, then

$$\sum_{h=1}^{J_m} \varepsilon_{m,h} \le (e\Delta_m)^{\Lambda - c}.$$

This series is summable over all integer dimensions when $c > \Lambda + 1$, and over every stretched-exponential sparse schedule when $c > \Lambda$. The matrix-tail part is precisely the Θ_{tc} summability calculation from the tail-profile corollary. □

Corollary 7.60 (Mixed sub-Weibull quantile-budget norm-tail selection)**.** *Work in the setting of Corollary 7.59. For each active increment suppose that, for deterministic $A_{m,h,i} \ge 1$, $\alpha_{m,h,i} > 0$, and $s_{m,h,i} > 0$,*

$$\mathbb{P}\{\|X_{m,h,i}\|_{\mathrm{op}} > s_{m,h,i} t \mid \mathcal{P}_{m,h,i}\} \le \min\{1, A_{m,h,i} e^{-t^{\alpha_{m,h,i}}}\} \qquad (t \ge 0).$$

Choose a stream cap budget $\varepsilon_{m,h} \in (0,1]$ and positive shares $\omega_{m,h,i}$ with $\sum_i \omega_{m,h,i} \le 1$. Put

$$\lambda_{m,h,i} := \log \frac{e A_{m,h,i}}{\varepsilon_{m,h} \omega_{m,h,i}}, \qquad b_{m,h,i} := s_{m,h,i} \lambda_{m,h,i}^{1/\alpha_{m,h,i}},$$

and

$$\widehat{\rho}_{m,h,i} := \frac{2A_{m,h,i}}{\alpha_{m,h,i}} \Gamma\left(\frac{2}{\alpha_{m,h,i}}\right) s_{m,h,i}^2.$$

Define the closed-form mixed sub-Weibull scale

$$q_{m,h} := \max\left\{ \max_i b_{m,h,i}, \left(\frac{\ell_m}{n_m} \sum_i \widehat{\rho}_{m,h,i}\right)^{1/2} \right\}.$$

Assume the truncated-tail drift bound from Corollary 7.58, with these caps, holds outside $E_{m,h}$. If $\ell_m \le B\sqrt{n_m}$ eventually,

$$\sum_m \sum_{h=1}^{J_m} \left[\mathbb{P}(E_{m,h}) + \varepsilon_{m,h}\right] < \infty,$$

and

$$\sum_m J_m \Delta_m (e\Delta_m)^{-\gamma_{\mathrm{tc}}(K,1,1,B)} < \infty,$$

then the original streams are eventually contraction-normalized by $\max\{1, K+\eta\}q_{m,h}$*, and the all-plus signing gives discrepancy at most* $\sqrt{n_m}$*, simultaneously for all active* h*.*

If $J_m\Delta_m \le (e\Delta_m)^{\Gamma}$*,* $J_m \le (e\Delta_m)^{\Lambda}$*,* $n_m \ge a\ell_m^2$ *eventually, and* $\varepsilon_{m,h} = (e\Delta_m)^{-c}$*, then the cap-budget part is summable for all integer dimensions when* $c > \Lambda + 1$*, and along every stretched-exponential sparse dimension schedule when* $c > \Lambda$*. The matrix-tail margins remain*

$$K > \Theta_{\mathrm{tc}}(\Gamma + 1, a, 1, 1)$$

and

$$K > \Theta_{\mathrm{tc}}(\Gamma, a, 1, 1),$$

respectively. In the conditionally centered independent case, or in the adapted martingale-difference case, the drift hypothesis is implied by

$$\sum_i s_{m,h,i}\left[\lambda_{m,h,i}^{1/\alpha_{m,h,i}}\varepsilon_{m,h}\omega_{m,h,i} + A_{m,h,i}\int_{\lambda_{m,h,i}^{1/\alpha_{m,h,i}}}^{\infty} e^{-t^{\alpha_{m,h,i}}}\,dt\right] \le \eta q_{m,h}\sqrt{n_m}$$

outside $E_{m,h}$*. The shares may be optimized to reduce the maximum cap term across mixed tail constants, scales, and exponents.*

Proof. Let

$$\pi_{m,h,i}(t) := \min\{1, A_{m,h,i}e^{-t^{\alpha_{m,h,i}}}\}.$$

With $u_{m,h,i} := \lambda_{m,h,i}^{1/\alpha_{m,h,i}}$,

$$\pi_{m,h,i}(u_{m,h,i}) \le A_{m,h,i}e^{-\lambda_{m,h,i}} = e^{-1}\varepsilon_{m,h}\omega_{m,h,i} \le \varepsilon_{m,h}\omega_{m,h,i}.$$

Thus these $u_{m,h,i}$ are admissible upper quantiles for the quantile-budget corollary. Its exact variance profile satisfies

$$2s_{m,h,i}^2\int_0^{u_{m,h,i}} t\,\pi_{m,h,i}(t)\,dt \le 2A_{m,h,i}s_{m,h,i}^2\int_0^{\infty} te^{-t^{\alpha_{m,h,i}}}\,dt = \widehat{\rho}_{m,h,i}.$$

Consequently the displayed $q_{m,h}$ dominates the exact tail-profile scale furnished by Corollary 7.59. Applying that corollary gives the general conclusion, and enlarging the normalization from the exact scale to $q_{m,h}$ preserves both contraction normalization and the $\sqrt{n_m}$ discrepancy bound.

The summability thresholds are the same cap-budget and matrix-tail calculations as in the quantile-budget corollary. Finally, in the centered or martingale case, the tail-drift sufficient condition from Corollary 7.58 is bounded above by the displayed mixed sub-Weibull drift sum because $\pi_{m,h,i}(u_{m,h,i}) \le \varepsilon_{m,h}\omega_{m,h,i}$ and $\pi_{m,h,i}(t) \le A_{m,h,i}e^{-t^{\alpha_{m,h,i}}}$. □

Corollary 7.61 (Optimized common-exponent sub-Weibull budget selection)**.** *Work in the setting of Corollary 7.60, but assume that the tail exponent is a fixed common* $\alpha > 0$*. For each active stream define the common-cap tail budget*

$$\Phi_{m,h}(r) := \sum_{i=1}^{n_m} A_{m,h,i}\exp\left\{-\left(\frac{r}{s_{m,h,i}}\right)^{\alpha}\right\} \qquad (r \ge 0).$$

Choose a stream cap budget $\varepsilon_{m,h} \in (0,1]$, and let $r_{m,h}$ be any deterministic cap satisfying

$$\Phi_{m,h}(r_{m,h}) \le e^{-1}\varepsilon_{m,h}.$$

In the optimized form, take the smallest such cap

$$r^*_{m,h} := \inf\{r \ge 0 : \Phi_{m,h}(r) \le e^{-1}\varepsilon_{m,h}\}.$$

Put $u_{m,h,i} := r_{m,h}/s_{m,h,i}$ and define the exact capped tail-energy

$$V_{m,h} := 2\sum_i s^2_{m,h,i}\int_0^{u_{m,h,i}} t\,\min\{1, A_{m,h,i}e^{-t^\alpha}\}\,dt.$$

It is always bounded by the closed-form full-tail energy

$$V_{m,h} \le \frac{2}{\alpha}\Gamma\!\left(\frac{2}{\alpha}\right)\sum_i A_{m,h,i}s^2_{m,h,i}.$$

Set

$$q_{m,h} := \max\left\{r_{m,h}, \left(\frac{\ell_m}{n_m}V_{m,h}\right)^{1/2}\right\}.$$

Assume the truncated-tail drift bound from Corollary 7.58, with the common caps $b_{m,h,i} = r_{m,h}$, holds outside $E_{m,h}$. If $\ell_m \le B\sqrt{n_m}$ eventually,

$$\sum_m \sum_{h=1}^{J_m}\left[\mathbb{P}(E_{m,h}) + \varepsilon_{m,h}\right] < \infty,$$

and

$$\sum_m J_m\Delta_m(e\Delta_m)^{-\gamma_{\mathrm{tc}}(K,1,1,B)} < \infty,$$

then the original streams are eventually contraction-normalized by $\max\{1, K+\eta\}q_{m,h}$, and the all-plus signing gives discrepancy at most $\sqrt{n_m}$, simultaneously for all active h.

If $J_m\Delta_m \le (e\Delta_m)^\Gamma$, $J_m \le (e\Delta_m)^\Lambda$, $n_m \ge a\ell_m^2$ eventually, and $\varepsilon_{m,h} = (e\Delta_m)^{-c}$, then the cap-budget part is summable for all integer dimensions when $c > \Lambda + 1$, and along every stretched-exponential sparse dimension schedule when $c > \Lambda$. The matrix-tail margins remain

$$K > \Theta_{\mathrm{tc}}(\Gamma + 1, a, 1, 1)$$

and

$$K > \Theta_{\mathrm{tc}}(\Gamma, a, 1, 1),$$

respectively.

If the stream is conditionally centered independent, or is an adapted martingale difference, the drift hypothesis is implied by

$$\sum_i\left[r_{m,h}A_{m,h,i}e^{-u^\alpha_{m,h,i}} + A_{m,h,i}s_{m,h,i}\int_{u_{m,h,i}}^\infty e^{-t^\alpha}\,dt\right] \le \eta q_{m,h}\sqrt{n_m}$$

outside $E_{m,h}$.

Moreover, $r^*_{m,h}$ is the optimal common cap among all deterministic budget-share choices in Corollary 7.60: a share choice with every mixed sub-Weibull cap at most r exists if and only if $\Phi_{m,h}(r) \le e^{-1}\varepsilon_{m,h}$.

Proof. For fixed m, h, set

$$\pi_i(t) := \min\{1, A_{m,h,i}e^{-t^\alpha}\}, \qquad u_i := r_{m,h}/s_{m,h,i}.$$

Then

$$\sum_i \pi_i(u_i) \le \sum_i A_{m,h,i}e^{-u_i^\alpha} = \Phi_{m,h}(r_{m,h}) \le e^{-1}\varepsilon_{m,h} \le \varepsilon_{m,h}.$$

The displayed $V_{m,h}$ is exactly the sum of the capped tail-energy terms from Corollary 7.58. That corollary therefore gives the general conclusion with $\beta = R = 1$. The closed-form bound on $V_{m,h}$ follows from

$$2s^2_{m,h,i}\int_0^{u_i} t\,\pi_i(t)\,dt \le 2A_{m,h,i}s^2_{m,h,i}\int_0^\infty te^{-t^\alpha}\,dt = \frac{2A_{m,h,i}}{\alpha}\Gamma\!\left(\frac{2}{\alpha}\right)s^2_{m,h,i}.$$

The summability thresholds are the same cap-budget and matrix-tail calculations as in the quantile-budget corollary. In the centered or martingale case, the stated drift condition dominates the tail-drift bound in Corollary 7.58, since $\pi_i(t) \le A_{m,h,i}e^{-t^\alpha}$ and $s_{m,h,i}u_i = r_{m,h}$.

It remains only to record optimality of the common cap. If shares $\omega_{m,h,i} > 0$ with $\sum_i \omega_{m,h,i} \le 1$ produce mixed sub-Weibull caps no larger than r, then

$$\log\frac{eA_{m,h,i}}{\varepsilon_{m,h}\omega_{m,h,i}} \le \left(\frac{r}{s_{m,h,i}}\right)^\alpha,$$

hence

$$\omega_{m,h,i} \ge \frac{eA_{m,h,i}}{\varepsilon_{m,h}}\exp\left\{-\left(\frac{r}{s_{m,h,i}}\right)^\alpha\right\}.$$

Summing in i gives $\Phi_{m,h}(r) \le e^{-1}\varepsilon_{m,h}$. Conversely, if $\Phi_{m,h}(r) \le e^{-1}\varepsilon_{m,h}$, choose

$$\omega_{m,h,i} := \frac{eA_{m,h,i}}{\varepsilon_{m,h}}\exp\left\{-\left(\frac{r}{s_{m,h,i}}\right)^\alpha\right\}.$$

Then $\sum_i \omega_{m,h,i} \le 1$, and the mixed sub-Weibull cap attached to each i is exactly r. Thus the infimum $r^*_{m,h}$ is the smallest common cap available to the budget-share rule. □

Corollary 7.62 (Closed-form common-exponent sub-Weibull schedules)**.** *Keep the hypotheses and notation of Corollary 7.61. Set*

$$A^\Sigma_{m,h} := \sum_i A_{m,h,i}, \qquad s^{\max}_{m,h} := \max_i s_{m,h,i}, \qquad C_\alpha := \frac{2}{\alpha}\Gamma\!\left(\frac{2}{\alpha}\right).$$

For a stream budget $\varepsilon_{m,h} \in (0,1]$, define the explicit envelope cap

$$r^{\mathrm{env}}_{m,h} := s^{\max}_{m,h}\left(\log\frac{eA^\Sigma_{m,h}}{\varepsilon_{m,h}}\right)^{1/\alpha}$$

and the closed variance envelope

$$V^{\mathrm{env}}_{m,h} := C_\alpha \sum_i A_{m,h,i} s^2_{m,h,i}.$$

Then the scale

$$q^{\mathrm{env}}_{m,h} := \max\left\{ r^{\mathrm{env}}_{m,h}, \left(\frac{\ell_m}{n_m} V^{\mathrm{env}}_{m,h}\right)^{1/2} \right\}$$

dominates the optimized common-exponent scale. Consequently, under the summability and matrix-tail hypotheses of Corollary 7.61, with the truncated-tail drift checked at the common cap $r^{\mathrm{env}}_{m,h}$, the original streams are eventually contraction-normalized by $\max\{1, K+\eta\} q^{\mathrm{env}}_{m,h}$, and the all-plus signing gives discrepancy at most $\sqrt{n_m}$, simultaneously for all active h.

If all scales in a stream are equal, $s_{m,h,i} = s_{m,h}$, then

$$r^{\mathrm{env}}_{m,h} = s_{m,h} \left(\log \frac{e A^\Sigma_{m,h}}{\varepsilon_{m,h}}\right)^{1/\alpha}$$

is the optimized common cap $r^*_{m,h}$. In particular, if $A^\Sigma_{m,h} \le (e\Delta_m)^\chi$ and $\varepsilon_{m,h} = (e\Delta_m)^{-c}$, then

$$r^{\mathrm{env}}_{m,h} \le s^{\max}_{m,h} \big((\chi + c + 1)\ell_m\big)^{1/\alpha}.$$

Proof. For every i,

$$\left(\frac{r^{\mathrm{env}}_{m,h}}{s_{m,h,i}}\right)^\alpha \ge \left(\frac{r^{\mathrm{env}}_{m,h}}{s^{\max}_{m,h}}\right)^\alpha = \log \frac{e A^\Sigma_{m,h}}{\varepsilon_{m,h}}.$$

Hence the common-cap budget from Corollary 7.61 satisfies

$$\Phi_{m,h}(r^{\mathrm{env}}_{m,h}) \le A^\Sigma_{m,h} \exp\left\{ -\log \frac{e A^\Sigma_{m,h}}{\varepsilon_{m,h}} \right\} = e^{-1}\varepsilon_{m,h}.$$

The full-tail estimate in that corollary gives

$$V_{m,h} \le V^{\mathrm{env}}_{m,h}.$$

Thus $q^{\mathrm{env}}_{m,h}$ dominates the admissible common-exponent tail-profile scale, and the stated conclusion is exactly the previous corollary applied with this explicit cap and enlarged variance envelope.

When all $s_{m,h,i}$ are equal to $s_{m,h}$,

$$\Phi_{m,h}(r) = A^\Sigma_{m,h} e^{-(r/s_{m,h})^\alpha},$$

so the equality $\Phi_{m,h}(r) = e^{-1}\varepsilon_{m,h}$ is solved by the displayed $r^{\mathrm{env}}_{m,h}$. This is the minimal admissible common cap by the optimality assertion in the previous corollary. The final growth bound follows from

$$\log \frac{e A^\Sigma_{m,h}}{\varepsilon_{m,h}} \le (\chi + c + 1)\log(e\Delta_m) = (\chi + c + 1)\ell_m.$$

□

Corollary 7.63 (Uniform common-exponent sub-Weibull schedules)**.** *Work in the setting of Corollary 7.61. Assume that the sub-Weibull exponent is a fixed common $\alpha > 0$, and that for each active stream there are deterministic $A_{m,h} \geq 1$ and $s_{m,h} > 0$ such that, for every increment and every $t \geq 0$,*

$$\mathbb{P}\{\|X_{m,h,i}\|_{\mathrm{op}} > t \mid \mathcal{P}_{m,h,i}\} \leq A_{m,h} \exp\{-(t/s_{m,h})^{\alpha}\}.$$

Put $C_\alpha = (2/\alpha)\Gamma(2/\alpha)$. For a stream budget $\varepsilon_{m,h} \in (0,1]$, define

$$r_{m,h}^{\psi,\mathrm{unif}} := s_{m,h} \left(\log \frac{e n_m A_{m,h}}{\varepsilon_{m,h}} \right)^{1/\alpha}$$

and

$$V_{m,h}^{\psi,\mathrm{unif}} := C_\alpha n_m A_{m,h} s_{m,h}^2.$$

Set

$$q_{m,h}^{\psi,\mathrm{unif}} := \max\left\{ r_{m,h}^{\psi,\mathrm{unif}}, \left(\frac{\ell_m}{n_m} V_{m,h}^{\psi,\mathrm{unif}} \right)^{1/2} \right\}.$$

Under the summability and matrix-tail hypotheses of Corollary 7.61, with the truncated-tail drift checked at the common cap $r_{m,h}^{\psi,\mathrm{unif}}$, the original streams are eventually contraction-normalized by $\max\{1, K+\eta\} q_{m,h}^{\psi,\mathrm{unif}}$, and the all-plus signing gives discrepancy at most $\sqrt{n_m}$, simultaneously for all active h.

In particular, if $\varepsilon_{m,h} = (e\Delta_m)^{-c}$,

$$A_{m,h} \leq (e\Delta_m)^{\chi}, \qquad n_m \leq (e\Delta_m)^{\nu},$$

then

$$r_{m,h}^{\psi,\mathrm{unif}} \leq s_{m,h} \big((\nu + \chi + c + 1)\ell_m \big)^{1/\alpha}$$

and

$$V_{m,h}^{\psi,\mathrm{unif}} \leq C_\alpha s_{m,h}^2 (e\Delta_m)^{\nu+\chi}.$$

Equivalently,

$$q_{m,h}^{\psi,\mathrm{unif}} \leq s_{m,h} \max\left\{ \big((\nu + \chi + c + 1)\ell_m \big)^{1/\alpha}, \Big(C_\alpha \ell_m (e\Delta_m)^{\nu+\chi} / n_m \Big)^{1/2} \right\}.$$

Proof. Apply Corollary 7.62 after choosing the admissible tail parameters $A_{m,h,i} = A_{m,h}$ and $s_{m,h,i} = s_{m,h}$ for every i. Then

$$A_{m,h}^{\Sigma} = n_m A_{m,h}, \qquad s_{m,h}^{\max} = s_{m,h},$$

and the closed-form envelope in that corollary is exactly the displayed $r_{m,h}^{\psi,\mathrm{unif}}$, $V_{m,h}^{\psi,\mathrm{unif}}$, and $q_{m,h}^{\psi,\mathrm{unif}}$. The eventual normalization and discrepancy statement follow.

The growth estimates follow from

$$\log \frac{e n_m A_{m,h}}{\varepsilon_{m,h}} \leq (\nu + \chi + c + 1)\ell_m$$

and $n_m A_{m,h} \leq (e\Delta_m)^{\nu+\chi}$. Substituting these bounds into the definition of $q_{m,h}^{\psi,\mathrm{unif}}$ gives the final display. □

Corollary 7.64 (Mixed finite-moment quantile-budget norm-tail selection)**.** *Work in the same independent or adapted setting as Corollary 7.59. For each active increment suppose that, for deterministic $A_{m,h,i} \geq 1$, $p_{m,h,i} > 0$, and $s_{m,h,i} > 0$,*

$$\mathbb{E}\left(\|X_{m,h,i}\|_{\mathrm{op}}^{p_{m,h,i}} \mid \mathcal{P}_{m,h,i}\right) \leq A_{m,h,i} s_{m,h,i}^{p_{m,h,i}}.$$

Choose a stream cap budget $\varepsilon_{m,h} \in (0,1]$ and positive shares $\omega_{m,h,i}$ with $\sum_i \omega_{m,h,i} \leq 1$. Put

$$\lambda_{m,h,i} := \frac{eA_{m,h,i}}{\varepsilon_{m,h}\omega_{m,h,i}}, \qquad b_{m,h,i} := s_{m,h,i}\lambda_{m,h,i}^{1/p_{m,h,i}},$$

and define the moment-energy proxy

$$\widehat{\rho}_{m,h,i} := \begin{cases} A_{m,h,i} s_{m,h,i}^2 \lambda_{m,h,i}^{(2-p_{m,h,i})/p_{m,h,i}}, & 0 < p_{m,h,i} < 2, \\ A_{m,h,i}^{2/p_{m,h,i}} s_{m,h,i}^2, & p_{m,h,i} \geq 2. \end{cases}$$

Set

$$q_{m,h} := \max\left\{\max_i b_{m,h,i}, \left(\frac{\ell_m}{n_m}\sum_i \widehat{\rho}_{m,h,i}\right)^{1/2}\right\}.$$

Assume the truncated-tail drift bound from Corollary 7.56, with these caps, holds outside $E_{m,h}$. If $\ell_m \leq B\sqrt{n_m}$ eventually,

$$\sum_m \sum_{h=1}^{J_m} \left[\mathbb{P}(E_{m,h}) + \varepsilon_{m,h}\right] < \infty,$$

and

$$\sum_m J_m \Delta_m (e\Delta_m)^{-\gamma_{\mathrm{tc}}(K,1,1,B)} < \infty,$$

then the original streams are eventually contraction-normalized by $\max\{1, K+\eta\}q_{m,h}$, and the all-plus signing gives discrepancy at most $\sqrt{n_m}$, simultaneously for all active h.

If $J_m\Delta_m \leq (e\Delta_m)^{\Gamma}$, $J_m \leq (e\Delta_m)^{\Lambda}$, $n_m \geq a\ell_m^2$ eventually, and $\varepsilon_{m,h} = (e\Delta_m)^{-c}$, then the cap-budget part is summable for all integer dimensions when $c > \Lambda + 1$, and along every stretched-exponential sparse dimension schedule when $c > \Lambda$. The matrix-tail margins remain

$$K > \Theta_{\mathrm{tc}}(\Gamma + 1, a, 1, 1)$$

and

$$K > \Theta_{\mathrm{tc}}(\Gamma, a, 1, 1),$$

respectively.

If every $p_{m,h,i} > 1$, then in the conditionally centered independent case, or in the adapted martingale-difference case, the drift hypothesis is implied by

$$\sum_i A_{m,h,i} s_{m,h,i} \lambda_{m,h,i}^{-(p_{m,h,i}-1)/p_{m,h,i}} \leq \eta q_{m,h}\sqrt{n_m}$$

outside $E_{m,h}$. The shares may be optimized to trade off the largest cap against the aggregate truncated second-moment energy for mixed moment orders.

Proof. Markov's inequality gives

$$\mathbb{P}\{\|X_{m,h,i}\|_{\mathrm{op}} > b_{m,h,i} \mid \mathcal{P}_{m,h,i}\} \le A_{m,h,i}\lambda_{m,h,i}^{-1} = e^{-1}\varepsilon_{m,h}\omega_{m,h,i} \le \varepsilon_{m,h}\omega_{m,h,i}.$$

Thus the total cap-failure probability is at most $\varepsilon_{m,h}$. Also $b_{m,h,i} \le q_{m,h}$, so the heterogeneous cap-profile theorem will apply with $R = 1$ once the uncentered second-moment profile is bounded.

For $0 < p_{m,h,i} < 2$, on the capped event,

$$\|T_{m,h,i}\|_{\mathrm{op}}^2 \le b_{m,h,i}^{2-p_{m,h,i}} \|X_{m,h,i}\|_{\mathrm{op}}^{p_{m,h,i}},$$

and hence

$$\mathbb{E}(\|T_{m,h,i}\|_{\mathrm{op}}^2 \mid \mathcal{P}_{m,h,i}) \le \widehat{\rho}_{m,h,i}.$$

For $p_{m,h,i} \ge 2$, conditional Lyapunov monotonicity gives the same bound:

$$\mathbb{E}(\|T_{m,h,i}\|_{\mathrm{op}}^2 \mid \mathcal{P}_{m,h,i}) \le \Big[\mathbb{E}(\|X_{m,h,i}\|_{\mathrm{op}}^{p_{m,h,i}} \mid \mathcal{P}_{m,h,i})\Big]^{2/p_{m,h,i}} \le \widehat{\rho}_{m,h,i}.$$

In the self-adjoint case $T_{m,h,i}^2 \preceq \|T_{m,h,i}\|_{\mathrm{op}}^2 I$, and in the rectangular case the same scalar bound applies to $T_{m,h,i}T_{m,h,i}^*$ and $T_{m,h,i}^*T_{m,h,i}$. Therefore every uncentered second-moment profile is bounded by

$$\sum_i \widehat{\rho}_{m,h,i} \le q_{m,h}^2 \, \frac{n_m}{\ell_m}.$$

Corollary 7.56, with $\beta = R = 1$, gives the general conclusion. The cap-budget and matrix-tail summability thresholds are the same as in the quantile-budget corollary.

Finally, if $p_{m,h,i} > 1$ and the original variables are centered relative to $\mathcal{P}_{m,h,i}$, then

$$\begin{aligned}\|\mu_{m,h,i}\|_{\mathrm{op}} &\le \mathbb{E}\Big(\|X_{m,h,i}\|_{\mathrm{op}}\mathbf{1}_{\{\|X_{m,h,i}\|_{\mathrm{op}} > b_{m,h,i}\}} \;\Big|\; \mathcal{P}_{m,h,i}\Big) \\ &\le A_{m,h,i}s_{m,h,i}\lambda_{m,h,i}^{-(p_{m,h,i}-1)/p_{m,h,i}}.\end{aligned}$$

Summing gives the displayed sufficient drift bound. □

Corollary 7.65 (Optimized common-moment quantile-budget selection)**.** *Work in the setting of Corollary 7.64, but assume that the moment order is a fixed common $p > 0$. For each active stream put*

$$S_{m,h} := \sum_{i=1}^{n_m} A_{m,h,i}s_{m,h,i}^p$$

and choose a stream cap budget $\varepsilon_{m,h} \in (0, 1]$. Define the equalized Markov cap

$$r_{m,h} := \left(\frac{eS_{m,h}}{\varepsilon_{m,h}}\right)^{1/p}$$

and the aggregate moment-energy proxy

$$V_{m,h} := \begin{cases} \left(\dfrac{e}{\varepsilon_{m,h}}\right)^{(2-p)/p} S_{m,h}^{2/p}, & 0 < p < 2, \\ \displaystyle\sum_i A_{m,h,i}^{2/p} s_{m,h,i}^2, & p \geq 2. \end{cases}$$

Set

$$q_{m,h} := \max\left\{ r_{m,h}, \left(\frac{\ell_m}{n_m} V_{m,h}\right)^{1/2} \right\}.$$

Assume the truncated-tail drift bound from Corollary 7.56, with the common caps $b_{m,h,i} = r_{m,h}$, holds outside $E_{m,h}$. If $\ell_m \leq B\sqrt{n_m}$ eventually,

$$\sum_m \sum_{h=1}^{J_m} \left[\mathbb{P}(E_{m,h}) + \varepsilon_{m,h}\right] < \infty,$$

and

$$\sum_m J_m \Delta_m (e\Delta_m)^{-\gamma_{\mathrm{tc}}(K,1,1,B)} < \infty,$$

then the original streams are eventually contraction-normalized by $\max\{1, K+\eta\} q_{m,h}$, and the all-plus signing gives discrepancy at most $\sqrt{n_m}$, simultaneously for all active h.

If $J_m \Delta_m \leq (e\Delta_m)^{\Gamma}$, $J_m \leq (e\Delta_m)^{\Lambda}$, $n_m \geq a\ell_m^2$ eventually, and $\varepsilon_{m,h} = (e\Delta_m)^{-c}$, then the cap-budget part is summable for all integer dimensions when $c > \Lambda + 1$, and along every stretched-exponential sparse dimension schedule when $c > \Lambda$. The matrix-tail margins remain

$$K > \Theta_{\mathrm{tc}}(\Gamma + 1, a, 1, 1)$$

and

$$K > \Theta_{\mathrm{tc}}(\Gamma, a, 1, 1),$$

respectively.

If $p > 1$, then in the conditionally centered independent case, or in the adapted martingale-difference case, the drift hypothesis is implied by

$$S_{m,h} r_{m,h}^{1-p} \leq \eta q_{m,h} \sqrt{n_m}$$

outside $E_{m,h}$. Equivalently, the optimal budget shares are

$$\omega_{m,h,i} := \frac{A_{m,h,i} s_{m,h,i}^p}{S_{m,h}},$$

for which every cap $b_{m,h,i}$ in Corollary 7.64 is equal to $r_{m,h}$.

Proof. Apply Corollary 7.64 with

$$\omega_{m,h,i} = \frac{A_{m,h,i} s_{m,h,i}^p}{S_{m,h}}.$$

Then $\sum_i \omega_{m,h,i} = 1$ and

$$\lambda_{m,h,i} = \frac{eA_{m,h,i}}{\varepsilon_{m,h}\omega_{m,h,i}} = \frac{eS_{m,h}}{\varepsilon_{m,h}s^p_{m,h,i}},$$

so

$$b_{m,h,i} = s_{m,h,i}\lambda^{1/p}_{m,h,i} = r_{m,h} \qquad (1 \le i \le n_m).$$

If $0 < p < 2$, the mixed moment-energy terms sum to

$$\begin{aligned}\sum_i A_{m,h,i}s^2_{m,h,i}\lambda^{(2-p)/p}_{m,h,i} &= \left(\frac{eS_{m,h}}{\varepsilon_{m,h}}\right)^{(2-p)/p}\sum_i A_{m,h,i}s^p_{m,h,i} \\ &= \left(\frac{e}{\varepsilon_{m,h}}\right)^{(2-p)/p} S^{2/p}_{m,h}.\end{aligned}$$

For $p \ge 2$, the mixed theorem gives the displayed sum $\sum_i A^{2/p}_{m,h,i}s^2_{m,h,i}$. Thus its scale is exactly the displayed $q_{m,h}$, and all conclusions and summability thresholds follow from that corollary. The drift bound is the mixed finite-moment drift specialized to the same shares:

$$\sum_i A_{m,h,i}s_{m,h,i}\lambda^{-(p-1)/p}_{m,h,i} = S_{m,h}r^{1-p}_{m,h}.$$

□

Corollary 7.66 (Closed-form common-moment growth schedules)**.** *Keep the hypotheses and notation of Corollary 7.65. Let $\bar{S}_{m,h}$ be any deterministic envelope with*

$$S_{m,h} \le \bar{S}_{m,h}.$$

When $p \ge 2$, also let $\bar{W}_{m,h}$ satisfy

$$\sum_i A^{2/p}_{m,h,i}s^2_{m,h,i} \le \bar{W}_{m,h}.$$

For a stream budget $\varepsilon_{m,h} \in (0,1]$, set

$$r^{\mathrm{env}}_{m,h} := \left(\frac{e\bar{S}_{m,h}}{\varepsilon_{m,h}}\right)^{1/p}$$

and

$$V^{\mathrm{env}}_{m,h} := \begin{cases}\left(\dfrac{e}{\varepsilon_{m,h}}\right)^{(2-p)/p}\bar{S}^{2/p}_{m,h}, & 0 < p < 2,\\ \bar{W}_{m,h}, & p \ge 2.\end{cases}$$

Define

$$q^{\mathrm{env}}_{m,h} := \max\left\{r^{\mathrm{env}}_{m,h}, \left(\frac{\ell_m}{n_m}V^{\mathrm{env}}_{m,h}\right)^{1/2}\right\}.$$

Under the summability and matrix-tail hypotheses of Corollary 7.65, with the truncated-tail drift checked for the exact cap $r_{m,h}$, the original streams are eventually contraction-normalized by $\max\{1, K+\eta\}q^{\mathrm{env}}_{m,h}$, and the all-plus

signing gives discrepancy at most $\sqrt{n_m}$, simultaneously for all active h. If $p > 1$, the centered independent or martingale-difference drift condition is implied by the envelope bound

$$\bar{S}_{m,h}(r^{\mathrm{env}}_{m,h})^{1-p} \le \eta q^{\mathrm{env}}_{m,h}\sqrt{n_m}.$$

In particular, if $\varepsilon_{m,h} = (e\Delta_m)^{-c}$ and $\bar{S}_{m,h} \le (e\Delta_m)^{\chi}$, then

$$r^{\mathrm{env}}_{m,h} \le (e\Delta_m)^{(\chi+c+1)/p}.$$

For $0 < p < 2$,

$$V^{\mathrm{env}}_{m,h} \le (e\Delta_m)^{((2-p)(c+1)+2\chi)/p},$$

whereas for $p \ge 2$, if $\bar{W}_{m,h} \le (e\Delta_m)^{\upsilon}$, then

$$V^{\mathrm{env}}_{m,h} \le (e\Delta_m)^{\upsilon}.$$

Proof. The exact common-moment cap from Corollary 7.65 satisfies

$$r_{m,h} = \left(\frac{eS_{m,h}}{\varepsilon_{m,h}}\right)^{1/p} \le r^{\mathrm{env}}_{m,h}.$$

For $0 < p < 2$, the exact energy is monotone in $S_{m,h}$, hence is at most the displayed $V^{\mathrm{env}}_{m,h}$. For $p \ge 2$, this is exactly the assumption defining $\bar{W}_{m,h}$. Thus $q^{\mathrm{env}}_{m,h}$ dominates the exact optimized common-moment scale, so the previous corollary gives the same eventual normalization after enlarging the scale.

If $p > 1$, then

$$S_{m,h}r_{m,h}^{1-p} = S_{m,h}^{1/p}\left(\frac{e}{\varepsilon_{m,h}}\right)^{-(p-1)/p} \le \bar{S}_{m,h}^{1/p}\left(\frac{e}{\varepsilon_{m,h}}\right)^{-(p-1)/p} = \bar{S}_{m,h}(r^{\mathrm{env}}_{m,h})^{1-p},$$

so the envelope drift condition implies the exact drift condition. The final growth displays follow from

$$\frac{e}{\varepsilon_{m,h}} \le (e\Delta_m)^{c+1} \qquad \text{and} \qquad \bar{S}_{m,h} \le (e\Delta_m)^{\chi}.$$

□

Corollary 7.67 (Uniform common-moment operator-tail schedules)**.** *Work in the setting of Corollary 7.65. Assume that the moment order is a fixed common $p > 0$, and that for each active stream there are deterministic $A_{m,h} \ge 1$ and $s_{m,h} > 0$ such that, for every increment,*

$$\mathbb{E}\left(\|X_{m,h,i}\|^p_{\mathrm{op}} \mid \mathcal{P}_{m,h,i}\right) \le A_{m,h}s^p_{m,h}.$$

For a stream budget $\varepsilon_{m,h} \in (0,1]$, define

$$r^{\mathrm{unif}}_{m,h} := s_{m,h}\left(\frac{en_mA_{m,h}}{\varepsilon_{m,h}}\right)^{1/p}$$

and

$$V_{m,h}^{\mathrm{unif}} := \begin{cases} s_{m,h}^2 \left(\dfrac{e}{\varepsilon_{m,h}}\right)^{(2-p)/p} (n_m A_{m,h})^{2/p}, & 0 < p < 2, \\ n_m A_{m,h}^{2/p} s_{m,h}^2, & p \geq 2. \end{cases}$$

Set

$$q_{m,h}^{\mathrm{unif}} := \max\left\{ r_{m,h}^{\mathrm{unif}}, \left(\frac{\ell_m}{n_m} V_{m,h}^{\mathrm{unif}}\right)^{1/2} \right\}.$$

Under the summability and matrix-tail hypotheses of Corollary 7.65, with the truncated-tail drift checked at the common cap $r_{m,h}^{\mathrm{unif}}$, the original streams are eventually contraction-normalized by $\max\{1, K+\eta\} q_{m,h}^{\mathrm{unif}}$, and the all-plus signing gives discrepancy at most $\sqrt{n_m}$, simultaneously for all active h.

If $p > 1$, then in the centered independent or martingale-difference case the drift condition is implied by

$$n_m A_{m,h} s_{m,h}^p \left(r_{m,h}^{\mathrm{unif}}\right)^{1-p} = s_{m,h} (n_m A_{m,h})^{1/p} \left(\frac{e}{\varepsilon_{m,h}}\right)^{-(p-1)/p} \leq \eta q_{m,h}^{\mathrm{unif}} \sqrt{n_m}.$$

In particular, if $\varepsilon_{m,h} = (e\Delta_m)^{-c}$,

$$A_{m,h} \leq (e\Delta_m)^{\chi}, \qquad n_m \leq (e\Delta_m)^{\nu},$$

then

$$r_{m,h}^{\mathrm{unif}} \leq s_{m,h} (e\Delta_m)^{(\nu+\chi+c+1)/p}.$$

For $0 < p < 2$,

$$V_{m,h}^{\mathrm{unif}} \leq s_{m,h}^2 (e\Delta_m)^{((2-p)(c+1)+2(\nu+\chi))/p},$$

whereas for $p \geq 2$,

$$V_{m,h}^{\mathrm{unif}} \leq s_{m,h}^2 (e\Delta_m)^{\nu+2\chi/p}.$$

Proof. Apply Corollary 7.66 after choosing the admissible moment parameters $A_{m,h,i} = A_{m,h}$ and $s_{m,h,i} = s_{m,h}$ for every i. Then

$$S_{m,h} = n_m A_{m,h} s_{m,h}^p.$$

When $p \geq 2$, also

$$\sum_i A_{m,h,i}^{2/p} s_{m,h,i}^2 = n_m A_{m,h}^{2/p} s_{m,h}^2.$$

Use these exact aggregates as $\bar{S}_{m,h}$ and $\bar{W}_{m,h}$ in that corollary. The definitions of $r_{m,h}^{\mathrm{unif}}$, $V_{m,h}^{\mathrm{unif}}$, and $q_{m,h}^{\mathrm{unif}}$ are exactly the envelope definitions from that corollary with these choices, so the eventual normalization and discrepancy conclusion follow.

For $p > 1$, the envelope drift condition from the previous corollary becomes

$$\bar{S}_{m,h} \left(r_{m,h}^{\mathrm{unif}}\right)^{1-p} \leq \eta q_{m,h}^{\mathrm{unif}} \sqrt{n_m}.$$

Substituting $\bar{S}_{m,h} = n_m A_{m,h} s_{m,h}^p$ gives the displayed identity. The three growth estimates follow from

$$\frac{e}{\varepsilon_{m,h}} \le (e\Delta_m)^{c+1}, \qquad n_m A_{m,h} \le (e\Delta_m)^{\nu+\chi}, \qquad A_{m,h}^{2/p} \le (e\Delta_m)^{2\chi/p}.$$

□

Corollary 7.68 (Mixed dominated operator-tail active libraries)**.** *For each $m \ge 1$, let $D_m, n_m, J_m \ge 1$, put $\Delta_m := D_m + n_m + 1$ and $\ell_m := \log(e\Delta_m)$, and let each active stream $1 \le h \le J_m$ be tagged either ψ or μ. Choose $\varepsilon_{m,h} \in (0,1]$, $A_{m,h} \ge 1$, and $s_{m,h} > 0$. If h is tagged ψ, assume that for some $\alpha_{m,h} > 0$,*

$$\mathbb{P}\{\|X_{m,h,i}\|_{\mathrm{op}} > t \mid \mathcal{P}_{m,h,i}\} \le A_{m,h} \exp\{-(t/s_{m,h})^{\alpha_{m,h}}\} \qquad (t \ge 0)$$

for every increment. Set

$$C_{\alpha_{m,h}} := \frac{2}{\alpha_{m,h}} \Gamma\left(\frac{2}{\alpha_{m,h}}\right), \qquad r_{m,h} := s_{m,h} \left(\log \frac{e n_m A_{m,h}}{\varepsilon_{m,h}}\right)^{1/\alpha_{m,h}},$$

and

$$V_{m,h} := C_{\alpha_{m,h}} n_m A_{m,h} s_{m,h}^2.$$

If h is tagged μ, assume that for some $p_{m,h} > 0$,

$$\mathbb{E}\left(\|X_{m,h,i}\|_{\mathrm{op}}^{p_{m,h}} \mid \mathcal{P}_{m,h,i}\right) \le A_{m,h} s_{m,h}^{p_{m,h}}$$

for every increment. Set

$$r_{m,h} := s_{m,h} \left(\frac{e n_m A_{m,h}}{\varepsilon_{m,h}}\right)^{1/p_{m,h}}$$

and

$$V_{m,h} := \begin{cases} s_{m,h}^2 \left(\dfrac{e}{\varepsilon_{m,h}}\right)^{(2-p_{m,h})/p_{m,h}} (n_m A_{m,h})^{2/p_{m,h}}, & 0 < p_{m,h} < 2, \\ n_m A_{m,h}^{2/p_{m,h}} s_{m,h}^2, & p_{m,h} \ge 2. \end{cases}$$

In both cases define

$$q_{m,h} := \max\left\{r_{m,h}, \left(\frac{\ell_m}{n_m} V_{m,h}\right)^{1/2}\right\}.$$

Assume that the corresponding truncated-tail drift hypothesis from Corollary 7.60 or Corollary 7.64, with the common cap $r_{m,h}$, holds outside $E_{m,h}$. If $\ell_m \le B\sqrt{n_m}$ eventually,

$$\sum_m \sum_{h=1}^{J_m} [\mathbb{P}(E_{m,h}) + \varepsilon_{m,h}] < \infty,$$

and

$$\sum_m J_m \Delta_m (e\Delta_m)^{-\gamma_{\mathrm{tc}}(K,1,1,B)} < \infty,$$

then the original streams are eventually contraction-normalized by $\max\{1, K+\eta\}q_{m,h}$*, and the all-plus signing gives discrepancy at most* $\sqrt{n_m}$*, simultaneously for all active* h*.*

If $J_m\Delta_m \le (e\Delta_m)^{\Gamma}$*,* $J_m \le (e\Delta_m)^{\Lambda}$*,* $n_m \ge a\ell_m^2$ *eventually, and* $\varepsilon_{m,h} = (e\Delta_m)^{-c}$*, then the cap-budget part is summable for all integer dimensions when* $c > \Lambda + 1$*, and along every stretched-exponential sparse dimension schedule when* $c > \Lambda$*. The matrix-tail margins remain*

$$K > \Theta_{\mathrm{tc}}(\Gamma + 1, a, 1, 1)$$

and

$$K > \Theta_{\mathrm{tc}}(\Gamma, a, 1, 1),$$

respectively.

In the centered independent or martingale-difference case, the drift hypothesis for a ψ*-tagged stream is implied by*

$$s_{m,h}\left[\varepsilon_{m,h}u_{m,h} + n_m A_{m,h}\int_{u_{m,h}}^{\infty} e^{-t^{\alpha_{m,h}}}\,dt\right] \le \eta q_{m,h}\sqrt{n_m},$$

$$u_{m,h} := \left(\log\frac{en_mA_{m,h}}{\varepsilon_{m,h}}\right)^{1/\alpha_{m,h}}.$$

For a μ*-tagged stream with* $p_{m,h} > 1$*, it is implied by*

$$n_m A_{m,h} s_{m,h}^{p_{m,h}} r_{m,h}^{1-p_{m,h}} \le \eta q_{m,h}\sqrt{n_m}.$$

Proof. For a ψ-tagged stream, apply Corollary 7.60 with

$$A_{m,h,i} = A_{m,h}, \qquad s_{m,h,i} = s_{m,h}, \qquad \alpha_{m,h,i} = \alpha_{m,h}, \qquad \omega_{m,h,i} = 1/n_m.$$

Then every cap $b_{m,h,i}$ equals $r_{m,h}$, and the mixed sub-Weibull energy is $V_{m,h}$. The displayed centered drift condition is exactly the drift sum in that corollary with these equal shares.

For a μ-tagged stream, apply Corollary 7.64 with

$$A_{m,h,i} = A_{m,h}, \qquad s_{m,h,i} = s_{m,h}, \qquad p_{m,h,i} = p_{m,h}, \qquad \omega_{m,h,i} = 1/n_m.$$

Again every cap equals $r_{m,h}$. If $0 < p_{m,h} < 2$, the aggregate moment-energy is

$$n_m A_{m,h} s_{m,h}^2 \left(\frac{en_mA_{m,h}}{\varepsilon_{m,h}}\right)^{(2-p_{m,h})/p_{m,h}}$$
$$= s_{m,h}^2\left(\frac{e}{\varepsilon_{m,h}}\right)^{(2-p_{m,h})/p_{m,h}} (n_mA_{m,h})^{2/p_{m,h}},$$

and if $p_{m,h} \ge 2$ it is $n_m A_{m,h}^{2/p_{m,h}} s_{m,h}^2$. The displayed moment drift condition is the same specialization of the mixed finite-moment drift bound.

Apply the appropriate selector to every active stream. The two selectors have the same cap-budget summability and matrix-tail requirement, so summing over the active menu gives the simultaneous eventual conclusion

by Borel–Cantelli. The all-integer and sparse dimension thresholds are the common thresholds from those two selector corollaries. □

Corollary 7.69 (Signed cores plus mixed dominated operator-tail libraries)**.** *For each $m \geq 1$, let $D_m, n_m, J_m \geq 1$, put $\Delta_m := D_m + n_m + 1$ and $\ell_m := \log(e\Delta_m)$, and let $\mathcal{H}_m$ be a row environment. For $h \geq 1$, let*

$$B_{m,h,1}, \dots, B_{m,h,n_m}, \qquad Y_{m,h,1}, \dots, Y_{m,h,n_m}$$

be self-adjoint random matrices in $\mathrm{M}_{D_m}(\mathbb{C})$. Assume that there are events $F_m^{\mathrm{core}} \in \mathcal{H}_m$ with $\sum_m \mathbb{P}(F_m^{\mathrm{core}}) < \infty$ such that, outside F_m^{core}, the active core tuples are contractions and admit one $\mathcal{H}_m$-measurable active-prefix signing:

$$\|B_{m,h,i}\|_{\mathrm{op}} \leq 1, \qquad \left\| \sum_{i=1}^{n_m} \varepsilon_{m,i}^{\star} B_{m,h,i} \right\|_{\mathrm{op}} \leq K_{\mathrm{core}} \sqrt{n_m}$$

for every $h \leq J_m$ and $i \leq n_m$.

For each active $h \leq J_m$, tag the perturbation stream either ψ or μ, choose the parameters $\varepsilon_{m,h}, A_{m,h}, s_{m,h}$, and define $r_{m,h}$, $V_{m,h}$, and $q_{m,h}$ exactly as in Corollary 7.68, with $Y_{m,h,i}$ in place of $X_{m,h,i}$. Assume that the corresponding conditional tail or moment hypotheses, and the matching truncated-tail drift hypotheses outside events $E_{m,h}$, hold after adjoining $\mathcal{H}_m$ to the predictable sigma-fields. Let $K_{\mathrm{tail}}, \eta > 0$, and put

$$\begin{aligned} \lambda_{m,h} &:= \max\{1, K_{\mathrm{tail}} + \eta\}\, q_{m,h}, \\ Z_{m,h,i} &:= \frac{Y_{m,h,i}}{\lambda_{m,h}}, \\ X_{m,h,i} &:= B_{m,h,i} + Z_{m,h,i}. \end{aligned}$$

If $\ell_m \leq B\sqrt{n_m}$ eventually,

$$\sum_m \left[\mathbb{P}(F_m^{\mathrm{core}}) + \sum_{h=1}^{J_m} \big(\mathbb{P}(E_{m,h}) + \varepsilon_{m,h} \big) \right] < \infty,$$

and

$$\sum_m J_m \Delta_m (e\Delta_m)^{-\gamma_{\mathrm{tc}}(K_{\mathrm{tail}},1,1,B)} < \infty,$$

then, with probability one, for all sufficiently large m and all $h \leq J_m$,

$$\max_{i \leq n_m} \|Z_{m,h,i}\|_{\mathrm{op}} \leq 1, \qquad \max_{i \leq n_m} \|X_{m,h,i}\|_{\mathrm{op}} \leq 2,$$

and the same active-prefix core signs satisfy

$$\left\| \sum_{i=1}^{n_m} \varepsilon_{m,i}^{\star} X_{m,h,i} \right\|_{\mathrm{op}} \leq (K_{\mathrm{core}} + 1)\sqrt{n_m}.$$

Consequently the normalized contractions $\widehat{X}_{m,h,i} := X_{m,h,i}/2$ have common active-prefix discrepancy at most $(K_{\mathrm{core}} + 1)\sqrt{n_m}/2$, and every fixed menu element is eventually covered whenever $J_m \to \infty$.

The all-integer and sparse active-menu thresholds are the same as in Corollary 7.68: after adding $\sum_m \mathbb{P}(F_m^{\mathrm{core}}) < \infty$, *the cap-budget part is summable under* $c > \Lambda + 1$ *for all integer dimensions and under* $c > \Lambda$ *on stretched-exponential sparse schedules, with the same* Θ_{tc} *margins for* K_{tail}.

Proof. Work outside F_m^{core}. The signing $\varepsilon_m^\star$ is $\mathcal{H}_m$-measurable, so after conditioning on the row environment the signed perturbation streams

$$\widetilde{Y}_{m,h,i} := \varepsilon_{m,i}^\star Y_{m,h,i}$$

have the same operator-norm tail and moment bounds as $Y_{m,h,i}$. The centered independent and martingale-difference drift conditions are also unchanged after multiplication by these predictable signs, because all tail-drift estimates use only operator norms.

Apply Corollary 7.68 to the active library $(\widetilde{Y}_{m,h,i})$. The displayed summability assumptions, together with Borel–Cantelli, imply that eventually

$$\max_i \|Z_{m,h,i}\|_{\mathrm{op}} \le 1 \quad \text{and} \quad \left\|\sum_i \varepsilon_{m,i}^\star Z_{m,h,i}\right\|_{\mathrm{op}} \le \sqrt{n_m}$$

simultaneously for all $h \le J_m$. Combining this with the core signing gives

$$\begin{aligned}\left\|\sum_i \varepsilon_{m,i}^\star X_{m,h,i}\right\|_{\mathrm{op}} &\le \left\|\sum_i \varepsilon_{m,i}^\star B_{m,h,i}\right\|_{\mathrm{op}} + \left\|\sum_i \varepsilon_{m,i}^\star Z_{m,h,i}\right\|_{\mathrm{op}} \\ &\le (K_{\mathrm{core}} + 1)\sqrt{n_m},\end{aligned}$$

and the contraction bound follows from

$$\|X_{m,h,i}\|_{\mathrm{op}} \le \|B_{m,h,i}\|_{\mathrm{op}} + \|Z_{m,h,i}\|_{\mathrm{op}} \le 2.$$

The final threshold statement is inherited verbatim from the mixed dominated operator-tail corollary. □

Corollary 7.70 (Raw noncentered core-plus-tail libraries)**.** *Keep the hypotheses and notation of Corollary 7.69, but regard*

$$W_{m,h,i} := B_{m,h,i} + Y_{m,h,i}$$

as the raw active random matrices. Define

$$\lambda_{m,h} := \max\{1, K_{\mathrm{tail}} + \eta\}\, q_{m,h}, \qquad \Lambda_{m,h} := \max\{1, \lambda_{m,h}\}.$$

Then, under the same summability and matrix-tail hypotheses, with probability one, for all sufficiently large m, *one* $\mathcal{H}_m$*-measurable sign vector* $\varepsilon_m^\star$ *satisfies, simultaneously for every active* $h \le J_m$,

$$\max_{i \le n_m} \left\|\frac{W_{m,h,i}}{\Lambda_{m,h}}\right\|_{\mathrm{op}} \le 2$$

and

$$\left\|\sum_{i=1}^{n_m} \varepsilon_{m,i}^\star \frac{W_{m,h,i}}{\Lambda_{m,h}}\right\|_{\mathrm{op}} \le (K_{\mathrm{core}} + 1)\sqrt{n_m}.$$

Consequently the raw normalized contractions

$$\widehat{W}_{m,h,i} := \frac{W_{m,h,i}}{2\Lambda_{m,h}}$$

have common active-prefix discrepancy at most $(K_{\mathrm{core}}+1)\sqrt{n_m}/2$, *and every fixed menu element is eventually covered whenever* $J_m \to \infty$.

In particular, one may take

$$B_{m,h,i} = \mathbb{E}(W_{m,h,i} \mid \mathcal{H}_m), \qquad Y_{m,h,i} = W_{m,h,i} - \mathbb{E}(W_{m,h,i} \mid \mathcal{H}_m),$$

whenever the conditional-mean core tuple satisfies the common-signing hypothesis and the centered fluctuations satisfy the same mixed dominated tail, moment, and drift assumptions. Thus noncentered random matrix libraries with structured conditional means reduce to the same per-stream $\Lambda_{m,h}$ *normalizations and the same all-integer and sparse thresholds.*

Proof. The proof of Corollary 7.69, applied to the signed perturbation streams $\widetilde{Y}_{m,h,i} = \varepsilon^{\star}_{m,i} Y_{m,h,i}$, gives eventually, simultaneously for every active $h \le J_m$,

$$\max_i \left\| \frac{Y_{m,h,i}}{\lambda_{m,h}} \right\|_{\mathrm{op}} \le 1, \qquad \left\| \sum_i \varepsilon^{\star}_{m,i} \frac{Y_{m,h,i}}{\lambda_{m,h}} \right\|_{\mathrm{op}} \le \sqrt{n_m}.$$

Equivalently,

$$\max_i \|Y_{m,h,i}\|_{\mathrm{op}} \le \lambda_{m,h}, \qquad \left\| \sum_i \varepsilon^{\star}_{m,i} Y_{m,h,i} \right\|_{\mathrm{op}} \le \lambda_{m,h}\sqrt{n_m}.$$

Since $\Lambda_{m,h} \ge 1$ and $\Lambda_{m,h} \ge \lambda_{m,h}$, the signed-core hypothesis gives

$$\left\| \sum_i \varepsilon^{\star}_{m,i} \frac{W_{m,h,i}}{\Lambda_{m,h}} \right\|_{\mathrm{op}} \le \frac{K_{\mathrm{core}}\sqrt{n_m}}{\Lambda_{m,h}} + \frac{\lambda_{m,h}\sqrt{n_m}}{\Lambda_{m,h}} \le (K_{\mathrm{core}}+1)\sqrt{n_m}.$$

Similarly,

$$\left\| \frac{W_{m,h,i}}{\Lambda_{m,h}} \right\|_{\mathrm{op}} \le \frac{\|B_{m,h,i}\|_{\mathrm{op}}}{\Lambda_{m,h}} + \frac{\|Y_{m,h,i}\|_{\mathrm{op}}}{\Lambda_{m,h}} \le 2.$$

The conditional-mean specialization is only the decomposition $W = B + Y$ with B chosen to be $\mathcal{H}_m$-measurable; the hypotheses are exactly those of the signed-core corollary for that decomposition. □

Corollary 7.71 (Scaled raw noncentered core-plus-tail libraries)**.** *In the setting of Corollary 7.70, replace the unit core bounds by the following scale-aware bounds. There are positive* $\mathcal{H}_m$*-measurable numbers* $\rho_{m,h}$ *such that, outside* F_m^{core}, *the same active-prefix sign vector* $\varepsilon^{\star}_m$ *satisfies*

$$\|B_{m,h,i}\|_{\mathrm{op}} \le \rho_{m,h}, \qquad \left\| \sum_{i=1}^{n_m} \varepsilon^{\star}_{m,i} B_{m,h,i} \right\|_{\mathrm{op}} \le K_{\mathrm{core}}\rho_{m,h}\sqrt{n_m}$$

for every $h \le J_m$ *and* $i \le n_m$. *Keep the same mixed dominated operator-tail hypotheses for* $Y_{m,h,i}$, *and define*

$$\lambda_{m,h} := \max\{1, K_{\mathrm{tail}} + \eta\}\, q_{m,h}, \qquad \Lambda^{\mathrm{sc}}_{m,h} := \max\{\rho_{m,h}, \lambda_{m,h}\}.$$

Then, under the same summability and matrix-tail assumptions, with probability one, for all sufficiently large m, the same $\mathcal{H}_m$-measurable sign vector $\varepsilon_m^\star$ satisfies, simultaneously for every active $h \le J_m$,

$$\max_{i\le n_m}\left\|\frac{W_{m,h,i}}{\Lambda^{\mathrm{sc}}_{m,h}}\right\|_{\mathrm{op}} \le 2$$

and

$$\left\|\sum_{i=1}^{n_m}\varepsilon^\star_{m,i}\frac{W_{m,h,i}}{\Lambda^{\mathrm{sc}}_{m,h}}\right\|_{\mathrm{op}} \le (K_{\mathrm{core}}+1)\sqrt{n_m}.$$

Thus the raw matrices

$$\widehat{W}^{\mathrm{sc}}_{m,h,i} := \frac{W_{m,h,i}}{2\Lambda^{\mathrm{sc}}_{m,h}}$$

are contractions with common active-prefix discrepancy at most $(K_{\mathrm{core}}+1)\sqrt{n_m}/2$, and every fixed menu element is eventually covered whenever $J_m \to \infty$.

In particular, for noncentered ensembles one may take $B_{m,h,i} = \mathbb{E}(W_{m,h,i} \mid \mathcal{H}_m)$ and $Y_{m,h,i} = W_{m,h,i} - \mathbb{E}(W_{m,h,i} \mid \mathcal{H}_m)$ whenever the conditional-mean tuple has a common signing at scale $\rho_{m,h}$. The centered fluctuation scale $\lambda_{m,h}$ and the structured-mean scale $\rho_{m,h}$ then compete only through the single realized normalizer $\Lambda^{\mathrm{sc}}_{m,h}$.

Proof. The mixed dominated perturbation proof used in Corollary 7.70 gives eventually, simultaneously for every active $h \le J_m$,

$$\max_i \|Y_{m,h,i}\|_{\mathrm{op}} \le \lambda_{m,h}, \qquad \left\|\sum_i \varepsilon^\star_{m,i}Y_{m,h,i}\right\|_{\mathrm{op}} \le \lambda_{m,h}\sqrt{n_m}.$$

Since $\Lambda^{\mathrm{sc}}_{m,h} \ge \rho_{m,h}$ and $\Lambda^{\mathrm{sc}}_{m,h} \ge \lambda_{m,h}$, the scaled core hypothesis gives

$$\left\|\sum_i \varepsilon^\star_{m,i}\frac{W_{m,h,i}}{\Lambda^{\mathrm{sc}}_{m,h}}\right\|_{\mathrm{op}} \le \frac{K_{\mathrm{core}}\rho_{m,h}\sqrt{n_m}}{\Lambda^{\mathrm{sc}}_{m,h}} + \frac{\lambda_{m,h}\sqrt{n_m}}{\Lambda^{\mathrm{sc}}_{m,h}} \le (K_{\mathrm{core}}+1)\sqrt{n_m}.$$

The contraction estimate is the same pathwise comparison:

$$\left\|\frac{W_{m,h,i}}{\Lambda^{\mathrm{sc}}_{m,h}}\right\|_{\mathrm{op}} \le \frac{\rho_{m,h}}{\Lambda^{\mathrm{sc}}_{m,h}} + \frac{\lambda_{m,h}}{\Lambda^{\mathrm{sc}}_{m,h}} \le 2.$$

No summability threshold changes, because the tail event for Y is exactly the event used in the raw noncentered corollary and the core event is still summable. □

Corollary 7.72 (Two-scale raw noncentered core-plus-tail libraries)**.** *In the setting of Corollary 7.70, suppose instead that the active cores have separate row and signing scales. Namely, outside F_m^{core}, the same active-prefix sign vector $\varepsilon_m^\star$ satisfies*

$$\|B_{m,h,i}\|_{\mathrm{op}} \le \rho_{m,h}, \qquad \left\|\sum_{i=1}^{n_m}\varepsilon^\star_{m,i}B_{m,h,i}\right\|_{\mathrm{op}} \le K_{\mathrm{core}}\sigma_{m,h}\sqrt{n_m}$$

for every $h \le J_m$ and $i \le n_m$, where $\rho_{m,h} > 0$ and $\sigma_{m,h} > 0$ are $\mathcal{H}_m$-measurable. Keep the same mixed dominated operator-tail hypotheses for $Y_{m,h,i}$, and define

$$\lambda_{m,h} := \max\{1, K_{\mathrm{tail}} + \eta\}\, q_{m,h}, \qquad \Lambda^{\mathrm{rs}}_{m,h} := \max\{\rho_{m,h}, \sigma_{m,h}, \lambda_{m,h}\}.$$

Then, under the same summability and matrix-tail assumptions, with probability one, for all sufficiently large m, the same signs satisfy, simultaneously for every active $h \le J_m$,

$$\max_{i \le n_m} \left\| \frac{W_{m,h,i}}{\Lambda^{\mathrm{rs}}_{m,h}} \right\|_{\mathrm{op}} \le 2$$

and

$$\left\| \sum_{i=1}^{n_m} \varepsilon^{\star}_{m,i} \frac{W_{m,h,i}}{\Lambda^{\mathrm{rs}}_{m,h}} \right\|_{\mathrm{op}} \le (K_{\mathrm{core}} + 1)\sqrt{n_m}.$$

Consequently

$$\widehat{W}^{\mathrm{rs}}_{m,h,i} := \frac{W_{m,h,i}}{2\Lambda^{\mathrm{rs}}_{m,h}}$$

is a contraction tuple with common active-prefix discrepancy at most $(K_{\mathrm{core}} + 1)\sqrt{n_m}/2$, and every fixed menu element is eventually covered whenever $J_m \to \infty$.

Thus the row-size estimate for a structured conditional mean and the scale of its available common signing need not be the same. Finite-lift, algebraic, Toeplitz/Hankel, group-convolution, or selected structured cores may be used with whichever realized row bound $\rho_{m,h}$ and signed-core certificate $\sigma_{m,h}$ their construction supplies; the centered mixed dominated fluctuation competes with both only through the maximum normalizer $\Lambda^{\mathrm{rs}}_{m,h}$.

Proof. As in Corollary 7.70, the mixed dominated perturbation event gives eventually, simultaneously for every active $h \le J_m$,

$$\max_i \|Y_{m,h,i}\|_{\mathrm{op}} \le \lambda_{m,h}, \qquad \left\| \sum_i \varepsilon^{\star}_{m,i} Y_{m,h,i} \right\|_{\mathrm{op}} \le \lambda_{m,h}\sqrt{n_m}.$$

Because $\Lambda^{\mathrm{rs}}_{m,h}$ dominates $\rho_{m,h}$, $\sigma_{m,h}$, and $\lambda_{m,h}$,

$$\left\| \sum_i \varepsilon^{\star}_{m,i} \frac{W_{m,h,i}}{\Lambda^{\mathrm{rs}}_{m,h}} \right\|_{\mathrm{op}} \le \frac{K_{\mathrm{core}}\sigma_{m,h}\sqrt{n_m}}{\Lambda^{\mathrm{rs}}_{m,h}} + \frac{\lambda_{m,h}\sqrt{n_m}}{\Lambda^{\mathrm{rs}}_{m,h}} \le (K_{\mathrm{core}} + 1)\sqrt{n_m},$$

while

$$\left\| \frac{W_{m,h,i}}{\Lambda^{\mathrm{rs}}_{m,h}} \right\|_{\mathrm{op}} \le \frac{\rho_{m,h}}{\Lambda^{\mathrm{rs}}_{m,h}} + \frac{\lambda_{m,h}}{\Lambda^{\mathrm{rs}}_{m,h}} \le 2.$$

The probabilistic input is unchanged from the raw noncentered theorem, so the same Borel–Cantelli and active-menu threshold conclusions apply. □

Corollary 7.73 (Two-scale raw structured cores plus mixed dominated tails)**.** *For each $m \geq 1$, let $D_m, n_m, J_m \geq 1$, put $\Delta_m := D_m + n_m + 1$ and $\ell_m := \log(e\Delta_m)$, and let $\mathcal{H}_m$ be a row environment. Let*

$$W_{m,h,i} := B_{m,h,i} + Y_{m,h,i}$$

be active raw self-adjoint matrices in $\mathrm{M}_{D_m}(\mathbb{C})$. Assume that there are events $F_m^{\mathrm{str}} \in \mathcal{H}_m$, with $\sum_m \mathbb{P}(F_m^{\mathrm{str}}) < \infty$, and positive $\mathcal{H}_m$-measurable scales $\rho_{m,h}$, $\sigma_{m,h}$ such that, outside F_m^{str},

$$\|B_{m,h,i}\|_{\mathrm{op}} \leq \rho_{m,h} \qquad (h \leq J_m,\ i \leq n_m).$$

Suppose moreover that one of the following active-prefix structural certificates holds for

$$C_{m,h,i} := \frac{B_{m,h,i}}{\sigma_{m,h}}.$$

(a) *The tuples $(C_{m,h,i})_{i \leq n_m}$ are contractions and admit finite block lifts whose active-prefix parameters in the sense of Corollary 3.2 obey*

$$M_m^{(J)} \leq M_0, \qquad E_m^{(J)} \leq C_E n_m.$$

Set $K_{\mathrm{str}} := CM_0\sqrt{\log(e + C_E)}$.

(b) *Each $(C_{m,h,i})_{i \leq n_m}$ is a contraction tuple in a finite-dimensional C^*-algebra $\mathcal{A}_{m,h} \subset \mathrm{M}_{D_m}(\mathbb{C})$, and*

$$\sum_{h=1}^{J_m} \dim_{\mathbb{C}} \mathcal{A}_{m,h} \leq C_0 n_m.$$

Set $K_{\mathrm{str}} := C(C_0)$, the Akbas–Sra constant from Theorem 4.2.

For each active $h \leq J_m$, tag the centered fluctuation stream either ψ or μ, choose the parameters $\varepsilon_{m,h}, A_{m,h}, s_{m,h}$, and define $r_{m,h}$, $V_{m,h}$, and $q_{m,h}$ as in Corollary 7.68, with $Y_{m,h,i}$ in place of $X_{m,h,i}$. Assume that the corresponding mixed dominated tail or moment hypotheses, and the matching truncated-tail drift hypotheses outside events $E_{m,h}$, hold after adjoining $\mathcal{H}_m$ to the predictable sigma-fields. Let $K_{\mathrm{tail}}, \eta > 0$, and put

$$\lambda_{m,h} := \max\{1, K_{\mathrm{tail}} + \eta\}\, q_{m,h}, \qquad \Lambda_{m,h}^{\mathrm{str}} := \max\{\rho_{m,h}, \sigma_{m,h}, \lambda_{m,h}\}.$$

If $\ell_m \leq B\sqrt{n_m}$ eventually,

$$\sum_m \left[\mathbb{P}(F_m^{\mathrm{str}}) + \sum_{h=1}^{J_m} \left(\mathbb{P}(E_{m,h}) + \varepsilon_{m,h}\right)\right] < \infty,$$

and

$$\sum_m J_m \Delta_m (e\Delta_m)^{-\gamma_{\mathrm{tc}}(K_{\mathrm{tail}},1,1,B)} < \infty,$$

then, with probability one, for all sufficiently large m, one $\mathcal{H}_m$-measurable sign vector $\varepsilon_m^\star$ satisfies, simultaneously for every active $h \leq J_m$,

$$\max_{i \leq n_m} \left\| \frac{W_{m,h,i}}{\Lambda_{m,h}^{\mathrm{str}}} \right\|_{\mathrm{op}} \leq 2$$

and

$$\left\|\sum_{i=1}^{n_m}\varepsilon^{\star}_{m,i}\frac{W_{m,h,i}}{\Lambda^{\mathrm{str}}_{m,h}}\right\|_{\mathrm{op}}\le (K_{\mathrm{str}}+1)\sqrt{n_m}.$$

Thus the raw matrices

$$\widehat{W}^{\mathrm{str}}_{m,h,i}:=\frac{W_{m,h,i}}{2\Lambda^{\mathrm{str}}_{m,h}}$$

are contractions with common active-prefix discrepancy at most $(K_{\mathrm{str}}+1)\sqrt{n_m}/2$*, and every fixed menu element is eventually covered whenever* $J_m\to\infty$*.*

The all-integer and sparse active-menu thresholds are unchanged from Corollary 7.68: if $J_m\Delta_m\le(e\Delta_m)^{\Gamma}$*,* $J_m\le(e\Delta_m)^{\Lambda}$*,* $n_m\ge a\ell_m^2$ *eventually, and* $\varepsilon_{m,h}=(e\Delta_m)^{-c}$*, then the cap-budget part is summable under* $c>\Lambda+1$ *for all integer dimensions and under* $c>\Lambda$ *on stretched-exponential sparse schedules, with the same* Θ_{tc} *margins for* K_{tail}*.*

Proof. Work outside F^{str}_m. In case (a), Corollary 3.2 gives signs satisfying

$$\left\|\sum_i\varepsilon^{\star}_{m,i}C_{m,h,i}\right\|_{\mathrm{op}}\le K_{\mathrm{str}}\sqrt{n_m}\qquad(h\le J_m).$$

In case (b), apply Theorem 4.2 to the direct-sum algebra $\bigoplus_{h\le J_m}\mathcal{A}_{m,h}$ and the direct-sum tuple $\bigoplus_{h\le J_m}C_{m,h,i}$ to get the same estimate. Choose the lexicographically first successful sign vector so that it is $\mathcal{H}_m$-measurable. Multiplying by $\sigma_{m,h}$ gives the two-scale core signing estimate

$$\left\|\sum_i\varepsilon^{\star}_{m,i}B_{m,h,i}\right\|_{\mathrm{op}}\le K_{\mathrm{str}}\sigma_{m,h}\sqrt{n_m}.$$

Together with the row bound $\|B_{m,h,i}\|_{\mathrm{op}}\le\rho_{m,h}$, this verifies the core hypotheses of Corollary 7.72 with $K_{\mathrm{core}}=K_{\mathrm{str}}$ and $F^{\mathrm{core}}_m=F^{\mathrm{str}}_m$. The mixed dominated assumptions and the displayed summability hypotheses are exactly the perturbation and Borel–Cantelli hypotheses required there, so the raw normalization, fixed-menu coverage, and all-integer/sparse threshold conclusions follow. □

Proposition 7.74 (Sign-symmetric random-tuple transfer)**.** *Let* $A=(A_1,\dots,A_n)$ *be a random self-adjoint tuple in* $\mathrm{M}_D(\mathbb{C})$*. Assume that its law is invariant under coordinatewise sign changes: for every* $\delta\in\{\pm1\}^n$*,*

$$(\delta_1A_1,\dots,\delta_nA_n)\quad\textit{has the same law as}\quad(A_1,\dots,A_n).$$

If H *is any event invariant under the same coordinatewise sign changes, then, for every* $t\ge0$*,*

$$\mathbb{P}\left(H\cap\{\mathrm{disc}(A_1,\dots,A_n)>t\}\right)\le\mathbb{P}\left(H\cap\left\{\left\|\sum_{i=1}^{n}A_i\right\|_{\mathrm{op}}>t\right\}\right).$$

In particular, if $H \subset \{\max_i \|A_i\|_{\mathrm{op}} \le 1\}$*, then any high-probability all-plus estimate*

$$\mathbb{P}\left(H^c \cup \left\{\left\|\sum_{i=1}^n A_i\right\|_{\mathrm{op}} > K\sqrt{n}\right\}\right) \le p$$

implies

$$\mathbb{P}\left\{\max_i \|A_i\|_{\mathrm{op}} \le 1 \text{ and } \operatorname{disc}(A_1,\dots,A_n) \le K\sqrt{n}\right\} \ge 1-p.$$

Thus, for coordinatewise sign-symmetric random matrix models, the Matrix Spencer problem is reduced to the all-plus operator-norm tail for the same realization-normalized ensemble.

Proof. Let $\eta = (\eta_1,\dots,\eta_n)$ be uniform on $\{\pm 1\}^n$, independent of A. On the event $H \cap \{\operatorname{disc}(A) > t\}$, every signing has norm larger than t. Hence

$$\mathbb{P}\left(H \cap \{\operatorname{disc}(A) > t\}\right) \le \mathbb{P}\left(H \cap \left\{\left\|\sum_i \eta_i A_i\right\|_{\mathrm{op}} > t\right\}\right).$$

By the sign-invariance of the law of A, and by the sign-invariance of H, the last probability is exactly

$$\mathbb{P}\left(H \cap \left\{\left\|\sum_i A_i\right\|_{\mathrm{op}} > t\right\}\right).$$

The final statement applies this inequality with $t = K\sqrt{n}$ and with H chosen to enforce the contraction normalization. □

Corollary 7.75 (Borel–Cantelli sign-symmetric all-plus menus)**.** *For each* $m \ge 1$*, let* $J_m \ge 1$ *be a deterministic active-menu size. For every* $1 \le h \le J_m$*, let*

$$A_{m,h} = (A_{m,h,1},\dots,A_{m,h,n_m})$$

be a random self-adjoint tuple in $\mathrm{M}_{D_m}(\mathbb{C})$ *whose law is invariant under coordinatewise sign changes, and let* $H_{m,h}$ *be a sign-invariant event such that*

$$H_{m,h} \subset \left\{\max_{1\le i\le n_m} \|A_{m,h,i}\|_{\mathrm{op}} \le 1\right\}.$$

Let $K_m \ge 0$ *be deterministic. If*

$$\sum_m \sum_{h=1}^{J_m} \mathbb{P}\left(H_{m,h}^c \cup \left\{\left\|\sum_{i=1}^{n_m} A_{m,h,i}\right\|_{\mathrm{op}} > K_m\sqrt{n_m}\right\}\right) < \infty,$$

then, with probability one, for all sufficiently large m*, every active menu element is a normalized Spencer-scale instance:*

$$\max_i \|A_{m,h,i}\|_{\mathrm{op}} \le 1, \qquad \operatorname{disc}(A_{m,h,1},\dots,A_{m,h,n_m}) \le K_m\sqrt{n_m} \qquad (1 \le h \le J_m).$$

Equivalently, let $X_{m,h,i}$ be coordinatewise sign-symmetric random self-adjoint matrices, let $H_{m,h}$ be sign-invariant events, and let $s_{m,h} > 0$ be deterministic scales. If

$$H_{m,h} \subset \left\{ \max_i \|X_{m,h,i}\|_{\mathrm{op}} \le s_{m,h} \right\}$$

and

$$\sum_m \sum_{h=1}^{J_m} \mathbb{P}\left(H_{m,h}^c \cup \left\{ \left\| \sum_{i=1}^{n_m} X_{m,h,i} \right\|_{\mathrm{op}} > K_m s_{m,h} \sqrt{n_m} \right\} \right) < \infty,$$

then, almost surely for all sufficiently large m and all $1 \le h \le J_m$,

$$\mathrm{disc}(X_{m,h,1}, \dots, X_{m,h,n_m}) \le K_m s_{m,h} \sqrt{n_m}.$$

If the streams are defined for all $h \ge 1$, $J_m \to \infty$, and the active menu is a deterministic exhaustion of a countable library, then every fixed sign-symmetric model in the library is covered for all sufficiently large m.

Proof. By Proposition 7.74, for each m, h the probability that $H_{m,h}$ holds but the discrepancy exceeds $K_m \sqrt{n_m}$ is at most the displayed all-plus tail. Hence the total probability of active-menu failures is summable, and the first Borel–Cantelli lemma gives the simultaneous eventual conclusion. The scaled version applies the first statement to $A_{m,h,i} = s_{m,h}^{-1} X_{m,h,i}$. If $J_m \to \infty$, each fixed library element belongs to the active menu eventually. □

Corollary 7.76 (Matrix-Bernstein sign-symmetric all-plus menus)**.** *For each $m \ge 1$, let $J_m \ge 1$ be a deterministic active-menu size. For every $1 \le h \le J_m$, let*

$$X_{m,h,1}, \dots, X_{m,h,n_m} \in \mathrm{M}_{D_m}(\mathbb{C})$$

be independent self-adjoint random matrices. Assume that each $X_{m,h,i}$ has symmetric law,

$$X_{m,h,i} \overset{d}{=} -X_{m,h,i},$$

and that, for a deterministic scale $s_{m,h} > 0$,

$$\|X_{m,h,i}\|_{\mathrm{op}} \le s_{m,h} \qquad \text{almost surely for all } i.$$

Put

$$\sigma_{m,h}^2 := \left\| \sum_{i=1}^{n_m} \mathbb{E} X_{m,h,i}^2 \right\|_{\mathrm{op}}.$$

For deterministic thresholds $u_{m,h} > 0$, suppose that

$$\sum_m \sum_{h=1}^{J_m} 2D_m \exp\left\{ -\frac{u_{m,h}^2}{2(\sigma_{m,h}^2 + s_{m,h} u_{m,h}/3)} \right\} < \infty.$$

Then, with probability one, for all sufficiently large m and all active $1 \le h \le J_m$,

$$\mathrm{disc}(X_{m,h,1}, \dots, X_{m,h,n_m}) \le u_{m,h}.$$

In particular, if $u_{m,h} = K_m s_{m,h}\sqrt{n_m}$, then the normalized tuples $s_{m,h}^{-1} X_{m,h,i}$ are contractions and have discrepancy at most $K_m\sqrt{n_m}$ simultaneously on the active menu, eventually almost surely. If the streams are defined for all $h \geq 1$, $J_m \to \infty$, and the active menu exhausts a countable library, then every fixed independent bounded sign-symmetric model in that library is eventually covered.

Proof. Fix m and h. Independence together with the individual symmetry $X_{m,h,i} \overset{d}{=} -X_{m,h,i}$ makes the whole tuple coordinatewise sign-symmetric, and also gives $\mathbb{E}X_{m,h,i} = 0$. Applying Proposition 7.74 with $H = \Omega$ gives

$$\mathbb{P}\left\{\operatorname{disc}(X_{m,h,1}, \dots, X_{m,h,n_m}) > u_{m,h}\right\} \leq \mathbb{P}\left\{\left\|\sum_i X_{m,h,i}\right\|_{\mathrm{op}} > u_{m,h}\right\}.$$

The two-sided matrix Bernstein inequality [25], with jump bound $s_{m,h}$ and variance proxy $\sigma_{m,h}^2$, bounds the last probability by the displayed summand. The summability assumption and the first Borel–Cantelli lemma give the simultaneous active-menu conclusion. The normalized and countable-exhaustion statements are immediate from the deterministic scale bound and from $J_m \to \infty$. □

Corollary 7.77 (Explicit sign-symmetric Bernstein schedules)**.** *In the setting of Corollary 7.76, put $\ell_m = \log(eD_m)$. Fix constants $\beta \geq 0$ and $K > 0$. Suppose that outside events E_m,*

$$\sigma_{m,h}^2 \leq \beta\, s_{m,h}^2\, \frac{n_m}{\ell_m} \qquad (1 \leq h \leq J_m),$$

and that $\sum_m \mathbb{P}(E_m) < \infty$. Then the following deterministic schedule conditions imply that, almost surely, for all sufficiently large m and every active $1 \leq h \leq J_m$,

$$\operatorname{disc}(X_{m,h,1}, \dots, X_{m,h,n_m}) \leq K\, s_{m,h}\sqrt{n_m}.$$

(a) *General dimension schedules. If there is $B > 0$ such that*

$$\ell_m \leq B\sqrt{n_m}$$

eventually, and

$$\sum_m J_m D_m (eD_m)^{-\gamma(K,\beta,B)} < \infty, \qquad \gamma(K,\beta,B) := \frac{K^2}{2(\beta + KB/3)},$$

then the conclusion holds.

(b) *All integer dimensions. Suppose the rows are indexed by $D = 2, 3, \dots$, with ambient dimension D, and*

$$J_D D \leq (eD)^{\Gamma}$$

eventually. If

$$n_D \geq a \log^2(eD)$$

eventually and

$$\frac{K^2}{2(\beta + K/(3\sqrt{a}))} > \Gamma + 1,$$

then the conclusion holds along all integer dimensions.

(c) *Sparse stretched-exponential dimensions. If*

$$D_m \geq \exp(cm^\kappa)$$

eventually for some $c, \kappa > 0$, if

$$J_m D_m \leq (eD_m)^\Gamma$$

eventually, if

$$n_m \geq a \, \log^2(eD_m)$$

eventually, and if

$$\frac{K^2}{2(\beta + K/(3\sqrt{a}))} > \Gamma,$$

then the conclusion holds along the sparse schedule.

In the two active-prefix cases, if $J_m \to \infty$, the same schedules cover every fixed member of the countable independent bounded sign-symmetric library eventually.

Proof. Apply Corollary 7.76 with $u_{m,h} = K s_{m,h} \sqrt{n_m}$. On E_m^c, the h-th active summand in that corollary is bounded by

$$2D_m \exp\left\{ -\frac{K^2 n_m}{2(\beta n_m/\ell_m + K\sqrt{n_m}/3)} \right\}.$$

If $\ell_m \leq B\sqrt{n_m}$, then

$$\frac{K^2 n_m}{2(\beta n_m/\ell_m + K\sqrt{n_m}/3)} \geq \gamma(K, \beta, B)\ell_m.$$

Thus the total active-row tail is at most

$$2J_m D_m (eD_m)^{-\gamma(K,\beta,B)}.$$

The first schedule is therefore summable, and Borel–Cantelli gives the eventual conclusion together with the summable variance-budget exceptions.

For all integer dimensions, $n_D \geq a\ell_D^2$ gives $B = 1/\sqrt{a}$, so the active-row tail is bounded by

$$2(eD)^{\Gamma - \gamma(K,\beta,1/\sqrt{a})}.$$

This is summable over $D = 2, 3, \ldots$ when $\gamma(K, \beta, 1/\sqrt{a}) > \Gamma + 1$. Along a sparse stretched-exponential schedule the same bound is summable as soon as the exponent is strictly negative, which is exactly the displayed sparse condition. If $J_m \to \infty$, each fixed countable-library element is in the active prefix eventually. □

Corollary 7.78 (Closed-form sign-symmetric Bernstein margins)**.** *In the setting of Corollary 7.77, for $A \geq 0$ define*

$$\Theta(A, a, \beta) := \frac{A}{3\sqrt{a}} + \sqrt{\frac{A^2}{9a} + 2A\beta}.$$

Assume the same variance-budget hypothesis

$$\sigma_{m,h}^2 \leq \beta\, s_{m,h}^2 \frac{n_m}{\log(eD_m)}$$

outside summable exceptions.

If the rows are indexed by all integer dimensions $D = 2, 3, \ldots$, if

$$J_D D \leq (eD)^{\Gamma}, \qquad n_D \geq a \log^2(eD)$$

eventually, and if

$$K > \Theta(\Gamma + 1, a, \beta),$$

then almost surely, for all sufficiently large D and all active $1 \leq h \leq J_D$,

$$\operatorname{disc}(X_{D,h,1}, \ldots, X_{D,h,n_D}) \leq K\, s_{D,h} \sqrt{n_D}.$$

Along any sparse stretched-exponential dimension schedule $D_m \geq \exp(cm^{\kappa})$, the same conclusion holds under

$$J_m D_m \leq (eD_m)^{\Gamma}, \qquad n_m \geq a \log^2(eD_m), \qquad K > \Theta(\Gamma, a, \beta).$$

If $J_m \to \infty$, both statements exhaust countable independent bounded sign-symmetric libraries.

Equivalently, for a prescribed K and target exponent $A > 0$, the condition

$$\frac{K^2}{2(\beta + K/(3\sqrt{a}))} > A$$

is guaranteed whenever $K^2 > 2A\beta$ and

$$a > \left(\frac{2AK}{3(K^2 - 2A\beta)} \right)^2.$$

Proof. The inequality

$$\frac{K^2}{2(\beta + K/(3\sqrt{a}))} > A$$

is equivalent, for $A > 0$, to

$$K^2 - \frac{2A}{3\sqrt{a}} K - 2A\beta > 0.$$

The positive root of the quadratic on the left is exactly $\Theta(A, a, \beta)$. Hence $K > \Theta(\Gamma + 1, a, \beta)$ implies the all-integer hypothesis in Corollary 7.77, and $K > \Theta(\Gamma, a, \beta)$ implies the sparse hypothesis. The case $A = 0$ is immediate for every $K > 0$. Solving the same inequality for a gives the final equivalent margin condition. □

Corollary 7.79 (Conditional sign-symmetric Bernstein menus)**.** *For each $m \geq 1$, let $J_m \geq 1$ be a deterministic active-menu size. For every $1 \leq h \leq J_m$, let*

$$X_{m,h,1}, \dots, X_{m,h,n_m} \in \mathrm{M}_{D_m}(\mathbb{C})$$

be self-adjoint random matrices and let $\mathcal{G}_{m,h}$ be a conditioning sigma-field. Assume that, conditionally on $\mathcal{G}_{m,h}$, the matrices are independent and each $X_{m,h,i}$ has symmetric conditional law. Let $E_{m,h} \in \mathcal{G}_{m,h}$, and let $s_{m,h} > 0$, $v_{m,h} \geq 0$, and $u_{m,h} > 0$ be deterministic. Suppose that on $E_{m,h}^c$,

$$\mathbb{P}\left\{ \max_i \|X_{m,h,i}\|_{\mathrm{op}} \leq s_{m,h} \,\middle|\, \mathcal{G}_{m,h} \right\} = 1$$

and

$$\left\| \sum_{i=1}^{n_m} \mathbb{E}\left(X_{m,h,i}^2 \mid \mathcal{G}_{m,h} \right) \right\|_{\mathrm{op}} \leq v_{m,h}.$$

If

$$\sum_m \sum_{h=1}^{J_m} \left[\mathbb{P}(E_{m,h}) + 2D_m \exp\left\{ -\frac{u_{m,h}^2}{2(v_{m,h} + s_{m,h}u_{m,h}/3)} \right\} \right] < \infty,$$

then, with probability one, for all sufficiently large m and all active $1 \leq h \leq J_m$,

$$\max_i \|X_{m,h,i}\|_{\mathrm{op}} \leq s_{m,h}, \qquad \mathrm{disc}(X_{m,h,1}, \dots, X_{m,h,n_m}) \leq u_{m,h}.$$

Consequently, put $\ell_m = \log(eD_m)$ and take $u_{m,h} = K s_{m,h}\sqrt{n_m}$. If, outside the same summable envelope failures,

$$v_{m,h} \leq \beta\, s_{m,h}^2\, \frac{n_m}{\ell_m},$$

then the three schedule conclusions of Corollary 7.77 hold with $\sigma_{m,h}^2$ replaced by $v_{m,h}$. In particular, under $J_D D \leq (eD)^\Gamma$ and $n_D \geq a \log^2(eD)$, all integer dimensions are covered for every

$$K > \Theta(\Gamma + 1, a, \beta),$$

while sparse stretched-exponential dimension schedules with $J_m D_m \leq (eD_m)^\Gamma$ are covered for every

$$K > \Theta(\Gamma, a, \beta).$$

If $J_m \to \infty$, these conditional schedules exhaust countable conditionally independent bounded sign-symmetric libraries.

In particular, the hypotheses apply to random-direction scalar-multiplier models $X_{m,h,i} = \xi_{m,h,i} B_{m,h,i}$, where the directions $B_{m,h,i}$ are $\mathcal{G}_{m,h}$-measurable self-adjoint matrices and the real multipliers $\xi_{m,h,i}$ are conditionally independent, conditionally symmetric, and bounded by 1 in absolute value, with

$$v_{m,h} = \left\| \sum_i \mathbb{E}(\xi_{m,h,i}^2 \mid \mathcal{G}_{m,h}) B_{m,h,i}^2 \right\|_{\mathrm{op}}.$$

Proof. Fix m, h and condition on $\mathcal{G}_{m,h}$. On $E_{m,h}^c$, the conditional law is coordinatewise sign-symmetric and is supported on the operator-norm ball of radius $s_{m,h}$. Applying the proof of Proposition 7.74 to the regular conditional law gives

$$\mathbb{P}\left\{\operatorname{disc}(X_{m,h,1},\dots,X_{m,h,n_m}) > u_{m,h} \,|\, \mathcal{G}_{m,h}\right\} \le \mathbb{P}\left\{\left\|\sum_i X_{m,h,i}\right\|_{\mathrm{op}} > u_{m,h} \,\middle|\, \mathcal{G}_{m,h}\right\}$$

on $E_{m,h}^c$. Conditional matrix Bernstein, with jump bound $s_{m,h}$ and conditional variance proxy bounded by $v_{m,h}$, bounds the right-hand side by the displayed exponential term. Integrating and adding the bad event $E_{m,h}$ gives a summable active-menu failure series, so the first Borel–Cantelli lemma gives the simultaneous eventual conclusion. Substituting $u_{m,h} = K s_{m,h}\sqrt{n_m}$ and the variance envelope gives exactly the estimates used in Corollary 7.77; the closed-form thresholds are Corollary 7.78. The scalar-multiplier specialization follows by conditioning on the directions. □

Corollary 7.80 (Symmetric scalar-multiplier Bernstein schedules)**.** *For each $m \ge 1$, let $D_m \ge 2$, let $J_m \ge 1$ be a deterministic active-menu size, and put $\ell_m = \log(eD_m)$. For every $1 \le h \le J_m$, let $\mathcal{G}_{m,h}$ be a conditioning sigma-field, let*

$$B_{m,h,1},\dots,B_{m,h,n_m} \in \mathrm{M}_{D_m}(\mathbb{C})$$

be $\mathcal{G}_{m,h}$-measurable self-adjoint contractions, and, conditionally on $\mathcal{G}_{m,h}$, let $\xi_{m,h,1},\dots,\xi_{m,h,n_m}$ be independent real random variables with symmetric conditional laws and $|\xi_{m,h,i}| \le 1$. Put

$$X_{m,h,i} := \xi_{m,h,i} B_{m,h,i}, \qquad q_{m,h,i} := \mathbb{E}(\xi_{m,h,i}^2 \mid \mathcal{G}_{m,h}),$$

and

$$\nu_{m,h} := \left\|\sum_{i=1}^{n_m} q_{m,h,i} B_{m,h,i}^2\right\|_{\mathrm{op}}.$$

Fix $\beta \ge 0$ and $K > 0$. Suppose that outside events $E_{m,h}$,

$$\nu_{m,h} \le \beta \frac{n_m}{\ell_m} \qquad (1 \le h \le J_m),$$

and that $\sum_m \sum_{h \le J_m} \mathbb{P}(E_{m,h}) < \infty$. Then the following deterministic schedule conditions imply that, almost surely, for all sufficiently large m and every active $1 \le h \le J_m$,

$$\operatorname{disc}(X_{m,h,1},\dots,X_{m,h,n_m}) \le K\sqrt{n_m}.$$

(a) *General dimension schedules. If there is $B > 0$ such that*

$$\ell_m \le B\sqrt{n_m}$$

eventually, and

$$\sum_m J_m D_m (eD_m)^{-\gamma(K,\beta,B)} < \infty, \qquad \gamma(K,\beta,B) := \frac{K^2}{2(\beta + KB/3)},$$

then the conclusion holds.

(b) *All integer dimensions. Suppose the rows are indexed by* $D = 2, 3, \ldots$, *with ambient dimension* D, *and*

$$J_D D \le (eD)^{\Gamma}$$

eventually. If

$$n_D \ge a \log^2(eD)$$

eventually and

$$K > \Theta(\Gamma + 1, a, \beta),$$

then the conclusion holds along all integer dimensions.

(c) *Sparse stretched-exponential dimensions. If*

$$D_m \ge \exp(cm^{\kappa})$$

eventually for some $c, \kappa > 0$, *if*

$$J_m D_m \le (eD_m)^{\Gamma}$$

eventually, if

$$n_m \ge a \log^2(eD_m)$$

eventually, and if

$$K > \Theta(\Gamma, a, \beta),$$

then the conclusion holds along the sparse schedule.

The variance-envelope hypothesis is automatic, with no $E_{m,h}$, *whenever*

$$q_{m,h,i} \le p_m \le \frac{\beta}{\ell_m} \qquad (1 \le h \le J_m,\ 1 \le i \le n_m),$$

because $0 \preceq B_{m,h,i}^2 \preceq I$. *If* $J_m \to \infty$, *the same schedules exhaust countable libraries of symmetric scalar-multiplier streams.*

Proof. The conditional hypotheses imply that the $X_{m,h,i}$'s are conditionally independent, conditionally sign-symmetric, and satisfy $\|X_{m,h,i}\|_{\mathrm{op}} \le 1$. Their conditional variance proxy is exactly $\nu_{m,h}$. Corollary 7.79 with $s_{m,h} = 1$, $v_{m,h} = \beta n_m/\ell_m$, and $u_{m,h} = K\sqrt{n_m}$ gives the general summability criterion. The three displayed schedules and the closed-form margins are precisely Corollaries 7.77 and 7.78 in this scalar-multiplier normalization.

If $q_{m,h,i} \le p_m$, then

$$\nu_{m,h} \le p_m \left\| \sum_i B_{m,h,i}^2 \right\|_{\mathrm{op}} \le p_m n_m \le \beta \frac{n_m}{\ell_m}.$$

The countable-library assertion follows from the active-prefix exhaustion. □

Corollary 7.81 (Sub-Gaussian symmetric scalar-multiplier normalizations)**.** *For each* $m \ge 1$, *let* $D_m \ge 2$, *let* $J_m \ge 1$ *be a deterministic active-menu size, and put* $\ell_m = \log(eD_m)$. *For every* $1 \le h \le J_m$, *let* $\mathcal{G}_{m,h}$ *be a conditioning sigma-field and let*

$$B_{m,h,1}, \ldots, B_{m,h,n_m} \in \mathrm{M}_{D_m}(\mathbb{C})$$

be $\mathcal{G}_{m,h}$-measurable self-adjoint contractions. Conditionally on $\mathcal{G}_{m,h}$, let $\xi_{m,h,1},\dots,\xi_{m,h,n_m}$ be independent real random variables with symmetric conditional laws and deterministic sub-Gaussian proxies $a_{m,h,i}\ge 0$:

$$\mathbb{E}\{\exp(\lambda\xi_{m,h,i})\mid \mathcal{G}_{m,h}\}\le \exp(\lambda^2 a_{m,h,i}^2/2)\qquad(\lambda\in\mathbb{R}).$$

Let $L_{m,h}>0$ be deterministic normalization caps and set

$$A_{m,h,i}:=L_{m,h}^{-1}\xi_{m,h,i}B_{m,h,i},\qquad \nu_{m,h}:=L_{m,h}^{-2}\left\|\sum_{i=1}^{n_m}a_{m,h,i}^2B_{m,h,i}^2\right\|_{\mathrm{op}}.$$

Fix $\beta>0$ and $K>0$. Suppose that outside events $E_{m,h}\in\mathcal{G}_{m,h}$,

$$\nu_{m,h}\le \beta\,\frac{n_m}{\ell_m}\qquad(1\le h\le J_m),$$

and that the variance-budget and cap-failure series

$$\sum_m\sum_{h=1}^{J_m}\left[\mathbb{P}(E_{m,h})+2\sum_{i=1}^{n_m}\exp\left\{-\frac{L_{m,h}^2}{2a_{m,h,i}^2}\right\}\right]<\infty$$

is finite, with the exponential term interpreted as 0 when $a_{m,h,i}=0$. Put

$$\gamma_{\mathrm{sg}}(K,\beta):=\frac{K^2}{2\beta}.$$

Then each of the following deterministic schedule conditions implies that, almost surely, for all sufficiently large m and every active $1\le h\le J_m$,

$$\max_i\|A_{m,h,i}\|_{\mathrm{op}}\le 1,\qquad \mathrm{disc}(A_{m,h,1},\dots,A_{m,h,n_m})\le K\sqrt{n_m}.$$

(a) *General dimension schedules. If*

$$\sum_m J_mD_m(eD_m)^{-\gamma_{\mathrm{sg}}(K,\beta)}<\infty,$$

then the conclusion holds.

(b) *All integer dimensions. Suppose the rows are indexed by $D=2,3,\dots$, with ambient dimension D, and*

$$J_DD\le(eD)^{\Gamma}$$

eventually. If

$$K>\sqrt{2\beta(\Gamma+1)},$$

then the conclusion holds along all integer dimensions.

(c) *Sparse stretched-exponential dimensions. If*

$$D_m\ge\exp(cm^{\kappa})$$

eventually for some $c,\kappa>0$, if

$$J_mD_m\le(eD_m)^{\Gamma}$$

eventually, and if

$$K>\sqrt{2\beta\Gamma},$$

then the conclusion holds along the sparse schedule.

If $J_m \to \infty$, the same schedules exhaust countable libraries of sub-Gaussian symmetric scalar-multiplier streams. In particular, for uniformly proxied multipliers $a_{m,h,i} \le a_{m,h}$, the cap-failure contribution is bounded by

$$2n_m \exp\left\{-\frac{L_{m,h}^2}{2a_{m,h}^2}\right\};$$

choosing $L_{m,h}$ above the active logarithmic union scale makes the unbounded-multiplier normalization exception summable.

Proof. Let

$$H_{m,h} := \left\{\max_i |\xi_{m,h,i}| \le L_{m,h}\right\}.$$

On $H_{m,h}$, the normalized matrices $A_{m,h,i}$ are contractions. The event $H_{m,h}$ is invariant under coordinatewise sign changes. Conditional on $\mathcal{G}_{m,h}$, the tuple $A_{m,h}$ is coordinatewise sign-symmetric, so Proposition 7.74, applied to the regular conditional law and to $H_{m,h}$, gives on $E_{m,h}^c$

$$\mathbb{P}\left(H_{m,h} \cap \{\mathrm{disc}(A_{m,h,1},\dots,A_{m,h,n_m}) > K\sqrt{n_m}\} \,|\, \mathcal{G}_{m,h}\right) \le \mathbb{P}\left\{\left\|\sum_i A_{m,h,i}\right\|_{\mathrm{op}} > K\sqrt{n_m} \,\middle|\, \mathcal{G}_{m,h}\right\}.$$

If $\nu_{m,h} = 0$, then $a_{m,h,i}B_{m,h,i} = 0$ for every i, hence $\xi_{m,h,i}B_{m,h,i} = 0$ almost surely under the conditional law. Otherwise the matrix Laplace bound of Tropp [25], as in Lemma 7.89, bounds the last probability by

$$2D_m \exp\left\{-\frac{K^2 n_m}{2\nu_{m,h}}\right\} \le 2D_m (eD_m)^{-\gamma_{\mathrm{sg}}(K,\beta)}$$

on $E_{m,h}^c$. Also, by the scalar sub-Gaussian tail bound and a union bound,

$$\mathbb{P}(H_{m,h}^c \mid \mathcal{G}_{m,h}) \le 2\sum_{i=1}^{n_m} \exp\left\{-\frac{L_{m,h}^2}{2a_{m,h,i}^2}\right\}.$$

After adding $\mathbb{P}(E_{m,h})$, summing over the active menu, and using the displayed summability hypotheses, Borel–Cantelli gives the simultaneous eventual contraction and discrepancy conclusion.

The all-integer and sparse assertions are immediate from the general schedule: under $J_D D \le (eD)^\Gamma$, the row tail is $O((eD)^{\Gamma-\gamma_{\mathrm{sg}}(K,\beta)})$, summable over all integer dimensions when $\gamma_{\mathrm{sg}}(K,\beta) > \Gamma+1$, and summable along every stretched-exponential schedule when $\gamma_{\mathrm{sg}}(K,\beta) > \Gamma$. The countable-library statement follows from active-prefix exhaustion. □

Corollary 7.82 (Tail-capped symmetric scalar-multiplier schedules)**.** *For each $m \ge 1$, let $D_m \ge 2$, let $J_m \ge 1$ be a deterministic active-menu size, and put $\ell_m = \log(eD_m)$. For every $1 \le h \le J_m$, let $\mathcal{G}_{m,h}$ be a conditioning sigma-field and let*

$$B_{m,h,1},\dots,B_{m,h,n_m} \in \mathrm{M}_{D_m}(\mathbb{C})$$

be $\mathcal{G}_{m,h}$-measurable self-adjoint contractions. Conditionally on $\mathcal{G}_{m,h}$, let $\xi_{m,h,1},\dots,\xi_{m,h,n_m}$ be independent real random variables with symmetric conditional laws. Let $L_{m,h}>0$ be deterministic caps and set

$$A_{m,h,i} := L_{m,h}^{-1}\xi_{m,h,i}B_{m,h,i},$$

$$\widehat{q}_{m,h,i} := L_{m,h}^{-2}\mathbb{E}\left(\xi_{m,h,i}^2\mathbf{1}_{\{|\xi_{m,h,i}|\le L_{m,h}\}} \mid \mathcal{G}_{m,h}\right),$$

and

$$\widehat{\nu}_{m,h} := \left\|\sum_{i=1}^{n_m}\widehat{q}_{m,h,i}B_{m,h,i}^2\right\|_{\mathrm{op}}.$$

Fix $\beta\ge 0$ and $K>0$. Suppose that outside events $E_{m,h}\in\mathcal{G}_{m,h}$,

$$\widehat{\nu}_{m,h}\le \beta\,\frac{n_m}{\ell_m}\qquad (1\le h\le J_m),$$

and that the cap and variance-exception series

$$\sum_m\sum_{h=1}^{J_m}\left[\mathbb{P}(E_{m,h})+\sum_{i=1}^{n_m}\mathbb{P}\{|\xi_{m,h,i}|>L_{m,h}\}\right]<\infty$$

is finite. Then each of the following deterministic schedule conditions implies that, almost surely, for all sufficiently large m and every active $1\le h\le J_m$,

$$\max_i\|A_{m,h,i}\|_{\mathrm{op}}\le 1,\qquad \mathrm{disc}(A_{m,h,1},\dots,A_{m,h,n_m})\le K\sqrt{n_m}.$$

(a) *General dimension schedules. If there is $B>0$ such that*

$$\ell_m\le B\sqrt{n_m}$$

eventually, and

$$\sum_m J_mD_m(eD_m)^{-\gamma(K,\beta,B)}<\infty,\qquad \gamma(K,\beta,B):=\frac{K^2}{2(\beta+KB/3)},$$

then the conclusion holds.

(b) *All integer dimensions. Suppose the rows are indexed by $D=2,3,\dots$, with ambient dimension D, and*

$$J_DD\le(eD)^{\Gamma}$$

eventually. If

$$n_D\ge a\log^2(eD)$$

eventually and

$$K>\Theta(\Gamma+1,a,\beta),$$

then the conclusion holds along all integer dimensions.

(c) *Sparse stretched-exponential dimensions. If*

$$D_m\ge\exp(cm^{\kappa})$$

eventually for some $c,\kappa>0$, if

$$J_mD_m\le(eD_m)^{\Gamma}$$

eventually, if

$$n_m \geq a \log^2(eD_m)$$

eventually, and if

$$K > \Theta(\Gamma, a, \beta),$$

then the conclusion holds along the sparse schedule.

If $J_m \to \infty$, the same schedules exhaust countable libraries of tail-capped symmetric scalar-multiplier streams. Moreover, suppose that for some $\alpha > 0$ the conditional tails obey the deterministic sub-Weibull bounds

$$\mathbb{P}\{|\xi_{m,h,i}| > t \mid \mathcal{G}_{m,h}\} \leq 2\exp\{-(t/a_{m,h,i})^\alpha\} \qquad (t > 0),$$

with the right-hand side interpreted as 0 for $a_{m,h,i} = 0$. Then the cap contribution is bounded by

$$2\sum_{i=1}^{n_m} \exp\{-(L_{m,h}/a_{m,h,i})^\alpha\},$$

and it suffices for the variance budget to have

$$C_\alpha L_{m,h}^{-2} \left\| \sum_{i=1}^{n_m} a_{m,h,i}^2 B_{m,h,i}^2 \right\|_{\mathrm{op}} \leq \beta \frac{n_m}{\ell_m}, \qquad C_\alpha := \frac{4}{\alpha}\Gamma(2/\alpha).$$

For uniformly proxied multipliers $a_{m,h,i} \leq a_{m,h}$, the cap contribution is at most

$$2n_m \exp\{-(L_{m,h}/a_{m,h})^\alpha\}.$$

Proof. Define the truncated multipliers and matrices

$$\widetilde{\xi}_{m,h,i} := \xi_{m,h,i}\mathbf{1}_{\{|\xi_{m,h,i}| \leq L_{m,h}\}}, \qquad \widetilde{A}_{m,h,i} := L_{m,h}^{-1}\widetilde{\xi}_{m,h,i}B_{m,h,i}.$$

Conditionally on $\mathcal{G}_{m,h}$, the variables $\widetilde{\xi}_{m,h,i}/L_{m,h}$ are independent, symmetric, and bounded by 1 in absolute value. The conditional variance proxy of the bounded scalar-multiplier tuple $(\widetilde{A}_{m,h,i})_i$ is exactly $\widehat{\nu}_{m,h}$. Hence Corollary 7.79, or equivalently Corollary 7.80 applied to the truncated tuple, gives the stated Bernstein active-menu failure bound under the three displayed schedule hypotheses.

Let

$$H_{m,h} := \{\max_i |\xi_{m,h,i}| \leq L_{m,h}\}.$$

The cap-failure series and the first Borel–Cantelli lemma imply that $H_{m,h}$ holds for all sufficiently large active pairs (m,h). On this event $\widetilde{A}_{m,h,i} = A_{m,h,i}$ for every i, and the $A_{m,h,i}$ are contractions. Combining this eventual cap event with the eventual discrepancy conclusion for the truncated tuple proves the assertion for $(A_{m,h,i})_i$.

It remains only to justify the advertised sub-Weibull reduction. Integrating the tail estimate gives, conditionally on $\mathcal{G}_{m,h}$,

$$\mathbb{E}(\xi_{m,h,i}^2 \mid \mathcal{G}_{m,h}) = \int_0^\infty 2t\,\mathbb{P}\{|\xi_{m,h,i}| > t \mid \mathcal{G}_{m,h}\}\,dt \leq \frac{4}{\alpha}\Gamma(2/\alpha)a_{m,h,i}^2.$$

Thus $\widehat{q}_{m,h,i} \le C_\alpha L_{m,h}^{-2} a_{m,h,i}^2$, while the same tail display at $t = L_{m,h}$ gives the cap contribution. The uniform proxy bound is immediate. □

Corollary 7.83 (Polynomial-moment tail-capped scalar-multiplier schedules)**.** *In the setting of Corollary 7.82, suppose that for some $p > 0$ there are deterministic proxies $a_{m,h,i} \ge 0$ such that*

$$\mathbb{E}\left(|\xi_{m,h,i}|^p \mid \mathcal{G}_{m,h}\right) \le a_{m,h,i}^p \qquad (1 \le h \le J_m,\ 1 \le i \le n_m).$$

Put $r := p \wedge 2$. If, outside events $E_{m,h} \in \mathcal{G}_{m,h}$,

$$L_{m,h}^{-r} \left\| \sum_{i=1}^{n_m} a_{m,h,i}^r B_{m,h,i}^2 \right\|_{\mathrm{op}} \le \beta \, \frac{n_m}{\ell_m},$$

and

$$\sum_m \sum_{h=1}^{J_m} \left[\mathbb{P}(E_{m,h}) + \sum_{i=1}^{n_m} \left(\frac{a_{m,h,i}}{L_{m,h}} \right)^p \right] < \infty,$$

then the three schedule conclusions of Corollary 7.82 hold. Thus the normalized matrices

$$A_{m,h,i} = L_{m,h}^{-1} \xi_{m,h,i} B_{m,h,i}$$

are eventually contractions and satisfy

$$\operatorname{disc}(A_{m,h,1}, \dots, A_{m,h,n_m}) \le K\sqrt{n_m}$$

simultaneously over the active menus under the same general, all-integer, and sparse stretched-exponential dimension schedules.

In particular, assume $a_{m,h,i} \le a_{m,h}$ for all i, take $\beta > 0$, and choose some $\delta > 0$. If

$$L_{m,h} \ge a_{m,h} \max\left\{ \left(\frac{\ell_m}{\beta} \right)^{1/r}, \left(J_m n_m (e+m)^{1+\delta} \right)^{1/p} \right\}$$

for every active h, then the variance and cap hypotheses are automatic with no $E_{m,h}$. Hence polynomial-moment symmetric multipliers of every tail index $p > 0$ have active-menu Matrix Spencer signings after this explicit deterministic cap normalization.

Proof. For the cap failures, Markov's inequality gives

$$\mathbb{P}\{|\xi_{m,h,i}| > L_{m,h} \mid \mathcal{G}_{m,h}\} \le \left(\frac{a_{m,h,i}}{L_{m,h}} \right)^p.$$

It remains to compare the truncated second-moment profile with the displayed moment profile. If $0 < p \le 2$, then on $\{|\xi_{m,h,i}| \le L_{m,h}\}$,

$$|\xi_{m,h,i}|^2 \le L_{m,h}^{2-p} |\xi_{m,h,i}|^p,$$

so

$$\widehat{q}_{m,h,i} \le L_{m,h}^{-p} a_{m,h,i}^p = L_{m,h}^{-r} a_{m,h,i}^r.$$

If $p > 2$, monotonicity of conditional L^q-norms gives

$$\mathbb{E}(\xi_{m,h,i}^2 \mid \mathcal{G}_{m,h}) \le a_{m,h,i}^2,$$

hence again

$$\widehat{q}_{m,h,i} \le L_{m,h}^{-2} a_{m,h,i}^2 = L_{m,h}^{-r} a_{m,h,i}^r.$$

Thus the hypotheses imply the variance and cap hypotheses of Corollary 7.82, which gives the first assertion.

For the uniform proxy statement, use $0 \preceq B_{m,h,i}^2 \preceq I$ to get

$$L_{m,h}^{-r} \left\| \sum_i a_{m,h,i}^r B_{m,h,i}^2 \right\|_{\mathrm{op}} \le n_m \left(\frac{a_{m,h}}{L_{m,h}} \right)^r \le \beta \, \frac{n_m}{\ell_m}.$$

The same lower bound on $L_{m,h}$ gives

$$\sum_{h=1}^{J_m} \sum_{i=1}^{n_m} \left(\frac{a_{m,h,i}}{L_{m,h}} \right)^p \le (e+m)^{-1-\delta},$$

which is summable in m. Applying the first part proves the explicit normalization claim. □

Corollary 7.84 (Environment-adaptive tail-capped scalar-multiplier schedules)**.** *In the setting of Corollary 7.82, replace the deterministic caps by positive $\mathcal{G}_{m,h}$-measurable caps $L_{m,h}$. Define*

$$A_{m,h,i} := L_{m,h}^{-1} \xi_{m,h,i} B_{m,h,i},$$

$$\widehat{q}_{m,h,i} := L_{m,h}^{-2} \mathbb{E}\left(\xi_{m,h,i}^2 \mathbf{1}_{\{|\xi_{m,h,i}| \le L_{m,h}\}} \mid \mathcal{G}_{m,h} \right),$$

and

$$\widehat{\nu}_{m,h} := \left\| \sum_{i=1}^{n_m} \widehat{q}_{m,h,i} B_{m,h,i}^2 \right\|_{\mathrm{op}}.$$

Fix $\beta \ge 0$ and $K > 0$. Suppose that, outside events $E_{m,h} \in \mathcal{G}_{m,h}$,

$$\widehat{\nu}_{m,h} \le \beta \, \frac{n_m}{\ell_m},$$

and that

$$\sum_m \sum_{h=1}^{J_m} \left[\mathbb{P}(E_{m,h}) + \mathbb{E} \sum_{i=1}^{n_m} \mathbb{P}\{|\xi_{m,h,i}| > L_{m,h} \mid \mathcal{G}_{m,h}\} \right] < \infty.$$

Then the three schedule conclusions of Corollary 7.82 remain valid. Thus, almost surely, all sufficiently large active normalized matrices are contractions and satisfy

$$\operatorname{disc}(A_{m,h,1}, \dots, A_{m,h,n_m}) \le K \sqrt{n_m}.$$

More concretely, let $p > 0$, put $r = p \wedge 2$, and let $a_{m,h,i}$ be nonnegative $\mathcal{G}_{m,h}$-measurable moment proxies with

$$\mathbb{E}(|\xi_{m,h,i}|^p \mid \mathcal{G}_{m,h}) \le a_{m,h,i}^p.$$

If, outside $E_{m,h}$,

$$L_{m,h}^{-r} \left\| \sum_i a_{m,h,i}^r B_{m,h,i}^2 \right\|_{\mathrm{op}} \le \beta \, \frac{n_m}{\ell_m},$$

and

$$\sum_{m}\sum_{h=1}^{J_m}\left[\mathbb{P}(E_{m,h})+\mathbb{E}\sum_{i}\left(\frac{a_{m,h,i}}{L_{m,h}}\right)^{p}\right]<\infty,$$

then the same conclusion holds.

In particular, if $\beta>0$ and $a_{m,h,i}\le a_{m,h}$ for an $\mathcal{G}_{m,h}$-measurable envelope $a_{m,h}$, then for any $\delta>0$ the conditions are automatic, with no $E_{m,h}$, whenever

$$L_{m,h}\ge a_{m,h}\max\left\{\left(\frac{\ell_m}{\beta}\right)^{1/r},\left(J_m n_m(e+m)^{1+\delta}\right)^{1/p}\right\}$$

almost surely for each active h. When $a_{m,h}=0$, any positive $\mathcal{G}_{m,h}$-measurable cap is allowed.

Proof. Fix m,h and condition on $\mathcal{G}_{m,h}$. The cap $L_{m,h}$, the directions $B_{m,h,i}$, and the event $E_{m,h}$ are then fixed. Truncate

$$\widetilde{\xi}_{m,h,i}:=\xi_{m,h,i}\mathbf{1}_{\{|\xi_{m,h,i}|\le L_{m,h}\}},\qquad \widetilde{A}_{m,h,i}:=L_{m,h}^{-1}\widetilde{\xi}_{m,h,i}B_{m,h,i}.$$

Conditionally on $\mathcal{G}_{m,h}$, the tuple $(\widetilde{A}_{m,h,i})_i$ is independent, sign-symmetric, and made of contractions, with conditional variance proxy $\widehat{\nu}_{m,h}$. On $E_{m,h}^{c}$, Corollary 7.79 therefore gives the same active-menu Bernstein failure estimate as in the deterministic-cap theorem. The displayed conditional cap-failure summability and Borel–Cantelli imply that eventually the truncated and untruncated normalized tuples agree over all active menus. This proves the adaptive-cap schedule conclusions.

For the moment-proxy specialization, conditional Markov gives

$$\mathbb{P}\{|\xi_{m,h,i}|>L_{m,h}\mid\mathcal{G}_{m,h}\}\le\left(\frac{a_{m,h,i}}{L_{m,h}}\right)^{p}.$$

As in the proof of Corollary 7.83, the truncated second moment obeys

$$\widehat{q}_{m,h,i}\le L_{m,h}^{-r}a_{m,h,i}^{r}.$$

This gives the second assertion. The uniform-envelope cap makes

$$L_{m,h}^{-r}\left\|\sum_{i}a_{m,h,i}^{r}B_{m,h,i}^{2}\right\|_{\mathrm{op}}\le\beta\,\frac{n_m}{\ell_m}$$

and

$$\sum_{h=1}^{J_m}\sum_{i=1}^{n_m}\left(\frac{a_{m,h,i}}{L_{m,h}}\right)^{p}\le(e+m)^{-1-\delta},$$

which is summable. □

Corollary 7.85 (Signed cores plus full random operator-tail noise)**.** *Let $B_1,\dots,B_n$ be random self-adjoint matrices in $\mathrm{M}_D(\mathbb{C})$, let E be an event, let $K\ge0$, and let $K_0\ge1$. Suppose that, outside E,*

$$\|B_i\|_{\mathrm{op}}\le1\quad(1\le i\le n)$$

and there are signs $\varepsilon_1^\star, \dots, \varepsilon_n^\star$, measurable with respect to $B_1, \dots, B_n$, such that

$$\left\| \sum_{i=1}^n \varepsilon_i^\star B_i \right\|_{\mathrm{op}} \le K\sqrt{n}.$$

Let $Y_1, \dots, Y_n$ be centered self-adjoint random matrices such that, conditionally on $B_1, \dots, B_n$, for every $a = (a_1, \dots, a_n) \in \mathbb{R}^n$ and every $t \ge 0$,

$$\mathbb{P}\left\{ \left\| \sum_{i=1}^n a_i Y_i \right\|_{\mathrm{op}} > K_0 \|a\|_2 (\sqrt{D} + t) \,\middle|\, B_1, \dots, B_n \right\} \le 2e^{-t^2}.$$

Put

$$s_{D,n} := \sqrt{D} + \sqrt{3\log(en)}, \qquad Z_i := \frac{Y_i}{K_0 s_{D,n}}, \qquad X_i := B_i + Z_i.$$

Then

$$\mathbb{P}\left\{ \begin{array}{c} \mathrm{disc}(X_1, \dots, X_n) > (K+1)\sqrt{n}, \\ \text{or } \max_i \|X_i\|_{\mathrm{op}} > 2 \end{array} \right\} \le \mathbb{P}(E) + 2(n+1)(en)^{-3}.$$

Consequently, for $\widehat{X}_i = X_i/2$,

$$\mathbb{P}\left\{ \begin{array}{c} \mathrm{disc}(\widehat{X}_1, \dots, \widehat{X}_n) > \dfrac{K+1}{2}\sqrt{n}, \\ \text{or } \max_i \|\widehat{X}_i\|_{\mathrm{op}} > 1 \end{array} \right\} \le \mathbb{P}(E) + 2(n+1)(en)^{-3}.$$

Thus any deterministic or random core class whose existing theorem supplies a $K\sqrt{n}$ signing–including finite-lift cores, scalar Toeplitz or Hankel cores, algebraic cores, and their random versions–remains Spencer-scale after adding full centered GOE/GUE, Wigner-type, correlated Gaussian, or other Gaussian-operator-tail noise at its natural spectral scale.

Proof. Condition on a core realization outside E. Then the signs $\varepsilon^\star$ are fixed from the point of view of the conditional tail. The conditional operator-tail hypothesis applied with $a = \varepsilon^\star$ gives

$$\mathbb{P}\left\{ \left\| \sum_i \varepsilon_i^\star Y_i \right\|_{\mathrm{op}} > K_0\sqrt{n}\, s_{D,n} \,\middle|\, B_1, \dots, B_n \right\} \le 2(en)^{-3}.$$

The same conditional tail applied with $a = e_i$, followed by a union bound over i, shows that all $\|Z_i\|_{\mathrm{op}} \le 1$ except with conditional probability at most $2n(en)^{-3}$. On the complementary noise event,

$$\left\| \sum_i \varepsilon_i^\star X_i \right\|_{\mathrm{op}} \le \left\| \sum_i \varepsilon_i^\star B_i \right\|_{\mathrm{op}} + \left\| \sum_i \varepsilon_i^\star Z_i \right\|_{\mathrm{op}} \le (K+1)\sqrt{n},$$

and $\|X_i\|_{\mathrm{op}} \le \|B_i\|_{\mathrm{op}} + \|Z_i\|_{\mathrm{op}} \le 2$ for every i. Averaging over the core realization gives the displayed probability bound. □

Corollary 7.86 (Finite-lift core menus plus full operator-tail noise)**.** *For each $m \geq 1$, let $D_m \geq 2$, let*

$$J_1 \leq J_2 \leq \cdots, \qquad J_m \to \infty,$$

be a deterministic active-prefix exhaustion, and let $\mathcal{G}_m$ be a row environment. For each $h \geq 1$, let

$$B_{m,1}^{(h)}, \ldots, B_{m,n_m}^{(h)}, \qquad Y_{m,1}^{(h)}, \ldots, Y_{m,n_m}^{(h)}$$

be self-adjoint random matrices in $\mathrm{M}_{D_m}(\mathbb{C})$. Put

$$\mathcal{H}_m = \sigma\left(\mathcal{G}_m,\, B_{m,i}^{(h)} : 1 \leq i \leq n_m,\ 1 \leq h \leq J_m\right).$$

Assume that there are events $F_m^{\mathrm{lift}} \in \mathcal{H}_m$ with $\sum_m \mathbb{P}(F_m^{\mathrm{lift}}) < \infty$ such that, outside F_m^{lift}, the active-prefix core tuples are contractions and admit finite block lifts with active-prefix loss and entropy

$$M_m^{(J)} \leq M_0, \qquad E_m^{(J)} \leq C_E n_m$$

in the sense of Corollary 3.2. Suppose also that for each $h \leq J_m$ there is $K_{0,m,h} \geq 1$ such that, conditionally on $\mathcal{H}_m$, for every $a \in \mathbb{R}^{n_m}$ and every $t \geq 0$,

$$\mathbb{P}\left\{\left\|\sum_{i=1}^{n_m} a_i Y_{m,i}^{(h)}\right\|_{\mathrm{op}} > K_{0,m,h}\|a\|_2(\sqrt{D_m} + t) \,\middle|\, \mathcal{H}_m\right\} \leq 2e^{-t^2}.$$

Let $\tau_m > 0$ be deterministic and define

$$Z_{m,i}^{(h)} = \frac{Y_{m,i}^{(h)}}{K_{0,m,h}(\sqrt{D_m} + \tau_m)}, \qquad X_{m,i}^{(h)} = B_{m,i}^{(h)} + Z_{m,i}^{(h)}.$$

If

$$\sum_m \left[\mathbb{P}(F_m^{\mathrm{lift}}) + 2J_m(n_m + 1)e^{-\tau_m^2}\right] < \infty,$$

then, with probability one, for all sufficiently large m, one $\mathcal{H}_m$-measurable sign vector $\varepsilon_m^\star$ satisfies

$$\left\|\sum_{i=1}^{n_m} \varepsilon_{m,i}^\star X_{m,i}^{(h)}\right\|_{\mathrm{op}} \leq (K_{\mathrm{fl}} + 1)\sqrt{n_m} \qquad (1 \leq h \leq J_m),$$

where $K_{\mathrm{fl}} = CM_0\sqrt{\log(e + C_E)}$. On the same event,

$$\max_{1 \leq h \leq J_m} \max_{1 \leq i \leq n_m} \|X_{m,i}^{(h)}\|_{\mathrm{op}} \leq 2.$$

Consequently the normalized contractions $\widehat{X}_{m,i}^{(h)} = X_{m,i}^{(h)}/2$ have a common active-prefix signing of size $(K_{\mathrm{fl}} + 1)\sqrt{n_m}/2$, and every fixed menu element is eventually covered.

Proof. On $(F_m^{\mathrm{lift}})^c$, Corollary 3.2 gives one sign vector $\varepsilon_m^\star$ satisfying

$$\left\|\sum_i \varepsilon_{m,i}^\star B_{m,i}^{(h)}\right\|_{\mathrm{op}} \le K_{\mathrm{fl}}\sqrt{n_m} \qquad (1 \le h \le J_m).$$

As before, choose the lexicographically first successful sign vector to make it $\mathcal{H}_m$-measurable. Conditional on $\mathcal{H}_m$, apply the operator-tail hypothesis for each $h \le J_m$ with $a = \varepsilon_m^\star$ and with $a = e_i$, $1 \le i \le n_m$, at $t = \tau_m$. A union bound gives conditional failure probability at most $2J_m(n_m+1)e^{-\tau_m^2}$. Outside this noise event,

$$\left\|\sum_i \varepsilon_{m,i}^\star Z_{m,i}^{(h)}\right\|_{\mathrm{op}} \le \sqrt{n_m}, \qquad \max_i \|Z_{m,i}^{(h)}\|_{\mathrm{op}} \le 1 \qquad (1 \le h \le J_m).$$

Combining the core and noise estimates gives the displayed active-prefix signing and norm bounds. The summability assumption and Borel–Cantelli give the almost-sure eventual statement; $J_m \to \infty$ gives the fixed-menu coverage. □

Corollary 7.87 (Operator-tail schedules for finite-lift core menus)**.** *In the setting of Corollary 7.86, put*

$$L_m^{\mathrm{ot}} := J_m(n_m+1).$$

If $\gamma > 0$ and

$$\tau_m^2 \ge \log(2L_m^{\mathrm{ot}}) + (1+\gamma)\log(e+m)$$

eventually, while $\sum_m \mathbb{P}(F_m^{\mathrm{lift}}) < \infty$, then the active-prefix and fixed-menu conclusions of Corollary 7.86 hold almost surely. Thus arbitrary dimension growth is allowed after scaling the h-th full operator-tail noise by

$$K_{0,m,h}\left(\sqrt{D_m} + \tau_m\right).$$

In particular, suppose $L_m^{\mathrm{ot}} \le (eD_m)^\Gamma$ eventually. Along all integer dimensions $D = 2, 3, \ldots$, if

$$\tau_D = \sqrt{R\log(eD)}, \qquad R > \Gamma + 1,$$

then the same active-prefix and fixed-menu conclusions hold. Along any sparse schedule $D_m \ge \exp(cm^\kappa)$, the same conclusions hold whenever

$$\tau_m = \sqrt{R\log(eD_m)}, \qquad R > \Gamma.$$

Proof. The arbitrary-growth hypothesis gives

$$2L_m^{\mathrm{ot}} e^{-\tau_m^2} \le (e+m)^{-(1+\gamma)},$$

which is summable. Therefore the summability condition in Corollary 7.86 follows from the lift-exception summability.

For the all-integer specialization,

$$2J_D(n_D+1)e^{-\tau_D^2} \le 2(eD)^{\Gamma-R},$$

which is summable over $D = 2, 3, \dots$ when $R - \Gamma > 1$. For the sparse schedule the same bound is a stretched-exponential summable sequence whenever $R - \Gamma > 0$. □

Corollary 7.88 (Finite-lift core menus plus mixed dominated operator-tail noise)**.** *Keep the finite-lift core assumptions and notation of Corollary 7.86, but replace the displayed full operator-tail hypothesis and its scaling as follows. Put $\Delta_m := D_m + n_m + 1$ and $\ell_m := \log(e\Delta_m)$. For each active $h \le J_m$, tag the perturbation stream either ψ or μ, choose $\varepsilon_{m,h} \in (0, 1]$, $A_{m,h} \ge 1$, and $s_{m,h} > 0$, and define $r_{m,h}$, $V_{m,h}$, and $q_{m,h}$ as in Corollary 7.68, with $Y^{(h)}_{m,i}$ in place of $X_{m,h,i}$. Assume that, after adjoining $\mathcal{H}_m$ to the predictable sigma-fields, the corresponding sub-Weibull or finite-moment dominated hypotheses and the matching truncated tail-drift hypotheses hold outside events $E_{m,h}$.*

Let $K_{\mathrm{tail}}, \eta > 0$ and set

$$\lambda_{m,h} := \max\{1, K_{\mathrm{tail}} + \eta\}\, q_{m,h},$$
$$Z^{(h)}_{m,i} := \frac{Y^{(h)}_{m,i}}{\lambda_{m,h}},$$
$$X^{(h)}_{m,i} := B^{(h)}_{m,i} + Z^{(h)}_{m,i}.$$

If $\ell_m \le B\sqrt{n_m}$ eventually,

$$\sum_m \left[\mathbb{P}(F^{\mathrm{lift}}_m) + \sum_{h=1}^{J_m} (\mathbb{P}(E_{m,h}) + \varepsilon_{m,h}) \right] < \infty,$$

and

$$\sum_m J_m \Delta_m (e\Delta_m)^{-\gamma_{\mathrm{tc}}(K_{\mathrm{tail}},1,1,B)} < \infty,$$

then, with probability one, for all sufficiently large m, one $\mathcal{H}_m$-measurable sign vector $\varepsilon^\star_m$ satisfies

$$\left\| \sum_{i=1}^{n_m} \varepsilon^\star_{m,i} X^{(h)}_{m,i} \right\|_{\mathrm{op}} \le (K_{\mathrm{fl}} + 1)\sqrt{n_m} \qquad (1 \le h \le J_m),$$

where $K_{\mathrm{fl}} = CM_0\sqrt{\log(e + C_E)}$. On the same event,

$$\max_{1 \le h \le J_m} \max_{1 \le i \le n_m} \|X^{(h)}_{m,i}\|_{\mathrm{op}} \le 2.$$

Consequently the normalized contractions $\widehat{X}^{(h)}_{m,i} = X^{(h)}_{m,i}/2$ have a common active-prefix signing of size $(K_{\mathrm{fl}} + 1)\sqrt{n_m}/2$, and every fixed menu element is eventually covered.

In particular, if $J_m\Delta_m \le (e\Delta_m)^\Gamma$, $J_m \le (e\Delta_m)^\Lambda$, $n_m \ge a\ell_m^2$ eventually, and $\varepsilon_{m,h} = (e\Delta_m)^{-c}$, then after adding $\sum_m \mathbb{P}(F^{\mathrm{lift}}_m) < \infty$ the cap-budget part is summable for all integer dimensions when $c > \Lambda + 1$, and along every stretched-exponential sparse dimension schedule when $c > \Lambda$. The matrix-tail margins are

$$K_{\mathrm{tail}} > \Theta_{\mathrm{tc}}(\Gamma + 1, a, 1, 1)$$

and

$$K_{\mathrm{tail}} > \Theta_{\mathrm{tc}}(\Gamma, a, 1, 1),$$

respectively.

Proof. Outside F_m^{lift}, Corollary 3.2 gives one active-prefix signing for the finite-lift core menu:

$$\left\| \sum_i \varepsilon^\star_{m,i} B^{(h)}_{m,i} \right\|_{\mathrm{op}} \le K_{\mathrm{fl}} \sqrt{n_m} \qquad (1 \le h \le J_m).$$

As in Corollary 7.86, choose the lexicographically first successful sign vector to make it $\mathcal{H}_m$-measurable. Now apply Corollary 7.69 with

$$B_{m,h,i} = B^{(h)}_{m,i}, \qquad Y_{m,h,i} = Y^{(h)}_{m,i},$$
$$K_{\mathrm{core}} = K_{\mathrm{fl}}, \qquad F_m^{\mathrm{core}} = F_m^{\mathrm{lift}}.$$

The finite-lift conclusion above supplies the signed-core hypothesis, while the displayed mixed dominated assumptions are exactly the perturbation hypotheses. The almost-sure active-prefix statement, fixed-menu coverage, and the all-integer and sparse schedule thresholds are therefore the corresponding conclusions of the signed-core mixed dominated corollary. □

Lemma 7.89 (Matrix-series sub-Gaussian tail)**.** *Let $\mathcal{K}$ be a sigma-field. Conditionally on $\mathcal{K}$, let $\theta_1, \dots, \theta_N$ be independent centered real variables satisfying*

$$\mathbb{E}\{\exp(\lambda\theta_r) \mid \mathcal{K}\} \le \exp(\lambda^2/2) \qquad (\lambda \in \mathbb{R}).$$

Let $A_1, \dots, A_N \in \mathrm{M}_D(\mathbb{C})$ be $\mathcal{K}$-measurable self-adjoint matrices, and put

$$v = \left\| \sum_{r=1}^N A_r^2 \right\|_{\mathrm{op}}.$$

Let

$$\rho = \max\left\{1, \operatorname{rank}\left(\sum_{r=1}^N A_r^2\right)\right\}.$$

There is a universal constant C_{ms} such that, conditionally on $\mathcal{K}$, for every $\tau \ge 0$,

$$\mathbb{P}\left\{ \left\| \sum_{r=1}^N \theta_r A_r \right\|_{\mathrm{op}} > C_{\mathrm{ms}} \sqrt{v}\, (\sqrt{\log(e\rho)} + \tau) \,\middle|\, \mathcal{K} \right\} \le 2e^{-\tau^2}.$$

The same conclusion applies to conditional Gaussian, Rademacher, or any uniformly scalar-sub-Gaussian coefficient family after changing the universal constant by the corresponding proxy.

Proof. Condition on $\mathcal{K}$. The matrix Laplace bound of Tropp [25], applied with $\mathbb{E}e^{\lambda\theta_r A_r} \preceq e^{\lambda^2 A_r^2/2}$, gives

$$\mathbb{P}\left\{ \left\| \sum_r \theta_r A_r \right\|_{\mathrm{op}} > s \right\} \le 2D \exp\left\{ -\frac{s^2}{2v} \right\}$$

when $v > 0$. If $v = 0$, then all $A_r = 0$ and the claim is trivial. When $v > 0$, let $V = \sum_r A_r^2$ and $E = \operatorname{ran} V$. Since

$$\ker V = \bigcap_r \ker A_r,$$

each A_r kills $E^\perp$, and self-adjointness makes E invariant. Thus the random series is supported on the ρ-dimensional space E. Applying the preceding Laplace bound to the restricted matrices on E replaces the factor D by ρ. Taking $s = C_{\mathrm{ms}}\sqrt{v}(\sqrt{\log(e\rho)} + \tau)$ with C_{ms} large enough absorbs this dimension factor and leaves $2e^{-\tau^2}$. □

Corollary 7.90 (Finite-lift cores plus matrix-series noise)**.** *In the setting of Corollary 7.86, keep the same finite-lift core assumptions and active-prefix exhaustion. Let $\mathcal{K}_m \supseteq \mathcal{H}_m$ be a sigma-field. For every active (m, h, i), let $\mathcal{R}_{m,h,i}$ be a finite index set and let*

$$G_{m,h,i,r} \in \mathrm{M}_{D_m}(\mathbb{C}) \qquad (r \in \mathcal{R}_{m,h,i})$$

be $\mathcal{K}_m$-measurable self-adjoint matrices. Conditionally on $\mathcal{K}_m$, let the scalar coefficients $\theta_{m,h,i,r}$ be independent, centered, and satisfy the scalar sub-Gaussian bound in Lemma 7.89. Set

$$Y_{m,i}^{(h)} = \sum_{r \in \mathcal{R}_{m,h,i}} \theta_{m,h,i,r} G_{m,h,i,r}.$$

Let $q_{m,h} > 0$, $\tau_m > 0$, and $1 \le \kappa_{m,h} \le D_m$ be deterministic. Let $F_m^{\mathrm{ms}} \in \mathcal{K}_m$ be the event that for some $h \le J_m$ at least one of the following four bounds fails:

$$\max_{i \le n_m} \left\| \sum_{r \in \mathcal{R}_{m,h,i}} G_{m,h,i,r}^2 \right\|_{\mathrm{op}} \le q_{m,h}^2,$$

$$\left\| \sum_{i=1}^{n_m} \sum_{r \in \mathcal{R}_{m,h,i}} G_{m,h,i,r}^2 \right\|_{\mathrm{op}} \le n_m q_{m,h}^2,$$

$$\max_{i \le n_m} \operatorname{rank} \left(\sum_{r \in \mathcal{R}_{m,h,i}} G_{m,h,i,r}^2 \right) \le \kappa_{m,h},$$

$$\operatorname{rank} \left(\sum_{i=1}^{n_m} \sum_{r \in \mathcal{R}_{m,h,i}} G_{m,h,i,r}^2 \right) \le \kappa_{m,h}.$$

Define

$$Z_{m,i}^{(h)} = \frac{Y_{m,i}^{(h)}}{C_{\mathrm{ms}} q_{m,h}(\sqrt{\log(e\kappa_{m,h})} + \tau_m)}, \qquad X_{m,i}^{(h)} = B_{m,i}^{(h)} + Z_{m,i}^{(h)}.$$

If

$$\sum_m \left[\mathbb{P}(F_m^{\mathrm{lift}}) + \mathbb{P}(F_m^{\mathrm{ms}}) + 2J_m(n_m + 1)e^{-\tau_m^2} \right] < \infty,$$

then, with probability one, for all sufficiently large m, the active-prefix and fixed-menu conclusions of Corollary 7.86 hold for these matrix-series perturbations. Thus finite-lift core libraries absorb conditionally Gaussian, Rademacher, or scalar-sub-Gaussian matrix-series noise at the sharper scale

$$q_{m,h}(\sqrt{\log(e\kappa_{m,h})}+\tau_m),$$

rather than the generic full-operator-tail scale $q_{m,h}(\sqrt{D_m}+\tau_m)$, whenever the realized coefficient square function satisfies the displayed variance and rank bounds outside a summable event. The full-rank choice $\kappa_{m,h}=D_m$ recovers the preceding log-dimensional matrix-series scale.

The same arbitrary-growth, all-integer, and sparse-dimension schedules as in Corollary 7.87 apply with this log-dimensional matrix-series scale: if $J_m(n_m+1)\le (eD_m)^{\Gamma}$, then $\tau_D^2=S\log(eD)$ with $S>\Gamma+1$ suffices along all integer dimensions, while sparse stretched-exponential dimension schedules need only $S>\Gamma$.

Proof. Work outside $F_m^{\mathrm{lift}}\cup F_m^{\mathrm{ms}}$. The finite-lift common-menu corollary gives one active-prefix core signing $\varepsilon_m^\star$, which is $\mathcal{H}_m$-measurable and hence $\mathcal{K}_m$-measurable. Conditional on $\mathcal{K}_m$, Lemma 7.89 applied to the signed series

$$\sum_{i=1}^{n_m}\varepsilon_{m,i}^\star Y_{m,i}^{(h)}=\sum_{i,r}\theta_{m,h,i,r}\varepsilon_{m,i}^\star G_{m,h,i,r}$$

has variance parameter at most $n_m q_{m,h}^2$ and rank parameter at most $\kappa_{m,h}$. Therefore, except with conditional probability at most $2e^{-\tau_m^2}$,

$$\left\|\sum_i \varepsilon_{m,i}^\star Z_{m,i}^{(h)}\right\|_{\mathrm{op}}\le\sqrt{n_m}.$$

The same lemma applied to each individual $Y_{m,i}^{(h)}$ has variance parameter at most $q_{m,h}^2$ and rank parameter at most $\kappa_{m,h}$, and gives $\|Z_{m,i}^{(h)}\|_{\mathrm{op}}\le 1$ except with conditional probability at most $2e^{-\tau_m^2}$. A union bound over $h\le J_m$ and $i=1,\dots,n_m$, together with the signed-sum events, gives the failure probability $2J_m(n_m+1)e^{-\tau_m^2}$. On the complementary event, the same estimates as in Corollary 7.86 give the active-prefix discrepancy and contraction bounds. Borel–Cantelli gives the almost-sure eventual conclusion, and the schedule paragraph is the same summability calculation as in Corollary 7.87. □

Corollary 7.91 (Low-rank finite-lift matrix-series noise)**.** *In the setting of Corollary 7.90, suppose that for each active $h\le J_m$ there is a $\mathcal{K}_m$-measurable orthogonal projection $P_{m,h}$ with*

$$\operatorname{rank}P_{m,h}\le\kappa_{m,h}$$

such that every active coefficient direction satisfies

$$G_{m,h,i,r}=P_{m,h}G_{m,h,i,r}P_{m,h}.$$

Let $F_m^{\mathrm{lr}} \in \mathcal{K}_m$ *be the event that for some* $h \le J_m$ *one of the two square-function bounds*

$$\max_{i\le n_m}\left\|\sum_{r\in\mathcal{R}_{m,h,i}} G_{m,h,i,r}^2\right\|_{\mathrm{op}} \le q_{m,h}^2$$

or

$$\left\|\sum_{i=1}^{n_m}\sum_{r\in\mathcal{R}_{m,h,i}} G_{m,h,i,r}^2\right\|_{\mathrm{op}} \le n_m q_{m,h}^2$$

fails. If

$$\sum_m \left[\mathbb{P}(F_m^{\mathrm{lift}}) + \mathbb{P}(F_m^{\mathrm{lr}}) + 2J_m(n_m+1)e^{-\tau_m^2}\right] < \infty,$$

then, with probability one, for all sufficiently large m*, the active-prefix and fixed-menu conclusions of Corollary 7.86 hold after scaling the* h*-th low-rank matrix-series noise stream by*

$$C_{\mathrm{ms}} q_{m,h}\left(\sqrt{\log(e\kappa_{m,h})} + \tau_m\right).$$

Thus finite-lift core libraries absorb conditionally Gaussian, Rademacher, or scalar-sub-Gaussian matrix-series noise supported on a common $\kappa_{m,h}$*-dimensional subspace at the rank-sensitive spectral scale. This covers spiked matrix-series perturbations, block-localized matrix-series noise, and random-subspace matrix-series perturbations after conditioning on the support projection. If the two square-function bounds hold deterministically, then* $F_m^{\mathrm{lr}} = \emptyset$*.*

The arbitrary-growth, all-integer, and sparse-dimension schedules are the same as in Corollary 7.87; only the scale changes from $\sqrt{\log(eD_m)}$ *to* $\sqrt{\log(e\kappa_{m,h})}$*.*

Proof. The common support projection gives

$$\operatorname{ran}\left(\sum_r G_{m,h,i,r}^2\right) \subseteq \operatorname{ran} P_{m,h} \qquad \text{and} \qquad \operatorname{ran}\left(\sum_{i,r} G_{m,h,i,r}^2\right) \subseteq \operatorname{ran} P_{m,h}.$$

Hence both rank bounds required in Corollary 7.90 are automatic with rank envelope $\kappa_{m,h}$. On $(F_m^{\mathrm{lr}})^c$, the two variance bounds required there also hold, so the corollary applies with $F_m^{\mathrm{ms}} = F_m^{\mathrm{lr}}$. The examples are exactly the displayed support-projection choices, and the schedule statement is unchanged. □

Corollary 7.92 (Atomic norm-budget low-rank matrix-series noise)**.** *In the setting of Corollary 7.91, suppose the common support projections* $P_{m,h}$ *are given as there. Suppose also that there are* $\mathcal{K}_m$*-measurable nonnegative amplitudes* $b_{m,h,i,r}$ *such that every active coefficient direction satisfies*

$$G_{m,h,i,r} = P_{m,h} G_{m,h,i,r} P_{m,h}, \qquad \|G_{m,h,i,r}\|_{\mathrm{op}} \le b_{m,h,i,r}.$$

Let $F_m^{\mathrm{at}} \in \mathcal{K}_m$ *be the event that for some* $h \le J_m$ *one of the two scalar atom-budget bounds*

$$\max_{i\le n_m}\sum_{r\in\mathcal{R}_{m,h,i}} b_{m,h,i,r}^2 \le q_{m,h}^2$$

or

$$\sum_{i=1}^{n_m} \sum_{r\in\mathcal{R}_{m,h,i}} b_{m,h,i,r}^2 \le n_m q_{m,h}^2$$

fails. If

$$\sum_m \left[\mathbb{P}(F_m^{\mathrm{lift}}) + \mathbb{P}(F_m^{\mathrm{at}}) + 2J_m(n_m+1)e^{-\tau_m^2}\right] < \infty,$$

then, with probability one, for all sufficiently large m, the active-prefix and fixed-menu conclusions of Corollary 7.86 hold after scaling the h-th atomic matrix-series noise stream by

$$C_{\mathrm{ms}} q_{m,h}(\sqrt{\log(e\kappa_{m,h})} + \tau_m).$$

Thus finite-lift core libraries absorb rank-one spike fields, self-adjoint edge-atom series, and finite frame/dictionary matrix-series noise inside a common low-rank support whenever the scalar squared-amplitude budgets are summable outside F_m^{at}. If the scalar budgets hold deterministically, then $F_m^{\mathrm{at}} = \emptyset$.

Proof. For a self-adjoint matrix supported on $P_{m,h}$,

$$G_{m,h,i,r}^2 \preceq \|G_{m,h,i,r}\|_{\mathrm{op}}^2 P_{m,h} \preceq b_{m,h,i,r}^2 P_{m,h}.$$

Therefore, outside F_m^{at},

$$\left\|\sum_r G_{m,h,i,r}^2\right\|_{\mathrm{op}} \le \sum_r b_{m,h,i,r}^2 \le q_{m,h}^2$$

for every active i, h, and

$$\left\|\sum_{i,r} G_{m,h,i,r}^2\right\|_{\mathrm{op}} \le \sum_{i,r} b_{m,h,i,r}^2 \le n_m q_{m,h}^2.$$

The common support projection gives the rank bounds exactly as in Corollary 7.91. Applying that corollary with $F_m^{\mathrm{lr}} = F_m^{\mathrm{at}}$ proves the conclusion. □

Corollary 7.93 (Block-load atomic matrix-series noise)**.** *In the setting of Corollary 7.90, suppose that for each active $h \le J_m$ there is a finite nonempty block set $\mathcal{B}_{m,h}$ and pairwise orthogonal $\mathcal{K}_m$-measurable projections*

$$P_{m,h,\beta} \qquad (\beta \in \mathcal{B}_{m,h})$$

such that

$$\operatorname{rank}\left(\sum_{\beta\in\mathcal{B}_{m,h}} P_{m,h,\beta}\right) \le \kappa_{m,h}.$$

For every active coefficient direction let $S_{m,h,i,r} \subseteq \mathcal{B}_{m,h}$ be a $\mathcal{K}_m$-measurable support set, put

$$P_{m,h,S_{m,h,i,r}} = \sum_{\beta\in S_{m,h,i,r}} P_{m,h,\beta},$$

and assume that for some $\mathcal{K}_m$-measurable nonnegative amplitudes $b_{m,h,i,r}$,

$$G_{m,h,i,r} = P_{m,h,S_{m,h,i,r}} G_{m,h,i,r} P_{m,h,S_{m,h,i,r}}, \qquad \|G_{m,h,i,r}\|_{\mathrm{op}} \le b_{m,h,i,r}.$$

Let $F_m^{\mathrm{bl}} \in \mathcal{K}_m$ be the event that for some $h \le J_m$ one of the two block-load bounds

$$\max_{i \le n_m} \max_{\beta \in \mathcal{B}_{m,h}} \sum_{\substack{r \in \mathcal{R}_{m,h,i} \\ \beta \in S_{m,h,i,r}}} b_{m,h,i,r}^2 \le q_{m,h}^2$$

or

$$\max_{\beta \in \mathcal{B}_{m,h}} \sum_{i=1}^{n_m} \sum_{\substack{r \in \mathcal{R}_{m,h,i} \\ \beta \in S_{m,h,i,r}}} b_{m,h,i,r}^2 \le n_m q_{m,h}^2$$

fails. If

$$\sum_m \left[\mathbb{P}(F_m^{\mathrm{lift}}) + \mathbb{P}(F_m^{\mathrm{bl}}) + 2J_m(n_m+1)e^{-\tau_m^2} \right] < \infty,$$

then, with probability one, for all sufficiently large m, the active-prefix and fixed-menu conclusions of Corollary 7.86 hold after scaling the h-th block-atomic matrix-series noise stream by

$$C_{\mathrm{ms}} q_{m,h} (\sqrt{\log(e\kappa_{m,h})} + \tau_m).$$

Thus block-diagonal atom libraries, self-adjoint off-diagonal edge atoms, and cluster/frame dictionaries supported on a low-total-rank orthogonal block family are controlled by their maximum incident squared-amplitude load, rather than by the total squared-amplitude load.

Proof. For each active atom, self-adjoint support on $P_{m,h,S_{m,h,i,r}}$ gives

$$G_{m,h,i,r}^2 \preceq b_{m,h,i,r}^2 P_{m,h,S_{m,h,i,r}}.$$

Hence

$$\sum_{r \in \mathcal{R}_{m,h,i}} G_{m,h,i,r}^2 \preceq \sum_{\beta \in \mathcal{B}_{m,h}} \left(\sum_{\substack{r \in \mathcal{R}_{m,h,i} \\ \beta \in S_{m,h,i,r}}} b_{m,h,i,r}^2 \right) P_{m,h,\beta}.$$

The projections are orthogonal, so the operator norm of the right side is the largest displayed block load, at most $q_{m,h}^2$ outside F_m^{bl}. The same calculation after summing over i gives

$$\left\| \sum_{i,r} G_{m,h,i,r}^2 \right\|_{\mathrm{op}} \le n_m q_{m,h}^2.$$

Both square functions are supported on $\sum_{\beta \in \mathcal{B}_{m,h}} P_{m,h,\beta}$, whose rank is at most $\kappa_{m,h}$. Therefore all four hypotheses defining $(F_m^{\mathrm{ms}})^c$ in Corollary 7.90 hold on $(F_m^{\mathrm{bl}})^c$. Apply that corollary with $F_m^{\mathrm{ms}} = F_m^{\mathrm{bl}}$. □

Corollary 7.94 (Bounded weighted block-design matrix-series noise)**.** *In the setting of Corollary 7.93, suppose that for every active $h \le J_m$ there are deterministic $q_{m,h} \ge 0$, $s_{m,h} > 0$, and weights $\omega_{m,h,i,r,\beta} \ge 0$. Let $F_m^{\mathrm{wbd}} \in \mathcal{K}_m$*

be the event that, for some $h \le J_m$, at least one of the following weighted block-design bounds fails:

$$b_{m,h,i,r}^2 \le \frac{q_{m,h}^2}{s_{m,h}}\,\omega_{m,h,i,r,\beta} \qquad \text{whenever } \beta \in S_{m,h,i,r},$$

$$\max_{i\le n_m}\max_{\beta\in\mathcal{B}_{m,h}} \sum_{\substack{r\in\mathcal{R}_{m,h,i}\\ \beta\in S_{m,h,i,r}}} \omega_{m,h,i,r,\beta} \le s_{m,h},$$

and

$$\max_{\beta\in\mathcal{B}_{m,h}} \sum_{i=1}^{n_m} \sum_{\substack{r\in\mathcal{R}_{m,h,i}\\ \beta\in S_{m,h,i,r}}} \omega_{m,h,i,r,\beta} \le n_m s_{m,h}.$$

If

$$\sum_m \left[\mathbb{P}(F_m^{\mathrm{lift}}) + \mathbb{P}(F_m^{\mathrm{wbd}}) + 2J_m(n_m+1)e^{-\tau_m^2}\right] < \infty,$$

then, with probability one, for all sufficiently large m, the active-prefix and fixed-menu conclusions of Corollary 7.86 hold after scaling the h-th weighted block-design matrix-series noise stream by

$$C_{\mathrm{ms}} q_{m,h}(\sqrt{\log(e\kappa_{m,h})} + \tau_m).$$

In particular, if the support law is almost surely supported on the displayed weighted block-design constraints, then $F_m^{\mathrm{wbd}} = \emptyset$. In the unweighted case, if every active block is incident to at most $d_{m,h}$ atoms from each row and at most $n_m d_{m,h}$ atoms in total, and if $b_{m,h,i,r} \le q_{m,h}/\sqrt{d_{m,h}}$ for $d_{m,h} \ge 1$, then the same conclusion holds with no support-concentration payment. Thus random regular block hypergraphs, matching and permutation block designs, balanced without-replacement assignments, and deterministic or randomly chosen weighted-degree block samplers enter at the same rank-sensitive matrix-series scale whenever their weighted block degrees are deterministic or fail only summably.

The explicit operator-tail schedules are exactly those of Corollary 7.87: if $J_m(n_m+1) \le (eD_m)^\Gamma$, then all integer dimensions are covered by $\tau_D^2 = S\log(eD)$ with $S > \Gamma + 1$, while sparse stretched-exponential dimension schedules need only $S > \Gamma$.

Proof. On $(F_m^{\mathrm{wbd}})^c$, for every active h, i, β,

$$\begin{aligned}\sum_{\substack{r\in\mathcal{R}_{m,h,i}\\ \beta\in S_{m,h,i,r}}} b_{m,h,i,r}^2 &\le \frac{q_{m,h}^2}{s_{m,h}} \sum_{\substack{r\in\mathcal{R}_{m,h,i}\\ \beta\in S_{m,h,i,r}}} \omega_{m,h,i,r,\beta}\\ &\le q_{m,h}^2.\end{aligned}$$

The same calculation after summing over all i gives, for every $\beta \in \mathcal{B}_{m,h}$,

$$\sum_{i=1}^{n_m} \sum_{\substack{r \in \mathcal{R}_{m,h,i} \\ \beta \in S_{m,h,i,r}}} b_{m,h,i,r}^2 \le n_m q_{m,h}^2.$$

Therefore the block-load failure event F_m^{bl} from Corollary 7.93 is contained in F_m^{wbd}. Applying that corollary gives the almost-sure active-prefix and fixed-menu conclusions. The examples are the corresponding choices of deterministic weights ω, and the schedule statement is Corollary 7.87. □

Corollary 7.95 (Bounded-overlap block-hypergraph matrix-series noise)**.** *In the setting of Corollary 7.93, suppose that for every active $h \le J_m$ there is a deterministic degree parameter $d_{m,h} \ge 1$. Let $F_m^{\mathrm{bhg}} \in \mathcal{K}_m$ be the event that, for some $h \le J_m$, at least one of the two block-overlap bounds*

$$\max_{i \le n_m} \max_{\beta \in \mathcal{B}_{m,h}} \#\{r \in \mathcal{R}_{m,h,i} : \beta \in S_{m,h,i,r}\} \le d_{m,h}$$

or

$$\max_{\beta \in \mathcal{B}_{m,h}} \sum_{i=1}^{n_m} \#\{r \in \mathcal{R}_{m,h,i} : \beta \in S_{m,h,i,r}\} \le n_m d_{m,h}$$

fails. Assume also that, on $(F_m^{\mathrm{bhg}})^c$,

$$b_{m,h,i,r} \le \frac{q_{m,h}}{\sqrt{d_{m,h}}} \qquad (r \in \mathcal{R}_{m,h,i})$$

for deterministic $q_{m,h} \ge 0$. If

$$\sum_m \left[\mathbb{P}(F_m^{\mathrm{lift}}) + \mathbb{P}(F_m^{\mathrm{bhg}}) + 2J_m(n_m+1)e^{-\tau_m^2} \right] < \infty,$$

then, with probability one, for all sufficiently large m, the active-prefix and fixed-menu conclusions of Corollary 7.86 hold after scaling the h-th bounded-overlap block-hypergraph matrix-series noise stream by

$$C_{\mathrm{ms}} q_{m,h} (\sqrt{\log(e\kappa_{m,h})} + \tau_m).$$

In particular, if the support law is supported on block hypergraphs with the displayed row and total block-overlap degrees, then $F_m^{\mathrm{bhg}} = \emptyset$. This includes deterministic block designs, random regular block hypergraphs, balanced block dictionaries, and without-replacement block-hyperedge samplers conditioned on their degree sequence. Their all-integer and sparse schedules are exactly the usual operator-tail schedules, with no block-load concentration payment.

Proof. Apply Corollary 7.94 with $s_{m,h} = d_{m,h}$ and

$$\omega_{m,h,i,r,\beta} = \mathbf{1}_{\{\beta \in S_{m,h,i,r}\}}.$$

On $(F_m^{\mathrm{bhg}})^c$, the two displayed overlap bounds are exactly the row and total weighted-degree constraints. The amplitude condition gives

$$b_{m,h,i,r}^2 \le q_{m,h}^2 / d_{m,h} = q_{m,h}^2 \omega_{m,h,i,r,\beta} / d_{m,h}$$

whenever $\beta \in S_{m,h,i,r}$. Hence the weighted block-design failure event is contained in F_m^{bhg}, and the cited corollary proves the claim. □

Corollary 7.96 (Heterogeneous bounded-overlap block-hypergraph matrix-series noise)**.** *In the setting of Corollary 7.93, suppose that for every active $h \le J_m$ there are deterministic local degree profiles*

$$\delta^{\mathrm{ind}}_{m,h,i,\beta} \ge 1, \qquad \Delta^{\mathrm{tot}}_{m,h,\beta} \ge 1 \qquad (i \le n_m,\ \beta \in \mathcal{B}_{m,h}).$$

Let $F_m^{\mathrm{hbhg}} \in \mathcal{K}_m$ be the event that, for some $h \le J_m$, at least one of the heterogeneous block-overlap bounds

$$\#\{r \in \mathcal{R}_{m,h,i} : \beta \in S_{m,h,i,r}\} \le \delta^{\mathrm{ind}}_{m,h,i,\beta} \qquad (i \le n_m,\ \beta \in \mathcal{B}_{m,h})$$

or

$$\sum_{i=1}^{n_m} \#\{r \in \mathcal{R}_{m,h,i} : \beta \in S_{m,h,i,r}\} \le \Delta^{\mathrm{tot}}_{m,h,\beta} \qquad (\beta \in \mathcal{B}_{m,h})$$

fails. Assume also that, on $(F_m^{\mathrm{hbhg}})^c$, every active incidence $\beta \in S_{m,h,i,r}$ satisfies

$$b^2_{m,h,i,r} \le \min\left\{ \frac{q^2_{m,h}}{\delta^{\mathrm{ind}}_{m,h,i,\beta}}, \frac{n_m q^2_{m,h}}{\Delta^{\mathrm{tot}}_{m,h,\beta}} \right\}$$

for deterministic $q_{m,h} \ge 0$. If

$$\sum_m \left[\mathbb{P}(F_m^{\mathrm{lift}}) + \mathbb{P}(F_m^{\mathrm{hbhg}}) + 2J_m(n_m+1)e^{-\tau_m^2} \right] < \infty,$$

then, with probability one, for all sufficiently large m, the active-prefix and fixed-menu conclusions of Corollary 7.86 hold after scaling the h-th heterogeneous bounded-overlap block-hypergraph matrix-series noise stream by

$$C_{\mathrm{ms}} q_{m,h}(\sqrt{\log(e\kappa_{m,h})} + \tau_m).$$

Thus irregular degree-certified block hypergraphs, stratified block dictionaries, and conditioned configuration-model block samplers are covered at the local-degree normalization scale, with no reduction to the worst active block degree and no support-concentration payment when the displayed degree profiles hold deterministically.

Proof. On $(F_m^{\mathrm{hbhg}})^c$, for every active h, i, β,

$$\begin{aligned} \sum_{\substack{r \in \mathcal{R}_{m,h,i} \\ \beta \in S_{m,h,i,r}}} b^2_{m,h,i,r} &\le \#\{r \in \mathcal{R}_{m,h,i} : \beta \in S_{m,h,i,r}\} \frac{q^2_{m,h}}{\delta^{\mathrm{ind}}_{m,h,i,\beta}} \\ &\le q^2_{m,h}. \end{aligned}$$

Similarly, for every active h, β,

$$\begin{aligned} \sum_{i=1}^{n_m} \sum_{\substack{r \in \mathcal{R}_{m,h,i} \\ \beta \in S_{m,h,i,r}}} b^2_{m,h,i,r} &\le \left(\sum_{i=1}^{n_m} \#\{r \in \mathcal{R}_{m,h,i} : \beta \in S_{m,h,i,r}\} \right) \frac{n_m q^2_{m,h}}{\Delta^{\mathrm{tot}}_{m,h,\beta}} \\ &\le n_m q^2_{m,h}. \end{aligned}$$

Therefore the block-load failure event F_m^{bl} from Corollary 7.93 is contained in F_m^{hbhg}, and that corollary gives the asserted finite-lift active-prefix and fixed-menu conclusions. □

Corollary 7.97 (Row-wise block-permutation matrix-series noise)**.** *In the setting of Corollary 7.93, suppose that every active atom has singleton block support*

$$S_{m,h,i,r} = \{\beta_{m,h,i,r}\} \subseteq \mathcal{B}_{m,h}.$$

For each active $h \le J_m$, let $d_{m,h} \ge 1$ be deterministic and suppose that each row index set $\mathcal{R}_{m,h,i}$ is decomposed into

$$\mathcal{R}_{m,h,i} = \mathcal{R}_{m,h,i}^{(1)} \cup \cdots \cup \mathcal{R}_{m,h,i}^{(d_{m,h})},$$

so that, outside an event $F_m^{\mathrm{perm}} \in \mathcal{K}_m$, each layer is block-injective:

$$\#\{r \in \mathcal{R}_{m,h,i}^{(\ell)} : \beta_{m,h,i,r} = \beta\} \le 1 \qquad (i \le n_m,\ \beta \in \mathcal{B}_{m,h},\ 1 \le \ell \le d_{m,h}).$$

Assume also that, on $(F_m^{\mathrm{perm}})^c$,

$$b_{m,h,i,r} \le \frac{q_{m,h}}{\sqrt{d_{m,h}}} \qquad (r \in \mathcal{R}_{m,h,i})$$

for deterministic $q_{m,h} \ge 0$. If

$$\sum_m \left[\mathbb{P}(F_m^{\mathrm{lift}}) + \mathbb{P}(F_m^{\mathrm{perm}}) + 2J_m(n_m+1)e^{-\tau_m^2}\right] < \infty,$$

then, with probability one, for all sufficiently large m, the active-prefix and fixed-menu conclusions of Corollary 7.86 hold after scaling the h-th row-wise block-permutation matrix-series noise stream by

$$C_{\mathrm{ms}} q_{m,h}(\sqrt{\log(e\kappa_{m,h})} + \tau_m).$$

In particular, $F_m^{\mathrm{perm}} = \emptyset$ for laws supported on row-wise block permutations, row-wise partial matchings, $d_{m,h}$-round balanced without-replacement block assignments, or mixtures of such laws. Thus these random block assignment models have the same active-prefix finite-lift schedule as deterministic block designs: if $J_m(n_m+1) \le (eD_m)^\Gamma$, all integer dimensions are covered by $\tau_D^2 = S\log(eD)$ with $S > \Gamma + 1$, while sparse stretched-exponential dimension schedules need only $S > \Gamma$.

Proof. Apply Corollary 7.94 with $s_{m,h} = d_{m,h}$ and

$$\omega_{m,h,i,r,\beta} = \mathbf{1}_{\{\beta = \beta_{m,h,i,r}\}}.$$

On $(F_m^{\mathrm{perm}})^c$, the block-injectivity of each layer gives row-block weighted degree at most $d_{m,h}$. Summing this row bound over $i \le n_m$ gives total block weighted degree at most $n_m d_{m,h}$. The amplitude cap is exactly

$$b_{m,h,i,r}^2 \le q_{m,h}^2/d_{m,h} = q_{m,h}^2 \omega_{m,h,i,r,\beta_{m,h,i,r}}/d_{m,h}.$$

Thus the weighted block-design failure event is contained in F_m^{perm}, and the cited corollary proves the claim. □

Corollary 7.98 (Random block-load atomic matrix-series noise)**.** *In the setting of Corollary 7.93, suppose that the block projections and amplitudes are measurable with respect to a sigma-field $\mathcal{L}_m \subseteq \mathcal{K}_m$. Write*

$$\eta_{m,h,i,r,\beta} = \mathbf{1}_{\{\beta \in S_{m,h,i,r}\}}.$$

Assume that, for each fixed active $h \le J_m$ and $\beta \in \mathcal{B}_{m,h}$, the variables $\eta_{m,h,i,r,\beta}$ are conditionally independent over the pairs (i,r), given $\mathcal{L}_m$, and that

$$\mathbb{P}\{\eta_{m,h,i,r,\beta} = 1 \mid \mathcal{L}_m\} \le p_{m,h,i,r,\beta}.$$

Here the $p_{m,h,i,r,\beta}$ are $\mathcal{L}_m$-measurable. Let deterministic envelopes $\mu^{\mathrm{ind}}_{m,h}$, $\sigma^{2,\mathrm{ind}}_{m,h}$, $\mu^{\mathrm{tot}}_{m,h}$, $\sigma^{2,\mathrm{tot}}_{m,h}$, and $\omega_{m,h}$ satisfy, almost surely,

$$\max_{i,\beta} \sum_{r \in \mathcal{R}_{m,h,i}} b^2_{m,h,i,r} p_{m,h,i,r,\beta} \le \mu^{\mathrm{ind}}_{m,h},$$

$$\max_{i,\beta} \sum_{r \in \mathcal{R}_{m,h,i}} b^4_{m,h,i,r} p_{m,h,i,r,\beta} \le \sigma^{2,\mathrm{ind}}_{m,h},$$

$$\max_{\beta} \sum_{i=1}^{n_m} \sum_{r \in \mathcal{R}_{m,h,i}} b^2_{m,h,i,r} p_{m,h,i,r,\beta} \le \mu^{\mathrm{tot}}_{m,h},$$

$$\max_{\beta} \sum_{i=1}^{n_m} \sum_{r \in \mathcal{R}_{m,h,i}} b^4_{m,h,i,r} p_{m,h,i,r,\beta} \le \sigma^{2,\mathrm{tot}}_{m,h}, \qquad \max_{i,r} b^2_{m,h,i,r} \le \omega_{m,h}.$$

Let $u^{\mathrm{ind}}_{m,h}, u^{\mathrm{tot}}_{m,h} \ge 0$ be deterministic and suppose

$$\mu^{\mathrm{ind}}_{m,h} + u^{\mathrm{ind}}_{m,h} \le q^2_{m,h}, \qquad \mu^{\mathrm{tot}}_{m,h} + u^{\mathrm{tot}}_{m,h} \le n_m q^2_{m,h}.$$

With the convention that a Bernstein term with zero denominator is omitted, put

$$\mathfrak{B}_m := \sum_{h=1}^{J_m} |\mathcal{B}_{m,h}| \left[n_m \exp\left\{ -\frac{(u^{\mathrm{ind}}_{m,h})^2}{2(\sigma^{2,\mathrm{ind}}_{m,h} + \omega_{m,h} u^{\mathrm{ind}}_{m,h}/3)} \right\} + \exp\left\{ -\frac{(u^{\mathrm{tot}}_{m,h})^2}{2(\sigma^{2,\mathrm{tot}}_{m,h} + \omega_{m,h} u^{\mathrm{tot}}_{m,h}/3)} \right\} \right].$$

If

$$\sum_m \left[\mathbb{P}(F^{\mathrm{lift}}_m) + \mathfrak{B}_m + 2J_m(n_m+1)e^{-\tau_m^2} \right] < \infty,$$

then, with probability one, for all sufficiently large m, the active-prefix and fixed-menu conclusions of Corollary 7.86 hold after scaling the h-th random block-atomic matrix-series noise stream by

$$C_{\mathrm{ms}} q_{m,h} (\sqrt{\log(e\kappa_{m,h})} + \tau_m).$$

In particular, independent random block choices, random edge-atom supports, and random clustered frame dictionaries are covered whenever their conditional incident block loads have summable scalar-Bernstein tails.

Proof. For fixed m, h, i, β, condition on $\mathcal{L}_m$ and apply scalar Bernstein to the independent variables

$$b_{m,h,i,r}^2 \left(\eta_{m,h,i,r,\beta} - \mathbb{E}[\eta_{m,h,i,r,\beta} \mid \mathcal{L}_m]\right).$$

Their conditional mean budget, variance proxy, and jump are bounded by $\mu_{m,h}^{\mathrm{ind}}$, $\sigma_{m,h}^{2,\mathrm{ind}}$, and $\omega_{m,h}$. Therefore the corresponding individual block load exceeds $q_{m,h}^2$ with conditional probability at most the displayed individual Bernstein exponential. A union bound over $i \le n_m$ and $\beta \in \mathcal{B}_{m,h}$ gives the first part of $\mathfrak{B}_m$.

The same conditional Bernstein argument, now summing over all pairs (i, r) for fixed h, β, shows that the total block load exceeds $n_m q_{m,h}^2$ with conditional probability at most the displayed total-load exponential. Union over β and $h \le J_m$ gives $\mathbb{P}(F_m^{\mathrm{bl}}) \le \mathfrak{B}_m$. The result follows from Corollary 7.93. □

Corollary 7.99 (Uniform random block-support matrix-series noise)**.** *In the setting of Corollary 7.93, suppose that the block projections and amplitudes are measurable with respect to a sigma-field $\mathcal{L}_m \subseteq \mathcal{K}_m$. Put*

$$B_{m,h}^{\#} = |\mathcal{B}_{m,h}|.$$

For each active atom let $s_{m,h,i,r} \in [0, B_{m,h}^{\#}]$ and $a_{m,h,i,r} \ge 0$ be $\mathcal{L}_m$-measurable quantities such that

$$b_{m,h,i,r} \le a_{m,h,i,r}.$$

Assume that, conditionally on $\mathcal{L}_m$, the support sets $S_{m,h,i,r}$ are independent over the pairs (i, r), and for every $\beta \in \mathcal{B}_{m,h}$,

$$\mathbb{P}\{\beta \in S_{m,h,i,r} \mid \mathcal{L}_m\} \le \frac{s_{m,h,i,r}}{B_{m,h}^{\#}}.$$

Let deterministic envelopes $\mu_{m,h}^{\mathrm{ind}}$, $\sigma_{m,h}^{2,\mathrm{ind}}$, $\mu_{m,h}^{\mathrm{tot}}$, $\sigma_{m,h}^{2,\mathrm{tot}}$, and $\omega_{m,h}$ satisfy, almost surely,

$$\max_{i \le n_m} \sum_{r \in \mathcal{R}_{m,h,i}} a_{m,h,i,r}^2 \frac{s_{m,h,i,r}}{B_{m,h}^{\#}} \le \mu_{m,h}^{\mathrm{ind}},$$

$$\max_{i \le n_m} \sum_{r \in \mathcal{R}_{m,h,i}} a_{m,h,i,r}^4 \frac{s_{m,h,i,r}}{B_{m,h}^{\#}} \le \sigma_{m,h}^{2,\mathrm{ind}},$$

$$\sum_{i=1}^{n_m} \sum_{r \in \mathcal{R}_{m,h,i}} a_{m,h,i,r}^2 \frac{s_{m,h,i,r}}{B_{m,h}^{\#}} \le \mu_{m,h}^{\mathrm{tot}},$$

$$\sum_{i=1}^{n_m} \sum_{r \in \mathcal{R}_{m,h,i}} a_{m,h,i,r}^4 \frac{s_{m,h,i,r}}{B_{m,h}^{\#}} \le \sigma_{m,h}^{2,\mathrm{tot}}, \qquad \max_{i,r} a_{m,h,i,r}^2 \le \omega_{m,h}.$$

Let $u_{m,h}^{\mathrm{ind}}, u_{m,h}^{\mathrm{tot}} \ge 0$ be deterministic and suppose

$$\mu_{m,h}^{\mathrm{ind}} + u_{m,h}^{\mathrm{ind}} \le q_{m,h}^2, \qquad \mu_{m,h}^{\mathrm{tot}} + u_{m,h}^{\mathrm{tot}} \le n_m q_{m,h}^2.$$

With the convention that a Bernstein term with zero denominator is omitted, put

$$\mathfrak{U}_m := \sum_{h=1}^{J_m} B_{m,h}^{\#} \left[n_m \exp\left\{ -\frac{(u_{m,h}^{\mathrm{ind}})^2}{2(\sigma_{m,h}^{2,\mathrm{ind}} + \omega_{m,h} u_{m,h}^{\mathrm{ind}}/3)} \right\} + \exp\left\{ -\frac{(u_{m,h}^{\mathrm{tot}})^2}{2(\sigma_{m,h}^{2,\mathrm{tot}} + \omega_{m,h} u_{m,h}^{\mathrm{tot}}/3)} \right\} \right].$$

If

$$\sum_m \left[\mathbb{P}(F_m^{\mathrm{lift}}) + \mathfrak{U}_m + 2J_m(n_m+1)e^{-\tau_m^2} \right] < \infty,$$

then, with probability one, for all sufficiently large m, the active-prefix and fixed-menu conclusions of Corollary 7.86 hold after scaling the h-th uniformly block-sampled matrix-series noise stream by

$$C_{\mathrm{ms}} q_{m,h} (\sqrt{\log(e\kappa_{m,h})} + \tau_m).$$

This covers atoms choosing a fixed number of blocks uniformly without replacement, independent Bernoulli block thinning, random block hyperedges, and randomly clustered dictionaries, with $s_{m,h,i,r}$ equal to the conditional expected support size or any upper bound for it.

Proof. Set

$$p_{m,h,i,r,\beta} = \frac{s_{m,h,i,r}}{B_{m,h}^{\#}}.$$

For each fixed h, β, the indicators $\mathbf{1}_{\{\beta \in S_{m,h,i,r}\}}$ are conditionally independent over (i,r), and their conditional probabilities are bounded by $p_{m,h,i,r,\beta}$. Since $b_{m,h,i,r} \le a_{m,h,i,r}$, the displayed envelopes imply the four mean and variance hypotheses and the jump bound in Corollary 7.98. The conclusion follows from that corollary, with $\mathfrak{B}_m \le \mathfrak{U}_m$. The examples have exactly the displayed uniform inclusion bound. □

Corollary 7.100 (Uniform block-support profile schedules)**.** *In the setting of Corollary 7.99, retain the deterministic envelopes and slacks*

$$\mu_{m,h}^{\mathrm{ind}},\ \sigma_{m,h}^{2,\mathrm{ind}},\ \mu_{m,h}^{\mathrm{tot}},\ \sigma_{m,h}^{2,\mathrm{tot}},\ \omega_{m,h},\ u_{m,h}^{\mathrm{ind}},\ u_{m,h}^{\mathrm{tot}}.$$

Define the two Bernstein exponents, with the zero-denominator convention interpreted as $+\infty$, by

$$\Phi_{m,h}^{\mathrm{ind}} = \begin{cases} \dfrac{(u_{m,h}^{\mathrm{ind}})^2}{2(\sigma_{m,h}^{2,\mathrm{ind}} + \omega_{m,h} u_{m,h}^{\mathrm{ind}}/3)}, & \sigma_{m,h}^{2,\mathrm{ind}} + \omega_{m,h} u_{m,h}^{\mathrm{ind}}/3 > 0, \\ +\infty, & \sigma_{m,h}^{2,\mathrm{ind}} + \omega_{m,h} u_{m,h}^{\mathrm{ind}}/3 = 0, \end{cases}$$

$$\Phi_{m,h}^{\mathrm{tot}} = \begin{cases} \dfrac{(u_{m,h}^{\mathrm{tot}})^2}{2(\sigma_{m,h}^{2,\mathrm{tot}} + \omega_{m,h} u_{m,h}^{\mathrm{tot}}/3)}, & \sigma_{m,h}^{2,\mathrm{tot}} + \omega_{m,h} u_{m,h}^{\mathrm{tot}}/3 > 0, \\ +\infty, & \sigma_{m,h}^{2,\mathrm{tot}} + \omega_{m,h} u_{m,h}^{\mathrm{tot}}/3 = 0. \end{cases}$$

Set

$$\Phi_m = \min_{1\le h\le J_m} \min\{\Phi^{\mathrm{ind}}_{m,h}, \Phi^{\mathrm{tot}}_{m,h}\}, \qquad L^{\mathrm{ubs}}_m = (n_m+1)\left(J_m + \sum_{h=1}^{J_m} B^{\#}_{m,h}\right).$$

Assume that the half-budget inequalities in Corollary 7.99 hold and that $\sum_m \mathbb{P}(F^{\mathrm{lift}}_m) < \infty$. *If, for some* $\gamma > 0$,

$$\Phi_m \ge \log(eL^{\mathrm{ubs}}_m) + (1+\gamma)\log(e+m)$$

and

$$\tau^2_m \ge \log(2L^{\mathrm{ubs}}_m) + (1+\gamma)\log(e+m)$$

eventually, then the active-prefix and fixed-menu conclusions of Corollary 7.86 hold almost surely after scaling the h*-th uniformly block-sampled matrix-series noise stream by*

$$C_{\mathrm{ms}} q_{m,h}\left(\sqrt{\log(e\kappa_{m,h})} + \tau_m\right).$$

In particular, suppose $L^{\mathrm{ubs}}_m \le (eD_m)^{\Gamma}$ *eventually. Along all integer dimensions* $D = 2, 3, \ldots$, *the same conclusion holds if*

$$\Phi_D \ge R\log(eD), \qquad \tau^2_D \ge S\log(eD),$$

eventually for constants $R > \Gamma + 1$ *and* $S > \Gamma + 1$. *Along any sparse schedule* $D_m \ge \exp(cm^{\alpha})$, $c, \alpha > 0$, *the same conclusion holds if*

$$\Phi_m \ge R\log(eD_m), \qquad \tau^2_m \ge S\log(eD_m),$$

eventually for constants $R > \Gamma$ *and* $S > \Gamma$. *Thus heterogeneous amplitudes and expected support sizes need only produce summable Bernstein profile exponents; no homogeneity is required.*

Proof. By the definition of Φ_m,

$$\mathfrak{U}_m \le (n_m+1)\sum_{h=1}^{J_m} B^{\#}_{m,h} e^{-\Phi_m} \le L^{\mathrm{ubs}}_m e^{-\Phi_m}.$$

The arbitrary-growth lower bound on Φ_m gives

$$\mathfrak{U}_m \le e^{-1}(e+m)^{-(1+\gamma)}.$$

Also $J_m(n_m+1) \le L^{\mathrm{ubs}}_m$, so the lower bound on τ^2_m gives

$$2J_m(n_m+1)e^{-\tau^2_m} \le (e+m)^{-(1+\gamma)}.$$

The summability hypothesis in Corollary 7.99 therefore follows.

For all integer dimensions,

$$\mathfrak{U}_D \le (eD)^{\Gamma-R}, \qquad 2J_D(n_D+1)e^{-\tau^2_D} \le 2(eD)^{\Gamma-S},$$

which is summable when $R - \Gamma > 1$ and $S - \Gamma > 1$. For sparse dimension schedules the same bounds are stretched-exponential summable as soon as $R > \Gamma$ and $S > \Gamma$. □

Corollary 7.101 (Homogeneous uniform block-support matrix-series noise)**.** *In the setting of Corollary 7.99, suppose that for each active $h \le J_m$ there are deterministic parameters*

$$N^{\mathrm{ind}}_{m,h}, \quad N^{\mathrm{tot}}_{m,h}, \quad a_{m,h} > 0, \quad s_{m,h} \in [0, B^{\#}_{m,h}]$$

such that, almost surely,

$$|\mathcal{R}_{m,h,i}| \le N^{\mathrm{ind}}_{m,h} \quad (i \le n_m), \qquad \sum_{i=1}^{n_m} |\mathcal{R}_{m,h,i}| \le N^{\mathrm{tot}}_{m,h},$$

and every active atom has $a_{m,h,i,r} \le a_{m,h}$ and $s_{m,h,i,r} \le s_{m,h}$. Put

$$\rho^{\mathrm{ind}}_{m,h} = N^{\mathrm{ind}}_{m,h} a^2_{m,h} \frac{s_{m,h}}{B^{\#}_{m,h}}, \qquad \rho^{\mathrm{tot}}_{m,h} = N^{\mathrm{tot}}_{m,h} a^2_{m,h} \frac{s_{m,h}}{B^{\#}_{m,h}}.$$

Assume the half-budget bounds

$$2\rho^{\mathrm{ind}}_{m,h} \le q^2_{m,h}, \qquad 2\rho^{\mathrm{tot}}_{m,h} \le n_m q^2_{m,h}.$$

With the convention that an exponential term with $\rho = 0$ is omitted, set

$$\mathfrak{H}_m = \sum_{h=1}^{J_m} B^{\#}_{m,h} \left[n_m \exp\left\{ -\frac{3\rho^{\mathrm{ind}}_{m,h}}{8a^2_{m,h}} \right\} + \exp\left\{ -\frac{3\rho^{\mathrm{tot}}_{m,h}}{8a^2_{m,h}} \right\} \right].$$

If

$$\sum_m \left[\mathbb{P}(F^{\mathrm{lift}}_m) + \mathfrak{H}_m + 2J_m(n_m+1)e^{-\tau_m^2} \right] < \infty,$$

then, with probability one, for all sufficiently large m, the active-prefix and fixed-menu conclusions of Corollary 7.86 hold after scaling the h-th homogeneous uniformly block-sampled matrix-series noise stream by

$$C_{\mathrm{ms}} q_{m,h} (\sqrt{\log(e\kappa_{m,h})} + \tau_m).$$

Equivalently, in this homogeneous model the block-load concentration exponent is controlled by the expected number of active atom incidences per block:

$$\frac{\rho^{\mathrm{ind}}_{m,h}}{a^2_{m,h}} = \frac{N^{\mathrm{ind}}_{m,h} s_{m,h}}{B^{\#}_{m,h}}, \qquad \frac{\rho^{\mathrm{tot}}_{m,h}}{a^2_{m,h}} = \frac{N^{\mathrm{tot}}_{m,h} s_{m,h}}{B^{\#}_{m,h}}.$$

Proof. Apply Corollary 7.99 with

$$\mu^{\mathrm{ind}}_{m,h} = \rho^{\mathrm{ind}}_{m,h}, \quad \sigma^{2,\mathrm{ind}}_{m,h} = a^2_{m,h}\rho^{\mathrm{ind}}_{m,h}, \quad \mu^{\mathrm{tot}}_{m,h} = \rho^{\mathrm{tot}}_{m,h}, \quad \sigma^{2,\mathrm{tot}}_{m,h} = a^2_{m,h}\rho^{\mathrm{tot}}_{m,h}, \quad \omega_{m,h} = a^2_{m,h},$$

and choose $u^{\mathrm{ind}}_{m,h} = \rho^{\mathrm{ind}}_{m,h}$ and $u^{\mathrm{tot}}_{m,h} = \rho^{\mathrm{tot}}_{m,h}$. The half-budget assumptions give the two required inequalities. When $\rho > 0$,

$$\frac{u^2}{2(\sigma^2 + \omega u/3)} = \frac{\rho^2}{2(a^2_{m,h}\rho + a^2_{m,h}\rho/3)} = \frac{3\rho}{8a^2_{m,h}},$$

which is the displayed $\mathfrak{H}_m$. The zero-load terms are deterministic and need no exceptional probability. □

Corollary 7.102 (Homogeneous uniform block-support schedules)**.** *In the setting of Corollary 7.101, put*

$$\lambda_{m,h}^{\mathrm{ind}} = \frac{N_{m,h}^{\mathrm{ind}} s_{m,h}}{B_{m,h}^{\#}}, \qquad \lambda_{m,h}^{\mathrm{tot}} = \frac{N_{m,h}^{\mathrm{tot}} s_{m,h}}{B_{m,h}^{\#}},$$

$$\lambda_m = \min_{1\le h\le J_m} \min\{\lambda_{m,h}^{\mathrm{ind}}, \lambda_{m,h}^{\mathrm{tot}}\}, \qquad L_m^{\mathrm{blk}} = \sum_{h=1}^{J_m} B_{m,h}^{\#}(n_m+1).$$

Assume the half-budget bounds of that corollary and $\sum_m \mathbb{P}(F_m^{\mathrm{lift}}) < \infty$*. If, for some* $\gamma > 0$*,*

$$\lambda_m \ge \frac{8}{3}\left[\log(eL_m^{\mathrm{blk}}) + (1+\gamma)\log(e+m)\right]$$

and

$$\tau_m^2 \ge \log(2J_m(n_m+1)) + (1+\gamma)\log(e+m)$$

eventually, then the active-prefix and fixed-menu conclusions of Corollary 7.86 hold almost surely after scaling the h*-th homogeneous uniformly block-sampled matrix-series noise stream by*

$$C_{\mathrm{ms}} q_{m,h}(\sqrt{\log(e\kappa_{m,h})} + \tau_m).$$

In particular, suppose $L_m^{\mathrm{blk}} \le (eD_m)^{\Gamma}$ *eventually. Along all integer dimensions* $D = 2, 3, \ldots$*, the same conclusion holds if*

$$\lambda_D \ge R\log(eD), \qquad \tau_D^2 \ge S\log(eD),$$

eventually for constants

$$R > \frac{8(\Gamma+1)}{3}, \qquad S > \Gamma + 1.$$

Along any sparse schedule $D_m \ge \exp(cm^{\alpha})$*,* $c, \alpha > 0$*, the same conclusion holds if*

$$\lambda_m \ge R\log(eD_m), \qquad \tau_m^2 \ge S\log(eD_m),$$

eventually for constants

$$R > \frac{8\Gamma}{3}, \qquad S > \Gamma.$$

Thus homogeneous block-sampled matrix-series perturbations are covered whenever the expected individual and total atom incidences per active block dominate the active block-count logarithm at these margins.

Proof. For every active h,

$$\frac{\rho_{m,h}^{\mathrm{ind}}}{a_{m,h}^2} = \lambda_{m,h}^{\mathrm{ind}}, \qquad \frac{\rho_{m,h}^{\mathrm{tot}}}{a_{m,h}^2} = \lambda_{m,h}^{\mathrm{tot}}.$$

Hence the homogeneous Bernstein budget in Corollary 7.101 satisfies

$$\mathfrak{H}_m \le L_m^{\mathrm{blk}} \exp\left\{-\frac{3\lambda_m}{8}\right\}.$$

The arbitrary-growth hypothesis gives

$$L_m^{\mathrm{blk}} \exp\left\{-\frac{3\lambda_m}{8}\right\} \le e^{-1}(e+m)^{-(1+\gamma)},$$

and the displayed lower bound on τ_m^2 gives

$$2J_m(n_m+1)e^{-\tau_m^2} \le (e+m)^{-(1+\gamma)}.$$

Together with the lift-exception summability, the summability hypothesis of Corollary 7.101 follows.

For all integer dimensions,

$$\mathfrak{H}_D \le (eD)^{\Gamma-3R/8}, \qquad 2J_D(n_D+1)e^{-\tau_D^2} \le 2(eD)^{\Gamma-S}.$$

These two series are summable when $3R/8-\Gamma>1$ and $S-\Gamma>1$. For a sparse schedule $D_m \ge \exp(cm^\alpha)$, the same bounds are stretched-exponential summable as soon as $3R/8>\Gamma$ and $S>\Gamma$. □

Corollary 7.103 (Random weighted block-load atomic matrix-series noise)**.** *In the setting of Corollary 7.93, let $\mathcal{L}_m \subseteq \mathcal{K}_m$ be a sigma-field with respect to which the block projections are measurable. The support sets and amplitudes may be $\mathcal{K}_m$-measurable random objects. Put*

$$W_{m,h,i,r,\beta} = b_{m,h,i,r}^2 \mathbf{1}_{\{\beta \in S_{m,h,i,r}\}}.$$

Assume that, for each fixed active $h \le J_m$ and $\beta \in \mathcal{B}_{m,h}$, the nonnegative variables $W_{m,h,i,r,\beta}$ are conditionally independent over the pairs (i,r), given $\mathcal{L}_m$. Let deterministic envelopes $\mu_{m,h}^{\mathrm{ind}}$, $\sigma_{m,h}^{2,\mathrm{ind}}$, $\mu_{m,h}^{\mathrm{tot}}$, $\sigma_{m,h}^{2,\mathrm{tot}}$, and $\omega_{m,h}$ satisfy, almost surely,

$$\max_{i,\beta} \sum_{r \in \mathcal{R}_{m,h,i}} \mathbb{E}[W_{m,h,i,r,\beta} \mid \mathcal{L}_m] \le \mu_{m,h}^{\mathrm{ind}},$$

$$\max_{i,\beta} \sum_{r \in \mathcal{R}_{m,h,i}} \mathbb{E}[W_{m,h,i,r,\beta}^2 \mid \mathcal{L}_m] \le \sigma_{m,h}^{2,\mathrm{ind}},$$

$$\max_{\beta} \sum_{i=1}^{n_m} \sum_{r \in \mathcal{R}_{m,h,i}} \mathbb{E}[W_{m,h,i,r,\beta} \mid \mathcal{L}_m] \le \mu_{m,h}^{\mathrm{tot}},$$

$$\max_{\beta} \sum_{i=1}^{n_m} \sum_{r \in \mathcal{R}_{m,h,i}} \mathbb{E}[W_{m,h,i,r,\beta}^2 \mid \mathcal{L}_m] \le \sigma_{m,h}^{2,\mathrm{tot}}, \qquad \max_{i,r,\beta} W_{m,h,i,r,\beta} \le \omega_{m,h}.$$

Let $u_{m,h}^{\mathrm{ind}}, u_{m,h}^{\mathrm{tot}} \ge 0$ be deterministic and suppose

$$\mu_{m,h}^{\mathrm{ind}} + u_{m,h}^{\mathrm{ind}} \le q_{m,h}^2, \qquad \mu_{m,h}^{\mathrm{tot}} + u_{m,h}^{\mathrm{tot}} \le n_m q_{m,h}^2.$$

With the convention that a Bernstein term with zero denominator is omitted, put

$$\mathfrak{W}_m := \sum_{h=1}^{J_m} |\mathcal{B}_{m,h}| \Bigg[n_m \exp\left\{ -\frac{(u_{m,h}^{\mathrm{ind}})^2}{2(\sigma_{m,h}^{2,\mathrm{ind}} + \omega_{m,h} u_{m,h}^{\mathrm{ind}}/3)} \right\} + \exp\left\{ -\frac{(u_{m,h}^{\mathrm{tot}})^2}{2(\sigma_{m,h}^{2,\mathrm{tot}} + \omega_{m,h} u_{m,h}^{\mathrm{tot}}/3)} \right\} \Bigg].$$

If

$$\sum_m \left[\mathbb{P}(F_m^{\mathrm{lift}}) + \mathfrak{W}_m + 2J_m(n_m+1)e^{-\tau_m^2} \right] < \infty,$$

then, with probability one, for all sufficiently large m, the active-prefix and fixed-menu conclusions of Corollary 7.86 hold after scaling the h-th random weighted block-atomic matrix-series noise stream by

$$C_{\mathrm{ms}} q_{m,h} (\sqrt{\log(e\kappa_{m,h})} + \tau_m).$$

This covers independent random amplitudes, random atom intensities, bounded compound-thinned atom fields, and random weighted dictionaries whose realized incident block weights have the displayed conditional Bernstein envelopes.

Proof. For fixed m, h, i, β, condition on $\mathcal{L}_m$ and apply scalar Bernstein to

$$\sum_{r \in \mathcal{R}_{m,h,i}} \left(W_{m,h,i,r,\beta} - \mathbb{E}[W_{m,h,i,r,\beta} \mid \mathcal{L}_m] \right).$$

The conditional variance is bounded by the sum of the conditional second moments, and each centered summand has jump at most $\omega_{m,h}$. Hence the individual block load exceeds $q_{m,h}^2$ only with the displayed individual Bernstein probability. Union over $i \le n_m$ and $\beta \in \mathcal{B}_{m,h}$ gives the first part of $\mathfrak{W}_m$.

The same argument applied to the sum over all pairs (i, r), for fixed h, β, gives the displayed total-load Bernstein probability, and union over $h \le J_m$ and $\beta \in \mathcal{B}_{m,h}$ yields $\mathbb{P}(F_m^{\mathrm{bl}}) \le \mathfrak{W}_m$. The conclusion is then exactly Corollary 7.93. □

Corollary 7.104 (Predictable weighted block-load atomic matrix-series noise)**.** *In the setting of Corollary 7.93, fix for each active m, h a deterministic enumeration*

$$\mathcal{I}_{m,h} = \{(i,r) : 1 \le i \le n_m,\ r \in \mathcal{R}_{m,h,i}\} = \{a = 1, \dots, N_{m,h}\},$$

and write $i(a), r(a)$ for the corresponding pair. Let $(\mathcal{L}_{m,h,a})_{a=0}^{N_{m,h}}$ be a filtration contained in $\mathcal{K}_m$, with the block projections $P_{m,h,\beta}$ measurable with respect to $\mathcal{L}_{m,h,0}$. Put

$$W_{m,h,a,\beta} = b_{m,h,i(a),r(a)}^2 \mathbf{1}_{\{\beta \in S_{m,h,i(a),r(a)}\}}.$$

Assume $W_{m,h,a,\beta}$ is $\mathcal{L}_{m,h,a}$-measurable for every a, β. Define the predictable first and second moment profiles

$$M_{m,h,a,\beta} = \mathbb{E}[W_{m,h,a,\beta} \mid \mathcal{L}_{m,h,a-1}], \qquad V_{m,h,a,\beta} = \mathbb{E}[W_{m,h,a,\beta}^2 \mid \mathcal{L}_{m,h,a-1}].$$

Let deterministic envelopes $\mu_{m,h}^{\mathrm{ind}}$, $\sigma_{m,h}^{2,\mathrm{ind}}$, $\mu_{m,h}^{\mathrm{tot}}$, $\sigma_{m,h}^{2,\mathrm{tot}}$, *and* $\omega_{m,h}$ *satisfy, almost surely,*

$$\max_{i,\beta} \sum_{\substack{a\le N_{m,h}\\ i(a)=i}} M_{m,h,a,\beta} \le \mu_{m,h}^{\mathrm{ind}},$$

$$\max_{i,\beta} \sum_{\substack{a\le N_{m,h}\\ i(a)=i}} V_{m,h,a,\beta} \le \sigma_{m,h}^{2,\mathrm{ind}},$$

$$\max_{\beta} \sum_{a=1}^{N_{m,h}} M_{m,h,a,\beta} \le \mu_{m,h}^{\mathrm{tot}},$$

$$\max_{\beta} \sum_{a=1}^{N_{m,h}} V_{m,h,a,\beta} \le \sigma_{m,h}^{2,\mathrm{tot}}, \qquad \max_{a,\beta} W_{m,h,a,\beta} \le \omega_{m,h}.$$

Let $u_{m,h}^{\mathrm{ind}}, u_{m,h}^{\mathrm{tot}} \ge 0$ *be deterministic and suppose*

$$\mu_{m,h}^{\mathrm{ind}} + u_{m,h}^{\mathrm{ind}} \le q_{m,h}^2, \qquad \mu_{m,h}^{\mathrm{tot}} + u_{m,h}^{\mathrm{tot}} \le n_m q_{m,h}^2.$$

With the convention that a Freedman term with zero denominator is omitted, put

$$\mathfrak{P}_m := \sum_{h=1}^{J_m} |\mathcal{B}_{m,h}| \left[n_m \exp\left\{ -\frac{(u_{m,h}^{\mathrm{ind}})^2}{2(\sigma_{m,h}^{2,\mathrm{ind}} + \omega_{m,h} u_{m,h}^{\mathrm{ind}}/3)} \right\} + \exp\left\{ -\frac{(u_{m,h}^{\mathrm{tot}})^2}{2(\sigma_{m,h}^{2,\mathrm{tot}} + \omega_{m,h} u_{m,h}^{\mathrm{tot}}/3)} \right\} \right].$$

If

$$\sum_m \left[\mathbb{P}(F_m^{\mathrm{lift}}) + \mathfrak{P}_m + 2J_m(n_m+1)e^{-\tau_m^2} \right] < \infty,$$

then, with probability one, for all sufficiently large m, *the active-prefix and fixed-menu conclusions of Corollary 7.86 hold after scaling the* h*-th predictable weighted block-atomic matrix-series noise stream by*

$$C_{\mathrm{ms}} q_{m,h}(\sqrt{\log(e\kappa_{m,h})} + \tau_m).$$

Thus the block-load matrix-series theorem also covers adaptive atom sampling, online random weighted dictionaries, sequential block choices, and without-replacement block exposure whenever the predictable incident-weight budgets have summable scalar-Freedman tails.

Proof. Fix m, h, i, β. Along the enumeration define

$$X_a = \mathbf{1}_{\{i(a)=i\}} (W_{m,h,a,\beta} - M_{m,h,a,\beta}).$$

Then $(X_a, \mathcal{L}_{m,h,a})$ is a scalar martingale-difference sequence, $|X_a| \le \omega_{m,h}$, and its predictable quadratic variation is bounded by

$$\sum_{\substack{a\le N_{m,h}\\ i(a)=i}} V_{m,h,a,\beta} \le \sigma_{m,h}^{2,\mathrm{ind}}.$$

Since the corresponding predictable mean is at most $\mu^{\mathrm{ind}}_{m,h}$, the individual block load can exceed $q^2_{m,h}$ only if $\sum_a X_a > u^{\mathrm{ind}}_{m,h}$. The scalar Freedman inequality therefore gives the displayed individual exponential. Union over $i \le n_m$ and $\beta \in \mathcal{B}_{m,h}$ gives the first part of $\mathfrak{P}_m$.

For the total load at fixed h, β, use the martingale differences $W_{m,h,a,\beta} - M_{m,h,a,\beta}$ over all a. The predictable mean, quadratic variation, and jump are bounded by $\mu^{\mathrm{tot}}_{m,h}$, $\sigma^{2,\mathrm{tot}}_{m,h}$, and $\omega_{m,h}$, respectively. Scalar Freedman gives the displayed total exponential, and union over h and β yields $\mathbb{P}(F^{\mathrm{bl}}_m) \le \mathfrak{P}_m$. Apply Corollary 7.93. □

Corollary 7.105 (Predictable block-support profile matrix-series noise)**.** *In the setting of Corollary 7.93, fix for each active m, h the deterministic enumeration and filtration from Corollary 7.104. Assume that the block projections are $\mathcal{L}_{m,h,0}$-measurable, that $S_{m,h,i(a),r(a)}$ is $\mathcal{L}_{m,h,a}$-measurable, and that there are predictable profiles*

$$c_{m,h,a} \ge 0, \qquad 0 \le p_{m,h,a,\beta} \le 1 \qquad (a \le N_{m,h},\ \beta \in \mathcal{B}_{m,h})$$

which are $\mathcal{L}_{m,h,a-1}$-measurable and satisfy

$$b_{m,h,i(a),r(a)} \le c_{m,h,a}, \qquad \mathbb{P}\{\beta \in S_{m,h,i(a),r(a)} \mid \mathcal{L}_{m,h,a-1}\} \le p_{m,h,a,\beta}.$$

Let deterministic envelopes $\mu^{\mathrm{ind}}_{m,h}$, $\sigma^{2,\mathrm{ind}}_{m,h}$, $\mu^{\mathrm{tot}}_{m,h}$, $\sigma^{2,\mathrm{tot}}_{m,h}$, and $\omega_{m,h}$ satisfy, almost surely,

$$\max_{i,\beta} \sum_{\substack{a \le N_{m,h} \\ i(a)=i}} c^2_{m,h,a} p_{m,h,a,\beta} \le \mu^{\mathrm{ind}}_{m,h},$$

$$\max_{i,\beta} \sum_{\substack{a \le N_{m,h} \\ i(a)=i}} c^4_{m,h,a} p_{m,h,a,\beta} \le \sigma^{2,\mathrm{ind}}_{m,h},$$

$$\max_{\beta} \sum_{a=1}^{N_{m,h}} c^2_{m,h,a} p_{m,h,a,\beta} \le \mu^{\mathrm{tot}}_{m,h},$$

$$\max_{\beta} \sum_{a=1}^{N_{m,h}} c^4_{m,h,a} p_{m,h,a,\beta} \le \sigma^{2,\mathrm{tot}}_{m,h}, \qquad \max_a c^2_{m,h,a} \le \omega_{m,h}.$$

Let $u^{\mathrm{ind}}_{m,h}, u^{\mathrm{tot}}_{m,h} \ge 0$ be deterministic and suppose

$$\mu^{\mathrm{ind}}_{m,h} + u^{\mathrm{ind}}_{m,h} \le q^2_{m,h}, \qquad \mu^{\mathrm{tot}}_{m,h} + u^{\mathrm{tot}}_{m,h} \le n_m q^2_{m,h}.$$

With the convention that a Freedman term with zero denominator is omitted, put

$$\mathfrak{A}_m := \sum_{h=1}^{J_m} |\mathcal{B}_{m,h}| \left[n_m \exp\left\{ -\frac{(u^{\mathrm{ind}}_{m,h})^2}{2(\sigma^{2,\mathrm{ind}}_{m,h} + \omega_{m,h} u^{\mathrm{ind}}_{m,h}/3)} \right\} + \exp\left\{ -\frac{(u^{\mathrm{tot}}_{m,h})^2}{2(\sigma^{2,\mathrm{tot}}_{m,h} + \omega_{m,h} u^{\mathrm{tot}}_{m,h}/3)} \right\} \right].$$

If

$$\sum_m \left[\mathbb{P}(F_m^{\mathrm{lift}}) + \mathfrak{A}_m + 2J_m(n_m+1)e^{-\tau_m^2}\right] < \infty,$$

then, with probability one, for all sufficiently large m, the active-prefix and fixed-menu conclusions of Corollary 7.86 hold after scaling the h-th predictable block-support matrix-series noise stream by

$$C_{\mathrm{ms}} q_{m,h}(\sqrt{\log(e\kappa_{m,h})} + \tau_m).$$

Thus adaptive block samplers, online block dictionaries, urn-type sampling, and without-replacement block exposure need no independence between different atoms: it is enough to control the one-step conditional inclusion probabilities and the predictable amplitude profile. The uniform predictable case is obtained by taking $p_{m,h,a,\beta} \le s_{m,h,a}/|\mathcal{B}_{m,h}|$.

Proof. Put

$$W_{m,h,a,\beta} = b_{m,h,i(a),r(a)}^2 \mathbf{1}_{\{\beta \in S_{m,h,i(a),r(a)}\}}.$$

Then $W_{m,h,a,\beta}$ is $\mathcal{L}_{m,h,a}$-measurable and

$$\mathbb{E}[W_{m,h,a,\beta} \mid \mathcal{L}_{m,h,a-1}] \le c_{m,h,a}^2 p_{m,h,a,\beta},$$

while

$$\mathbb{E}[W_{m,h,a,\beta}^2 \mid \mathcal{L}_{m,h,a-1}] \le c_{m,h,a}^4 p_{m,h,a,\beta}, \qquad W_{m,h,a,\beta} \le c_{m,h,a}^2 \le \omega_{m,h}.$$

The displayed deterministic envelopes therefore imply all predictable first moment, second moment, and jump hypotheses in Corollary 7.104. The half-budget inequalities are the same, and its Freedman budget $\mathfrak{P}_m$ is bounded by $\mathfrak{A}_m$. Applying that corollary gives the result. □

Corollary 7.106 (Predictable block-support profile schedules)**.** *In the setting of Corollary 7.105, retain the deterministic envelopes and slacks*

$$\mu_{m,h}^{\mathrm{ind}},\ \sigma_{m,h}^{2,\mathrm{ind}},\ \mu_{m,h}^{\mathrm{tot}},\ \sigma_{m,h}^{2,\mathrm{tot}},\ \omega_{m,h},\ u_{m,h}^{\mathrm{ind}},\ u_{m,h}^{\mathrm{tot}}.$$

Define the two predictable-support Freedman exponents by

$$\Theta_{m,h}^{\mathrm{ind}} = \begin{cases} \dfrac{(u_{m,h}^{\mathrm{ind}})^2}{2(\sigma_{m,h}^{2,\mathrm{ind}} + \omega_{m,h}u_{m,h}^{\mathrm{ind}}/3)}, & \sigma_{m,h}^{2,\mathrm{ind}} + \omega_{m,h}u_{m,h}^{\mathrm{ind}}/3 > 0, \\ +\infty, & \sigma_{m,h}^{2,\mathrm{ind}} + \omega_{m,h}u_{m,h}^{\mathrm{ind}}/3 = 0, \end{cases}$$

$$\Theta_{m,h}^{\mathrm{tot}} = \begin{cases} \dfrac{(u_{m,h}^{\mathrm{tot}})^2}{2(\sigma_{m,h}^{2,\mathrm{tot}} + \omega_{m,h}u_{m,h}^{\mathrm{tot}}/3)}, & \sigma_{m,h}^{2,\mathrm{tot}} + \omega_{m,h}u_{m,h}^{\mathrm{tot}}/3 > 0, \\ +\infty, & \sigma_{m,h}^{2,\mathrm{tot}} + \omega_{m,h}u_{m,h}^{\mathrm{tot}}/3 = 0. \end{cases}$$

Set

$$\Theta_m = \min_{1\le h\le J_m} \min\{\Theta_{m,h}^{\mathrm{ind}}, \Theta_{m,h}^{\mathrm{tot}}\}, \qquad L_m^{\mathrm{pbs}} = (n_m+1)\left(J_m + \sum_{h=1}^{J_m} |\mathcal{B}_{m,h}|\right).$$

Assume the half-budget inequalities from Corollary 7.105 and $\sum_m \mathbb{P}(F_m^{\mathrm{lift}}) < \infty$. *If, for some* $\gamma > 0$,

$$\Theta_m \geq \log(eL_m^{\mathrm{pbs}}) + (1+\gamma)\log(e+m)$$

and

$$\tau_m^2 \geq \log(2L_m^{\mathrm{pbs}}) + (1+\gamma)\log(e+m)$$

eventually, then the active-prefix and fixed-menu conclusions of Corollary 7.86 hold almost surely after scaling the h*-th predictable block-support matrix-series noise stream by*

$$C_{\mathrm{ms}} q_{m,h}(\sqrt{\log(e\kappa_{m,h})} + \tau_m).$$

In particular, suppose $L_m^{\mathrm{pbs}} \leq (eD_m)^{\Gamma}$ *eventually. Along all integer dimensions* $D = 2, 3, \ldots$, *the same conclusion holds if*

$$\Theta_D \geq R\log(eD), \qquad \tau_D^2 \geq S\log(eD),$$

eventually for constants $R > \Gamma + 1$ *and* $S > \Gamma + 1$. *Along any sparse schedule* $D_m \geq \exp(cm^{\alpha})$, $c, \alpha > 0$, *the same conclusion holds if*

$$\Theta_m \geq R\log(eD_m), \qquad \tau_m^2 \geq S\log(eD_m),$$

eventually for constants $R > \Gamma$ *and* $S > \Gamma$. *Thus heterogeneous adaptive block samplers need only make their predictable support Freedman exponents dominate the active block-count logarithm.*

Proof. By the definition of Θ_m,

$$\mathfrak{A}_m \leq (n_m+1)\sum_{h=1}^{J_m} |\mathcal{B}_{m,h}| e^{-\Theta_m} \leq L_m^{\mathrm{pbs}} e^{-\Theta_m}.$$

The arbitrary-growth lower bound on Θ_m gives

$$\mathfrak{A}_m \leq e^{-1}(e+m)^{-(1+\gamma)}.$$

Also $J_m(n_m+1) \leq L_m^{\mathrm{pbs}}$, so the lower bound on τ_m^2 gives

$$2J_m(n_m+1)e^{-\tau_m^2} \leq (e+m)^{-(1+\gamma)}.$$

The summability hypothesis in Corollary 7.105 therefore follows.

For all integer dimensions,

$$\mathfrak{A}_D \leq (eD)^{\Gamma-R}, \qquad 2J_D(n_D+1)e^{-\tau_D^2} \leq 2(eD)^{\Gamma-S},$$

which is summable when $R - \Gamma > 1$ and $S - \Gamma > 1$. For sparse dimension schedules the same bounds are stretched-exponential summable as soon as $R > \Gamma$ and $S > \Gamma$. □

Corollary 7.107 (Inhomogeneous predictable block-support incidence schedules)**.** *In the setting of Corollary 7.105, suppose that for each active* $h \leq J_m$ *there are deterministic parameters*

$$a_{m,h} > 0, \qquad \lambda_{m,h}^{\mathrm{ind}} > 0, \qquad \lambda_{m,h}^{\mathrm{tot}} > 0$$

such that, almost surely,

$$c_{m,h,a} \le a_{m,h} \qquad (a \le N_{m,h}),$$
$$\max_{i \le n_m} \max_{\beta \in \mathcal{B}_{m,h}} \sum_{\substack{a \le N_{m,h} \\ i(a)=i}} p_{m,h,a,\beta} \le \lambda^{\mathrm{ind}}_{m,h},$$

and

$$\max_{\beta \in \mathcal{B}_{m,h}} \sum_{a=1}^{N_{m,h}} p_{m,h,a,\beta} \le \lambda^{\mathrm{tot}}_{m,h}.$$

Assume the half-budget bounds

$$2a^2_{m,h}\lambda^{\mathrm{ind}}_{m,h} \le q^2_{m,h}, \qquad 2a^2_{m,h}\lambda^{\mathrm{tot}}_{m,h} \le n_m q^2_{m,h}.$$

Put

$$\lambda_m = \min_{1 \le h \le J_m} \min\{\lambda^{\mathrm{ind}}_{m,h}, \lambda^{\mathrm{tot}}_{m,h}\}, \qquad L^{\mathrm{pinh}}_m = \sum_{h=1}^{J_m} |\mathcal{B}_{m,h}|(n_m+1).$$

Assume $\sum_m \mathbb{P}(F^{\mathrm{lift}}_m) < \infty$. *If, for some* $\gamma > 0$,

$$\lambda_m \ge \frac{8}{3}\left[\log(eL^{\mathrm{pinh}}_m) + (1+\gamma)\log(e+m)\right]$$

and

$$\tau^2_m \ge \log\bigl(2J_m(n_m+1)\bigr) + (1+\gamma)\log(e+m)$$

eventually, then the active-prefix and fixed-menu conclusions of Corollary 7.86 hold almost surely after scaling the h*-th inhomogeneous predictable block-support matrix-series noise stream by*

$$C_{\mathrm{ms}} q_{m,h}\bigl(\sqrt{\log(e\kappa_{m,h})} + \tau_m\bigr).$$

In particular, suppose $L^{\mathrm{pinh}}_m \le (eD_m)^{\Gamma}$ *eventually. Along all integer dimensions* $D = 2, 3, \ldots$, *the same conclusion holds if*

$$\lambda_D \ge R\log(eD), \qquad \tau^2_D \ge S\log(eD),$$

eventually for constants

$$R > \frac{8(\Gamma+1)}{3}, \qquad S > \Gamma + 1.$$

Along any sparse schedule $D_m \ge \exp(cm^{\alpha})$, $c, \alpha > 0$, *the same conclusion holds if*

$$\lambda_m \ge R\log(eD_m), \qquad \tau^2_m \ge S\log(eD_m),$$

eventually for constants

$$R > \frac{8\Gamma}{3}, \qquad S > \Gamma.$$

Thus biased adaptive categorical block samplers, inhomogeneous urn exposure, and nonuniform without-replacement block choices are covered once their predictable row-block and total-block incidence masses dominate the active block-count logarithm.

Proof. The displayed bounds imply the deterministic envelopes in Corollary 7.105 with

$$\mu^{\mathrm{ind}}_{m,h} = a^2_{m,h}\lambda^{\mathrm{ind}}_{m,h}, \quad \sigma^{2,\mathrm{ind}}_{m,h} = a^4_{m,h}\lambda^{\mathrm{ind}}_{m,h}, \quad \mu^{\mathrm{tot}}_{m,h} = a^2_{m,h}\lambda^{\mathrm{tot}}_{m,h}, \quad \sigma^{2,\mathrm{tot}}_{m,h} = a^4_{m,h}\lambda^{\mathrm{tot}}_{m,h}, \quad \omega_{m,h} = a^2_{m,h}.$$

Choose

$$u^{\mathrm{ind}}_{m,h} = a^2_{m,h}\lambda^{\mathrm{ind}}_{m,h}, \qquad u^{\mathrm{tot}}_{m,h} = a^2_{m,h}\lambda^{\mathrm{tot}}_{m,h}.$$

The half-budget assumptions give the two budget inequalities. For either load parameter λ,

$$\frac{(a^2_{m,h}\lambda)^2}{2(a^4_{m,h}\lambda + a^2_{m,h}a^2_{m,h}\lambda/3)} = \frac{3\lambda}{8}.$$

Hence the predictable support budget is bounded by

$$\mathfrak{A}_m \le L^{\mathrm{pinh}}_m \exp\left\{-\frac{3\lambda_m}{8}\right\}.$$

The arbitrary-growth, all-integer, and sparse conclusions now follow exactly as in Corollary 7.108. □

Corollary 7.108 (Homogeneous predictable uniform block-support schedules)**.** *In the setting of Corollary 7.105, put $B^{\#}_{m,h} = |\mathcal{B}_{m,h}|$. Suppose that for each active $h \le J_m$ there are deterministic parameters*

$$N^{\mathrm{ind}}_{m,h}, \quad N^{\mathrm{tot}}_{m,h}, \quad a_{m,h} > 0, \quad s_{m,h} \in [0, B^{\#}_{m,h}]$$

such that

$$\#\{a \le N_{m,h} : i(a) = i\} \le N^{\mathrm{ind}}_{m,h} \quad (i \le n_m), \qquad N_{m,h} \le N^{\mathrm{tot}}_{m,h},$$

and the predictable profiles satisfy, almost surely,

$$c_{m,h,a} \le a_{m,h}, \qquad p_{m,h,a,\beta} \le \frac{s_{m,h}}{B^{\#}_{m,h}} \qquad (a \le N_{m,h},\ \beta \in \mathcal{B}_{m,h}).$$

Define

$$\lambda^{\mathrm{ind}}_{m,h} = \frac{N^{\mathrm{ind}}_{m,h}s_{m,h}}{B^{\#}_{m,h}}, \qquad \lambda^{\mathrm{tot}}_{m,h} = \frac{N^{\mathrm{tot}}_{m,h}s_{m,h}}{B^{\#}_{m,h}},$$

and assume the half-budget bounds

$$2a^2_{m,h}\lambda^{\mathrm{ind}}_{m,h} \le q^2_{m,h}, \qquad 2a^2_{m,h}\lambda^{\mathrm{tot}}_{m,h} \le n_m q^2_{m,h}.$$

Set

$$\lambda_m = \min_{1\le h\le J_m} \min\{\lambda^{\mathrm{ind}}_{m,h}, \lambda^{\mathrm{tot}}_{m,h}\}, \qquad L^{\mathrm{pblk}}_m = \sum_{h=1}^{J_m} B^{\#}_{m,h}(n_m+1).$$

Assume $\sum_m \mathbb{P}(F^{\mathrm{lift}}_m) < \infty$. If, for some $\gamma > 0$,

$$\lambda_m \ge \frac{8}{3}\left[\log(eL^{\mathrm{pblk}}_m) + (1+\gamma)\log(e+m)\right]$$

and

$$\tau^2_m \ge \log(2J_m(n_m+1)) + (1+\gamma)\log(e+m)$$

eventually, then the active-prefix and fixed-menu conclusions of Corollary 7.86 hold almost surely after scaling the h-th homogeneous predictable uniformly block-sampled matrix-series noise stream by

$$C_{\mathrm{ms}} q_{m,h}(\sqrt{\log(e\kappa_{m,h})} + \tau_m).$$

In particular, suppose $L_m^{\mathrm{pblk}} \le (eD_m)^{\Gamma}$ eventually. Along all integer dimensions $D = 2, 3, \dots$, the same conclusion holds if

$$\lambda_D \ge R \log(eD), \qquad \tau_D^2 \ge S \log(eD),$$

eventually for constants

$$R > \frac{8(\Gamma+1)}{3}, \qquad S > \Gamma + 1.$$

Along any sparse schedule $D_m \ge \exp(cm^{\alpha})$, $c, \alpha > 0$, the same conclusion holds if

$$\lambda_m \ge R \log(eD_m), \qquad \tau_m^2 \ge S \log(eD_m),$$

eventually for constants

$$R > \frac{8\Gamma}{3}, \qquad S > \Gamma.$$

Thus adaptive uniform block samplers, online urn models, and without-replacement block exposures have the same incidence thresholds as the independent homogeneous block-support model; independence is replaced by predictability of the one-step inclusion profile.

Proof. For every active h, set

$$\rho_{m,h}^{\mathrm{ind}} = a_{m,h}^2 \lambda_{m,h}^{\mathrm{ind}}, \qquad \rho_{m,h}^{\mathrm{tot}} = a_{m,h}^2 \lambda_{m,h}^{\mathrm{tot}}.$$

The homogeneous bounds imply the deterministic envelopes in Corollary 7.105 with

$$\mu_{m,h}^{\mathrm{ind}} = \rho_{m,h}^{\mathrm{ind}}, \quad \sigma_{m,h}^{2,\mathrm{ind}} = a_{m,h}^2 \rho_{m,h}^{\mathrm{ind}}, \quad \mu_{m,h}^{\mathrm{tot}} = \rho_{m,h}^{\mathrm{tot}}, \quad \sigma_{m,h}^{2,\mathrm{tot}} = a_{m,h}^2 \rho_{m,h}^{\mathrm{tot}}, \quad \omega_{m,h} = a_{m,h}^2.$$

Choose $u_{m,h}^{\mathrm{ind}} = \rho_{m,h}^{\mathrm{ind}}$ and $u_{m,h}^{\mathrm{tot}} = \rho_{m,h}^{\mathrm{tot}}$. The half-budget assumptions are exactly the two required budget inequalities, and the Freedman exponent for a nonzero load ρ is

$$\frac{\rho^2}{2(a_{m,h}^2 \rho + a_{m,h}^2 \rho/3)} = \frac{3\rho}{8a_{m,h}^2}.$$

Zero-load terms are deterministic and omitted. Hence the predictable block-support budget satisfies

$$\mathfrak{A}_m \le L_m^{\mathrm{pblk}} \exp\left\{-\frac{3\lambda_m}{8}\right\}.$$

The arbitrary-growth lower bound on λ_m makes this budget at most $e^{-1}(e+m)^{-(1+\gamma)}$, while the lower bound on τ_m^2 gives

$$2J_m(n_m+1)e^{-\tau_m^2} \le (e+m)^{-(1+\gamma)}.$$

Thus the summability hypothesis of Corollary 7.105 holds.

For all integer dimensions,

$$\mathfrak{A}_D \le (eD)^{\Gamma-3R/8}, \qquad 2J_D(n_D+1)e^{-\tau_D^2} \le 2(eD)^{\Gamma-S},$$

which is summable when $3R/8-\Gamma>1$ and $S-\Gamma>1$. Along a sparse dimension schedule the same estimates are stretched-exponential summable once $3R/8>\Gamma$ and $S>\Gamma$. □

Corollary 7.109 (Weighted block-load profile schedules)**.** *Work in the triangular-array setting of either Corollary 7.103 or Corollary 7.104. Retain the deterministic envelopes and slacks*

$$\mu_{m,h}^{\mathrm{ind}},\ \sigma_{m,h}^{2,\mathrm{ind}},\ \mu_{m,h}^{\mathrm{tot}},\ \sigma_{m,h}^{2,\mathrm{tot}},\ \omega_{m,h},\ u_{m,h}^{\mathrm{ind}},\ u_{m,h}^{\mathrm{tot}}.$$

Define

$$\Psi_{m,h}^{\mathrm{ind}} = \begin{cases} \dfrac{(u_{m,h}^{\mathrm{ind}})^2}{2(\sigma_{m,h}^{2,\mathrm{ind}}+\omega_{m,h}u_{m,h}^{\mathrm{ind}}/3)}, & \sigma_{m,h}^{2,\mathrm{ind}}+\omega_{m,h}u_{m,h}^{\mathrm{ind}}/3>0,\\ +\infty, & \sigma_{m,h}^{2,\mathrm{ind}}+\omega_{m,h}u_{m,h}^{\mathrm{ind}}/3=0, \end{cases}$$

$$\Psi_{m,h}^{\mathrm{tot}} = \begin{cases} \dfrac{(u_{m,h}^{\mathrm{tot}})^2}{2(\sigma_{m,h}^{2,\mathrm{tot}}+\omega_{m,h}u_{m,h}^{\mathrm{tot}}/3)}, & \sigma_{m,h}^{2,\mathrm{tot}}+\omega_{m,h}u_{m,h}^{\mathrm{tot}}/3>0,\\ +\infty, & \sigma_{m,h}^{2,\mathrm{tot}}+\omega_{m,h}u_{m,h}^{\mathrm{tot}}/3=0. \end{cases}$$

Put

$$\Psi_m = \min_{1\le h\le J_m} \min\{\Psi_{m,h}^{\mathrm{ind}},\Psi_{m,h}^{\mathrm{tot}}\}, \qquad L_m^{\mathrm{wbl}} = (n_m+1)\left(J_m+\sum_{h=1}^{J_m}|\mathcal{B}_{m,h}|\right).$$

Assume the relevant half-budget inequalities and $\sum_m \mathbb{P}(F_m^{\mathrm{lift}})<\infty$*. If, for some* $\gamma>0$*,*

$$\Psi_m \ge \log(eL_m^{\mathrm{wbl}}) + (1+\gamma)\log(e+m)$$

and

$$\tau_m^2 \ge \log(2L_m^{\mathrm{wbl}}) + (1+\gamma)\log(e+m)$$

eventually, then the active-prefix and fixed-menu conclusions of Corollary 7.86 hold almost surely for the corresponding random weighted or predictable weighted block-atomic matrix-series perturbations, after scaling the h-th noise stream by

$$C_{\mathrm{ms}}q_{m,h}(\sqrt{\log(e\kappa_{m,h})}+\tau_m).$$

In particular, suppose $L_m^{\mathrm{wbl}} \le (eD_m)^{\Gamma}$ *eventually. Along all integer dimensions* $D=2,3,\dots$*, the same conclusion holds if*

$$\Psi_D \ge R\log(eD), \qquad \tau_D^2 \ge S\log(eD),$$

eventually for constants $R>\Gamma+1$ *and* $S>\Gamma+1$*. Along any sparse schedule* $D_m \ge \exp(cm^\alpha)$*,* $c,\alpha>0$*, the same conclusion holds if*

$$\Psi_m \ge R\log(eD_m), \qquad \tau_m^2 \ge S\log(eD_m),$$

eventually for constants $R > \Gamma$ and $S > \Gamma$. Thus random amplitudes, random atom intensities, adaptive atom sampling, and without-replacement block exposure all obey the same active-prefix schedule once their scalar Bernstein or Freedman exponents dominate the active block count.

Proof. Let $\mathfrak{Q}_m$ denote the budget term $\mathfrak{W}_m$ in the independent weighted case and $\mathfrak{P}_m$ in the predictable case. By the definition of Ψ_m,

$$\mathfrak{Q}_m \le (n_m+1)\sum_{h=1}^{J_m} |\mathcal{B}_{m,h}| e^{-\Psi_m} \le L_m^{\mathrm{wbl}} e^{-\Psi_m}.$$

The arbitrary-growth lower bound on Ψ_m gives

$$\mathfrak{Q}_m \le e^{-1}(e+m)^{-(1+\gamma)}.$$

Also $J_m(n_m+1) \le L_m^{\mathrm{wbl}}$, so the lower bound on τ_m^2 gives

$$2J_m(n_m+1)e^{-\tau_m^2} \le (e+m)^{-(1+\gamma)}.$$

The summability hypothesis in the relevant weighted block-load corollary follows.

For all integer dimensions,

$$\mathfrak{Q}_D \le (eD)^{\Gamma-R}, \qquad 2J_D(n_D+1)e^{-\tau_D^2} \le 2(eD)^{\Gamma-S},$$

which is summable when $R-\Gamma>1$ and $S-\Gamma>1$. For sparse dimension schedules, the same bounds are stretched-exponential summable as soon as $R>\Gamma$ and $S>\Gamma$. □

Corollary 7.110 (Bounded weighted block-load incidence schedules)**.** *Work in the triangular-array setting of Corollary 7.109. Suppose that, for each active $h \le J_m$, there are deterministic parameters*

$$\omega_{m,h}^{\mathrm{bd}} > 0, \qquad \rho_{m,h}^{\mathrm{ind}} > 0, \qquad \rho_{m,h}^{\mathrm{tot}} > 0$$

such that the deterministic envelopes may be chosen to satisfy

$$\mu_{m,h}^{\mathrm{ind}} \le \rho_{m,h}^{\mathrm{ind}}, \qquad \sigma_{m,h}^{2,\mathrm{ind}} \le \omega_{m,h}^{\mathrm{bd}}\rho_{m,h}^{\mathrm{ind}}, \qquad \omega_{m,h} \le \omega_{m,h}^{\mathrm{bd}},$$

and

$$\mu_{m,h}^{\mathrm{tot}} \le \rho_{m,h}^{\mathrm{tot}}, \qquad \sigma_{m,h}^{2,\mathrm{tot}} \le \omega_{m,h}^{\mathrm{bd}}\rho_{m,h}^{\mathrm{tot}}.$$

Assume the half-budget bounds

$$2\rho_{m,h}^{\mathrm{ind}} \le q_{m,h}^2, \qquad 2\rho_{m,h}^{\mathrm{tot}} \le n_m q_{m,h}^2.$$

Put

$$\Lambda_m^{\mathrm{bw}} = \min_{1\le h\le J_m} \min\left\{\frac{\rho_{m,h}^{\mathrm{ind}}}{\omega_{m,h}^{\mathrm{bd}}}, \frac{\rho_{m,h}^{\mathrm{tot}}}{\omega_{m,h}^{\mathrm{bd}}}\right\}, \qquad L_m^{\mathrm{bw}} = (n_m+1)\left(J_m + \sum_{h=1}^{J_m}|\mathcal{B}_{m,h}|\right).$$

Assume $\sum_m \mathbb{P}(F_m^{\mathrm{lift}}) < \infty$. If, for some $\gamma > 0$,

$$\Lambda_m^{\mathrm{bw}} \ge \frac{8}{3}\left[\log(eL_m^{\mathrm{bw}}) + (1+\gamma)\log(e+m)\right]$$

and

$$\tau_m^2 \geq \log(2L_m^{\mathrm{bw}}) + (1+\gamma)\log(e+m)$$

eventually, then the active-prefix and fixed-menu conclusions of Corollary 7.86 hold almost surely for the corresponding random weighted or predictable weighted block-atomic matrix-series perturbations, after scaling the h-th noise stream by

$$C_{\mathrm{ms}} q_{m,h}\bigl(\sqrt{\log(e\kappa_{m,h})} + \tau_m\bigr).$$

In particular, suppose $L_m^{\mathrm{bw}} \leq (eD_m)^{\Gamma}$ eventually. Along all integer dimensions $D = 2, 3, \ldots$, the same conclusion holds if

$$\Lambda_D^{\mathrm{bw}} \geq R\log(eD), \qquad \tau_D^2 \geq S\log(eD),$$

eventually for constants

$$R > \frac{8(\Gamma+1)}{3}, \qquad S > \Gamma + 1.$$

Along any sparse schedule $D_m \geq \exp(cm^{\alpha})$, $c, \alpha > 0$, the same conclusion holds if

$$\Lambda_m^{\mathrm{bw}} \geq R\log(eD_m), \qquad \tau_m^2 \geq S\log(eD_m),$$

eventually for constants

$$R > \frac{8\Gamma}{3}, \qquad S > \Gamma.$$

Thus bounded random intensities, bounded compound-thinned atom fields, adaptive weighted dictionaries, and without-replacement weighted exposure are covered once their normalized row-block and total-block mean loads dominate the active block-count logarithm.

Proof. Use the weighted block-load profile schedule with the deterministic slacks

$$u_{m,h}^{\mathrm{ind}} = \rho_{m,h}^{\mathrm{ind}}, \qquad u_{m,h}^{\mathrm{tot}} = \rho_{m,h}^{\mathrm{tot}}.$$

The displayed half-budget assumptions give $\mu_{m,h}^{\mathrm{ind}} + u_{m,h}^{\mathrm{ind}} \leq q_{m,h}^2$ and $\mu_{m,h}^{\mathrm{tot}} + u_{m,h}^{\mathrm{tot}} \leq n_m q_{m,h}^2$. Moreover

$$\Psi_{m,h}^{\mathrm{ind}} \geq \frac{(\rho_{m,h}^{\mathrm{ind}})^2}{2(\omega_{m,h}^{\mathrm{bd}}\rho_{m,h}^{\mathrm{ind}} + \omega_{m,h}^{\mathrm{bd}}\rho_{m,h}^{\mathrm{ind}}/3)} = \frac{3\rho_{m,h}^{\mathrm{ind}}}{8\omega_{m,h}^{\mathrm{bd}}},$$

and the same estimate gives

$$\Psi_{m,h}^{\mathrm{tot}} \geq \frac{3\rho_{m,h}^{\mathrm{tot}}}{8\omega_{m,h}^{\mathrm{bd}}}.$$

Hence $\Psi_m \geq 3\Lambda_m^{\mathrm{bw}}/8$. The arbitrary-growth, all-integer, and sparse conclusions are exactly those of Corollary 7.109 with this lower bound for Ψ_m. □

Lemma 7.111 (Quadratic-form tails imply operator tails)**.** *Let $Y_1, \dots, Y_n$ be conditionally centered self-adjoint random matrices in $\mathrm{M}_D(\mathbb{C})$, relative to a sigma-field $\mathcal{H}$. Suppose that there is $q > 0$ such that, conditionally on $\mathcal{H}$, for every $a \in \mathbb{R}^n$, every deterministic unit vector $u \in \mathbb{C}^D$, and every $s \geq 0$,*

$$\mathbb{P}\left\{ \left| \left\langle u, \sum_{i=1}^n a_i Y_i u \right\rangle \right| > q\|a\|_2 s \,\middle|\, \mathcal{H} \right\} \leq 2e^{-s^2}.$$

Then, conditionally on $\mathcal{H}$, for every $a \in \mathbb{R}^n$ and every $t \geq 0$,

$$\mathbb{P}\left\{ \left\| \sum_{i=1}^n a_i Y_i \right\|_{\mathrm{op}} > 5q\|a\|_2(\sqrt{D} + t) \,\middle|\, \mathcal{H} \right\} \leq 2e^{-t^2}.$$

Proof. Fix a and t, and put $S = \sum_i a_i Y_i$. Let $\mathcal{N}$ be a deterministic $1/4$-net of the unit sphere of $\mathbb{C}^D$, viewed as a real $2D$-dimensional sphere, with $|\mathcal{N}| \leq 9^{2D}$. For self-adjoint S,

$$\|S\|_{\mathrm{op}} \leq 2 \max_{v \in \mathcal{N}} |\langle v, Sv \rangle|.$$

Therefore the conditional failure probability is at most

$$2|\mathcal{N}| \exp\left(-\frac{25}{4}(\sqrt{D} + t)^2 \right) \leq 2 \exp\left(2D \log 9 - \frac{25}{4} D - \frac{25}{4} t^2 \right) \leq 2e^{-t^2},$$

since $25/4 > 2 \log 9$. □

Corollary 7.112 (Quadratic-form noise for finite-lift core menus)**.** *In the setting of Corollary 7.86, keep the same finite-lift core assumptions and the same active-prefix exhaustion. Replace the operator-tail hypothesis by the following quadratic-form hypothesis: for each $h \leq J_m$ there is $q_{m,h} > 0$ such that, conditionally on $\mathcal{H}_m$, for every $a \in \mathbb{R}^{n_m}$, every deterministic unit vector $u \in \mathbb{C}^{D_m}$, and every $s \geq 0$,*

$$\mathbb{P}\left\{ \left| \left\langle u, \sum_{i=1}^{n_m} a_i Y_{m,i}^{(h)} u \right\rangle \right| > q_{m,h}\|a\|_2 s \,\middle|\, \mathcal{H}_m \right\} \leq 2e^{-s^2}.$$

For deterministic $\tau_m > 0$, define

$$Z_{m,i}^{(h)} = \frac{Y_{m,i}^{(h)}}{5q_{m,h}(\sqrt{D_m} + \tau_m)}, \qquad X_{m,i}^{(h)} = B_{m,i}^{(h)} + Z_{m,i}^{(h)}.$$

If

$$\sum_m \left[\mathbb{P}(F_m^{\mathrm{lift}}) + 2J_m(n_m + 1)e^{-\tau_m^2} \right] < \infty,$$

then the active-prefix and fixed-menu conclusions of Corollary 7.86 hold almost surely for these $X_{m,i}^{(h)}$. In particular, the explicit schedules of Corollary 7.87 apply with noise scale $5q_{m,h}(\sqrt{D_m} + \tau_m)$.

Consequently the finite-lift core menu theorem applies directly to conditionally Gaussian matrix series whose scalar quadratic forms have conditional variance at most $q_{m,h}^2\|a\|_2^2$, after enlarging $q_{m,h}$ by an absolute constant. The

same conclusion applies to conditionally sub-Gaussian Wigner-type or correlated entry models whenever their conditional quadratic-form sub-Gaussian proxy is uniformly bounded by $q_{m,h}\|a\|_2$.

Proof. Lemma 7.111, applied conditionally on $\mathcal{H}_m$ for each active-prefix h, gives the operator-tail hypothesis of Corollary 7.86 with $K_{0,m,h} = 5q_{m,h}$. The displayed conclusion is therefore exactly that corollary, and the schedule statement is Corollary 7.87.

For the Gaussian specialization, the displayed variance bound gives the quadratic-form tail by the one-dimensional Gaussian estimate, with only an absolute change of constants. The sub-Gaussian version is the same one-dimensional linear-form estimate applied to each fixed quadratic form. □

Corollary 7.113 (Entrywise sub-Gaussian noise menus)**.** *In the setting of Corollary 7.86, keep the same finite-lift core assumptions and the same active-prefix exhaustion. Suppose that, conditionally on* $\mathcal{H}_m$, *for each* $h \le J_m$ *the upper-triangular entries of* $\{Y^{(h)}_{m,i} : 1 \le i \le n_m\}$ *are independent and centered, with*

$$Y^{(h)}_{m,i} = \sum_{p=1}^{D_m} \xi^{(h)}_{m,i,p} E_{pp} + \sum_{1\le p<q\le D_m} \left(\zeta^{(h)}_{m,i,pq} E_{pq} + \overline{\zeta^{(h)}_{m,i,pq}} E_{qp} \right).$$

Assume that, for every $h \le J_m$, *there is a positive deterministic envelope* $Q_{m,h}$ *such that for every* $i \le n_m$ *and every unit* $u \in \mathbb{C}^{D_m}$,

$$\left[\sum_p \alpha^{(h)2}_{m,i,p} |u_p|^4 + 4 \sum_{p<q} \beta^{(h)2}_{m,i,pq} |u_p|^2 |u_q|^2 \right]^{1/2} \le Q_{m,h},$$

where conditionally on $\mathcal{H}_m$, $\xi^{(h)}_{m,i,p}$ *is real sub-Gaussian with proxy* $\alpha^{(h)}_{m,i,p}$, *and every real projection* $\mathrm{Re}(e^{\sqrt{-1}\theta}\zeta^{(h)}_{m,i,pq})$ *is sub-Gaussian with proxy* $\beta^{(h)}_{m,i,pq}$, *uniformly in* $\theta \in \mathbb{R}$.

Then there is a universal constant C_{ent} *such that, for every deterministic* $\tau_m > 0$, *the matrices*

$$Z^{(h)}_{m,i} = \frac{Y^{(h)}_{m,i}}{C_{\mathrm{ent}} Q_{m,h}(\sqrt{D_m} + \tau_m)}, \qquad X^{(h)}_{m,i} = B^{(h)}_{m,i} + Z^{(h)}_{m,i}$$

have the active-prefix and fixed-menu conclusions of Corollary 7.86 whenever

$$\sum_m \left[\mathbb{P}(F^{\mathrm{lift}}_m) + 2J_m(n_m+1)e^{-\tau_m^2} \right] < \infty.$$

The explicit schedules of Corollary 7.87 therefore apply with entrywise noise scale

$$C_{\mathrm{ent}} Q_{m,h}(\sqrt{D_m} + \tau_m).$$

In particular, if all displayed entry proxies are bounded by $\sigma_{m,h}$, *then one may take* $Q_{m,h} = \sqrt{2}\,\sigma_{m,h}$. *Thus normalized Wigner, GOE, GUE, and variance-profile Gaussian or sub-Gaussian active-prefix noise libraries with*

entry scale $\sigma_{m,h}$ are covered by the same finite-lift core signs at the natural spectral scale $\sigma_{m,h}(\sqrt{D_m}+\tau_m)$.

Proof. Fix m, h, i, a unit vector u, and condition on $\mathcal{H}_m$. The quadratic form is

$$\left\langle u, Y_{m,i}^{(h)} u \right\rangle = \sum_p \xi_{m,i,p}^{(h)} |u_p|^2 + 2 \sum_{p<q} \mathrm{Re}\left(\zeta_{m,i,pq}^{(h)} \overline{u_p} u_q\right).$$

By independence of the upper-triangular entries and the standard linear combination rule for sub-Gaussian variables, this scalar is conditionally sub-Gaussian with proxy at most a universal constant times the displayed square-function, hence at most $CQ_{m,h}$. Independence across the matrices i, conditionally on $\mathcal{H}_m$, gives for every $a \in \mathbb{R}^{n_m}$ the proxy $CQ_{m,h}\|a\|_2$ for $\langle u, \sum_i a_i Y_{m,i}^{(h)} u\rangle$. After increasing the universal constant so that the tail is in the $2e^{-s^2}$ normalization, the hypothesis of Corollary 7.112 holds. That corollary gives the asserted finite-lift menu conclusion and schedules.

If all entry proxies are at most $\sigma_{m,h}$, then

$$\sum_p |u_p|^4 + 4 \sum_{p<q} |u_p|^2 |u_q|^2 = 2 - \sum_p |u_p|^4 \le 2,$$

so $Q_{m,h} = \sqrt{2}\,\sigma_{m,h}$ is admissible. □

Corollary 7.114 (Row-variance entry envelopes)**.** *In the setting and notation of Corollary 7.113, suppose that for every active-prefix $h \le J_m$ there is $R_{m,h} > 0$ such that, for all $i \le n_m$ and all $1 \le p \le D_m$,*

$$\alpha_{m,i,p}^{(h)2} + 2 \sum_{q<p} \beta_{m,i,qp}^{(h)2} + 2 \sum_{p<q} \beta_{m,i,pq}^{(h)2} \le R_{m,h}^2.$$

Then $Q_{m,h} = R_{m,h}$ is admissible in Corollary 7.113. Hence the active-prefix finite-lift core menu theorem, and all schedules of Corollary 7.87, hold after scaling the h-th entrywise noise stream by

$$C_{\mathrm{ent}} R_{m,h}(\sqrt{D_m}+\tau_m).$$

In particular, if the off-diagonal proxies in the h-th stream are supported on graphs of maximum degree $\Delta_{m,h}$, while $\alpha_{m,i,p}^{(h)} \le \alpha_{m,h}$ and $\beta_{m,i,pq}^{(h)} \le \beta_{m,h}$, then one may take

$$R_{m,h} = \left(\alpha_{m,h}^2 + 2\Delta_{m,h}\beta_{m,h}^2\right)^{1/2}.$$

Thus sparse or banded Wigner/GOE/GUE-type active-prefix noise with degree $\Delta_{m,h}$ and off-diagonal proxy $\beta_{m,h}$ is covered at scale

$$\left(\alpha_{m,h}^2 + 2\Delta_{m,h}\beta_{m,h}^2\right)^{1/2} (\sqrt{D_m}+\tau_m),$$

up to the universal constant C_{ent}. In the normalized sparse case $\beta_{m,h} \lesssim \Delta_{m,h}^{-1/2}$ and $\alpha_{m,h} = O(1)$, this is the same $O(\sqrt{D_m}+\tau_m)$ operator-tail scale as in the dense normalized models.

Proof. Fix m, h, i and put $x_p = |u_p|^2$, so $x_p \geq 0$ and $\sum_p x_p = 1$. The square-function in Corollary 7.113 is

$$\sum_p \alpha_{m,i,p}^{(h)2} x_p^2 + 4 \sum_{p<q} \beta_{m,i,pq}^{(h)2} x_p x_q.$$

This is $x^T V x$, where $V_{pp} = \alpha_{m,i,p}^{(h)2}$ and $V_{pq} = 2\beta_{m,i,pq}^{(h)2}$ for $p < q$, symmetrically extended. Since all entries of V are nonnegative and every row sum is at most $R_{m,h}^2$,

$$x^T V x \leq \sum_p x_p R_{m,h}^2 = R_{m,h}^2.$$

Therefore $Q_{m,h} = R_{m,h}$ satisfies the entrywise square-function hypothesis, and Corollary 7.113 gives the conclusion.

The graph-degree specialization is the displayed row-sum bound with at most $\Delta_{m,h}$ nonzero off-diagonal proxies touching each row. □

Corollary 7.115 (Sparse and banded entry-profile schedules)**.** *In the setting of Corollary 7.114, suppose that for each active-prefix $h \leq J_m$ the off-diagonal support of every $Y_{m,i}^{(h)}$ is contained in a deterministic graph $G_{m,h}$ on $\{1, \dots, D_m\}$ with maximum degree $\Delta_{m,h}$. Put $d_{m,h} = \max(1, \Delta_{m,h})$. Suppose also that for deterministic $a_{m,h}, b_{m,h} \geq 0$, with $a_{m,h}^2 + b_{m,h}^2 > 0$,*

$$\alpha_{m,i,p}^{(h)} \leq a_{m,h}, \qquad \beta_{m,i,pq}^{(h)} \leq \frac{b_{m,h}}{\sqrt{d_{m,h}}}$$

for every active i, p, q, h. Then all active-prefix and fixed-menu conclusions of Corollary 7.86 hold, with the same Borel–Cantelli schedules as Corollary 7.87, after scaling the h-th entrywise noise stream by

$$C_{\mathrm{ent}}(a_{m,h}^2 + 2b_{m,h}^2)^{1/2}(\sqrt{D_m} + \tau_m).$$

Equivalently, if $a_{m,h}$ and $b_{m,h}$ are uniformly bounded on the active prefixes, sparse, banded, and variance-profile Wigner/GOE/GUE-type noise libraries at the normalized graph scale $d_{m,h}^{-1/2}$ are absorbed at the same $O(\sqrt{D_m} + \tau_m)$ spectral scale as dense normalized ensembles.

More explicitly, if

$$L_m^{\mathrm{ot}} = J_m(n_m + 1) \leq (eD_m)^\Gamma$$

eventually, then along all integer dimensions $D = 2, 3, \dots$ it is enough to take $\tau_D = \sqrt{R \log(eD)}$ with $R > \Gamma + 1$. Along sparse schedules $D_m \geq \exp(cm^\kappa)$, the same conclusion holds for $\tau_m = \sqrt{R \log(eD_m)}$ whenever $R > \Gamma$.

For a one-dimensional bandwidth-$w_{m,h}$ profile $|p - q| \leq w_{m,h}$, one may take $\Delta_{m,h} \leq 2w_{m,h}$. Thus off-diagonal proxy $b_{m,h}/\sqrt{\max(1, 2w_{m,h})}$ gives the displayed scale with no additional dependence on the bandwidth.

Proof. The row-variance envelope in Corollary 7.114 is bounded by

$$a_{m,h}^2 + 2\Delta_{m,h} \frac{b_{m,h}^2}{d_{m,h}} \leq a_{m,h}^2 + 2b_{m,h}^2.$$

That corollary gives the active-prefix finite-lift conclusion at the displayed entry-profile scale, and Corollary 7.87 gives the explicit all-integer and sparse-dimension schedules. The banded specialization uses the elementary degree bound $\Delta_{m,h} \le 2w_{m,h}$. □

Corollary 7.116 (Random graph-mask entry schedules)**.** *In the setting of Corollary 7.113, suppose that the off-diagonal support is random. More precisely, for each active (m,h,i) and each $p<q$, let $\eta_{m,h,i,pq} \in \{0,1\}$ be a support indicator. Assume that, conditionally on the core environment, these indicators are independent within each fixed (m,h,i) and satisfy*

$$\mathbb{P}\{\eta_{m,h,i,pq}=1 \mid \mathcal{H}_m\} \le p_{m,h,i,pq}.$$

Put

$$\lambda_{m,h} = \max_{i,p}\left(\sum_{q<p} p_{m,h,i,qp} + \sum_{p<q} p_{m,h,i,pq}\right), \qquad s_{m,h} = \max(1, \lambda_{m,h} + u_{m,h}),$$

where $u_{m,h} > 0$ is deterministic. Conditional on the core environment and on all support indicators, assume that the upper-triangular entries are independent and centered, the diagonal proxies satisfy $\alpha^{(h)}_{m,i,p} \le a_{m,h}$, and the off-diagonal entries vanish when $\eta_{m,h,i,pq}=0$ while, when $\eta_{m,h,i,pq}=1$, their sub-Gaussian proxies satisfy

$$\beta^{(h)}_{m,i,pq} \le \frac{b_{m,h}}{\sqrt{s_{m,h}}}.$$

If

$$\sum_m \left[\mathbb{P}(F_m^{\mathrm{lift}}) + \sum_{h\le J_m} n_m D_m \exp\left\{-\frac{u_{m,h}^2}{2(\lambda_{m,h}+u_{m,h}/3)}\right\} + 2J_m(n_m+1)e^{-\tau_m^2}\right] < \infty,$$

then all active-prefix and fixed-menu conclusions of Corollary 7.86 hold, after scaling the h-th random-mask entrywise noise stream by

$$C_{\mathrm{ent}}(a_{m,h}^2 + 2b_{m,h}^2)^{1/2}(\sqrt{D_m}+\tau_m).$$

The same conclusion holds without the Bernstein degree-exception term for any dependent random mask law which is almost surely supported on graphs of maximum degree at most $\Delta_{m,h}$, after setting $s_{m,h} = \max(1, \Delta_{m,h})$. Thus random d-regular masks, random matchings, random bounded-degree subgraphs of a band, and other dependent bounded-degree support laws are covered with no support-probability loss.

In particular, for inhomogeneous Erdos–Renyi masks with $p_{m,h,i,pq} \le p_{m,h}$, one has $\lambda_{m,h} \le (D_m-1)p_{m,h}$. If

$$L_m^{\deg} = \log(eJ_m n_m D_m) + (1+\gamma)\log(e+m), \qquad u_{m,h} = \max\{\lambda_{m,h}, AL_m^{\deg}\},$$

with $A > 8/3$, then the degree-exception series is summable. Thus random graph-supported Wigner/GOE/GUE-type active-prefix libraries are covered at the same displayed scale with off-diagonal normalization $s_{m,h}^{-1/2}$. When the

expected degree dominates the union logarithm, this is the usual expected-degree normalization, up to an absolute constant.

Proof. Let $F_m^{\deg}$ be the event that for some $h \le J_m$, $i \le n_m$, and $1 \le p \le D_m$, the random support degree

$$N_{m,h,i,p} = \sum_{q<p} \eta_{m,h,i,qp} + \sum_{p<q} \eta_{m,h,i,pq}$$

exceeds $\lambda_{m,h} + u_{m,h}$. For fixed m, h, i, p, this is a sum of independent $[0,1]$-valued variables with conditional mean at most $\lambda_{m,h}$. Bernstein's inequality gives

$$\mathbb{P}\{N_{m,h,i,p} > \lambda_{m,h} + u_{m,h} \mid \mathcal{H}_m\} \le \exp\left\{-\frac{u_{m,h}^2}{2(\lambda_{m,h} + u_{m,h}/3)}\right\}.$$

A union bound over h, i, p gives the displayed degree-exception term.

Adjoin the support indicators to the conditioning sigma-field. On $(F_m^{\deg})^c$, every active off-diagonal support graph has maximum degree at most $s_{m,h}$, and the finite-lift core hypotheses are unchanged. Hence the row-variance envelope is bounded by

$$a_{m,h}^2 + 2s_{m,h}\frac{b_{m,h}^2}{s_{m,h}} = a_{m,h}^2 + 2b_{m,h}^2.$$

Apply Corollary 7.114 on this event, with the exceptional event enlarged from F_m^{lift} to $F_m^{\mathrm{lift}} \cup F_m^{\deg}$, and then use the displayed summability condition and Borel–Cantelli.

For the homogeneous schedule, $u_{m,h} \ge \lambda_{m,h}$, so

$$\frac{u_{m,h}^2}{2(\lambda_{m,h} + u_{m,h}/3)} \ge \frac{3}{8}u_{m,h} \ge \frac{3A}{8}L_m^{\deg}.$$

Since $A > 8/3$, the union-count factor is summable in m.

For the deterministic-degree variant, skip the Bernstein step and take $F_m^{\deg} = \emptyset$. The same row-variance bound holds with $s_{m,h} = \max(1, \Delta_{m,h})$. □

Corollary 7.117 (Weighted random-mask variance profiles)**.** *In the setting of Corollary 7.113, suppose that for each active (m,h,i) and each $p<q$ there is a random support indicator $\eta_{m,h,i,pq} \in \{0,1\}$. Assume that, conditionally on $\mathcal{H}_m$, these indicators are independent within each fixed (m,h,i) and satisfy*

$$\mathbb{P}\{\eta_{m,h,i,pq} = 1 \mid \mathcal{H}_m\} \le p_{m,h,i,pq}.$$

Let $a_{m,h} \ge 0$, $c_{m,h,i,pq} \ge 0$, and $t_{m,h} > 0$ be deterministic. Conditional on the core environment and on the support indicators, assume that the upper-triangular entries are independent and centered, the diagonal proxies satisfy $\alpha_{m,i,p}^{(h)} \le a_{m,h}$, and the off-diagonal entries vanish when $\eta_{m,h,i,pq} = 0$, while when $\eta_{m,h,i,pq} = 1$ their proxies satisfy

$$\beta_{m,i,pq}^{(h)} \le c_{m,h,i,pq}.$$

Put

$$\mu_{m,h} = \max_{i,p} 2\left(\sum_{q<p} p_{m,h,i,qp} c^2_{m,h,i,qp} + \sum_{p<q} p_{m,h,i,pq} c^2_{m,h,i,pq}\right),$$
$$\sigma^2_{m,h} = \max_{i,p} 4\left(\sum_{q<p} p_{m,h,i,qp} c^4_{m,h,i,qp} + \sum_{p<q} p_{m,h,i,pq} c^4_{m,h,i,pq}\right),$$
$$b_{m,h} = 2\max_{i,p<q} c^2_{m,h,i,pq}, \qquad R^2_{m,h} = a^2_{m,h} + \mu_{m,h} + t_{m,h}.$$

If

$$\sum_m \left[\mathbb{P}(F_m^{\mathrm{lift}}) + \sum_{h\le J_m} n_m D_m \exp\left\{-\frac{t^2_{m,h}}{2(\sigma^2_{m,h} + b_{m,h}t_{m,h}/3)}\right\} + 2J_m(n_m+1)e^{-\tau_m^2}\right] < \infty,$$

with the Bernstein term omitted when $\sigma^2_{m,h} + b_{m,h}t_{m,h}/3 = 0$*, then all active-prefix and fixed-menu conclusions of Corollary 7.86 hold, after scaling the* h*-th weighted random-mask entrywise noise stream by*

$$C_{\mathrm{ent}} R_{m,h}(\sqrt{D_m} + \tau_m).$$

Thus the finite-lift core signs absorb inhomogeneous random variance-profile Wigner/GOE/GUE-type masks whenever the random weighted row-variance $2\sum_q \eta_{pq} c^2_{pq}$ *is eventually bounded by a summable Bernstein budget. This criterion can be strictly sharper than a maximum-degree bound when the potential entries have highly nonuniform variance proxies.*

Proof. For fixed m, h, i, p, set

$$W_{m,h,i,p} = 2\sum_{q<p} \eta_{m,h,i,qp} c^2_{m,h,i,qp} + 2\sum_{p<q} \eta_{m,h,i,pq} c^2_{m,h,i,pq}.$$

Conditionally on $\mathcal{H}_m$, this is a sum of independent $[0, b_{m,h}]$-valued variables, with conditional mean at most $\mu_{m,h}$ and total variance at most $\sigma^2_{m,h}$. Bernstein's inequality gives

$$\mathbb{P}\{W_{m,h,i,p} > \mu_{m,h} + t_{m,h} \mid \mathcal{H}_m\} \le \exp\left\{-\frac{t^2_{m,h}}{2(\sigma^2_{m,h} + b_{m,h}t_{m,h}/3)}\right\},$$

with the stated zero-variance convention. A union bound over h, i, p gives the displayed support-profile exception.

Adjoin the support indicators to the conditioning sigma-field. Outside this exception, every active row satisfies

$$\alpha^{(h)2}_{m,i,p} + 2\sum_{q<p} \beta^{(h)2}_{m,i,qp} + 2\sum_{p<q} \beta^{(h)2}_{m,i,pq} \le a^2_{m,h} + \mu_{m,h} + t_{m,h} = R^2_{m,h}.$$

Corollary 7.114 therefore applies on the complement of the enlarged exceptional event, and Borel–Cantelli gives the asserted active-prefix and fixed-menu conclusions. □

Corollary 7.118 (Weighted random-mask Bernstein schedules)**.** *In the setting of Corollary 7.117, put*

$$L_m^{\mathrm{var}} = J_m n_m D_m, \qquad L_m^{\mathrm{ot}} = J_m(n_m + 1),$$

and, when $\sigma_{m,h}^2 + b_{m,h}t_{m,h}/3 > 0$,

$$\Phi_{m,h} = \frac{t_{m,h}^2}{2(\sigma_{m,h}^2 + b_{m,h}t_{m,h}/3)}.$$

If $\sum_m \mathbb{P}(F_m^{\mathrm{lift}}) < \infty$, $\gamma > 0$, *and eventually*

$$\Phi_{m,h} \geq \log L_m^{\mathrm{var}} + (1+\gamma)\log(e+m) \qquad (1 \leq h \leq J_m),$$

with the support term omitted in the zero-variance cases, while

$$\tau_m^2 \geq \log(2L_m^{\mathrm{ot}}) + (1+\gamma)\log(e+m),$$

then the active-prefix and fixed-menu conclusions of Corollary 7.117 hold almost surely.

In particular, suppose

$$\max(L_m^{\mathrm{var}}, L_m^{\mathrm{ot}}) \leq (eD_m)^{\Gamma}$$

eventually. Along all integer dimensions $D = 2, 3, \ldots$, *the same conclusion holds if*

$$\Phi_{D,h} \geq R\log(eD), \qquad \tau_D^2 \geq R\log(eD) \qquad (1 \leq h \leq J_D)$$

eventually for some $R > \Gamma + 1$. *Along sparse schedules* $D_m \geq \exp(cm^{\kappa})$, *the same conclusion holds under the same inequalities whenever* $R > \Gamma$. *The resulting entrywise noise scale is*

$$C_{\mathrm{ent}}(a_{m,h}^2 + \mu_{m,h} + t_{m,h})^{1/2}(\sqrt{D_m} + \tau_m).$$

Proof. The arbitrary-growth inequalities give

$$\sum_{h \leq J_m} n_m D_m e^{-\Phi_{m,h}} \leq (e+m)^{-(1+\gamma)}$$

and

$$2J_m(n_m+1)e^{-\tau_m^2} \leq (e+m)^{-(1+\gamma)}.$$

Together with the lift-exception summability, this is the summability condition in Corollary 7.117.

For all integer dimensions, both exceptional row factors are bounded by constant multiples of $(eD)^{\Gamma - R}$, which is summable when $R > \Gamma + 1$. For sparse stretched-exponential dimension schedules, the same bound is summable as soon as $R > \Gamma$. □

Corollary 7.119 (Bounded weighted random-mask incidence schedules)**.** *In the setting of Corollary 7.118, suppose that, for each active* $h \leq J_m$, *deterministic parameters*

$$A_{m,h} \geq 0, \qquad \rho_{m,h} > 0, \qquad \omega_{m,h} > 0$$

satisfy

$$a_{m,h}^2 \le A_{m,h}^2, \qquad \mu_{m,h} \le \rho_{m,h}, \qquad \sigma_{m,h}^2 \le \omega_{m,h}\rho_{m,h}, \qquad b_{m,h} \le \omega_{m,h}.$$

Put

$$\Lambda_m^{\mathrm{mask}} = \min_{1\le h\le J_m} \frac{\rho_{m,h}}{\omega_{m,h}}, \qquad L_m^{\mathrm{var}} = J_m n_m D_m, \qquad L_m^{\mathrm{ot}} = J_m(n_m+1).$$

Assume $\sum_m \mathbb{P}(F_m^{\mathrm{lift}}) < \infty$. *If, for some* $\gamma > 0$,

$$\Lambda_m^{\mathrm{mask}} \ge \frac{8}{3}\left[\log(eL_m^{\mathrm{var}}) + (1+\gamma)\log(e+m)\right]$$

and

$$\tau_m^2 \ge \log(2L_m^{\mathrm{ot}}) + (1+\gamma)\log(e+m)$$

eventually, then the active-prefix and fixed-menu conclusions of Corollary 7.86 hold after scaling the h-th bounded weighted random-mask entrywise noise stream by

$$C_{\mathrm{ent}}(A_{m,h}^2 + 2\rho_{m,h})^{1/2}(\sqrt{D_m} + \tau_m).$$

In particular, suppose $\max\{L_m^{\mathrm{var}}, L_m^{\mathrm{ot}}\} \le (eD_m)^{\Gamma}$ *eventually. Along all integer dimensions* $D = 2, 3, \ldots$, *the same conclusion holds if*

$$\Lambda_D^{\mathrm{mask}} \ge R\log(eD), \qquad \tau_D^2 \ge S\log(eD),$$

eventually for constants

$$R > \frac{8(\Gamma+1)}{3}, \qquad S > \Gamma + 1.$$

Along any sparse schedule $D_m \ge \exp(cm^{\alpha})$, $c, \alpha > 0$, *the same conclusion holds if*

$$\Lambda_m^{\mathrm{mask}} \ge R\log(eD_m), \qquad \tau_m^2 \ge S\log(eD_m),$$

eventually for constants

$$R > \frac{8\Gamma}{3}, \qquad S > \Gamma.$$

The same statement holds in the predictable-profile setting of Corollary 7.120 after replacing $a_{m,h}, \mu_{m,h}, \sigma_{m,h}, b_{m,h}$ *by the deterministic envelopes* $\bar{a}_{m,h}, \bar{\mu}_{m,h}, \bar{\sigma}_{m,h}, \bar{b}_{m,h}$ *and adding* $\sum_m \mathbb{P}(F_m^{\mathrm{pr}}) < \infty$. *Thus independent or environment-predictable inhomogeneous Wigner/GOE/GUE-type masks are covered by the normalized mean row-variance ratio* ρ/ω, *rather than by maximum degree.*

Proof. Apply Corollary 7.118 with $t_{m,h} = \rho_{m,h}$. The resulting row-variance scale satisfies

$$a_{m,h}^2 + \mu_{m,h} + t_{m,h} \le A_{m,h}^2 + 2\rho_{m,h}.$$

The support exponent obeys

$$\Phi_{m,h} = \frac{\rho_{m,h}^2}{2(\sigma_{m,h}^2 + b_{m,h}\rho_{m,h}/3)} \ge \frac{\rho_{m,h}^2}{2(\omega_{m,h}\rho_{m,h} + \omega_{m,h}\rho_{m,h}/3)} = \frac{3\rho_{m,h}}{8\omega_{m,h}}.$$

Hence $\Phi_{m,h} \geq 3\Lambda_m^{\mathrm{mask}}/8$ for every active h. The arbitrary-growth, all-integer, and sparse estimates are the corresponding estimates in Corollary 7.118 with this lower bound. The predictable-profile variant is identical after working on $(F_m^{\mathrm{pr}})^c$ and using the deterministic barred envelopes. □

Corollary 7.120 (Predictable weighted random-mask profiles)**.** *In the setting of Corollary 7.113, keep the same finite-lift core assumptions and active-prefix exhaustion. Allow the random-mask intensity and proxy profile to be chosen from the core environment: for each active $h \leq J_m$, let*

$$a_{m,h}, \qquad p_{m,h,i,pq} \in [0,1], \qquad c_{m,h,i,pq} \geq 0$$

be $\mathcal{H}_m$-measurable. Conditionally on $\mathcal{H}_m$, assume that the support indicators $\eta_{m,h,i,pq}$ are independent within each fixed (m,h,i) and satisfy

$$\mathbb{P}\{\eta_{m,h,i,pq} = 1 \mid \mathcal{H}_m\} \leq p_{m,h,i,pq}.$$

Conditional on $\mathcal{H}_m$ and on the supports, assume that the upper-triangular entries are independent and centered, the diagonal proxies are bounded by $a_{m,h}$, and the off-diagonal entries vanish when $\eta_{m,h,i,pq} = 0$, while their proxies are bounded by $c_{m,h,i,pq}$ when $\eta_{m,h,i,pq} = 1$.

Define the conditional profile budgets

$$\mu_{m,h}^{\mathrm{pr}} = \max_{i,p} 2\left(\sum_{q<p} p_{m,h,i,qp} c_{m,h,i,qp}^2 + \sum_{p<q} p_{m,h,i,pq} c_{m,h,i,pq}^2\right),$$

$$(\sigma_{m,h}^{\mathrm{pr}})^2 = \max_{i,p} 4\left(\sum_{q<p} p_{m,h,i,qp} c_{m,h,i,qp}^4 + \sum_{p<q} p_{m,h,i,pq} c_{m,h,i,pq}^4\right),$$

$$b_{m,h}^{\mathrm{pr}} = 2\max_{i,p<q} c_{m,h,i,pq}^2.$$

Let $\bar{a}_{m,h}, \bar{\mu}_{m,h}, \bar{\sigma}_{m,h}, \bar{b}_{m,h} \geq 0$, $t_{m,h} > 0$, and $\tau_m > 0$ be deterministic, and let $F_m^{\mathrm{pr}} \in \mathcal{H}_m$ be the event that for some $h \leq J_m$,

$$a_{m,h}^2 > \bar{a}_{m,h}^2, \qquad \mu_{m,h}^{\mathrm{pr}} > \bar{\mu}_{m,h}, \qquad (\sigma_{m,h}^{\mathrm{pr}})^2 > \bar{\sigma}_{m,h}^2, \quad \text{or} \quad b_{m,h}^{\mathrm{pr}} > \bar{b}_{m,h}.$$

If

$$\sum_m \Bigg[\mathbb{P}(F_m^{\mathrm{lift}}) + \mathbb{P}(F_m^{\mathrm{pr}}) + \sum_{h \leq J_m} n_m D_m \exp\left\{-\frac{t_{m,h}^2}{2(\bar{\sigma}_{m,h}^2 + \bar{b}_{m,h} t_{m,h}/3)}\right\} + 2J_m(n_m+1)e^{-\tau_m^2}\Bigg] < \infty,$$

with the Bernstein term omitted when $\bar{\sigma}_{m,h}^2 + \bar{b}_{m,h} t_{m,h}/3 = 0$, then all active-prefix and fixed-menu conclusions of Corollary 7.86 hold, after scaling the h-th predictable random-mask entrywise noise stream by

$$C_{\mathrm{ent}}(\bar{a}_{m,h}^2 + \bar{\mu}_{m,h} + t_{m,h})^{1/2}(\sqrt{D_m} + \tau_m).$$

Equivalently, the deterministic-profile theorem remains valid after replacing the fixed variance profile by an $\mathcal{H}_m$-measurable random or adaptive profile, provided the conditional Bernstein envelopes exceed their prescribed deterministic budgets only summably. If

$$\bar{\Phi}_{m,h} = \frac{t_{m,h}^2}{2(\bar{\sigma}_{m,h}^2 + \bar{b}_{m,h} t_{m,h}/3)}$$

and $L_m^{\mathrm{var}} = J_m n_m D_m$, $L_m^{\mathrm{ot}} = J_m(n_m + 1)$, then the arbitrary-growth, all-integer, and sparse-dimension schedule conclusions of Corollary 7.118 hold with $\Phi_{m,h}$ replaced by $\bar{\Phi}_{m,h}$, as long as $\sum_m \mathbb{P}(F_m^{\mathrm{pr}}) < \infty$ is added to the lift-exception summability.

Proof. Work on $(F_m^{\mathrm{pr}})^c$ and condition on $\mathcal{H}_m$. For fixed m, h, i, p, the weighted row profile

$$W_{m,h,i,p} = 2 \sum_{q<p} \eta_{m,h,i,qp} c_{m,h,i,qp}^2 + 2 \sum_{p<q} \eta_{m,h,i,pq} c_{m,h,i,pq}^2$$

is a sum of independent variables bounded by $\bar{b}_{m,h}$, with conditional mean at most $\bar{\mu}_{m,h}$ and total conditional variance at most $\bar{\sigma}_{m,h}^2$. Bernstein's inequality gives

$$\mathbb{P}\{W_{m,h,i,p} > \bar{\mu}_{m,h} + t_{m,h} \mid \mathcal{H}_m\} \le \exp\left\{ -\frac{t_{m,h}^2}{2(\bar{\sigma}_{m,h}^2 + \bar{b}_{m,h} t_{m,h}/3)} \right\},$$

with the stated zero-variance convention. A union bound over h, i, p gives the displayed predictable-profile exception.

Adjoin the supports to the conditioning sigma-field. Outside the predictable profile exception, every active row satisfies

$$\alpha_{m,i,p}^{(h)2} + 2 \sum_{q<p} \beta_{m,i,qp}^{(h)2} + 2 \sum_{p<q} \beta_{m,i,pq}^{(h)2} \le \bar{a}_{m,h}^2 + \bar{\mu}_{m,h} + t_{m,h}.$$

Corollary 7.114 therefore applies on the complement of $F_m^{\mathrm{lift}} \cup F_m^{\mathrm{pr}}$ and the support-profile exception, while Corollary 7.86 supplies the same active-prefix conclusion after the operator-tail union term is included. Borel–Cantelli gives the almost-sure eventual statement.

The schedule paragraph is exactly the proof of Corollary 7.118, with $\Phi_{m,h}$ replaced by $\bar{\Phi}_{m,h}$ and the summable events F_m^{pr} added to F_m^{lift}. □

Corollary 7.121 (Dependent weighted-budget mask laws)**.** *In the setting of Corollary 7.113, keep the same finite-lift core assumptions and active-prefix exhaustion. For each active (m, h, i) and each $p < q$, let $\eta_{m,h,i,pq} \in \{0, 1\}$ be an arbitrary random support indicator; no independence, product structure, or identical distribution is assumed for these supports. Let $a_{m,h} \ge 0$, $c_{m,h,i,pq} \ge 0$, $R_{m,h} > 0$, and $\tau_m > 0$ be deterministic. Conditional on the core environment and on all active supports, assume that the upper-triangular entries are independent and centered, the diagonal proxies satisfy*

$\alpha^{(h)}_{m,i,p} \le a_{m,h}$, *and the off-diagonal entries vanish when* $\eta_{m,h,i,pq} = 0$, *while when* $\eta_{m,h,i,pq} = 1$ *their proxies satisfy*

$$\beta^{(h)}_{m,i,pq} \le c_{m,h,i,pq}.$$

Let F^{wb}_m *be the weighted-budget failure event that for some* $h \le J_m$, $i \le n_m$, *and* $1 \le p \le D_m$,

$$a^2_{m,h} + 2\sum_{q<p} \eta_{m,h,i,qp} c^2_{m,h,i,qp} + 2\sum_{p<q} \eta_{m,h,i,pq} c^2_{m,h,i,pq} > R^2_{m,h}.$$

If

$$\sum_m \left[\mathbb{P}(F^{\mathrm{lift}}_m) + \mathbb{P}(F^{\mathrm{wb}}_m) + 2J_m(n_m+1)e^{-\tau_m^2}\right] < \infty,$$

then, with probability one, for all sufficiently large m, *the active-prefix and fixed-menu conclusions of Corollary 7.86 hold after scaling the* h*-th dependent-mask entrywise noise stream by*

$$C_{\mathrm{ent}} R_{m,h}(\sqrt{D_m} + \tau_m).$$

The common sign vector may be chosen from the finite-lift core environment; it does not need to observe the random support masks.

Consequently every support law whose realized weighted row budgets are bounded deterministically has no support-probability cost. This includes random regular or biregular masks, matchings, sampling without replacement, random bounded-degree subgraphs of a band, determinantal or negatively dependent support samplers, and mixtures of such laws, whenever the displayed weighted row budget is bounded by $R^2_{m,h}$ *outside a summable event. The explicit operator-tail schedules of Corollary 7.87 apply with no additional change: if* $J_m(n_m+1) \le (eD_m)^\Gamma$, *then* $\tau_D^2 = S\log(eD)$ *with* $S > \Gamma+1$ *suffices along all integer dimensions, while sparse stretched-exponential dimension schedules need only* $S > \Gamma$.

Proof. Enlarge the conditioning sigma-field by all active support indicators. On $(F^{\mathrm{wb}}_m)^c$, every active row satisfies the row-variance hypothesis

$$\alpha^{(h)2}_{m,i,p} + 2\sum_{q<p} \beta^{(h)2}_{m,i,qp} + 2\sum_{p<q} \beta^{(h)2}_{m,i,pq} \le R^2_{m,h}.$$

Thus Corollary 7.114 applies conditionally on the supports, with $Q_{m,h} = R_{m,h}$. The finite-lift core signs are obtained before revealing the supports, so the same $\mathcal{H}_m$-measurable sign vector is admissible after the conditioning is enlarged. The only remaining random failures are F^{lift}_m, F^{wb}_m, and the usual operator-tail union event of size $2J_m(n_m+1)e^{-\tau_m^2}$. Borel–Cantelli gives the active-prefix and fixed-menu almost-sure conclusion, and the schedule statement is Corollary 7.87. □

Corollary 7.122 (Bounded weighted-degree dependent masks)**.** *In the setting of Corollary 7.121, suppose that for every active* $h \le J_m$ *there are*

deterministic $a_{m,h}, b_{m,h} \geq 0$, $s_{m,h} > 0$, *and weights* $\omega_{m,h,i,pq} \geq 0$ *such that the diagonal proxies are at most* $a_{m,h}$ *and, whenever* $\eta_{m,h,i,pq} = 1$,

$$\beta^{(h)}_{m,i,pq} \leq b_{m,h} \left(\frac{\omega_{m,h,i,pq}}{s_{m,h}} \right)^{1/2}.$$

Let F_m^{wdeg} *be the event that for some* $h \leq J_m$, $i \leq n_m$, *and* $1 \leq p \leq D_m$,

$$\sum_{q<p} \eta_{m,h,i,qp} \omega_{m,h,i,qp} + \sum_{p<q} \eta_{m,h,i,pq} \omega_{m,h,i,pq} > s_{m,h}.$$

If

$$\sum_m \left[\mathbb{P}(F_m^{\mathrm{lift}}) + \mathbb{P}(F_m^{\mathrm{wdeg}}) + 2J_m(n_m+1)e^{-\tau_m^2} \right] < \infty,$$

then, with probability one, for all sufficiently large m, *all active-prefix and fixed-menu conclusions of Corollary 7.86 hold after scaling the* h*-th dependent weighted-degree noise stream by*

$$C_{\mathrm{ent}}(a_{m,h}^2 + 2b_{m,h}^2)^{1/2}(\sqrt{D_m} + \tau_m).$$

In particular, when the support law is almost surely supported on the displayed weighted-degree constraint, $F_m^{\mathrm{wdeg}} = \emptyset$. *For unweighted degree-bounded masks this says: if every active support graph has maximum degree at most* $\Delta_{m,h}$ *and the off-diagonal proxies are bounded by* $b_{m,h}/\sqrt{\max(1, \Delta_{m,h})}$, *then the finite-lift core signs absorb the noise at scale*

$$C_{\mathrm{ent}}(a_{m,h}^2 + 2b_{m,h}^2)^{1/2}(\sqrt{D_m} + \tau_m)$$

with no support exception. Thus random d*-regular and biregular masks, matchings,* k*-nearest-neighbor masks sampled without replacement, bounded band subgraphs, and weighted-degree graph samplers all have the same normalized finite-lift scale whenever their weighted degree is deterministically bounded or fails only summably.*

The explicit operator-tail schedules are exactly those of Corollary 7.87: if $J_m(n_m+1) \leq (eD_m)^{\Gamma}$, *then all integer dimensions are covered by* $\tau_D^2 = S\log(eD)$ *with* $S > \Gamma + 1$, *and sparse stretched-exponential dimension schedules need only* $S > \Gamma$.

Proof. On $(F_m^{\mathrm{wdeg}})^c$, put

$$W^{\omega}_{m,h,i,p} = \sum_{q<p} \eta_{m,h,i,qp} \omega_{m,h,i,qp} + \sum_{p<q} \eta_{m,h,i,pq} \omega_{m,h,i,pq} \leq s_{m,h}.$$

Then every active row satisfies

$$\begin{aligned} a_{m,h}^2 + 2\sum_{q<p} \eta_{m,h,i,qp} \beta^{(h)2}_{m,i,qp} + 2\sum_{p<q} \eta_{m,h,i,pq} \beta^{(h)2}_{m,i,pq} \\ \leq a_{m,h}^2 + \frac{2b_{m,h}^2}{s_{m,h}} W^{\omega}_{m,h,i,p} \leq a_{m,h}^2 + 2b_{m,h}^2. \end{aligned}$$

Therefore Corollary 7.121 applies with $R_{m,h} = (a_{m,h}^2 + 2b_{m,h}^2)^{1/2}$ and $F_m^{\mathrm{wb}} = F_m^{\mathrm{wdeg}}$. The examples are the corresponding choices of the deterministic weights ω. The final schedule statement again comes from Corollary 7.87. □

Corollary 7.123 (Algebraic cores plus full random operator-tail noise)**.** *For every $C_0 \geq 1$ there is $C(C_0)$ with the following property. Let $B_1, \dots, B_n$ be random self-adjoint matrices in $\mathrm{M}_D(\mathbb{C})$. Suppose that, outside an event E, there is a finite-dimensional C^*-subalgebra $\mathcal{A}_\omega \subset \mathrm{M}_D(\mathbb{C})$ such that*

$$\dim_{\mathbb{C}} \mathcal{A}_\omega \leq C_0 n, \qquad B_i(\omega) \in \mathcal{A}_\omega, \qquad \|B_i(\omega)\|_{\mathrm{op}} \leq 1$$

for every i. Let $Y_1, \dots, Y_n$ be centered self-adjoint random matrices, independent of $B_1, \dots, B_n$, satisfying the operator-tail hypothesis of Corollary 7.12 with constant K_0. Put

$$s_{D,n} := \sqrt{D} + \sqrt{3\log(en)}, \qquad Z_i := \frac{Y_i}{K_0 s_{D,n}}, \qquad X_i := B_i + Z_i.$$

Then, with probability at least $1 - \mathbb{P}(E) - 2(n+1)(en)^{-3}$,

$$\mathrm{disc}(X_1, \dots, X_n) \leq C(C_0)\sqrt{n}.$$

On the same event, all X_i have operator norm at most 2. Hence the normalized contractions

$$\widehat{X}_i := \frac{1}{2} X_i$$

satisfy

$$\mathrm{disc}(\widehat{X}_1, \dots, \widehat{X}_n) \leq C(C_0)\sqrt{n}.$$

Thus dimension-controlled random algebraic cores remain in the Matrix Spencer regime after adding full centered GOE/GUE, Wigner-type, or other Gaussian-operator-tail noise in arbitrary ambient dimension, once that noise is put at its natural spectral scale. The noise matrices themselves may be dependent, provided the displayed scalar-combination tail holds.

Proof. On a realization outside E, apply Theorem 4.2 to $B_1, \dots, B_n$ and choose signs $\varepsilon_i^\star$ such that

$$\left\| \sum_i \varepsilon_i^\star B_i \right\|_{\mathrm{op}} \leq C(C_0)\sqrt{n}.$$

The signs are measurable with respect to the algebraic-core data and are independent of the noise. Conditional on the core realization, the operator-tail hypothesis applied with $a = \varepsilon^\star$ gives

$$\mathbb{P}\left\{ \left\| \sum_i \varepsilon_i^\star Y_i \right\|_{\mathrm{op}} > K_0 \sqrt{n}\, s_{D,n} \,\middle|\, B_1, \dots, B_n \right\} \leq 2(en)^{-3}.$$

Also, applying the same tail to $a = e_i$ and union-bounding over i shows that all $\|Z_i\|_{\mathrm{op}} \leq 1$ except with conditional probability at most $2n(en)^{-3}$.

Therefore, outside a further event of probability at most $2(n+1)(en)^{-3}$,

$$\left\|\sum_i \varepsilon_i^\star X_i\right\|_{\mathrm{op}} \le C(C_0)\sqrt{n}+\sqrt{n} \le C'(C_0)\sqrt{n},$$

and all $\|X_i\|_{\mathrm{op}} \le 2$. Absorb the harmless additive 1 into $C(C_0)$ and divide by 2 for the normalized contraction statement. □

Corollary 7.124 (Algebraic core menus plus mixed dominated operator-tail noise)**.** *For each $m \ge 1$, let $D_m, n_m, J_m \ge 1$, put $\Delta_m := D_m + n_m + 1$ and $\ell_m := \log(e\Delta_m)$, and let $\mathcal{H}_m$ be a row environment. For $h \ge 1$, let*

$$B_{m,h,1}, \dots, B_{m,h,n_m}, \qquad Y_{m,h,1}, \dots, Y_{m,h,n_m}$$

be self-adjoint random matrices in $\mathrm{M}_{D_m}(\mathbb{C})$. Fix $C_0 \ge 1$, and let $K_{\mathrm{alg}} = C(C_0)$ be the Akbas–Sra constant from Theorem 4.2. Assume that there are events $F_m^{\mathrm{alg}} \in \mathcal{H}_m$ with $\sum_m \mathbb{P}(F_m^{\mathrm{alg}}) < \infty$ such that, outside F_m^{alg}, for every active $h \le J_m$ there is a finite-dimensional C^*-subalgebra $\mathcal{A}_{m,h} \subset \mathrm{M}_{D_m}(\mathbb{C})$ with

$$B_{m,h,i} \in \mathcal{A}_{m,h}, \qquad \|B_{m,h,i}\|_{\mathrm{op}} \le 1 \quad (1 \le i \le n_m),$$

and the active-prefix intrinsic dimension obeys

$$\sum_{h=1}^{J_m} \dim_{\mathbb{C}} \mathcal{A}_{m,h} \le C_0 n_m.$$

For each active $h \le J_m$, tag the perturbation stream either ψ or μ, choose $\varepsilon_{m,h} \in (0,1]$, $A_{m,h} \ge 1$, and $s_{m,h} > 0$, and define $r_{m,h}$, $V_{m,h}$, and $q_{m,h}$ as in Corollary 7.68, with $Y_{m,h,i}$ in place of $X_{m,h,i}$. Assume that, after adjoining $\mathcal{H}_m$ to the predictable sigma-fields, the corresponding sub-Weibull or finite-moment dominated hypotheses and the matching truncated tail-drift hypotheses hold outside events $E_{m,h}$.

Let $K_{\mathrm{tail}}, \eta > 0$ and set

$$\begin{aligned} \lambda_{m,h} &:= \max\{1, K_{\mathrm{tail}} + \eta\}\, q_{m,h}, \\ Z_{m,h,i} &:= \frac{Y_{m,h,i}}{\lambda_{m,h}}, \\ X_{m,h,i} &:= B_{m,h,i} + Z_{m,h,i}. \end{aligned}$$

If $\ell_m \le B\sqrt{n_m}$ eventually,

$$\sum_m \left[\mathbb{P}(F_m^{\mathrm{alg}}) + \sum_{h=1}^{J_m} \left(\mathbb{P}(E_{m,h}) + \varepsilon_{m,h}\right)\right] < \infty,$$

and

$$\sum_m J_m \Delta_m (e\Delta_m)^{-\gamma_{\mathrm{tc}}(K_{\mathrm{tail}},1,1,B)} < \infty,$$

then, with probability one, for all sufficiently large m, one $\mathcal{H}_m$-measurable sign vector $\varepsilon_m^\star$ satisfies

$$\left\|\sum_{i=1}^{n_m} \varepsilon_{m,i}^\star X_{m,h,i}\right\|_{\mathrm{op}} \le (K_{\mathrm{alg}}+1)\sqrt{n_m} \qquad (1 \le h \le J_m).$$

On the same event,

$$\max_{h\le J_m} \max_{i\le n_m} \|X_{m,h,i}\|_{\mathrm{op}} \le 2.$$

Consequently the normalized contractions $\widehat{X}_{m,h,i} := X_{m,h,i}/2$ have common active-prefix discrepancy at most $(K_{\mathrm{alg}}+1)\sqrt{n_m}/2$, and every fixed menu element is eventually covered whenever $J_m \to \infty$.

In particular, if $J_m\Delta_m \le (e\Delta_m)^\Gamma$, $J_m \le (e\Delta_m)^\Lambda$, $n_m \ge a\ell_m^2$ eventually, and $\varepsilon_{m,h} = (e\Delta_m)^{-c}$, then after adding $\sum_m \mathbb{P}(F_m^{\mathrm{alg}}) < \infty$ the cap-budget part is summable for all integer dimensions when $c > \Lambda + 1$, and along every stretched-exponential sparse dimension schedule when $c > \Lambda$. The matrix-tail margins are

$$K_{\mathrm{tail}} > \Theta_{\mathrm{tc}}(\Gamma+1, a, 1, 1)$$

and

$$K_{\mathrm{tail}} > \Theta_{\mathrm{tc}}(\Gamma, a, 1, 1),$$

respectively.

Proof. Work outside F_m^{alg}. Form the direct-sum algebra

$$\mathcal{A}_m^{(J)} := \bigoplus_{h=1}^{J_m} \mathcal{A}_{m,h}$$

and the direct-sum core tuple

$$\widetilde{B}_{m,i} := \bigoplus_{h=1}^{J_m} B_{m,h,i} \qquad (1 \le i \le n_m).$$

Then $\dim_{\mathbb{C}} \mathcal{A}_m^{(J)} \le C_0 n_m$, each $\widetilde{B}_{m,i}$ is a self-adjoint contraction in $\mathcal{A}_m^{(J)}$, and Theorem 4.2 gives signs $\varepsilon_m^\star$ such that

$$\max_{h\le J_m} \left\|\sum_i \varepsilon_{m,i}^\star B_{m,h,i}\right\|_{\mathrm{op}} = \left\|\sum_i \varepsilon_{m,i}^\star \widetilde{B}_{m,i}\right\|_{\mathrm{op}} \le K_{\mathrm{alg}}\sqrt{n_m}.$$

Choose the lexicographically first successful sign vector to make $\varepsilon_m^\star$ $\mathcal{H}_m$-measurable.

Apply Corollary 7.69 to the present arrays $B_{m,h,i}$ and $Y_{m,h,i}$, with $K_{\mathrm{core}} = K_{\mathrm{alg}}$ and $F_m^{\mathrm{core}} = F_m^{\mathrm{alg}}$. The direct-sum Akbas–Sra signing verifies the signed-core hypothesis, and the displayed mixed dominated assumptions are exactly the perturbation hypotheses. The active-prefix conclusion, fixed-menu coverage, and all-integer/sparse thresholds are the corresponding conclusions of the signed-core mixed dominated corollary. □

Corollary 7.125 (Structured core catalogue with mixed dominated noise)**.** *The mixed dominated operator-tail active-menu conclusion applies, with the same normalizations and all-integer/sparse thresholds, to each of the following structured random-matrix core catalogues.*

(a) *Finite-lift structured cores satisfying the active-prefix finite-lift budget in Corollary 7.88. This includes scalar Toeplitz and Hankel cores, one-level fixed-block Toeplitz/Hankel cores, fixed-level multilevel block Toeplitz/Hankel cores when the active-prefix symbol dimension is $O(n_m)$, and banded multilevel block Toeplitz/Hankel cores when the active-prefix sum of anisotropic bandwidth products is $O(n_m)$.*

(b) *Intrinsic algebraic cores satisfying the active-prefix dimension budget in Corollary 7.124. This includes common finite-dimensional C*-algebra cores and finite group convolution cores whenever the active-prefix sums of $\dim_{\mathbb{C}} \mathcal{A}_{m,h}$, or respectively $|G_{m,h}|$, are $O(n_m)$.*

In either case, each active perturbation stream may be tagged ψ or μ and scaled by its own $\lambda_{m,h} = \max\{1, K_{\mathrm{tail}} + \eta\} q_{m,h}$. Under the corresponding cap-budget, drift, and matrix-tail summability assumptions, the same environment-measurable signs balance all active normalized sums

$$\left\| \sum_i \varepsilon^{\star}_{m,i} \left(B_{m,h,i} + \frac{Y_{m,h,i}}{\lambda_{m,h}} \right) \right\|_{\mathrm{op}} = O(\sqrt{n_m}) \qquad (1 \le h \le J_m)$$

eventually almost surely, and the normalized contractions have the same active-prefix Matrix Spencer bound.

Proof. For the finite-lift catalogue, the cited Toeplitz/Hankel and banded multilevel theorems give finite block lifts with active-prefix entropy equal, up to the fixed structural constants, to the displayed symbol or bandwidth budget. Corollary 7.88 therefore applies.

For the algebraic catalogue, Theorem 4.2 applies to the direct sum of the active finite-dimensional C*-algebras, and Corollary 4.9 identifies the group convolution case with the dimension budget $|G|$. Hence Corollary 7.124 applies. The displayed normalization and schedule assertions are exactly the conclusions of those two mixed dominated corollaries. □

Corollary 7.126 (Concrete structured size budgets with mixed dominated noise)**.** *Fix $r_0, b_0, C_0 \ge 1$. In Corollary 7.88, the abstract active-prefix finite-lift assumption may be replaced by the following concrete structural condition. Outside a summable structural event, every active core tuple is a self-adjoint contraction tuple of one of the following types, with $r \le r_0$ and $b \le b_0$:*

(i) *scalar Toeplitz or Hankel on $d_{m,h}$ coordinates, with $S_{m,h} = d_{m,h}$;*

(ii) *one-level $b \times b$ block Toeplitz or Hankel on ambient dimension $D_{m,h}$, with $S_{m,h} = D_{m,h}$;*

(iii) *full-bandwidth r-level $b \times b$ block Toeplitz or Hankel on ambient dimension $D_{m,h}$, with $S_{m,h} = D_{m,h}$;*

(iv) *banded r-level $b \times b$ block Toeplitz or Hankel with anisotropic half-bandwidths $w_{m,h,1}, \dots, w_{m,h,r}$ and boxes satisfying the hypotheses of Corollary 5.17, with*

$$S_{m,h} := \prod_{j=1}^{r} (2w_{m,h,j} + 1).$$

If

$$\sum_{h=1}^{J_m} S_{m,h} \le C_0 n_m$$

eventually, then the full active-prefix conclusion, fixed-menu coverage, and all-integer/sparse thresholds of Corollary 7.88 hold, with constants depending only on r_0, b_0, C_0.

Likewise, in Corollary 7.124, the active algebraic dimension condition is satisfied whenever, outside a summable structural event, every active core tuple is either contained in a finite-dimensional C-algebra $\mathcal{A}_{m,h}$ or is a finite group-convolution core on a group $G_{m,h}$, and*

$$\sum_{h=1}^{J_m} s_{m,h} \le C_0 n_m.$$

Here

$$s_{m,h} := \begin{cases} \dim_{\mathbb{C}} \mathcal{A}_{m,h}, & \text{in the C*-algebra case,} \\ |G_{m,h}|, & \text{in the group-convolution case.} \end{cases}$$

Thus those algebraic and group-convolution active menus have the same mixed dominated perturbation conclusion and the same thresholds.

Proof. For the finite-lift alternatives, the proofs of Theorem 5.2, Corollary 5.4, Theorem 5.9, Theorem 5.22, and Corollary 5.17 supply finite block lifts with ΛK bounded in terms of r_0, b_0 and active-prefix entropy bounded by $C(r_0, b_0) \sum_{h \le J_m} S_{m,h}$. The displayed size budget therefore implies the active-prefix finite-lift hypothesis with M_0 and C_E depending only on r_0, b_0, C_0, and Corollary 7.88 applies.

For the algebraic alternatives, finite-dimensional C*-algebra cores are already in the hypothesis of Corollary 7.124. For group convolution, the reduced convolution C*-algebra has complex dimension $\sum_{\pi \in \widehat{G}} d_\pi^2 = |G|$, so the same size budget is precisely the active-prefix intrinsic-dimension budget. Applying the algebraic mixed dominated corollary gives the stated conclusion. □

Corollary 7.127 (Random orbit closures of structured mixed menus)**.** *Fix $r_0, b_0, c_0, L_0, C_0 \ge 1$. In the finite-lift part of Corollary 7.126, the concrete structural hypothesis may be replaced by the following closure hypothesis. Outside a summable structural event, for every active $h \le J_m$ there are finitely many components $a \in I_{m,h}$, integers $1 \le c_{m,h,a} \le c_0$, and size parameters $S_{m,h,a}$. For each component, the base structured space is one of*

the four finite-lift spaces listed in Corollary 7.126, with levels and block sizes bounded by r_0, b_0, and with base size $S_{m,h,a}$. The component tuple

$$C_{m,h,a,1}, \dots, C_{m,h,a,n_m}$$

is a self-adjoint contraction tuple in the $c_{m,h,a}$-matrix amplification of that base space. The active core tuple is obtained from the component direct sum by an $\mathcal{H}_m$-measurable self-adjoint linear image $\Phi_{m,h}$, with $\|\Phi_{m,h}\| \le L_0$:

$$B_{m,h,i} = \Phi_{m,h}\left(\bigoplus_{a \in I_{m,h}} C_{m,h,a,i}\right), \qquad \|B_{m,h,i}\|_{\mathrm{op}} \le 1.$$

If the active-prefix charge satisfies

$$\sum_{h=1}^{J_m} \sum_{a \in I_{m,h}} c_{m,h,a}^2 S_{m,h,a} \le C_0 n_m$$

eventually, then the active-prefix signing, fixed-menu coverage, and integer or sparse schedule thresholds of Corollary 7.88 hold. The constants depend only on the five displayed structural parameters.

In particular, the structured finite-lift mixed-tail result is stable under environment-measurable unitary or permutation conjugacy, orthogonal corners and compressions, fixed-size matrix-valued tensor amplification, block direct sums, and bounded self-adjoint images, provided the displayed active-prefix charge is $O(n_m)$.

Likewise, in the algebraic part of Corollary 7.126, one may replace each active algebraic or group-convolution component by an environment-measurable unitary conjugate, by a finite direct sum of such components, or by a c-matrix amplification with $c \le c_0$. The charge is

$$c^2 \dim_{\mathbb{C}} \mathcal{A} \qquad \text{or} \qquad c^2 |G|,$$

respectively. If the active-prefix sum of these algebraic charges is at most $C_0 n_m$ eventually, then the conclusions and thresholds of Corollary 7.124 hold with constants depending only on c_0, C_0.

Proof. For a base finite-lift structured component, the proof of Corollary 7.126 gives a finite block lift with ΛK and maximum block size bounded in terms of r_0, b_0, and entropy bounded by $C(r_0, b_0) S_{m,h,a}$. Proposition 6.5 shows that $c_{m,h,a}$-matrix amplification multiplies the entropy by $c_{m,h,a}^2$ and keeps the lift constants bounded in terms of r_0, b_0, c_0. Taking the direct sum over a adds the reduced dimensions with the maximum block size still bounded in terms of r_0, b_0, c_0. Composing the reconstruction map with $\Phi_{m,h}$ only multiplies the lift map norm by at most L_0. Hence the displayed charge bound gives the active-prefix finite-lift hypothesis with M_0 and C_E depending only on r_0, b_0, c_0, L_0, C_0. The finite-lift mixed dominated corollary applies. Unitary and permutation conjugacy, corners, and compressions are the special case in which $\Phi_{m,h}$ is a norm-one self-adjoint image.

For the algebraic closure statement, unitary conjugacy preserves the finite-dimensional C*-algebra dimension, direct sums add dimensions, and tensor amplification by $\mathrm{M}_c(\mathbb{C})$ multiplies dimension by c^2. A group convolution component has reduced C*-algebra dimension $|G|$, so its c-amplification has dimension $c^2|G|$. The active-prefix charge bound is therefore exactly the direct-sum intrinsic-dimension hypothesis of Corollary 7.124. □

Corollary 7.128 (Randomly selected structured submenus with mixed dominated noise)**.** *Fix structural constants $r_0, b_0, c_0, L_0, C_0 \geq 1$. For each m, let $\mathcal{H}_m$ be a row environment and let $\mathscr{S}_m$ be an $\mathcal{H}_m$-measurable finite set of labels chosen from a countable structured catalogue. Assume that there are deterministic $J_m \geq 1$ and events $F_m^{\mathrm{sel}} \in \mathcal{H}_m$ with $\sum_m \mathbb{P}(F_m^{\mathrm{sel}}) < \infty$ such that, outside F_m^{sel},*

$$|\mathscr{S}_m| \leq J_m.$$

Order $\mathscr{S}_m$ by a fixed countable total order and write the selected labels as $\xi_{m,1}, \dots, \xi_{m,N_m}$, $N_m := |\mathscr{S}_m|$. For each selected label $h \leq N_m$, let

$$B_{m,h,1}, \dots, B_{m,h,n_m}, \qquad Y_{m,h,1}, \dots, Y_{m,h,n_m}$$

be the selected self-adjoint core and perturbation streams in $\mathrm{M}_{D_m}(\mathbb{C})$. Assume outside F_m^{sel} that all $B_{m,h,i}$ are contractions and that each selected core has either a finite-lift orbit-closure certificate as in Corollary 7.127, or an algebraic/group certificate from the same corollary. Let $s_{m,h}$ denote the corresponding charge: $c^2 S$ for a finite-lift component, $c^2 \dim_{\mathbb{C}} \mathcal{A}$ for an algebraic component, and $c^2|G|$ for a group-convolution component, summed over the finitely many components of the selected label. Suppose that

$$\sum_{h=1}^{N_m} s_{m,h} \leq C_0 n_m$$

eventually outside F_m^{sel}.

For every selected perturbation stream, choose a ψ or μ tag and the parameters $\varepsilon_{m,h}, A_{m,h}, s'_{m,h}$ of Corollary 7.68; define $q_{m,h}$ there with $Y_{m,h,i}$ in place of $X_{m,h,i}$. Assume that the corresponding conditional tail or moment hypotheses and truncated-tail drift hypotheses hold after adjoining $\mathcal{H}_m$, outside events $E_{m,h}^{\mathrm{tail}}$, and put

$$\lambda_{m,h} := \max\{1, K_{\mathrm{tail}} + \eta\} q_{m,h}.$$

If $\ell_m := \log(e(D_m + n_m + 1)) \leq B\sqrt{n_m}$ eventually,

$$\sum_m \left[\mathbb{P}(F_m^{\mathrm{sel}}) + \sum_{h=1}^{J_m} \left(\mathbb{P}(E_{m,h}^{\mathrm{tail}}) + \varepsilon_{m,h} \right) \right] < \infty,$$

with dummy zero terms for $h > N_m$, and

$$\sum_m J_m (D_m + n_m + 1)(e(D_m + n_m + 1))^{-\gamma_{\mathrm{tc}}(K_{\mathrm{tail}},1,1,B)} < \infty,$$

then, with probability one, for all sufficiently large m, one $\mathcal{H}_m$-measurable sign vector $\varepsilon_m^\star$ satisfies

$$\left\|\sum_{i=1}^{n_m}\varepsilon_{m,i}^\star\left(B_{m,h,i}+\frac{Y_{m,h,i}}{\lambda_{m,h}}\right)\right\|_{\mathrm{op}}\le C\sqrt{n_m}\qquad(1\le h\le N_m),$$

where C depends only on $r_0,b_0,c_0,L_0,C_0,K_{\mathrm{tail}},\eta$. On the same event all selected normalized perturbations are contractions, all selected core-plus-tail tuples have operator norm at most 2, and the normalized tuples have the corresponding common Matrix Spencer bound. The all-integer and sparse threshold statements are the same as in Corollary 7.68, with J_m as the deterministic selected-cardinality envelope.

Proof. Work outside F_m^{sel}. For every selected finite-lift orbit-closure component, the proof of Corollary 7.127 produces a finite-dimensional certificate algebra whose complex dimension is bounded by a constant depending only on r_0,b_0,c_0 times its charge, with lift and image norms bounded by the structural constants. Algebraic and group components are already contained in certificate algebras of dimensions $c^2\dim_{\mathbb{C}}\mathcal{A}$ and $c^2|G|$. Direct-sum all selected certificates. The total certificate dimension is at most $C(r_0,b_0,c_0,L_0,C_0)n_m$.

After normalizing the certificate representatives by the structural lift and image constants, Theorem 4.2 gives one $\mathcal{H}_m$-measurable sign vector whose pushed-forward discrepancy is at most $K_{\mathrm{sel}}\sqrt{n_m}$ for every selected core stream, with K_{sel} depending only on the displayed structural constants. The lexicographically first successful sign vector is measurable because the selected catalogue and certificates are $\mathcal{H}_m$-measurable.

Pad the selected list by zero streams up to the deterministic length J_m. Apply Corollary 7.69 to this ordered active prefix, with $K_{\mathrm{core}}=K_{\mathrm{sel}}$ and $F_m^{\mathrm{core}}=F_m^{\mathrm{sel}}$. The displayed tail summability and matrix-tail summability are exactly the hypotheses of that corollary after the padding, and its conclusion gives the selected-submenu estimate. The integer and sparse thresholds are inherited from the mixed dominated active-library selector. □

Corollary 7.129 (Bernstein-selected structured submenus with mixed dominated noise)**.** *Fix structural constants $r_0,b_0,c_0,L_0,C_0\ge 1$. For each m, let $\mathcal{G}_m$ be a base environment and let*

$$1,\dots,M_m$$

be a finite candidate list of structured labels. The k-th candidate has a charge $\mathfrak{s}_{m,k}\in[0,b_m]$, core streams $B_{m,k,1},\dots,B_{m,k,n_m}$, and perturbation streams $Y_{m,k,1},\dots,Y_{m,k,n_m}$. Let $\eta_{m,1},\dots,\eta_{m,M_m}\in\{0,1\}$ be conditionally independent given $\mathcal{G}_m$, with

$$\mathbb{P}\{\eta_{m,k}=1\mid\mathcal{G}_m\}\le p_{m,k}\qquad(1\le k\le M_m),$$

where the $p_{m,k}\in[0,1]$ are deterministic. Put

$$\mathscr{S}_m:=\{k:\eta_{m,k}=1\},\qquad\mathcal{H}_m:=\mathcal{G}_m\vee\sigma(\eta_{m,1},\dots,\eta_{m,M_m}),$$

and define

$$\mu_m^{\mathrm{ch}} := \sum_{k=1}^{M_m} p_{m,k}\mathfrak{s}_{m,k}, \qquad v_m^{\mathrm{ch}} := \sum_{k=1}^{M_m} p_{m,k}\mathfrak{s}_{m,k}^2, \qquad \mu_m^{\#} := \sum_{k=1}^{M_m} p_{m,k}.$$

Assume that outside events $F_m^{\mathrm{cert}} \in \mathcal{G}_m$, with $\sum_m \mathbb{P}(F_m^{\mathrm{cert}}) < \infty$, every selected core is a contraction and has one of the finite-lift orbit-closure or algebraic/group certificates from Corollary 7.128, with charge bounded by the displayed $\mathfrak{s}_{m,k}$. Choose $t_m, u_m > 0$ and a deterministic cardinality envelope $J_m \geq 1$ such that

$$\mu_m^{\mathrm{ch}} + t_m \leq C_0 n_m, \qquad \mu_m^{\#} + u_m \leq J_m.$$

If

$$\sum_m \left[\mathbb{P}(F_m^{\mathrm{cert}}) + \exp\left\{-\frac{t_m^2}{2(v_m^{\mathrm{ch}} + b_m t_m/3)}\right\} + \exp\left\{-\frac{u_m^2}{2(\mu_m^{\#} + u_m/3)}\right\}\right] < \infty,$$

then the selection event in Corollary 7.128 is summable: almost surely, eventually

$$|\mathscr{S}_m| \leq J_m, \qquad \sum_{k \in \mathscr{S}_m} \mathfrak{s}_{m,k} \leq C_0 n_m.$$

Consequently, suppose each candidate perturbation stream is tagged ψ or μ, with parameters $\varepsilon_{m,k}, A_{m,k}, s'_{m,k}$, and satisfies the corresponding mixed dominated tail and drift hypotheses after adjoining $\mathcal{H}_m$. If

$$\ell_m := \log(e(D_m + n_m + 1)) \leq B\sqrt{n_m}$$

eventually,

$$\sum_m \sum_{k=1}^{M_m} \left(\mathbb{P}(E_{m,k}^{\mathrm{tail}}) + \varepsilon_{m,k}\right) < \infty,$$

and

$$\sum_m J_m (D_m + n_m + 1)(e(D_m + n_m + 1))^{-\gamma_{\mathrm{tc}}(K_{\mathrm{tail}},1,1,B)} < \infty,$$

then the selected structured submenus have the same eventual common signing, contraction normalization, fixed-menu coverage, and all-integer/sparse thresholds as in Corollary 7.128.

Proof. Condition on $\mathcal{G}_m$. For the selected charge,

$$Z_k := \eta_{m,k}\mathfrak{s}_{m,k}$$

are independent, satisfy $0 \leq Z_k \leq b_m$, and have conditional means at most $p_{m,k}\mathfrak{s}_{m,k}$. Their conditional variance sum is at most v_m^{ch}. The scalar Bernstein inequality gives

$$\mathbb{P}\left\{\sum_k \eta_{m,k}\mathfrak{s}_{m,k} > C_0 n_m \,\middle|\, \mathcal{G}_m\right\} \leq \exp\left\{-\frac{t_m^2}{2(v_m^{\mathrm{ch}} + b_m t_m/3)}\right\}.$$

The same argument with $Z_k = \eta_{m,k}$, jump 1, and variance sum at most $\mu_m^{\#}$ gives

$$\mathbb{P}\{|\mathscr{S}_m| > J_m \,|\, \mathcal{G}_m\} \le \exp\left\{-\frac{u_m^2}{2(\mu_m^{\#} + u_m/3)}\right\}.$$

After averaging and adding F_m^{cert}, the displayed summability gives a summable selection failure event. On its complement, the selected set satisfies the cardinality and charge hypotheses of Corollary 7.128. Ordering the selected labels and padding by zero streams converts the candidate tail summability into the active-prefix tail summability required there. The final conclusions are exactly that corollary's conclusions. □

Corollary 7.130 (Laplace-dominated selected structured submenus with mixed dominated noise)**.** *Fix structural constants $r_0, b_0, c_0, L_0, C_0 \ge 1$. For each m, let $\mathcal{G}_m$ be a base environment and let $1, \dots, M_m$ be a finite candidate list of structured labels. The k-th candidate has charge $\mathfrak{s}_{m,k} \in [0, b_m]$, with $b_m > 0$, core streams $B_{m,k,1}, \dots, B_{m,k,n_m}$, and perturbation streams $Y_{m,k,1}, \dots, Y_{m,k,n_m}$. Let $\eta_{m,1}, \dots, \eta_{m,M_m} \in \{0,1\}$ be $\mathcal{H}_m$-measurable selection indicators, where*

$$\mathcal{H}_m := \mathcal{G}_m \vee \sigma(\eta_{m,1}, \dots, \eta_{m,M_m}), \qquad \mathscr{S}_m := \{k : \eta_{m,k} = 1\}.$$

Assume there are deterministic $p_{m,k} \in [0,1]$ such that, conditionally on $\mathcal{G}_m$, the selected cardinality and charge satisfy the Laplace domination bounds

$$\mathbb{E}\left[\exp\left\{\theta \sum_{k=1}^{M_m} \eta_{m,k}\mathfrak{s}_{m,k}\right\} \,\middle|\, \mathcal{G}_m\right] \le \exp\left\{\sum_{k=1}^{M_m} p_{m,k}(e^{\theta \mathfrak{s}_{m,k}} - 1)\right\}$$

and

$$\mathbb{E}\left[\exp\left\{\theta \sum_{k=1}^{M_m} \eta_{m,k}\right\} \,\middle|\, \mathcal{G}_m\right] \le \exp\left\{\sum_{k=1}^{M_m} p_{m,k}(e^{\theta} - 1)\right\}$$

for every $\theta \ge 0$. Put

$$\mu_m^{\mathrm{ch}} := \sum_{k=1}^{M_m} p_{m,k}\mathfrak{s}_{m,k}, \qquad v_m^{\mathrm{ch}} := \sum_{k=1}^{M_m} p_{m,k}\mathfrak{s}_{m,k}^2, \qquad \mu_m^{\#} := \sum_{k=1}^{M_m} p_{m,k}.$$

Assume that outside events $F_m^{\mathrm{cert}} \in \mathcal{G}_m$, with $\sum_m \mathbb{P}(F_m^{\mathrm{cert}}) < \infty$, every selected core is a contraction and has one of the finite-lift orbit-closure or algebraic/group certificates from Corollary 7.128, with charge bounded by the displayed $\mathfrak{s}_{m,k}$. Choose $t_m, u_m > 0$ and a deterministic cardinality envelope $J_m \ge 1$ such that

$$\mu_m^{\mathrm{ch}} + t_m \le C_0 n_m, \qquad \mu_m^{\#} + u_m \le J_m.$$

If

$$\sum_m \left[\mathbb{P}(F_m^{\mathrm{cert}}) + \exp\left\{-\frac{t_m^2}{2(v_m^{\mathrm{ch}} + b_m t_m/3)}\right\} + \exp\left\{-\frac{u_m^2}{2(\mu_m^{\#} + u_m/3)}\right\}\right] < \infty,$$

then the selection event in Corollary 7.128 is summable: almost surely, eventually

$$|\mathscr{S}_m| \le J_m, \qquad \sum_{k\in\mathscr{S}_m} \mathfrak{s}_{m,k} \le C_0 n_m.$$

Consequently, suppose each candidate perturbation stream is tagged ψ or μ, with parameters $\varepsilon_{m,k}, A_{m,k}, s'_{m,k}$, and satisfies the corresponding mixed dominated tail and drift hypotheses after adjoining $\mathcal{H}_m$. If

$$\ell_m := \log(e(D_m + n_m + 1)) \le B\sqrt{n_m}$$

eventually,

$$\sum_m \sum_{k=1}^{M_m} \big(\mathbb{P}(E^{\mathrm{tail}}_{m,k}) + \varepsilon_{m,k}\big) < \infty,$$

and

$$\sum_m J_m (D_m + n_m + 1)(e(D_m + n_m + 1))^{-\gamma_{\mathrm{tc}}(K_{\mathrm{tail}},1,1,B)} < \infty,$$

then the Laplace-dominated selected structured submenus have the same eventual common signing, contraction normalization, fixed-menu coverage, and all-integer/sparse thresholds as in Corollary 7.128.

The displayed Laplace domination is automatic, for example, whenever conditionally on $\mathcal{G}_m$ the indicators are negatively associated and $\mathbb{E}[\eta_{m,k} \mid \mathcal{G}_m] \le p_{m,k}$ for every k. Here negative association is meant conditionally on $\mathcal{G}_m$, in the usual sense that increasing functions of disjoint coordinate sets have nonpositive conditional covariance.

Proof. Condition on $\mathcal{G}_m$. The first Laplace domination hypothesis and the standard Bennett bound

$$e^{\theta x} - 1 - \theta x \le \frac{x^2}{b_m^2}(e^{\theta b_m} - 1 - \theta b_m) \qquad (0 \le x \le b_m)$$

give, for $S^{\mathrm{ch}}_m := \sum_k \eta_{m,k}\mathfrak{s}_{m,k}$,

$$\log \mathbb{E}\left[e^{\theta(S^{\mathrm{ch}}_m - \mu^{\mathrm{ch}}_m)} \,\middle|\, \mathcal{G}_m\right] \le \frac{v^{\mathrm{ch}}_m}{b_m^2}(e^{\theta b_m} - 1 - \theta b_m).$$

Chernoff optimization gives

$$\mathbb{P}\{S^{\mathrm{ch}}_m > \mu^{\mathrm{ch}}_m + t_m \mid \mathcal{G}_m\} \le \exp\left\{-\frac{t_m^2}{2(v^{\mathrm{ch}}_m + b_m t_m/3)}\right\}.$$

The second Laplace domination hypothesis gives the same estimate for $|\mathscr{S}_m|$, with $b_m = 1$, v^{ch}_m replaced by $\mu^{\#}_m$, and t_m replaced by u_m:

$$\mathbb{P}\{|\mathscr{S}_m| > \mu^{\#}_m + u_m \mid \mathcal{G}_m\} \le \exp\left\{-\frac{u_m^2}{2(\mu^{\#}_m + u_m/3)}\right\}.$$

After averaging, adding F^{cert}_m, and applying Borel–Cantelli, the selected set eventually satisfies the deterministic cardinality and charge hypotheses of Corollary 7.128. Ordering the selected labels and padding by zero streams

converts the candidate tail summability into the active-prefix tail summability required there, and the deterministic matrix-tail series is unchanged.

For the final sufficient condition, fix $\theta \geq 0$ and nonnegative weights a_k, and put

$$L_m(\theta, a) := \mathbb{E}\left[\exp\left\{\theta \sum_k a_k \eta_{m,k}\right\} \middle| \mathcal{G}_m\right].$$

Negative association gives

$$\begin{aligned} L_m(\theta, a) &\leq \prod_k \mathbb{E}[e^{\theta a_k \eta_{m,k}} \mid \mathcal{G}_m] \\ &\leq \prod_k \left(1 + p_{m,k}(e^{\theta a_k} - 1)\right) \\ &\leq \exp\left\{\sum_k p_{m,k}(e^{\theta a_k} - 1)\right\}. \end{aligned}$$

Taking $a_k = \mathfrak{s}_{m,k}$ and $a_k = 1$ gives the two Laplace domination bounds. □

Corollary 7.131 (Random-envelope Laplace-dominated selected structured submenus)**.** *Fix structural constants $r_0, b_0, c_0, L_0, C_0 \geq 1$. For each m, let $\mathcal{G}_m$ be a base environment and let $1, \ldots, M_m$ be a finite candidate list of structured labels. The k-th candidate has charge $\mathfrak{s}_{m,k} \in [0, b_m]$, with $b_m > 0$, core streams $B_{m,k,1}, \ldots, B_{m,k,n_m}$, and perturbation streams $Y_{m,k,1}, \ldots, Y_{m,k,n_m}$. Let $\eta_{m,1}, \ldots, \eta_{m,M_m} \in \{0,1\}$ be selection indicators, set*

$$\mathcal{H}_m := \mathcal{G}_m \vee \sigma(\eta_{m,1}, \ldots, \eta_{m,M_m}), \qquad \mathscr{S}_m := \{k : \eta_{m,k} = 1\},$$

and suppose there are $\mathcal{G}_m$-measurable envelopes $P_{m,k} \in [0,1]$ such that, conditionally on $\mathcal{G}_m$, for every $\theta \geq 0$,

$$\mathbb{E}\left[\exp\left\{\theta \sum_{k=1}^{M_m} \eta_{m,k}\mathfrak{s}_{m,k}\right\} \middle| \mathcal{G}_m\right] \leq \exp\left\{\sum_{k=1}^{M_m} P_{m,k}(e^{\theta \mathfrak{s}_{m,k}} - 1)\right\}$$

and

$$\mathbb{E}\left[\exp\left\{\theta \sum_{k=1}^{M_m} \eta_{m,k}\right\} \middle| \mathcal{G}_m\right] \leq \exp\left\{\sum_{k=1}^{M_m} P_{m,k}(e^{\theta} - 1)\right\}.$$

Put

$$M_m^{\mathrm{ch}} := \sum_{k=1}^{M_m} P_{m,k}\mathfrak{s}_{m,k}, \qquad V_m^{\mathrm{ch}} := \sum_{k=1}^{M_m} P_{m,k}\mathfrak{s}_{m,k}^2, \qquad M_m^{\#} := \sum_{k=1}^{M_m} P_{m,k}.$$

Assume that there are events $F_m^{\mathrm{env}} \in \mathcal{G}_m$, with $\sum_m \mathbb{P}(F_m^{\mathrm{env}}) < \infty$, and deterministic bounds $\bar{\mu}_m^{\mathrm{ch}}, \bar{v}_m^{\mathrm{ch}}, \bar{\mu}_m^{\#} \geq 0$ such that outside F_m^{env},

$$M_m^{\mathrm{ch}} \leq \bar{\mu}_m^{\mathrm{ch}}, \qquad V_m^{\mathrm{ch}} \leq \bar{v}_m^{\mathrm{ch}}, \qquad M_m^{\#} \leq \bar{\mu}_m^{\#}.$$

Assume also that outside events $F_m^{\mathrm{cert}} \in \mathcal{G}_m$, with $\sum_m \mathbb{P}(F_m^{\mathrm{cert}}) < \infty$, every selected core is a contraction and has one of the finite-lift orbit-closure or algebraic/group certificates from Corollary 7.128, with charge bounded by the

displayed $\mathfrak{s}_{m,k}$. *Choose* $t_m, u_m > 0$ *and a deterministic cardinality envelope* $J_m \geq 1$ *such that*

$$\bar{\mu}_m^{\mathrm{ch}} + t_m \leq C_0 n_m, \qquad \bar{\mu}_m^{\#} + u_m \leq J_m.$$

Put

$$\Lambda_m := \mathbb{P}(F_m^{\mathrm{env}}) + \mathbb{P}(F_m^{\mathrm{cert}}) + \exp\left\{-\frac{t_m^2}{2(\bar{v}_m^{\mathrm{ch}} + b_m t_m/3)}\right\} + \exp\left\{-\frac{u_m^2}{2(\bar{\mu}_m^{\#} + u_m/3)}\right\}.$$

If

$$\sum_m \Lambda_m < \infty,$$

then the selection event in Corollary 7.128 is summable: almost surely, eventually

$$|\mathscr{S}_m| \leq J_m, \qquad \sum_{k \in \mathscr{S}_m} \mathfrak{s}_{m,k} \leq C_0 n_m.$$

Consequently, suppose each candidate perturbation stream is tagged ψ *or* μ, *with parameters* $\varepsilon_{m,k}, A_{m,k}, s'_{m,k}$, *and satisfies the corresponding mixed dominated tail and drift hypotheses after adjoining* $\mathcal{H}_m$. *If*

$$\ell_m := \log(e(D_m + n_m + 1)) \leq B\sqrt{n_m}$$

eventually,

$$\sum_m \sum_{k=1}^{M_m} \left(\mathbb{P}(E_{m,k}^{\mathrm{tail}}) + \varepsilon_{m,k}\right) < \infty,$$

and

$$\sum_m J_m (D_m + n_m + 1)(e(D_m + n_m + 1))^{-\gamma_{\mathrm{tc}}(K_{\mathrm{tail}},1,1,B)} < \infty,$$

then the random-envelope Laplace-dominated selected structured submenus have the same eventual common signing, contraction normalization, fixed-menu coverage, and all-integer/sparse thresholds as in Corollary 7.128.

The hypotheses hold, for example, under conditional negative association with $\mathbb{E}[\eta_{m,k} \mid \mathcal{G}_m] \leq P_{m,k}$, *followed by any summable high-probability control of the three aggregate profiles. Thus the deterministic envelopes in Corollary 7.130 may be replaced by data-dependent or environment-random probabilities whenever their total expected charge, quadratic charge, and cardinality are certified.*

Proof. Work on $(F_m^{\mathrm{env}})^c$ and condition on $\mathcal{G}_m$. For $S_m^{\mathrm{ch}} := \sum_k \eta_{m,k}\mathfrak{s}_{m,k}$, the Laplace domination and the Bennett bound

$$e^{\theta x} - 1 - \theta x \leq \frac{x^2}{b_m^2}(e^{\theta b_m} - 1 - \theta b_m) \qquad (0 \leq x \leq b_m)$$

give

$$\log \mathbb{E}\left[e^{\theta(S_m^{\mathrm{ch}} - M_m^{\mathrm{ch}})} \,\middle|\, \mathcal{G}_m\right] \leq \frac{V_m^{\mathrm{ch}}}{b_m^2}(e^{\theta b_m} - 1 - \theta b_m).$$

Since $M_m^{\mathrm{ch}} \le \bar{\mu}_m^{\mathrm{ch}}$ and $V_m^{\mathrm{ch}} \le \bar{v}_m^{\mathrm{ch}}$ on the present event, Chernoff optimization gives

$$\mathbb{P}\{S_m^{\mathrm{ch}} > \bar{\mu}_m^{\mathrm{ch}} + t_m \mid \mathcal{G}_m\} \le \exp\left\{-\frac{t_m^2}{2(\bar{v}_m^{\mathrm{ch}} + b_m t_m/3)}\right\}.$$

The second Laplace domination bound, with $b_m = 1$, gives

$$\mathbb{P}\{|\mathscr{S}_m| > \bar{\mu}_m^{\#} + u_m \mid \mathcal{G}_m\} \le \exp\left\{-\frac{u_m^2}{2(\bar{\mu}_m^{\#} + u_m/3)}\right\}.$$

After averaging, adding $F_m^{\mathrm{env}} \cup F_m^{\mathrm{cert}}$, and applying Borel–Cantelli, the selected catalogue eventually satisfies the deterministic cardinality and charge hypotheses of Corollary 7.128. Ordering the selected labels and padding by zero streams converts the candidate tail summability into the active-prefix tail summability required there.

The negative-association sufficient condition is the same product-mgf argument as in Corollary 7.130, with $P_{m,k}$ in place of $p_{m,k}$. □

Corollary 7.132 (Random-envelope Poisson-count selected structured submenus)**.** *Fix structural constants $r_0, b_0, c_0, L_0, C_0 \ge 1$. For each m, let $\mathcal{G}_m$ be a base environment and let $1, \dots, M_m$ be a finite candidate list of structured labels. The k-th candidate has charge $\mathfrak{s}_{m,k} \in [0, b_m]$, with $b_m > 0$, core streams $B_{m,k,1}, \dots, B_{m,k,n_m}$, and perturbation streams $Y_{m,k,1}, \dots, Y_{m,k,n_m}$. Let $N_{m,k} \in \mathbb{Z}_{\ge 0}$ be random proposal counts and select the support of the proposal field,*

$$\eta_{m,k} := \mathbf{1}_{\{N_{m,k} \ge 1\}}, \qquad \mathscr{S}_m := \{k : \eta_{m,k} = 1\},$$

with

$$\mathcal{H}_m := \mathcal{G}_m \vee \sigma(N_{m,1}, \dots, N_{m,M_m}).$$

Assume that there are $\mathcal{G}_m$-measurable intensity envelopes $\Lambda_{m,k} \ge 0$ such that, conditionally on $\mathcal{G}_m$, for every $\theta \ge 0$,

$$\mathbb{E}\left[\exp\left\{\theta \sum_{k=1}^{M_m} N_{m,k}\mathfrak{s}_{m,k}\right\} \,\middle|\, \mathcal{G}_m\right] \le \exp\left\{\sum_{k=1}^{M_m} \Lambda_{m,k}(e^{\theta \mathfrak{s}_{m,k}} - 1)\right\}$$

and

$$\mathbb{E}\left[\exp\left\{\theta \sum_{k=1}^{M_m} N_{m,k}\right\} \,\middle|\, \mathcal{G}_m\right] \le \exp\left\{\sum_{k=1}^{M_m} \Lambda_{m,k}(e^{\theta} - 1)\right\}.$$

Define the random count profiles

$$M_m^{\mathrm{cnt,ch}} := \sum_{k=1}^{M_m} \Lambda_{m,k}\mathfrak{s}_{m,k},$$

$$V_m^{\mathrm{cnt,ch}} := \sum_{k=1}^{M_m} \Lambda_{m,k}\mathfrak{s}_{m,k}^2,$$

$$M_m^{\mathrm{cnt},\#} := \sum_{k=1}^{M_m} \Lambda_{m,k}.$$

Assume that there are events $F_m^{\mathrm{env}} \in \mathcal{G}_m$, *with* $\sum_m \mathbb{P}(F_m^{\mathrm{env}}) < \infty$, *and deterministic bounds* $\bar{\mu}_m^{\mathrm{cnt,ch}}, \bar{v}_m^{\mathrm{cnt,ch}}, \bar{\mu}_m^{\mathrm{cnt},\#} \geq 0$ *such that outside* F_m^{env},

$$M_m^{\mathrm{cnt,ch}} \leq \bar{\mu}_m^{\mathrm{cnt,ch}}, \qquad V_m^{\mathrm{cnt,ch}} \leq \bar{v}_m^{\mathrm{cnt,ch}}, \qquad M_m^{\mathrm{cnt},\#} \leq \bar{\mu}_m^{\mathrm{cnt},\#}.$$

Assume also that outside events $F_m^{\mathrm{cert}} \in \mathcal{G}_m$, *with* $\sum_m \mathbb{P}(F_m^{\mathrm{cert}}) < \infty$, *every selected core is a contraction and has one of the finite-lift orbit-closure or algebraic/group certificates from Corollary 7.128, with charge bounded by the displayed* $\mathfrak{s}_{m,k}$. *Choose* $t_m, u_m > 0$ *and a deterministic cardinality envelope* $J_m \geq 1$ *such that*

$$\bar{\mu}_m^{\mathrm{cnt,ch}} + t_m \leq C_0 n_m, \qquad \bar{\mu}_m^{\mathrm{cnt},\#} + u_m \leq J_m.$$

Put

$$\Lambda_m^{\mathrm{cnt}} := \mathbb{P}(F_m^{\mathrm{env}}) + \mathbb{P}(F_m^{\mathrm{cert}}) + \exp\left\{-\frac{t_m^2}{2(\bar{v}_m^{\mathrm{cnt,ch}} + b_m t_m/3)}\right\} + \exp\left\{-\frac{u_m^2}{2(\bar{\mu}_m^{\mathrm{cnt},\#} + u_m/3)}\right\}.$$

If

$$\sum_m \Lambda_m^{\mathrm{cnt}} < \infty,$$

then the selection event in Corollary 7.128 is summable: almost surely, eventually

$$|\mathscr{S}_m| \leq J_m, \qquad \sum_{k \in \mathscr{S}_m} \mathfrak{s}_{m,k} \leq C_0 n_m.$$

Consequently, suppose each candidate perturbation stream is tagged ψ *or* μ, *with parameters* $\varepsilon_{m,k}, A_{m,k}, s'_{m,k}$, *and satisfies the corresponding mixed dominated tail and drift hypotheses after adjoining* $\mathcal{H}_m$. *If*

$$\ell_m := \log(e(D_m + n_m + 1)) \leq B\sqrt{n_m}$$

eventually,

$$\sum_m \sum_{k=1}^{M_m} (\mathbb{P}(E_{m,k}^{\mathrm{tail}}) + \varepsilon_{m,k}) < \infty,$$

and

$$\sum_m J_m (D_m + n_m + 1)(e(D_m + n_m + 1))^{-\gamma_{\mathrm{tc}}(K_{\mathrm{tail}},1,1,B)} < \infty,$$

then the Poisson-count selected structured submenus have the same eventual common signing, contraction normalization, fixed-menu coverage, and all-integer/sparse thresholds as in Corollary 7.128.

This applies to independent Poisson counts, Cox or environment-random Poisson counts after summable certification of their aggregate intensity profiles, and more generally to any proposal-count field whose cardinality and charge transforms are dominated by the displayed compound-Poisson envelopes. It covers Poissonized model proposals, occupancy libraries, random hyperparameter proposal clouds, and repeated-draw structured catalogues after passing from counts to their occupied support.

Proof. Since $N_{m,k}$ are nonnegative integers,

$$\eta_{m,k} \le N_{m,k}, \qquad |\mathscr{S}_m| \le \sum_k N_{m,k}, \qquad \sum_{k\in\mathscr{S}_m} \mathfrak{s}_{m,k} \le \sum_k N_{m,k}\mathfrak{s}_{m,k}.$$

Work on $(F_m^{\mathrm{env}})^c$ and condition on $\mathcal{G}_m$. For

$$T_m^{\mathrm{ch}} := \sum_k N_{m,k}\mathfrak{s}_{m,k},$$

the first compound-Poisson domination hypothesis and the Bennett bound

$$e^{\theta x} - 1 - \theta x \le \frac{x^2}{b_m^2}(e^{\theta b_m} - 1 - \theta b_m) \qquad (0 \le x \le b_m)$$

give

$$\log \mathbb{E}\left[e^{\theta(T_m^{\mathrm{ch}} - M_m^{\mathrm{cnt,ch}})} \,\middle|\, \mathcal{G}_m\right] \le \frac{V_m^{\mathrm{cnt,ch}}}{b_m^2}(e^{\theta b_m} - 1 - \theta b_m).$$

Using the deterministic bounds on the profiles and optimizing in θ,

$$\mathbb{P}\left\{(F_m^{\mathrm{env}})^c,\ T_m^{\mathrm{ch}} > \bar{\mu}_m^{\mathrm{cnt,ch}} + t_m\right\} \le \exp\left\{-\frac{t_m^2}{2(\bar{v}_m^{\mathrm{cnt,ch}} + b_m t_m/3)}\right\}.$$

The second compound-Poisson domination bound, with atom size 1, gives

$$\mathbb{P}\left\{(F_m^{\mathrm{env}})^c,\ \sum_k N_{m,k} > \bar{\mu}_m^{\mathrm{cnt},\#} + u_m\right\} \le \exp\left\{-\frac{u_m^2}{2(\bar{\mu}_m^{\mathrm{cnt},\#} + u_m/3)}\right\}.$$

Adding $F_m^{\mathrm{env}} \cup F_m^{\mathrm{cert}}$ and summing over m, the Borel–Cantelli lemma gives the eventual selected-support cardinality and charge bounds displayed above.

On the resulting good event, order the occupied labels by candidate order and pad by zero streams up to J_m. The candidate tail summability dominates the selected active-prefix tail summability required by Corollary 7.128, and the deterministic matrix-tail series is unchanged. Applying that corollary gives the signing, normalization, coverage, and threshold conclusions. The listed Poisson and Cox examples have equality in the conditional transform display after conditioning on their intensity field; the stated domination variant is the same argument with the inequality in place of equality. □

Corollary 7.133 (Cox-Poisson occupied-support selected structured submenus)**.** *Fix structural constants $r_0, b_0, c_0, L_0, C_0 \ge 1$. In the setting of Corollary 7.132, assume more specifically that, conditionally on $\mathcal{G}_m$, the proposal counts $N_{m,1}, \dots, N_{m,M_m}$ are independent Poisson variables with $\mathcal{G}_m$-measurable intensities $\Lambda_{m,k} \ge 0$. Select the occupied support*

$$\eta_{m,k} := \mathbf{1}_{\{N_{m,k}\ge 1\}}, \qquad \mathscr{S}_m := \{k : \eta_{m,k} = 1\},$$

and put

$$Q_{m,k} := 1 - e^{-\Lambda_{m,k}}.$$

Define the occupied-support profiles

$$M_m^{\mathrm{occ,ch}} := \sum_{k=1}^{M_m} Q_{m,k}\mathfrak{s}_{m,k},$$

$$V_m^{\mathrm{occ,ch}} := \sum_{k=1}^{M_m} Q_{m,k}\mathfrak{s}_{m,k}^2,$$

$$M_m^{\mathrm{occ},\#} := \sum_{k=1}^{M_m} Q_{m,k}.$$

Assume that there are events $F_m^{\mathrm{env}} \in \mathcal{G}_m$, with $\sum_m \mathbb{P}(F_m^{\mathrm{env}}) < \infty$, and deterministic bounds $\bar{\mu}_m^{\mathrm{occ,ch}}, \bar{v}_m^{\mathrm{occ,ch}}, \bar{\mu}_m^{\mathrm{occ},\#} \geq 0$ such that outside F_m^{env},

$$M_m^{\mathrm{occ,ch}} \leq \bar{\mu}_m^{\mathrm{occ,ch}}, \qquad V_m^{\mathrm{occ,ch}} \leq \bar{v}_m^{\mathrm{occ,ch}}, \qquad M_m^{\mathrm{occ},\#} \leq \bar{\mu}_m^{\mathrm{occ},\#}.$$

Assume the same candidate certificate condition as in Corollary 7.132, outside summable events $F_m^{\mathrm{cert}} \in \mathcal{G}_m$. Choose $t_m, u_m > 0$ and a deterministic cardinality envelope $J_m \geq 1$ so that

$$\bar{\mu}_m^{\mathrm{occ,ch}} + t_m \leq C_0 n_m, \qquad \bar{\mu}_m^{\mathrm{occ},\#} + u_m \leq J_m.$$

If

$$\sum_m \Bigg[\mathbb{P}(F_m^{\mathrm{env}}) + \mathbb{P}(F_m^{\mathrm{cert}}) + \exp\left\{ -\frac{t_m^2}{2(\bar{v}_m^{\mathrm{occ,ch}} + b_m t_m/3)} \right\} + \exp\left\{ -\frac{u_m^2}{2(\bar{\mu}_m^{\mathrm{occ},\#} + u_m/3)} \right\} \Bigg] < \infty,$$

then the selection event in Corollary 7.128 is summable: almost surely, eventually

$$|\mathscr{S}_m| \leq J_m, \qquad \sum_{k \in \mathscr{S}_m} \mathfrak{s}_{m,k} \leq C_0 n_m.$$

Consequently, under the same candidate perturbation, mixed dominated tail, drift, cap-budget, and matrix-tail summability assumptions as in Corollary 7.132, with this deterministic envelope J_m, the Cox-Poisson occupied-support submenus have the same eventual common signing, contraction normalization, fixed-menu coverage, and all-integer/sparse thresholds as in Corollary 7.128.

Thus Poissonized repeated-draw catalogues may be charged by occupied labels rather than by total proposal multiplicity. When some intensities are large, the budget uses $1 - e^{-\Lambda_{m,k}} \leq 1$, improving the count profile $\Lambda_{m,k}$ from Corollary 7.132.

Proof. Condition on $\mathcal{G}_m$. The occupied indicators are independent Bernoulli variables and

$$\mathbb{P}\{\eta_{m,k} = 1 \mid \mathcal{G}_m\} = 1 - e^{-\Lambda_{m,k}} = Q_{m,k}.$$

Therefore, for every $\theta \geq 0$,

$$\mathbb{E}\left[\exp\left\{\theta \sum_k \eta_{m,k}\mathfrak{s}_{m,k}\right\} \middle| \mathcal{G}_m\right] = \prod_k \left(1 + Q_{m,k}(e^{\theta \mathfrak{s}_{m,k}} - 1)\right) \leq \exp\left\{\sum_k Q_{m,k}(e^{\theta \mathfrak{s}_{m,k}} - 1)\right\},$$

and the same calculation with $\mathfrak{s}_{m,k} = 1$ gives the cardinality Laplace domination. Hence Corollary 7.131 applies with $P_{m,k} = Q_{m,k}$. Its conclusion is exactly the displayed selection event and the inherited mixed-tail Matrix Spencer conclusion. □

Corollary 7.134 (Product-categorical selected structured submenus with mixed dominated noise)**.** *Fix structural constants $r_0, b_0, c_0, L_0, C_0 \geq 1$. For each m, let $\mathcal{G}_m$ be a base environment and let the finite candidate list be partitioned into deterministic slots*

$$\mathcal{I}_{m,1}, \ldots, \mathcal{I}_{m,L_m}.$$

The k-th candidate has charge $\mathfrak{s}_{m,k} \in [0, b_m]$, with $b_m > 0$, core streams $B_{m,k,1}, \ldots, B_{m,k,n_m}$, and perturbation streams $Y_{m,k,1}, \ldots, Y_{m,k,n_m}$. In slot g, choose at most one label by an $\mathcal{I}_{m,g} \cup \{\varnothing\}$-valued random variable $\xi_{m,g}$. Conditionally on $\mathcal{G}_m$, assume that $\xi_{m,1}, \ldots, \xi_{m,L_m}$ are independent and that deterministic probability envelopes $\pi_{m,k} \in [0,1]$ satisfy

$$\mathbb{P}\{\xi_{m,g} = k \mid \mathcal{G}_m\} \leq \pi_{m,k} \qquad (k \in \mathcal{I}_{m,g}).$$

Let

$$\eta_{m,k} := \mathbf{1}_{\{\xi_{m,g}=k\}} \quad (k \in \mathcal{I}_{m,g}), \qquad \mathscr{S}_m := \{k : \eta_{m,k} = 1\},$$

and put

$$\mathcal{H}_m := \mathcal{G}_m \vee \sigma(\xi_{m,1}, \ldots, \xi_{m,L_m}).$$

Define

$$\mu_m^{\mathrm{ch}} := \sum_k \pi_{m,k}\mathfrak{s}_{m,k}, \qquad v_m^{\mathrm{ch}} := \sum_k \pi_{m,k}\mathfrak{s}_{m,k}^2.$$

Assume that outside events $F_m^{\mathrm{cert}} \in \mathcal{G}_m$, with $\sum_m \mathbb{P}(F_m^{\mathrm{cert}}) < \infty$, every selected core is a contraction and has one of the finite-lift orbit-closure or algebraic/group certificates from Corollary 7.128, with charge bounded by the displayed $\mathfrak{s}_{m,k}$. Choose $t_m > 0$ and a deterministic cardinality envelope $J_m \geq L_m$ such that

$$\mu_m^{\mathrm{ch}} + t_m \leq C_0 n_m.$$

If

$$\sum_m \left[\mathbb{P}(F_m^{\mathrm{cert}}) + \exp\left\{-\frac{t_m^2}{2(v_m^{\mathrm{ch}} + b_m t_m/3)}\right\}\right] < \infty,$$

then the selection event in Corollary 7.128 is summable: almost surely, eventually

$$|\mathscr{S}_m| \leq J_m, \qquad \sum_{k \in \mathscr{S}_m} \mathfrak{s}_{m,k} \leq C_0 n_m.$$

Consequently, suppose each candidate perturbation stream is tagged ψ or μ, with parameters $\varepsilon_{m,k}, A_{m,k}, s'_{m,k}$, and satisfies the corresponding mixed dominated tail and drift hypotheses after adjoining $\mathcal{H}_m$. If

$$\ell_m := \log(e(D_m + n_m + 1)) \le B\sqrt{n_m}$$

eventually,

$$\sum_m \sum_k \bigl(\mathbb{P}(E^{\mathrm{tail}}_{m,k}) + \varepsilon_{m,k}\bigr) < \infty,$$

and

$$\sum_m J_m (D_m + n_m + 1)(e(D_m + n_m + 1))^{-\gamma_{\mathrm{tc}}(K_{\mathrm{tail}},1,1,B)} < \infty,$$

then the product-categorical selected structured submenus have the same eventual common signing, contraction normalization, fixed-menu coverage, and all-integer/sparse thresholds as in Corollary 7.128.

This covers independent randomized choice of one structured template per block, product mixtures of finite-lift and algebraic core templates, randomized hyperparameter menus, and sampling with replacement when each draw is treated as a distinct candidate copy.

Proof. Condition on $\mathcal{G}_m$ and put

$$X_{m,g} := \begin{cases} \mathfrak{s}_{m,\xi_{m,g}}, & \xi_{m,g} \neq \varnothing, \\ 0, & \xi_{m,g} = \varnothing. \end{cases}$$

Then $0 \le X_{m,g} \le b_m$, the variables $X_{m,g}$ are conditionally independent, and

$$\sum_g \mathbb{E}[X_{m,g} \mid \mathcal{G}_m] \le \mu^{\mathrm{ch}}_m, \qquad \sum_g \mathbb{E}[X^2_{m,g} \mid \mathcal{G}_m] \le v^{\mathrm{ch}}_m.$$

Hence their conditional variance sum is at most v^{ch}_m. Bernstein's inequality gives

$$\mathbb{P}\left\{ \sum_{k \in \mathscr{S}_m} \mathfrak{s}_{m,k} > \mu^{\mathrm{ch}}_m + t_m \,\middle|\, \mathcal{G}_m \right\} \le \exp\left\{ -\frac{t_m^2}{2(v^{\mathrm{ch}}_m + b_m t_m/3)} \right\}.$$

Also $|\mathscr{S}_m| \le L_m \le J_m$ deterministically. After averaging, adding F^{cert}_m, and applying Borel–Cantelli, the selected catalogue eventually satisfies the deterministic hypotheses of Corollary 7.128. Ordering the selected labels and padding by zero streams again converts the candidate tail summability into the active-prefix summability required there. The final assertions are exactly the conclusions of the selected-submenu corollary. □

Corollary 7.135 (Adaptive-categorical selected structured submenus with mixed dominated noise)**.** *Fix structural constants $r_0, b_0, c_0, L_0, C_0 \ge 1$. For each m, let*

$$\mathcal{G}_m = \mathcal{F}_{m,0} \subset \mathcal{F}_{m,1} \subset \cdots \subset \mathcal{F}_{m,L_m}$$

be a filtration and let the finite candidate list be partitioned into deterministic slots $\mathcal{I}_{m,1}, \dots, \mathcal{I}_{m,L_m}$. The k-th candidate has charge $\mathfrak{s}_{m,k} \in [0, b_m]$,

with $b_m > 0$, core streams $B_{m,k,1}, \dots, B_{m,k,n_m}$, and perturbation streams $Y_{m,k,1}, \dots, Y_{m,k,n_m}$. In slot g, choose at most one label by an $\mathcal{I}_{m,g} \cup \{\varnothing\}$-valued random variable $\xi_{m,g}$ which is $\mathcal{F}_{m,g}$-measurable. Assume deterministic probability envelopes $\pi_{m,k} \in [0,1]$ satisfy

$$\mathbb{P}\{\xi_{m,g} = k \mid \mathcal{F}_{m,g-1}\} \le \pi_{m,k} \qquad (k \in \mathcal{I}_{m,g}).$$

Let

$$\eta_{m,k} := \mathbf{1}_{\{\xi_{m,g}=k\}} \quad (k \in \mathcal{I}_{m,g}),$$
$$\mathscr{S}_m := \{k : \eta_{m,k} = 1\},$$
$$\mathcal{H}_m := \mathcal{F}_{m,L_m},$$

and define

$$\mu_m^{\mathrm{ch}} := \sum_k \pi_{m,k}\mathfrak{s}_{m,k}, \qquad v_m^{\mathrm{ch}} := \sum_k \pi_{m,k}\mathfrak{s}_{m,k}^2.$$

Assume that outside events $F_m^{\mathrm{cert}} \in \mathcal{G}_m$, with $\sum_m \mathbb{P}(F_m^{\mathrm{cert}}) < \infty$, every candidate core is a contraction and has one of the finite-lift orbit-closure or algebraic/group certificates from Corollary 7.128, with charge bounded by the displayed $\mathfrak{s}_{m,k}$. Choose $t_m > 0$ and a deterministic cardinality envelope $J_m \ge L_m$ such that

$$\mu_m^{\mathrm{ch}} + t_m \le C_0 n_m.$$

If

$$\sum_m \left[\mathbb{P}(F_m^{\mathrm{cert}}) + \exp\left\{-\frac{t_m^2}{2(v_m^{\mathrm{ch}} + b_m t_m/3)}\right\}\right] < \infty,$$

then the selection event in Corollary 7.128 is summable: almost surely, eventually

$$|\mathscr{S}_m| \le J_m, \qquad \sum_{k \in \mathscr{S}_m} \mathfrak{s}_{m,k} \le C_0 n_m.$$

Consequently, suppose each candidate perturbation stream is tagged ψ or μ, with parameters $\varepsilon_{m,k}, A_{m,k}, s'_{m,k}$, and satisfies the corresponding mixed dominated tail and drift hypotheses after adjoining $\mathcal{H}_m$. If

$$\ell_m := \log(e(D_m + n_m + 1)) \le B\sqrt{n_m}$$

eventually,

$$\sum_m \sum_k \left(\mathbb{P}(E_{m,k}^{\mathrm{tail}}) + \varepsilon_{m,k}\right) < \infty,$$

and

$$\sum_m J_m(D_m + n_m + 1)(e(D_m + n_m + 1))^{-\gamma_{\mathrm{tc}}(K_{\mathrm{tail}},1,1,B)} < \infty,$$

then the adaptively categorical selected structured submenus have the same eventual common signing, contraction normalization, fixed-menu coverage, and all-integer/sparse thresholds as in Corollary 7.128.

This covers online or state-dependent randomized template choices, Markov hyperparameter menus, adaptive blockwise model selection, and sequential architecture/library choices whenever each stage chooses at most one structured candidate and has deterministic predictable probability envelopes.

Proof. For each slot put

$$X_{m,g} := \begin{cases} \mathfrak{s}_{m,\xi_{m,g}}, & \xi_{m,g} \neq \varnothing, \\ 0, & \xi_{m,g} = \varnothing. \end{cases}$$

Then $0 \le X_{m,g} \le b_m$ and

$$\sum_g \mathbb{E}[X_{m,g} \mid \mathcal{F}_{m,g-1}] \le \mu_m^{\mathrm{ch}}, \qquad \sum_g \mathbb{E}[X_{m,g}^2 \mid \mathcal{F}_{m,g-1}] \le v_m^{\mathrm{ch}}.$$

Hence the martingale

$$M_j := \sum_{g \le j} (X_{m,g} - \mathbb{E}[X_{m,g} \mid \mathcal{F}_{m,g-1}])$$

has increments bounded above by b_m and predictable quadratic variation at most v_m^{ch}. If $\sum_{k \in \mathscr{S}_m} \mathfrak{s}_{m,k} > \mu_m^{\mathrm{ch}} + t_m$, then $M_{L_m} > t_m$. The scalar Freedman inequality gives

$$\mathbb{P}\left\{ \sum_{k \in \mathscr{S}_m} \mathfrak{s}_{m,k} > \mu_m^{\mathrm{ch}} + t_m \right\} \le \exp\left\{ -\frac{t_m^2}{2(v_m^{\mathrm{ch}} + b_m t_m/3)} \right\}.$$

Also $|\mathscr{S}_m| \le L_m \le J_m$ deterministically. Adding F_m^{cert} and applying Borel–Cantelli gives the deterministic selection hypotheses eventually. Order the selected labels by slot and then by candidate order inside each slot, pad by zero streams up to J_m, and apply Corollary 7.128. □

Corollary 7.136 (Random-envelope adaptive-categorical selected structured submenus)**.** *Fix structural constants $r_0, b_0, c_0, L_0, C_0 \ge 1$. For each m, let*

$$\mathcal{G}_m = \mathcal{F}_{m,0} \subset \mathcal{F}_{m,1} \subset \cdots \subset \mathcal{F}_{m,L_m}$$

be a filtration and let the finite candidate list be partitioned into deterministic slots $\mathcal{I}_{m,1}, \dots, \mathcal{I}_{m,L_m}$. The k-th candidate has charge $\mathfrak{s}_{m,k} \in [0, b_m]$, with $b_m > 0$, core streams $B_{m,k,1}, \dots, B_{m,k,n_m}$, and perturbation streams $Y_{m,k,1}, \dots, Y_{m,k,n_m}$. In slot g, choose at most one label by an $\mathcal{I}_{m,g} \cup \{\varnothing\}$-valued random variable $\xi_{m,g}$ which is $\mathcal{F}_{m,g}$-measurable. For each $k \in \mathcal{I}_{m,g}$, let $\Pi_{m,k} \in [0,1]$ be $\mathcal{F}_{m,g-1}$-measurable and assume

$$\mathbb{P}\{\xi_{m,g} = k \mid \mathcal{F}_{m,g-1}\} \le \Pi_{m,k}.$$

Let

$$\eta_{m,k} := \mathbf{1}_{\{\xi_{m,g}=k\}} \quad (k \in \mathcal{I}_{m,g}),$$
$$\mathscr{S}_m := \{k : \eta_{m,k} = 1\},$$
$$\mathcal{H}_m := \mathcal{F}_{m,L_m},$$

and define the predictable random charge profiles

$$M_m^{\mathrm{ch}} := \sum_k \Pi_{m,k} \mathfrak{s}_{m,k}, \qquad V_m^{\mathrm{ch}} := \sum_k \Pi_{m,k} \mathfrak{s}_{m,k}^2.$$

Assume that there are events $F_m^{\mathrm{env}} \in \mathcal{F}_{m,L_m}$, with $\sum_m \mathbb{P}(F_m^{\mathrm{env}}) < \infty$, and deterministic bounds $\bar{\mu}_m^{\mathrm{ch}}, \bar{v}_m^{\mathrm{ch}} \ge 0$ such that outside F_m^{env},

$$M_m^{\mathrm{ch}} \le \bar{\mu}_m^{\mathrm{ch}}, \qquad V_m^{\mathrm{ch}} \le \bar{v}_m^{\mathrm{ch}}.$$

Assume also that outside events $F_m^{\mathrm{cert}} \in \mathcal{G}_m$, with $\sum_m \mathbb{P}(F_m^{\mathrm{cert}}) < \infty$, every candidate core is a contraction and has one of the finite-lift orbit-closure or algebraic/group certificates from Corollary 7.128, with charge bounded by the displayed $\mathfrak{s}_{m,k}$. Choose $t_m > 0$ and a deterministic cardinality envelope $J_m \geq L_m$ such that

$$\bar{\mu}_m^{\mathrm{ch}} + t_m \leq C_0 n_m.$$

If

$$\sum_m \left[\mathbb{P}(F_m^{\mathrm{env}}) + \mathbb{P}(F_m^{\mathrm{cert}}) + \exp\left\{ -\frac{t_m^2}{2(\bar{v}_m^{\mathrm{ch}} + b_m t_m/3)} \right\} \right] < \infty,$$

then the selection event in Corollary 7.128 is summable: almost surely, eventually

$$|\mathscr{S}_m| \leq J_m, \qquad \sum_{k \in \mathscr{S}_m} \mathfrak{s}_{m,k} \leq C_0 n_m.$$

Consequently, suppose each candidate perturbation stream is tagged ψ or μ, with parameters $\varepsilon_{m,k}, A_{m,k}, s'_{m,k}$, and satisfies the corresponding mixed dominated tail and drift hypotheses after adjoining $\mathcal{H}_m$. If

$$\ell_m := \log(e(D_m + n_m + 1)) \leq B\sqrt{n_m}$$

eventually,

$$\sum_m \sum_k \left(\mathbb{P}(E_{m,k}^{\mathrm{tail}}) + \varepsilon_{m,k} \right) < \infty,$$

and

$$\sum_m J_m (D_m + n_m + 1)(e(D_m + n_m + 1))^{-\gamma_{\mathrm{tc}}(K_{\mathrm{tail}},1,1,B)} < \infty,$$

then the random-envelope adaptive-categorical selected structured submenus have the same eventual common signing, contraction normalization, fixed-menu coverage, and all-integer/sparse thresholds as in Corollary 7.128.

This covers state-dependent template probabilities, posterior or online hyperparameter menus, adaptive blockwise architecture choices, and sequential model-selection sublibraries whose predictable aggregate charge profiles are certified with summable failure probability.

Proof. For each slot put

$$X_{m,g} := \begin{cases} \mathfrak{s}_{m,\xi_{m,g}}, & \xi_{m,g} \neq \varnothing, \\ 0, & \xi_{m,g} = \varnothing. \end{cases}$$

Let $q_{m,g} := \mathbb{E}[X_{m,g} \mid \mathcal{F}_{m,g-1}]$. Then

$$q_{m,g} \leq \sum_{k \in \mathcal{I}_{m,g}} \Pi_{m,k} \mathfrak{s}_{m,k}, \qquad \mathbb{E}[X_{m,g}^2 \mid \mathcal{F}_{m,g-1}] \leq \sum_{k \in \mathcal{I}_{m,g}} \Pi_{m,k} \mathfrak{s}_{m,k}^2.$$

Hence

$$M_j := \sum_{g \leq j} (X_{m,g} - q_{m,g})$$

is a martingale with increments bounded above by b_m, and its predictable quadratic variation is at most V_m^{ch}. On $(F_m^{\mathrm{env}})^c$, if

$$\sum_{k\in\mathscr{S}_m} \mathfrak{s}_{m,k} > \bar{\mu}_m^{\mathrm{ch}} + t_m,$$

then $M_{L_m} > t_m$ while the predictable quadratic variation is at most $\bar{v}_m^{\mathrm{ch}}$. The scalar Freedman inequality therefore gives

$$\mathbb{P}\left\{(F_m^{\mathrm{env}})^c, \sum_{k\in\mathscr{S}_m} \mathfrak{s}_{m,k} > \bar{\mu}_m^{\mathrm{ch}} + t_m\right\} \le \exp\left\{-\frac{t_m^2}{2(\bar{v}_m^{\mathrm{ch}} + b_m t_m/3)}\right\}.$$

Together with $\mathbb{P}(F_m^{\mathrm{env}}) + \mathbb{P}(F_m^{\mathrm{cert}})$, the displayed series is summable. Since $|\mathscr{S}_m| \le L_m \le J_m$ deterministically, Borel–Cantelli gives the deterministic cardinality and charge hypotheses eventually.

On that event, order selected labels by slot and candidate order, pad by zero streams up to J_m, and apply Corollary 7.128. The candidate tail summability dominates the selected active-prefix tail summability, and the deterministic matrix-tail series is unchanged. □

Corollary 7.137 (Freedman-selected adaptive structured submenus with mixed dominated noise)**.** *Fix structural constants $r_0, b_0, c_0, L_0, C_0 \ge 1$. For each m, let*

$$\mathcal{G}_m = \mathcal{F}_{m,0} \subset \mathcal{F}_{m,1} \subset \cdots \subset \mathcal{F}_{m,M_m}$$

be a filtration exposing a finite candidate list $1, \dots, M_m$. The k-th candidate has charge $\mathfrak{s}_{m,k} \in [0, b_m]$, with $b_m > 0$, core streams $B_{m,k,1}, \dots, B_{m,k,n_m}$, and perturbation streams $Y_{m,k,1}, \dots, Y_{m,k,n_m}$. Let $\eta_{m,k} \in \{0,1\}$ be $\mathcal{F}_{m,k}$-measurable selection indicators satisfying the predictable envelope

$$\mathbb{E}[\eta_{m,k} \mid \mathcal{F}_{m,k-1}] \le p_{m,k} \qquad (1 \le k \le M_m),$$

where the $p_{m,k} \in [0,1]$ are deterministic. No conditional independence is assumed. Put

$$\mathscr{S}_m := \{k : \eta_{m,k} = 1\}, \qquad \mathcal{H}_m := \mathcal{F}_{m,M_m},$$

and define

$$\mu_m^{\mathrm{ch}} := \sum_{k=1}^{M_m} p_{m,k}\mathfrak{s}_{m,k}, \qquad v_m^{\mathrm{ch}} := \sum_{k=1}^{M_m} p_{m,k}\mathfrak{s}_{m,k}^2, \qquad \mu_m^{\#} := \sum_{k=1}^{M_m} p_{m,k}.$$

Assume that outside events $F_m^{\mathrm{cert}} \in \mathcal{G}_m$, with $\sum_m \mathbb{P}(F_m^{\mathrm{cert}}) < \infty$, every selected core is a contraction and has one of the finite-lift orbit-closure or algebraic/group certificates from Corollary 7.128, with charge bounded by the displayed $\mathfrak{s}_{m,k}$. Choose $t_m, u_m > 0$ and a deterministic cardinality envelope $J_m \ge 1$ such that

$$\mu_m^{\mathrm{ch}} + t_m \le C_0 n_m, \qquad \mu_m^{\#} + u_m \le J_m.$$

If

$$\sum_m \left[\mathbb{P}(F_m^{\mathrm{cert}}) + \exp\left\{ -\frac{t_m^2}{2(v_m^{\mathrm{ch}} + b_m t_m/3)} \right\} + \exp\left\{ -\frac{u_m^2}{2(\mu_m^{\#} + u_m/3)} \right\} \right] < \infty,$$

then the selection event in Corollary 7.128 is summable: almost surely, eventually

$$|\mathscr{S}_m| \le J_m, \qquad \sum_{k \in \mathscr{S}_m} \mathfrak{s}_{m,k} \le C_0 n_m.$$

Consequently, suppose each candidate perturbation stream is tagged ψ or μ, with parameters $\varepsilon_{m,k}, A_{m,k}, s'_{m,k}$, and satisfies the corresponding mixed dominated tail and drift hypotheses after adjoining $\mathcal{H}_m$. If

$$\ell_m := \log(e(D_m + n_m + 1)) \le B\sqrt{n_m}$$

eventually,

$$\sum_m \sum_{k=1}^{M_m} \left(\mathbb{P}(E_{m,k}^{\mathrm{tail}}) + \varepsilon_{m,k} \right) < \infty,$$

and

$$\sum_m J_m (D_m + n_m + 1)(e(D_m + n_m + 1))^{-\gamma_{\mathrm{tc}}(K_{\mathrm{tail}},1,1,B)} < \infty,$$

then the adaptively selected structured submenus have the same eventual common signing, contraction normalization, fixed-menu coverage, and all-integer/sparse thresholds as in Corollary 7.128.

Proof. For each k, set

$$\pi_{m,k} := \mathbb{E}[\eta_{m,k} \mid \mathcal{F}_{m,k-1}].$$

The process

$$M_j^{\mathrm{ch}} := \sum_{k \le j} (\eta_{m,k} - \pi_{m,k}) \mathfrak{s}_{m,k}$$

is a martingale with increments bounded above by b_m, and its predictable quadratic variation is at most

$$\sum_k \pi_{m,k} \mathfrak{s}_{m,k}^2 \le v_m^{\mathrm{ch}}.$$

Since $\sum_k (\pi_{m,k} - p_{m,k}) \mathfrak{s}_{m,k} \le 0$, the event

$$\sum_k \eta_{m,k} \mathfrak{s}_{m,k} > \mu_m^{\mathrm{ch}} + t_m$$

implies $M_{M_m}^{\mathrm{ch}} > t_m$. The scalar Freedman inequality therefore gives

$$\mathbb{P}\left\{ \sum_k \eta_{m,k} \mathfrak{s}_{m,k} > C_0 n_m \right\} \le \exp\left\{ -\frac{t_m^2}{2(v_m^{\mathrm{ch}} + b_m t_m/3)} \right\}.$$

The same argument with increments $\eta_{m,k} - \pi_{m,k}$, jump bound 1, and predictable variance at most $\sum_k p_{m,k} = \mu_m^{\#}$ gives

$$\mathbb{P}\{|\mathscr{S}_m| > J_m\} \le \exp\left\{-\frac{u_m^2}{2(\mu_m^{\#} + u_m/3)}\right\}.$$

Adding F_m^{cert}, the displayed series is summable, so Borel–Cantelli gives the required eventual cardinality and charge bounds.

On the resulting good event, order the selected labels by the candidate order and pad by zero streams up to the deterministic length J_m. The candidate tail summability dominates the selected tail summability required by Corollary 7.128, and the deterministic matrix-tail series is unchanged. Applying that corollary gives all stated conclusions. □

Corollary 7.138 (Random-envelope Freedman-selected adaptive structured submenus)**.** *Fix structural constants $r_0, b_0, c_0, L_0, C_0 \ge 1$. For each m, let*

$$\mathcal{G}_m = \mathcal{F}_{m,0} \subset \mathcal{F}_{m,1} \subset \cdots \subset \mathcal{F}_{m,M_m}$$

be a filtration exposing a finite candidate list $1, \dots, M_m$. The k-th candidate has charge $\mathfrak{s}_{m,k} \in [0, b_m]$, with $b_m > 0$, core streams $B_{m,k,1}, \dots, B_{m,k,n_m}$, and perturbation streams $Y_{m,k,1}, \dots, Y_{m,k,n_m}$. Let $\eta_{m,k} \in \{0,1\}$ be $\mathcal{F}_{m,k}$-measurable selection indicators. Assume there are predictable random envelopes $P_{m,k} \in [0,1]$, with $P_{m,k}$ $\mathcal{F}_{m,k-1}$-measurable, such that

$$\mathbb{E}[\eta_{m,k} \mid \mathcal{F}_{m,k-1}] \le P_{m,k} \qquad (1 \le k \le M_m).$$

Put

$$\mathscr{S}_m := \{k : \eta_{m,k} = 1\}, \qquad \mathcal{H}_m := \mathcal{F}_{m,M_m},$$

and define the random predictable profiles

$$M_m^{\mathrm{ch}} := \sum_{k=1}^{M_m} P_{m,k}\mathfrak{s}_{m,k}, \qquad V_m^{\mathrm{ch}} := \sum_{k=1}^{M_m} P_{m,k}\mathfrak{s}_{m,k}^2, \qquad M_m^{\#} := \sum_{k=1}^{M_m} P_{m,k}.$$

Assume that there are events $F_m^{\mathrm{env}} \in \mathcal{F}_{m,M_m}$, with $\sum_m \mathbb{P}(F_m^{\mathrm{env}}) < \infty$, and deterministic bounds $\bar{\mu}_m^{\mathrm{ch}}, \bar{v}_m^{\mathrm{ch}}, \bar{\mu}_m^{\#} \ge 0$ such that outside F_m^{env},

$$M_m^{\mathrm{ch}} \le \bar{\mu}_m^{\mathrm{ch}}, \qquad V_m^{\mathrm{ch}} \le \bar{v}_m^{\mathrm{ch}}, \qquad M_m^{\#} \le \bar{\mu}_m^{\#}.$$

Assume also that outside events $F_m^{\mathrm{cert}} \in \mathcal{G}_m$, with $\sum_m \mathbb{P}(F_m^{\mathrm{cert}}) < \infty$, every selected core is a contraction and has one of the finite-lift orbit-closure or algebraic/group certificates from Corollary 7.128, with charge bounded by the displayed $\mathfrak{s}_{m,k}$. Choose $t_m, u_m > 0$ and a deterministic cardinality envelope $J_m \ge 1$ such that

$$\bar{\mu}_m^{\mathrm{ch}} + t_m \le C_0 n_m, \qquad \bar{\mu}_m^{\#} + u_m \le J_m.$$

Put

$$\Lambda_m := \mathbb{P}(F_m^{\mathrm{env}}) + \mathbb{P}(F_m^{\mathrm{cert}}) + \exp\left\{-\frac{t_m^2}{2(\bar{v}_m^{\mathrm{ch}} + b_m t_m/3)}\right\} + \exp\left\{-\frac{u_m^2}{2(\bar{\mu}_m^{\#} + u_m/3)}\right\}.$$

If

$$\sum_m \Lambda_m < \infty,$$

then the selection event in Corollary 7.128 is summable: almost surely, eventually

$$|\mathscr{S}_m| \le J_m, \qquad \sum_{k\in\mathscr{S}_m} \mathfrak{s}_{m,k} \le C_0 n_m.$$

Consequently, suppose each candidate perturbation stream is tagged ψ or μ, with parameters $\varepsilon_{m,k}, A_{m,k}, s'_{m,k}$, and satisfies the corresponding mixed dominated tail and drift hypotheses after adjoining $\mathcal{H}_m$. If

$$\ell_m := \log(e(D_m + n_m + 1)) \le B\sqrt{n_m}$$

eventually,

$$\sum_m \sum_{k=1}^{M_m} \big(\mathbb{P}(E^{\mathrm{tail}}_{m,k}) + \varepsilon_{m,k}\big) < \infty,$$

and

$$\sum_m J_m(D_m + n_m + 1)(e(D_m + n_m + 1))^{-\gamma_{\mathrm{tc}}(K_{\mathrm{tail}},1,1,B)} < \infty,$$

then the random-envelope Freedman-selected structured submenus have the same eventual common signing, contraction normalization, fixed-menu coverage, and all-integer/sparse thresholds as in Corollary 7.128.

This covers adaptive thinning, stopping-rule catalogues, online selected structured libraries, and general sequential model-selection submenus with state-dependent predictable inclusion intensities, once their realized envelope profiles are certified outside summable exceptional events.

Proof. Set

$$\pi_{m,k} := \mathbb{E}[\eta_{m,k} \mid \mathcal{F}_{m,k-1}].$$

For the charge process,

$$M^{\mathrm{ch}}_j := \sum_{k\le j} (\eta_{m,k} - \pi_{m,k})\mathfrak{s}_{m,k}$$

is a martingale with increments bounded above by b_m, and predictable quadratic variation at most

$$\sum_k \pi_{m,k}\mathfrak{s}^2_{m,k} \le V^{\mathrm{ch}}_m.$$

On $(F^{\mathrm{env}}_m)^c$, the event

$$\sum_k \eta_{m,k}\mathfrak{s}_{m,k} > \bar{\mu}^{\mathrm{ch}}_m + t_m$$

implies $M^{\mathrm{ch}}_{M_m} > t_m$, while the predictable quadratic variation is at most $\bar{v}^{\mathrm{ch}}_m$. The scalar Freedman inequality gives

$$\mathbb{P}\left\{(F^{\mathrm{env}}_m)^c, \sum_k \eta_{m,k}\mathfrak{s}_{m,k} > \bar{\mu}^{\mathrm{ch}}_m + t_m\right\} \le \exp\left\{-\frac{t_m^2}{2(\bar{v}^{\mathrm{ch}}_m + b_m t_m/3)}\right\}.$$

The same argument applied to

$$M^{\#}_j := \sum_{k\le j}(\eta_{m,k} - \pi_{m,k})$$

uses jump bound 1 and predictable quadratic variation at most $M^{\#}_m$. Hence

$$\mathbb{P}\left\{(F^{\mathrm{env}}_m)^c,\ |\mathscr{S}_m| > \bar{\mu}^{\#}_m + u_m\right\} \le \exp\left\{-\frac{u_m^2}{2(\bar{\mu}^{\#}_m + u_m/3)}\right\}.$$

After adding $F^{\mathrm{env}}_m \cup F^{\mathrm{cert}}_m$, the displayed series is summable, so Borel–Cantelli gives the eventual deterministic cardinality and charge bounds.

On the resulting good event, order the selected labels by candidate order and pad by zero streams up to J_m. The candidate tail summability dominates the selected tail summability required by Corollary 7.128, and the deterministic matrix-tail series is unchanged. Applying that corollary gives all stated conclusions. □

Corollary 7.139 (Serfling-selected fixed-size structured submenus with mixed dominated noise)**.** *Fix structural constants $r_0, b_0, c_0, L_0, C_0 \ge 1$. For each m, let $\mathcal{G}_m$ be a base environment. Let a finite candidate catalogue be partitioned into deterministic strata*

$$\mathcal{I}_{m,1}, \dots, \mathcal{I}_{m,A_m}, \qquad |\mathcal{I}_{m,a}| = M_{m,a}.$$

From stratum a, choose exactly $R_{m,a} \le M_{m,a}$ labels uniformly without replacement, conditionally on $\mathcal{G}_m$. Write the chosen subset as $\mathscr{S}_{m,a}$, put

$$\mathscr{S}_m := \bigcup_{a=1}^{A_m} \mathscr{S}_{m,a}, \qquad J_m := \sum_{a=1}^{A_m} R_{m,a},$$

and let

$$\mathcal{H}_m := \mathcal{G}_m \vee \sigma(\mathscr{S}_{m,1}, \dots, \mathscr{S}_{m,A_m}).$$

The k-th candidate in stratum a has charge $\mathfrak{s}_{m,a,k} \in [0, b_{m,a}]$, core streams $B_{m,a,k,1}, \dots, B_{m,a,k,n_m}$, and perturbation streams $Y_{m,a,k,1}, \dots, Y_{m,a,k,n_m}$. Define the finite-population mean charge

$$\bar{\mathfrak{s}}_{m,a} := \frac{1}{M_{m,a}} \sum_{k\in\mathcal{I}_{m,a}} \mathfrak{s}_{m,a,k}.$$

Assume that outside events $F^{\mathrm{cert}}_m \in \mathcal{G}_m$, with $\sum_m \mathbb{P}(F^{\mathrm{cert}}_m) < \infty$, every candidate core is a contraction and has one of the finite-lift orbit-closure or algebraic/group certificates from Corollary 7.128, with charge bounded by the

displayed $\mathfrak{s}_{m,a,k}$. *Choose numbers* $t_{m,a} > 0$ *for all strata with* $R_{m,a} > 0$ *so that*

$$\sum_{a=1}^{A_m} (R_{m,a}\bar{\mathfrak{s}}_{m,a} + t_{m,a}) \le C_0 n_m,$$

where the a*-term is read as* 0 *when* $R_{m,a} = 0$. *If*

$$\sum_m \left[\mathbb{P}(F_m^{\mathrm{cert}}) + \sum_{\substack{1 \le a \le A_m \\ R_{m,a} > 0}} \exp\left\{ -\frac{2t_{m,a}^2}{R_{m,a} b_{m,a}^2} \right\} \right] < \infty,$$

then the selection event in Corollary 7.128 is summable: almost surely, eventually

$$|\mathscr{S}_m| = J_m, \qquad \sum_{a=1}^{A_m} \sum_{k \in \mathscr{S}_{m,a}} \mathfrak{s}_{m,a,k} \le C_0 n_m.$$

Consequently, suppose each candidate perturbation stream is tagged ψ *or* μ, *with parameters* $\varepsilon_{m,a,k}, A_{m,a,k}, s'_{m,a,k}$, *and satisfies the corresponding mixed dominated tail and drift hypotheses after adjoining* $\mathcal{H}_m$. *If*

$$\ell_m := \log(e(D_m + n_m + 1)) \le B\sqrt{n_m}$$

eventually,

$$\sum_m \sum_{a=1}^{A_m} \sum_{k \in \mathcal{I}_{m,a}} \left(\mathbb{P}(E_{m,a,k}^{\mathrm{tail}}) + \varepsilon_{m,a,k} \right) < \infty,$$

and

$$\sum_m J_m (D_m + n_m + 1)(e(D_m + n_m + 1))^{-\gamma_{\mathrm{tc}}(K_{\mathrm{tail}},1,1,B)} < \infty,$$

then the fixed-size or stratified without-replacement selected submenus have the same eventual common signing, contraction normalization, fixed-menu coverage, and all-integer/sparse thresholds as in Corollary 7.128.

Proof. The selected cardinality is deterministically J_m. Condition on $\mathcal{G}_m$. In stratum a, the selected charges are sampled without replacement from the finite population $\{\mathfrak{s}_{m,a,k} : k \in \mathcal{I}_{m,a}\} \subset [0, b_{m,a}]$. Hoeffding's comparison for sampling without replacement gives, for $R_{m,a} > 0$,

$$\mathbb{P}\left\{ \sum_{k \in \mathscr{S}_{m,a}} \mathfrak{s}_{m,a,k} > R_{m,a}\bar{\mathfrak{s}}_{m,a} + t_{m,a} \,\middle|\, \mathcal{G}_m \right\} \le \exp\left\{ -\frac{2t_{m,a}^2}{R_{m,a} b_{m,a}^2} \right\}.$$

For $R_{m,a} = 0$ the stratum contributes nothing. A union bound over strata shows that the total selected charge exceeds $C_0 n_m$ with conditional probability bounded by the displayed stratum-tail sum. After averaging and adding F_m^{cert}, the assumed summability and Borel–Cantelli give the summable selection event required by Corollary 7.128.

On the eventual good event, order all selected labels by the fixed stratum order and the catalogue order inside each stratum, then pad by zero streams

up to length J_m. The candidate tail summability dominates the selected tail summability in the selected-submenu corollary, and the deterministic matrix-tail series is the same one with cardinality envelope J_m. Applying that corollary gives all stated signing, normalization, coverage, and threshold conclusions. □

Corollary 7.140 (Hybrid selected structured submenus with mixed dominated noise)**.** *Fix structural constants $r_0, b_0, c_0, L_0, C_0 \geq 1$. For each m, let $\mathcal{G}_m$ be a base environment and partition a finite candidate catalogue into deterministic components*

$$\mathcal{I}_{m,1}, \dots, \mathcal{I}_{m,R_m}.$$

The candidate $k \in \mathcal{I}_{m,h}$ has charge $\mathfrak{s}_{m,h,k} \geq 0$, core stream $(B_{m,h,k,i})_{i \leq n_m}$, and perturbation stream $(Y_{m,h,k,i})_{i \leq n_m}$. In component h, select a random subset $\mathscr{S}_{m,h} \subset \mathcal{I}_{m,h}$, by any mechanism, and put

$$\mathscr{S}_m := \bigcup_{h=1}^{R_m} \mathscr{S}_{m,h}, \qquad \mathcal{H}_m := \mathcal{G}_m \vee \sigma(\mathscr{S}_{m,1}, \dots, \mathscr{S}_{m,R_m}).$$

Assume that for every component there are deterministic envelopes

$$J_{m,h} \in \mathbb{Z}_{\geq 0}, \qquad U_{m,h} \geq 0,$$

and selection-failure events $F^{\mathrm{sel}}_{m,h}$, such that

$$\sum_m \sum_{h=1}^{R_m} \mathbb{P}(F^{\mathrm{sel}}_{m,h}) < \infty$$

and, outside $F^{\mathrm{sel}}_{m,h}$,

$$|\mathscr{S}_{m,h}| \leq J_{m,h}, \qquad \sum_{k \in \mathscr{S}_{m,h}} \mathfrak{s}_{m,h,k} \leq U_{m,h}.$$

Let

$$J_m := \max\left\{1, \sum_{h=1}^{R_m} J_{m,h}\right\}, \qquad \sum_{h=1}^{R_m} U_{m,h} \leq C_0 n_m.$$

Assume also that outside events $F^{\mathrm{cert}}_m \in \mathcal{G}_m$, with $\sum_m \mathbb{P}(F^{\mathrm{cert}}_m) < \infty$, every selected core is a contraction and has one of the finite-lift orbit-closure or algebraic/group certificates from Corollary 7.128, with charge bounded by the displayed $\mathfrak{s}_{m,h,k}$.

Then the selection event in Corollary 7.128 is summable: almost surely, eventually

$$|\mathscr{S}_m| \leq J_m, \qquad \sum_{h=1}^{R_m} \sum_{k \in \mathscr{S}_{m,h}} \mathfrak{s}_{m,h,k} \leq C_0 n_m.$$

Consequently, suppose each candidate perturbation stream is tagged ψ or μ, with parameters $\varepsilon_{m,h,k}, A_{m,h,k}, s'_{m,h,k}$, and satisfies the corresponding mixed dominated tail and drift hypotheses after adjoining $\mathcal{H}_m$. If

$$\ell_m := \log(e(D_m + n_m + 1)) \leq B\sqrt{n_m}$$

eventually,

$$\sum_m \sum_{h=1}^{R_m} \sum_{k\in\mathcal{I}_{m,h}} \left(\mathbb{P}(E^{\mathrm{tail}}_{m,h,k}) + \varepsilon_{m,h,k}\right) < \infty,$$

and

$$\sum_m J_m(D_m+n_m+1)(e(D_m+n_m+1))^{-\gamma_{\mathrm{tc}}(K_{\mathrm{tail}},1,1,B)} < \infty,$$

then the hybrid selected structured submenus have the same eventual common signing, contraction normalization, fixed-menu coverage, and schedule thresholds as in Corollary 7.128.

Each component may be certified by any one of the preceding Bernstein, Laplace-dominated, random-envelope, Poisson/Cox, categorical, Freedman, or Serfling selected-submenu corollaries, or by a deterministic pathwise check. No independence, negative association, or filtration relation between different components is required beyond the displayed summability of the component selection failures.

Proof. Let

$$F_m := F_m^{\mathrm{cert}} \cup \bigcup_{h=1}^{R_m} F_{m,h}^{\mathrm{sel}}.$$

By the assumed summability, $\sum_m \mathbb{P}(F_m) < \infty$. On F_m^c, the component bounds give

$$|\mathscr{S}_m| \le \sum_{h=1}^{R_m} |\mathscr{S}_{m,h}| \le J_m$$

and

$$\sum_{h=1}^{R_m} \sum_{k\in\mathscr{S}_{m,h}} \mathfrak{s}_{m,h,k} \le \sum_{h=1}^{R_m} U_{m,h} \le C_0 n_m.$$

Thus Borel–Cantelli gives the eventual deterministic selected-submenu cardinality and charge bounds.

On that eventual good event, order all selected labels first by component index and then by the fixed catalogue order inside the component, padding by zero streams up to J_m. The candidate tail summability dominates the selected tail summability required by Corollary 7.128, and the deterministic matrix-tail series is exactly the displayed one with cardinality envelope J_m. Applying that corollary gives the stated signing, normalization, coverage, and threshold conclusions. The proof used only the component good events, so cross-component dependence is irrelevant. □

Corollary 7.141 (Random-budget hybrid selected structured submenus)**.** *In the setting and notation of Corollary 7.140, keep the candidate catalogue, component partition, selections $\mathscr{S}_{m,h}$, and filtration $\mathcal{H}_m$. Replace the deterministic component envelopes by $\mathcal{G}_m$-measurable random envelopes*

$$\mathcal{J}_{m,h} \in \mathbb{Z}_{\ge 0}, \qquad \mathcal{U}_{m,h} \ge 0.$$

Assume there are selection-failure events $F^{\mathrm{sel}}_{m,h}$, aggregate budget-failure events $F^{\mathrm{bud}}_m \in \mathcal{G}_m$, and certificate-failure events $F^{\mathrm{cert}}_m \in \mathcal{G}_m$ such that

$$\sum_m \left[\mathbb{P}(F^{\mathrm{bud}}_m) + \mathbb{P}(F^{\mathrm{cert}}_m) + \sum_{h=1}^{R_m} \mathbb{P}(F^{\mathrm{sel}}_{m,h}) \right] < \infty.$$

Outside $F^{\mathrm{sel}}_{m,h}$, the component selection obeys

$$|\mathscr{S}_{m,h}| \le \mathcal{J}_{m,h}, \qquad \sum_{k \in \mathscr{S}_{m,h}} \mathfrak{s}_{m,h,k} \le \mathcal{U}_{m,h}.$$

Outside F^{bud}_m, deterministic envelopes $J_m \ge 1$ and $C_0 n_m$ certify the aggregate random budgets:

$$\sum_{h=1}^{R_m} \mathcal{J}_{m,h} \le J_m, \qquad \sum_{h=1}^{R_m} \mathcal{U}_{m,h} \le C_0 n_m.$$

Outside F^{cert}_m, every selected core is a contraction and has one of the finite-lift orbit-closure or algebraic/group certificates from Corollary 7.128, with charge bounded by $\mathfrak{s}_{m,h,k}$.

Then the selection event in Corollary 7.128 is summable: almost surely, eventually

$$|\mathscr{S}_m| \le J_m, \qquad \sum_{h=1}^{R_m} \sum_{k \in \mathscr{S}_{m,h}} \mathfrak{s}_{m,h,k} \le C_0 n_m.$$

Consequently, under the same mixed dominated tail, drift, candidate-tail summability, and matrix-tail summability hypotheses as in Corollary 7.140, with this deterministic aggregate J_m, the random-budget hybrid submenus have the same eventual common signing, contraction normalization, fixed-menu coverage, and schedule thresholds as in Corollary 7.128.

Thus the mixture weights, posterior component masses, Cox intensities, adaptive slot budgets, or stratum counts may be random and data-dependent. It is enough to certify their total realized cardinality and charge budgets summably before applying the common Matrix Spencer signing.

Proof. Let

$$F_m := F^{\mathrm{bud}}_m \cup F^{\mathrm{cert}}_m \cup \bigcup_{h=1}^{R_m} F^{\mathrm{sel}}_{m,h}.$$

The displayed series gives $\sum_m \mathbb{P}(F_m) < \infty$. On F_m^c,

$$|\mathscr{S}_m| \le \sum_{h=1}^{R_m} |\mathscr{S}_{m,h}| \le \sum_{h=1}^{R_m} \mathcal{J}_{m,h} \le J_m$$

and

$$\sum_{h=1}^{R_m} \sum_{k \in \mathscr{S}_{m,h}} \mathfrak{s}_{m,h,k} \le \sum_{h=1}^{R_m} \mathcal{U}_{m,h} \le C_0 n_m.$$

Borel–Cantelli gives the eventual selected-submenu event. Ordering selected labels by component and catalogue order, then padding by zero streams up to J_m, reduces the conclusion to Corollary 7.128 exactly as in Corollary 7.140. □

Corollary 7.142 (Hierarchical random-budget selected structured submenus)**.** *Fix structural constants $r_0, b_0, c_0, L_0, C_0 \geq 1$. For each m, let $\mathfrak{T}_m$ be a countable rooted tree with a fixed depth-first order. Each vertex $v \in \mathfrak{T}_m$ carries a charge $\mathfrak{s}_{m,v} \geq 0$, core stream $(B_{m,v,i})_{i \leq n_m}$, and perturbation stream $(Y_{m,v,i})_{i \leq n_m}$. A random recursive procedure, possibly depending on a base environment $\mathcal{G}_m$, produces an $\mathcal{H}_m$-measurable terminal antichain*

$$\mathscr{A}_m \subset \mathfrak{T}_m, \qquad \mathcal{H}_m := \mathcal{G}_m \vee \sigma(\mathscr{A}_m)$$

such that no two selected vertices are in an ancestor–descendant relation. The fixed depth-first order is used only to list the selected vertices; $\mathscr{A}_m$ is the realized frontier of the random tree or adaptive search.

Assume that there are deterministic envelopes $J_m \geq 1$, tree-budget failure events $F_m^{\mathrm{tree}} \in \mathcal{H}_m$, and certificate-failure events $F_m^{\mathrm{cert}} \in \mathcal{H}_m$ such that

$$\sum_m \bigl(\mathbb{P}(F_m^{\mathrm{tree}}) + \mathbb{P}(F_m^{\mathrm{cert}})\bigr) < \infty$$

and, outside F_m^{tree},

$$|\mathscr{A}_m| \leq J_m, \qquad \sum_{v \in \mathscr{A}_m} \mathfrak{s}_{m,v} \leq C_0 n_m.$$

Outside F_m^{cert}, every selected core is a contraction and has one of the finite-lift orbit-closure or algebraic/group certificates from Corollary 7.128, with charge bounded by $\mathfrak{s}_{m,v}$.

For every candidate vertex v, choose a tail type in $\{\psi, \mu\}$. Choose parameters

$$\varepsilon_{m,v}, \qquad A_{m,v}, \qquad s'_{m,v},$$

and let $q_{m,v}$ be the corresponding normalization from Corollary 7.68. Suppose the conditional mixed dominated tail and truncated-drift hypotheses hold for $(Y_{m,v,i})_{i \leq n_m}$ after adjoining $\mathcal{H}_m$, outside events $E_{m,v}^{\mathrm{tail}}$, and that

$$\ell_m := \log(e(D_m + n_m + 1)) \leq B\sqrt{n_m}$$

eventually,

$$\sum_m \sum_{v \in \mathfrak{T}_m} \bigl(\mathbb{P}(E_{m,v}^{\mathrm{tail}}) + \varepsilon_{m,v}\bigr) < \infty,$$

and

$$\sum_m J_m (D_m + n_m + 1)(e(D_m + n_m + 1))^{-\gamma_{\mathrm{tc}}(K_{\mathrm{tail}},1,1,B)} < \infty.$$

Then the realized frontiers $\mathscr{A}_m$ have the same eventual common signing, contraction normalization, fixed-menu coverage, and all-integer and sparse schedule thresholds as in Corollary 7.128, with deterministic selected-cardinality envelope J_m.

In particular, finite-depth or countably generated random trees, branching Poisson/Cox catalogues, adaptive beam-search libraries, hierarchical mixtures, random forests of structured cores, and recursive random basis/corner choices are covered whenever their realized frontier size and charge have a summable budget certificate. No independence between generations, siblings, or different branches is required beyond the displayed budget and candidate-tail summability.

Proof. On $(F_m^{\mathrm{tree}} \cup F_m^{\mathrm{cert}})^c$, the terminal frontier $\mathscr{A}_m$ is an $\mathcal{H}_m$-measurable finite selected set from the countable catalogue $\mathfrak{T}_m$, ordered by the fixed depth-first order. It satisfies the deterministic cardinality and charge bounds required by Corollary 7.128, and its selected cores have the required certificates. The displayed summability gives $\sum_m \mathbb{P}(F_m^{\mathrm{tree}} \cup F_m^{\mathrm{cert}}) < \infty$, so Borel–Cantelli gives the selected-submenu structural event eventually.

The candidate-tail series over the whole tree dominates the selected-tail series after ordering the frontier and padding by zero streams up to J_m. The deterministic matrix-tail series is exactly the selected-submenu series with cardinality envelope J_m. Applying Corollary 7.128 therefore gives the claimed common signing and threshold conclusions. The antichain condition only records that the selected labels are terminal models; the proof uses no independence between the recursive branching decisions. □

Proposition 7.143 (Square-function Matrix Spencer certificate)**.** *There is a universal constant C such that, for every self-adjoint tuple $A_1, \dots, A_n \in \mathrm{M}_D(\mathbb{C})$,*

$$\operatorname{disc}(A_1, \dots, A_n) \le C\sqrt{\log(eD)}\left\|\sum_{i=1}^n A_i^2\right\|_{\mathrm{op}}^{1/2}.$$

More precisely, if

$$V := \sum_{i=1}^n A_i^2, \qquad \rho(A) := \max\{1, \operatorname{rank} V\},$$

then the ambient dimension may be replaced by the realized joint square-function rank:

$$\operatorname{disc}(A_1, \dots, A_n) \le C\sqrt{\log(e\rho(A))}\left\|\sum_{i=1}^n A_i^2\right\|_{\mathrm{op}}^{1/2}.$$

Consequently, if $A_1, \dots, A_n$ are random self-adjoint contractions and

$$F_{C_0} := \left\{\left\|\sum_{i=1}^n A_i^2\right\|_{\mathrm{op}} \le \frac{C_0 n}{\log(eD)}\right\},$$

then

$$\mathbb{P}\left\{\operatorname{disc}(A_1, \dots, A_n) > C\sqrt{C_0 n}\right\} \le \mathbb{P}(F_{C_0}^c).$$

Thus any random-matrix model whose realized noncommutative variance profile is at most $O(n/\log(eD))$ is in the Matrix Spencer regime, independently of the ambient dimension D.

Proof. Put $V = \sum_i A_i^2$ and $E = \operatorname{ran} V$. If $x \in \ker V$, then

$$0 = \langle Vx, x\rangle = \sum_i \|A_i x\|^2,$$

so every A_i kills $\ker V$. Since the A_i's are self-adjoint, E is invariant for every A_i. Restricting the tuple to E and applying the finite-dimensional noncommutative Khintchine inequality [18, 23] in dimension $\rho(A)$ gives, for independent Rademacher signs $\delta_1, \dots, \delta_n$,

$$\mathbb{E}_\delta \left\| \sum_{i=1}^n \delta_i A_i \right\|_{\mathrm{op}} \le C\sqrt{\log(e\rho(A))} \left\| \left(\sum_{i=1}^n A_i^2 \right)^{1/2} \right\|_{\mathrm{op}}.$$

The minimum over sign vectors is bounded by this expectation, which proves the rank-sensitive deterministic estimate. Since $\rho(A) \le D$, this also gives the displayed ambient-dimensional form. The probability statement is the same bound applied pathwise on the event F_{C_0}. □

Corollary 7.144 (Random sign search under a square profile)**.** *There is a universal constant C with the following property. Let $A_1, \dots, A_n \in \mathrm{M}_D(\mathbb{C})$ be self-adjoint, and let $\eta^{(1)}, \dots, \eta^{(m)}$ be independent uniform sign vectors in $\{\pm 1\}^n$. Then, for every $m \ge 1$,*

$$\mathbb{P}_\eta \left\{ \min_{1\le r\le m} \left\| \sum_{i=1}^n \eta_i^{(r)} A_i \right\|_{\mathrm{op}} > 2C\sqrt{\log(e\rho(A))} \left\| \sum_{i=1}^n A_i^2 \right\|_{\mathrm{op}}^{1/2} \right\} \le 2^{-m},$$

where $\rho(A) = \max\{1, \operatorname{rank}(\sum_i A_i^2)\}$. In particular,

$$\mathbb{P}_\eta \left\{ \min_{1\le r\le m} \left\| \sum_{i=1}^n \eta_i^{(r)} A_i \right\|_{\mathrm{op}} > 2C\sqrt{\log(eD)} \left\| \sum_{i=1}^n A_i^2 \right\|_{\mathrm{op}}^{1/2} \right\} \le 2^{-m}.$$

Consequently, if $A_1, \dots, A_n$ are random self-adjoint contractions, independent of the search signs, and F_{C_0} is the square-profile event from Proposition 7.143, then

$$\mathbb{P}\left\{ \min_{1\le r\le m} \left\| \sum_{i=1}^n \eta_i^{(r)} A_i \right\|_{\mathrm{op}} > 2C\sqrt{C_0 n} \right\} \le \mathbb{P}(F_{C_0}^c) + 2^{-m}.$$

Thus every random ensemble covered by the square-profile route also has a simple randomized signing certificate: m independent trials lose only an additional 2^{-m} in failure probability.

Proof. For one uniform sign vector η, the noncommutative Khintchine estimate used in the rank-sensitive part of Proposition 7.143 gives

$$\mathbb{E}_\eta \left\| \sum_i \eta_i A_i \right\|_{\mathrm{op}} \le C\sqrt{\log(e\rho(A))} \left\| \sum_i A_i^2 \right\|_{\mathrm{op}}^{1/2}.$$

Markov's inequality therefore makes one trial successful with probability at least $1/2$. The m trials are independent, so all of them fail with probability at most 2^{-m}. Since $\rho(A) \le D$, the displayed ambient-dimensional search bound follows. The random-tuple statement follows by conditioning on $A_1, \dots, A_n$ and then restricting to F_{C_0}. □

Corollary 7.145 (Common signs for self-adjoint square-profile menus)**.** *Fix $C_0 \ge 1$. For each $m \ge 1$, let $J_m \ge 1$, $D_m, n_m \ge 1$, and*

$$D_m^{\mathrm{com}} := J_m D_m.$$

For every $1 \le h \le J_m$, let

$$A_{m,h,1}, \dots, A_{m,h,n_m} \in \mathrm{M}_{D_m}(\mathbb{C})$$

be arbitrary random self-adjoint matrices. Assume that there are events $E_{m,h}^{\mathrm{sq}}$ such that, outside $E_{m,h}^{\mathrm{sq}}$,

$$\left\| \sum_{i=1}^{n_m} A_{m,h,i}^2 \right\|_{\mathrm{op}} \le \frac{C_0 n_m}{\log(e D_m^{\mathrm{com}})}.$$

If

$$\sum_m \sum_{h=1}^{J_m} \mathbb{P}(E_{m,h}^{\mathrm{sq}}) < \infty,$$

then, almost surely, for all sufficiently large m, one sign vector $\varepsilon^{(m)} \in \{\pm 1\}^{n_m}$ balances the whole active self-adjoint menu:

$$\max_{1 \le h \le J_m} \left\| \sum_{i=1}^{n_m} \varepsilon_i^{(m)} A_{m,h,i} \right\|_{\mathrm{op}} \le C(C_0) \sqrt{n_m}.$$

If M_m independent random sign vectors are sampled for the whole active menu and

$$\sum_m 2^{-M_m} < \infty,$$

then the sampled list almost surely eventually contains such a common certificate. If $J_m \to \infty$, active prefixes may exhaust a countable self-adjoint random-matrix library, and every fixed library element is eventually covered by the common active-prefix signing.

Thus any self-adjoint active library whose realized noncommutative variance profiles are summably certified at the common-sign scale has a single simultaneous Matrix-Spencer signing.

Proof. Outside the union of the events $E_{m,h}^{\mathrm{sq}}$, set

$$\widetilde{A}_{m,i} := \bigoplus_{h=1}^{J_m} A_{m,h,i} \in \mathrm{M}_{D_m^{\mathrm{com}}}(\mathbb{C}).$$

Then

$$\left\| \sum_i \widetilde{A}_{m,i}^2 \right\|_{\mathrm{op}} = \max_{h \le J_m} \left\| \sum_i A_{m,h,i}^2 \right\|_{\mathrm{op}} \le \frac{C_0 n_m}{\log(e D_m^{\mathrm{com}})}.$$

Proposition 7.143 applied in $\mathrm{M}_{D_m^{\mathrm{com}}}(\mathbb{C})$ gives one signing of the block-diagonal tuple, which is exactly one sign vector balancing every active stream. Since

$$\sum_m \mathbb{P}\left(\bigcup_{h\le J_m} E_{m,h}^{\mathrm{sq}}\right) \le \sum_m \sum_{h\le J_m} \mathbb{P}(E_{m,h}^{\mathrm{sq}}) < \infty,$$

Borel–Cantelli gives the eventual deterministic certificates. The sampled statement follows from Corollary 7.144 applied to the same block-diagonal tuple, adding only $\sum_m 2^{-M_m}$. If $J_m \to \infty$, each fixed library stream belongs to all sufficiently large active prefixes. □

Corollary 7.146 (Common signs for rank-sensitive self-adjoint square-profile menus)**.** *Fix $C_0 \ge 1$. For each $m \ge 1$, let $J_m \ge 1$, $D_m, n_m \ge 1$, and let $R_m \ge 1$. For every $1 \le h \le J_m$, let*

$$A_{m,h,1}, \dots, A_{m,h,n_m} \in \mathrm{M}_{D_m}(\mathbb{C})$$

be arbitrary random self-adjoint matrices, and put

$$V_{m,h} := \sum_{i=1}^{n_m} A_{m,h,i}^2, \qquad \rho_{m,h}^{\mathrm{sq}} := \max\{1, \operatorname{rank} V_{m,h}\}.$$

Assume that there are events E_m^{rsq} such that, outside E_m^{rsq},

$$\sum_{h=1}^{J_m} \rho_{m,h}^{\mathrm{sq}} \le R_m, \qquad \max_{1\le h\le J_m} \|V_{m,h}\|_{\mathrm{op}} \le \frac{C_0 n_m}{\log(eR_m)}.$$

If

$$\sum_m \mathbb{P}(E_m^{\mathrm{rsq}}) < \infty,$$

then, almost surely, for all sufficiently large m, one sign vector $\varepsilon^{(m)} \in \{\pm 1\}^{n_m}$ balances the whole active menu:

$$\max_{1\le h\le J_m} \left\| \sum_{i=1}^{n_m} \varepsilon_i^{(m)} A_{m,h,i} \right\|_{\mathrm{op}} \le C(C_0)\sqrt{n_m}.$$

If M_m independent random sign vectors are sampled for the whole active menu and

$$\sum_m 2^{-M_m} < \infty,$$

then the sampled list almost surely eventually contains such a common certificate.

In particular, if each active stream is supported on a possibly random $\kappa_{m,h}$-dimensional corner, or more generally $\operatorname{rank} V_{m,h} \le \kappa_{m,h}$, one may take

$$R_m = \sum_{h=1}^{J_m} \max\{1, \kappa_{m,h}\}.$$

Thus common signs for low-rank active self-adjoint libraries pay the logarithm of total active square-function rank, not the ambient block dimension $J_m D_m$.

Proof. On $(E_m^{\mathrm{rsq}})^c$, define the block-diagonal tuple

$$\widetilde{A}_{m,i} := \bigoplus_{h=1}^{J_m} A_{m,h,i} \in \mathrm{M}_{J_m D_m}(\mathbb{C}).$$

Its square function is

$$\widetilde{V}_m := \sum_i \widetilde{A}_{m,i}^2 = \bigoplus_{h=1}^{J_m} V_{m,h}.$$

Hence

$$\|\widetilde{V}_m\|_{\mathrm{op}} = \max_{h \le J_m} \|V_{m,h}\|_{\mathrm{op}} \le \frac{C_0 n_m}{\log(eR_m)}$$

and, because $R_m \ge 1$,

$$\max\{1, \operatorname{rank} \widetilde{V}_m\} \le \sum_{h=1}^{J_m} \rho_{m,h}^{\mathrm{sq}} \le R_m.$$

The rank-sensitive part of Proposition 7.143 gives

$$\operatorname{disc}(\widetilde{A}_{m,1}, \dots, \widetilde{A}_{m,n_m}) \le C\sqrt{\log(eR_m)}\,\|\widetilde{V}_m\|_{\mathrm{op}}^{1/2} \le C(C_0)\sqrt{n_m}.$$

A signing of the block-diagonal tuple is exactly one sign vector balancing every active stream. The summability of $\mathbb{P}(E_m^{\mathrm{rsq}})$ and Borel–Cantelli give the eventual almost-sure statement.

For sampled signs, condition on the active menu on $(E_m^{\mathrm{rsq}})^c$ and apply Corollary 7.144 to the same block-diagonal tuple in its rank-sensitive form. The sampled failure for the whole menu is 2^{-M_m}, so the displayed summability and Borel–Cantelli complete the proof. The final specialization follows from $\operatorname{rank} V_{m,h} \le \kappa_{m,h}$. □

Corollary 7.147 (Rectangular square-profile certificate)**.** *Let $X_1, \dots, X_n \in \mathrm{M}_{d_1 \times d_2}(\mathbb{C})$. Put*

$$V_L := \sum_{i=1}^n X_i X_i^*, \qquad V_R := \sum_{i=1}^n X_i^* X_i, \qquad D_{\mathrm{rec}} := d_1 + d_2,$$

and

$$\rho_{\mathrm{rec}} := \max\{1, \operatorname{rank} V_L + \operatorname{rank} V_R\}.$$

Then

$$\min_{\varepsilon \in \{\pm 1\}^n} \left\| \sum_{i=1}^n \varepsilon_i X_i \right\|_{\mathrm{op}} \le C\sqrt{\log(e\rho_{\mathrm{rec}})}\,\max\{\|V_L\|_{\mathrm{op}}, \|V_R\|_{\mathrm{op}}\}^{1/2}.$$

In particular the rank factor may always be replaced by D_{rec}.

Moreover, if $\eta^{(1)}, \dots, \eta^{(m)}$ are independent uniform sign vectors, then

$$\mathbb{P}_\eta \left\{ \min_{1 \le r \le m} \left\| \sum_{i=1}^n \eta_i^{(r)} X_i \right\|_{\mathrm{op}} > 2C\sqrt{\log(e\rho_{\mathrm{rec}})}\,\max\{\|V_L\|_{\mathrm{op}}, \|V_R\|_{\mathrm{op}}\}^{1/2} \right\} \le 2^{-m}.$$

Consequently, rectangular random contractions are in the Matrix-Spencer scale whenever their realized left and right square profiles are both $O(n/\log(eD_{\mathrm{rec}}))$.

Proof. Apply the Hermitian dilation

$$\mathscr{H}(X_i) = \begin{pmatrix} 0 & X_i \\ X_i^* & 0 \end{pmatrix} \in \mathrm{M}_{D_{\mathrm{rec}}}(\mathbb{C}).$$

Then

$$\mathscr{H}(X_i)^2 = \begin{pmatrix} X_i X_i^* & 0 \\ 0 & X_i^* X_i \end{pmatrix},$$

so

$$\left\| \sum_i \mathscr{H}(X_i)^2 \right\|_{\mathrm{op}} = \max\{\|V_L\|_{\mathrm{op}}, \|V_R\|_{\mathrm{op}}\},$$

and the rank of this square function is at most $\operatorname{rank} V_L + \operatorname{rank} V_R$. Also

$$\left\| \sum_i \varepsilon_i \mathscr{H}(X_i) \right\|_{\mathrm{op}} = \left\| \sum_i \varepsilon_i X_i \right\|_{\mathrm{op}}.$$

The deterministic and randomized claims are therefore exactly Proposition 7.143 and Corollary 7.144 applied to the dilated self-adjoint tuple. □

Corollary 7.148 (Bernstein square-profile ensembles)**.** *Let $D \geq 2$, and let $A_1, \ldots, A_n \in \mathrm{M}_D(\mathbb{C})$ be independent random self-adjoint contractions. Put*

$$\mu := \left\| \sum_{i=1}^n \mathbb{E} A_i^2 \right\|_{\mathrm{op}}.$$

Then, for every $C_0 \geq 1$, there is $C(C_0)$ such that, for every $t > 0$ satisfying

$$\mu + t \leq \frac{C_0 n}{\log(eD)},$$

one has

$$\mathbb{P}\left\{\operatorname{disc}(A_1, \ldots, A_n) > C(C_0)\sqrt{n}\right\} \leq D \exp\left\{-\frac{t^2}{2(\mu + t/3)}\right\}.$$

In particular, if $\mu \leq C_0 n/(2\log(eD))$, then

$$\mathbb{P}\left\{\operatorname{disc}(A_1, \ldots, A_n) > C(C_0)\sqrt{n}\right\} \leq D \exp\left\{-\frac{3C_0 n}{16\log(eD)}\right\}.$$

Consequently, let $T_1, \ldots, T_n$ be arbitrary, possibly random, self-adjoint contractions, let $U_1, \ldots, U_n$ be independent Haar unitary matrices independent of the T_i, and set $A_i = U_i T_i U_i^*$. If

$$G := \left\{ \frac{1}{D} \sum_{i=1}^n \operatorname{tr}(T_i^2) \leq \frac{C_0 n}{2\log(eD)} \right\},$$

then

$$\mathbb{P}\left\{\operatorname{disc}(A_1, \ldots, A_n) > C(C_0)\sqrt{n}\right\} \leq \mathbb{P}(G^c) + D \exp\left\{-\frac{3C_0 n}{16\log(eD)}\right\}.$$

Thus independent Haar-random rotations of fixed spectra, or of conditionally random spectra, satisfy Matrix Spencer whenever their average Frobenius-square energy is $O(D/\log(eD))$. This includes low-stable-rank and soft-spectrum full-rank contractions, not only hard subspace-supported matrices.

Proof. Write $B_i = A_i^2$. Since A_i is a self-adjoint contraction, $0 \le B_i \le I$. Let $X_i = B_i - \mathbb{E}B_i$. Then $\mathbb{E}X_i = 0$, $\lambda_{\max}(X_i) \le 1$, and

$$\sum_i \mathbb{E}X_i^2 = \sum_i \left(\mathbb{E}B_i^2 - (\mathbb{E}B_i)^2\right) \le \sum_i \mathbb{E}B_i^2 \le \sum_i \mathbb{E}B_i.$$

Hence the variance parameter in the one-sided matrix Bernstein inequality [25] is at most μ. Therefore

$$\begin{aligned}\mathbb{P}\left\{\left\|\sum_i A_i^2\right\|_{\mathrm{op}} > \mu + t\right\} &\le \mathbb{P}\left\{\lambda_{\max}\left(\sum_i X_i\right) > t\right\} \\ &\le D\exp\left\{-\frac{t^2}{2(\mu + t/3)}\right\}.\end{aligned}$$

On the complementary event, Proposition 7.143 gives the discrepancy bound. If $\mu \le C_0 n/(2\log(eD))$, take $t = C_0 n/(2\log(eD))$, so that $\mu \le t$ and

$$\frac{t^2}{2(\mu + t/3)} \ge \frac{t^2}{2(t + t/3)} = \frac{3C_0 n}{16\log(eD)}.$$

For the Haar-rotation specialization, condition on $T_1, \dots, T_n$. Haar invariance gives

$$\mathbb{E}(A_i^2 \mid T_i) = \mathbb{E}(U_i T_i^2 U_i^* \mid T_i) = \frac{\operatorname{tr}(T_i^2)}{D} I.$$

Thus, on G, the conditional value of μ is at most $C_0 n/(2\log(eD))$. Applying the preceding bound conditionally and then averaging proves the displayed estimate. □

Corollary 7.149 (Conditional square-twirl ensembles)**.** *Let $D \ge 2$. Let $T_1, \dots, T_n \in \mathrm{M}_D(\mathbb{C})$ be arbitrary, possibly dependent, random self-adjoint contractions. Conditional on $T = (T_1, \dots, T_n)$, let $U_1, \dots, U_n$ be independent random unitaries and set*

$$A_i = U_i T_i U_i^*.$$

Define the conditional square profile

$$\mu(T) := \left\|\sum_{i=1}^n \mathbb{E}\left(U_i T_i^2 U_i^* \,\middle|\, T\right)\right\|_{\mathrm{op}}.$$

Then, for every $C_0 \ge 1$ there is $C(C_0)$ such that, whenever $v, u > 0$ satisfy

$$v + u \le \frac{C_0 n}{\log(eD)},$$

one has

$$\mathbb{P}\left\{\operatorname{disc}(A_1,\dots,A_n) > C(C_0)\sqrt{n}\right\} \le \mathbb{P}\{\mu(T) > v\} + D\exp\left\{-\frac{u^2}{2(v+u/3)}\right\}.$$

Consequently, any independent random conjugation scheme whose conditional square twirl has operator norm $O(n/\log(eD))$ is Spencer-scale with the displayed high probability. The conjugating laws may be Haar, finite, or sample-dependent; only the conditional independence and the square profile enter the proof.

Proof. Condition on T. The matrices A_i are then independent self-adjoint contractions and

$$\left\|\sum_i \mathbb{E}(A_i^2 \mid T)\right\|_{\mathrm{op}} = \mu(T).$$

On the event $\{\mu(T) \le v\}$, the proof of Corollary 7.148, applied conditionally with $t = u$, gives conditional failure probability at most

$$D\exp\left\{-\frac{u^2}{2(v+u/3)}\right\}.$$

Averaging over T and adding the event $\{\mu(T) > v\}$ proves the claim. □

Corollary 7.150 (Finite-design and signed-permutation rotations)**.** *Let $D \ge 2$, and let $T_1,\dots,T_n \in \mathrm{M}_D(\mathbb{C})$ be arbitrary, possibly dependent, random self-adjoint contractions. Conditional on $T = (T_1,\dots,T_n)$, let $U_1,\dots,U_n$ be independent random unitaries such that, for every deterministic matrix $X \in \mathrm{M}_D(\mathbb{C})$,*

$$\mathbb{E}\left(U_i X U_i^* \mid T\right) = \frac{\operatorname{tr} X}{D} I.$$

Set $A_i = U_i T_i U_i^*$ and

$$\Phi(T) := \frac{1}{D}\sum_{i=1}^{n} \operatorname{tr}(T_i^2).$$

Then, for every $C_0 \ge 1$ there is $C(C_0)$ such that, whenever $v, u > 0$ satisfy $v + u \le C_0 n/\log(eD)$,

$$\mathbb{P}\left\{\operatorname{disc}(A_1,\dots,A_n) > C(C_0)\sqrt{n}\right\} \le \mathbb{P}\{\Phi(T) > v\} + D\exp\left\{-\frac{u^2}{2(v+u/3)}\right\}.$$

In particular, if $\Phi(T) \le C_0 n/(2\log(eD))$ outside an event E, then

$$\mathbb{P}\left\{\operatorname{disc}(A_1,\dots,A_n) > C(C_0)\sqrt{n}\right\} \le \mathbb{P}(E) + D\exp\left\{-\frac{3C_0 n}{16\log(eD)}\right\}.$$

This applies not only to Haar unitary or orthogonal rotations, but also to uniform draws from any finite irreducible unitary group, to exact conjugation unitary 1-designs, and in particular to the signed-permutation/monomial twirl. Thus discrete random bases give the same Matrix Spencer conclusion as Haar bases under the same average Frobenius-square profile.

Proof. The displayed twirl identity gives

$$\sum_i \mathbb{E}\left(U_i T_i^2 U_i^* \,\middle|\, T\right) = \frac{1}{D}\sum_i \operatorname{tr}(T_i^2) I,$$

so the conditional square profile in Corollary 7.149 is $\Phi(T)$. That corollary gives the first estimate, and the second follows by taking $v = u = C_0 n/(2\log(eD))$.

For a finite irreducible unitary group the twirl identity is Schur's lemma. For the signed-permutation group it is also immediate: random signs kill the off-diagonal entries of X, and the random permutation averages the diagonal entries to $(\operatorname{tr} X/D)I$. □

Corollary 7.151 (Block-design twirl profile)**.** *Let $D \geq 2$. Let $\mathcal{G}$ be an environment, and let $T_1, \dots, T_n \in \mathrm{M}_D(\mathbb{C})$ be $\mathcal{G}$-measurable self-adjoint contractions. For each i, let*

$$I = \sum_{\alpha \in I_i} P_{i,\alpha}$$

be a $\mathcal{G}$-measurable orthogonal decomposition, and put $d_{i,\alpha} := \operatorname{rank} P_{i,\alpha}$. Conditional on $\mathcal{G}$, let $U_1, \dots, U_n$ be independent random unitaries satisfying the block twirl identity

$$\mathbb{E}\,(U_i X U_i^* \,|\, \mathcal{G}) = \sum_{\alpha \in I_i} \frac{\operatorname{tr}(P_{i,\alpha} X P_{i,\alpha})}{d_{i,\alpha}} P_{i,\alpha} \qquad (X \in \mathrm{M}_D(\mathbb{C})).$$

Set $A_i = U_i T_i U_i^*$, and define the block square profile

$$\Psi(\mathcal{G}) := \left\| \sum_{i=1}^{n} \sum_{\alpha \in I_i} \frac{\operatorname{tr}(P_{i,\alpha} T_i^2 P_{i,\alpha})}{d_{i,\alpha}} P_{i,\alpha} \right\|_{\mathrm{op}}.$$

Then, for every $C_0 \geq 1$ there is $C(C_0)$ such that, whenever $v, u > 0$ satisfy $v + u \leq C_0 n/\log(eD)$,

$$\mathbb{P}\left\{\operatorname{disc}(A_1, \dots, A_n) > C(C_0)\sqrt{n}\right\} \leq \mathbb{P}\{\Psi(\mathcal{G}) > v\} + D \exp\left\{-\frac{u^2}{2(v + u/3)}\right\}.$$

In particular, if $\Psi(\mathcal{G}) \leq C_0 n/(2\log(eD))$ outside an event E, then

$$\mathbb{P}\left\{\operatorname{disc}(A_1, \dots, A_n) > C(C_0)\sqrt{n}\right\} \leq \mathbb{P}(E) + D \exp\left\{-\frac{3C_0 n}{16\log(eD)}\right\}.$$

Thus independent block-Haar rotations, block finite-design rotations, and block signed-permutation or monomial randomizations satisfy Matrix Spencer whenever their block-energy profile is $O(n/\log(eD))$. The decomposition may vary with i, and may itself be random as part of the environment.

Proof. Apply the block twirl identity with $X = T_i^2$. Since $A_i^2 = U_i T_i^2 U_i^*$, the conditional square profile in Corollary 7.149 is exactly $\Psi(\mathcal{G})$. That corollary gives the first estimate; the second follows by taking $v = u = C_0 n/(2\log(eD))$.

The listed examples satisfy the displayed identity block by block. Haar measure on each block and uniform draws from finite irreducible unitary groups give the scalar block expectation by Schur's lemma; signed-permutation or monomial randomization inside a block first kills off-diagonal entries and then averages the diagonal entries over the block. □

Corollary 7.152 (Dominated approximate twirl profiles)**.** *Let $D \geq 2$. Let $\mathcal{G}$ be an environment, and let $T_1, \dots, T_n \in \mathrm{M}_D(\mathbb{C})$ be $\mathcal{G}$-measurable self-adjoint contractions. Conditional on $\mathcal{G}$, let $U_1, \dots, U_n$ be independent random unitaries and set $A_i = U_i T_i U_i^*$. Suppose that there are $\mathcal{G}$-measurable positive semidefinite matrices $V_1, \dots, V_n$ such that*

$$\mathbb{E}\left(U_i T_i^2 U_i^* \,\middle|\, \mathcal{G}\right) \preceq V_i \qquad (1 \leq i \leq n).$$

Put

$$\Xi(\mathcal{G}) := \left\| \sum_{i=1}^n V_i \right\|_{\mathrm{op}}.$$

Then, for every $C_0 \geq 1$ there is $C(C_0)$ such that, whenever $v, u > 0$ satisfy $v + u \leq C_0 n / \log(eD)$,

$$\mathbb{P}\left\{\mathrm{disc}(A_1, \dots, A_n) > C(C_0)\sqrt{n}\right\} \leq \mathbb{P}\{\Xi(\mathcal{G}) > v\} + D \exp\left\{-\frac{u^2}{2(v + u/3)}\right\}.$$

Consequently, if $\Xi(\mathcal{G}) \leq C_0 n/(2\log(eD))$ outside an event E, then

$$\mathbb{P}\left\{\mathrm{disc}(A_1, \dots, A_n) > C(C_0)\sqrt{n}\right\} \leq \mathbb{P}(E) + D \exp\left\{-\frac{3C_0 n}{16 \log(eD)}\right\}.$$

Two useful specializations make the approximate-design cost explicit. First, if $\Gamma_i \succeq 0$ are $\mathcal{G}$-measurable and, for every $0 \preceq X \preceq I$,

$$\mathbb{E}\left(U_i X U_i^* \,|\, \mathcal{G}\right) \preceq \frac{\mathrm{tr}\, X}{D} I + \Gamma_i,$$

then the theorem applies with

$$V_i = \frac{\mathrm{tr}(T_i^2)}{D} I + \Gamma_i.$$

Thus approximate scalar unitary 1-designs are Spencer-scale whenever

$$\left\| \frac{1}{D} \sum_i \mathrm{tr}(T_i^2) I + \sum_i \Gamma_i \right\|_{\mathrm{op}} = O\left(\frac{n}{\log(eD)}\right).$$

Second, for $\mathcal{G}$-measurable decompositions $I = \sum_{\alpha \in I_i} P_{i,\alpha}$, if

$$\mathbb{E}\left(U_i X U_i^* \,|\, \mathcal{G}\right) \preceq \sum_{\alpha \in I_i} \frac{\mathrm{tr}(P_{i,\alpha} X P_{i,\alpha})}{d_{i,\alpha}} P_{i,\alpha} + \Gamma_i \qquad (0 \preceq X \preceq I),$$

then the theorem applies with the block profile from Corollary 7.151 plus the aggregate error $\sum_i \Gamma_i$. Hence approximate block designs and finite random-basis samplers only pay their total positive semidefinite twirl error.

Proof. Since

$$\sum_i \mathbb{E}(A_i^2 \mid \mathcal{G}) = \sum_i \mathbb{E}(U_i T_i^2 U_i^* \mid \mathcal{G}) \preceq \sum_i V_i,$$

the conditional square profile in Corollary 7.149 is at most $\Xi(\mathcal{G})$. Applying that corollary gives the finite-probability bounds. The two displayed specializations follow by taking $X = T_i^2$, which satisfies $0 \preceq T_i^2 \preceq I$, in the corresponding approximate twirl inequality. □

Corollary 7.153 (Predictable square-profile ensembles)**.** *Let $D \geq 2$, let $(\mathcal{F}_i)_{i=0}^n$ be a filtration, and let $A_1, \dots, A_n \in \mathrm{M}_D(\mathbb{C})$ be self-adjoint contractions such that A_i is $\mathcal{F}_i$-measurable. Put*

$$\nu := \left\| \sum_{i=1}^n \mathbb{E}(A_i^2 \mid \mathcal{F}_{i-1}) \right\|_{\mathrm{op}}.$$

Then, for every $C_0 \geq 1$, there is $C(C_0)$ such that, for every $v, t > 0$ satisfying

$$v + t \leq \frac{C_0 n}{\log(eD)},$$

one has

$$\mathbb{P}\left\{\mathrm{disc}(A_1, \dots, A_n) > C(C_0)\sqrt{n}\right\} \leq \mathbb{P}\{\nu > v\} + D \exp\left\{-\frac{t^2}{2(v + t/3)}\right\}.$$

Moreover, if $\eta^{(1)}, \dots, \eta^{(m)}$ are independent uniform sign vectors, independent of $A_1, \dots, A_n$, then, with possibly larger $C(C_0)$,

$$\mathbb{P}\left\{ \min_{1 \leq r \leq m} \left\| \sum_{i=1}^n \eta_i^{(r)} A_i \right\|_{\mathrm{op}} > C(C_0)\sqrt{n} \right\} \leq \mathbb{P}\{\nu > v\} + D \exp\left\{-\frac{t^2}{2(v + t/3)}\right\} + 2^{-m}.$$

In particular, if $\nu \leq C_0 n/(2\log(eD))$ almost surely, then both statements hold with failure at most

$$D \exp\left\{-\frac{3C_0 n}{16 \log(eD)}\right\}$$

and, for the randomized signing search, with the additional 2^{-m} term. Thus the square-profile route covers adapted dependent random matrix streams, not only independent ensembles, whenever their predictable noncommutative variance profile is $O(n/\log(eD))$.

Proof. Set $B_i = A_i^2$ and $M_i = \mathbb{E}(B_i \mid \mathcal{F}_{i-1})$. Then $0 \leq B_i \leq I$, and

$$X_i := B_i - M_i$$

is a self-adjoint matrix martingale difference with $\lambda_{\max}(X_i) \le 1$. Its predictable quadratic variation satisfies

$$\sum_i \mathbb{E}(X_i^2 \mid \mathcal{F}_{i-1}) = \sum_i \left(\mathbb{E}(B_i^2 \mid \mathcal{F}_{i-1}) - M_i^2\right) \le \sum_i \mathbb{E}(B_i^2 \mid \mathcal{F}_{i-1}) \le \sum_i M_i,$$

because $0 \le B_i \le I$ implies $B_i^2 \le B_i$. On the event $\{\nu \le v\}$, the variance process is therefore bounded by v. The matrix Freedman inequality [25] gives

$$\mathbb{P}\left\{\lambda_{\max}\left(\sum_i X_i\right) > t \text{ and } \nu \le v\right\} \le D \exp\left\{-\frac{t^2}{2(v + t/3)}\right\}.$$

On the complementary high-probability event,

$$\left\|\sum_i A_i^2\right\|_{\mathrm{op}} = \lambda_{\max}\left(\sum_i B_i\right) \le v + t.$$

The first claim follows from Proposition 7.143. For the random-search claim, condition on the realized tuple and apply Corollary 7.144 on the event $\|\sum_i A_i^2\|_{\mathrm{op}} \le v + t$. The final displayed specialization is the choice $v = t = C_0 n/(2\log(eD))$, as in Corollary 7.148. □

Corollary 7.154 (Tail-capped Hermitian entry-variance profiles)**.** *Fix $C_0 \ge 1$, and put $c_0 := 3C_0/16$. For each $m \ge 1$, let $D_m \ge 2$, $J_m \ge 1$, and $\ell_m = \log(eD_m)$. For each active stream $1 \le h \le J_m$, let*

$$A_{m,h,i} \in \mathrm{M}_{D_m}(\mathbb{C}), \qquad 1 \le i \le n_m,$$

be self-adjoint and adapted to a filtration $(\mathcal{F}_{m,h,i})_{i=0}^{n_m}$. Write

$$A_{m,h,i} = \sum_{p=1}^{D_m} \xi_{m,h,i,p} E_{pp} + \sum_{1 \le p < q \le D_m} \left(\zeta_{m,h,i,pq} E_{pq} + \overline{\zeta_{m,h,i,pq}} E_{qp}\right).$$

Conditionally on $\mathcal{F}_{m,h,i-1}$, assume that the upper-triangular variables $\{\xi_{m,h,i,p}\}_p$ and $\{\zeta_{m,h,i,pq}\}_{p<q}$ are independent and centered, with $\xi_{m,h,i,p}$ real-valued. Let deterministic variance envelopes satisfy

$$\mathbb{E}(\xi_{m,h,i,p}^2 \mid \mathcal{F}_{m,h,i-1}) \le \alpha_{m,h,i,p}^2, \qquad \mathbb{E}(|\zeta_{m,h,i,pq}|^2 \mid \mathcal{F}_{m,h,i-1}) \le \beta_{m,h,i,pq}^2.$$

Let $K_{m,h} > 0$ be deterministic operator-norm caps and let $\eta_{m,h,i} \ge 0$ satisfy

$$\mathbb{P}\{\|A_{m,h,i}\|_{\mathrm{op}} > K_{m,h}\} \le \eta_{m,h,i}.$$

Define the cumulative Hermitian row-variance profile

$$W_{m,h}^{\mathrm{herm}} := \max_{1 \le p \le D_m} \sum_{i=1}^{n_m} \left(\alpha_{m,h,i,p}^2 + \sum_{q<p} \beta_{m,h,i,qp}^2 + \sum_{p<q} \beta_{m,h,i,pq}^2\right).$$

Assume

$$W_{m,h}^{\mathrm{herm}} \le \frac{C_0 K_{m,h}^2 n_m}{2\ell_m} \qquad (1 \le h \le J_m)$$

and

$$\sum_{m}\sum_{h=1}^{J_m}\left[\sum_{i=1}^{n_m}\eta_{m,h,i}+D_m\exp\left\{-c_0\frac{n_m}{\ell_m}\right\}\right]<\infty.$$

Then, almost surely, for all sufficiently large m, every active Hermitian entry-variance stream has a whole-matrix signing

$$\operatorname{disc}(A_{m,h,1},\dots,A_{m,h,n_m})\le C(C_0)K_{m,h}\sqrt{n_m}\qquad(1\le h\le J_m).$$

Independent or common sampled-sign batches are certified under the same 2^{-M}-summability hypotheses as in Corollary 7.219.

The bounded case is the specialization $\eta_{m,h,i}=0$. More generally, put $L_m=R\ell_m$, and let $S_{m,h}>0$. If, for fixed $\alpha>0$ and $A_{\mathrm{tail}}\ge 1$,

$$\mathbb{P}\{\|A_{m,h,i}\|_{\mathrm{op}}>S_{m,h}u\}\le A_{\mathrm{tail}}e^{-u^\alpha}\qquad(u\ge 0),$$

choose $K_{m,h}=S_{m,h}L_m^{1/\alpha}$. If instead, for fixed $p_0>0$,

$$\mathbb{E}\|A_{m,h,i}\|_{\mathrm{op}}^{p_0}\le S_{m,h}^{p_0},$$

choose $K_{m,h}=S_{m,h}\exp\{L_m/p_0\}$. In either case the cap-tail contribution for stream (m,h) is at most

$$A_*n_me^{-L_m},\qquad A_*=\begin{cases}A_{\mathrm{tail}}, & \textit{sub-Weibull case},\\ 1, & \textit{moment case}.\end{cases}$$

Consequently, if $J_mn_m\le(eD_m)^\Gamma$ and $n_m\ge a\log^2(eD_m)$, then the full summability condition holds over all integer dimensions $D_m=m$ whenever

$$R>\Gamma+1,\qquad c_0a>\Gamma+2,$$

and along stretched-exponential sparse dimension schedules whenever

$$R>\Gamma,\qquad c_0a>\Gamma+1.$$

Thus Wigner, GOE/GUE, random band, sparse Hermitian, and variance-profile upper-triangular ensembles are covered whenever their cumulative row variance is $O(K^2n/\log(eD))$, with one discrepancy sign assigned to each whole Hermitian matrix.

Proof. For each stream set

$$Y_{m,h,i}:=A_{m,h,i}\mathbf{1}_{\{\|A_{m,h,i}\|_{\mathrm{op}}\le K_{m,h}\}}.$$

The cap-exception probability for stream (m,h) is at most $\sum_i\eta_{m,h,i}$, and outside that event $Y_{m,h,i}=A_{m,h,i}$ for all i. The normalized matrices $K_{m,h}^{-1}Y_{m,h,i}$ are self-adjoint contractions, and

$$Y_{m,h,i}^2\preceq A_{m,h,i}^2.$$

By conditional independence and centering of the upper-triangular variables, $\mathbb{E}(A_{m,h,i}^2\mid\mathcal{F}_{m,h,i-1})$ is diagonal. Its p-th diagonal entry is bounded by

$$\alpha_{m,h,i,p}^2+\sum_{q<p}\beta_{m,h,i,qp}^2+\sum_{p<q}\beta_{m,h,i,pq}^2.$$

Therefore the predictable square profile of the capped normalized stream is at most $W^{\mathrm{herm}}_{m,h}/K^2_{m,h}$. The half-budget hypothesis and Corollary 7.153 give failure probability at most

$$D_m \exp\left\{-c_0 \frac{n_m}{\ell_m}\right\}$$

for each active stream, plus the sampled-sign 2^{-M} term when sampled search is used. Summability and Borel–Cantelli prove the eventual signing statement.

The sub-Weibull cap gives $\eta_{m,h,i} \le A_{\mathrm{tail}} e^{-L_m}$, while Markov's inequality gives $\eta_{m,h,i} \le e^{-L_m}$ in the moment case. The all-integer and sparse thresholds are the same series tests for

$$J_m n_m e^{-R\ell_m} \quad \text{and} \quad J_m D_m e^{-c_0 n_m/\ell_m}$$

used in the rectangular tail-capped schedules below. □

Corollary 7.155 (Predictable rectangular square-profile ensembles)**.** *Let $d_1, d_2 \ge 1$, put $D_{\mathrm{rec}} = d_1 + d_2$, and let $(\mathcal{F}_i)_{i=0}^n$ be a filtration. Let*

$$X_1, \dots, X_n \in \mathrm{M}_{d_1 \times d_2}(\mathbb{C})$$

be contractions such that X_i is $\mathcal{F}_i$-measurable. Define the predictable row and column square profiles

$$\nu_L := \left\| \sum_{i=1}^n \mathbb{E}(X_i X_i^* \mid \mathcal{F}_{i-1}) \right\|_{\mathrm{op}}, \qquad \nu_R := \left\| \sum_{i=1}^n \mathbb{E}(X_i^* X_i \mid \mathcal{F}_{i-1}) \right\|_{\mathrm{op}},$$

and put $\nu_{\mathrm{rec}} := \max\{\nu_L, \nu_R\}$. Then, for every $C_0 \ge 1$, there is $C(C_0)$ such that, for every $v, t > 0$ satisfying

$$v + t \le \frac{C_0 n}{\log(eD_{\mathrm{rec}})},$$

one has

$$\mathbb{P}\left\{ \min_{\varepsilon \in \{\pm 1\}^n} \left\| \sum_{i=1}^n \varepsilon_i X_i \right\|_{\mathrm{op}} > C(C_0)\sqrt{n} \right\} \le \mathbb{P}\{\nu_{\mathrm{rec}} > v\} + D_{\mathrm{rec}} \exp\left\{ -\frac{t^2}{2(v + t/3)} \right\}.$$

If $\eta^{(1)}, \dots, \eta^{(m)}$ are independent uniform sign vectors, independent of $X_1, \dots, X_n$, then, with possibly larger $C(C_0)$,

$$\mathbb{P}\left\{ \min_{1 \le r \le m} \left\| \sum_{i=1}^n \eta_i^{(r)} X_i \right\|_{\mathrm{op}} > C(C_0)\sqrt{n} \right\}$$
$$\le \mathbb{P}\{\nu_{\mathrm{rec}} > v\} + D_{\mathrm{rec}} \exp\left\{ -\frac{t^2}{2(v + t/3)} \right\} + 2^{-m}.$$

In particular, if $\nu_{\mathrm{rec}} \le C_0 n/(2\log(eD_{\mathrm{rec}}))$ almost surely, then both statements hold with structural failure at most

$$D_{\mathrm{rec}} \exp\left\{ -\frac{3C_0 n}{16 \log(eD_{\mathrm{rec}})} \right\},$$

plus 2^{-m} for sampled sign search. For independent rectangular contractions the same conclusion holds with

$$\nu_{\mathrm{rec}} = \max\left\{\left\|\sum_i \mathbb{E}X_iX_i^*\right\|_{\mathrm{op}}, \left\|\sum_i \mathbb{E}X_i^*X_i\right\|_{\mathrm{op}}\right\}.$$

Thus rectangular random operator models, bipartite support models, and non-self-adjoint matrix-series models are Spencer-scale once both left and right variance profiles are $O(n/\log(e(d_1+d_2)))$.

Proof. For each i, let

$$Y_i := \mathscr{H}(X_i) = \begin{pmatrix} 0 & X_i \\ X_i^* & 0 \end{pmatrix}.$$

Then Y_i is an adapted self-adjoint contraction in $\mathrm{M}_{D_{\mathrm{rec}}}(\mathbb{C})$, and

$$\left\|\sum_i \mathbb{E}(Y_i^2 \mid \mathcal{F}_{i-1})\right\|_{\mathrm{op}} = \max\{\nu_L, \nu_R\} = \nu_{\mathrm{rec}}.$$

Also

$$\left\|\sum_i \varepsilon_i Y_i\right\|_{\mathrm{op}} = \left\|\sum_i \varepsilon_i X_i\right\|_{\mathrm{op}}$$

for every sign vector. Applying Corollary 7.153 to the self-adjoint tuple (Y_i) gives the two probability bounds. The half-budget specialization is the choice $v = t = C_0 n/(2\log(eD_{\mathrm{rec}}))$, and independence makes the predictable profiles deterministic after conditioning on the past. □

Corollary 7.156 (Centered independent-entry rectangular ensembles)**.** *Fix $C_0 \geq 1$, and put $c_0 := 3C_0/16$. For each $m \geq 1$, let $J_m \geq 1$, $d_{m,1}, d_{m,2} \geq 1$, $D_m^{\mathrm{rec}} = d_{m,1} + d_{m,2}$, and $\ell_m = \log(eD_m^{\mathrm{rec}})$. For each active stream $1 \leq h \leq J_m$, let*

$$X_{m,h,i} = (x_{m,h,i,r,s})_{r,s} \in \mathrm{M}_{d_{m,1}\times d_{m,2}}(\mathbb{C}), \qquad 1 \leq i \leq n_m,$$

be adapted to a filtration $(\mathcal{F}_{m,h,i})_{i=0}^{n_m}$. Assume that $\|X_{m,h,i}\|_{\mathrm{op}} \leq K_{m,h}$ almost surely and that, conditionally on $\mathcal{F}_{m,h,i-1}$, the entries of $X_{m,h,i}$ are independent and centered. Suppose deterministic variance envelopes $\sigma^2_{m,h,i,r,s} \geq 0$ satisfy

$$\mathbb{E}\{|x_{m,h,i,r,s}|^2 \mid \mathcal{F}_{m,h,i-1}\} \leq \sigma^2_{m,h,i,r,s}$$

almost surely, and define

$$V_{m,h}^{\mathrm{ent}} := \max\left\{\begin{array}{l} \displaystyle\max_{1\leq r\leq d_{m,1}} \sum_{i=1}^{n_m}\sum_{s=1}^{d_{m,2}} \sigma^2_{m,h,i,r,s}, \\ \displaystyle\max_{1\leq s\leq d_{m,2}} \sum_{i=1}^{n_m}\sum_{r=1}^{d_{m,1}} \sigma^2_{m,h,i,r,s} \end{array}\right\}.$$

If

$$V_{m,h}^{\mathrm{ent}} \leq \frac{C_0 K_{m,h}^2 n_m}{2\ell_m} \qquad (1 \leq h \leq J_m)$$

and

$$\sum_{m}\sum_{h=1}^{J_m} D_m^{\mathrm{rec}} \exp\left\{-c_0\frac{n_m}{\ell_m}\right\} < \infty,$$

then, almost surely, for all sufficiently large m, every active independent-entry rectangular stream has a per-matrix signing

$$\min_{\varepsilon\in\{\pm1\}^{n_m}} \left\|\sum_{i=1}^{n_m} \varepsilon_i X_{m,h,i}\right\|_{\mathrm{op}} \le C(C_0)K_{m,h}\sqrt{n_m} \qquad (1\le h\le J_m).$$

Independent or common sampled-sign batches are certified under the same 2^{-M}-summability hypotheses as in Corollary 7.158.

In particular, if $K_{m,h}\le K_*$, $J_m\le (eD_m^{\mathrm{rec}})^{\Gamma}$, and

$$n_m \ge a\log^2(eD_m^{\mathrm{rec}}),$$

then the structural series is summable over all integer dimensions $D_m^{\mathrm{rec}} = m$ whenever

$$c_0 a > \Gamma + 2,$$

and along every stretched-exponential sparse dimension schedule $D_m^{\mathrm{rec}} \ge \exp(cm^{\kappa})$ whenever

$$c_0 a > \Gamma + 1.$$

Thus centered sparse or dense rectangular random matrices with independent entries are Spencer-scale, with one sign per matrix, whenever their maximum cumulative row and column variance is $O(K^2 n/\log(eD_{\mathrm{rec}}))$.

Proof. Apply Corollary 7.155 to $\widetilde{X}_{m,h,i} := K_{m,h}^{-1}X_{m,h,i}$. Conditional independence and centering make

$$\mathbb{E}(X_{m,h,i}X_{m,h,i}^{*} \mid \mathcal{F}_{m,h,i-1})$$

diagonal, with r-th diagonal entry at most $\sum_s \sigma^2_{m,h,i,r,s}$. Similarly, $\mathbb{E}(X^{*}_{m,h,i}X_{m,h,i} \mid \mathcal{F}_{m,h,i-1})$ is diagonal, with s-th diagonal entry at most $\sum_r \sigma^2_{m,h,i,r,s}$. Hence the predictable rectangular square profile of the normalized stream is at most $V^{\mathrm{ent}}_{m,h}/K^2_{m,h}$, which is bounded by $C_0 n_m/(2\ell_m)$. The half-budget form of Corollary 7.155 gives failure probability at most

$$D_m^{\mathrm{rec}} \exp\left\{-c_0\frac{n_m}{\ell_m}\right\}$$

for each active stream, plus 2^{-M} for sampled sign search. Summing over m, h and applying Borel–Cantelli gives the eventual simultaneous statement. The all-integer and sparse schedule thresholds are the same p-series and stretched-exponential tests for

$$J_m D_m^{\mathrm{rec}} \exp\{-c_0 n_m/\ell_m\}.$$

□

Corollary 7.157 (Tail-capped independent-entry rectangular ensembles)**.** *Fix $C_0 \geq 1$, and put $c_0 := 3C_0/16$. For each $m \geq 1$, let $J_m \geq 1$, $d_{m,1}, d_{m,2} \geq 1$, $D_m^{\mathrm{rec}} = d_{m,1} + d_{m,2}$, and $\ell_m = \log(eD_m^{\mathrm{rec}})$. For each active stream $1 \leq h \leq J_m$, let*

$$X_{m,h,i} = (x_{m,h,i,r,s})_{r,s} \in \mathrm{M}_{d_{m,1} \times d_{m,2}}(\mathbb{C}), \qquad 1 \leq i \leq n_m,$$

be adapted to a filtration $(\mathcal{F}_{m,h,i})_{i=0}^{n_m}$. Conditionally on $\mathcal{F}_{m,h,i-1}$, assume the entries of $X_{m,h,i}$ are independent and centered, and let deterministic variance envelopes satisfy

$$\mathbb{E}\{|x_{m,h,i,r,s}|^2 \mid \mathcal{F}_{m,h,i-1}\} \leq \sigma_{m,h,i,r,s}^2.$$

Let $K_{m,h} > 0$ be deterministic operator-norm caps and let $\eta_{m,h,i} \geq 0$ satisfy

$$\mathbb{P}\{\|X_{m,h,i}\|_{\mathrm{op}} > K_{m,h}\} \leq \eta_{m,h,i}.$$

Define $V_{m,h}^{\mathrm{ent}}$ as in Corollary 7.156. Assume

$$V_{m,h}^{\mathrm{ent}} \leq \frac{C_0 K_{m,h}^2 n_m}{2\ell_m} \qquad (1 \leq h \leq J_m)$$

and

$$\sum_m \sum_{h=1}^{J_m} \left[\sum_{i=1}^{n_m} \eta_{m,h,i} + D_m^{\mathrm{rec}} \exp\left\{ -c_0 \frac{n_m}{\ell_m} \right\} \right] < \infty.$$

Then, almost surely, for all sufficiently large m, every active independent-entry rectangular stream has a per-matrix signing

$$\min_{\varepsilon \in \{\pm 1\}^{n_m}} \left\| \sum_{i=1}^{n_m} \varepsilon_i X_{m,h,i} \right\|_{\mathrm{op}} \leq C(C_0) K_{m,h} \sqrt{n_m} \qquad (1 \leq h \leq J_m).$$

Independent or common sampled-sign batches are certified under the same 2^{-M}-summability hypotheses as in Corollary 7.158.

The cap may be chosen from standard whole-matrix tails. Put $L_m = R\ell_m$, and let $S_{m,h} > 0$. If, for fixed $\alpha > 0$ and $A_{\mathrm{tail}} \geq 1$,

$$\mathbb{P}\{\|X_{m,h,i}\|_{\mathrm{op}} > S_{m,h} u\} \leq A_{\mathrm{tail}} e^{-u^\alpha} \qquad (u \geq 0),$$

choose $K_{m,h} = S_{m,h} L_m^{1/\alpha}$. If instead, for fixed $p > 0$,

$$\mathbb{E}\|X_{m,h,i}\|_{\mathrm{op}}^p \leq S_{m,h}^p,$$

choose $K_{m,h} = S_{m,h} \exp\{L_m/p\}$. In the two cases the cap-tail contribution for stream (m,h) is at most

$$A_* n_m e^{-L_m}, \qquad A_* = \begin{cases} A_{\mathrm{tail}}, & \text{sub-Weibull case,} \\ 1, & \text{moment case.} \end{cases}$$

Consequently, if $J_m n_m \leq (eD_m^{\mathrm{rec}})^\Gamma$ and $n_m \geq a \log^2(eD_m^{\mathrm{rec}})$, then the full summability condition holds over all integer dimensions $D_m^{\mathrm{rec}} = m$ whenever

$$R > \Gamma + 1, \qquad c_0 a > \Gamma + 2,$$

and along stretched-exponential sparse dimension schedules whenever

$$R > \Gamma, \qquad c_0 a > \Gamma + 1.$$

Proof. For each stream set

$$Y_{m,h,i} := X_{m,h,i}\mathbf{1}_{\{\|X_{m,h,i}\|_{\mathrm{op}} \le K_{m,h}\}}.$$

The cap-exception probability for stream (m,h) is at most $\sum_i \eta_{m,h,i}$, and outside that event $Y_{m,h,i} = X_{m,h,i}$ for all i. The normalized matrices $K_{m,h}^{-1} Y_{m,h,i}$ are contractions. Moreover,

$$Y_{m,h,i}Y_{m,h,i}^* \preceq X_{m,h,i}X_{m,h,i}^*, \qquad Y_{m,h,i}^* Y_{m,h,i} \preceq X_{m,h,i}^* X_{m,h,i}.$$

By the same conditional independence and centering calculation as in Corollary 7.156, the left and right predictable square profiles of the capped normalized stream are therefore at most $V_{m,h}^{\mathrm{ent}}/K_{m,h}^2$. The half-budget hypothesis and Corollary 7.155 give the displayed structural failure term, plus the sampled-sign 2^{-M} term when sampled search is used. Summability and Borel–Cantelli prove the eventual signing statement.

The sub-Weibull cap gives $\eta_{m,h,i} \le A_{\mathrm{tail}} e^{-L_m}$, while Markov's inequality gives $\eta_{m,h,i} \le e^{-L_m}$ in the moment case. The all-integer and sparse thresholds are the same series tests for

$$J_m n_m e^{-R\ell_m} \quad \text{and} \quad J_m D_m^{\mathrm{rec}} e^{-c_0 n_m/\ell_m}$$

used in Corollary 7.196. □

Corollary 7.158 (Active menus of rectangular square-profile streams)**.** *For each $m \ge 1$, let $J_m \ge 1$, $d_{m,1}, d_{m,2} \ge 1$, $D_m^{\mathrm{rec}} = d_{m,1} + d_{m,2}$, and $n_m \ge 1$. For every $1 \le h \le J_m$, let*

$$X_{m,h,1}, \dots, X_{m,h,n_m} \in \mathrm{M}_{d_{m,1}\times d_{m,2}}(\mathbb{C})$$

be an adapted rectangular-contraction stream with predictable rectangular profile $\nu_{m,h}^{\mathrm{rec}}$ as in Corollary 7.155. Assume that, outside events $E_{m,h}$,

$$\nu_{m,h}^{\mathrm{rec}} \le \frac{C_0 n_m}{2\log(eD_m^{\mathrm{rec}})} \qquad (1 \le h \le J_m),$$

and that

$$\sum_m \sum_{h=1}^{J_m} \mathbb{P}(E_{m,h}) < \infty, \qquad \sum_m J_m D_m^{\mathrm{rec}} \exp\left\{-\frac{3C_0 n_m}{16\log(eD_m^{\mathrm{rec}})}\right\} < \infty.$$

Then, almost surely, for all sufficiently large m, every active rectangular stream has a Spencer-scale signing:

$$\min_{\varepsilon\in\{\pm1\}^{n_m}} \left\| \sum_{i=1}^{n_m} \varepsilon_i X_{m,h,i} \right\|_{\mathrm{op}} \le C(C_0)\sqrt{n_m} \qquad (1 \le h \le J_m).$$

Sampled certificates are also eventual almost surely if either

$$\sum_m \sum_{h=1}^{J_m} 2^{-M_{m,h}} < \infty$$

for independent per-stream batches, or

$$\sum_m J_m 2^{-M_m} < \infty$$

for one common active-menu batch. If $J_m \to \infty$, the active prefixes may exhaust a countable library of rectangular random-matrix models; every fixed library element is then covered eventually.

Proof. Apply Corollary 7.155 to each active stream with $v = t = C_0 n_m/(2\log(eD_m^{\mathrm{rec}}))$, then sum the displayed failure probabilities over m and h. The first Borel–Cantelli lemma gives the simultaneous eventual signings. The sampled-search statements add the corresponding $2^{-M_{m,h}}$ or common $J_m 2^{-M_m}$ losses before the same Borel–Cantelli argument. If $J_m \to \infty$, each fixed library stream is active for all sufficiently large m. □

Corollary 7.159 (Common signs for rectangular square-profile menus)**.** *Fix $C_0 \geq 1$. For each $m \geq 1$, let $J_m \geq 1$, $d_{m,1}, d_{m,2} \geq 1$, $D_m^{\mathrm{rec}} = d_{m,1} + d_{m,2}$, and*

$$D_m^{\mathrm{com}} := J_m D_m^{\mathrm{rec}}.$$

For every $1 \leq h \leq J_m$, let

$$X_{m,h,1}, \dots, X_{m,h,n_m} \in \mathrm{M}_{d_{m,1}\times d_{m,2}}(\mathbb{C})$$

be arbitrary random rectangular matrices. Assume that there are events $E_{m,h}^{\mathrm{sq}}$ such that, outside $E_{m,h}^{\mathrm{sq}}$,

$$\max\left\{\left\|\sum_{i=1}^{n_m} X_{m,h,i}X_{m,h,i}^*\right\|_{\mathrm{op}}, \left\|\sum_{i=1}^{n_m} X_{m,h,i}^* X_{m,h,i}\right\|_{\mathrm{op}}\right\} \leq \frac{C_0 n_m}{\log(eD_m^{\mathrm{com}})}.$$

If

$$\sum_m \sum_{h=1}^{J_m} \mathbb{P}(E_{m,h}^{\mathrm{sq}}) < \infty,$$

then, almost surely, for all sufficiently large m, one sign vector $\varepsilon^{(m)} \in \{\pm 1\}^{n_m}$ balances the whole active rectangular menu:

$$\max_{1\leq h\leq J_m} \left\|\sum_{i=1}^{n_m} \varepsilon_i^{(m)} X_{m,h,i}\right\|_{\mathrm{op}} \leq C(C_0)\sqrt{n_m}.$$

If M_m independent random sign vectors are sampled for the whole active menu and

$$\sum_m 2^{-M_m} < \infty,$$

then the sampled list almost surely eventually contains such a common certificate. If $J_m \to \infty$, active prefixes may exhaust a countable library of rectangular or non-self-adjoint random-matrix models, and every fixed library element is eventually covered by the common active-prefix signing.

Thus any finite or countably exhausted rectangular model library whose realized left and right square profiles are summably certified at the common-sign scale has a single simultaneous Matrix-Spencer signing. Its rank-sensitive

form replaces $J_m(d_{m,1}+d_{m,2})$ by the total active left-plus-right square-profile rank, preserving the same shared sampled-sign loss 2^{-M_m}.

Proof. Outside the union of the events $E^{\mathrm{sq}}_{m,h}$, define

$$\widetilde{X}_{m,i} := \bigoplus_{h=1}^{J_m} X_{m,h,i} \in \mathrm{M}_{J_m d_{m,1}\times J_m d_{m,2}}(\mathbb{C}).$$

Then

$$\left\|\sum_i \widetilde{X}_{m,i}\widetilde{X}^*_{m,i}\right\|_{\mathrm{op}} = \max_{h\le J_m}\left\|\sum_i X_{m,h,i}X^*_{m,h,i}\right\|_{\mathrm{op}},$$
$$\left\|\sum_i \widetilde{X}^*_{m,i}\widetilde{X}_{m,i}\right\|_{\mathrm{op}} = \max_{h\le J_m}\left\|\sum_i X^*_{m,h,i}X_{m,h,i}\right\|_{\mathrm{op}}.$$

Corollary 7.147 in dimension D^{com}_m gives one signing of the block-direct-sum tuple, which is the displayed common signing. Since

$$\sum_m \mathbb{P}\left(\bigcup_{h\le J_m} E^{\mathrm{sq}}_{m,h}\right) \le \sum_m \sum_{h\le J_m} \mathbb{P}(E^{\mathrm{sq}}_{m,h}) < \infty,$$

Borel–Cantelli gives the eventual deterministic certificates. The sampled statement applies the sampled part of Corollary 7.147 to the same direct-sum tuple and adds only the failure series $\sum_m 2^{-M_m}$. If $J_m \to \infty$, each fixed library stream belongs to all sufficiently large active prefixes. □

Corollary 7.160 (Common signs for rank-sensitive rectangular square-profile menus)**.** *Fix $C_0 \ge 1$. For each $m \ge 1$, let $J_m \ge 1$, $d_{m,1}, d_{m,2}, n_m \ge 1$, and let $R_m \ge 1$. For every $1 \le h \le J_m$, let*

$$X_{m,h,1}, \dots, X_{m,h,n_m} \in \mathrm{M}_{d_{m,1}\times d_{m,2}}(\mathbb{C})$$

be arbitrary random rectangular matrices. Put

$$V^L_{m,h} := \sum_{i=1}^{n_m} X_{m,h,i}X^*_{m,h,i}, \qquad V^R_{m,h} := \sum_{i=1}^{n_m} X^*_{m,h,i}X_{m,h,i},$$

and

$$\rho^{\mathrm{rec}}_{m,h} := \max\{1, \operatorname{rank} V^L_{m,h} + \operatorname{rank} V^R_{m,h}\}.$$

Assume that there are events E^{rrec}_m such that, outside E^{rrec}_m,

$$\sum_{h=1}^{J_m} \rho^{\mathrm{rec}}_{m,h} \le R_m, \qquad \max_{1\le h\le J_m} \max\{\|V^L_{m,h}\|_{\mathrm{op}}, \|V^R_{m,h}\|_{\mathrm{op}}\} \le \frac{C_0 n_m}{\log(eR_m)}.$$

If

$$\sum_m \mathbb{P}(E^{\mathrm{rrec}}_m) < \infty,$$

then, almost surely, for all sufficiently large m, one sign vector $\varepsilon^{(m)} \in \{\pm 1\}^{n_m}$ balances the whole active rectangular menu:

$$\max_{1\le h\le J_m} \left\| \sum_{i=1}^{n_m} \varepsilon_i^{(m)} X_{m,h,i} \right\|_{\mathrm{op}} \le C(C_0)\sqrt{n_m}.$$

If M_m independent random sign vectors are sampled for the whole active menu and

$$\sum_m 2^{-M_m} < \infty,$$

then the sampled list almost surely eventually contains such a common certificate.

In particular, if each active stream has certified realized profile ranks

$$\operatorname{rank} V_{m,h}^L \le \kappa_{m,h}^L, \qquad \operatorname{rank} V_{m,h}^R \le \kappa_{m,h}^R,$$

one may take

$$R_m = \sum_{h=1}^{J_m} \max\{1, \kappa_{m,h}^L + \kappa_{m,h}^R\}.$$

Thus common signs for low-rank rectangular or non-self-adjoint active libraries pay the logarithm of total active left-plus-right square-function rank, not the ambient block dimension $J_m(d_{m,1} + d_{m,2})$.

Proof. On $(E_m^{\mathrm{rrec}})^c$, form the block-direct-sum rectangular tuple

$$\widetilde{X}_{m,i} := \bigoplus_{h=1}^{J_m} X_{m,h,i} \in \mathrm{M}_{J_m d_{m,1} \times J_m d_{m,2}}(\mathbb{C}).$$

Its left and right square profiles are

$$\sum_i \widetilde{X}_{m,i}\widetilde{X}_{m,i}^* = \bigoplus_{h=1}^{J_m} V_{m,h}^L, \qquad \sum_i \widetilde{X}_{m,i}^*\widetilde{X}_{m,i} = \bigoplus_{h=1}^{J_m} V_{m,h}^R.$$

Therefore

$$\max\left\{ \left\| \sum_i \widetilde{X}_{m,i}\widetilde{X}_{m,i}^* \right\|_{\mathrm{op}}, \left\| \sum_i \widetilde{X}_{m,i}^*\widetilde{X}_{m,i} \right\|_{\mathrm{op}} \right\} \le \frac{C_0 n_m}{\log(eR_m)}$$

and the rectangular square-profile rank of the direct-sum tuple is at most

$$\sum_{h=1}^{J_m} \left(\operatorname{rank} V_{m,h}^L + \operatorname{rank} V_{m,h}^R \right) \le \sum_{h=1}^{J_m} \rho_{m,h}^{\mathrm{rec}} \le R_m.$$

The rank-sensitive part of Corollary 7.147 gives one signing of the direct-sum tuple with norm at most $C(C_0)\sqrt{n_m}$, and this is exactly one sign vector balancing every active rectangular stream. Borel–Cantelli applied to $\sum_m \mathbb{P}(E_m^{\mathrm{rrec}})$ gives the eventual almost-sure statement.

For sampled signs, condition on the active menu on $(E_m^{\mathrm{rrec}})^c$ and apply the sampled rank-sensitive part of Corollary 7.147 to the same direct-sum tuple. The sampled failure for the whole active menu is 2^{-M_m}, so the displayed

summability and Borel–Cantelli complete the proof. The final specialization is immediate from the rank bounds. □

Corollary 7.161 (Rectangular low-overlap support ensembles)**.** *For every $C_0 \geq 1$ there is $C(C_0)$ with the following property. Let*

$$X_1, \dots, X_n \in \mathrm{M}_{d_1 \times d_2}(\mathbb{C})$$

be random rectangular matrices and put $D_{\mathrm{rec}} = d_1 + d_2$. Suppose that, outside an event E, there are orthogonal projections

$$P_i^L \in \mathrm{M}_{d_1}(\mathbb{C}), \qquad P_i^R \in \mathrm{M}_{d_2}(\mathbb{C}), \qquad 0 \leq \rho_i \leq 1,$$

such that

$$X_i = P_i^L X_i P_i^R, \qquad \|X_i\|_{\mathrm{op}} \leq \rho_i \qquad (1 \leq i \leq n),$$

and

$$\max\left\{ \left\| \sum_{i=1}^n \rho_i^2 P_i^L \right\|_{\mathrm{op}}, \left\| \sum_{i=1}^n \rho_i^2 P_i^R \right\|_{\mathrm{op}} \right\} \leq \frac{C_0 n}{\log(eD_{\mathrm{rec}})}.$$

Then

$$\mathbb{P}\left\{ \min_{\varepsilon \in \{\pm 1\}^n} \left\| \sum_{i=1}^n \varepsilon_i X_i \right\|_{\mathrm{op}} > C(C_0)\sqrt{n} \right\} \leq \mathbb{P}(E).$$

If m independent random sign trials are used, the same conclusion holds with an additional 2^{-m} failure probability. Thus rectangular sparse, bipartite, row-column localized, and rectangular random-subspace models reduce to the two weighted overlap operators on the range and domain sides.

Proof. On a realization outside E, the support and norm assumptions give

$$0 \preceq X_i X_i^* \preceq \rho_i^2 P_i^L, \qquad 0 \preceq X_i^* X_i \preceq \rho_i^2 P_i^R.$$

Hence the two rectangular square profiles in Corollary 7.147 are at most $C_0 n / \log(eD_{\mathrm{rec}})$. That corollary gives the deterministic signing and the sampled-search statement on this realization. □

Corollary 7.162 (Active menus with dependent rectangular overlap budgets)**.** *Fix $C_0 \geq 1$. For each $m \geq 1$, let $J_m \geq 1$, $d_{m,1}, d_{m,2} \geq 1$, $D_m^{\mathrm{rec}} = d_{m,1} + d_{m,2}$, and $n_m \geq 1$. For every $1 \leq h \leq J_m$, let*

$$X_{m,h,1}, \dots, X_{m,h,n_m} \in \mathrm{M}_{d_{m,1} \times d_{m,2}}(\mathbb{C})$$

be arbitrary random rectangular matrices. Assume that there are events $F_{m,h}^{\mathrm{ov}}$, random orthogonal projections

$$P_{m,h,i}^L \in \mathrm{M}_{d_{m,1}}(\mathbb{C}), \qquad P_{m,h,i}^R \in \mathrm{M}_{d_{m,2}}(\mathbb{C}),$$

and random amplitudes $0 \leq \rho_{m,h,i} \leq 1$ such that, outside $F_{m,h}^{\mathrm{ov}}$,

$$X_{m,h,i} = P_{m,h,i}^L X_{m,h,i} P_{m,h,i}^R, \qquad \|X_{m,h,i}\|_{\mathrm{op}} \leq \rho_{m,h,i} \qquad (1 \leq i \leq n_m),$$

and

$$\max\left\{\left\|\sum_{i=1}^{n_m}\rho_{m,h,i}^2P_{m,h,i}^L\right\|_{\mathrm{op}},\left\|\sum_{i=1}^{n_m}\rho_{m,h,i}^2P_{m,h,i}^R\right\|_{\mathrm{op}}\right\}\le\frac{C_0n_m}{\log(eD_m^{\mathrm{rec}})}.$$

If

$$\sum_m\sum_{h=1}^{J_m}\mathbb{P}(F_{m,h}^{\mathrm{ov}})<\infty,$$

then, almost surely, for all sufficiently large m, every active stream has a Spencer-scale rectangular signing:

$$\min_{\varepsilon\in\{\pm1\}^{n_m}}\left\|\sum_{i=1}^{n_m}\varepsilon_iX_{m,h,i}\right\|_{\mathrm{op}}\le C(C_0)\sqrt{n_m}\qquad(1\le h\le J_m).$$

Sampled certificates are also eventual almost surely under the same per-stream or common-batch 2^{-M}-summability hypotheses as in Corollary 7.158. If $J_m\to\infty$, active prefixes may exhaust a countable library.

Thus arbitrary dependent rectangular support laws, random regular or biregular bipartite supports, matchings, without-replacement row/column choices, determinantal or negatively dependent supports, and adaptive block/subspace frontiers are covered once their realized left and right weighted overlap budgets fail only summably.

Proof. Apply Corollary 7.161 to each active stream on $(F_{m,h}^{\mathrm{ov}})^c$. Summability of the overlap-budget failures and the first Borel–Cantelli lemma give the simultaneous eventual signings. The sampled-certificate statements add the same per-stream $\sum_{m,h}2^{-M_{m,h}}$ or common-batch $\sum_m J_m2^{-M_m}$ failure series used in Corollary 7.158. If $J_m\to\infty$, each fixed library stream belongs to the active prefix for all sufficiently large m. □

Corollary 7.163 (Tail-capped dependent rectangular overlap budgets)**.** *Fix $C_0\ge1$. In the active-menu setting of Corollary 7.162, replace the pointwise norm bounds by deterministic caps*

$$0\le\rho_{m,h,i}\le1\qquad(1\le h\le J_m,\ 1\le i\le n_m).$$

Assume that there are events $F_{m,h}^{\mathrm{ov}}$, random orthogonal projections

$$P_{m,h,i}^L\in\mathrm{M}_{d_{m,1}}(\mathbb{C}),\qquad P_{m,h,i}^R\in\mathrm{M}_{d_{m,2}}(\mathbb{C}),$$

and cap-tail budgets $\delta_{m,h,i}\ge0$ such that

$$\mathbb{P}\{\|X_{m,h,i}\|_{\mathrm{op}}>\rho_{m,h,i}\}\le\delta_{m,h,i},$$

and, outside $F_{m,h}^{\mathrm{ov}}$,

$$X_{m,h,i}=P_{m,h,i}^LX_{m,h,i}P_{m,h,i}^R\qquad(1\le i\le n_m)$$

and

$$\max\left\{\left\|\sum_{i=1}^{n_m}\rho_{m,h,i}^2P_{m,h,i}^L\right\|_{\mathrm{op}},\left\|\sum_{i=1}^{n_m}\rho_{m,h,i}^2P_{m,h,i}^R\right\|_{\mathrm{op}}\right\}\le\frac{C_0n_m}{\log(eD_m^{\mathrm{rec}})}.$$

If

$$\sum_{m}\sum_{h=1}^{J_m}\left[\mathbb{P}(F_{m,h}^{\mathrm{ov}})+\sum_{i=1}^{n_m}\delta_{m,h,i}\right]<\infty,$$

then, almost surely, for all sufficiently large m, every active stream has a Spencer-scale rectangular signing:

$$\min_{\varepsilon\in\{\pm1\}^{n_m}}\left\|\sum_{i=1}^{n_m}\varepsilon_i X_{m,h,i}\right\|_{\mathrm{op}}\le C(C_0)\sqrt{n_m}\qquad(1\le h\le J_m).$$

Sampled certificates and countable-library exhaustions follow under the same additional summability hypotheses as in Corollary 7.162.

For example, if

$$\mathbb{P}\{\|X_{m,h,i}\|_{\mathrm{op}}>s_{m,h,i}u\}\le A_*e^{-u^\alpha}\qquad(u\ge0)$$

with fixed $\alpha>0$, choose $\rho_{m,h,i}=s_{m,h,i}\ell_{m,h,i}^{1/\alpha}\le1$ and $\delta_{m,h,i}=A_*e^{-\ell_{m,h,i}}$. If instead $\mathbb{E}\|X_{m,h,i}\|_{\mathrm{op}}^p\le s_{m,h,i}^p$, choose $\rho_{m,h,i}=s_{m,h,i}e^{\ell_{m,h,i}/p}\le1$ and $\delta_{m,h,i}=e^{-\ell_{m,h,i}}$. These deterministic caps feed the same realized left/right overlap budget, so dependent heavy-tailed rectangular supports are reduced to cap-tail summability plus a capped overlap certificate.

Proof. For every active stream put

$$\Omega_{m,h}^{\mathrm{cap}}:=\{\|X_{m,h,i}\|_{\mathrm{op}}\le\rho_{m,h,i}\text{ for all }i\le n_m\}.$$

On $(F_{m,h}^{\mathrm{ov}})^c\cap\Omega_{m,h}^{\mathrm{cap}}$, the hypotheses of Corollary 7.162 hold with the deterministic amplitudes $\rho_{m,h,i}$. The exceptional probability for stream (m,h) is at most

$$\mathbb{P}(F_{m,h}^{\mathrm{ov}})+\sum_{i=1}^{n_m}\delta_{m,h,i}.$$

The displayed summability and Borel–Cantelli give the eventual active-menu signings; sampled certificates add the same sampled-sign failure series as in the dependent-overlap corollary. The sub-Weibull and moment cap choices follow from the displayed tail bound and Markov's inequality, respectively. □

Corollary 7.164 (Common signs for dependent rectangular overlap menus)**.** *Fix $C_0\ge1$. For each $m\ge1$, let $J_m\ge1$, $d_{m,1},d_{m,2}\ge1$, $D_m^{\mathrm{rec}}=d_{m,1}+d_{m,2}$, and*

$$D_m^{\mathrm{com}}:=J_mD_m^{\mathrm{rec}}.$$

For every $1\le h\le J_m$, let

$$X_{m,h,1},\dots,X_{m,h,n_m}\in\mathrm{M}_{d_{m,1}\times d_{m,2}}(\mathbb{C})$$

be arbitrary random rectangular matrices. Assume that there are events $F_{m,h}^{\mathrm{ov}}$, random orthogonal projections

$$P_{m,h,i}^L\in\mathrm{M}_{d_{m,1}}(\mathbb{C}),\qquad P_{m,h,i}^R\in\mathrm{M}_{d_{m,2}}(\mathbb{C}),$$

and random amplitudes $0 \le \rho_{m,h,i} \le 1$ *such that, outside* $F^{\mathrm{ov}}_{m,h}$,

$$X_{m,h,i} = P^L_{m,h,i} X_{m,h,i} P^R_{m,h,i}, \qquad \|X_{m,h,i}\|_{\mathrm{op}} \le \rho_{m,h,i} \qquad (1 \le i \le n_m),$$

and

$$\max\left\{ \left\| \sum_{i=1}^{n_m} \rho^2_{m,h,i} P^L_{m,h,i} \right\|_{\mathrm{op}}, \left\| \sum_{i=1}^{n_m} \rho^2_{m,h,i} P^R_{m,h,i} \right\|_{\mathrm{op}} \right\} \le \frac{C_0 n_m}{\log(e D^{\mathrm{com}}_m)}.$$

If

$$\sum_m \sum_{h=1}^{J_m} \mathbb{P}(F^{\mathrm{ov}}_{m,h}) < \infty,$$

then, almost surely, for all sufficiently large m, *there is one sign vector* $\varepsilon^{(m)} \in \{\pm 1\}^{n_m}$ *such that*

$$\max_{1 \le h \le J_m} \left\| \sum_{i=1}^{n_m} \varepsilon^{(m)}_i X_{m,h,i} \right\|_{\mathrm{op}} \le C(C_0)\sqrt{n_m}.$$

If, for each m, M_m *independent random sign vectors are sampled and tested against the whole active menu, and*

$$\sum_m 2^{-M_m} < \infty,$$

then the sampled list almost surely eventually contains such a common certificate. If $J_m \to \infty$, *active prefixes may exhaust a countable library of dependent rectangular models; every fixed library element is then covered eventually by the common active-prefix signing.*

Thus arbitrary dependent bipartite supports, random regular or biregular rectangular supports, matchings, without-replacement row/column choices, and adaptive rectangular block or subspace frontiers have a single simultaneous Matrix-Spencer signing whenever their realized left and right overlap budgets are summably certified at the common-sign scale.

Proof. Outside the union of the events $F^{\mathrm{ov}}_{m,h}$, define the rectangular block-direct-sum tuple

$$\widetilde{X}_{m,i} := \bigoplus_{h=1}^{J_m} X_{m,h,i} \in \mathrm{M}_{J_m d_{m,1} \times J_m d_{m,2}}(\mathbb{C}) \qquad (1 \le i \le n_m).$$

Then

$$\sum_i \widetilde{X}_{m,i} \widetilde{X}^*_{m,i} = \bigoplus_{h=1}^{J_m} \sum_i X_{m,h,i} X^*_{m,h,i},$$

$$\sum_i \widetilde{X}^*_{m,i} \widetilde{X}_{m,i} = \bigoplus_{h=1}^{J_m} \sum_i X^*_{m,h,i} X_{m,h,i}.$$

The support and norm bounds give

$$X_{m,h,i} X^*_{m,h,i} \preceq \rho^2_{m,h,i} P^L_{m,h,i}, \qquad X^*_{m,h,i} X_{m,h,i} \preceq \rho^2_{m,h,i} P^R_{m,h,i},$$

so the two square profiles of the direct-sum tuple are both bounded by $C_0 n_m/\log(eD_m^{\mathrm{com}})$. Corollary 7.147 applied in rectangular dimension D_m^{com} gives one signing of the direct-sum tuple, which is exactly one sign vector balancing every active stream. The first Borel–Cantelli lemma applies because

$$\sum_m \mathbb{P}\left(\bigcup_{h\le J_m} F_{m,h}^{\mathrm{ov}}\right) \le \sum_m \sum_{h\le J_m} \mathbb{P}(F_{m,h}^{\mathrm{ov}}) < \infty.$$

The sampled statement follows by applying the sampled part of Corollary 7.147 to the same direct-sum tuple, adding only the failure series $\sum_m 2^{-M_m}$. If $J_m \to \infty$, each fixed library element belongs to every sufficiently large active prefix. □

Corollary 7.165 (Common signs for support-rank rectangular overlap menus)**.** *Fix $C_0 \ge 1$. For each $m \ge 1$, let $J_m \ge 1$, $d_{m,1}, d_{m,2}, n_m \ge 1$, and let $R_m \ge 1$. For every $1 \le h \le J_m$, let*

$$X_{m,h,1}, \dots, X_{m,h,n_m} \in \mathrm{M}_{d_{m,1}\times d_{m,2}}(\mathbb{C})$$

be arbitrary random rectangular matrices. Assume that there are events F_m^{sr}, random orthogonal projections

$$P_{m,h,i}^L \in \mathrm{M}_{d_{m,1}}(\mathbb{C}), \qquad P_{m,h,i}^R \in \mathrm{M}_{d_{m,2}}(\mathbb{C}),$$

random support-envelope projections

$$Q_{m,h}^L \in \mathrm{M}_{d_{m,1}}(\mathbb{C}), \qquad Q_{m,h}^R \in \mathrm{M}_{d_{m,2}}(\mathbb{C}),$$

and amplitudes $0 \le \rho_{m,h,i} \le 1$ such that, outside F_m^{sr},

$$X_{m,h,i} = P_{m,h,i}^L X_{m,h,i} P_{m,h,i}^R, \qquad \|X_{m,h,i}\|_{\mathrm{op}} \le \rho_{m,h,i},$$

$$P_{m,h,i}^L \le Q_{m,h}^L, \qquad P_{m,h,i}^R \le Q_{m,h}^R \qquad (1 \le h \le J_m,\ 1 \le i \le n_m),$$

and

$$\sum_{h=1}^{J_m} \max\{1, \operatorname{rank} Q_{m,h}^L + \operatorname{rank} Q_{m,h}^R\} \le R_m.$$

Assume also that, outside F_m^{sr},

$$\max_{1\le h\le J_m} \max\left\{ \left\| \sum_{i=1}^{n_m} \rho_{m,h,i}^2 P_{m,h,i}^L \right\|_{\mathrm{op}}, \left\| \sum_{i=1}^{n_m} \rho_{m,h,i}^2 P_{m,h,i}^R \right\|_{\mathrm{op}} \right\} \le \frac{C_0 n_m}{\log(eR_m)}.$$

If

$$\sum_m \mathbb{P}(F_m^{\mathrm{sr}}) < \infty,$$

then, almost surely, for all sufficiently large m, there is one sign vector $\varepsilon^{(m)} \in \{\pm 1\}^{n_m}$ such that

$$\max_{1\le h\le J_m} \left\| \sum_{i=1}^{n_m} \varepsilon_i^{(m)} X_{m,h,i} \right\|_{\mathrm{op}} \le C(C_0)\sqrt{n_m}.$$

If M_m independent random sign vectors are sampled and tested against the whole active menu, and

$$\sum_m 2^{-M_m} < \infty,$$

then the sampled list almost surely eventually contains such a common certificate. If $J_m \to \infty$, active prefixes may exhaust a countable library and every fixed library element is covered eventually.

Thus dependent rectangular support models with small active support-envelope rank pay $\log(eR_m)$, *not* $\log(eJ_m(d_{m,1}+d_{m,2}))$.

Proof. On $(F_m^{\mathrm{sr}})^c$, put

$$V_{m,h}^L = \sum_i X_{m,h,i}X_{m,h,i}^*, \qquad V_{m,h}^R = \sum_i X_{m,h,i}^* X_{m,h,i}.$$

The support and norm hypotheses give

$$0 \preceq V_{m,h}^L \preceq \sum_i \rho_{m,h,i}^2 P_{m,h,i}^L, \qquad 0 \preceq V_{m,h}^R \preceq \sum_i \rho_{m,h,i}^2 P_{m,h,i}^R,$$

and $V_{m,h}^L = Q_{m,h}^L V_{m,h}^L Q_{m,h}^L$, $V_{m,h}^R = Q_{m,h}^R V_{m,h}^R Q_{m,h}^R$. Hence

$$\operatorname{rank} V_{m,h}^L + \operatorname{rank} V_{m,h}^R \le \operatorname{rank} Q_{m,h}^L + \operatorname{rank} Q_{m,h}^R$$

and the left/right square-profile norms are bounded by $C_0 n_m/\log(eR_m)$. Therefore the hypotheses of Corollary 7.160 hold with exceptional event F_m^{sr}. That corollary gives the common deterministic signing and the sampled-sign conclusion. The summability of $\mathbb{P}(F_m^{\mathrm{sr}})$ and Borel–Cantelli make the conclusions eventual almost surely. □

Corollary 7.166 (Tail-capped support-rank common signs for rectangular overlap menus)**.** *Fix $C_0 \ge 1$. For each $m \ge 1$, let $J_m \ge 1$, $d_{m,1}, d_{m,2}, n_m \ge 1$, and let $R_m \ge 1$. For every $1 \le h \le J_m$, let*

$$X_{m,h,1}, \dots, X_{m,h,n_m} \in \mathrm{M}_{d_{m,1}\times d_{m,2}}(\mathbb{C})$$

be arbitrary random rectangular matrices. Assume that there are events F_m^{sr}, random orthogonal projections

$$P_{m,h,i}^L \in \mathrm{M}_{d_{m,1}}(\mathbb{C}), \qquad P_{m,h,i}^R \in \mathrm{M}_{d_{m,2}}(\mathbb{C}),$$

random support-envelope projections

$$Q_{m,h}^L \in \mathrm{M}_{d_{m,1}}(\mathbb{C}), \qquad Q_{m,h}^R \in \mathrm{M}_{d_{m,2}}(\mathbb{C}),$$

deterministic caps $0 \le \rho_{m,h,i} \le 1$, and cap-tail budgets $\delta_{m,h,i} \ge 0$ such that

$$\mathbb{P}\{\|X_{m,h,i}\|_{\mathrm{op}} > \rho_{m,h,i}\} \le \delta_{m,h,i}.$$

Outside F_m^{sr}, assume that

$$X_{m,h,i} = P_{m,h,i}^L X_{m,h,i} P_{m,h,i}^R, \qquad P_{m,h,i}^L \le Q_{m,h}^L, \qquad P_{m,h,i}^R \le Q_{m,h}^R$$

for all $h \le J_m$ and $i \le n_m$, and

$$\sum_{h=1}^{J_m} \max\{1, \operatorname{rank} Q_{m,h}^L + \operatorname{rank} Q_{m,h}^R\} \le R_m.$$

Assume also that, outside F_m^{sr},

$$\max_{1\le h\le J_m} \max\left\{ \left\| \sum_{i=1}^{n_m} \rho_{m,h,i}^2 P_{m,h,i}^L \right\|_{\mathrm{op}}, \left\| \sum_{i=1}^{n_m} \rho_{m,h,i}^2 P_{m,h,i}^R \right\|_{\mathrm{op}} \right\} \le \frac{C_0 n_m}{\log(eR_m)}.$$

If

$$\sum_m \left[\mathbb{P}(F_m^{\mathrm{sr}}) + \sum_{h=1}^{J_m} \sum_{i=1}^{n_m} \delta_{m,h,i} \right] < \infty,$$

then, almost surely, for all sufficiently large m*, there is one sign vector* $\varepsilon^{(m)} \in \{\pm 1\}^{n_m}$ *such that*

$$\max_{1\le h\le J_m} \left\| \sum_{i=1}^{n_m} \varepsilon_i^{(m)} X_{m,h,i} \right\|_{\mathrm{op}} \le C(C_0)\sqrt{n_m}.$$

If M_m *independent random sign vectors are sampled and tested against the whole active menu, and*

$$\sum_m 2^{-M_m} < \infty,$$

then the sampled list almost surely eventually contains such a common certificate. If $J_m \to \infty$*, active prefixes may exhaust a countable library and every fixed library element is covered eventually.*

Thus unbounded localized rectangular libraries retain the support-envelope logarithm $\log(eR_m)$ *after paying only summable cap-tail failures.*

Proof. Let

$$\Omega_m^{\mathrm{cap}} := \{ \|X_{m,h,i}\|_{\mathrm{op}} \le \rho_{m,h,i} \text{ for all } h \le J_m,\ i \le n_m \}.$$

The union bound gives

$$\sum_m \mathbb{P}\big((\Omega_m^{\mathrm{cap}})^c\big) \le \sum_m \sum_{h=1}^{J_m} \sum_{i=1}^{n_m} \delta_{m,h,i} < \infty.$$

On the complement of

$$G_m := F_m^{\mathrm{sr}} \cup (\Omega_m^{\mathrm{cap}})^c$$

the support-rank rectangular corollary applies with amplitudes $\rho_{m,h,i}$. Since the exceptional events G_m are summable, Borel–Cantelli gives the eventual deterministic common signings and the sampled-sign conclusion. The countable-library conclusion is inherited from the same corollary. □

Corollary 7.167 (Common signs for rectangular coordinate-window Bernstein menus)**.** *Fix* $C_0 \ge 1$*. For each* $m \ge 1$*, let* $J_m \ge 1$*,* $d_{m,1}, d_{m,2}, n_m \ge 1$*, and let* $R_m \ge 1$*. For every* $1 \le h \le J_m$*, let*

$$E_{m,h}^L \subset \{1, \dots, d_{m,1}\}, \qquad E_{m,h}^R \subset \{1, \dots, d_{m,2}\}$$

be deterministic coordinate windows satisfying

$$\sum_{h=1}^{J_m} \max\{1, |E_{m,h}^L| + |E_{m,h}^R|\} \le R_m.$$

For each h, i, let

$$R_{m,h,i} \subset E^L_{m,h}, \qquad S_{m,h,i} \subset E^R_{m,h}$$

be random row and column supports. Assume that, for every $a \in E^L_{m,h}$, the indicators $\mathbf{1}_{\{a \in R_{m,h,i}\}}$, $1 \le i \le n_m$, are independent and

$$\mathbb{P}\{a \in R_{m,h,i}\} \le p^L_{m,h,i,a},$$

and similarly, for every $b \in E^R_{m,h}$,

$$\mathbb{P}\{b \in S_{m,h,i}\} \le p^R_{m,h,i,b}$$

with independence in i. Conditional on the supports, let

$$X_{m,h,i} \in \mathrm{M}_{d_{m,1} \times d_{m,2}}(\mathbb{C})$$

be arbitrary random rectangular matrices satisfying

$$X_{m,h,i} = P_{R_{m,h,i}} X_{m,h,i} P_{S_{m,h,i}}.$$

Let deterministic caps $0 \le \rho_{m,h,i} \le 1$ and cap-tail budgets $\delta_{m,h,i} \ge 0$ satisfy

$$\mathbb{P}\{\|X_{m,h,i}\|_{\mathrm{op}} > \rho_{m,h,i}\} \le \delta_{m,h,i}.$$

Put

$$b_{m,h} := \max_i \rho^2_{m,h,i},$$

$$\mu_{m,h} := \max\left\{ \max_{a \in E^L_{m,h}} \sum_i \rho^2_{m,h,i} p^L_{m,h,i,a}, \max_{b \in E^R_{m,h}} \sum_i \rho^2_{m,h,i} p^R_{m,h,i,b} \right\},$$

and

$$\sigma^2_{m,h} := \max\left\{ \max_{a \in E^L_{m,h}} \sum_i \rho^4_{m,h,i} p^L_{m,h,i,a}, \max_{b \in E^R_{m,h}} \sum_i \rho^4_{m,h,i} p^R_{m,h,i,b} \right\},$$

where a maximum over an empty window is interpreted as 0. Suppose that there are $t_{m,h} > 0$ such that

$$\mu_{m,h} + t_{m,h} \le \frac{C_0 n_m}{\log(eR_m)} \qquad (1 \le h \le J_m),$$

and

$$\sum_m \left[\sum_{h=1}^{J_m} (|E^L_{m,h}| + |E^R_{m,h}|) \exp\left\{ -\frac{t^2_{m,h}}{2(\sigma^2_{m,h} + b_{m,h} t_{m,h}/3)} \right\} + \sum_{h=1}^{J_m} \sum_{i=1}^{n_m} \delta_{m,h,i} \right] < \infty,$$

with zero denominator interpreted as zero contribution. Then, almost surely, for all sufficiently large m, there is one sign vector $\varepsilon^{(m)} \in \{\pm 1\}^{n_m}$ such that

$$\max_{1 \le h \le J_m} \left\| \sum_{i=1}^{n_m} \varepsilon^{(m)}_i X_{m,h,i} \right\|_{\mathrm{op}} \le C(C_0) \sqrt{n_m}.$$

The same sampled common-certificate and countable-library conclusions as in Corollary 7.166 hold.

In particular, if

$$\mu_{m,h} \le \frac{C_0 n_m}{2\log(eR_m)} \qquad (1 \le h \le J_m),$$

then one may take $t_{m,h} = C_0 n_m/(2\log(eR_m))$*, and the displayed Bernstein summand is bounded by*

$$\sum_{h=1}^{J_m} (|E^L_{m,h}| + |E^R_{m,h}|) \exp\left\{-\frac{3C_0 n_m}{16\log(eR_m)}\right\}.$$

Proof. For each row coordinate define

$$N^L_{m,h,a} = \sum_i \rho^2_{m,h,i} \mathbf{1}_{\{a \in R_{m,h,i}\}},$$

and define $N^R_{m,h,b}$ analogously on the column side. Scalar Bernstein and the union bound over the displayed windows show that the probability that some active coordinate has load above $\mu_{m,h} + t_{m,h}$ is bounded by the displayed Bernstein summand. Off this event, with $Q^L_{m,h} = P_{E^L_{m,h}}$, $Q^R_{m,h} = P_{E^R_{m,h}}$, $P^L_{m,h,i} = P_{R_{m,h,i}}$, and $P^R_{m,h,i} = P_{S_{m,h,i}}$, the support-envelope rank and left/right overlap hypotheses of Corollary 7.166 hold. The cap-tail summability is exactly the remaining summand, so that corollary gives the common signing, sampled certificates, and countable-library conclusion.

For the final display, $\sigma^2_{m,h} \le \mu_{m,h} \le t_{m,h}$ and $b_{m,h} \le 1$, so

$$\frac{t^2_{m,h}}{2(\sigma^2_{m,h} + b_{m,h} t_{m,h}/3)} \ge \frac{3t_{m,h}}{8} = \frac{3C_0 n_m}{16\log(eR_m)}.$$

□

Corollary 7.168 (Common signs for rectangular predictable coordinate-window Freedman menus)**.** *Fix* $C_0 \ge 1$*. For each* $m \ge 1$*, let* $J_m \ge 1$*,* $d_{m,1}, d_{m,2}, n_m \ge 1$*, and let* $R_m \ge 1$*. For every* $1 \le h \le J_m$*, let*

$$E^L_{m,h} \subset \{1, \dots, d_{m,1}\}, \qquad E^R_{m,h} \subset \{1, \dots, d_{m,2}\}$$

be deterministic coordinate windows satisfying

$$\sum_{h=1}^{J_m} \max\{1, |E^L_{m,h}| + |E^R_{m,h}|\} \le R_m.$$

For each m, h*, let*

$$\mathcal{F}_{m,h,0} \subset \mathcal{F}_{m,h,1} \subset \cdots \subset \mathcal{F}_{m,h,n_m}$$

be a filtration. For each h, i*, let*

$$R_{m,h,i} \subset E^L_{m,h}, \qquad S_{m,h,i} \subset E^R_{m,h}$$

be $\mathcal{F}_{m,h,i}$*-measurable random row and column supports. Assume that there are deterministic envelopes* $p^L_{m,h,i,a}, p^R_{m,h,i,b} \in [0,1]$ *such that, for every* $a \in E^L_{m,h}$

and $b \in E^R_{m,h}$,

$$\mathbb{P}\{a \in R_{m,h,i} \mid \mathcal{F}_{m,h,i-1}\} \le p^L_{m,h,i,a}, \qquad \mathbb{P}\{b \in S_{m,h,i} \mid \mathcal{F}_{m,h,i-1}\} \le p^R_{m,h,i,b}.$$

Conditional on the supports, let

$$X_{m,h,i} \in \mathrm{M}_{d_{m,1}\times d_{m,2}}(\mathbb{C})$$

be arbitrary random rectangular matrices satisfying

$$X_{m,h,i} = P_{R_{m,h,i}} X_{m,h,i} P_{S_{m,h,i}}.$$

Let deterministic caps $0 \le \rho_{m,h,i} \le 1$ *and cap-tail budgets* $\delta_{m,h,i} \ge 0$ *satisfy*

$$\mathbb{P}\{\|X_{m,h,i}\|_{\mathrm{op}} > \rho_{m,h,i}\} \le \delta_{m,h,i}.$$

Put

$$b_{m,h} := \max_i \rho^2_{m,h,i},$$

$$\mu_{m,h} := \max\left\{ \max_{a\in E^L_{m,h}} \sum_i \rho^2_{m,h,i} p^L_{m,h,i,a}, \max_{b\in E^R_{m,h}} \sum_i \rho^2_{m,h,i} p^R_{m,h,i,b} \right\},$$

and

$$\sigma^2_{m,h} := \max\left\{ \max_{a\in E^L_{m,h}} \sum_i \rho^4_{m,h,i} p^L_{m,h,i,a}, \max_{b\in E^R_{m,h}} \sum_i \rho^4_{m,h,i} p^R_{m,h,i,b} \right\},$$

where a maximum over an empty window is interpreted as 0. *Suppose that there are* $t_{m,h} > 0$ *such that*

$$\mu_{m,h} + t_{m,h} \le \frac{C_0 n_m}{\log(eR_m)} \qquad (1 \le h \le J_m),$$

and

$$\sum_m \left[\sum_{h=1}^{J_m} (|E^L_{m,h}| + |E^R_{m,h}|) \exp\left\{ -\frac{t^2_{m,h}}{2(\sigma^2_{m,h} + b_{m,h} t_{m,h}/3)} \right\} + \sum_{h=1}^{J_m} \sum_{i=1}^{n_m} \delta_{m,h,i} \right] < \infty,$$

with zero denominator interpreted as zero contribution. Then, almost surely, for all sufficiently large m, *there is one sign vector* $\varepsilon^{(m)} \in \{\pm 1\}^{n_m}$ *such that*

$$\max_{1\le h\le J_m} \left\| \sum_{i=1}^{n_m} \varepsilon^{(m)}_i X_{m,h,i} \right\|_{\mathrm{op}} \le C(C_0)\sqrt{n_m}.$$

The same sampled common-certificate and countable-library conclusions as in Corollary 7.166 hold.

In particular, if

$$\mu_{m,h} \le \frac{C_0 n_m}{2\log(eR_m)} \qquad (1 \le h \le J_m),$$

then one may take $t_{m,h} = C_0 n_m/(2\log(eR_m))$*, and the displayed Freedman summand is bounded by*

$$\sum_{h=1}^{J_m}(|E^L_{m,h}| + |E^R_{m,h}|)\exp\left\{-\frac{3C_0 n_m}{16\log(eR_m)}\right\}.$$

Proof. For each row coordinate set

$$N^L_{m,h,a} = \sum_i \rho^2_{m,h,i}\mathbf{1}_{\{a\in R_{m,h,i}\}},$$

and define $N^R_{m,h,b}$ analogously on the column side. For fixed m, h, a, the process

$$\sum_{i'\le i}\rho^2_{m,h,i'}\left(\mathbf{1}_{\{a\in R_{m,h,i'}\}} - \mathbb{E}(\mathbf{1}_{\{a\in R_{m,h,i'}\}} \mid \mathcal{F}_{m,h,i'-1})\right)$$

is a scalar martingale with increments bounded above by $b_{m,h}$ and predictable quadratic variation at most $\sum_i \rho^4_{m,h,i}p^L_{m,h,i,a}$. Scalar Freedman and the union bound over all row and column coordinates in the active windows show that the probability that some active coordinate load exceeds $\mu_{m,h} + t_{m,h}$ is bounded by the displayed Freedman summand. Off this event, with $Q^L_{m,h} = P_{E^L_{m,h}}$, $Q^R_{m,h} = P_{E^R_{m,h}}$, $P^L_{m,h,i} = P_{R_{m,h,i}}$, and $P^R_{m,h,i} = P_{S_{m,h,i}}$, the support-envelope rank and left/right overlap hypotheses of Corollary 7.166 hold. The cap-tail summability is exactly the remaining summand, so that corollary gives the common signing, sampled certificates, and countable-library conclusion. The half-budget estimate follows from $\sigma^2_{m,h} \le \mu_{m,h} \le t_{m,h}$ and $b_{m,h} \le 1$, giving exponent at least $3t_{m,h}/8$. □

Corollary 7.169 (Common-sign rectangular predictable coordinate-window schedules)**.** *In the setting of Corollary 7.168, suppose that*

$$\mu_{m,h} \le \frac{C_0 n_m}{2\log(eR_m)} \qquad (1 \le h \le J_m)$$

eventually, and that

$$\sum_m \sum_{h=1}^{J_m}\sum_{i=1}^{n_m}\delta_{m,h,i} < \infty.$$

Then the same common-sign, sampled-certificate, and countable-library conclusions hold whenever

$$\sum_m R_m \exp\left\{-\frac{3C_0 n_m}{16\log(eR_m)}\right\} < \infty.$$

In particular, after the displayed half-budget and cap-tail assumptions, the structural tail is summable in each of the following support-rank schedules.

(i) *Arbitrary growth. If, for some* $\gamma > 0$*,*

$$n_m \ge \frac{16}{3C_0}\log(eR_m)\big(\log(eR_m) + (1+\gamma)\log(e+m)\big)$$

eventually, then the almost-sure common-sign conclusion holds. If a shared batch of M_m *independent random sign vectors is used for*

the whole active menu, the sampled conclusion follows from $M_m \geq (1+\delta)\log_2(e+m)$ *eventually.*

(ii) *All support ranks. Suppose the rows are indexed by* $R = 2, 3, \ldots$ *and the* R*-th row has active coordinate-window rank parameter at most* R*. If*

$$n_R \geq a \log^2(eR)$$

eventually for some $a > 32/(3C_0)$*, then the common-sign conclusion holds along all support ranks. The sampled conclusion follows if* $M_R \geq (1+\delta)\log_2(eR)$ *eventually.*

(iii) *Sparse support ranks. If* $R_m \geq \exp(cm^\kappa)$ *eventually for some* $c, \kappa > 0$*, and*

$$n_m \geq a \log^2(eR_m)$$

eventually for some $a > 16/(3C_0)$*, then the common-sign conclusion holds. The sampled conclusion follows from* $M_m \geq (1+\delta)\log_2(e+m)$ *eventually.*

Proof. Under the half-budget hypothesis, the final estimate in Corollary 7.168 bounds the structural Freedman tail by

$$\sum_{h=1}^{J_m} (|E_{m,h}^L| + |E_{m,h}^R|) \exp\left\{-\frac{3C_0 n_m}{16\log(eR_m)}\right\} \leq R_m \exp\left\{-\frac{3C_0 n_m}{16\log(eR_m)}\right\}.$$

The cap-tail series is summable by assumption, so the preceding corollary and Borel–Cantelli give the common signing and the inherited sampled and countable-library assertions.

For arbitrary growth, the lower bound on n_m makes the displayed structural tail at most $e^{-1}(e+m)^{-(1+\gamma)}$. For all support ranks, the R-th tail is bounded by

$$R \exp\left\{-\frac{3C_0 a}{16}\log(eR)\right\},$$

which is summable when $3C_0 a/16 > 2$. For sparse support ranks the same estimate is $O(R_m^{1-3C_0 a/16})$, summable under the stretched-exponential growth of R_m as soon as $3C_0 a/16 > 1$. The sampled thresholds are exactly the summability of 2^{-M_m} or 2^{-M_R}. □

Corollary 7.170 (Common signs for rectangular subspace-envelope Bernstein menus)**.** *Fix* $C_0 \geq 1$*. For each* $m \geq 1$*, let* $J_m \geq 1$*,* $d_{m,1}, d_{m,2}, n_m \geq 1$*, and let* $R_m \geq 1$*. For every* $1 \leq h \leq J_m$*, let*

$$Q_{m,h}^L \in \mathrm{M}_{d_{m,1}}(\mathbb{C}), \qquad Q_{m,h}^R \in \mathrm{M}_{d_{m,2}}(\mathbb{C})$$

be deterministic orthogonal projections satisfying

$$\sum_{h=1}^{J_m} \max\{1, \operatorname{rank} Q_{m,h}^L + \operatorname{rank} Q_{m,h}^R\} \leq R_m.$$

For each h, i*, let*

$$P_{m,h,i}^L \leq Q_{m,h}^L, \qquad P_{m,h,i}^R \leq Q_{m,h}^R$$

be random orthogonal projections, with the pairs $(P^L_{m,h,i}, P^R_{m,h,i})$, $1 \le i \le n_m$, *independent for each fixed* (m,h). *Conditional on these projections, let*

$$X_{m,h,i} \in \mathrm{M}_{d_{m,1}\times d_{m,2}}(\mathbb{C})$$

be arbitrary random rectangular matrices satisfying

$$X_{m,h,i} = P^L_{m,h,i} X_{m,h,i} P^R_{m,h,i}.$$

Let deterministic caps $0 \le \rho_{m,h,i} \le 1$ *and cap-tail budgets* $\delta_{m,h,i} \ge 0$ *satisfy*

$$\mathbb{P}\{\|X_{m,h,i}\|_{\mathrm{op}} > \rho_{m,h,i}\} \le \delta_{m,h,i}.$$

Put

$$b_{m,h} := \max_i \rho^2_{m,h,i},$$

and

$$\mu_{m,h} := \max\left\{ \left\| \sum_i \rho^2_{m,h,i} \mathbb{E} P^L_{m,h,i} \right\|_{\mathrm{op}}, \left\| \sum_i \rho^2_{m,h,i} \mathbb{E} P^R_{m,h,i} \right\|_{\mathrm{op}} \right\}.$$

Suppose that there are $t_{m,h} > 0$ *such that*

$$\mu_{m,h} + t_{m,h} \le \frac{C_0 n_m}{\log(eR_m)} \qquad (1 \le h \le J_m),$$

and

$$\sum_m \left[\sum_{h=1}^{J_m} (\operatorname{rank} Q^L_{m,h} + \operatorname{rank} Q^R_{m,h}) \exp\left\{ -\frac{t^2_{m,h}}{2(\mu_{m,h} + b_{m,h} t_{m,h}/3)} \right\} + \sum_{h=1}^{J_m} \sum_{i=1}^{n_m} \delta_{m,h,i} \right] < \infty,$$

with zero denominator interpreted as zero contribution. Then, almost surely, for all sufficiently large m, *there is one sign vector* $\varepsilon^{(m)} \in \{\pm 1\}^{n_m}$ *such that*

$$\max_{1\le h\le J_m} \left\| \sum_{i=1}^{n_m} \varepsilon^{(m)}_i X_{m,h,i} \right\|_{\mathrm{op}} \le C(C_0)\sqrt{n_m}.$$

The same sampled common-certificate and countable-library conclusions as in Corollary 7.166 hold.

In particular, if

$$\mu_{m,h} \le \frac{C_0 n_m}{2\log(eR_m)} \qquad (1 \le h \le J_m),$$

then one may take $t_{m,h} = C_0 n_m/(2\log(eR_m))$, *and the displayed Bernstein summand is bounded by*

$$\sum_{h=1}^{J_m} (\operatorname{rank} Q^L_{m,h} + \operatorname{rank} Q^R_{m,h}) \exp\left\{ -\frac{3C_0 n_m}{16\log(eR_m)} \right\}.$$

For Haar-distributed rank-$r^L_{m,h,i}$ and rank-$r^R_{m,h,i}$ projections inside the ranges of $Q^L_{m,h}$ and $Q^R_{m,h}$, the half-budget condition is

$$\max\left\{\frac{\sum_i \rho^2_{m,h,i} r^L_{m,h,i}}{\operatorname{rank} Q^L_{m,h}}, \frac{\sum_i \rho^2_{m,h,i} r^R_{m,h,i}}{\operatorname{rank} Q^R_{m,h}}\right\} \le \frac{C_0 n_m}{2\log(eR_m)},$$

omitting a ratio whose denominator is 0.

Proof. Apply matrix Bernstein separately in the row and column envelope subspaces. On the row side, set

$$Y^L_{m,h,i} = \rho^2_{m,h,i}(P^L_{m,h,i} - \mathbb{E}P^L_{m,h,i})$$

as an operator on the range of $Q^L_{m,h}$. Then $\mathbb{E}Y^L_{m,h,i} = 0$, $\|Y^L_{m,h,i}\|_{\mathrm{op}} \le b_{m,h}$, and

$$\sum_i \mathbb{E}(Y^L_{m,h,i})^2 \preceq \sum_i \rho^2_{m,h,i}\mathbb{E}P^L_{m,h,i},$$

so the variance parameter is at most $\mu_{m,h}$. The same argument applies on the column side. Union over h and over the two envelope dimensions gives the displayed Bernstein failure series for the event

$$\max_h \max\left\{\left\|\sum_i \rho^2_{m,h,i}P^L_{m,h,i}\right\|_{\mathrm{op}}, \left\|\sum_i \rho^2_{m,h,i}P^R_{m,h,i}\right\|_{\mathrm{op}}\right\} \le \frac{C_0 n_m}{\log(eR_m)}.$$

Off this event, the support-envelope rank and overlap hypotheses of Corollary 7.166 hold with envelopes $Q^L_{m,h}, Q^R_{m,h}$. The cap-tail summability is the remaining summand, so that corollary gives the common signing, sampled certificates, and countable-library conclusion. The half-budget and Haar specializations follow from the same computation as in Corollary 7.176. □

Corollary 7.171 (Common signs for rectangular predictable subspace-envelope Freedman menus)**.** *Fix $C_0 \ge 1$. For each $m \ge 1$, let $J_m \ge 1$, $d_{m,1}, d_{m,2}, n_m \ge 1$, and let $R_m \ge 1$. For every $1 \le h \le J_m$, let*

$$Q^L_{m,h} \in \mathrm{M}_{d_{m,1}}(\mathbb{C}), \qquad Q^R_{m,h} \in \mathrm{M}_{d_{m,2}}(\mathbb{C})$$

be deterministic orthogonal projections satisfying

$$\sum_{h=1}^{J_m} \max\{1, \operatorname{rank} Q^L_{m,h} + \operatorname{rank} Q^R_{m,h}\} \le R_m.$$

For each m, h, let

$$\mathcal{F}_{m,h,0} \subset \mathcal{F}_{m,h,1} \subset \cdots \subset \mathcal{F}_{m,h,n_m}$$

be a filtration. For each h, i, let

$$P^L_{m,h,i} \le Q^L_{m,h}, \qquad P^R_{m,h,i} \le Q^R_{m,h}$$

be $\mathcal{F}_{m,h,i}$-measurable random orthogonal projections. Conditional on these projections, let

$$X_{m,h,i} \in \mathrm{M}_{d_{m,1}\times d_{m,2}}(\mathbb{C})$$

be arbitrary random rectangular matrices satisfying

$$X_{m,h,i} = P^L_{m,h,i} X_{m,h,i} P^R_{m,h,i}.$$

Let deterministic caps $0 \le \rho_{m,h,i} \le 1$, *cap-tail budgets* $\delta_{m,h,i} \ge 0$, *deterministic predictable-overlap budgets* $v_{m,h} \ge 0$, *and events* F^{pr}_m *be such that*

$$\mathbb{P}\{\|X_{m,h,i}\|_{\mathrm{op}} > \rho_{m,h,i}\} \le \delta_{m,h,i},$$

and, outside F^{pr}_m, *for every* $h \le J_m$,

$$\left\| \sum_i \rho^2_{m,h,i} \mathbb{E}(P^L_{m,h,i} \mid \mathcal{F}_{m,h,i-1}) \right\|_{\mathrm{op}} \le v_{m,h},$$

$$\left\| \sum_i \rho^2_{m,h,i} \mathbb{E}(P^R_{m,h,i} \mid \mathcal{F}_{m,h,i-1}) \right\|_{\mathrm{op}} \le v_{m,h}.$$

Put

$$b_{m,h} := \max_i \rho^2_{m,h,i}.$$

Suppose that there are $t_{m,h} > 0$ *such that*

$$v_{m,h} + t_{m,h} \le \frac{C_0 n_m}{\log(eR_m)} \qquad (1 \le h \le J_m),$$

and

$$\sum_m \left[\mathbb{P}(F^{\mathrm{pr}}_m) + \sum_{h=1}^{J_m} 2(\operatorname{rank} Q^L_{m,h} + \operatorname{rank} Q^R_{m,h}) \exp\left\{ -\frac{t^2_{m,h}}{2(v_{m,h} + b_{m,h} t_{m,h}/3)} \right\} + \sum_{h=1}^{J_m} \sum_{i=1}^{n_m} \delta_{m,h,i} \right] < \infty,$$

with zero denominator interpreted as zero contribution. Then, almost surely, for all sufficiently large m, *there is one sign vector* $\varepsilon^{(m)} \in \{\pm 1\}^{n_m}$ *such that*

$$\max_{1 \le h \le J_m} \left\| \sum_{i=1}^{n_m} \varepsilon^{(m)}_i X_{m,h,i} \right\|_{\mathrm{op}} \le C(C_0) \sqrt{n_m}.$$

The same sampled common-certificate and countable-library conclusions as in Corollary 7.166 hold.

In particular, if

$$v_{m,h} \le \frac{C_0 n_m}{2 \log(eR_m)} \qquad (1 \le h \le J_m),$$

then one may take $t_{m,h} = C_0 n_m / (2 \log(eR_m))$, *and the Freedman summand is bounded by*

$$2 \sum_{h=1}^{J_m} (\operatorname{rank} Q^L_{m,h} + \operatorname{rank} Q^R_{m,h}) \exp\left\{ -\frac{3 C_0 n_m}{16 \log(eR_m)} \right\}.$$

If, conditionally on $\mathcal{F}_{m,h,i-1}$, the row and column projections are Haar-distributed inside the ranges of $Q^L_{m,h}$ and $Q^R_{m,h}$ with predictable ranks $r^L_{m,h,i}$ and $r^R_{m,h,i}$, the half-budget condition is

$$\max\left\{\frac{\sum_i \rho^2_{m,h,i} r^L_{m,h,i}}{\operatorname{rank} Q^L_{m,h}}, \frac{\sum_i \rho^2_{m,h,i} r^R_{m,h,i}}{\operatorname{rank} Q^R_{m,h}}\right\} \le \frac{C_0 n_m}{2\log(eR_m)},$$

omitting a ratio whose denominator is 0.

Proof. Apply the two-sided matrix Freedman inequality separately in the row and column envelope subspaces. On the row side, set

$$Z^L_{m,h,i} = \rho^2_{m,h,i}\left(P^L_{m,h,i} - \mathbb{E}(P^L_{m,h,i} \mid \mathcal{F}_{m,h,i-1})\right)$$

as an operator on the range of $Q^L_{m,h}$. Then $(Z^L_{m,h,i}, \mathcal{F}_{m,h,i})$ is a self-adjoint martingale-difference sequence, $\|Z^L_{m,h,i}\|_{\mathrm{op}} \le b_{m,h}$, and

$$\sum_i \mathbb{E}((Z^L_{m,h,i})^2 \mid \mathcal{F}_{m,h,i-1}) \preceq \sum_i \rho^2_{m,h,i}\mathbb{E}(P^L_{m,h,i} \mid \mathcal{F}_{m,h,i-1}).$$

The same construction applies on the column side. Hence, off F^{pr}_m, the displayed Freedman series bounds the probability that some active row or column realized overlap exceeds $v_{m,h} + t_{m,h}$. Off that event and the cap-failure event, the support-envelope rank and overlap hypotheses of Corollary 7.166 hold with envelopes $Q^L_{m,h}, Q^R_{m,h}$. The summability assumptions and Borel–Cantelli therefore give the common signing, sampled certificates, and countable-library conclusion. The half-budget estimate uses $b_{m,h} \le 1$ and $v_{m,h} \le t_{m,h}$. Conditional Haar invariance gives

$$\mathbb{E}(P^L_{m,h,i} \mid \mathcal{F}_{m,h,i-1}) = \frac{r^L_{m,h,i}}{\operatorname{rank} Q^L_{m,h}} Q^L_{m,h}$$

and the analogous column identity, which gives the final specialization. □

Corollary 7.172 (Common-sign rectangular predictable subspace-envelope schedules)**.** *In the setting of Corollary 7.171, suppose that*

$$v_{m,h} \le \frac{C_0 n_m}{2\log(eR_m)} \qquad (1 \le h \le J_m)$$

eventually, and that

$$\sum_m \mathbb{P}(F^{\mathrm{pr}}_m) < \infty, \qquad \sum_m \sum_{h=1}^{J_m} \sum_{i=1}^{n_m} \delta_{m,h,i} < \infty.$$

Then the same common-sign, sampled-certificate, and countable-library conclusions hold whenever

$$\sum_m 2R_m \exp\left\{-\frac{3C_0 n_m}{16\log(eR_m)}\right\} < \infty.$$

In particular, after the displayed half-budget, predictable-exception, and cap-tail assumptions, the structural tail is summable in each of the following support-rank schedules.

(i) *Arbitrary growth. If, for some* $\gamma > 0$,

$$n_m \geq \frac{16}{3C_0} \log(eR_m)\big(\log(2eR_m) + (1+\gamma)\log(e+m)\big)$$

eventually, then the almost-sure common-sign conclusion holds. If a shared batch of M_m *independent random sign vectors is used for the whole active menu, the sampled conclusion follows from* $M_m \geq (1+\delta)\log_2(e+m)$ *eventually.*

(ii) *All support ranks. Suppose the rows are indexed by* $R = 2, 3, \ldots$ *and the* R*-th row has active subspace-envelope rank parameter at most* R*. If*

$$n_R \geq a \log^2(eR)$$

eventually for some $a > 32/(3C_0)$*, then the common-sign conclusion holds along all support ranks. The sampled conclusion follows if* $M_R \geq (1+\delta)\log_2(eR)$ *eventually.*

(iii) *Sparse support ranks. If* $R_m \geq \exp(cm^\kappa)$ *eventually for some* $c, \kappa > 0$*, and*

$$n_m \geq a \log^2(eR_m)$$

eventually for some $a > 16/(3C_0)$*, then the common-sign conclusion holds. The sampled conclusion follows from* $M_m \geq (1+\delta)\log_2(e+m)$ *eventually.*

Proof. Under the half-budget hypothesis, the final estimate in Corollary 7.171 bounds the structural Freedman tail by

$$2\sum_{h=1}^{J_m} (\operatorname{rank} Q_{m,h}^L + \operatorname{rank} Q_{m,h}^R) \exp\left\{-\frac{3C_0 n_m}{16\log(eR_m)}\right\} \leq 2R_m \exp\left\{-\frac{3C_0 n_m}{16\log(eR_m)}\right\}.$$

The predictable-exception and cap-tail series are summable by assumption, so the preceding corollary and Borel–Cantelli give the common signing and the inherited sampled and countable-library assertions.

For arbitrary growth, the lower bound on n_m makes the displayed structural tail at most $e^{-1}(e+m)^{-(1+\gamma)}$. For all support ranks, the R-th tail is bounded by

$$2R \exp\left\{-\frac{3C_0 a}{16}\log(eR)\right\},$$

which is summable when $3C_0 a/16 > 2$. For sparse support ranks the same estimate is $O(R_m^{1-3C_0 a/16})$, summable under the stretched-exponential growth of R_m as soon as $3C_0 a/16 > 1$. The sampled thresholds are exactly the summability of 2^{-M_m} or 2^{-M_R}. □

Corollary 7.173 (Tail-capped common signs for dependent rectangular overlap menus)**.** *In the setting of Corollary 7.164, replace the pointwise norm bounds by deterministic caps*

$$0 \le \rho_{m,h,i} \le 1 \qquad (1 \le h \le J_m,\ 1 \le i \le n_m).$$

Assume that there are events $F^{\mathrm{ov}}_{m,h}$, random orthogonal projections

$$P^{L}_{m,h,i} \in \mathrm{M}_{d_{m,1}}(\mathbb{C}), \qquad P^{R}_{m,h,i} \in \mathrm{M}_{d_{m,2}}(\mathbb{C}),$$

and cap-tail budgets $\delta_{m,h,i} \ge 0$ such that

$$\mathbb{P}\{\|X_{m,h,i}\|_{\mathrm{op}} > \rho_{m,h,i}\} \le \delta_{m,h,i},$$

and, outside $F^{\mathrm{ov}}_{m,h}$,

$$X_{m,h,i} = P^{L}_{m,h,i} X_{m,h,i} P^{R}_{m,h,i} \qquad (1 \le i \le n_m)$$

and

$$\max\left\{ \left\| \sum_{i=1}^{n_m} \rho^2_{m,h,i} P^{L}_{m,h,i} \right\|_{\mathrm{op}}, \left\| \sum_{i=1}^{n_m} \rho^2_{m,h,i} P^{R}_{m,h,i} \right\|_{\mathrm{op}} \right\} \le \frac{C_0 n_m}{\log(eD^{\mathrm{com}}_m)}.$$

If

$$\sum_{m} \sum_{h=1}^{J_m} \left[\mathbb{P}(F^{\mathrm{ov}}_{m,h}) + \sum_{i=1}^{n_m} \delta_{m,h,i} \right] < \infty,$$

then, almost surely, for all sufficiently large m, one sign vector $\varepsilon^{(m)}$ balances the whole active rectangular menu:

$$\max_{1 \le h \le J_m} \left\| \sum_{i=1}^{n_m} \varepsilon^{(m)}_i X_{m,h,i} \right\|_{\mathrm{op}} \le C(C_0)\sqrt{n_m}.$$

The sampled common-certificate and countable-library conclusions of Corollary 7.164 hold as well. The sub-Weibull and finite-moment cap choices in Corollary 7.163 apply verbatim, with the realized overlap budget checked at the common-sign scale $\log(eD^{\mathrm{com}}_m)$.

Proof. Let

$$\Omega^{\mathrm{cap}}_{m,h} := \{\|X_{m,h,i}\|_{\mathrm{op}} \le \rho_{m,h,i} \text{ for all } i \le n_m\}.$$

The summability hypothesis implies

$$\sum_{m} \sum_{h=1}^{J_m} \mathbb{P}\big((\Omega^{\mathrm{cap}}_{m,h})^c\big) < \infty.$$

Outside $F^{\mathrm{ov}}_{m,h} \cup (\Omega^{\mathrm{cap}}_{m,h})^c$, the hypotheses of Corollary 7.164 hold with deterministic amplitudes $\rho_{m,h,i}$. Borel–Cantelli therefore reduces the assertion to that corollary. The displayed cap choices are the same sub-Weibull tail estimate and Markov estimate used in Corollary 7.163. □

Corollary 7.174 (Rectangular coordinate-support Bernstein)**.** *Let $d_1, d_2 \geq 1$, put $D_{\mathrm{rec}} = d_1 + d_2$, and let $\rho_1, \dots, \rho_n \in [0,1]$ be deterministic. For each i, let*

$$R_i \subset \{1, \dots, d_1\}, \qquad S_i \subset \{1, \dots, d_2\}$$

be random row and column supports. Assume that, for every row coordinate a, the indicators $\mathbf{1}_{\{a \in R_i\}}$, $i = 1, \dots, n$, are independent and

$$\mathbb{P}\{a \in R_i\} \leq p_{ia}^L,$$

and that, for every column coordinate b, the indicators $\mathbf{1}_{\{b \in S_i\}}$, $i = 1, \dots, n$, are independent and

$$\mathbb{P}\{b \in S_i\} \leq p_{ib}^R.$$

Conditional on the supports, let $X_1, \dots, X_n$ be arbitrary random rectangular matrices satisfying

$$X_i = P_{R_i} X_i P_{S_i}, \qquad \|X_i\|_{\mathrm{op}} \leq \rho_i.$$

Put

$$b := \max_i \rho_i^2,$$

$$\mu := \max\left\{ \max_{1 \leq a \leq d_1} \sum_i \rho_i^2 p_{ia}^L, \max_{1 \leq b \leq d_2} \sum_i \rho_i^2 p_{ib}^R \right\},$$

and

$$\sigma^2 := \max\left\{ \max_{1 \leq a \leq d_1} \sum_i \rho_i^4 p_{ia}^L, \max_{1 \leq b \leq d_2} \sum_i \rho_i^4 p_{ib}^R \right\}.$$

Then, for every $t > 0$ and every $C_0 \geq 1$ satisfying

$$\mu + t \leq \frac{C_0 n}{\log(e D_{\mathrm{rec}})},$$

one has

$$\mathbb{P}\left\{ \min_{\varepsilon \in \{\pm 1\}^n} \left\| \sum_{i=1}^n \varepsilon_i X_i \right\|_{\mathrm{op}} > C(C_0)\sqrt{n} \right\} \leq D_{\mathrm{rec}} \exp\left\{ -\frac{t^2}{2(\sigma^2 + bt/3)} \right\},$$

with the convention that the exponential term is zero when $\sigma^2 + bt/3 = 0$.

In particular, if $p_{ia}^L \leq p_L$, $p_{ib}^R \leq p_R$, and

$$\max\{p_L, p_R\} \sum_i \rho_i^2 \leq \frac{C_0 n}{2 \log(e D_{\mathrm{rec}})},$$

then

$$\mathbb{P}\left\{ \min_{\varepsilon \in \{\pm 1\}^n} \left\| \sum_i \varepsilon_i X_i \right\|_{\mathrm{op}} > C(C_0)\sqrt{n} \right\} \leq D_{\mathrm{rec}} \exp\left\{ -\frac{3 C_0 n}{16 \log(e D_{\mathrm{rec}})} \right\}.$$

Uniformly chosen fixed-size row and column supports of sizes r_i and s_i are covered by the half-budget condition

$$\max\left\{ \frac{1}{d_1} \sum_i \rho_i^2 r_i, \frac{1}{d_2} \sum_i \rho_i^2 s_i \right\} \leq \frac{C_0 n}{2 \log(e D_{\mathrm{rec}})}.$$

Proof. For each row coordinate define

$$N_a^L := \sum_i \rho_i^2 \mathbf{1}_{\{a \in R_i\}},$$

and for each column coordinate define

$$N_b^R := \sum_i \rho_i^2 \mathbf{1}_{\{b \in S_i\}}.$$

By the one-coordinate independence hypotheses, scalar Bernstein gives for every row a and every column b

$$\mathbb{P}\{N_a^L > \mu + t\}, \mathbb{P}\{N_b^R > \mu + t\} \le \exp\left\{-\frac{t^2}{2(\sigma^2 + bt/3)}\right\}.$$

Union over the $d_1 + d_2$ coordinates. On the complementary event,

$$\left\|\sum_i \rho_i^2 P_{R_i}\right\|_{\mathrm{op}} \le \mu + t, \qquad \left\|\sum_i \rho_i^2 P_{S_i}\right\|_{\mathrm{op}} \le \mu + t,$$

so Corollary 7.161 gives the signing bound. For the homogeneous display, take $t = C_0 n/(2\log(eD_{\mathrm{rec}}))$. Then $\mu \le t$, $\sigma^2 \le \mu \le t$, and $b \le 1$, so the exponent is at least $3C_0 n/(16\log(eD_{\mathrm{rec}}))$. Uniform fixed-size supports have one-coordinate probabilities r_i/d_1 and s_i/d_2. □

Corollary 7.175 (Tail-capped rectangular coordinate-support Bernstein)**.** *Let $d_1, d_2 \ge 1$, put $D_{\mathrm{rec}} = d_1 + d_2$, and let $\rho_1, \dots, \rho_n \in [0,1]$ be deterministic caps. For each i, let*

$$R_i \subset \{1, \dots, d_1\}, \qquad S_i \subset \{1, \dots, d_2\}$$

be random row and column supports satisfying the coordinatewise independence and inclusion-probability hypotheses of Corollary 7.174. Conditional on the supports, let $X_1, \dots, X_n$ be arbitrary random rectangular matrices satisfying

$$X_i = P_{R_i} X_i P_{S_i}.$$

Assume that cap-tail budgets $\delta_i \ge 0$ satisfy

$$\mathbb{P}\{\|X_i\|_{\mathrm{op}} > \rho_i\} \le \delta_i.$$

Define b, μ_{cap}, and σ_{cap}^2 from the displays in Corollary 7.174, with the caps ρ_i. Then, for every $t > 0$ and every $C_0 \ge 1$ satisfying

$$\mu_{\mathrm{cap}} + t \le \frac{C_0 n}{\log(eD_{\mathrm{rec}})},$$

one has

$$\mathbb{P}\left\{\min_{\varepsilon \in \{\pm 1\}^n} \left\|\sum_i \varepsilon_i X_i\right\|_{\mathrm{op}} > C(C_0)\sqrt{n}\right\} \le \sum_{i=1}^n \delta_i + D_{\mathrm{rec}} \exp\left\{-\frac{t^2}{2(\sigma_{\mathrm{cap}}^2 + bt/3)}\right\},$$

with the same zero-denominator convention as before.

In particular, if $p_{ia}^L \le p_L$, $p_{ib}^R \le p_R$, *and*

$$\max\{p_L, p_R\} \sum_i \rho_i^2 \le \frac{C_0 n}{2\log(eD_{\mathrm{rec}})},$$

then the Bernstein term is at most

$$D_{\mathrm{rec}} \exp\left\{-\frac{3C_0 n}{16\log(eD_{\mathrm{rec}})}\right\}.$$

Uniformly chosen fixed-size supports have the same row/column half-budget condition as in Corollary 7.174, now with the deterministic caps ρ_i. *Thus unbounded sparse rectangular support masks are reduced to deterministic cap tails plus the capped row/column occupancy budget.*

Proof. Let

$$\Omega_{\mathrm{cap}} := \{\|X_i\|_{\mathrm{op}} \le \rho_i \text{ for all } i \le n\}.$$

On Ω_{cap}, the hypotheses of Corollary 7.174 hold with the caps ρ_i. Add the probability of Ω_{cap}^c, which is at most $\sum_i \delta_i$, and use the bounded coordinate-support Bernstein bound. The homogeneous and fixed-size support specializations are the corresponding specializations of that corollary with $t = C_0 n/(2\log(eD_{\mathrm{rec}}))$. □

Corollary 7.176 (Rectangular random-subspace Bernstein)**.** *Let* $d_1, d_2 \ge 1$, *put* $D_{\mathrm{rec}} = d_1 + d_2$, *and let* $\rho_1, \dots, \rho_n \in [0,1]$ *be deterministic. For each* i, *let*

$$P_i^L \in \mathrm{M}_{d_1}(\mathbb{C}), \qquad P_i^R \in \mathrm{M}_{d_2}(\mathbb{C})$$

be random orthogonal projections, with the pairs (P_i^L, P_i^R) *independent over* i. *Conditional on these projections, let*

$$X_1, \dots, X_n \in \mathrm{M}_{d_1 \times d_2}(\mathbb{C})$$

be arbitrary random rectangular matrices satisfying

$$X_i = P_i^L X_i P_i^R, \qquad \|X_i\|_{\mathrm{op}} \le \rho_i.$$

Put

$$b := \max_i \rho_i^2,$$

and

$$\mu := \max\left\{\left\|\sum_i \rho_i^2 \mathbb{E} P_i^L\right\|_{\mathrm{op}}, \left\|\sum_i \rho_i^2 \mathbb{E} P_i^R\right\|_{\mathrm{op}}\right\}.$$

Then, for every $t > 0$ *and every* $C_0 \ge 1$ *satisfying*

$$\mu + t \le \frac{C_0 n}{\log(eD_{\mathrm{rec}})},$$

one has

$$\mathbb{P}\left\{\min_{\varepsilon \in \{\pm 1\}^n} \left\|\sum_i \varepsilon_i X_i\right\|_{\mathrm{op}} > C(C_0)\sqrt{n}\right\} \le D_{\mathrm{rec}} \exp\left\{-\frac{t^2}{2(\mu + bt/3)}\right\},$$

with zero contribution when the denominator is zero.

In particular, if P_i^L and P_i^R are Haar-distributed projections of ranks r_i^L and r_i^R, and

$$\max\left\{\frac{1}{d_1}\sum_i \rho_i^2 r_i^L, \frac{1}{d_2}\sum_i \rho_i^2 r_i^R\right\} \le \frac{C_0 n}{2\log(eD_{\mathrm{rec}})},$$

then

$$\mathbb{P}\left\{\min_{\varepsilon\in\{\pm1\}^n}\left\|\sum_i \varepsilon_i X_i\right\|_{\mathrm{op}} > C(C_0)\sqrt{n}\right\} \le D_{\mathrm{rec}}\exp\left\{-\frac{3C_0 n}{16\log(eD_{\mathrm{rec}})}\right\}.$$

Thus rectangular random low-rank, row-column subspace-localized, and rectangular GOE/GUE/Wigner-block models are Spencer-scale whenever the weighted average row and column ranks are $O(d_1/\log(eD_{\mathrm{rec}}))$ and $O(d_2/\log(eD_{\mathrm{rec}}))$, respectively.

Proof. Apply matrix Bernstein separately on the row and column sides. For the row side set

$$Y_i^L := \rho_i^2(P_i^L - \mathbb{E}P_i^L).$$

Then $\mathbb{E}Y_i^L = 0$, $\|Y_i^L\|_{\mathrm{op}} \le b$, and

$$\sum_i \mathbb{E}(Y_i^L)^2 = \sum_i \rho_i^4\left(\mathbb{E}P_i^L - (\mathbb{E}P_i^L)^2\right) \preceq \sum_i \rho_i^2 \mathbb{E}P_i^L,$$

so the row variance parameter is at most μ. The same argument applies to the column projections. A union bound over the row and column Bernstein tails therefore gives

$$\mathbb{P}\left\{\max\left[\left\|\sum_i \rho_i^2 P_i^L\right\|_{\mathrm{op}}, \left\|\sum_i \rho_i^2 P_i^R\right\|_{\mathrm{op}}\right] > \mu + t\right\} \le D_{\mathrm{rec}}\exp\left\{-\frac{t^2}{2(\mu + bt/3)}\right\}.$$

On the complementary event, Corollary 7.161 applies. For Haar ranks,

$$\mathbb{E}P_i^L = \frac{r_i^L}{d_1} I_{d_1}, \qquad \mathbb{E}P_i^R = \frac{r_i^R}{d_2} I_{d_2}.$$

Taking $t = C_0 n/(2\log(eD_{\mathrm{rec}}))$ and using $\mu \le t$, $b \le 1$, gives the final displayed tail. □

Corollary 7.177 (Tail-capped rectangular random subspaces)**.** *Let $d_1, d_2 \ge 1$, put $D_{\mathrm{rec}} = d_1 + d_2$, and let $\rho_1, \dots, \rho_n \in [0,1]$ be deterministic caps. For each i, let*

$$P_i^L \in \mathrm{M}_{d_1}(\mathbb{C}), \qquad P_i^R \in \mathrm{M}_{d_2}(\mathbb{C})$$

be random orthogonal projections, with the pairs (P_i^L, P_i^R) independent over i. Conditional on these projections, let

$$X_1, \dots, X_n \in \mathrm{M}_{d_1\times d_2}(\mathbb{C})$$

be arbitrary random rectangular matrices satisfying

$$X_i = P_i^L X_i P_i^R.$$

Put

$$b := \max_i \rho_i^2,$$

and

$$\mu_{\mathrm{cap}} := \max\left\{ \left\| \sum_i \rho_i^2 \mathbb{E} P_i^L \right\|_{\mathrm{op}}, \left\| \sum_i \rho_i^2 \mathbb{E} P_i^R \right\|_{\mathrm{op}} \right\}.$$

Then, for every $t > 0$ and every $C_0 \geq 1$ satisfying

$$\mu_{\mathrm{cap}} + t \leq \frac{C_0 n}{\log(eD_{\mathrm{rec}})},$$

one has

$$\mathbb{P}\left\{ \min_{\varepsilon \in \{\pm 1\}^n} \left\| \sum_i \varepsilon_i X_i \right\|_{\mathrm{op}} > C(C_0)\sqrt{n} \right\} \leq \mathbb{P}\{\exists i : \ \|X_i\|_{\mathrm{op}} > \rho_i\} + D_{\mathrm{rec}} \exp\left\{ -\frac{t^2}{2(\mu_{\mathrm{cap}} + bt/3)} \right\}.$$

In particular, the first term may be bounded by $\sum_i \mathbb{P}\{\|X_i\|_{\mathrm{op}} > \rho_i\}$, with no independence assumption on the matrix amplitudes. If P_i^L and P_i^R are Haar-distributed projections of ranks r_i^L and r_i^R, and

$$\max\left\{ \frac{1}{d_1} \sum_i \rho_i^2 r_i^L, \frac{1}{d_2} \sum_i \rho_i^2 r_i^R \right\} \leq \frac{C_0 n}{2\log(eD_{\mathrm{rec}})},$$

then

$$\mathbb{P}\left\{ \min_{\varepsilon \in \{\pm 1\}^n} \left\| \sum_i \varepsilon_i X_i \right\|_{\mathrm{op}} > C(C_0)\sqrt{n} \right\} \leq \sum_i \mathbb{P}\{\|X_i\|_{\mathrm{op}} > \rho_i\} + D_{\mathrm{rec}} \exp\left\{ -\frac{3C_0 n}{16\log(eD_{\mathrm{rec}})} \right\}.$$

Thus rectangular random low-rank, random row/column subspace-localized, and random block-factor models remain Spencer-scale with unbounded internal coefficients whenever the operator-norm cap tails and capped average row/column rank profiles are controlled.

Proof. On the event

$$\Omega_{\mathrm{cap}} := \{\|X_i\|_{\mathrm{op}} \leq \rho_i \text{ for all } i\},$$

Corollary 7.176 applies with the same projections and caps. Add $\mathbb{P}(\Omega_{\mathrm{cap}}^c)$, then use the union bound for the displayed tail-sum variant. The Haar-rank specialization is the corresponding specialization in Corollary 7.176. □

Corollary 7.178 (Random rectangular dyads from frame laws)**.** *Let $d_1, d_2 \geq 1$, put $D_{\mathrm{rec}} = d_1 + d_2$, and let $\rho_1, \dots, \rho_n \in [0,1]$ be deterministic. For each i, let $(u_i, v_i) \in \mathbb{C}^{d_1} \times \mathbb{C}^{d_2}$ be a random pair of unit vectors, with the pairs independent over i. Conditional on these vectors, let a_i be an arbitrary scalar random variable with $|a_i| \leq \rho_i$, and set*

$$X_i := a_i u_i v_i^* \in \mathrm{M}_{d_1 \times d_2}(\mathbb{C}).$$

Put

$$b := \max_i \rho_i^2, \qquad \mu_{\mathrm{dyad}} := \max\left\{ \left\| \sum_i \rho_i^2 \mathbb{E} u_i u_i^* \right\|_{\mathrm{op}}, \left\| \sum_i \rho_i^2 \mathbb{E} v_i v_i^* \right\|_{\mathrm{op}} \right\}.$$

Then, for every $t > 0$ and $C_0 \geq 1$ satisfying

$$\mu_{\mathrm{dyad}} + t \leq \frac{C_0 n}{\log(eD_{\mathrm{rec}})},$$

one has

$$\mathbb{P}\left\{ \min_{\varepsilon \in \{\pm 1\}^n} \left\| \sum_i \varepsilon_i X_i \right\|_{\mathrm{op}} > C(C_0)\sqrt{n} \right\} \leq D_{\mathrm{rec}} \exp\left\{ -\frac{t^2}{2(\mu_{\mathrm{dyad}} + bt/3)} \right\}.$$

In the Haar-isotropic case,

$$\mathbb{E} u_i u_i^* = \frac{1}{d_1} I_{d_1}, \qquad \mathbb{E} v_i v_i^* = \frac{1}{d_2} I_{d_2},$$

so the half-budget condition becomes

$$\max\left\{ \frac{1}{d_1} \sum_i \rho_i^2, \frac{1}{d_2} \sum_i \rho_i^2 \right\} \leq \frac{C_0 n}{2\log(eD_{\mathrm{rec}})},$$

and under this condition the failure probability is at most

$$D_{\mathrm{rec}} \exp\left\{ -\frac{3C_0 n}{16 \log(eD_{\mathrm{rec}})} \right\}.$$

Thus random rank-one measurement matrices, randomized feature dyads, and nonuniformly sampled rectangular atom dictionaries are Spencer-scale whenever their two weighted expected frame operators are $O(n/\log(eD_{\mathrm{rec}}))$.

Proof. Let

$$P_i^L := u_i u_i^*, \qquad P_i^R := v_i v_i^*.$$

These are rank-one orthogonal projections, independent as pairs over i, and $X_i = P_i^L X_i P_i^R$ with $\|X_i\|_{\mathrm{op}} = |a_i| \leq \rho_i$. The assertion is therefore Corollary 7.176. The Haar-isotropic displays are the rank-one Haar specialization of that corollary. □

Corollary 7.179 (Tail-capped rectangular dyads)**.** *Let $d_1, d_2 \geq 1$, put $D_{\mathrm{rec}} = d_1 + d_2$, and let $\rho_1, \dots, \rho_n \in [0,1]$ be deterministic caps. For each i, let $(u_i, v_i) \in \mathbb{C}^{d_1} \times \mathbb{C}^{d_2}$ be a random pair of unit vectors, with the pairs independent over i. Conditional on these vectors, let a_i be an arbitrary scalar random variable, and put*

$$X_i := a_i u_i v_i^*.$$

Set

$$b := \max_i \rho_i^2, \qquad \mu_{\mathrm{cap}} := \max\left\{ \left\| \sum_i \rho_i^2 \mathbb{E} u_i u_i^* \right\|_{\mathrm{op}}, \left\| \sum_i \rho_i^2 \mathbb{E} v_i v_i^* \right\|_{\mathrm{op}} \right\}.$$

Then, for every $t > 0$ and $C_0 \geq 1$ satisfying

$$\mu_{\mathrm{cap}} + t \leq \frac{C_0 n}{\log(eD_{\mathrm{rec}})},$$

one has

$$\mathbb{P}\left\{ \min_{\varepsilon \in \{\pm 1\}^n} \left\| \sum_i \varepsilon_i X_i \right\|_{\mathrm{op}} > C(C_0)\sqrt{n} \right\} \leq \mathbb{P}\{\exists i : \ |a_i| > \rho_i\} + D_{\mathrm{rec}} \exp\left\{ -\frac{t^2}{2(\mu_{\mathrm{cap}} + bt/3)} \right\}.$$

In particular, the first term may be bounded by $\sum_i \mathbb{P}\{|a_i| > \rho_i\}$, with no independence assumption on the amplitudes. If the dyad directions are Haar-isotropic and

$$\max\left\{ \frac{1}{d_1} \sum_i \rho_i^2, \ \frac{1}{d_2} \sum_i \rho_i^2 \right\} \leq \frac{C_0 n}{2\log(eD_{\mathrm{rec}})},$$

then

$$\mathbb{P}\left\{ \min_{\varepsilon \in \{\pm 1\}^n} \left\| \sum_i \varepsilon_i X_i \right\|_{\mathrm{op}} > C(C_0)\sqrt{n} \right\} \leq \sum_i \mathbb{P}\{|a_i| > \rho_i\} + D_{\mathrm{rec}} \exp\left\{ -\frac{3C_0 n}{16\log(eD_{\mathrm{rec}})} \right\}.$$

Thus Gaussian, sub-Gaussian, sub-exponential, or heavy-tailed scalar weights enter rectangular dyad and randomized-feature models through any deterministic cap profile whose tail cost is acceptable and whose capped expected frame operators have the Matrix Spencer scale.

Proof. On the event $\{|a_i| \leq \rho_i \text{ for all } i\}$, apply Corollary 7.178 with the same dyad directions and caps. Add the probability of the complementary event. The union bound gives the displayed summable-tail variant, and the Haar-isotropic case is the homogeneous specialization from Corollary 7.178. □

Corollary 7.180 (Active menus of rectangular coordinate-support streams)**.** *Fix $C_0 \geq 1$. For each $m \geq 1$, let $J_m \geq 1$, $d_{m,1}, d_{m,2} \geq 1$, $D_m^{\mathrm{rec}} = d_{m,1} + d_{m,2}$, and $n_m \geq 1$. For every $1 \leq h \leq J_m$, let*

$$X_{m,h,1}, \dots, X_{m,h,n_m} \in \mathrm{M}_{d_{m,1} \times d_{m,2}}(\mathbb{C})$$

be a rectangular coordinate-support model as in Corollary 7.174, with parameters $\rho_{m,h,i}$, $b_{m,h}$, $\mu_{m,h}$, and $\sigma_{m,h}^2$. Suppose there are $t_{m,h} > 0$ such that

$$\mu_{m,h} + t_{m,h} \leq \frac{C_0 n_m}{\log(eD_m^{\mathrm{rec}})} \qquad (1 \leq h \leq J_m),$$

and

$$\sum_m \sum_{h=1}^{J_m} D_m^{\mathrm{rec}} \exp\left\{ -\frac{t_{m,h}^2}{2(\sigma_{m,h}^2 + b_{m,h} t_{m,h}/3)} \right\} < \infty,$$

again with zero denominator interpreted as zero contribution. Then, almost surely, for all sufficiently large m, every active rectangular coordinate-support stream has a Spencer-scale signing:

$$\min_{\varepsilon\in\{\pm1\}^{n_m}} \left\| \sum_{i=1}^{n_m} \varepsilon_i X_{m,h,i} \right\|_{\mathrm{op}} \le C(C_0)\sqrt{n_m} \qquad (1 \le h \le J_m).$$

Sampled certificates are eventual almost surely under the same additional summability conditions as in Corollary 7.158. If $J_m \to \infty$, active prefixes exhaust countable libraries of rectangular bipartite sparse-support models.

Proof. Apply Corollary 7.174 to each active stream and sum the displayed failure probabilities. The first Borel–Cantelli lemma gives the eventual simultaneous signing. The sampled-search and countable-library statements are the same additions as in Corollary 7.158. □

Corollary 7.181 (Active menus of tail-capped rectangular coordinate-support streams)**.** *Fix $C_0 \ge 1$. For each $m \ge 1$, let $J_m \ge 1$, $d_{m,1}, d_{m,2} \ge 1$, $D_m^{\mathrm{rec}} = d_{m,1} + d_{m,2}$, and $n_m \ge 1$. For every $1 \le h \le J_m$, let*

$$X_{m,h,1}, \dots, X_{m,h,n_m} \in \mathrm{M}_{d_{m,1}\times d_{m,2}}(\mathbb{C})$$

be a tail-capped rectangular coordinate-support model as in Corollary 7.175, with deterministic caps $\rho_{m,h,i}$, parameters $b_{m,h}$, $\mu_{m,h}^{\mathrm{cap}}$, $\sigma_{m,h,\mathrm{cap}}^2$, and tail budgets $\delta_{m,h,i} \ge 0$. Suppose there are $t_{m,h} > 0$ such that

$$\mu_{m,h}^{\mathrm{cap}} + t_{m,h} \le \frac{C_0 n_m}{\log(eD_m^{\mathrm{rec}})} \qquad (1 \le h \le J_m),$$

and

$$\sum_m \sum_{h=1}^{J_m} \left[\sum_{i=1}^{n_m} \delta_{m,h,i} + D_m^{\mathrm{rec}} \exp\left\{ -\frac{t_{m,h}^2}{2(\sigma_{m,h,\mathrm{cap}}^2 + b_{m,h}t_{m,h}/3)} \right\} \right] < \infty,$$

again with zero denominator interpreted as zero contribution. Then, almost surely, for all sufficiently large m, every active tail-capped rectangular coordinate-support stream has a Spencer-scale signing:

$$\min_{\varepsilon\in\{\pm1\}^{n_m}} \left\| \sum_{i=1}^{n_m} \varepsilon_i X_{m,h,i} \right\|_{\mathrm{op}} \le C(C_0)\sqrt{n_m} \qquad (1 \le h \le J_m).$$

Sampled certificates and countable-library exhaustions follow under the same additional summability hypotheses as in Corollary 7.180.

In the half-budget case

$$\max\left\{ \max_a \sum_i \rho_{m,h,i}^2 p_{m,h,i,a}^L, \max_b \sum_i \rho_{m,h,i}^2 p_{m,h,i,b}^R \right\} \le \frac{C_0 n_m}{2\log(eD_m^{\mathrm{rec}})},$$

the matrix part of the active-menu failure is bounded by

$$J_m D_m^{\mathrm{rec}} \exp\left\{ -\frac{3C_0 n_m}{16\log(eD_m^{\mathrm{rec}})} \right\}.$$

Consequently the arbitrary-growth, all-integer, sparse, and sampled-certificate schedules of Corollary 7.184 apply verbatim after adding the cap-tail summability

$$\sum_{m}\sum_{h=1}^{J_m}\sum_{i=1}^{n_m}\delta_{m,h,i}<\infty.$$

Proof. Apply Corollary 7.175 to each active stream and sum its cap-tail and Bernstein failure probabilities. The first Borel–Cantelli lemma gives eventual simultaneous signings. Sampled signs add the same conditional search failures as in the bounded coordinate-support case, because on the cap event the proof reduces to Corollary 7.161. Under the half-budget hypothesis, take $t_{m,h}=C_0n_m/(2\log(eD_m^{\mathrm{rec}}))$; then $\sigma^2_{m,h,\mathrm{cap}}\le\mu^{\mathrm{cap}}_{m,h}\le t_{m,h}$ and $b_{m,h}\le 1$, giving the displayed Bernstein exponent and the inherited growth schedules. □

Corollary 7.182 (Sub-Weibull and moment caps for coordinate-support menus)**.** *Fix $C_0\ge 1$. Work in the active-menu coordinate-support setting of Corollary 7.181, and write $p^L_{m,h,i,a}$ and $p^R_{m,h,i,b}$ for the row and column inclusion-probability envelopes. For each active triple (m,h,i), choose a level $\ell_{m,h,i}>0$ and a scale $s_{m,h,i}>0$. Assume that the deterministic cap $\rho_{m,h,i}\le 1$ is chosen by one of the following two rules.*

(a) Sub-Weibull operator tails: for some fixed $\alpha>0$ and $A_{\mathrm{tail}}\ge 1$,

$$\mathbb{P}\{\|X_{m,h,i}\|_{\mathrm{op}}>s_{m,h,i}u\}\le A_{\mathrm{tail}}\exp\{-u^\alpha\}\qquad(u\ge 0),$$

and

$$\rho_{m,h,i}:=s_{m,h,i}\ell_{m,h,i}^{1/\alpha}\le 1.$$

(b) Moment operator tails: for some fixed $p>0$,

$$\mathbb{E}\|X_{m,h,i}\|_{\mathrm{op}}^p\le s_{m,h,i}^p,$$

and

$$\rho_{m,h,i}:=s_{m,h,i}\exp\{\ell_{m,h,i}/p\}\le 1.$$

In case (a) put $A_*=A_{\mathrm{tail}}$, and in case (b) put $A_*=1$. Let $b_{m,h}:=\max_i\rho^2_{m,h,i}$, and define the capped coordinate budgets

$$\mu^{\mathrm{csel}}_{m,h}:=\max\left\{\max_a\sum_i\rho^2_{m,h,i}p^L_{m,h,i,a},\max_b\sum_i\rho^2_{m,h,i}p^R_{m,h,i,b}\right\},$$

and

$$(\sigma^{\mathrm{csel}}_{m,h})^2:=\max\left\{\max_a\sum_i\rho^4_{m,h,i}p^L_{m,h,i,a},\max_b\sum_i\rho^4_{m,h,i}p^R_{m,h,i,b}\right\}.$$

If there are $t_{m,h}>0$ such that

$$\mu^{\mathrm{csel}}_{m,h}+t_{m,h}\le\frac{C_0n_m}{\log(eD_m^{\mathrm{rec}})}\qquad(1\le h\le J_m),$$

and

$$\sum_m \sum_{h=1}^{J_m} \left[A_* \sum_{i=1}^{n_m} \exp\{-\ell_{m,h,i}\} + D_m^{\mathrm{rec}} \exp\left\{ -\frac{t_{m,h}^2}{2((\sigma_{m,h}^{\mathrm{csel}})^2 + b_{m,h} t_{m,h}/3)} \right\} \right] < \infty,$$

with zero denominator interpreted as zero contribution, then, almost surely, for all sufficiently large m, every active coordinate-support stream has a Spencer-scale signing:

$$\min_{\varepsilon \in \{\pm 1\}^{n_m}} \left\| \sum_{i=1}^{n_m} \varepsilon_i X_{m,h,i} \right\|_{\mathrm{op}} \le C(C_0)\sqrt{n_m} \qquad (1 \le h \le J_m).$$

Sampled certificates and countable-library exhaustions follow under the same additional summability hypotheses as in Corollary 7.181.

For uniformly chosen fixed-size row and column supports, the two displayed budgets are obtained by substituting

$$p_{m,h,i,a}^L = r_{m,h,i}^L / d_{m,1}, \qquad p_{m,h,i,b}^R = r_{m,h,i}^R / d_{m,2}.$$

Thus sub-Weibull sparse rectangular blocks pay $\ell^{2/\alpha}$ in the coordinate-load budget and $\ell^{4/\alpha}$ in the Bernstein variance proxy, while finite p-moment blocks pay $\exp\{2\ell/p\}$ and $\exp\{4\ell/p\}$, respectively.

Consequently, if $J_m n_m \le (eD_m^{\mathrm{rec}})^{\Gamma}$, the constants in the tail bounds are uniform, and one takes the common level $\ell_{m,h,i} = R\log(eD_m^{\mathrm{rec}})$, then the cap-tail part of the series is summable along the full integer dimension schedule $D_m^{\mathrm{rec}} = m$ whenever $R > \Gamma + 1$, and along every stretched-exponential sparse schedule whenever $R > \Gamma$. The remaining condition is exactly the displayed capped coordinate-occupancy Bernstein summability.

Proof. In the sub-Weibull case,

$$\mathbb{P}\{\|X_{m,h,i}\|_{\mathrm{op}} > \rho_{m,h,i}\} \le A_{\mathrm{tail}} \exp\{-\ell_{m,h,i}\}.$$

In the moment case, Markov's inequality gives

$$\mathbb{P}\{\|X_{m,h,i}\|_{\mathrm{op}} > \rho_{m,h,i}\} \le \frac{s_{m,h,i}^p}{s_{m,h,i}^p \exp\{\ell_{m,h,i}\}} = \exp\{-\ell_{m,h,i}\}.$$

Use these quantities as the tail budgets $\delta_{m,h,i}$ in Corollary 7.181. The fixed-size support formulas are the coordinate-support probabilities $r_{m,h,i}^L/d_{m,1}$ and $r_{m,h,i}^R/d_{m,2}$. The final all-integer and sparse cap-tail thresholds follow from summing $(eD_m^{\mathrm{rec}})^{\Gamma-R}$. □

Corollary 7.183 (Active menus of rectangular random-subspace streams)**.** *Fix $C_0 \ge 1$. For each $m \ge 1$, let $J_m \ge 1$, $d_{m,1}, d_{m,2} \ge 1$, $D_m^{\mathrm{rec}} = d_{m,1} + d_{m,2}$, and $n_m \ge 1$. For every $1 \le h \le J_m$, let*

$$X_{m,h,1}, \dots, X_{m,h,n_m} \in \mathrm{M}_{d_{m,1} \times d_{m,2}}(\mathbb{C})$$

be a rectangular random-subspace model as in Corollary 7.176, with parameters $b_{m,h}$ and $\mu_{m,h}$. Suppose there are $t_{m,h} > 0$ such that

$$\mu_{m,h} + t_{m,h} \le \frac{C_0 n_m}{\log(eD_m^{\mathrm{rec}})} \qquad (1 \le h \le J_m),$$

and

$$\sum_m \sum_{h=1}^{J_m} D_m^{\mathrm{rec}} \exp\left\{ -\frac{t_{m,h}^2}{2(\mu_{m,h} + b_{m,h} t_{m,h}/3)} \right\} < \infty,$$

with zero denominator interpreted as zero contribution. Then, almost surely, for all sufficiently large m, every active rectangular random-subspace stream has a Spencer-scale signing:

$$\min_{\varepsilon \in \{\pm 1\}^{n_m}} \left\| \sum_{i=1}^{n_m} \varepsilon_i X_{m,h,i} \right\|_{\mathrm{op}} \le C(C_0)\sqrt{n_m} \qquad (1 \le h \le J_m).$$

Sampled certificates and countable-library exhaustions follow under the same additional summability hypotheses as in Corollary 7.158.

Proof. Apply Corollary 7.176 to each active stream and sum the displayed failure probabilities. The first Borel–Cantelli lemma gives the eventual simultaneous signing, and the sampled and countable-library additions are the same as before. □

Corollary 7.184 (Explicit rectangular weighted-support schedules)**.** *Fix $C_0 \ge 1$, put $c_0 := 3C_0/16$, and, for each row m, let*

$$J_m \ge 1, \qquad d_{m,1}, d_{m,2} \ge 1, \qquad n_m \ge 1,$$

and let

$$X_{m,h,1}, \dots, X_{m,h,n_m} \in \mathrm{M}_{d_{m,1} \times d_{m,2}}(\mathbb{C}) \qquad (1 \le h \le J_m).$$

Set

$$D_m^{\mathrm{rec}} := d_{m,1} + d_{m,2}, \qquad \ell_m := \log(eD_m^{\mathrm{rec}}).$$

For every active stream $1 \le h \le J_m$, assume that one of the following half-budget models holds.

(a) *Rectangular coordinate supports as in Corollary 7.174, with deterministic amplitudes $\rho_{m,h,i} \in [0,1]$, and*

$$\max\left\{ \max_a \sum_i \rho_{m,h,i}^2 p_{m,h,i,a}^L, \max_b \sum_i \rho_{m,h,i}^2 p_{m,h,i,b}^R \right\} \le \frac{C_0 n_m}{2\ell_m}.$$

(b) *Rectangular random row/column subspaces as in Corollary 7.176, with independent projection pairs and deterministic amplitudes $\rho_{m,h,i} \in [0,1]$, and*

$$\max\left\{ \left\| \sum_i \rho_{m,h,i}^2 \mathbb{E} P_{m,h,i}^L \right\|_{\mathrm{op}}, \left\| \sum_i \rho_{m,h,i}^2 \mathbb{E} P_{m,h,i}^R \right\|_{\mathrm{op}} \right\} \le \frac{C_0 n_m}{2\ell_m}.$$

Then, for all large m,

$$\mathbb{P}\left\{\exists h \le J_m : \min_{\varepsilon\in\{\pm1\}^{n_m}} \left\|\sum_{i=1}^{n_m} \varepsilon_i X_{m,h,i}\right\|_{\mathrm{op}} > C(C_0)\sqrt{n_m}\right\} \le J_m D_m^{\mathrm{rec}} \exp\left\{-c_0\,\frac{n_m}{\ell_m}\right\}.$$

Consequently, if

$$\sum_m J_m D_m^{\mathrm{rec}} \exp\left\{-c_0\,\frac{n_m}{\ell_m}\right\} < \infty,$$

then, almost surely, for all sufficiently large m, every active rectangular coordinate-support or random-subspace stream has a Spencer-scale signing.

The structural series is summable in each of the following explicit schedules.

(i) *Arbitrary growth: for some $\gamma > 0$,*

$$n_m \ge \frac{1}{c_0}\,\ell_m\big(\log(eJ_m D_m^{\mathrm{rec}}) + (1+\gamma)\log(e+m)\big)$$

eventually.

(ii) *All rectangular dimensions: if $D_m^{\mathrm{rec}} = D = 2, 3, \ldots$, $J_D \le (eD)^\Gamma$, and*

$$n_D \ge a\log^2(eD)$$

eventually for some $a > (\Gamma+2)/c_0$, then the conclusion holds along all integer D.

(iii) *Sparse rectangular dimensions: if $D_m^{\mathrm{rec}} \ge \exp(cm^\kappa)$ eventually, $J_m \le (eD_m^{\mathrm{rec}})^\Gamma$, and*

$$n_m \ge a\log^2(eD_m^{\mathrm{rec}})$$

eventually for some $a > (\Gamma+1)/c_0$, then the conclusion holds along the sparse schedule.

The same assertions hold with sampled sign certificates. More precisely, for each active stream draw $M_{m,h}$ independent uniform sign vectors, independent of the matrix array. If

$$\sum_m \left[J_m D_m^{\mathrm{rec}} \exp\left\{-c_0\,\frac{n_m}{\ell_m}\right\} + \sum_{h=1}^{J_m} 2^{-M_{m,h}}\right] < \infty,$$

then, almost surely for all sufficiently large m, each active stream has one sampled sign vector certifying the same $C(C_0)\sqrt{n_m}$ bound. In the common batch-size case $M_{m,h} = M_m$, it is enough to add the summability of $\sum_m J_m 2^{-M_m}$. Thus the arbitrary-growth schedule is sampled-certified whenever

$$M_m \ge \log_2(eJ_m) + (1+\delta)\log_2(e+m)$$

eventually, and, under $J_D \le (eD)^\Gamma$, the all-dimension schedule is sampled-certified whenever $M_D \ge (\Gamma+1+\delta)\log_2(eD)$ eventually. Under the sparse schedule it is enough to take

$$M_m \ge \log_2(eJ_m) + (1+\delta)\log_2(e+m)$$

eventually, or the displayed bound with $\log_2(eJ_m)$ *replaced by* $\Gamma \log_2(eD_m^{\mathrm{rec}})$.

Uniform fixed-size row/column supports are included by the half-budget condition

$$\max\left\{\frac{1}{d_{m,1}}\sum_i \rho_{m,h,i}^2 r_{m,h,i}^L, \frac{1}{d_{m,2}}\sum_i \rho_{m,h,i}^2 r_{m,h,i}^R\right\} \le \frac{C_0 n_m}{2\ell_m},$$

and Haar row/column subspaces use the identical formula with r^L, r^R *the projection ranks.*

Proof. For a fixed active stream, take

$$t_m = \frac{C_0 n_m}{2\ell_m}.$$

In the coordinate-support model the half-budget hypothesis gives $\mu_{m,h} \le t_m$, while $\sigma_{m,h}^2 \le \mu_{m,h} \le t_m$ and $b_{m,h} \le 1$. Hence Corollary 7.174 gives failure at most

$$D_m^{\mathrm{rec}} \exp\left\{-c_0 \frac{n_m}{\ell_m}\right\}.$$

The random-subspace model is the same calculation with Corollary 7.176. Union over $h \le J_m$ gives the displayed active-menu tail, and Borel–Cantelli gives the summable-array conclusion.

For arbitrary growth, the lower bound on n_m makes the active-menu tail at most $e^{-1}(e+m)^{-(1+\gamma)}$. In the all-dimension schedule it is bounded by

$$(eD)^{\Gamma+1-c_0 a},$$

which is summable over $D \ge 2$ when $c_0 a > \Gamma + 2$. In the sparse schedule the same bound is stretched-exponentially summable when $c_0 a > \Gamma + 1$.

For sampled certificates, condition on the complement of the coordinate or projection overlap-failure event for a fixed active stream. The proof of Corollary 7.161 gives the deterministic rectangular square-profile certificate, including sampled-search failure $2^{-M_{m,h}}$ after enlarging $C(C_0)$. Summing these failures over the active streams gives the sampled Borel–Cantelli statement. The displayed choices of M_m make $J_m 2^{-M_m}$ summable in the arbitrary-growth, all-dimension, and sparse cases. The fixed-size support and Haar-rank formulas are the corresponding one-coordinate probabilities and projection expectations. □

Corollary 7.185 (Active menus of tail-capped rectangular random-subspace streams)**.** *Fix* $C_0 \ge 1$. *For each* $m \ge 1$, *let* $J_m \ge 1$, $d_{m,1}, d_{m,2} \ge 1$, $D_m^{\mathrm{rec}} = d_{m,1} + d_{m,2}$, *and* $n_m \ge 1$. *For every* $1 \le h \le J_m$, *let*

$$X_{m,h,1}, \dots, X_{m,h,n_m} \in \mathrm{M}_{d_{m,1}\times d_{m,2}}(\mathbb{C})$$

be a rectangular random-subspace model as in Corollary 7.177, with deterministic caps $\rho_{m,h,i}$, *parameters* $b_{m,h}$ *and* $\mu_{m,h}^{\mathrm{cap}}$, *and tail budgets* $\delta_{m,h,i} \ge 0$ *satisfying*

$$\mathbb{P}\{\|X_{m,h,i}\|_{\mathrm{op}} > \rho_{m,h,i}\} \le \delta_{m,h,i}.$$

Suppose there are $t_{m,h} > 0$ such that

$$\mu^{\mathrm{cap}}_{m,h} + t_{m,h} \le \frac{C_0 n_m}{\log(eD^{\mathrm{rec}}_m)} \qquad (1 \le h \le J_m),$$

and

$$\sum_m \sum_{h=1}^{J_m} \left[\sum_{i=1}^{n_m} \delta_{m,h,i} + D^{\mathrm{rec}}_m \exp\left\{ -\frac{t^2_{m,h}}{2(\mu^{\mathrm{cap}}_{m,h} + b_{m,h} t_{m,h}/3)} \right\} \right] < \infty,$$

with zero denominator interpreted as zero contribution. Then, almost surely, for all sufficiently large m, every active tail-capped rectangular random-subspace stream has a Spencer-scale signing:

$$\min_{\varepsilon \in \{\pm 1\}^{n_m}} \left\| \sum_{i=1}^{n_m} \varepsilon_i X_{m,h,i} \right\|_{\mathrm{op}} \le C(C_0)\sqrt{n_m} \qquad (1 \le h \le J_m).$$

Sampled certificates and countable-library exhaustions follow under the same additional summability hypotheses as in Corollary 7.158.

Proof. Apply Corollary 7.177 to each active stream, bound its cap-exception event by $\sum_i \delta_{m,h,i}$, and sum the displayed probabilities. The first Borel–Cantelli lemma gives eventual simultaneous signings. The sampled and countable-library additions are the same as in Corollary 7.183. □

Corollary 7.186 (Sub-Weibull and moment caps for random-subspace menus)**.** *Fix $C_0 \ge 1$. Work in the active-menu setting of Corollary 7.185. For each active triple (m, h, i), choose a level $\ell_{m,h,i} > 0$ and a scale $s_{m,h,i} > 0$. Assume that the deterministic cap $\rho_{m,h,i} \le 1$ is chosen by one of the following two rules.*

(a) Sub-Weibull operator tails: for some fixed $\alpha > 0$ and $A_{\mathrm{tail}} \ge 1$,

$$\mathbb{P}\{\|X_{m,h,i}\|_{\mathrm{op}} > s_{m,h,i} u\} \le A_{\mathrm{tail}} \exp\{-u^\alpha\} \qquad (u \ge 0),$$

and

$$\rho_{m,h,i} := s_{m,h,i} \ell^{1/\alpha}_{m,h,i} \le 1.$$

(b) Moment operator tails: for some fixed $p > 0$,

$$\mathbb{E}\|X_{m,h,i}\|^p_{\mathrm{op}} \le s^p_{m,h,i},$$

and

$$\rho_{m,h,i} := s_{m,h,i} \exp\{\ell_{m,h,i}/p\} \le 1.$$

In case (a) put $A_* = A_{\mathrm{tail}}$, and in case (b) put $A_* = 1$. Let $b_{m,h} := \max_i \rho^2_{m,h,i}$, and let $\mu^{\mathrm{sel}}_{m,h}$ be the capped row/column projection budget

$$\mu^{\mathrm{sel}}_{m,h} := \max\left\{ \left\| \sum_i \rho^2_{m,h,i} \mathbb{E} P^L_{m,h,i} \right\|_{\mathrm{op}}, \left\| \sum_i \rho^2_{m,h,i} \mathbb{E} P^R_{m,h,i} \right\|_{\mathrm{op}} \right\}.$$

If there are $t_{m,h} > 0$ such that

$$\mu^{\mathrm{sel}}_{m,h} + t_{m,h} \le \frac{C_0 n_m}{\log(eD^{\mathrm{rec}}_m)} \qquad (1 \le h \le J_m),$$

and

$$\sum_{m}\sum_{h=1}^{J_m}\left[A_*\sum_{i=1}^{n_m}\exp\{-\ell_{m,h,i}\}+D_m^{\mathrm{rec}}\exp\left\{-\frac{t_{m,h}^2}{2(\mu_{m,h}^{\mathrm{sel}}+b_{m,h}t_{m,h}/3)}\right\}\right]<\infty,$$

with zero denominator interpreted as zero contribution, then, almost surely, for all sufficiently large m, every active random-subspace stream has a Spencer-scale signing:

$$\min_{\varepsilon\in\{\pm1\}^{n_m}}\left\|\sum_{i=1}^{n_m}\varepsilon_i X_{m,h,i}\right\|_{\mathrm{op}}\le C(C_0)\sqrt{n_m}\qquad(1\le h\le J_m).$$

Sampled certificates and countable-library exhaustions follow under the same additional summability hypotheses as in Corollary 7.185.

In the Haar row/column rank specialization, $\mu_{m,h}^{\mathrm{sel}}$ is obtained by replacing $\rho_{m,h,i}^2$ with

$$s_{m,h,i}^2\ell_{m,h,i}^{2/\alpha}\quad\text{in case (a)},\qquad s_{m,h,i}^2\exp\{2\ell_{m,h,i}/p\}\quad\text{in case (b)},$$

inside the two average-rank sums. Thus sub-Weibull low-rank coefficients need only lose the logarithmic cap factor $\ell^{2/\alpha}$, while finite p-moment coefficients are allowed whenever their normalization absorbs the factor $\exp\{2\ell/p\}$.

Consequently, if $J_m n_m\le(eD_m^{\mathrm{rec}})^{\Gamma}$, the constants in the tail bounds are uniform, and one takes the common level $\ell_{m,h,i}=R\log(eD_m^{\mathrm{rec}})$, then the cap-tail part of the series is summable along the full integer dimension schedule $D_m^{\mathrm{rec}}=m$ whenever $R>\Gamma+1$, and along every stretched-exponential sparse schedule whenever $R>\Gamma$. The remaining condition is exactly the displayed capped row/column projection Bernstein summability.

Proof. In the sub-Weibull case,

$$\mathbb{P}\{\|X_{m,h,i}\|_{\mathrm{op}}>\rho_{m,h,i}\}\le A_{\mathrm{tail}}\exp\{-\ell_{m,h,i}\}.$$

In the moment case, Markov's inequality gives

$$\mathbb{P}\{\|X_{m,h,i}\|_{\mathrm{op}}>\rho_{m,h,i}\}\le\frac{s_{m,h,i}^p}{s_{m,h,i}^p\exp\{\ell_{m,h,i}\}}=\exp\{-\ell_{m,h,i}\}.$$

Use these quantities as the tail budgets $\delta_{m,h,i}$ in Corollary 7.185. The Haar-rank formulas are the corresponding specialization of $\mathbb{E}P_{m,h,i}^L=(r_{m,h,i}^L/d_{m,1})I$ and $\mathbb{E}P_{m,h,i}^R=(r_{m,h,i}^R/d_{m,2})I$. The final all-integer and sparse cap-tail thresholds follow from summing $(eD_m^{\mathrm{rec}})^{\Gamma-R}$. □

Corollary 7.187 (Active menus of rectangular dyad/frame streams)**.** *Fix $C_0\ge1$. For each $m\ge1$, let $J_m\ge1$, $d_{m,1},d_{m,2}\ge1$, $D_m^{\mathrm{rec}}=d_{m,1}+d_{m,2}$, and $n_m\ge1$. For each active stream $1\le h\le J_m$, let*

$$X_{m,h,i}=a_{m,h,i}u_{m,h,i}v_{m,h,i}^*\in\mathrm{M}_{d_{m,1}\times d_{m,2}}(\mathbb{C}),\qquad1\le i\le n_m,$$

be as in Corollary 7.178: the unit-vector pairs $(u_{m,h,i},v_{m,h,i})$ are independent over i, and conditional on them the amplitudes satisfy $|a_{m,h,i}|\le\rho_{m,h,i}\le1$.

Put

$$b_{m,h} := \max_i \rho_{m,h,i}^2$$

and

$$\mu_{m,h}^{\mathrm{dyad}} := \max\left\{ \left\| \sum_i \rho_{m,h,i}^2 \mathbb{E} u_{m,h,i} u_{m,h,i}^* \right\|_{\mathrm{op}}, \left\| \sum_i \rho_{m,h,i}^2 \mathbb{E} v_{m,h,i} v_{m,h,i}^* \right\|_{\mathrm{op}} \right\}.$$

Suppose there are $t_{m,h} > 0$ such that

$$\mu_{m,h}^{\mathrm{dyad}} + t_{m,h} \le \frac{C_0 n_m}{\log(e D_m^{\mathrm{rec}})} \qquad (1 \le h \le J_m),$$

and

$$\sum_m \sum_{h=1}^{J_m} D_m^{\mathrm{rec}} \exp\left\{ -\frac{t_{m,h}^2}{2(\mu_{m,h}^{\mathrm{dyad}} + b_{m,h} t_{m,h}/3)} \right\} < \infty,$$

with zero denominator interpreted as zero contribution. Then, almost surely, for all sufficiently large m, every active rectangular dyad/frame stream has a Spencer-scale signing:

$$\min_{\varepsilon \in \{\pm 1\}^{n_m}} \left\| \sum_{i=1}^{n_m} \varepsilon_i X_{m,h,i} \right\|_{\mathrm{op}} \le C(C_0) \sqrt{n_m} \qquad (1 \le h \le J_m).$$

The same statement holds for active prefixes exhausting countable libraries of randomized feature maps or sampled rectangular atom dictionaries. Sampled certificates follow under the same additional summability hypotheses as in Corollary 7.158.

Proof. Apply Corollary 7.178 to each active stream and sum the displayed failure probabilities. The first Borel–Cantelli lemma gives the eventual simultaneous signing. The active-prefix and sampled-search additions are the same as in Corollary 7.183. □

Corollary 7.188 (Active menus of tail-capped rectangular dyads)**.** *Fix $C_0 \ge 1$. For each $m \ge 1$, let $J_m \ge 1$, $d_{m,1}, d_{m,2} \ge 1$, $D_m^{\mathrm{rec}} = d_{m,1} + d_{m,2}$, and $n_m \ge 1$. For each active stream $1 \le h \le J_m$, let*

$$X_{m,h,i} = a_{m,h,i} u_{m,h,i} v_{m,h,i}^*, \qquad 1 \le i \le n_m,$$

where the unit-vector pairs $(u_{m,h,i}, v_{m,h,i})$ are independent over i, and conditional on them the scalar amplitudes $a_{m,h,i}$ are arbitrary. Let deterministic caps $\rho_{m,h,i} \in [0,1]$ and tail budgets $\delta_{m,h,i} \ge 0$ satisfy

$$\mathbb{P}\{|a_{m,h,i}| > \rho_{m,h,i}\} \le \delta_{m,h,i}.$$

Put

$$b_{m,h} := \max_i \rho_{m,h,i}^2$$

and

$$\mu_{m,h}^{\mathrm{cap}} := \max\left\{ \left\| \sum_i \rho_{m,h,i}^2 \mathbb{E} u_{m,h,i} u_{m,h,i}^* \right\|_{\mathrm{op}}, \left\| \sum_i \rho_{m,h,i}^2 \mathbb{E} v_{m,h,i} v_{m,h,i}^* \right\|_{\mathrm{op}} \right\}.$$

Suppose there are $t_{m,h} > 0$ such that

$$\mu^{\mathrm{cap}}_{m,h} + t_{m,h} \le \frac{C_0 n_m}{\log(eD^{\mathrm{rec}}_m)} \qquad (1 \le h \le J_m),$$

and

$$\sum_m \sum_{h=1}^{J_m} \left[\sum_{i=1}^{n_m} \delta_{m,h,i} + D^{\mathrm{rec}}_m \exp\left\{ -\frac{t^2_{m,h}}{2(\mu^{\mathrm{cap}}_{m,h} + b_{m,h} t_{m,h}/3)} \right\} \right] < \infty,$$

with zero denominator interpreted as zero contribution. Then, almost surely, for all sufficiently large m, every active tail-capped rectangular dyad/frame stream has a Spencer-scale signing:

$$\min_{\varepsilon \in \{\pm 1\}^{n_m}} \left\| \sum_{i=1}^{n_m} \varepsilon_i X_{m,h,i} \right\|_{\mathrm{op}} \le C(C_0)\sqrt{n_m} \qquad (1 \le h \le J_m).$$

Consequently, countable libraries of Gaussian, sub-Gaussian, sub-exponential, or heavy-tailed randomized-feature and sampled-atom streams are covered whenever their cap-tail costs and capped frame-Bernstein costs are summable. Sampled certificates follow under the same additional summability hypotheses as in Corollary 7.158.

Proof. Apply Corollary 7.179 to each active stream, bound the cap-exception event by $\sum_i \delta_{m,h,i}$, and sum over m, h. The first Borel–Cantelli lemma gives eventual simultaneous signings. The active-prefix and sampled-search additions are the same as in Corollary 7.187. □

Corollary 7.189 (Sub-Weibull and moment caps for dyad/frame menus)**.** *Fix $C_0 \ge 1$. Work in the active-menu dyad setting of Corollary 7.188. For each active triple (m, h, i), choose a level $\ell_{m,h,i} > 0$ and a scalar scale $s_{m,h,i} > 0$. Assume that the deterministic cap $\rho_{m,h,i} \le 1$ is chosen by one of the following two rules.*

(a) Sub-Weibull scalar tails: for some fixed $\alpha > 0$ and $A_{\mathrm{tail}} \ge 1$,

$$\mathbb{P}\{|a_{m,h,i}| > s_{m,h,i} u\} \le A_{\mathrm{tail}} \exp\{-u^\alpha\} \qquad (u \ge 0),$$

and

$$\rho_{m,h,i} := s_{m,h,i} \ell^{1/\alpha}_{m,h,i} \le 1.$$

(b) Moment scalar tails: for some fixed $p > 0$,

$$\mathbb{E}|a_{m,h,i}|^p \le s^p_{m,h,i},$$

and

$$\rho_{m,h,i} := s_{m,h,i} \exp\{\ell_{m,h,i}/p\} \le 1.$$

In case (a) put $A_* = A_{\mathrm{tail}}$, and in case (b) put $A_* = 1$. Let $b_{m,h} := \max_i \rho^2_{m,h,i}$, and put

$$\mu^{\mathrm{dsel}}_{m,h} := \max\left\{ \left\| \sum_i \rho^2_{m,h,i} \mathbb{E} u_{m,h,i} u^*_{m,h,i} \right\|_{\mathrm{op}}, \left\| \sum_i \rho^2_{m,h,i} \mathbb{E} v_{m,h,i} v^*_{m,h,i} \right\|_{\mathrm{op}} \right\}.$$

If there are $t_{m,h} > 0$ such that

$$\mu^{\mathrm{dsel}}_{m,h} + t_{m,h} \le \frac{C_0 n_m}{\log(eD^{\mathrm{rec}}_m)} \qquad (1 \le h \le J_m),$$

and

$$\sum_m \sum_{h=1}^{J_m} \left[A_* \sum_{i=1}^{n_m} \exp\{-\ell_{m,h,i}\} + D^{\mathrm{rec}}_m \exp\left\{ -\frac{t^2_{m,h}}{2(\mu^{\mathrm{dsel}}_{m,h} + b_{m,h} t_{m,h}/3)} \right\} \right] < \infty,$$

with zero denominator interpreted as zero contribution, then, almost surely, for all sufficiently large m, every active dyad/frame stream has a Spencer-scale signing:

$$\min_{\varepsilon \in \{\pm 1\}^{n_m}} \left\| \sum_{i=1}^{n_m} \varepsilon_i a_{m,h,i} u_{m,h,i} v^*_{m,h,i} \right\|_{\mathrm{op}} \le C(C_0)\sqrt{n_m} \qquad (1 \le h \le J_m).$$

Sampled certificates and countable-library exhaustions follow under the same additional summability hypotheses as in Corollary 7.188.

In the Haar-isotropic dyad case, $\mu^{\mathrm{dsel}}_{m,h}$ is the maximum of

$$\frac{1}{d_{m,1}} \sum_i \rho^2_{m,h,i}, \qquad \frac{1}{d_{m,2}} \sum_i \rho^2_{m,h,i}.$$

Thus sub-Weibull feature weights pay the logarithmic frame factor $\ell^{2/\alpha}$, while finite p-moment feature weights pay $\exp\{2\ell/p\}$, inside the two expected frame budgets.

Consequently, if $J_m n_m \le (eD^{\mathrm{rec}}_m)^{\Gamma}$, the constants in the tail bounds are uniform, and one takes the common level $\ell_{m,h,i} = R\log(eD^{\mathrm{rec}}_m)$, then the cap-tail part of the series is summable along the full integer dimension schedule $D^{\mathrm{rec}}_m = m$ whenever $R > \Gamma + 1$, and along every stretched-exponential sparse schedule whenever $R > \Gamma$. The remaining condition is the displayed capped frame-Bernstein summability.

Proof. The sub-Weibull tail rule gives

$$\mathbb{P}\{|a_{m,h,i}| > \rho_{m,h,i}\} \le A_{\mathrm{tail}} \exp\{-\ell_{m,h,i}\}.$$

The moment rule gives the same bound with $A_{\mathrm{tail}} = 1$ by Markov's inequality:

$$\mathbb{P}\{|a_{m,h,i}| > \rho_{m,h,i}\} \le \frac{s^p_{m,h,i}}{s^p_{m,h,i} \exp\{\ell_{m,h,i}\}} = \exp\{-\ell_{m,h,i}\}.$$

Use these as the $\delta_{m,h,i}$ in Corollary 7.188. The Haar-isotropic formula is the specialization $\mathbb{E} u_{m,h,i} u^*_{m,h,i} = d^{-1}_{m,1} I$ and $\mathbb{E} v_{m,h,i} v^*_{m,h,i} = d^{-1}_{m,2} I$. The all-integer and sparse cap-tail thresholds follow by summing $(eD^{\mathrm{rec}}_m)^{\Gamma - R}$. □

Corollary 7.190 (Homogeneous isotropic dyad cap schedules)**.** *Fix $C_0 \ge 1$, and put $c_0 := 3C_0/16$. For each $m \ge 1$, let $J_m \ge 1$, $d_{m,1}, d_{m,2} \ge 1$, $D^{\mathrm{rec}}_m = d_{m,1} + d_{m,2}$, $d_{m,*} := \min\{d_{m,1}, d_{m,2}\}$, and $n_m \ge 1$. For each active stream $1 \le h \le J_m$, let*

$$X_{m,h,i} = a_{m,h,i} u_{m,h,i} v^*_{m,h,i}, \qquad 1 \le i \le n_m,$$

where the unit-vector pairs are independent over i and isotropic:

$$\mathbb{E}u_{m,h,i}u_{m,h,i}^{*}=\frac{1}{d_{m,1}}I_{d_{m,1}},\qquad \mathbb{E}v_{m,h,i}v_{m,h,i}^{*}=\frac{1}{d_{m,2}}I_{d_{m,2}}.$$

Let $L_m:=R\log(eD_m^{\mathrm{rec}})$. Suppose that, for every active stream, one of the following two homogeneous tail models holds with scale $s_{m,h}>0$.

(a) Sub-Weibull weights: for fixed $\alpha>0$ and $A_{\mathrm{tail}}\geq 1$,

$$\mathbb{P}\{|a_{m,h,i}|>s_{m,h}u\}\leq A_{\mathrm{tail}}\exp\{-u^{\alpha}\}\qquad (u\geq 0,\ 1\leq i\leq n_m),$$

and

$$q_{m,h}:=s_{m,h}L_m^{1/\alpha}.$$

(b) Moment weights: for fixed $p>0$,

$$\mathbb{E}|a_{m,h,i}|^{p}\leq s_{m,h}^{p}\qquad (1\leq i\leq n_m),$$

and

$$q_{m,h}:=s_{m,h}\exp\{L_m/p\}.$$

In case (a) put $A_*=A_{\mathrm{tail}}$, and in case (b) put $A_*=1$. Assume the homogeneous cap and frame half-budget bounds

$$q_{m,h}\leq 1,\qquad q_{m,h}^{2}\leq\frac{C_0d_{m,*}}{2\log(eD_m^{\mathrm{rec}})}\qquad (1\leq h\leq J_m).$$

If

$$\sum_m\sum_{h=1}^{J_m}\left[A_*n_m\exp\{-L_m\}+D_m^{\mathrm{rec}}\exp\left\{-\frac{c_0n_m}{\log(eD_m^{\mathrm{rec}})}\right\}\right]<\infty,$$

then, almost surely, for all sufficiently large m, every active homogeneous isotropic dyad stream has a Spencer-scale signing:

$$\min_{\varepsilon\in\{\pm1\}^{n_m}}\left\|\sum_{i=1}^{n_m}\varepsilon_ia_{m,h,i}u_{m,h,i}v_{m,h,i}^{*}\right\|_{\mathrm{op}}\leq C(C_0)\sqrt{n_m}\qquad (1\leq h\leq J_m).$$

In particular, assume

$$D_m^{\mathrm{rec}}=m,\qquad J_mn_m\leq(eD_m^{\mathrm{rec}})^{\Gamma},\qquad n_m\geq a\log^{2}(eD_m^{\mathrm{rec}}).$$

Then the displayed summability holds whenever

$$R>\Gamma+1,\qquad c_0a>\Gamma+2.$$

Along any stretched-exponential sparse dimension schedule $D_m^{\mathrm{rec}}\geq\exp(cm^{\kappa})$, it is enough that

$$R>\Gamma,\qquad c_0a>\Gamma+1.$$

Proof. Apply Corollary 7.189 with the common levels $\ell_{m,h,i}=L_m$ and common caps $\rho_{m,h,i}=q_{m,h}$. The sub-Weibull or moment tail contribution is bounded by $A_*n_m\exp\{-L_m\}$ for stream (m,h). By isotropy,

$$\mu_{m,h}^{\mathrm{dsel}}=\max\left\{\frac{n_mq_{m,h}^{2}}{d_{m,1}},\frac{n_mq_{m,h}^{2}}{d_{m,2}}\right\}\leq\frac{C_0n_m}{2\log(eD_m^{\mathrm{rec}})}.$$

Taking $t_{m,h} = C_0 n_m/(2\log(eD_m^{\mathrm{rec}}))$ and using $b_{m,h} \le 1$ gives

$$D_m^{\mathrm{rec}} \exp\left\{-\frac{t_{m,h}^2}{2(\mu_{m,h}^{\mathrm{dsel}} + b_{m,h}t_{m,h}/3)}\right\} \le D_m^{\mathrm{rec}} \exp\left\{-\frac{c_0 n_m}{\log(eD_m^{\mathrm{rec}})}\right\}.$$

Summability gives the almost-sure conclusion by Borel–Cantelli. The all-integer and sparse displays follow from

$$\sum_{h \le J_m} A_* n_m \exp\{-L_m\} \le A_* (eD_m^{\mathrm{rec}})^{\Gamma - R}$$

and

$$\sum_{h \le J_m} D_m^{\mathrm{rec}} \exp\left\{-\frac{c_0 n_m}{\log(eD_m^{\mathrm{rec}})}\right\} \le (eD_m^{\mathrm{rec}})^{\Gamma + 1 - c_0 a}.$$

□

Corollary 7.191 (Coordinate-sampled dyad cap schedules)**.** *Fix $C_0 \ge 1$, and put $c_0 := 3C_0/16$. For each $m \ge 1$, let $J_m \ge 1$, $d_{m,1}, d_{m,2} \ge 1$, $D_m^{\mathrm{rec}} = d_{m,1} + d_{m,2}$, and $n_m \ge 1$. For each active stream $1 \le h \le J_m$, let*

$$X_{m,h,i} = a_{m,h,i} e_{I_{m,h,i}} e_{J_{m,h,i}}^* \in \mathrm{M}_{d_{m,1} \times d_{m,2}}(\mathbb{C}), \qquad 1 \le i \le n_m,$$

where the coordinate pairs $(I_{m,h,i}, J_{m,h,i})$ are independent over i. Assume that their row and column marginals have deterministic envelopes

$$\mathbb{P}\{I_{m,h,i} = r\} \le \lambda_{m,h}^L, \qquad \mathbb{P}\{J_{m,h,i} = s\} \le \lambda_{m,h}^R$$

for every admissible i, r, s, and put

$$\lambda_{m,h} := \max\{\lambda_{m,h}^L, \lambda_{m,h}^R\}.$$

Let $L_m := R\log(eD_m^{\mathrm{rec}})$. Suppose that, for every active stream, one of the following two homogeneous scalar-tail models holds with scale $s_{m,h} > 0$.

(a) Sub-Weibull weights: for fixed $\alpha > 0$ and $A_{\mathrm{tail}} \ge 1$,

$$\mathbb{P}\{|a_{m,h,i}| > s_{m,h}u\} \le A_{\mathrm{tail}} \exp\{-u^\alpha\} \qquad (u \ge 0,\ 1 \le i \le n_m),$$

and

$$q_{m,h} := s_{m,h} L_m^{1/\alpha}.$$

(b) Moment weights: for fixed $p > 0$,

$$\mathbb{E}|a_{m,h,i}|^p \le s_{m,h}^p \qquad (1 \le i \le n_m),$$

and

$$q_{m,h} := s_{m,h} \exp\{L_m/p\}.$$

In case (a) put $A_* = A_{\mathrm{tail}}$, and in case (b) put $A_* = 1$. Assume the coordinate cap and marginal-load half-budget bounds

$$q_{m,h} \le 1, \qquad q_{m,h}^2 \lambda_{m,h} \le \frac{C_0}{2\log(eD_m^{\mathrm{rec}})} \qquad (1 \le h \le J_m).$$

If

$$\sum_m \sum_{h=1}^{J_m} \left[A_* n_m \exp\{-L_m\} + D_m^{\mathrm{rec}} \exp\left\{-\frac{c_0 n_m}{\log(eD_m^{\mathrm{rec}})}\right\}\right] < \infty,$$

then, almost surely, for all sufficiently large m, every active coordinate-sampled stream has a Spencer-scale signing:

$$\min_{\varepsilon\in\{\pm1\}^{n_m}} \left\| \sum_{i=1}^{n_m} \varepsilon_i a_{m,h,i} e_{I_{m,h,i}} e_{J_{m,h,i}}^* \right\|_{\mathrm{op}} \le C(C_0)\sqrt{n_m} \qquad (1 \le h \le J_m).$$

In particular, assume

$$D_m^{\mathrm{rec}} = m, \qquad J_m n_m \le (eD_m^{\mathrm{rec}})^{\Gamma}, \qquad n_m \ge a \log^2(eD_m^{\mathrm{rec}}).$$

Then the displayed summability holds whenever

$$R > \Gamma + 1, \qquad c_0 a > \Gamma + 2.$$

Along any stretched-exponential sparse dimension schedule $D_m^{\mathrm{rec}} \ge \exp(cm^{\kappa})$, it is enough that

$$R > \Gamma, \qquad c_0 a > \Gamma + 1.$$

For uniform sampled entries, $\lambda_{m,h} = 1/d_{m,*}$, where $d_{m,*} = \min\{d_{m,1}, d_{m,2}\}$, and the half-budget condition reduces to

$$q_{m,h}^2 \le \frac{C_0 d_{m,*}}{2\log(eD_m^{\mathrm{rec}})}.$$

Proof. Apply Corollary 7.189 with

$$u_{m,h,i} = e_{I_{m,h,i}}, \qquad v_{m,h,i} = e_{J_{m,h,i}},$$

common levels $\ell_{m,h,i} = L_m$, and common caps $\rho_{m,h,i} = q_{m,h}$. The sub-Weibull or moment rule gives cap-tail cost at most $A_* n_m \exp\{-L_m\}$ for stream (m,h). The marginal envelopes give

$$\left\| \mathbb{E} e_{I_{m,h,i}} e_{I_{m,h,i}}^* \right\|_{\mathrm{op}} \le \lambda_{m,h}^L, \qquad \left\| \mathbb{E} e_{J_{m,h,i}} e_{J_{m,h,i}}^* \right\|_{\mathrm{op}} \le \lambda_{m,h}^R,$$

and hence

$$\mu_{m,h}^{\mathrm{dsel}} \le n_m q_{m,h}^2 \lambda_{m,h} \le \frac{C_0 n_m}{2\log(eD_m^{\mathrm{rec}})}.$$

Taking $t_{m,h} = C_0 n_m/(2\log(eD_m^{\mathrm{rec}}))$ and using $q_{m,h} \le 1$ gives the displayed Bernstein term with exponent $c_0 n_m/\log(eD_m^{\mathrm{rec}})$, exactly as in Corollary 7.190. Borel–Cantelli gives the almost-sure conclusion, and the all-integer and sparse schedule thresholds are the same two summability computations from that corollary. In the uniform sampled-entry case the row and column marginal loads are $d_{m,1}^{-1}$ and $d_{m,2}^{-1}$. □

Corollary 7.192 (Predictable coordinate-dyad cap schedules)**.** *Fix $C_0 \ge 1$, and put $c_0 := 3C_0/16$. For each $m \ge 1$, let $J_m \ge 1$, $d_{m,1}, d_{m,2} \ge 1$, $D_m^{\mathrm{rec}} = d_{m,1} + d_{m,2}$, and $n_m \ge 1$. For each active stream $1 \le h \le J_m$, let $(\mathcal{F}_{m,h,i})_{i=0}^{n_m}$ be a filtration and let*

$$X_{m,h,i} = a_{m,h,i} e_{I_{m,h,i}} e_{J_{m,h,i}}^* \in \mathrm{M}_{d_{m,1}\times d_{m,2}}(\mathbb{C}), \qquad 1 \le i \le n_m,$$

be $\mathcal{F}_{m,h,i}$-measurable. Assume that the one-step coordinate marginals satisfy

$$\mathbb{P}\{I_{m,h,i} = r \mid \mathcal{F}_{m,h,i-1}\} \le \lambda_{m,h}^L, \qquad \mathbb{P}\{J_{m,h,i} = s \mid \mathcal{F}_{m,h,i-1}\} \le \lambda_{m,h}^R$$

almost surely for every admissible i, r, s, and put

$$\lambda_{m,h} := \max\{\lambda^L_{m,h}, \lambda^R_{m,h}\}.$$

Let $L_m := R\log(eD^{\mathrm{rec}}_m)$. Suppose that, for every active stream, one of the following two scalar-tail models holds with scale $s_{m,h} > 0$.

(a) Sub-Weibull weights: for fixed $\alpha > 0$ and $A_{\mathrm{tail}} \geq 1$,

$$\mathbb{P}\{|a_{m,h,i}| > s_{m,h}u\} \leq A_{\mathrm{tail}}\exp\{-u^\alpha\} \qquad (u \geq 0,\ 1 \leq i \leq n_m),$$

and

$$q_{m,h} := s_{m,h}L_m^{1/\alpha}.$$

(b) Moment weights: for fixed $p > 0$,

$$\mathbb{E}|a_{m,h,i}|^p \leq s^p_{m,h} \qquad (1 \leq i \leq n_m),$$

and

$$q_{m,h} := s_{m,h}\exp\{L_m/p\}.$$

In case (a) put $A_* = A_{\mathrm{tail}}$, and in case (b) put $A_* = 1$. Assume

$$q_{m,h} \leq 1, \qquad q^2_{m,h}\lambda_{m,h} \leq \frac{C_0}{2\log(eD^{\mathrm{rec}}_m)} \qquad (1 \leq h \leq J_m).$$

If

$$\sum_m \sum_{h=1}^{J_m}\left[A_* n_m \exp\{-L_m\} + D^{\mathrm{rec}}_m \exp\left\{-\frac{c_0 n_m}{\log(eD^{\mathrm{rec}}_m)}\right\}\right] < \infty,$$

then, almost surely, for all sufficiently large m, every active predictable coordinate-dyad stream has a Spencer-scale signing:

$$\min_{\varepsilon\in\{\pm1\}^{n_m}}\left\|\sum_{i=1}^{n_m}\varepsilon_i a_{m,h,i}e_{I_{m,h,i}}e^*_{J_{m,h,i}}\right\|_{\mathrm{op}} \leq C(C_0)\sqrt{n_m} \qquad (1 \leq h \leq J_m).$$

In particular, if

$$D^{\mathrm{rec}}_m = m, \qquad J_m n_m \leq (eD^{\mathrm{rec}}_m)^\Gamma, \qquad n_m \geq a\log^2(eD^{\mathrm{rec}}_m),$$

then the displayed summability holds whenever

$$R > \Gamma + 1, \qquad c_0 a > \Gamma + 2.$$

Along any stretched-exponential sparse dimension schedule $D^{\mathrm{rec}}_m \geq \exp(cm^\kappa)$, it is enough that

$$R > \Gamma, \qquad c_0 a > \Gamma + 1.$$

Thus adaptive sampled-entry streams, without-replacement coordinate exposure, and permutation or matching windows obey the same schedule whenever their one-step row and column marginal envelopes satisfy the displayed half-budget.

Proof. For stream (m,h), define the capped adapted contraction

$$Y_{m,h,i} := a_{m,h,i}\mathbf{1}_{\{|a_{m,h,i}|\le q_{m,h}\}}e_{I_{m,h,i}}e_{J_{m,h,i}}^{*}.$$

By the sub-Weibull tail bound, or by Markov's inequality in the moment case,

$$\mathbb{P}\{\exists i:\ |a_{m,h,i}|>q_{m,h}\}\le A_{*}n_m\exp\{-L_m\}.$$

On the complement of this cap-exception event, $X_{m,h,i}=Y_{m,h,i}$ for all i.

For the capped stream,

$$Y_{m,h,i}Y_{m,h,i}^{*}\preceq q_{m,h}^{2}e_{I_{m,h,i}}e_{I_{m,h,i}}^{*},\qquad Y_{m,h,i}^{*}Y_{m,h,i}\preceq q_{m,h}^{2}e_{J_{m,h,i}}e_{J_{m,h,i}}^{*}.$$

Taking conditional expectations and using the marginal envelopes gives

$$\left\|\sum_i \mathbb{E}(Y_{m,h,i}Y_{m,h,i}^{*}\mid\mathcal{F}_{m,h,i-1})\right\|_{\mathrm{op}}\le n_m q_{m,h}^{2}\lambda_{m,h}^{L},$$

and similarly on the column side. Hence the predictable rectangular profile of the capped stream is at most

$$n_m q_{m,h}^{2}\lambda_{m,h}\le\frac{C_0 n_m}{2\log(eD_m^{\mathrm{rec}})}.$$

Apply Corollary 7.155 to the capped stream with $v=t=C_0n_m/(2\log(eD_m^{\mathrm{rec}}))$. The structural failure is at most

$$D_m^{\mathrm{rec}}\exp\left\{-\frac{c_0 n_m}{\log(eD_m^{\mathrm{rec}})}\right\}.$$

Adding the cap-exception probability and summing over m,h, the first Borel–Cantelli lemma gives the eventual simultaneous conclusion for the original streams. The all-integer and sparse thresholds are the same two summability computations as in Corollary 7.191. □

Corollary 7.193 (Profile-capped predictable coordinate dyads)**.** *Fix $C_0\ge 1$. For each $m\ge 1$, let $J_m\ge 1$, $d_{m,1},d_{m,2}\ge 1$, $D_m^{\mathrm{rec}}=d_{m,1}+d_{m,2}$, and $n_m\ge 1$. For each active stream $1\le h\le J_m$, let $(\mathcal{F}_{m,h,i})_{i=0}^{n_m}$ be a filtration and let*

$$X_{m,h,i}=a_{m,h,i}e_{I_{m,h,i}}e_{J_{m,h,i}}^{*}\in\mathrm{M}_{d_{m,1}\times d_{m,2}}(\mathbb{C}),\qquad 1\le i\le n_m,$$

be $\mathcal{F}_{m,h,i}$-measurable. Let deterministic caps $\rho_{m,h,i}\in[0,1]$ and tail budgets $\delta_{m,h,i}\ge 0$ satisfy

$$\mathbb{P}\{|a_{m,h,i}|>\rho_{m,h,i}\}\le\delta_{m,h,i}.$$

Assume that deterministic one-step coordinate envelopes $\pi_{m,h,i,r}^{L}$ and $\pi_{m,h,i,s}^{R}$ satisfy, almost surely,

$$\mathbb{P}\{I_{m,h,i}=r\mid\mathcal{F}_{m,h,i-1}\}\le\pi_{m,h,i,r}^{L},\qquad\mathbb{P}\{J_{m,h,i}=s\mid\mathcal{F}_{m,h,i-1}\}\le\pi_{m,h,i,s}^{R}.$$

Define the capped row/column coordinate profile

$$\Lambda_{m,h}^{\mathrm{coord}}:=\max\left\{\max_{1\le r\le d_{m,1}}\sum_{i=1}^{n_m}\rho_{m,h,i}^{2}\pi_{m,h,i,r}^{L},\max_{1\le s\le d_{m,2}}\sum_{i=1}^{n_m}\rho_{m,h,i}^{2}\pi_{m,h,i,s}^{R}\right\}.$$

Suppose there are $t_{m,h} > 0$ such that

$$\Lambda^{\mathrm{coord}}_{m,h} + t_{m,h} \le \frac{C_0 n_m}{\log(eD^{\mathrm{rec}}_m)} \qquad (1 \le h \le J_m),$$

and

$$\sum_m \sum_{h=1}^{J_m} \left[\sum_{i=1}^{n_m} \delta_{m,h,i} + D^{\mathrm{rec}}_m \exp\left\{ -\frac{t^2_{m,h}}{2(\Lambda^{\mathrm{coord}}_{m,h} + t_{m,h}/3)} \right\} \right] < \infty.$$

Then, almost surely, for all sufficiently large m, every active profile-capped predictable coordinate stream has a Spencer-scale signing:

$$\min_{\varepsilon \in \{\pm 1\}^{n_m}} \left\| \sum_{i=1}^{n_m} \varepsilon_i a_{m,h,i} e_{I_{m,h,i}} e^*_{J_{m,h,i}} \right\|_{\mathrm{op}} \le C(C_0)\sqrt{n_m} \qquad (1 \le h \le J_m).$$

Consequently, sub-Weibull or finite-moment amplitudes may be inserted by choosing caps exactly as in Corollary 7.189: sub-Weibull tails use $\rho_{m,h,i} = s_{m,h,i}\ell^{1/\alpha}_{m,h,i} \le 1$ and $\delta_{m,h,i} = A_{\mathrm{tail}} e^{-\ell_{m,h,i}}$, while finite p-moments use $\rho_{m,h,i} = s_{m,h,i}\exp\{\ell_{m,h,i}/p\} \le 1$ and $\delta_{m,h,i} = e^{-\ell_{m,h,i}}$. Thus nonuniform sampled-entry, adaptive sparse-observation, and weighted without-replacement coordinate models are charged only by their maximum capped predictable row or column load.

Proof. For each active stream set

$$Y_{m,h,i} := a_{m,h,i} \mathbf{1}_{\{|a_{m,h,i}| \le \rho_{m,h,i}\}} e_{I_{m,h,i}} e^*_{J_{m,h,i}}.$$

The cap-exception event has probability at most $\sum_i \delta_{m,h,i}$, and outside it $X_{m,h,i} = Y_{m,h,i}$ for all i. Also $\|Y_{m,h,i}\|_{\mathrm{op}} \le \rho_{m,h,i} \le 1$, and

$$Y_{m,h,i}Y^*_{m,h,i} \preceq \rho^2_{m,h,i} e_{I_{m,h,i}} e^*_{I_{m,h,i}}, \qquad Y^*_{m,h,i}Y_{m,h,i} \preceq \rho^2_{m,h,i} e_{J_{m,h,i}} e^*_{J_{m,h,i}}.$$

Taking conditional expectations gives diagonal row and column profiles whose r-th and s-th diagonal entries are bounded by

$$\sum_i \rho^2_{m,h,i} \pi^L_{m,h,i,r}, \qquad \sum_i \rho^2_{m,h,i} \pi^R_{m,h,i,s}.$$

Hence the predictable rectangular square profile of the capped stream is at most $\Lambda^{\mathrm{coord}}_{m,h}$. Apply Corollary 7.155 with $v = \Lambda^{\mathrm{coord}}_{m,h}$ and $t = t_{m,h}$, add the cap-exception probability, and sum over m, h. Borel–Cantelli gives the almost-sure simultaneous conclusion. The displayed sub-Weibull and moment caps give the claimed tail budgets by the same calculation as in Corollary 7.189. □

Corollary 7.194 (Finite-menu profile-capped coordinate schedules)**.** *Fix $C_0 \ge 1$, and put $\gamma_0 := 3C_0/16$. For each $m \ge 1$, let $J_m \ge 1$, $d_{m,1}, d_{m,2} \ge 1$, $D^{\mathrm{rec}}_m = d_{m,1} + d_{m,2}$, and*

$$\ell_m := \log(eD^{\mathrm{rec}}_m).$$

For every active $1 \le h \le J_m$, let

$$X_{m,h,i} = a_{m,h,i} e_{I_{m,h,i}} e^*_{J_{m,h,i}} \in \mathrm{M}_{d_{m,1} \times d_{m,2}}(\mathbb{C}), \qquad 1 \le i \le n_m,$$

be an adapted coordinate-dyad stream satisfying the hypotheses of Corollary 7.193, with deterministic caps $\rho_{m,h,i}$, tail budgets $\delta_{m,h,i}$, one-step envelopes $\pi^{L}_{m,h,i,r}$, $\pi^{R}_{m,h,i,s}$, and capped coordinate profile $\Lambda^{\mathrm{coord}}_{m,h}$. Assume the half-budget profile bound

$$\Lambda^{\mathrm{coord}}_{m,h} \le \frac{C_0 n_m}{2\ell_m} \qquad (1 \le h \le J_m)$$

and the two summability conditions

$$\sum_m \sum_{h=1}^{J_m} \sum_{i=1}^{n_m} \delta_{m,h,i} < \infty, \qquad \sum_m J_m D_m^{\mathrm{rec}} \exp\left\{-\gamma_0 \frac{n_m}{\ell_m}\right\} < \infty.$$

Then, almost surely, for all sufficiently large m, every active profile-capped predictable coordinate stream has a Spencer-scale signing:

$$\min_{\varepsilon \in \{\pm 1\}^{n_m}} \left\| \sum_{i=1}^{n_m} \varepsilon_i a_{m,h,i} e_{I_{m,h,i}} e^*_{J_{m,h,i}} \right\|_{\mathrm{op}} \le C(C_0)\sqrt{n_m} \qquad (1 \le h \le J_m).$$

If each active stream is supplied with an independent batch of $M_{m,h}$ uniform random sign vectors, independent of the stream, then the same eventual statement holds for sampled sign certificates provided

$$\sum_m \sum_{h=1}^{J_m} 2^{-M_{m,h}} < \infty.$$

If a common batch of M_m sign vectors is reused across the active menu and is independent of all active streams, it is enough that

$$\sum_m J_m 2^{-M_m} < \infty.$$

The structural summability follows from each of the following concrete schedules.

(a) *General schedules. If, for some $\delta > 0$,*

$$n_m \ge \gamma_0^{-1} \ell_m \big(\log(eJ_m D_m^{\mathrm{rec}}) + (1+\delta)\log(e+m)\big)$$

eventually, then the structural term is summable. A common sampled batch is summable if, for some $\delta' > 0$,

$$M_m \ge \log_2(eJ_m) + (1+\delta')\log_2(e+m)$$

eventually.

(b) *Full integer dimensions. If the rows are indexed by $D_m^{\mathrm{rec}} = D = 2, 3, \ldots,$ $J_D D \le (eD)^{\Gamma}$ eventually, and*

$$n_D \ge a \log^2(eD)$$

eventually, then the structural term is summable whenever

$$\gamma_0 a > \Gamma + 1.$$

(c) *Sparse stretched-exponential dimensions. If* $D_m^{\mathrm{rec}} \geq \exp(cm^{\kappa})$ *eventually for some* $c, \kappa > 0$, $J_m D_m^{\mathrm{rec}} \leq (eD_m^{\mathrm{rec}})^{\Gamma}$ *eventually, and*

$$n_m \geq a \log^2(eD_m^{\mathrm{rec}})$$

eventually, then the structural term is summable whenever

$$\gamma_0 a > \Gamma.$$

In particular, under the common sample-menu size control $J_m n_m \leq (eD_m^{\mathrm{rec}})^{\Gamma}$, *the structural term is summable on all integer dimensions once* $\gamma_0 a > \Gamma + 2$, *and on sparse stretched-exponential dimension schedules once* $\gamma_0 a > \Gamma + 1$. *If, in addition, the caps are selected with* $\delta_{m,h,i} \leq A_* e^{-R\ell_m}$, *as in the sub-Weibull and moment cap-selection rules above, then the cap-tail term is summable under the same sample-menu control whenever* $R > \Gamma + 1$ *on all integer dimensions and whenever* $R > \Gamma$ *on sparse stretched-exponential dimension schedules. If the streams are defined for all* $h \geq 1$ *and* $J_m \to \infty$, *then every fixed stream in the countable coordinate library is included for all sufficiently large* m. *Thus finite active menus, deterministic exhaustions of countable coordinate libraries, and sampled-sign certificate searches inherit explicit profile-capped schedules.*

Proof. For fixed m and $h \leq J_m$, apply Corollary 7.193 with

$$t_{m,h} = \frac{C_0 n_m}{2\ell_m}.$$

The half-budget assumption gives $\Lambda_{m,h}^{\mathrm{coord}} + t_{m,h} \leq C_0 n_m / \ell_m$. Moreover

$$\frac{t_{m,h}^2}{2(\Lambda_{m,h}^{\mathrm{coord}} + t_{m,h}/3)} \geq \frac{3C_0 n_m}{16\ell_m} = \gamma_0 \frac{n_m}{\ell_m}.$$

Hence the failure probability for the active stream is at most

$$\sum_{i=1}^{n_m} \delta_{m,h,i} + D_m^{\mathrm{rec}} \exp\left\{-\gamma_0 \frac{n_m}{\ell_m}\right\}.$$

Union over $h \leq J_m$ and sum over m; the two displayed summability hypotheses and Borel–Cantelli give the simultaneous eventual signing conclusion. The sampled-sign assertions use the sampled-search part of Corollary 7.155 in the proof of Corollary 7.193, then add the displayed $2^{-M_{m,h}}$ or $J_m 2^{-M_m}$ union cost.

For the general schedule,

$$J_m D_m^{\mathrm{rec}} \exp\left\{-\gamma_0 \frac{n_m}{\ell_m}\right\} \leq e^{-1}(e+m)^{-1-\delta}.$$

The common sampled-batch estimate is identical. In full integer dimensions,

$$J_D D \exp\left\{-\gamma_0 \frac{n_D}{\log(eD)}\right\} \leq (eD)^{\Gamma - \gamma_0 a},$$

which is summable when $\gamma_0 a > \Gamma + 1$. In the sparse schedule the same bound is a stretched-exponential summable sequence when $\gamma_0 a > \Gamma$. If

$J_m n_m \le (eD_m^{\mathrm{rec}})^{\Gamma}$, then $J_m D_m^{\mathrm{rec}} \le (eD_m^{\mathrm{rec}})^{\Gamma+1}$, giving the stated $\Gamma+2$ and $\Gamma+1$ structural margins. Finally,

$$\sum_{h\le J_m}\sum_{i\le n_m}\delta_{m,h,i} \le A_* J_m n_m e^{-R\ell_m} \le A_*(eD_m^{\mathrm{rec}})^{\Gamma-R},$$

so the cap-tail summability is the usual integer p-series or sparse stretched-exponential series test. □

Corollary 7.195 (Predictable inhomogeneous entry-intensity schedules)**.** *Fix $C_0 \ge 1$, and put $\gamma_0 := 3C_0/16$. For each $m \ge 1$, let $J_m \ge 1$, $d_{m,1}, d_{m,2} \ge 1$, $D_m^{\mathrm{rec}} = d_{m,1} + d_{m,2}$, and*

$$\ell_m := \log(eD_m^{\mathrm{rec}}).$$

For each active stream $1 \le h \le J_m$, let $(\mathcal{F}_{m,h,i})_{i=0}^{n_m}$ be a filtration and let

$$X_{m,h,i} = a_{m,h,i} e_{I_{m,h,i}} e_{J_{m,h,i}}^* \in \mathrm{M}_{d_{m,1}\times d_{m,2}}(\mathbb{C}), \qquad 1 \le i \le n_m,$$

be $\mathcal{F}_{m,h,i}$-measurable. Let deterministic caps $\rho_{m,h,i} \in [0,1]$, tail budgets $\delta_{m,h,i} \ge 0$, and entry-intensity envelopes $p_{m,h,i,r,s} \ge 0$ satisfy

$$\mathbb{P}\{|a_{m,h,i}| > \rho_{m,h,i}\} \le \delta_{m,h,i},$$
$$\mathbb{P}\{I_{m,h,i} = r,\ J_{m,h,i} = s \mid \mathcal{F}_{m,h,i-1}\} \le p_{m,h,i,r,s}$$

almost surely for every admissible i, r, s. Define the capped predictable row/column entry-load profile

$$\Theta_{m,h}^{\mathrm{ent}} := \max\left\{ \max_{1\le r\le d_{m,1}} \sum_{i=1}^{n_m} \rho_{m,h,i}^2 \sum_{s=1}^{d_{m,2}} p_{m,h,i,r,s},\ \max_{1\le s\le d_{m,2}} \sum_{i=1}^{n_m} \rho_{m,h,i}^2 \sum_{r=1}^{d_{m,1}} p_{m,h,i,r,s} \right\}.$$

Assume the half-budget condition

$$\Theta_{m,h}^{\mathrm{ent}} \le \frac{C_0 n_m}{2\ell_m} \qquad (1 \le h \le J_m)$$

and the two summability conditions

$$\sum_m \sum_{h=1}^{J_m}\sum_{i=1}^{n_m} \delta_{m,h,i} < \infty, \qquad \sum_m J_m D_m^{\mathrm{rec}} \exp\left\{-\gamma_0 \frac{n_m}{\ell_m}\right\} < \infty.$$

Then, almost surely, for all sufficiently large m, every active inhomogeneous entry-intensity stream has a Spencer-scale signing:

$$\min_{\varepsilon\in\{\pm1\}^{n_m}} \left\| \sum_{i=1}^{n_m} \varepsilon_i a_{m,h,i} e_{I_{m,h,i}} e_{J_{m,h,i}}^* \right\|_{\mathrm{op}} \le C(C_0)\sqrt{n_m} \qquad (1 \le h \le J_m).$$

Independent or common sampled-sign batches are certified under the same 2^{-M}-summability hypotheses as in Corollary 7.194. The general, full-integer, sparse stretched-exponential, countable-exhaustion, and cap-selection schedules of that corollary apply verbatim with $\Theta_{m,h}^{\mathrm{ent}}$ in place of $\Lambda_{m,h}^{\mathrm{coord}}$.

In particular, adaptive inhomogeneous sampled-entry matrices, bipartite edge streams, Poissonized sparse observations, weighted without-replacement edge exposure, and permutation or matching windows are covered as soon as their predictable weighted expected row and column degrees obey the displayed half-budget and their cap tails are summable.

Proof. Set

$$\pi^{L}_{m,h,i,r} := \sum_{s=1}^{d_{m,2}} p_{m,h,i,r,s}, \qquad \pi^{R}_{m,h,i,s} := \sum_{r=1}^{d_{m,1}} p_{m,h,i,r,s}.$$

The pairwise envelope implies the one-step marginal bounds required in Corollary 7.193, and the resulting capped coordinate profile is at most $\Theta^{\mathrm{ent}}_{m,h}$. Therefore the half-budget and summability hypotheses are exactly those of Corollary 7.194. Applying that corollary gives the signing, sampled-certificate, and schedule claims. □

Corollary 7.196 (Closed-form entry-intensity cap schedules)**.** *Fix $C_0 \geq 1$, and put $c_0 := 3C_0/16$. For each $m \geq 1$, let $J_m \geq 1$, $d_{m,1}, d_{m,2} \geq 1$, $D^{\mathrm{rec}}_m = d_{m,1} + d_{m,2}$, and $n_m \geq 1$. For each active stream $1 \leq h \leq J_m$, let*

$$X_{m,h,i} = a_{m,h,i} e_{I_{m,h,i}} e^{*}_{J_{m,h,i}} \in \mathrm{M}_{d_{m,1}\times d_{m,2}}(\mathbb{C}), \qquad 1 \leq i \leq n_m,$$

be adapted to a filtration $(\mathcal{F}_{m,h,i})_{i=0}^{n_m}$. Assume that deterministic entry-intensity envelopes $p_{m,h,i,r,s} \geq 0$ satisfy

$$\mathbb{P}\{I_{m,h,i} = r,\ J_{m,h,i} = s \mid \mathcal{F}_{m,h,i-1}\} \leq p_{m,h,i,r,s}$$

almost surely for every admissible i, r, s. Put

$$\Delta^{\mathrm{ent}}_{m,h} := \max\left\{ \begin{array}{l} \displaystyle\max_{1\leq r\leq d_{m,1}} \sum_{i=1}^{n_m}\sum_{s=1}^{d_{m,2}} p_{m,h,i,r,s}, \\ \displaystyle\max_{1\leq s\leq d_{m,2}} \sum_{i=1}^{n_m}\sum_{r=1}^{d_{m,1}} p_{m,h,i,r,s} \end{array} \right\}.$$

Let $L_m := R\log(eD^{\mathrm{rec}}_m)$. Suppose that, for every active stream, one of the following two scalar-tail models holds with scale $s_{m,h} > 0$.

(a) Sub-Weibull weights: for fixed $\alpha > 0$ and $A_{\mathrm{tail}} \geq 1$,

$$\mathbb{P}\{|a_{m,h,i}| > s_{m,h}u\} \leq A_{\mathrm{tail}} \exp\{-u^{\alpha}\} \qquad (u \geq 0,\ 1 \leq i \leq n_m),$$

and

$$q_{m,h} := s_{m,h} L_m^{1/\alpha}.$$

(b) Moment weights: for fixed $p > 0$,

$$\mathbb{E}|a_{m,h,i}|^{p} \leq s^{p}_{m,h} \qquad (1 \leq i \leq n_m),$$

and

$$q_{m,h} := s_{m,h} \exp\{L_m/p\}.$$

In case (a) put $A_* := A_{\mathrm{tail}}$, and in case (b) put $A_* := 1$. Assume

$$q_{m,h} \le 1, \qquad q_{m,h}^2 \Delta_{m,h}^{\mathrm{ent}} \le \frac{C_0 n_m}{2\log(eD_m^{\mathrm{rec}})} \qquad (1 \le h \le J_m).$$

If

$$\sum_m \sum_{h=1}^{J_m} \left[A_* n_m \exp\{-L_m\} + D_m^{\mathrm{rec}} \exp\left\{ -\frac{c_0 n_m}{\log(eD_m^{\mathrm{rec}})} \right\} \right] < \infty,$$

then, almost surely, for all sufficiently large m, every active entry-intensity stream has a Spencer-scale signing:

$$\min_{\varepsilon \in \{\pm 1\}^{n_m}} \left\| \sum_{i=1}^{n_m} \varepsilon_i a_{m,h,i} e_{I_{m,h,i}} e_{J_{m,h,i}}^* \right\|_{\mathrm{op}} \le C(C_0)\sqrt{n_m} \qquad (1 \le h \le J_m).$$

Sampled certificates follow under the same additional 2^{-M}-summability hypotheses as in Corollary 7.194.

In particular, if

$$D_m^{\mathrm{rec}} = m, \qquad J_m n_m \le (eD_m^{\mathrm{rec}})^{\Gamma}, \qquad n_m \ge a \log^2(eD_m^{\mathrm{rec}}),$$

then the displayed summability holds whenever

$$R > \Gamma + 1, \qquad c_0 a > \Gamma + 2.$$

Along any stretched-exponential sparse dimension schedule $D_m^{\mathrm{rec}} \ge \exp(cm^{\kappa})$, it is enough that

$$R > \Gamma, \qquad c_0 a > \Gamma + 1.$$

Thus inhomogeneous random bipartite-edge, sparse observation, and adaptive sampled-entry models have a closed-form criterion: after scalar capping, the maximum predictable expected row or column degree must be at most $C_0 n_m/(2q_{m,h}^2 \log(eD_m^{\mathrm{rec}}))$.

Proof. Use the deterministic cap $\rho_{m,h,i} = q_{m,h}$. The sub-Weibull tail model gives

$$\mathbb{P}\{|a_{m,h,i}| > q_{m,h}\} \le A_{\mathrm{tail}} e^{-L_m},$$

and the moment model gives the same bound with $A_{\mathrm{tail}} = 1$ by Markov's inequality. Hence the cap-exception contribution for stream (m,h) is at most $A_* n_m e^{-L_m}$. With these caps, the entry-intensity profile in Corollary 7.195 is

$$\Theta_{m,h}^{\mathrm{ent}} = q_{m,h}^2 \Delta_{m,h}^{\mathrm{ent}} \le \frac{C_0 n_m}{2\log(eD_m^{\mathrm{rec}})}.$$

Applying that corollary with the half-budget choice gives the structural term

$$D_m^{\mathrm{rec}} \exp\left\{ -\frac{c_0 n_m}{\log(eD_m^{\mathrm{rec}})} \right\}.$$

Summability and Borel–Cantelli give the eventual simultaneous signing conclusion, and the sampled-certificate statement is inherited from the same corollary. Under $J_m n_m \le (eD_m^{\mathrm{rec}})^{\Gamma}$, summing the cap tails over $h \le J_m$ gives at most $A_*(eD_m^{\mathrm{rec}})^{\Gamma - R}$. The structural active-menu term is at most $(eD_m^{\mathrm{rec}})^{\Gamma + 1 - c_0 a}$. These are summable over all integer dimensions under $R >$

$\Gamma+1$ and $c_0 a > \Gamma+2$, and along stretched-exponential sparse schedules under $R>\Gamma$ and $c_0 a > \Gamma+1$. □

Corollary 7.197 (Position-dependent entry-intensity caps)**.** *Fix $C_0 \ge 1$, and put $c_0 := 3C_0/16$. For each $m \ge 1$, let $J_m \ge 1$, $d_{m,1}, d_{m,2} \ge 1$, $D_m^{\mathrm{rec}} = d_{m,1}+d_{m,2}$, and $\ell_m = \log(eD_m^{\mathrm{rec}})$. For each active stream $1 \le h \le J_m$, let*

$$X_{m,h,i} = a_{m,h,i} e_{I_{m,h,i}} e_{J_{m,h,i}}^* \in \mathrm{M}_{d_{m,1}\times d_{m,2}}(\mathbb{C}), \qquad 1 \le i \le n_m,$$

be adapted to a filtration $(\mathcal{F}_{m,h,i})_{i=0}^{n_m}$. Assume that deterministic pairwise entry envelopes $p_{m,h,i,r,s} \ge 0$ satisfy

$$\mathbb{P}\{I_{m,h,i} = r,\ J_{m,h,i} = s \mid \mathcal{F}_{m,h,i-1}\} \le p_{m,h,i,r,s}$$

almost surely. Let deterministic site caps $q_{m,h,i,r,s} \in [0,1]$ and tail budgets $\delta_{m,h,i,r,s} \ge 0$ satisfy

$$\mathbb{P}\{|a_{m,h,i}| > q_{m,h,i,r,s},\ I_{m,h,i} = r,\ J_{m,h,i} = s\} \le \delta_{m,h,i,r,s}.$$

Define the site-capped weighted entry profile

$$\Xi_{m,h}^{\mathrm{ent}} := \max\left\{ \max_{1\le r\le d_{m,1}} \sum_{i=1}^{n_m}\sum_{s=1}^{d_{m,2}} q_{m,h,i,r,s}^2 p_{m,h,i,r,s},\ \max_{1\le s\le d_{m,2}} \sum_{i=1}^{n_m}\sum_{r=1}^{d_{m,1}} q_{m,h,i,r,s}^2 p_{m,h,i,r,s} \right\}.$$

Assume the half-budget condition

$$\Xi_{m,h}^{\mathrm{ent}} \le \frac{C_0 n_m}{2\ell_m} \qquad (1 \le h \le J_m)$$

and the summability condition

$$\sum_m \sum_{h=1}^{J_m} \left[\sum_{i=1}^{n_m}\sum_{r=1}^{d_{m,1}}\sum_{s=1}^{d_{m,2}} \delta_{m,h,i,r,s} + D_m^{\mathrm{rec}} \exp\left\{ -c_0 \frac{n_m}{\ell_m} \right\} \right] < \infty.$$

Then, almost surely, for all sufficiently large m, every active stream has a Spencer-scale signing:

$$\min_{\varepsilon\in\{\pm1\}^{n_m}} \left\| \sum_{i=1}^{n_m} \varepsilon_i a_{m,h,i} e_{I_{m,h,i}} e_{J_{m,h,i}}^* \right\|_{\mathrm{op}} \le C(C_0)\sqrt{n_m} \qquad (1 \le h \le J_m).$$

Sampled certificates follow under the same additional 2^{-M}-summability hypotheses as in Corollary 7.194.

Moreover, the caps may be selected entry by entry. Let $L_m = R\ell_m$. If, for fixed $\alpha > 0$ and $A_{\mathrm{tail}} \ge 1$,

$$\mathbb{P}\{|a_{m,h,i}| > s_{m,h,i,r,s} u,\ I_{m,h,i} = r,\ J_{m,h,i} = s\} \le A_{\mathrm{tail}} e^{-u^\alpha} p_{m,h,i,r,s},$$

then choose

$$q_{m,h,i,r,s} = s_{m,h,i,r,s} L_m^{1/\alpha}.$$

If instead, for a fixed $p_0 > 0$,

$$\mathbb{E}\left[|a_{m,h,i}|^{p_0}\mathbf{1}_{\{I_{m,h,i}=r,\ J_{m,h,i}=s\}}\right] \le s_{m,h,i,r,s}^{p_0} p_{m,h,i,r,s},$$

then choose

$$q_{m,h,i,r,s} = s_{m,h,i,r,s}\exp\{L_m/p_0\}.$$

In either case, assuming all chosen caps are at most 1, the cap-tail contribution for stream (m,h) is at most

$$A_*\exp\{-L_m\}B_{m,h}^{\mathrm{pair}}, \qquad B_{m,h}^{\mathrm{pair}} := \sum_{i=1}^{n_m}\sum_{r=1}^{d_{m,1}}\sum_{s=1}^{d_{m,2}} p_{m,h,i,r,s},$$

where $A_* = A_{\mathrm{tail}}$ in the sub-Weibull case and $A_* = 1$ in the moment case. Thus, whenever $B_{m,h}^{\mathrm{pair}} \le n_m$, the same all-integer and sparse stretched-exponential schedule thresholds as in Corollary 7.196 apply, but with the sharper site-weighted profile $\Xi_{m,h}^{\mathrm{ent}}$ in place of the coarser product $q_{m,h}^2\Delta_{m,h}^{\mathrm{ent}}$.

Proof. For stream (m,h), define the capped adapted contraction

$$Y_{m,h,i} := a_{m,h,i}\mathbf{1}_{\{|a_{m,h,i}|\le q_{m,h,i,I_{m,h,i},J_{m,h,i}}\}}e_{I_{m,h,i}}e_{J_{m,h,i}}^*.$$

The cap-exception event has probability at most the displayed sum of $\delta_{m,h,i,r,s}$, and outside this event $Y_{m,h,i} = X_{m,h,i}$ for all i. Since all caps are at most 1, the $Y_{m,h,i}$ are contractions. For each row coordinate,

$$\mathbb{E}(Y_{m,h,i}Y_{m,h,i}^* \mid \mathcal{F}_{m,h,i-1})_{r,r} \le \sum_{s=1}^{d_{m,2}} q_{m,h,i,r,s}^2 p_{m,h,i,r,s},$$

and the off-diagonal entries vanish. The column profile is bounded similarly by the sums over r. Hence the predictable rectangular square profile of the capped stream is at most $\Xi_{m,h}^{\mathrm{ent}}$. Apply Corollary 7.155 with

$$v = \Xi_{m,h}^{\mathrm{ent}}, \qquad t = \frac{C_0 n_m}{2\ell_m}.$$

The half-budget gives $v \le t$, so the structural exponent is at least

$$\frac{t^2}{2(v+t/3)} \ge \frac{3t}{8} = c_0\frac{n_m}{\ell_m}.$$

Adding the cap-exception probability and summing over m,h, Borel–Cantelli gives the eventual simultaneous conclusion; the sampled-certificate statement comes from the sampled version of the same rectangular square-profile corollary.

For the sub-Weibull site-tail model, the chosen cap gives $\delta_{m,h,i,r,s} = A_{\mathrm{tail}}e^{-L_m}p_{m,h,i,r,s}$. For the moment site-tail model, Markov's inequality gives the same estimate with $A_* = 1$. Summing these bounds over i,r,s gives the displayed cap-tail contribution. When $B_{m,h}^{\mathrm{pair}} \le n_m$, the cap-tail summability and structural summability are exactly the two series tests computed in Corollary 7.196. □

Corollary 7.198 (Rank-sensitive predictable square-profile ensembles)**.** *Let $D \geq 2$, let $1 \leq \kappa \leq D$, and let $(\mathcal{F}_i)_{i=0}^n$ be a filtration. Let P be an $\mathcal{F}_0$-measurable orthogonal projection with*

$$\operatorname{rank} P \leq \kappa.$$

Let $A_1, \dots, A_n \in \mathrm{M}_D(\mathbb{C})$ be self-adjoint contractions such that

$$A_i \text{ is } \mathcal{F}_i\text{-measurable}, \qquad A_i = PA_iP.$$

Put

$$\nu_\kappa := \left\| \sum_{i=1}^n \mathbb{E}(A_i^2 \mid \mathcal{F}_{i-1}) \right\|_{\mathrm{op}}.$$

Then, for every $C_0 \geq 1$, there is $C(C_0)$ such that, for every $v, t > 0$ satisfying

$$v + t \leq \frac{C_0 n}{\log(e\kappa)},$$

one has

$$\mathbb{P}\left\{\operatorname{disc}(A_1, \dots, A_n) > C(C_0)\sqrt{n}\right\} \leq \mathbb{P}\{\nu_\kappa > v\} + \kappa \exp\left\{-\frac{t^2}{2(v + t/3)}\right\}.$$

Moreover, if $\eta^{(1)}, \dots, \eta^{(m)}$ are independent uniform sign vectors independent of $\mathcal{F}_n$, then, after increasing $C(C_0)$ if needed,

$$\mathbb{P}\left\{ \min_{1 \leq r \leq m} \left\| \sum_{i=1}^n \eta_i^{(r)} A_i \right\|_{\mathrm{op}} > C(C_0)\sqrt{n} \right\} \leq \mathbb{P}\{\nu_\kappa > v\} + \kappa \exp\left\{-\frac{t^2}{2(v + t/3)}\right\} + 2^{-m}.$$

In particular, if

$$\nu_\kappa \leq \frac{C_0 n}{2\log(e\kappa)}$$

almost surely, then both bounds hold with structural failure at most

$$\kappa \exp\left\{-\frac{3C_0 n}{16\log(e\kappa)}\right\},$$

with the additional 2^{-m} term for sampled sign search. Thus adapted random contractions supported in a common deterministic or random κ-dimensional corner obey the square-profile Matrix Spencer theorem with D replaced everywhere by the joint support rank.

Proof. Set $B_i = A_i^2$ and $M_i = \mathbb{E}(B_i \mid \mathcal{F}_{i-1})$. Because $P \in \mathcal{F}_0 \subseteq \mathcal{F}_{i-1}$ and $B_i = PB_iP$, also $M_i = PM_iP$. Hence the martingale differences

$$X_i := B_i - M_i$$

are supported on $\operatorname{ran} P$. Restrict the matrices to this space, whose dimension is at most κ. The proof of Corollary 7.153, now in dimension $\operatorname{rank} P$, gives on $\{\nu_\kappa \le v\}$

$$\mathbb{P}\left\{\left\|\sum_i A_i^2\right\|_{\mathrm{op}} > v + t \text{ and } \nu_\kappa \le v\right\} \le \kappa \exp\left\{-\frac{t^2}{2(v + t/3)}\right\}.$$

On the complementary event the realized square function has rank at most κ and norm at most $v + t$. The rank-sensitive part of Proposition 7.143 therefore yields

$$\operatorname{disc}(A_1, \dots, A_n) \le C\sqrt{\log(e\kappa)}\sqrt{v + t} \le C(C_0)\sqrt{n}.$$

This proves the first estimate. The sampled-search estimate follows by conditioning on the realized tuple and applying Corollary 7.144 in its rank-sensitive form. The half-budget specialization is the choice $v = t = C_0 n/(2\log(e\kappa))$. □

Proposition 7.199 (Two-scale square-function certificate)**.** *There is a universal constant C with the following property. Let $D \ge 2$, let $1 \le \kappa \le D$, and let P be an orthogonal projection on $\mathbb{C}^D$ with $\operatorname{rank} P \le \kappa$. Suppose*

$$A_i = B_i + R_i, \qquad 1 \le i \le n,$$

where all A_i, B_i, R_i are self-adjoint and $B_i = PB_iP$. Put

$$\Theta(B, R) := \sqrt{\log(e\kappa)}\left\|\sum_{i=1}^n B_i^2\right\|_{\mathrm{op}}^{1/2} + \sqrt{\log(eD)}\left\|\sum_{i=1}^n R_i^2\right\|_{\mathrm{op}}^{1/2}.$$

Then

$$\operatorname{disc}(A_1, \dots, A_n) \le C\,\Theta(B, R).$$

Moreover, if $\eta^{(1)}, \dots, \eta^{(m)}$ are independent uniform sign vectors in $\{\pm 1\}^n$, then, conditionally on the fixed decomposition,

$$\mathbb{P}_\eta\left\{\min_{1 \le r \le m}\left\|\sum_{i=1}^n \eta_i^{(r)} A_i\right\|_{\mathrm{op}} > 2C\,\Theta(B, R)\right\} \le 2^{-m}.$$

Consequently, a tuple with a κ-dimensional square-function core and a small ambient square-function residual has a single Spencer-scale signing: if

$$\left\|\sum_i B_i^2\right\|_{\mathrm{op}} \le \frac{C_{\mathrm{lo}} n}{\log(e\kappa)}, \qquad \left\|\sum_i R_i^2\right\|_{\mathrm{op}} \le \frac{C_{\mathrm{hi}} n}{\log(eD)},$$

then

$$\operatorname{disc}(A_1, \dots, A_n) \le C(C_{\mathrm{lo}}, C_{\mathrm{hi}})\sqrt{n},$$

and m sampled sign vectors fail to find such a signing with probability at most 2^{-m}.

Proof. For one uniform sign vector η, the triangle inequality gives

$$\mathbb{E}_\eta \left\| \sum_i \eta_i A_i \right\|_{\mathrm{op}} \le \mathbb{E}_\eta \left\| \sum_i \eta_i B_i \right\|_{\mathrm{op}} + \mathbb{E}_\eta \left\| \sum_i \eta_i R_i \right\|_{\mathrm{op}}.$$

The first tuple acts on $\operatorname{ran} P$, whose dimension is at most κ, so the rank-sensitive noncommutative Khintchine estimate used in Proposition 7.143 bounds the first term by

$$C\sqrt{\log(e\kappa)} \left\| \sum_i B_i^2 \right\|_{\mathrm{op}}^{1/2}.$$

The same inequality in the ambient dimension D bounds the residual term by

$$C\sqrt{\log(eD)} \left\| \sum_i R_i^2 \right\|_{\mathrm{op}}^{1/2}.$$

Hence the expected norm of one signed sum is at most $C\Theta(B,R)$, and the minimum over all signs is bounded by that expectation. Markov's inequality makes one sampled sign vector successful with probability at least $1/2$; independence gives the 2^{-m} failure bound for m trials. The final display follows by substituting the two square-function budgets into $\Theta(B,R)$. □

Corollary 7.200 (Two-scale predictable square-profile ensembles)**.** *Let $D \ge 2$, $1 \le \kappa \le D$, and let $(\mathcal{F}_i)_{i=0}^n$ be a filtration. Let P be an $\mathcal{F}_0$-measurable orthogonal projection with* $\operatorname{rank} P \le \kappa$. *Suppose*

$$A_i = B_i + R_i, \qquad 1 \le i \le n,$$

where B_i and R_i are $\mathcal{F}_i$-measurable self-adjoint contractions and $B_i = PB_iP$. Define the predictable core and residual square profiles by

$$\nu_{\mathrm{lo}} := \left\| \sum_{i=1}^n \mathbb{E}(B_i^2 \mid \mathcal{F}_{i-1}) \right\|_{\mathrm{op}}, \qquad \nu_{\mathrm{hi}} := \left\| \sum_{i=1}^n \mathbb{E}(R_i^2 \mid \mathcal{F}_{i-1}) \right\|_{\mathrm{op}}.$$

Then, for every $C_{\mathrm{lo}}, C_{\mathrm{hi}} \ge 1$, there is $C(C_{\mathrm{lo}}, C_{\mathrm{hi}})$ such that, whenever

$$v_{\mathrm{lo}} + t_{\mathrm{lo}} \le \frac{C_{\mathrm{lo}} n}{\log(e\kappa)}, \qquad v_{\mathrm{hi}} + t_{\mathrm{hi}} \le \frac{C_{\mathrm{hi}} n}{\log(eD)},$$

one has

$$\begin{aligned} &\mathbb{P}\left\{ \operatorname{disc}(A_1, \dots, A_n) > C(C_{\mathrm{lo}}, C_{\mathrm{hi}})\sqrt{n} \right\} \\ &\quad \le \mathbb{P}\{\nu_{\mathrm{lo}} > v_{\mathrm{lo}}\} + \kappa \exp\left\{ -\frac{t_{\mathrm{lo}}^2}{2(v_{\mathrm{lo}} + t_{\mathrm{lo}}/3)} \right\} \\ &\qquad + \mathbb{P}\{\nu_{\mathrm{hi}} > v_{\mathrm{hi}}\} + D \exp\left\{ -\frac{t_{\mathrm{hi}}^2}{2(v_{\mathrm{hi}} + t_{\mathrm{hi}}/3)} \right\}. \end{aligned}$$

If $\eta^{(1)},\dots,\eta^{(m)}$ are independent uniform sign vectors independent of $\mathcal{F}_n$, then the same bound holds for sampled sign search after adding 2^{-m}. In particular, if the half-budget bounds

$$\nu_{\mathrm{lo}} \le \frac{C_{\mathrm{lo}} n}{2\log(e\kappa)}, \qquad \nu_{\mathrm{hi}} \le \frac{C_{\mathrm{hi}} n}{2\log(eD)}$$

hold outside events E_{lo} and E_{hi}, respectively, then the failure probability is at most

$$\mathbb{P}(E_{\mathrm{lo}} \cup E_{\mathrm{hi}}) + \kappa \exp\left\{-\frac{3C_{\mathrm{lo}} n}{16\log(e\kappa)}\right\} + D\exp\left\{-\frac{3C_{\mathrm{hi}} n}{16\log(eD)}\right\},$$

with an additional 2^{-m} for randomized search.

Thus the exact common-corner theorem is stable under a full ambient residual, provided that residual has its own logarithmic square-function budget. This covers random tuples which are not themselves low-rank but decompose into a low-rank adaptive core plus diffuse martingale, twirl, scalar-multiplier, or localized noise.

Proof. Apply the proof of Corollary 7.198 to the adapted core tuple B_i. On $\{\nu_{\mathrm{lo}} \le v_{\mathrm{lo}}\}$, except with probability at most

$$\kappa \exp\left\{-\frac{t_{\mathrm{lo}}^2}{2(v_{\mathrm{lo}} + t_{\mathrm{lo}}/3)}\right\},$$

one has

$$\left\|\sum_i B_i^2\right\|_{\mathrm{op}} \le v_{\mathrm{lo}} + t_{\mathrm{lo}}.$$

Similarly, the proof of Corollary 7.153 applied to the residual tuple R_i gives, on $\{\nu_{\mathrm{hi}} \le v_{\mathrm{hi}}\}$, except with probability at most

$$D\exp\left\{-\frac{t_{\mathrm{hi}}^2}{2(v_{\mathrm{hi}} + t_{\mathrm{hi}}/3)}\right\},$$

the bound

$$\left\|\sum_i R_i^2\right\|_{\mathrm{op}} \le v_{\mathrm{hi}} + t_{\mathrm{hi}}.$$

On the intersection of these two square-function events, Proposition 7.199 gives

$$\operatorname{disc}(A_1,\dots,A_n) \le C\sqrt{\log(e\kappa)}\sqrt{v_{\mathrm{lo}} + t_{\mathrm{lo}}} + C\sqrt{\log(eD)}\sqrt{v_{\mathrm{hi}} + t_{\mathrm{hi}}} \le C(C_{\mathrm{lo}}, C_{\mathrm{hi}})\sqrt{n}.$$

The sampled-search statement follows by conditioning on the realized decomposition and applying the sampled part of Proposition 7.199. The half-budget form is the choice

$$v_{\mathrm{lo}} = t_{\mathrm{lo}} = \frac{C_{\mathrm{lo}} n}{2\log(e\kappa)}, \qquad v_{\mathrm{hi}} = t_{\mathrm{hi}} = \frac{C_{\mathrm{hi}} n}{2\log(eD)}.$$

□

Corollary 7.201 (Finite-menu two-scale predictable square-profile schedules)**.** *Fix $C_{\mathrm{lo}}, C_{\mathrm{hi}} \geq 1$, and put*

$$\gamma_{\mathrm{lo}} := \frac{3C_{\mathrm{lo}}}{16}, \qquad \gamma_{\mathrm{hi}} := \frac{3C_{\mathrm{hi}}}{16}.$$

For each $m \geq 1$, let $D_m \geq 2$, let $J_m \geq 1$, and, for $1 \leq h \leq J_m$, let $1 \leq \kappa_{m,h} \leq D_m$. Put

$$\ell^{\kappa}_{m,h} := \log(e\kappa_{m,h}), \qquad \ell^{D}_{m} := \log(eD_m).$$

For every active pair (m,h), suppose that

$$A_{m,h,i} = B_{m,h,i} + R_{m,h,i}, \qquad 1 \leq i \leq n_m,$$

is a two-scale predictable stream as in Corollary 7.200, with core rank envelope $\kappa_{m,h}$. Let

$$\nu^{\mathrm{lo}}_{m,h} := \left\| \sum_i \mathbb{E}(B^2_{m,h,i} \mid \mathcal{F}_{m,h,i-1}) \right\|_{\mathrm{op}}, \quad \nu^{\mathrm{hi}}_{m,h} := \left\| \sum_i \mathbb{E}(R^2_{m,h,i} \mid \mathcal{F}_{m,h,i-1}) \right\|_{\mathrm{op}}.$$

Assume that

$$\nu^{\mathrm{lo}}_{m,h} \leq \frac{C_{\mathrm{lo}} n_m}{2\ell^{\kappa}_{m,h}}, \qquad \nu^{\mathrm{hi}}_{m,h} \leq \frac{C_{\mathrm{hi}} n_m}{2\ell^{D}_{m}} \qquad (1 \leq h \leq J_m)$$

outside events $E^{\mathrm{lo}}_{m,h}$ and $E^{\mathrm{hi}}_{m,h}$, and

$$\sum_m \sum_{h=1}^{J_m} \mathbb{P}(E^{\mathrm{lo}}_{m,h} \cup E^{\mathrm{hi}}_{m,h}) < \infty,$$

$$\sum_m \sum_{h=1}^{J_m} \kappa_{m,h} \exp\left\{ -\gamma_{\mathrm{lo}} \frac{n_m}{\ell^{\kappa}_{m,h}} \right\} < \infty, \qquad \sum_m J_m D_m \exp\left\{ -\gamma_{\mathrm{hi}} \frac{n_m}{\ell^{D}_{m}} \right\} < \infty.$$

Then, with probability one, for all sufficiently large m, every active two-scale stream has a Spencer-scale signing:

$$\operatorname{disc}(A_{m,h,1}, \ldots, A_{m,h,n_m}) \leq C(C_{\mathrm{lo}}, C_{\mathrm{hi}})\sqrt{n_m} \qquad (1 \leq h \leq J_m).$$

If the streams are supplied with independent sampled sign batches and

$$\sum_m \sum_{h=1}^{J_m} 2^{-M_{m,h}} < \infty,$$

then the same eventual statement holds for the sampled certificates. If one common batch of M_m sign vectors is reused across the active menu, it is enough that

$$\sum_m J_m 2^{-M_m} < \infty.$$

A concrete sufficient growth condition is the following. Define

$$\Lambda^{\kappa}_{m} := \sum_{h=1}^{J_m} \kappa_{m,h}, \qquad \ell^{\kappa}_{m} := \max_{1 \leq h \leq J_m} \ell^{\kappa}_{m,h}.$$

If, for some $\delta > 0$,

$$n_m \geq \max\left\{\gamma_{\mathrm{lo}}^{-1}\ell_m^{\kappa}\big(\log(e\Lambda_m^{\kappa}) + (1+\delta)\log(e+m)\big), \gamma_{\mathrm{hi}}^{-1}\ell_m^{D}\big(\log(eJ_mD_m) + (1+\delta)\log(e+m)\big)\right\}$$

eventually, then the two structural series above are summable. With a common sampled batch, it is enough in addition that

$$M_m \geq \log_2(eJ_m) + (1+\delta')\log_2(e+m)$$

eventually for some $\delta' > 0$. *If the streams are defined for all* $h \geq 1$, $J_m \to \infty$, *and* $h \leq J_m$ *is a deterministic exhaustion, then every fixed two-scale predictable random-matrix model in the countable library is eventually covered.*

Proof. For a fixed active pair (m,h), apply Corollary 7.200 with half-budget choices. The failure probability is at most

$$\mathbb{P}(E_{m,h}^{\mathrm{lo}} \cup E_{m,h}^{\mathrm{hi}}) + \kappa_{m,h}\exp\left\{-\gamma_{\mathrm{lo}}\frac{n_m}{\ell_{m,h}^{\kappa}}\right\} + D_m\exp\left\{-\gamma_{\mathrm{hi}}\frac{n_m}{\ell_m^{D}}\right\}.$$

Union over $1 \leq h \leq J_m$, then apply the first Borel–Cantelli lemma. The sampled-sign assertions add either $\sum_h 2^{-M_{m,h}}$ or the common-batch union bound $J_m 2^{-M_m}$.

For the concrete schedule, the low-rank structural cost is bounded by

$$\Lambda_m^{\kappa}\exp\left\{-\gamma_{\mathrm{lo}}\frac{n_m}{\ell_m^{\kappa}}\right\} \leq e^{-1}(e+m)^{-(1+\delta)},$$

and the ambient residual cost is bounded by

$$J_mD_m\exp\left\{-\gamma_{\mathrm{hi}}\frac{n_m}{\ell_m^{D}}\right\} \leq e^{-1}(e+m)^{-(1+\delta)}.$$

The sampled-batch condition is summable by the displayed lower bound on M_m. If $J_m \to \infty$, every fixed countable-menu element is active eventually. □

Proposition 7.202 (Finite multiscale square-function certificate)**.** *There is a universal constant* C *with the following property. Let* $D \geq 2$, *let* $Q \geq 1$, *let* $1 \leq \kappa_q \leq D$, *and, for* $1 \leq q \leq Q$, *let* $P^{(q)}$ *be an orthogonal projection on* $\mathbb{C}^D$ *with* $\operatorname{rank} P^{(q)} \leq \kappa_q$. *Suppose*

$$A_i = R_i + \sum_{q=1}^{Q} B_i^{(q)}, \qquad 1 \leq i \leq n,$$

where all matrices are self-adjoint and

$$B_i^{(q)} = P^{(q)}B_i^{(q)}P^{(q)} \qquad (1 \leq q \leq Q,\ 1 \leq i \leq n).$$

Put

$$\Theta_Q(B,R) := \sum_{q=1}^{Q}\sqrt{\log(e\kappa_q)}\left\|\sum_{i=1}^{n}(B_i^{(q)})^2\right\|_{\mathrm{op}}^{1/2} + \sqrt{\log(eD)}\left\|\sum_{i=1}^{n}R_i^2\right\|_{\mathrm{op}}^{1/2}.$$

Then

$$\operatorname{disc}(A_1,\dots,A_n) \leq C\,\Theta_Q(B,R).$$

Moreover, if $\eta^{(1)}, \dots, \eta^{(m)}$ are independent uniform sign vectors in $\{\pm 1\}^n$, then, conditionally on the fixed decomposition,

$$\mathbb{P}_\eta \left\{ \min_{1 \le r \le m} \left\| \sum_{i=1}^n \eta_i^{(r)} A_i \right\|_{\mathrm{op}} > 2C\, \Theta_Q(B, R) \right\} \le 2^{-m}.$$

In particular, if

$$\left\| \sum_i (B_i^{(q)})^2 \right\|_{\mathrm{op}} \le \frac{a_q n}{\log(e\kappa_q)} \quad (1 \le q \le Q), \qquad \left\| \sum_i R_i^2 \right\|_{\mathrm{op}} \le \frac{a_* n}{\log(eD)},$$

then

$$\operatorname{disc}(A_1, \dots, A_n) \le C \left(\sum_{q=1}^Q \sqrt{a_q} + \sqrt{a_*} \right) \sqrt{n},$$

and the same threshold is found by m sampled sign vectors except with probability at most 2^{-m}.

Proof. For one uniform sign vector η, the triangle inequality gives

$$\mathbb{E}_\eta \left\| \sum_i \eta_i A_i \right\|_{\mathrm{op}} \le \sum_{q=1}^Q \mathbb{E}_\eta \left\| \sum_i \eta_i B_i^{(q)} \right\|_{\mathrm{op}} + \mathbb{E}_\eta \left\| \sum_i \eta_i R_i \right\|_{\mathrm{op}}.$$

For each q, the tuple $B_i^{(q)}$ acts on $\operatorname{ran} P^{(q)}$, whose dimension is at most κ_q. The rank-sensitive noncommutative Khintchine estimate from Proposition 7.143 therefore bounds the q-th term by

$$C \sqrt{\log(e\kappa_q)} \left\| \sum_i (B_i^{(q)})^2 \right\|_{\mathrm{op}}^{1/2}.$$

The residual term is bounded in the ambient dimension D by the same inequality. Thus the expected signed norm is at most $C\Theta_Q(B, R)$, so some signing has that norm. Markov's inequality gives success probability at least $1/2$ for one sampled sign vector at threshold $2C\Theta_Q(B, R)$, and independence gives 2^{-m} for m trials. The final display follows by substituting the square-function budgets into $\Theta_Q(B, R)$. □

Corollary 7.203 (Finite multiscale predictable square-profile ensembles)**.** *Let $D \ge 2$, let $Q \ge 1$, let $1 \le \kappa_q \le D$, and let $(\mathcal{F}_i)_{i=0}^n$ be a filtration. For $1 \le q \le Q$, let $P^{(q)}$ be an $\mathcal{F}_0$-measurable orthogonal projection with* $\operatorname{rank} P^{(q)} \le \kappa_q$. *Suppose*

$$A_i = R_i + \sum_{q=1}^Q B_i^{(q)}, \qquad 1 \le i \le n,$$

where R_i and $B_i^{(q)}$ are $\mathcal{F}_i$-measurable self-adjoint contractions and $B_i^{(q)} = P^{(q)} B_i^{(q)} P^{(q)}$. Define

$$\nu_q := \left\| \sum_{i=1}^{n} \mathbb{E}((B_i^{(q)})^2 \mid \mathcal{F}_{i-1}) \right\|_{\mathrm{op}}, \qquad \nu_* := \left\| \sum_{i=1}^{n} \mathbb{E}(R_i^2 \mid \mathcal{F}_{i-1}) \right\|_{\mathrm{op}}.$$

Let $a_1, \dots, a_Q, a_* > 0$. Whenever

$$v_q + t_q \le \frac{a_q n}{\log(e\kappa_q)} \quad (1 \le q \le Q), \qquad v_* + t_* \le \frac{a_* n}{\log(eD)},$$

one has

$$\begin{aligned} \mathbb{P}\left\{ \operatorname{disc}(A_1, \dots, A_n) > C \left(\sum_{q=1}^{Q} \sqrt{a_q} + \sqrt{a_*} \right) \sqrt{n} \right\} \\ \le \sum_{q=1}^{Q} \left[\mathbb{P}\{\nu_q > v_q\} + \kappa_q \exp\left\{ -\frac{t_q^2}{2(v_q + t_q/3)} \right\} \right] \\ + \mathbb{P}\{\nu_* > v_*\} + D \exp\left\{ -\frac{t_*^2}{2(v_* + t_*/3)} \right\}. \end{aligned}$$

If $\eta^{(1)}, \dots, \eta^{(m)}$ are independent uniform sign vectors independent of $\mathcal{F}_n$, then the same bound holds for sampled sign search after adding 2^{-m}.

In particular, if the half-budget inequalities

$$\nu_q \le \frac{a_q n}{2\log(e\kappa_q)} \quad (1 \le q \le Q), \qquad \nu_* \le \frac{a_* n}{2\log(eD)}$$

hold outside events E_q and E_*, then the failure probability is at most

$$\mathbb{P}\left(E_* \cup \bigcup_{q=1}^{Q} E_q \right) + \sum_{q=1}^{Q} \kappa_q \exp\left\{ -\frac{3a_q n}{16\log(e\kappa_q)} \right\} + D \exp\left\{ -\frac{3a_* n}{16\log(eD)} \right\},$$

with an additional 2^{-m} for sampled sign search.

Thus finite-depth mixtures of adaptive low-rank cores and one full ambient residual are Matrix-Spencer scale whenever the total multiscale budget $\sum_q \sqrt{a_q} + \sqrt{a_*}$ is bounded.

Proof. For each q, apply the proof of Corollary 7.198 to the adapted tuple $B_i^{(q)}$ on the κ_q-dimensional support $\operatorname{ran} P^{(q)}$. On $\{\nu_q \le v_q\}$, except with probability at most

$$\kappa_q \exp\left\{ -\frac{t_q^2}{2(v_q + t_q/3)} \right\},$$

one has

$$\left\| \sum_i (B_i^{(q)})^2 \right\|_{\mathrm{op}} \le v_q + t_q.$$

Apply the proof of Corollary 7.153 to the ambient residual tuple R_i; on $\{\nu_* \le v_*\}$, except with probability at most

$$D \exp\left\{-\frac{t_*^2}{2(v_* + t_*/3)}\right\},$$

one has $\|\sum_i R_i^2\|_{\mathrm{op}} \le v_* + t_*$. On the intersection of all these square-function events, Proposition 7.202 gives the displayed discrepancy bound. The sampled-search statement follows by conditioning on the realized multiscale decomposition and applying the sampled part of that proposition. The half-budget form is obtained by taking

$$v_q = t_q = \frac{a_q n}{2\log(e\kappa_q)} \quad (1 \le q \le Q), \qquad v_* = t_* = \frac{a_* n}{2\log(eD)}.$$

□

Corollary 7.204 (Finite-menu multiscale predictable schedules)**.** *For each $m \ge 1$, let $D_m \ge 2$ and $J_m \ge 1$. For every active (m,h), let $Q_{m,h} \ge 1$, let*

$$1 \le \kappa_{m,h,q} \le D_m \qquad (1 \le q \le Q_{m,h}),$$

and suppose

$$A_{m,h,i} = R_{m,h,i} + \sum_{q=1}^{Q_{m,h}} B_{m,h,i}^{(q)}, \qquad 1 \le i \le n_m,$$

is a finite multiscale predictable stream as in Corollary 7.203. Let

$$\nu_{m,h,q} := \left\| \sum_i \mathbb{E}((B_{m,h,i}^{(q)})^2 \mid \mathcal{F}_{m,h,i-1}) \right\|_{\mathrm{op}}, \qquad \nu_{m,h,*} := \left\| \sum_i \mathbb{E}(R_{m,h,i}^2 \mid \mathcal{F}_{m,h,i-1}) \right\|_{\mathrm{op}}.$$

Let

$$a_{m,h,q} > 0, \qquad a_{m,h,*} > 0,$$

and assume the uniform multiscale size bound

$$S_* := \sup_{m,h} \left(\sum_{q=1}^{Q_{m,h}} \sqrt{a_{m,h,q}} + \sqrt{a_{m,h,*}} \right) < \infty.$$

Assume that the half-budget profile inequalities

$$\nu_{m,h,q} \le \frac{a_{m,h,q} n_m}{2\log(e\kappa_{m,h,q})} \qquad (1 \le q \le Q_{m,h}), \qquad \nu_{m,h,*} \le \frac{a_{m,h,*} n_m}{2\log(eD_m)}$$

hold outside events $E_{m,h,q}$ and $E_{m,h,*}$, and that

$$\sum_m \sum_{h=1}^{J_m} \mathbb{P}\left(E_{m,h,*} \cup \bigcup_{q=1}^{Q_{m,h}} E_{m,h,q} \right) < \infty,$$

$$\sum_m \sum_{h=1}^{J_m} \sum_{q=1}^{Q_{m,h}} \kappa_{m,h,q} \exp\left\{-\frac{3 a_{m,h,q} n_m}{16\log(e\kappa_{m,h,q})}\right\} < \infty,$$

and

$$\sum_{m}\sum_{h=1}^{J_m} D_m \exp\left\{-\frac{3a_{m,h,*}n_m}{16\log(eD_m)}\right\} < \infty.$$

Then, with probability one, for all sufficiently large m, every active multiscale stream has a Spencer-scale signing:

$$\operatorname{disc}(A_{m,h,1},\dots,A_{m,h,n_m}) \le CS_*\sqrt{n_m} \qquad (1 \le h \le J_m).$$

Sampled sign certificates are eventual almost surely if independent batches satisfy

$$\sum_{m}\sum_{h=1}^{J_m} 2^{-M_{m,h}} < \infty,$$

or, for one common active-menu batch, if

$$\sum_{m} J_m 2^{-M_m} < \infty.$$

For a concrete schedule, suppose $a_{m,h,q} \ge a_{\mathrm{lo}} > 0$ and $a_{m,h,*} \ge a_{\mathrm{hi}} > 0$. Put

$$\Lambda_m^{\mathrm{ms}} := \sum_{h=1}^{J_m}\sum_{q=1}^{Q_{m,h}} \kappa_{m,h,q}, \qquad \ell_m^{\mathrm{ms}} := \max_{h,q} \log(e\kappa_{m,h,q}), \qquad \ell_m^{D} := \log(eD_m).$$

Then the two structural series are summable whenever, for some $\delta > 0$,

$$n_m \ge \max\left\{\frac{16}{3a_{\mathrm{lo}}}\,\ell_m^{\mathrm{ms}}\big(\log(e\Lambda_m^{\mathrm{ms}}) + (1+\delta)\log(e+m)\big), \frac{16}{3a_{\mathrm{hi}}}\,\ell_m^{D}\big(\log(eJ_mD_m) + (1+\delta)\log(e+m)\big)\right\}$$

eventually. A common sampled batch is summable under

$$M_m \ge \log_2(eJ_m) + (1+\delta')\log_2(e+m)$$

eventually. If $J_m \to \infty$, the active menus exhaust a countable library of finite-depth multiscale random-matrix models.

Proof. Apply Corollary 7.203 to each active pair (m,h) with half-budget choices and union over $1 \le h \le J_m$. The three displayed summability hypotheses make the resulting active-row failure probabilities summable, so Borel–Cantelli gives the eventual discrepancy bound. Sampled signs add the displayed search failure series.

Under the concrete schedule,

$$\Lambda_m^{\mathrm{ms}} \exp\left\{-\frac{3a_{\mathrm{lo}}n_m}{16\ell_m^{\mathrm{ms}}}\right\} \le e^{-1}(e+m)^{-(1+\delta)}$$

and

$$J_mD_m \exp\left\{-\frac{3a_{\mathrm{hi}}n_m}{16\ell_m^{D}}\right\} \le e^{-1}(e+m)^{-(1+\delta)}.$$

The common sampled-batch estimate is identical to the two-scale schedule. □

Proposition 7.205 (Summable multiscale square-function certificate)**.** *There is a universal constant C with the following property. Let $D \geq 2$. For $q \geq 1$, let $1 \leq \kappa_q \leq D$, and let $P^{(q)}$ be an orthogonal projection on $\mathbb{C}^D$ with* $\operatorname{rank} P^{(q)} \leq \kappa_q$*. Suppose*

$$A_i = R_i + \sum_{q \geq 1} B_i^{(q)}, \qquad 1 \leq i \leq n,$$

where all matrices are self-adjoint, the series converges in operator norm for each i, and

$$B_i^{(q)} = P^{(q)} B_i^{(q)} P^{(q)} \qquad (q \geq 1,\ 1 \leq i \leq n).$$

If

$$\Theta_\infty(B, R) := \sum_{q \geq 1} \sqrt{\log(e\kappa_q)} \left\| \sum_{i=1}^n (B_i^{(q)})^2 \right\|_{\mathrm{op}}^{1/2} + \sqrt{\log(eD)} \left\| \sum_{i=1}^n R_i^2 \right\|_{\mathrm{op}}^{1/2} < \infty,$$

then

$$\operatorname{disc}(A_1, \dots, A_n) \leq C\, \Theta_\infty(B, R).$$

Moreover, if $\eta^{(1)}, \dots, \eta^{(m)}$ are independent uniform sign vectors in $\{\pm 1\}^n$, then, conditionally on the fixed decomposition,

$$\mathbb{P}_\eta \left\{ \min_{1 \leq r \leq m} \left\| \sum_{i=1}^n \eta_i^{(r)} A_i \right\|_{\mathrm{op}} > 2C\, \Theta_\infty(B, R) \right\} \leq 2^{-m}.$$

In particular, if

$$\left\| \sum_i (B_i^{(q)})^2 \right\|_{\mathrm{op}} \leq \frac{a_q n}{\log(e\kappa_q)} \quad (q \geq 1), \qquad \left\| \sum_i R_i^2 \right\|_{\mathrm{op}} \leq \frac{a_* n}{\log(eD)}$$

with $\sum_q \sqrt{a_q} < \infty$, then

$$\operatorname{disc}(A_1, \dots, A_n) \leq C \left(\sum_{q \geq 1} \sqrt{a_q} + \sqrt{a_*} \right) \sqrt{n},$$

and m sampled sign vectors fail to find that threshold with probability at most 2^{-m}.

Proof. For $Q \geq 1$, put

$$A_i^{(Q)} = R_i + \sum_{q=1}^{Q} B_i^{(q)}.$$

By Proposition 7.202,

$$\mathbb{E}_\eta \left\| \sum_i \eta_i A_i^{(Q)} \right\|_{\mathrm{op}} \leq C \left[\sum_{q=1}^{Q} \sqrt{\log(e\kappa_q)} \left\| \sum_i (B_i^{(q)})^2 \right\|_{\mathrm{op}}^{1/2} + \sqrt{\log(eD)} \left\| \sum_i R_i^2 \right\|_{\mathrm{op}}^{1/2} \right].$$

For every fixed sign vector, the operator-norm convergence of the layer series implies

$$\sum_i \eta_i A_i^{(Q)} \longrightarrow \sum_i \eta_i A_i.$$

Fatou's lemma therefore gives

$$\mathbb{E}_\eta \left\| \sum_i \eta_i A_i \right\|_{\mathrm{op}} \le C\,\Theta_\infty(B,R).$$

The minimum over all signs is bounded by this expectation. Markov's inequality gives one-sample success probability at least $1/2$ at threshold $2C\Theta_\infty(B,R)$, and independent trials give the 2^{-m} failure bound. The final displayed estimate is the square-profile specialization of $\Theta_\infty(B,R)$. □

Corollary 7.206 (Summable multiscale predictable square-profile ensembles)**.** *Let $D \ge 2$, and let $(\mathcal{F}_i)_{i=0}^n$ be a filtration. For $q \ge 1$, let $1 \le \kappa_q \le D$, and let $P^{(q)}$ be an $\mathcal{F}_0$-measurable orthogonal projection with* $\operatorname{rank} P^{(q)} \le \kappa_q$*. Suppose*

$$A_i = R_i + \sum_{q\ge 1} B_i^{(q)}, \qquad 1 \le i \le n,$$

where R_i and $B_i^{(q)}$ are $\mathcal{F}_i$-measurable self-adjoint contractions, the series converges in operator norm for each i, and

$$B_i^{(q)} = P^{(q)} B_i^{(q)} P^{(q)}.$$

Define

$$\nu_q := \left\| \sum_{i=1}^n \mathbb{E}\big((B_i^{(q)})^2 \mid \mathcal{F}_{i-1}\big) \right\|_{\mathrm{op}}, \qquad \nu_* := \left\| \sum_{i=1}^n \mathbb{E}(R_i^2 \mid \mathcal{F}_{i-1}) \right\|_{\mathrm{op}}.$$

Let $a_* > 0$ and $a_q > 0$, $q \ge 1$, with

$$A_{\mathrm{ms}} := \sum_{q\ge 1} \sqrt{a_q} + \sqrt{a_*} < \infty.$$

Whenever

$$v_q + t_q \le \frac{a_q n}{\log(e\kappa_q)} \quad (q \ge 1), \qquad v_* + t_* \le \frac{a_* n}{\log(eD)},$$

one has

$$\begin{aligned} &\mathbb{P}\left\{\operatorname{disc}(A_1,\dots,A_n) > C A_{\mathrm{ms}} \sqrt{n}\right\} \\ &\quad \le \sum_{q\ge 1} \left[\mathbb{P}\{\nu_q > v_q\} + \kappa_q \exp\left\{ -\frac{t_q^2}{2(v_q + t_q/3)} \right\} \right] \\ &\qquad + \mathbb{P}\{\nu_* > v_*\} + D \exp\left\{ -\frac{t_*^2}{2(v_* + t_*/3)} \right\}. \end{aligned}$$

If $\eta^{(1)},\dots,\eta^{(m)}$ are independent uniform sign vectors independent of $\mathcal{F}_n$, then the same bound holds for sampled sign search after adding 2^{-m}.

In particular, if the half-budget inequalities

$$\nu_q \le \frac{a_q n}{2\log(e\kappa_q)} \quad (q \ge 1), \qquad \nu_* \le \frac{a_* n}{2\log(eD)}$$

hold outside events E_q and E_*, then the failure probability is at most

$$\mathbb{P}\left(E_* \cup \bigcup_{q\ge 1} E_q\right) + \sum_{q\ge 1} \kappa_q \exp\left\{-\frac{3a_q n}{16\log(e\kappa_q)}\right\} + D\exp\left\{-\frac{3a_* n}{16\log(eD)}\right\},$$

with an additional 2^{-m} for sampled sign search.

Proof. For every fixed q, the proof of Corollary 7.198 gives, on $\{\nu_q \le v_q\}$, except with probability at most

$$\kappa_q \exp\left\{-\frac{t_q^2}{2(v_q + t_q/3)}\right\},$$

the bound

$$\left\|\sum_i (B_i^{(q)})^2\right\|_{\mathrm{op}} \le v_q + t_q.$$

The proof of Corollary 7.153 gives the analogous residual event

$$\left\|\sum_i R_i^2\right\|_{\mathrm{op}} \le v_* + t_*$$

outside the displayed ambient tail. A countable union bound controls the complement of the intersection of all layer and residual square-function events. On that intersection, Proposition 7.205 gives the discrepancy bound. The sampled-search statement follows by conditioning on the realized decomposition. The half-budget specialization is the choice

$$v_q = t_q = \frac{a_q n}{2\log(e\kappa_q)} \quad (q \ge 1), \qquad v_* = t_* = \frac{a_* n}{2\log(eD)}.$$

□

Corollary 7.207 (Active menus of summable multiscale streams)**.** *For each $m \ge 1$, let $D_m \ge 2$ and $J_m \ge 1$. For each active (m,h), suppose*

$$A_{m,h,i} = R_{m,h,i} + \sum_{q\ge 1} B_{m,h,i}^{(q)}, \qquad 1 \le i \le n_m,$$

is a summable multiscale predictable stream as in Corollary 7.206, with rank envelopes $\kappa_{m,h,q}$. Let

$$\nu_{m,h,q} := \left\|\sum_i \mathbb{E}\big((B_{m,h,i}^{(q)})^2 \mid \mathcal{F}_{m,h,i-1}\big)\right\|_{\mathrm{op}}, \qquad \nu_{m,h,*} := \left\|\sum_i \mathbb{E}(R_{m,h,i}^2 \mid \mathcal{F}_{m,h,i-1})\right\|_{\mathrm{op}}.$$

Let

$$a_{m,h,q} > 0, \qquad a_{m,h,*} > 0,$$

and assume

$$S_\infty := \sup_{m,h}\left(\sum_{q\ge 1}\sqrt{a_{m,h,q}} + \sqrt{a_{m,h,*}}\right) < \infty.$$

Assume that the half-budget profile inequalities hold outside events $E_{m,h,q}$ and $E_{m,h,*}$:

$$\nu_{m,h,q} \le \frac{a_{m,h,q} n_m}{2\log(e\kappa_{m,h,q})} \quad (q \ge 1), \qquad \nu_{m,h,*} \le \frac{a_{m,h,*} n_m}{2\log(eD_m)}.$$

If

$$\sum_m \sum_{h=1}^{J_m} \mathbb{P}\left(E_{m,h,*} \cup \bigcup_{q\ge 1} E_{m,h,q} \right) < \infty,$$

$$\sum_m \sum_{h=1}^{J_m} \sum_{q\ge 1} \kappa_{m,h,q} \exp\left\{ -\frac{3a_{m,h,q} n_m}{16\log(e\kappa_{m,h,q})} \right\} < \infty,$$

and

$$\sum_m \sum_{h=1}^{J_m} D_m \exp\left\{ -\frac{3a_{m,h,*} n_m}{16\log(eD_m)} \right\} < \infty,$$

then, with probability one, for all sufficiently large m, every active summable-depth stream has a Spencer-scale signing:

$$\operatorname{disc}(A_{m,h,1}, \dots, A_{m,h,n_m}) \le C S_\infty \sqrt{n_m} \qquad (1 \le h \le J_m).$$

Sampled certificates are eventual almost surely if

$$\sum_m \sum_{h=1}^{J_m} 2^{-M_{m,h}} < \infty,$$

or, for one common active-menu batch, if

$$\sum_m J_m 2^{-M_m} < \infty.$$

If $J_m \to \infty$, the active menus may exhaust a countable library of summable-depth random-matrix models.

Proof. Apply Corollary 7.206 to each active pair (m,h) with the half-budget choices, then union over $1 \le h \le J_m$. The three displayed summability assumptions make the active-row failure probabilities summable. Borel–Cantelli gives the eventual simultaneous discrepancy bound. The sampled-sign assertions add either the independent-batch search series or the common-batch union bound $J_m 2^{-M_m}$. □

Corollary 7.208 (Flag-generated summable multiscale certificates)**.** *There is a universal constant C with the following property. Let $D \ge 2$, let*

$$0 = P^{(0)} \preceq P^{(1)} \preceq P^{(2)} \preceq \cdots$$

be an increasing sequence of orthogonal projections on $\mathbb{C}^D$, and suppose

$$\operatorname{rank} P^{(q)} \le \kappa_q \qquad (q \ge 1).$$

Let $P^{(\infty)}$ be the supremum projection. For self-adjoint $A_1, \dots, A_n \in \mathrm{M}_D(\mathbb{C})$, define the flag shells and the residual by

$$\Delta_i^{(q)} := P^{(q)} A_i P^{(q)} - P^{(q-1)} A_i P^{(q-1)}, \qquad R_i := A_i - P^{(\infty)} A_i P^{(\infty)}.$$

If

$$\Theta_{\mathrm{flag}} := \sum_{q\ge 1} \sqrt{\log(e\kappa_q)} \left\| \sum_{i=1}^{n} (\Delta_i^{(q)})^2 \right\|_{\mathrm{op}}^{1/2} + \sqrt{\log(eD)} \left\| \sum_{i=1}^{n} R_i^2 \right\|_{\mathrm{op}}^{1/2} < \infty,$$

then

$$\mathrm{disc}(A_1, \dots, A_n) \le C\, \Theta_{\mathrm{flag}}.$$

Moreover, if $\eta^{(1)}, \dots, \eta^{(m)}$ *are independent uniform sign vectors, then*

$$\mathbb{P}_\eta \left\{ \min_{1\le r\le m} \left\| \sum_{i=1}^{n} \eta_i^{(r)} A_i \right\|_{\mathrm{op}} > 2C\, \Theta_{\mathrm{flag}} \right\} \le 2^{-m}.$$

Consequently, whenever

$$\left\| \sum_i (\Delta_i^{(q)})^2 \right\|_{\mathrm{op}} \le \frac{a_q n}{\log(e\kappa_q)} \quad (q \ge 1), \qquad \left\| \sum_i R_i^2 \right\|_{\mathrm{op}} \le \frac{a_* n}{\log(eD)}$$

and $\sum_q \sqrt{a_q} < \infty$*, one has*

$$\mathrm{disc}(A_1, \dots, A_n) \le C \left(\sum_{q\ge 1} \sqrt{a_q} + \sqrt{a_*} \right) \sqrt{n},$$

with sampled-search failure at most 2^{-m} *at twice that threshold.*

Proof. For every $Q \ge 1$, the shells telescope:

$$P^{(Q)} A_i P^{(Q)} = \sum_{q=1}^{Q} \Delta_i^{(q)}.$$

Since the projections are increasing in finite dimension, $P^{(Q)}$ converges in operator norm to $P^{(\infty)}$. Hence

$$A_i = R_i + \sum_{q\ge 1} \Delta_i^{(q)}$$

with operator-norm convergence. Also $\Delta_i^{(q)} = P^{(q)} \Delta_i^{(q)} P^{(q)}$, so the q-th shell is supported in a κ_q-dimensional corner. Proposition 7.205 applied to this decomposition gives the discrepancy and sampled-search estimates. The square-profile specialization is immediate from the definition of Θ_{flag}. □

Corollary 7.209 (Predictable flag-compressible square-profile streams)**.** *Let* $D \ge 2$*, let* $(\mathcal{F}_i)_{i=0}^n$ *be a filtration, and let*

$$0 = P^{(0)} \preceq P^{(1)} \preceq P^{(2)} \preceq \cdots$$

be $\mathcal{F}_0$*-measurable orthogonal projections with* $\operatorname{rank} P^{(q)} \le \kappa_q$*. Let* $P^{(\infty)} = \sup_q P^{(q)}$*. Suppose* $A_1, \dots, A_n \in \mathrm{M}_D(\mathbb{C})$ *are self-adjoint contractions and* A_i *is* $\mathcal{F}_i$*-measurable. Define*

$$\Delta_i^{(q)} := P^{(q)} A_i P^{(q)} - P^{(q-1)} A_i P^{(q-1)}, \qquad R_i := A_i - P^{(\infty)} A_i P^{(\infty)},$$

and the predictable shell profiles

$$\nu_q := \left\| \sum_{i=1}^{n} \mathbb{E}\big((\Delta_i^{(q)})^2 \mid \mathcal{F}_{i-1}\big) \right\|_{\mathrm{op}}, \qquad \nu_* := \left\| \sum_{i=1}^{n} \mathbb{E}(R_i^2 \mid \mathcal{F}_{i-1}) \right\|_{\mathrm{op}}.$$

Let $a_* > 0$ and $a_q > 0$, $q \geq 1$, with

$$A_{\mathrm{flag}} := \sum_{q \geq 1} \sqrt{a_q} + \sqrt{a_*} < \infty.$$

Whenever

$$v_q + t_q \leq \frac{a_q n}{\log(e\kappa_q)} \quad (q \geq 1), \qquad v_* + t_* \leq \frac{a_* n}{\log(eD)},$$

one has

$$\begin{aligned} &\mathbb{P}\left\{\mathrm{disc}(A_1, \dots, A_n) > C A_{\mathrm{flag}} \sqrt{n}\right\} \\ &\quad \leq \sum_{q \geq 1} \left[\mathbb{P}\{\nu_q > v_q\} + \kappa_q \exp\left\{ -\frac{t_q^2}{8(v_q + t_q/3)} \right\} \right] \\ &\qquad + \mathbb{P}\{\nu_* > v_*\} + D \exp\left\{ -\frac{t_*^2}{8(v_* + t_*/3)} \right\}. \end{aligned}$$

If $\eta^{(1)}, \dots, \eta^{(m)}$ are independent uniform sign vectors independent of $\mathcal{F}_n$, then the same estimate holds for sampled sign search after adding 2^{-m}.

In particular, if the half-budget inequalities

$$\nu_q \leq \frac{a_q n}{2\log(e\kappa_q)} \quad (q \geq 1), \qquad \nu_* \leq \frac{a_* n}{2\log(eD)}$$

hold outside events E_q and E_*, then the failure probability is at most

$$\mathbb{P}\left(E_* \cup \bigcup_{q \geq 1} E_q \right) + \sum_{q \geq 1} \kappa_q \exp\left\{ -\frac{3a_q n}{64 \log(e\kappa_q)} \right\} + D \exp\left\{ -\frac{3a_* n}{64 \log(eD)} \right\},$$

with an additional 2^{-m} for sampled sign search.

Thus a predictable stream is Spencer-scale whenever its energy becomes summable across a fixed or random initial flag: one pays $\log(e\kappa_q)$ for the q-th compressed shell and the ambient logarithm only for the part outside the terminal flag subspace.

Proof. The shell and residual matrices are adapted, self-adjoint, and have operator norm at most 2. Fix a shell q, and write $Y_i = \Delta_i^{(q)}$, $B_i = Y_i^2$, and $M_i = \mathbb{E}(B_i \mid \mathcal{F}_{i-1})$. On the range of $P^{(q)}$, the martingale differences $X_i = B_i - M_i$ satisfy

$$\lambda_{\max}(X_i) \leq 4, \qquad \sum_i \mathbb{E}(X_i^2 \mid \mathcal{F}_{i-1}) \preceq 4 \sum_i M_i,$$

because $0 \preceq B_i \preceq 4I$ implies $B_i^2 \preceq 4B_i$. Matrix Freedman in dimension at most κ_q gives

$$\mathbb{P}\left\{\left\|\sum_i (\Delta_i^{(q)})^2\right\|_{\mathrm{op}} > v_q + t_q \text{ and } \nu_q \le v_q\right\} \le \kappa_q \exp\left\{-\frac{t_q^2}{8(v_q + t_q/3)}\right\}.$$

The same argument in the ambient dimension D, with $Y_i = R_i$, gives the residual tail. A countable union bound controls the complement of the event on which all shell and residual square functions are bounded by $v_q + t_q$ and $v_* + t_*$. On that event, Corollary 7.208 gives the discrepancy bound. Conditioning on the realized tuple and applying the sampled-search part of that corollary adds 2^{-m}. The half-budget form is obtained by taking

$$v_q = t_q = \frac{a_q n}{2\log(e\kappa_q)} \quad (q \ge 1), \qquad v_* = t_* = \frac{a_* n}{2\log(eD)}.$$

□

Corollary 7.210 (Active menus of predictable flag-compressible streams)**.** *For each $m \ge 1$, let $D_m \ge 2$ and $J_m \ge 1$. For every active (m,h), let*

$$0 = P_{m,h}^{(0)} \preceq P_{m,h}^{(1)} \preceq P_{m,h}^{(2)} \preceq \cdots$$

be an $\mathcal{F}_{m,h,0}$-measurable flag in $\mathrm{M}_{D_m}(\mathbb{C})$, with

$$\operatorname{rank} P_{m,h}^{(q)} \le \kappa_{m,h,q}.$$

Let $A_{m,h,1}, \dots, A_{m,h,n_m}$ be adapted self-adjoint contractions, and let $\nu_{m,h,q}$ and $\nu_{m,h,*}$ be the flag-shell and residual predictable profiles from Corollary 7.209. Let

$$a_{m,h,q} > 0, \qquad a_{m,h,*} > 0,$$

and assume

$$S_{\mathrm{flag}} := \sup_{m,h}\left(\sum_{q\ge1}\sqrt{a_{m,h,q}} + \sqrt{a_{m,h,*}}\right) < \infty.$$

Assume that the half-budget profile inequalities hold outside events $E_{m,h,q}$ and $E_{m,h,*}$:

$$\nu_{m,h,q} \le \frac{a_{m,h,q} n_m}{2\log(e\kappa_{m,h,q})} \quad (q \ge 1), \qquad \nu_{m,h,*} \le \frac{a_{m,h,*} n_m}{2\log(eD_m)}.$$

If

$$\sum_m \sum_{h=1}^{J_m} \mathbb{P}\left(E_{m,h,*} \cup \bigcup_{q\ge1} E_{m,h,q}\right) < \infty,$$

$$\sum_m \sum_{h=1}^{J_m} \sum_{q\ge1} \kappa_{m,h,q} \exp\left\{-\frac{3a_{m,h,q} n_m}{64\log(e\kappa_{m,h,q})}\right\} < \infty,$$

and

$$\sum_m \sum_{h=1}^{J_m} D_m \exp\left\{-\frac{3a_{m,h,*} n_m}{64\log(eD_m)}\right\} < \infty,$$

then, with probability one, for all sufficiently large m, every active flag-compressible stream has a Spencer-scale signing:

$$\operatorname{disc}(A_{m,h,1},\dots,A_{m,h,n_m}) \le CS_{\mathrm{flag}}\sqrt{n_m} \qquad (1 \le h \le J_m).$$

Sampled certificates are eventual almost surely if

$$\sum_m \sum_{h=1}^{J_m} 2^{-M_{m,h}} < \infty,$$

or, for one common active-menu batch, if

$$\sum_m J_m 2^{-M_m} < \infty.$$

If $J_m \to \infty$, the active menus may exhaust a countable library of flag-compressible random-matrix models.

Proof. Apply Corollary 7.209 to each active pair (m,h) with the half-budget choices, then union over $1 \le h \le J_m$. The displayed summability assumptions make the active-row failure probabilities summable, so Borel–Cantelli gives the eventual simultaneous discrepancy bound. The sampled-sign assertions add either the independent-batch series or the common-batch union bound $J_m 2^{-M_m}$. □

Corollary 7.211 (Compression-error flag certificates)**.** *There is a universal constant C with the following property. Let $D \ge 2$, and let*

$$0 = P^{(0)} \preceq P^{(1)} \preceq P^{(2)} \preceq \cdots$$

be an increasing sequence of orthogonal projections on $\mathbb{C}^D$, with

$$\operatorname{rank} P^{(q)} \le \kappa_q \qquad (q \ge 1).$$

Let $P^{(\infty)} = \sup_q P^{(q)}$. Let $A_1,\dots,A_n \in \mathrm{M}_D(\mathbb{C})$ be self-adjoint contractions. Suppose that nonnegative numbers $\tau_{i,q}$ and $\tau_{i,*}$ satisfy

$$\left\|A_i - P^{(q)}A_iP^{(q)}\right\|_{\mathrm{op}} \le \tau_{i,q} \quad (q \ge 0), \qquad \left\|A_i - P^{(\infty)}A_iP^{(\infty)}\right\|_{\mathrm{op}} \le \tau_{i,*}.$$

Put

$$\Theta_{\mathrm{ce}} := \sum_{q\ge1} \sqrt{\log(e\kappa_q)} \left(\sum_{i=1}^n (\tau_{i,q-1}+\tau_{i,q})^2\right)^{1/2} + \sqrt{\log(eD)} \left(\sum_{i=1}^n \tau_{i,*}^2\right)^{1/2}.$$

If $\Theta_{\mathrm{ce}} < \infty$, then

$$\operatorname{disc}(A_1,\dots,A_n) \le C\,\Theta_{\mathrm{ce}},$$

and m independent sampled sign vectors fail to find the threshold $2C\Theta_{\mathrm{ce}}$ with probability at most 2^{-m}.

In particular, if $P^{(\infty)} = I$ and there are deterministic shell-error levels b_q such that $\tau_{i,q} \le b_q$ for every i,q, then

$$\operatorname{disc}(A_1,\dots,A_n) \le C\sqrt{n}\sum_{q\ge1}\sqrt{\log(e\kappa_q)}(b_{q-1}+b_q).$$

Thus any tuple that is uniformly compressible along a common flag is Spencer-scale whenever its weighted approximation-error increments are summable.

Proof. Let

$$\Delta_i^{(q)} = P^{(q)}A_iP^{(q)} - P^{(q-1)}A_iP^{(q-1)}, \qquad R_i = A_i - P^{(\infty)}A_iP^{(\infty)}.$$

Then

$$\|\Delta_i^{(q)}\|_{\mathrm{op}} \le \left\|A_i - P^{(q-1)}A_iP^{(q-1)}\right\|_{\mathrm{op}} + \left\|A_i - P^{(q)}A_iP^{(q)}\right\|_{\mathrm{op}} \le \tau_{i,q-1} + \tau_{i,q}.$$

Since $(\Delta_i^{(q)})^2 \preceq \|\Delta_i^{(q)}\|_{\mathrm{op}}^2 P^{(q)}$,

$$\left\|\sum_i (\Delta_i^{(q)})^2\right\|_{\mathrm{op}}^{1/2} \le \left(\sum_i (\tau_{i,q-1} + \tau_{i,q})^2\right)^{1/2}.$$

Similarly, $\|\sum_i R_i^2\|_{\mathrm{op}}^{1/2} \le (\sum_i \tau_{i,*}^2)^{1/2}$. Corollary 7.208 therefore gives the discrepancy and sampled-search bounds. The uniform compressibility specialization substitutes $\tau_{i,q} \le b_q$ and $\tau_{i,*} = 0$. □

Corollary 7.212 (Predictable compression-error flag streams)**.** *Let $D \ge 2$, let $(\mathcal{F}_i)_{i=0}^n$ be a filtration, and let*

$$0 = P^{(0)} \preceq P^{(1)} \preceq P^{(2)} \preceq \cdots$$

be an $\mathcal{F}_0$-measurable flag with $\operatorname{rank} P^{(q)} \le \kappa_q$. Let $P^{(\infty)} = \sup_q P^{(q)}$. Let $A_1, \dots, A_n$ be adapted self-adjoint contractions. Suppose that nonnegative $\mathcal{F}_{i-1}$-measurable variables $\tau_{i,q}$ and $\tau_{i,*}$ satisfy

$$\left\|A_i - P^{(q)}A_iP^{(q)}\right\|_{\mathrm{op}} \le \tau_{i,q} \quad (q \ge 0), \qquad \left\|A_i - P^{(\infty)}A_iP^{(\infty)}\right\|_{\mathrm{op}} \le \tau_{i,*}.$$

Define the predictable compression-error budgets

$$U_q := \sum_{i=1}^n (\tau_{i,q-1} + \tau_{i,q})^2, \qquad U_* := \sum_{i=1}^n \tau_{i,*}^2.$$

Let $a_* > 0$ and $a_q > 0$, $q \ge 1$, with

$$A_{\mathrm{ce}} := \sum_{q \ge 1} \sqrt{a_q} + \sqrt{a_*} < \infty.$$

Assume that the half-budget inequalities

$$U_q \le \frac{a_q n}{2\log(e\kappa_q)} \quad (q \ge 1), \qquad U_* \le \frac{a_* n}{2\log(eD)}$$

hold outside events E_q and E_*. Then

$$\mathbb{P}\left\{\operatorname{disc}(A_1, \dots, A_n) > CA_{\mathrm{ce}}\sqrt{n}\right\}$$
$$\le \mathbb{P}\left(E_* \cup \bigcup_{q \ge 1} E_q\right) + \sum_{q \ge 1} \kappa_q \exp\left\{-\frac{3a_q n}{64\log(e\kappa_q)}\right\} + D\exp\left\{-\frac{3a_* n}{64\log(eD)}\right\}.$$

If $\eta^{(1)}, \dots, \eta^{(m)}$ are independent uniform sign vectors independent of $\mathcal{F}_n$, then the same bound holds for sampled sign search after adding 2^{-m}.

Proof. Use the notation $\Delta_i^{(q)}$ and R_i from the previous proof. The same triangle inequality gives $\|\Delta_i^{(q)}\|_{\mathrm{op}} \le \tau_{i,q-1} + \tau_{i,q}$, hence

$$\mathbb{E}\big((\Delta_i^{(q)})^2 \mid \mathcal{F}_{i-1}\big) \preceq (\tau_{i,q-1} + \tau_{i,q})^2 P^{(q)}.$$

Therefore the shell profile ν_q in Corollary 7.209 is at most U_q, and the residual profile ν_* is at most U_*. Applying that corollary with

$$v_q = t_q = \frac{a_q n}{2\log(e\kappa_q)}, \qquad v_* = t_* = \frac{a_* n}{2\log(eD)}$$

gives the displayed estimate and the sampled-search variant. □

Corollary 7.213 (Active menus of predictable compression-error flag streams)**.** *For each $m \ge 1$, let $D_m \ge 2$ and $J_m \ge 1$. For every active (m,h), let $A_{m,h,1}, \dots, A_{m,h,n_m}$ be an adapted self-adjoint contraction stream with an initial flag*

$$0 = P_{m,h}^{(0)} \preceq P_{m,h}^{(1)} \preceq P_{m,h}^{(2)} \preceq \cdots, \qquad \operatorname{rank} P_{m,h}^{(q)} \le \kappa_{m,h,q}.$$

Let $U_{m,h,q}$ and $U_{m,h,*}$ be predictable compression-error budgets as in Corollary 7.212. Let $a_{m,h,q} > 0$ and $a_{m,h,*} > 0$, and assume

$$S_{\mathrm{ce}} := \sup_{m,h} \left(\sum_{q\ge1} \sqrt{a_{m,h,q}} + \sqrt{a_{m,h,*}} \right) < \infty.$$

Suppose the half-budget inequalities

$$U_{m,h,q} \le \frac{a_{m,h,q} n_m}{2\log(e\kappa_{m,h,q})} \quad (q \ge 1), \qquad U_{m,h,*} \le \frac{a_{m,h,*} n_m}{2\log(eD_m)}$$

hold outside events $E_{m,h,q}$ and $E_{m,h,*}$. If

$$\sum_m \sum_{h=1}^{J_m} \mathbb{P}\left(E_{m,h,*} \cup \bigcup_{q\ge1} E_{m,h,q} \right) < \infty,$$

$$\sum_m \sum_{h=1}^{J_m} \sum_{q\ge1} \kappa_{m,h,q} \exp\left\{ -\frac{3 a_{m,h,q} n_m}{64\log(e\kappa_{m,h,q})} \right\} < \infty,$$

and

$$\sum_m \sum_{h=1}^{J_m} D_m \exp\left\{ -\frac{3 a_{m,h,*} n_m}{64\log(eD_m)} \right\} < \infty,$$

then, with probability one, for all sufficiently large m, every active compression-error flag stream has a Spencer-scale signing:

$$\operatorname{disc}(A_{m,h,1}, \dots, A_{m,h,n_m}) \le C S_{\mathrm{ce}} \sqrt{n_m} \qquad (1 \le h \le J_m).$$

Sampled certificates are eventual almost surely if

$$\sum_m \sum_{h=1}^{J_m} 2^{-M_{m,h}} < \infty,$$

or, for one common active-menu batch, if

$$\sum_m J_m 2^{-M_m} < \infty.$$

Proof. Apply Corollary 7.212 to each active pair (m,h) and union over $1 \le h \le J_m$. The displayed summability assumptions make the active-row failure probabilities summable, so Borel–Cantelli gives the simultaneous eventual bound. The sampled-sign assertions add the independent-batch or common-batch search series. □

Corollary 7.214 (Common-flag tail envelopes)**.** *There is a universal constant C with the following property. Let $D \ge 2$, let*

$$0 = P^{(0)} \preceq P^{(1)} \preceq P^{(2)} \preceq \cdots$$

be an increasing flag on $\mathbb{C}^D$, and suppose $\operatorname{rank} P^{(q)} \le \kappa_q$. *Let $P^{(\infty)} = \sup_q P^{(q)}$. Let $b_q \ge 0$, and put*

$$B_{\mathrm{flag}}(b,\kappa) := \sum_{q\ge 1} \sqrt{\log(e\kappa_q)}(b_{q-1} + b_q).$$

Let $A_1, \dots, A_n \in \mathrm{M}_D(\mathbb{C})$ be self-adjoint contractions. Suppose that there are nonnegative amplitudes L_i and residual amplitudes r_i such that

$$\left\|A_i - P^{(q)} A_i P^{(q)}\right\|_{\mathrm{op}} \le L_i b_q \quad (q \ge 0), \qquad \left\|A_i - P^{(\infty)} A_i P^{(\infty)}\right\|_{\mathrm{op}} \le r_i.$$

If $B_{\mathrm{flag}}(b,\kappa) < \infty$, then

$$\operatorname{disc}(A_1, \dots, A_n) \le C \left[B_{\mathrm{flag}}(b,\kappa) \left(\sum_i L_i^2 \right)^{1/2} + \sqrt{\log(eD)} \left(\sum_i r_i^2 \right)^{1/2} \right].$$

The same threshold, enlarged by a factor 2, is found by m independent sampled sign vectors with failure probability at most 2^{-m}.

Consequently, if

$$\sum_i L_i^2 \le L^2 n, \qquad \sum_i r_i^2 \le \frac{R^2 n}{\log(eD)},$$

then

$$\operatorname{disc}(A_1, \dots, A_n) \le C\big(L B_{\mathrm{flag}}(b,\kappa) + R\big)\sqrt{n}.$$

In particular the flag sum is finite in each of the following common regimes:

$$\kappa_q \le \exp(\gamma q^s), \qquad b_q \le M(1+q)^{-p}, \qquad p > 1 + \frac{s}{2},$$

and

$$\kappa_q \le K_0 \rho^q, \qquad b_q \le M\rho^{-\alpha q}, \qquad \rho > 1,\ \alpha > 0.$$

Thus polynomial spectral-window tails beat stretched-exponential flag growth when $p > 1 + s/2$, and any positive power tail in geometrically growing rank windows gives a Spencer-scale common-flag model.

Proof. Apply Corollary 7.211 with

$$\tau_{i,q} = L_i b_q, \qquad \tau_{i,*} = r_i.$$

Then

$$\left(\sum_i (\tau_{i,q-1} + \tau_{i,q})^2\right)^{1/2} = (b_{q-1} + b_q)\left(\sum_i L_i^2\right)^{1/2},$$

and the displayed bound follows. The sampled-search statement is inherited from the same corollary.

For the polynomial example, $\sqrt{\log(e\kappa_q)} \le C_{\gamma,s}(1+q)^{s/2}$, so the flag sum is dominated by a constant multiple of $\sum_q (1+q)^{s/2-p}$, which converges exactly under $p > 1 + s/2$. For the geometric rank/power-tail example, $\sqrt{\log(e\kappa_q)} \le C(1+q)^{1/2}$, while $b_{q-1} + b_q \le C_{\rho,\alpha,M}\rho^{-\alpha q}$; the resulting series is geometric up to a polynomial factor and therefore converges. □

Corollary 7.215 (Active menus of common-flag tail-envelope streams)**.** *For each $m \ge 1$, let $D_m \ge 2$ and $J_m \ge 1$. For every active (m,h), let*

$$0 = P_{m,h}^{(0)} \preceq P_{m,h}^{(1)} \preceq P_{m,h}^{(2)} \preceq \cdots$$

be a flag in $\mathrm{M}_{D_m}(\mathbb{C})$, with $\operatorname{rank} P_{m,h}^{(q)} \le \kappa_{m,h,q}$. Let $P_{m,h}^{(\infty)} = \sup_q P_{m,h}^{(q)}$, and let $b_{m,h,q} \ge 0$. Put

$$B_{m,h}^{\mathrm{flag}} := \sum_{q \ge 1} \sqrt{\log(e\kappa_{m,h,q})}\,(b_{m,h,q-1} + b_{m,h,q}).$$

Suppose that outside an event $E_{m,h}$, the active stream $A_{m,h,1}, \dots, A_{m,h,n_m}$ consists of self-adjoint contractions and there are amplitudes $L_{m,h,i}$ and $r_{m,h,i}$ such that

$$\left\| A_{m,h,i} - P_{m,h}^{(q)} A_{m,h,i} P_{m,h}^{(q)} \right\|_{\mathrm{op}} \le L_{m,h,i} b_{m,h,q} \quad (q \ge 0),$$

$$\left\| A_{m,h,i} - P_{m,h}^{(\infty)} A_{m,h,i} P_{m,h}^{(\infty)} \right\|_{\mathrm{op}} \le r_{m,h,i},$$

and

$$B_{m,h}^{\mathrm{flag}} \left(n_m^{-1} \sum_i L_{m,h,i}^2\right)^{1/2} + \sqrt{\log(eD_m)} \left(n_m^{-1} \sum_i r_{m,h,i}^2\right)^{1/2} \le S.$$

If

$$\sum_m \sum_{h=1}^{J_m} \mathbb{P}(E_{m,h}) < \infty,$$

then, with probability one, for all sufficiently large m, every active common-flag tail-envelope stream has a Spencer-scale signing:

$$\operatorname{disc}(A_{m,h,1}, \dots, A_{m,h,n_m}) \le CS\sqrt{n_m} \qquad (1 \le h \le J_m).$$

Sampled certificates are eventual almost surely if

$$\sum_{m}\sum_{h=1}^{J_m} 2^{-M_{m,h}} < \infty,$$

or, for one common active-menu batch, if

$$\sum_{m} J_m 2^{-M_m} < \infty.$$

If $J_m \to \infty$, the active prefixes may exhaust a countable library of common-flag spectral-truncation, wavelet-tail, or low-rank-approximation random-matrix models. The polynomial and geometric tail regimes in Corollary 7.214 give checkable uniform choices of S.

Proof. On $E^c_{m,h}$, Corollary 7.214 gives

$$\operatorname{disc}(A_{m,h,1},\dots,A_{m,h,n_m}) \le CS\sqrt{n_m}$$

and the corresponding sampled-search statement. The assumed summability of the active bad-event probabilities, together with Borel–Cantelli, gives the eventual simultaneous deterministic-signing conclusion. The sampled-sign assertions add the independent-batch or common-batch search series. □

Corollary 7.216 (Rank-sensitive dominated square-twirl ensembles)**.** *Let $D \ge 2$, let $1 \le \kappa \le D$, and let $\mathcal{G}$ be an environment. Let P be a $\mathcal{G}$-measurable orthogonal projection with*

$$1 \le \operatorname{rank} P \le \kappa.$$

Let $A_1,\dots,A_n \in \mathrm{M}_D(\mathbb{C})$ be conditionally independent self-adjoint contractions over $\mathcal{G}$ such that

$$A_i = PA_iP.$$

Suppose that there are $\mathcal{G}$-measurable positive semidefinite matrices $V_1,\dots,V_n$ satisfying

$$\mathbb{E}(A_i^2 \mid \mathcal{G}) \preceq V_i \qquad (1 \le i \le n),$$

and put

$$\Xi_\kappa(\mathcal{G}) := \left\| \sum_{i=1}^{n} V_i \right\|_{\mathrm{op}}.$$

Then, for every $C_0 \ge 1$, there is $C(C_0)$ such that, whenever

$$v + t \le \frac{C_0 n}{\log(e\kappa)},$$

one has

$$\mathbb{P}\left\{\operatorname{disc}(A_1,\dots,A_n) > C(C_0)\sqrt{n}\right\} \le \mathbb{P}\{\Xi_\kappa(\mathcal{G}) > v\} + \kappa \exp\left\{-\frac{t^2}{2(v+t/3)}\right\}.$$

If $\eta^{(1)}, \dots, \eta^{(m)}$ are independent uniform sign vectors, independent of the tuple, then the same bound holds for sampled sign search after adding 2^{-m}. In particular, if

$$\Xi_\kappa(\mathcal{G}) \le \frac{C_0 n}{2\log(e\kappa)}$$

outside an event E, then

$$\mathbb{P}\left\{\operatorname{disc}(A_1, \dots, A_n) > C(C_0)\sqrt{n}\right\} \le \mathbb{P}(E) + \kappa \exp\left\{-\frac{3C_0 n}{16\log(e\kappa)}\right\},$$

with the additional 2^{-m} term for sampled sign search.

Equivalently, the ambient dimension is irrelevant for random-basis ensembles that live inside a common random κ-dimensional corner. For example, let $T_i = PT_iP$ be $\mathcal{G}$-measurable self-adjoint contractions, and, conditional on $\mathcal{G}$, let U_i be independent Haar unitaries, exact finite unitary 1-designs, or signed-permutation/monomial designs on $\operatorname{ran} P$*, extended by the identity on $P^\perp$. With $A_i = U_i T_i U_i^*$ and $r = \operatorname{rank} P$,*

$$\mathbb{E}(A_i^2 \mid \mathcal{G}) = \frac{\operatorname{tr}(T_i^2)}{r} P,$$

so the theorem applies with

$$\Xi_\kappa(\mathcal{G}) = \frac{1}{r}\sum_{i=1}^{n} \operatorname{tr}(T_i^2).$$

More generally, approximate scalar designs in the corner satisfying, for $0 \preceq X \preceq P$,

$$\mathbb{E}(U_i X U_i^* \mid \mathcal{G}) \preceq \frac{\operatorname{tr} X}{r} P + \Gamma_i$$

are covered with

$$V_i = \frac{\operatorname{tr}(T_i^2)}{r} P + \Gamma_i,$$

and block designs inside $P = \sum_{\alpha \in I_i} P_{i,\alpha}$ are covered by replacing the scalar term by

$$\sum_{\alpha \in I_i} \frac{\operatorname{tr}(P_{i,\alpha} T_i^2 P_{i,\alpha})}{\operatorname{rank} P_{i,\alpha}} P_{i,\alpha}$$

plus any positive semidefinite block-twirl error. Thus random low-rank Haar subspace rotations, finite-design rotations, signed-permutation rotations, and their dominated approximate versions are Spencer-scale under the κ-dimensional square profile $O(n/\log(e\kappa))$, rather than the ambient D-dimensional profile.

Proof. Let

$$\mathcal{F}_i := \sigma(\mathcal{G}, A_1, \dots, A_i) \qquad (0 \le i \le n),$$

with $\mathcal{F}_0 = \mathcal{G}$. Conditional independence over $\mathcal{G}$ gives

$$\mathbb{E}(A_i^2 \mid \mathcal{F}_{i-1}) = \mathbb{E}(A_i^2 \mid \mathcal{G}) \preceq V_i.$$

Therefore the predictable square profile ν_κ in Corollary 7.198 is at most $\Xi_\kappa(\mathcal{G})$. Applying that corollary proves the discrepancy and sampled-search estimates, and the half-budget form follows by taking $v = t = C_0 n/(2\log(e\kappa))$.

For the scalar twirl specializations, the Haar, finite 1-design, and signed-permutation laws on $\operatorname{ran} P$ satisfy

$$\mathbb{E}(U_i X U_i^* \mid \mathcal{G}) = \frac{\operatorname{tr} X}{r} P \qquad (X = PXP),$$

so substituting $X = T_i^2$ gives the displayed profile. The approximate scalar and block forms follow in the same way from the assumed Loewner domination, again with $X = T_i^2$. □

Corollary 7.217 (Finite-menu rank-sensitive dominated square-twirl schedules)**.** *Fix $C_0 \geq 1$ and put $\gamma_0 := 3C_0/16$. For each $m \geq 1$, let $D_m \geq 2$, let $J_m \geq 1$, and, for every $1 \leq h \leq J_m$, let*

$$1 \leq \kappa_{m,h} \leq D_m, \qquad \ell_{m,h} := \log(e\kappa_{m,h}).$$

Let $\mathcal{G}_{m,h}$ be an environment, and let $P_{m,h}$ be a $\mathcal{G}_{m,h}$-measurable orthogonal projection with

$$1 \leq \operatorname{rank} P_{m,h} \leq \kappa_{m,h}.$$

Let

$$A_{m,h,1}, \dots, A_{m,h,n_m} \in \mathrm{M}_{D_m}(\mathbb{C})$$

be conditionally independent self-adjoint contractions over $\mathcal{G}_{m,h}$ such that

$$A_{m,h,i} = P_{m,h} A_{m,h,i} P_{m,h}.$$

Suppose that $\mathcal{G}_{m,h}$-measurable positive semidefinite matrices $V_{m,h,1}, \dots, V_{m,h,n_m}$ satisfy

$$\mathbb{E}(A_{m,h,i}^2 \mid \mathcal{G}_{m,h}) \preceq V_{m,h,i},$$

and put

$$\Xi_{m,h}^{\kappa} := \left\| \sum_{i=1}^{n_m} V_{m,h,i} \right\|_{\mathrm{op}}.$$

Assume that

$$\Xi_{m,h}^{\kappa} \leq \frac{C_0 n_m}{2\ell_{m,h}} \qquad (1 \leq h \leq J_m)$$

outside events $E_{m,h}$, and that

$$\sum_m \sum_{h=1}^{J_m} \mathbb{P}(E_{m,h}) < \infty, \qquad \sum_m \sum_{h=1}^{J_m} \kappa_{m,h} \exp\left\{ -\gamma_0 \frac{n_m}{\ell_{m,h}} \right\} < \infty.$$

Then, with probability one, for all sufficiently large m, every active rank-sensitive dominated-twirl stream has a Spencer-scale signing:

$$\operatorname{disc}(A_{m,h,1}, \dots, A_{m,h,n_m}) \leq C(C_0)\sqrt{n_m} \qquad (1 \leq h \leq J_m).$$

If the streams are supplied with independent sampled sign batches and

$$\sum_m \sum_{h=1}^{J_m} 2^{-M_{m,h}} < \infty,$$

then, after enlarging $C(C_0)$, almost surely every active stream has a sampled signing certificate for all sufficiently large m. If one common batch of M_m signs is reused across the active menu and is independent of all active streams, it is enough that

$$\sum_m J_m 2^{-M_m} < \infty.$$

The concrete rank-sensitive schedule from Corollary 7.218 applies verbatim. Namely, with

$$\Lambda_m^\kappa := \sum_{h=1}^{J_m} \kappa_{m,h}, \qquad \ell_m^\kappa := \max_{1\le h\le J_m} \ell_{m,h},$$

the structural tail is summable whenever, for some $\delta > 0$,

$$n_m \ge \gamma_0^{-1} \ell_m^\kappa \left(\log(e\Lambda_m^\kappa) + (1+\delta)\log(e+m)\right)$$

eventually. With a common sign batch, the sampled conclusion follows if

$$M_m \ge \log_2(eJ_m) + (1+\delta')\log_2(e+m)$$

eventually for some $\delta' > 0$. If the streams are defined for all $h \ge 1$, $J_m \to \infty$, and $h \le J_m$ is a deterministic exhaustion, then every fixed low-rank dominated-twirl model in the countable library is covered for all sufficiently large m.

In particular, the statement applies to active libraries of low-rank Haar, finite-design, signed-permutation, block-design, and approximate random-basis ensembles. In the scalar corner-twirl case $A_{m,h,i} = U_{m,h,i} T_{m,h,i} U_{m,h,i}^*$, with $T_{m,h,i} = P_{m,h} T_{m,h,i} P_{m,h}$ and $r_{m,h} = \operatorname{rank} P_{m,h}$, the half-budget exception may be taken as

$$\frac{1}{r_{m,h}} \sum_{i=1}^{n_m} \operatorname{tr}(T_{m,h,i}^2) > \frac{C_0 n_m}{2\ell_{m,h}},$$

with the corresponding block-energy or aggregate PSD-error dominated profile for block and approximate designs.

Proof. For a fixed active pair (m,h), apply Corollary 7.216 with

$$v = t = \frac{C_0 n_m}{2\ell_{m,h}}.$$

This bounds its discrepancy failure by

$$\mathbb{P}(E_{m,h}) + \kappa_{m,h} \exp\left\{-\gamma_0 \frac{n_m}{\ell_{m,h}}\right\},$$

and adds $2^{-M_{m,h}}$, or $J_m 2^{-M_m}$ after union over the active menu, for sampled sign search. The assumed summability and Borel–Cantelli prove the almost-sure simultaneous conclusions. The displayed concrete schedule is the same estimate as in Corollary 7.218:

$$\sum_{h=1}^{J_m} \kappa_{m,h} \exp\left\{-\gamma_0 \frac{n_m}{\ell_{m,h}}\right\} \le \Lambda_m^\kappa \exp\left\{-\gamma_0 \frac{n_m}{\ell_m^\kappa}\right\}.$$

The low-rank random-basis examples are exactly the specialization recorded in Corollary 7.216. □

Corollary 7.218 (Finite-menu rank-sensitive square-profile schedules)**.** *Fix $C_0 \geq 1$ and put $\gamma_0 := 3C_0/16$. For each $m \geq 1$, let $D_m \geq 2$, let $J_m \geq 1$, and, for every $1 \leq h \leq J_m$, let*

$$1 \leq \kappa_{m,h} \leq D_m, \qquad \ell_{m,h} := \log(e\kappa_{m,h}).$$

Let $(\mathcal{F}_{m,h,i})_{i=0}^{n_m}$ be a filtration, let $P_{m,h}$ be an $\mathcal{F}_{m,h,0}$-measurable orthogonal projection with

$$\operatorname{rank} P_{m,h} \leq \kappa_{m,h},$$

and let

$$A_{m,h,1}, \dots, A_{m,h,n_m} \in \mathrm{M}_{D_m}(\mathbb{C})$$

be self-adjoint contractions with $A_{m,h,i}$ $\mathcal{F}_{m,h,i}$-measurable and

$$A_{m,h,i} = P_{m,h} A_{m,h,i} P_{m,h}.$$

Put

$$\nu_{m,h}^{\kappa} := \left\| \sum_{i=1}^{n_m} \mathbb{E}(A_{m,h,i}^2 \mid \mathcal{F}_{m,h,i-1}) \right\|_{\mathrm{op}}.$$

Assume that

$$\nu_{m,h}^{\kappa} \leq \frac{C_0 n_m}{2\ell_{m,h}} \qquad (1 \leq h \leq J_m)$$

outside events $E_{m,h}$, and that

$$\sum_m \sum_{h=1}^{J_m} \mathbb{P}(E_{m,h}) < \infty, \qquad \sum_m \sum_{h=1}^{J_m} \kappa_{m,h} \exp\left\{ -\gamma_0 \frac{n_m}{\ell_{m,h}} \right\} < \infty.$$

Then, with probability one, for all sufficiently large m, every active rank-sensitive stream has a Spencer-scale signing:

$$\operatorname{disc}(A_{m,h,1}, \dots, A_{m,h,n_m}) \leq C(C_0)\sqrt{n_m} \qquad (1 \leq h \leq J_m).$$

If the streams are supplied with independent sampled sign batches and

$$\sum_m \sum_{h=1}^{J_m} 2^{-M_{m,h}} < \infty,$$

then, after enlarging $C(C_0)$, almost surely every active stream has a sampled certificate for all sufficiently large m. If one common batch of M_m signs is reused across the active menu and is independent of all active streams, it is enough that

$$\sum_m J_m 2^{-M_m} < \infty.$$

In particular, define

$$\Lambda_m^{\kappa} := \sum_{h=1}^{J_m} \kappa_{m,h}, \qquad \ell_m^{\kappa} := \max_{1 \leq h \leq J_m} \ell_{m,h}.$$

The structural tail is summable whenever, for some $\delta > 0$,

$$n_m \geq \gamma_0^{-1} \ell_m^\kappa \left(\log(e\Lambda_m^\kappa) + (1+\delta)\log(e+m)\right)$$

eventually, and the exceptional events are summable. With a common sign batch, the sampled conclusion follows if

$$M_m \geq \log_2(eJ_m) + (1+\delta')\log_2(e+m)$$

eventually for some $\delta' > 0$. *If the streams are defined for all* $h \geq 1$, $J_m \to \infty$, *and the active menu is a deterministic exhaustion, then every fixed rank-sensitive common-corner stream is covered for all sufficiently large* m.

Proof. For a fixed active pair (m,h), apply Corollary 7.198 with

$$v = t = \frac{C_0 n_m}{2\ell_{m,h}}.$$

This gives failure probability at most

$$\mathbb{P}(E_{m,h}) + \kappa_{m,h} \exp\left\{-\gamma_0 \frac{n_m}{\ell_{m,h}}\right\}.$$

Summing over active h and over m, then applying the first Borel–Cantelli lemma, proves the simultaneous almost-sure signing conclusion. The sampled-sign assertions add the displayed $2^{-M_{m,h}}$ failures, or the common-batch union bound $J_m 2^{-M_m}$.

For the concrete schedule, use $\ell_{m,h} \leq \ell_m^\kappa$ and

$$\sum_{h=1}^{J_m} \kappa_{m,h} \exp\left\{-\gamma_0 \frac{n_m}{\ell_{m,h}}\right\} \leq \Lambda_m^\kappa \exp\left\{-\gamma_0 \frac{n_m}{\ell_m^\kappa}\right\} \leq e^{-1}(e+m)^{-1-\delta}.$$

The common sampled-batch estimate is identical. If $J_m \to \infty$, each fixed countable-menu element belongs to the active prefix eventually. □

Corollary 7.219 (Finite-menu predictable square-profile schedules)**.** *Fix* $C_0 \geq 1$ *and put*

$$\gamma_0 := \frac{3C_0}{16}.$$

For each $m \geq 1$, *let* $D_m \geq 2$, *put*

$$\ell_m := \log(eD_m),$$

and let $J_m \geq 1$ *be a deterministic active-menu size. For every* $1 \leq h \leq J_m$, *let* $(\mathcal{F}_{m,h,i})_{i=0}^{n_m}$ *be a filtration and let*

$$A_{m,h,1}, \ldots, A_{m,h,n_m} \in \mathrm{M}_{D_m}(\mathbb{C})$$

be self-adjoint contractions with $A_{m,h,i}$ $\mathcal{F}_{m,h,i}$*-measurable. Put*

$$\nu_{m,h} := \left\| \sum_{i=1}^{n_m} \mathbb{E}(A_{m,h,i}^2 \mid \mathcal{F}_{m,h,i-1}) \right\|_{\mathrm{op}}, \qquad L_m^{\mathrm{sq}} := J_m D_m.$$

Assume that

$$\nu_{m,h} \leq \frac{C_0 n_m}{2\ell_m} \qquad (1 \leq h \leq J_m)$$

outside events $E_{m,h}$, and that

$$\sum_m \sum_{h=1}^{J_m} \mathbb{P}(E_{m,h}) < \infty, \qquad \sum_m L_m^{\mathrm{sq}} \exp\left\{-\gamma_0 \frac{n_m}{\ell_m}\right\} < \infty.$$

Then, with probability one, for all sufficiently large m, every active adapted stream has a Spencer-scale signing:

$$\operatorname{disc}(A_{m,h,1}, \dots, A_{m,h,n_m}) \le C(C_0)\sqrt{n_m} \qquad (1 \le h \le J_m).$$

If the menu elements are supplied with independent sampled sign batches and

$$\sum_m \sum_{h=1}^{J_m} 2^{-M_{m,h}} < \infty,$$

then, after enlarging $C(C_0)$, almost surely every active menu element has a sampled certificate for all sufficiently large m. If instead one common batch of M_m sign vectors is reused for the entire active menu and is independent of all active streams, it is enough that

$$\sum_m J_m 2^{-M_m} < \infty.$$

In particular, under the displayed half-budget exception hypothesis, the structural tail is summable in each of the following schedules.

(a) *General schedules. If, for some $\delta > 0$,*

$$n_m \ge \gamma_0^{-1} \ell_m (\log(eL_m^{\mathrm{sq}}) + (1+\delta)\log(e+m))$$

eventually, then the simultaneous active-menu conclusion holds. With a common sign batch, the sampled conclusion follows if

$$M_m \ge \log_2(eJ_m) + (1+\delta')\log_2(e+m)$$

eventually for some $\delta' > 0$.

(b) *Full integer dimensions. Suppose the rows are indexed by $D = 2, 3, \dots$, with $D_m = D$, and $L_D^{\mathrm{sq}} \le (eD)^\Gamma$ eventually for some $\Gamma \ge 0$. If*

$$n_D \ge a \log^2(eD)$$

eventually for some

$$a > \frac{16(\Gamma+1)}{3C_0},$$

then the simultaneous active-menu conclusion holds along all integer dimensions.

(c) *Sparse stretched-exponential dimensions. If there are $c, \kappa > 0$ such that $D_m \ge \exp(cm^\kappa)$ eventually, $L_m^{\mathrm{sq}} \le (eD_m)^\Gamma$ eventually, and*

$$n_m \ge a \log^2(eD_m)$$

eventually for some

$$a > \frac{16\Gamma}{3C_0},$$

then the simultaneous active-menu conclusion holds along the sparse schedule.

If the streams are defined for all $h \geq 1$, $J_m \to \infty$, and the active menu is a deterministic exhaustion of a countable library, then every fixed adapted square-profile model is covered for all sufficiently large m. This applies directly to finite or countable libraries of predictable twirl, predictable scalar-multiplier, predictable localized, frame-localized, and other adapted random-matrix streams once their predictable square profiles obey the displayed half-budget condition.

Proof. For fixed m and $h \leq J_m$, the half-budget specialization of Corollary 7.153 gives failure probability at most

$$\mathbb{P}(E_{m,h}) + D_m \exp\left\{-\gamma_0 \frac{n_m}{\ell_m}\right\}.$$

Union over $1 \leq h \leq J_m$, then apply the first Borel–Cantelli lemma. The sampled-sign assertions add the corresponding $2^{-M_{m,h}}$ failures, or the common-batch union bound $J_m 2^{-M_m}$.

For the general schedule,

$$L_m^{\mathrm{sq}} \exp\left\{-\gamma_0 \frac{n_m}{\ell_m}\right\} \leq e^{-1}(e+m)^{-(1+\delta)}.$$

The common sampled-batch condition gives $J_m 2^{-M_m} \leq e^{-1}(e+m)^{-(1+\delta')}$. In the all-integer schedule,

$$L_D^{\mathrm{sq}} \exp\left\{-\gamma_0 \frac{n_D}{\log(eD)}\right\} \leq (eD)^{\Gamma - \gamma_0 a},$$

which is summable over D when $\gamma_0 a > \Gamma + 1$. In the sparse schedule the same estimate is bounded by $\exp\{-(\gamma_0 a - \Gamma)\log(eD_m)\}$, which is summable under the stretched-exponential lower bound on D_m when $\gamma_0 a > \Gamma$. If $J_m \to \infty$, then every fixed countable-menu element lies in the active menu eventually. □

Corollary 7.220 (Predictable dominated twirl streams)**.** *Let $D \geq 2$, let $(\mathcal{F}_i)_{i=0}^n$ be a filtration, and let $T_1, \dots, T_n \in \mathrm{M}_D(\mathbb{C})$ be self-adjoint contractions such that T_i is $\mathcal{F}_{i-1}$-measurable. Let U_i be an $\mathcal{F}_i$-measurable random unitary, set*

$$A_i = U_i T_i U_i^*,$$

and assume that there are $\mathcal{F}_{i-1}$-measurable positive semidefinite matrices V_i such that

$$\mathbb{E}\left(U_i T_i^2 U_i^* \,\middle|\, \mathcal{F}_{i-1}\right) \preceq V_i \qquad (1 \leq i \leq n).$$

Put

$$\Xi = \left\| \sum_{i=1}^n V_i \right\|_{\mathrm{op}}.$$

Then, for every $C_0 \geq 1$, there is $C(C_0)$ such that, for every $v, t > 0$ satisfying

$$v + t \leq \frac{C_0 n}{\log(eD)},$$

one has

$$\mathbb{P}\left\{\operatorname{disc}(A_1, \dots, A_n) > C(C_0)\sqrt{n}\right\} \leq \mathbb{P}\{\Xi > v\} + D \exp\left\{-\frac{t^2}{2(v + t/3)}\right\}.$$

Moreover, if $\eta^{(1)}, \dots, \eta^{(m)}$ are independent uniform sign vectors independent of $\mathcal{F}_n$, then, with possibly larger $C(C_0)$,

$$\mathbb{P}\left\{\min_{1 \leq r \leq m} \left\|\sum_{i=1}^{n} \eta_i^{(r)} A_i\right\|_{\mathrm{op}} > C(C_0)\sqrt{n}\right\}$$
$$\leq \mathbb{P}\{\Xi > v\} + D \exp\left\{-\frac{t^2}{2(v + t/3)}\right\} + 2^{-m}.$$

In particular, if $\Xi \leq C_0 n/(2\log(eD))$ outside an event E, then the discrepancy failure is at most

$$\mathbb{P}(E) + D \exp\left\{-\frac{3C_0 n}{16 \log(eD)}\right\},$$

with an additional 2^{-m} for the randomized search.

This covers online adaptive random-basis samplers. For example, suppose $\Gamma_i \succeq 0$ are $\mathcal{F}_{i-1}$-measurable and the one-step conditional law satisfies, for every $\mathcal{F}_{i-1}$-measurable positive contraction X,

$$\mathbb{E}\left(U_i X U_i^* \mid \mathcal{F}_{i-1}\right) \preceq \frac{\operatorname{tr} X}{D} I + \Gamma_i.$$

Then the theorem applies with

$$V_i = \frac{\operatorname{tr}(T_i^2)}{D} I + \Gamma_i.$$

Likewise, if $I = \sum_{\alpha \in I_i} P_{i,\alpha}$ is an $\mathcal{F}_{i-1}$-measurable orthogonal decomposition, $d_{i,\alpha} = \operatorname{rank} P_{i,\alpha}$, and, for every $\mathcal{F}_{i-1}$-measurable positive contraction X,

$$\mathbb{E}\left(U_i X U_i^* \mid \mathcal{F}_{i-1}\right) \preceq \sum_{\alpha \in I_i} \frac{\operatorname{tr}(P_{i,\alpha} X P_{i,\alpha})}{d_{i,\alpha}} P_{i,\alpha} + \Gamma_i$$

then the theorem applies with the predictable block profile

$$V_i = \sum_{\alpha \in I_i} \frac{\operatorname{tr}(P_{i,\alpha} T_i^2 P_{i,\alpha})}{d_{i,\alpha}} P_{i,\alpha} + \Gamma_i.$$

Thus exact or approximate scalar designs, block designs, signed-permutation samplers, and finite random-basis samplers may be selected adaptively from the past; conditional independence between different times is not required.

Proof. The matrices A_i are $\mathcal{F}_i$-measurable self-adjoint contractions and

$$A_i^2 = U_i T_i^2 U_i^*.$$

Therefore the predictable square profile ν in Corollary 7.153 satisfies

$$\nu = \left\| \sum_i \mathbb{E}(A_i^2 \mid \mathcal{F}_{i-1}) \right\|_{\mathrm{op}} \le \left\| \sum_i V_i \right\|_{\mathrm{op}} = \Xi.$$

Applying Corollary 7.153 gives the discrepancy and randomized-search estimates. The final specialization follows by taking $v = t = C_0 n/(2\log(eD))$. The scalar and block forms are obtained by applying the corresponding one-step domination inequality with $X = T_i^2$. □

Corollary 7.221 (Rank-sensitive predictable dominated twirl streams)**.** *Let $D \ge 2$, let $1 \le \kappa \le D$, and let $(\mathcal{F}_i)_{i=0}^n$ be a filtration. Let P be an $\mathcal{F}_0$-measurable orthogonal projection with*

$$1 \le \operatorname{rank} P \le \kappa.$$

Let $T_1, \dots, T_n \in \mathrm{M}_D(\mathbb{C})$ be self-adjoint contractions such that

$$T_i \text{ is } \mathcal{F}_{i-1}\text{-measurable}, \qquad T_i = P T_i P.$$

Let U_i be an $\mathcal{F}_i$-measurable random unitary satisfying

$$U_i P = P U_i,$$

and set

$$A_i = U_i T_i U_i^*.$$

Assume that there are $\mathcal{F}_{i-1}$-measurable positive semidefinite matrices V_i such that

$$\mathbb{E}\left(U_i T_i^2 U_i^* \,\middle|\, \mathcal{F}_{i-1} \right) \preceq V_i \qquad (1 \le i \le n),$$

and put

$$\Xi_\kappa := \left\| \sum_{i=1}^n V_i \right\|_{\mathrm{op}}.$$

Then, for every $C_0 \ge 1$, there is $C(C_0)$ such that, whenever

$$v + t \le \frac{C_0 n}{\log(e\kappa)},$$

one has

$$\mathbb{P}\left\{ \operatorname{disc}(A_1, \dots, A_n) > C(C_0)\sqrt{n} \right\} \le \mathbb{P}\{\Xi_\kappa > v\} + \kappa \exp\left\{ -\frac{t^2}{2(v + t/3)} \right\}.$$

Moreover, if $\eta^{(1)}, \dots, \eta^{(m)}$ are independent uniform sign vectors independent of $\mathcal{F}_n$, then the same bound holds for sampled sign search after adding 2^{-m}. In particular, if

$$\Xi_\kappa \le \frac{C_0 n}{2\log(e\kappa)}$$

outside an event E, then the discrepancy failure is at most

$$\mathbb{P}(E) + \kappa \exp\left\{-\frac{3C_0 n}{16 \log(e\kappa)}\right\},$$

with an additional 2^{-m} for randomized search.

This covers online adaptive low-rank random-basis samplers. If $r = \operatorname{rank} P$, $\Gamma_i \succeq 0$ are $\mathcal{F}_{i-1}$-measurable, and the one-step conditional law on $\operatorname{ran} P$ satisfies, for every $\mathcal{F}_{i-1}$-measurable positive contraction $X = PXP$,

$$\mathbb{E}(U_i X U_i^* \mid \mathcal{F}_{i-1}) \preceq \frac{\operatorname{tr} X}{r} P + \Gamma_i,$$

then the theorem applies with

$$V_i = \frac{\operatorname{tr}(T_i^2)}{r} P + \Gamma_i.$$

Likewise, if $P = \sum_{\alpha \in I_i} P_{i,\alpha}$ is an $\mathcal{F}_{i-1}$-measurable orthogonal decomposition and the one-step law satisfies

$$\mathbb{E}(U_i X U_i^* \mid \mathcal{F}_{i-1}) \preceq \sum_{\alpha \in I_i} \frac{\operatorname{tr}(P_{i,\alpha} X P_{i,\alpha})}{\operatorname{rank} P_{i,\alpha}} P_{i,\alpha} + \Gamma_i \qquad (0 \preceq X = PXP \preceq P),$$

then the theorem applies with the corresponding predictable block profile plus Γ_i. Thus exact or approximate scalar designs, block designs, signed-permutation samplers, and finite random-basis samplers may be selected adaptively from the past inside a common random low-rank corner; conditional independence between different times is not required.

Proof. Because U_i preserves $\operatorname{ran} P$ and $T_i = PT_iP$, the matrices A_i are $\mathcal{F}_i$-measurable self-adjoint contractions satisfying

$$A_i = PA_iP.$$

Also $A_i^2 = U_i T_i^2 U_i^*$, so the predictable square profile in Corollary 7.198 satisfies

$$\left\| \sum_i \mathbb{E}(A_i^2 \mid \mathcal{F}_{i-1}) \right\|_{\mathrm{op}} \le \left\| \sum_i V_i \right\|_{\mathrm{op}} = \Xi_\kappa.$$

Applying Corollary 7.198 gives the discrepancy and sampled-search estimates. The half-budget form follows from $v = t = C_0 n/(2\log(e\kappa))$. The scalar and block specializations are obtained by substituting $X = T_i^2$ in the displayed one-step Loewner domination assumptions. □

Corollary 7.222 (Finite-menu rank-sensitive predictable twirl schedules)**.** *Fix $C_0 \ge 1$ and put $\gamma_0 := 3C_0/16$. For each $m \ge 1$, let $D_m \ge 2$, let $J_m \ge 1$, and, for $1 \le h \le J_m$, let*

$$1 \le \kappa_{m,h} \le D_m, \qquad \ell_{m,h} := \log(e\kappa_{m,h}).$$

For every active pair (m,h), let

$$A_{m,h,i} = U_{m,h,i} T_{m,h,i} U_{m,h,i}^*, \qquad 1 \le i \le n_m,$$

be a rank-sensitive predictable dominated-twirl stream as in Corollary 7.221, with filtration $(\mathcal{F}_{m,h,i})_{i=0}^{n_m}$*, common support projection* $P_{m,h}$*, and rank envelope* $\kappa_{m,h}$*. Thus* $P_{m,h}$ *is* $\mathcal{F}_{m,h,0}$*-measurable,* $\operatorname{rank} P_{m,h} \le \kappa_{m,h}$*,* $T_{m,h,i} = P_{m,h}T_{m,h,i}P_{m,h}$*,* $U_{m,h,i}P_{m,h} = P_{m,h}U_{m,h,i}$*, and there are predictable positive semidefinite profiles* $V_{m,h,i}$ *satisfying*

$$\mathbb{E}\left(U_{m,h,i}T_{m,h,i}^2U_{m,h,i}^* \,\middle|\, \mathcal{F}_{m,h,i-1}\right) \preceq V_{m,h,i}.$$

Put

$$\Xi_{m,h}^{\kappa} := \left\|\sum_{i=1}^{n_m} V_{m,h,i}\right\|_{\mathrm{op}}.$$

Assume that

$$\Xi_{m,h}^{\kappa} \le \frac{C_0 n_m}{2\ell_{m,h}} \qquad (1 \le h \le J_m)$$

outside events $E_{m,h}$*, and that*

$$\sum_m \sum_{h=1}^{J_m} \mathbb{P}(E_{m,h}) < \infty, \qquad \sum_m \sum_{h=1}^{J_m} \kappa_{m,h} \exp\left\{-\gamma_0 \frac{n_m}{\ell_{m,h}}\right\} < \infty.$$

Then, with probability one, for all sufficiently large m*, every active online low-rank twirl stream has a Spencer-scale signing:*

$$\operatorname{disc}(A_{m,h,1}, \dots, A_{m,h,n_m}) \le C(C_0)\sqrt{n_m} \qquad (1 \le h \le J_m).$$

If the streams are supplied with independent sampled sign batches and

$$\sum_m \sum_{h=1}^{J_m} 2^{-M_{m,h}} < \infty,$$

then, after enlarging $C(C_0)$*, almost surely every active stream has a sampled certificate for all sufficiently large* m*. If one common batch of* M_m *signs is reused across the active menu and is independent of all active streams, it is enough that*

$$\sum_m J_m 2^{-M_m} < \infty.$$

Equivalently, with

$$\Lambda_m^{\kappa} := \sum_{h=1}^{J_m} \kappa_{m,h}, \qquad \ell_m^{\kappa} := \max_{1 \le h \le J_m} \ell_{m,h},$$

the structural tail is summable whenever, for some $\delta > 0$*,*

$$n_m \ge \gamma_0^{-1}\ell_m^{\kappa}\left(\log(e\Lambda_m^{\kappa}) + (1+\delta)\log(e+m)\right)$$

eventually. With a common sign batch, the sampled conclusion follows if

$$M_m \ge \log_2(eJ_m) + (1+\delta')\log_2(e+m)$$

eventually for some $\delta' > 0$*. If the streams are defined for all* $h \ge 1$*,* $J_m \to \infty$*, and* $h \le J_m$ *is a deterministic exhaustion, then every fixed adaptive low-rank random-basis stream in the countable library is covered for all sufficiently large* m*.*

Proof. For a fixed active pair (m,h), apply Corollary 7.221 with

$$v = t = \frac{C_0 n_m}{2\ell_{m,h}}.$$

Its failure probability is at most

$$\mathbb{P}(E_{m,h}) + \kappa_{m,h} \exp\left\{-\gamma_0 \frac{n_m}{\ell_{m,h}}\right\},$$

with the additional sampled-search terms displayed in the statement. The Borel–Cantelli lemma gives the almost-sure simultaneous conclusions. The explicit schedule follows from

$$\sum_{h=1}^{J_m} \kappa_{m,h} \exp\left\{-\gamma_0 \frac{n_m}{\ell_{m,h}}\right\} \le \Lambda_m^{\kappa} \exp\left\{-\gamma_0 \frac{n_m}{\ell_m^{\kappa}}\right\} \le e^{-1}(e+m)^{-1-\delta}.$$

The common sampled-batch estimate is identical. □

Corollary 7.223 (Almost-sure predictable twirl arrays)**.** *Fix $C_0 \ge 1$. For each $m \ge 1$, let $(\mathcal{F}_{m,i})_{i=0}^{n_m}$ be a filtration and let*

$$A_{m,i} = U_{m,i} T_{m,i} U_{m,i}^* \in \mathrm{M}_{D_m}(\mathbb{C}), \qquad 1 \le i \le n_m,$$

be a predictable dominated-twirl stream as in Corollary 7.220. Thus $T_{m,i}$ is $\mathcal{F}_{m,i-1}$-measurable, $U_{m,i}$ is $\mathcal{F}_{m,i}$-measurable, and there are $\mathcal{F}_{m,i-1}$-measurable positive semidefinite matrices $V_{m,i}$ such that

$$\mathbb{E}\left(U_{m,i} T_{m,i}^2 U_{m,i}^* \,\middle|\, \mathcal{F}_{m,i-1}\right) \preceq V_{m,i}.$$

Put

$$\Xi_m = \left\| \sum_{i=1}^{n_m} V_{m,i} \right\|_{\mathrm{op}}.$$

Assume that there are deterministic $v_m, t_m > 0$ satisfying

$$v_m + t_m \le \frac{C_0 n_m}{\log(eD_m)}$$

and

$$\sum_m \left[\mathbb{P}\{\Xi_m > v_m\} + D_m \exp\left\{-\frac{t_m^2}{2(v_m + t_m/3)}\right\} \right] < \infty.$$

Then, with probability one, for all sufficiently large m,

$$\operatorname{disc}(A_{m,1}, \dots, A_{m,n_m}) \le C(C_0)\sqrt{n_m}.$$

If, in addition, for each m we sample M_m independent uniform sign vectors $\eta_m^{(1)}, \dots, \eta_m^{(M_m)} \in \{\pm 1\}^{n_m}$, independent of the m-th matrix row, and

$$\sum_m 2^{-M_m} < \infty,$$

then, with probability one, for all sufficiently large m at least one sampled sign vector satisfies

$$\left\| \sum_{i=1}^{n_m} \eta_{m,i}^{(r)} A_{m,i} \right\|_{\mathrm{op}} \le C(C_0)\sqrt{n_m}.$$

In particular, if

$$\Xi_m \le \frac{C_0 n_m}{2\log(eD_m)}$$

outside an event E_m, and

$$\sum_m \left[\mathbb{P}(E_m) + D_m \exp\left\{ -\frac{3C_0 n_m}{16\log(eD_m)} \right\} \right] < \infty,$$

then the same almost-sure conclusions hold, with the sampled-certificate conclusion again requiring $\sum_m 2^{-M_m} < \infty$. This applies in particular to predictable exact scalar designs, predictable approximate scalar designs, predictable block designs, and predictable approximate block designs by using the one-step profiles in Corollary 7.220.

Proof. For the m-th row, Corollary 7.220 gives failure probability at most

$$\mathbb{P}\{\Xi_m > v_m\} + D_m \exp\left\{ -\frac{t_m^2}{2(v_m + t_m/3)} \right\}.$$

The displayed summability and the first Borel–Cantelli lemma imply that only finitely many row failures occur. The sampled-sign statement follows from the same finite-row corollary with the additional 2^{-M_m} term and another application of Borel–Cantelli. The final specialization is obtained by taking $v_m = t_m = C_0 n_m/(2\log(eD_m))$. □

Corollary 7.224 (Explicit predictable twirl-array schedules)**.** *Fix $C_0 \ge 1$ and put*

$$\gamma_0 := \frac{3C_0}{16}.$$

In the setting of Corollary 7.223, assume that

$$\Xi_m \le \frac{C_0 n_m}{2\log(eD_m)}$$

outside events E_m with

$$\sum_m \mathbb{P}(E_m) < \infty.$$

Then the following deterministic schedule conditions imply that, almost surely,

$$\mathrm{disc}(A_{m,1}, \dots, A_{m,n_m}) \le C(C_0)\sqrt{n_m}$$

for all sufficiently large m. The same conclusions hold for sampled sign certificates whenever $\sum_m 2^{-M_m} < \infty$.

(a) *General schedules. If, for some* $\delta > 0$,

$$n_m \geq \gamma_0^{-1} \log(eD_m)\big(\log(eD_m) + (1+\delta)\log(e+m)\big)$$

eventually, then the conclusion holds.

(b) *Full integer dimensions. Suppose the rows are indexed by* $D = 2, 3, \ldots$, *with ambient dimension* D. *If*

$$n_D \geq a \, \log^2(eD)$$

eventually for some

$$a > \frac{32}{3C_0},$$

then the conclusion holds along all integer dimensions.

(c) *Sparse stretched-exponential dimensions. If there are* $c, \kappa > 0$ *such that*

$$D_m \geq \exp(cm^\kappa)$$

eventually, and

$$n_m \geq a \, \log^2(eD_m)$$

eventually for some

$$a > \frac{16}{3C_0},$$

then the conclusion holds along the sparse schedule.

In each case, the same statement applies to predictable exact scalar designs, predictable approximate scalar designs, predictable block designs, and predictable approximate block designs by inserting the one-step profiles from Corollary 7.220.

Proof. By Corollary 7.223, it remains only to check

$$\sum_m D_m \exp\left\{-\gamma_0 \frac{n_m}{\log(eD_m)}\right\} < \infty.$$

For part (a), write $\ell_m = \log(eD_m)$. Since $D_m \leq e^{\ell_m}$, the summand is bounded by

$$\exp\{\ell_m - \ell_m - (1+\delta)\log(e+m)\} = (e+m)^{-1-\delta},$$

which is summable.

For part (b), choose $\delta > 0$ so small that $a > \gamma_0^{-1}(2+\delta)$. With $m = D$, part (a) applies because $\log(e+m) = \log(eD)$.

For part (c), choose $\theta > 0$ with $\gamma_0 a - 1 > \theta$. Then, eventually,

$$D_m \exp\left\{-\gamma_0 \frac{n_m}{\log(eD_m)}\right\} \leq \exp\{-\theta \log(eD_m)\} \leq \exp\{-\theta c m^\kappa\},$$

which is summable. □

Corollary 7.225 (Finite-menu predictable twirl-array schedules)**.** *Fix $C_0 \geq 1$ and put*

$$\gamma_0 := \frac{3C_0}{16}.$$

For each $m \geq 1$, let $D_m \geq 2$, put

$$\ell_m := \log(eD_m),$$

and let $J_m \geq 1$ be a deterministic active-menu size. For every $1 \leq h \leq J_m$, let

$$A_{m,i}^{(h)} = U_{m,h,i} T_{m,h,i} U_{m,h,i}^* \in \mathrm{M}_{D_m}(\mathbb{C}), \qquad 1 \leq i \leq n_m,$$

be a predictable dominated-twirl stream as in Corollary 7.220, with its own filtration $(\mathcal{F}_{m,h,i})_{i=0}^{n_m}$ allowed. Thus there are predictable positive semidefinite profiles $V_{m,h,i}$ such that

$$\mathbb{E}\left(U_{m,h,i} T_{m,h,i}^2 U_{m,h,i}^* \,\middle|\, \mathcal{F}_{m,h,i-1}\right) \preceq V_{m,h,i}.$$

Put

$$\Xi_{m,h} := \left\| \sum_{i=1}^{n_m} V_{m,h,i} \right\|_{\mathrm{op}}, \qquad L_m^{\mathrm{tw}} := J_m D_m.$$

Assume that

$$\Xi_{m,h} \leq \frac{C_0 n_m}{2\ell_m} \qquad (1 \leq h \leq J_m)$$

outside events $E_{m,h}$, and that

$$\sum_m \sum_{h=1}^{J_m} \mathbb{P}(E_{m,h}) < \infty, \qquad \sum_m L_m^{\mathrm{tw}} \exp\left\{-\gamma_0 \frac{n_m}{\ell_m}\right\} < \infty.$$

Then, with probability one, for all sufficiently large m, every active menu element has a Spencer-scale signing:

$$\mathrm{disc}(A_{m,1}^{(h)}, \ldots, A_{m,n_m}^{(h)}) \leq C(C_0)\sqrt{n_m} \qquad (1 \leq h \leq J_m).$$

If the menu elements are supplied with independent sampled sign batches and

$$\sum_m \sum_{h=1}^{J_m} 2^{-M_{m,h}} < \infty,$$

then, after enlarging $C(C_0)$, almost surely every active menu element has a sampled certificate for all sufficiently large m. If instead one common batch of M_m sign vectors is reused for the entire active menu and is independent of all active streams, it is enough that

$$\sum_m J_m 2^{-M_m} < \infty.$$

In particular, under the displayed half-budget exception hypothesis, the structural tail is summable in each of the following schedules.

(a) *General schedules. If, for some* $\delta > 0$,

$$n_m \geq \gamma_0^{-1} \ell_m \big(\log(eL_m^{\mathrm{tw}}) + (1+\delta)\log(e+m)\big)$$

eventually, then the simultaneous active-menu conclusion holds. With a common sign batch, the sampled conclusion follows if

$$M_m \geq \log_2(eJ_m) + (1+\delta')\log_2(e+m)$$

eventually for some $\delta' > 0$.

(b) *Full integer dimensions. Suppose the rows are indexed by* $D = 2, 3, \ldots$, *with* $D_m = D$, *and* $L_D^{\mathrm{tw}} \leq (eD)^{\Gamma}$ *eventually for some* $\Gamma \geq 0$. *If*

$$n_D \geq a \log^2(eD)$$

eventually for some

$$a > \frac{16(\Gamma+1)}{3C_0},$$

then the simultaneous active-menu conclusion holds along all integer dimensions.

(c) *Sparse stretched-exponential dimensions. If there are* $c, \kappa > 0$ *such that* $D_m \geq \exp(cm^{\kappa})$ *eventually,* $L_m^{\mathrm{tw}} \leq (eD_m)^{\Gamma}$ *eventually, and*

$$n_m \geq a \log^2(eD_m)$$

eventually for some

$$a > \frac{16\Gamma}{3C_0},$$

then the simultaneous active-menu conclusion holds along the sparse schedule.

The same statements apply to finite active menus of predictable exact scalar designs, predictable approximate scalar designs, predictable block designs, and predictable approximate block designs by inserting the one-step profiles from Corollary 7.220. If the streams are defined for all $h \geq 1$, $J_m \to \infty$, *and the active menu is a deterministic exhaustion of a countable library, then every fixed library element is covered for all sufficiently large* m.

Proof. For fixed m and $h \leq J_m$, the half-budget specialization of Corollary 7.223 gives failure probability at most

$$\mathbb{P}(E_{m,h}) + D_m \exp\left\{-\gamma_0 \frac{n_m}{\ell_m}\right\}.$$

Union over $1 \leq h \leq J_m$, then apply the first Borel–Cantelli lemma. The sampled-sign assertions add the corresponding $2^{-M_{m,h}}$ failures, or the common-batch union bound $J_m 2^{-M_m}$.

For the general schedule,

$$L_m^{\mathrm{tw}} \exp\left\{-\gamma_0 \frac{n_m}{\ell_m}\right\} \leq e^{-1}(e+m)^{-(1+\delta)}.$$

The common sampled-batch condition gives $J_m 2^{-M_m} \le e^{-1}(e+m)^{-(1+\delta')}$. In the all-integer schedule,

$$L_D^{\mathrm{tw}} \exp\left\{-\gamma_0 \frac{n_D}{\log(eD)}\right\} \le (eD)^{\Gamma-\gamma_0 a},$$

which is summable over D when $\gamma_0 a > \Gamma + 1$. In the sparse schedule the same estimate is bounded by $\exp\{-(\gamma_0 a - \Gamma)\log(eD_m)\}$, which is summable under the stretched-exponential lower bound on D_m when $\gamma_0 a > \Gamma$. The design specializations are exactly the dominated-twirl specializations already recorded in Corollary 7.220. If $J_m \to \infty$, then every fixed countable-menu element lies in the active menu eventually. □

Corollary 7.226 (Predictable subspace-localized ensembles)**.** *Let $D \ge 2$, let $(\mathcal{F}_i)_{i=0}^n$ be a filtration, and let $P_1, \dots, P_n \in \mathrm{M}_D(\mathbb{C})$ be random orthogonal projections such that P_i is $\mathcal{F}_i$-measurable. Let A_i be $\mathcal{F}_i$-measurable self-adjoint contractions satisfying*

$$A_i = P_i A_i P_i,$$

and put

$$\mu := \left\| \sum_{i=1}^n \mathbb{E}(P_i \mid \mathcal{F}_{i-1}) \right\|_{\mathrm{op}}.$$

Then, for every $C_0 \ge 1$, there is $C(C_0)$ such that, for every $v, t > 0$ satisfying

$$v + t \le \frac{C_0 n}{\log(eD)},$$

one has

$$\mathbb{P}\left\{\mathrm{disc}(A_1, \dots, A_n) > C(C_0)\sqrt{n}\right\} \le \mathbb{P}\{\mu > v\} + D \exp\left\{-\frac{t^2}{2(v + t/3)}\right\}.$$

Moreover, if $\eta^{(1)}, \dots, \eta^{(m)}$ are independent uniform sign vectors independent of $\mathcal{F}_n$, then, with possibly larger $C(C_0)$,

$$\mathbb{P}\left\{ \min_{1 \le r \le m} \left\| \sum_{i=1}^n \eta_i^{(r)} A_i \right\|_{\mathrm{op}} > C(C_0)\sqrt{n} \right\} \le \mathbb{P}\{\mu > v\} + D \exp\left\{-\frac{t^2}{2(v + t/3)}\right\} + 2^{-m}.$$

In particular, if

$$\sum_{i=1}^n \mathbb{E}(P_i \mid \mathcal{F}_{i-1}) \preceq pn\, I \qquad \text{almost surely}$$

and $p \le C_0/(2\log(eD))$, then the discrepancy failure is at most

$$D \exp\left\{-\frac{3C_0 n}{16 \log(eD)}\right\},$$

with the additional 2^{-m} term for randomized search. This covers adaptive coordinate supports, online random subspaces, and conditionally sampled subspace-localized Wigner/GOE/GUE blocks whenever their predictable overlap operator is $O(n/\log(eD))$.

Proof. Since $A_i = P_i A_i P_i$ and $\|A_i\|_{\mathrm{op}} \le 1$, we have

$$0 \le A_i^2 \preceq P_i.$$

Indeed, for every vector x,

$$\langle A_i^2 x, x\rangle = \|A_i x\|^2 = \|A_i P_i x\|^2 \le \|P_i x\|^2 = \langle P_i x, x\rangle.$$

Therefore

$$\sum_i \mathbb{E}(A_i^2 \mid \mathcal{F}_{i-1}) \preceq \sum_i \mathbb{E}(P_i \mid \mathcal{F}_{i-1}),$$

so the predictable square-profile ν from Corollary 7.153 satisfies $\nu \le \mu$. The first two estimates follow by applying that corollary. If the displayed predictable overlap bound holds, then $\mu \le pn \le C_0 n/(2\log(eD))$; choose $v = t = C_0 n/(2\log(eD))$, as in Corollary 7.153. □

Corollary 7.227 (Finite-menu predictable subspace-localized schedules)**.** *Fix $C_0 \ge 1$ and put*

$$\gamma_0 := \frac{3C_0}{16}.$$

For each $m \ge 1$, let $D_m \ge 2$, put

$$\ell_m := \log(eD_m),$$

and let $J_m \ge 1$ be a deterministic active-menu size. For every $1 \le h \le J_m$, let $(\mathcal{F}_{m,h,i})_{i=0}^{n_m}$ be a filtration, let

$$P_{m,h,1}, \dots, P_{m,h,n_m} \in \mathrm{M}_{D_m}(\mathbb{C})$$

be orthogonal projections with $P_{m,h,i}$ $\mathcal{F}_{m,h,i}$-measurable, and let $A_{m,h,i}$ be $\mathcal{F}_{m,h,i}$-measurable self-adjoint contractions satisfying

$$A_{m,h,i} = P_{m,h,i} A_{m,h,i} P_{m,h,i}.$$

Put

$$\mu_{m,h} := \left\| \sum_{i=1}^{n_m} \mathbb{E}(P_{m,h,i} \mid \mathcal{F}_{m,h,i-1}) \right\|_{\mathrm{op}}, \qquad L_m^{\mathrm{loc}} := J_m D_m.$$

Assume that

$$\mu_{m,h} \le \frac{C_0 n_m}{2\ell_m} \qquad (1 \le h \le J_m)$$

outside events $E_{m,h}$, and that

$$\sum_m \sum_{h=1}^{J_m} \mathbb{P}(E_{m,h}) < \infty, \qquad \sum_m L_m^{\mathrm{loc}} \exp\left\{-\gamma_0 \frac{n_m}{\ell_m}\right\} < \infty.$$

Then, with probability one, for all sufficiently large m, every active localized stream has a Spencer-scale signing:

$$\mathrm{disc}(A_{m,h,1}, \dots, A_{m,h,n_m}) \le C(C_0)\sqrt{n_m} \qquad (1 \le h \le J_m).$$

If the menu elements are supplied with independent sampled sign batches and

$$\sum_{m}\sum_{h=1}^{J_m} 2^{-M_{m,h}} < \infty,$$

then, after enlarging $C(C_0)$, almost surely every active menu element has a sampled certificate for all sufficiently large m. If instead one common batch of M_m sign vectors is reused for the entire active menu and is independent of all active streams, it is enough that

$$\sum_{m} J_m 2^{-M_m} < \infty.$$

In particular, under the displayed half-budget exception hypothesis, the structural tail is summable in each of the following schedules.

(a) *General schedules. If, for some $\delta > 0$,*

$$n_m \geq \gamma_0^{-1}\ell_m\big(\log(eL_m^{\mathrm{loc}}) + (1+\delta)\log(e+m)\big)$$

eventually, then the simultaneous active-menu conclusion holds. With a common sign batch, the sampled conclusion follows if

$$M_m \geq \log_2(eJ_m) + (1+\delta')\log_2(e+m)$$

eventually for some $\delta' > 0$.

(b) *Full integer dimensions. Suppose the rows are indexed by $D = 2, 3, \ldots$, with $D_m = D$, and $L_D^{\mathrm{loc}} \leq (eD)^{\Gamma}$ eventually for some $\Gamma \geq 0$. If*

$$n_D \geq a\log^2(eD)$$

eventually for some

$$a > \frac{16(\Gamma+1)}{3C_0},$$

then the simultaneous active-menu conclusion holds along all integer dimensions.

(c) *Sparse stretched-exponential dimensions. If there are $c, \kappa > 0$ such that $D_m \geq \exp(cm^{\kappa})$ eventually, $L_m^{\mathrm{loc}} \leq (eD_m)^{\Gamma}$ eventually, and*

$$n_m \geq a\log^2(eD_m)$$

eventually for some

$$a > \frac{16\Gamma}{3C_0},$$

then the simultaneous active-menu conclusion holds along the sparse schedule.

The half-budget hypothesis is automatic, with no $E_{m,h}$, whenever

$$\sum_{i=1}^{n_m} \mathbb{E}(P_{m,h,i} \mid \mathcal{F}_{m,h,i-1}) \preceq p_m n_m I, \qquad p_m \leq \frac{C_0}{2\ell_m},$$

for all active h. In the adaptive coordinate-support case this is the condition

$$\max_j \sum_i \mathbb{P}\{j \in S_{m,h,i} \mid \mathcal{F}_{m,h,i-1}\} \le \frac{C_0 n_m}{2\ell_m},$$

and for conditional Haar rank-$r_{m,h,i}$ subspaces it is the average-rank condition

$$\sum_i \frac{r_{m,h,i}}{D_m} \le \frac{C_0 n_m}{2\ell_m}.$$

If the streams are defined for all $h \ge 1$, $J_m \to \infty$, and the active menu is a deterministic exhaustion of a countable library, then every fixed adaptive localized model is covered for all sufficiently large m.

Proof. For fixed m and $h \le J_m$, the half-budget specialization of Corollary 7.226 gives failure probability at most

$$\mathbb{P}(E_{m,h}) + D_m \exp\left\{-\gamma_0 \frac{n_m}{\ell_m}\right\}.$$

Union over $1 \le h \le J_m$, then apply the first Borel–Cantelli lemma. The sampled-sign assertions add the corresponding $2^{-M_{m,h}}$ failures, or the common-batch union bound $J_m 2^{-M_m}$.

The schedule estimates are identical to those in Corollary 7.225, with L_m^{tw} replaced by L_m^{loc}. The coordinate and Haar specializations are the displayed forms of the predictable overlap operator: coordinate projections are diagonal with conditional diagonal entries given by the conditional inclusion probabilities, while conditional Haar rank-r projections satisfy $\mathbb{E}(P_{m,h,i} \mid \mathcal{F}_{m,h,i-1}) = (r_{m,h,i}/D_m)I$. If $J_m \to \infty$, then every fixed countable-menu element lies in the active menu eventually. □

Corollary 7.228 (Random scalar-multiplier ensembles)**.** *Let $D \ge 2$. Let $B_1, \dots, B_n \in \mathrm{M}_D(\mathbb{C})$ be arbitrary, possibly dependent, random self-adjoint contractions. Conditional on $B = (B_1, \dots, B_n)$, let $\theta_1, \dots, \theta_n$ be independent real random variables satisfying $|\theta_i| \le 1$, and set*

$$A_i = \theta_i B_i, \qquad q_i = \mathbb{E}(\theta_i^2 \mid B), \qquad \nu = \left\| \sum_{i=1}^n q_i B_i^2 \right\|_{\mathrm{op}}.$$

Then, for every $C_0 \ge 1$, there is $C(C_0)$ such that, for all deterministic $v \ge 0$ and $t > 0$ satisfying

$$v + t \le \frac{C_0 n}{\log(eD)},$$

one has

$$\mathbb{P}\left\{\operatorname{disc}(A_1, \dots, A_n) > C(C_0)\sqrt{n}\right\} \le \mathbb{P}\{\nu > v\} + D \exp\left\{-\frac{t^2}{2(v + t/3)}\right\}.$$

In particular, if $q_i \le p$ almost surely for every i and

$$p \le \frac{C_0}{2\log(eD)},$$

then

$$\mathbb{P}\left\{\operatorname{disc}(A_1,\dots,A_n) > C(C_0)\sqrt{n}\right\} \le D\exp\left\{-\frac{3C_0 n}{16\log(eD)}\right\}.$$

Thus arbitrary deterministic or random self-adjoint contraction tuples become Spencer-scale after independent Bernoulli thinning of density $O(1/\log(eD))$, or after any independent bounded scalar amplitudes with the same conditional second-moment bound. No support, commutation, or low-rank assumption is imposed on the underlying directions B_i.

Proof. Condition on the environment B. The matrices $A_i = \theta_i B_i$ are then independent self-adjoint contractions and

$$\left\|\sum_i \mathbb{E}(A_i^2 \mid B)\right\|_{\mathrm{op}} = \left\|\sum_i q_i B_i^2\right\|_{\mathrm{op}} = \nu.$$

On the event $\{\nu \le v\}$, Corollary 7.148 applied conditionally gives

$$\mathbb{P}\left(\operatorname{disc}(A_1,\dots,A_n) > C(C_0)\sqrt{n} \,\middle|\, B\right) \le D\exp\left\{-\frac{t^2}{2(v+t/3)}\right\},$$

because $\nu + t \le v + t \le C_0 n/\log(eD)$. Averaging over B and adding the exceptional event $\{\nu > v\}$ proves the first display.

If $q_i \le p$, then $0 \le B_i^2 \le I$, so

$$\nu \le p\left\|\sum_i B_i^2\right\|_{\mathrm{op}} \le pn.$$

Take $v = pn$ and $t = C_0 n/(2\log(eD))$. The assumption gives $v \le t$, and the same elementary calculation as above gives

$$\frac{t^2}{2(v+t/3)} \ge \frac{3C_0 n}{16\log(eD)}.$$

□

Corollary 7.229 (Predictable scalar-multiplier ensembles)**.** *Let $D \ge 2$, let $(\mathcal{F}_i)_{i=0}^n$ be a filtration, and let $H_1,\dots,H_n \in \mathrm{M}_D(\mathbb{C})$ be self-adjoint contractions such that H_i is $\mathcal{F}_{i-1}$-measurable. Let θ_i be real $\mathcal{F}_i$-measurable random variables with $|\theta_i| \le 1$, and put*

$$A_i = \theta_i H_i, \qquad q_i = \mathbb{E}(\theta_i^2 \mid \mathcal{F}_{i-1}), \qquad \nu = \left\|\sum_{i=1}^n q_i H_i^2\right\|_{\mathrm{op}}.$$

Then, for every $C_0 \ge 1$, there is $C(C_0)$ such that, for all deterministic $v \ge 0$ and $t > 0$ satisfying

$$v + t \le \frac{C_0 n}{\log(eD)},$$

one has

$$\mathbb{P}\left\{\operatorname{disc}(A_1,\dots,A_n) > C(C_0)\sqrt{n}\right\} \le \mathbb{P}\{\nu > v\} + D\exp\left\{-\frac{t^2}{2(v+t/3)}\right\}.$$

If, in addition, $\eta^{(1)},\dots,\eta^{(m)}$ are independent uniform sign vectors independent of $\mathcal{F}_n$, then, with possibly larger $C(C_0)$,

$$\mathbb{P}\left\{\min_{1\le r\le m}\left\|\sum_{i=1}^{n}\eta_i^{(r)}A_i\right\|_{\mathrm{op}} > C(C_0)\sqrt{n}\right\} \le \mathbb{P}\{\nu > v\} + D\exp\left\{-\frac{t^2}{2(v+t/3)}\right\} + 2^{-m}.$$

In particular, if $q_i \le p$ almost surely for every i and

$$p \le \frac{C_0}{2\log(eD)},$$

then the discrepancy failure is at most

$$D\exp\left\{-\frac{3C_0 n}{16\log(eD)}\right\},$$

with the additional 2^{-m} term for the randomized search. Thus the scalar-multiplier theorem remains valid for online or adaptive matrix directions and bounded scalar amplitudes; conditional independence of the amplitudes is not needed once the one-step conditional second moments are controlled.

Proof. The matrices A_i are self-adjoint contractions and are $\mathcal{F}_i$-measurable. Since H_i is $\mathcal{F}_{i-1}$-measurable and θ_i is scalar,

$$\mathbb{E}(A_i^2 \mid \mathcal{F}_{i-1}) = \mathbb{E}(\theta_i^2 \mid \mathcal{F}_{i-1})H_i^2 = q_i H_i^2.$$

The first two bounds are therefore exactly Corollary 7.153. If $q_i \le p$, then

$$\nu \le p\left\|\sum_i H_i^2\right\|_{\mathrm{op}} \le pn,$$

and the final estimate follows by taking $v = pn$ and $t = C_0 n/(2\log(eD))$, as in Corollary 7.228. □

Corollary 7.230 (Almost-sure predictable scalar-multiplier arrays)**.** *Fix $C_0 \ge 1$. For each $m \ge 1$, let $D_m \ge 2$, let $(\mathcal{F}_{m,i})_{i=0}^{n_m}$ be a filtration, and let*

$$A_{m,i} = \theta_{m,i}H_{m,i} \in \mathrm{M}_{D_m}(\mathbb{C}), \qquad 1 \le i \le n_m,$$

be a predictable scalar-multiplier stream as in Corollary 7.229. Thus $H_{m,i}$ is an $\mathcal{F}_{m,i-1}$-measurable self-adjoint contraction, $\theta_{m,i}$ is real, $\mathcal{F}_{m,i}$-measurable, and $|\theta_{m,i}| \le 1$. Put

$$q_{m,i} := \mathbb{E}(\theta_{m,i}^2 \mid \mathcal{F}_{m,i-1}), \qquad \nu_m := \left\|\sum_{i=1}^{n_m} q_{m,i}H_{m,i}^2\right\|_{\mathrm{op}}.$$

Assume that there are deterministic $v_m, t_m > 0$ satisfying

$$v_m + t_m \le \frac{C_0 n_m}{\log(eD_m)}$$

and

$$\sum_m \left[\mathbb{P}\{\nu_m > v_m\} + D_m \exp\left\{ -\frac{t_m^2}{2(v_m + t_m/3)} \right\} \right] < \infty.$$

Then, with probability one, for all sufficiently large m,

$$\operatorname{disc}(A_{m,1}, \dots, A_{m,n_m}) \le C(C_0)\sqrt{n_m}.$$

If, in addition, for each m we sample M_m independent uniform sign vectors $\eta_m^{(1)}, \dots, \eta_m^{(M_m)} \in \{\pm 1\}^{n_m}$, independent of the m-th matrix row, and

$$\sum_m 2^{-M_m} < \infty,$$

then, with probability one, for all sufficiently large m at least one sampled sign vector satisfies

$$\left\| \sum_{i=1}^{n_m} \eta_{m,i}^{(r)} A_{m,i} \right\|_{\mathrm{op}} \le C(C_0)\sqrt{n_m}.$$

In particular, if

$$\nu_m \le \frac{C_0 n_m}{2\log(eD_m)}$$

outside an event E_m, and

$$\sum_m \left[\mathbb{P}(E_m) + D_m \exp\left\{ -\frac{3C_0 n_m}{16\log(eD_m)} \right\} \right] < \infty,$$

then the same almost-sure conclusions hold. This half-budget hypothesis is implied by the pointwise conditional second-moment bound

$$q_{m,i} \le p_m \le \frac{C_0}{2\log(eD_m)} \qquad (1 \le i \le n_m)$$

outside E_m, because $0 \le H_{m,i}^2 \le I$.

Proof. For the m-th row, Corollary 7.229 gives failure probability at most

$$\mathbb{P}\{\nu_m > v_m\} + D_m \exp\left\{ -\frac{t_m^2}{2(v_m + t_m/3)} \right\}.$$

The summability assumption and the first Borel–Cantelli lemma imply that only finitely many discrepancy failures occur. The sampled-sign statement follows from the same finite-row corollary with the additional 2^{-M_m} term. The half-budget specialization takes $v_m = t_m = C_0 n_m/(2\log(eD_m))$. Finally, on the event $q_{m,i} \le p_m$ for all i,

$$\nu_m \le p_m \left\| \sum_{i=1}^{n_m} H_{m,i}^2 \right\|_{\mathrm{op}} \le p_m n_m \le \frac{C_0 n_m}{2\log(eD_m)}.$$

□

Corollary 7.231 (Explicit predictable scalar-multiplier schedules)**.** *Fix $C_0 \geq 1$ and put $\gamma_0 := 3C_0/16$. In the setting of Corollary 7.230, assume that the half-budget event*

$$\nu_m \leq \frac{C_0 n_m}{2\log(eD_m)}$$

holds outside events E_m with $\sum_m \mathbb{P}(E_m) < \infty$. Then the following deterministic schedule conditions imply that, almost surely,

$$\operatorname{disc}(A_{m,1}, \dots, A_{m,n_m}) \leq C(C_0)\sqrt{n_m}$$

for all sufficiently large m. The same conclusions hold for sampled sign certificates whenever $\sum_m 2^{-M_m} < \infty$.

(a) *General schedules. If, for some $\delta > 0$,*

$$n_m \geq \gamma_0^{-1}\log(eD_m)\big(\log(eD_m) + (1+\delta)\log(e+m)\big)$$

eventually, then the conclusion holds.

(b) *Full integer dimensions. Suppose the rows are indexed by $D = 2, 3, \dots$, with ambient dimension D. If*

$$n_D \geq a\,\log^2(eD)$$

eventually for some $a > 32/(3C_0)$, then the conclusion holds along all integer dimensions.

(c) *Sparse stretched-exponential dimensions. If there are $c, \kappa > 0$ such that $D_m \geq \exp(cm^\kappa)$ eventually, and*

$$n_m \geq a\,\log^2(eD_m)$$

eventually for some $a > 16/(3C_0)$, then the conclusion holds along the sparse schedule.

In particular, the same explicit schedules apply whenever the adaptive amplitudes satisfy $q_{m,i} \leq p_m \leq C_0/(2\log(eD_m))$ outside summable events. Thus arbitrary adaptive matrix directions become eventually Spencer-scale after bounded online thinning or scalar amplitudes at the logarithmic conditional second-moment density.

Proof. By Corollary 7.230, it remains only to check

$$\sum_m D_m \exp\left\{-\gamma_0 \frac{n_m}{\log(eD_m)}\right\} < \infty.$$

The verification is identical to the proof of Corollary 7.224. For part (a), with $\ell_m = \log(eD_m)$, the summand is at most

$$\exp\{\ell_m - \ell_m - (1+\delta)\log(e+m)\} = (e+m)^{-1-\delta}.$$

For part (b), choose $\delta > 0$ so small that $a > \gamma_0^{-1}(2+\delta)$; then part (a) applies because along the full integer sequence $\log(e+D) \leq \log(eD)$. For part (c), if $a > \gamma_0^{-1}$, then

$$D_m \exp\left\{-\gamma_0 \frac{n_m}{\log(eD_m)}\right\} \leq \exp\{-(\gamma_0 a - 1)\log(eD_m)\},$$

which is summable along every stretched-exponential dimension schedule. □

Corollary 7.232 (Finite menus of predictable scalar-multiplier schedules)**.** *Fix $C_0 \geq 1$. For each $m \geq 1$, let $D_m \geq 2$, put $\ell_m = \log(eD_m)$, and let $\mathcal{H}_m$ be a nonempty finite menu. For each $h \in \mathcal{H}_m$, let*

$$A_{m,i}^{(h)} = \theta_{m,i}^{(h)} H_{m,i}^{(h)} \in \mathrm{M}_{D_m}(\mathbb{C}), \qquad 1 \leq i \leq n_m,$$

be a predictable scalar-multiplier stream as in Corollary 7.229, with its own filtration $(\mathcal{F}_{m,i}^{(h)})_{i=0}^{n_m}$ allowed. Put

$$q_{m,i}^{(h)} = \mathbb{E}\big((\theta_{m,i}^{(h)})^2 \mid \mathcal{F}_{m,i-1}^{(h)}\big), \qquad \nu_{m,h} = \left\| \sum_i q_{m,i}^{(h)} (H_{m,i}^{(h)})^2 \right\|_{\mathrm{op}},$$

and set

$$L_m := |\mathcal{H}_m| D_m.$$

Assume that the half-budget condition

$$\nu_{m,h} \leq \frac{C_0 n_m}{2\ell_m} \qquad (h \in \mathcal{H}_m)$$

holds outside events $E_{m,h}$, and that

$$\sum_m \sum_{h \in \mathcal{H}_m} \mathbb{P}(E_{m,h}) < \infty, \qquad \sum_m L_m \exp\left\{ -\frac{3C_0 n_m}{16\ell_m} \right\} < \infty.$$

Then, with probability one, for all sufficiently large m, every $h \in \mathcal{H}_m$ has a Spencer-scale signing:

$$\mathrm{disc}(A_{m,1}^{(h)}, \dots, A_{m,n_m}^{(h)}) \leq C(C_0)\sqrt{n_m}.$$

If each menu element is also supplied with $M_{m,h}$ independent uniform sign vectors in $\{\pm 1\}^{n_m}$, independent of the corresponding row, and

$$\sum_m \sum_{h \in \mathcal{H}_m} 2^{-M_{m,h}} < \infty,$$

then after enlarging $C(C_0)$, almost surely for all sufficiently large m every menu element has one sampled sign certifying the same bound. If a common batch of M_m signs is reused for all $h \in \mathcal{H}_m$, independent of the whole m-th menu, it is enough that

$$\sum_m |\mathcal{H}_m| \, 2^{-M_m} < \infty.$$

In particular, under the displayed half-budget exception hypothesis, the structural menu tail is summable in each of the following schedules.

(a) *General schedules. If, for some $\delta > 0$,*

$$n_m \geq \frac{16}{3C_0} \ell_m \big(\log(eL_m) + (1+\delta) \log(e+m) \big)$$

eventually, then the simultaneous almost-sure signing conclusion holds. With a common sign batch, the simultaneous sampled conclusion follows if

$$M_m \geq \log_2(e|\mathcal{H}_m|) + (1+\delta')\log_2(e+m)$$

eventually for some $\delta' > 0$.

(b) *Full integer dimensions. Suppose the rows are indexed by* $D = 2, 3, \ldots$, *with* $D_m = D$, *and* $L_D \leq (eD)^\Gamma$ *eventually for some* $\Gamma \geq 0$. *If*

$$n_D \geq a \log^2(eD)$$

eventually for some $a > 16(\Gamma+1)/(3C_0)$, *then the simultaneous almost-sure signing conclusion holds.*

(c) *Sparse stretched-exponential dimensions. If there are* $c, \kappa > 0$ *such that* $D_m \geq \exp(cm^\kappa)$ *eventually,* $L_m \leq (eD_m)^\Gamma$ *eventually, and*

$$n_m \geq a \log^2(eD_m)$$

eventually for some $a > 16\Gamma/(3C_0)$, *then the simultaneous almost-sure signing conclusion holds.*

The half-budget hypothesis is automatic, with no $E_{m,h}$, *whenever*

$$q_{m,i}^{(h)} \leq p_m \leq \frac{C_0}{2\ell_m} \qquad (h \in \mathcal{H}_m,\ 1 \leq i \leq n_m),$$

because $0 \leq (H_{m,i}^{(h)})^2 \leq I$.

Proof. For each fixed m and h, Corollary 7.230 with $v_m = t_m = C_0 n_m/(2\ell_m)$ gives failure probability at most

$$\mathbb{P}(E_{m,h}) + D_m \exp\left\{-\frac{3C_0 n_m}{16\ell_m}\right\}.$$

Union over $h \in \mathcal{H}_m$, and apply the first Borel–Cantelli lemma. The sampled-sign statements follow from Corollary 7.229 by adding the corresponding $2^{-M_{m,h}}$ failures, or by union bounding the reused common batch as $|\mathcal{H}_m|2^{-M_m}$.

For the general schedule, the lower bound on n_m gives

$$L_m \exp\left\{-\frac{3C_0 n_m}{16\ell_m}\right\} \leq e^{-1}(e+m)^{-(1+\delta)}.$$

The common sampled-batch condition gives $|\mathcal{H}_m|2^{-M_m} \leq e^{-1}(e+m)^{-(1+\delta')}$. The all-integer and sparse cases are the same estimates with $L_m \leq (eD_m)^\Gamma$. Finally, the pointwise q-bound implies

$$\nu_{m,h} \leq p_m \left\| \sum_i (H_{m,i}^{(h)})^2 \right\|_{\mathrm{op}} \leq p_m n_m \leq \frac{C_0 n_m}{2\ell_m}.$$

□

Corollary 7.233 (Exhaustive countable menus of predictable scalar-multiplier schedules)**.** *Fix $C_0 \geq 1$. For each $m \geq 1$, let $D_m \geq 2$, put $\ell_m = \log(eD_m)$, and let*

$$J_1 \leq J_2 \leq \cdots, \qquad J_m \to \infty,$$

be a deterministic exhaustion of the countable menu $\mathbb{N} = \{1, 2, \ldots\}$. For each m and $h \geq 1$, let

$$A_{m,i}^{(h)} = \theta_{m,i}^{(h)} H_{m,i}^{(h)} \in \mathrm{M}_{D_m}(\mathbb{C}), \qquad 1 \leq i \leq n_m,$$

be a predictable scalar-multiplier stream as in Corollary 7.229, with its own filtration $(\mathcal{F}_{m,i}^{(h)})_{i=0}^{n_m}$ allowed. Put

$$q_{m,i}^{(h)} = \mathbb{E}\big((\theta_{m,i}^{(h)})^2 \mid \mathcal{F}_{m,i-1}^{(h)}\big), \qquad \nu_{m,h} = \left\| \sum_i q_{m,i}^{(h)} (H_{m,i}^{(h)})^2 \right\|_{\mathrm{op}},$$

and set

$$L_m^{(J)} := J_m D_m.$$

Assume that the half-budget condition

$$\nu_{m,h} \leq \frac{C_0 n_m}{2\ell_m} \qquad (1 \leq h \leq J_m)$$

holds outside events $E_{m,h}$, and that

$$\sum_m \sum_{1 \leq h \leq J_m} \mathbb{P}(E_{m,h}) < \infty, \qquad \sum_m L_m^{(J)} \exp\left\{ -\frac{3C_0 n_m}{16 \ell_m} \right\} < \infty.$$

Then, with probability one, for all sufficiently large m, every active prefix stream $1 \leq h \leq J_m$ has a Spencer-scale signing:

$$\mathrm{disc}(A_{m,1}^{(h)}, \ldots, A_{m,n_m}^{(h)}) \leq C(C_0)\sqrt{n_m} \qquad (1 \leq h \leq J_m).$$

Consequently, for each fixed h, the same bound holds for all sufficiently large m. If the active-prefix streams are supplied with independent sampled signs and

$$\sum_m \sum_{1 \leq h \leq J_m} 2^{-M_{m,h}} < \infty,$$

then after enlarging $C(C_0)$, almost surely every active-prefix stream has a sampled certificate for all sufficiently large m. If one common batch of M_m signs is reused for all $1 \leq h \leq J_m$, independent of the whole active prefix, it is enough that

$$\sum_m J_m \, 2^{-M_m} < \infty.$$

In particular, under the displayed half-budget exception hypothesis, the structural prefix tail is summable in each of the following schedules.

(a) *General schedules. If, for some $\delta > 0$,*

$$n_m \geq \frac{16}{3C_0} \ell_m \big(\log(eL_m^{(J)}) + (1+\delta) \log(e+m) \big)$$

eventually, then the active-prefix and fixed-stream almost-sure signing conclusions hold. With a common sign batch, the sampled conclusion follows if

$$M_m \geq \log_2(eJ_m) + (1+\delta')\log_2(e+m)$$

eventually for some $\delta' > 0$.

(b) *Full integer dimensions. Suppose the rows are indexed by* $D = 2, 3, \ldots$, *with* $D_m = D$, $J_D \to \infty$, *and* $L_D^{(J)} \leq (eD)^\Gamma$ *eventually for some* $\Gamma \geq 0$. *If*

$$n_D \geq a \log^2(eD)$$

eventually for some $a > 16(\Gamma+1)/(3C_0)$, *then the active-prefix and fixed-stream almost-sure signing conclusions hold.*

(c) *Sparse stretched-exponential dimensions. If there are* $c, \kappa > 0$ *such that* $D_m \geq \exp(cm^\kappa)$ *eventually,* $J_m \to \infty$, $L_m^{(J)} \leq (eD_m)^\Gamma$ *eventually, and*

$$n_m \geq a \log^2(eD_m)$$

eventually for some $a > 16\Gamma/(3C_0)$, *then the active-prefix and fixed-stream almost-sure signing conclusions hold.*

The half-budget hypothesis is automatic, with no $E_{m,h}$, *whenever*

$$q_{m,i}^{(h)} \leq p_m \leq \frac{C_0}{2\ell_m} \qquad (1 \leq h \leq J_m,\ 1 \leq i \leq n_m),$$

because $0 \leq (H_{m,i}^{(h)})^2 \leq I$.

Proof. Apply Corollary 7.232 to the finite active menu $\{1, \ldots, J_m\}$ in the m-th row. The displayed exception and structural summability hypotheses are exactly the finite-menu hypotheses for those active prefixes, so Borel–Cantelli gives the simultaneous active-prefix conclusion. Since $J_m \to \infty$, each fixed h lies in the active prefix for all sufficiently large m. The sampled-sign claims are the finite-menu sampled claims summed over $1 \leq h \leq J_m$, or the common batch union bound $J_m 2^{-M_m}$.

For the general schedule,

$$L_m^{(J)} \exp\left\{-\frac{3C_0 n_m}{16\ell_m}\right\} \leq e^{-1}(e+m)^{-(1+\delta)}.$$

The common sampled-batch condition gives $J_m 2^{-M_m} \leq e^{-1}(e+m)^{-(1+\delta')}$. The all-integer and sparse dimension cases are the same estimates with $L_m^{(J)} \leq (eD_m)^\Gamma$. Finally, the pointwise q-bound implies

$$\nu_{m,h} \leq p_m \left\| \sum_i (H_{m,i}^{(h)})^2 \right\|_{\mathrm{op}} \leq p_m n_m \leq \frac{C_0 n_m}{2\ell_m}$$

for every active-prefix stream. □

Corollary 7.234 (Core signings with centered noise)**.** *Let $D \geq 2$. Let $B_i, H_i \in \mathrm{M}_D(\mathbb{C})$, $i = 1, \ldots, n$, be arbitrary, possibly dependent, random self-adjoint contractions. Write B, H for the two tuples. Conditional on (B, H), let $\theta_1, \ldots, \theta_n$ be independent real random variables satisfying*

$$\mathbb{E}(\theta_i \mid B, H) = 0, \qquad |\theta_i| \leq 1,$$

and set

$$X_i = B_i + \theta_i H_i, \qquad q_i = \mathbb{E}(\theta_i^2 \mid B, H), \qquad \nu = \left\| \sum_{i=1}^n q_i H_i^2 \right\|_{\mathrm{op}}.$$

Assume that, outside an event E, there are signs $\varepsilon_i^\star = \varepsilon_i^\star(B, H) \in \{-1, 1\}$ such that

$$\left\| \sum_{i=1}^n \varepsilon_i^\star B_i \right\|_{\mathrm{op}} \leq K\sqrt{n}.$$

Then for every deterministic $v \geq 0$ and $s > 0$,

$$\mathbb{P}\left\{\mathrm{disc}(X_1, \ldots, X_n) > K\sqrt{n} + s\right\} \leq \mathbb{P}(E) + \mathbb{P}\{\nu > v\} + 2D \exp\left\{ -\frac{s^2}{2(v + s/3)} \right\}.$$

Since $\|X_i\|_{\mathrm{op}} \leq 2$, the normalized contractions $X_i/2$ satisfy the same estimate with the right-hand discrepancy threshold divided by 2.

In particular, if $q_i \leq p$ almost surely for every i, then

$$\mathbb{P}\left\{\mathrm{disc}(X_1, \ldots, X_n) > K\sqrt{n} + s\right\} \leq \mathbb{P}(E) + 2D \exp\left\{ -\frac{s^2}{2(pn + s/3)} \right\}.$$

Thus algebraic cores, finite-lift cores, Toeplitz/Hankel cores, or any other class already carrying a $K\sqrt{n}$ signing remain Spencer-scale after adding centered independent Bernoulli or bounded scalar-amplitude noise in arbitrary self-adjoint contraction directions, whenever the displayed Bernstein tail is small at $s = O(\sqrt{n})$.

Proof. Work on E^c and condition on the environment (B, H). For the environment-measurable signing $\varepsilon^\star$, set

$$Z_i = \varepsilon_i^\star \theta_i H_i.$$

Then Z_i are conditionally independent centered self-adjoint matrices with $\|Z_i\|_{\mathrm{op}} \leq 1$, and

$$\left\| \sum_i \mathbb{E}(Z_i^2 \mid B, H) \right\|_{\mathrm{op}} = \left\| \sum_i q_i H_i^2 \right\|_{\mathrm{op}} = \nu.$$

The two-sided matrix Bernstein inequality [25] gives, on the event $\{\nu \leq v\}$,

$$\mathbb{P}\left(\left\| \sum_i Z_i \right\|_{\mathrm{op}} > s \,\middle|\, B, H \right) \leq 2D \exp\left\{ -\frac{s^2}{2(v + s/3)} \right\}.$$

On the complementary good event,

$$\left\|\sum_i \varepsilon_i^\star X_i\right\|_{\mathrm{op}} \le \left\|\sum_i \varepsilon_i^\star B_i\right\|_{\mathrm{op}} + \left\|\sum_i Z_i\right\|_{\mathrm{op}} \le K\sqrt{n} + s.$$

Taking expectations over the environment and adding $E \cup \{\nu > v\}$ proves the first display. If $q_i \le p$, then $0 \le H_i^2 \le I$, so $\nu \le p\|\sum_i H_i^2\|_{\mathrm{op}} \le pn$, which gives the final form. □

Corollary 7.235 (Core signings with matrix martingale noise)**.** *Let $D \ge 2$. Let $\mathcal{G}$ be a sigma-field, and let $B_1, \dots, B_n \in \mathrm{M}_D(\mathbb{C})$ be $\mathcal{G}$-measurable random self-adjoint contractions. Let $(\mathcal{F}_i)_{i=0}^n$ be a filtration with $\mathcal{G} \subset \mathcal{F}_0$, and let $Y_i \in \mathrm{M}_D(\mathbb{C})$ be self-adjoint $\mathcal{F}_i$-measurable random matrices satisfying*

$$\mathbb{E}(Y_i \mid \mathcal{F}_{i-1}) = 0, \qquad \|Y_i\|_{\mathrm{op}} \le 1.$$

Set

$$X_i = B_i + Y_i, \qquad \nu = \left\|\sum_{i=1}^n \mathbb{E}(Y_i^2 \mid \mathcal{F}_{i-1})\right\|_{\mathrm{op}}.$$

Assume that, outside an event $E \in \mathcal{G}$, there are $\mathcal{G}$-measurable signs $\varepsilon_i^\star \in \{-1, 1\}$ such that

$$\left\|\sum_{i=1}^n \varepsilon_i^\star B_i\right\|_{\mathrm{op}} \le K\sqrt{n}.$$

Then, for every deterministic $v \ge 0$ and $s > 0$,

$$\mathbb{P}\left\{\mathrm{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \le \mathbb{P}(E) + \mathbb{P}\{\nu > v\} + 2D \exp\left\{-\frac{s^2}{2(v + s/3)}\right\}.$$

Since $\|X_i\|_{\mathrm{op}} \le 2$, the normalized contractions $X_i/2$ satisfy the same estimate with the discrepancy threshold divided by 2. Thus any random core class with an environment-measurable $K\sqrt{n}$ signing admits the displayed tail after arbitrary bounded self-adjoint matrix-valued martingale-difference noise. Equivalently, the random noise has the usual Freedman signing scale $s \asymp \sqrt{v \log(eD)} + \log(eD)$; this is $O(\sqrt{n})$ when $v = O(n/\log(eD))$ and $\log(eD) = O(\sqrt{n})$.

Proof. Work on E^c. For the $\mathcal{G}$-measurable signing $\varepsilon^\star$, set

$$Z_i = \varepsilon_i^\star Y_i.$$

Since $\mathcal{G} \subset \mathcal{F}_{i-1}$, the matrices Z_i are self-adjoint martingale differences with $\|Z_i\|_{\mathrm{op}} \le 1$, and

$$\mathbb{E}(Z_i^2 \mid \mathcal{F}_{i-1}) = \mathbb{E}(Y_i^2 \mid \mathcal{F}_{i-1}).$$

On $\{\nu \le v\}$, the predictable quadratic variation of the martingale $\sum_i Z_i$ has operator norm at most v. Applying the matrix Freedman inequality [25] to

Z_i and to $-Z_i$, and taking a union bound, gives

$$\mathbb{P}\left\{\left\|\sum_i Z_i\right\|_{\mathrm{op}} > s \text{ and } \nu \le v\right\} \le 2D\exp\left\{-\frac{s^2}{2(v+s/3)}\right\}.$$

On the complementary good event,

$$\left\|\sum_i \varepsilon_i^\star X_i\right\|_{\mathrm{op}} \le \left\|\sum_i \varepsilon_i^\star B_i\right\|_{\mathrm{op}} + \left\|\sum_i Z_i\right\|_{\mathrm{op}} \le K\sqrt{n} + s.$$

Adding the exceptional events E and $\{\nu > v\}$ proves the estimate. □

Corollary 7.236 (Almost-sure signed-core matrix martingale arrays)**.** *For each $m \ge 1$, let $D_m \ge 2$, let $\mathcal{G}_m$ be a sigma-field, and let $(\mathcal{F}_{m,i})_{i=0}^{n_m}$ be a filtration with $\mathcal{G}_m \subset \mathcal{F}_{m,0}$. Let*

$$X_{m,i} = B_{m,i} + Y_{m,i} \in \mathrm{M}_{D_m}(\mathbb{C}), \qquad 1 \le i \le n_m,$$

where $B_{m,i}$ is a $\mathcal{G}_m$-measurable random self-adjoint contraction and $Y_{m,i}$ is a self-adjoint $\mathcal{F}_{m,i}$-measurable martingale difference satisfying

$$\mathbb{E}(Y_{m,i} \mid \mathcal{F}_{m,i-1}) = 0, \qquad \|Y_{m,i}\|_{\mathrm{op}} \le 1.$$

Put

$$\nu_m := \left\|\sum_{i=1}^{n_m} \mathbb{E}(Y_{m,i}^2 \mid \mathcal{F}_{m,i-1})\right\|_{\mathrm{op}}.$$

Assume that, outside an event $E_m \in \mathcal{G}_m$, there are $\mathcal{G}_m$-measurable signs $\varepsilon_{m,i}^\star$ such that

$$\left\|\sum_{i=1}^{n_m} \varepsilon_{m,i}^\star B_{m,i}\right\|_{\mathrm{op}} \le K_m\sqrt{n_m}.$$

Let $v_m \ge 0$ and $s_m > 0$ be deterministic. If

$$\sum_m \left[\mathbb{P}(E_m) + \mathbb{P}\{\nu_m > v_m\} + 2D_m\exp\left\{-\frac{s_m^2}{2(v_m + s_m/3)}\right\}\right] < \infty,$$

then, with probability one, for all sufficiently large m,

$$\mathrm{disc}(X_{m,1}, \dots, X_{m,n_m}) \le K_m\sqrt{n_m} + s_m.$$

In particular, put $\ell_m = \log(eD_m)$. Suppose that $K_m \le K$, that

$$\nu_m \le Q\frac{n_m}{\ell_m}$$

outside events F_m, that

$$\sum_m [\mathbb{P}(E_m) + \mathbb{P}(F_m)] < \infty,$$

and that, for some $A, B > 0$,

$$\sum_m (eD_m)^{-A} < \infty, \qquad \ell_m \le B\sqrt{n_m}$$

eventually. Choose $L > 0$ with

$$\frac{L^2}{2(Q + LB/3)} \geq A + 1.$$

Then, almost surely, for all sufficiently large m,

$$\operatorname{disc}(X_{m,1}, \dots, X_{m,n_m}) \leq (K + L)\sqrt{n_m}.$$

Consequently:

(a) *Along all integer dimensions $D = 2, 3, \dots$, if $K_D \leq K$, $\nu_D \leq Qn_D/\log(eD)$ outside summable events, and*

$$n_D \geq a \log^2(eD)$$

eventually for some $a > 0$, then for every $A > 1$ and every L satisfying

$$\frac{L^2}{2(Q + L/(3\sqrt{a}))} \geq A + 1$$

one has

$$\operatorname{disc}(X_{D,1}, \dots, X_{D,n_D}) \leq (K + L)\sqrt{n_D}$$

eventually almost surely.

(b) *If $D_m \geq \exp(cm^\kappa)$ eventually for some $c, \kappa > 0$, if $K_m \leq K$, if $\nu_m \leq Qn_m/\log(eD_m)$ outside summable events, and if*

$$n_m \geq a \log^2(eD_m)$$

eventually for some $a > 0$, then the same conclusion holds along the sparse schedule for every $A > 0$ and every L satisfying

$$\frac{L^2}{2(Q + L/(3\sqrt{a}))} \geq A + 1.$$

The normalized contractions $X_{m,i}/2$ satisfy the same statements with the discrepancy thresholds divided by 2.

Proof. For the m-th row, write $\Delta_m = \operatorname{disc}(X_{m,1}, \dots, X_{m,n_m})$. Corollary 7.235 gives

$$\mathbb{P}\{\Delta_m > K_m\sqrt{n_m} + s_m\} \leq \mathbb{P}(E_m) + \mathbb{P}\{\nu_m > v_m\} + 2D_m \exp\left\{-\frac{s_m^2}{2(v_m + s_m/3)}\right\}.$$

The displayed summability and the first Borel–Cantelli lemma prove the first claim.

For the concrete profile, take $s_m = L\sqrt{n_m}$ and $v_m = Qn_m/\ell_m$. On F_m^c,

$$\frac{s_m^2}{2(v_m + s_m/3)} = \frac{L^2 n_m}{2(Qn_m/\ell_m + L\sqrt{n_m}/3)} \geq \frac{L^2}{2(Q + LB/3)}\,\ell_m \geq (A + 1)\ell_m.$$

Therefore

$$2D_m \exp\left\{-\frac{s_m^2}{2(v_m + s_m/3)}\right\} \leq 2(eD_m)^{-A},$$

which is summable by assumption.

For part (a), take $B = 1/\sqrt{a}$ and any $A > 1$, since $\sum_{D\ge 2}(eD)^{-A} < \infty$. For part (b), again take $B = 1/\sqrt{a}$; the stretched-exponential lower bound on D_m makes $\sum_m (eD_m)^{-A}$ finite for every $A > 0$. The final normalized statement is obtained by homogeneity. □

Corollary 7.237 (Exhaustive countable menus of signed-core matrix martingale-noise arrays)**.** *For each $m \ge 1$, let $D_m \ge 2$, put $\ell_m = \log(eD_m)$, and let*

$$J_1 \le J_2 \le \cdots, \qquad J_m \to \infty,$$

be a deterministic exhaustion of the countable menu $\mathbb{N} = \{1, 2, \ldots\}$. For each m and $h \ge 1$, let $\mathcal{G}_m^{(h)}$ be a sigma-field and let $(\mathcal{F}_{m,i}^{(h)})_{i=0}^{n_m}$ be a filtration with $\mathcal{G}_m^{(h)} \subset \mathcal{F}_{m,0}^{(h)}$. Let

$$X_{m,i}^{(h)} = B_{m,i}^{(h)} + Y_{m,i}^{(h)} \in \mathrm{M}_{D_m}(\mathbb{C}), \qquad 1 \le i \le n_m,$$

where $B_{m,i}^{(h)}$ is a $\mathcal{G}_m^{(h)}$-measurable self-adjoint contraction and $Y_{m,i}^{(h)}$ is a self-adjoint $\mathcal{F}_{m,i}^{(h)}$-measurable martingale difference satisfying

$$\mathbb{E}(Y_{m,i}^{(h)} \mid \mathcal{F}_{m,i-1}^{(h)}) = 0, \qquad \|Y_{m,i}^{(h)}\|_{\mathrm{op}} \le 1.$$

Put

$$\nu_{m,h} := \left\| \sum_{i=1}^{n_m} \mathbb{E}\big((Y_{m,i}^{(h)})^2 \mid \mathcal{F}_{m,i-1}^{(h)}\big) \right\|_{\mathrm{op}}.$$

Assume that there is $K < \infty$ such that, outside events $E_{m,h} \in \mathcal{G}_m^{(h)}$, there are $\mathcal{G}_m^{(h)}$-measurable signs $\varepsilon_{m,i}^{(h),\star}$ with

$$\left\| \sum_{i=1}^{n_m} \varepsilon_{m,i}^{(h),\star} B_{m,i}^{(h)} \right\|_{\mathrm{op}} \le K\sqrt{n_m} \qquad (1 \le h \le J_m).$$

Assume also that, for some $Q, A, B_0 > 0$, the variance budgets obey

$$\nu_{m,h} \le Q\frac{n_m}{\ell_m} \qquad (1 \le h \le J_m)$$

outside events $F_{m,h}$, and that

$$\sum_m \sum_{1\le h\le J_m} \big[\mathbb{P}(E_{m,h}) + \mathbb{P}(F_{m,h})\big] < \infty, \qquad \sum_m J_m (eD_m)^{-A} < \infty,$$

while

$$\ell_m \le B_0\sqrt{n_m}$$

eventually. If $L > 0$ satisfies

$$\frac{L^2}{2(Q + LB_0/3)} \ge A + 1,$$

then, with probability one, for all sufficiently large m, every active prefix member $1 \le h \le J_m$ satisfies

$$\operatorname{disc}(X_{m,1}^{(h)}, \dots, X_{m,n_m}^{(h)}) \le (K+L)\sqrt{n_m}.$$

Consequently, for every fixed h, the same bound holds for all sufficiently large m. The normalized contractions $X_{m,i}^{(h)}/2$ satisfy the same statements with the discrepancy threshold divided by 2. The active-prefix conclusion also applies to arbitrary row-dependent finite menus with J_m members; only the final fixed-h conclusion uses the countable exhaustion.

In particular, if $n_m \ge a\log^2(eD_m)$ eventually, then one may take $B_0 = a^{-1/2}$. Along all integer dimensions $D = 2, 3, \dots$, if $J_D \le (eD)^\Gamma$ eventually and $n_D \ge a\log^2(eD)$, then the conclusion holds for every $A > \Gamma + 1$ and every L satisfying

$$\frac{L^2}{2(Q + L/(3\sqrt{a}))} \ge A + 1.$$

Along a sparse schedule with $D_m \ge \exp(cm^\kappa)$ eventually, $J_m \le (eD_m)^\Gamma$ eventually, and $n_m \ge a\log^2(eD_m)$, the same conclusion holds for every $A > \Gamma$ and every L satisfying the same display.

Proof. For fixed m and $h \le J_m$, apply Corollary 7.235 with

$$v_m = Q\frac{n_m}{\ell_m}, \qquad s_m = L\sqrt{n_m}.$$

Outside $F_{m,h}$, the variance parameter is at most v_m. Therefore

$$\begin{aligned}\mathbb{P}\left\{\operatorname{disc}(X_{m,1}^{(h)}, \dots, X_{m,n_m}^{(h)}) > (K+L)\sqrt{n_m}\right\} \\ \le \mathbb{P}(E_{m,h}) + \mathbb{P}(F_{m,h}) + 2D_m \exp\left\{-\frac{L^2 n_m}{2(Qn_m/\ell_m + L\sqrt{n_m}/3)}\right\}.\end{aligned}$$

For all sufficiently large m, $\ell_m \le B_0\sqrt{n_m}$, so the exponent is at least $(A+1)\ell_m$. Hence the last term is at most

$$2D_m(eD_m)^{-(A+1)} \le 2(eD_m)^{-A}.$$

Summing over $1 \le h \le J_m$ and then over m, the displayed hypotheses make the total failure probability finite. The first Borel–Cantelli lemma gives the simultaneous active-prefix conclusion. Since $J_m \to \infty$, each fixed h belongs to the active prefix eventually. The normalized statement is homogeneity, and the finite-menu statement is the same union bound over the row's finite menu.

If $n_m \ge a\log^2(eD_m)$, then $\ell_m \le a^{-1/2}\sqrt{n_m}$. For all integer dimensions, the prefix series is bounded by a constant multiple of $\sum_D (eD)^{\Gamma - A}$, which converges when $A > \Gamma + 1$. For the sparse case it is bounded by $\sum_m \exp\{-(A - \Gamma)\log(eD_m)\}$, which converges under the stretched-exponential lower bound whenever $A > \Gamma$. □

Corollary 7.238 (Scalar-dominated matrix martingale noise with random amplitudes)**.** *Let $D \geq 2$. Let $\mathcal{G}$ be a sigma-field, and let $B_1, \dots, B_n \in \mathrm{M}_D(\mathbb{C})$ be $\mathcal{G}$-measurable random self-adjoint contractions. Assume that, outside an event $E \in \mathcal{G}$, there are $\mathcal{G}$-measurable signs $\varepsilon_i^\star \in \{-1, 1\}$ such that*

$$\left\| \sum_{i=1}^n \varepsilon_i^\star B_i \right\|_{\mathrm{op}} \leq K\sqrt{n}.$$

Let (θ_i, ρ_i), $i = 1, \dots, n$, be independent pairs, independent of $\mathcal{G}$, with $\theta_i, \rho_i \in [0, 1]$. Let $\theta_* \in [0, 1]$ be deterministic with $\theta_i \leq \theta_*$ almost surely for every i. Suppose that $(\mathcal{F}_i)_{i=0}^n$ is a filtration with $\mathcal{G}, (\theta_i, \rho_i)_{i=1}^n \subset \mathcal{F}_0$. Let Y_i be self-adjoint and $\mathcal{F}_i$-measurable with

$$\mathbb{E}(Y_i \mid \mathcal{F}_{i-1}) = 0, \qquad \|Y_i\|_{\mathrm{op}} \leq \rho_i, \qquad \mathbb{E}(Y_i^2 \mid \mathcal{F}_{i-1}) \preceq \theta_i \rho_i^2 I,$$

and set $X_i = B_i + Y_i$. Define

$$\mu := \sum_{i=1}^n \mathbb{E}(\theta_i \rho_i^2), \qquad \sigma^2 := \sum_{i=1}^n \mathbb{E}(\theta_i^2 \rho_i^2).$$

Then, for every $u \geq 0$ and $s > 0$,

$$\mathbb{P}\left\{\mathrm{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \leq \mathbb{P}(E) + \exp\left\{-\frac{u^2}{2(\sigma^2 + \theta_* u/3)}\right\} + 2D \exp\left\{-\frac{s^2}{2(\mu + u + s/3)}\right\},$$

with the convention that the first exponential term is omitted when $\sigma^2 + \theta_* u/3 = 0$. In particular, if $\theta_i \leq \theta$ almost surely for all i and $A = \sum_i \mathbb{E}\rho_i^2$, then

$$\mathbb{P}\left\{\mathrm{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \leq \mathbb{P}(E) + \exp\left\{-\frac{u^2}{2(\theta^2 A + \theta u/3)}\right\} + 2D \exp\left\{-\frac{s^2}{2(\theta A + u + s/3)}\right\}.$$

This covers bounded matrix martingale noise whose conditional second moment is scalar-dominated, without requiring a coordinate, block, or projection support model.

Proof. The variance-domination assumption gives

$$\sum_i \mathbb{E}(Y_i^2 \mid \mathcal{F}_{i-1}) \preceq \left(\sum_i \theta_i \rho_i^2\right) I.$$

Thus the variance parameter ν in Corollary 7.235 is bounded by $W := \sum_i \theta_i \rho_i^2$. Applying that corollary with $v = \mu + u$ reduces the exceptional term to

$\mathbb{P}\{W > \mu + u\}$. For $Z_i := \theta_i \rho_i^2$, independence gives

$$0 \le Z_i \le \theta_*, \qquad \sum_i \mathbb{E} Z_i = \mu,$$

$$\sum_i \operatorname{Var}(Z_i) \le \sum_i \mathbb{E} Z_i^2 \le \sum_i \mathbb{E}(\theta_i^2 \rho_i^2) = \sigma^2.$$

Scalar Bernstein gives the budget-exception bound. The homogeneous display follows from $\mu \le \theta A$, $\sigma^2 \le \theta^2 A$, and $\theta_* \le \theta$. □

Corollary 7.239 (Matrix-profile dominated martingale noise with random amplitudes)**.** *Let $D \ge 2$. Let $\mathcal{G}$, B_i, and the event $E \in \mathcal{G}$ be as in Corollary 7.235, and assume that outside E there are $\mathcal{G}$-measurable signs $\varepsilon_i^\star \in \{-1, 1\}$ such that*

$$\left\| \sum_{i=1}^n \varepsilon_i^\star B_i \right\|_{\mathrm{op}} \le K\sqrt{n}.$$

Let (V_i, ρ_i), $i = 1, \dots, n$, be independent pairs, independent of $\mathcal{G}$, where $V_i \in \mathrm{M}_D(\mathbb{C})$ is positive semidefinite, $\rho_i \in [0, 1]$, and

$$0 \preceq V_i \preceq \tau_* I$$

almost surely for a deterministic $\tau_* \in [0, 1]$. Suppose that $(\mathcal{F}_i)_{i=0}^n$ is a filtration with $\mathcal{G}, (V_i, \rho_i)_{i=1}^n \subset \mathcal{F}_0$. Let Y_i be self-adjoint and $\mathcal{F}_i$-measurable with

$$\mathbb{E}(Y_i \mid \mathcal{F}_{i-1}) = 0, \qquad \|Y_i\|_{\mathrm{op}} \le \rho_i, \qquad \mathbb{E}(Y_i^2 \mid \mathcal{F}_{i-1}) \preceq \rho_i^2 V_i,$$

and set $X_i = B_i + Y_i$. Define

$$\mu := \left\| \sum_{i=1}^n \mathbb{E}(\rho_i^2 V_i) \right\|_{\mathrm{op}}, \qquad \omega^2 := \left\| \sum_{i=1}^n \mathbb{E}(\rho_i^4 V_i^2) \right\|_{\mathrm{op}}.$$

Then, for every $u \ge 0$ and $s > 0$,

$$\mathbb{P}\left\{ \operatorname{disc}(X_1, \dots, X_n) > K\sqrt{n} + s \right\} \le \mathbb{P}(E) + D \exp\left\{ -\frac{u^2}{2(\omega^2 + \tau_* u/3)} \right\} + 2D \exp\left\{ -\frac{s^2}{2(\mu + u + s/3)} \right\},$$

with the convention that the first exponential term is omitted when $\omega^2 + \tau_* u/3 = 0$. In particular, if $V_i \preceq \tau I$ almost surely for all i and $A = \sum_i \mathbb{E}\rho_i^2$, then

$$\mathbb{P}\left\{ \operatorname{disc}(X_1, \dots, X_n) > K\sqrt{n} + s \right\} \le \mathbb{P}(E) + D \exp\left\{ -\frac{u^2}{2(\tau^2 A + \tau u/3)} \right\} + 2D \exp\left\{ -\frac{s^2}{2(\tau A + u + s/3)} \right\}.$$

This covers anisotropic random conditional-variance profiles, random covariance mixtures, and leverage-profile matrix martingale noise, without asking the variance profile to be scalar.

Proof. The variance-domination assumption gives

$$\sum_i \mathbb{E}(Y_i^2 \mid \mathcal{F}_{i-1}) \preceq \sum_i \rho_i^2 V_i.$$

Hence the variation parameter ν in Corollary 7.235 is bounded by

$$W := \left\| \sum_i \rho_i^2 V_i \right\|_{\mathrm{op}}.$$

Applying that corollary with $v = \mu + u$ leaves only the budget exception $\mathbb{P}\{W > \mu + u\}$. Put $Z_i := \rho_i^2 V_i$. Then the Z_i are independent positive semidefinite matrices, $0 \preceq Z_i \preceq \tau_* I$, and

$$\left\| \sum_i \mathbb{E} Z_i \right\|_{\mathrm{op}} = \mu.$$

Moreover, the one-sided matrix Bernstein variance proxy for $Z_i - \mathbb{E}Z_i$ is at most

$$\left\| \sum_i \mathbb{E}(Z_i - \mathbb{E}Z_i)^2 \right\|_{\mathrm{op}} \le \left\| \sum_i \mathbb{E} Z_i^2 \right\|_{\mathrm{op}} = \omega^2.$$

Tropp's matrix Bernstein inequality gives

$$\mathbb{P}\{W > \mu + u\} \le D \exp\left\{ -\frac{u^2}{2(\omega^2 + \tau_* u/3)} \right\}.$$

For the homogeneous display, use

$$\mu \le \tau A, \qquad \omega^2 \le \tau^2 A,$$

and take the deterministic jump bound to be $\tau_* = \tau$. □

Corollary 7.240 (Trace-budget dominated martingale noise with random amplitudes)**.** *Let $D \ge 2$. Let $\mathcal{G}$, B_i, and the event $E \in \mathcal{G}$ be as in Corollary 7.235, and assume that outside E there are $\mathcal{G}$-measurable signs $\varepsilon_i^\star \in \{-1, 1\}$ such that*

$$\left\| \sum_{i=1}^n \varepsilon_i^\star B_i \right\|_{\mathrm{op}} \le K\sqrt{n}.$$

Let (V_i, ρ_i), $i = 1, \dots, n$, be independent pairs, independent of $\mathcal{G}$, where $V_i \in \mathrm{M}_D(\mathbb{C})$ is positive semidefinite and $\rho_i \in [0, 1]$. Assume that

$$\operatorname{tr} V_i \le \gamma_*$$

almost surely for a deterministic $\gamma_* \ge 0$. Suppose that $(\mathcal{F}_i)_{i=0}^n$ is a filtration with $\mathcal{G}, (V_i, \rho_i)_{i=1}^n \subset \mathcal{F}_0$. Let Y_i be self-adjoint and $\mathcal{F}_i$-measurable with

$$\mathbb{E}(Y_i \mid \mathcal{F}_{i-1}) = 0, \qquad \|Y_i\|_{\mathrm{op}} \le \rho_i, \qquad \mathbb{E}(Y_i^2 \mid \mathcal{F}_{i-1}) \preceq \rho_i^2 V_i,$$

and set $X_i = B_i + Y_i$. *Define*

$$\mu_{\mathrm{tr}} := \sum_{i=1}^{n} \mathbb{E}(\rho_i^2 \operatorname{tr} V_i), \qquad \sigma_{\mathrm{tr}}^2 := \sum_{i=1}^{n} \mathbb{E}(\rho_i^4 (\operatorname{tr} V_i)^2).$$

Then, for every $u \geq 0$ *and* $s > 0$,

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \leq \mathbb{P}(E) + \exp\left\{-\frac{u^2}{2(\sigma_{\mathrm{tr}}^2 + \gamma_* u/3)}\right\} + 2D \exp\left\{-\frac{s^2}{2(\mu_{\mathrm{tr}} + u + s/3)}\right\},$$

with the convention that the first exponential term is omitted when $\sigma_{\mathrm{tr}}^2 + \gamma_* u/3 = 0$. *In particular, if* $\operatorname{tr} V_i \leq \gamma$ *almost surely for all* i *and* $A = \sum_i \mathbb{E}\rho_i^2$, *then*

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \leq \mathbb{P}(E) + \exp\left\{-\frac{u^2}{2(\gamma^2 A + \gamma u/3)}\right\} + 2D \exp\left\{-\frac{s^2}{2(\gamma A + u + s/3)}\right\}.$$

Thus low-total-trace random variance profiles give a dimension-free budget exception; the price is that the Freedman variance parameter is controlled by the trace budget rather than the sharper operator-norm budget.

Proof. As in the matrix-profile corollary,

$$\sum_i \mathbb{E}(Y_i^2 \mid \mathcal{F}_{i-1}) \preceq \sum_i \rho_i^2 V_i.$$

Because the right-hand side is positive semidefinite,

$$\left\| \sum_i \rho_i^2 V_i \right\|_{\mathrm{op}} \leq \operatorname{tr}\left(\sum_i \rho_i^2 V_i \right) = \sum_i \rho_i^2 \operatorname{tr} V_i.$$

Hence Corollary 7.235 applies with the random scalar budget

$$T := \sum_i \rho_i^2 \operatorname{tr} V_i.$$

Taking $v = \mu_{\mathrm{tr}} + u$, it remains to bound $\mathbb{P}\{T > \mu_{\mathrm{tr}} + u\}$. For $Z_i := \rho_i^2 \operatorname{tr} V_i$, the variables are independent and satisfy

$$0 \leq Z_i \leq \gamma_*, \qquad \sum_i \mathbb{E} Z_i = \mu_{\mathrm{tr}},$$

$$\sum_i \operatorname{Var}(Z_i) \leq \sum_i \mathbb{E} Z_i^2 = \sum_i \mathbb{E}(\rho_i^4 (\operatorname{tr} V_i)^2) = \sigma_{\mathrm{tr}}^2.$$

Scalar Bernstein gives the first display. The homogeneous display follows from $\mu_{\mathrm{tr}} \leq \gamma A$ and $\sigma_{\mathrm{tr}}^2 \leq \gamma^2 A$. □

Corollary 7.241 (Trace-envelope and low-rank variance profiles)**.** *Let $D \geq 2$. Let $\mathcal{G}$, B_i, and the event $E \in \mathcal{G}$ be as in Corollary 7.235, and assume that outside E there are $\mathcal{G}$-measurable signs $\varepsilon_i^\star \in \{-1, 1\}$ such that*

$$\left\| \sum_{i=1}^n \varepsilon_i^\star B_i \right\|_{\mathrm{op}} \leq K\sqrt{n}.$$

Let (V_i, Γ_i, ρ_i), $i = 1, \dots, n$, be independent triples, independent of $\mathcal{G}$, where $V_i \in \mathrm{M}_D(\mathbb{C})$ is positive semidefinite, $\rho_i \in [0, 1]$, and

$$0 \leq \Gamma_i \leq \gamma_*, \qquad \operatorname{tr} V_i \leq \Gamma_i$$

almost surely for a deterministic $\gamma_* \geq 0$. Suppose that $(\mathcal{F}_i)_{i=0}^n$ is a filtration with $\mathcal{G}, (V_i, \Gamma_i, \rho_i)_{i=1}^n \subset \mathcal{F}_0$. Let Y_i be self-adjoint and $\mathcal{F}_i$-measurable with

$$\mathbb{E}(Y_i \mid \mathcal{F}_{i-1}) = 0, \qquad \|Y_i\|_{\mathrm{op}} \leq \rho_i, \qquad \mathbb{E}(Y_i^2 \mid \mathcal{F}_{i-1}) \preceq \rho_i^2 V_i,$$

and set $X_i = B_i + Y_i$. Define

$$\mu_\Gamma := \sum_{i=1}^n \mathbb{E}(\rho_i^2 \Gamma_i), \qquad \sigma_\Gamma^2 := \sum_{i=1}^n \mathbb{E}(\rho_i^4 \Gamma_i^2).$$

Then, for every $u \geq 0$ and $s > 0$,

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \leq \mathbb{P}(E) + \exp\left\{-\frac{u^2}{2(\sigma_\Gamma^2 + \gamma_* u/3)}\right\} + 2D \exp\left\{-\frac{s^2}{2(\mu_\Gamma + u + s/3)}\right\},$$

with the convention that the first exponential term is omitted when $\sigma_\Gamma^2 + \gamma_* u/3 = 0$.

In particular, suppose that the envelope is supplied by random low-rank data: there are independent tuples $(V_i, R_i, \tau_i, \rho_i)$, independent of $\mathcal{G}$, such that $V_i \succeq 0$,

$$\operatorname{rank} V_i \leq R_i, \qquad \|V_i\|_{\mathrm{op}} \leq \tau_i, \qquad R_i \tau_i \leq \gamma_*$$

almost surely. Then the same conclusion holds with

$$\mu_{\mathrm{lr}} := \sum_{i=1}^n \mathbb{E}(R_i \tau_i \rho_i^2), \qquad \sigma_{\mathrm{lr}}^2 := \sum_{i=1}^n \mathbb{E}(R_i^2 \tau_i^2 \rho_i^4)$$

in place of $\mu_\Gamma, \sigma_\Gamma^2$. If moreover $R_i \leq r$ and $\tau_i \leq \tau$ almost surely for every i, and $A = \sum_i \mathbb{E}\rho_i^2$, then

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \leq \mathbb{P}(E) + \exp\left\{-\frac{u^2}{2(r^2\tau^2 A + r\tau u/3)}\right\} + 2D \exp\left\{-\frac{s^2}{2(r\tau A + u + s/3)}\right\}.$$

Thus trace-enveloped low-rank and spiked variance profiles are controlled by their scalar envelope rather than by the ambient dimension.

Proof. As in the proof of Corollary 7.240, the variation in Corollary 7.235 is bounded by the random scalar trace budget. Since $\operatorname{tr} V_i \le \Gamma_i$, that budget satisfies

$$T = \sum_i \rho_i^2 \operatorname{tr} V_i \le \sum_i \rho_i^2 \Gamma_i.$$

For $Z_i := \rho_i^2 \Gamma_i$, the variables are independent, $0 \le Z_i \le \gamma_*$, and

$$\sum_i \mathbb{E} Z_i = \mu_\Gamma, \qquad \sum_i \operatorname{Var}(Z_i) \le \sum_i \mathbb{E} Z_i^2 = \sigma_\Gamma^2.$$

Scalar Bernstein therefore gives

$$\mathbb{P}\{T > \mu_\Gamma + u\} \le \exp\left\{ -\frac{u^2}{2(\sigma_\Gamma^2 + \gamma_* u/3)} \right\}.$$

Combining this with Corollary 7.235 at $v = \mu_\Gamma + u$ proves the trace-envelope display.

For the low-rank specialization, take $\Gamma_i = R_i \tau_i$. The trace bound follows from

$$\operatorname{tr} V_i \le \operatorname{rank}(V_i) \|V_i\|_{\mathrm{op}} \le R_i \tau_i.$$

The last display uses

$$\mu_{\mathrm{lr}} \le r\tau A, \qquad \sigma_{\mathrm{lr}}^2 \le r^2 \tau^2 A, \qquad \gamma_* = r\tau,$$

where $\rho_i^4 \le \rho_i^2$. □

Corollary 7.242 (Schatten-envelope variance profiles with random amplitudes)**.** *Let $1 \le p \le \infty$, and put $\beta = 1 - 1/p$, with the convention $\beta = 1$ when $p = \infty$. Keep the signed-core, filtration, and martingale-noise hypotheses of Corollary 7.241, but replace the scalar trace-envelope assumption by the following Schatten envelope. Let (V_i, R_i, L_i, ρ_i), $i = 1, \dots, n$, be independent tuples, independent of $\mathcal{G}$, where $V_i \succeq 0$, $\rho_i \in [0,1]$, $1 \le R_i \le D$, and*

$$\operatorname{rank} V_i \le R_i, \qquad \|V_i\|_{S_p} \le L_i, \qquad R_i^\beta L_i \le \gamma_*$$

almost surely. Define

$$\mu_p := \sum_{i=1}^n \mathbb{E}(R_i^\beta L_i \rho_i^2), \qquad \sigma_p^2 := \sum_{i=1}^n \mathbb{E}(R_i^{2\beta} L_i^2 \rho_i^4).$$

Then, for every $u \ge 0$ and $s > 0$,

$$\mathbb{P}\left\{ \operatorname{disc}(X_1, \dots, X_n) > K\sqrt{n} + s \right\} \le \mathbb{P}(E) + \exp\left\{ -\frac{u^2}{2(\sigma_p^2 + \gamma_* u/3)} \right\} + 2D \exp\left\{ -\frac{s^2}{2(\mu_p + u + s/3)} \right\}.$$

In particular, if $R_i \le r$, $L_i \le L$ almost surely, and $A = \sum_i \mathbb{E}\rho_i^2$, then

$$\mathbb{P}\left\{\operatorname{disc}(X_1,\dots,X_n) > K\sqrt{n}+s\right\} \le \mathbb{P}(E) + \exp\left\{-\frac{u^2}{2(r^{2\beta}L^2A + r^\beta Lu/3)}\right\} + 2D\exp\left\{-\frac{s^2}{2(r^\beta LA + u + s/3)}\right\}.$$

Thus trace control ($p=1$), Frobenius-plus-rank control ($p=2$), and operator-norm-plus-rank control ($p=\infty$) are all instances of the same signed-core random variance-profile theorem.

Proof. For a positive semidefinite matrix V_i with at most R_i nonzero singular values, Holder's inequality for the singular-value vector gives

$$\operatorname{tr} V_i = \|V_i\|_{S_1} \le R_i^{1-1/p}\|V_i\|_{S_p} \le R_i^\beta L_i.$$

Apply Corollary 7.241 with $\Gamma_i = R_i^\beta L_i$. The homogeneous display follows from $\rho_i^4 \le \rho_i^2$. □

Corollary 7.243 (Quantile trace-envelope variance profiles)**.** *Let $D \ge 2$. Keep the signed-core, filtration, and martingale-noise hypotheses of Corollary 7.241, but replace the boundedness and independence assumptions on the trace envelopes by the following confidence-envelope assumption. The triples (V_i, Γ_i, ρ_i) are $\mathcal{F}_0$-measurable, $V_i \succeq 0$, $\rho_i \in [0,1]$, $\Gamma_i \ge 0$, and*

$$\operatorname{tr} V_i \le \Gamma_i \qquad \text{almost surely.}$$

Let S be any nonnegative random variable such that

$$\sum_{i=1}^n \rho_i^2\Gamma_i \le S \qquad \text{almost surely.}$$

Then, for every deterministic $v \ge 0$ and $s > 0$,

$$\mathbb{P}\left\{\operatorname{disc}(X_1,\dots,X_n) > K\sqrt{n}+s\right\} \le \mathbb{P}(E) + \mathbb{P}\{S > v\} + 2D\exp\left\{-\frac{s^2}{2(v+s/3)}\right\}.$$

Consequently, if q_δ is any deterministic upper δ-quantile of S, meaning $\mathbb{P}\{S > q_\delta\} \le \delta$, then

$$\mathbb{P}\left\{\operatorname{disc}(X_1,\dots,X_n) > K\sqrt{n}+s\right\} \le \mathbb{P}(E) + \delta + 2D\exp\left\{-\frac{s^2}{2(q_\delta+s/3)}\right\}.$$

If, more concretely, $\mathbb{E}S^r \le M_r < \infty$ for some $r > 0$, then for every $0 < \delta \le 1$,

$$\mathbb{P}\left\{\operatorname{disc}(X_1,\dots,X_n) > K\sqrt{n}+s\right\} \le \mathbb{P}(E) + \delta + 2D\exp\left\{-\frac{s^2}{2((M_r/\delta)^{1/r}+s/3)}\right\}.$$

Thus heavy-tailed trace-class random covariance, spike, low-rank, and Schatten variance profiles still admit the same signed-core Matrix Spencer reduction at any confidence level for which the total weighted trace envelope has a usable upper quantile. In the Schatten setting one may take $S = \sum_i \rho_i^2 R_i^{1-1/p} L_i$ *whenever* $\operatorname{rank} V_i \le R_i$ *and* $\|V_i\|_{S_p} \le L_i$*; no independence between these envelope data is needed for this quantile form.*

Proof. The proof of Corollary 7.241 uses only the trace domination to bound the predictable variation in Corollary 7.235. Here the same argument gives

$$\nu = \left\| \sum_i \mathbb{E}(Y_i^2 \mid \mathcal{F}_{i-1}) \right\|_{\mathrm{op}} \le \sum_i \rho_i^2 \operatorname{tr} V_i \le \sum_i \rho_i^2 \Gamma_i \le S.$$

Applying Corollary 7.235 with the deterministic parameter v and using $\{\nu > v\} \subseteq \{S > v\}$ proves the first display. The quantile form is the same estimate with $v = q_\delta$. The moment form follows from Markov's inequality,

$$\mathbb{P}\{S > (M_r/\delta)^{1/r}\} \le \delta,$$

with the evident interpretation when $M_r = 0$. The Schatten sentence follows from Holder's inequality for the singular-value vector, as in Corollary 7.242. □

Corollary 7.244 (Almost-sure quantile trace-envelope families)**.** *For each* $m \ge 1$*, let*

$$X_{m,1}, \dots, X_{m,n_m} \in \mathrm{M}_{D_m}(\mathbb{C}), \qquad D_m \ge 2,$$

be a random instance satisfying the hypotheses of Corollary 7.243, with parameters K_m*, exceptional event* E_m*, and dominating trace budget* S_m*. Let* q_m, δ_m, s_m *be deterministic numbers with* $q_m, s_m \ge 0$*,* $0 \le \delta_m \le 1$*, and*

$$\mathbb{P}\{S_m > q_m\} \le \delta_m.$$

If

$$\sum_{m=1}^{\infty} \left[\mathbb{P}(E_m) + \delta_m + 2D_m \exp\left\{ -\frac{s_m^2}{2(q_m + s_m/3)} \right\} \right] < \infty,$$

then, almost surely, for all sufficiently large m*,*

$$\operatorname{disc}(X_{m,1}, \dots, X_{m,n_m}) \le K_m \sqrt{n_m} + s_m.$$

In particular, put $\ell_m = \log(eD_m)$*. Suppose that* $K_m \le K$*, that*

$$q_m \le Q \frac{n_m}{\ell_m}, \qquad \ell_m \le B\sqrt{n_m}$$

for fixed $Q, B < \infty$*, and that for some* $A > 0$*,*

$$\sum_{m=1}^{\infty} \left[\mathbb{P}(E_m) + \delta_m + (eD_m)^{-A} \right] < \infty.$$

Choose $L > 0$ *so that*

$$\frac{L^2}{2(Q + LB/3)} \ge A + 1.$$

Then almost surely, for all sufficiently large m,

$$\operatorname{disc}(X_{m,1},\dots,X_{m,n_m}) \le (K+L)\sqrt{n_m}.$$

The same criterion may be checked from moments: if $\mathbb{E}S_m^r \le M_{m,r}$, one may take $q_m = (M_{m,r}/\delta_m)^{1/r}$. Thus summable quantile or finite-moment control of heavy-tailed trace budgets at scale $n_m/\log(eD_m)$ gives eventual Matrix Spencer bounds for triangular arrays, with no independence assumption across the dimensions m.

Proof. Corollary 7.243 gives, for each m, with $\Delta_m = \operatorname{disc}(X_{m,1},\dots,X_{m,n_m})$,

$$\mathbb{P}\{\Delta_m > K_m\sqrt{n_m} + s_m\} \le \mathbb{P}(E_m) + \delta_m + 2D_m \exp\left\{-\frac{s_m^2}{2(q_m + s_m/3)}\right\}.$$

The displayed summability hypothesis and the first Borel–Cantelli lemma give the almost-sure eventual estimate.

For the concrete profile, take $s_m = L\sqrt{n_m}$. Since $\ell_m \le B\sqrt{n_m}$, we have

$$L\sqrt{n_m}/3 \le (LB/3)\, n_m/\ell_m.$$

Together with $q_m \le Qn_m/\ell_m$, this gives

$$\frac{s_m^2}{2(q_m + s_m/3)} \ge \frac{L^2}{2(Q + LB/3)}\,\ell_m \ge (A+1)\ell_m.$$

Therefore

$$2D_m \exp\left\{-\frac{s_m^2}{2(q_m + s_m/3)}\right\} \le 2D_m(eD_m)^{-(A+1)} \le 2(eD_m)^{-A}.$$

The assumed summability and the first part apply. The moment assertion is Markov's inequality, exactly as in Corollary 7.243. □

Corollary 7.245 (Moment-summable heavy-tail trace-envelope families)**.** *In the setting of Corollary 7.244, put $\ell_m = \log(eD_m)$. Suppose that $K_m \le K$ and $\ell_m \le B\sqrt{n_m}$ for fixed $K, B < \infty$. Let $r > 0$, $Q > 0$, and $A > 0$. If*

$$\sum_{m=1}^{\infty}\left[\mathbb{P}(E_m) + \mathbb{E}\left(\frac{\ell_m S_m}{Qn_m}\right)^r + (eD_m)^{-A}\right] < \infty,$$

then, for every $L > 0$ satisfying

$$\frac{L^2}{2(Q + LB/3)} \ge A + 1,$$

almost surely, for all sufficiently large m,

$$\operatorname{disc}(X_{m,1},\dots,X_{m,n_m}) \le (K+L)\sqrt{n_m}.$$

Equivalently, it is enough that

$$\mathbb{E}S_m^r \le Q^r\left(\frac{n_m}{\ell_m}\right)^r a_m \qquad \text{with} \qquad \sum_m a_m < \infty,$$

along with the summability of $\mathbb{P}(E_m)$ *and* $(eD_m)^{-A}$*. Thus heavy-tailed trace envelopes whose normalized* r*-th moments are summable still give eventual* $O(\sqrt{n_m})$ *Matrix Spencer bounds; no exponential tail or independence across the dimensions is required.*

Proof. Set

$$q_m = Q\frac{n_m}{\ell_m}, \qquad \delta_m = \mathbb{E}\left(\frac{\ell_m S_m}{Qn_m}\right)^r.$$

Markov's inequality gives

$$\mathbb{P}\{S_m > q_m\} \le \delta_m.$$

The displayed moment summability is exactly the summability hypothesis in the concrete profile of Corollary 7.244. The stated almost-sure conclusion follows. The equivalent form is just the same criterion after bounding the normalized moment by a_m. □

Corollary 7.246 (Termwise-moment heavy-tail trace-envelope families)**.** *In the setting of Corollary 7.245, suppose that, for each* m*, the dominating trace budget has an almost-sure termwise bound*

$$S_m \le \sum_{i=1}^{n_m} W_{m,i}, \qquad W_{m,i} \ge 0.$$

This includes the basic trace-envelope choice $W_{m,i} = \rho_{m,i}^2 \Gamma_{m,i}$*, and in the Schatten-envelope case the choice* $W_{m,i} = \rho_{m,i}^2 R_{m,i}^{1-1/p} L_{m,i}$*. Assume that* $K_m \le K$ *and* $\ell_m = \log(eD_m) \le B\sqrt{n_m}$*, and fix* $Q, A > 0$*. If, for some* $0 < r \le 1$*,*

$$\sum_{m=1}^{\infty}\left[\mathbb{P}(E_m) + \left(\frac{\ell_m}{Qn_m}\right)^r \sum_{i=1}^{n_m} \mathbb{E}W_{m,i}^r + (eD_m)^{-A}\right] < \infty,$$

then, for every $L > 0$ *satisfying*

$$\frac{L^2}{2(Q + LB/3)} \ge A + 1,$$

almost surely, for all sufficiently large m*,*

$$\operatorname{disc}(X_{m,1}, \dots, X_{m,n_m}) \le (K + L)\sqrt{n_m}.$$

The same conclusion holds for $r \ge 1$ *under the alternative summability condition*

$$\sum_{m=1}^{\infty}\left[\mathbb{P}(E_m) + \left(\frac{\ell_m}{Qn_m}\right)^r \left(\sum_{i=1}^{n_m} (\mathbb{E}W_{m,i}^r)^{1/r}\right)^r + (eD_m)^{-A}\right] < \infty.$$

Thus random spike, low-rank, trace-class, and Schatten variance profiles with only termwise heavy-tail moments still give almost-sure eventual Matrix Spencer bounds, without any independence among the terms $W_{m,i}$*.*

Proof. Put $\widetilde{S}_m = \sum_i W_{m,i}$. Since $S_m \le \widetilde{S}_m$, it is enough to verify the normalized-moment hypothesis of Corollary 7.245 with $\widetilde{S}_m$ in place of S_m. If $0 < r \le 1$, then $(\sum_i w_i)^r \le \sum_i w_i^r$ for nonnegative w_i, so

$$\mathbb{E}\widetilde{S}_m^r \le \sum_{i=1}^{n_m} \mathbb{E}W_{m,i}^r.$$

If $r \ge 1$, Minkowski's inequality in L^r gives

$$\left(\mathbb{E}\widetilde{S}_m^r\right)^{1/r} \le \sum_{i=1}^{n_m} \left(\mathbb{E}W_{m,i}^r\right)^{1/r}.$$

Each displayed summability condition therefore implies the corresponding normalized-moment summability for $\widetilde{S}_m$, and the preceding corollary applies. □

Corollary 7.247 (Termwise-tail heavy-tail trace-envelope families)**.** *In the setting of Corollary 7.246, suppose again that*

$$S_m \le \sum_{i=1}^{n_m} W_{m,i}, \qquad W_{m,i} \ge 0.$$

Let $K_m \le K$, $\ell_m = \log(eD_m) \le B\sqrt{n_m}$, and fix $Q, A > 0$. For each m, let $b_{m,1}, \dots, b_{m,n_m} \ge 0$ and $\delta_{m,1}, \dots, \delta_{m,n_m} \ge 0$ be deterministic numbers such that

$$\sum_{i=1}^{n_m} b_{m,i} \le Q\frac{n_m}{\ell_m}, \qquad \mathbb{P}\{W_{m,i} > b_{m,i}\} \le \delta_{m,i} \quad (1 \le i \le n_m).$$

If

$$\sum_{m=1}^{\infty} \left[\mathbb{P}(E_m) + \sum_{i=1}^{n_m} \delta_{m,i} + (eD_m)^{-A}\right] < \infty,$$

then, for every $L > 0$ satisfying

$$\frac{L^2}{2(Q + LB/3)} \ge A + 1,$$

almost surely, for all sufficiently large m,

$$\operatorname{disc}(X_{m,1}, \dots, X_{m,n_m}) \le (K + L)\sqrt{n_m}.$$

Thus it is enough to truncate each heavy-tailed spike, low-rank, trace-class, or Schatten atom at deterministic levels whose total mass is $O(n_m/\log(eD_m))$, with summable total truncation failure; no independence among the atoms is required.

Proof. Set

$$q_m = Q\frac{n_m}{\ell_m}, \qquad \delta_m = \sum_{i=1}^{n_m} \delta_{m,i}.$$

If $W_{m,i} \le b_{m,i}$ for every i, then

$$S_m \le \sum_i W_{m,i} \le \sum_i b_{m,i} \le q_m.$$

Therefore, by the union bound,

$$\mathbb{P}\{S_m > q_m\} \le \sum_{i=1}^{n_m} \mathbb{P}\{W_{m,i} > b_{m,i}\} \le \delta_m.$$

The concrete profile in Corollary 7.244 now applies with these q_m and δ_m. □

Corollary 7.248 (Scale-mixture termwise-tail trace-envelope families)**.** *In the setting of Corollary 7.247, suppose that*

$$S_m \le \sum_{i=1}^{n_m} W_{m,i}, \qquad W_{m,i} \le a_{m,i} Z_{m,i}$$

almost surely, where $a_{m,i} \ge 0$ are deterministic and $Z_{m,i} \ge 0$ are arbitrary random multipliers. No independence is assumed among the $Z_{m,i}$. Let $K_m \le K$, $\ell_m = \log(eD_m) \le B\sqrt{n_m}$, and fix $Q, A > 0$. If there are deterministic thresholds $\tau_m \ge 0$ such that

$$\tau_m \sum_{i=1}^{n_m} a_{m,i} \le Q \frac{n_m}{\ell_m}$$

and

$$\sum_{m=1}^{\infty} \left[\mathbb{P}(E_m) + \sum_{i=1}^{n_m} \mathbb{P}\{Z_{m,i} > \tau_m\} + (eD_m)^{-A} \right] < \infty,$$

then, for every $L > 0$ satisfying

$$\frac{L^2}{2(Q + LB/3)} \ge A + 1,$$

almost surely, for all sufficiently large m,

$$\operatorname{disc}(X_{m,1}, \dots, X_{m,n_m}) \le (K + L)\sqrt{n_m}.$$

In particular, put $A_m = \sum_i a_{m,i}$. If, for some $\alpha > 0$, the multipliers satisfy the uniform polynomial tail

$$\mathbb{P}\{Z_{m,i} > t\} \le C_m t^{-\alpha} \qquad (t > 0,\ 1 \le i \le n_m),$$

then the conclusion follows from

$$\sum_{m=1}^{\infty} \left[\mathbb{P}(E_m) + (eD_m)^{-A} \right] + \sum_{m : A_m > 0} n_m C_m \left(\frac{\ell_m A_m}{Q n_m} \right)^{\alpha} < \infty,$$

with the zero-scale rows $A_m = 0$ carrying no multiplier-tail term. Thus deterministic trace, rank, or Schatten atom weights may carry arbitrary dependent heavy-tailed multipliers as long as their aggregate upper tail is summable at the $n_m/\log(eD_m)$ budget scale.

Proof. Apply Corollary 7.247 with $b_{m,i} = \tau_m a_{m,i}$ and $\delta_{m,i} = \mathbb{P}\{Z_{m,i} > \tau_m\}$. Indeed, if $Z_{m,i} \le \tau_m$, then $W_{m,i} \le b_{m,i}$, while if $a_{m,i} = 0$ the bound $W_{m,i} \le 0 = b_{m,i}$ is automatic. Hence

$$\mathbb{P}\{W_{m,i} > b_{m,i}\} \le \mathbb{P}\{Z_{m,i} > \tau_m\}.$$

The mass condition gives $\sum_i b_{m,i} \le Qn_m/\ell_m$, and the displayed summability hypothesis is precisely the termwise-tail hypothesis.

For the polynomial-tail specialization, take $\tau_m = Qn_m/(\ell_m A_m)$ when $A_m > 0$. Then $\tau_m A_m = Qn_m/\ell_m$, and

$$\sum_{i=1}^{n_m} \mathbb{P}\{Z_{m,i} > \tau_m\} \le n_m C_m \left(\frac{\ell_m A_m}{Qn_m}\right)^{\alpha}.$$

When $A_m = 0$, all $W_{m,i}$ vanish almost surely, so no multiplier tail term is needed. □

Corollary 7.249 (Common-scale heavy-tail trace-envelope families)**.** *In the setting of Corollary 7.244, put $\ell_m = \log(eD_m)$. Suppose that $K_m \le K$ and $\ell_m \le B\sqrt{n_m}$ for fixed $K, B < \infty$. Let $Q, A > 0$. Assume that there are deterministic weights $a_{m,i} \ge 0$ and a nonnegative random multiplier Z_m such that, with $A_m = \sum_i a_{m,i}$,*

$$S_m \le Z_m A_m \qquad \text{almost surely.}$$

This includes the atomwise situation $S_m \le \sum_i W_{m,i}$ with $W_{m,i} \le a_{m,i} Z_m$, where the same random scale multiplies every atom. If

$$\sum_{m=1}^{\infty} \left[\mathbb{P}(E_m) + (eD_m)^{-A}\right] + \sum_{m:A_m>0} \mathbb{P}\left\{Z_m > Q\frac{n_m}{\ell_m A_m}\right\} < \infty,$$

then, for every $L > 0$ satisfying

$$\frac{L^2}{2(Q + LB/3)} \ge A + 1,$$

almost surely, for all sufficiently large m,

$$\operatorname{disc}(X_{m,1}, \dots, X_{m,n_m}) \le (K + L)\sqrt{n_m}.$$

In particular, if for some $\alpha > 0$

$$\mathbb{P}\{Z_m > t\} \le C_m t^{-\alpha} \qquad (t > 0),$$

then the conclusion follows from

$$\sum_{m=1}^{\infty} \left[\mathbb{P}(E_m) + (eD_m)^{-A}\right] + \sum_{m:A_m>0} C_m \left(\frac{\ell_m A_m}{Qn_m}\right)^{\alpha} < \infty.$$

Thus global random volatility, elliptical covariance scales, and other common-multiplier trace-envelope models pay only the tail of the shared scale, not a union bound over all atoms.

Proof. Put $q_m = Qn_m/\ell_m$. If $A_m = 0$, then $S_m = 0$ almost surely. If $A_m > 0$, then

$$\mathbb{P}\{S_m > q_m\} \le \mathbb{P}\{Z_m A_m > q_m\} = \mathbb{P}\left\{Z_m > Q\frac{n_m}{\ell_m A_m}\right\}.$$

Thus the concrete profile in Corollary 7.244 applies with these deterministic quantiles. The polynomial tail specialization is the same estimate after inserting $t = Qn_m/(\ell_m A_m)$ for rows with $A_m > 0$. □

Corollary 7.250 (Grouped-scale heavy-tail trace-envelope families)**.** *In the setting of Corollary 7.244, put* $\ell_m = \log(eD_m)$*. Suppose that* $K_m \le K$ *and* $\ell_m \le B\sqrt{n_m}$ *for fixed* $K, B < \infty$*. Let* $Q, A > 0$*. For each* m*, let* G_m *be a finite set, let* $A_{m,g} \ge 0$ *be deterministic group masses, and let* $Z_{m,g} \ge 0$ *be arbitrary random group multipliers. Assume that*

$$S_m \le \sum_{g \in G_m} A_{m,g} Z_{m,g} \qquad \text{almost surely.}$$

If there are deterministic thresholds $\tau_{m,g} \ge 0$ *such that*

$$\sum_{g \in G_m} \tau_{m,g} A_{m,g} \le Q \frac{n_m}{\ell_m}$$

and

$$\sum_{m=1}^{\infty} \left[\mathbb{P}(E_m) + \sum_{g \in G_m} \mathbb{P}\{Z_{m,g} > \tau_{m,g}\} + (eD_m)^{-A} \right] < \infty,$$

then, for every $L > 0$ *satisfying*

$$\frac{L^2}{2(Q + LB/3)} \ge A + 1,$$

almost surely, for all sufficiently large m*,*

$$\operatorname{disc}(X_{m,1}, \dots, X_{m,n_m}) \le (K + L)\sqrt{n_m}.$$

In particular, suppose that, for some $\alpha > 0$*,*

$$\mathbb{P}\{Z_{m,g} > t\} \le C_{m,g} t^{-\alpha} \qquad (t > 0,\ g \in G_m),$$

and that there are deterministic budget shares $\lambda_{m,g} > 0$ *for every* g *with* $A_{m,g} > 0$*, satisfying* $\sum_{g : A_{m,g} > 0} \lambda_{m,g} \le 1$*. Then the conclusion follows from*

$$\sum_{m=1}^{\infty} \left[\mathbb{P}(E_m) + (eD_m)^{-A} \right] + \sum_{m} \sum_{\substack{g \in G_m \\ A_{m,g} > 0}} C_{m,g} \left(\frac{\ell_m A_{m,g}}{Q n_m \lambda_{m,g}} \right)^{\alpha} < \infty.$$

Thus block-volatility, sector-volatility, and multiscale elliptical trace-envelope models pay one heavy-tail cost per group, interpolating between the atomwise scale-mixture and common-scale criteria.

Proof. Put $q_m = Q n_m / \ell_m$. If $Z_{m,g} \le \tau_{m,g}$ for every $g \in G_m$, then

$$S_m \le \sum_{g \in G_m} A_{m,g} Z_{m,g} \le \sum_{g \in G_m} \tau_{m,g} A_{m,g} \le q_m.$$

The union bound therefore gives

$$\mathbb{P}\{S_m > q_m\} \le \sum_{g \in G_m} \mathbb{P}\{Z_{m,g} > \tau_{m,g}\}.$$

The concrete Borel profile in Corollary 7.244 applies.

For the polynomial-tail specialization, take

$$\tau_{m,g} = Q \frac{n_m}{\ell_m} \frac{\lambda_{m,g}}{A_{m,g}} \qquad (A_{m,g} > 0).$$

Rows or groups with $A_{m,g} = 0$ contribute no mass. The budget-share condition gives $\sum_g \tau_{m,g} A_{m,g} \le Q n_m/\ell_m$, and the displayed tail sum follows after inserting this value of $\tau_{m,g}$. □

Corollary 7.251 (Countable grouped-scale heavy-tail trace-envelope families)**.** *In the setting of Corollary 7.244, put $\ell_m = \log(eD_m)$. Suppose that $K_m \le K$ and $\ell_m \le B\sqrt{n_m}$ for fixed $K, B < \infty$. Let $Q, A > 0$. For each m, let G_m be finite or countable, let $A_{m,g} \ge 0$ be deterministic group masses, and let $Z_{m,g} \ge 0$ be arbitrary random group multipliers. Assume that*

$$S_m \le \sum_{g \in G_m} A_{m,g} Z_{m,g} \qquad \text{almost surely.}$$

If there are deterministic thresholds $\tau_{m,g} \ge 0$ such that

$$\sum_{g \in G_m} \tau_{m,g} A_{m,g} \le Q \frac{n_m}{\ell_m}$$

and

$$\sum_{m=1}^{\infty} \left[\mathbb{P}(E_m) + \sum_{g \in G_m} \mathbb{P}\{Z_{m,g} > \tau_{m,g}\} + (eD_m)^{-A} \right] < \infty,$$

then, for every $L > 0$ satisfying

$$\frac{L^2}{2(Q + LB/3)} \ge A + 1,$$

almost surely, for all sufficiently large m,

$$\operatorname{disc}(X_{m,1}, \dots, X_{m,n_m}) \le (K + L)\sqrt{n_m}.$$

In particular, suppose that, for some $\alpha > 0$,

$$\mathbb{P}\{Z_{m,g} > t\} \le C_{m,g} t^{-\alpha} \qquad (t > 0,\ g \in G_m),$$

and that there are deterministic budget shares $\lambda_{m,g} > 0$ for every g with $A_{m,g} > 0$, satisfying $\sum_{g : A_{m,g} > 0} \lambda_{m,g} \le 1$. Then the conclusion follows from

$$\sum_{m=1}^{\infty} \left[\mathbb{P}(E_m) + (eD_m)^{-A} \right] + \sum_m \sum_{\substack{g \in G_m \\ A_{m,g} > 0}} C_{m,g} \left(\frac{\ell_m A_{m,g}}{Q n_m \lambda_{m,g}} \right)^{\alpha} < \infty.$$

Thus countable sector, frequency, scale, and trace-class volatility decompositions can be handled with one truncation failure per active group and no independence assumptions.

Proof. Put $q_m = Q n_m/\ell_m$. If $Z_{m,g} \le \tau_{m,g}$ for every $g \in G_m$, then monotone convergence for nonnegative series gives

$$S_m \le \sum_{g \in G_m} A_{m,g} Z_{m,g} \le \sum_{g \in G_m} \tau_{m,g} A_{m,g} \le q_m.$$

The countable union bound therefore gives

$$\mathbb{P}\{S_m > q_m\} \le \sum_{g \in G_m} \mathbb{P}\{Z_{m,g} > \tau_{m,g}\}.$$

The concrete Borel profile in Corollary 7.244 applies.

For the polynomial-tail specialization, take

$$\tau_{m,g} = Q\frac{n_m}{\ell_m}\frac{\lambda_{m,g}}{A_{m,g}} \qquad (A_{m,g} > 0),$$

and ignore the zero-mass groups. The share condition gives $\sum_g \tau_{m,g}A_{m,g} \le Qn_m/\ell_m$, and substituting this $\tau_{m,g}$ into the countable tail sum gives the displayed criterion. □

Corollary 7.252 (Optimized tail-profile grouped trace envelopes)**.** *In the setting of Corollary 7.251, let G_m be finite or countable and write $\ell_m = \log(eD_m)$. Suppose that $K_m \le K$, $\ell_m \le B\sqrt{n_m}$, and*

$$S_m \le \sum_{g\in G_m} A_{m,g}Z_{m,g} \qquad \text{almost surely,}$$

where $A_{m,g} \ge 0$ are deterministic and $Z_{m,g} \ge 0$ are arbitrary random multipliers. For each positive-mass group let $\pi_{m,g} : (0,\infty) \to [0,\infty]$ be a deterministic tail profile such that

$$\mathbb{P}\{Z_{m,g} > t\} \le \pi_{m,g}(t) \qquad (t > 0).$$

For fixed $Q > 0$, define the optimized profile cost

$$\Theta_m(Q) = \inf_{\lambda} \sum_{\substack{g\in G_m\\ A_{m,g}>0}} \pi_{m,g}\left(Q\frac{n_m}{\ell_m}\frac{\lambda_g}{A_{m,g}}\right),$$

where the infimum is over all deterministic $\lambda_g > 0$ on the positive-mass groups with $\sum_{g:A_{m,g}>0}\lambda_g \le 1$; if there are no positive-mass groups, the empty infimum is 0. If, for some $A > 0$,

$$\sum_{m=1}^{\infty}\Big[\mathbb{P}(E_m) + (eD_m)^{-A} + \Theta_m(Q)\Big] < \infty,$$

then, for every $L > 0$ satisfying

$$\frac{L^2}{2(Q + LB/3)} \ge A + 1,$$

almost surely, for all sufficiently large m,

$$\operatorname{disc}(X_{m,1},\dots,X_{m,n_m}) \le (K + L)\sqrt{n_m}.$$

In particular, it is enough to exhibit deterministic shares $\lambda_{m,g} > 0$, $\sum_{g:A_{m,g}>0}\lambda_{m,g} \le 1$, for which

$$\sum_{m=1}^{\infty}\Big[\mathbb{P}(E_m) + (eD_m)^{-A}\Big] + \sum_{m}\sum_{\substack{g\in G_m\\ A_{m,g}>0}} \pi_{m,g}\left(Q\frac{n_m}{\ell_m}\frac{\lambda_{m,g}}{A_{m,g}}\right) < \infty.$$

Thus one may use exact survival bounds, distribution-specific quantile estimates, polynomial tails, stretched-exponential tails, or log-Weibull tails such as $C\exp\{-c(\log(e+t))^{\beta}\}$ *in the same grouped trace-budget criterion. In particular, lognormal and multiplicative volatility envelopes fit the same almost-sure Matrix Spencer route whenever this optimized profile cost is summable.*

Proof. For a fixed family of shares set, for $A_{m,g} > 0$,

$$\tau_{m,g} = Q\frac{n_m}{\ell_m}\frac{\lambda_{m,g}}{A_{m,g}}.$$

Then

$$\sum_g \tau_{m,g}A_{m,g} = Q\frac{n_m}{\ell_m}\sum_{g:A_{m,g}>0}\lambda_{m,g} \le Q\frac{n_m}{\ell_m},$$

and

$$\mathbb{P}\{Z_{m,g} > \tau_{m,g}\} \le \pi_{m,g}(\tau_{m,g}).$$

The displayed fixed-share condition is therefore exactly the countable grouped threshold criterion of Corollary 7.251.

For the optimized criterion, the summability hypothesis implies $\Theta_m(Q) < \infty$ for every m. If positive-mass groups are present, choose deterministic shares $\lambda_{m,g} > 0$ with $\sum_g \lambda_{m,g} \le 1$ and

$$\sum_{\substack{g\in G_m\\ A_{m,g}>0}} \pi_{m,g}\left(Q\frac{n_m}{\ell_m}\frac{\lambda_{m,g}}{A_{m,g}}\right) \le \Theta_m(Q) + 2^{-m}.$$

This is possible by the definition of the infimum; when there are no positive-mass groups there is no multiplier-tail cost. The resulting fixed share series is summable, and the fixed-share criterion gives the conclusion. □

Corollary 7.253 (Profile-menu grouped trace envelopes)**.** *In the setting of Corollary 7.252, suppose that, for each* m*, the positive-mass groups are partitioned into a finite or countable family of profile classes* $G_{m,h}$*,* $h \in H_m$*. For each class there is a deterministic shape* $\phi_{m,h} : (0,\infty) \to [0,\infty]$ *and constants* $C_{m,g} \ge 0$ *such that*

$$\mathbb{P}\{Z_{m,g} > t\} \le C_{m,g}\phi_{m,h}(t) \qquad (t > 0,\ g \in G_{m,h},\ A_{m,g} > 0).$$

Put

$$\mathcal{R}_{m,h} = \sum_{\substack{g\in G_{m,h}\\ A_{m,g}>0}} A_{m,g}, \qquad \mathcal{C}_{m,h} = \sum_{\substack{g\in G_{m,h}\\ A_{m,g}>0}} C_{m,g},$$

and assume that $\mathcal{R}_{m,h} < \infty$ *whenever* $\mathcal{R}_{m,h} > 0$*. Suppose there are deterministic class shares* $\eta_{m,h} > 0$*, defined for active classes* $\mathcal{R}_{m,h} > 0$*, with*

$$\sum_{h:\mathcal{R}_{m,h}>0} \eta_{m,h} \le 1.$$

If, for some $Q, A > 0$,

$$\sum_{m=1}^{\infty} \left[\mathbb{P}(E_m) + (eD_m)^{-A}\right] + \sum_{m} \sum_{h:\mathcal{R}_{m,h}>0} \mathcal{C}_{m,h}\, \phi_{m,h}\left(Q\frac{n_m}{\ell_m}\frac{\eta_{m,h}}{\mathcal{R}_{m,h}}\right) < \infty,$$

then, for every $L > 0$ satisfying

$$\frac{L^2}{2(Q + LB/3)} \geq A + 1,$$

almost surely, for all sufficiently large m,

$$\operatorname{disc}(X_{m,1}, \ldots, X_{m,n_m}) \leq (K + L)\sqrt{n_m}.$$

Thus mixed volatility menus, for instance a finite or countable collection of exponential, log-Weibull, regularly varying, and distribution-specific tail classes, can be checked using only class-level total trace masses $\mathcal{R}_{m,h}$, aggregate coefficients $\mathcal{C}_{m,h}$, and a class-level budget allocation. No independence between the group multipliers is needed.

Proof. For each active class and each positive-mass group $g \in G_{m,h}$, set

$$\lambda_{m,g} = \eta_{m,h}\frac{A_{m,g}}{\mathcal{R}_{m,h}}.$$

Then

$$\sum_{\substack{g \in G_m \\ A_{m,g}>0}} \lambda_{m,g} = \sum_{h:\mathcal{R}_{m,h}>0} \eta_{m,h} \leq 1.$$

Use the deterministic profile $\pi_{m,g}(t) = C_{m,g}\phi_{m,h}(t)$ for $g \in G_{m,h}$. The corresponding fixed-share threshold is constant inside each profile class:

$$Q\frac{n_m}{\ell_m}\frac{\lambda_{m,g}}{A_{m,g}} = Q\frac{n_m}{\ell_m}\frac{\eta_{m,h}}{\mathcal{R}_{m,h}}.$$

Therefore the fixed-share cost in Corollary 7.252 is bounded by the displayed class-level aggregate series. The fixed-share criterion gives the conclusion. □

Corollary 7.254 (Optimized profile-menu grouped trace envelopes)**.** *In the setting of Corollary 7.253, define, for $Q > 0$, the optimized class-profile cost*

$$\Xi_m(Q) = \inf_{\eta} \sum_{h:\mathcal{R}_{m,h}>0} \mathcal{C}_{m,h}\, \phi_{m,h}\left(Q\frac{n_m}{\ell_m}\frac{\eta_h}{\mathcal{R}_{m,h}}\right),$$

where the infimum is over all deterministic $\eta_h > 0$ on the active profile classes with $\sum_{h:\mathcal{R}_{m,h}>0} \eta_h \leq 1$; if there are no active classes, the empty infimum is 0. If, for some $A > 0$,

$$\sum_{m=1}^{\infty} \left[\mathbb{P}(E_m) + (eD_m)^{-A} + \Xi_m(Q)\right] < \infty,$$

then, for every $L > 0$ satisfying

$$\frac{L^2}{2(Q + LB/3)} \geq A + 1,$$

almost surely, for all sufficiently large m,

$$\operatorname{disc}(X_{m,1}, \dots, X_{m,n_m}) \leq (K + L)\sqrt{n_m}.$$

Thus mixed tail menus can be checked at the best deterministic allocation of the trace budget among tail classes, while still avoiding optimization over individual groups inside a class.

Proof. The summability hypothesis implies $\Xi_m(Q) < \infty$ for every m. When active profile classes are present, choose deterministic class shares $\eta_{m,h} > 0$ with $\sum_h \eta_{m,h} \leq 1$ and

$$\sum_{h:\mathcal{R}_{m,h}>0} \mathcal{C}_{m,h}\, \phi_{m,h}\left(Q\frac{n_m}{\ell_m}\frac{\eta_{m,h}}{\mathcal{R}_{m,h}}\right) \leq \Xi_m(Q) + 2^{-m}.$$

If there are no active profile classes, there is no multiplier-tail cost. The resulting class-level series is summable, so Corollary 7.253 applies. □

Corollary 7.255 (Finite-menu heterogeneous polynomial profile schedules)**.** *In the setting of Corollary 7.254, assume that, for each m, the active profile classes are finite and every active class has a polynomial tail shape*

$$\phi_{m,h}(t) = t^{-\alpha_{m,h}}, \qquad \alpha_{m,h} \geq \alpha_0 > 0.$$

Let

$$\mathcal{N}_m = \#\{h : \mathcal{R}_{m,h} > 0\}, \qquad \Lambda_m = \log(e + m).$$

Assume that the base exceptional series is finite and that there are constants $N_0, C_0, R_0 < \infty$, $w, p, u \geq 0$, and $z, q, v \in \mathbb{R}$ such that, for all sufficiently large m,

$$\mathcal{N}_m \leq N_0 m^w \Lambda_m^z,$$

and, for every active class h,

$$\mathcal{C}_{m,h} \leq C_0 m^p \Lambda_m^q, \qquad \mathcal{R}_{m,h} \leq R_0 m^u \Lambda_m^v.$$

If there are $a > 0$ and $b, c \in \mathbb{R}$ such that, for all sufficiently large m,

$$Q\frac{n_m}{\ell_m} \geq a m^b \Lambda_m^c,$$

then the same almost-sure conclusion holds whenever either

$$\alpha_0 b > p + \alpha_0 u + (\alpha_0 + 1)w + 1,$$

or

$$\alpha_0 b = p + \alpha_0 u + (\alpha_0 + 1)w + 1 \qquad \text{and} \qquad q + \alpha_0 v + (\alpha_0 + 1)z - \alpha_0 c < -1.$$

Thus finitely many polynomial profile classes with different Pareto indices are controlled by the worst tail exponent α_0, the active class count, and the worst aggregate coefficient and trace-mass growth.

Proof. If there are no active classes, there is no profile-menu cost. Otherwise set

$$T_m = Q\frac{n_m}{\ell_m}, \qquad \eta_{m,h} = \frac{1}{\mathcal{N}_m}$$

on the active classes. The growth hypotheses and either displayed exponent condition imply $b > w + u$, so for every active class and all sufficiently large m,

$$\frac{\mathcal{N}_m\mathcal{R}_{m,h}}{T_m} \le \frac{N_0R_0}{a}\, m^{w+u-b}\Lambda_m^{z+v-c} \le 1.$$

Since $\alpha_{m,h} \ge \alpha_0$, this gives

$$\begin{aligned}\mathcal{C}_{m,h}\left(T_m\frac{\eta_{m,h}}{\mathcal{R}_{m,h}}\right)^{-\alpha_{m,h}} &= \mathcal{C}_{m,h}\left(\frac{\mathcal{N}_m\mathcal{R}_{m,h}}{T_m}\right)^{\alpha_{m,h}} \\ &\le \mathcal{C}_{m,h}\left(\frac{\mathcal{N}_m\mathcal{R}_{m,h}}{T_m}\right)^{\alpha_0}.\end{aligned}$$

Summing over the at most $\mathcal{N}_m$ active classes,

$$\begin{aligned}\Xi_m(Q) &\le \sum_{h:\mathcal{R}_{m,h}>0} \mathcal{C}_{m,h}\left(\frac{\mathcal{N}_m\mathcal{R}_{m,h}}{T_m}\right)^{\alpha_0} \\ &\le C\, m^{p+\alpha_0u+(\alpha_0+1)w-\alpha_0b}\Lambda_m^{q+\alpha_0v+(\alpha_0+1)z-\alpha_0c},\end{aligned}$$

with $C = C_0N_0^{\alpha_0+1}R_0^{\alpha_0}a^{-\alpha_0}$. This upper bound is summable under the stated strict inequality, and at the endpoint precisely under the displayed logarithmic inequality. The optimized profile-menu corollary now applies. □

Corollary 7.256 (Countable weighted heterogeneous polynomial profile schedules)**.** *In the setting of Corollary 7.254, suppose the active profile classes have been indexed by positive integers and every active class has a polynomial tail shape*

$$\phi_{m,h}(t) = t^{-\alpha_{m,h}}, \qquad \alpha_{m,h} \ge \alpha_0 > 0.$$

Let $(\omega_h)_{h\ge1}$ be positive weights with

$$W = \sum_{h=1}^{\infty}\omega_h < \infty.$$

Assume that the base exceptional series

$$\sum_{m=1}^{\infty}\left[\mathbb{P}(E_m) + (eD_m)^{-A}\right]$$

is finite and that, for all sufficiently large m and every active class h,

$$W\,\frac{\mathcal{R}_{m,h}}{\omega_h} \le Q\frac{n_m}{\ell_m}.$$

If

$$\sum_{m=1}^{\infty}\left(\frac{\ell_mW}{Qn_m}\right)^{\alpha_0}\sum_{h:\mathcal{R}_{m,h}>0}\mathcal{C}_{m,h}\left(\frac{\mathcal{R}_{m,h}}{\omega_h}\right)^{\alpha_0} < \infty,$$

then, for every $L > 0$ satisfying

$$\frac{L^2}{2(Q + LB/3)} \geq A + 1,$$

almost surely, for all sufficiently large m,

$$\operatorname{disc}(X_{m,1}, \dots, X_{m,n_m}) \leq (K + L)\sqrt{n_m}.$$

In particular, put $\Lambda_m = \log(e + m)$. Suppose that there are constants $C_0, R_0 < \infty$, $p, u \geq 0$, $q, v \in \mathbb{R}$, $r, s \in \mathbb{R}$, and $\tau > 1$ such that $\tau < s$,

$$r + \alpha_0(s - \tau) > 1,$$

and, for all sufficiently large m and every active class h,

$$\mathcal{C}_{m,h} \leq C_0 m^p \Lambda_m^q h^{-r}, \qquad \mathcal{R}_{m,h} \leq R_0 m^u \Lambda_m^v h^{-s}.$$

If there are $a > 0$ and $b, c \in \mathbb{R}$ such that, for all sufficiently large m,

$$Q\frac{n_m}{\ell_m} \geq a m^b \Lambda_m^c,$$

then the same almost-sure conclusion holds whenever either

$$\alpha_0 b > p + \alpha_0 u + 1,$$

or

$$\alpha_0 b = p + \alpha_0 u + 1 \qquad \text{and} \qquad q + \alpha_0 v - \alpha_0 c < -1.$$

Equivalently, the displayed power-decay condition can be arranged whenever $s > 1$ and $r + \alpha_0(s - 1) > 1$. Thus countably many mixed-Pareto profile classes are covered by a fixed summable class-share distribution whenever the trace masses decay slightly faster than that distribution and the weighted coefficient–trace tail is summable.

Proof. Set

$$T_m = Q\frac{n_m}{\ell_m}.$$

If there are no active profile classes, there is no profile-menu cost. Otherwise choose deterministic class shares

$$\eta_{m,h} = \frac{\omega_h}{W}$$

on the active classes; their total is at most one. By the size hypothesis,

$$x_{m,h} := \frac{W\mathcal{R}_{m,h}}{T_m \omega_h} \leq 1$$

for all sufficiently large m and every active class h. Since $\alpha_{m,h} \geq \alpha_0$,

$$x_{m,h}^{\alpha_{m,h}} \leq x_{m,h}^{\alpha_0}.$$

Hence the optimized profile-menu cost satisfies, for all large m,

$$\Xi_m(Q) \le \sum_{h:\mathcal{R}_{m,h}>0} \mathcal{C}_{m,h} \left(\frac{W\mathcal{R}_{m,h}}{T_m\omega_h}\right)^{\alpha_{m,h}} \le \left(\frac{\ell_m W}{Qn_m}\right)^{\alpha_0} \sum_{h:\mathcal{R}_{m,h}>0} \mathcal{C}_{m,h} \left(\frac{\mathcal{R}_{m,h}}{\omega_h}\right)^{\alpha_0}.$$

The displayed summability hypothesis therefore implies the summability of $\sum_m \Xi_m(Q)$, and the optimized profile-menu corollary applies.

For the power-decaying specialization, take $\omega_h = h^{-\tau}$ and $W = \sum_h h^{-\tau}$. The strict or endpoint exponent condition implies $b > u$. Since $\tau < s$,

$$\frac{W\mathcal{R}_{m,h}}{T_m\omega_h} \le \frac{WR_0}{a}\, m^{u-b}\Lambda_m^{v-c}h^{\tau-s} \le \frac{WR_0}{a}\, m^{u-b}\Lambda_m^{v-c},$$

which is at most one for all sufficiently large m. Also

$$\sum_{h:\mathcal{R}_{m,h}>0} \mathcal{C}_{m,h} \left(\frac{\mathcal{R}_{m,h}}{\omega_h}\right)^{\alpha_0} \le C_0 R_0^{\alpha_0} m^{p+\alpha_0 u}\Lambda_m^{q+\alpha_0 v} \sum_{h=1}^{\infty} h^{-r-\alpha_0(s-\tau)}.$$

The last series is finite by $r + \alpha_0(s-\tau) > 1$. Therefore

$$\Xi_m(Q) \le C\, m^{p+\alpha_0 u-\alpha_0 b}\Lambda_m^{q+\alpha_0 v-\alpha_0 c}$$

for a finite constant C. This is summable under the strict displayed inequality, and at equality precisely under the displayed logarithmic endpoint condition. Finally, if $s > 1$ and $r + \alpha_0(s-1) > 1$, choose $\tau > 1$ close enough to 1 so that $\tau < s$ and $r + \alpha_0(s-\tau) > 1$. □

Corollary 7.257 (Finite-menu heterogeneous stretched-exponential profile schedules)**.** *In the setting of Corollary 7.254, assume that, for each m, the active profile classes are finite and every active class has a stretched-exponential tail shape*

$$\phi_{m,h}(t) = \exp\{-c_{m,h}t^{\beta_{m,h}}\}, \qquad c_{m,h} \ge c_0 > 0, \qquad \beta_{m,h} \ge \beta_0 > 0.$$

Let

$$\mathcal{N}_m = \#\{h : \mathcal{R}_{m,h} > 0\}, \qquad \mathcal{R}_m^* = \max_{h:\mathcal{R}_{m,h}>0} \mathcal{R}_{m,h},$$

with $\mathcal{R}_m^* = 0$ if there are no active classes. Put

$$\Lambda_m = \log(e+m), \qquad M_m = \log(e+\Lambda_m).$$

Assume that the base exceptional series

$$\sum_{m=1}^{\infty} \left[\mathbb{P}(E_m) + (eD_m)^{-A}\right]$$

is finite and that there are constants $N_0, C_0 < \infty$, $w, p \ge 0$, and $z, q \in \mathbb{R}$ such that, for all sufficiently large m,

$$\mathcal{N}_m \le N_0 m^w \Lambda_m^z,$$

and, for every active class h,

$$\mathcal{C}_{m,h} \le C_0 m^p \Lambda_m^q.$$

Suppose that there are $a > 0$ and $b \in \mathbb{R}$ such that, for all sufficiently large m with $\mathcal{R}_m^* > 0$,

$$Q\frac{n_m}{\ell_m}\frac{1}{\mathcal{N}_m \mathcal{R}_m^*} \ge a(\Lambda_m + bM_m)^{1/\beta_0}.$$

Let $\delta = c_0 a^{\beta_0}$. Then the same almost-sure conclusion holds whenever either

$$\delta > p + w + 1,$$

or

$$\delta = p + w + 1 \qquad \text{and} \qquad q + z - \delta b < -1.$$

Equivalently, if $\mathcal{R}_m^* \le R_0 m^u \Lambda_m^v$ eventually and

$$Q\frac{n_m}{\ell_m} \ge a_0 m^{u+w} \Lambda_m^{v+z} (\Lambda_m + bM_m)^{1/\beta_0},$$

then the same conclusion follows with $\delta = c_0(a_0/(N_0 R_0))^{\beta_0}$. Thus finitely many mixed exponential, Weibull, and sub-Weibull profile classes are absorbed by a logarithmic trace-budget margin whose polynomial exponent only pays for the class count and aggregate coefficient growth.

Proof. If there are no active classes, there is no profile-menu cost. Otherwise set

$$T_m = Q\frac{n_m}{\ell_m}, \qquad \eta_{m,h} = \frac{1}{\mathcal{N}_m}$$

on the active classes. The budget hypothesis gives, for every active class and all large m,

$$T_m \frac{\eta_{m,h}}{\mathcal{R}_{m,h}} \ge T_m \frac{1}{\mathcal{N}_m \mathcal{R}_m^*} \ge a(\Lambda_m + bM_m)^{1/\beta_0}.$$

The right-hand side tends to infinity, so it is at least 1 for all sufficiently large m. For $t \ge 1$, the lower bounds $c_{m,h} \ge c_0$ and $\beta_{m,h} \ge \beta_0$ imply

$$\exp\{-c_{m,h} t^{\beta_{m,h}}\} \le \exp\{-c_0 t^{\beta_0}\}.$$

Hence, with $\delta = c_0 a^{\beta_0}$,

$$\begin{aligned}
\Xi_m(Q) &\le \sum_{h:\mathcal{R}_{m,h}>0} \mathcal{C}_{m,h} \exp\left\{-c_{m,h}\left(T_m \frac{\eta_{m,h}}{\mathcal{R}_{m,h}}\right)^{\beta_{m,h}}\right\} \\
&\le \mathcal{N}_m C_0 m^p \Lambda_m^q \exp\{-\delta(\Lambda_m + bM_m)\} \\
&\le C\, m^{p+w-\delta} \Lambda_m^{q+z-\delta b}
\end{aligned}$$

for a finite constant C. This is summable under the strict inequality, and at equality precisely when $q + z - \delta b < -1$. The optimized profile-menu corollary gives the asserted almost-sure Matrix Spencer bound.

The absolute-budget variant follows because

$$\frac{T_m}{\mathcal{N}_m \mathcal{R}_m^*} \ge \frac{a_0}{N_0 R_0}(\Lambda_m + bM_m)^{1/\beta_0}$$

for all sufficiently large active m. □

Corollary 7.258 (Countable weighted heterogeneous stretched-exponential profile schedules)**.** *In the setting of Corollary 7.254, suppose the active profile classes have been indexed by positive integers and every active class has a stretched-exponential tail shape*

$$\phi_{m,h}(t) = \exp\{-c_{m,h}t^{\beta_{m,h}}\}, \qquad c_{m,h} \geq c_0 > 0, \qquad \beta_{m,h} \geq \beta_0 > 0.$$

Let $(\omega_h)_{h\geq 1}$ be positive weights with

$$W = \sum_{h=1}^{\infty} \omega_h < \infty.$$

Assume that the base exceptional series

$$\sum_{m=1}^{\infty} \left[\mathbb{P}(E_m) + (eD_m)^{-A}\right]$$

is finite and that, for all sufficiently large m and every active class h,

$$Q\frac{n_m}{\ell_m}\frac{\omega_h}{W\mathcal{R}_{m,h}} \geq 1.$$

If

$$\sum_{m=1}^{\infty} \sum_{h:\mathcal{R}_{m,h}>0} \mathcal{C}_{m,h} \exp\left\{-c_0\left(Q\frac{n_m}{\ell_m}\frac{\omega_h}{W\mathcal{R}_{m,h}}\right)^{\beta_0}\right\} < \infty,$$

then, for every $L > 0$ satisfying

$$\frac{L^2}{2(Q + LB/3)} \geq A + 1,$$

almost surely, for all sufficiently large m,

$$\operatorname{disc}(X_{m,1}, \ldots, X_{m,n_m}) \leq (K + L)\sqrt{n_m}.$$

In particular, put $\Lambda_m = \log(e + m)$ and $M_m = \log(e + \Lambda_m)$. Suppose that there are constants $C_0, R_0 < \infty$, $p, u \geq 0$, $q, v, r \in \mathbb{R}$, and $s, \tau \in \mathbb{R}$ such that $\tau > 1$, $s > \tau$, and, for all sufficiently large m and every active class h,

$$\mathcal{C}_{m,h} \leq C_0 m^p \Lambda_m^q h^{-r}, \qquad \mathcal{R}_{m,h} \leq R_0 m^u \Lambda_m^v h^{-s}.$$

If there are $a > 0$ and $b \in \mathbb{R}$ such that, for all sufficiently large m,

$$Q\frac{n_m}{\ell_m} \geq a m^u \Lambda_m^v (\Lambda_m + bM_m)^{1/\beta_0},$$

then the same almost-sure conclusion holds with

$$\delta = c_0\left(\frac{a}{WR_0}\right)^{\beta_0}, \qquad W = \sum_{h=1}^{\infty} h^{-\tau},$$

whenever either

$$\delta > p + 1,$$

or

$$\delta = p + 1 \qquad \text{and} \qquad q - \delta b < -1.$$

Thus countably many mixed exponential, Weibull, and sub-Weibull profile classes are allowed when the deterministic trace masses decay faster than a fixed summable class-share distribution; no class-count factor appears in the final triangular-array budget rule.

Proof. If there are no active classes, there is no profile-menu cost. Otherwise choose deterministic class shares

$$\eta_{m,h} = \frac{\omega_h}{W}.$$

Their total over active classes is at most one. For all sufficiently large m, the displayed size hypothesis makes

$$t_{m,h} = Q\frac{n_m}{\ell_m}\frac{\eta_{m,h}}{\mathcal{R}_{m,h}} = Q\frac{n_m}{\ell_m}\frac{\omega_h}{W\mathcal{R}_{m,h}} \geq 1$$

on every active class. Hence

$$\phi_{m,h}(t_{m,h}) = \exp\{-c_{m,h}t_{m,h}^{\beta_{m,h}}\} \leq \exp\{-c_0 t_{m,h}^{\beta_0}\}.$$

The fixed class-share criterion in Corollary 7.253 therefore gives the first assertion.

For the power-decaying specialization, take $\omega_h = h^{-\tau}$. Since $s > \tau$, the budget hypothesis gives, for every active class and all large m,

$$t_{m,h} \geq \frac{a}{WR_0}(\Lambda_m + bM_m)^{1/\beta_0}h^{s-\tau}.$$

The right-hand side is eventually at least 1. With

$$\delta = c_0\left(\frac{a}{WR_0}\right)^{\beta_0}, \qquad \gamma = \beta_0(s-\tau) > 0,$$

the class-profile cost is bounded by

$$\sum_{h:\mathcal{R}_{m,h}>0} \mathcal{C}_{m,h}\phi_{m,h}(t_{m,h}) \leq C_0 m^p \Lambda_m^q \sum_{h=1}^{\infty} h^{-r}\exp\{-\delta(\Lambda_m + bM_m)h^\gamma\}.$$

For all sufficiently large m, $\Lambda_m + bM_m \geq 1$, and

$$\sum_{h=1}^{\infty} h^{-r}\exp\{-\delta(\Lambda_m + bM_m)h^\gamma\} \leq C_{r,\gamma,\delta}\exp\{-\delta(\Lambda_m + bM_m)\}.$$

Indeed, after factoring the $h = 1$ exponential term, the remaining summand is bounded by

$$h^{-r}\exp\{-\delta(h^\gamma - 1)\},$$

whose series is finite. Thus

$$\Xi_m(Q) \leq C\, m^{p-\delta}\Lambda_m^{q-\delta b}.$$

This series is summable if $\delta > p+1$, and at equality precisely when $q-\delta b < -1$. The optimized profile-menu corollary now gives the asserted conclusion. □

Corollary 7.259 (Finite-menu heterogeneous log-Weibull profile schedules)**.** *In the setting of Corollary 7.254, assume that, for each m, the active profile classes are finite and every active class has a log-Weibull tail shape*

$$\phi_{m,h}(t) = \exp\{-c_{m,h}(\log(e+t))^{\beta_{m,h}}\},$$
$$c_{m,h} \geq c_0 > 0, \qquad \beta_{m,h} \geq \beta_0 > 0.$$

Let

$$\mathcal{N}_m = \#\{h : \mathcal{R}_{m,h} > 0\}, \qquad \mathcal{R}_m^* = \max_{h:\mathcal{R}_{m,h}>0} \mathcal{R}_{m,h},$$

with $\mathcal{R}_m^* = 0$ if there are no active classes. Put

$$\Lambda_m = \log(e+m), \qquad M_m = \log(e+\Lambda_m).$$

Assume that the base exceptional series

$$\sum_{m=1}^{\infty} \left[\mathbb{P}(E_m) + (eD_m)^{-A}\right]$$

is finite and that there are constants $N_0, C_0 < \infty$, $w, p \geq 0$, and $z, q \in \mathbb{R}$ such that, for all sufficiently large m,

$$\mathcal{N}_m \leq N_0 m^w \Lambda_m^z,$$

and, for every active class h,

$$\mathcal{C}_{m,h} \leq C_0 m^p \Lambda_m^q.$$

Suppose that there are $a > 0$ and $b \in \mathbb{R}$ such that, for all sufficiently large m with $\mathcal{R}_m^* > 0$,

$$Q\frac{n_m}{\ell_m}\frac{1}{\mathcal{N}_m \mathcal{R}_m^*} \geq \exp\{a(\Lambda_m + bM_m)^{1/\beta_0}\}.$$

Let $\delta = c_0 a^{\beta_0}$. Then the same almost-sure conclusion holds whenever either

$$\delta > p + w + 1,$$

or

$$\delta = p + w + 1 \qquad \text{and} \qquad q + z - \delta b < -1.$$

Equivalently, if $\mathcal{R}_m^* \leq R_0 m^u \Lambda_m^v$ eventually and

$$Q\frac{n_m}{\ell_m} \geq N_0 R_0 m^{u+w} \Lambda_m^{v+z} \exp\{a(\Lambda_m + bM_m)^{1/\beta_0}\},$$

then the same conclusion follows. Thus finitely many mixed log-Weibull, including lognormal-type, profile classes are absorbed by a subpolynomial trace-budget margin; the class-count and aggregate coefficient growth enter only through the polynomial summability exponent $p + w$.

Proof. If there are no active classes, there is no profile-menu cost. Otherwise set

$$T_m = Q\frac{n_m}{\ell_m}, \qquad \eta_{m,h} = \frac{1}{\mathcal{N}_m}$$

on the active classes. For all sufficiently large m, $\Lambda_m + bM_m \geq 1$. The budget hypothesis gives, for every active class and all large m,

$$T_m \frac{\eta_{m,h}}{\mathcal{R}_{m,h}} \geq T_m \frac{1}{\mathcal{N}_m \mathcal{R}_m^*} \geq \exp\{a(\Lambda_m + bM_m)^{1/\beta_0}\}.$$

Therefore

$$\log\left(e + T_m \frac{\eta_{m,h}}{\mathcal{R}_{m,h}}\right) \geq a(\Lambda_m + bM_m)^{1/\beta_0}$$

for all large m. Since $\log(e+t) \geq 1$, the lower bounds $c_{m,h} \geq c_0$ and $\beta_{m,h} \geq \beta_0$ imply

$$\exp\{-c_{m,h}(\log(e+t))^{\beta_{m,h}}\} \leq \exp\{-c_0(\log(e+t))^{\beta_0}\}.$$

Hence, with $\delta = c_0 a^{\beta_0}$,

$$\begin{aligned}
\Xi_m(Q) &\leq \sum_{h:\mathcal{R}_{m,h}>0} \mathcal{C}_{m,h} \exp\left\{-c_{m,h}\left(\log\left(e + T_m \frac{\eta_{m,h}}{\mathcal{R}_{m,h}}\right)\right)^{\beta_{m,h}}\right\} \\
&\leq \mathcal{N}_m C_0 m^p \Lambda_m^q \exp\{-\delta(\Lambda_m + bM_m)\} \\
&\leq C\, m^{p+w-\delta} \Lambda_m^{q+z-\delta b}
\end{aligned}$$

for a finite constant C. This is summable under the strict inequality, and at equality precisely when $q + z - \delta b < -1$. The optimized profile-menu corollary gives the asserted almost-sure Matrix Spencer bound.

The absolute-budget variant follows immediately from the displayed bounds on $\mathcal{N}_m$ and $\mathcal{R}_m^*$. □

Corollary 7.260 (Countable weighted heterogeneous log-Weibull profile schedules)**.** *In the setting of Corollary 7.254, suppose the active profile classes have been indexed by positive integers and every active class has a log-Weibull tail shape*

$$\phi_{m,h}(t) = \exp\{-c_{m,h}(\log(e+t))^{\beta_{m,h}}\},$$
$$c_{m,h} \geq c_0 > 0, \qquad \beta_{m,h} \geq \beta_0 > 0.$$

Let $(\omega_h)_{h\geq 1}$ be positive weights with

$$W = \sum_{h=1}^{\infty} \omega_h < \infty.$$

Assume that the base exceptional series

$$\sum_{m=1}^{\infty} \left[\mathbb{P}(E_m) + (eD_m)^{-A}\right]$$

is finite. If

$$\sum_{m=1}^{\infty} \sum_{h:\mathcal{R}_{m,h}>0} \mathcal{C}_{m,h} \exp\left\{-c_0\left(\log\left(e + Q\frac{n_m}{\ell_m}\frac{\omega_h}{W\mathcal{R}_{m,h}}\right)\right)^{\beta_0}\right\} < \infty,$$

then, for every $L > 0$ satisfying

$$\frac{L^2}{2(Q + LB/3)} \geq A + 1,$$

almost surely, for all sufficiently large m,

$$\operatorname{disc}(X_{m,1}, \dots, X_{m,n_m}) \leq (K + L)\sqrt{n_m}.$$

In particular, put $\Lambda_m = \log(e + m)$ and $M_m = \log(e + \Lambda_m)$. Suppose that there are constants $C_0, R_0 < \infty$, $p, u \geq 0$, $q, v, r \in \mathbb{R}$, and $s, \tau \in \mathbb{R}$ such that $\tau > 1$, $s > \tau$, and, for all sufficiently large m and every active class h,

$$\mathcal{C}_{m,h} \leq C_0 m^p \Lambda_m^q h^{-r}, \qquad \mathcal{R}_{m,h} \leq R_0 m^u \Lambda_m^v h^{-s}.$$

Let

$$\kappa_{\beta_0} = \min\{1, 2^{\beta_0 - 1}\}, \qquad d = s - \tau,$$

and assume

$$H_{\beta_0,d,r} = \sum_{h=1}^{\infty} h^{-r} \exp\{-c_0 \kappa_{\beta_0} d^{\beta_0} (\log h)^{\beta_0}\} < \infty.$$

If there are $a > 0$ and $b \in \mathbb{R}$ such that, for all sufficiently large m,

$$Q \frac{n_m}{\ell_m} \geq W R_0 m^u \Lambda_m^v \exp\{a(\Lambda_m + bM_m)^{1/\beta_0}\},$$

then the same almost-sure conclusion holds with

$$\delta = c_0 \kappa_{\beta_0} a^{\beta_0}$$

whenever either

$$\delta > p + 1,$$

or

$$\delta = p + 1 \qquad \text{and} \qquad q - \delta b < -1.$$

The class-index summability hypothesis is automatic for every r when $\beta_0 > 1$, becomes $r + c_0 d > 1$ when $\beta_0 = 1$, and reduces to $r > 1$ when $0 < \beta_0 < 1$. Thus countable mixed log-Weibull and lognormal-type profile classes are controlled by a subpolynomial trace-budget margin whenever their trace masses decay faster than a summable class-share distribution and the residual class-index tail is summable.

Proof. If there are no active classes, there is no profile-menu cost. Otherwise choose deterministic class shares

$$\eta_{m,h} = \frac{\omega_h}{W}.$$

Their total over active classes is at most one. With

$$t_{m,h} = Q \frac{n_m}{\ell_m} \frac{\eta_{m,h}}{\mathcal{R}_{m,h}} = Q \frac{n_m}{\ell_m} \frac{\omega_h}{W \mathcal{R}_{m,h}},$$

the lower bounds on $c_{m,h}$ and $\beta_{m,h}$, together with $\log(e + t_{m,h}) \geq 1$, give

$$\phi_{m,h}(t_{m,h}) \leq \exp\{-c_0 (\log(e + t_{m,h}))^{\beta_0}\}.$$

The fixed class-share criterion in Corollary 7.253 therefore gives the first assertion.

For the power-decaying specialization, take $\omega_h = h^{-\tau}$. For all sufficiently large m, $\Lambda_m + bM_m \geq 1$. The budget and trace-mass hypotheses give, for every active class,

$$t_{m,h} \geq \exp\{a(\Lambda_m + bM_m)^{1/\beta_0}\} h^{s-\tau}.$$

Hence

$$\log(e + t_{m,h}) \geq a(\Lambda_m + bM_m)^{1/\beta_0} + d \log h.$$

For $x, y \geq 0$,

$$(x+y)^{\beta_0} \geq \kappa_{\beta_0}(x^{\beta_0} + y^{\beta_0}),$$

where $\kappa_{\beta_0} = \min\{1, 2^{\beta_0 - 1}\}$. Therefore the class-profile cost is bounded by

$$\sum_{h:\mathcal{R}_{m,h}>0} \mathcal{C}_{m,h}\phi_{m,h}(t_{m,h}) \leq C_0 m^p \Lambda_m^q \exp\{-\delta(\Lambda_m + bM_m)\} H_{\beta_0,d,r}$$

$$\leq C\, m^{p-\delta} \Lambda_m^{q-\delta b}.$$

This series is summable if $\delta > p+1$, and at equality precisely when $q - \delta b < -1$. The optimized profile-menu corollary gives the asserted almost-sure Matrix Spencer bound. The displayed descriptions of when $H_{\beta_0,d,r}$ is finite follow by comparing $\exp\{-C(\log h)^{\beta_0}\}$ with powers of h. □

Corollary 7.261 (Common-index polynomial profile-menu trace envelopes)**.** *In the setting of Corollary 7.254, assume that, for a fixed $\alpha > 0$, every active profile class has the polynomial shape*

$$\phi_{m,h}(t) = t^{-\alpha}.$$

Define the class-level effective polynomial mass

$$\mathcal{M}_m = \sum_{h:\mathcal{R}_{m,h}>0} \mathcal{C}_{m,h}^{1/(\alpha+1)} \mathcal{R}_{m,h}^{\alpha/(\alpha+1)}$$

with the convention that $\mathcal{M}_m = 0$ if there are no active classes. Then

$$\Xi_m(Q) = \left(\frac{\ell_m}{Qn_m}\right)^{\alpha} \mathcal{M}_m^{\alpha+1}.$$

Consequently, if, for some $A, Q > 0$,

$$\sum_{m=1}^{\infty} \left[\mathbb{P}(E_m) + (eD_m)^{-A} + \left(\frac{\ell_m}{Qn_m}\right)^{\alpha} \mathcal{M}_m^{\alpha+1}\right] < \infty,$$

then, for every $L > 0$ satisfying

$$\frac{L^2}{2(Q + LB/3)} \geq A + 1,$$

almost surely, for all sufficiently large m,

$$\operatorname{disc}(X_{m,1}, \ldots, X_{m,n_m}) \leq (K+L)\sqrt{n_m}.$$

This is the class-level analogue of the optimized grouped polynomial-tail criterion: it needs only aggregate class coefficients and total deterministic trace mass inside each profile class, not individual group-tail data.

Proof. Fix m, suppress the index m, and put

$$a_h = \mathcal{C}_h \mathcal{R}_h^{\alpha}, \qquad w_h = a_h^{1/(\alpha+1)} = \mathcal{C}_h^{1/(\alpha+1)} \mathcal{R}_h^{\alpha/(\alpha+1)}.$$

For every admissible class-share family (η_h),

$$\begin{aligned} \sum_h w_h &= \sum_h (a_h \eta_h^{-\alpha})^{1/(\alpha+1)} \eta_h^{\alpha/(\alpha+1)} \\ &\le \left(\sum_h a_h \eta_h^{-\alpha} \right)^{1/(\alpha+1)} \left(\sum_h \eta_h \right)^{\alpha/(\alpha+1)} \\ &\le \left(\sum_h a_h \eta_h^{-\alpha} \right)^{1/(\alpha+1)}. \end{aligned}$$

Thus

$$\sum_h a_h \eta_h^{-\alpha} \ge \mathcal{M}_m^{\alpha+1}.$$

Conversely, if $0 < \mathcal{M}_m < \infty$, the proportional shares

$$\eta_h = \frac{w_h}{\mathcal{M}_m}$$

on the positive-weight classes have total one and give equality. If zero-weight active classes are also present, assign them total share $\varepsilon > 0$ and scale the positive-weight shares by $1 - \varepsilon$; their cost increases by the factor $(1 - \varepsilon)^{-\alpha}$, and letting $\varepsilon \downarrow 0$ gives the same infimum. If $\mathcal{M}_m = 0$, all terms have zero cost. If $\mathcal{M}_m = \infty$, the lower bound applied to finite subfamilies forces the infimum to be infinite. Multiplying by $(\ell_m/(Qn_m))^{\alpha}$ gives the displayed formula for $\Xi_m(Q)$. The summability criterion is now Corollary 7.254. □

Corollary 7.262 (Common-index polynomial profile-menu schedules)**.** *In the setting of Corollary 7.261, assume that the base exceptional series*

$$\sum_{m=1}^{\infty} \left[\mathbb{P}(E_m) + (eD_m)^{-A} \right]$$

is finite. Put $\Lambda_m = \log(e + m)$. Suppose that there are constants $M_0 < \infty$, $p \ge 0$, $q \in \mathbb{R}$, $a > 0$, and $b, c \in \mathbb{R}$ such that, for all sufficiently large m,

$$\mathcal{M}_m \le M_0 m^p \Lambda_m^q, \qquad Q \frac{n_m}{\ell_m} \ge a m^b \Lambda_m^c.$$

If either

$$\alpha b > (\alpha + 1)p + 1,$$

or

$$\alpha b = (\alpha + 1)p + 1 \qquad \text{and} \qquad (\alpha + 1)q - \alpha c < -1,$$

then, for every $L > 0$ satisfying

$$\frac{L^2}{2(Q + LB/3)} \geq A + 1,$$

almost surely, for all sufficiently large m,

$$\operatorname{disc}(X_{m,1}, \dots, X_{m,n_m}) \leq (K + L)\sqrt{n_m}.$$

Thus a triangular array with polynomially many effective profile classes, or with polynomially growing aggregate class coefficients and trace masses, is covered as soon as the total trace budget Qn_m/ℓ_m outruns the effective class mass at the exponent dictated by the optimized polynomial allocation.

Proof. By Corollary 7.261, it is enough to prove the summability of

$$\sum_m \left(\frac{\ell_m}{Qn_m}\right)^{\alpha} \mathcal{M}_m^{\alpha+1}.$$

For all sufficiently large m, the growth hypotheses give

$$\left(\frac{\ell_m}{Qn_m}\right)^{\alpha} \mathcal{M}_m^{\alpha+1} \leq a^{-\alpha} M_0^{\alpha+1} m^{(\alpha+1)p-\alpha b} \Lambda_m^{(\alpha+1)q-\alpha c}.$$

The latter series converges if $\alpha b > (\alpha+1)p+1$, and at equality precisely under $(\alpha+1)q - \alpha c < -1$. The preceding corollary gives the asserted almost-sure Matrix Spencer bound. □

Corollary 7.263 (Finite-menu common-index polynomial profile schedules)**.** *In the setting of Corollary 7.262, assume that the active profile classes are finite for each m, and let*

$$\mathcal{N}_m = \#\{h : \mathcal{R}_{m,h} > 0\}.$$

Let $\Lambda_m = \log(e + m)$. Assume that the base exceptional series is finite and that there are constants $N_0, C_0, R_0 < \infty$, $w, p, u \geq 0$, and $z, q, v \in \mathbb{R}$ such that, for all sufficiently large m,

$$\mathcal{N}_m \leq N_0 m^w \Lambda_m^z,$$

and, for every active class h,

$$\mathcal{C}_{m,h} \leq C_0 m^p \Lambda_m^q, \qquad \mathcal{R}_{m,h} \leq R_0 m^u \Lambda_m^v.$$

If there are $a > 0$ and $b, c \in \mathbb{R}$ such that, for all sufficiently large m,

$$Q\frac{n_m}{\ell_m} \geq a m^b \Lambda_m^c,$$

then the same almost-sure conclusion holds whenever either

$$\alpha b > p + \alpha u + (\alpha + 1)w + 1,$$

or

$$\alpha b = p + \alpha u + (\alpha + 1)w + 1 \qquad \text{and} \qquad q + \alpha v + (\alpha + 1)z - \alpha c < -1.$$

Thus polynomially many common-index polynomial tail classes can be checked from only worst-class aggregate coefficient and trace-mass bounds, plus the class count.

Proof. For all sufficiently large m,

$$\begin{aligned}\mathcal{M}_m &= \sum_{h:\mathcal{R}_{m,h}>0} \mathcal{C}_{m,h}^{1/(\alpha+1)} \mathcal{R}_{m,h}^{\alpha/(\alpha+1)} \\ &\le N_0 C_0^{1/(\alpha+1)} R_0^{\alpha/(\alpha+1)} m^{w+(p+\alpha u)/(\alpha+1)} \Lambda_m^{z+(q+\alpha v)/(\alpha+1)}.\end{aligned}$$

Corollary 7.262 therefore applies with

$$p' = w + \frac{p+\alpha u}{\alpha+1}, \qquad q' = z + \frac{q+\alpha v}{\alpha+1}.$$

The conditions $\alpha b > (\alpha+1)p' + 1$ and $(\alpha+1)q' - \alpha c < -1$ are exactly the displayed inequalities. □

Corollary 7.264 (Power-decaying common-index polynomial profile menus)**.** *In the setting of Corollary 7.262, suppose the active profile classes have been indexed by positive integers. Let $\Lambda_m = \log(e+m)$. Assume that the base exceptional series is finite and that there are constants $C_0, R_0 < \infty$, $p, u \ge 0$, $q, v \in \mathbb{R}$, and $r, s \in \mathbb{R}$ such that, for all sufficiently large m and every active class h,*

$$\mathcal{C}_{m,h} \le C_0 m^p \Lambda_m^q h^{-r}, \qquad \mathcal{R}_{m,h} \le R_0 m^u \Lambda_m^v h^{-s}.$$

If

$$r + \alpha s > \alpha + 1,$$

and there are $a > 0$ and $b, c \in \mathbb{R}$ such that, for all sufficiently large m,

$$Q \frac{n_m}{\ell_m} \ge a m^b \Lambda_m^c,$$

then the same almost-sure conclusion holds whenever either

$$\alpha b > p + \alpha u + 1,$$

or

$$\alpha b = p + \alpha u + 1 \qquad \text{and} \qquad q + \alpha v - \alpha c < -1.$$

Thus countably many polynomial-tail profile classes are allowed when their aggregate coefficients and deterministic trace masses have enough class-index decay; only the aggregate growth exponents p, u, q, v enter the final triangular-array budget rule.

Proof. The effective class mass from Corollary 7.261 satisfies, for all large m,

$$\begin{aligned}\mathcal{M}_m &= \sum_{h:\mathcal{R}_{m,h}>0} \mathcal{C}_{m,h}^{1/(\alpha+1)} \mathcal{R}_{m,h}^{\alpha/(\alpha+1)} \\ &\le C_0^{1/(\alpha+1)} R_0^{\alpha/(\alpha+1)} m^{(p+\alpha u)/(\alpha+1)} \Lambda_m^{(q+\alpha v)/(\alpha+1)} \sum_{h=1}^{\infty} h^{-(r+\alpha s)/(\alpha+1)}.\end{aligned}$$

The last series is finite exactly under $r + \alpha s > \alpha + 1$. Hence Corollary 7.262 applies with effective growth exponents

$$p' = \frac{p+\alpha u}{\alpha+1}, \qquad q' = \frac{q+\alpha v}{\alpha+1}.$$

Its conditions $\alpha b > (\alpha+1)p' + 1$ and $(\alpha+1)q' - \alpha c < -1$ are precisely the displayed inequalities. □

Corollary 7.265 (Common-profile grouped trace envelopes)**.** *In the setting of Corollary 7.252, suppose that the group tails have a common deterministic shape: for each m there is $\phi_m : (0,\infty) \to [0,\infty]$ and constants $C_{m,g} \geq 0$ such that, for every positive-mass group,*

$$\mathbb{P}\{Z_{m,g} > t\} \leq C_{m,g}\phi_m(t) \qquad (t > 0).$$

Put

$$\mathcal{R}_m = \sum_{\substack{g \in G_m \\ A_{m,g} > 0}} A_{m,g},$$

and assume that $\mathcal{R}_m < \infty$ whenever positive-mass groups are present. If, for some $Q, A > 0$,

$$\sum_{m=1}^{\infty} \left[\mathbb{P}(E_m) + (eD_m)^{-A}\right] + \sum_{m:\mathcal{R}_m>0} \sum_{\substack{g \in G_m \\ A_{m,g} > 0}} C_{m,g}\, \phi_m\left(Q\frac{n_m}{\ell_m \mathcal{R}_m}\right) < \infty,$$

then, for every $L > 0$ satisfying

$$\frac{L^2}{2(Q + LB/3)} \geq A + 1,$$

almost surely, for all sufficiently large m,

$$\operatorname{disc}(X_{m,1}, \ldots, X_{m,n_m}) \leq (K+L)\sqrt{n_m}.$$

In particular, if

$$\mathbb{P}\{Z_{m,g} > t\} \leq C_{m,g} \exp\{-c_m(\log(e+t))^{\beta_m}\} \qquad (t > 0),$$

then it is enough that

$$\sum_{m=1}^{\infty} \left[\mathbb{P}(E_m) + (eD_m)^{-A}\right] + \sum_{m:\mathcal{R}_m>0} \sum_{\substack{g \in G_m \\ A_{m,g} > 0}} C_{m,g} \exp\left\{-c_m\left(\log\left(e + Q\frac{n_m}{\ell_m \mathcal{R}_m}\right)\right)^{\beta_m}\right\} < \infty.$$

Thus countable grouped volatility decompositions with a common log-Weibull, including lognormal-type, multiplier tail are controlled by one deterministic total trace mass $\mathcal{R}_m$ and one aggregate group-tail coefficient, without solving the optimized share problem.

Proof. If there are no positive-mass groups, there is no multiplier-tail cost. When $\mathcal{R}_m > 0$, choose

$$\lambda_{m,g} = \frac{A_{m,g}}{\mathcal{R}_m} \qquad (A_{m,g} > 0).$$

These shares are positive and sum to one. With $\pi_{m,g}(t) = C_{m,g}\phi_m(t)$, the corresponding threshold in Corollary 7.252 is

$$Q\frac{n_m}{\ell_m}\frac{\lambda_{m,g}}{A_{m,g}} = Q\frac{n_m}{\ell_m\mathcal{R}_m},$$

independent of g. The fixed-share criterion in that corollary is exactly the displayed aggregate-tail summability. The log-Weibull specialization is the same estimate with the displayed choice of ϕ_m. □

Corollary 7.266 (Log-Weibull common-profile schedules)**.** *In the setting of Corollary 7.265, assume that, for fixed $c, \beta > 0$,*

$$\mathbb{P}\{Z_{m,g} > t\} \le C_{m,g}\exp\{-c(\log(e+t))^{\beta}\} \qquad (t > 0)$$

for every positive-mass group, and put

$$\mathcal{C}_m = \sum_{\substack{g\in G_m\\ A_{m,g}>0}} C_{m,g}.$$

Assume also that the base exceptional series

$$\sum_{m=1}^{\infty}\left[\mathbb{P}(E_m) + (eD_m)^{-A}\right]$$

is finite. If there are constants $C_0 < \infty$, $p \ge 0$, and $a > 0$ such that, for all sufficiently large m with $\mathcal{R}_m > 0$,

$$\mathcal{C}_m \le C_0 m^p, \qquad Q\frac{n_m}{\ell_m\mathcal{R}_m} \ge \exp\{a(\log m)^{1/\beta}\},$$

and

$$ca^{\beta} > p + 1,$$

then, for every $L > 0$ satisfying

$$\frac{L^2}{2(Q + LB/3)} \ge A + 1,$$

almost surely, for all sufficiently large m,

$$\operatorname{disc}(X_{m,1}, \dots, X_{m,n_m}) \le (K + L)\sqrt{n_m}.$$

In particular, lognormal-type grouped volatility, corresponding to $\beta = 2$, is covered whenever the trace-budget ratio $Qn_m/(\ell_m\mathcal{R}_m)$ is at least $\exp\{a\sqrt{\log m}\}$ and $ca^2 > p + 1$. Thus a subpolynomial safety margin in the total deterministic trace mass is already enough to absorb polynomially many common lognormal-type volatility groups.

Proof. By Corollary 7.265 it is enough to show

$$\sum_{m:\mathcal{R}_m>0} \mathcal{C}_m \exp\left\{-c\left(\log\left(e + Q\frac{n_m}{\ell_m\mathcal{R}_m}\right)\right)^{\beta}\right\} < \infty.$$

For all sufficiently large m with $\mathcal{R}_m > 0$, the growth hypothesis gives

$$\log\left(e + Q\frac{n_m}{\ell_m \mathcal{R}_m}\right) \geq a(\log m)^{1/\beta}.$$

Hence the m-th summand is at most

$$C_0 m^p \exp\{-ca^\beta \log m\} = C_0 m^{p-ca^\beta}.$$

This is summable because $ca^\beta > p+1$. The common-profile corollary now gives the asserted almost-sure Matrix Spencer bound. □

Corollary 7.267 (Stretched-exponential common-profile schedules)**.** *In the setting of Corollary 7.265, assume that, for fixed $c, \beta > 0$,*

$$\mathbb{P}\{Z_{m,g} > t\} \leq C_{m,g} \exp\{-ct^\beta\} \qquad (t > 0)$$

for every positive-mass group, and put

$$\mathcal{C}_m = \sum_{\substack{g \in G_m \\ A_{m,g} > 0}} C_{m,g}.$$

Assume also that

$$\sum_{m=1}^{\infty} \left[\mathbb{P}(E_m) + (eD_m)^{-A}\right] < \infty.$$

Let $\Lambda_m = \log(e+m)$ and $M_m = \log(e+\Lambda_m)$. Suppose that there are constants $C_0 < \infty$, $p \geq 0$, $q \in \mathbb{R}$, $a > 0$, and $b \in \mathbb{R}$ such that, for all sufficiently large m with $\mathcal{R}_m > 0$,

$$\mathcal{C}_m \leq C_0 m^p \Lambda_m^q, \qquad Q\frac{n_m}{\ell_m \mathcal{R}_m} \geq a(\Lambda_m + bM_m)^{1/\beta}.$$

If either

$$ca^\beta > p+1,$$

or

$$ca^\beta = p+1 \qquad \text{and} \qquad q - ca^\beta b < -1,$$

then, for every $L > 0$ satisfying

$$\frac{L^2}{2(Q + LB/3)} \geq A + 1,$$

almost surely, for all sufficiently large m,

$$\operatorname{disc}(X_{m,1}, \dots, X_{m,n_m}) \leq (K+L)\sqrt{n_m}.$$

Thus common exponential, Weibull, and sub-Weibull volatility groups require only logarithmic growth of the trace-budget ratio against polynomially many aggregate tail coefficients. At the critical balance $ca^\beta = p+1$, the log-log budget correction b and the aggregate logarithmic coefficient q decide summability through $q - ca^\beta b < -1$.

Proof. By Corollary 7.265 it is enough to show that

$$\sum_{m:\mathcal{R}_m>0} \mathcal{C}_m \exp\left\{-c\left(Q\frac{n_m}{\ell_m\mathcal{R}_m}\right)^{\beta}\right\} < \infty.$$

For all sufficiently large m, the quantity $\Lambda_m + bM_m$ is positive. With $\delta = ca^{\beta}$, the growth hypotheses give

$$\mathcal{C}_m \exp\left\{-c\left(Q\frac{n_m}{\ell_m\mathcal{R}_m}\right)^{\beta}\right\} \le C_0 m^p \Lambda_m^q \exp\{-\delta\Lambda_m - \delta b M_m\}.$$

Since $e^{-\delta\Lambda_m} = (e+m)^{-\delta}$ and $e^{-\delta b M_m} = (e+\Lambda_m)^{-\delta b}$, the m-th summand is at most

$$C m^{p-\delta}\Lambda_m^{q-\delta b}.$$

This is summable if $\delta > p+1$, and in the endpoint case $\delta = p+1$ precisely when $q - \delta b < -1$. The common-profile corollary therefore gives the asserted almost-sure Matrix Spencer bound. □

Corollary 7.268 (Regularly varying common-profile schedules)**.** *In the setting of Corollary 7.265, assume that, for fixed $\alpha > 0$ and $\gamma \in \mathbb{R}$,*

$$\mathbb{P}\{Z_{m,g} > t\} \le C_{m,g} t^{-\alpha}(\log(e+t))^{\gamma} \qquad (t>0)$$

for every positive-mass group, and put

$$\mathcal{C}_m = \sum_{\substack{g\in G_m \\ A_{m,g}>0}} C_{m,g}.$$

Assume also that

$$\sum_{m=1}^{\infty}\left[\mathbb{P}(E_m) + (eD_m)^{-A}\right] < \infty.$$

Let $\Lambda_m = \log(e+m)$. Suppose that there are constants $C_0 < \infty$, $p \ge 0$, $q \in \mathbb{R}$, $a > 0$, and $b \in \mathbb{R}$ such that, for all sufficiently large m with $\mathcal{R}_m > 0$,

$$\mathcal{C}_m \le C_0 m^p \Lambda_m^q, \qquad Q\frac{n_m}{\ell_m\mathcal{R}_m} \ge m^a \Lambda_m^b.$$

If either

$$\alpha a > p+1,$$

or

$$\alpha a = p+1 \qquad \text{and} \qquad q + \gamma - \alpha b < -1,$$

then, for every $L > 0$ satisfying

$$\frac{L^2}{2(Q + LB/3)} \ge A + 1,$$

almost surely, for all sufficiently large m,

$$\operatorname{disc}(X_{m,1},\dots,X_{m,n_m}) \le (K+L)\sqrt{n_m}.$$

Thus common regularly varying volatility tails with logarithmic corrections are controlled by the polynomial growth of the aggregate tail coefficient and by the polynomial/logarithmic safety margin in the deterministic trace budget.

At the critical polynomial balance $\alpha a = p+1$, the logarithmic inequality $q+\gamma-\alpha b<-1$ is the remaining summability condition.

Proof. By Corollary 7.265 it is enough to prove the summability of

$$\sum_{m:\mathcal{R}_m>0} \mathcal{C}_m T_m^{-\alpha}(\log(e+T_m))^{\gamma}, \qquad T_m = Q\frac{n_m}{\ell_m \mathcal{R}_m}.$$

The function $t \mapsto t^{-\alpha}(\log(e+t))^{\gamma}$ is decreasing for all sufficiently large t. Since $T_m \geq m^a \Lambda_m^b$ and $a>0$, this monotonicity applies for all sufficiently large active m. Hence, after enlarging the implicit constant,

$$T_m^{-\alpha}(\log(e+T_m))^{\gamma} \leq C m^{-\alpha a}\Lambda_m^{\gamma-\alpha b}.$$

The m-th summand is therefore bounded by

$$C m^{p-\alpha a}\Lambda_m^{q+\gamma-\alpha b}.$$

This series converges if $\alpha a > p+1$, and in the endpoint case $\alpha a = p+1$ precisely under $q+\gamma-\alpha b<-1$. The common-profile corollary gives the asserted almost-sure Matrix Spencer bound. □

Corollary 7.269 (Heterogeneous polynomial grouped trace envelopes)**.** *In the setting of Corollary 7.251, allow G_m to be finite or countable. Suppose that $K_m \leq K$ and $\ell_m = \log(eD_m) \leq B\sqrt{n_m}$ for fixed $K, B < \infty$, and let $Q, A > 0$. Assume that*

$$S_m \leq \sum_{g\in G_m} A_{m,g} Z_{m,g} \qquad \text{almost surely,}$$

with deterministic $A_{m,g} \geq 0$ and arbitrary random $Z_{m,g} \geq 0$. For every g with $A_{m,g}>0$, suppose that there are constants $\alpha_{m,g}>0$ and $C_{m,g}>0$ such that

$$\mathbb{P}\{Z_{m,g} > t\} \leq C_{m,g} t^{-\alpha_{m,g}} \qquad (t>0),$$

and deterministic budget shares $\lambda_{m,g}>0$ satisfying

$$\sum_{g:A_{m,g}>0} \lambda_{m,g} \leq 1.$$

If

$$\sum_{m=1}^{\infty}\Big[\mathbb{P}(E_m) + (eD_m)^{-A}\Big] + \sum_m \sum_{\substack{g\in G_m \\ A_{m,g}>0}} C_{m,g}\left(\frac{\ell_m A_{m,g}}{Q n_m \lambda_{m,g}}\right)^{\alpha_{m,g}} < \infty,$$

then, for every $L>0$ satisfying

$$\frac{L^2}{2(Q+LB/3)} \geq A+1,$$

almost surely, for all sufficiently large m,

$$\operatorname{disc}(X_{m,1},\dots,X_{m,n_m}) \leq (K+L)\sqrt{n_m}.$$

Thus countable grouped trace-envelope models may mix tail indices across sectors, scales, frequencies, or trace-class atoms, rather than using a single polynomial exponent.

Proof. For each positive-mass group set

$$\tau_{m,g} = Q\frac{n_m}{\ell_m}\frac{\lambda_{m,g}}{A_{m,g}}.$$

Then

$$\sum_{g\in G_m} \tau_{m,g}A_{m,g} = Q\frac{n_m}{\ell_m}\sum_{g:A_{m,g}>0}\lambda_{m,g} \le Q\frac{n_m}{\ell_m}.$$

Moreover

$$\mathbb{P}\{Z_{m,g} > \tau_{m,g}\} \le C_{m,g}\left(\frac{\ell_m A_{m,g}}{Qn_m\lambda_{m,g}}\right)^{\alpha_{m,g}}.$$

The displayed summability condition is therefore exactly the countable threshold criterion of Corollary 7.251 with these thresholds. □

Corollary 7.270 (Optimized heterogeneous polynomial grouped trace envelopes)**.** *In the setting of Corollary 7.269, let G_m be finite or countable and write $\ell_m = \log(eD_m)$. For fixed $Q > 0$, define the optimized share cost*

$$H_m(Q) = \inf_{\lambda}\sum_{\substack{g\in G_m\\ A_{m,g}>0}} C_{m,g}\left(\frac{\ell_m A_{m,g}}{Qn_m\lambda_g}\right)^{\alpha_{m,g}},$$

where the infimum is over all deterministic $\lambda_g > 0$ on the positive-mass groups with $\sum_{g:A_{m,g}>0}\lambda_g \le 1$; if there are no positive-mass groups, the empty infimum is 0. Suppose that $K_m \le K$, $\ell_m \le B\sqrt{n_m}$, and

$$\sum_{m=1}^{\infty}\Big[\mathbb{P}(E_m) + (eD_m)^{-A} + H_m(Q)\Big] < \infty.$$

Then, for every $L > 0$ satisfying

$$\frac{L^2}{2(Q + LB/3)} \ge A + 1,$$

almost surely, for all sufficiently large m,

$$\operatorname{disc}(X_{m,1},\dots,X_{m,n_m}) \le (K+L)\sqrt{n_m}.$$

Thus heterogeneous grouped polynomial tails can be checked at the best available deterministic allocation of the $n_m/\log(eD_m)$ trace budget.

Proof. The summability hypothesis implies $H_m(Q) < \infty$ for every m. If there are no positive-mass groups, there is no multiplier-tail contribution. When positive-mass groups are present, choose deterministic shares $\lambda_{m,g} > 0$ with $\sum_g \lambda_{m,g} \le 1$ and

$$\sum_{\substack{g\in G_m\\ A_{m,g}>0}} C_{m,g}\left(\frac{\ell_m A_{m,g}}{Qn_m\lambda_{m,g}}\right)^{\alpha_{m,g}} \le H_m(Q) + 2^{-m}.$$

This is possible by the definition of the infimum. Summing over m, the right-hand side is finite. Applying Corollary 7.269 with these shares gives the conclusion. □

Corollary 7.271 (Optimized sub-Weibull grouped trace envelopes)**.** *In the setting of Corollary 7.251, let G_m be finite or countable and write $\ell_m = \log(eD_m)$. Suppose that $K_m \le K$, $\ell_m \le B\sqrt{n_m}$, and*

$$S_m \le \sum_{g\in G_m} A_{m,g} Z_{m,g} \qquad \text{almost surely,}$$

where $A_{m,g} \ge 0$ are deterministic and $Z_{m,g} \ge 0$ are arbitrary random multipliers. For each positive-mass group assume that there are constants $C_{m,g}, c_{m,g}, \beta_{m,g} > 0$ such that

$$\mathbb{P}\{Z_{m,g} > t\} \le C_{m,g} \exp\{-c_{m,g} t^{\beta_{m,g}}\} \qquad (t > 0).$$

For fixed $Q > 0$, define the optimized sub-Weibull share cost

$$\Psi_m(Q) = \inf_{\lambda} \sum_{\substack{g\in G_m \\ A_{m,g}>0}} C_{m,g} \exp\left\{-c_{m,g}\left(\frac{Q n_m \lambda_g}{\ell_m A_{m,g}}\right)^{\beta_{m,g}}\right\},$$

where the infimum is over all deterministic $\lambda_g > 0$ on the positive-mass groups with $\sum_{g:A_{m,g}>0} \lambda_g \le 1$; if there are no positive-mass groups, the empty infimum is 0. If, for some $A > 0$,

$$\sum_{m=1}^{\infty} \left[\mathbb{P}(E_m) + (eD_m)^{-A} + \Psi_m(Q)\right] < \infty,$$

then, for every $L > 0$ satisfying

$$\frac{L^2}{2(Q + LB/3)} \ge A + 1,$$

almost surely, for all sufficiently large m,

$$\operatorname{disc}(X_{m,1}, \dots, X_{m,n_m}) \le (K + L)\sqrt{n_m}.$$

In particular, it is enough to exhibit deterministic shares $\lambda_{m,g} > 0$, $\sum_{g:A_{m,g}>0} \lambda_{m,g} \le 1$, for which

$$\sum_{m=1}^{\infty} \left[\mathbb{P}(E_m) + (eD_m)^{-A}\right] + \sum_{m} \sum_{\substack{g\in G_m \\ A_{m,g}>0}} C_{m,g} \exp\left\{-c_{m,g}\left(\frac{Q n_m \lambda_{m,g}}{\ell_m A_{m,g}}\right)^{\beta_{m,g}}\right\} < \infty.$$

Thus sector, scale, frequency, and trace-class volatility decompositions with mixed stretched-exponential tail indices still give almost-sure eventual Spencer-scale discrepancy at the trace-envelope threshold.

Proof. The displayed fixed-share criterion is the threshold criterion of Corollary 7.251 with

$$\tau_{m,g} = Q\frac{n_m}{\ell_m}\frac{\lambda_{m,g}}{A_{m,g}} \qquad (A_{m,g} > 0),$$

zero-mass groups being ignored.

For the optimized criterion, the summability hypothesis implies $\Psi_m(Q) < \infty$ for every m. If positive-mass groups are present, choose deterministic shares $\lambda_{m,g} > 0$ with $\sum_g \lambda_{m,g} \le 1$ and

$$\sum_{\substack{g\in G_m\\ A_{m,g}>0}} C_{m,g}\exp\left\{-c_{m,g}\left(\frac{Qn_m\lambda_{m,g}}{\ell_m A_{m,g}}\right)^{\beta_{m,g}}\right\} \le \Psi_m(Q) + 2^{-m}.$$

This is possible by the definition of the infimum; when there are no positive-mass groups there is no multiplier-tail cost. The resulting fixed share series is summable, and the fixed-share criterion gives the conclusion. □

Corollary 7.272 (Optimized grouped polynomial-tail trace envelopes)**.** *In the setting of Corollary 7.244, put $\ell_m = \log(eD_m)$. Suppose that $K_m \le K$ and $\ell_m \le B\sqrt{n_m}$ for fixed $K, B < \infty$. Let $\alpha, Q, A > 0$. For each m, let G_m be finite, let $A_{m,g} \ge 0$ be deterministic, and let $Z_{m,g} \ge 0$ be arbitrary random multipliers such that*

$$S_m \le \sum_{g\in G_m} A_{m,g}Z_{m,g} \qquad \text{almost surely.}$$

Assume that, for every g with $A_{m,g} > 0$, there is a constant $C_{m,g} > 0$ such that

$$\mathbb{P}\{Z_{m,g} > t\} \le C_{m,g}t^{-\alpha} \qquad (t > 0).$$

Set

$$R_m = \sum_{\substack{g\in G_m\\ A_{m,g}>0}} C_{m,g}^{1/(\alpha+1)}A_{m,g}^{\alpha/(\alpha+1)}.$$

If

$$\sum_{m=1}^{\infty}\left[\mathbb{P}(E_m) + (eD_m)^{-A} + \left(\frac{\ell_m}{Qn_m}\right)^{\alpha}R_m^{\alpha+1}\right] < \infty,$$

then, for every $L > 0$ satisfying

$$\frac{L^2}{2(Q + LB/3)} \ge A + 1,$$

almost surely, for all sufficiently large m,

$$\operatorname{disc}(X_{m,1}, \dots, X_{m,n_m}) \le (K + L)\sqrt{n_m}.$$

Thus the polynomial grouped-scale condition can be checked by a single effective group-tail mass, without choosing budget shares by hand.

Proof. If $R_m = 0$, then all $A_{m,g}$ vanish and the grouped-scale corollary has no multiplier-tail contribution. Otherwise set

$$\lambda_{m,g} = \frac{C_{m,g}^{1/(\alpha+1)} A_{m,g}^{\alpha/(\alpha+1)}}{R_m} \qquad (A_{m,g} > 0).$$

These shares are positive and sum to one. The polynomial-tail specialization of Corollary 7.250 then requires the summability of

$$\sum_m \sum_{\substack{g \in G_m \\ A_{m,g} > 0}} C_{m,g} \left(\frac{\ell_m A_{m,g}}{Q n_m \lambda_{m,g}} \right)^{\alpha},$$

besides the base exceptional series. For the displayed choice of $\lambda_{m,g}$, this double sum is

$$\sum_m \left(\frac{\ell_m}{Q n_m} \right)^{\alpha} R_m^{\alpha} \sum_{\substack{g \in G_m \\ A_{m,g} > 0}} C_{m,g}^{1/(\alpha+1)} A_{m,g}^{\alpha/(\alpha+1)} = \sum_m \left(\frac{\ell_m}{Q n_m} \right)^{\alpha} R_m^{\alpha+1}.$$

The grouped-scale corollary gives the asserted almost-sure Spencer-scale bound. □

Corollary 7.273 (Countable optimized grouped polynomial-tail trace envelopes)**.** *In the setting of Corollary 7.244, put $\ell_m = \log(eD_m)$. Suppose that $K_m \le K$ and $\ell_m \le B\sqrt{n_m}$ for fixed $K, B < \infty$. Let $\alpha, Q, A > 0$. For each m, let G_m be finite or countable, let $A_{m,g} \ge 0$ be deterministic, and let $Z_{m,g} \ge 0$ be arbitrary random multipliers such that*

$$S_m \le \sum_{g \in G_m} A_{m,g} Z_{m,g} \qquad \text{almost surely.}$$

Assume that, for every g with $A_{m,g} > 0$, there is a constant $C_{m,g} > 0$ such that

$$\mathbb{P}\{Z_{m,g} > t\} \le C_{m,g} t^{-\alpha} \qquad (t > 0).$$

Define

$$R_m = \sum_{\substack{g \in G_m \\ A_{m,g} > 0}} C_{m,g}^{1/(\alpha+1)} A_{m,g}^{\alpha/(\alpha+1)} \in [0, \infty].$$

If

$$\sum_{m=1}^{\infty} \left[\mathbb{P}(E_m) + (eD_m)^{-A} + \left(\frac{\ell_m}{Q n_m} \right)^{\alpha} R_m^{\alpha+1} \right] < \infty,$$

then, for every $L > 0$ satisfying

$$\frac{L^2}{2(Q + LB/3)} \ge A + 1,$$

almost surely, for all sufficiently large m,

$$\operatorname{disc}(X_{m,1}, \dots, X_{m,n_m}) \le (K + L)\sqrt{n_m}.$$

Consequently, countable sector, frequency, scale, or trace-class decompositions with polynomial group scales are controlled by the same effective group-tail mass as the finite grouped model.

Proof. Put $q_m = Qn_m/\ell_m$. If $R_m = 0$, then there are no positive-mass groups and $S_m = 0$ almost surely. Otherwise the summability hypothesis forces $R_m < \infty$ for the relevant m. Set

$$\lambda_{m,g} = \frac{C_{m,g}^{1/(\alpha+1)} A_{m,g}^{\alpha/(\alpha+1)}}{R_m} \qquad (A_{m,g} > 0),$$

so that $\sum_{g:A_{m,g}>0} \lambda_{m,g} = 1$. Define

$$\tau_{m,g} = Q\frac{n_m}{\ell_m}\frac{\lambda_{m,g}}{A_{m,g}} \qquad (A_{m,g} > 0).$$

If $Z_{m,g} \le \tau_{m,g}$ for every positive-mass group, monotone convergence for the nonnegative series gives

$$S_m \le \sum_{g\in G_m} A_{m,g}Z_{m,g} \le \sum_{A_{m,g}>0} A_{m,g}\tau_{m,g} = q_m.$$

Hence, by the countable union bound and the polynomial tails,

$$\begin{aligned}\mathbb{P}\{S_m > q_m\} &\le \sum_{A_{m,g}>0} \mathbb{P}\{Z_{m,g} > \tau_{m,g}\} \\ &\le \sum_{A_{m,g}>0} C_{m,g}\left(\frac{\ell_m A_{m,g}}{Qn_m\lambda_{m,g}}\right)^{\alpha} \\ &= \left(\frac{\ell_m}{Qn_m}\right)^{\alpha} R_m^{\alpha+1}.\end{aligned}$$

The concrete Borel profile in Corollary 7.244 applies with this deterministic quantile-failure bound. □

Corollary 7.274 (Sub-gamma trace-envelope variance profiles)**.** *Let $D \ge 2$. Keep the signed-core, filtration, and martingale-noise hypotheses of Corollary 7.241, but do not assume that the trace envelopes are uniformly bounded. Thus (V_i, Γ_i, ρ_i), $i = 1, \dots, n$, are independent triples, independent of $\mathcal{G}$, with $V_i \succeq 0$, $\rho_i \in [0, 1]$, and*

$$\operatorname{tr} V_i \le \Gamma_i \qquad \textit{almost surely.}$$

Put

$$Z_i = \rho_i^2\Gamma_i, \qquad \mu = \sum_{i=1}^{n} \mathbb{E}Z_i.$$

Assume that the centered envelope sum is sub-gamma: for some $\sigma^2, c \ge 0$,

$$\log \mathbb{E}\exp\left\{\lambda\sum_{i=1}^{n}(Z_i - \mathbb{E}Z_i)\right\} \le \frac{\lambda^2\sigma^2}{2(1-c\lambda)} \qquad (0 \le \lambda < 1/c),$$

with the usual interpretation $1/c = \infty$ *when* $c = 0$. *Then, for every* $u \geq 0$ *and* $s > 0$,

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \leq \mathbb{P}(E) + \exp\left\{-\frac{u^2}{2(\sigma^2 + cu)}\right\} + 2D \exp\left\{-\frac{s^2}{2(\mu + u + s/3)}\right\},$$

with the convention that the first exponential term is omitted when $\sigma^2 + cu = 0$. *Thus the signed-core theorem also covers unbounded random trace envelopes whenever their total weighted envelope has a Bernstein-type upper tail.*

Proof. As before,

$$\left\| \sum_i \mathbb{E}(Y_i^2 \mid \mathcal{F}_{i-1}) \right\|_{\mathrm{op}} \leq T := \sum_i \rho_i^2 \operatorname{tr} V_i \leq \sum_i Z_i.$$

The sub-gamma assumption and Chernoff's bound imply

$$\mathbb{P}\left\{\sum_i Z_i > \mu + u\right\} \leq \exp\left\{-\frac{u^2}{2(\sigma^2 + cu)}\right\}.$$

Indeed, when $\sigma^2 + cu > 0$, take $\lambda = u/(\sigma^2 + cu)$ if $\sigma^2 > 0$. If $\sigma^2 = 0 < c$, use $\lambda = (1-\eta)/c$ and let $\eta \downarrow 0$. The displayed moment-generating-function bound gives the stated exponent in either case. Combining this budget-exception estimate with Corollary 7.235 at $v = \mu + u$ proves the claim. □

Corollary 7.275 (Termwise sub-gamma trace envelopes)**.** *Keep the hypotheses and notation of Corollary 7.274, but replace the global sub-gamma assumption by the following termwise one. For each* i, *assume that there are* $\sigma_i^2, c_i \geq 0$ *such that, for* $Z_i = \rho_i^2 \Gamma_i$,

$$\log \mathbb{E} \exp\{\lambda(Z_i - \mathbb{E}Z_i)\} \leq \frac{\lambda^2 \sigma_i^2}{2(1 - c_i\lambda)} \qquad (0 \leq \lambda < 1/c_i).$$

Put

$$\mu = \sum_i \mathbb{E}Z_i, \qquad \sigma^2 = \sum_i \sigma_i^2, \qquad c = \max_{1 \leq i \leq n} c_i.$$

Then, for every $u \geq 0$ *and* $s > 0$,

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \leq \mathbb{P}(E) + \exp\left\{-\frac{u^2}{2(\sigma^2 + cu)}\right\} + 2D \exp\left\{-\frac{s^2}{2(\mu + u + s/3)}\right\}.$$

The same conclusion therefore applies whenever the weighted trace envelopes, or the low-rank/Schatten envelopes $R_i \tau_i \rho_i^2$ *and* $R_i^{\beta} L_i \rho_i^2$, *are independently sub-gamma with summable variance proxies and bounded scale.*

Proof. Independence gives, for $0 \le \lambda < 1/c$,

$$\log \mathbb{E} \exp\left\{\lambda \sum_i (Z_i - \mathbb{E}Z_i)\right\} = \sum_i \log \mathbb{E} \exp\{\lambda(Z_i - \mathbb{E}Z_i)\} \le \sum_i \frac{\lambda^2 \sigma_i^2}{2(1 - c_i\lambda)} \le \frac{\lambda^2 \sigma^2}{2(1 - c\lambda)}.$$

This is exactly the sub-gamma hypothesis of Corollary 7.274. □

Corollary 7.276 (Gamma-dominated trace-envelope variance profiles)**.** *Keep the signed-core, filtration, and martingale-noise hypotheses of Corollary 7.274. Assume that there are independent random variables W_i, independent of $\mathcal{G}$, such that*

$$\rho_i^2 \Gamma_i \le W_i \qquad \text{almost surely}$$

and W_i has the Gamma law with shape $\kappa_i \ge 0$ and scale $a_i \ge 0$, with the convention that $W_i = 0$ when $a_i\kappa_i = 0$. Define

$$\mu_\gamma = \sum_i \kappa_i a_i, \qquad \sigma_\gamma^2 = \sum_i \kappa_i a_i^2, \qquad a_* = \max_i a_i.$$

Then, for every $u \ge 0$ and $s > 0$,

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \le \mathbb{P}(E) + \exp\left\{-\frac{u^2}{2(\sigma_\gamma^2 + a_* u)}\right\} + 2D \exp\left\{-\frac{s^2}{2(\mu_\gamma + u + s/3)}\right\}.$$

with the convention that the first exponential term is omitted when $\sigma_\gamma^2 + a_* u = 0$. Equivalently, if $W_i = (a_i/2)\chi^2_{\nu_i}$, then $\kappa_i = \nu_i/2$,

$$\mu_\gamma = \frac{1}{2}\sum_i \nu_i a_i, \qquad \sigma_\gamma^2 = \frac{1}{2}\sum_i \nu_i a_i^2.$$

Thus chi-square and Wishart-type random trace envelopes give an explicit signed-core Matrix Spencer tail.

Proof. The weighted trace budget satisfies

$$T = \sum_i \rho_i^2 \operatorname{tr} V_i \le \sum_i \rho_i^2 \Gamma_i \le \sum_i W_i.$$

For $W_i \sim \operatorname{Gamma}(\kappa_i, a_i)$,

$$\log \mathbb{E} \exp\{\lambda(W_i - \kappa_i a_i)\} = \kappa_i\left[-\log(1 - a_i\lambda) - a_i\lambda\right] \le \frac{\lambda^2 \kappa_i a_i^2}{2(1 - a_i\lambda)}$$

for $0 \le \lambda < 1/a_i$; terms with $a_i = 0$ are deterministic and are omitted. Here we used the elementary bound $-\log(1 - x) - x \le x^2/(2(1 - x))$. Summing over i gives the global sub-gamma bound with parameters σ_γ^2 and a_*. Hence

$$\mathbb{P}\{T > \mu_\gamma + u\} \le \exp\left\{-\frac{u^2}{2(\sigma_\gamma^2 + a_* u)}\right\}.$$

Combining this with Corollary 7.235 at $v = \mu_\gamma + u$ proves the displayed tail. The chi-square specialization is the identity $\chi^2_\nu \sim \mathrm{Gamma}(\nu/2, 2)$. □

Corollary 7.277 (Gamma-atom trace-envelope variance profiles)**.** *Keep the signed-core, filtration, and martingale-noise hypotheses of Corollary 7.274. Assume that there are independent nonnegative random variables $H_{i\ell}$, independent of $\mathcal{G}$, such that*

$$\rho_i^2 \Gamma_i \le W_i := \sum_{\ell=1}^{m_i} H_{i\ell} \qquad \text{almost surely}$$

and $H_{i\ell}$ has the Gamma law with shape $\kappa_{i\ell} \ge 0$ and scale $a_{i\ell} \ge 0$, with the convention that $H_{i\ell} = 0$ when $a_{i\ell}\kappa_{i\ell} = 0$. Define

$$\mu_{\Gamma\text{-at}} = \sum_{i,\ell} \kappa_{i\ell} a_{i\ell}, \qquad \sigma^2_{\Gamma\text{-at}} = \sum_{i,\ell} \kappa_{i\ell} a_{i\ell}^2, \qquad a_* = \max_{i,\ell} a_{i\ell}.$$

Then, for every $u \ge 0$ and $s > 0$,

$$\mathbb{P}\left\{\mathrm{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \le \mathbb{P}(E) + \exp\left\{-\frac{u^2}{2(\sigma^2_{\Gamma\text{-at}} + a_* u)}\right\} + 2D \exp\left\{-\frac{s^2}{2(\mu_{\Gamma\text{-at}} + u + s/3)}\right\},$$

with the convention that the first exponential term is omitted when $\sigma^2_{\Gamma\text{-at}} + a_* u = 0$. In particular, independent exponential trace atoms have $\kappa_{i\ell} = 1$, while independent chi-square atoms $(a_{i\ell}/2)\chi^2_{\nu_{i\ell}}$ have $\kappa_{i\ell} = \nu_{i\ell}/2$. Thus heterogeneous gamma, exponential, and chi-square atom decompositions give explicit signed-core Matrix Spencer tails with the largest atom scale as the jump parameter.

Proof. The weighted trace budget satisfies

$$T = \sum_i \rho_i^2 \operatorname{tr} V_i \le \sum_i \rho_i^2 \Gamma_i \le \sum_{i,\ell} H_{i\ell}.$$

For $H_{i\ell} \sim \mathrm{Gamma}(\kappa_{i\ell}, a_{i\ell})$,

$$\log \mathbb{E} \exp\{\lambda(H_{i\ell} - \kappa_{i\ell} a_{i\ell})\} \le \frac{\lambda^2 \kappa_{i\ell} a_{i\ell}^2}{2(1 - a_{i\ell}\lambda)}$$

for $0 \le \lambda < 1/a_{i\ell}$; deterministic terms with $a_{i\ell} = 0$ are omitted. This is the same elementary Gamma mgf bound used in Corollary 7.276. Summing over all atoms gives the sub-gamma bound with parameters $\sigma^2_{\Gamma\text{-at}}$ and a_*. Hence

$$\mathbb{P}\{T > \mu_{\Gamma\text{-at}} + u\} \le \exp\left\{-\frac{u^2}{2(\sigma^2_{\Gamma\text{-at}} + a_* u)}\right\}.$$

Combining this with Corollary 7.235 at $v = \mu_{\Gamma\text{-at}} + u$ proves the displayed tail. □

Corollary 7.278 (Countable gamma-atom trace-envelope variance profiles)**.** *Keep the signed-core, filtration, and martingale-noise hypotheses of Corollary 7.274. Assume that there are independent nonnegative random variables $H_{i\ell}$, independent of $\mathcal{G}$, such that*

$$\rho_i^2\Gamma_i \le W_i := \sum_{\ell\ge 1} H_{i\ell} \qquad \text{almost surely}$$

and $H_{i\ell}$ has the Gamma law with shape $\kappa_{i\ell}\ge 0$ and scale $a_{i\ell}\ge 0$, with the convention that $H_{i\ell}=0$ when $a_{i\ell}\kappa_{i\ell}=0$. Suppose

$$\mu_{\Gamma,\infty} := \sum_{i,\ell}\kappa_{i\ell}a_{i\ell} < \infty, \qquad \sigma^2_{\Gamma,\infty} := \sum_{i,\ell}\kappa_{i\ell}a_{i\ell}^2 < \infty, \qquad a_* := \sup_{i,\ell} a_{i\ell} < \infty.$$

Then, for every $u\ge 0$ and $s>0$,

$$\mathbb{P}\left\{\operatorname{disc}(X_1,\dots,X_n) > K\sqrt{n}+s\right\} \le \mathbb{P}(E) + \exp\left\{-\frac{u^2}{2(\sigma^2_{\Gamma,\infty}+a_*u)}\right\} + 2D\exp\left\{-\frac{s^2}{2(\mu_{\Gamma,\infty}+u+s/3)}\right\},$$

with the convention that the first exponential term is omitted when $\sigma^2_{\Gamma,\infty}+a_*u=0$. Thus countable exponential and chi-square trace-atom expansions, including trace-class real and complex Gaussian quadratic-form/Wishart envelopes, inherit the same signed-core Matrix Spencer tail whenever the total mean trace budget and the largest atom scale are finite.

Proof. The assumption $\mu_{\Gamma,\infty}<\infty$ implies $\sum_{i,\ell}H_{i\ell}<\infty$ almost surely, so the weighted trace budget satisfies

$$T = \sum_i \rho_i^2 \operatorname{tr} V_i \le \sum_i \rho_i^2\Gamma_i \le \sum_{i,\ell} H_{i\ell}.$$

If $a_*=0$, then all atoms vanish almost surely and the claim follows directly from Corollary 7.235. Assume $a_*>0$. For a finite set F of index pairs put

$$S_F = \sum_{(i,\ell)\in F} H_{i\ell}, \qquad \mu_F = \sum_{(i,\ell)\in F}\kappa_{i\ell}a_{i\ell}, \qquad \sigma_F^2 = \sum_{(i,\ell)\in F}\kappa_{i\ell}a_{i\ell}^2.$$

By the finite gamma-atom bound in Corollary 7.277, for $0\le\lambda<1/a_*$,

$$\mathbb{E}\exp\{\lambda(S_F-\mu_F)\} \le \exp\left\{\frac{\lambda^2\sigma_F^2}{2(1-a_*\lambda)}\right\} \le \exp\left\{\frac{\lambda^2\sigma^2_{\Gamma,\infty}}{2(1-a_*\lambda)}\right\}.$$

Since $\mu_F\le\mu_{\Gamma,\infty}$, the same bound holds with $S_F-\mu_{\Gamma,\infty}$ in place of $S_F-\mu_F$. Letting F increase to all index pairs and using monotone convergence gives the same bound for $S=\sum_{i,\ell}H_{i\ell}$ centered at $\mu_{\Gamma,\infty}$. Hence

$$\mathbb{P}\{T>\mu_{\Gamma,\infty}+u\} \le \exp\left\{-\frac{u^2}{2(\sigma^2_{\Gamma,\infty}+a_*u)}\right\}.$$

Combining this with Corollary 7.235 at $v=\mu_{\Gamma,\infty}+u$ proves the displayed tail. □

Corollary 7.279 (Compound-Poisson dominated trace-envelope variance profiles)**.** *Keep the signed-core, filtration, and martingale-noise hypotheses of Corollary 7.274. Assume that there are independent random variables W_i, independent of $\mathcal{G}$, such that*

$$\rho_i^2 \Gamma_i \le W_i \qquad \text{almost surely}$$

and each W_i has a compound-Poisson representation

$$W_i = \sum_{\ell=1}^{N_i} Z_{i\ell}.$$

Here $N_i \sim \mathrm{Poisson}(\lambda_i)$, the variables $Z_{i\ell}$ are conditionally iid given N_i, independent of N_i, nonnegative, and satisfy $Z_{i\ell} \le a_i$ almost surely. Put

$$\mu_{\mathrm{P}} = \sum_i \lambda_i \mathbb{E} Z_{i1}, \qquad \sigma_{\mathrm{P}}^2 = \sum_i \lambda_i \mathbb{E} Z_{i1}^2, \qquad a_* = \max_i a_i.$$

Then, for every $u \ge 0$ and $s > 0$,

$$\mathbb{P}\left\{\mathrm{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \le \mathbb{P}(E) + \exp\left\{-\frac{u^2}{2(\sigma_{\mathrm{P}}^2 + a_* u/3)}\right\} + 2D \exp\left\{-\frac{s^2}{2(\mu_{\mathrm{P}} + u + s/3)}\right\},$$

with the convention that the first exponential term is omitted when $\sigma_{\mathrm{P}}^2 + a_* u/3 = 0$. In particular, if $Z_{i\ell} = a_i$ almost surely, then $W_i = a_i N_i$,

$$\mu_{\mathrm{P}} = \sum_i \lambda_i a_i, \qquad \sigma_{\mathrm{P}}^2 = \sum_i \lambda_i a_i^2.$$

Thus Poisson random counts of bounded trace-envelope atoms, including random bounded spike-count variance profiles, give an explicit signed-core Matrix Spencer tail.

Proof. The weighted trace budget is bounded by

$$T = \sum_i \rho_i^2 \operatorname{tr} V_i \le \sum_i \rho_i^2 \Gamma_i \le \sum_i W_i.$$

For $t \ge 0$,

$$\log \mathbb{E} \exp\{t(W_i - \mathbb{E} W_i)\} = \lambda_i \mathbb{E}\big(e^{tZ_{i1}} - 1 - tZ_{i1}\big).$$

The elementary bound $e^x - 1 - x \le x^2/(2(1 - x/3))$, valid for $0 \le x < 3$, gives

$$\log \mathbb{E} \exp\{t(W_i - \mathbb{E} W_i)\} \le \frac{t^2 \lambda_i \mathbb{E} Z_{i1}^2}{2(1 - a_i t/3)}$$

for $0 \le t < 3/a_i$; terms with $a_i = 0$ are deterministic and are omitted. Summing over i gives the sub-gamma bound with parameters σ_{P}^2 and $a_*/3$. Hence

$$\mathbb{P}\{T > \mu_{\mathrm{P}} + u\} \le \exp\left\{-\frac{u^2}{2(\sigma_{\mathrm{P}}^2 + a_* u/3)}\right\}.$$

Combining this with Corollary 7.235 at $v = \mu_{\mathrm{P}} + u$ proves the displayed tail. □

Corollary 7.280 (Atomized bounded trace-envelope variance profiles)**.** *Keep the signed-core, filtration, and martingale-noise hypotheses of Corollary 7.274. Assume that there are independent nonnegative random variables $Z_{i\ell}$, independent of $\mathcal{G}$, such that*

$$0 \le Z_{i\ell} \le a_{i\ell} \qquad \text{almost surely}$$

and

$$\rho_i^2 \Gamma_i \le W_i := \sum_{\ell=1}^{m_i} Z_{i\ell} \qquad \text{almost surely.}$$

Define

$$\mu_{\mathrm{A}} = \sum_{i,\ell} \mathbb{E} Z_{i\ell}, \qquad \sigma_{\mathrm{A}}^2 = \sum_{i,\ell} \mathbb{E} Z_{i\ell}^2, \qquad a_* = \max_{i,\ell} a_{i\ell}.$$

Then, for every $u \ge 0$ and $s > 0$,

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \le \mathbb{P}(E) + \exp\left\{-\frac{u^2}{2(\sigma_{\mathrm{A}}^2 + a_* u/3)}\right\} + 2D \exp\left\{-\frac{s^2}{2(\mu_{\mathrm{A}} + u + s/3)}\right\},$$

with the convention that the first exponential term is omitted when $\sigma_{\mathrm{A}}^2 + a_* u/3 = 0$. In particular, if $Z_{i\ell} = a_{i\ell} B_{i\ell}$ with independent Bernoulli variables $B_{i\ell}$ of means $p_{i\ell}$, then

$$\mu_{\mathrm{A}} = \sum_{i,\ell} a_{i\ell} p_{i\ell}, \qquad \sigma_{\mathrm{A}}^2 = \sum_{i,\ell} a_{i\ell}^2 p_{i\ell}.$$

Thus finite Bernoulli/binomial random sums of bounded trace-envelope atoms, including bounded random spike and low-rank atom variance profiles, give an explicit signed-core Matrix Spencer tail with atom-level jump parameter.

Proof. The weighted trace budget satisfies

$$T = \sum_i \rho_i^2 \operatorname{tr} V_i \le \sum_i \rho_i^2 \Gamma_i \le \sum_{i,\ell} Z_{i\ell}.$$

For a nonnegative $Z \le a$,

$$\log \mathbb{E} e^{t(Z - \mathbb{E}Z)} \le \mathbb{E}(e^{tZ} - 1 - tZ) \le \frac{t^2 \mathbb{E} Z^2}{2(1 - at/3)}$$

for $0 \le t < 3/a$, using $\log x \le x - 1$ and $e^x - 1 - x \le x^2/(2(1 - x/3))$; terms with $a = 0$ are deterministic and are omitted. Summing over all atoms gives the sub-gamma bound with parameters σ_{A}^2 and $a_*/3$. Therefore

$$\mathbb{P}\{T > \mu_{\mathrm{A}} + u\} \le \exp\left\{-\frac{u^2}{2(\sigma_{\mathrm{A}}^2 + a_* u/3)}\right\}.$$

Combining this with Corollary 7.235 at $v = \mu_{\mathrm{A}} + u$ proves the displayed tail. □

Corollary 7.281 (Countable bounded-atom trace-envelope variance profiles)**.** *Keep the signed-core, filtration, and martingale-noise hypotheses of Corollary 7.274. Assume that there are independent nonnegative random variables $Z_{i\ell}$, independent of $\mathcal{G}$, such that*

$$0 \le Z_{i\ell} \le a_{i\ell} \qquad \text{almost surely,} \qquad \rho_i^2 \Gamma_i \le W_i := \sum_{\ell \ge 1} Z_{i\ell} \qquad \text{almost surely.}$$

Suppose

$$\mu_{\mathrm{A},\infty} := \sum_{i,\ell} \mathbb{E} Z_{i\ell} < \infty, \qquad \sigma^2_{\mathrm{A},\infty} := \sum_{i,\ell} \mathbb{E} Z_{i\ell}^2 < \infty, \qquad a_* := \sup_{i,\ell} a_{i\ell} < \infty.$$

Then, for every $u \ge 0$ and $s > 0$,

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \le \mathbb{P}(E) + \exp\left\{-\frac{u^2}{2(\sigma^2_{\mathrm{A},\infty} + a_* u/3)}\right\} + 2D \exp\left\{-\frac{s^2}{2(\mu_{\mathrm{A},\infty} + u + s/3)}\right\},$$

with the convention that the first exponential term is omitted when $\sigma^2_{\mathrm{A},\infty} + a_* u/3 = 0$. In particular, if $Z_{i\ell} = a_{i\ell} B_{i\ell}$ with independent Bernoulli variables $B_{i\ell}$ of means $p_{i\ell}$, then

$$\mu_{\mathrm{A},\infty} = \sum_{i,\ell} a_{i\ell} p_{i\ell}, \qquad \sigma^2_{\mathrm{A},\infty} = \sum_{i,\ell} a_{i\ell}^2 p_{i\ell}.$$

Thus countable Bernoulli and bounded-atom trace-envelope expansions, including trace-class random spike and low-rank atom variance profiles, give explicit signed-core Matrix Spencer tails with the supremal atom size as the jump parameter.

Proof. The assumption $\mu_{\mathrm{A},\infty} < \infty$ implies $\sum_{i,\ell} Z_{i\ell} < \infty$ almost surely. Hence

$$T = \sum_i \rho_i^2 \operatorname{tr} V_i \le \sum_i \rho_i^2 \Gamma_i \le \sum_{i,\ell} Z_{i\ell}.$$

If $a_* = 0$, then all atoms vanish almost surely and the claim follows directly from Corollary 7.235. Assume $a_* > 0$. For a finite set F of index pairs put

$$S_F = \sum_{(i,\ell) \in F} Z_{i\ell}, \qquad \mu_F = \sum_{(i,\ell) \in F} \mathbb{E} Z_{i\ell}, \qquad \sigma_F^2 = \sum_{(i,\ell) \in F} \mathbb{E} Z_{i\ell}^2.$$

The finite-atom mgf estimate in the proof of Corollary 7.280 gives, for $0 \le t < 3/a_*$,

$$\mathbb{E} \exp\{t(S_F - \mu_F)\} \le \exp\left\{\frac{t^2 \sigma_F^2}{2(1 - a_* t/3)}\right\} \le \exp\left\{\frac{t^2 \sigma^2_{\mathrm{A},\infty}}{2(1 - a_* t/3)}\right\}.$$

Since $\mu_F \le \mu_{\mathrm{A},\infty}$, the same bound holds with $S_F - \mu_{\mathrm{A},\infty}$ in place of $S_F - \mu_F$. Letting F increase to all index pairs and using monotone convergence gives the

same sub-gamma mgf bound for $S = \sum_{i,\ell} Z_{i\ell}$ centered at $\mu_{\mathrm{A},\infty}$. Chernoff's bound yields

$$\mathbb{P}\{T > \mu_{\mathrm{A},\infty} + u\} \le \exp\left\{-\frac{u^2}{2(\sigma^2_{\mathrm{A},\infty} + a_* u/3)}\right\}.$$

Combining this with Corollary 7.235 at $v = \mu_{\mathrm{A},\infty} + u$ proves the displayed tail. □

Corollary 7.282 (Mixed atom trace-envelope variance profiles)**.** *Keep the signed-core, filtration, and martingale-noise hypotheses of Corollary 7.274. Let $\mathcal{I}_\Gamma$, $\mathcal{I}_\mathrm{A}$, and $\mathcal{I}_\mathrm{P}$ be finite or countable index sets. Assume that all random variables below are mutually independent and independent of $\mathcal{G}$, and that*

$$\sum_i \rho_i^2 \Gamma_i \le \sum_{\alpha \in \mathcal{I}_\Gamma} H_\alpha + \sum_{\beta \in \mathcal{I}_\mathrm{A}} Z_\beta + \sum_{j \in \mathcal{I}_\mathrm{P}} C_j \qquad \text{almost surely.}$$

Here $H_\alpha \sim \mathrm{Gamma}(\kappa_\alpha, a_\alpha)$ with $\kappa_\alpha, a_\alpha \ge 0$, $0 \le Z_\beta \le b_\beta$ almost surely, and

$$C_j = \sum_{\ell=1}^{N_j} U_{j\ell}, \qquad N_j \sim \mathrm{Poisson}(\lambda_j), \qquad 0 \le U_{j\ell} \le d_j \quad \text{almost surely,}$$

with the $U_{j\ell}$ iid for fixed j and independent of N_j. Suppose

$$\mu_{\mathrm{mix}} := \sum_\alpha \kappa_\alpha a_\alpha + \sum_\beta \mathbb{E} Z_\beta + \sum_j \lambda_j \mathbb{E} U_{j1} < \infty,$$
$$\sigma^2_{\mathrm{mix}} := \sum_\alpha \kappa_\alpha a_\alpha^2 + \sum_\beta \mathbb{E} Z_\beta^2 + \sum_j \lambda_j \mathbb{E} U_{j1}^2 < \infty,$$

and put

$$c_{\mathrm{mix}} := \max\left\{\sup_\alpha a_\alpha,\ \frac{1}{3}\sup_\beta b_\beta,\ \frac{1}{3}\sup_j d_j\right\} < \infty,$$

where the supremum over an empty family is 0. Then, for every $u \ge 0$ and $s > 0$,

$$\mathbb{P}\left\{\mathrm{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \le \mathbb{P}(E) + \exp\left\{-\frac{u^2}{2(\sigma^2_{\mathrm{mix}} + c_{\mathrm{mix}} u)}\right\} + 2D \exp\left\{-\frac{s^2}{2(\mu_{\mathrm{mix}} + u + s/3)}\right\},$$

with the convention that the first exponential term is omitted when $\sigma^2_{\mathrm{mix}} + c_{\mathrm{mix}} u = 0$. Thus a single signed-core Matrix Spencer tail covers mixed trace envelopes containing, at the same time, bounded Bernoulli/spike atoms, gamma or chi-square/Wishart atoms, and compound-Poisson random counts of bounded atoms.

Proof. Let

$$S = \sum_\alpha H_\alpha + \sum_\beta Z_\beta + \sum_j C_j.$$

The assumption $\mu_{\mathrm{mix}} < \infty$ implies $S < \infty$ almost surely, and the weighted trace budget satisfies

$$T = \sum_i \rho_i^2 \operatorname{tr} V_i \le \sum_i \rho_i^2 \Gamma_i \le S.$$

If $c_{\mathrm{mix}} = 0$, then all components vanish almost surely and the claim follows directly from Corollary 7.235. Assume $c_{\mathrm{mix}} > 0$. For finite subfamilies F_Γ, F_A, F_P, write S_F, μ_F, and σ_F^2 for the corresponding truncated sum, mean, and variance proxy. The gamma, bounded-atom, and compound-Poisson estimates proved above give, for $0 \le t < 1/c_{\mathrm{mix}}$,

$$\mathbb{E} \exp\{t(S_F - \mu_F)\} \le \exp\left\{ \frac{t^2 \sigma_F^2}{2(1 - c_{\mathrm{mix}} t)} \right\} \le \exp\left\{ \frac{t^2 \sigma_{\mathrm{mix}}^2}{2(1 - c_{\mathrm{mix}} t)} \right\}.$$

Indeed the denominator $1 - c_{\mathrm{mix}} t$ is no larger than the denominator coming from any individual gamma, bounded, or compound-Poisson component. Since $\mu_F \le \mu_{\mathrm{mix}}$, the same bound holds with $S_F - \mu_{\mathrm{mix}}$ in place of $S_F - \mu_F$. Letting the finite subfamilies increase to all indices and using monotone convergence gives the same mgf bound for $S - \mu_{\mathrm{mix}}$. Chernoff's bound yields

$$\mathbb{P}\{T > \mu_{\mathrm{mix}} + u\} \le \exp\left\{ -\frac{u^2}{2(\sigma_{\mathrm{mix}}^2 + c_{\mathrm{mix}} u)} \right\}.$$

Combining this budget-exception estimate with Corollary 7.235 at $v = \mu_{\mathrm{mix}} + u$ proves the displayed tail. □

Corollary 7.283 (Weighted chi-square trace-envelope variance profiles)**.** *Keep the signed-core, filtration, and martingale-noise hypotheses of Corollary 7.274. Assume that there are independent random variables W_i, independent of $\mathcal{G}$, such that*

$$\rho_i^2 \Gamma_i \le W_i \qquad \text{almost surely}$$

and

$$W_i = \sum_{\ell=1}^{r_i} a_{i\ell} g_{i\ell}^2,$$

where $a_{i\ell} \ge 0$ and the $g_{i\ell}$ are independent standard real Gaussians. Define

$$\mu_\chi = \sum_{i,\ell} a_{i\ell}, \qquad \sigma_\chi^2 = 2 \sum_{i,\ell} a_{i\ell}^2, \qquad a_* = \max_{i,\ell} a_{i\ell}.$$

Then, for every $u \ge 0$ and $s > 0$,

$$\mathbb{P}\left\{ \operatorname{disc}(X_1, \dots, X_n) > K\sqrt{n} + s \right\} \le \mathbb{P}(E) + \exp\left\{ -\frac{u^2}{2(\sigma_\chi^2 + 2a_* u)} \right\} + 2D \exp\left\{ -\frac{s^2}{2(\mu_\chi + u + s/3)} \right\},$$

with the convention that the first exponential term is omitted when $\sigma_\chi^2+2a_*u=0$. *In particular, if* G_i *is a* $d_i\times m_i$ *matrix with independent standard real Gaussian entries and* $A_i\succeq 0$, *then*

$$W_i=\operatorname{tr}(G_i^*A_iG_i)$$

is covered by taking the eigenvalues of A_i, *each repeated* m_i *times. The corresponding parameters are*

$$\mu_\chi=\sum_i m_i\operatorname{tr}A_i,\qquad \sigma_\chi^2=2\sum_i m_i\operatorname{tr}(A_i^2),\qquad a_*=\max_i\|A_i\|_{\mathrm{op}}.$$

Thus weighted chi-square and real Wishart trace envelopes give explicit signed-core Matrix Spencer tails.

Proof. The weighted trace budget satisfies

$$T=\sum_i\rho_i^2\operatorname{tr}V_i\le\sum_i\rho_i^2\Gamma_i\le\sum_{i,\ell}a_{i\ell}g_{i\ell}^2.$$

For a standard real Gaussian g and $a\ge 0$,

$$\log\mathbb{E}\exp\{t(ag^2-a)\}=-\frac{1}{2}\log(1-2at)-at\le\frac{t^2a^2}{1-2at}=\frac{t^2(2a^2)}{2(1-2at)}$$

for $0\le t<1/(2a)$; terms with $a=0$ are deterministic and are omitted. The inequality is the bound $-\log(1-x)-x\le x^2/(2(1-x))$ with $x=2at$. Summing over all Gaussian coordinates gives the sub-gamma bound with parameters σ_χ^2 and $2a_*$. Therefore

$$\mathbb{P}\{T>\mu_\chi+u\}\le\exp\left\{-\frac{u^2}{2(\sigma_\chi^2+2a_*u)}\right\}.$$

Combining this with Corollary 7.235 at $v=\mu_\chi+u$ proves the displayed tail. The Wishart specialization follows by diagonalizing each A_i and writing $\operatorname{tr}(G_i^*A_iG_i)$ as the corresponding sum of independent weighted squares. □

Corollary 7.284 (Weighted exponential and complex-Wishart trace envelopes)**.** *Keep the signed-core, filtration, and martingale-noise hypotheses of Corollary 7.274. Assume that there are independent random variables* W_i, *independent of* $\mathcal{G}$, *such that*

$$\rho_i^2\Gamma_i\le W_i\qquad\text{almost surely}$$

and

$$W_i=\sum_{\ell=1}^{r_i}a_{i\ell}|z_{i\ell}|^2,$$

where $a_{i\ell}\ge 0$ *and the* $z_{i\ell}$ *are independent standard complex Gaussians, normalized so that* $\mathbb{E}|z_{i\ell}|^2=1$. *Define*

$$\mu_{\mathbb{C}}=\sum_{i,\ell}a_{i\ell},\qquad\sigma_{\mathbb{C}}^2=\sum_{i,\ell}a_{i\ell}^2,\qquad a_*=\max_{i,\ell}a_{i\ell}.$$

Then, for every $u \geq 0$ and $s > 0$,

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \leq \mathbb{P}(E) + \exp\left\{-\frac{u^2}{2(\sigma_{\mathbb{C}}^2 + a_* u)}\right\} + 2D \exp\left\{-\frac{s^2}{2(\mu_{\mathbb{C}} + u + s/3)}\right\},$$

with the convention that the first exponential term is omitted when $\sigma_{\mathbb{C}}^2 + a_* u = 0$. In particular, if G_i is a $d_i \times m_i$ matrix with independent standard complex Gaussian entries and $A_i \succeq 0$, then

$$W_i = \operatorname{tr}(G_i^* A_i G_i)$$

is covered by taking the eigenvalues of A_i, each repeated m_i times, and the parameters are

$$\mu_{\mathbb{C}} = \sum_i m_i \operatorname{tr} A_i, \qquad \sigma_{\mathbb{C}}^2 = \sum_i m_i \operatorname{tr}(A_i^2), \qquad a_* = \max_i \|A_i\|_{\mathrm{op}}.$$

Thus complex Wishart and weighted exponential trace envelopes give explicit signed-core Matrix Spencer tails.

Proof. The weighted trace budget satisfies

$$T = \sum_i \rho_i^2 \operatorname{tr} V_i \leq \sum_i \rho_i^2 \Gamma_i \leq \sum_{i,\ell} a_{i\ell} |z_{i\ell}|^2.$$

Since $|z|^2$ is exponential with mean one,

$$\log \mathbb{E} \exp\{t(a|z|^2 - a)\} = -\log(1 - at) - at \leq \frac{t^2 a^2}{2(1 - at)}$$

for $0 \leq t < 1/a$; terms with $a = 0$ are deterministic and are omitted. Summing over all complex Gaussian coordinates gives the sub-gamma bound with parameters $\sigma_{\mathbb{C}}^2$ and a_*. Therefore

$$\mathbb{P}\{T > \mu_{\mathbb{C}} + u\} \leq \exp\left\{-\frac{u^2}{2(\sigma_{\mathbb{C}}^2 + a_* u)}\right\}.$$

Combining this with Corollary 7.235 at $v = \mu_{\mathbb{C}} + u$ proves the displayed tail. The complex-Wishart specialization follows by diagonalizing each A_i. □

Corollary 7.285 (Noncentral Gaussian and Wishart trace envelopes)**.** *Keep the signed-core, filtration, and martingale-noise hypotheses of Corollary 7.274. Assume that there are independent random variables W_i, independent of $\mathcal{G}$, such that*

$$\rho_i^2 \Gamma_i \leq W_i \qquad \text{almost surely.}$$

Suppose either that

$$W_i = \delta_i + \sum_{\ell=1}^{r_i} a_{i\ell}(g_{i\ell} + b_{i\ell})^2,$$

where $\delta_i \ge 0$, $a_{i\ell} \ge 0$, $b_{i\ell} \in \mathbb{R}$ are deterministic, and the $g_{i\ell}$ are independent standard real Gaussians, or that

$$W_i = \delta_i + \sum_{\ell=1}^{r_i} a_{i\ell} |z_{i\ell} + b_{i\ell}|^2,$$

where $\delta_i \ge 0$, $a_{i\ell} \ge 0$, $b_{i\ell} \in \mathbb{C}$ are deterministic, and the $z_{i\ell}$ are independent standard complex Gaussians normalized by $\mathbb{E}|z_{i\ell}|^2 = 1$. In the real case put

$$\mu_{\mathrm{R,nc}} = \sum_i \delta_i + \sum_{i,\ell} a_{i\ell}(1 + b_{i\ell}^2), \qquad \sigma^2_{\mathrm{R,nc}} = 2 \sum_{i,\ell} a_{i\ell}^2 (1 + 2b_{i\ell}^2), \qquad c_{\mathrm{R}} = 2a_*,$$

and in the complex case put

$$\mu_{\mathbb{C},\mathrm{nc}} = \sum_i \delta_i + \sum_{i,\ell} a_{i\ell}(1 + |b_{i\ell}|^2), \qquad \sigma^2_{\mathbb{C},\mathrm{nc}} = \sum_{i,\ell} a_{i\ell}^2 (1 + 2|b_{i\ell}|^2), \qquad c_{\mathbb{C}} = a_*,$$

where $a_* = \max_{i,\ell} a_{i\ell}$, with $a_* = 0$ if all weights vanish. Let (μ, σ^2, c) denote the corresponding real or complex triple. Then, for every $u \ge 0$ and $s > 0$,

$$\mathbb{P}\left\{\mathrm{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \le \mathbb{P}(E) + \exp\left\{-\frac{u^2}{2(\sigma^2 + cu)}\right\} + 2D \exp\left\{-\frac{s^2}{2(\mu + u + s/3)}\right\},$$

with the convention that the first exponential term is omitted when $\sigma^2 + cu = 0$.

In particular, if $G_i = M_i + Z_i$ is a real $d_i \times m_i$ Gaussian matrix with deterministic mean M_i, independent standard real Gaussian noise Z_i, and $A_i \succeq 0$, then

$$W_i = \mathrm{tr}(G_i^* A_i G_i)$$

is covered with

$$\mu_{\mathrm{R,nc}} = \sum_i \left(m_i \,\mathrm{tr}\, A_i + \mathrm{tr}(M_i^* A_i M_i)\right),$$

$$\sigma^2_{\mathrm{R,nc}} = 2 \sum_i \Big(m_i \,\mathrm{tr}(A_i^2) + 2\,\mathrm{tr}(M_i^* A_i^2 M_i)\Big), \qquad c_{\mathrm{R}} = 2 \max_i \|A_i\|_{\mathrm{op}}.$$

If $G_i = M_i + Z_i$ is instead complex Gaussian with the same normalization as above, then the same trace envelope is covered with

$$\mu_{\mathbb{C},\mathrm{nc}} = \sum_i \left(m_i \,\mathrm{tr}\, A_i + \mathrm{tr}(M_i^* A_i M_i)\right),$$

$$\sigma^2_{\mathbb{C},\mathrm{nc}} = \sum_i \Big(m_i \,\mathrm{tr}(A_i^2) + 2\,\mathrm{tr}(M_i^* A_i^2 M_i)\Big), \qquad c_{\mathbb{C}} = \max_i \|A_i\|_{\mathrm{op}}.$$

Thus nonzero-mean Gaussian directions and noncentral real or complex Wishart trace envelopes give explicit signed-core Matrix Spencer tails.

More generally, if

$$W_i = \|C_i \xi_i + h_i\|_2^2$$

with deterministic real or complex C_i, h_i and a standard real or complex Gaussian vector ξ_i, then the same conclusion holds after taking a singular-value decomposition of C_i. The weights are the nonzero squared singular values, the shifts are the coordinates of the projection of h_i to $\operatorname{Ran} C_i$ *divided by the corresponding singular values, and δ_i is the squared norm of the orthogonal residual. Hence correlated Gaussian vector, matrix-normal, and correlated noncentral Wishart trace envelopes are included after vectorizing the quadratic form.*

Proof. Let $T = \sum_i \rho_i^2 \operatorname{tr} V_i$. By the trace-envelope hypothesis,

$$T \le \sum_i \rho_i^2 \Gamma_i \le \sum_i W_i.$$

For a standard real Gaussian g, $b \in \mathbb{R}$, $a \ge 0$, and $0 \le t < 1/(2a)$,

$$\begin{aligned} &\log \mathbb{E} \exp\{t(a(g+b)^2 - a(1+b^2))\} \\ &\quad = -\frac{1}{2}\log(1-2at) - at + \frac{2a^2b^2t^2}{1-2at} \\ &\quad \le \frac{a^2(1+2b^2)t^2}{1-2at} = \frac{2a^2(1+2b^2)t^2}{2(1-2at)}. \end{aligned}$$

For a standard complex Gaussian z with $\mathbb{E}|z|^2 = 1$, $b \in \mathbb{C}$, $a \ge 0$, and $0 \le t < 1/a$,

$$\begin{aligned} &\log \mathbb{E} \exp\{t(a|z+b|^2 - a(1+|b|^2))\} \\ &\quad = -\log(1-at) - at + \frac{a^2|b|^2t^2}{1-at} \\ &\quad \le \frac{a^2(1+2|b|^2)t^2}{2(1-at)}. \end{aligned}$$

Summing over the independent coordinates gives a sub-gamma bound for $\sum_i W_i - \mu$ with variance proxy σ^2 and scale c. Therefore

$$\mathbb{P}\{T > \mu + u\} \le \exp\left\{-\frac{u^2}{2(\sigma^2 + cu)}\right\}.$$

Combining this with Corollary 7.235 at $v = \mu + u$ proves the displayed tail.

The Wishart specializations follow by diagonalizing each A_i. Orthogonal or unitary invariance preserves the law of the centered Gaussian noise, while the deterministic mean is transformed to the same eigenbasis. The displayed parameters are exactly the sums of $a(1+|b|^2)$ and of $a^2(1+2|b|^2)$, with the extra factor 2 in the real variance proxy. The final correlated Gaussian statement is the singular-value decomposition of C_i, followed by the orthogonal decomposition of h_i into the range of C_i and its orthogonal complement; the latter contributes exactly the deterministic residual δ_i. □

Corollary 7.286 (Core signings with localized matrix martingale noise)**.** *Let $D \ge 2$. Let $\mathcal{G}$ be a sigma-field, and let $B_1, \dots, B_n \in \mathrm{M}_D(\mathbb{C})$ be $\mathcal{G}$-measurable random self-adjoint contractions. Let $(\mathcal{F}_i)_{i=0}^n$ be a filtration with $\mathcal{G} \subset \mathcal{F}_0$.*

For each i, let P_i be an $\mathcal{F}_{i-1}$-measurable orthogonal projection, and let Y_i be a self-adjoint $\mathcal{F}_i$-measurable matrix satisfying

$$\mathbb{E}(Y_i \mid \mathcal{F}_{i-1}) = 0, \qquad \|Y_i\|_{\mathrm{op}} \le 1, \qquad Y_i = P_i Y_i P_i.$$

Set $X_i = B_i + Y_i$ and

$$\mu = \left\| \sum_{i=1}^{n} P_i \right\|_{\mathrm{op}}.$$

Assume that, outside an event $E \in \mathcal{G}$, there are $\mathcal{G}$-measurable signs $\varepsilon_i^\star \in \{-1, 1\}$ such that

$$\left\| \sum_{i=1}^{n} \varepsilon_i^\star B_i \right\|_{\mathrm{op}} \le K\sqrt{n}.$$

Then, for every deterministic $v \ge 0$ and $s > 0$,

$$\mathbb{P}\left\{\mathrm{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \le \mathbb{P}(E) + \mathbb{P}\{\mu > v\} + 2D \exp\left\{-\frac{s^2}{2(v + s/3)}\right\}.$$

Thus algebraic, finite-lift, Toeplitz/Hankel, and other already-signed random cores obey the same Freedman tail after online coordinate-supported, subspace-localized, or frame-localized matrix martingale-difference noise, with variation controlled by the predictable overlap $\|\sum_i P_i\|_{\mathrm{op}}$.

Proof. Since $Y_i = P_i Y_i P_i$ and $\|Y_i\|_{\mathrm{op}} \le 1$, we have $0 \le Y_i^2 \preceq P_i$. As P_i is $\mathcal{F}_{i-1}$-measurable,

$$\sum_i \mathbb{E}(Y_i^2 \mid \mathcal{F}_{i-1}) \preceq \sum_i P_i.$$

Therefore the variation parameter ν in Corollary 7.235 satisfies $\nu \le \mu$, and the displayed bound follows from that corollary. □

Corollary 7.287 (Core signings with conditionally sampled localized matrix noise)**.** *Let $D \ge 2$. Let $\mathcal{G}$ be a sigma-field, and let $B_1, \dots, B_n \in \mathrm{M}_D(\mathbb{C})$ be $\mathcal{G}$-measurable random self-adjoint contractions. Let $(\mathcal{F}_i)_{i=0}^n$ be a filtration with $\mathcal{G} \subset \mathcal{F}_0$. For each i, let P_i be an $\mathcal{F}_i$-measurable orthogonal projection, and let Y_i be a self-adjoint $\mathcal{F}_i$-measurable matrix satisfying*

$$\mathbb{E}(Y_i \mid \mathcal{F}_{i-1}) = 0, \qquad \|Y_i\|_{\mathrm{op}} \le 1, \qquad Y_i = P_i Y_i P_i.$$

Set $X_i = B_i + Y_i$ and

$$\mu = \left\| \sum_{i=1}^{n} \mathbb{E}(P_i \mid \mathcal{F}_{i-1}) \right\|_{\mathrm{op}}.$$

Assume that, outside an event $E \in \mathcal{G}$, there are $\mathcal{G}$-measurable signs $\varepsilon_i^\star \in \{-1, 1\}$ such that

$$\left\| \sum_{i=1}^{n} \varepsilon_i^\star B_i \right\|_{\mathrm{op}} \le K\sqrt{n}.$$

Then, for every deterministic $v \geq 0$ and $s > 0$,

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \leq \mathbb{P}(E) + \mathbb{P}\{\mu > v\} + 2D \exp\left\{-\frac{s^2}{2(v + s/3)}\right\}.$$

In particular, if

$$\sum_{i=1}^{n} \mathbb{E}(P_i \mid \mathcal{F}_{i-1}) \preceq pn\, I \qquad \text{almost surely,}$$

then

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \leq \mathbb{P}(E) + 2D \exp\left\{-\frac{s^2}{2(pn + s/3)}\right\}.$$

Thus a signed core remains balanced when the matrix martingale noise is supported on coordinate sets, random subspaces, or any other sampled projection model at time i, as long as the predictable conditional overlap is small.

Proof. Since $Y_i = P_i Y_i P_i$ and $\|Y_i\|_{\mathrm{op}} \leq 1$, the same compression estimate as above gives $0 \leq Y_i^2 \preceq P_i$. Hence

$$\sum_i \mathbb{E}(Y_i^2 \mid \mathcal{F}_{i-1}) \preceq \sum_i \mathbb{E}(P_i \mid \mathcal{F}_{i-1}),$$

so the variation parameter ν in Corollary 7.235 satisfies $\nu \leq \mu$. The first bound follows from that corollary. The final display is the same estimate with $v = pn$. □

Corollary 7.288 (Core signings with weighted sampled localized matrix noise)**.** *Let $D \geq 2$. Let $\mathcal{G}$ be a sigma-field, and let $B_1, \dots, B_n \in \mathrm{M}_D(\mathbb{C})$ be $\mathcal{G}$-measurable random self-adjoint contractions. Let $(\mathcal{F}_i)_{i=0}^n$ be a filtration with $\mathcal{G} \subset \mathcal{F}_0$. For each i, let P_i be an $\mathcal{F}_i$-measurable orthogonal projection and let $\rho_i \in [0, 1]$ be $\mathcal{F}_{i-1}$-measurable. Let Y_i be a self-adjoint $\mathcal{F}_i$-measurable matrix satisfying*

$$\mathbb{E}(Y_i \mid \mathcal{F}_{i-1}) = 0, \qquad \|Y_i\|_{\mathrm{op}} \leq \rho_i, \qquad Y_i = P_i Y_i P_i.$$

Set $X_i = B_i + Y_i$ and

$$\mu_\rho = \left\| \sum_{i=1}^{n} \rho_i^2\, \mathbb{E}(P_i \mid \mathcal{F}_{i-1}) \right\|_{\mathrm{op}}.$$

Assume that, outside an event $E \in \mathcal{G}$, there are $\mathcal{G}$-measurable signs $\varepsilon_i^\star \in \{-1, 1\}$ such that

$$\left\| \sum_{i=1}^{n} \varepsilon_i^\star B_i \right\|_{\mathrm{op}} \leq K\sqrt{n}.$$

Then, for every deterministic $v \geq 0$ and $s > 0$,

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \leq \mathbb{P}(E) + \mathbb{P}\{\mu_\rho > v\} + 2D \exp\left\{-\frac{s^2}{2(v + s/3)}\right\}.$$

In particular, if

$$\sum_{i=1}^{n} \rho_i^2 \, \mathbb{E}(P_i \mid \mathcal{F}_{i-1}) \preceq VI \qquad \text{almost surely}$$

for deterministic $V \geq 0$, then

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \leq \mathbb{P}(E) + 2D \exp\left\{-\frac{s^2}{2(V + s/3)}\right\}.$$

Proof. Since $Y_i = P_i Y_i P_i$ and $\|Y_i\|_{\mathrm{op}} \leq \rho_i$, we have

$$0 \leq Y_i^2 \preceq \rho_i^2 P_i.$$

Because ρ_i is $\mathcal{F}_{i-1}$-measurable,

$$\sum_i \mathbb{E}(Y_i^2 \mid \mathcal{F}_{i-1}) \preceq \sum_i \rho_i^2 \mathbb{E}(P_i \mid \mathcal{F}_{i-1}).$$

Thus the variation parameter ν in Corollary 7.235 satisfies $\nu \leq \mu_\rho$, and that corollary gives the first display. The deterministic weighted-overlap specialization follows by taking $v = V$. □

Corollary 7.289 (Signed cores with weighted sampled coordinate or Haar–subspace matrix noise)**.** *Let $D \geq 2$. Let $\mathcal{G}$, B_i, $(\mathcal{F}_i)$, P_i, ρ_i, Y_i, and $X_i = B_i + Y_i$ be as in Corollary 7.288. Assume that, outside an event $E \in \mathcal{G}$, there are $\mathcal{G}$-measurable signs $\varepsilon_i^\star \in \{-1, 1\}$ such that*

$$\left\| \sum_{i=1}^{n} \varepsilon_i^\star B_i \right\|_{\mathrm{op}} \leq K\sqrt{n}.$$

Suppose that one of the following two sampled-projection hypotheses holds.

(a) *Weighted coordinate supports. $P_i = P_{S_i}$ is the coordinate projection onto an $\mathcal{F}_i$-measurable random set $S_i \subset \{1, \dots, D\}$. There are $\mathcal{F}_{i-1}$-measurable random variables $q_{ij} \in [0, 1]$ such that*

$$\mathbb{P}\{j \in S_i \mid \mathcal{F}_{i-1}\} \leq q_{ij} \qquad \text{for every } i, j.$$

In this case put

$$W = \max_{1 \leq j \leq D} \sum_{i=1}^{n} \rho_i^2 q_{ij}.$$

(b) *Weighted Haar subspaces. There are $\mathcal{F}_{i-1}$-measurable ranks $r_i \in \{0, \dots, D\}$ such that, conditionally on $\mathcal{F}_{i-1}$, P_i is Haar-distributed*

on the Grassmannian of rank-r_i orthogonal projections. In this case put

$$W = \sum_{i=1}^{n} \rho_i^2 \frac{r_i}{D}.$$

Then, for every deterministic $v \geq 0$ and $s > 0$,

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \ldots, X_n) > K\sqrt{n} + s\right\} \leq \mathbb{P}(E) + \mathbb{P}\{W > v\} + 2D \exp\left\{-\frac{s^2}{2(v + s/3)}\right\}.$$

In particular, if $W \leq V$ almost surely for a deterministic $V \geq 0$, then for every $s > 0$,

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \ldots, X_n) > K\sqrt{n} + s\right\} \leq \mathbb{P}(E) + 2D \exp\left\{-\frac{s^2}{2(V + s/3)}\right\}.$$

Consequently, adaptive coordinate-supported or conditional Haar-subspace matrix martingale noise preserves the core signing at the usual weighted Freedman scale. The previous unweighted form is recovered by taking $\rho_i = 1$ and $q_{ij} = p_i$. If $q_{ij} \leq p_i$ deterministically and $\sum_i \rho_i^2 p_i \leq V$ almost surely, or if $\sum_i \rho_i^2 r_i / D \leq V$ in the Haar-subspace case, then the same tail holds.

Proof. It is enough to prove that the weighted predictable conditional overlap in Corollary 7.288 is at most W. In the coordinate case, $\mathbb{E}(P_i \mid \mathcal{F}_{i-1})$ is diagonal with j-th diagonal entry $\mathbb{P}\{j \in S_i \mid \mathcal{F}_{i-1}\}$, hence

$$\sum_i \rho_i^2 \mathbb{E}(P_i \mid \mathcal{F}_{i-1}) \preceq \operatorname{diag}\left(\sum_i \rho_i^2 q_{ij}\right)_{j=1}^{D} \preceq WI.$$

In the Haar-subspace case, Haar invariance gives

$$\mathbb{E}(P_i \mid \mathcal{F}_{i-1}) = \frac{r_i}{D} I,$$

and therefore the same bound is exactly the definition of W. Thus $\mu_\rho \leq W$. Apply Corollary 7.288; since $\{W \leq v\} \subseteq \{\mu_\rho \leq v\}$, its exceptional term $\mathbb{P}\{\mu_\rho > v\}$ is bounded by $\mathbb{P}\{W > v\}$. The deterministic budget case follows by taking $v = V$. □

Corollary 7.290 (Almost-sure signed-core sampled localized matrix-noise arrays)**.** *For each $m \geq 1$, let $D_m \geq 2$, let $\mathcal{G}_m$ be a sigma-field, and let $(\mathcal{F}_{m,i})_{i=0}^{n_m}$ be a filtration with $\mathcal{G}_m \subset \mathcal{F}_{m,0}$. Let*

$$X_{m,i} = B_{m,i} + Y_{m,i} \in \mathrm{M}_{D_m}(\mathbb{C}) \qquad (1 \leq i \leq n_m)$$

be a signed-core weighted sampled localized matrix-noise row as in Corollary 7.288: the $B_{m,i}$ are $\mathcal{G}_m$-measurable self-adjoint contractions, $\rho_{m,i} \in [0,1]$ is $\mathcal{F}_{m,i-1}$-measurable, $P_{m,i}$ is an $\mathcal{F}_{m,i}$-measurable orthogonal projection, and

$$\mathbb{E}(Y_{m,i} \mid \mathcal{F}_{m,i-1}) = 0, \qquad \|Y_{m,i}\|_{\mathrm{op}} \leq \rho_{m,i}, \qquad Y_{m,i} = P_{m,i} Y_{m,i} P_{m,i}.$$

Put

$$\mu_m = \left\| \sum_{i=1}^{n_m} \rho_{m,i}^2 \, \mathbb{E}(P_{m,i} \mid \mathcal{F}_{m,i-1}) \right\|_{\mathrm{op}}.$$

Assume that, outside an event $E_m \in \mathcal{G}_m$, *there are* $\mathcal{G}_m$*-measurable signs* $\varepsilon_{m,i}^\star \in \{-1,1\}$ *such that*

$$\left\| \sum_{i=1}^{n_m} \varepsilon_{m,i}^\star B_{m,i} \right\|_{\mathrm{op}} \le K_m \sqrt{n_m}.$$

Let $v_m \ge 0$ *and* $s_m > 0$ *be deterministic. If*

$$\sum_m \left[\mathbb{P}(E_m) + \mathbb{P}\{\mu_m > v_m\} + 2D_m \exp\left\{ -\frac{s_m^2}{2(v_m + s_m/3)} \right\} \right] < \infty,$$

then, with probability one, for all sufficiently large m,

$$\operatorname{disc}(X_{m,1}, \dots, X_{m,n_m}) \le K_m \sqrt{n_m} + s_m.$$

In particular, put $\ell_m = \log(eD_m)$. *Suppose that* $K_m \le K$, *that*

$$\mu_m \le Q \frac{n_m}{\ell_m}$$

outside events F_m, *that*

$$\sum_m [\mathbb{P}(E_m) + \mathbb{P}(F_m)] < \infty,$$

and that, for some $A, B > 0$,

$$\sum_m (eD_m)^{-A} < \infty, \qquad \ell_m \le B\sqrt{n_m}$$

eventually. Choose $L > 0$ *with*

$$\frac{L^2}{2(Q + LB/3)} \ge A + 1.$$

Then, almost surely, for all sufficiently large m,

$$\operatorname{disc}(X_{m,1}, \dots, X_{m,n_m}) \le (K + L)\sqrt{n_m}.$$

The same conclusions hold with μ_m *replaced by any random budget* W_m *that dominates the predictable weighted overlap. In particular, this applies to the two model classes in Corollary 7.289. For coordinate supports* $P_{m,i} = P_{S_{m,i}}$, *it is enough to have predictable bounds*

$$\mathbb{P}\{j \in S_{m,i} \mid \mathcal{F}_{m,i-1}\} \le q_{m,ij}, \qquad W_m = \max_{1 \le j \le D_m} \sum_{i=1}^{n_m} \rho_{m,i}^2 q_{m,ij}.$$

For conditional Haar subspaces of predictable rank $r_{m,i}$, *it is enough to take*

$$W_m = \sum_{i=1}^{n_m} \rho_{m,i}^2 \frac{r_{m,i}}{D_m}.$$

Thus, along all integer dimensions $D = 2, 3, \dots$, if $K_D \le K$, if either $\mu_D \le Qn_D/\log(eD)$ or the corresponding coordinate/Haar budget $W_D \le Qn_D/\log(eD)$ outside summable events, and if

$$n_D \ge a \log^2(eD)$$

eventually for some $a > 0$, then for every $A > 1$ and every L satisfying

$$\frac{L^2}{2(Q + L/(3\sqrt{a}))} \ge A + 1$$

one has

$$\operatorname{disc}(X_{D,1}, \dots, X_{D,n_D}) \le (K + L)\sqrt{n_D}$$

eventually almost surely. If $D_m \ge \exp(cm^\kappa)$ eventually for some $c, \kappa > 0$, the same conclusion holds along the sparse schedule for every $A > 0$ under the same logarithmic-square sample lower bound. The normalized contractions $X_{m,i}/2$ inherit the same almost-sure conclusions with the displayed thresholds divided by 2.

Proof. For the m-th row, write $\Delta_m = \operatorname{disc}(X_{m,1}, \dots, X_{m,n_m})$. Corollary 7.288 gives

$$\mathbb{P}\{\Delta_m > K_m\sqrt{n_m} + s_m\} \le \mathbb{P}(E_m) + \mathbb{P}\{\mu_m > v_m\} + 2D_m \exp\left\{-\frac{s_m^2}{2(v_m + s_m/3)}\right\}.$$

The first assertion is the first Borel–Cantelli lemma.

For the concrete profile, take $s_m = L\sqrt{n_m}$ and $v_m = Qn_m/\ell_m$. On F_m^c,

$$\frac{s_m^2}{2(v_m + s_m/3)} = \frac{L^2 n_m}{2(Qn_m/\ell_m + L\sqrt{n_m}/3)} \ge \frac{L^2}{2(Q + LB/3)}\,\ell_m \ge (A+1)\ell_m.$$

Therefore the exponential term is at most $2(eD_m)^{-A}$, and the assumed summability again gives Borel–Cantelli.

The coordinate and Haar-subspace budgets dominate μ_m by the proof of Corollary 7.289. The all-integer case is the preceding profile with $m = D$, $B = 1/\sqrt{a}$, and $\sum_D (eD)^{-A} < \infty$ for $A > 1$. The sparse case uses $\sum_m \exp(-Acm^\kappa) < \infty$, valid for every $A > 0$. Finally $\|X_{m,i}\|_{\mathrm{op}} \le 2$, and discrepancy is homogeneous. □

Corollary 7.291 (Exhaustive countable menus of signed-core sampled localized matrix-noise arrays)**.** *For each $m \ge 1$, let $D_m \ge 2$, put $\ell_m = \log(eD_m)$, and let*

$$J_1 \le J_2 \le \cdots, \qquad J_m \to \infty,$$

be a deterministic exhaustion of the countable menu $\mathbb{N} = \{1, 2, \dots\}$. For each m and $h \ge 1$, let $\mathcal{G}_m^{(h)}$ be a sigma-field and let $(\mathcal{F}_{m,i}^{(h)})_{i=0}^{n_m}$ be a filtration with $\mathcal{G}_m^{(h)} \subset \mathcal{F}_{m,0}^{(h)}$. Let

$$X_{m,i}^{(h)} = B_{m,i}^{(h)} + Y_{m,i}^{(h)} \in \mathrm{M}_{D_m}(\mathbb{C}) \qquad (1 \le i \le n_m)$$

be a signed-core weighted sampled localized matrix-noise row as in Corollary 7.288, with its own filtration allowed. Thus $B_{m,i}^{(h)}$ *is* $\mathcal{G}_m^{(h)}$*-measurable,* $\rho_{m,i}^{(h)} \in [0,1]$ *is* $\mathcal{F}_{m,i-1}^{(h)}$*-measurable,* $P_{m,i}^{(h)}$ *is* $\mathcal{F}_{m,i}^{(h)}$*-measurable, and*

$$\mathbb{E}(Y_{m,i}^{(h)} \mid \mathcal{F}_{m,i-1}^{(h)}) = 0, \qquad \|Y_{m,i}^{(h)}\|_{\mathrm{op}} \le \rho_{m,i}^{(h)}, \qquad Y_{m,i}^{(h)} = P_{m,i}^{(h)} Y_{m,i}^{(h)} P_{m,i}^{(h)}.$$

Put

$$\mu_{m,h} := \left\| \sum_{i=1}^{n_m} (\rho_{m,i}^{(h)})^2 \, \mathbb{E}(P_{m,i}^{(h)} \mid \mathcal{F}_{m,i-1}^{(h)}) \right\|_{\mathrm{op}}.$$

Assume that there is $K < \infty$ *such that, outside events* $E_{m,h} \in \mathcal{G}_m^{(h)}$*, there are* $\mathcal{G}_m^{(h)}$*-measurable signs* $\varepsilon_{m,i}^{(h),\star}$ *with*

$$\left\| \sum_{i=1}^{n_m} \varepsilon_{m,i}^{(h),\star} B_{m,i}^{(h)} \right\|_{\mathrm{op}} \le K\sqrt{n_m} \qquad (1 \le h \le J_m).$$

Assume also that, for some $Q, A, B_0 > 0$*,*

$$\mu_{m,h} \le Q \frac{n_m}{\ell_m} \qquad (1 \le h \le J_m)$$

outside events $F_{m,h}$*, and that*

$$\sum_m \sum_{1 \le h \le J_m} \left[\mathbb{P}(E_{m,h}) + \mathbb{P}(F_{m,h}) \right] < \infty, \qquad \sum_m J_m (eD_m)^{-A} < \infty,$$

while

$$\ell_m \le B_0 \sqrt{n_m}$$

eventually. If $L > 0$ *satisfies*

$$\frac{L^2}{2(Q + LB_0/3)} \ge A + 1,$$

then, with probability one, for all sufficiently large m*, every active prefix member* $1 \le h \le J_m$ *satisfies*

$$\mathrm{disc}(X_{m,1}^{(h)}, \dots, X_{m,n_m}^{(h)}) \le (K + L)\sqrt{n_m}.$$

Consequently, for every fixed h*, the same bound holds for all sufficiently large* m*. The normalized contractions* $X_{m,i}^{(h)}/2$ *satisfy the same statements with the discrepancy threshold divided by* 2*.*

The same conclusions hold with $\mu_{m,h}$ *replaced by any random budget* $W_{m,h}$ *that dominates the predictable weighted overlap. In particular, for active-prefix coordinate supports it is enough to have predictable bounds*

$$\mathbb{P}\{j \in S_{m,i}^{(h)} \mid \mathcal{F}_{m,i-1}^{(h)}\} \le q_{m,ij}^{(h)}, \qquad W_{m,h} = \max_{1 \le j \le D_m} \sum_{i=1}^{n_m} (\rho_{m,i}^{(h)})^2 q_{m,ij}^{(h)}.$$

For active-prefix conditional Haar subspaces of predictable ranks $r_{m,i}^{(h)}$, it is enough to take

$$W_{m,h} = \sum_{i=1}^{n_m} (\rho_{m,i}^{(h)})^2 \frac{r_{m,i}^{(h)}}{D_m}.$$

In particular, if $n_m \geq a \log^2(eD_m)$ eventually, then one may take $B_0 = a^{-1/2}$. Along all integer dimensions $D = 2, 3, \ldots$, if $J_D \leq (eD)^\Gamma$ eventually and $n_D \geq a \log^2(eD)$, then the conclusion holds for every $A > \Gamma + 1$ and every L satisfying

$$\frac{L^2}{2(Q + L/(3\sqrt{a}))} \geq A + 1.$$

Along a sparse schedule with $D_m \geq \exp(cm^\kappa)$ eventually, $J_m \leq (eD_m)^\Gamma$ eventually, and $n_m \geq a \log^2(eD_m)$, the same conclusion holds for every $A > \Gamma$ and every L satisfying the same display.

Proof. For fixed m and $h \leq J_m$, apply Corollary 7.288 with

$$v_m = Q \frac{n_m}{\ell_m}, \qquad s_m = L\sqrt{n_m}.$$

Outside $F_{m,h}$, the weighted-overlap parameter is at most v_m. Hence

$$\mathbb{P}\left\{\operatorname{disc}(X_{m,1}^{(h)}, \ldots, X_{m,n_m}^{(h)}) > (K + L)\sqrt{n_m}\right\} \leq \mathbb{P}(E_{m,h}) + \mathbb{P}(F_{m,h}) + 2D_m \exp\left\{-\frac{L^2 n_m}{2(Qn_m/\ell_m + L\sqrt{n_m}/3)}\right\}.$$

For all sufficiently large m, the exponent is at least $(A+1)\ell_m$, so the last term is bounded by $2(eD_m)^{-A}$. Summing over $1 \leq h \leq J_m$ and then over m, the hypotheses make the total failure probability finite. Borel–Cantelli gives the active-prefix conclusion, and $J_m \to \infty$ gives the fixed-h statement.

The coordinate and Haar budgets dominate $\mu_{m,h}$ by the proof of Corollary 7.289. The normalized statement is homogeneity. If $n_m \geq a \log^2(eD_m)$, then $\ell_m \leq a^{-1/2}\sqrt{n_m}$. The all-integer and sparse schedule summability checks are the same as in Corollary 7.237. □

Corollary 7.292 (Signed cores with coordinate matrix noise and random amplitudes)**.** *Let $D \geq 2$. Let $\mathcal{G}$, B_i, and the event $E \in \mathcal{G}$ be as in Corollary 7.288, and assume that outside E there are $\mathcal{G}$-measurable signs $\varepsilon_i^\star \in \{-1, 1\}$ such that*

$$\left\| \sum_{i=1}^{n} \varepsilon_i^\star B_i \right\|_{\mathrm{op}} \leq K\sqrt{n}.$$

Let $\rho_1, \ldots, \rho_n$ be independent random variables in $[0, 1]$, independent of $\mathcal{G}$, revealed in $\mathcal{F}_0$, and put

$$a_i := \mathbb{E}\rho_i^2.$$

Let $p_{ij} \in [0,1]$ be deterministic numbers, and suppose that $(\mathcal{F}_i)_{i=0}^n$ is a filtration with $\mathcal{G}, \rho_1, \dots, \rho_n \subset \mathcal{F}_0$. For each i, let $S_i \subset \{1, \dots, D\}$ be $\mathcal{F}_i$-measurable and set $P_i = P_{S_i}$. Assume

$$\mathbb{P}\{j \in S_i \mid \mathcal{F}_{i-1}\} \le p_{ij} \qquad \text{for every } i, j.$$

Let Y_i be self-adjoint and $\mathcal{F}_i$-measurable with

$$\mathbb{E}(Y_i \mid \mathcal{F}_{i-1}) = 0, \qquad \|Y_i\|_{\mathrm{op}} \le \rho_i, \qquad Y_i = P_i Y_i P_i,$$

and set $X_i = B_i + Y_i$. Define

$$\mu := \max_{1 \le j \le D} \sum_{i=1}^n p_{ij} a_i, \qquad \sigma^2 := \max_{1 \le j \le D} \sum_{i=1}^n p_{ij}^2 a_i, \qquad p_* := \max_{i,j} p_{ij}.$$

Then, for every $u \ge 0$ and $s > 0$,

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \le \mathbb{P}(E) + D \exp\left\{-\frac{u^2}{2(\sigma^2 + p_* u/3)}\right\} + 2D \exp\left\{-\frac{s^2}{2(\mu + u + s/3)}\right\},$$

with the convention that the first exponential term is omitted when $\sigma^2 + p_* u/3 = 0$. In particular, if the amplitudes are deterministic, or more generally if $\max_j \sum_i p_{ij} \rho_i^2 \le V$ almost surely, the first exceptional term may be omitted and one recovers the deterministic budget V.

Proof. Condition on the amplitudes $\rho = (\rho_1, \dots, \rho_n)$. Since ρ is independent of $\mathcal{G}$, the core exception contributes $\mathbb{P}(E)$ after averaging. The preceding corollary applies in the coordinate-support case with

$$W(\rho) = \max_{1 \le j \le D} \sum_{i=1}^n p_{ij} \rho_i^2.$$

Thus, with $v = \mu + u$,

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \le \mathbb{P}(E) + \mathbb{P}\{W > \mu + u\} + 2D \exp\left\{-\frac{s^2}{2(\mu + u + s/3)}\right\}.$$

It remains to bound the budget tail. For each fixed j, the variables $Z_{ij} := p_{ij} \rho_i^2$ are independent, satisfy

$$0 \le Z_{ij} \le p_*, \qquad \sum_i \mathbb{E} Z_{ij} \le \mu,$$

and

$$\sum_i \operatorname{Var}(Z_{ij}) \le \sum_i \mathbb{E} Z_{ij}^2 \le \sum_i p_{ij}^2 a_i \le \sigma^2.$$

Scalar Bernstein gives

$$\mathbb{P}\left\{\sum_i Z_{ij} > \mu + u\right\} \le \exp\left\{-\frac{u^2}{2(\sigma^2 + p_* u/3)}\right\}.$$

Taking a union bound over $j = 1, \dots, D$ gives the displayed estimate. The deterministic-budget specialization is just the $W \le V$ case of Corollary 7.289. □

Corollary 7.293 (Coordinate matrix noise with random profiles and amplitudes)**.** *Let $D \ge 2$. Let $\mathcal{G}$, B_i, and the event $E \in \mathcal{G}$ be as in Corollary 7.288, and assume that outside E there are $\mathcal{G}$-measurable signs $\varepsilon_i^\star \in \{-1, 1\}$ such that*

$$\left\| \sum_{i=1}^{n} \varepsilon_i^\star B_i \right\|_{\mathrm{op}} \le K\sqrt{n}.$$

Let (Q_i, ρ_i), $i = 1, \dots, n$, be independent pairs, independent of $\mathcal{G}$, where $Q_i = (Q_{i1}, \dots, Q_{iD}) \in [0,1]^D$ and $\rho_i \in [0,1]$. Let $q_* \in [0,1]$ be deterministic with $Q_{ij} \le q_*$ almost surely for every i, j. Suppose that $(\mathcal{F}_i)_{i=0}^n$ is a filtration with $\mathcal{G}, (Q_i, \rho_i)_{i=1}^n \subset \mathcal{F}_0$. For each i, let $S_i \subset \{1, \dots, D\}$ be $\mathcal{F}_i$-measurable, set $P_i = P_{S_i}$, and assume

$$\mathbb{P}\{j \in S_i \mid \mathcal{F}_{i-1}\} \le Q_{ij} \qquad \text{for every } i, j.$$

Let Y_i be self-adjoint and $\mathcal{F}_i$-measurable with

$$\mathbb{E}(Y_i \mid \mathcal{F}_{i-1}) = 0, \qquad \|Y_i\|_{\mathrm{op}} \le \rho_i, \qquad Y_i = P_i Y_i P_i,$$

and set $X_i = B_i + Y_i$. Define

$$\mu := \max_{1 \le j \le D} \sum_{i=1}^{n} \mathbb{E}(Q_{ij}\rho_i^2), \qquad \sigma^2 := \max_{1 \le j \le D} \sum_{i=1}^{n} \mathbb{E}(Q_{ij}^2 \rho_i^2).$$

Then, for every $u \ge 0$ and $s > 0$,

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \le \mathbb{P}(E) + D \exp\left\{-\frac{u^2}{2(\sigma^2 + q_* u/3)}\right\} + 2D \exp\left\{-\frac{s^2}{2(\mu + u + s/3)}\right\},$$

with the convention that the first exponential term is omitted when $\sigma^2 + q_* u/3 = 0$. In particular, if $Q_{ij} \le p$ almost surely for all i, j and $A = \sum_i \mathbb{E}\rho_i^2$, then

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \le \mathbb{P}(E) + D \exp\left\{-\frac{u^2}{2(p^2 A + pu/3)}\right\} + 2D \exp\left\{-\frac{s^2}{2(pA + u + s/3)}\right\}.$$

Proof. Condition on the random profiles and amplitudes. Since they are independent of $\mathcal{G}$, the core exception contributes $\mathbb{P}(E)$ after averaging. The coordinate case of Corollary 7.289 applies with

$$W = \max_{1 \le j \le D} \sum_{i=1}^{n} Q_{ij}\rho_i^2.$$

Thus, with $v = \mu + u$,

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \le \mathbb{P}(E) + \mathbb{P}\{W > \mu + u\} + 2D \exp\left\{-\frac{s^2}{2(\mu + u + s/3)}\right\}.$$

For each fixed coordinate j, the variables $Z_{ij} := Q_{ij}\rho_i^2$ are independent and satisfy

$$0 \le Z_{ij} \le q_*, \qquad \sum_i \mathbb{E}Z_{ij} \le \mu,$$

$$\sum_i \operatorname{Var}(Z_{ij}) \le \sum_i \mathbb{E}Z_{ij}^2 \le \sum_i \mathbb{E}(Q_{ij}^2\rho_i^2) \le \sigma^2.$$

Scalar Bernstein gives

$$\mathbb{P}\left\{\sum_i Z_{ij} > \mu + u\right\} \le \exp\left\{-\frac{u^2}{2(\sigma^2 + q_* u/3)}\right\}.$$

Union over $j = 1, \dots, D$. The homogeneous-profile display follows by choosing the jump bound $q_* = p$, and using $\mu \le pA$ and $\sigma^2 \le p^2 A$. □

Corollary 7.294 (Block-coordinate matrix noise with random profiles and amplitudes)**.** *Let $D \ge 2$. Let $\Pi_1, \dots, \Pi_m$ be nonzero pairwise orthogonal projections on $\mathbb{C}^D$ with $\sum_{\ell=1}^m \Pi_\ell = I$. Let $\mathcal{G}$, B_i, and the event $E \in \mathcal{G}$ be as in Corollary 7.288, and assume that outside E there are $\mathcal{G}$-measurable signs $\varepsilon_i^\star \in \{-1, 1\}$ such that*

$$\left\|\sum_{i=1}^n \varepsilon_i^\star B_i\right\|_{\mathrm{op}} \le K\sqrt{n}.$$

Let (Q_i, ρ_i), $i = 1, \dots, n$, be independent pairs, independent of $\mathcal{G}$, where $Q_i = (Q_{i1}, \dots, Q_{im}) \in [0,1]^m$ and $\rho_i \in [0,1]$. Let $q_* \in [0,1]$ be deterministic with $Q_{i\ell} \le q_*$ almost surely for every i, ℓ. Suppose that $(\mathcal{F}_i)_{i=0}^n$ is a filtration with $\mathcal{G}, (Q_i, \rho_i)_{i=1}^n \subset \mathcal{F}_0$. For each i, let $T_i \subset \{1, \dots, m\}$ be $\mathcal{F}_i$-measurable and put

$$P_i = \sum_{\ell \in T_i} \Pi_\ell.$$

Assume

$$\mathbb{P}\{\ell \in T_i \mid \mathcal{F}_{i-1}\} \le Q_{i\ell} \qquad \text{for every } i, \ell.$$

Let Y_i be self-adjoint and $\mathcal{F}_i$-measurable with

$$\mathbb{E}(Y_i \mid \mathcal{F}_{i-1}) = 0, \qquad \|Y_i\|_{\mathrm{op}} \le \rho_i, \qquad Y_i = P_i Y_i P_i,$$

and set $X_i = B_i + Y_i$. Define

$$\mu := \max_{1 \le \ell \le m} \sum_{i=1}^n \mathbb{E}(Q_{i\ell}\rho_i^2), \qquad \sigma^2 := \max_{1 \le \ell \le m} \sum_{i=1}^n \mathbb{E}(Q_{i\ell}^2\rho_i^2).$$

Then, for every $u \geq 0$ and $s > 0$,

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \leq \mathbb{P}(E) + m \exp\left\{-\frac{u^2}{2(\sigma^2 + q_* u/3)}\right\} + 2D \exp\left\{-\frac{s^2}{2(\mu + u + s/3)}\right\},$$

with the convention that the first exponential term is omitted when $\sigma^2 + q_* u/3 = 0$. In particular, if $Q_{i\ell} \leq p$ almost surely for all i, ℓ and $A = \sum_i \mathbb{E}\rho_i^2$, then

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \leq \mathbb{P}(E) + m \exp\left\{-\frac{u^2}{2(p^2 A + pu/3)}\right\} + 2D \exp\left\{-\frac{s^2}{2(pA + u + s/3)}\right\}.$$

Proof. Set

$$W = \max_{1 \leq \ell \leq m} \sum_{i=1}^{n} Q_{i\ell} \rho_i^2.$$

Since the Π_ℓ are orthogonal projections,

$$\mathbb{E}(P_i \mid \mathcal{F}_{i-1}) = \sum_{\ell=1}^{m} \mathbb{P}\{\ell \in T_i \mid \mathcal{F}_{i-1}\} \Pi_\ell \preceq \sum_{\ell=1}^{m} Q_{i\ell} \Pi_\ell.$$

Therefore

$$\sum_i \rho_i^2 \mathbb{E}(P_i \mid \mathcal{F}_{i-1}) \preceq \sum_{\ell=1}^{m} \left(\sum_i Q_{i\ell} \rho_i^2 \right) \Pi_\ell \preceq W I.$$

Corollary 7.288 gives, with $v = \mu + u$,

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \leq \mathbb{P}(E) + \mathbb{P}\{W > \mu + u\} + 2D \exp\left\{-\frac{s^2}{2(\mu + u + s/3)}\right\}.$$

For fixed ℓ, the variables $Z_{i\ell} := Q_{i\ell} \rho_i^2$ are independent and satisfy

$$0 \leq Z_{i\ell} \leq q_*, \qquad \sum_i \mathbb{E} Z_{i\ell} \leq \mu,$$

$$\sum_i \operatorname{Var}(Z_{i\ell}) \leq \sum_i \mathbb{E} Z_{i\ell}^2 \leq \sum_i \mathbb{E}(Q_{i\ell}^2 \rho_i^2) \leq \sigma^2.$$

Scalar Bernstein and a union bound over the m blocks give the displayed estimate. The homogeneous display follows from $\mu \leq pA$, $\sigma^2 \leq p^2 A$, and $q_* \leq p$. □

Corollary 7.295 (Homogeneous coordinate matrix noise with random amplitudes)**.** *In the setting of Corollary 7.292, suppose in addition that*

$p_{ij} \le p$ for all i, j, and put

$$A := \sum_{i=1}^{n} \mathbb{E}\rho_i^2.$$

Then, for every $u \ge 0$ and $s > 0$,

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \le \mathbb{P}(E) + D \exp\left\{-\frac{u^2}{2(p^2 A + pu/3)}\right\} + 2D \exp\left\{-\frac{s^2}{2(pA + u + s/3)}\right\}.$$

with the same convention that the first exponential term is omitted when $p^2 A + pu/3 = 0$. Thus, if the average amplitude energy is small, then the coordinate sampling budget is governed by pA rather than by pn. In particular, taking $u = pA$ gives the simpler bound

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \le \mathbb{P}(E) + D \exp\{-3A/8\} + 2D \exp\left\{-\frac{s^2}{2(2pA + s/3)}\right\}.$$

Proof. In Corollary 7.292 we have

$$\mu \le pA, \qquad \sigma^2 \le p^2 A, \qquad p_* \le p.$$

Substitute these bounds in the displayed estimate there. The last display is the specialization $u = pA$; when $pA = 0$, the first budget-exception term can be omitted, and the displayed $D \exp\{-3A/8\}$ term is only a harmless weaker upper bound. □

Corollary 7.296 (Haar-subspace matrix noise with random amplitudes)**.** *Let $D \ge 2$. Let $\mathcal{G}$, B_i, and the event $E \in \mathcal{G}$ be as in Corollary 7.288, and assume that outside E there are $\mathcal{G}$-measurable signs $\varepsilon_i^\star \in \{-1, 1\}$ such that*

$$\left\| \sum_{i=1}^{n} \varepsilon_i^\star B_i \right\|_{\mathrm{op}} \le K\sqrt{n}.$$

Let $\rho_1, \dots, \rho_n$ be independent random variables in $[0, 1]$, independent of $\mathcal{G}$, revealed in $\mathcal{F}_0$, and put

$$a_i := \mathbb{E}\rho_i^2.$$

Let $r_i \in \{0, \dots, D\}$ be deterministic ranks and write $\alpha_i = r_i/D$. Suppose that $(\mathcal{F}_i)_{i=0}^n$ is a filtration with $\mathcal{G}, \rho_1, \dots, \rho_n \subset \mathcal{F}_0$. For each i, let P_i be $\mathcal{F}_i$-measurable and, conditionally on $\mathcal{F}_{i-1}$, Haar-distributed on the Grassmannian of rank-r_i orthogonal projections. Let Y_i be self-adjoint and $\mathcal{F}_i$-measurable with

$$\mathbb{E}(Y_i \mid \mathcal{F}_{i-1}) = 0, \qquad \|Y_i\|_{\mathrm{op}} \le \rho_i, \qquad Y_i = P_i Y_i P_i,$$

and set $X_i = B_i + Y_i$. Define

$$\mu := \sum_{i=1}^{n} \alpha_i a_i, \qquad \sigma^2 := \sum_{i=1}^{n} \alpha_i^2 a_i, \qquad \alpha_* := \max_i \alpha_i.$$

Then, for every $u \geq 0$ and $s > 0$,

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \ldots, X_n) > K\sqrt{n} + s\right\} \leq \mathbb{P}(E) + \exp\left\{-\frac{u^2}{2(\sigma^2 + \alpha_* u/3)}\right\} + 2D \exp\left\{-\frac{s^2}{2(\mu + u + s/3)}\right\},$$

with the convention that the first exponential term is omitted when $\sigma^2 + \alpha_* u/3 = 0$. In particular, if $\alpha_i \leq \alpha$ for all i and $A = \sum_i \mathbb{E}\rho_i^2$, then

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \ldots, X_n) > K\sqrt{n} + s\right\} \leq \mathbb{P}(E) + \exp\left\{-\frac{u^2}{2(\alpha^2 A + \alpha u/3)}\right\} + 2D \exp\left\{-\frac{s^2}{2(\alpha A + u + s/3)}\right\}.$$

Taking $u = \alpha A$ gives the simpler bound

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \ldots, X_n) > K\sqrt{n} + s\right\} \leq \mathbb{P}(E) + \exp\{-3A/8\} + 2D \exp\left\{-\frac{s^2}{2(2\alpha A + s/3)}\right\}.$$

Proof. Condition on the amplitudes $\rho = (\rho_1, \ldots, \rho_n)$. The Haar-subspace case of Corollary 7.289 applies with

$$W(\rho) = \sum_{i=1}^{n} \alpha_i \rho_i^2.$$

Thus, with $v = \mu + u$,

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \ldots, X_n) > K\sqrt{n} + s\right\} \leq \mathbb{P}(E) + \mathbb{P}\{W > \mu + u\} + 2D \exp\left\{-\frac{s^2}{2(\mu + u + s/3)}\right\}.$$

The variables $Z_i := \alpha_i \rho_i^2$ are independent and satisfy

$$0 \leq Z_i \leq \alpha_*, \qquad \sum_i \mathbb{E}Z_i = \mu, \qquad \sum_i \operatorname{Var}(Z_i) \leq \sum_i \mathbb{E}Z_i^2 \leq \sigma^2.$$

Scalar Bernstein gives

$$\mathbb{P}\{W > \mu + u\} \leq \exp\left\{-\frac{u^2}{2(\sigma^2 + \alpha_* u/3)}\right\}.$$

This proves the first display. The homogeneous-rank display follows from $\mu \leq \alpha A$, $\sigma^2 \leq \alpha^2 A$, and $\alpha_* \leq \alpha$; the last display is the specialization $u = \alpha A$, with the same harmless zero-denominator convention as above. □

Corollary 7.297 (Random-rank Haar-subspace matrix noise with random amplitudes)**.** *Let $D \geq 2$. Let $\mathcal{G}$, B_i, and the event $E \in \mathcal{G}$ be as in*

Corollary 7.288, and assume that outside E there are $\mathcal{G}$-measurable signs $\varepsilon_i^\star \in \{-1,1\}$ such that

$$\left\| \sum_{i=1}^n \varepsilon_i^\star B_i \right\|_{\mathrm{op}} \le K\sqrt{n}.$$

Let (R_i,ρ_i), $i=1,\dots,n$, be independent pairs, independent of $\mathcal{G}$, with $R_i \in \{0,\dots,D\}$ and $\rho_i \in [0,1]$. Put $\alpha_i = R_i/D$, and let $\alpha_* \in [0,1]$ be deterministic with $\alpha_i \le \alpha_*$ almost surely for every i. Suppose that $(\mathcal{F}_i)_{i=0}^n$ is a filtration with $\mathcal{G}, (R_i,\rho_i)_{i=1}^n \subset \mathcal{F}_0$. For each i, let P_i be $\mathcal{F}_i$-measurable and, conditionally on $\mathcal{F}_{i-1}$, Haar-distributed on the Grassmannian of rank-R_i orthogonal projections. Let Y_i be self-adjoint and $\mathcal{F}_i$-measurable with

$$\mathbb{E}(Y_i \mid \mathcal{F}_{i-1}) = 0, \qquad \|Y_i\|_{\mathrm{op}} \le \rho_i, \qquad Y_i = P_i Y_i P_i,$$

and set $X_i = B_i + Y_i$. Define

$$\mu := \sum_{i=1}^n \mathbb{E}(\alpha_i \rho_i^2), \qquad \sigma^2 := \sum_{i=1}^n \mathbb{E}(\alpha_i^2 \rho_i^2).$$

Then, for every $u \ge 0$ and $s > 0$,

$$\mathbb{P}\left\{\mathrm{disc}(X_1,\dots,X_n) > K\sqrt{n} + s\right\} \le \mathbb{P}(E) + \exp\left\{-\frac{u^2}{2(\sigma^2 + \alpha_* u/3)}\right\} + 2D\exp\left\{-\frac{s^2}{2(\mu + u + s/3)}\right\},$$

with the convention that the first exponential term is omitted when $\sigma^2 + \alpha_* u/3 = 0$. In particular, if $\alpha_i \le \alpha$ almost surely for all i and $A = \sum_i \mathbb{E}\rho_i^2$, then

$$\mathbb{P}\left\{\mathrm{disc}(X_1,\dots,X_n) > K\sqrt{n} + s\right\} \le \mathbb{P}(E) + \exp\left\{-\frac{u^2}{2(\alpha^2 A + \alpha u/3)}\right\} + 2D\exp\left\{-\frac{s^2}{2(\alpha A + u + s/3)}\right\}.$$

Proof. Condition on the random ranks and amplitudes. The Haar-subspace case of Corollary 7.289 applies with

$$W = \sum_{i=1}^n \alpha_i \rho_i^2.$$

Therefore, with $v = \mu + u$,

$$\mathbb{P}\left\{\mathrm{disc}(X_1,\dots,X_n) > K\sqrt{n} + s\right\} \le \mathbb{P}(E) + \mathbb{P}\{W > \mu + u\} + 2D\exp\left\{-\frac{s^2}{2(\mu + u + s/3)}\right\}.$$

The variables $Z_i := \alpha_i \rho_i^2$ are independent and satisfy

$$0 \le Z_i \le \alpha_*, \qquad \sum_i \mathbb{E} Z_i = \mu,$$
$$\sum_i \operatorname{Var}(Z_i) \le \sum_i \mathbb{E} Z_i^2 \le \sum_i \mathbb{E}(\alpha_i^2 \rho_i^2) = \sigma^2.$$

Scalar Bernstein gives the displayed budget-exception bound. The final display follows from $\mu \le \alpha A$, $\sigma^2 \le \alpha^2 A$, and $\alpha_* \le \alpha$. □

Corollary 7.298 (Isotropic projection matrix noise with random amplitudes)**.** *Let $D \ge 2$. Let $\mathcal{G}$, B_i, and the event $E \in \mathcal{G}$ be as in Corollary 7.288, and assume that outside E there are $\mathcal{G}$-measurable signs $\varepsilon_i^\star \in \{-1, 1\}$ such that*

$$\left\| \sum_{i=1}^n \varepsilon_i^\star B_i \right\|_{\mathrm{op}} \le K\sqrt{n}.$$

Let (α_i, ρ_i), $i = 1, \dots, n$, be independent pairs, independent of $\mathcal{G}$, with $\alpha_i, \rho_i \in [0, 1]$. Let $\alpha_* \in [0, 1]$ be deterministic with $\alpha_i \le \alpha_*$ almost surely for every i. Suppose that $(\mathcal{F}_i)_{i=0}^n$ is a filtration with $\mathcal{G}, (\alpha_i, \rho_i)_{i=1}^n \subset \mathcal{F}_0$. For each i, let P_i be an $\mathcal{F}_i$-measurable orthogonal projection such that

$$\mathbb{E}(P_i \mid \mathcal{F}_{i-1}) = \alpha_i I.$$

Let Y_i be self-adjoint and $\mathcal{F}_i$-measurable with

$$\mathbb{E}(Y_i \mid \mathcal{F}_{i-1}) = 0, \qquad \|Y_i\|_{\mathrm{op}} \le \rho_i, \qquad Y_i = P_i Y_i P_i,$$

and set $X_i = B_i + Y_i$. Define

$$\mu := \sum_{i=1}^n \mathbb{E}(\alpha_i \rho_i^2), \qquad \sigma^2 := \sum_{i=1}^n \mathbb{E}(\alpha_i^2 \rho_i^2).$$

Then, for every $u \ge 0$ and $s > 0$,

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \le \mathbb{P}(E) + \exp\left\{-\frac{u^2}{2(\sigma^2 + \alpha_* u/3)}\right\} + 2D \exp\left\{-\frac{s^2}{2(\mu + u + s/3)}\right\},$$

with the convention that the first exponential term is omitted when $\sigma^2 + \alpha_* u/3 = 0$. In particular, if $\alpha_i \le \alpha$ almost surely for all i and $A = \sum_i \mathbb{E}\rho_i^2$, then

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \le \mathbb{P}(E) + \exp\left\{-\frac{u^2}{2(\alpha^2 A + \alpha u/3)}\right\} + 2D \exp\left\{-\frac{s^2}{2(\alpha A + u + s/3)}\right\}.$$

Proof. The conditional isotropy assumption gives

$$\sum_i \rho_i^2 \mathbb{E}(P_i \mid \mathcal{F}_{i-1}) = \left(\sum_i \alpha_i \rho_i^2\right) I.$$

Therefore Corollary 7.288 applies with

$$W = \sum_{i=1}^n \alpha_i \rho_i^2.$$

Taking $v = \mu + u$ reduces the matter to a scalar budget tail for W. The variables $Z_i := \alpha_i \rho_i^2$ are independent and satisfy

$$0 \le Z_i \le \alpha_*, \qquad \sum_i \mathbb{E} Z_i = \mu,$$

$$\sum_i \operatorname{Var}(Z_i) \le \sum_i \mathbb{E} Z_i^2 \le \sum_i \mathbb{E}(\alpha_i^2 \rho_i^2) = \sigma^2.$$

Scalar Bernstein gives the first display. The homogeneous display follows from $\mu \le \alpha A$, $\sigma^2 \le \alpha^2 A$, and $\alpha_* \le \alpha$. □

Corollary 7.299 (Spectrally dominated projection matrix noise with random amplitudes)**.** *Let $D \ge 2$. Let $\mathcal{G}$, B_i, and the event $E \in \mathcal{G}$ be as in Corollary 7.288, and assume that outside E there are $\mathcal{G}$-measurable signs $\varepsilon_i^\star \in \{-1, 1\}$ such that*

$$\left\| \sum_{i=1}^n \varepsilon_i^\star B_i \right\|_{\mathrm{op}} \le K\sqrt{n}.$$

Let (α_i, ρ_i), $i = 1, \dots, n$, be independent pairs, independent of $\mathcal{G}$, with $\alpha_i, \rho_i \in [0, 1]$. Let $\alpha_* \in [0, 1]$ be deterministic with $\alpha_i \le \alpha_*$ almost surely for every i. Suppose that $(\mathcal{F}_i)_{i=0}^n$ is a filtration with $\mathcal{G}, (\alpha_i, \rho_i)_{i=1}^n \subset \mathcal{F}_0$. For each i, let P_i be an $\mathcal{F}_i$-measurable orthogonal projection such that

$$\mathbb{E}(P_i \mid \mathcal{F}_{i-1}) \preceq \alpha_i I.$$

Let Y_i be self-adjoint and $\mathcal{F}_i$-measurable with

$$\mathbb{E}(Y_i \mid \mathcal{F}_{i-1}) = 0, \qquad \|Y_i\|_{\mathrm{op}} \le \rho_i, \qquad Y_i = P_i Y_i P_i,$$

and set $X_i = B_i + Y_i$. Define

$$\mu := \sum_{i=1}^n \mathbb{E}(\alpha_i \rho_i^2), \qquad \sigma^2 := \sum_{i=1}^n \mathbb{E}(\alpha_i^2 \rho_i^2).$$

Then, for every $u \ge 0$ and $s > 0$,

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \le \mathbb{P}(E) + \exp\left\{-\frac{u^2}{2(\sigma^2 + \alpha_* u/3)}\right\} + 2D \exp\left\{-\frac{s^2}{2(\mu + u + s/3)}\right\},$$

with the convention that the first exponential term is omitted when $\sigma^2 + \alpha_* u/3 = 0$. *In particular, if* $\alpha_i \le \alpha$ *almost surely for all* i *and* $A = \sum_i \mathbb{E}\rho_i^2$, *then*

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \le \mathbb{P}(E) + \exp\left\{-\frac{u^2}{2(\alpha^2 A + \alpha u/3)}\right\} + 2D \exp\left\{-\frac{s^2}{2(\alpha A + u + s/3)}\right\}.$$

Proof. The domination assumption gives

$$\sum_i \rho_i^2 \mathbb{E}(P_i \mid \mathcal{F}_{i-1}) \preceq \left(\sum_i \alpha_i \rho_i^2\right) I.$$

Thus Corollary 7.288 applies with $W = \sum_i \alpha_i \rho_i^2$. Taking $v = \mu + u$, it remains only to bound $\mathbb{P}\{W > \mu + u\}$. The variables $Z_i := \alpha_i \rho_i^2$ are independent and satisfy

$$0 \le Z_i \le \alpha_*, \qquad \sum_i \mathbb{E}Z_i = \mu,$$

$$\sum_i \operatorname{Var}(Z_i) \le \sum_i \mathbb{E}Z_i^2 \le \sum_i \mathbb{E}(\alpha_i^2 \rho_i^2) = \sigma^2.$$

Scalar Bernstein gives the first display. The homogeneous display follows from $\mu \le \alpha A$, $\sigma^2 \le \alpha^2 A$, and $\alpha_* \le \alpha$. □

Corollary 7.300 (Mixed coordinate or Haar-subspace matrix noise with random amplitudes)**.** *Let* $D \ge 2$. *Let* $\mathcal{G}$, B_i, *and the event* $E \in \mathcal{G}$ *be as in Corollary 7.288, and assume that outside* E *there are* $\mathcal{G}$*-measurable signs* $\varepsilon_i^\star \in \{-1, 1\}$ *such that*

$$\left\| \sum_{i=1}^n \varepsilon_i^\star B_i \right\|_{\mathrm{op}} \le K\sqrt{n}.$$

Let (M_i, Q_i, R_i, ρ_i), $i = 1, \dots, n$, *be independent tuples, independent of* $\mathcal{G}$, *where*

$$M_i \in \{0, 1\}, \qquad R_i \in \{0, \dots, D\}, \qquad \rho_i \in [0, 1],$$

$$Q_i = (Q_{i1}, \dots, Q_{iD}) \in [0, 1]^D.$$

Here $M_i = 1$ *denotes the coordinate-support branch and* $M_i = 0$ *denotes the Haar-subspace branch. Put*

$$\beta_{ij} := M_i Q_{ij} + (1 - M_i)\frac{R_i}{D},$$

and let $b_* \in [0, 1]$ *be deterministic with* $\beta_{ij} \le b_*$ *almost surely for every* i, j. *Suppose that* $(\mathcal{F}_i)_{i=0}^n$ *is a filtration with* $\mathcal{G}, (M_i, Q_i, R_i, \rho_i)_{i=1}^n \subset \mathcal{F}_0$. *For each* i, *let* P_i *be an* $\mathcal{F}_i$*-measurable orthogonal projection. On the event* $\{M_i = 1\}$, *assume that there is an* $\mathcal{F}_i$*-measurable set* $S_i \subset \{1, \dots, D\}$ *such that* $P_i = P_{S_i}$ *and*

$$\mathbb{P}\{j \in S_i \mid \mathcal{F}_{i-1}\} \le Q_{ij} \qquad \text{for every } j.$$

On the event $\{M_i = 0\}$*, assume that, conditionally on* $\mathcal{F}_{i-1}$*,* P_i *is Haar-distributed on the Grassmannian of rank-*R_i *orthogonal projections. Let* Y_i *be self-adjoint and* $\mathcal{F}_i$*-measurable with*

$$\mathbb{E}(Y_i \mid \mathcal{F}_{i-1}) = 0, \qquad \|Y_i\|_{\mathrm{op}} \le \rho_i, \qquad Y_i = P_i Y_i P_i,$$

and set $X_i = B_i + Y_i$*. Define*

$$\mu := \max_{1\le j\le D} \sum_{i=1}^{n} \mathbb{E}(\beta_{ij}\rho_i^2), \qquad \sigma^2 := \max_{1\le j\le D} \sum_{i=1}^{n} \mathbb{E}(\beta_{ij}^2\rho_i^2).$$

Then, for every $u \ge 0$ *and* $s > 0$*,*

$$\mathbb{P}\left\{\mathrm{disc}(X_1,\dots,X_n) > K\sqrt{n} + s\right\} \le \mathbb{P}(E) + D\exp\left\{-\frac{u^2}{2(\sigma^2 + b_* u/3)}\right\} + 2D\exp\left\{-\frac{s^2}{2(\mu + u + s/3)}\right\},$$

with the convention that the first exponential term is omitted when $\sigma^2 + b_* u/3 = 0$*. In particular, if* $\beta_{ij} \le b$ *almost surely for all* i, j *and* $A = \sum_i \mathbb{E}\rho_i^2$*, then*

$$\mathbb{P}\left\{\mathrm{disc}(X_1,\dots,X_n) > K\sqrt{n} + s\right\} \le \mathbb{P}(E) + D\exp\left\{-\frac{u^2}{2(b^2 A + bu/3)}\right\} + 2D\exp\left\{-\frac{s^2}{2(bA + u + s/3)}\right\}.$$

Proof. Set

$$W = \max_{1\le j\le D} \sum_{i=1}^{n} \beta_{ij}\rho_i^2.$$

We first show that W controls the weighted conditional overlap in Corollary 7.288. Because (M_i, Q_i, R_i, ρ_i) is $\mathcal{F}_{i-1}$-measurable, the coordinate branch gives

$$\mathbb{E}(P_i \mid \mathcal{F}_{i-1}) \preceq \mathrm{diag}(Q_{ij})_{j=1}^{D} \qquad \text{on } \{M_i = 1\},$$

while Haar invariance gives

$$\mathbb{E}(P_i \mid \mathcal{F}_{i-1}) = \frac{R_i}{D} I \qquad \text{on } \{M_i = 0\}.$$

Thus, pointwise,

$$\mathbb{E}(P_i \mid \mathcal{F}_{i-1}) \preceq \mathrm{diag}(\beta_{ij})_{j=1}^{D}.$$

Consequently

$$\sum_i \rho_i^2 \mathbb{E}(P_i \mid \mathcal{F}_{i-1}) \preceq \mathrm{diag}\left(\sum_i \beta_{ij}\rho_i^2\right)_{j=1}^{D} \preceq W I.$$

Corollary 7.288 therefore applies with $v = \mu + u$ and gives

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \le \mathbb{P}(E) + \mathbb{P}\{W > \mu + u\} + 2D \exp\left\{-\frac{s^2}{2(\mu + u + s/3)}\right\}.$$

It remains to control the random budget. For fixed j, the variables $Z_{ij} := \beta_{ij}\rho_i^2$ are independent and satisfy

$$0 \le Z_{ij} \le b_*, \qquad \sum_i \mathbb{E}Z_{ij} \le \mu,$$

$$\sum_i \operatorname{Var}(Z_{ij}) \le \sum_i \mathbb{E}Z_{ij}^2 \le \sum_i \mathbb{E}(\beta_{ij}^2\rho_i^2) \le \sigma^2.$$

Scalar Bernstein gives

$$\mathbb{P}\left\{\sum_i Z_{ij} > \mu + u\right\} \le \exp\left\{-\frac{u^2}{2(\sigma^2 + b_* u/3)}\right\}.$$

Union over $j = 1, \dots, D$. The homogeneous display follows from $\mu \le bA$, $\sigma^2 \le b^2 A$, and $b_* \le b$. □

Corollary 7.301 (Almost-sure random-amplitude localized matrix-noise menus)**.** *For each $m \ge 1$, let $D_m \ge 2$, and let*

$$X_{m,i} = B_{m,i} + Y_{m,i} \in \mathrm{M}_{D_m}(\mathbb{C}), \qquad 1 \le i \le n_m,$$

be a signed-core localized matrix-noise row with independent random profile or amplitude data as in one of Corollaries 7.292, 7.293, 7.294, 7.296, 7.297, 7.298, 7.299, or 7.300. Thus, outside an event E_m, the core matrices have $\mathcal{G}_m$-measurable signs with

$$\left\|\sum_{i=1}^{n_m} \varepsilon_{m,i}^\star B_{m,i}\right\|_{\mathrm{op}} \le K_m\sqrt{n_m}.$$

Let $N_m \ge 1$, $\mu_m \ge 0$, $\sigma_m^2 \ge 0$, and $b_m \in [0,1]$ be the budget-union count, mean budget, Bernstein variance proxy, and jump parameter in the corresponding one-row corollary. Concretely, $N_m = D_m$ for coordinate-profile and mixed coordinate/Haar menus, N_m is the number of structural blocks in the block-profile menu, and $N_m = 1$ for Haar-rank, isotropic, or spectrally dominated projection menus.

Let $u_m \ge 0$ and $s_m > 0$ be deterministic. If

$$\sum_m \left[\mathbb{P}(E_m) + N_m \exp\left\{-\frac{u_m^2}{2(\sigma_m^2 + b_m u_m/3)}\right\} + 2D_m \exp\left\{-\frac{s_m^2}{2(\mu_m + u_m + s_m/3)}\right\}\right] < \infty,$$

with the usual convention that the Bernstein term is omitted when its denominator is zero, then, with probability one, for all sufficiently large m,

$$\operatorname{disc}(X_{m,1}, \dots, X_{m,n_m}) \le K_m\sqrt{n_m} + s_m.$$

In particular, put $\ell_m = \log(eD_m)$. *Suppose that* $K_m \le K$, *that*

$$\mu_m + u_m \le Q\frac{n_m}{\ell_m}$$

eventually, and that, for some $A, B > 0$,

$$\sum_m \left[\mathbb{P}(E_m) + N_m \exp\left\{-\frac{u_m^2}{2(\sigma_m^2 + b_m u_m/3)}\right\} + (eD_m)^{-A}\right] < \infty, \qquad \ell_m \le B\sqrt{n_m}$$

eventually. Choose $L > 0$ *with*

$$\frac{L^2}{2(Q + LB/3)} \ge A + 1.$$

Then, almost surely, for all sufficiently large m,

$$\operatorname{disc}(X_{m,1}, \dots, X_{m,n_m}) \le (K + L)\sqrt{n_m}.$$

A fully explicit half-budget schedule is obtained as follows. Assume $\sum_m \mathbb{P}(E_m) < \infty$ *and*

$$\mu_m \le \frac{Qn_m}{2\ell_m}, \qquad \sigma_m^2 \le S\frac{n_m}{\ell_m}, \qquad b_m \le b, \qquad N_m \le (eD_m)^\Gamma$$

eventually, and set

$$u_m = \frac{Qn_m}{2\ell_m}.$$

If $n_m \ge a\ell_m^2$ *eventually, then the random-budget Bernstein exception is bounded by*

$$(eD_m)^{\Gamma - \delta}, \qquad \delta = \frac{Q^2 a}{8S + \frac{4}{3}bQ}.$$

Consequently, along all integer dimensions $D = 2, 3, \dots$, *if* $\delta - \Gamma > 1$, *then for every* $A > 1$ *and every* L *satisfying*

$$\frac{L^2}{2(Q + L/(3\sqrt{a}))} \ge A + 1$$

one has

$$\operatorname{disc}(X_{D,1}, \dots, X_{D,n_D}) \le (K + L)\sqrt{n_D}$$

eventually almost surely. Along any sparse schedule $D_m \ge \exp(cm^\kappa)$, *the same conclusion holds whenever* $\delta - \Gamma > 0$, *for every* $A > 0$ *and every* L *satisfying the same display. These criteria apply directly to homogeneous coordinate noise* $(\mu, \sigma^2, b) = (pA, p^2A, p)$, *Haar or dominated projection noise* $(\mu, \sigma^2, b) = (\alpha A, \alpha^2 A, \alpha)$, *block-profile menus with the block count in* N_m, *and the mixed coordinate/Haar menu with* $(\mu, \sigma^2, b) = (bA, b^2A, b)$ *after the corresponding homogeneous bounds.*

Proof. Each listed one-row corollary has the common form

$$\mathbb{P}\left\{\operatorname{disc}(X_{m,1},\dots,X_{m,n_m}) > K_m\sqrt{n_m}+s_m\right\} \le \mathbb{P}(E_m) + N_m \exp\left\{-\frac{u_m^2}{2(\sigma_m^2+b_m u_m/3)}\right\} + 2D_m \exp\left\{-\frac{s_m^2}{2(\mu_m+u_m+s_m/3)}\right\}.$$

The first assertion is the first Borel–Cantelli lemma.

For the profile form, take $s_m = L\sqrt{n_m}$. On the event $\mu_m + u_m \le Qn_m/\ell_m$,

$$\frac{s_m^2}{2(\mu_m+u_m+s_m/3)} \ge \frac{L^2 n_m}{2(Qn_m/\ell_m + L\sqrt{n_m}/3)} \ge \frac{L^2}{2(Q+LB/3)}\,\ell_m \ge (A+1)\ell_m.$$

Thus the matrix-Freedman term is at most $2(eD_m)^{-A}$, and the stated summability gives the almost-sure conclusion.

For the explicit half-budget schedule, the choice $u_m = Qn_m/(2\ell_m)$ gives

$$\frac{u_m^2}{2(\sigma_m^2+b_m u_m/3)} \ge \frac{Q^2 n_m}{(8S+\frac{4}{3}bQ)\ell_m} \ge \delta\,\ell_m.$$

Hence the budget exception is at most $(eD_m)^{\Gamma} e^{-\delta\ell_m} = (eD_m)^{\Gamma-\delta}$. The all-integer and sparse summability conditions follow from the usual p-series and stretched-exponential series tests. The final examples are obtained by substituting the homogeneous parameter bounds from the corresponding one-row corollaries. □

Corollary 7.302 (Exhaustive countable random-amplitude localized matrix-noise menus)**.** *For each $m \ge 1$, let $D_m \ge 2$, put $\ell_m = \log(eD_m)$, and let*

$$J_1 \le J_2 \le \cdots, \qquad J_m \to \infty,$$

be a deterministic exhaustion of the countable menu $\mathbb{N} = \{1,2,\dots\}$. For each m and $h \ge 1$, let

$$X_{m,i}^{(h)} = B_{m,i}^{(h)} + Y_{m,i}^{(h)} \in \mathrm{M}_{D_m}(\mathbb{C}), \qquad 1 \le i \le n_m,$$

be a signed-core localized matrix-noise row with independent random profile or amplitude data as in one of the one-row corollaries listed in Corollary 7.301, with its own filtration and random environment allowed. Thus, outside an event $E_{m,h}$, the h-th core has environment-measurable signs satisfying

$$\left\|\sum_{i=1}^{n_m} \varepsilon_{m,i}^{(h),\star} B_{m,i}^{(h)}\right\|_{\mathrm{op}} \le K_{m,h}\sqrt{n_m}.$$

Let $N_{m,h} \ge 1$, $\mu_{m,h} \ge 0$, $\sigma_{m,h}^2 \ge 0$, and $b_{m,h} \in [0,1]$ be the budget-union count, mean budget, Bernstein variance proxy, and jump parameter supplied by the corresponding one-row corollary. Let $u_{m,h} \ge 0$ and $s_m > 0$ be deterministic. If

$$\sum_m \sum_{1\le h\le J_m} \left[\mathbb{P}(E_{m,h}) + N_{m,h}\exp\left\{-\frac{u_{m,h}^2}{2(\sigma_{m,h}^2+b_{m,h}u_{m,h}/3)}\right\} + 2D_m\exp\left\{-\frac{s_m^2}{2(\mu_{m,h}+u_{m,h}+s_m/3)}\right\}\right] <$$

with the usual convention that the Bernstein term is omitted when its denominator is zero, then, with probability one, for all sufficiently large m, every active-prefix element $1 \le h \le J_m$ satisfies

$$\operatorname{disc}(X_{m,1}^{(h)}, \dots, X_{m,n_m}^{(h)}) \le K_{m,h}\sqrt{n_m} + s_m.$$

Consequently, each fixed h is signed at that scale for all sufficiently large m.

In particular, suppose that $K_{m,h} \le K$ for $1 \le h \le J_m$, that

$$\mu_{m,h} + u_{m,h} \le Q\frac{n_m}{\ell_m} \qquad (1 \le h \le J_m)$$

eventually, and that, for some $A, B > 0$,

$$\sum_m \sum_{1\le h\le J_m} \left[\mathbb{P}(E_{m,h}) + N_{m,h}\exp\left\{-\frac{u_{m,h}^2}{2(\sigma_{m,h}^2 + b_{m,h}u_{m,h}/3)}\right\}\right] + \sum_m J_m(eD_m)^{-A} < \infty,$$

while $\ell_m \le B\sqrt{n_m}$ eventually. If $L > 0$ satisfies

$$\frac{L^2}{2(Q + LB/3)} \ge A + 1,$$

then, almost surely, for all sufficiently large m,

$$\operatorname{disc}(X_{m,1}^{(h)}, \dots, X_{m,n_m}^{(h)}) \le (K + L)\sqrt{n_m} \qquad (1 \le h \le J_m),$$

and every fixed menu element is eventually covered.

For an explicit half-budget schedule, set

$$\Lambda_m^{(J)} := \sum_{1\le h\le J_m} N_{m,h}.$$

Assume that, eventually for every $1 \le h \le J_m$,

$$\mu_{m,h} \le \frac{Qn_m}{2\ell_m}, \qquad \sigma_{m,h}^2 \le S\frac{n_m}{\ell_m}, \qquad b_{m,h} \le b,$$

that $K_{m,h} \le K$, $\sum_m \sum_{h\le J_m} \mathbb{P}(E_{m,h}) < \infty$, $\Lambda_m^{(J)} \le (eD_m)^\Gamma$, and $n_m \ge a\ell_m^2$. Put

$$u_{m,h} = \frac{Qn_m}{2\ell_m}, \qquad \delta = \frac{Q^2 a}{8S + \frac{4}{3}bQ}.$$

Along all integer dimensions $D = 2, 3, \dots$, if $\delta > \Gamma + 1$, then the active-prefix and fixed-menu almost-sure conclusions hold for every $A > \Gamma + 1$ and every L satisfying

$$\frac{L^2}{2(Q + L/(3\sqrt{a}))} \ge A + 1.$$

Along any sparse schedule $D_m \ge \exp(cm^\kappa)$, if $\delta > \Gamma$, the same conclusions hold for every $A > \Gamma$ and every L satisfying the same display.

Proof. For each fixed m and h, the corresponding one-row corollary has the common tail bound displayed in the proof of Corollary 7.301. Summing that bound over $1 \le h \le J_m$ and over m gives a finite total failure probability by hypothesis, so the first Borel–Cantelli lemma proves the active-prefix

conclusion. Since $J_m \to \infty$, every fixed h belongs to the active prefix eventually.

For the profile form, take $s_m = L\sqrt{n_m}$. On the active prefix,

$$\frac{s_m^2}{2(\mu_{m,h} + u_{m,h} + s_m/3)} \geq \frac{L^2 n_m}{2(Qn_m/\ell_m + L\sqrt{n_m}/3)} \geq \frac{L^2}{2(Q + LB/3)}\,\ell_m \geq (A{+}1)\ell_m,$$

so the matrix-Freedman term is at most $2(eD_m)^{-A}$ for each active h. The displayed $J_m(eD_m)^{-A}$ summability controls these matrix tails, while the assumed double series controls the core and random-budget exceptions.

For the half-budget schedule, the same computation as in Corollary 7.301 gives

$$N_{m,h} \exp\left\{-\frac{u_{m,h}^2}{2(\sigma_{m,h}^2 + b_{m,h}u_{m,h}/3)}\right\} \leq N_{m,h}(eD_m)^{-\delta}.$$

Summing over $h \leq J_m$ bounds the active-prefix budget exception by $\Lambda_m^{(J)}(eD_m)^{-\delta} \leq (eD_m)^{\Gamma-\delta}$. Also $J_m \leq \Lambda_m^{(J)}$, since $N_{m,h} \geq 1$, so the matrix tails are summable under the stated conditions on A. The all-integer and sparse criteria are the usual p-series and stretched-exponential summability checks. □

Corollary 7.303 (Homogeneous random-amplitude localized schedules)**.** *In the setting of Corollary 7.301, suppose that each row belongs to one of the homogeneous localized models below. Let*

$$H_m = \sum_{i=1}^{n_m} \mathbb{E}\rho_{m,i}^2$$

be the amplitude energy, let $\beta_m \in [0,1]$ be a deterministic homogeneous intensity bound, and let $N_m \geq 1$ be the corresponding budget union count:

(a) *coordinate or random-coordinate-profile supports, where $p_{m,ij} \leq \beta_m$ or $Q_{m,ij} \leq \beta_m$ and $N_m = D_m$;*
(b) *block-profile supports over N_m fixed orthogonal blocks, where $Q_{m,i\ell} \leq \beta_m$;*
(c) *fixed-rank Haar, random-rank Haar, isotropic, or spectrally dominated projection laws, where the rank fraction or intensity satisfies $\alpha_{m,i} \leq \beta_m$ and $N_m = 1$;*
(d) *mixed coordinate/Haar menus, where $M_{m,i}Q_{m,ij}+(1-M_{m,i})R_{m,i}/D_m \leq \beta_m$ and $N_m = D_m$.*

Assume the same signed-core event E_m and core constant K_m as in Corollary 7.301. Then, for every $s_m > 0$,

$$\mathbb{P}\{\mathrm{disc}(X_{m,1},\dots,X_{m,n_m}) > K_m\sqrt{n_m} + s_m\} \leq \mathbb{P}(E_m) + N_m \exp\{-3H_m/8\} + 2D_m \exp\left\{-\frac{s_m^2}{2(2\beta_m H_m + s_m/3)}\right\}.$$

Consequently, if

$$\sum_m \left[\mathbb{P}(E_m) + N_m \exp\{-3H_m/8\} + 2D_m \exp\left\{-\frac{s_m^2}{2(2\beta_m H_m + s_m/3)}\right\}\right] < \infty,$$

then, with probability one, for all sufficiently large m,

$$\operatorname{disc}(X_{m,1}, \dots, X_{m,n_m}) \le K_m\sqrt{n_m} + s_m.$$

For a concrete $n/\log(eD)$ schedule, put $\ell_m = \log(eD_m)$. Suppose that $K_m \le K$, that

$$2\beta_m H_m \le Q\frac{n_m}{\ell_m}, \qquad H_m \ge R\ell_m, \qquad N_m \le (eD_m)^\Gamma$$

eventually, and that $\sum_m \mathbb{P}(E_m) < \infty$. If $\ell_m \le B\sqrt{n_m}$ eventually and

$$\sum_m (eD_m)^{\Gamma - 3R/8} < \infty,$$

then, for every $A > 0$ with $\sum_m (eD_m)^{-A} < \infty$ and every $L > 0$ satisfying

$$\frac{L^2}{2(Q + LB/3)} \ge A + 1,$$

one has, almost surely for all sufficiently large m,

$$\operatorname{disc}(X_{m,1}, \dots, X_{m,n_m}) \le (K + L)\sqrt{n_m}.$$

In particular, along all integer dimensions $D = 2, 3, \dots$, it is enough that

$$n_D \ge a\log^2(eD), \qquad 2\beta_D H_D \le Q\frac{n_D}{\log(eD)}, \qquad H_D \ge R\log(eD),$$

$N_D \le (eD)^\Gamma$, and $R > 8(\Gamma + 1)/3$. Then the same conclusion holds for every $A > 1$ and every L with

$$\frac{L^2}{2(Q + L/(3\sqrt{a}))} \ge A + 1.$$

Along any sparse schedule $D_m \ge \exp(cm^\kappa)$, the same conclusion holds under the same n_m, budget, and energy hypotheses whenever $R > 8\Gamma/3$, for every $A > 0$ and the same displayed condition on L.

Proof. In each listed homogeneous case, the corresponding one-row corollary gives

$$\mu_m \le \beta_m H_m, \qquad \sigma_m^2 \le \beta_m^2 H_m, \qquad b_m \le \beta_m,$$

with budget-union count N_m. Taking $u_m = \beta_m H_m$ in Corollary 7.301, or directly in the one-row tail, gives when $\beta_m H_m > 0$,

$$N_m \exp\left\{-\frac{\beta_m^2 H_m^2}{2(\beta_m^2 H_m + \beta_m^2 H_m/3)}\right\} \le N_m e^{-3H_m/8},$$

while the zero-intensity case has the Bernstein term omitted. Also $\mu_m + u_m \le 2\beta_m H_m$, which proves the first displayed tail and the Borel–Cantelli consequence.

For the schedule form, take $s_m = L\sqrt{n_m}$. The budget hypothesis gives

$$\frac{s_m^2}{2(2\beta_m H_m + s_m/3)} \geq \frac{L^2 n_m}{2(Qn_m/\ell_m + L\sqrt{n_m}/3)} \geq \frac{L^2}{2(Q + LB/3)}\,\ell_m \geq (A{+}1)\ell_m.$$

Thus the matrix tail is at most $2(eD_m)^{-A}$. The scalar-budget exception is bounded by $N_m e^{-3H_m/8} \leq (eD_m)^{\Gamma - 3R/8}$, which is summable by assumption. The all-integer and sparse criteria are the corresponding p-series and stretched-exponential summability tests, with $\ell_D/\sqrt{n_D} \leq 1/\sqrt{a}$. □

Corollary 7.304 (Common-sign countable random-amplitude localized menus)**.** *For each $m \geq 1$, let $D_m \geq 2$, put $\ell_m = \log(eD_m)$, and let*

$$J_1 \leq J_2 \leq \cdots, \qquad J_m \to \infty,$$

be a deterministic exhaustion of the countable menu $\mathbb{N} = \{1, 2, \ldots\}$. Let $\mathcal{G}_m$ be a row environment. For each m and $h \geq 1$, let

$$X_{m,i}^{(h)} = B_{m,i}^{(h)} + Y_{m,i}^{(h)} \in \mathrm{M}_{D_m}(\mathbb{C}), \qquad 1 \leq i \leq n_m,$$

be a signed-core localized matrix-noise row with independent random profile or amplitude data as in one of the one-row corollaries listed in Corollary 7.301. The menu elements may be dependent on one another, but for each fixed h the corresponding one-row hypotheses hold relative to a filtration containing $\mathcal{G}_m$ at time zero. Suppose that there is an event $E_m^{\mathrm{com}} \in \mathcal{G}_m$ such that, outside E_m^{com}, one $\mathcal{G}_m$-measurable sign vector $\varepsilon_m^\star \in \{-1, 1\}^{n_m}$ satisfies

$$\left\| \sum_{i=1}^{n_m} \varepsilon_{m,i}^\star B_{m,i}^{(h)} \right\|_{\mathrm{op}} \leq K_{m,h}\sqrt{n_m} \qquad (1 \leq h \leq J_m).$$

Let $N_{m,h} \geq 1$, $\mu_{m,h} \geq 0$, $\sigma_{m,h}^2 \geq 0$, and $b_{m,h} \in [0, 1]$ be the budget-union count, mean budget, Bernstein variance proxy, and jump parameter supplied by the corresponding one-row corollary for the h-th noise model. Let $u_{m,h} \geq 0$ and $s_m > 0$ be deterministic. If

$$\sum_m \left[\mathbb{P}(E_m^{\mathrm{com}}) + \sum_{1 \leq h \leq J_m} \left(N_{m,h} \exp\left\{ -\frac{u_{m,h}^2}{2(\sigma_{m,h}^2 + b_{m,h}u_{m,h}/3)} \right\} + 2D_m \exp\left\{ -\frac{s_m^2}{2(\mu_{m,h} + u_{m,h} + s_m/3)} \right\} \right) \right.$$

with the usual convention that the Bernstein term is omitted when its denominator is zero, then, with probability one, for all sufficiently large m, the same sign vector $\varepsilon_m^\star$ satisfies

$$\left\| \sum_{i=1}^{n_m} \varepsilon_{m,i}^\star X_{m,i}^{(h)} \right\|_{\mathrm{op}} \leq K_{m,h}\sqrt{n_m} + s_m \qquad (1 \leq h \leq J_m).$$

Consequently, every fixed h is eventually covered by the row's common active-prefix signing.

In particular, suppose that $K_{m,h} \leq K$ for $1 \leq h \leq J_m$, that

$$\mu_{m,h} + u_{m,h} \leq Q\frac{n_m}{\ell_m} \qquad (1 \leq h \leq J_m)$$

eventually, and that, for some $A, B > 0$,

$$\sum_m \left[\mathbb{P}(E_m^{\mathrm{com}}) + \sum_{1 \le h \le J_m} N_{m,h} \exp\left\{ -\frac{u_{m,h}^2}{2(\sigma_{m,h}^2 + b_{m,h} u_{m,h}/3)} \right\} \right] + \sum_m J_m (eD_m)^{-A} < \infty,$$

while $\ell_m \le B\sqrt{n_m}$ *eventually. If* $L > 0$ *satisfies*

$$\frac{L^2}{2(Q + LB/3)} \ge A + 1,$$

then, almost surely, for all sufficiently large m, *one common sign vector* $\varepsilon_m^\star$ *obeys*

$$\left\| \sum_{i=1}^{n_m} \varepsilon_{m,i}^\star X_{m,i}^{(h)} \right\|_{\mathrm{op}} \le (K + L)\sqrt{n_m} \qquad (1 \le h \le J_m),$$

and every fixed menu element is eventually covered by these common row signs.

For an explicit half-budget schedule, set

$$\Lambda_m^{(J)} := \sum_{1 \le h \le J_m} N_{m,h}.$$

Assume that, eventually for every $1 \le h \le J_m$,

$$\mu_{m,h} \le \frac{Qn_m}{2\ell_m}, \qquad \sigma_{m,h}^2 \le S\frac{n_m}{\ell_m}, \qquad b_{m,h} \le b,$$

that $K_{m,h} \le K$, $\sum_m \mathbb{P}(E_m^{\mathrm{com}}) < \infty$, $\Lambda_m^{(J)} \le (eD_m)^\Gamma$, *and* $n_m \ge a\ell_m^2$. *Put*

$$u_{m,h} = \frac{Qn_m}{2\ell_m}, \qquad \delta = \frac{Q^2 a}{8S + \frac{4}{3}bQ}.$$

Along all integer dimensions $D = 2, 3, \ldots$, *if* $\delta > \Gamma + 1$, *then the active-prefix common-sign and fixed-menu conclusions hold for every* $A > \Gamma + 1$ *and every* L *satisfying*

$$\frac{L^2}{2(Q + L/(3\sqrt{a}))} \ge A + 1.$$

Along any sparse schedule $D_m \ge \exp(cm^\kappa)$, *if* $\delta > \Gamma$, *the same conclusions hold for every* $A > \Gamma$ *and every* L *satisfying the same display.*

Proof. Work on the common core event $(E_m^{\mathrm{com}})^c$, and fix the common signs $\varepsilon_m^\star$. For each $h \le J_m$, the proof of the corresponding one-row random-amplitude localized corollary applies with these prescribed signs: after the random budget is at most $\mu_{m,h} + u_{m,h}$, matrix Freedman bounds

$$\left\| \sum_i \varepsilon_{m,i}^\star Y_{m,i}^{(h)} \right\|_{\mathrm{op}}$$

by s_m except with probability at most the displayed $2D_m$-term, while scalar Bernstein gives the displayed budget-exception term. Hence

$$\mathbb{P}\left\{\exists h \le J_m : \left\|\sum_i \varepsilon^\star_{m,i} X^{(h)}_{m,i}\right\|_{\mathrm{op}} > K_{m,h}\sqrt{n_m} + s_m\right\}$$
$$\le \mathbb{P}(E^{\mathrm{com}}_m) + \sum_{h\le J_m}\left[N_{m,h}\exp\left\{-\frac{u^2_{m,h}}{2(\sigma^2_{m,h} + b_{m,h}u_{m,h}/3)}\right\} + 2D_m\exp\left\{-\frac{s^2_m}{2(\mu_{m,h} + u_{m,h} + s_m/3)}\right\}\right].$$

The first assertion follows from the first Borel–Cantelli lemma. Since $J_m \to \infty$, every fixed h is in the active prefix for all large m.

The profile and half-budget schedule proofs are the same summability estimates as in Corollary 7.302. Indeed, with $s_m = L\sqrt{n_m}$, the active-prefix matrix tail is bounded by $2(eD_m)^{-A}$ under the displayed condition on L, so the $J_m(eD_m)^{-A}$ series controls all matrix tails. In the half-budget setting,

$$N_{m,h}\exp\left\{-\frac{u^2_{m,h}}{2(\sigma^2_{m,h} + b_{m,h}u_{m,h}/3)}\right\} \le N_{m,h}(eD_m)^{-\delta},$$

and summing over $h \le J_m$ gives $\Lambda^{(J)}_m(eD_m)^{-\delta} \le (eD_m)^{\Gamma-\delta}$. Since $J_m \le \Lambda^{(J)}_m$, the stated all-integer and sparse thresholds are the usual p-series and stretched-exponential summability tests. □

Corollary 7.305 (Common-sign homogeneous random-amplitude localized schedules)**.** *For each $m \ge 1$, let $D_m \ge 2$, put $\ell_m = \log(eD_m)$, and let*

$$J_1 \le J_2 \le \cdots, \qquad J_m \to \infty,$$

be a deterministic exhaustion of the countable menu $\mathbb{N} = \{1, 2, \ldots\}$. Let $\mathcal{G}_m$ be a row environment. For each m and $h \ge 1$, let

$$X^{(h)}_{m,i} = B^{(h)}_{m,i} + Y^{(h)}_{m,i} \in \mathrm{M}_{D_m}(\mathbb{C}), \qquad 1 \le i \le n_m,$$

be one of the homogeneous localized random-amplitude models listed in Corollary 7.303, with energy

$$H_{m,h} = \sum_{i=1}^{n_m} \mathbb{E}(\rho^{(h)}_{m,i})^2,$$

homogeneous intensity $\beta_{m,h} \in [0,1]$, and budget-union count $N_{m,h} \ge 1$. Suppose that there is an event $E^{\mathrm{com}}_m \in \mathcal{G}_m$ such that, outside E^{com}_m, one $\mathcal{G}_m$-measurable sign vector $\varepsilon^\star_m \in \{-1,1\}^{n_m}$ satisfies

$$\left\|\sum_{i=1}^{n_m} \varepsilon^\star_{m,i} B^{(h)}_{m,i}\right\|_{\mathrm{op}} \le K_{m,h}\sqrt{n_m} \qquad (1 \le h \le J_m).$$

Then, for every deterministic $s_m > 0$, if

$$\sum_m \left[\mathbb{P}(E^{\mathrm{com}}_m) + \sum_{1\le h\le J_m}\left(N_{m,h}e^{-3H_{m,h}/8} + 2D_m\exp\left\{-\frac{s^2_m}{2(2\beta_{m,h}H_{m,h} + s_m/3)}\right\}\right)\right] < \infty,$$

then, with probability one, for all sufficiently large m, the same sign vector $\varepsilon_m^\star$ satisfies

$$\left\|\sum_{i=1}^{n_m}\varepsilon_{m,i}^\star X_{m,i}^{(h)}\right\|_{\mathrm{op}} \le K_{m,h}\sqrt{n_m}+s_m \qquad (1\le h\le J_m).$$

Consequently, every fixed homogeneous localized random-amplitude model is eventually covered by the row's common active-prefix signing.

Set

$$\Lambda_m^{(J)} := \sum_{1\le h\le J_m} N_{m,h}.$$

Suppose that $K_{m,h}\le K$ for $1\le h\le J_m$, and that there are constants $Q,R,\Gamma\ge 0$ such that, eventually on the active prefix,

$$2\beta_{m,h}H_{m,h}\le Q\frac{n_m}{\ell_m}, \qquad H_{m,h}\ge R\ell_m, \qquad \Lambda_m^{(J)}\le (eD_m)^\Gamma,$$

while $\sum_m \mathbb{P}(E_m^{\mathrm{com}})<\infty$. If $\ell_m\le B\sqrt{n_m}$ eventually and, for some $A>0$,

$$\sum_m (eD_m)^{\Gamma-3R/8}<\infty, \qquad \sum_m J_m(eD_m)^{-A}<\infty,$$

then for every $L>0$ satisfying

$$\frac{L^2}{2(Q+LB/3)}\ge A+1,$$

one has, almost surely for all sufficiently large m, one common sign vector $\varepsilon_m^\star$ obeys

$$\left\|\sum_{i=1}^{n_m}\varepsilon_{m,i}^\star X_{m,i}^{(h)}\right\|_{\mathrm{op}} \le (K+L)\sqrt{n_m} \qquad (1\le h\le J_m),$$

and each fixed menu element is eventually covered by these common row signs.

Along all integer dimensions $D=2,3,\dots$, it is enough that

$$n_D\ge a\log^2(eD), \qquad 2\beta_{D,h}H_{D,h}\le Q\frac{n_D}{\log(eD)}, \qquad H_{D,h}\ge R\log(eD)$$

for $h\le J_D$, that $\Lambda_D^{(J)}\le (eD)^\Gamma$, and that

$$R>\frac{8(\Gamma+1)}{3}.$$

Then the active-prefix common-sign and fixed-model conclusions hold for every $A>\Gamma+1$ and every L satisfying

$$\frac{L^2}{2(Q+L/(3\sqrt{a}))}\ge A+1.$$

Along any sparse schedule $D_m\ge \exp(cm^\kappa)$, the same conclusions hold under the same active-prefix budget and energy hypotheses whenever

$$R>\frac{8\Gamma}{3},$$

for every $A > \Gamma$ and the same displayed condition on L.

Proof. For each active-prefix model, the homogeneous one-row bounds give

$$\mu_{m,h} \le \beta_{m,h} H_{m,h}, \qquad \sigma^2_{m,h} \le \beta^2_{m,h} H_{m,h}, \qquad b_{m,h} \le \beta_{m,h}.$$

Apply Corollary 7.304 with $u_{m,h} = \beta_{m,h} H_{m,h}$. When $\beta_{m,h} H_{m,h} > 0$,

$$N_{m,h} \exp\left\{ -\frac{u^2_{m,h}}{2(\sigma^2_{m,h} + b_{m,h} u_{m,h}/3)} \right\} \le N_{m,h} e^{-3H_{m,h}/8},$$

and in the zero-intensity case the Bernstein term is omitted. Also $\mu_{m,h} + u_{m,h} \le 2\beta_{m,h} H_{m,h}$, which proves the first assertion.

For the profile form, take $s_m = L\sqrt{n_m}$. On the active prefix,

$$\frac{s^2_m}{2(2\beta_{m,h} H_{m,h} + s_m/3)} \ge \frac{L^2 n_m}{2(Q n_m/\ell_m + L\sqrt{n_m}/3)} \ge \frac{L^2}{2(Q + LB/3)}\, \ell_m \ge (A{+}1)\ell_m.$$

Thus each matrix tail is at most $2(eD_m)^{-A}$, and the $J_m (eD_m)^{-A}$ series controls all active-prefix matrix tails. The homogeneous scalar-budget exceptions satisfy

$$\sum_{h \le J_m} N_{m,h} e^{-3H_{m,h}/8} \le \Lambda^{(J)}_m e^{-3R\ell_m/8} \le (eD_m)^{\Gamma - 3R/8},$$

which is summable by assumption. The common core exceptions are summable by hypothesis.

The all-integer and sparse schedule checks are the same p-series and stretched-exponential estimates: in integer dimensions $(eD)^{\Gamma - 3R/8}$ is summable when $3R/8 - \Gamma > 1$, while $J_D (eD)^{-A} \le (eD)^{\Gamma - A}$ is summable when $A > \Gamma + 1$; in sparse dimensions the corresponding stretched-exponential series only require $3R/8 - \Gamma > 0$ and $A > \Gamma$. Finally $\ell_D/\sqrt{n_D} \le 1/\sqrt{a}$ under $n_D \ge a \log^2(eD)$. □

Corollary 7.306 (Finite-lift core libraries with random-amplitude localized noise)**.** *In the triangular-array setting of Corollary 7.304, replace the common-core signing hypothesis by the following finite-lift menu hypothesis. There are events $F^{\mathrm{lift}}_m \in \mathcal{G}_m$ with $\sum_m \mathbb{P}(F^{\mathrm{lift}}_m) < \infty$ such that, outside F^{lift}_m, the active-prefix core tuples*

$$B^{(h)}_{m,1}, \dots, B^{(h)}_{m,n_m} \in \mathrm{M}_{D_m}(\mathbb{C}), \qquad 1 \le h \le J_m,$$

are contractions and admit finite block lifts

$$\mathcal{B}_{m,h} = \bigoplus_{\alpha=1}^{q_{m,h}} \mathrm{M}_{r_{m,h,\alpha}}(\mathbb{C})$$

with lift constants $K^{\mathrm{lift}}_{m,h}, \Lambda^{\mathrm{lift}}_{m,h}$. Put

$$M^{(J)}_m = \max_{1 \le h \le J_m} \Lambda^{\mathrm{lift}}_{m,h} K^{\mathrm{lift}}_{m,h},$$

$$h^{(J)}_m = \max_{\substack{1 \le h \le J_m \\ 1 \le \alpha \le q_{m,h}}} r_{m,h,\alpha}, \qquad R^{(J)}_m = \sum_{1 \le h \le J_m} \sum_{\alpha=1}^{q_{m,h}} r_{m,h,\alpha},$$

and $E_m^{(J)} = h_m^{(J)} R_m^{(J)}$. *If*

$$M_m^{(J)} \le M_0, \qquad E_m^{(J)} \le C_E n_m$$

eventually outside F_m^{lift}, *then every conclusion of Corollary 7.304 holds with*

$$E_m^{\mathrm{com}} = F_m^{\mathrm{lift}}, \qquad K_{m,h} = K_{\mathrm{fl}} := C M_0 \sqrt{\log(e + C_E)}.$$

In particular, if the lift condition is deterministic then there is no core exception. In the homogeneous localized subcases, every conclusion and explicit all-integer or sparse schedule of Corollary 7.305 holds with the same substitution $K = K_{\mathrm{fl}}$ *and* $E_m^{\mathrm{com}} = F_m^{\mathrm{lift}}$.

Proof. On $(F_m^{\mathrm{lift}})^c$, apply Corollary 3.2 to the finite menu $\{B_m^{(h)} : 1 \le h \le J_m\}$. The displayed bounds on $M_m^{(J)}$ and $E_m^{(J)}$ give one sign vector $\varepsilon_m^\star$ such that

$$\left\| \sum_{i=1}^{n_m} \varepsilon_{m,i}^\star B_{m,i}^{(h)} \right\|_{\mathrm{op}} \le K_{\mathrm{fl}} \sqrt{n_m} \qquad (1 \le h \le J_m).$$

Since the set of sign vectors is finite, choosing the lexicographically first successful sign vector makes $\varepsilon_m^\star$ $\mathcal{G}_m$-measurable. This is exactly the common-core hypothesis in the two common-sign random-amplitude corollaries, with $E_m^{\mathrm{com}} = F_m^{\mathrm{lift}}$. Applying those corollaries gives the stated conclusions. □

Corollary 7.307 (Exhaustive homogeneous random-amplitude localized schedules)**.** *For each* $m \ge 1$, *let* $D_m \ge 2$, *put* $\ell_m = \log(eD_m)$, *and let*

$$J_1 \le J_2 \le \cdots, \qquad J_m \to \infty,$$

be a deterministic exhaustion of the countable menu $\mathbb{N} = \{1, 2, \ldots\}$. *For each* m *and* $h \ge 1$, *let*

$$X_{m,i}^{(h)} = B_{m,i}^{(h)} + Y_{m,i}^{(h)} \in \mathrm{M}_{D_m}(\mathbb{C}), \qquad 1 \le i \le n_m,$$

be one of the homogeneous localized random-amplitude models listed in Corollary 7.303, with its own signed-core event $E_{m,h}$, *core constant* $K_{m,h}$, *energy*

$$H_{m,h} = \sum_{i=1}^{n_m} \mathbb{E}(\rho_{m,i}^{(h)})^2,$$

homogeneous intensity $\beta_{m,h} \in [0,1]$, *and budget-union count* $N_{m,h} \ge 1$. *Set*

$$\Lambda_m^{(J)} := \sum_{1 \le h \le J_m} N_{m,h}.$$

Then, for every deterministic $s_m > 0$, *if*

$$\sum_m \sum_{1 \le h \le J_m} \left[\mathbb{P}(E_{m,h}) + N_{m,h} e^{-3H_{m,h}/8} + 2D_m \exp\left\{ -\frac{s_m^2}{2(2\beta_{m,h} H_{m,h} + s_m/3)} \right\} \right] < \infty,$$

then, with probability one, for all sufficiently large m, *every active prefix element* $1 \le h \le J_m$ *satisfies*

$$\mathrm{disc}(X_{m,1}^{(h)}, \ldots, X_{m,n_m}^{(h)}) \le K_{m,h} \sqrt{n_m} + s_m.$$

Consequently, every fixed homogeneous localized random-amplitude model is covered for all sufficiently large m.

In particular, suppose that $K_{m,h} \le K$ for $1 \le h \le J_m$, and that there are constants $Q, R, \Gamma \ge 0$ such that, eventually on the active prefix,

$$2\beta_{m,h}H_{m,h} \le Q\frac{n_m}{\ell_m}, \qquad H_{m,h} \ge R\ell_m, \qquad \Lambda_m^{(J)} \le (eD_m)^{\Gamma},$$

while

$$\sum_m \sum_{1\le h\le J_m} \mathbb{P}(E_{m,h}) < \infty.$$

If $\ell_m \le B\sqrt{n_m}$ eventually and, for some $A > 0$,

$$\sum_m (eD_m)^{\Gamma - 3R/8} < \infty, \qquad \sum_m J_m (eD_m)^{-A} < \infty,$$

then for every $L > 0$ satisfying

$$\frac{L^2}{2(Q + LB/3)} \ge A + 1,$$

one has, almost surely for all sufficiently large m,

$$\operatorname{disc}(X_{m,1}^{(h)}, \dots, X_{m,n_m}^{(h)}) \le (K + L)\sqrt{n_m} \qquad (1 \le h \le J_m),$$

and each fixed menu element is eventually covered.

Along all integer dimensions $D = 2, 3, \dots$, it is enough that

$$n_D \ge a\log^2(eD), \qquad 2\beta_{D,h}H_{D,h} \le Q\frac{n_D}{\log(eD)}, \qquad H_{D,h} \ge R\log(eD)$$

for $h \le J_D$, that $\Lambda_D^{(J)} \le (eD)^{\Gamma}$, and that

$$R > \frac{8(\Gamma + 1)}{3}.$$

Then the active-prefix and fixed-model conclusions hold for every $A > \Gamma + 1$ and every L satisfying

$$\frac{L^2}{2(Q + L/(3\sqrt{a}))} \ge A + 1.$$

Along any sparse schedule $D_m \ge \exp(cm^{\kappa})$, the same conclusions hold under the same active-prefix budget and energy hypotheses whenever

$$R > \frac{8\Gamma}{3},$$

for every $A > \Gamma$ and the same displayed condition on L.

Proof. For each fixed m and h, Corollary 7.303 gives the one-row failure bound with parameters $E_{m,h}$, $K_{m,h}$, $H_{m,h}$, $\beta_{m,h}$, and $N_{m,h}$. Summing this bound over $1 \le h \le J_m$ and then over m gives a finite total failure probability, so the first Borel–Cantelli lemma proves the active-prefix statement. Since $J_m \to \infty$, each fixed h lies in the active prefix for all large m.

For the profile form, take $s_m = L\sqrt{n_m}$. On the active prefix,

$$\frac{s_m^2}{2(2\beta_{m,h}H_{m,h} + s_m/3)} \geq \frac{L^2 n_m}{2(Qn_m/\ell_m + L\sqrt{n_m}/3)} \geq \frac{L^2}{2(Q + LB/3)}\,\ell_m \geq (A{+}1)\ell_m.$$

Thus each matrix tail is at most $2(eD_m)^{-A}$, and the active-prefix matrix tails are summable by the displayed $J_m(eD_m)^{-A}$ hypothesis. The homogeneous scalar-budget exceptions satisfy

$$\sum_{h \leq J_m} N_{m,h} e^{-3H_{m,h}/8} \leq \Lambda_m^{(J)} e^{-3R\ell_m/8} \leq (eD_m)^{\Gamma - 3R/8},$$

which is summable by assumption. The core exceptions are summable by hypothesis.

The all-integer case follows because $(eD)^{\Gamma - 3R/8}$ is summable when $3R/8 - \Gamma > 1$, while $J_D(eD)^{-A} \leq (eD)^{\Gamma - A}$ is summable when $A > \Gamma + 1$. Also $\ell_D/\sqrt{n_D} \leq 1/\sqrt{a}$. The sparse case is the same calculation with stretched-exponential summability, requiring only $3R/8 - \Gamma > 0$ and $A > \Gamma$. □

Corollary 7.308 (Core signings with predictable martingale scalar noise)**.** *Let $D \geq 2$. Let $\mathcal{G}$ be a sigma-field, and let $B_i \in \mathrm{M}_D(\mathbb{C})$, $i = 1, \ldots, n$, be $\mathcal{G}$-measurable random self-adjoint contractions. Let $(\mathcal{F}_i)_{i=0}^n$ be a filtration with $\mathcal{G} \subset \mathcal{F}_0$, and let $H_i \in \mathrm{M}_D(\mathbb{C})$ be self-adjoint contractions such that H_i is $\mathcal{F}_{i-1}$-measurable. Let θ_i be real $\mathcal{F}_i$-measurable random variables satisfying*

$$\mathbb{E}(\theta_i \mid \mathcal{F}_{i-1}) = 0, \qquad |\theta_i| \leq 1.$$

Set

$$X_i = B_i + \theta_i H_i, \qquad q_i = \mathbb{E}(\theta_i^2 \mid \mathcal{F}_{i-1}), \qquad \nu = \left\| \sum_{i=1}^n q_i H_i^2 \right\|_{\mathrm{op}}.$$

Assume that, outside an event $E \in \mathcal{G}$, there are $\mathcal{G}$-measurable signs $\varepsilon_i^\star \in \{-1, 1\}$ such that

$$\left\| \sum_{i=1}^n \varepsilon_i^\star B_i \right\|_{\mathrm{op}} \leq K\sqrt{n}.$$

Then, for every deterministic $v \geq 0$ and $s > 0$,

$$\mathbb{P}\left\{\mathrm{disc}(X_1, \ldots, X_n) > K\sqrt{n} + s\right\} \leq \mathbb{P}(E) + \mathbb{P}\{\nu > v\} + 2D \exp\left\{-\frac{s^2}{2(v + s/3)}\right\}.$$

Since $\|X_i\|_{\mathrm{op}} \leq 2$, the normalized contractions $X_i/2$ satisfy the same estimate with the discrepancy threshold divided by 2. In particular, if $q_i \leq p$ almost surely for every i, then

$$\mathbb{P}\left\{\mathrm{disc}(X_1, \ldots, X_n) > K\sqrt{n} + s\right\} \leq \mathbb{P}(E) + 2D \exp\left\{-\frac{s^2}{2(pn + s/3)}\right\}.$$

Thus the signed-core noise theorem does not require conditional independence of the scalar amplitudes, nor does it require the noise directions to be fixed at

time zero: after the core signing is fixed, predictable matrix directions with bounded martingale-difference scalar amplitudes are absorbed by the same core signing.

Proof. Work on E^c. For the $\mathcal{G}$-measurable signing $\varepsilon^\star$, define

$$Z_i = \varepsilon_i^\star \theta_i H_i.$$

Since $\mathcal{G} \subset \mathcal{F}_{i-1}$ and H_i is $\mathcal{F}_{i-1}$-measurable, the matrices $\varepsilon_i^\star H_i$ are $\mathcal{F}_{i-1}$-measurable. Hence Z_i is a self-adjoint matrix martingale difference and $\|Z_i\|_{\mathrm{op}} \le 1$. Moreover,

$$\mathbb{E}(Z_i^2 \mid \mathcal{F}_{i-1}) = q_i H_i^2.$$

On $\{\nu \le v\}$, the predictable quadratic variation therefore has operator norm at most v. Applying the matrix Freedman inequality [25] to Z_i and to $-Z_i$, and taking a union bound, gives

$$\mathbb{P}\left\{ \left\| \sum_i Z_i \right\|_{\mathrm{op}} > s \text{ and } \nu \le v \right\} \le 2D \exp\left\{ -\frac{s^2}{2(v + s/3)} \right\}.$$

On the complementary good event,

$$\left\| \sum_i \varepsilon_i^\star X_i \right\|_{\mathrm{op}} \le \left\| \sum_i \varepsilon_i^\star B_i \right\|_{\mathrm{op}} + \left\| \sum_i Z_i \right\|_{\mathrm{op}} \le K\sqrt{n} + s.$$

Adding the exceptional events E and $\{\nu > v\}$ proves the first estimate. If $q_i \le p$, then $0 \le H_i^2 \le I$, so $\nu \le p\| \sum_i H_i^2 \|_{\mathrm{op}} \le pn$, which gives the final form. □

Corollary 7.309 (Core signings with predictable localized martingale noise)**.** *Let $D \ge 2$. Let $\mathcal{G}$ be a sigma-field, and let $B_i \in \mathrm{M}_D(\mathbb{C})$, $i = 1, \dots, n$, be $\mathcal{G}$-measurable random self-adjoint contractions. Let $(\mathcal{F}_i)_{i=0}^n$ be a filtration with $\mathcal{G} \subset \mathcal{F}_0$. For each i, let P_i be an $\mathcal{F}_{i-1}$-measurable orthogonal projection and let H_i be an $\mathcal{F}_{i-1}$-measurable self-adjoint contraction satisfying*

$$H_i = P_i H_i P_i.$$

Let θ_i be real $\mathcal{F}_i$-measurable random variables satisfying

$$\mathbb{E}(\theta_i \mid \mathcal{F}_{i-1}) = 0, \qquad |\theta_i| \le 1,$$

and set

$$X_i = B_i + \theta_i H_i, \qquad q_i = \mathbb{E}(\theta_i^2 \mid \mathcal{F}_{i-1}), \qquad \mu = \left\| \sum_{i=1}^n q_i P_i \right\|_{\mathrm{op}}.$$

Assume that, outside an event $E \in \mathcal{G}$, there are $\mathcal{G}$-measurable signs $\varepsilon_i^\star \in \{-1, 1\}$ with

$$\left\| \sum_{i=1}^n \varepsilon_i^\star B_i \right\|_{\mathrm{op}} \le K\sqrt{n}.$$

Then, for every deterministic $v \geq 0$ and $s > 0$,

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \leq \mathbb{P}(E) + \mathbb{P}\{\mu > v\} + 2D \exp\left\{-\frac{s^2}{2(v + s/3)}\right\}.$$

In particular, if $q_i \leq p$ almost surely for every i, then

$$\mathbb{P}\left\{\operatorname{disc}(X_1, \dots, X_n) > K\sqrt{n} + s\right\} \leq \mathbb{P}(E) + \mathbb{P}\left\{\left\|\sum_i P_i\right\|_{\mathrm{op}} > M\right\} + 2D \exp\left\{-\frac{s^2}{2(pM + s/3)}\right\}$$

for every deterministic $M \geq 0$. Thus already-signed random cores admit an explicit tail bound after online coordinate-supported, subspace-localized, or frame-localized martingale scalar noise, and they remain Spencer-scale at $s = O(\sqrt{n})$ when the predictable weighted overlap $\|\sum_i q_i P_i\|_{\mathrm{op}}$ is $O(n/\log(eD))$.

Proof. Since $H_i = P_i H_i P_i$ and $\|H_i\|_{\mathrm{op}} \leq 1$, we have $0 \leq H_i^2 \preceq P_i$. Hence

$$\left\|\sum_i q_i H_i^2\right\|_{\mathrm{op}} \leq \left\|\sum_i q_i P_i\right\|_{\mathrm{op}} = \mu.$$

The first estimate follows from Corollary 7.308. If $q_i \leq p$, then $\mu \leq p\|\sum_i P_i\|_{\mathrm{op}}$; applying the first estimate on the event $\|\sum_i P_i\|_{\mathrm{op}} \leq M$ gives the displayed specialization. □

Corollary 7.310 (Almost-sure signed-core predictable scalar-noise arrays)**.** *For each $m \geq 1$, let $D_m \geq 2$, let $\mathcal{G}_m$ be a sigma-field, let $(\mathcal{F}_{m,i})_{i=0}^{n_m}$ be a filtration with $\mathcal{G}_m \subset \mathcal{F}_{m,0}$, and let*

$$X_{m,i} = B_{m,i} + \theta_{m,i} H_{m,i} \in \mathrm{M}_{D_m}(\mathbb{C}) \qquad (1 \leq i \leq n_m)$$

be a signed-core predictable scalar-noise row as in Corollary 7.308. Thus $B_{m,i}$ is $\mathcal{G}_m$-measurable, $H_{m,i}$ is $\mathcal{F}_{m,i-1}$-measurable, $\mathbb{E}(\theta_{m,i} \mid \mathcal{F}_{m,i-1}) = 0$, and $|\theta_{m,i}| \leq 1$. Put

$$q_{m,i} := \mathbb{E}(\theta_{m,i}^2 \mid \mathcal{F}_{m,i-1}), \qquad \nu_m := \left\|\sum_{i=1}^{n_m} q_{m,i} H_{m,i}^2\right\|_{\mathrm{op}}.$$

Assume that, outside an event $E_m \in \mathcal{G}_m$, there are $\mathcal{G}_m$-measurable signs $\varepsilon_{m,i}^\star$ such that

$$\left\|\sum_{i=1}^{n_m} \varepsilon_{m,i}^\star B_{m,i}\right\|_{\mathrm{op}} \leq K_m \sqrt{n_m}.$$

Let $v_m \geq 0$ and $s_m > 0$ be deterministic. If

$$\sum_m \left[\mathbb{P}(E_m) + \mathbb{P}\{\nu_m > v_m\} + 2D_m \exp\left\{-\frac{s_m^2}{2(v_m + s_m/3)}\right\}\right] < \infty,$$

then, with probability one, for all sufficiently large m,

$$\operatorname{disc}(X_{m,1},\dots,X_{m,n_m}) \le K_m\sqrt{n_m} + s_m.$$

In particular, put $\ell_m = \log(eD_m)$. Suppose that $K_m \le K$, that

$$\nu_m \le Q\frac{n_m}{\ell_m}$$

outside events F_m, that

$$\sum_m [\mathbb{P}(E_m) + \mathbb{P}(F_m)] < \infty,$$

and that, for some $A, B > 0$,

$$\sum_m (eD_m)^{-A} < \infty, \qquad \ell_m \le B\sqrt{n_m}$$

eventually. Choose $L > 0$ with

$$\frac{L^2}{2(Q + LB/3)} \ge A + 1.$$

Then, almost surely, for all sufficiently large m,

$$\operatorname{disc}(X_{m,1},\dots,X_{m,n_m}) \le (K + L)\sqrt{n_m}.$$

The variance-budget hypothesis is implied by the pointwise condition $q_{m,i} \le Q/\ell_m$ outside F_m.

Consequently:

(a) *Along all integer dimensions $D = 2, 3, \dots$, if $K_D \le K$, $\nu_D \le Qn_D/\log(eD)$ outside summable events, and*

$$n_D \ge a\,\log^2(eD)$$

eventually for some $a > 0$, then for every $A > 1$ and every L satisfying

$$\frac{L^2}{2(Q + L/(3\sqrt{a}))} \ge A + 1$$

one has

$$\operatorname{disc}(X_{D,1},\dots,X_{D,n_D}) \le (K + L)\sqrt{n_D}$$

eventually almost surely.

(b) *If $D_m \ge \exp(cm^\kappa)$ eventually for some $c, \kappa > 0$, if $K_m \le K$, if $\nu_m \le Qn_m/\log(eD_m)$ outside summable events, and if*

$$n_m \ge a\,\log^2(eD_m)$$

eventually for some $a > 0$, then the same conclusion holds along the sparse schedule for every $A > 0$ and every L satisfying

$$\frac{L^2}{2(Q + L/(3\sqrt{a}))} \ge A + 1.$$

Proof. For the m-th row, write $\Delta_m = \operatorname{disc}(X_{m,1}, \dots, X_{m,n_m})$. Corollary 7.308 gives

$$\mathbb{P}\{\Delta_m > K_m\sqrt{n_m} + s_m\} \le \mathbb{P}(E_m) + \mathbb{P}\{\nu_m > v_m\} + 2D_m \exp\left\{-\frac{s_m^2}{2(v_m + s_m/3)}\right\}.$$

The displayed summability and the first Borel–Cantelli lemma prove the first claim.

For the concrete profile, take $s_m = L\sqrt{n_m}$ and $v_m = Qn_m/\ell_m$. On F_m^c,

$$\frac{s_m^2}{2(v_m + s_m/3)} = \frac{L^2 n_m}{2(Qn_m/\ell_m + L\sqrt{n_m}/3)} \ge \frac{L^2}{2(Q + LB/3)}\,\ell_m \ge (A+1)\ell_m.$$

Therefore

$$2D_m \exp\left\{-\frac{s_m^2}{2(v_m + s_m/3)}\right\} \le 2(eD_m)^{-A},$$

which is summable by assumption. If $q_{m,i} \le Q/\ell_m$, then

$$\nu_m \le \frac{Q}{\ell_m}\left\|\sum_{i=1}^{n_m} H_{m,i}^2\right\|_{\mathrm{op}} \le Q\frac{n_m}{\ell_m}.$$

For part (a), take $B = 1/\sqrt{a}$ and any $A > 1$, since $\sum_{D\ge 2}(eD)^{-A} < \infty$. For part (b), again take $B = 1/\sqrt{a}$; the stretched-exponential lower bound on D_m makes $\sum_m (eD_m)^{-A}$ finite for every $A > 0$. □

Corollary 7.311 (Weighted random low-overlap support ensembles)**.** *For every $C_0 \ge 1$ there is $C(C_0)$ with the following property. Let $A_1, \dots, A_n$ be random self-adjoint matrices in $\mathrm{M}_D(\mathbb{C})$. Suppose that, outside an event E, there are orthogonal projections $P_1, \dots, P_n$ and numbers $\rho_1, \dots, \rho_n \in [0,1]$ such that*

$$A_i = P_i A_i P_i, \qquad \|A_i\|_{\mathrm{op}} \le \rho_i, \qquad \left\|\sum_{i=1}^{n} \rho_i^2 P_i\right\|_{\mathrm{op}} \le \frac{C_0 n}{\log(eD)}$$

for every i. Then

$$\mathbb{P}\left\{\operatorname{disc}(A_1, \dots, A_n) > C(C_0)\sqrt{n}\right\} \le \mathbb{P}(E).$$

In particular, the Matrix Spencer bound holds with the same probability as the displayed weighted-overlap event for coordinate-supported random matrices, random localized Wigner/GOE/GUE blocks, and random projection or frame ensembles. The unweighted form is recovered by taking $\rho_i = 1$, in which case the coordinate condition is maximum occupancy $O(n/\log(eD))$.

Proof. On a realization outside E, each A_i is self-adjoint and supported on the range of P_i, with $\|A_i\|_{\mathrm{op}} \le \rho_i$. Hence $0 \le A_i^2 \le \rho_i^2 P_i$, and therefore

$$\left\|\sum_i A_i^2\right\|_{\mathrm{op}} \le \left\|\sum_i \rho_i^2 P_i\right\|_{\mathrm{op}} \le \frac{C_0 n}{\log(eD)}.$$

Proposition 7.143 gives the claimed discrepancy bound on that realization. For coordinate supports, P_i is the coordinate projection and $\|\sum_i \rho_i^2 P_i\|_{\mathrm{op}}$ is exactly the maximum weighted coordinate occupancy; for subspace or frame models it is the weighted empirical frame-operator norm. □

Corollary 7.312 (Active menus with dependent self-adjoint overlap budgets)**.** *Fix $C_0 \geq 1$. For each $m \geq 1$, let $D_m \geq 2$, $n_m \geq 1$, and $J_m \geq 1$. For each active index $1 \leq h \leq J_m$, let*

$$A_{m,h,1}, \dots, A_{m,h,n_m} \in \mathrm{M}_{D_m}(\mathbb{C})$$

be random self-adjoint matrices. Suppose that there are events $F^{\mathrm{ov}}_{m,h}$, random orthogonal projections $P_{m,h,i}$, and random amplitudes $0 \leq \rho_{m,h,i} \leq 1$ such that, outside $F^{\mathrm{ov}}_{m,h}$,

$$A_{m,h,i} = P_{m,h,i} A_{m,h,i} P_{m,h,i}, \qquad \|A_{m,h,i}\|_{\mathrm{op}} \leq \rho_{m,h,i} \qquad (1 \leq i \leq n_m),$$

and

$$\left\| \sum_{i=1}^{n_m} \rho_{m,h,i}^2 P_{m,h,i} \right\|_{\mathrm{op}} \leq \frac{C_0 n_m}{\log(eD_m)}.$$

If

$$\sum_m \sum_{h=1}^{J_m} \mathbb{P}(F^{\mathrm{ov}}_{m,h}) < \infty,$$

then, almost surely, for all sufficiently large m, every active stream has a Matrix Spencer signing:

$$\mathrm{disc}(A_{m,h,1}, \dots, A_{m,h,n_m}) \leq C(C_0)\sqrt{n_m} \qquad (1 \leq h \leq J_m).$$

Sampled sign certificates are eventual almost surely if the per-stream candidate lists also satisfy $\sum_{m,h} 2^{-M_{m,h}} < \infty$; a common list length M_m for all active streams only requires $\sum_m J_m 2^{-M_m} < \infty$. If $J_m \to \infty$, active prefixes may exhaust a countable library.

Thus arbitrary dependent self-adjoint support laws, random regular graphs, matchings, without-replacement coordinate choices, determinantal or negatively dependent supports, and adaptive projection/frontier models are covered once their realized weighted overlap budgets fail only summably.

Proof. On $(F^{\mathrm{ov}}_{m,h})^c$, the pathwise hypotheses of Corollary 7.311 hold for stream (m,h). Summability of the overlap-budget failures and the first Borel–Cantelli lemma give the simultaneous eventual signings. The sampled-certificate assertion is the same conditional random-sign argument as in Corollary 7.340, with the displayed per-stream or common-list failure series. If $J_m \to \infty$, each fixed library stream belongs to the active prefix for all sufficiently large m. □

Corollary 7.313 (Tail-capped dependent self-adjoint overlap budgets)**.** *Fix $C_0 \geq 1$. In the active-menu setting of Corollary 7.312, replace the pointwise norm bounds by deterministic caps*

$$0 \leq \rho_{m,h,i} \leq 1 \qquad (1 \leq h \leq J_m,\ 1 \leq i \leq n_m).$$

Assume that there are events $F^{\mathrm{ov}}_{m,h}$, random orthogonal projections $P_{m,h,i}$, and cap-tail budgets $\delta_{m,h,i} \geq 0$ such that

$$\mathbb{P}\{\|A_{m,h,i}\|_{\mathrm{op}} > \rho_{m,h,i}\} \leq \delta_{m,h,i},$$

and, outside $F^{\mathrm{ov}}_{m,h}$,

$$A_{m,h,i} = P_{m,h,i} A_{m,h,i} P_{m,h,i} \qquad (1 \leq i \leq n_m)$$

and

$$\left\| \sum_{i=1}^{n_m} \rho_{m,h,i}^2 P_{m,h,i} \right\|_{\mathrm{op}} \leq \frac{C_0 n_m}{\log(eD_m)}.$$

If

$$\sum_m \sum_{h=1}^{J_m} \left[\mathbb{P}(F^{\mathrm{ov}}_{m,h}) + \sum_{i=1}^{n_m} \delta_{m,h,i} \right] < \infty,$$

then, almost surely, for all sufficiently large m, every active stream has

$$\mathrm{disc}(A_{m,h,1}, \ldots, A_{m,h,n_m}) \leq C(C_0)\sqrt{n_m} \qquad (1 \leq h \leq J_m).$$

Sampled certificates and countable-library exhaustions follow under the same additional summability hypotheses as in Corollary 7.312.

For example, if

$$\mathbb{P}\{\|A_{m,h,i}\|_{\mathrm{op}} > s_{m,h,i} u\} \leq A_* e^{-u^\alpha} \qquad (u \geq 0)$$

with fixed $\alpha > 0$, choose $\rho_{m,h,i} = s_{m,h,i} \ell_{m,h,i}^{1/\alpha} \leq 1$ and $\delta_{m,h,i} = A_* e^{-\ell_{m,h,i}}$. If instead $\mathbb{E}\|A_{m,h,i}\|_{\mathrm{op}}^p \leq s_{m,h,i}^p$, choose $\rho_{m,h,i} = s_{m,h,i} e^{\ell_{m,h,i}/p} \leq 1$ and $\delta_{m,h,i} = e^{-\ell_{m,h,i}}$. These deterministic caps feed the same realized self-adjoint overlap budget, so dependent heavy-tailed localized models reduce to cap-tail summability plus a capped overlap certificate.

Proof. For each active stream set

$$\Omega^{\mathrm{cap}}_{m,h} := \{\|A_{m,h,i}\|_{\mathrm{op}} \leq \rho_{m,h,i} \text{ for all } i \leq n_m\}.$$

On $(F^{\mathrm{ov}}_{m,h})^c \cap \Omega^{\mathrm{cap}}_{m,h}$, the hypotheses of Corollary 7.312 hold with the deterministic caps $\rho_{m,h,i}$. The exceptional probability for stream (m,h) is at most

$$\mathbb{P}(F^{\mathrm{ov}}_{m,h}) + \sum_{i=1}^{n_m} \delta_{m,h,i}.$$

The displayed summability and Borel–Cantelli give the eventual active-menu signings; sampled certificates add the same sampled-sign failure series as in the dependent-overlap corollary. The sub-Weibull and moment cap choices follow from the displayed tail bound and Markov's inequality, respectively. □

Corollary 7.314 (Common signs for dependent self-adjoint overlap menus)**.** *Fix $C_0 \geq 1$. For each $m \geq 1$, let $D_m \geq 2$, $n_m \geq 1$, $J_m \geq 1$, and put*

$$D_m^{\mathrm{com}} := J_m D_m.$$

For $1 \le h \le J_m$, let

$$A_{m,h,1}, \dots, A_{m,h,n_m} \in \mathrm{M}_{D_m}(\mathbb{C})$$

be random self-adjoint matrices. Suppose that there are events $F^{\mathrm{ov}}_{m,h}$, random orthogonal projections $P_{m,h,i}$, and random amplitudes $0 \le \rho_{m,h,i} \le 1$ such that, outside $F^{\mathrm{ov}}_{m,h}$,

$$A_{m,h,i} = P_{m,h,i} A_{m,h,i} P_{m,h,i}, \qquad \|A_{m,h,i}\|_{\mathrm{op}} \le \rho_{m,h,i} \qquad (1 \le i \le n_m),$$

and

$$\left\| \sum_{i=1}^{n_m} \rho_{m,h,i}^2 P_{m,h,i} \right\|_{\mathrm{op}} \le \frac{C_0 n_m}{\log(e D_m^{\mathrm{com}})}.$$

If

$$\sum_m \sum_{h=1}^{J_m} \mathbb{P}(F^{\mathrm{ov}}_{m,h}) < \infty,$$

then, almost surely, for all sufficiently large m, there is one sign vector $\varepsilon^{(m)} \in \{\pm 1\}^{n_m}$ such that

$$\max_{1 \le h \le J_m} \left\| \sum_{i=1}^{n_m} \varepsilon_i^{(m)} A_{m,h,i} \right\|_{\mathrm{op}} \le C(C_0) \sqrt{n_m}.$$

If M_m independent random sign vectors are sampled for the whole active menu and $\sum_m 2^{-M_m} < \infty$, then the sampled list contains such a common certificate eventually almost surely. If $J_m \to \infty$, the common signs cover each active prefix in a countable library eventually.

Proof. Outside the union of the events $F^{\mathrm{ov}}_{m,h}$, define block-diagonal self-adjoint matrices

$$\widetilde{A}_{m,i} := \bigoplus_{h=1}^{J_m} A_{m,h,i} \in \mathrm{M}_{D_m^{\mathrm{com}}}(\mathbb{C}).$$

Then

$$\left\| \sum_i \widetilde{A}_{m,i}^2 \right\|_{\mathrm{op}} = \max_{h \le J_m} \left\| \sum_i A_{m,h,i}^2 \right\|_{\mathrm{op}} \le \frac{C_0 n_m}{\log(e D_m^{\mathrm{com}})}.$$

Proposition 7.143 applied in $\mathrm{M}_{D_m^{\mathrm{com}}}(\mathbb{C})$ gives one signing of the block-diagonal tuple, which is exactly the displayed common signing. The failure event is summable by assumption, so Borel–Cantelli gives the eventual deterministic common signing. The sampled version follows from Corollary 7.144 applied to the same block-diagonal tuple, adding only the failure series $\sum_m 2^{-M_m}$. □

Corollary 7.315 (Common signs for support-rank self-adjoint overlap menus)**.** *Fix $C_0 \ge 1$. For each $m \ge 1$, let $D_m \ge 2$, $n_m \ge 1$, $J_m \ge 1$, and let $R_m \ge 1$. For $1 \le h \le J_m$, let*

$$A_{m,h,1}, \dots, A_{m,h,n_m} \in \mathrm{M}_{D_m}(\mathbb{C})$$

be random self-adjoint matrices. Suppose that there are events F_m^{sr}, random orthogonal projections $P_{m,h,i}$, random support-envelope projections $Q_{m,h}$, and random amplitudes $0 \le \rho_{m,h,i} \le 1$ such that, outside F_m^{sr},

$$A_{m,h,i} = P_{m,h,i} A_{m,h,i} P_{m,h,i}, \qquad \|A_{m,h,i}\|_{\mathrm{op}} \le \rho_{m,h,i}, \qquad P_{m,h,i} \le Q_{m,h},$$

for all $h \le J_m$ and $i \le n_m$, and

$$\sum_{h=1}^{J_m} \max\{1, \operatorname{rank} Q_{m,h}\} \le R_m.$$

Assume also that, outside F_m^{sr},

$$\max_{1 \le h \le J_m} \left\| \sum_{i=1}^{n_m} \rho_{m,h,i}^2 P_{m,h,i} \right\|_{\mathrm{op}} \le \frac{C_0 n_m}{\log(eR_m)}.$$

If

$$\sum_m \mathbb{P}(F_m^{\mathrm{sr}}) < \infty,$$

then, almost surely, for all sufficiently large m, there is one sign vector $\varepsilon^{(m)} \in \{\pm 1\}^{n_m}$ such that

$$\max_{1 \le h \le J_m} \left\| \sum_{i=1}^{n_m} \varepsilon_i^{(m)} A_{m,h,i} \right\|_{\mathrm{op}} \le C(C_0) \sqrt{n_m}.$$

If M_m independent random sign vectors are sampled for the whole active menu and

$$\sum_m 2^{-M_m} < \infty,$$

then the sampled list contains such a common certificate eventually almost surely. If $J_m \to \infty$, the common signs cover each active prefix in a countable library eventually.

Thus dependent localized self-adjoint models with small active support-envelope rank pay $\log(eR_m)$, not $\log(eJ_m D_m)$.

Proof. On $(F_m^{\mathrm{sr}})^c$, put

$$V_{m,h} := \sum_i A_{m,h,i}^2.$$

The support and norm hypotheses give

$$0 \preceq V_{m,h} \preceq \sum_i \rho_{m,h,i}^2 P_{m,h,i}, \qquad V_{m,h} = Q_{m,h} V_{m,h} Q_{m,h}.$$

Thus

$$\operatorname{rank} V_{m,h} \le \operatorname{rank} Q_{m,h}$$

and the realized square-profile norms are bounded by $C_0 n_m / \log(eR_m)$. The hypotheses of Corollary 7.146 therefore hold with exceptional event F_m^{sr}. That corollary gives the common deterministic signing and the sampled-sign conclusion. The summability of $\mathbb{P}(F_m^{\mathrm{sr}})$ and Borel–Cantelli make the conclusions eventual almost surely. □

Corollary 7.316 (Tail-capped support-rank common signs for self-adjoint overlap menus)**.** *Fix $C_0 \geq 1$. For each $m \geq 1$, let $D_m \geq 2$, $n_m \geq 1$, $J_m \geq 1$, and let $R_m \geq 1$. For $1 \leq h \leq J_m$, let*

$$A_{m,h,1}, \dots, A_{m,h,n_m} \in \mathrm{M}_{D_m}(\mathbb{C})$$

be random self-adjoint matrices. Suppose that there are events F_m^{sr}, random orthogonal projections $P_{m,h,i}$, random support-envelope projections $Q_{m,h}$, deterministic caps $0 \leq \rho_{m,h,i} \leq 1$, and cap-tail budgets $\delta_{m,h,i} \geq 0$ such that

$$\mathbb{P}\{\|A_{m,h,i}\|_{\mathrm{op}} > \rho_{m,h,i}\} \leq \delta_{m,h,i}.$$

Outside F_m^{sr}, assume that

$$A_{m,h,i} = P_{m,h,i} A_{m,h,i} P_{m,h,i}, \qquad P_{m,h,i} \leq Q_{m,h}$$

for all $h \leq J_m$ and $i \leq n_m$, and

$$\sum_{h=1}^{J_m} \max\{1, \operatorname{rank} Q_{m,h}\} \leq R_m.$$

Assume also that, outside F_m^{sr},

$$\max_{1 \leq h \leq J_m} \left\| \sum_{i=1}^{n_m} \rho_{m,h,i}^2 P_{m,h,i} \right\|_{\mathrm{op}} \leq \frac{C_0 n_m}{\log(eR_m)}.$$

If

$$\sum_m \left[\mathbb{P}(F_m^{\mathrm{sr}}) + \sum_{h=1}^{J_m} \sum_{i=1}^{n_m} \delta_{m,h,i} \right] < \infty,$$

then, almost surely, for all sufficiently large m, there is one sign vector $\varepsilon^{(m)} \in \{\pm 1\}^{n_m}$ such that

$$\max_{1 \leq h \leq J_m} \left\| \sum_{i=1}^{n_m} \varepsilon_i^{(m)} A_{m,h,i} \right\|_{\mathrm{op}} \leq C(C_0) \sqrt{n_m}.$$

If M_m independent random sign vectors are sampled for the whole active menu and

$$\sum_m 2^{-M_m} < \infty,$$

then the sampled list contains such a common certificate eventually almost surely. If $J_m \to \infty$, the common signs cover each active prefix in a countable library eventually.

Thus unbounded localized self-adjoint libraries retain the support-envelope logarithm $\log(eR_m)$ after paying only summable cap-tail failures.

Proof. For each m, set

$$\Omega_m^{\mathrm{cap}} := \{\|A_{m,h,i}\|_{\mathrm{op}} \leq \rho_{m,h,i} \text{ for all } h \leq J_m,\ i \leq n_m\}.$$

Then

$$\sum_m \mathbb{P}\big((\Omega_m^{\mathrm{cap}})^c\big) \leq \sum_m \sum_{h=1}^{J_m} \sum_{i=1}^{n_m} \delta_{m,h,i} < \infty.$$

On the complement of

$$G_m := F_m^{\mathrm{sr}} \cup (\Omega_m^{\mathrm{cap}})^c$$

the hypotheses of Corollary 7.315 hold with amplitudes $\rho_{m,h,i}$. The summability of $\mathbb{P}(G_m)$ lets that corollary and Borel–Cantelli supply the eventual deterministic common signings, sampled common certificates, and countable-library conclusion. □

Corollary 7.317 (Common signs for self-adjoint coordinate-window Bernstein menus)**.** *Fix $C_0 \geq 1$. For each $m \geq 1$, let $D_m \geq 2$, $J_m, n_m \geq 1$, and let $R_m \geq 1$. For every $1 \leq h \leq J_m$, let*

$$E_{m,h} \subset \{1, \dots, D_m\}$$

be a deterministic coordinate window satisfying

$$\sum_{h=1}^{J_m} \max\{1, |E_{m,h}|\} \leq R_m.$$

For each h, i, let $S_{m,h,i} \subset E_{m,h}$ be a random support. Assume that, for every coordinate $j \in E_{m,h}$, the indicators $\mathbf{1}_{\{j \in S_{m,h,i}\}}$, $1 \leq i \leq n_m$, are independent and

$$\mathbb{P}\{j \in S_{m,h,i}\} \leq p_{m,h,i,j}.$$

Conditional on the supports, let

$$A_{m,h,i} \in \mathrm{M}_{D_m}(\mathbb{C})$$

be arbitrary random self-adjoint matrices satisfying

$$A_{m,h,i} = P_{S_{m,h,i}} A_{m,h,i} P_{S_{m,h,i}}.$$

Let deterministic caps $0 \leq \rho_{m,h,i} \leq 1$ and cap-tail budgets $\delta_{m,h,i} \geq 0$ satisfy

$$\mathbb{P}\{\|A_{m,h,i}\|_{\mathrm{op}} > \rho_{m,h,i}\} \leq \delta_{m,h,i}.$$

Put

$$b_{m,h} := \max_i \rho_{m,h,i}^2, \qquad \mu_{m,h} := \max_{j \in E_{m,h}} \sum_i \rho_{m,h,i}^2 p_{m,h,i,j},$$

and

$$\sigma_{m,h}^2 := \max_{j \in E_{m,h}} \sum_i \rho_{m,h,i}^4 p_{m,h,i,j},$$

with maxima over empty windows interpreted as 0. Suppose that there are $t_{m,h} > 0$ such that

$$\mu_{m,h} + t_{m,h} \leq \frac{C_0 n_m}{\log(eR_m)} \qquad (1 \leq h \leq J_m),$$

and

$$\sum_m \left[\sum_{h=1}^{J_m} |E_{m,h}| \exp\left\{ -\frac{t_{m,h}^2}{2(\sigma_{m,h}^2 + b_{m,h} t_{m,h}/3)} \right\} + \sum_{h=1}^{J_m} \sum_{i=1}^{n_m} \delta_{m,h,i} \right] < \infty,$$

with zero denominator interpreted as zero contribution. Then, almost surely, for all sufficiently large m, there is one sign vector $\varepsilon^{(m)} \in \{\pm 1\}^{n_m}$ such that

$$\max_{1\le h\le J_m}\left\|\sum_{i=1}^{n_m}\varepsilon_i^{(m)}A_{m,h,i}\right\|_{\mathrm{op}} \le C(C_0)\sqrt{n_m}.$$

The same sampled common-certificate and countable-library conclusions as in Corollary 7.316 hold.

In particular, if

$$\mu_{m,h} \le \frac{C_0 n_m}{2\log(eR_m)} \qquad (1 \le h \le J_m),$$

then one may take $t_{m,h} = C_0 n_m/(2\log(eR_m))$, and the displayed Bernstein summand is bounded by

$$\sum_{h=1}^{J_m}|E_{m,h}|\exp\left\{-\frac{3C_0 n_m}{16\log(eR_m)}\right\}.$$

Proof. For each active coordinate define

$$N_{m,h,j} := \sum_i \rho_{m,h,i}^2 \mathbf{1}_{\{j\in S_{m,h,i}\}}.$$

Scalar Bernstein and the union bound over the windows $E_{m,h}$ show that the probability that some $N_{m,h,j}$ exceeds $\mu_{m,h}+t_{m,h}$ is bounded by the displayed Bernstein summand. Off this event, with $Q_{m,h} = P_{E_{m,h}}$ and $P_{m,h,i} = P_{S_{m,h,i}}$, the support-envelope rank and overlap hypotheses of Corollary 7.316 hold. The cap-tail summability is the remaining summand, so that corollary gives the common signing, sampled certificates, and countable-library conclusion.

For the final display, $\sigma_{m,h}^2 \le \mu_{m,h} \le t_{m,h}$ and $b_{m,h} \le 1$, hence the Bernstein exponent is at least $3t_{m,h}/8$, as claimed. □

Corollary 7.318 (Common signs for self-adjoint predictable coordinate-window Freedman menus)**.** *Fix $C_0 \ge 1$. For each $m \ge 1$, let $D_m \ge 2$, $J_m, n_m \ge 1$, and let $R_m \ge 1$. For every $1 \le h \le J_m$, let*

$$E_{m,h} \subset \{1,\dots,D_m\}$$

be a deterministic coordinate window satisfying

$$\sum_{h=1}^{J_m}\max\{1,|E_{m,h}|\} \le R_m.$$

For each m, h, let

$$\mathcal{F}_{m,h,0} \subset \mathcal{F}_{m,h,1} \subset \cdots \subset \mathcal{F}_{m,h,n_m}$$

be a filtration. For each h, i, let

$$S_{m,h,i} \subset E_{m,h}$$

be an $\mathcal{F}_{m,h,i}$-measurable random support. Assume that there are deterministic envelopes $p_{m,h,i,j} \in [0,1]$ such that, for every $j \in E_{m,h}$,

$$\mathbb{P}\{j \in S_{m,h,i} \mid \mathcal{F}_{m,h,i-1}\} \le p_{m,h,i,j}.$$

Conditional on the supports, let

$$A_{m,h,i} \in \mathrm{M}_{D_m}(\mathbb{C})$$

be arbitrary random self-adjoint matrices satisfying

$$A_{m,h,i} = P_{S_{m,h,i}} A_{m,h,i} P_{S_{m,h,i}}.$$

Let deterministic caps $0 \le \rho_{m,h,i} \le 1$ and cap-tail budgets $\delta_{m,h,i} \ge 0$ satisfy

$$\mathbb{P}\{\|A_{m,h,i}\|_{\mathrm{op}} > \rho_{m,h,i}\} \le \delta_{m,h,i}.$$

Put

$$b_{m,h} := \max_i \rho_{m,h,i}^2, \qquad \mu_{m,h} := \max_{j \in E_{m,h}} \sum_i \rho_{m,h,i}^2 p_{m,h,i,j},$$

and

$$\sigma_{m,h}^2 := \max_{j \in E_{m,h}} \sum_i \rho_{m,h,i}^4 p_{m,h,i,j},$$

with maxima over empty windows interpreted as 0. Suppose that there are $t_{m,h} > 0$ such that

$$\mu_{m,h} + t_{m,h} \le \frac{C_0 n_m}{\log(eR_m)} \qquad (1 \le h \le J_m),$$

and

$$\sum_m \left[\sum_{h=1}^{J_m} |E_{m,h}| \exp\left\{ -\frac{t_{m,h}^2}{2(\sigma_{m,h}^2 + b_{m,h} t_{m,h}/3)} \right\} + \sum_{h=1}^{J_m} \sum_{i=1}^{n_m} \delta_{m,h,i} \right] < \infty,$$

with zero denominator interpreted as zero contribution. Then, almost surely, for all sufficiently large m, there is one sign vector $\varepsilon^{(m)} \in \{\pm 1\}^{n_m}$ such that

$$\max_{1 \le h \le J_m} \left\| \sum_{i=1}^{n_m} \varepsilon_i^{(m)} A_{m,h,i} \right\|_{\mathrm{op}} \le C(C_0) \sqrt{n_m}.$$

The same sampled common-certificate and countable-library conclusions as in Corollary 7.316 hold.

In particular, if

$$\mu_{m,h} \le \frac{C_0 n_m}{2 \log(eR_m)} \qquad (1 \le h \le J_m),$$

then one may take $t_{m,h} = C_0 n_m / (2 \log(eR_m))$, and the displayed Freedman summand is bounded by

$$\sum_{h=1}^{J_m} |E_{m,h}| \exp\left\{ -\frac{3 C_0 n_m}{16 \log(eR_m)} \right\}.$$

Proof. For each active coordinate define

$$N_{m,h,j} := \sum_i \rho_{m,h,i}^2 \mathbf{1}_{\{j \in S_{m,h,i}\}}.$$

For fixed m, h, j, the centered partial sums

$$\sum_{i' \le i} \rho_{m,h,i'}^2 \left(\mathbf{1}_{\{j \in S_{m,h,i'}\}} - \mathbb{E}(\mathbf{1}_{\{j \in S_{m,h,i'}\}} \mid \mathcal{F}_{m,h,i'-1}) \right)$$

form a scalar martingale with increments bounded above by $b_{m,h}$ and predictable quadratic variation at most $\sum_i \rho_{m,h,i}^4 p_{m,h,i,j}$. Scalar Freedman and the union bound over the windows $E_{m,h}$ show that the probability that some $N_{m,h,j}$ exceeds $\mu_{m,h} + t_{m,h}$ is bounded by the displayed Freedman summand. Off this event, with $Q_{m,h} = P_{E_{m,h}}$ and $P_{m,h,i} = P_{S_{m,h,i}}$, the support-envelope rank and overlap hypotheses of Corollary 7.316 hold. The cap-tail summability is the remaining summand, so that corollary gives the common signing, sampled certificates, and countable-library conclusion. The half-budget estimate follows from $\sigma_{m,h}^2 \le \mu_{m,h} \le t_{m,h}$ and $b_{m,h} \le 1$, giving exponent at least $3t_{m,h}/8$. □

Corollary 7.319 (Common-sign self-adjoint predictable coordinate-window schedules)**.** *In the setting of Corollary 7.318, suppose that*

$$\mu_{m,h} \le \frac{C_0 n_m}{2 \log(eR_m)} \qquad (1 \le h \le J_m)$$

eventually, and that

$$\sum_m \sum_{h=1}^{J_m} \sum_{i=1}^{n_m} \delta_{m,h,i} < \infty.$$

Then the same common-sign, sampled-certificate, and countable-library conclusions hold whenever

$$\sum_m R_m \exp\left\{ -\frac{3C_0 n_m}{16 \log(eR_m)} \right\} < \infty.$$

In particular, after the displayed half-budget and cap-tail assumptions, the structural tail is summable in each of the following support-rank schedules.

(i) *Arbitrary growth. If, for some $\gamma > 0$,*

$$n_m \ge \frac{16}{3C_0} \log(eR_m)(\log(eR_m) + (1+\gamma)\log(e+m))$$

eventually, then the almost-sure common-sign conclusion holds. If a shared batch of M_m independent random sign vectors is used for the whole active menu, the sampled conclusion follows from $M_m \ge (1+\delta)\log_2(e+m)$ eventually.

(ii) *All support ranks. Suppose the rows are indexed by $R = 2, 3, \ldots$ and the R-th row has active coordinate-window rank parameter at most R. If*

$$n_R \ge a \log^2(eR)$$

eventually for some $a > 32/(3C_0)$, then the common-sign conclusion holds along all support ranks. The sampled conclusion follows if $M_R \geq (1+\delta)\log_2(eR)$ eventually.

(iii) *Sparse support ranks. If $R_m \geq \exp(cm^\kappa)$ eventually for some $c, \kappa > 0$, and*

$$n_m \geq a \log^2(eR_m)$$

eventually for some $a > 16/(3C_0)$, then the common-sign conclusion holds. The sampled conclusion follows from $M_m \geq (1+\delta)\log_2(e+m)$ eventually.

Proof. Under the half-budget hypothesis, the final estimate in Corollary 7.318 bounds the structural Freedman tail by

$$\sum_{h=1}^{J_m} |E_{m,h}| \exp\left\{-\frac{3C_0 n_m}{16\log(eR_m)}\right\} \leq R_m \exp\left\{-\frac{3C_0 n_m}{16\log(eR_m)}\right\}.$$

The cap-tail series is summable by assumption, so the preceding corollary and Borel–Cantelli give the common signing and the inherited sampled and countable-library assertions.

For arbitrary growth, the lower bound on n_m makes the displayed structural tail at most $e^{-1}(e+m)^{-(1+\gamma)}$. For all support ranks, the R-th tail is bounded by

$$R \exp\left\{-\frac{3C_0 a}{16}\log(eR)\right\},$$

which is summable when $3C_0 a/16 > 2$. For sparse support ranks the same estimate is $O(R_m^{1-3C_0 a/16})$, summable under the stretched-exponential growth of R_m as soon as $3C_0 a/16 > 1$. The sampled thresholds are exactly the summability of 2^{-M_m} or 2^{-M_R}. □

Corollary 7.320 (Common signs for self-adjoint subspace-envelope Bernstein menus)**.** *Fix $C_0 \geq 1$. For each $m \geq 1$, let $D_m \geq 2$, $J_m, n_m \geq 1$, and let $R_m \geq 1$. For every $1 \leq h \leq J_m$, let*

$$Q_{m,h} \in \mathrm{M}_{D_m}(\mathbb{C})$$

be a deterministic orthogonal projection satisfying

$$\sum_{h=1}^{J_m} \max\{1, \operatorname{rank} Q_{m,h}\} \leq R_m.$$

For each h, i, let $P_{m,h,i} \leq Q_{m,h}$ be random orthogonal projections, with $P_{m,h,1}, \ldots, P_{m,h,n_m}$ independent for each fixed (m,h). Conditional on these projections, let

$$A_{m,h,i} \in \mathrm{M}_{D_m}(\mathbb{C})$$

be arbitrary random self-adjoint matrices satisfying

$$A_{m,h,i} = P_{m,h,i} A_{m,h,i} P_{m,h,i}.$$

Let deterministic caps $0 \le \rho_{m,h,i} \le 1$ and cap-tail budgets $\delta_{m,h,i} \ge 0$ satisfy

$$\mathbb{P}\{\|A_{m,h,i}\|_{\mathrm{op}} > \rho_{m,h,i}\} \le \delta_{m,h,i}.$$

Put

$$b_{m,h} := \max_i \rho_{m,h,i}^2, \qquad \mu_{m,h} := \left\| \sum_i \rho_{m,h,i}^2 \mathbb{E}P_{m,h,i} \right\|_{\mathrm{op}}.$$

Suppose that there are $t_{m,h} > 0$ such that

$$\mu_{m,h} + t_{m,h} \le \frac{C_0 n_m}{\log(eR_m)} \qquad (1 \le h \le J_m),$$

and

$$\sum_m \left[\sum_{h=1}^{J_m} \operatorname{rank} Q_{m,h} \exp\left\{ -\frac{t_{m,h}^2}{2(\mu_{m,h} + b_{m,h} t_{m,h}/3)} \right\} + \sum_{h=1}^{J_m} \sum_{i=1}^{n_m} \delta_{m,h,i} \right] < \infty,$$

with zero denominator interpreted as zero contribution. Then, almost surely, for all sufficiently large m, there is one sign vector $\varepsilon^{(m)} \in \{\pm 1\}^{n_m}$ such that

$$\max_{1 \le h \le J_m} \left\| \sum_{i=1}^{n_m} \varepsilon_i^{(m)} A_{m,h,i} \right\|_{\mathrm{op}} \le C(C_0)\sqrt{n_m}.$$

The same sampled common-certificate and countable-library conclusions as in Corollary 7.316 hold.

In particular, if

$$\mu_{m,h} \le \frac{C_0 n_m}{2\log(eR_m)} \qquad (1 \le h \le J_m),$$

then one may take $t_{m,h} = C_0 n_m/(2\log(eR_m))$, and the displayed Bernstein summand is bounded by

$$\sum_{h=1}^{J_m} \operatorname{rank} Q_{m,h} \exp\left\{ -\frac{3C_0 n_m}{16\log(eR_m)} \right\}.$$

For Haar-distributed rank-$r_{m,h,i}$ projections inside the range of $Q_{m,h}$, the half-budget condition is

$$\frac{1}{\operatorname{rank} Q_{m,h}} \sum_i \rho_{m,h,i}^2 r_{m,h,i} \le \frac{C_0 n_m}{2\log(eR_m)},$$

with the zero-rank case omitted.

Proof. For fixed m, h, apply matrix Bernstein on the range of $Q_{m,h}$ to

$$Y_{m,h,i} := \rho_{m,h,i}^2 (P_{m,h,i} - \mathbb{E}P_{m,h,i}).$$

Then $\mathbb{E}Y_{m,h,i} = 0$, $\|Y_{m,h,i}\|_{\mathrm{op}} \le b_{m,h}$, and

$$\sum_i \mathbb{E}Y_{m,h,i}^2 \preceq \sum_i \rho_{m,h,i}^2 \mathbb{E}P_{m,h,i},$$

so the variance parameter is at most $\mu_{m,h}$. The displayed Bernstein series therefore bounds the probability that some active h violates

$$\left\|\sum_i \rho_{m,h,i}^2 P_{m,h,i}\right\|_{\mathrm{op}} \le \frac{C_0 n_m}{\log(eR_m)}.$$

Off this event, the support-envelope rank and overlap hypotheses of Corollary 7.316 hold with envelopes $Q_{m,h}$. The cap-tail summability is the remaining summand, so that corollary gives the common signing, sampled certificates, and countable-library conclusion. The half-budget and Haar specializations follow from the same computation as in Corollary 7.325. □

Corollary 7.321 (Common signs for self-adjoint predictable subspace-envelope Freedman menus)**.** *Fix $C_0 \ge 1$. For each $m \ge 1$, let $D_m \ge 2$, $J_m, n_m \ge 1$, and let $R_m \ge 1$. For every $1 \le h \le J_m$, let*

$$Q_{m,h} \in \mathrm{M}_{D_m}(\mathbb{C})$$

be a deterministic orthogonal projection satisfying

$$\sum_{h=1}^{J_m} \max\{1, \operatorname{rank} Q_{m,h}\} \le R_m.$$

For each m, h, let

$$\mathcal{F}_{m,h,0} \subset \mathcal{F}_{m,h,1} \subset \cdots \subset \mathcal{F}_{m,h,n_m}$$

be a filtration. For each h, i, let $P_{m,h,i} \le Q_{m,h}$ be an $\mathcal{F}_{m,h,i}$-measurable random orthogonal projection. Conditional on these projections, let

$$A_{m,h,i} \in \mathrm{M}_{D_m}(\mathbb{C})$$

be arbitrary random self-adjoint matrices satisfying

$$A_{m,h,i} = P_{m,h,i} A_{m,h,i} P_{m,h,i}.$$

Let deterministic caps $0 \le \rho_{m,h,i} \le 1$, cap-tail budgets $\delta_{m,h,i} \ge 0$, deterministic predictable-overlap budgets $v_{m,h} \ge 0$, and events F_m^{pr} be such that

$$\mathbb{P}\{\|A_{m,h,i}\|_{\mathrm{op}} > \rho_{m,h,i}\} \le \delta_{m,h,i},$$

and, outside F_m^{pr}, for every $h \le J_m$,

$$\left\|\sum_i \rho_{m,h,i}^2 \mathbb{E}(P_{m,h,i} \mid \mathcal{F}_{m,h,i-1})\right\|_{\mathrm{op}} \le v_{m,h}.$$

Put

$$b_{m,h} := \max_i \rho_{m,h,i}^2.$$

Suppose that there are $t_{m,h} > 0$ such that

$$v_{m,h} + t_{m,h} \le \frac{C_0 n_m}{\log(eR_m)} \qquad (1 \le h \le J_m),$$

and

$$\sum_m \left[\mathbb{P}(F_m^{\mathrm{pr}}) + \sum_{h=1}^{J_m} 2 \operatorname{rank} Q_{m,h} \exp\left\{ -\frac{t_{m,h}^2}{2(v_{m,h} + b_{m,h} t_{m,h}/3)} \right\} + \sum_{h=1}^{J_m} \sum_{i=1}^{n_m} \delta_{m,h,i} \right] < \infty,$$

with zero denominator interpreted as zero contribution. Then, almost surely, for all sufficiently large m, there is one sign vector $\varepsilon^{(m)} \in \{\pm 1\}^{n_m}$ such that

$$\max_{1 \le h \le J_m} \left\| \sum_{i=1}^{n_m} \varepsilon_i^{(m)} A_{m,h,i} \right\|_{\mathrm{op}} \le C(C_0) \sqrt{n_m}.$$

The same sampled common-certificate and countable-library conclusions as in Corollary 7.316 hold.

In particular, if

$$v_{m,h} \le \frac{C_0 n_m}{2 \log(e R_m)} \qquad (1 \le h \le J_m),$$

then one may take $t_{m,h} = C_0 n_m / (2 \log(e R_m))$, and the Freedman summand is bounded by

$$2 \sum_{h=1}^{J_m} \operatorname{rank} Q_{m,h} \exp\left\{ -\frac{3 C_0 n_m}{16 \log(e R_m)} \right\}.$$

If, conditionally on $\mathcal{F}_{m,h,i-1}$, the projections are Haar-distributed inside the range of $Q_{m,h}$ with predictable ranks $r_{m,h,i}$, the half-budget condition is

$$\frac{1}{\operatorname{rank} Q_{m,h}} \sum_i \rho_{m,h,i}^2 r_{m,h,i} \le \frac{C_0 n_m}{2 \log(e R_m)},$$

with the zero-rank case omitted.

Proof. For fixed m, h, apply the two-sided matrix Freedman inequality on the range of $Q_{m,h}$ to

$$Z_{m,h,i} := \rho_{m,h,i}^2 \left(P_{m,h,i} - \mathbb{E}(P_{m,h,i} \mid \mathcal{F}_{m,h,i-1}) \right).$$

Then $(Z_{m,h,i}, \mathcal{F}_{m,h,i})$ is a self-adjoint martingale-difference sequence. Also $\|Z_{m,h,i}\|_{\mathrm{op}} \le b_{m,h}$, and

$$\sum_i \mathbb{E}(Z_{m,h,i}^2 \mid \mathcal{F}_{m,h,i-1}) \preceq \sum_i \rho_{m,h,i}^2 \mathbb{E}(P_{m,h,i} \mid \mathcal{F}_{m,h,i-1}).$$

Thus, off F_m^{pr}, the displayed Freedman series bounds the probability that some active realized overlap exceeds $v_{m,h} + t_{m,h}$. Off that event and the cap-failure event, the support-envelope rank and overlap hypotheses of Corollary 7.316 hold with envelopes $Q_{m,h}$. The summability assumptions and Borel–Cantelli

therefore give the common signing, sampled certificates, and countable-library conclusion. The half-budget estimate uses $b_{m,h} \le 1$ and $v_{m,h} \le t_{m,h}$. Conditional Haar invariance gives

$$\mathbb{E}(P_{m,h,i} \mid \mathcal{F}_{m,h,i-1}) = \frac{r_{m,h,i}}{\operatorname{rank} Q_{m,h}} Q_{m,h},$$

which is the displayed specialization. □

Corollary 7.322 (Common-sign self-adjoint predictable subspace-envelope schedules)**.** *In the setting of Corollary 7.321, suppose that*

$$v_{m,h} \le \frac{C_0 n_m}{2\log(eR_m)} \qquad (1 \le h \le J_m)$$

eventually, and that

$$\sum_m \mathbb{P}(F_m^{\mathrm{pr}}) < \infty, \qquad \sum_m \sum_{h=1}^{J_m} \sum_{i=1}^{n_m} \delta_{m,h,i} < \infty.$$

Then the same common-sign, sampled-certificate, and countable-library conclusions hold whenever

$$\sum_m 2R_m \exp\left\{-\frac{3C_0 n_m}{16\log(eR_m)}\right\} < \infty.$$

In particular, after the displayed half-budget, predictable-exception, and captail assumptions, the structural tail is summable in each of the following support-rank schedules.

(i) *Arbitrary growth. If, for some $\gamma > 0$,*

$$n_m \ge \frac{16}{3C_0}\log(eR_m)\big(\log(2eR_m) + (1+\gamma)\log(e+m)\big)$$

eventually, then the almost-sure common-sign conclusion holds. If a shared batch of M_m independent random sign vectors is used for the whole active menu, the sampled conclusion follows from $M_m \ge (1+\delta)\log_2(e+m)$ eventually.

(ii) *All support ranks. Suppose the rows are indexed by $R = 2, 3, \dots$ and the R-th row has active subspace-envelope rank parameter at most R. If*

$$n_R \ge a\log^2(eR)$$

eventually for some $a > 32/(3C_0)$, then the common-sign conclusion holds along all support ranks. The sampled conclusion follows if $M_R \ge (1+\delta)\log_2(eR)$ eventually.

(iii) *Sparse support ranks. If $R_m \ge \exp(cm^\kappa)$ eventually for some $c, \kappa > 0$, and*

$$n_m \ge a\log^2(eR_m)$$

eventually for some $a > 16/(3C_0)$, then the common-sign conclusion holds. The sampled conclusion follows from $M_m \ge (1+\delta)\log_2(e+m)$ eventually.

Proof. Under the half-budget hypothesis, the final estimate in Corollary 7.321 bounds the structural Freedman tail by

$$2\sum_{h=1}^{J_m}\operatorname{rank} Q_{m,h}\exp\left\{-\frac{3C_0n_m}{16\log(eR_m)}\right\}\le 2R_m\exp\left\{-\frac{3C_0n_m}{16\log(eR_m)}\right\}.$$

The predictable-exception and cap-tail series are summable by assumption, so the preceding corollary and Borel–Cantelli give the common signing and the inherited sampled and countable-library assertions.

For arbitrary growth, the lower bound on n_m makes the displayed structural tail at most $e^{-1}(e+m)^{-(1+\gamma)}$. For all support ranks, the R-th tail is bounded by

$$2R\exp\left\{-\frac{3C_0a}{16}\log(eR)\right\},$$

which is summable when $3C_0a/16>2$. For sparse support ranks the same estimate is $O(R_m^{1-3C_0a/16})$, summable under the stretched-exponential growth of R_m as soon as $3C_0a/16>1$. The sampled thresholds are exactly the summability of 2^{-M_m} or 2^{-M_R}. □

Corollary 7.323 (Tail-capped common signs for dependent self-adjoint overlap menus)**.** *Fix $C_0\ge 1$. In the setting of Corollary 7.314, replace the pointwise norm bounds by deterministic caps*

$$0\le\rho_{m,h,i}\le 1.$$

Assume that there are cap-tail budgets $\delta_{m,h,i}\ge 0$ such that

$$\mathbb{P}\{\|A_{m,h,i}\|_{\mathrm{op}}>\rho_{m,h,i}\}\le\delta_{m,h,i},$$

and, outside $F^{\mathrm{ov}}_{m,h}$,

$$A_{m,h,i}=P_{m,h,i}A_{m,h,i}P_{m,h,i}\qquad(1\le i\le n_m)$$

and

$$\left\|\sum_{i=1}^{n_m}\rho_{m,h,i}^2P_{m,h,i}\right\|_{\mathrm{op}}\le\frac{C_0n_m}{\log(eD_m^{\mathrm{com}})}.$$

If

$$\sum_m\sum_{h=1}^{J_m}\left[\mathbb{P}(F^{\mathrm{ov}}_{m,h})+\sum_{i=1}^{n_m}\delta_{m,h,i}\right]<\infty,$$

then, almost surely, for all sufficiently large m, one sign vector balances the whole active menu:

$$\max_{1\le h\le J_m}\left\|\sum_{i=1}^{n_m}\varepsilon_i^{(m)}A_{m,h,i}\right\|_{\mathrm{op}}\le C(C_0)\sqrt{n_m}.$$

The same sampled common-certificate and countable-library conclusions as in Corollary 7.314 hold.

The sub-Weibull and finite-moment cap choices in Corollary 7.313 apply verbatim, with the overlap budget measured at the common-sign logarithmic scale $\log(eD_m^{\mathrm{com}})$.

Proof. For each active stream set

$$\Omega_{m,h}^{\mathrm{cap}} := \{\|A_{m,h,i}\|_{\mathrm{op}} \le \rho_{m,h,i} \text{ for all } i \le n_m\}.$$

On the complement of $F_{m,h}^{\mathrm{ov}}$ and on $\Omega_{m,h}^{\mathrm{cap}}$, the hypotheses of Corollary 7.314 hold for stream h at the common dimension D_m^{com}. The exceptional probabilities are bounded by the displayed cap-tail sum, so Borel–Cantelli and the preceding corollary give the eventual common signings. The sampled and cap specializations are inherited from the preceding corollary and the one-stream tail-capped overlap corollary. □

Corollary 7.324 (Weighted inhomogeneous Bernstein coordinate-support ensembles)**.** *Let $D \ge 2$, let $\rho_1, \dots, \rho_n \in [0,1]$ be deterministic, and let $S_1, \dots, S_n$ be random subsets of $\{1, \dots, D\}$. Suppose that for every coordinate j the indicators $\mathbf{1}_{\{j \in S_i\}}$, $i = 1, \dots, n$, are independent and satisfy*

$$\mathbb{P}\{j \in S_i\} \le p_{ij} \qquad (1 \le i \le n,\ 1 \le j \le D)$$

for deterministic $p_{ij} \in [0,1]$. Conditional on these supports, let $A_1, \dots, A_n \in \mathrm{M}_D(\mathbb{C})$ be arbitrary random self-adjoint matrices such that $A_i = P_{S_i} A_i P_{S_i}$ and $\|A_i\|_{\mathrm{op}} \le \rho_i$, where P_{S_i} is the coordinate projection onto S_i. Put

$$b := \max_i \rho_i^2, \qquad \mu := \max_{1 \le j \le D} \sum_{i=1}^n \rho_i^2 p_{ij}, \qquad \sigma^2 := \max_{1 \le j \le D} \sum_{i=1}^n \rho_i^4 p_{ij}.$$

Then for every $t > 0$ and every $C_0 \ge 1$ satisfying

$$\mu + t \le \frac{C_0 n}{\log(eD)},$$

one has

$$\mathbb{P}\left\{\mathrm{disc}(A_1, \dots, A_n) > C(C_0)\sqrt{n}\right\} \le D \exp\left\{-\frac{t^2}{2(\sigma^2 + bt/3)}\right\},$$

with the convention that the exponential term is zero when $\sigma^2 + bt/3 = 0$. In particular, if $p_{ij} \le p$ for all i, j and

$$p \sum_{i=1}^n \rho_i^2 \le \frac{C_0 n}{2 \log(eD)},$$

then

$$\mathbb{P}\left\{\mathrm{disc}(A_1, \dots, A_n) > C(C_0)\sqrt{n}\right\} \le D \exp\left\{-\frac{3C_0 n}{16 \log(eD)}\right\}.$$

Thus inhomogeneous random coordinate-supported self-adjoint matrices with weighted coordinate load $O(n/\log(eD))$ satisfy the Matrix Spencer bound with high probability as soon as the displayed Bernstein tail is small. The model includes Bernoulli supports with nonidentical probabilities, uniformly chosen fixed-size supports of sizes r_i satisfying $D^{-1} \sum_i \rho_i^2 r_i \le C_0 n/(2 \log(eD))$, and the older homogeneous case $p_{ij} = p$, $\rho_i = 1$.

Proof. Let

$$N_j = \sum_i \rho_i^2 \mathbf{1}_{\{j \in S_i\}}.$$

For each fixed j, the summands in N_j are independent, lie in $[0, b]$, have total mean at most μ, and have total variance at most σ^2. Bernstein's inequality gives

$$\mathbb{P}\{N_j > \mu + t\} \le \exp\left\{-\frac{t^2}{2(\sigma^2 + bt/3)}\right\}.$$

A union bound over $j = 1, \dots, D$ shows that, outside the displayed exceptional event,

$$\left\|\sum_i \rho_i^2 P_{S_i}\right\|_{\mathrm{op}} = \max_{1 \le j \le D} N_j \le \mu + t \le \frac{C_0 n}{\log(eD)}.$$

Corollary 7.311 then gives the discrepancy bound. For the final display take $t = C_0 n/(2\log(eD))$. The assumptions give $\mu \le t$, while $\sigma^2 \le p\sum_i \rho_i^4 \le p\sum_i \rho_i^2 \le t$ and $b \le 1$. Hence

$$\frac{t^2}{2(\sigma^2 + bt/3)} \ge \frac{t^2}{2(t + t/3)} = \frac{3t}{8} = \frac{3C_0 n}{16\log(eD)}.$$

For uniformly chosen fixed-size supports, each coordinate is present in each support with probability r_i/D, and independence holds across i. □

Corollary 7.325 (Weighted Matrix-Bernstein random subspace ensembles)**.** *Let $P_1, \dots, P_n$ be independent random orthogonal projections in $\mathrm{M}_D(\mathbb{C})$, let $\rho_1, \dots, \rho_n \in [0, 1]$ be deterministic, and put*

$$b := \max_i \rho_i^2, \qquad \mu := \left\|\sum_{i=1}^n \rho_i^2 \mathbb{E}P_i\right\|_{\mathrm{op}}.$$

Conditional on these projections, let $A_1, \dots, A_n \in \mathrm{M}_D(\mathbb{C})$ be arbitrary random self-adjoint matrices satisfying $A_i = P_i A_i P_i$ and $\|A_i\|_{\mathrm{op}} \le \rho_i$. Then for every $t > 0$ and every $C_0 \ge 1$ satisfying

$$\mu + t \le \frac{C_0 n}{\log(eD)},$$

one has

$$\mathbb{P}\left\{\mathrm{disc}(A_1, \dots, A_n) > C(C_0)\sqrt{n}\right\} \le D \exp\left\{-\frac{t^2}{2(\mu + bt/3)}\right\}.$$

In particular, if P_i are independent Haar-distributed projections of ranks r_i and

$$\frac{1}{D}\sum_{i=1}^n \rho_i^2 r_i \le \frac{C_0 n}{2\log(eD)},$$

then

$$\mathbb{P}\left\{\mathrm{disc}(A_1, \dots, A_n) > C(C_0)\sqrt{n}\right\} \le D \exp\left\{-\frac{3C_0 n}{16\log(eD)}\right\}.$$

Thus random subspace-localized Wigner, GOE/GUE, or arbitrary self-adjoint blocks are Spencer-scale with high probability whenever the weighted average random subspace rank is $O(D/\log(eD))$, uniformly in the ambient dimension apart from the displayed tail. The old unweighted rank condition is the special case $\rho_i = 1$.

Proof. Set $Y_i = \rho_i^2(P_i - \mathbb{E}P_i)$. Then $\mathbb{E}Y_i = 0$, $\|Y_i\|_{\mathrm{op}} \le b$, and

$$\sum_i \mathbb{E}Y_i^2 = \sum_i \rho_i^4 \left(\mathbb{E}P_i - (\mathbb{E}P_i)^2\right) \preceq \sum_i \rho_i^2 \mathbb{E}P_i,$$

so the matrix variance parameter is at most μ. The matrix Bernstein inequality [25] gives

$$\mathbb{P}\left\{\left\|\sum_i \rho_i^2 P_i\right\|_{\mathrm{op}} > \mu + t\right\} \le D \exp\left\{-\frac{t^2}{2(\mu + bt/3)}\right\}.$$

On the complementary event,

$$\left\|\sum_i \rho_i^2 P_i\right\|_{\mathrm{op}} \le \mu + t \le \frac{C_0 n}{\log(eD)},$$

and Corollary 7.311 gives the discrepancy bound. For Haar rank-r_i projections, $\mathbb{E}P_i = (r_i/D)I$, hence $\mu = D^{-1}\sum_i \rho_i^2 r_i$. Taking $t = C_0 n/(2\log(eD))$ and using $\mu \le t$ gives the final displayed failure bound, since $b \le 1$. □

Corollary 7.326 (Random-amplitude weighted support ensembles)**.** *Fix $C_0 \ge 1$. Let $D \ge 2$.*

(a) *Coordinate supports. Let (ρ_i, S_i), $i = 1, \dots, n$, be independent pairs, where $0 \le \rho_i \le 1$ and $S_i \subset \{1, \dots, D\}$. Assume that for deterministic $p_{ij} \in [0,1]$,*

$$\mathbb{P}\{j \in S_i \mid \rho_i\} \le p_{ij} \qquad \text{a.s.}$$

Conditional on the amplitudes and supports, let $A_1, \dots, A_n \in \mathrm{M}_D(\mathbb{C})$ be arbitrary random self-adjoint matrices such that $A_i = P_{S_i} A_i P_{S_i}$ and $\|A_i\|_{\mathrm{op}} \le \rho_i$. Put

$$\mu_c := \max_j \sum_i p_{ij}\mathbb{E}\rho_i^2, \qquad \sigma_c^2 := \max_j \sum_i p_{ij}\mathbb{E}\rho_i^4.$$

Then, for every $t > 0$ with

$$\mu_c + t \le \frac{C_0 n}{\log(eD)},$$

one has

$$\mathbb{P}\left\{\mathrm{disc}(A_1, \dots, A_n) > C(C_0)\sqrt{n}\right\} \le D \exp\left\{-\frac{t^2}{2(\sigma_c^2 + t/3)}\right\}.$$

(b) *Random subspaces. Let (ρ_i, P_i), $i = 1, \dots, n$, be independent pairs, where $0 \le \rho_i \le 1$ and $P_i \in \mathrm{M}_D(\mathbb{C})$ is an orthogonal projection. Conditional on the amplitudes and projections, let $A_1, \dots, A_n \in \mathrm{M}_D(\mathbb{C})$*

be arbitrary random self-adjoint matrices such that $A_i = P_i A_i P_i$ *and* $\|A_i\|_{\mathrm{op}} \le \rho_i$. *Put*

$$\mu_s := \left\| \sum_i \mathbb{E}(\rho_i^2 P_i) \right\|_{\mathrm{op}}.$$

Then, for every $t > 0$ *with*

$$\mu_s + t \le \frac{C_0 n}{\log(eD)},$$

one has

$$\mathbb{P}\left\{ \mathrm{disc}(A_1, \dots, A_n) > C(C_0)\sqrt{n} \right\} \le D \exp\left\{ -\frac{t^2}{2(\mu_s + t/3)} \right\}.$$

In either case, the half-budget condition

$$\mu_c \le \frac{C_0 n}{2\log(eD)} \qquad \text{or} \qquad \mu_s \le \frac{C_0 n}{2\log(eD)}$$

gives the common bound

$$\mathbb{P}\left\{ \mathrm{disc}(A_1, \dots, A_n) > C(C_0)\sqrt{n} \right\} \le D \exp\left\{ -\frac{3C_0 n}{16\log(eD)} \right\}.$$

Consequently, the Borel–Cantelli, all-dimension, growth-rate, and sampled certificate conclusions of Corollaries 7.338, 7.339, 7.341, and 7.342 remain valid with the deterministic overlap budgets replaced by μ_c *or* μ_s. *For uniformly chosen random supports of random sizes* R_i, *or Haar projections of random ranks* R_i, *the homogeneous budget is*

$$\frac{1}{D} \sum_i \mathbb{E}(\rho_i^2 R_i).$$

Proof. In the coordinate case, set

$$W_j = \sum_i \rho_i^2 \mathbf{1}_{\{j \in S_i\}}.$$

For fixed j, the summands are independent, lie in $[0,1]$, have total mean at most $\sum_i p_{ij} \mathbb{E}\rho_i^2$, and have total variance at most

$$\sum_i \mathbb{E}[\rho_i^4 \mathbf{1}_{\{j \in S_i\}}] \le \sum_i p_{ij} \mathbb{E}\rho_i^4.$$

Bernstein's inequality and a union bound over j give

$$\mathbb{P}\left\{ \max_j W_j > \mu_c + t \right\} \le D \exp\left\{ -\frac{t^2}{2(\sigma_c^2 + t/3)} \right\}.$$

On the complementary event,

$$\left\| \sum_i \rho_i^2 P_{S_i} \right\|_{\mathrm{op}} = \max_j W_j \le \frac{C_0 n}{\log(eD)},$$

so Corollary 7.311 gives the discrepancy bound.

In the subspace case, put $X_i = \rho_i^2 P_i$. Then $0 \preceq X_i \preceq I$ and

$$\left\| \sum_i \mathbb{E} X_i \right\|_{\mathrm{op}} = \mu_s.$$

For $Y_i = X_i - \mathbb{E}X_i$, the summands are independent, self-adjoint, and $\|Y_i\|_{\mathrm{op}} \le 1$. Moreover,

$$\sum_i \mathbb{E}Y_i^2 = \sum_i (\mathbb{E}X_i^2 - (\mathbb{E}X_i)^2) \preceq \sum_i \mathbb{E}X_i^2 \preceq \sum_i \mathbb{E}X_i,$$

because $X_i^2 = \rho_i^4 P_i \preceq \rho_i^2 P_i = X_i$. Matrix Bernstein gives

$$\mathbb{P}\left\{ \left\| \sum_i X_i \right\|_{\mathrm{op}} > \mu_s + t \right\} \le D \exp\left\{ -\frac{t^2}{2(\mu_s + t/3)} \right\}.$$

On the complementary event, Corollary 7.311 again applies.

For the half-budget display, take $t = C_0 n/(2\log(eD))$. In the coordinate case $\sigma_c^2 \le \mu_c \le t$, and in the subspace case $\mu_s \le t$, so both exponents are at least $3t/8 = 3C_0 n/(16\log(eD))$. The array and sampled-certificate assertions are the same Borel–Cantelli applications as in the cited corollaries, using this identical row-failure bound. □

Corollary 7.327 (Random-amplitude block-support ensembles)**.** *Fix $C_0 \ge 1$. Let $D \ge 2$, and let $\Pi_1, \dots, \Pi_B$ be pairwise orthogonal projections in $\mathrm{M}_D(\mathbb{C})$ with $\sum_{\alpha=1}^B \Pi_\alpha = I$. Let (ρ_i, T_i), $i = 1, \dots, n$, be independent pairs, where $0 \le \rho_i \le 1$ and $T_i \subset \{1, \dots, B\}$. Assume that for deterministic $q_{i\alpha} \in [0,1]$,*

$$\mathbb{P}\{\alpha \in T_i \mid \rho_i\} \le q_{i\alpha} \qquad a.s.$$

Put $P_{T_i} = \sum_{\alpha \in T_i} \Pi_\alpha$. Conditional on the amplitudes and selected blocks, let $A_1, \dots, A_n \in \mathrm{M}_D(\mathbb{C})$ be arbitrary random self-adjoint matrices such that

$$A_i = P_{T_i} A_i P_{T_i}, \qquad \|A_i\|_{\mathrm{op}} \le \rho_i.$$

Define

$$\mu_B := \max_{\alpha \le B} \sum_i q_{i\alpha} \mathbb{E}\rho_i^2, \qquad \sigma_B^2 := \max_{\alpha \le B} \sum_i q_{i\alpha} \mathbb{E}\rho_i^4.$$

Then, for every $t > 0$ with

$$\mu_B + t \le \frac{C_0 n}{\log(eD)},$$

one has

$$\mathbb{P}\left\{ \mathrm{disc}(A_1, \dots, A_n) > C(C_0)\sqrt{n} \right\} \le B \exp\left\{ -\frac{t^2}{2(\sigma_B^2 + t/3)} \right\}.$$

In particular, if

$$\mu_B \le \frac{C_0 n}{2\log(eD)},$$

then

$$\mathbb{P}\left\{\operatorname{disc}(A_1,\dots,A_n) > C(C_0)\sqrt{n}\right\} \le B\exp\left\{-\frac{3C_0 n}{16\log(eD)}\right\}.$$

For uniformly chosen block sets of random sizes R_i, the homogeneous budget is

$$\frac{1}{B}\sum_i \mathbb{E}(\rho_i^2 R_i).$$

Proof. Set

$$W_\alpha = \sum_i \rho_i^2 \mathbf{1}_{\{\alpha\in T_i\}}.$$

For fixed α, the summands are independent, lie in $[0,1]$, have total mean at most $\sum_i q_{i\alpha}\mathbb{E}\rho_i^2$, and have total variance at most

$$\sum_i \mathbb{E}[\rho_i^4 \mathbf{1}_{\{\alpha\in T_i\}}] \le \sum_i q_{i\alpha}\mathbb{E}\rho_i^4.$$

Scalar Bernstein and a union bound over the B blocks give

$$\mathbb{P}\left\{\max_{\alpha\le B} W_\alpha > \mu_B + t\right\} \le B\exp\left\{-\frac{t^2}{2(\sigma_B^2 + t/3)}\right\}.$$

On the complementary event,

$$\left\|\sum_i \rho_i^2 P_{T_i}\right\|_{\mathrm{op}} = \left\|\sum_{\alpha=1}^{B} W_\alpha \Pi_\alpha\right\|_{\mathrm{op}} = \max_{\alpha\le B} W_\alpha \le \frac{C_0 n}{\log(eD)}.$$

Corollary 7.311 gives the discrepancy bound. The half-budget display follows with $t = C_0 n/(2\log(eD))$, since $\sigma_B^2 \le \mu_B \le t$. If, conditional on R_i and ρ_i, the set T_i is uniformly chosen among block sets of size R_i, then $\mathbb{P}\{\alpha\in T_i \mid R_i,\rho_i\} = R_i/B$, giving the final budget. □

Corollary 7.328 (Random-profile random-amplitude support ensembles)**.** *Fix $C_0 \ge 1$ and $D \ge 2$. The following two random-profile variants of the preceding support theorems hold.*

(a) *Coordinate profiles. Let (ρ_i, Q_i, S_i), $i = 1,\dots,n$, be independent triples, where $0 \le \rho_i \le 1$, $Q_i = (Q_{ij})_{j=1}^D \in [0,1]^D$, and $S_i \subset \{1,\dots,D\}$. Assume that*

$$\mathbb{P}\{j\in S_i \mid \rho_i, Q_i\} \le Q_{ij} \qquad a.s.$$

Conditional on the amplitudes and supports, let $A_1,\dots,A_n \in \mathrm{M}_D(\mathbb{C})$ be arbitrary random self-adjoint matrices such that $A_i = P_{S_i}A_iP_{S_i}$ and $\|A_i\|_{\mathrm{op}} \le \rho_i$. Put

$$\mu_c^{\mathrm{prof}} := \max_j \sum_i \mathbb{E}(\rho_i^2 Q_{ij}), \qquad (\sigma_c^{\mathrm{prof}})^2 := \max_j \sum_i \mathbb{E}(\rho_i^4 Q_{ij}).$$

If $t > 0$ and

$$\mu_c^{\mathrm{prof}} + t \le \frac{C_0 n}{\log(eD)},$$

then

$$\mathbb{P}\left\{\operatorname{disc}(A_1,\dots,A_n) > C(C_0)\sqrt{n}\right\} \le D\exp\left\{-\frac{t^2}{2((\sigma_c^{\mathrm{prof}})^2 + t/3)}\right\}.$$

(b) *Block profiles. Let* $\Pi_1,\dots,\Pi_B$ *be pairwise orthogonal projections in* $\mathrm{M}_D(\mathbb{C})$ *with* $\sum_{\alpha=1}^{B}\Pi_\alpha = I$. *Let* (ρ_i, Q_i, T_i), $i = 1,\dots,n$, *be independent triples, where* $0 \le \rho_i \le 1$, $Q_i = (Q_{i\alpha})_{\alpha=1}^{B} \in [0,1]^B$, *and* $T_i \subset \{1,\dots,B\}$. *Assume that*

$$\mathbb{P}\{\alpha \in T_i \mid \rho_i, Q_i\} \le Q_{i\alpha} \qquad a.s.$$

Put $P_{T_i} = \sum_{\alpha\in T_i}\Pi_\alpha$. *Conditional on the amplitudes and selected blocks, let* $A_1,\dots,A_n \in \mathrm{M}_D(\mathbb{C})$ *be arbitrary random self-adjoint matrices such that* $A_i = P_{T_i}A_iP_{T_i}$ *and* $\|A_i\|_{\mathrm{op}} \le \rho_i$. *Put*

$$\mu_B^{\mathrm{prof}} := \max_{\alpha\le B}\sum_i \mathbb{E}(\rho_i^2 Q_{i\alpha}), \qquad (\sigma_B^{\mathrm{prof}})^2 := \max_{\alpha\le B}\sum_i \mathbb{E}(\rho_i^4 Q_{i\alpha}).$$

If $t > 0$ *and*

$$\mu_B^{\mathrm{prof}} + t \le \frac{C_0 n}{\log(eD)},$$

then

$$\mathbb{P}\left\{\operatorname{disc}(A_1,\dots,A_n) > C(C_0)\sqrt{n}\right\} \le B\exp\left\{-\frac{t^2}{2((\sigma_B^{\mathrm{prof}})^2 + t/3)}\right\}.$$

In either part, the half-budget condition

$$\mu_c^{\mathrm{prof}} \le \frac{C_0 n}{2\log(eD)} \qquad \text{or} \qquad \mu_B^{\mathrm{prof}} \le \frac{C_0 n}{2\log(eD)}$$

gives the corresponding row failure

$$N\exp\left\{-\frac{3C_0 n}{16\log(eD)}\right\},$$

where $N = D$ *in the coordinate case and* $N = B$ *in the block case. Therefore the all-dimension, sparse, arbitrary-growth, and sampled-certificate schedules in Corollaries 7.343 and 7.344 remain valid after replacing the deterministic profile budgets by* μ_c^{prof} *or* μ_B^{prof}. *In particular, uniformly chosen coordinate supports of random sizes* R_i, *or uniformly chosen block sets of random sizes* R_i, *use the effective budgets*

$$\frac{1}{D}\sum_i \mathbb{E}(\rho_i^2 R_i) \qquad \text{and} \qquad \frac{1}{B}\sum_i \mathbb{E}(\rho_i^2 R_i),$$

respectively, with no independence required between R_i *and* ρ_i *inside a row variable.*

Proof. In the coordinate case, set

$$W_j = \sum_i \rho_i^2 \mathbf{1}_{\{j\in S_i\}}.$$

For fixed j, the summands are independent, lie in $[0,1]$, and satisfy

$$\mathbb{E}[\rho_i^2 \mathbf{1}_{\{j\in S_i\}}] \le \mathbb{E}(\rho_i^2 Q_{ij}), \qquad \mathbb{E}[\rho_i^4 \mathbf{1}_{\{j\in S_i\}}] \le \mathbb{E}(\rho_i^4 Q_{ij}).$$

Scalar Bernstein and a union bound over j give

$$\mathbb{P}\left\{\max_j W_j > \mu_c^{\mathrm{prof}} + t\right\} \le D\exp\left\{-\frac{t^2}{2((\sigma_c^{\mathrm{prof}})^2 + t/3)}\right\}.$$

On the complementary event,

$$\left\|\sum_i \rho_i^2 P_{S_i}\right\|_{\mathrm{op}} = \max_j W_j \le \frac{C_0 n}{\log(eD)},$$

and Corollary 7.311 gives the discrepancy bound.

The block case is identical with

$$W_\alpha = \sum_i \rho_i^2 \mathbf{1}_{\{\alpha\in T_i\}}$$

and the union bound over $\alpha = 1,\dots,B$; on the good event

$$\left\|\sum_i \rho_i^2 P_{T_i}\right\|_{\mathrm{op}} = \left\|\sum_{\alpha=1}^B W_\alpha \Pi_\alpha\right\|_{\mathrm{op}} = \max_{\alpha\le B} W_\alpha.$$

For the half-budget statement take $t = C_0 n/(2\log(eD))$ and use $\rho_i^4 \le \rho_i^2$. The schedule and sampled-certificate assertions are the same Borel–Cantelli and conditional sign-search arguments as in the cited corollaries, using these identical row tails. For uniform random sizes, take $Q_{ij} = R_i/D$ or $Q_{i\alpha} = R_i/B$. □

Corollary 7.329 (Predictable random-profile support ensembles)**.** *Fix $C_0 \ge 1$, $D \ge 2$, and a filtration $(\mathcal{F}_i)_{i=0}^n$. The following two predictable support models have the same Spencer-scale conclusion.*

(a) *Coordinate profiles. Let ρ_i and $Q_i = (Q_{ij})_{j=1}^D$ be $\mathcal{F}_{i-1}$-measurable, with $0 \le \rho_i \le 1$ and $Q_i \in [0,1]^D$. Let $S_i \subset \{1,\dots,D\}$ be $\mathcal{F}_i$-measurable and assume that*

$$\mathbb{P}\{j \in S_i \mid \mathcal{F}_{i-1}\} \le Q_{ij} \qquad a.s.$$

Let $A_1,\dots,A_n \in \mathrm{M}_D(\mathbb{C})$ be arbitrary random self-adjoint matrices such that

$$A_i = P_{S_i} A_i P_{S_i}, \qquad \|A_i\|_{\mathrm{op}} \le \rho_i.$$

Put

$$U = \max_{j\le D}\sum_i \rho_i^2 Q_{ij}, \qquad V = \max_{j\le D}\sum_i \rho_i^4 Q_{ij}, \qquad N = D.$$

(b) *Block profiles. Let $\Pi_1,\dots,\Pi_B$ be pairwise orthogonal projections in $\mathrm{M}_D(\mathbb{C})$ with $\sum_{\alpha=1}^B \Pi_\alpha = I$. Let ρ_i and $Q_i = (Q_{i\alpha})_{\alpha=1}^B$ be $\mathcal{F}_{i-1}$-measurable, with $0 \le \rho_i \le 1$ and $Q_i \in [0,1]^B$. Let $T_i \subset \{1,\dots,B\}$ be $\mathcal{F}_i$-measurable and assume that*

$$\mathbb{P}\{\alpha \in T_i \mid \mathcal{F}_{i-1}\} \le Q_{i\alpha} \qquad a.s.$$

Put $P_{T_i} = \sum_{\alpha \in T_i} \Pi_\alpha$, *and let* $A_1, \dots, A_n \in \mathrm{M}_D(\mathbb{C})$ *be arbitrary random self-adjoint matrices such that*

$$A_i = P_{T_i} A_i P_{T_i}, \qquad \|A_i\|_{\mathrm{op}} \le \rho_i.$$

Put

$$U = \max_{\alpha \le B} \sum_i \rho_i^2 Q_{i\alpha}, \qquad V = \max_{\alpha \le B} \sum_i \rho_i^4 Q_{i\alpha}, \qquad N = B.$$

In either part, if deterministic μ, σ^2, t *satisfy*

$$\mu + t \le \frac{C_0 n}{\log(eD)},$$

then

$$\mathbb{P}\left\{\mathrm{disc}(A_1, \dots, A_n) > C(C_0)\sqrt{n}\right\} \le \mathbb{P}\{U > \mu \text{ or } V > \sigma^2\} + N \exp\left\{-\frac{t^2}{2(\sigma^2 + t/3)}\right\}.$$

In particular, if

$$U \le \frac{C_0 n}{2\log(eD)} \qquad \text{a.s.},$$

then

$$\mathbb{P}\left\{\mathrm{disc}(A_1, \dots, A_n) > C(C_0)\sqrt{n}\right\} \le N \exp\left\{-\frac{3C_0 n}{16\log(eD)}\right\}.$$

If $\eta^{(1)}, \dots, \eta^{(M)}$ *are independent uniform sign vectors in* $\{\pm 1\}^n$, *independent of the matrix array, then after enlarging* $C(C_0)$ *if necessary, the same right-hand sides gain only* $+2^{-M}$ *for the event that no sampled sign is a* $C(C_0)\sqrt{n}$ *certificate. For predictably chosen uniform coordinate supports, or predictably chosen uniform block sets, of sizes* R_i, *the realized half-budget is respectively*

$$\frac{1}{D}\sum_i \rho_i^2 R_i \qquad \text{or} \qquad \frac{1}{B}\sum_i \rho_i^2 R_i.$$

Consequently, triangular arrays of either predictable model have eventual almost-sure signings, and eventual sampled certificates, whenever the displayed row-failure bounds and the sampled-search terms are summable.

Proof. We prove the coordinate case; the block case is identical after replacing coordinates by blocks. For fixed j, set

$$X_i = \rho_i^2 \mathbf{1}_{\{j \in S_i\}}, \qquad m_i = \mathbb{E}(X_i \mid \mathcal{F}_{i-1}).$$

Then $0 \le X_i \le 1$, $m_i \le \rho_i^2 Q_{ij}$, and

$$\sum_i \mathbb{E}[(X_i - m_i)^2 \mid \mathcal{F}_{i-1}] \le \sum_i \mathbb{E}(X_i^2 \mid \mathcal{F}_{i-1}) \le \sum_i \rho_i^4 Q_{ij}.$$

On $\{U \le \mu,\ V \le \sigma^2\}$, if $\sum_i X_i > \mu + t$, then $\sum_i (X_i - m_i) > t$, while the scalar predictable quadratic variation is at most σ^2. The scalar Freedman inequality, as the one-dimensional case of [25], gives

$$\mathbb{P}\left\{\sum_i X_i > \mu + t,\ U \le \mu,\ V \le \sigma^2\right\} \le \exp\left\{-\frac{t^2}{2(\sigma^2 + t/3)}\right\}.$$

Union over j. On the complementary event to the resulting occupancy failure and the budget exception,

$$\left\|\sum_i \rho_i^2 P_{S_i}\right\|_{\mathrm{op}} = \max_j \sum_i \rho_i^2 \mathbf{1}_{\{j \in S_i\}} \le \frac{C_0 n}{\log(eD)}.$$

Corollary 7.311 gives the discrepancy bound.

In the block case the same argument controls

$$W_\alpha = \sum_i \rho_i^2 \mathbf{1}_{\{\alpha \in T_i\}},$$

and the orthogonality of the blocks gives

$$\left\|\sum_i \rho_i^2 P_{T_i}\right\|_{\mathrm{op}} = \left\|\sum_{\alpha=1}^{B} W_\alpha \Pi_\alpha\right\|_{\mathrm{op}} = \max_{\alpha \le B} W_\alpha.$$

For the half-budget estimate take $\mu = t = C_0 n/(2\log(eD))$ and use $V \le U$. The sampled-sign statement follows by conditioning on the weighted-overlap event and applying Corollary 7.144. The triangular-array assertion is the first Borel–Cantelli lemma applied to the row failures. The uniform-size budgets follow from $Q_{ij} = R_i/D$ and $Q_{i\alpha} = R_i/B$. □

Corollary 7.330 (Predictable-profile support schedules)**.** *Fix $C_0 \ge 1$. For each $m \ge 1$, let $D_m \ge 2$, put $\ell_m = \log(eD_m)$, and let $A_{m,1}, \dots, A_{m,n_m} \in \mathrm{M}_{D_m}(\mathbb{C})$ be a row satisfying one of the two predictable support models of Corollary 7.329. In the coordinate case put $N_m = D_m$ and let U_m be the coordinate predictable overlap budget; in the block case put $N_m = B_m$ and let U_m be the block predictable overlap budget. Assume that*

$$\sum_m \mathbb{P}\left\{U_m > \frac{C_0 n_m}{2\ell_m}\right\} < \infty$$

and

$$\sum_m N_m \exp\left\{-\frac{3C_0 n_m}{16\ell_m}\right\} < \infty.$$

Then, with probability one, for all sufficiently large m,

$$\operatorname{disc}(A_{m,1}, \dots, A_{m,n_m}) \le C(C_0)\sqrt{n_m}.$$

If each row is also supplied with M_m independent uniform sign vectors in $\{\pm 1\}^{n_m}$, independent of the matrix array, and $\sum_m 2^{-M_m} < \infty$, then after enlarging $C(C_0)$, almost surely for all sufficiently large m one sampled sign certifies the same bound.

In particular, the structural tail is summable in each of the following schedules, provided the displayed predictable-budget exception is summable.

(i) *Arbitrary growth. If, for some* $\gamma > 0$,

$$n_m \geq \frac{16}{3C_0}\,\ell_m\big(\log(eN_m) + (1+\gamma)\log(e+m)\big)$$

eventually, then the almost-sure signing conclusion holds. The sampled conclusion follows if $M_m \geq (1+\delta)\log_2(e+m)$ *eventually for some* $\delta > 0$.

(ii) *All integer dimensions. Suppose the rows are indexed by* $D = 2, 3, \ldots$, *with* $D_m = D$, *and* $N_D \leq (eD)^{\Gamma}$ *eventually for some* $\Gamma \geq 0$. *If*

$$n_D \geq a\log^2(eD)$$

eventually for some $a > 16(\Gamma+1)/(3C_0)$, *then the almost-sure signing conclusion holds. The sampled conclusion follows if* $M_D \geq (1+\delta)\log_2(eD)$ *eventually.*

(iii) *Sparse dimensions. If* $D_m \geq \exp(cm^{\kappa})$ *eventually for some* $c, \kappa > 0$, $N_m \leq (eD_m)^{\Gamma}$ *eventually, and*

$$n_m \geq a\log^2(eD_m)$$

eventually for some $a > 16\Gamma/(3C_0)$, *then the almost-sure signing conclusion holds. The sampled conclusion follows if* $M_m \geq (1+\delta)\log_2(e+m)$ *eventually.*

For predictably chosen uniform coordinate supports or uniform block sets of sizes $R_{m,i}$, *the predictable-budget exception above is respectively*

$$\left\{\frac{1}{D_m}\sum_i \rho_{m,i}^2 R_{m,i} > \frac{C_0 n_m}{2\ell_m}\right\} \qquad \text{or} \qquad \left\{\frac{1}{B_m}\sum_i \rho_{m,i}^2 R_{m,i} > \frac{C_0 n_m}{2\ell_m}\right\}.$$

Proof. On the event $U_m \leq C_0 n_m/(2\ell_m)$, the corresponding variance budget satisfies $V_m \leq U_m$. Applying Corollary 7.329 to the m-th row with

$$\mu = t = \frac{C_0 n_m}{2\ell_m}, \qquad \sigma^2 = \frac{C_0 n_m}{2\ell_m},$$

gives the row failure

$$\mathbb{P}\left\{U_m > \frac{C_0 n_m}{2\ell_m}\right\} + N_m \exp\left\{-\frac{3C_0 n_m}{16\ell_m}\right\}.$$

The first Borel–Cantelli lemma proves the almost-sure signing conclusion, and the sampled version follows by adding the summable 2^{-M_m} row failures from the same corollary.

For arbitrary growth, the displayed lower bound gives

$$N_m \exp\left\{-\frac{3C_0 n_m}{16\ell_m}\right\} \leq e^{-1}(e+m)^{-(1+\gamma)}.$$

For all integer dimensions,

$$N_D \exp\left\{-\frac{3C_0 n_D}{16\log(eD)}\right\} \leq (eD)^{\Gamma - 3C_0 a/16},$$

which is summable when $3C_0a/16 > \Gamma + 1$. Along sparse schedules the same bound is at most

$$\exp\left\{-(3C_0a/16 - \Gamma)\log(eD_m)\right\},$$

which is summable under the stated stretched-exponential lower bound on D_m. The sampled-search thresholds are immediate from the summability of $(e+m)^{-(1+\delta)}$ or $(eD)^{-(1+\delta)}$. The final display is just the uniform-size specialization from Corollary 7.329. □

Corollary 7.331 (Finite menus of predictable-profile support schedules)**.** *Fix $C_0 \geq 1$. For each $m \geq 1$, let $D_m \geq 2$, put $\ell_m = \log(eD_m)$, and let $\mathcal{H}_m$ be a finite menu. For each $h \in \mathcal{H}_m$, let*

$$A_{m,1}^{(h)}, \dots, A_{m,n_m}^{(h)} \in \mathrm{M}_{D_m}(\mathbb{C})$$

be a row satisfying one of the two predictable support models of Corollary 7.329, with its own filtration allowed. In the coordinate case put $N_{m,h} = D_m$ and let $U_{m,h}$ be the coordinate predictable overlap budget. In the block case put $N_{m,h} = B_{m,h}$, the number of structural blocks for that menu element, and let $U_{m,h}$ be the block predictable overlap budget. Set

$$L_m := \sum_{h \in \mathcal{H}_m} N_{m,h}.$$

Assume that

$$\sum_m \sum_{h \in \mathcal{H}_m} \mathbb{P}\left\{U_{m,h} > \frac{C_0 n_m}{2\ell_m}\right\} < \infty$$

and

$$\sum_m L_m \exp\left\{-\frac{3C_0 n_m}{16\ell_m}\right\} < \infty.$$

Then, with probability one, for all sufficiently large m, every $h \in \mathcal{H}_m$ has a Spencer-scale signing:

$$\operatorname{disc}(A_{m,1}^{(h)}, \dots, A_{m,n_m}^{(h)}) \leq C(C_0)\sqrt{n_m}.$$

If each menu element is also supplied with $M_{m,h}$ independent uniform sign vectors in $\{\pm 1\}^{n_m}$, independent of the corresponding row, and

$$\sum_m \sum_{h \in \mathcal{H}_m} 2^{-M_{m,h}} < \infty,$$

then after enlarging $C(C_0)$, almost surely for all sufficiently large m every menu element has one sampled sign certifying the same bound. If a common batch of M_m signs is reused for all $h \in \mathcal{H}_m$, independent of the whole m-th menu, it is enough that

$$\sum_m |\mathcal{H}_m|\, 2^{-M_m} < \infty.$$

In particular, once the displayed predictable-budget exceptions are summable, the structural menu tail is summable in each of the following schedules.

(i) *Arbitrary growth. If, for some* $\gamma > 0$,

$$n_m \geq \frac{16}{3C_0}\,\ell_m\big(\log(eL_m) + (1+\gamma)\log(e+m)\big)$$

eventually, then the simultaneous almost-sure signing conclusion holds. With a common sign batch, the simultaneous sampled conclusion follows if

$$M_m \geq \log_2(e|\mathcal{H}_m|) + (1+\delta)\log_2(e+m)$$

eventually for some $\delta > 0$.

(ii) *All integer dimensions. Suppose the rows are indexed by* $D = 2, 3, \ldots$, *with* $D_m = D$, *and* $L_D \leq (eD)^\Gamma$ *eventually for some* $\Gamma \geq 0$. *If*

$$n_D \geq a\log^2(eD)$$

eventually for some $a > 16(\Gamma+1)/(3C_0)$, *then the simultaneous almost-sure signing conclusion holds.*

(iii) *Sparse dimensions. If* $D_m \geq \exp(cm^\kappa)$ *eventually for some* $c, \kappa > 0$, $L_m \leq (eD_m)^\Gamma$ *eventually, and*

$$n_m \geq a\log^2(eD_m)$$

eventually for some $a > 16\Gamma/(3C_0)$, *then the simultaneous almost-sure signing conclusion holds.*

For predictably chosen uniform coordinate supports or uniform block sets of sizes $R_{m,i}^{(h)}$, *the corresponding menu budget exception is respectively*

$$\left\{\frac{1}{D_m}\sum_i (\rho_{m,i}^{(h)})^2 R_{m,i}^{(h)} > \frac{C_0 n_m}{2\ell_m}\right\} \qquad \text{or} \qquad \left\{\frac{1}{B_{m,h}}\sum_i (\rho_{m,i}^{(h)})^2 R_{m,i}^{(h)} > \frac{C_0 n_m}{2\ell_m}\right\}.$$

Proof. For fixed m and h, Corollary 7.329 with

$$\mu = t = \sigma^2 = \frac{C_0 n_m}{2\ell_m}$$

gives row failure at most

$$\mathbb{P}\left\{U_{m,h} > \frac{C_0 n_m}{2\ell_m}\right\} + N_{m,h}\exp\left\{-\frac{3C_0 n_m}{16\ell_m}\right\}.$$

Union over $h \in \mathcal{H}_m$, then apply the first Borel–Cantelli lemma. The sampled-sign statements follow by adding the conditional search failure $2^{-M_{m,h}}$ for each menu element, or by union bounding a reused common batch as $|\mathcal{H}_m|2^{-M_m}$.

For arbitrary growth, the displayed lower bound gives

$$L_m\exp\left\{-\frac{3C_0 n_m}{16\ell_m}\right\} \leq e^{-1}(e+m)^{-(1+\gamma)}.$$

The common sampled-batch condition gives $|\mathcal{H}_m|2^{-M_m} \leq e^{-1}(e+m)^{-(1+\delta)}$. The all-integer and sparse cases are the same estimates with $L_m \leq (eD_m)^\Gamma$. The final display is the uniform-size specialization from Corollary 7.329. □

Corollary 7.332 (Exhaustive countable menus of predictable-profile support schedules)**.** *Fix $C_0 \geq 1$. For each $m \geq 1$, let $D_m \geq 2$, put $\ell_m = \log(eD_m)$, and let*

$$J_1 \leq J_2 \leq \cdots, \qquad J_m \to \infty,$$

be a deterministic exhaustion of the countable menu $\mathbb{N} = \{1, 2, \ldots\}$. For each m and $h \geq 1$, let

$$A^{(h)}_{m,1}, \ldots, A^{(h)}_{m,n_m} \in \mathrm{M}_{D_m}(\mathbb{C})$$

be a row satisfying one of the two predictable support models of Corollary 7.329, with its own filtration allowed. In the coordinate case put $N_{m,h} = D_m$ and let $U_{m,h}$ be the coordinate predictable overlap budget. In the block case put $N_{m,h} = B_{m,h}$, the number of structural blocks for that menu element, and let $U_{m,h}$ be the block predictable overlap budget. Set

$$L^{(J)}_m := \sum_{1 \leq h \leq J_m} N_{m,h}.$$

Assume that

$$\sum_m \sum_{1 \leq h \leq J_m} \mathbb{P}\left\{U_{m,h} > \frac{C_0 n_m}{2\ell_m}\right\} < \infty$$

and

$$\sum_m L^{(J)}_m \exp\left\{-\frac{3C_0 n_m}{16\ell_m}\right\} < \infty.$$

Then, with probability one, for all sufficiently large m, every model in the active prefix $1 \leq h \leq J_m$ has a Spencer-scale signing:

$$\operatorname{disc}(A^{(h)}_{m,1}, \ldots, A^{(h)}_{m,n_m}) \leq C(C_0)\sqrt{n_m} \qquad (1 \leq h \leq J_m).$$

Consequently, for each fixed h, the same bound holds for all sufficiently large m. If the h-th menu element is also supplied with $M_{m,h}$ independent uniform sign vectors in $\{\pm 1\}^{n_m}$, independent of the corresponding row, and

$$\sum_m \sum_{1 \leq h \leq J_m} 2^{-M_{m,h}} < \infty,$$

then after enlarging $C(C_0)$, almost surely every active prefix element has one sampled sign certifying the same bound for all sufficiently large m. If a common batch of M_m signs is reused for all $1 \leq h \leq J_m$, independent of the whole active prefix, it is enough that

$$\sum_m J_m\, 2^{-M_m} < \infty.$$

In particular, once the displayed predictable-budget exceptions over the active prefixes are summable, the structural prefix tail is summable in each of the following schedules.

(i) *Arbitrary growth. If, for some $\gamma > 0$,*

$$n_m \geq \frac{16}{3C_0}\,\ell_m\big(\log(eL^{(J)}_m) + (1+\gamma)\log(e+m)\big)$$

eventually, then the active-prefix and fixed-model almost-sure signing conclusions hold. With a common sign batch, the sampled conclusion follows if

$$M_m \geq \log_2(eJ_m) + (1+\delta)\log_2(e+m)$$

eventually for some $\delta > 0$.

(ii) *All integer dimensions. Suppose the rows are indexed by $D = 2, 3, \ldots$, with $D_m = D$, $J_D \to \infty$, and $L_D^{(J)} \leq (eD)^\Gamma$ eventually for some $\Gamma \geq 0$. If*

$$n_D \geq a \log^2(eD)$$

eventually for some $a > 16(\Gamma+1)/(3C_0)$, then the active-prefix and fixed-model almost-sure signing conclusions hold.

(iii) *Sparse dimensions. If $D_m \geq \exp(cm^\kappa)$ eventually for some $c, \kappa > 0$, $J_m \to \infty$, $L_m^{(J)} \leq (eD_m)^\Gamma$ eventually, and*

$$n_m \geq a \log^2(eD_m)$$

eventually for some $a > 16\Gamma/(3C_0)$, then the active-prefix and fixed-model almost-sure signing conclusions hold.

For predictably chosen uniform coordinate supports or uniform block sets of sizes $R_{m,i}^{(h)}$, the prefix budget exceptions are the corresponding events

$$\left\{ \frac{1}{D_m} \sum_i (\rho_{m,i}^{(h)})^2 R_{m,i}^{(h)} > \frac{C_0 n_m}{2\ell_m} \right\} \qquad \text{or} \qquad \left\{ \frac{1}{B_{m,h}} \sum_i (\rho_{m,i}^{(h)})^2 R_{m,i}^{(h)} > \frac{C_0 n_m}{2\ell_m} \right\},$$

summed over m and $1 \leq h \leq J_m$.

Proof. Apply Corollary 7.331 to the finite active menu $\{1, \ldots, J_m\}$ in the m-th row. The two displayed summability assumptions are exactly the finite-menu budget-exception series and the finite-menu structural series for these active prefixes, so Borel–Cantelli gives the simultaneous active-prefix conclusion. Since $J_m \to \infty$, each fixed h belongs to the active prefix for all sufficiently large m, giving the fixed-model statement. The sampled-sign claims are the finite-menu sampled claims summed over $1 \leq h \leq J_m$, or the common-batch union bound $J_m 2^{-M_m}$.

The arbitrary-growth estimate is

$$L_m^{(J)} \exp\left\{ -\frac{3C_0 n_m}{16\ell_m} \right\} \leq e^{-1}(e+m)^{-(1+\gamma)}.$$

The common sampled-batch condition gives $J_m 2^{-M_m} \leq e^{-1}(e+m)^{-(1+\delta)}$. The all-integer and sparse dimension cases are the same estimates with $L_m^{(J)} \leq (eD_m)^\Gamma$. The uniform-size events are inherited from Corollary 7.329. □

Corollary 7.333 (Adaptive deterministic-envelope support ensembles)**.** *Fix $C_0 \geq 1$, $D \geq 2$, and a filtration $(\mathcal{F}_i)_{i=0}^n$. The following adaptive deterministic-envelope models satisfy Spencer-scale tails with no budget-exception term.*

(a) *Coordinate supports. Let* $a_i \in [0,1]$ *and* $p_{ij} \in [0,1]$ *be deterministic. Let* ρ_i *be* $\mathcal{F}_{i-1}$*-measurable with* $0 \le \rho_i \le a_i$*, and let* $S_i \subset \{1,\dots,D\}$ *be* $\mathcal{F}_i$*-measurable with*

$$\mathbb{P}\{j \in S_i \mid \mathcal{F}_{i-1}\} \le p_{ij} \qquad a.s.$$

Conditional on the amplitudes and supports, let $A_1,\dots,A_n \in \mathrm{M}_D(\mathbb{C})$ *be arbitrary random self-adjoint matrices such that* $A_i = P_{S_i} A_i P_{S_i}$ *and* $\|A_i\|_{\mathrm{op}} \le \rho_i$*. Put*

$$\mu_c^{\mathrm{env}} := \max_{j \le D} \sum_i a_i^2 p_{ij}.$$

If $t > 0$ *and*

$$\mu_c^{\mathrm{env}} + t \le \frac{C_0 n}{\log(eD)},$$

then

$$\mathbb{P}\left\{\mathrm{disc}(A_1,\dots,A_n) > C(C_0)\sqrt{n}\right\} \le D \exp\left\{-\frac{t^2}{2(\mu_c^{\mathrm{env}} + t/3)}\right\}.$$

(b) *Block supports. Let* $\Pi_1,\dots,\Pi_B$ *be pairwise orthogonal projections in* $\mathrm{M}_D(\mathbb{C})$ *with* $\sum_{\alpha=1}^B \Pi_\alpha = I$*. Let* $a_i \in [0,1]$ *and* $q_{i\alpha} \in [0,1]$ *be deterministic. Let* ρ_i *be* $\mathcal{F}_{i-1}$*-measurable with* $0 \le \rho_i \le a_i$*, and let* $T_i \subset \{1,\dots,B\}$ *be* $\mathcal{F}_i$*-measurable with*

$$\mathbb{P}\{\alpha \in T_i \mid \mathcal{F}_{i-1}\} \le q_{i\alpha} \qquad a.s.$$

Put $P_{T_i} = \sum_{\alpha \in T_i} \Pi_\alpha$*. Conditional on the amplitudes and selected blocks, let* $A_1,\dots,A_n \in \mathrm{M}_D(\mathbb{C})$ *be arbitrary random self-adjoint matrices such that* $A_i = P_{T_i} A_i P_{T_i}$ *and* $\|A_i\|_{\mathrm{op}} \le \rho_i$*. Put*

$$\mu_B^{\mathrm{env}} := \max_{\alpha \le B} \sum_i a_i^2 q_{i\alpha}.$$

If $t > 0$ *and*

$$\mu_B^{\mathrm{env}} + t \le \frac{C_0 n}{\log(eD)},$$

then

$$\mathbb{P}\left\{\mathrm{disc}(A_1,\dots,A_n) > C(C_0)\sqrt{n}\right\} \le B \exp\left\{-\frac{t^2}{2(\mu_B^{\mathrm{env}} + t/3)}\right\}.$$

In either part, the half-budget condition

$$\mu_c^{\mathrm{env}} \le \frac{C_0 n}{2\log(eD)} \qquad \text{or} \qquad \mu_B^{\mathrm{env}} \le \frac{C_0 n}{2\log(eD)}$$

gives the corresponding row failure

$$N \exp\left\{-\frac{3C_0 n}{16\log(eD)}\right\},$$

where $N = D$ *in the coordinate case and* $N = B$ *in the block case. Therefore the schedule and sampled-certificate conclusions of Corollary 7.330 apply with zero predictable-budget exception. In particular, if the row supports are*

predictably uniform of sizes $R_i \le r_i$, then the deterministic envelope budgets are

$$\frac{1}{D}\sum_i a_i^2 r_i \qquad \text{and} \qquad \frac{1}{B}\sum_i a_i^2 r_i$$

for coordinate supports and block supports, respectively.

Proof. In the coordinate case apply Corollary 7.329 with the predictable profile $Q_{ij} = p_{ij}$. Then

$$U = \max_j \sum_i \rho_i^2 p_{ij} \le \mu_c^{\mathrm{env}}, \qquad V = \max_j \sum_i \rho_i^4 p_{ij} \le \mu_c^{\mathrm{env}},$$

so the finite tail follows with $\mu = \sigma^2 = \mu_c^{\mathrm{env}}$. The block case is identical with $Q_{i\alpha} = q_{i\alpha}$. The half-budget estimate uses $t = C_0 n/(2\log(eD))$, and the schedule and sampled-certificate assertions are the corresponding zero-exception specialization of Corollary 7.330. For predictably uniform support sizes, take $p_{ij} = r_i/D$ or $q_{i\alpha} = r_i/B$. □

Corollary 7.334 (Adaptive deterministic-envelope support schedules)**.** *Fix $C_0 \ge 1$. For each $m \ge 1$, let $D_m \ge 2$, put $\ell_m = \log(eD_m)$, and let $A_{m,1}, \dots, A_{m,n_m} \in \mathrm{M}_{D_m}(\mathbb{C})$ be a row satisfying one of the two deterministic-envelope models of Corollary 7.333. In the coordinate case set $N_m = D_m$ and*

$$\mu_m^{\mathrm{env}} = \max_{j \le D_m} \sum_i a_{m,i}^2 p_{m,ij}.$$

In the block case, with B_m structural blocks, set $N_m = B_m$ and

$$\mu_m^{\mathrm{env}} = \max_{\alpha \le B_m} \sum_i a_{m,i}^2 q_{m,i\alpha}.$$

Assume that the deterministic half-budget condition

$$\mu_m^{\mathrm{env}} \le \frac{C_0 n_m}{2\ell_m}$$

holds eventually, and that

$$\sum_m N_m \exp\left\{-\frac{3C_0 n_m}{16\ell_m}\right\} < \infty.$$

Then, with probability one, for all sufficiently large m,

$$\operatorname{disc}(A_{m,1}, \dots, A_{m,n_m}) \le C(C_0)\sqrt{n_m}.$$

If each row is also supplied with M_m independent uniform sign vectors in $\{\pm 1\}^{n_m}$, independent of the matrix array, and $\sum_m 2^{-M_m} < \infty$, then after enlarging $C(C_0)$, almost surely for all sufficiently large m one sampled sign certifies the same bound.

In particular, under the deterministic half-budget condition, the following growth schedules imply the corresponding almost-sure conclusions.

(i) *Arbitrary growth. If, for some* $\gamma > 0$,

$$n_m \geq \frac{16}{3C_0}\,\ell_m\big(\log(eN_m) + (1+\gamma)\log(e+m)\big)$$

eventually, then the almost-sure signing conclusion holds. The sampled conclusion follows if $M_m \geq (1+\delta)\log_2(e+m)$ *eventually for some* $\delta > 0$.

(ii) *All integer dimensions. Suppose the rows are indexed by* $D = 2, 3, \ldots$, *with* $D_m = D$, *and* $N_D \leq (eD)^{\Gamma}$ *eventually for some* $\Gamma \geq 0$. *If*

$$n_D \geq a\log^2(eD)$$

eventually for some $a > 16(\Gamma+1)/(3C_0)$, *then the almost-sure signing conclusion holds. The sampled conclusion follows if* $M_D \geq (1+\delta)\log_2(eD)$ *eventually.*

(iii) *Sparse dimensions. If* $D_m \geq \exp(cm^{\kappa})$ *eventually for some* $c, \kappa > 0$, $N_m \leq (eD_m)^{\Gamma}$ *eventually, and*

$$n_m \geq a\log^2(eD_m)$$

eventually for some $a > 16\Gamma/(3C_0)$, *then the almost-sure signing conclusion holds. The sampled conclusion follows if* $M_m \geq (1+\delta)\log_2(e+m)$ *eventually.*

For predictably uniform coordinate supports or uniform block sets with deterministic size caps $R_{m,i} \leq r_{m,i}$, *the half-budget condition is respectively*

$$\frac{1}{D_m}\sum_i a_{m,i}^2 r_{m,i} \leq \frac{C_0 n_m}{2\ell_m} \qquad \text{or} \qquad \frac{1}{B_m}\sum_i a_{m,i}^2 r_{m,i} \leq \frac{C_0 n_m}{2\ell_m}.$$

Proof. Corollary 7.333, with $t = C_0 n_m/(2\ell_m)$, gives the row failure

$$N_m \exp\left\{-\frac{3C_0 n_m}{16\ell_m}\right\}$$

whenever the deterministic half-budget holds. The first Borel–Cantelli lemma proves the almost-sure signing conclusion, and the sampled version follows by adding the summable 2^{-M_m} row failures. The three growth schedules are exactly the summability estimates in Corollary 7.330, now with no predictable-budget exception term. The final display is the uniform-size specialization from Corollary 7.333. □

Corollary 7.335 (Finite menus of adaptive deterministic-envelope schedules)**.** *Fix* $C_0 \geq 1$. *For each* $m \geq 1$, *let* $D_m \geq 2$, *put* $\ell_m = \log(eD_m)$, *and let* $\mathcal{H}_m$ *be a finite menu. For each* $h \in \mathcal{H}_m$, *let*

$$A_{m,1}^{(h)}, \ldots, A_{m,n_m}^{(h)} \in \mathrm{M}_{D_m}(\mathbb{C})$$

be a random self-adjoint row satisfying one of the two deterministic-envelope models of Corollary 7.333, with its own filtration allowed. If h *is a coordinate-support model, put*

$$N_{m,h} = D_m, \qquad \mu_{m,h}^{\mathrm{env}} = \max_{j \leq D_m}\sum_i a_{m,i,h}^2 p_{m,ij,h}.$$

If h is a block-support model with $B_{m,h}$ structural blocks, put

$$N_{m,h} = B_{m,h}, \qquad \mu^{\mathrm{env}}_{m,h} = \max_{\alpha \le B_{m,h}} \sum_i a^2_{m,i,h} q_{m,i\alpha,h}.$$

Finally set

$$L_m = \sum_{h \in \mathcal{H}_m} N_{m,h}.$$

Assume that the deterministic half-budget condition

$$\max_{h \in \mathcal{H}_m} \mu^{\mathrm{env}}_{m,h} \le \frac{C_0 n_m}{2\ell_m}$$

holds eventually, and that

$$\sum_m L_m \exp\left\{-\frac{3C_0 n_m}{16\ell_m}\right\} < \infty.$$

Then, with probability one, for all sufficiently large m, every $h \in \mathcal{H}_m$ has a Spencer-scale signing:

$$\mathrm{disc}(A^{(h)}_{m,1}, \dots, A^{(h)}_{m,n_m}) \le C(C_0)\sqrt{n_m}.$$

If each menu element is also supplied with $M_{m,h}$ independent uniform sign vectors in $\{\pm 1\}^{n_m}$, independent of the corresponding row, and

$$\sum_m \sum_{h \in \mathcal{H}_m} 2^{-M_{m,h}} < \infty,$$

then after enlarging $C(C_0)$, almost surely for all sufficiently large m every $h \in \mathcal{H}_m$ has one sampled sign certifying the same bound. If a common batch of M_m signs is reused for all $h \in \mathcal{H}_m$, the same conclusion follows from

$$\sum_m |\mathcal{H}_m|\, 2^{-M_m} < \infty.$$

In particular, under the deterministic half-budget condition, the structural menu tail is summable in each of the following schedules.

(i) *Arbitrary growth. If, for some $\gamma > 0$,*

$$n_m \ge \frac{16}{3C_0}\,\ell_m\big(\log(eL_m) + (1+\gamma)\log(e+m)\big)$$

eventually, then the simultaneous almost-sure signing conclusion holds. With a common sign batch, the simultaneous sampled conclusion follows if

$$M_m \ge \log_2(e|\mathcal{H}_m|) + (1+\delta)\log_2(e+m)$$

eventually for some $\delta > 0$.

(ii) *All integer dimensions. Suppose the rows are indexed by $D = 2, 3, \dots$, with $D_m = D$, and $L_D \le (eD)^\Gamma$ eventually for some $\Gamma \ge 0$. If*

$$n_D \ge a \log^2(eD)$$

eventually for some $a > 16(\Gamma+1)/(3C_0)$, then the simultaneous almost-sure signing conclusion holds.

(iii) *Sparse dimensions. If* $D_m \geq \exp(cm^\kappa)$ *eventually for some* $c, \kappa > 0$, $L_m \leq (eD_m)^\Gamma$ *eventually, and*

$$n_m \geq a \log^2(eD_m)$$

eventually for some $a > 16\Gamma/(3C_0)$*, then the simultaneous almost-sure signing conclusion holds.*

Proof. For each fixed m and h, Corollary 7.334 gives the row failure

$$N_{m,h} \exp\left\{-\frac{3C_0 n_m}{16\ell_m}\right\}$$

under the deterministic half-budget condition. Union over $h \in \mathcal{H}_m$; the structural failure is at most

$$L_m \exp\left\{-\frac{3C_0 n_m}{16\ell_m}\right\}.$$

The first Borel–Cantelli lemma gives the simultaneous eventual signing statement. Conditional sign search gives an additional $2^{-M_{m,h}}$ failure for each menu element, and union over the menu gives either the displayed double sum or, for a reused common batch, the bound $|\mathcal{H}_m|2^{-M_m}$.

The arbitrary-growth lower bound gives

$$L_m \exp\left\{-\frac{3C_0 n_m}{16\ell_m}\right\} \leq e^{-1}(e+m)^{-(1+\gamma)}.$$

The common sampled-batch condition similarly gives $|\mathcal{H}_m|2^{-M_m} \leq e^{-1}(e+m)^{-(1+\delta)}$. The all-integer and sparse dimension estimates are the same summability computations with L_m in place of N_m. □

Corollary 7.336 (Exhaustive countable menus of adaptive deterministic-envelope schedules)**.** *Fix* $C_0 \geq 1$*. For each* $m \geq 1$*, let* $D_m \geq 2$*, put* $\ell_m = \log(eD_m)$*, and let*

$$J_1 \leq J_2 \leq \cdots, \qquad J_m \to \infty,$$

be a deterministic exhaustion of the countable menu $\mathbb{N} = \{1, 2, \ldots\}$*. For each* m *and* $h \geq 1$*, let*

$$A_{m,1}^{(h)}, \ldots, A_{m,n_m}^{(h)} \in \mathrm{M}_{D_m}(\mathbb{C})$$

be a random self-adjoint row satisfying one of the two deterministic-envelope models of Corollary 7.333, with its own filtration allowed. If h *is a coordinate-support model, put*

$$N_{m,h} = D_m, \qquad \mu_{m,h}^{\mathrm{env}} = \max_{j \leq D_m} \sum_i a_{m,i,h}^2 p_{m,ij,h}.$$

If h *is a block-support model with* $B_{m,h}$ *structural blocks, put*

$$N_{m,h} = B_{m,h}, \qquad \mu_{m,h}^{\mathrm{env}} = \max_{\alpha \leq B_{m,h}} \sum_i a_{m,i,h}^2 q_{m,i\alpha,h}.$$

Set

$$L_m^{(J)} := \sum_{1 \leq h \leq J_m} N_{m,h}.$$

Assume that the deterministic prefix half-budget condition

$$\max_{1\le h\le J_m} \mu^{\mathrm{env}}_{m,h} \le \frac{C_0 n_m}{2\ell_m}$$

holds eventually, and that

$$\sum_m L^{(J)}_m \exp\left\{-\frac{3C_0 n_m}{16\ell_m}\right\} < \infty.$$

Then, with probability one, for all sufficiently large m, every active prefix model $1 \le h \le J_m$ has a Spencer-scale signing:

$$\operatorname{disc}(A^{(h)}_{m,1},\dots,A^{(h)}_{m,n_m}) \le C(C_0)\sqrt{n_m} \qquad (1 \le h \le J_m).$$

Consequently, for each fixed h, the same bound holds for all sufficiently large m. If the active-prefix models are supplied with independent sampled signs and

$$\sum_m \sum_{1\le h\le J_m} 2^{-M_{m,h}} < \infty,$$

then after enlarging $C(C_0)$, almost surely every active-prefix model has a sampled certificate for all sufficiently large m. If one common batch of M_m signs is reused for all $1 \le h \le J_m$, independent of the whole active prefix, it is enough that

$$\sum_m J_m\, 2^{-M_m} < \infty.$$

In particular, under the deterministic prefix half-budget condition, the structural prefix tail is summable in each of the following schedules.

(i) *Arbitrary growth. If, for some $\gamma > 0$,*

$$n_m \ge \frac{16}{3C_0}\,\ell_m\big(\log(eL^{(J)}_m) + (1+\gamma)\log(e+m)\big)$$

eventually, then the active-prefix and fixed-model almost-sure signing conclusions hold. With a common sign batch, the sampled conclusion follows if

$$M_m \ge \log_2(eJ_m) + (1+\delta)\log_2(e+m)$$

eventually for some $\delta > 0$.

(ii) *All integer dimensions. Suppose the rows are indexed by $D = 2, 3, \dots$, with $D_m = D$, $J_D \to \infty$, and $L^{(J)}_D \le (eD)^\Gamma$ eventually for some $\Gamma \ge 0$. If*

$$n_D \ge a\log^2(eD)$$

eventually for some $a > 16(\Gamma+1)/(3C_0)$, then the active-prefix and fixed-model almost-sure signing conclusions hold.

(iii) *Sparse dimensions. If $D_m \ge \exp(cm^\kappa)$ eventually for some $c, \kappa > 0$, $J_m \to \infty$, $L^{(J)}_m \le (eD_m)^\Gamma$ eventually, and*

$$n_m \ge a\log^2(eD_m)$$

eventually for some $a > 16\Gamma/(3C_0)$, then the active-prefix and fixed-model almost-sure signing conclusions hold.

For predictably uniform coordinate supports or uniform block sets with deterministic size caps $R_{m,i}^{(h)} \le r_{m,i}^{(h)}$, the active-prefix half-budget condition is implied by

$$\max_{1\le h\le J_m} \frac{1}{D_m}\sum_i (a_{m,i,h})^2 r_{m,i}^{(h)} \le \frac{C_0 n_m}{2\ell_m}$$

in the coordinate case, or by

$$\max_{1\le h\le J_m} \frac{1}{B_{m,h}}\sum_i (a_{m,i,h})^2 r_{m,i}^{(h)} \le \frac{C_0 n_m}{2\ell_m}$$

in the block case.

Proof. Apply Corollary 7.335 to the finite active menu $\{1,\dots,J_m\}$ in the m-th row. The displayed prefix half-budget and structural summability hypotheses are exactly the finite-menu hypotheses for those active prefixes, so Borel–Cantelli gives the simultaneous active-prefix conclusion. Since $J_m \to \infty$, each fixed h is in the active prefix for all sufficiently large m. The sampled-sign statements are the finite-menu sampled statements summed over $1 \le h \le J_m$, or the common-batch union bound $J_m 2^{-M_m}$.

The arbitrary-growth estimate is

$$L_m^{(J)} \exp\left\{-\frac{3C_0 n_m}{16\ell_m}\right\} \le e^{-1}(e+m)^{-(1+\gamma)}.$$

The common sampled-batch condition gives $J_m 2^{-M_m} \le e^{-1}(e+m)^{-(1+\delta)}$. The all-integer and sparse dimension cases are the same estimates with $L_m^{(J)} \le (eD_m)^{\Gamma}$. The uniform-size budgets are the deterministic uniform-support specialization in Corollary 7.333. □

Corollary 7.337 (Almost-sure random-profile support arrays)**.** *Fix $C_0 \ge 1$. For each $m \ge 1$, let $D_m \ge 2$, put $\ell_m = \log(eD_m)$, and let $A_{m,1},\dots,A_{m,n_m}$ be random self-adjoint matrices in $\mathrm{M}_{D_m}(\mathbb{C})$. Suppose that, for each row m, one of the following two random-profile models holds.*

(a) *Coordinate profiles. The triples $(\rho_{m,i}, Q_{m,i}, S_{m,i})$, $i = 1,\dots,n_m$, are independent, $0 \le \rho_{m,i} \le 1$, $Q_{m,i} = (Q_{m,ij})_{j=1}^{D_m} \in [0,1]^{D_m}$, and $S_{m,i} \subset \{1,\dots,D_m\}$, with*

$$\mathbb{P}\{j \in S_{m,i} \mid \rho_{m,i}, Q_{m,i}\} \le Q_{m,ij} \qquad a.s.$$

Conditional on the amplitudes, profiles, and supports,

$$A_{m,i} = P_{S_{m,i}} A_{m,i} P_{S_{m,i}}, \qquad \|A_{m,i}\|_{\mathrm{op}} \le \rho_{m,i}.$$

Put

$$N_m = D_m, \qquad \mu_m = \max_{j\le D_m} \sum_i \mathbb{E}(\rho_{m,i}^2 Q_{m,ij}),$$

and

$$\sigma_m^2 = \max_{j \le D_m} \sum_i \mathbb{E}(\rho_{m,i}^4 Q_{m,ij}).$$

(b) *Block profiles. There are pairwise orthogonal* $\Pi_{m,\alpha} \in \mathrm{M}_{D_m}(\mathbb{C})$, $1 \le \alpha \le B_m$, *with* $\sum_{\alpha=1}^{B_m} \Pi_{m,\alpha} = I$. *The triples* $(\rho_{m,i}, Q_{m,i}, T_{m,i})$, $i = 1, \dots, n_m$, *are independent,* $0 \le \rho_{m,i} \le 1$, $Q_{m,i} = (Q_{m,i\alpha})_{\alpha=1}^{B_m} \in [0,1]^{B_m}$, *and* $T_{m,i} \subset \{1, \dots, B_m\}$, *with*

$$\mathbb{P}\{\alpha \in T_{m,i} \mid \rho_{m,i}, Q_{m,i}\} \le Q_{m,i\alpha} \qquad \text{a.s.}$$

Put $P_{m,T_{m,i}} = \sum_{\alpha \in T_{m,i}} \Pi_{m,\alpha}$. *Conditional on the amplitudes, profiles, and selected blocks,*

$$A_{m,i} = P_{m,T_{m,i}} A_{m,i} P_{m,T_{m,i}}, \qquad \|A_{m,i}\|_{\mathrm{op}} \le \rho_{m,i}.$$

Put

$$N_m = B_m, \qquad \mu_m = \max_{\alpha \le B_m} \sum_i \mathbb{E}(\rho_{m,i}^2 Q_{m,i\alpha}),$$

and

$$\sigma_m^2 = \max_{\alpha \le B_m} \sum_i \mathbb{E}(\rho_{m,i}^4 Q_{m,i\alpha}).$$

Assume that there are $t_m > 0$ *such that*

$$\mu_m + t_m \le \frac{C_0 n_m}{\ell_m}$$

and

$$\sum_m N_m \exp\left\{-\frac{t_m^2}{2(\sigma_m^2 + t_m/3)}\right\} < \infty.$$

Then, with probability one, for all sufficiently large m,

$$\mathrm{disc}(A_{m,1}, \dots, A_{m,n_m}) \le C(C_0)\sqrt{n_m}.$$

In particular, under the half-budget condition

$$\mu_m \le \frac{C_0 n_m}{2\ell_m},$$

the same conclusion holds whenever

$$\sum_m N_m \exp\left\{-\frac{3C_0 n_m}{16\ell_m}\right\} < \infty.$$

This summability is implied, for any $\gamma > 0$, *by the direct growth rule*

$$n_m \ge \frac{16}{3C_0}\,\ell_m\big(\log(eN_m) + (1+\gamma)\log(e+m)\big)$$

eventually. If, in addition, each row is supplied with M_m *independent uniform sign vectors in* $\{\pm 1\}^{n_m}$, *independent of the matrix array, and* $\sum_m 2^{-M_m} < \infty$, *then after enlarging* $C(C_0)$ *if necessary, almost surely for all sufficiently large* m *one sampled sign certifies the same* $C(C_0)\sqrt{n_m}$ *bound.*

Proof. Corollary 7.328 applied to the m-th row gives the displayed one-row failure probability in either model. The summability assumption and the first Borel–Cantelli lemma give the almost-sure discrepancy conclusion. Under the half-budget assumption take $t_m = C_0 n_m/(2\ell_m)$; since $\rho_{m,i}^4 \le \rho_{m,i}^2$, one has $\sigma_m^2 \le \mu_m \le t_m$, and the exponent is at least $3C_0 n_m/(16\ell_m)$.

The direct growth rule gives

$$N_m \exp\left\{-\frac{3C_0 n_m}{16\ell_m}\right\} \le e^{-1}(e+m)^{-(1+\gamma)},$$

which is summable. For sampled signs, condition on the realized row and on the weighted-overlap event used in the proof of Corollary 7.328. On that event, $\|\sum_i A_{m,i}^2\|_{\mathrm{op}} \le C_0 n_m/\ell_m$, so Corollary 7.144 gives conditional sampled-search failure at most 2^{-M_m}, after enlarging $C(C_0)$. Another application of Borel–Cantelli proves the sampled-certificate statement. □

Corollary 7.338 (Almost-sure weighted support arrays)**.** *Fix $C_0 \ge 1$. For each $m \ge 1$, let $A_{m,1}, \dots, A_{m,n_m}$ be random self-adjoint matrices in $\mathrm{M}_{D_m}(\mathbb{C})$, with deterministic amplitudes $\rho_{m,i} \in [0,1]$. Suppose that, for each m, one of the following two models holds.*

(a) *Coordinate supports. There are random sets $S_{m,i} \subset \{1, \dots, D_m\}$ such that, for every coordinate j, the indicators $\mathbf{1}_{\{j \in S_{m,i}\}}$, $i = 1, \dots, n_m$, are independent and satisfy*

$$\mathbb{P}\{j \in S_{m,i}\} \le p_{m,ij}.$$

Conditional on the supports, $A_{m,i} = P_{S_{m,i}} A_{m,i} P_{S_{m,i}}$ and $\|A_{m,i}\|_{\mathrm{op}} \le \rho_{m,i}$. Put

$$b_m = \max_i \rho_{m,i}^2, \qquad \mu_m = \max_j \sum_i \rho_{m,i}^2 p_{m,ij}, \qquad \sigma_m^2 = \max_j \sum_i \rho_{m,i}^4 p_{m,ij}.$$

Assume that there are $t_m > 0$ with

$$\mu_m + t_m \le \frac{C_0 n_m}{\log(eD_m)}$$

and

$$\sum_m D_m \exp\left\{-\frac{t_m^2}{2(\sigma_m^2 + b_m t_m/3)}\right\} < \infty,$$

with the zero-denominator convention of Corollary 7.324.

(b) *Random subspaces. There are independent random orthogonal projections $P_{m,1}, \dots, P_{m,n_m}$ such that, conditional on the projections, $A_{m,i} = P_{m,i} A_{m,i} P_{m,i}$ and $\|A_{m,i}\|_{\mathrm{op}} \le \rho_{m,i}$. Put*

$$b_m = \max_i \rho_{m,i}^2, \qquad \mu_m = \left\| \sum_i \rho_{m,i}^2 \mathbb{E} P_{m,i} \right\|_{\mathrm{op}}.$$

Assume that there are $t_m > 0$ with

$$\mu_m + t_m \le \frac{C_0 n_m}{\log(eD_m)}$$

and

$$\sum_m D_m \exp\left\{-\frac{t_m^2}{2(\mu_m + b_m t_m/3)}\right\} < \infty.$$

Then, with probability one, for all sufficiently large m,

$$\operatorname{disc}(A_{m,1}, \dots, A_{m,n_m}) \le C(C_0)\sqrt{n_m}.$$

In particular, the almost-sure conclusion holds in the homogeneous coordinate case $p_{m,ij} \le p_m$ whenever

$$p_m \sum_i \rho_{m,i}^2 \le \frac{C_0 n_m}{2\log(eD_m)} \qquad \text{and} \qquad \sum_m D_m \exp\left\{-\frac{3C_0 n_m}{16\log(eD_m)}\right\} < \infty,$$

and it holds for Haar rank-$r_{m,i}$ projections under the same summability condition whenever

$$\frac{1}{D_m}\sum_i \rho_{m,i}^2 r_{m,i} \le \frac{C_0 n_m}{2\log(eD_m)}.$$

Proof. In the coordinate case, apply Corollary 7.324 to the m-th row. In the random-subspace case, apply Corollary 7.325 to the m-th row. The displayed summability assumptions make the resulting failure probabilities summable, so the first Borel–Cantelli lemma gives the eventual almost-sure bound. The homogeneous coordinate and Haar specializations are exactly the "in particular" estimates in those two finite-dimensional corollaries, with $t_m = C_0 n_m/(2\log(eD_m))$. □

Corollary 7.339 (All-dimension weighted random-support schedules)**.** *Fix $C_0 \ge 1$. For each row m, let $\ell_m = \log(eD_m)$, let $D_m \ge 2$, and let $A_{m,1}, \dots, A_{m,n_m}$ be random self-adjoint matrices in $\mathrm{M}_{D_m}(\mathbb{C})$ with deterministic amplitudes $\rho_{m,i} \in [0,1]$. Assume that one of the following two half-budget models holds for all large m.*

(a) *Coordinate supports. There are random supports $S_{m,i} \subset \{1, \dots, D_m\}$ such that, for every coordinate j, the indicators $\mathbf{1}_{\{j \in S_{m,i}\}}$, $i = 1, \dots, n_m$, are independent and satisfy*

$$\mathbb{P}\{j \in S_{m,i}\} \le p_{m,ij}, \qquad \max_j \sum_{i=1}^{n_m} \rho_{m,i}^2 p_{m,ij} \le \frac{C_0 n_m}{2\ell_m}.$$

Conditional on the supports, $A_{m,i} = P_{S_{m,i}} A_{m,i} P_{S_{m,i}}$ and $\|A_{m,i}\|_{\mathrm{op}} \le \rho_{m,i}$.

(b) *Random subspaces. There are independent random orthogonal projections $P_{m,1}, \dots, P_{m,n_m}$ such that, conditional on the projections, $A_{m,i} = P_{m,i} A_{m,i} P_{m,i}$, $\|A_{m,i}\|_{\mathrm{op}} \le \rho_{m,i}$, and*

$$\left\| \sum_{i=1}^{n_m} \rho_{m,i}^2 \mathbb{E} P_{m,i} \right\|_{\mathrm{op}} \le \frac{C_0 n_m}{2\ell_m}.$$

Then

$$\mathbb{P}\left\{\operatorname{disc}(A_{m,1},\dots,A_{m,n_m}) > C(C_0)\sqrt{n_m}\right\} \le D_m \exp\left\{-\frac{3C_0 n_m}{16\ell_m}\right\}$$

for all large m. Consequently, if

$$\sum_m D_m \exp\left\{-\frac{3C_0 n_m}{16\log(eD_m)}\right\} < \infty,$$

then, almost surely for all sufficiently large m,

$$\operatorname{disc}(A_{m,1},\dots,A_{m,n_m}) \le C(C_0)\sqrt{n_m}.$$

In particular, along all integer dimensions $D = 2, 3, \dots$, the almost-sure conclusion holds whenever the half-budget coordinate or subspace hypothesis above holds with $D_m = D$, $n_m = n_D$, and

$$n_D \ge a\log^2(eD)$$

for some $a > 32/(3C_0)$. Along any sparse schedule $D_m \ge \exp(cm^\kappa)$, it is enough that

$$n_m \ge a\log^2(eD_m)$$

eventually for some $a > 16/(3C_0)$.

For uniformly chosen fixed-size coordinate supports of sizes $r_{m,i}$, or for Haar-distributed projections of ranks $r_{m,i}$, the half-budget condition is

$$\frac{1}{D_m}\sum_i \rho_{m,i}^2 r_{m,i} \le \frac{C_0 n_m}{2\log(eD_m)}.$$

Thus the unweighted homogeneous coordinate case is covered by

$$p_m \le \frac{C_0}{2\log(eD_m)},$$

and the unweighted Haar-rank case is covered by

$$\frac{1}{n_m}\sum_i r_{m,i} \le \frac{C_0 D_m}{2\log(eD_m)},$$

with the same all-integer and sparse almost-sure schedules.

Proof. For the coordinate model, apply Corollary 7.324 with

$$t_m = \frac{C_0 n_m}{2\ell_m}.$$

The half-budget hypothesis gives $\mu_m \le t_m$, and because $\rho_{m,i} \le 1$,

$$\sigma_m^2 = \max_j \sum_i \rho_{m,i}^4 p_{m,ij} \le \max_j \sum_i \rho_{m,i}^2 p_{m,ij} \le t_m, \qquad b_m \le 1.$$

Hence

$$\frac{t_m^2}{2(\sigma_m^2 + b_m t_m/3)} \ge \frac{t_m^2}{2(t_m + t_m/3)} = \frac{3C_0 n_m}{16\ell_m}.$$

The random-subspace model is identical, using Corollary 7.325; its variance parameter is $\mu_m \le t_m$, and again $b_m \le 1$. This proves the displayed one-row tail, and the first Borel–Cantelli lemma gives the summable-array conclusion.

If $n_D \geq a \log^2(eD)$, then

$$D \exp\left\{-\frac{3C_0 n_D}{16 \log(eD)}\right\} \leq (eD)^{1-3C_0 a/16}.$$

The series over all integer $D \geq 2$ converges when $3C_0 a/16 > 2$. Along $D_m \geq \exp(cm^\kappa)$, the same bound is stretched-exponentially summable as soon as $3C_0 a/16 > 1$. The fixed-size and Haar-rank specializations are the corresponding expectations $\mathbb{P}\{j \in S_{m,i}\} = r_{m,i}/D_m$ and $\mathbb{E}P_{m,i} = (r_{m,i}/D_m)I$. □

Corollary 7.340 (Sampled sign certificates for weighted support ensembles)**.** *Fix $C_0 \geq 1$, and let $C(C_0)$ be sufficiently large. Let $\eta^{(1)}, \dots, \eta^{(M)}$ be independent uniform sign vectors in $\{\pm 1\}^n$, independent of the random matrices below.*

(a) *If $A_1, \dots, A_n$ satisfy the hypotheses of Corollary 7.311 with exceptional event E, then*

$$\mathbb{P}\left\{\min_{1 \leq r \leq M} \left\|\sum_{i=1}^n \eta_i^{(r)} A_i\right\|_{\mathrm{op}} > C(C_0)\sqrt{n}\right\} \leq \mathbb{P}(E) + 2^{-M}.$$

(b) *Under the hypotheses of Corollary 7.324, the same probability is bounded by*

$$D \exp\left\{-\frac{t^2}{2(\sigma^2 + bt/3)}\right\} + 2^{-M},$$

with the same zero-denominator convention; under the hypotheses of Corollary 7.325, it is bounded by

$$D \exp\left\{-\frac{t^2}{2(\mu + bt/3)}\right\} + 2^{-M}.$$

(c) *In the triangular-array setting of Corollary 7.338, let $\eta_m^{(1)}, \dots, \eta_m^{(M_m)}$ be independent uniform sign vectors in $\{\pm 1\}^{n_m}$, independent across m and independent of the matrix array. If, in addition to the summability hypotheses in that corollary,*

$$\sum_m 2^{-M_m} < \infty,$$

then, with probability one, for all sufficiently large m there is $1 \leq r \leq M_m$ such that

$$\left\|\sum_{i=1}^{n_m} \eta_{m,i}^{(r)} A_{m,i}\right\|_{\mathrm{op}} \leq C(C_0)\sqrt{n_m}.$$

Thus every weighted low-overlap coordinate, projection, frame, or random subspace ensemble covered above has an explicit randomized certificate: a constant number of independent sign trials gives a constant success probability,

and any choice of M_m with $\sum_m 2^{-M_m} < \infty$ makes the algorithmic failure summable along triangular arrays.

Proof. On the weighted-overlap event in Corollary 7.311, its proof shows

$$\left\| \sum_i A_i^2 \right\|_{\mathrm{op}} \le \frac{C_0 n}{\log(eD)}.$$

Conditioning on the realized tuple and applying Corollary 7.144 gives conditional failure at most 2^{-M} for the sampled signs, after enlarging $C(C_0)$. This proves part (a). Parts (b) and (c) are obtained by combining the same conditional 2^{-M} (or 2^{-M_m}) sign-search failure with the matrix failure bounds in Corollaries 7.324, 7.325, and 7.338. The array statement follows from the first Borel–Cantelli lemma applied to the sum of the matrix-row failures and the sign-search failures. □

Corollary 7.341 (All-dimension sampled certificates for weighted random supports)**.** *In the triangular-array setting of Corollary 7.339, let $\eta_m^{(1)}, \dots, \eta_m^{(M_m)}$ be independent uniform sign vectors in $\{\pm 1\}^{n_m}$, independent across m and independent of the matrix array. Under either half-budget coordinate-support or random-subspace model in that corollary, for all large m,*

$$\mathbb{P}\left\{ \min_{1 \le r \le M_m} \left\| \sum_{i=1}^{n_m} \eta_{m,i}^{(r)} A_{m,i} \right\|_{\mathrm{op}} > C(C_0)\sqrt{n_m} \right\} \le D_m \exp\left\{ -\frac{3C_0 n_m}{16 \log(eD_m)} \right\} + 2^{-M_m}.$$

Consequently, if

$$\sum_m \left[D_m \exp\left\{ -\frac{3C_0 n_m}{16 \log(eD_m)} \right\} + 2^{-M_m} \right] < \infty,$$

then, with probability one, for all sufficiently large m, one of the sampled sign vectors certifies

$$\left\| \sum_{i=1}^{n_m} \eta_{m,i}^{(r_m)} A_{m,i} \right\|_{\mathrm{op}} \le C(C_0)\sqrt{n_m}.$$

In particular:

(a) *Along all integer dimensions $D = 2, 3, \dots$, the sampled-certificate conclusion holds under the half-budget hypotheses of Corollary 7.339 if $n_D \ge a \log^2(eD)$ for some $a > 32/(3C_0)$ and, for some $\delta > 0$,*

$$M_D \ge (1+\delta)\log_2(eD)$$

eventually.

(b) *Along any sparse schedule $D_m \ge \exp(cm^\kappa)$, the same conclusion holds if $n_m \ge a \log^2(eD_m)$ eventually for some $a > 16/(3C_0)$ and, for some $\delta > 0$,*

$$M_m \ge (1+\delta)\log_2(e+m)$$

eventually.

Thus the existence statement for weighted coordinate and random-subspace ensembles can be made into an eventual randomized certificate with only logarithmically many independent sign trials per row.

Proof. Apply Corollary 7.340 to the m-th row with $t_m = C_0 n_m/(2\log(eD_m))$. The half-budget hypotheses and the proof of Corollary 7.339 give the matrix failure term

$$D_m \exp\left\{-\frac{3C_0 n_m}{16\log(eD_m)}\right\},$$

and Corollary 7.340 adds the independent sign-search failure 2^{-M_m}. The first Borel–Cantelli lemma gives the summable-array conclusion.

For integer dimensions, the matrix tail is summable by Corollary 7.339. The sign-search tail satisfies

$$2^{-M_D} \le (eD)^{-(1+\delta)},$$

which is summable over $D \ge 2$. In the sparse case, the matrix tail is again summable by Corollary 7.339, and

$$2^{-M_m} \le (e+m)^{-(1+\delta)}$$

is summable over m. □

Corollary 7.342 (Growth-rate weighted random-support schedules)**.** *In the half-budget coordinate-support or random-subspace setting of Corollary 7.339, put*

$$\ell_m = \log(eD_m), \qquad L_m = \log(e+m).$$

Assume that, for some $\gamma > 0$,

$$n_m \ge \frac{16}{3C_0}\,\ell_m(\ell_m + (1+\gamma)L_m)$$

eventually. Then, almost surely for all sufficiently large m,

$$\operatorname{disc}(A_{m,1},\dots,A_{m,n_m}) \le C(C_0)\sqrt{n_m}.$$

Moreover, if $\eta_m^{(1)},\dots,\eta_m^{(M_m)}$ are independent uniform sign vectors as in Corollary 7.341 and, for some $\delta > 0$,

$$M_m \ge (1+\delta)\log_2(e+m)$$

eventually, then, almost surely for all sufficiently large m, one sampled sign vector certifies

$$\left\|\sum_{i=1}^{n_m} \eta_{m,i}^{(r_m)} A_{m,i}\right\|_{\mathrm{op}} \le C(C_0)\sqrt{n_m}.$$

Thus the weighted coordinate-support and random-subspace models have eventual Spencer signings, and eventual sampled certificates, for every dimension growth regime after paying one logarithmic-in-row-index summability margin.

Proof. By Corollary 7.339, the row failure probability is at most

$$D_m \exp\left\{-\frac{3C_0 n_m}{16\ell_m}\right\}.$$

The growth hypothesis gives

$$\frac{3C_0 n_m}{16\ell_m} \geq \ell_m + (1+\gamma)L_m,$$

and hence

$$D_m \exp\left\{-\frac{3C_0 n_m}{16\ell_m}\right\} \leq e^{-1}(e+m)^{-(1+\gamma)}.$$

This is summable, so the first Borel–Cantelli lemma proves the discrepancy conclusion. With sampled signs, Corollary 7.341 adds the failure probability 2^{-M_m}, and

$$2^{-M_m} \leq (e+m)^{-(1+\delta)}$$

is summable. Another application of Borel–Cantelli gives the sampled certificate conclusion. □

Corollary 7.343 (Random-amplitude weighted-support schedules)**.** *Fix $C_0 \geq 1$. For each row m, put $\ell_m = \log(eD_m)$, let $D_m \geq 2$, and let $A_{m,1}, \dots, A_{m,n_m}$ be random self-adjoint matrices in $\mathrm{M}_{D_m}(\mathbb{C})$. Assume that, for all large m, one of the following random-amplitude half-budget models holds.*

(a) *Coordinate supports. The pairs $(\rho_{m,i}, S_{m,i})$, $1 \leq i \leq n_m$, are independent, $0 \leq \rho_{m,i} \leq 1$, $S_{m,i} \subset \{1, \dots, D_m\}$, and for deterministic $p_{m,ij} \in [0,1]$,*

$$\mathbb{P}\{j \in S_{m,i} \mid \rho_{m,i}\} \leq p_{m,ij} \qquad a.s.$$

Conditional on the amplitudes and supports, $A_{m,i} = P_{S_{m,i}} A_{m,i} P_{S_{m,i}}$ and $\|A_{m,i}\|_{\mathrm{op}} \leq \rho_{m,i}$. Suppose that

$$\mu_{m,c} := \max_j \sum_{i=1}^{n_m} p_{m,ij} \mathbb{E}\rho_{m,i}^2 \leq \frac{C_0 n_m}{2\ell_m}.$$

(b) *Random subspaces. The pairs $(\rho_{m,i}, P_{m,i})$, $1 \leq i \leq n_m$, are independent, $0 \leq \rho_{m,i} \leq 1$, and $P_{m,i}$ is an orthogonal projection in $\mathrm{M}_{D_m}(\mathbb{C})$. Conditional on the amplitudes and projections, $A_{m,i} = P_{m,i} A_{m,i} P_{m,i}$ and $\|A_{m,i}\|_{\mathrm{op}} \leq \rho_{m,i}$. Suppose that*

$$\mu_{m,s} := \left\| \sum_{i=1}^{n_m} \mathbb{E}(\rho_{m,i}^2 P_{m,i}) \right\|_{\mathrm{op}} \leq \frac{C_0 n_m}{2\ell_m}.$$

Then, for all large m,

$$\mathbb{P}\left\{\mathrm{disc}(A_{m,1}, \dots, A_{m,n_m}) > C(C_0)\sqrt{n_m}\right\} \leq D_m \exp\left\{-\frac{3C_0 n_m}{16\ell_m}\right\}.$$

Consequently, if

$$\sum_m D_m \exp\left\{-\frac{3C_0 n_m}{16\log(eD_m)}\right\} < \infty,$$

then, almost surely for all sufficiently large m,

$$\operatorname{disc}(A_{m,1},\dots,A_{m,n_m}) \le C(C_0)\sqrt{n_m}.$$

Let $\eta_m^{(1)},\dots,\eta_m^{(M_m)}$ be independent uniform sign vectors in $\{\pm1\}^{n_m}$, independent across m and independent of the matrix array. After enlarging $C(C_0)$ if necessary,

$$\mathbb{P}\left\{\min_{1\le r\le M_m}\left\|\sum_{i=1}^{n_m}\eta_{m,i}^{(r)}A_{m,i}\right\|_{\mathrm{op}} > C(C_0)\sqrt{n_m}\right\} \le D_m\exp\left\{-\frac{3C_0 n_m}{16\ell_m}\right\} + 2^{-M_m}.$$

If the right-hand side is summable in m, then almost surely for all sufficiently large m one of the sampled signs certifies the same $C(C_0)\sqrt{n_m}$ bound.

In particular:

(i) *Along all integer dimensions $D = 2,3,\dots$, the almost-sure signing conclusion holds whenever the random-amplitude half-budget hypothesis holds with $D_m = D$, $n_m = n_D$, and*

$$n_D \ge a\log^2(eD)$$

eventually for some $a > 32/(3C_0)$. The sampled-certificate conclusion holds as well if, for some $\delta > 0$,

$$M_D \ge (1+\delta)\log_2(eD)$$

eventually.

(ii) *Along any sparse schedule $D_m \ge \exp(cm^\kappa)$, the same almost-sure signing conclusion holds whenever*

$$n_m \ge a\log^2(eD_m)$$

eventually for some $a > 16/(3C_0)$. The sampled-certificate conclusion holds as well if, for some $\delta > 0$,

$$M_m \ge (1+\delta)\log_2(e+m)$$

eventually.

(iii) *For an arbitrary dimension schedule, if for some $\gamma > 0$*

$$n_m \ge \frac{16}{3C_0}\,\ell_m(\ell_m + (1+\gamma)\log(e+m))$$

eventually, then the almost-sure signing conclusion holds; the same $M_m \ge (1+\delta)\log_2(e+m)$ condition gives eventual sampled certificates.

For uniformly chosen coordinate supports of random sizes $R_{m,i}$, or Haar-distributed projections of random ranks $R_{m,i}$, the half-budget condition in either model is

$$\frac{1}{D_m}\sum_i \mathbb{E}(\rho_{m,i}^2 R_{m,i}) \le \frac{C_0 n_m}{2\log(eD_m)}.$$

Proof. Apply Corollary 7.326 to the m-th row with $t_m = C_0 n_m/(2\ell_m)$. In the coordinate model,

$$\max_j \sum_i p_{m,ij}\mathbb{E}\rho_{m,i}^4 \le \max_j \sum_i p_{m,ij}\mathbb{E}\rho_{m,i}^2 \le t_m,$$

and in the subspace model $\mu_{m,s} \le t_m$. The half-budget estimate in that corollary gives the displayed one-row matrix tail. The first Borel–Cantelli lemma gives the almost-sure signing conclusion under summability.

For sampled signs, on the overlap event used in the proof of Corollary 7.326, Corollary 7.311 gives $\|\sum_i A_{m,i}^2\|_{\mathrm{op}} \le C_0 n_m/\ell_m$. Conditioning on the realized tuple and applying Corollary 7.144 gives conditional sampled-sign failure at most 2^{-M_m}, after enlarging $C(C_0)$. Combining this with the matrix-row failure proves the sampled tail and the sampled Borel–Cantelli statement.

The integer and sparse schedule thresholds are exactly the summability calculations in Corollary 7.339. The growth-rate condition gives

$$D_m \exp\left\{-\frac{3C_0 n_m}{16\ell_m}\right\} \le e^{-1}(e+m)^{-(1+\gamma)},$$

and $2^{-M_m} \le (e+m)^{-(1+\delta)}$. For uniform random support sizes and Haar random ranks, the budget identity follows from $\mathbb{P}\{j \in S_{m,i} \mid R_{m,i}, \rho_{m,i}\} = R_{m,i}/D_m$ and $\mathbb{E}(P_{m,i} \mid R_{m,i}, \rho_{m,i}) = (R_{m,i}/D_m)I$. □

Corollary 7.344 (Block-count random-amplitude support schedules)**.** *Fix $C_0 \ge 1$. For each row m, put $\ell_m = \log(eD_m)$. Let $\Pi_{m,1}, \dots, \Pi_{m,B_m}$ be pairwise orthogonal projections in $\mathrm{M}_{D_m}(\mathbb{C})$ summing to I. Let $(\rho_{m,i}, T_{m,i})$, $1 \le i \le n_m$, be independent pairs, where $0 \le \rho_{m,i} \le 1$ and $T_{m,i} \subset \{1, \dots, B_m\}$. Assume that for deterministic $q_{m,i\alpha} \in [0,1]$,*

$$\mathbb{P}\{\alpha \in T_{m,i} \mid \rho_{m,i}\} \le q_{m,i\alpha} \qquad a.s.$$

Put $P_{m,T_{m,i}} = \sum_{\alpha \in T_{m,i}} \Pi_{m,\alpha}$, and suppose that, conditional on the amplitudes and selected blocks,

$$A_{m,i} = P_{m,T_{m,i}} A_{m,i} P_{m,T_{m,i}}, \qquad \|A_{m,i}\|_{\mathrm{op}} \le \rho_{m,i}.$$

If the block half-budget condition

$$\mu_{m,B} := \max_{\alpha \le B_m} \sum_{i=1}^{n_m} q_{m,i\alpha}\mathbb{E}\rho_{m,i}^2 \le \frac{C_0 n_m}{2\ell_m}$$

holds for all large m, then

$$\mathbb{P}\left\{\mathrm{disc}(A_{m,1}, \dots, A_{m,n_m}) > C(C_0)\sqrt{n_m}\right\} \le B_m \exp\left\{-\frac{3C_0 n_m}{16\ell_m}\right\}$$

for all large m. Hence the Spencer-scale conclusion holds almost surely eventually whenever

$$\sum_m B_m \exp\left\{-\frac{3C_0 n_m}{16\log(eD_m)}\right\} < \infty.$$

Moreover, if $\eta_m^{(1)}, \dots, \eta_m^{(M_m)}$ are independent uniform sign vectors in $\{\pm 1\}^{n_m}$, independent across m and independent of the matrix array, then after enlarging $C(C_0)$ if necessary,

$$\mathbb{P}\left\{\min_{1\le r\le M_m}\left\|\sum_{i=1}^{n_m}\eta_{m,i}^{(r)}A_{m,i}\right\|_{\mathrm{op}} > C(C_0)\sqrt{n_m}\right\} \le B_m\exp\left\{-\frac{3C_0 n_m}{16\ell_m}\right\} + 2^{-M_m}.$$

Thus sampled certificates are eventual almost surely whenever the displayed right-hand side is summable.

Concrete summable forms are as follows.

(a) *If the rows are indexed by all integer dimensions $D = 2, 3, \dots$, if $B_D \le (eD)^\Gamma$ eventually, and if*

$$n_D \ge a\log^2(eD)$$

eventually for some $a > 16(\Gamma+1)/(3C_0)$, then the almost-sure signing conclusion holds. The sampled-certificate conclusion holds as well if $M_D \ge (1+\delta)\log_2(eD)$ eventually for some $\delta > 0$.

(b) *If $D_m \ge \exp(cm^\kappa)$ eventually, if $B_m \le (eD_m)^\Gamma$ eventually, and if*

$$n_m \ge a\log^2(eD_m)$$

eventually for some $a > 16\Gamma/(3C_0)$, then the almost-sure signing conclusion holds. The sampled-certificate conclusion holds as well if $M_m \ge (1+\delta)\log_2(e+m)$ eventually for some $\delta > 0$.

(c) *For an arbitrary dimension and block-count schedule, if for some $\gamma > 0$,*

$$n_m \ge \frac{16}{3C_0}\,\ell_m\big(\log(eB_m) + (1+\gamma)\log(e+m)\big)$$

eventually, then the almost-sure signing conclusion holds. The same $M_m \ge (1+\delta)\log_2(e+m)$ condition gives eventual sampled certificates.

For uniformly chosen block sets of random sizes $R_{m,i}$, the half-budget condition is

$$\frac{1}{B_m}\sum_i \mathbb{E}(\rho_{m,i}^2 R_{m,i}) \le \frac{C_0 n_m}{2\log(eD_m)}.$$

Proof. Apply Corollary 7.327 to the m-th row with $t_m = C_0 n_m/(2\ell_m)$. Since $\mathbb{E}\rho_{m,i}^4 \le \mathbb{E}\rho_{m,i}^2$, its half-budget estimate gives the displayed block-count row failure. The first Borel–Cantelli lemma proves the summable-array conclusion.

For sampled signs, condition on the realized matrices and on the block-overlap event in the proof of Corollary 7.327. There, $\|\sum_i A_{m,i}^2\|_{\mathrm{op}} \le C_0 n_m/\ell_m$. Corollary 7.144 therefore gives conditional sampled-sign failure at most 2^{-M_m}, after enlarging $C(C_0)$. This proves the sampled tail and its Borel–Cantelli consequence.

For all integer dimensions,

$$B_D \exp\left\{-\frac{3C_0 n_D}{16\log(eD)}\right\} \le (eD)^{\Gamma - 3C_0 a/16},$$

which is summable when $3C_0 a/16 > \Gamma + 1$. Along stretched-exponential sparse schedules the same bound is summable when $3C_0 a/16 > \Gamma$. The arbitrary-growth condition gives

$$B_m \exp\left\{-\frac{3C_0 n_m}{16\ell_m}\right\} \le e^{-1}(e+m)^{-(1+\gamma)}.$$

The sampled-sign summability follows from the displayed lower bounds on M_D or M_m. The uniform block-size identity follows from $\mathbb{P}\{\alpha \in T_{m,i} \mid R_{m,i}, \rho_{m,i}\} = R_{m,i}/B_m$. □

Remark 7.345. These pathwise statements are not the full Matrix Spencer conjecture for generic $D \times D$ random matrix ensembles when $D^2 \gg n$, unless the realization has one of the structural decompositions above or one uses an average-case input such as Corollary 7.12, or a variance-profile input such as Proposition 7.143. The entropy obstruction below applies to full-unit-ball or full-span faithful finite-block reductions; it does not rule out tuple-dependent lifts, variance-sensitive arguments, or genuinely average-case discrepancy methods for random inputs. Average-case matrix discrepancy for such ensembles is a separate line of work; see, for example, Kunisky–Zhang [17], Maillard [19], and the later combined work of Kunisky, Oertel, Wengiel and Zhang [26].

8. Boundary of the finite-lift method

The framework gives a precise limitation for exact finite block lifts.

Proposition 8.1 (Entropy obstruction for the full matrix algebra)**.** *Suppose the full algebra* $\mathrm{M}_D(\mathbb{C})$ *admits an exact finite block lift through* $\mathcal{B} = \bigoplus_\alpha \mathrm{M}_{r_\alpha}(\mathbb{C})$ *in the sense that there is a linear map from* $\mathcal{B}$ *onto* $\mathrm{M}_D(\mathbb{C})$*. Then*

$$\mathsf{E}(\mathcal{B}) = \Big(\max_\alpha r_\alpha\Big)\sum_\alpha r_\alpha \ge D^2.$$

Consequently, Theorem 1.3 cannot yield better than $C\sqrt{n\log(e + D^2/n)}$ *when applied through an exact finite block lift that is onto the full algebra* $\mathrm{M}_D(\mathbb{C})$*.*

Proof. Surjectivity onto the D^2-dimensional vector space $\mathrm{M}_D(\mathbb{C})$ forces

$$\dim \mathcal{B} = \sum_\alpha r_\alpha^2 \ge D^2.$$

Since

$$\sum_\alpha r_\alpha^2 \le \Big(\max_\alpha r_\alpha\Big) \sum_\alpha r_\alpha = \mathsf{E}(\mathcal{B}),$$

the claim follows. □

Lemma 8.2 (Proper subspaces miss a unit vector at distance one)**.** *Let X be a finite-dimensional real or complex normed space and let $L \subsetneq X$ be a proper linear subspace. Then there is $x \in X$ with $\|x\| = 1$ and*

$$\operatorname{dist}(x, L) := \inf_{\ell \in L} \|x - \ell\| = 1.$$

Proof. By Hahn–Banach, choose a nonzero continuous linear functional $\varphi \in X^*$ that vanishes on L. Normalize it so that $\|\varphi\|_{X^*} = 1$. Since X is finite-dimensional, the compact unit ball contains a point x with $\|x\| = 1$ and $|\varphi(x)| = 1$. For every $\ell \in L$,

$$\|x - \ell\| \ge |\varphi(x - \ell)| = |\varphi(x)| = 1.$$

The opposite inequality follows from $\ell = 0$, so $\operatorname{dist}(x, L) = 1$. □

Proposition 8.3 (Robust entropy obstruction for approximate faithful lifts)**.** *Let $\mathcal{B} = \bigoplus_\alpha \mathrm{M}_{r_\alpha}(\mathbb{C})$.*

(1) Suppose $W \subseteq \mathcal{B}$ is a complex linear subspace and $\Pi : W \to \mathrm{M}_D(\mathbb{C})$ is complex-linear. If for some fixed $\delta < 1$, every complex contraction $A \in \mathrm{M}_D(\mathbb{C})$, $\|A\|_{\mathrm{op}} \le 1$, is within operator-norm distance δ of $\Pi(W)$, then

$$\mathsf{E}(\mathcal{B}) \ge D^2.$$

(2) Suppose $W_{\mathrm{sa}} \subseteq \mathcal{B}_{\mathrm{sa}}$ is a real linear subspace and $\Pi : W_{\mathrm{sa}} \to \mathrm{M}_D(\mathbb{C})_{\mathrm{sa}}$ is real-linear. If for some fixed $\delta < 1$, every self-adjoint contraction $A = A^*$, $\|A\|_{\mathrm{op}} \le 1$, is within operator-norm distance δ of $\Pi(W_{\mathrm{sa}})$, then again

$$\mathsf{E}(\mathcal{B}) \ge D^2.$$

Thus the obstruction in Proposition 8.1 is not merely an exact surjectivity obstruction: any fixed-error faithful reduction of the full matrix unit ball through finite block lifts already pays quadratic entropy.

Proof. For (1), set $L = \Pi(W) \subseteq \mathrm{M}_D(\mathbb{C})$. If L were a proper complex subspace, Lemma 8.2, applied to $X = \mathrm{M}_D(\mathbb{C})$ with the operator norm, would produce a contraction A whose distance to L is 1, contradicting the assumed $\delta < 1$ approximation. Hence $L = \mathrm{M}_D(\mathbb{C})$, so

$$\sum_\alpha r_\alpha^2 = \dim_{\mathbb{C}} \mathcal{B} \ge \dim_{\mathbb{C}} W \ge \dim_{\mathbb{C}} L = D^2.$$

Since $\sum_\alpha r_\alpha^2 \le \mathsf{E}(\mathcal{B})$, the entropy bound follows.

The self-adjoint case is identical over the real vector spaces $X = \mathrm{M}_D(\mathbb{C})_{\mathrm{sa}}$ and $L = \Pi(W_{\mathrm{sa}})$. If L is proper, the lemma gives a self-adjoint contraction at distance 1 from L, contradicting $\delta < 1$. Therefore $L = \mathrm{M}_D(\mathbb{C})_{\mathrm{sa}}$, and

$$\sum_\alpha r_\alpha^2 = \dim_{\mathbb{R}} \mathcal{B}_{\mathrm{sa}} \ge \dim_{\mathbb{R}} W_{\mathrm{sa}} \ge \dim_{\mathbb{R}} \mathrm{M}_D(\mathbb{C})_{\mathrm{sa}} = D^2.$$

Again $\mathsf{E}(\mathcal{B}) \geq \sum_\alpha r_\alpha^2$. □

Corollary 8.4 (No cheap universal finite-lift reduction)**.** *Suppose a proposed reduction of the full $D \times D$ Matrix Spencer problem assigns, to every tuple of contractions $A_1, \dots, A_n \in \mathrm{M}_D(\mathbb{C})$, a finite block lift $Y_i \in \mathcal{B} = \bigoplus_\alpha \mathrm{M}_{r_\alpha}(\mathbb{C})$ and a linear map Π as in Theorem 1.3, with constants independent of D. Then, applied to any tuple whose span is all of $\mathrm{M}_D(\mathbb{C})$, the reduction must use*

$$\mathsf{E}(\mathcal{B}) \geq D^2.$$

The same conclusion holds for any fixed-error version in which, for the lifted subspace $W \subseteq \mathcal{B}$, the image $\Pi(W)$ δ-approximates the operator-norm unit ball of $\mathrm{M}_D(\mathbb{C})$, or of its self-adjoint part, for some fixed $\delta < 1$. Consequently, within this exact or fixed-error faithful finite-lift route, no structured target family whose relevant finite-block realizations all have entropy $o(D^2)$ can linearly realize all $D \times D$ matrix contractions. In particular, same-scale diagonal, circulant, Toeplitz, and bounded-block models are not universal for the full matrix unit ball. They could serve as universal targets in this sense only after enlarging the target until the associated finite-block entropy is at least quadratic in D; thus their existing structured Matrix Spencer bounds do not by themselves settle the full square conjecture at the class/full-unit-ball level.

Proof. Let $A_1, \dots, A_n$ be any tuple whose complex span is $\mathrm{M}_D(\mathbb{C})$, and set $W = \operatorname{span}\{Y_1, \dots, Y_n\} \subseteq \mathcal{B}$. Since $\Pi(Y_i) = A_i$, we have $\Pi(W) = \mathrm{M}_D(\mathbb{C})$. Extending Π linearly from W to all of $\mathcal{B}$ gives a surjection $\mathcal{B} \to \mathrm{M}_D(\mathbb{C})$, so Proposition 8.1 gives $\mathsf{E}(\mathcal{B}) \geq D^2$. The same argument applies over real spans in the self-adjoint exact setting. The fixed-error claim is precisely Proposition 8.3, applied to W or W_{sa}. The final assertions are only about structured target families used through this same exact or fixed-error faithful finite-lift mechanism. □

The preceding corollary suggests isolating the complexity parameter actually seen by Theorem 1.3.

Definition 8.5 (Finite-lift entropy of a tuple)**.** Let $A_1, \dots, A_n \in \mathrm{M}_D(\mathbb{C})$ and let $K, \Lambda \geq 1$. Define

$$\mathsf{L}_{K,\Lambda}(A_1, \dots, A_n) = \inf \mathsf{E}(\mathcal{B}),$$

where the infimum runs over all finite block algebras $\mathcal{B} = \bigoplus_\alpha \mathrm{M}_{r_\alpha}(\mathbb{C})$ for which there are elements $Y_i \in \mathcal{B}$ and a fixed linear map

$$\Pi : \operatorname{span}\{Y_1, \dots, Y_n\} \longrightarrow \mathrm{M}_D(\mathbb{C})$$

satisfying

$$\Pi(Y_i) = A_i, \qquad \|Y_i\|_{\mathcal{B}} \leq K, \qquad \|\Pi(Y)\|_{\mathrm{op}} \leq \Lambda \, \|Y\|_{\mathcal{B}} \quad (Y \in \operatorname{span}\{Y_i\}).$$

If no such lift exists, set $\mathsf{L}_{K,\Lambda}(A_1, \dots, A_n) = +\infty$.

Proposition 8.6 (Discrepancy in terms of lift entropy)**.** *For every tuple of operator-norm contractions and every $K, \Lambda \geq 1$,*

$$\operatorname{disc}_{\mathrm{op}}(A_1, \ldots, A_n) \leq C\Lambda K \sqrt{n \log\left(e + \frac{\mathsf{L}_{K,\Lambda}(A_1, \ldots, A_n)}{n}\right)}$$

whenever the right-hand side is finite. Moreover:

(1) $\mathsf{L}_{1,1}(A_1, \ldots, A_n) \leq D^2$ *for every tuple, by the identity lift through* $\mathrm{M}_D(\mathbb{C})$.

(2) If $\operatorname{span}_{\mathbb{C}}\{A_1, \ldots, A_n\} = \mathrm{M}_D(\mathbb{C})$, *then* $\mathsf{L}_{K,\Lambda}(A_1, \ldots, A_n) \geq D^2$ *for all fixed* K, Λ. *The same lower bound holds for self-adjoint tuples whose real span is* $\mathrm{M}_D(\mathbb{C})_{\mathrm{sa}}$ *when one restricts to self-adjoint lifts and real-linear maps.*

(3) If the tuple lies in a common finite-dimensional C-algebra with reduced entropy hR, then* $\mathsf{L}_{1,1}(A_1, \ldots, A_n) \leq hR$. *If all matrices are supported on a common ρ-dimensional reducing subspace, then* $\mathsf{L}_{1,1} \leq \rho^2$. *For scalar Toeplitz or Hankel contractions,* $\mathsf{L}_{C,1} \leq CD$, *and for fixed-level banded multilevel block Toeplitz or Hankel tuples it is bounded by the corresponding sampled-symbol entropy* $O_{r,b}(\prod_j (2w_j + 1))$.

Thus L *is small on precisely the structured classes handled by the finite-lift theorem, while full-span generic matrix tuples have maximal finite-lift entropy* D^2.

Proof. For any $E > \mathsf{L}_{K,\Lambda}(A_1, \ldots, A_n)$, choose a lift with $\mathsf{E}(\mathcal{B}) \leq E$ and apply Theorem 1.3; then let $E \downarrow \mathsf{L}_{K,\Lambda}$. Item (1) uses $\mathcal{B} = \mathrm{M}_D(\mathbb{C})$, $Y_i = A_i$, and $\Pi = \mathrm{id}$. Item (2) is Proposition 8.1 applied to the span of the lifted tuple; in the self-adjoint formulation it is the real self-adjoint part of Proposition 8.3 with exact image. Item (3) records the lifts already proved in Theorem 4.1, the identity lift on the common $\rho \times \rho$ corner, Theorem 5.2 and Corollary 5.4, and the sampled multilevel estimates of Theorem 5.13 and Corollary 5.17. □

This explains why Toeplitz and Hankel classes are tractable by this method: the relevant finite Fourier data has only $O(D)$ scalar degrees of freedom and admits a bounded L^∞ extension. It also explains why common block diagonalization with bounded blocks is tractable: the reduced noncommutative entropy is linear in D. Conversely, the full matrix algebra has entropy D^2, and Proposition 8.3 shows that allowing a fixed operator-norm approximation error below 1 for the full unit ball does not lower that entropy floor.

Question 8.7 (Bounded finite-entropy lift problem)**.** *Classify structured operator systems $S_D \subset \mathrm{M}_D(\mathbb{C})$ for which every finite tuple in S_D has a common uniformly bounded linear lift on its span either to a finite block algebra $\bigoplus_\alpha \mathrm{M}_{r_\alpha}(\mathbb{C})$ with*

$$\Big(\max_\alpha r_\alpha\Big) \sum_\alpha r_\alpha = O(D),$$

or to a matrix-valued multiplier model with an $O(D)$-size norming set and bounded block size. Every such class satisfies Matrix Spencer in the regime $D \le n$.

This remains a useful classification problem for nonalgebraic structured operator systems, multiplier models, and compression/extension mechanisms. For the special case where the tuple is already contained in a finite-dimensional C*-algebra, Theorem 4.2 is the sharper current input: it asks for intrinsic dimension $O(n)$, not finite-lift entropy $O(n)$.

The Toeplitz, Hankel, one-level block Toeplitz, fixed-level multilevel block Toeplitz, banded multilevel Toeplitz, finite abelian convolution, bounded-irrep convolution, simultaneous diagonalization, bounded-block diagonalization, and commutative association-scheme examples are instances of this program. The full-unit-ball/full-span faithful finite-lift route would require either a way to beat the entropy obstruction for generic matrix algebras or a method not reducible to exact or fixed-error approximate faithful lifts of the full matrix unit ball.

9. Extension to AF and von Neumann algebras

The finite-lift principle is a statement about a single finite-dimensional C*-algebra: Theorem 4.1 signs contractions that lie in a common $\mathcal{A} \simeq \bigoplus_\alpha (\mathrm{M}_{r_\alpha} \otimes I_{m_\alpha})$ at the rate set by the reduced entropy $\mathsf{E} = hR$, $h = \max_\alpha r_\alpha$, $R = \sum_\alpha r_\alpha$. It is natural to ask whether anything survives in the inductive limit, that is, for contractions in an infinite-dimensional approximately finite-dimensional (AF) C*-algebra, or more generally in a von Neumann algebra. The answer is a clean reduction together with an equally clean obstruction, and the two together delimit what the finite-stage/full-span lift route proves and where it runs into the stated obstruction.

Throughout, A is a separable unital AF C*-algebra: the norm closure

$$A = \overline{\bigcup_{k\ge 1} A_k}^{\,\|\cdot\|}, \qquad A_1 \subseteq A_2 \subseteq \cdots, \qquad A_k = \bigoplus_{\alpha=1}^{m_k} \mathrm{M}_{r_{k,\alpha}}(\mathbb{C}),$$

of an increasing tower of finite-dimensional C*-subalgebras with unital inclusions, as encoded by a Bratteli diagram [6, 10]. We write

$$h_k = \max_\alpha r_{k,\alpha}, \qquad R_k = \sum_\alpha r_{k,\alpha}, \qquad \mathsf{E}(A_k) = h_k R_k,$$

and

$$d_k = \dim_{\mathbb{C}} A_k = \sum_\alpha r_{k,\alpha}^2$$

for the entropy of the kth stage, computed by Theorem 4.1 from the reduced algebra $\bigoplus_\alpha \mathrm{M}_{r_{k,\alpha}}(\mathbb{C})$; multiplicities in the embedding $A_k \hookrightarrow A$ do not enter, exactly as in Theorem 4.1. The UHF algebra $M_{2^\infty} = \varinjlim \mathrm{M}_{2^k}(\mathbb{C})$, with $A_k \simeq \mathrm{M}_{2^k}(\mathbb{C})$ a single block of size 2^k, is the running example; there $h_k = R_k = 2^k$ and $\mathsf{E}(A_k) = d_k = 4^k$. Always $d_k \le \mathsf{E}(A_k)$, and the inequality can be strict

for heterogeneous direct sums. The elementary finite-lift theorem sees $\mathsf{E}(A_k)$; the Akbas–Sra theorem sees d_k.

The point of this section is sixfold.

(1) For contractions that already lie in a fixed stage A_k, the bound is immediate from Theorem 4.1: the supporting stage controls the discrepancy (Lemma 9.1).
(2) The bound is *stable* under the inductive limit: contractions only ε-approximable by a common finite stage obey the same bound up to an $O(n\varepsilon)$ additive error, which vanishes as $\varepsilon \to 0$ (Lemma 9.2, Theorem 9.4).
(3) Whether this is a $\sqrt{n}$ bound or a weaker bound is decided entirely by the entropy of the *supporting* stage, which is small only for special classes of elements (§9.4); non-additivity of entropy along the tower (Remark 6.3) prevents a stage-uniform bound from this entropy accounting without restrictions on the supporting stage.
(4) Using the external theorem of Akbas–Sra, exact or approximate support on a stage with $d_k = O(n)$ already gives the Spencer-scale bound, even if the finite-lift entropy $h_k R_k$ is larger.
(5) The perturbative theorem of Akbas–Sra also survives the AF reduction in finite-dimensional representations: low-Frobenius self-adjoint tails may be added to a dimension-controlled AF-stage core (Corollary 9.8).
(6) For diffuse elements with no finite-dimensional supporting subalgebra, the exact finite-stage route gives no exact-stage statement (Proposition 9.10); moreover any approximate finite-stage route must pay at least the spectral packing number of the element (Proposition 9.11).

9.1. **Reduction to the supporting stage.** The first statement requires no new idea; it is recorded to fix the form of the bound.

Lemma 9.1 (Fixed-stage bound)**.** *Let $A_1, \dots, A_n$ be contractions, $\|A_i\|_{\mathrm{op}} \le 1$, all lying in a common finite stage A_k of the tower. Then*

$$\mathrm{disc}_{\mathrm{op}}(A_1, \dots, A_n) \le C\sqrt{n \log\left(e + \frac{\mathsf{E}(A_k)}{n}\right)},$$

with C the universal constant of Theorem 4.1; the self-adjoint version holds for self-adjoint A_i. The discrepancy is computed in A, and the signs achieving it are the same as those achieving it inside A_k.

Proof. A_k is a finite-dimensional C*-algebra; apply Theorem 4.1 to its reduced form $\bigoplus_\alpha \mathrm{M}_{r_{k,\alpha}}(\mathbb{C})$, of entropy $h_k R_k = \mathsf{E}(A_k)$. The inclusion $A_k \hookrightarrow A$ is an isometric $*$-homomorphism, so the operator norm of any fixed signed sum $\sum_i \varepsilon_i A_i$ is the same whether computed in A_k or in A. Hence the minimum over signs agrees as well. □

The content is the word "common." A signing is a single choice of $\varepsilon \in \{-1, 1\}^n$ that must work for all i simultaneously, so the relevant object is the smallest stage containing the *whole tuple*, and its entropy is the cost.

9.2. **A truncation lemma.** To pass from the algebraic union to its norm closure we need a finite-stage approximant that is itself a contraction. Norm-density gives an approximant in some A_k, but it need not be a contraction; the following clamp fixes this with no loss, using only the spectral mapping theorem. We avoid any operator-Lipschitz inequality: the naive bound "f is 1-Lipschitz on $\mathbb{R}$, hence $\|f(x)-f(y)\| \le \|x-y\|$" is *false* for non-commuting self-adjoints (the absolute value is the standard counterexample), so we use a triangle-inequality argument that costs only a harmless factor.

Lemma 9.2 (Contraction approximant)**.** *Let $a = a^* \in A$ with $\|a\|_{\mathrm{op}} \le 1$, and suppose $b_0 = b_0^* \in A_k$ satisfies $\|a - b_0\|_{\mathrm{op}} < \varepsilon$. Let $f(t) = \max(-1, \min(1, t))$ and put $b = f(b_0)$, by continuous functional calculus. Then*

$$b = b^* \in A_k, \qquad \|b\|_{\mathrm{op}} \le 1, \qquad \|a - b\|_{\mathrm{op}} < 2\varepsilon.$$

Proof. Since $b_0 = b_0^* \in A_k$ and A_k is a C*-subalgebra, the continuous functional calculus of b_0 stays in A_k, so $b = f(b_0) \in A_k$ and $b = b^*$. By the spectral mapping theorem $\mathrm{sp}(b) = f(\mathrm{sp}(b_0)) \subseteq [-1, 1]$, hence $\|b\|_{\mathrm{op}} \le 1$. For the error, set $g(t) = t - f(t)$, so that $g \equiv 0$ on $[-1, 1]$ and $|g(t)| = \mathrm{dist}(t, [-1, 1])$ for all t. Then $b_0 - b = g(b_0)$ and

$$\|b_0 - b\|_{\mathrm{op}} = \max_{t \in \mathrm{sp}(b_0)} \mathrm{dist}(t, [-1, 1]).$$

For each $t \in \mathrm{sp}(b_0)$, self-adjoint spectral continuity (Weyl) gives $\mathrm{dist}(t, \mathrm{sp}(a)) \le \|a - b_0\| < \varepsilon$, and $\mathrm{sp}(a) \subseteq [-1, 1]$ because a is a self-adjoint contraction, so $\mathrm{dist}(t, [-1, 1]) \le \mathrm{dist}(t, \mathrm{sp}(a)) < \varepsilon$. Thus $\|b_0 - b\| < \varepsilon$, and $\|a - b\| \le \|a - b_0\| + \|b_0 - b\| < 2\varepsilon$. □

Remark 9.3. The factor 2 is immaterial: one applies Lemma 9.2 with $\varepsilon/2$ in place of ε, or simply absorbs it, since in Theorem 9.4 the approximation parameter is sent to 0. For non-self-adjoint a one truncates the self-adjoint and anti-self-adjoint parts $\mathrm{Re}\, a = \frac{1}{2}(a + a^*)$ and $\mathrm{Im}\, a = \frac{1}{2i}(a - a^*)$ separately and recombines, losing only a further universal constant; this is the same realification already used in the proof of Theorem 1.3. We state the self-adjoint case and note the general case in Theorem 9.4.

9.3. **The AF stability theorem.** We now combine Lemma 9.1 with the truncation. The hypothesis is that the tuple is jointly ε-close to a single stage. This is exactly the local characterization of AF algebras: for a separable AF algebra, every finite subset and every ε are captured by some finite-dimensional C*-subalgebra, and that subalgebra may be taken inside the tower [6, 10], so the hypothesis can always be met for some $k = k(\varepsilon)$.

Theorem 9.4 (AF stability)**.** *Let $A = \overline{\bigcup_k A_k}$ be a separable unital AF C*-algebra and let $A_1, \dots, A_n \in A$ be contractions. Fix $\varepsilon > 0$ and a stage index k for which there are contractions $B_i \in A_k$ with $\|A_i - B_i\|_{\mathrm{op}} < \varepsilon$ for all i (in the self-adjoint case the B_i may be taken self-adjoint, by Lemma 9.2).*

Then

$$\operatorname{disc}_{\mathrm{op}}(A_1,\dots,A_n) \le C\sqrt{n\log\!\left(e+\frac{\mathsf{E}(A_k)}{n}\right)}+2n\varepsilon.$$

Consequently

$$\operatorname{disc}_{\mathrm{op}}(A_1,\dots,A_n) \le \inf_{k\ge 1}\left\{C\sqrt{n\log\!\left(e+\frac{\mathsf{E}(A_k)}{n}\right)}+2n\,\delta_k(A_1,\dots,A_n)\right\},$$

where $\delta_k(A_1,\dots,A_n)=\max_i \operatorname{dist}_{\mathrm{op}}(A_i,A_k^{\le 1})$ *is the joint distance of the tuple to the contractions of the* k*th stage, and* $\delta_k\to 0$ *as* $k\to\infty$*. The same statements hold for* disc *in the self-adjoint case.*

Proof. By Lemma 9.1 there are signs $\varepsilon^\star\in\{-1,1\}^n$ with

$$\Big\|\sum_i \varepsilon_i^\star B_i\Big\|_{\mathrm{op}} \le C\sqrt{n\log\!\left(e+\frac{\mathsf{E}(A_k)}{n}\right)}.$$

For these same signs, the triangle inequality and $\|A_i-B_i\|<\varepsilon$ give

$$\Big\|\sum_i \varepsilon_i^\star A_i\Big\|_{\mathrm{op}} \le \Big\|\sum_i \varepsilon_i^\star B_i\Big\|_{\mathrm{op}}+\sum_{i=1}^n \|A_i-B_i\|_{\mathrm{op}} < C\sqrt{n\log\!\left(e+\frac{\mathsf{E}(A_k)}{n}\right)}+n\varepsilon.$$

Since the B_i produced by Lemma 9.2 from an $(\varepsilon/2)$-approximant satisfy $\|A_i-B_i\|<\varepsilon$, the displayed bound with the stated $2n\varepsilon$ (from running Lemma 9.2 at parameter ε, giving $\|A_i-B_i\|<2\varepsilon$) follows; either normalization is the claim after relabelling ε. Taking the infimum over k, and over the choice of B_i realizing δ_k up to an arbitrarily small slack, gives the second display. Finally $\delta_k\to 0$ is the local characterization of AF algebras: the union $\bigcup_k A_k$ is norm-dense, so each A_i is approximated to within any ε by some stage, and after Lemma 9.2 the approximant is a contraction; taking the largest stage among the n approximants makes the approximation joint. The general (non-self-adjoint) case truncates real and imaginary parts separately and absorbs the resulting constant into C. □

The estimate is genuinely an infimum over a *trade-off*. Lowering the approximation error $\varepsilon=\delta_k$ forces a higher stage k, hence a larger $\mathsf{E}(A_k)$; the optimal k balances the additive sampling-type error $2n\delta_k$ against the entropy term. Two regimes make this concrete.

Corollary 9.5 (Elements of finite type)**.** *If the tuple lies exactly in some stage,* $\{A_1,\dots,A_n\}\subseteq A_k$ *with* $\delta_k=0$*, then the additive error vanishes and*

$$\operatorname{disc}_{\mathrm{op}}(A_1,\dots,A_n) \le C\sqrt{n\log\!\left(e+\frac{\mathsf{E}(A_k)}{n}\right)}.$$

In particular, if moreover $\mathsf{E}(A_k)=O(n)$*, the tuple satisfies the conjectured Matrix Spencer bound* $\operatorname{disc}_{\mathrm{op}}=O(\sqrt{n})$*, with a constant independent of every later stage and of the ambient (infinite) algebra* A*.*

Corollary 9.6 (Quantitative limit elements)**.** *Suppose the tower has stage entropies* $\mathsf{E}(A_k)$ *and the tuple has joint approximation rate* δ_k*. If there is a stage* $k^\star = k^\star(n)$ *with* $\mathsf{E}(A_{k^\star}) = O(n)$ *and* $\delta_{k^\star} = O(n^{-1/2})$*, then*

$$\operatorname{disc}_{\mathrm{op}}(A_1, \dots, A_n) = O(\sqrt{n}).$$

Proof. Take $k = k^\star$ in Theorem 9.4. The entropy term is $C\sqrt{n \log(e + O(1))} = O(\sqrt{n})$ and the additive term is $2n \cdot O(n^{-1/2}) = O(\sqrt{n})$. □

Corollary 9.6 is the precise sense in which the bound is "stable under the inductive limit": one never needs the element to sit in a stage, only to be approximable by a controlled-entropy stage at the scale $n^{-1/2}$ that the $\sqrt{n}$ target already tolerates.

Corollary 9.7 (Intrinsic-dimension AF stage bound)**.** *For every* $C_0 \geq 1$ *there is* $C(C_0)$ *such that the following holds. Let* $A = \overline{\bigcup_k A_k}$ *be a separable unital AF algebra and let* $A_1, \dots, A_n \in A$ *be contractions. Suppose that for some stage* k *with*

$$d_k = \dim_{\mathbb{C}} A_k \leq C_0 n$$

there are contractions $B_i \in A_k$ *satisfying* $\|A_i - B_i\|_{\mathrm{op}} < \varepsilon$ *for all* i*. Then*

$$\operatorname{disc}_{\mathrm{op}}(A_1, \dots, A_n) \leq C(C_0)\sqrt{n} + 2n\varepsilon.$$

Consequently, if the tuple is jointly approximable at scale $O(n^{-1/2})$ *by stages with intrinsic dimension* $O(n)$*, then it satisfies the Spencer-scale bound* $O(\sqrt{n})$*. The self-adjoint version holds for self-adjoint tuples.*

Proof. Apply Theorem 4.2 to the stage tuple $B_1, \dots, B_n \in A_k$, and then use the same triangle-inequality perturbation as in Theorem 9.4. The factor 2 is kept to match the normalization of Theorem 9.4, where contraction approximants may be obtained by the truncation lemma. □

Corollary 9.8 (Low-Frobenius perturbations of AF-stage cores)**.** *For every* $C_0 \geq 1$ *there is* $C(C_0)$ *such that the following holds. Let* $A = \overline{\bigcup_k A_k}$ *be a separable unital AF algebra and let* $\pi : A \to \mathrm{M}_D(\mathbb{C})$ *be a unital *-representation. Let*

$$X_i = H_i + L_i, \qquad i = 1, \dots, n,$$

be self-adjoint matrices, where $H_i \in \pi(A)$ *are self-adjoint contractions and the perturbations* $L_i = L_i^*$ *satisfy*

$$\|L_i\|_{\mathrm{op}} \leq 1, \qquad \|L_i\|_{\mathrm{F}}^2 \leq \frac{n}{\log^3(en)}$$

for all i*. Suppose that for some stage* k *with* $\dim_{\mathbb{C}} A_k \leq C_0 n$ *there are self-adjoint contractions* $B_i \in \pi(A_k)$ *satisfying* $\|H_i - B_i\|_{\mathrm{op}} < \varepsilon$ *for all* i*. Then*

$$\operatorname{disc}(X_1, \dots, X_n) \leq C(C_0)\sqrt{n} + 2n\varepsilon.$$

Consequently, an AF algebraic core that is jointly captured at scale $O(n^{-1/2})$ *by stages of intrinsic dimension* $O(n)$ *remains in the Spencer regime after such*

low-Frobenius self-adjoint perturbations. In particular, the same conclusion holds if the Frobenius hypothesis on the tails is replaced by

$$\operatorname{rank}(L_i) \le \frac{n}{\log^3(en)}$$

together with $\|L_i\|_{\mathrm{op}} \le 1$.

Proof. Apply Theorem 4.3 to the self-adjoint matrices $B_i + L_i$. For the resulting signs,

$$\left\|\sum_i \varepsilon_i X_i\right\|_{\mathrm{op}} \le \left\|\sum_i \varepsilon_i (B_i + L_i)\right\|_{\mathrm{op}} + \sum_i \|H_i - B_i\|_{\mathrm{op}} \le C(C_0)\sqrt{n} + n\varepsilon,$$

and the displayed $2n\varepsilon$ form again matches the truncation normalization used above. □

9.4. **Which AF elements have a controlled supporting stage.** Corollary 9.5 is only as useful as the supply of tuples whose supporting stage has entropy $O(n)$. Here the structure of the Bratteli diagram is decisive, and the obstruction of §9.5 is the reason a *uniform* statement from supporting-stage entropy alone is unavailable.

The favourable case: bounded-block towers. Suppose the tower has uniformly bounded block size, $h_k = \max_\alpha r_{k,\alpha} \le b$ for all k and some fixed b. Then $\mathsf{E}(A_k) = h_k R_k \le b R_k$ is linear in the reduced dimension R_k, and any tuple supported on a stage with $R_k = O(n)$ is in the Spencer regime by Corollary 9.5. Such AF algebras are exactly the ones whose Bratteli diagram has bounded vertex labels; the commutative AF algebras (all $r_{k,\alpha} = 1$, $b = 1$) are the extreme case, where $\mathsf{E}(A_k) = R_k$ is the number of minimal projections of the stage. A self-adjoint contraction supported on a commutative stage with $s = O(n)$ atoms is signed at the rate $\sqrt{n \log(e + s/n)}$, recovering the association-scheme and simultaneous-diagonalization corollaries of the main text in the limit.

The current larger favourable case: dimension-controlled stages. The bounded-block hypothesis is no longer the limit of what is known for exact finite-dimensional algebraic support. By Corollary 9.7, any tuple supported, or $O(n^{-1/2})$-approximated, on a stage with

$$d_k = \sum_\alpha r_{k,\alpha}^2 = O(n)$$

has discrepancy $O(\sqrt{n})$. This can be strictly stronger than the finite-lift criterion $\mathsf{E}(A_k) = h_k R_k = O(n)$ when a stage is a heterogeneous direct sum: a few larger blocks may make $h_k R_k$ large by multiplying the whole reduced dimension, while their actual square-sum contribution remains small. For a single full matrix block, however, $d_k = \mathsf{E}(A_k) = D^2$, so the new theorem does not change the full-block threshold.

The unfavourable case: UHF and growing blocks. For M_{2^∞} the block size doubles at every stage, $h_k = R_k = 2^k$, so $\mathsf{E}(A_k) = 4^k$ grows quadratically in the stage dimension 2^k; here the Akbas–Sra parameter d_k is the same 4^k. An arbitrary tuple controlled only through the full-stage bound lies in $\mathrm{M}_{2^k}(\mathbb{C})$ and gets the rate $\sqrt{n \log(e + 4^k/n)}$; to keep this at $\sqrt{n}$ one needs $4^k = O(n)$, i.e. $2^k = O(\sqrt{n})$, so the full-stage entropy criterion reaches arbitrary full-span tuples only at *logarithmically shallow* stages $k \le \frac{1}{2}\log_2 n + O(1)$. The full unit ball of $\mathrm{M}_{2^k}(\mathbb{C})$, and tuples whose compressed span is the whole block, are the full-matrix instances against which Proposition 8.1 warns: a single block of size 2^k carries 4^k scalar degrees of freedom, and no full-span faithful finite block lift of it can have entropy below 4^k.

Entropy is not additive along the tower. The two cases are not reconciled by summing. If a tuple decomposes as $A_i = \sum_\ell A_i^{(\ell)}$ with the ℓth piece supported on a sub-block $B_\ell \subseteq A_k$ of entropy $\mathsf{E}(B_\ell) = h(B_\ell)R(B_\ell)$, then by Proposition 6.2 and Remark 6.3 the entropy that governs the signed sum is

$$\mathsf{E}\Big(\bigoplus_\ell B_\ell\Big) = \Big(\max_\ell h(B_\ell)\Big) \sum_\ell R(B_\ell),$$

not $\sum_\ell \mathsf{E}(B_\ell)$. A single large block anywhere in the support raises $\max_\ell h(B_\ell)$ and multiplies the entire reduced dimension by it. Along a UHF tower this is the quantitative obstruction to a stage-uniform bound: the maximal block size $h_k = 2^k$ is a prefactor on *all* of R_k, so the entropy cannot be kept $O(n)$ for the full supporting stage once the support reaches depth $k > \frac{1}{2}\log_2 n$, however the tuple is decomposed.

Commutative decompositions do not evade the full-block floor. There is one useful special case of the direct-sum accounting. Let $S_j \subset \mathrm{M}_D(\mathbb{C})_{\mathrm{sa}}$ be the self-adjoint part of a maximal abelian subalgebra. If

$$S_1 + \cdots + S_s = \mathrm{M}_D(\mathbb{C})_{\mathrm{sa}},$$

then $s \ge D + 1$. Indeed, each S_j contains the common scalar direction $\mathbb{R}I$, and $S_j \cap \{X : \operatorname{tr} X = 0\}$ has real dimension $D - 1$. Projecting away $\mathbb{R}I$, the traceless space has dimension $D^2 - 1$, so

$$D^2 - 1 \le s(D - 1).$$

Thus $s \ge D + 1$. Consequently a faithful decomposition of a full self-adjoint block into commutative D-dimensional pieces, followed by the direct-sum lift of Proposition 6.2, has combined entropy

$$1 \cdot sD \ge D(D + 1) > D^2.$$

Splitting a full block across many MASAs therefore preserves, rather than removes, the quadratic entropy floor.

9.5. **The obstruction: von Neumann algebras and generic AF elements.** Theorem 9.4 requires a *finite*-dimensional approximating stage. Two distinct failures of this hypothesis bound the method from outside.

The first is internal to the finite-lift mechanism and is Proposition 8.3 read along the tower: if the compressed span of the tuple on a full matrix block $\mathrm{M}_D(\mathbb{C})$ is all of that block, or even if a proposed finite-block image approximates the full unit ball in that block to a fixed operator-norm error < 1, then the lift algebra has entropy at least D^2. This is a ceiling on full-span faithful finite-block reductions along an AF tower.

Proposition 9.9 (Robust entropy ceiling along an AF tower)**.** *Let $A = \varinjlim A_k$ be an AF algebra and suppose a supporting stage contains a full matrix block $\mathrm{M}_D(\mathbb{C})$, with central projection P. Suppose a finite block lift through $\mathcal{B} = \bigoplus_\alpha \mathrm{M}_{r_\alpha}(\mathbb{C})$ is used for the compressed tuple PA_iP. If the image of the lifted span is all of $\mathrm{M}_D(\mathbb{C})$, or more generally if for some fixed $\delta < 1$ it is within operator-norm distance δ of every contraction in $\mathrm{M}_D(\mathbb{C})$, then*

$$\mathsf{E}(\mathcal{B}) \geq D^2.$$

The same conclusion holds in the self-adjoint setting when the real image δ-approximates every self-adjoint contraction. Consequently the entropy term in Theorem 1.3, applied through any such full-span faithful lift, cannot be better than the scale

$$\sqrt{n \log\left(e + \frac{D^2}{n}\right)}.$$

In the regime $D = \Theta(\sqrt{n})$ this is the conjectured $O(\sqrt{n})$; for $D \gg \sqrt{n}$ it is strictly weaker, and improving it within this route would require leaving the full-span faithful finite-block setup or avoiding the entropy dependence.

Proof. Apply Proposition 8.3 to the image of the proposed lift on the compressed block $PA_kP \simeq \mathrm{M}_D(\mathbb{C})$, using the complex version in the non-self-adjoint case and the real self-adjoint version for self-adjoint tuples. It gives

$$\sum_\alpha r_\alpha^2 \geq D^2.$$

Since $\sum_\alpha r_\alpha^2 \leq \mathsf{E}(\mathcal{B})$, the entropy bound follows, and the stated scale is the entropy contribution in Theorem 1.3. □

The second failure is external to the exact finite-stage route: it removes *any* finite supporting stage. It applies to von Neumann algebras that are not AF, and to AF algebras for elements whose spectral data is genuinely infinite.

Proposition 9.10 (No finite stage for diffuse elements)**.** *Let $a = a^*$ be a self-adjoint element of a C*-algebra M whose spectrum $\mathrm{sp}(a)$ is an infinite set. Then a lies in no finite-dimensional C*-subalgebra of M. Consequently:*

(1) In any von Neumann algebra without minimal projections (a diffuse algebra; in particular any II_1 factor), there exist self-adjoint contractions a – for instance the real part $\frac{1}{2}(u + u^*)$ of a Haar unitary, with atomless arcsine spectral distribution on $[-1, 1]$ – that lie in no finite-dimensional subalgebra. For such an element neither Theorem 4.1 nor Theorem 9.4

applies, and the finite-lift method gives no exact-stage signing theorem; this does not rule out tuple-dependent lifts or the trivial one-element signing bound.

(2) *In an infinite-dimensional AF algebra* $A = \overline{\bigcup_k A_k}$ *every self-adjoint element has a finite-spectrum approximant in some stage (Lemma 9.2), but a generic self-adjoint element of* A *has infinite spectrum and therefore lies in no single stage; only the* approximate *statement of Theorem 9.4 is available, at the entropy of the stage that realizes the approximation.*

Proof. By the continuous functional calculus for a single self-adjoint element, $C^*(a, 1) \cong C(\mathrm{sp}(a))$. If a were contained in a finite-dimensional C*-subalgebra $C \subseteq M$, then $C^*(a, 1) \subseteq C$ would be finite-dimensional, forcing $C(\mathrm{sp}(a))$ finite-dimensional, i.e. $\mathrm{sp}(a)$ finite. This proves the displayed claim. For (1): a diffuse von Neumann algebra contains a copy of $L^\infty[0, 1]$, hence a self-adjoint element with Lebesgue (atomless) spectral measure and continuous spectrum; equivalently any II_1 factor contains a Haar unitary u, and $\mathrm{Re}\, u = \frac{1}{2}(u + u^*)$ has the arcsine law on $[-1, 1]$, an infinite (indeed uncountable, atomless) spectrum. Such a lies in no finite-dimensional subalgebra by the displayed claim, so it is contained in no stage of any tower and is contained in no finite-dimensional C*-subalgebra/stage; the finite-lift method has no finite-dimensional object to feed Theorem 4.1. For (2): in an infinite-dimensional AF algebra, finite-spectrum self-adjoints are dense, but they are meagre: for each fixed m, the set of self-adjoint elements with at most m spectral values is closed with empty interior. Hence a Baire-generic self-adjoint element has infinite spectrum, while the approximate bound is exactly Theorem 9.4. □

Proposition 9.11 (Spectral floor for finite-stage approximation)**.** *Let* $a = a^*$ *be a self-adjoint contraction in a C*-algebra* M*, and let*

$$B \simeq \bigoplus_{\alpha=1}^{m} \mathrm{M}_{r_\alpha}(\mathbb{C})$$

be a unital finite-dimensional C-subalgebra of* M*. Put* $R(B) = \sum_\alpha r_\alpha$ *and* $\mathsf{E}(B) = (\max_\alpha r_\alpha) R(B)$*. If* $b = b^* \in B$ *and* $\|a - b\|_{\mathrm{op}} < \eta$*, then*

$$R(B) \geq P_\eta(\mathrm{sp}(a)), \qquad \mathsf{E}(B) \geq P_\eta(\mathrm{sp}(a)),$$

where $P_\eta(E)$ *denotes the largest cardinality of a subset of* $E \subset \mathbb{R}$ *whose distinct points are separated by more than* 2η*. Consequently, if* $\mathrm{sp}(a)$ *is infinite, then the entropy of any finite-dimensional subalgebra containing self-adjoint approximants* $b_\eta \to a$ *must tend to infinity. If* $\mathrm{sp}(a)$ *contains a nontrivial interval, then this lower bound is* $\Omega(\eta^{-1})$ *as* $\eta \downarrow 0$*.*

Proof. For self-adjoint elements, spectra move by at most the operator-norm perturbation: the Hausdorff distance between $\mathrm{sp}(a)$ and $\mathrm{sp}(b)$ is at most $\|a - b\|_{\mathrm{op}} < \eta$. Let $\lambda_1, \ldots, \lambda_p \in \mathrm{sp}(a)$ be pairwise separated by more than 2η, with $p = P_\eta(\mathrm{sp}(a))$. For each j choose $\mu_j \in \mathrm{sp}(b)$ with $|\lambda_j - \mu_j| < \eta$. If

$\mu_i = \mu_j$ for $i \neq j$, then $|\lambda_i - \lambda_j| < 2\eta$, a contradiction. Hence $\mathrm{sp}(b)$ has at least p points.

On the other hand, a self-adjoint element of $\bigoplus_\alpha \mathrm{M}_{r_\alpha}(\mathbb{C})$ has at most $\sum_\alpha r_\alpha = R(B)$ distinct spectral values, since each block contributes at most its matrix size. Thus $R(B) \geq p$, and $\mathsf{E}(B) \geq R(B) \geq p$. If $\mathrm{sp}(a)$ is infinite, its packing numbers $P_\eta(\mathrm{sp}(a))$ are unbounded as $\eta \downarrow 0$; if it contains an interval of positive length, the elementary packing bound on an interval gives $P_\eta(\mathrm{sp}(a)) \geq c\,\eta^{-1}$ for small η. □

Remark 9.12 (What the spectral floor does and does not prove). Proposition 9.11 is a lower bound on the finite-stage entropy required to approximate a diffuse self-adjoint element; it is not a Matrix Spencer lower bound. For example, if $\mathrm{sp}(a) = [-1, 1]$ and one wants the approximation term in Theorem 9.4 to be $O(\sqrt{n})$, then one takes $\eta = O(n^{-1/2})$, and Proposition 9.11 only forces $\mathsf{E}(B) \gtrsim \sqrt{n}$. This is still compatible with the Spencer-scale upper bound. The proposition's role is narrower: it rules out a dimension-free finite-stage model for diffuse spectral data and makes the approximation/entropy trade-off in Theorem 9.4 unavoidable.

Corollary 9.13 (Commutative AF partition trade-off). *Let $A = C(X)$ be a commutative AF algebra, so finite stages are algebras of functions constant on finite clopen partitions. Let $f_1, \ldots, f_n \in C(X)$ be real-valued contractions, and let $\mathcal{P}$ be a finite clopen partition of X with $M = \#\mathcal{P}$. Suppose*

$$\mathrm{osc}_{\mathcal{P}}(f_i) := \max_{P \in \mathcal{P}} \sup_{x,y \in P} |f_i(x) - f_i(y)| \leq \eta \qquad (1 \leq i \leq n).$$

Then

$$\mathrm{disc}(f_1, \ldots, f_n) \leq C\sqrt{n \log\left(e + \frac{M}{n}\right)} + 2n\eta.$$

Consequently, if for each n the tuple is captured by a clopen partition with $M = O(n)$ and $\eta = O(n^{-1/2})$, then $\mathrm{disc}(f_1, \ldots, f_n) = O(\sqrt{n})$.

Conversely, if one $f_i(X)$ contains an interval of positive length, then any finite clopen partition that supports an η-approximant to that f_i must have $M = \Omega(\eta^{-1})$ atoms as $\eta \downarrow 0$. In particular, any commutative finite-stage certificate for the $O(n^{-1/2})$ approximation scale must use $M = \Omega(\sqrt{n})$ atoms. This is a lower bound on the stage complexity required by the finite-stage certificate, not a lower bound on the discrepancy itself.

Proof. For each $P \in \mathcal{P}$ and each i, choose a point $x_P \in P$ and let g_i be the function that is identically $f_i(x_P)$ on P. Then g_i lies in the finite-dimensional algebra $B_{\mathcal{P}} \simeq \mathbb{C}^M$, $\|f_i - g_i\|_\infty \leq \eta$, and $\mathsf{E}(B_{\mathcal{P}}) = M$. The displayed upper bound is Theorem 9.4 in the self-adjoint/commutative case.

For the lower statement, let $f = f_i$ and suppose $f(X)$ contains an interval I of length $\ell > 0$. If $g \in B_{\mathcal{P}}$ and $\|f - g\|_\infty < \eta$, then Proposition 9.11 gives

$$M = \mathsf{E}(B_{\mathcal{P}}) \geq P_\eta(f(X)).$$

The interval I contains at least $c_\ell \eta^{-1}$ points separated by more than 2η, for all sufficiently small η. Hence $M \geq c_\ell \eta^{-1}$. Taking $\eta \asymp n^{-1/2}$ gives the $\Omega(\sqrt{n})$ certificate-size floor. □

Corollary 9.14 (Metric clopen partitions)**.** *Let X be a compact zero-dimensional metric space and, for $r > 0$, let $M_X(r)$ be the least cardinality of a finite clopen partition of X whose atoms all have diameter at most r. Let $f_1, \dots, f_n \in C(X)$ be real-valued contractions, and put*

$$\omega(r) := \max_{1 \leq i \leq n} \sup_{d(x,y) \leq r} |f_i(x) - f_i(y)|.$$

Then, for every $r > 0$ with $M_X(r) < \infty$,

$$\operatorname{disc}(f_1, \dots, f_n) \leq C \sqrt{n \log\left(e + \frac{M_X(r)}{n}\right)} + 2n\omega(r).$$

In particular, if $M_X(r) \leq A r^{-d}$ for $0 < r \leq 1$ and the tuple is uniformly (L, α)-Hölder, meaning $\omega(r) \leq L r^\alpha$, then

$$\operatorname{disc}(f_1, \dots, f_n) \leq C_{A,d,L,\alpha} \sqrt{n \log\left(e + n^{d/(2\alpha)-1}\right)}.$$

Thus polynomial clopen partition growth gives an $O(\sqrt{n \log n})$ bound for fixed d and α, and the dimension-regularity threshold $d \leq 2\alpha$ gives the Spencer-scale $O(\sqrt{n})$ bound.

Proof. If every atom of $\mathcal{P}$ has diameter at most r, then $\operatorname{osc}_{\mathcal{P}}(f_i) \leq \omega(r)$ for every i. The first display is Corollary 9.13. Under the polynomial partition-growth and Hölder hypotheses, take $r = \min(1, (L\sqrt{n})^{-1/\alpha})$. The additive term is $O_{L,\alpha}(\sqrt{n})$, and

$$\frac{M_X(r)}{n} \leq C_{A,d,L,\alpha} n^{d/(2\alpha)-1}$$

after increasing the constant to absorb the finitely many small n. This gives the displayed bound and its two stated consequences. □

Lemma 9.15 (Clopen shrinkings of finite covers)**.** *Let X be a compact zero-dimensional metric space. If $U_1, \dots, U_m$ is a finite open cover of X, then, after allowing empty sets, there are pairwise disjoint clopen sets $V_1, \dots, V_m$ such that*

$$V_j \subset U_j \qquad (1 \leq j \leq m), \qquad \bigcup_{j=1}^{m} V_j = X.$$

Consequently, if $N_X(\rho)$ is the least number of open ρ-balls needed to cover X, then

$$M_X(2\rho) \leq N_X(\rho) \qquad (\rho > 0).$$

Proof. We first recall the elementary separation property used below. If $F \subset U \subset X$, F is closed and U is open, then zero dimensionality gives, for every $x \in F$, a clopen neighborhood C_x with $x \in C_x \subset U$. Compactness of

F gives a finite subcover, and the union of the selected C_x's is clopen; hence there is a clopen C with $F \subset C \subset U$.

The cover refinement follows by induction on m. For $m = 1$ there is nothing to prove. For $m > 1$, set

$$F_1 = X \setminus \bigcup_{j=2}^{m} U_j \subset U_1.$$

Choose a clopen V_1 with $F_1 \subset V_1 \subset U_1$. Then $X \setminus V_1$ is compact, zero-dimensional, and covered by $U_2, \dots, U_m$. Applying the induction hypothesis in this clopen subspace gives $V_2, \dots, V_m$, which are also clopen in X.

For the last assertion, cover X by $N_X(\rho)$ open ρ-balls and apply the first part. Each resulting clopen atom lies in one of those balls and therefore has diameter at most 2ρ. □

Corollary 9.16 (Ordinary covering entropy)**.** *Let X be a compact zero-dimensional metric space, let $N_X(\rho)$ be the least number of open ρ-balls needed to cover X, and let $f_1, \dots, f_n \in C(X)$ be real-valued contractions with common modulus*

$$\omega(r) := \max_{1 \le i \le n} \sup_{d(x,y) \le r} |f_i(x) - f_i(y)|.$$

Then, for every $\rho > 0$,

$$\operatorname{disc}(f_1, \dots, f_n) \le C \sqrt{n \log\left(e + \frac{N_X(\rho)}{n}\right)} + 2n\omega(2\rho).$$

In particular, if

$$N_X(\rho) \le A\rho^{-d} \qquad (0 < \rho \le 1)$$

and the tuple is uniformly (L, α)-Holder, then

$$\operatorname{disc}(f_1, \dots, f_n) \le C_{A,d,L,\alpha} \sqrt{n \log\left(e + n^{d/(2\alpha)-1}\right)}.$$

Thus ordinary polynomial metric entropy gives the same $O(\sqrt{n \log n})$ bound as clopen partition growth, and the threshold $d \le 2\alpha$ gives the Spencer-scale bound $O(\sqrt{n})$. More generally, if

$$N_X(\rho) \le A\rho^{-2\alpha} \left(\log(e/\rho)\right)^{\beta} \qquad (0 < \rho \le 1),$$

then

$$\operatorname{disc}(f_1, \dots, f_n) \le \begin{cases} C_{A,L,\alpha,\beta}\sqrt{n}, & \beta \le 0, \\ C_{A,L,\alpha,\beta}\sqrt{n \log\log(e^e + n)}, & \beta > 0. \end{cases}$$

Proof. Lemma 9.15 gives $M_X(2\rho) \le N_X(\rho)$, so the first display is Corollary 9.14 applied at scale 2ρ. If $N_X(\rho) \le A\rho^{-d}$ and $\omega(r) \le Lr^{\alpha}$, take

$$\rho = \min\left(1, \frac{1}{2}(L\sqrt{n})^{-1/\alpha}\right).$$

Then $2n\omega(2\rho) \le 2nL(2\rho)^\alpha = O_{L,\alpha}(\sqrt{n})$, while

$$\frac{N_X(\rho)}{n} \le C_{A,d,L,\alpha} n^{d/(2\alpha)-1}$$

after enlarging the constant for the finitely many small n. The logarithmic statement is identical, with $N_X(\rho)/n \le C_{A,L,\alpha}(\log(e+n))^\beta$. □

Definition 9.17 (Ordinary metric Minkowski data)**.** For a compact metric space X, let

$$\overline{\dim}_{\mathrm{M}}(X) := \limsup_{\rho \downarrow 0} \frac{\log N_X(\rho)}{\log(1/\rho)}$$

be its upper metric Minkowski dimension, where $N_X(\rho)$ denotes the least number of open ρ-balls needed to cover X. For $s \ge 0$, let

$$\overline{\mathcal{M}}^s_{\mathrm{M}}(X) := \limsup_{\rho \downarrow 0} \rho^s N_X(\rho)$$

be the corresponding upper metric s-dimensional Minkowski content.

Corollary 9.18 (Ordinary Minkowski threshold)**.** *Let X be a compact zero-dimensional metric space and let $f_1, \dots, f_n \in C(X)$ be real-valued contractions with common (L,α)-Hölder modulus. If $\overline{\dim}_{\mathrm{M}}(X) < \infty$, then*

$$\operatorname{disc}(f_1, \dots, f_n) \le C_{X,L,\alpha} \sqrt{n \log(e+n)}.$$

If, in addition,

$$\overline{\dim}_{\mathrm{M}}(X) < 2\alpha,$$

then

$$\operatorname{disc}(f_1, \dots, f_n) \le C_{X,L,\alpha} \sqrt{n}.$$

At the critical exponent, the same Spencer-scale conclusion holds whenever

$$\overline{\mathcal{M}}^{2\alpha}_{\mathrm{M}}(X) < \infty.$$

Thus the metric AF bound can be read from the ordinary covering dimension and ordinary critical content of X, without separately estimating clopen partition growth.

Proof. If $D > \overline{\dim}_{\mathrm{M}}(X)$, then there are constants $C_D < \infty$ and $\rho_D > 0$ such that

$$N_X(\rho) \le C_D \rho^{-D} \qquad (0 < \rho \le \rho_D).$$

After enlarging C_D to cover the finitely many larger scales, and after the same harmless metric rescaling used below, Corollary 9.16 gives

$$\operatorname{disc}(f_1, \dots, f_n) \le C_{D,L,\alpha,X} \sqrt{n \log\bigl(e + n^{D/(2\alpha)-1}\bigr)}.$$

This is $O_{X,L,\alpha}(\sqrt{n \log(e+n)})$ for finite D. If $\overline{\dim}_{\mathrm{M}}(X) < 2\alpha$, choose D strictly between $\overline{\dim}_{\mathrm{M}}(X)$ and 2α, making the logarithm bounded.

Finally, finite upper metric 2α-content gives

$$N_X(\rho) \le C_0 \rho^{-2\alpha} \qquad (0 < \rho \le \rho_0)$$

for all sufficiently small ρ. Enlarging the constant for larger scales puts the hypotheses of Corollary 9.16 in force with $d = 2\alpha$, hence the logarithmic factor is bounded and the Spencer-scale estimate follows. □

Definition 9.19 (Upper Assouad growth)**.** Let X be a compact metric space with $\operatorname{diam} X \le 1$. We say that X has upper Assouad growth (A, s) if every closed ball in X of radius $R \le 1$ can be covered by at most

$$A\left(\frac{R}{\rho}\right)^s$$

open balls of radius ρ, for all $0 < \rho \le R \le 1$.

Definition 9.20 (Logarithmic upper Assouad growth)**.** Let X be a compact metric space with $\operatorname{diam} X \le 1$. We say that X has logarithmic upper Assouad growth (A, s, γ) if every closed ball in X of radius $R \le 1$ can be covered by at most

$$A\left(\frac{R}{\rho}\right)^s \left(\log(eR/\rho)\right)^\gamma$$

open balls of radius ρ, for all $0 < \rho \le R \le 1$.

Corollary 9.21 (Doubling and Assouad entropy)**.** *Let X be a compact zero-dimensional metric space with* $\operatorname{diam} X \le 1$*, and let $f_1, \dots, f_n \in C(X)$ be real-valued contractions with common (L, α)-Hölder modulus. If X has upper Assouad growth (A, s), then*

$$\operatorname{disc}(f_1, \dots, f_n) \le C_{A,s,L,\alpha} \sqrt{n \log\left(e + n^{s/(2\alpha)-1}\right)}.$$

In particular, $s \le 2\alpha$ gives the Spencer-scale bound

$$\operatorname{disc}(f_1, \dots, f_n) \le C_{A,s,L,\alpha} \sqrt{n}.$$

Consequently, if X is D-doubling, meaning that every closed ball of radius $R \le 1$ is covered by at most D open balls of radius $R/2$, then the same display holds with $s = \log_2 D$ and constants depending on D, L, α. Thus a D-doubling zero-dimensional model is Spencer-scale for (L, α)-Hölder tuples whenever $\log_2 D \le 2\alpha$.

Proof. Fix $x_0 \in X$. Since $\operatorname{diam} X \le 1$, X is contained in the closed ball $\overline{B}(x_0, 1)$. The upper Assouad growth assumption with $R = 1$ gives

$$N_X(\rho) \le A\rho^{-s} \qquad (0 < \rho \le 1).$$

The first two assertions are therefore Corollary 9.16.

If X is D-doubling, iterate the doubling cover. For each integer $k \ge 0$, X is covered by at most D^k balls of radius 2^{-k}. Given $0 < \rho \le 1$, choose k with $2^{-k} \le \rho < 2^{-k+1}$. Then

$$N_X(\rho) \le D^k \le D\, \rho^{-\log_2 D}.$$

Thus X has the global polynomial covering bound with exponent $\log_2 D$, and another application of Corollary 9.16 gives the stated doubling consequence. □

Corollary 9.22 (Finite products add ordinary entropy)**.** *Let X_j, $1 \le j \le m$, be compact zero-dimensional metric spaces and let $X = \prod_{j=1}^m X_j$ with the max product metric. Suppose*

$$N_{X_j}(\rho) \le A_j \rho^{-d_j} \qquad (0 < \rho \le 1),$$

and put $d_* = \sum_{j=1}^m d_j$. If $f_1, \dots, f_n \in C(X)$ are real-valued contractions with common (L, α)-Hölder modulus, then

$$\operatorname{disc}(f_1, \dots, f_n) \le C_{\{A_j, d_j\}, L, \alpha, m} \sqrt{n \log\bigl(e + n^{d_*/(2\alpha)-1}\bigr)}.$$

In particular, $d_* \le 2\alpha$ gives the Spencer-scale bound

$$\operatorname{disc}(f_1, \dots, f_n) \le C_{\{A_j, d_j\}, L, \alpha, m} \sqrt{n}.$$

Equivalently, finite ordinary upper metric dimensions add under finite max-metric products: if

$$\sum_{j=1}^m \overline{\dim}_{\mathrm{M}}(X_j) < 2\alpha,$$

then every common (L, α)-Hölder tuple on X is Spencer-scale. At the critical endpoint, the same conclusion holds if

$$\sum_{j=1}^m s_j = 2\alpha \quad \text{and} \quad \overline{\mathcal{M}}_{\mathrm{M}}^{s_j}(X_j) < \infty \quad (1 \le j \le m).$$

Likewise, if each X_j has upper Assouad growth (A_j, s_j) and $\operatorname{diam} X_j \le 1$, then X has upper Assouad growth $(\prod_j A_j, \sum_j s_j)$, so $\sum_j s_j \le 2\alpha$ gives Spencer scale. In particular, a finite max product of D_j-doubling factors with $\operatorname{diam} X_j \le 1$ is Spencer-scale whenever

$$\sum_{j=1}^m \log_2 D_j \le 2\alpha.$$

Proof. Finite products of compact zero-dimensional metric spaces are again compact and zero-dimensional. A product of open ρ-balls in the factors is an open ρ-ball for the max product metric. Hence

$$N_X(\rho) \le \prod_{j=1}^m N_{X_j}(\rho) \le \left(\prod_{j=1}^m A_j \right) \rho^{-d_*}, \qquad 0 < \rho \le 1.$$

The displayed discrepancy estimate and the $d_* \le 2\alpha$ consequence follow from Corollary 9.16.

For the dimension statement, choose $D_j > \overline{\dim}_{\mathrm{M}}(X_j)$ with $\sum_j D_j < 2\alpha$ and apply the preceding paragraph. For the critical content endpoint, finite upper metric s_j-content gives $N_{X_j}(\rho) \le C_j \rho^{-s_j}$ at small scales, and enlarging constants handles the larger scales; the product bound then has exponent $\sum_j s_j = 2\alpha$.

For the Assouad statement, a closed R-ball in the max product metric is the product of the closed R-balls in the factors. Cover each factor ball

by at most $A_j(R/\rho)^{s_j}$ open ρ-balls and multiply the covers. The doubling consequence is the special case $s_j = \log_2 D_j$ from Corollary 9.21. □

Definition 9.23 (Zero-dimensional Hölder resolution)**.** Let Y be a compact metric space. An (A, s, H, β)-zero-dimensional Hölder resolution of Y consists of a compact zero-dimensional metric space X and a continuous surjection $\pi : X \to Y$ such that

$$N_X(\rho) \le A\rho^{-s} \qquad (0 < \rho \le 1)$$

and

$$d_Y(\pi x, \pi y) \le H\, d_X(x, y)^{\beta} \qquad (x, y \in X).$$

Corollary 9.24 (Hölder resolutions transfer the entropy bound)**.** *Let Y admit an (A, s, H, β)-zero-dimensional Hölder resolution. If $g_1, \dots, g_n \in C(Y)$ are real-valued contractions with common (L, α)-Hölder modulus, then*

$$\operatorname{disc}(g_1, \dots, g_n) \le C_{A,s,H,L,\alpha,\beta} \sqrt{n \log\bigl(e + n^{s/(2\alpha\beta)-1}\bigr)}.$$

In particular, $s \le 2\alpha\beta$ gives the Spencer-scale bound

$$\operatorname{disc}(g_1, \dots, g_n) \le C_{A,s,H,L,\alpha,\beta} \sqrt{n}.$$

Proof. Let $f_i = g_i \circ \pi$. Since π is onto, pullback is isometric:

$$\left\| \sum_i \varepsilon_i g_i \right\|_{C(Y)} = \left\| \sum_i \varepsilon_i f_i \right\|_{C(X)} \qquad (\varepsilon_i \in \{-1, 1\}).$$

Thus the discrepancies of the two tuples are equal. Moreover

$$|f_i(x) - f_i(y)| \le L\, d_Y(\pi x, \pi y)^{\alpha} \le L H^{\alpha} d_X(x, y)^{\alpha\beta}.$$

Corollary 9.16, applied on X with exponent $\alpha\beta$, gives the result. □

Proposition 9.25 (Entropy-profile zero-dimensional resolution)**.** *Let Y be a compact metric space with $\operatorname{diam} Y \le 1$, and let $0 < \theta < 1$. Then there are a compact zero-dimensional metric space X and a continuous surjection $\pi : X \to Y$ such that*

$$d_Y(\pi x, \pi x') \le \frac{2}{1-\theta} d_X(x, x') \qquad (x, x' \in X)$$

and

$$N_X(\rho) \le N_Y(\theta\rho) \qquad (0 < \rho \le 1).$$

In particular, if $N_Y(\rho) \le A\rho^{-d}$ for $0 < \rho \le 1$, then, for every $0 < \theta < 1$, Y admits a $(\theta^{-d}A, d, 2/(1-\theta), 1)$-zero-dimensional Hölder resolution.

Proof. Put $r_k = \theta^k$. For each $k \ge 0$, choose a set $C_k \subset Y$ with

$$|C_k| = N_Y(r_k) \qquad \text{and} \qquad Y \subset \bigcup_{c \in C_k} B(c, r_k).$$

For each $c \in C_{k+1}$, choose a parent $p(c) \in C_k$ with $d_Y(c, p(c)) < r_k$. Let X be the space of all infinite parent chains

$$x = (c_0, c_1, \dots), \qquad c_k \in C_k, \qquad p(c_{k+1}) = c_k.$$

As a closed subspace of the product of finite discrete spaces, X is compact and zero-dimensional.

For $x = (c_k) \in X$, the estimate

$$d_Y(c_\ell, c_k) \le \sum_{j=k}^{\ell-1} r_j < \frac{r_k}{1-\theta} \qquad (\ell > k)$$

shows that (c_k) converges in Y. Define $\pi(x) = \lim_k c_k$. This map is continuous.

To see that π is onto, fix $y \in Y$. For every m, choose $a_m \in C_m$ with $d_Y(y, a_m) < r_m$, and take the ancestor chain of a_m down to level 0. A diagonal subsequence stabilizes at every fixed level k, producing a chain $x = (c_k) \in X$. Along the stabilizing subsequence,

$$d_Y(y, c_k) \le r_m + \sum_{j=k}^{m-1} r_j < r_m + \frac{r_k}{1-\theta},$$

so $\pi(x) = y$.

Define an ultrametric on X by setting $d_X(x, x') = 0$ if $x = x'$, and otherwise

$$d_X(x, x') = \begin{cases} 1, & c_0 \ne c'_0, \\ \theta^k, & k = \max\{j : c_j = c'_j\}. \end{cases}$$

If x and x' agree through level k, then both limits are within $r_k/(1-\theta)$ of the common center c_k. Hence

$$d_Y(\pi x, \pi x') \le \frac{2r_k}{1-\theta} = \frac{2}{1-\theta} d_X(x, x')$$

whenever $d_X(x, x') = \theta^k$, and the same bound is trivial when $d_X(x, x') = 1$ because $\operatorname{diam} Y \le 1$.

It remains to check the covering entropy of X. Given $0 < \rho \le 1$, let k be minimal with $\theta^k < \rho$. The level-k cylinders cover X, are contained in open ρ-balls for d_X, and their number is at most $|C_k|$. Since $\theta^k \ge \theta\rho$ and covering numbers decrease with the radius,

$$N_X(\rho) \le |C_k| = N_Y(\theta^k) \le N_Y(\theta\rho).$$

The polynomial consequence follows by applying the assumed bound at scale $\theta\rho$. □

Corollary 9.26 (Compact metric entropy without zero-dimensionality)**.** *Let Y be a compact metric space with* $\operatorname{diam} Y \le 1$*, let $N_Y(\rho)$ be its ordinary covering number, and let $g_1, \dots, g_n \in C(Y)$ be real-valued contractions with common modulus*

$$\omega(r) := \max_{1 \le i \le n} \sup_{d_Y(y,z) \le r} |g_i(y) - g_i(z)|.$$

Then, for every $0 < \theta < 1$ *and* $0 < s \le 1$,

$$\operatorname{disc}(g_1, \dots, g_n) \le C\sqrt{n \log\left(e + \frac{N_Y(\theta s)}{n}\right)} + 2n\omega\left(\frac{4s}{1-\theta}\right).$$

In particular, for every $0 < \rho \le 1/2$,

$$\operatorname{disc}(g_1, \dots, g_n) \le C\sqrt{n \log\left(e + \frac{N_Y(\rho)}{n}\right)} + 2n\omega(16\rho).$$

In particular, if

$$N_Y(\rho) \le A\rho^{-d} \qquad (0 < \rho \le 1),$$

and the tuple is uniformly (L, α)*-Hölder, then*

$$\operatorname{disc}(g_1, \dots, g_n) \le C_{A,d,L,\alpha}\sqrt{n \log\bigl(e + n^{d/(2\alpha)-1}\bigr)}.$$

In particular, $d \le 2\alpha$ *gives*

$$\operatorname{disc}(g_1, \dots, g_n) \le C_{A,d,L,\alpha}\sqrt{n}.$$

More generally, if

$$N_Y(\rho) \le A\rho^{-2\alpha}\bigl(\log(e/\rho)\bigr)^{\beta} \qquad (0 < \rho \le 1),$$

then

$$\operatorname{disc}(g_1, \dots, g_n) \le \begin{cases} C_{A,L,\alpha,\beta}\sqrt{n}, & \beta \le 0, \\ C_{A,L,\alpha,\beta}\sqrt{n \log\log(e^e + n)}, & \beta > 0. \end{cases}$$

Consequently, the following hold for an arbitrary compact metric Y*. Finite upper metric Minkowski dimension gives*

$$\operatorname{disc}(g_1, \dots, g_n) \le C_{Y,L,\alpha}\sqrt{n \log(e + n)}.$$

Upper metric dimension $< 2\alpha$ *gives Spencer scale, and finite upper metric* 2α*-content gives the critical Spencer-scale endpoint.*

Proof. Fix $0 < \theta < 1$. Let X and $\pi : X \to Y$ be the resolution from Proposition 9.25 with this value of θ, and put $f_i = g_i \circ \pi$. As in Corollary 9.24, the discrepancies of (g_i) and (f_i) are equal. The common modulus ω_X of the pulled-back tuple satisfies $\omega_X(r) \le \omega(2r/(1-\theta))$. Applying Corollary 9.16 on X at scale s gives

$$\operatorname{disc}(g_1, \dots, g_n) \le C\sqrt{n \log\left(e + \frac{N_Y(\theta s)}{n}\right)} + 2n\omega\left(\frac{4s}{1-\theta}\right).$$

Taking $\theta = 1/2$ and $s = 2\rho$ gives the displayed direct profile bound.

If $N_Y(\rho) \le A\rho^{-d}$ and $\omega(r) \le Lr^{\alpha}$, take

$$\rho = \min\left(\frac{1}{2}, (16L\sqrt{n})^{-1/\alpha}\right).$$

Then the approximation term is $O_{L,\alpha}(\sqrt{n})$, while

$$\frac{N_Y(\rho)}{n} \le C_{A,d,L,\alpha} n^{d/(2\alpha)-1}.$$

This proves the polynomial estimate. The logarithmic critical-window estimate is identical, using

$$\frac{N_Y(\rho)}{n} \le C_{A,L,\alpha}\big(\log(e+n)\big)^{\beta}$$

at the same choice of ρ.

For the dimension and content consequences, first rescale the metric by a constant depending only on Y so that $\operatorname{diam} Y \le 1$. The upper metric dimension and critical-content consequences are exactly the same enlarging-of-constants argument as in Corollary 9.18: choose any exponent strictly above $\overline{\dim}_{\mathrm{M}}(Y)$, or use the finite 2α-content bound at small scales and enlarge constants for the remaining scales. □

Corollary 9.27 (Qualitative compact equicontinuity)**.** *Let Y be a compact metric space with $\operatorname{diam} Y \le 1$, and let ω be a modulus with $\omega(r) \to 0$ as $r \downarrow 0$. For each $n \ge 1$, let $g_{n,1}, \dots, g_{n,n} \in C(Y)$ be real-valued contractions such that*

$$|g_{n,i}(y) - g_{n,i}(z)| \le \omega(d_Y(y,z)) \qquad (y,z \in Y,\ 1 \le i \le n).$$

Then

$$\operatorname{disc}(g_{n,1}, \dots, g_{n,n}) = o(n).$$

More quantitatively, if $\rho_n \in (0, 1/2]$ satisfies

$$\log N_Y(\rho_n) = o(n) \qquad \text{and} \qquad \omega(16\rho_n) = o(1),$$

then the same conclusion holds.

Proof. The quantitative statement follows directly from Corollary 9.26:

$$\operatorname{disc}(g_{n,1}, \dots, g_{n,n}) \le C\sqrt{n \log\left(e + \frac{N_Y(\rho_n)}{n}\right)} + 2n\omega(16\rho_n).$$

If $E_n = \log N_Y(\rho_n)$, then

$$\frac{1}{n}\sqrt{n \log\left(e + \frac{N_Y(\rho_n)}{n}\right)} \le \sqrt{\frac{1+E_n}{n}} = o(1),$$

and the approximation term is $o(n)$ by assumption.

It remains only to note that such a scale sequence exists for every compact Y and every fixed modulus ω. Choose $\rho_k \in (0, 1/16]$ with $\rho_k \downarrow 0$ and $\omega(16\rho_k) \le 1/k$. Put $E_k = \log N_Y(\rho_k)$ and

$$m_k = \lceil k \max\{1, E_k\} \rceil.$$

For each sufficiently large n, let $k(n)$ be the largest k with $m_k \le n$, and set $\rho_n = \rho_{k(n)}$. Since $m_k \ge k$ and each m_k is finite, $k(n) \to \infty$. Moreover

$$\frac{\log N_Y(\rho_n)}{n} \le \frac{1}{k(n)} \qquad \text{and} \qquad \omega(16\rho_n) \le \frac{1}{k(n)}.$$

The quantitative statement applies. □

Corollary 9.28 (Precompact function classes)**.** *Let Y be a compact metric space with* $\operatorname{diam} Y \le 1$*, and let $\mathcal{F} \subset C(Y)$ be a relatively compact family of real-valued contractions. Define*

$$\Delta_n(\mathcal{F}) := \sup_{f_1,\dots,f_n \in \mathcal{F}} \operatorname{disc}(f_1,\dots,f_n).$$

Then

$$\Delta_n(\mathcal{F}) = o(n).$$

Proof. Let $K = \overline{\mathcal{F}}$ in $C(Y)$. Compactness of K gives a common modulus of continuity. Indeed, given $\varepsilon > 0$, choose an $\varepsilon/3$-net $h_1, \dots, h_m$ for K in the uniform norm, then choose $\delta > 0$ so that

$$|h_j(y) - h_j(z)| < \varepsilon/3 \qquad (d_Y(y,z) < \delta,\ 1 \le j \le m).$$

If $f \in K$ and $\|f - h_j\|_\infty < \varepsilon/3$, then $|f(y) - f(z)| < \varepsilon$ whenever $d_Y(y,z) < \delta$. Thus

$$\omega(r) := \sup_{f \in K} \sup_{d_Y(y,z) \le r} |f(y) - f(z)|$$

satisfies $\omega(r) \to 0$ as $r \downarrow 0$.

Apply Corollary 9.27 with this common modulus. Its proof gives scales ρ_n, depending only on Y and K, for which

$$\operatorname{disc}(f_1,\dots,f_n) \le C\sqrt{n \log\left(e + \frac{N_Y(\rho_n)}{n}\right)} + 2n\omega(16\rho_n) = o(n)$$

uniformly for all $f_1, \dots, f_n \in K$. Taking the supremum over $\mathcal{F}^n$ gives the claim. □

Corollary 9.29 (Precompact Banach classes)**.** *Let E be a normed space and let $\mathcal{K}$ be a totally bounded subset of the unit ball of E. Set*

$$\Delta_n^E(\mathcal{K}) := \sup_{x_1,\dots,x_n \in \mathcal{K}} \min_{\varepsilon_i \in \{-1,1\}} \left\| \sum_{i=1}^n \varepsilon_i x_i \right\|_E.$$

Let $N_{\mathcal{K}}(\eta)$ denote the least size of an η-net for $\mathcal{K}$ whose centers lie in $\mathcal{K}$. Then

$$\Delta_n^E(\mathcal{K}) \le N_{\mathcal{K}}(\eta) + n\eta \qquad (\eta > 0).$$

Consequently

$$\Delta_n^E(\mathcal{K}) = o(n).$$

Proof. Choose an η-net $u_1, \dots, u_M$ for $\mathcal{K}$, with $M = N_{\mathcal{K}}(\eta)$, and take $x_1, \dots, x_n \in \mathcal{K}$. For each i, choose $a(i)$ with $\|x_i - u_{a(i)}\|_E \le \eta$. For every $1 \le j \le M$, choose signs on the fiber

$$I_j = \{i : a(i) = j\}$$

so that

$$\left| \sum_{i \in I_j} \varepsilon_i \right| \le 1.$$

Then

$$\left\|\sum_i \varepsilon_i u_{a(i)}\right\|_E \le \sum_{j=1}^{M} \left|\sum_{i\in I_j} \varepsilon_i\right| \|u_j\|_E \le M,$$

while

$$\left\|\sum_i \varepsilon_i (x_i - u_{a(i)})\right\|_E \le n\eta.$$

This proves the quantitative estimate.

For the qualitative consequence, choose $\eta_k \downarrow 0$ with $\eta_k \le 1/k$, put $M_k = N_{\mathcal{K}}(\eta_k)$, and set

$$m_k = \lceil k \max\{1, M_k\} \rceil .$$

For sufficiently large n, let $k(n)$ be the largest k with $m_k \le n$. Then $k(n) \to \infty$, and the quantitative estimate with $\eta = \eta_{k(n)}$ gives

$$\frac{\Delta_n^E(\mathcal{K})}{n} \le \frac{1}{k(n)} + \eta_{k(n)} \le \frac{2}{k(n)} \to 0.$$

□

Corollary 9.30 (Banach entropy rates)**.** *Let E, $\mathcal{K}$, and $N_{\mathcal{K}}$ be as in Corollary 9.29. Then*

$$\Delta_n^E(\mathcal{K}) \le \inf_{0<\eta\le 1} \{N_{\mathcal{K}}(\eta) + n\eta\}.$$

Consequently:

(i) *If $N_{\mathcal{K}}(\eta) \le A\eta^{-d}$ for $0 < \eta \le 1$, with $A \ge 1$ and $d > 0$, then*

$$\Delta_n^E(\mathcal{K}) \le 2A^{1/(d+1)}\, n^{d/(d+1)} \qquad (n \ge 1).$$

(ii) *If $N_{\mathcal{K}}(\eta) \le \exp(A\eta^{-q})$ for $0 < \eta \le 1$, with $A, q > 0$, then*

$$\Delta_n^E(\mathcal{K}) \le C_{A,q} \frac{n}{(\log(e+n))^{1/q}} \qquad (n \ge 1).$$

(iii) *More generally, if*

$$N_{\mathcal{K}}(\eta) \le \exp\Big(A\eta^{-q}(\log(e/\eta))^{\gamma}\Big) \qquad (0 < \eta \le 1/2),$$

with $A, q > 0$ and $\gamma \in \mathbb{R}$, then

$$\Delta_n^E(\mathcal{K}) \le C_{A,q,\gamma}\, n\, (\log(e+n))^{-1/q} (\log\log(e^e+n))^{\gamma/q} \qquad (n \ge 2).$$

Proof. The infimum estimate is Corollary 9.29, restricted to $0 < \eta \le 1$.

For the polynomial bound, use the trivial estimate $\Delta_n^E(\mathcal{K}) \le n$ when $n < A$. Since

$$n \le A^{1/(d+1)} n^{d/(d+1)} \qquad (n < A),$$

this is enough in the small-n range. When $n \ge A$, choose $\eta = (A/n)^{1/(d+1)}$. Then $\eta \le 1$, and the infimum estimate gives

$$\Delta_n^E(\mathcal{K}) \le A\eta^{-d} + n\eta = 2A^{1/(d+1)} n^{d/(d+1)}.$$

For the stretched-exponential bound, enlarge $C_{A,q}$ to cover the bounded range where $(2A/\log(e+n))^{1/q} > 1$. In the remaining range take

$$\eta = \left(\frac{2A}{\log(e+n)}\right)^{1/q}.$$

Then $0 < \eta \le 1$ and

$$N_{\mathcal{K}}(\eta) \le \exp(A\eta^{-q}) = \exp\left(\frac{1}{2}\log(e+n)\right) \le C_q \frac{n}{(\log(e+n))^{1/q}},$$

after increasing C_q for small n. The term $n\eta$ has the displayed order with constant depending on A and q, so the claim follows.

For the signed-log refinement, write

$$L_n = \log(e+n), \qquad M_n = \log\log(e^e+n).$$

After enlarging constants, assume

$$\eta = BL_n^{-1/q} M_n^{\gamma/q} \le \frac{1}{2}$$

with $B \ge 1$ to be chosen. Then

$$c_{q,\gamma} M_n \le \log(e/\eta) \le C_{q,\gamma} M_n$$

for all sufficiently large n, and hence

$$A\eta^{-q}(\log(e/\eta))^{\gamma} \le C_{A,q,\gamma} B^{-q} L_n.$$

Choosing B large enough makes the last quantity at most $L_n/2$. Thus

$$N_{\mathcal{K}}(\eta) \le e^{L_n/2} \le C\sqrt{n}.$$

Corollary 9.29 gives

$$\Delta_n^E(\mathcal{K}) \le C\sqrt{n} + nBL_n^{-1/q} M_n^{\gamma/q}.$$

The second term dominates $\sqrt{n}$ for large n, and bounded n are absorbed into the constant. □

Corollary 9.31 (Compact operator images)**.** *Let X and E be normed spaces, and let $T : X \to E$ be a compact linear operator with $\|T\| \le 1$. Set*

$$\Delta_n(T) := \sup_{\|x_1\|_X,\dots,\|x_n\|_X \le 1} \min_{\varepsilon_i \in \{-1,1\}} \left\| \sum_{i=1}^n \varepsilon_i T x_i \right\|_E.$$

Let $N_T(\eta)$ be the least size of an η-net for $T(B_X)$, with centers in $T(B_X)$, where $B_X = \{x : \|x\|_X \le 1\}$. Then

$$\Delta_n(T) \le N_T(\eta) + n\eta \qquad (\eta > 0),$$

and in particular

$$\Delta_n(T) = o(n).$$

Moreover, all rate estimates in Corollary 9.30 apply with $N_{\mathcal{K}}$ replaced by N_T.

Proof. The set $\mathcal{K} = T(B_X)$ is totally bounded because T is compact, and it is contained in the unit ball of E because $\|T\| \le 1$. Applying Corollaries 9.29 and 9.30 to this $\mathcal{K}$ gives all assertions. □

Corollary 9.32 (Dyadic entropy-number rates for compact images)**.** *Let E be a real normed space and let $\mathcal{K} \subset E$ be totally bounded. Put*

$$R_{\mathcal{K}} := \sup_{x \in \mathcal{K}} \|x\|_E < \infty$$

and define $\Delta_n^E(\mathcal{K})$ as in Corollary 9.29. Define the internal dyadic entropy numbers

$$e_k^{\mathrm{int}}(\mathcal{K}) := \inf\{\eta > 0 : N_{\mathcal{K}}(\eta) \le 2^{k-1}\}, \qquad k \ge 1.$$

Then, for every $k \ge 1$,

$$\Delta_n^E(\mathcal{K}) \le R_{\mathcal{K}} 2^{k-1} + n\, e_k^{\mathrm{int}}(\mathcal{K}).$$

Consequently:

(i) *If $R_{\mathcal{K}} \le R$ and $e_k^{\mathrm{int}}(\mathcal{K}) \le A k^{-s}$ for all $k \ge 1$, with $A, R, s > 0$, then*

$$\Delta_n^E(\mathcal{K}) \le C_{A,R,s}\, n(\log(e+n))^{-s} \qquad (n \ge 2).$$

(ii) *More generally, if $R_{\mathcal{K}} \le R$ and*

$$e_k^{\mathrm{int}}(\mathcal{K}) \le A k^{-s} \big(\log(e+k)\big)^{\beta} \qquad (k \ge 1),$$

with $A, R, s > 0$ and $\beta \in \mathbb{R}$, then

$$\Delta_n^E(\mathcal{K}) \le C_{A,R,s,\beta}\, n(\log(e+n))^{-s} \big(\log\log(e^e+n)\big)^{\beta} \qquad (n \ge 2).$$

(iii) *If $R_{\mathcal{K}} \le R$ and $e_k^{\mathrm{int}}(\mathcal{K}) \le A e^{-bk}$ for all $k \ge 1$, with $A, R, b > 0$, then, with $\alpha_b = \log 2/(b + \log 2) \in (0,1)$,*

$$\Delta_n^E(\mathcal{K}) \le C_{A,R,b}\, n^{\alpha_b} \qquad (n \ge 2).$$

For a compact operator $T : X \to E$, the same first estimate holds with $\mathcal{K} = T(B_X)$, $R_{\mathcal{K}} \le \|T\|$, and $e_k^{\mathrm{int}}(T) := e_k^{\mathrm{int}}(T(B_X))$; in particular the three rates hold uniformly for compact operators with $\|T\| \le R$.

Proof. Choose any $\eta > e_k^{\mathrm{int}}(\mathcal{K})$ and an internal η-net $u_1, \dots, u_M \in \mathcal{K}$ with $M \le 2^{k-1}$. Given $x_1, \dots, x_n \in \mathcal{K}$, assign x_i to a net point $u_{a(i)}$ within distance η. On each fiber choose signs with imbalance at most one. Then

$$\Big\| \sum_i \varepsilon_i u_{a(i)} \Big\|_E \le \sum_{j=1}^{M} \Big| \sum_{i : a(i) = j} \varepsilon_i \Big| \, \|u_j\|_E \le R_{\mathcal{K}} M,$$

and the approximation error contributes at most $n\eta$. Letting $\eta \downarrow e_k^{\mathrm{int}}(\mathcal{K})$ proves the first display.

For the polynomial and log-refined bounds, put

$$k_n = \left\lceil \frac{\log(e+n)}{2 \log 2} \right\rceil.$$

Then $R_{\mathcal{K}} 2^{k_n-1} \le C_R\sqrt{e+n}$ and $k_n \asymp \log(e+n)$. Substituting $k = k_n$ in the first display gives the two asserted logarithmic rates; the $R_{\mathcal{K}}2^{k_n-1}$ term is absorbed because

$$\sqrt{n} = o\Big(n(\log(e+n))^{-s}\big(\log\log(e^e+n)\big)^{\beta}\Big)$$

for every fixed $s > 0$ and $\beta \in \mathbb{R}$.

For the exponential entropy-number bound, take, for large n,

$$k_n = \left\lfloor \frac{\log(e+n)}{b+\log 2} \right\rfloor,$$

and enlarge the constant to cover the finite range where $k_n < 1$. Then

$$R_{\mathcal{K}} 2^{k_n} \le C_{R,b} n^{\alpha_b}, \qquad ne^{-bk_n} \le C_b n^{\alpha_b}, \qquad \alpha_b = \frac{\log 2}{b+\log 2}.$$

The first display gives the claimed n^{α_b} rate. The operator version is the specialization $\mathcal{K} = T(B_X)$. □

Corollary 9.33 (External entropy numbers, unions, and sums)**.** *Let E be a real normed space and let $\mathcal{K} \subset E$ be totally bounded. Define the usual external dyadic entropy numbers*

$$e_k(\mathcal{K};E) := \inf\Big\{\eta > 0 : \mathcal{K} \subset \bigcup_{j=1}^{2^{k-1}} (y_j + \eta B_E) \text{ for some } y_j \in E\Big\}.$$

Then

$$e_k^{\mathrm{int}}(\mathcal{K}) \le 2e_k(\mathcal{K};E) \qquad (k \ge 1),$$

so all estimates of Corollary 9.32 remain valid, up to constant changes, with the standard external entropy numbers $e_k(\mathcal{K};E)$ in place of the internal ones.

Moreover, if $\mathcal{K} = \bigcup_{\ell=1}^r \mathcal{K}_\ell$, then

$$e_{k+\lceil \log_2 r\rceil}(\mathcal{K};E) \le \max_{1\le\ell\le r} e_k(\mathcal{K}_\ell;E) \qquad (k \ge 1).$$

If instead

$$\mathcal{K} \subset \mathcal{K}_1 + \cdots + \mathcal{K}_r := \{x_1 + \cdots + x_r : x_\ell \in \mathcal{K}_\ell\},$$

then, for every $k_1, \dots, k_r \ge 1$,

$$e_{1+\sum_{\ell=1}^r (k_\ell - 1)}(\mathcal{K};E) \le \sum_{\ell=1}^r e_{k_\ell}(\mathcal{K}_\ell;E).$$

Consequently finite unions and finite Minkowski sums of bounded compact classes preserve the polynomial and log-refined dyadic entropy decay classes, hence the corresponding discrepancy rates above; exponential dyadic entropy decay is also preserved, with the exponent for a finite Minkowski sum allowed to depend on the number of summands.

Proof. For the internal–external comparison, fix $\eta > e_k(\mathcal{K};E)$ and cover $\mathcal{K}$ by at most 2^{k-1} balls $y_j + \eta B_E$ with arbitrary centers $y_j \in E$. Discard empty balls. For every remaining ball choose one point $u_j \in \mathcal{K} \cap (y_j + \eta B_E)$. If $x \in \mathcal{K} \cap (y_j + \eta B_E)$, then $\|x - u_j\|_E \le 2\eta$. Thus these chosen points

form an internal 2η-net of the same cardinality. Letting $\eta \downarrow e_k(\mathcal{K};E)$ proves $e_k^{\mathrm{int}}(\mathcal{K}) \le 2e_k(\mathcal{K};E)$.

For a finite union, choose $e_k(\mathcal{K}_\ell;E)$-near covers of each $\mathcal{K}_\ell$. Combining the r covers uses at most

$$r\,2^{k-1} \le 2^{k+\lceil \log_2 r\rceil -1}$$

balls and radius $\max_\ell e_k(\mathcal{K}_\ell;E)$, up to an arbitrary positive enlargement. Letting that enlargement tend to zero proves the union bound.

For a finite Minkowski sum, choose covers of $\mathcal{K}_\ell$ with radii arbitrarily close to $e_{k_\ell}(\mathcal{K}_\ell;E)$. Sums of the cover centers cover $\mathcal{K}_1+\cdots+\mathcal{K}_r$ with radius equal to the sum of the radii and with at most

$$\prod_{\ell=1}^{r} 2^{k_\ell-1} = 2^{\sum_\ell (k_\ell-1)} = 2^{(1+\sum_\ell (k_\ell-1))-1}$$

balls. This proves the displayed sum bound. The final sentence follows by taking the same dyadic scale in each summand, up to fixed integer shifts. □

Corollary 9.34 (Summable Minkowski series)**.** *Let E be a real Banach space. For $j \ge 1$, let $\mathcal{K}_j \subset E$ be totally bounded and put*

$$R_j := \sup_{x\in\mathcal{K}_j} \|x\|_E < \infty.$$

Let $a_j \ge 0$ satisfy $\sum_{j\ge1} a_j R_j < \infty$, and set

$$\mathcal{K} := \left\{ \sum_{j=1}^{\infty} a_j x_j : x_j \in \mathcal{K}_j \right\}.$$

Then the displayed series converge absolutely, $\mathcal{K}$ is totally bounded, and for every $r \ge 1$ and $k_1,\dots,k_r \ge 1$,

$$e_{1+\sum_{j=1}^r (k_j-1)}(\mathcal{K};E) \le \sum_{j=1}^{r} a_j e_{k_j}(\mathcal{K}_j;E) + \sum_{j>r} a_j R_j.$$

Consequently, suppose in addition that $R_j \le R$ and $a_j \le A_0 e^{-cj}$ for all j, with $A_0, R, c > 0$.

(i) *If, uniformly in j,*

$$e_k(\mathcal{K}_j;E) \le B k^{-s}\big(\log(e+k)\big)^{\beta} \qquad (k\ge1),$$

where $B, s > 0$ and $\beta \in \mathbb{R}$, and $\beta_+ = \max(\beta,0)$, then

$$e_m(\mathcal{K};E) \le C m^{-s}\big(\log(e+m)\big)^{s+\beta_+} \qquad (m\ge2),$$

and

$$\Delta_n^E(\mathcal{K}) \le C'\, n(\log(e+n))^{-s}\big(\log\log(e^e+n)\big)^{s+\beta_+} \qquad (n\ge2).$$

(ii) *If, uniformly in j,*

$$e_k(\mathcal{K}_j;E) \le B e^{-bk} \qquad (k\ge1),$$

with $B, b > 0$, then there are $\gamma, \gamma' > 0$ such that

$$e_m(\mathcal{K}; E) \le Ce^{-\gamma\sqrt{m}} \qquad (m \ge 2),$$

and

$$\Delta_n^E(\mathcal{K}) \le C'\, n \exp(-\gamma'\sqrt{\log(e+n)}) \qquad (n \ge 2).$$

All constants depend only on the displayed parameters.

Proof. The absolute convergence follows from

$$\sum_{j\ge1} \|a_j x_j\|_E \le \sum_{j\ge1} a_j R_j < \infty,$$

and completeness of E. Fix r and $k_1, \dots, k_r$. For each $1 \le j \le r$, choose an external cover of $\mathcal{K}_j$ by at most 2^{k_j-1} balls with radius arbitrarily close to $e_{k_j}(\mathcal{K}_j; E)$. Sums of the first r chosen centers cover all partial sums $\sum_{j=1}^r a_j x_j$ with radius arbitrarily close to $\sum_{j=1}^r a_j e_{k_j}(\mathcal{K}_j; E)$, and the number of such centers is

$$\prod_{j=1}^{r} 2^{k_j-1} = 2^{(1+\sum_{j=1}^r (k_j-1))-1}.$$

The remaining tail has norm at most $\sum_{j>r} a_j R_j$. Letting the cover enlargements tend to zero gives the entropy-number estimate. Taking r so that the tail is small and then taking the finitely many k_j large proves total boundedness of $\mathcal{K}$.

Assume now $R_j \le R$ and $a_j \le A_0 e^{-cj}$. For the polynomial/log case, choose

$$r = \lceil C_0 \log(e+m) \rceil, \qquad k = 1 + \left\lfloor \frac{m-1}{r} \right\rfloor,$$

with C_0 large enough that $\sum_{j>r} a_j R_j \le Cm^{-s}$. For bounded m the claim is absorbed into the constant, so assume m is large enough that $k \ge 1$ and $k \asymp m/r$. The entropy estimate with $k_1 = \cdots = k_r = k$ gives an index at most m, hence

$$e_m(\mathcal{K}; E) \le Ck^{-s}\big(\log(e+k)\big)^{\beta} + Cm^{-s}.$$

If $\beta \ge 0$, then

$$k^{-s}\big(\log(e+k)\big)^{\beta} \le Cm^{-s}\big(\log(e+m)\big)^{s+\beta}.$$

If $\beta < 0$, the logarithmic factor is at most a constant and $k^{-s} \le Cm^{-s}(\log(e+m))^s$. This proves the displayed entropy bound with exponent $s + \beta_+$. The discrepancy estimate follows from Corollaries 9.32 and 9.33.

For the exponential component-entropy case, choose

$$r = \lceil C_1\sqrt{m} \rceil, \qquad k = 1 + \left\lfloor \frac{m-1}{r} \right\rfloor.$$

For large m, $k \asymp \sqrt{m}$, while the tail is $O(e^{-cr})$. The same entropy estimate therefore gives

$$e_m(\mathcal{K}; E) \le Ce^{-\gamma\sqrt{m}}$$

after decreasing $\gamma > 0$ and enlarging C to cover small m. Also

$$R_{\mathcal{K}} := \sup_{x \in \mathcal{K}} \|x\|_E \le \sum_{j \ge 1} a_j R_j < \infty.$$

The external-to-internal comparison gives $e_q^{\mathrm{int}}(\mathcal{K}) \le Ce^{-\gamma\sqrt{q}}$. Insert

$$q = \left\lfloor \frac{\log(e+n)}{2\log 2} \right\rfloor$$

in the first display of Corollary 9.32; after changing constants and covering bounded n,

$$\Delta_n^E(\mathcal{K}) \le C\sqrt{n} + Cne^{-\gamma\sqrt{q}} \le C'n\exp(-\gamma'\sqrt{\log(e+n)}).$$

□

Corollary 9.35 (Profile form for weighted Minkowski series)**.** *In the setting of Corollary 9.34, suppose that there are numbers $b_j \ge 0$ and a function $\varphi : \mathbb{N} \to [0, \infty)$ such that*

$$e_k(\mathcal{K}_j; E) \le b_j \varphi(k) \qquad (j, k \ge 1).$$

Put

$$S_r := \sum_{j=1}^{r} a_j b_j, \qquad T_r := \sum_{j > r} a_j R_j, \qquad R_{\mathcal{K}} := \sum_{j \ge 1} a_j R_j.$$

Then, for every $m \ge 2$,

$$e_m(\mathcal{K}; E) \le \inf_{1 \le r \le m} \left\{ S_r \, \varphi\left(1 + \left\lfloor \frac{m-1}{r} \right\rfloor\right) + T_r \right\}.$$

Consequently, for every $q \ge 1$,

$$\Delta_n^E(\mathcal{K}) \le R_{\mathcal{K}} 2^{q-1} + 2n \inf_{1 \le r \le q} \left\{ S_r \, \varphi\left(1 + \left\lfloor \frac{q-1}{r} \right\rfloor\right) + T_r \right\}.$$

In particular, with

$$q_n = \left\lfloor \frac{\log(e+n)}{2\log 2} \right\rfloor \qquad (n \ge 2),$$

one has

$$\Delta_n^E(\mathcal{K}) \le CR_{\mathcal{K}}\sqrt{n} + 2n \inf_{1 \le r \le q_n} \left\{ S_r \, \varphi\left(1 + \left\lfloor \frac{q_n - 1}{r} \right\rfloor\right) + T_r \right\}.$$

Proof. Fix $m \ge 2$ and $1 \le r \le m$, and set

$$k = 1 + \left\lfloor \frac{m-1}{r} \right\rfloor.$$

Then $1 + r(k-1) \le m$. Applying Corollary 9.34 with $k_1 = \cdots = k_r = k$, and using monotonicity of entropy numbers in the index, gives

$$e_m(\mathcal{K}; E) \le e_{1+r(k-1)}(\mathcal{K}; E) \le \sum_{j=1}^{r} a_j e_k(\mathcal{K}_j; E) + T_r \le S_r \varphi(k) + T_r.$$

Taking the infimum over r proves the entropy profile bound.

The external-to-internal comparison from Corollary 9.33 gives

$$e_q^{\mathrm{int}}(\mathcal{K}) \le 2e_q(\mathcal{K}; E).$$

Substitute the profile estimate for $e_q(\mathcal{K}; E)$ in the first display of Corollary 9.32. Since $\sup_{x\in\mathcal{K}} \|x\|_E \le R_{\mathcal{K}}$, the displayed discrepancy bound follows. The final estimate is the same bound with $q = q_n$, using $2^{q_n-1} \le C\sqrt{n}$. □

Corollary 9.36 (Polynomial profile optimization for weighted series)**.** *In the setting of Corollary 9.35, with S_r and T_r as there, assume that, for some $A > 0$, $\gamma > 0$, $\delta \ge 0$, and $\eta, \tau \in \mathbb{R}$,*

$$S_r \le Ar^{\delta}\big(\log(e+r)\big)^{\eta}, \qquad T_r \le Ar^{-\gamma}\big(\log(e+r)\big)^{-\tau} \qquad (r \ge 1).$$

Then the following hold.

(i) *If*

$$\varphi(k) \le Bk^{-s}\big(\log(e+k)\big)^{\beta} \qquad (k \ge 1),$$

with $B, s > 0$ and $\beta \in \mathbb{R}$, put

$$D = s + \delta + \gamma, \qquad \sigma = \frac{s\gamma}{D}, \qquad \lambda = \frac{\gamma(\beta+\eta) - \tau(s+\delta)}{D}.$$

Then

$$e_m(\mathcal{K}; E) \le Cm^{-\sigma}\big(\log(e+m)\big)^{\lambda} \qquad (m \ge 2),$$

and

$$\Delta_n^E(\mathcal{K}) \le C'\, n(\log(e+n))^{-\sigma}\big(\log\log(e^e+n)\big)^{\lambda} \qquad (n \ge 2).$$

(ii) *If*

$$\varphi(k) \le Be^{-bk} \qquad (k \ge 1)$$

with $B, b > 0$, then

$$e_m(\mathcal{K}; E) \le Cm^{-\gamma}\big(\log(e+m)\big)^{\gamma-\tau} \qquad (m \ge 2),$$

and

$$\Delta_n^E(\mathcal{K}) \le C'\, n(\log(e+n))^{-\gamma}\big(\log\log(e^e+n)\big)^{\gamma-\tau} \qquad (n \ge 2).$$

The constants depend only on the displayed parameters and on $R_{\mathcal{K}}$ in the discrepancy bounds.

Proof. For the polynomial/log profile, write $L_m = \log(e+m)$ and $D = s+\delta+\gamma$. For large m, choose

$$r = \left\lceil m^{s/D} L_m^{-(\beta+\eta+\tau)/D} \right\rceil, \qquad k = 1 + \left\lfloor \frac{m-1}{r} \right\rfloor.$$

The finite range of m is absorbed into the constant. Since $0 < s/D < 1$, for large m one has $1 \le r \le m$, $k \asymp m/r$, and

$$\log(e+r) \asymp L_m, \qquad \log(e+k) \asymp L_m.$$

Corollary 9.35 gives

$$e_m(\mathcal{K}; E) \le Cm^{-s} r^{s+\delta} L_m^{\beta+\eta} + Cr^{-\gamma} L_m^{-\tau}.$$

With the displayed choice of r, both terms are bounded by

$$Cm^{-s\gamma/D}L_m^{(\gamma(\beta+\eta)-\tau(s+\delta))/D},$$

which proves the entropy estimate. The discrepancy estimate follows from Corollaries 9.32 and 9.33; the term $R_{\mathcal{K}}2^{q-1}$ is absorbed by taking $q \asymp \log(e+n)$, since $\sigma > 0$.

For the exponential profile, choose $D_0 > 2(\gamma+\delta)/b$ and set

$$r = \left\lceil \frac{m}{D_0 \log(e+m)} \right\rceil, \qquad k = 1 + \left\lfloor \frac{m-1}{r} \right\rfloor.$$

For large m, $k \geq (D_0/2)\log(e+m)$, and the profile bound gives

$$e_m(\mathcal{K};E) \leq Cr^{\delta}\big(\log(e+r)\big)^{\eta}e^{-bk} + Cr^{-\gamma}\big(\log(e+r)\big)^{-\tau}.$$

The second term is at most

$$Cm^{-\gamma}\big(\log(e+m)\big)^{\gamma-\tau}.$$

The first is

$$\leq Cm^{\delta-bD_0/2}\big(\log(e+m)\big)^{\eta-\delta},$$

and is absorbed by the same tail scale because $bD_0/2 > \gamma+\delta$. The entropy estimate follows, and the discrepancy estimate is again the dyadic entropy-number bound and the external-to-internal comparison. □

Corollary 9.37 (Polynomial weights with growing component amplitudes)**.** *Let E be a real Banach space. For $j \geq 1$, let $\mathcal{K}_j \subset E$ be totally bounded, put*

$$R_j := \sup_{x\in\mathcal{K}_j} \|x\|_E,$$

and assume $R_j \leq R$. Let $a_j \geq 0$ satisfy

$$a_j \leq A_0 j^{-u}\big(\log(e+j)\big)^{-v} \qquad (j \geq 1)$$

for some $A_0, R > 0$, $u > 1$, and $v \in \mathbb{R}$. Define

$$\mathcal{K} := \left\{ \sum_{j=1}^{\infty} a_j x_j : x_j \in \mathcal{K}_j \right\}.$$

Let $\rho > u-1$ and $\eta \in \mathbb{R}$. Then $\mathcal{K}$ is totally bounded. Moreover:

(i) *If*

$$e_k(\mathcal{K}_j;E) \leq Bj^{\rho}\big(\log(e+j)\big)^{\eta}k^{-s}\big(\log(e+k)\big)^{\beta} \qquad (j,k \geq 1),$$

where $B, s > 0$ and $\beta \in \mathbb{R}$, then, with

$$\sigma = \frac{s(u-1)}{s+\rho}, \qquad \lambda = \frac{(u-1)(\beta+\eta) - v(s+\rho)}{s+\rho},$$

one has

$$e_m(\mathcal{K};E) \leq Cm^{-\sigma}\big(\log(e+m)\big)^{\lambda} \qquad (m \geq 2),$$

and therefore

$$\Delta_n^E(\mathcal{K}) \leq C'\, n(\log(e+n))^{-\sigma}\big(\log\log(e^e+n)\big)^{\lambda} \qquad (n \geq 2).$$

(ii) *If*

$$e_k(\mathcal{K}_j;E) \le Bj^{\rho}\big(\log(e+j)\big)^{\eta}e^{-bk} \qquad (j,k\ge 1),$$

with $B,b>0$, *then*

$$e_m(\mathcal{K};E) \le Cm^{-(u-1)}\big(\log(e+m)\big)^{u-1-v} \qquad (m\ge 2),$$

and hence

$$\Delta_n^E(\mathcal{K}) \le C'\,n(\log(e+n))^{-(u-1)}\big(\log\log(e^e+n)\big)^{u-1-v} \qquad (n\ge 2).$$

All constants depend only on the displayed parameters.

Proof. Since $u>1$ and the radii are uniformly bounded,

$$\sum_{j\ge 1} a_jR_j<\infty,$$

so Corollary 9.34 gives total boundedness. The regularly varying tail bound gives

$$T_r=\sum_{j>r} a_jR_j \le Cr^{-(u-1)}\big(\log(e+r)\big)^{-v}.$$

For the profile theorem take

$$b_j = Bj^{\rho}\big(\log(e+j)\big)^{\eta}.$$

Since $\rho>u-1$,

$$S_r=\sum_{j=1}^{r} a_jb_j \le Cr^{\rho-u+1}\big(\log(e+r)\big)^{\eta-v}.$$

Apply Corollary 9.36 with

$$\gamma=u-1, \qquad \delta=\rho-u+1, \qquad \eta_0=\eta-v, \qquad \tau=v.$$

For the polynomial/log component profile this gives

$$D=s+\delta+\gamma=s+\rho, \qquad \sigma=\frac{s(u-1)}{s+\rho},$$

and logarithmic power

$$\frac{(u-1)(\beta+\eta-v)-v(s+\rho-u+1)}{s+\rho}=\frac{(u-1)(\beta+\eta)-v(s+\rho)}{s+\rho}.$$

This proves the first pair of estimates. The exponential component profile is the second part of Corollary 9.36, which gives the tail-limited entropy and discrepancy rates displayed above. □

Corollary 9.38 (Stretched-exponentially weighted Minkowski series)**.** *Let* E *be a real Banach space. For* $j\ge 1$, *let* $\mathcal{K}_j\subset E$ *be totally bounded, put*

$$R_j:=\sup_{x\in\mathcal{K}_j}\|x\|_E,$$

and assume $R_j\le R$. *Let* $a_j\ge 0$ *satisfy*

$$a_j\le A_0\exp(-cj^{\theta}) \qquad (j\ge 1)$$

for some $A_0, R, c, \theta > 0$, and define

$$\mathcal{K} := \left\{ \sum_{j=1}^{\infty} a_j x_j : x_j \in \mathcal{K}_j \right\}.$$

Then $\mathcal{K}$ is totally bounded. Moreover:

(i) *If, uniformly in j,*

$$e_k(\mathcal{K}_j; E) \le B k^{-s} \big(\log(e+k)\big)^{\beta} \qquad (k \ge 1),$$

where $B, s > 0$ and $\beta \in \mathbb{R}$, and $\beta_+ = \max(\beta, 0)$, then

$$e_m(\mathcal{K}; E) \le C m^{-s} \big(\log(e+m)\big)^{s/\theta + \beta_+} \qquad (m \ge 2),$$

and

$$\Delta_n^E(\mathcal{K}) \le C'\, n (\log(e+n))^{-s} \big(\log\log(e^e+n)\big)^{s/\theta+\beta_+} \qquad (n \ge 2).$$

(ii) *If, uniformly in j,*

$$e_k(\mathcal{K}_j; E) \le B e^{-bk} \qquad (k \ge 1),$$

with $B, b > 0$, then there are $\gamma, \gamma' > 0$ such that

$$e_m(\mathcal{K}; E) \le C \exp\big(-\gamma m^{\theta/(1+\theta)}\big) \qquad (m \ge 2),$$

and

$$\Delta_n^E(\mathcal{K}) \le C'\, n \exp\big(-\gamma' (\log(e+n))^{\theta/(1+\theta)}\big) \qquad (n \ge 2).$$

All constants depend only on the displayed parameters.

Proof. Since $\sum_j e^{-cj^\theta} < \infty$, the series defining $\mathcal{K}$ is covered by Corollary 9.34. We shall also use the tail bound

$$\sum_{j>r} a_j R_j \le C \exp(-c_1 r^\theta) \qquad (r \ge 1),$$

for some $c_1 > 0$, which follows from the integral estimate for $\sum_{j>r} e^{-cj^\theta}$ after decreasing c and increasing C.

For the polynomial/log component entropy, choose C_0 large and set

$$r = \left\lceil C_0 (\log(e+m))^{1/\theta} \right\rceil, \qquad k = 1 + \left\lfloor \frac{m-1}{r} \right\rfloor.$$

For bounded m the claim is absorbed into the constant; for large m, $k \asymp m/r$ and $1 + r(k-1) \le m$. Choosing C_0 large enough makes $\sum_{j>r} a_j R_j \le C m^{-s}$. The finite part is bounded by

$$C k^{-s} \big(\log(e+k)\big)^{\beta} \le C m^{-s} \big(\log(e+m)\big)^{s/\theta+\beta_+},$$

with the same convention as above for negative β. The entropy estimate follows, and the discrepancy estimate is Corollaries 9.32 and 9.33.

For exponential component entropy, choose

$$r = \left\lceil C_1 m^{1/(1+\theta)} \right\rceil, \qquad k = 1 + \left\lfloor \frac{m-1}{r} \right\rfloor.$$

Then $k \asymp m^{\theta/(1+\theta)}$ and $r^\theta \asymp m^{\theta/(1+\theta)}$. The finite part Ce^{-bk} and the tail $Ce^{-c_1 r^\theta}$ are both bounded by $C\exp(-\gamma m^{\theta/(1+\theta)})$, after decreasing $\gamma > 0$ and covering small m. Finally, the first display of Corollary 9.32, applied with

$$q = \left\lfloor \frac{\log(e+n)}{2\log 2} \right\rfloor,$$

gives

$$\Delta_n^E(\mathcal{K}) \le C\sqrt{n} + Cn\exp(-\gamma q^{\theta/(1+\theta)}),$$

and the $\sqrt{n}$ term is absorbed by the displayed bound after decreasing γ' and enlarging C'. □

Corollary 9.39 (Polynomially weighted Minkowski series)**.** *Let E be a real Banach space. For $j \ge 1$, let $\mathcal{K}_j \subset E$ be totally bounded, let*

$$R_j := \sup_{x\in\mathcal{K}_j} \|x\|_E,$$

and assume $R_j \le R$. Let $a_j \ge 0$ satisfy

$$a_j \le A_0 j^{-u} \qquad (j \ge 1)$$

for some $A_0, R > 0$ and $u > 1$, and define

$$\mathcal{K} := \left\{ \sum_{j=1}^{\infty} a_j x_j : x_j \in \mathcal{K}_j \right\}.$$

Then $\mathcal{K}$ is totally bounded. Moreover:

(i) *If, uniformly in j,*

$$e_k(\mathcal{K}_j; E) \le Bk^{-s}\big(\log(e+k)\big)^{\beta} \qquad (k \ge 1),$$

where $B, s > 0$ and $\beta \in \mathbb{R}$, then, with

$$\sigma = \frac{s(u-1)}{s+u-1}, \qquad \beta_+ = \max(\beta, 0),$$

one has

$$e_m(\mathcal{K}; E) \le Cm^{-\sigma}\big(\log(e+m)\big)^{\beta_+} \qquad (m \ge 2),$$

and therefore

$$\Delta_n^E(\mathcal{K}) \le C'\, n(\log(e+n))^{-\sigma}\big(\log\log(e^e+n)\big)^{\beta_+} \qquad (n \ge 2).$$

(ii) *If, uniformly in j,*

$$e_k(\mathcal{K}_j; E) \le Be^{-bk} \qquad (k \ge 1),$$

with $B, b > 0$, then

$$e_m(\mathcal{K}; E) \le Cm^{-(u-1)}\big(\log(e+m)\big)^{u-1} \qquad (m \ge 2),$$

and hence

$$\Delta_n^E(\mathcal{K}) \le C'\, n(\log(e+n))^{-(u-1)}\big(\log\log(e^e+n)\big)^{u-1} \qquad (n \ge 2).$$

All constants depend only on the displayed parameters.

Proof. Since $u > 1$,

$$\sum_{j\geq 1} a_j R_j \leq A_0 R \sum_{j\geq 1} j^{-u} < \infty.$$

Thus Corollary 9.34 gives total boundedness and the truncation estimate. We shall use

$$\sum_{j=1}^{r} a_j \leq C_{A_0,u}, \qquad \sum_{j>r} a_j R_j \leq C_{A_0,R,u} r^{1-u}.$$

For the polynomial/log component entropy, put

$$r = \left\lceil m^{s/(s+u-1)} \right\rceil, \qquad k = 1 + \left\lfloor \frac{m-1}{r} \right\rfloor.$$

After enlarging constants to cover bounded m, $1 \leq r \leq m$ and $k \asymp m/r$. The truncation estimate with $k_1 = \cdots = k_r = k$ has entropy index $1+r(k-1) \leq m$. Therefore

$$e_m(\mathcal{K}; E) \leq C k^{-s} \big(\log(e+k)\big)^{\beta} + C r^{1-u}.$$

If $\beta \geq 0$, then $(\log(e+k))^{\beta} \leq (\log(e+m))^{\beta}$; if $\beta < 0$, this logarithmic factor is at most 1. Hence

$$k^{-s} \big(\log(e+k)\big)^{\beta} \leq C m^{-s} r^{s} \big(\log(e+m)\big)^{\beta_+}.$$

The chosen r makes $m^{-s} r^{s}$ and r^{1-u} both bounded by $C m^{-\sigma}$, which proves the entropy estimate. The discrepancy bound then follows from Corollaries 9.32 and 9.33.

For exponential component entropy, choose $D > 2(u-1)/b$ and set

$$r = \left\lceil \frac{m}{D \log(e+m)} \right\rceil, \qquad k = 1 + \left\lfloor \frac{m-1}{r} \right\rfloor.$$

For large m, $k \geq (D/2)\log(e+m)$, while

$$r^{1-u} \leq C m^{-(u-1)} \big(\log(e+m)\big)^{u-1}.$$

The finite part is

$$C e^{-bk} \leq C m^{-bD/2},$$

which is absorbed by the displayed tail bound because $bD/2 > u-1$. The entropy estimate follows, and the discrepancy bound is again Corollaries 9.32 and 9.33, now applied with polynomial exponent $u-1$ and logarithmic power $u-1$. □

Corollary 9.40 (Log-balanced polynomially weighted Minkowski series)**.** *Let E be a real Banach space. For $j \geq 1$, let $\mathcal{K}_j \subset E$ be totally bounded, put*

$$R_j := \sup_{x \in \mathcal{K}_j} \|x\|_E,$$

and assume $R_j \leq R$. Let $a_j \geq 0$ satisfy

$$a_j \leq A_0 j^{-u} \big(\log(e+j)\big)^{-v} \qquad (j \geq 1)$$

for some $A_0, R > 0$, $u > 1$, and $v \in \mathbb{R}$, and define

$$\mathcal{K} := \left\{ \sum_{j=1}^{\infty} a_j x_j : x_j \in \mathcal{K}_j \right\}.$$

Put $\gamma = u - 1$. Then $\mathcal{K}$ is totally bounded. Moreover:

(i) *If, uniformly in j,*

$$e_k(\mathcal{K}_j; E) \le B k^{-s} \big(\log(e + k)\big)^{\beta} \qquad (k \ge 1),$$

where $B, s > 0$ and $\beta \in \mathbb{R}$, then, with

$$\sigma = \frac{s\gamma}{s + \gamma}, \qquad \lambda = \frac{\gamma\beta - sv}{s + \gamma},$$

one has

$$e_m(\mathcal{K}; E) \le C m^{-\sigma} \big(\log(e + m)\big)^{\lambda} \qquad (m \ge 2),$$

and therefore

$$\Delta_n^E(\mathcal{K}) \le C'\, n (\log(e + n))^{-\sigma} \big(\log\log(e^e + n)\big)^{\lambda} \qquad (n \ge 2).$$

(ii) *If, uniformly in j,*

$$e_k(\mathcal{K}_j; E) \le B e^{-bk} \qquad (k \ge 1),$$

with $B, b > 0$, then

$$e_m(\mathcal{K}; E) \le C m^{-\gamma} \big(\log(e + m)\big)^{\gamma - v} \qquad (m \ge 2),$$

and hence

$$\Delta_n^E(\mathcal{K}) \le C'\, n (\log(e + n))^{-\gamma} \big(\log\log(e^e + n)\big)^{\gamma - v} \qquad (n \ge 2).$$

All constants depend only on the displayed parameters.

Proof. The dyadic, or integral, estimate for regularly varying tails gives, for $\gamma = u - 1 > 0$,

$$\sum_{j=1}^{r} a_j \le C, \qquad \sum_{j>r} a_j R_j \le C r^{-\gamma} \big(\log(e + r)\big)^{-v} \qquad (r \ge 1).$$

In particular $\sum_j a_j R_j < \infty$, so Corollary 9.34 gives total boundedness. We use the profile form in Corollary 9.35.

For the polynomial/log component entropy, let $L_m = \log(e + m)$ and, for large m, choose

$$r = \left\lceil m^{s/(s+\gamma)} L_m^{-(\beta+v)/(s+\gamma)} \right\rceil, \qquad k = 1 + \left\lfloor \frac{m - 1}{r} \right\rfloor.$$

The finite range of m is absorbed into the constant. For large m, $1 \le r \le m$, $k \asymp m/r$, and both $\log(e + r)$ and $\log(e + k)$ are comparable to L_m. Hence the profile bound gives

$$e_m(\mathcal{K}; E) \le C m^{-s} r^{s} L_m^{\beta} + C r^{-\gamma} L_m^{-v}.$$

With the displayed choice of r, both terms are bounded by

$$Cm^{-s\gamma/(s+\gamma)}L_m^{(\gamma\beta-sv)/(s+\gamma)}.$$

This proves the first entropy estimate. The discrepancy bound follows from Corollaries 9.32 and 9.33, applied with polynomial exponent σ and logarithmic power λ.

For exponential component entropy, choose $D > 2\gamma/b$ and set

$$r = \left\lceil \frac{m}{D\log(e+m)} \right\rceil, \qquad k = 1 + \left\lfloor \frac{m-1}{r} \right\rfloor.$$

For large m, $k \geq (D/2)\log(e+m)$, and therefore

$$e_m(\mathcal{K};E) \leq Ce^{-bk} + Cr^{-\gamma}\big(\log(e+r)\big)^{-v} \leq Cm^{-bD/2} + Cm^{-\gamma}\big(\log(e+m)\big)^{\gamma-v}.$$

Since $bD/2 > \gamma$, the first term is absorbed by the second after enlarging the constant. This proves the second entropy estimate, and the corresponding discrepancy estimate follows from the same dyadic entropy-number corollaries with exponent γ and logarithmic power $\gamma - v$. □

Corollary 9.41 (Hölder images of Banach compact classes)**.** *Let E, F be real normed spaces, let $\mathcal{K} \subset E$ be totally bounded, and let $\Phi : \mathcal{K} \to F$ be (L,α)-Hölder,*

$$\|\Phi(x) - \Phi(y)\|_F \leq L\|x-y\|_E^{\alpha} \qquad (x, y \in \mathcal{K}),$$

with $0 < \alpha \leq 1$. Then

$$e_k(\Phi(\mathcal{K});F) \leq L\,2^{\alpha} e_k(\mathcal{K};E)^{\alpha} \qquad (k \geq 1).$$

Consequently, if $R_\Phi := \sup_{x\in\mathcal{K}} \|\Phi(x)\|_F < \infty$, the following discrepancy bounds hold for $\Delta_n^F(\Phi(\mathcal{K}))$.

(i) *If $e_k(\mathcal{K};E) \leq Ak^{-s}$ for all $k \geq 1$, with $A, s > 0$, then*

$$\Delta_n^F(\Phi(\mathcal{K})) \leq C_{A,L,R_\Phi,s,\alpha}\, n(\log(e+n))^{-\alpha s} \qquad (n \geq 2).$$

(ii) *If*

$$e_k(\mathcal{K};E) \leq Ak^{-s}\big(\log(e+k)\big)^{\beta} \qquad (k \geq 1),$$

with $A, s > 0$ and $\beta \in \mathbb{R}$, then

$$\Delta_n^F(\Phi(\mathcal{K})) \leq C\, n(\log(e+n))^{-\alpha s}\big(\log\log(e^e+n)\big)^{\alpha\beta} \qquad (n \geq 2),$$

where C depends on $A, L, R_\Phi, s, \alpha, \beta$.

(iii) *If $e_k(\mathcal{K};E) \leq Ae^{-bk}$ for all $k \geq 1$, with $A, b > 0$, then*

$$\Delta_n^F(\Phi(\mathcal{K})) \leq C_{A,L,R_\Phi,b,\alpha}\, n^{\log 2/(\alpha b + \log 2)} \qquad (n \geq 2).$$

Proof. Fix $\eta > e_k(\mathcal{K};E)$ and cover $\mathcal{K}$ by at most 2^{k-1} external balls $y_j + \eta B_E$. As in the proof of Corollary 9.33, choose one point $u_j \in \mathcal{K}$ from each nonempty ball. These points form an internal 2η-net for $\mathcal{K}$. Hence the points $\Phi(u_j)$ cover $\Phi(\mathcal{K})$ in F with radius $L(2\eta)^{\alpha}$. Letting $\eta \downarrow e_k(\mathcal{K};E)$ proves the displayed entropy-number transfer.

The three rate estimates are now Corollary 9.32, together with the external-to-internal comparison in Corollary 9.33, applied to the compact class $\Phi(\mathcal{K})$. The polynomial exponent s, logarithmic power β, and exponential rate b are replaced respectively by αs, $\alpha\beta$, and αb. □

Corollary 9.42 (Finite-rank approximation envelope)**.** *Let E be a real normed space and let $\mathcal{K}$ be a totally bounded subset of the unit ball of E. For $m \geq 0$, put*

$$a_m(\mathcal{K}) := \inf_{\dim F \leq m} \sup_{x \in \mathcal{K}} \inf_{y \in F} \|x - y\|_E,$$

where the infimum is over real linear subspaces $F \subset E$ of dimension at most m. Then, for every $m \geq 0$ and $0 < \eta \leq 1$,

$$\Delta_n^E(\mathcal{K}) \leq 3\left(1 + \frac{6}{\eta}\right)^m + n(\eta + a_m(\mathcal{K})).$$

Consequently

$$\Delta_n^E(\mathcal{K}) \leq \inf_{\substack{m \geq 0 \\ 0 < \eta \leq 1}} \left\{3\left(1 + \frac{6}{\eta}\right)^m + n(\eta + a_m(\mathcal{K}))\right\}.$$

If X, E are real normed spaces and $T : X \to E$ is a compact operator with $\|T\| \leq 1$, the same estimate holds for $\Delta_n(T)$ with

$$a_m(T) := \inf_{\operatorname{rank} S \leq m} \|T - S\|_{X \to E}$$

in place of $a_m(\mathcal{K})$.

Proof. Fix $0 < \delta \leq 1$, and choose F with $\dim F \leq m$ and

$$\sup_{x \in \mathcal{K}} \inf_{y \in F} \|x - y\|_E \leq a_m(\mathcal{K}) + \delta.$$

For each $x \in \mathcal{K}$ choose $y_x \in F$ with $\|x - y_x\|_E \leq a_m(\mathcal{K}) + \delta$. Since $\mathcal{K}$ is contained in the unit ball, $a_m(\mathcal{K}) \leq 1$, and $\delta \leq 1$, all such y_x lie in $3B_E \cap F$. The standard volumetric covering bound in an m-dimensional normed space gives an η-net of $3B_E \cap F$ with at most $(1 + 6/\eta)^m$ points. Thus

$$\mathcal{K} \subset \bigcup_{j=1}^{M} (z_j + (a_m(\mathcal{K}) + \delta + \eta)B_E), \qquad M \leq \left(1 + \frac{6}{\eta}\right)^m,$$

with centers z_j lying in $3B_E \cap F$. Given $x_1, \ldots, x_n \in \mathcal{K}$, assign each x_i to a center $z_{a(i)}$ within distance $a_m(\mathcal{K}) + \delta + \eta$. On every fiber $\{i : a(i) = j\}$, choose signs with imbalance at most one. Then

$$\left\| \sum_i \varepsilon_i z_{a(i)} \right\|_E \leq 3\left(1 + \frac{6}{\eta}\right)^m$$

and the approximation error contributes at most $n(a_m(\mathcal{K}) + \delta + \eta)$. Letting $\delta \downarrow 0$ proves the first two displays.

For the operator version, take $\mathcal{K} = T(B_X)$. If S has rank at most m, then $T(B_X)$ lies within $\|T - S\|$ of the m-dimensional space range S. Hence $a_m(\mathcal{K}) \leq a_m(T)$, and the preceding estimate gives the claim. □

Corollary 9.43 (Approximation-number rates for compact images)**.** *Let X, E be real normed spaces and let $T : X \to E$ be a compact operator with $\|T\| \le 1$. With $a_m(T)$ as in Corollary 9.42, the following hold.*

(i) *If $a_m(T) \le Am^{-s}$ for all $m \ge 1$, with $A, s > 0$, then*

$$\Delta_n(T) \le C_{A,s}\, n \left(\frac{\log\log(e^e + n)}{\log(e+n)} \right)^s \qquad (n \ge 2).$$

(ii) *More generally, if*

$$a_m(T) \le Am^{-s}\big(\log(e+m)\big)^\beta \qquad (m \ge 1),$$

with $A, s > 0$ and $\beta \in \mathbb{R}$, then

$$\Delta_n(T) \le C_{A,s,\beta}\, n \left(\frac{\log\log(e^e + n)}{\log(e+n)} \right)^s \big(\log\log(e^e + n)\big)^\beta \qquad (n \ge 2).$$

(iii) *If $a_m(T) \le A\exp(-bm^\theta)$ for all $m \ge 1$, with $A, b, \theta > 0$, then*

$$\Delta_n(T) \le C_{A,b,\theta}\, n \exp\Big(-c_{b,\theta}\big(\log(e+n)\big)^{\theta/(1+\theta)}\Big) \qquad (n \ge 2).$$

For $\theta = 1$ this is $\Delta_n(T) \le C_{A,b} n \exp(-c_b\sqrt{\log(e+n)})$.

The same conclusions hold for any totally bounded $\mathcal{K} \subset B_E$ whose widths $a_m(\mathcal{K})$ satisfy the corresponding decay.

Proof. The final assertion follows from the same proof with $a_m(\mathcal{K})$ in place of $a_m(T)$, so we prove the operator statements.

Assume first that $a_m(T) \le Am^{-s}$. For bounded n the claim follows from the trivial estimate $\Delta_n(T) \le n$. For large n, set

$$m = \left\lfloor \frac{\log(e+n)}{4s\,\log\log(e^e+n)} \right\rfloor, \qquad \eta = Am^{-s},$$

enlarging constants to cover the range where $m < 1$ or $\eta > 1$. Then

$$m\log\left(1 + \frac{6}{\eta}\right) \le m(C_{A,s} + s\log m) \le \frac{1}{2}\log(e+n)$$

for all sufficiently large n. Hence the finite-rank net term in Corollary 9.42 is at most $C\sqrt{n}$, while

$$n(\eta + a_m(T)) \le 2Anm^{-s} \le C_{A,s} n \left(\frac{\log\log(e^e+n)}{\log(e+n)} \right)^s.$$

This term dominates $C\sqrt{n}$ after increasing the constant, giving (i).

The log-refined polynomial case is identical. Put

$$L_n = \log(e+n), \qquad M_n = \log\log(e^e+n),$$

and, for large n,

$$m = \left\lfloor c_{s,\beta} \frac{L_n}{M_n} \right\rfloor, \qquad \eta = Am^{-s}\big(\log(e+m)\big)^\beta,$$

where $c_{s,\beta} > 0$ is small. Enlarging constants covers the finite range where $m < 1$ or $\eta > 1$. Since

$$\log(1/\eta) \le C_{A,s,\beta} + s \log m + |\beta| \log\log(e+m),$$

the choice of $c_{s,\beta}$ ensures

$$m \log\left(1 + \frac{6}{\eta}\right) \le \frac{1}{2} L_n$$

for all large n. The net term is therefore $O(\sqrt{n})$, while

$$n(\eta + a_m(T)) \le C_{A,s,\beta} n m^{-s} \big(\log(e+m)\big)^{\beta} \le C_{A,s,\beta} n \left(\frac{M_n}{L_n}\right)^{s} M_n^{\beta}.$$

This proves the second item, after again absorbing the $O(\sqrt{n})$ net term.

If $a_m(T) \le A \exp(-b m^{\theta})$, take for large n

$$m = \left\lfloor c_{b,\theta} (\log(e+n))^{1/(1+\theta)} \right\rfloor, \qquad \eta = A \exp(-b m^{\theta}),$$

where $c_{b,\theta} > 0$ is small enough that

$$m \log\left(1 + \frac{6}{\eta}\right) \le \frac{1}{2} \log(e+n)$$

for all large n. Corollary 9.42 then gives a net term $O(\sqrt{n})$ and a tail term

$$n(\eta + a_m(T)) \le 2An e^{-b m^{\theta}} \le C_{A,b,\theta} n \exp\Big(-c_{b,\theta} (\log(e+n))^{\theta/(1+\theta)}\Big).$$

Again the latter term dominates after enlarging the constant for small n. □

Corollary 9.44 (Hilbert singular-value rates)**.** *Let H_1, H_2 be real Hilbert spaces and let $T : H_1 \to H_2$ be a compact operator with $\|T\| \le 1$. Let*

$$s_1(T) \ge s_2(T) \ge \cdots$$

be its singular values, repeated with multiplicity and continued by zeros if T has finite rank. Then the following hold.

(i) *If $s_j(T) \le A j^{-s}$ for all $j \ge 1$, with $A, s > 0$, then*

$$\Delta_n(T) \le C_{A,s}\, n \left(\frac{\log\log(e^e + n)}{\log(e+n)}\right)^{s} \qquad (n \ge 2).$$

(ii) *More generally, if*

$$s_j(T) \le A j^{-s} \big(\log(e+j)\big)^{\beta} \qquad (j \ge 1),$$

with $A, s > 0$ and $\beta \in \mathbb{R}$, then

$$\Delta_n(T) \le C_{A,s,\beta}\, n \left(\frac{\log\log(e^e + n)}{\log(e+n)}\right)^{s} \big(\log\log(e^e + n)\big)^{\beta} \qquad (n \ge 2).$$

(iii) *If $s_j(T) \le A \exp(-b j^{\theta})$ for all $j \ge 1$, with $A, b, \theta > 0$, then*

$$\Delta_n(T) \le C_{A,b,\theta}\, n \exp\Big(-c_{b,\theta} (\log(e+n))^{\theta/(1+\theta)}\Big) \qquad (n \ge 2).$$

For $\theta = 1$ this gives $\Delta_n(T) \le C_{A,b} n \exp(-c_b \sqrt{\log(e+n)})$.

Proof. For compact operators between Hilbert spaces, the approximation numbers in Corollary 9.42 satisfy

$$a_m(T) = s_{m+1}(T) \qquad (m \geq 0),$$

by the standard Schmidt approximation theorem. The displayed rates are therefore exactly the three cases of Corollary 9.43, after shifting the index by one and absorbing constants. □

Corollary 9.45 (Schatten-class Hilbert images)**.** *Let H_1, H_2 be real Hilbert spaces and let $T : H_1 \to H_2$ be a compact operator with $\|T\| \leq 1$. If for some $p, S > 0$ its singular values satisfy*

$$\left(\sum_{j\geq 1} s_j(T)^p\right)^{1/p} \leq S,$$

then

$$\Delta_n(T) \leq C_{S,p}\, n \left(\frac{\log\log(e^e + n)}{\log(e+n)}\right)^{1/p} \qquad (n \geq 2).$$

The same conclusion holds under the weak Schatten bound $\sup_{j\geq 1} j^{1/p} s_j(T) \leq S$.

Proof. Since the singular values are nonincreasing,

$$j\, s_j(T)^p \leq \sum_{\ell=1}^{j} s_\ell(T)^p \leq S^p,$$

and hence $s_j(T) \leq S j^{-1/p}$. The weak Schatten hypothesis is exactly the same pointwise estimate. Apply Corollary 9.44 with $s = 1/p$. □

Corollary 9.46 (Compact Assouad and logarithmic Assouad targets)**.** *Let Y be a compact metric space with* $\operatorname{diam} Y \leq 1$*, and let $g_1, \dots, g_n \in C(Y)$ be real-valued contractions with common (L, α)-Hölder modulus. If Y has upper Assouad growth (A, s), then*

$$\operatorname{disc}(g_1, \dots, g_n) \leq C_{A,s,L,\alpha} \sqrt{n \log\bigl(e + n^{s/(2\alpha)-1}\bigr)}.$$

In particular, $s \leq 2\alpha$ gives Spencer scale. Thus every compact D-doubling target is Spencer-scale whenever $\log_2 D \leq 2\alpha$.

More generally, if Y has logarithmic upper Assouad growth (A, s, γ), then

$$\operatorname{disc}(g_1, \dots, g_n) \leq C \sqrt{n \log\Bigl(e + n^{s/(2\alpha)-1} \bigl(\log(e+n)\bigr)^{\gamma}\Bigr)}.$$

Consequently $s < 2\alpha$ is Spencer-scale. At $s = 2\alpha$, the target is still Spencer-scale if $\gamma \leq 0$, and otherwise has only the critical-window loss

$$\operatorname{disc}(g_1, \dots, g_n) \leq C \sqrt{n \log\log(e^e + n)}.$$

Proof. Fix $y_0 \in Y$. Since $\operatorname{diam} Y \le 1$, the whole space lies in $\overline{B}(y_0, 1)$. Upper Assouad growth with $R = 1$ gives

$$N_Y(\rho) \le A\rho^{-s} \qquad (0 < \rho \le 1),$$

and Corollary 9.26 gives the first estimate. The doubling assertion follows from the standard iteration of the doubling cover, which gives global covering exponent $\log_2 D$.

For logarithmic upper Assouad growth, the same $R = 1$ specialization gives

$$N_Y(\rho) \le A\rho^{-s}\big(\log(e/\rho)\big)^{\gamma}.$$

Apply the direct compact-metric profile bound from Corollary 9.26 with

$$\rho = \min\left(\frac{1}{2}, (16L\sqrt{n})^{-1/\alpha}\right).$$

The approximation term is $O_{L,\alpha}(\sqrt{n})$, while

$$\frac{N_Y(\rho)}{n} \le Cn^{s/(2\alpha)-1}\big(\log(e+n)\big)^{\gamma}.$$

This proves the displayed logarithmic estimate, and the endpoint conclusions follow from the sign of the polynomial and logarithmic powers. □

Corollary 9.47 (Stretched-exponential compact entropy)**.** *Let Y be a compact metric space with $\operatorname{diam} Y \le 1$, and let $g_1, \dots, g_n \in C(Y)$ be real-valued contractions with common (L, α)-Hölder modulus. Suppose that, for some $A < \infty$, $q > 0$, and $\gamma \in \mathbb{R}$,*

$$\log N_Y(\rho) \le A\rho^{-q}\big(\log(e/\rho)\big)^{\gamma} \qquad (0 < \rho \le 1).$$

Then

$$\operatorname{disc}(g_1, \dots, g_n) \le C\, n^{(\alpha+q)/(2\alpha+q)}\big(\log(e+n)\big)^{\alpha\gamma/(2\alpha+q)}.$$

In particular, if $\log N_Y(\rho) = O(\rho^{-q})$ for some $q > 0$, then $\operatorname{disc}(g_1, \dots, g_n) = o(n)$ for every $\alpha > 0$.

Proof. It is enough to treat large n; enlarging C handles the remaining finite range. Put

$$\rho = n^{-1/(2\alpha+q)}\big(\log(e+n)\big)^{\gamma/(2\alpha+q)}.$$

For large n, $0 < \rho \le 1/2$, and $\log(e/\rho) \asymp_{\alpha,q,\gamma} \log(e+n)$. The direct compact profile bound in Corollary 9.26 gives

$$\operatorname{disc}(g_1, \dots, g_n) \le C\sqrt{n \log\left(e + \frac{N_Y(\rho)}{n}\right)} + C_{L,\alpha} n\rho^{\alpha}.$$

Writing

$$T = A\rho^{-q}\big(\log(e/\rho)\big)^{\gamma},$$

we have

$$\log\left(e + \frac{N_Y(\rho)}{n}\right) \le \log(e + \exp T) \le C(1+T).$$

The choice of ρ gives

$$\sqrt{nT} \le Cn^{(\alpha+q)/(2\alpha+q)}\big(\log(e+n)\big)^{\alpha\gamma/(2\alpha+q)}$$

and the same bound for $n\rho^\alpha$. This proves the estimate. □

Corollary 9.48 (Hölder quotients of compact metric models)**.** *Let X and Y be compact metric spaces with diameter at most 1, and let $\pi : X \to Y$ be a continuous surjection such that*

$$d_Y(\pi x, \pi x') \le H\, d_X(x,x')^\beta \qquad (x,x' \in X), \qquad 0 < \beta \le 1.$$

Assume that

$$N_X(\rho) \le A\rho^{-s} \qquad (0 < \rho \le 1).$$

If $g_1, \dots, g_n \in C(Y)$ are real-valued contractions with common (L,α)-Hölder modulus on Y, then

$$\operatorname{disc}(g_1,\dots,g_n) \le C_{A,s,H,L,\alpha,\beta}\sqrt{n\log\big(e+n^{s/(2\alpha\beta)-1}\big)}.$$

In particular, $s \le 2\alpha\beta$ gives the Spencer-scale estimate

$$\operatorname{disc}(g_1,\dots,g_n) \le C_{A,s,H,L,\alpha,\beta}\sqrt{n}.$$

Equivalently, if Y is a β-Hölder quotient of a compact model X with upper metric dimension $< 2\alpha\beta$, then every common (L,α)-Hölder tuple on Y is Spencer-scale. The endpoint is also Spencer-scale when the model has finite upper metric $2\alpha\beta$-content. Likewise, if X has upper Assouad growth (A,s), then the same estimate holds with that exponent s. In particular, a β-Hölder quotient of a compact D-doubling model is Spencer-scale whenever

$$\log_2 D \le 2\alpha\beta.$$

Proof. Let $f_i = g_i \circ \pi$. Since π is onto, pullback is an isometry for signed sums:

$$\left\|\sum_i \varepsilon_i g_i\right\|_{C(Y)} = \left\|\sum_i \varepsilon_i f_i\right\|_{C(X)} \qquad (\varepsilon_i \in \{-1,1\}).$$

Thus the two tuples have the same discrepancy. Moreover

$$|f_i(x) - f_i(x')| \le L\, d_Y(\pi x, \pi x')^\alpha \le LH^\alpha d_X(x,x')^{\alpha\beta}.$$

Apply Corollary 9.26 to X, with Hölder exponent $\alpha\beta$. The dimension and endpoint statements follow by choosing an exponent above the model's upper metric dimension, respectively by using finite upper metric $2\alpha\beta$-content at small scales and enlarging constants for the remaining scales. If X has upper Assouad growth (A,s), then, after fixing one point in X, the whole space is contained in a closed unit ball and $N_X(\rho) \le A\rho^{-s}$; the first part applies. The doubling conclusion follows from the standard iteration of the doubling cover, which gives upper Assouad exponent $\log_2 D$. □

Corollary 9.49 (Logarithmic windows for Hölder quotients)**.** *In the setting of Corollary 9.48, replace the covering hypothesis on X by*

$$N_X(\rho) \le A\rho^{-s}\big(\log(e/\rho)\big)^{\gamma} \qquad (0<\rho\le 1),$$

where $\gamma\in\mathbb{R}$. If $g_1,\dots,g_n\in C(Y)$ are real-valued contractions with common (L,α)-Hölder modulus on Y, then

$$\operatorname{disc}(g_1,\dots,g_n) \le C\sqrt{n\log\Big(e+n^{s/(2\alpha\beta)-1}\big(\log(e+n)\big)^{\gamma}\Big)}.$$

Consequently $s<2\alpha\beta$ is Spencer-scale. At $s=2\alpha\beta$, the quotient is still Spencer-scale if $\gamma\le 0$, and otherwise has only the critical-window loss

$$\operatorname{disc}(g_1,\dots,g_n) \le C\sqrt{n\log\log(e^e+n)}.$$

Proof. As in Corollary 9.48, pull back to $f_i=g_i\circ\pi$. The discrepancy is unchanged and the pulled-back tuple is common $(LH^{\alpha},\alpha\beta)$-Hölder on X. Apply the direct compact metric profile bound from Corollary 9.26 at

$$\rho=\min\left(\frac12,(16\max(1,LH^{\alpha})\sqrt{n})^{-1/(\alpha\beta)}\right).$$

The approximation term is $O\sqrt{n}$, while

$$\frac{N_X(\rho)}{n}\le Cn^{s/(2\alpha\beta)-1}\big(\log(e+n)\big)^{\gamma}.$$

This proves the displayed estimate. The subcritical and critical conclusions are immediate from the sign of the polynomial exponent and, at equality, from the sign of the logarithmic power. □

Corollary 9.50 (Snowflaked compact targets)**.** *Let (Y,d) be a compact metric space with* $\operatorname{diam} Y\le 1$*, and let $0<\vartheta\le 1$. Write*

$$d_\vartheta(y,z)=d(y,z)^{\vartheta}.$$

Assume that

$$N_{(Y,d)}(\rho)\le A\rho^{-s} \qquad (0<\rho\le 1).$$

If $g_1,\dots,g_n\in C(Y)$ are real-valued contractions with common (L,α)-Hölder modulus with respect to d_ϑ, then

$$\operatorname{disc}(g_1,\dots,g_n)\le C_{A,s,L,\alpha,\vartheta}\sqrt{n\log\big(e+n^{s/(2\alpha\vartheta)-1}\big)}.$$

In particular, $s\le 2\alpha\vartheta$ gives the Spencer-scale estimate

$$\operatorname{disc}(g_1,\dots,g_n)\le C_{A,s,L,\alpha,\vartheta}\sqrt{n}.$$

Equivalently, if Y has finite upper metric dimension $<2\alpha\vartheta$ in the original metric, then every common (L,α)-Hölder tuple for the snowflaked metric d^{ϑ} is Spencer-scale. The endpoint $s=2\alpha\vartheta$ is also Spencer-scale under finite upper metric s-content in the original metric.

Proof. For $0 < r \le 1$, an r-ball for the snowflaked metric d_ϑ is the same set as an $r^{1/\vartheta}$-ball for d. Hence

$$N_{(Y,d_\vartheta)}(r) = N_{(Y,d)}(r^{1/\vartheta}) \le A r^{-s/\vartheta}.$$

Apply Corollary 9.26 to the compact metric space (Y, d_ϑ). The displayed estimate is exactly the polynomial-covering estimate there with covering exponent s/ϑ. The dimension and content statements follow by the same enlarging-of-constants argument used in Corollary 9.26. □

Corollary 9.51 (Logarithmic windows for snowflaked compact targets)**.** *In the setting of Corollary 9.50, replace the covering hypothesis on the original metric by*

$$N_{(Y,d)}(\rho) \le A\rho^{-s}\big(\log(e/\rho)\big)^\gamma \qquad (0 < \rho \le 1),$$

where $\gamma \in \mathbb{R}$. If $g_1, \dots, g_n \in C(Y)$ are real-valued contractions with common (L,α)-Hölder modulus with respect to $d_\vartheta = d^\vartheta$, then

$$\operatorname{disc}(g_1, \dots, g_n) \le C\sqrt{n \log\Big(e + n^{s/(2\alpha\vartheta)-1}\big(\log(e+n)\big)^\gamma\Big)}.$$

Consequently $s < 2\alpha\vartheta$ is Spencer-scale. At $s = 2\alpha\vartheta$, the snowflaked target is still Spencer-scale if $\gamma \le 0$, and otherwise has only the critical-window loss

$$\operatorname{disc}(g_1, \dots, g_n) \le C\sqrt{n \log\log(e^e + n)}.$$

Proof. For $0 < r \le 1$,

$$N_{(Y,d_\vartheta)}(r) = N_{(Y,d)}(r^{1/\vartheta}) \le A r^{-s/\vartheta}\big(\log(e/r^{1/\vartheta})\big)^\gamma.$$

Since

$$\log(e/r) \le \log(e/r^{1/\vartheta}) \le \vartheta^{-1}\log(e/r) \qquad (0 < r \le 1),$$

the signed logarithmic power is bounded by a constant times $(\log(e/r))^\gamma$. Thus

$$N_{(Y,d_\vartheta)}(r) \le C r^{-s/\vartheta}\big(\log(e/r)\big)^\gamma.$$

Apply the direct compact-metric profile bound from Corollary 9.26 to (Y, d_ϑ), using

$$r = \min\left(\frac{1}{2}, (16\max(1,L)\sqrt{n})^{-1/\alpha}\right).$$

The approximation term is $O\sqrt{n}$, while

$$\frac{N_{(Y,d_\vartheta)}(r)}{n} \le C n^{s/(2\alpha\vartheta)-1}\big(\log(e+n)\big)^\gamma.$$

This proves the displayed estimate. The subcritical and critical conclusions follow from the polynomial exponent and, at equality, the sign of the logarithmic power. □

Corollary 9.52 (Finite compact-metric unions keep the largest exponent)**.** *Let Y be a compact metric space with* $\operatorname{diam} Y \le 1$*, and suppose $Y = \bigcup_{\ell=1}^m Y_\ell$ is a finite union of compact subsets. Assume that, for each ℓ,*

$$N_{Y_\ell}(\rho) \le A_\ell \rho^{-d_\ell} \qquad (0 < \rho \le 1),$$

where covering numbers are computed in the inherited metric. Put $d_* = \max_\ell d_\ell$. *If* $g_1, \dots, g_n \in C(Y)$ *are real-valued contractions with common* (L,α)*-Hölder modulus, then*

$$\operatorname{disc}(g_1, \dots, g_n) \le C_{\{A_\ell, d_\ell\}, L, \alpha, m} \sqrt{n \log\bigl(e + n^{d_*/(2\alpha)-1}\bigr)}.$$

In particular, $d_* \le 2\alpha$ *gives the Spencer-scale bound*

$$\operatorname{disc}(g_1, \dots, g_n) \le C_{\{A_\ell, d_\ell\}, L, \alpha, m} \sqrt{n}.$$

Equivalently, finite upper metric dimensions take the maximum under finite unions. Thus if

$$\max_\ell \overline{\dim}_{\mathrm{M}}(Y_\ell) < 2\alpha,$$

then every common (L,α)*-Hölder tuple on* Y *is Spencer-scale. At the critical endpoint, the same conclusion holds if each* Y_ℓ *has finite upper metric* 2α*-content. Likewise, if each* Y_ℓ *has upper Assouad growth* (A_ℓ, s_ℓ)*, then the same conclusion holds with* $d_* = \max_\ell s_\ell$. *In particular, if each* Y_ℓ *is* D_ℓ*-doubling, then the finite union is Spencer-scale whenever*

$$\max_\ell \log_2 D_\ell \le 2\alpha.$$

Proof. For $0 < \rho \le 1$, choose an inherited-metric ρ-cover of each Y_ℓ. Replacing each subspace ball by the ambient ball with the same center gives a cover of Y, so

$$N_Y(\rho) \le \sum_{\ell=1}^m N_{Y_\ell}(\rho) \le \sum_{\ell=1}^m A_\ell \rho^{-d_\ell} \le A_* \rho^{-d_*}, \qquad A_* := \sum_{\ell=1}^m A_\ell,$$

because $\rho \le 1$. Corollary 9.26 gives the displayed estimate and the Spencer-scale consequence.

For the dimension statement, write $d_\ell^{\mathrm{M}} := \overline{\dim}_{\mathrm{M}}(Y_\ell)$. Choose $D_\ell > d_\ell^{\mathrm{M}}$ with $\max_\ell D_\ell < 2\alpha$ and apply the preceding paragraph. If every Y_ℓ has finite upper metric 2α-content, then $N_{Y_\ell}(\rho) \le C_\ell \rho^{-2\alpha}$ at small scales; enlarging constants for the remaining scales gives the critical endpoint.

If Y_ℓ has upper Assouad growth (A_ℓ, s_ℓ), then, after fixing one point in Y_ℓ, the whole piece is contained in its closed unit-radius ball, so $N_{Y_\ell}(\rho) \le A_\ell \rho^{-s_\ell}$. The first paragraph applies. The doubling assertion follows from the standard iteration of the doubling cover, giving upper Assouad exponent $\log_2 D_\ell$ for each piece. □

Corollary 9.53 (Logarithmic windows for finite compact-metric unions)**.** *In the setting of Corollary 9.52, replace the covering hypothesis on each piece by*

$$N_{Y_\ell}(\rho) \le A_\ell \rho^{-d_\ell} \bigl(\log(e/\rho)\bigr)^{\gamma_\ell} \qquad (0 < \rho \le 1),$$

where $\gamma_\ell \in \mathbb{R}$. *Put*

$$d_* = \max_\ell d_\ell, \qquad \gamma_* = \max_{\ell:\, d_\ell = d_*} \gamma_\ell.$$

If $g_1, \dots, g_n \in C(Y)$ are real-valued contractions with common (L, α)-Hölder modulus, then

$$\operatorname{disc}(g_1, \dots, g_n) \le C\sqrt{n \log\Big(e + n^{d_*/(2\alpha)-1}\big(\log(e+n)\big)^{\gamma_*}\Big)}.$$

Consequently $d_* < 2\alpha$ is Spencer-scale. At $d_* = 2\alpha$, the finite union is still Spencer-scale if $\gamma_* \le 0$, and otherwise has only the critical-window loss

$$\operatorname{disc}(g_1, \dots, g_n) \le C\sqrt{n \log\log(e^e + n)}.$$

Proof. The inherited covers of the pieces give

$$N_Y(\rho) \le \sum_{\ell=1}^{m} A_\ell \rho^{-d_\ell} \big(\log(e/\rho)\big)^{\gamma_\ell}.$$

For $d_\ell = d_*$, the definition of γ_* dominates the logarithmic factor. For $d_\ell < d_*$, the function

$$\rho^{d_* - d_\ell} \big(\log(e/\rho)\big)^{\gamma_\ell - \gamma_*}$$

is bounded on $0 < \rho \le 1$. Hence

$$N_Y(\rho) \le C' \rho^{-d_*} \big(\log(e/\rho)\big)^{\gamma_*}.$$

Apply Corollary 9.26 with

$$\rho = \min\left(\frac{1}{2}, (16L\sqrt{n})^{-1/\alpha}\right),$$

exactly as in Corollary 9.58. □

Corollary 9.54 (Compact metric products add ordinary entropy)**.** *Let Y_j, $1 \le j \le m$, be compact metric spaces with $\operatorname{diam} Y_j \le 1$, and let $Y = \prod_{j=1}^m Y_j$ with the max product metric. Suppose*

$$N_{Y_j}(\rho) \le A_j \rho^{-d_j} \qquad (0 < \rho \le 1),$$

and put $d_* = \sum_{j=1}^m d_j$. If $g_1, \dots, g_n \in C(Y)$ are real-valued contractions with common (L, α)-Hölder modulus, then

$$\operatorname{disc}(g_1, \dots, g_n) \le C_{\{A_j, d_j\}, L, \alpha, m} \sqrt{n \log\big(e + n^{d_*/(2\alpha)-1}\big)}.$$

In particular, $d_* \le 2\alpha$ gives the Spencer-scale bound

$$\operatorname{disc}(g_1, \dots, g_n) \le C_{\{A_j, d_j\}, L, \alpha, m} \sqrt{n}.$$

Equivalently, finite ordinary upper metric dimensions add for arbitrary compact metric targets: if

$$\sum_{j=1}^{m} \overline{\dim}_{\mathrm{M}}(Y_j) < 2\alpha,$$

then every common (L, α)-Hölder tuple on Y is Spencer-scale. At the critical endpoint, the same conclusion holds if

$$\sum_{j=1}^{m} s_j = 2\alpha \quad \text{and} \quad \overline{\mathcal{M}}_{\mathrm{M}}^{s_j}(Y_j) < \infty \quad (1 \le j \le m).$$

Likewise, if each Y_j has upper Assouad growth (A_j, s_j), then Y has upper Assouad growth $(\prod_j A_j, \sum_j s_j)$, so $\sum_j s_j \le 2\alpha$ gives Spencer scale. In particular, a finite max product of compact D_j-doubling metric spaces is Spencer-scale whenever

$$\sum_{j=1}^{m} \log_2 D_j \le 2\alpha.$$

Proof. A product of open ρ-balls in the factors is an open ρ-ball in the max product metric. Hence

$$N_Y(\rho) \le \prod_{j=1}^{m} N_{Y_j}(\rho) \le \left(\prod_{j=1}^{m} A_j\right) \rho^{-d_*}, \qquad 0 < \rho \le 1.$$

The displayed discrepancy estimate and the $d_* \le 2\alpha$ consequence follow from Corollary 9.26.

The dimension statement follows by choosing $D_j > \overline{\dim}_{\mathrm{M}}(Y_j)$ with $\sum_j D_j < 2\alpha$ and applying the preceding paragraph. Finite upper metric s_j-content gives $N_{Y_j}(\rho) \le C_j \rho^{-s_j}$ at small scales; enlarging constants to cover the remaining scales gives the critical endpoint.

For the Assouad statement, a closed R-ball in the max product metric is the product of the closed R-balls in the factors. Cover each factor ball by at most $A_j(R/\rho)^{s_j}$ open ρ-balls and multiply the covers. The doubling consequence is the special case $s_j = \log_2 D_j$, after the standard iteration of the doubling cover. □

Corollary 9.55 (Logarithmic windows for compact metric products)**.** *In the setting of Corollary 9.54, replace the factor covering hypotheses by*

$$N_{Y_j}(\rho) \le A_j \rho^{-d_j} \big(\log(e/\rho)\big)^{\gamma_j} \qquad (0 < \rho \le 1),$$

where $\gamma_j \in \mathbb{R}$. Put

$$d_* = \sum_{j=1}^{m} d_j, \qquad \Gamma_* = \sum_{j=1}^{m} \gamma_j.$$

If $g_1, \dots, g_n \in C(Y)$ are real-valued contractions with common (L, α)-Hölder modulus, then

$$\operatorname{disc}(g_1, \dots, g_n) \le C \sqrt{n \log\Big(e + n^{d_*/(2\alpha)-1} \big(\log(e+n)\big)^{\Gamma_*}\Big)}.$$

Thus $d_* < 2\alpha$ gives Spencer scale. At $d_* = 2\alpha$, products with $\Gamma_* \le 0$ remain Spencer-scale, while $\Gamma_* > 0$ gives the $\sqrt{n \log\log n}$ critical-window loss.

Proof. A product of factor ρ-covers gives

$$N_Y(\rho) \le \left(\prod_{j=1}^{m} A_j\right) \rho^{-d_*} \big(\log(e/\rho)\big)^{\Gamma_*}.$$

The discrepancy estimate and endpoint consequences are the same direct application of Corollary 9.26 used in Corollary 9.53. □

Corollary 9.56 (Finite compact product strata keep the largest stratum exponent)**.** *Let Y be a compact metric space with* $\operatorname{diam} Y \le 1$*, and suppose*

$$Y = \bigcup_{\ell=1}^{M} Z_\ell$$

is a finite union of compact subsets. For each ℓ, let

$$P_\ell = \prod_{j=1}^{m_\ell} Y_{\ell,j}$$

with the max product metric, where each $Y_{\ell,j}$ is compact and $\operatorname{diam} Y_{\ell,j} \le 1$*. Assume that there is a B_ℓ-Lipschitz surjection $\pi_\ell : P_\ell \to Z_\ell$, where Z_ℓ carries the metric inherited from Y and $B_\ell \ge 1$. Suppose*

$$N_{Y_{\ell,j}}(\rho) \le A_{\ell,j}\rho^{-d_{\ell,j}} \qquad (0 < \rho \le 1),$$

and put

$$s_\ell = \sum_{j=1}^{m_\ell} d_{\ell,j}, \qquad s_* = \max_{1\le\ell\le M} s_\ell.$$

If $g_1, \dots, g_n \in C(Y)$ are real-valued contractions with common (L,α)-Hölder modulus, then

$$\operatorname{disc}(g_1, \dots, g_n) \le C\sqrt{n \log\bigl(e + n^{s_*/(2\alpha)-1}\bigr)},$$

where C depends only on L, α and the finite stratum data $\{A_{\ell,j}, d_{\ell,j}, B_\ell, m_\ell\}_{\ell,j}$. In particular, $s_* \le 2\alpha$ gives the Spencer-scale bound

$$\operatorname{disc}(g_1, \dots, g_n) \le C\sqrt{n}.$$

Equivalently, a finite union of product-parametrized compact strata is governed by the largest stratum sum: if

$$\max_{\ell} \sum_{j=1}^{m_\ell} \overline{\dim}_{\mathrm{M}}(Y_{\ell,j}) < 2\alpha,$$

then every common (L,α)-Hölder tuple on Y is Spencer-scale. At the critical endpoint, the same conclusion holds if, for every ℓ, there are $s_{\ell,j} \ge 0$ with

$$\sum_{j=1}^{m_\ell} s_{\ell,j} \le 2\alpha \quad \text{and} \quad \overline{\mathcal{M}}_{\mathrm{M}}^{s_{\ell,j}}(Y_{\ell,j}) < \infty \quad (1 \le j \le m_\ell).$$

Likewise, if each $Y_{\ell,j}$ has upper Assouad growth $(A_{\ell,j}, a_{\ell,j})$, then the same conclusion holds with

$$s_* = \max_{\ell} \sum_{j=1}^{m_\ell} a_{\ell,j}.$$

In particular, if each $Y_{\ell,j}$ is $D_{\ell,j}$-doubling, then Y is Spencer-scale whenever

$$\max_{\ell} \sum_{j=1}^{m_\ell} \log_2 D_{\ell,j} \le 2\alpha.$$

Proof. Fix a stratum ℓ. A product of open ρ/B_ℓ-balls in the factors is an open ρ/B_ℓ-ball in P_ℓ, and its image under π_ℓ is contained in an inherited-metric ρ-ball in Z_ℓ. Therefore, for $0 < \rho \le 1$,

$$N_{Z_\ell}(\rho) \le N_{P_\ell}(\rho/B_\ell) \le \prod_{j=1}^{m_\ell} N_{Y_{\ell,j}}(\rho/B_\ell) \le A'_\ell \rho^{-s_\ell},$$

where

$$A'_\ell = \prod_{j=1}^{m_\ell} A_{\ell,j} B_\ell^{d_{\ell,j}}.$$

Corollary 9.52 applied to the pieces Z_ℓ gives the displayed estimate and the $s_* \le 2\alpha$ consequence.

For the dimension assertion, choose $D_{\ell,j} > \overline{\dim}_{\mathrm{M}}(Y_{\ell,j})$ so that

$$\max_\ell \sum_{j=1}^{m_\ell} D_{\ell,j} < 2\alpha,$$

and use the preceding paragraph with $d_{\ell,j} = D_{\ell,j}$. The critical-content statement is the same argument with

$$N_{Y_{\ell,j}}(\rho) \le C_{\ell,j} \rho^{-s_{\ell,j}}$$

at small scales, after enlarging constants for the remaining scales. The Assouad and doubling assertions follow by the same global-covering reduction used in Corollaries 9.52 and 9.54. □

Corollary 9.57 (Hölder product strata divide the parametrizing exponent)**.** *Let Y be a compact metric space with* $\operatorname{diam} Y \le 1$*, and suppose*

$$Y = \bigcup_{\ell=1}^{M} Z_\ell$$

is a finite union of compact subsets. For each ℓ, let

$$P_\ell = \prod_{j=1}^{m_\ell} Y_{\ell,j}$$

with the max product metric, where each $Y_{\ell,j}$ is compact and $\operatorname{diam} Y_{\ell,j} \le 1$*. Assume that there is a (B_ℓ, β_ℓ)-Hölder surjection $\pi_\ell : P_\ell \to Z_\ell$, where $0 < \beta_\ell \le 1$, $B_\ell \ge 1$, and*

$$d_Y(\pi_\ell x, \pi_\ell y) \le B_\ell d_{P_\ell}(x, y)^{\beta_\ell}.$$

Suppose

$$N_{Y_{\ell,j}}(\rho) \le A_{\ell,j} \rho^{-d_{\ell,j}} \qquad (0 < \rho \le 1),$$

and put

$$s_\ell = \sum_{j=1}^{m_\ell} d_{\ell,j}, \qquad t_\ell = \frac{s_\ell}{\beta_\ell}, \qquad t_* = \max_{1 \le \ell \le M} t_\ell.$$

If $g_1, \dots, g_n \in C(Y)$ are real-valued contractions with common (L, α)-Hölder modulus, then

$$\operatorname{disc}(g_1, \dots, g_n) \le C\sqrt{n \log\bigl(e + n^{t_*/(2\alpha)-1}\bigr)},$$

where C depends only on L, α and on the finite collection of parameters

$$A_{\ell,j},\ d_{\ell,j},\ B_\ell,\ \beta_\ell,\ m_\ell \qquad (1 \le \ell \le M,\ 1 \le j \le m_\ell).$$

In particular, $t_* \le 2\alpha$ gives the Spencer-scale bound

$$\operatorname{disc}(g_1, \dots, g_n) \le C\sqrt{n}.$$

Equivalently, a finite union of Hölder product-parametrized compact strata is governed by

$$\max_\ell \frac{1}{\beta_\ell} \sum_{j=1}^{m_\ell} \overline{\dim}_{\mathrm{M}}(Y_{\ell,j}).$$

If this quantity is $< 2\alpha$, then every common (L, α)-Hölder tuple on Y is Spencer-scale. At the critical endpoint, the same conclusion holds if, for every ℓ, there are $s_{\ell,j} \ge 0$ with

$$\frac{1}{\beta_\ell} \sum_{j=1}^{m_\ell} s_{\ell,j} \le 2\alpha \quad \text{and} \quad \overline{\mathcal{M}}_{\mathrm{M}}^{s_{\ell,j}}(Y_{\ell,j}) < \infty \quad (1 \le j \le m_\ell).$$

Likewise, if each $Y_{\ell,j}$ has upper Assouad growth $(A_{\ell,j}, a_{\ell,j})$, then the same conclusion holds with

$$t_* = \max_\ell \frac{1}{\beta_\ell} \sum_{j=1}^{m_\ell} a_{\ell,j}.$$

In particular, if each $Y_{\ell,j}$ is $D_{\ell,j}$-doubling, then Y is Spencer-scale whenever

$$\max_\ell \frac{1}{\beta_\ell} \sum_{j=1}^{m_\ell} \log_2 D_{\ell,j} \le 2\alpha.$$

Proof. Fix a stratum ℓ. Put

$$r_\ell(\rho) = \left(\frac{\rho}{B_\ell}\right)^{1/\beta_\ell}.$$

Since $0 < \rho \le 1$ and $B_\ell \ge 1$, one has $0 < r_\ell(\rho) \le 1$. A product of open $r_\ell(\rho)$-balls in the factors is an open $r_\ell(\rho)$-ball in P_ℓ, and its image under π_ℓ is contained in an inherited-metric ρ-ball in Z_ℓ. Hence

$$N_{Z_\ell}(\rho) \le N_{P_\ell}(r_\ell(\rho)) \le \prod_{j=1}^{m_\ell} N_{Y_{\ell,j}}(r_\ell(\rho)) \le A''_\ell \rho^{-s_\ell/\beta_\ell},$$

where

$$A''_\ell = \prod_{j=1}^{m_\ell} A_{\ell,j} B_\ell^{d_{\ell,j}/\beta_\ell}.$$

Corollary 9.52 applied to the pieces Z_ℓ gives the displayed estimate and the $t_* \le 2\alpha$ consequence.

The ordinary dimension statement follows by choosing $D_{\ell,j} > \overline{\dim}_{\mathrm{M}}(Y_{\ell,j})$ so that

$$\max_{\ell} \frac{1}{\beta_\ell} \sum_{j=1}^{m_\ell} D_{\ell,j} < 2\alpha$$

and applying the preceding paragraph. The finite-content endpoint, Assouad statement, and doubling statement are the same covering reduction with $d_{\ell,j} = s_{\ell,j}$, $d_{\ell,j} = a_{\ell,j}$, and $a_{\ell,j} = \log_2 D_{\ell,j}$, respectively. □

Corollary 9.58 (Logarithmic windows for Hölder product strata)**.** *In the setting of Corollary 9.57, replace the factor covering hypothesis by*

$$N_{Y_{\ell,j}}(\rho) \le A_{\ell,j}\rho^{-d_{\ell,j}}\big(\log(e/\rho)\big)^{\gamma_{\ell,j}} \qquad (0 < \rho \le 1),$$

with $\gamma_{\ell,j} \in \mathbb{R}$. Put

$$t_\ell = \frac{1}{\beta_\ell}\sum_{j=1}^{m_\ell} d_{\ell,j}, \qquad \Gamma_\ell = \sum_{j=1}^{m_\ell} \gamma_{\ell,j}, \qquad t_* = \max_{\ell} t_\ell, \qquad \Gamma_* = \max_{\ell:\, t_\ell = t_*} \Gamma_\ell.$$

If $g_1, \dots, g_n \in C(Y)$ are real-valued contractions with common (L,α)-Hölder modulus, then

$$\operatorname{disc}(g_1, \dots, g_n) \le C\sqrt{n \log\Big(e + n^{t_*/(2\alpha)-1}\big(\log(e+n)\big)^{\Gamma_*}\Big)},$$

where C depends only on L, α and the finite logarithmic stratum data. Consequently $t_* < 2\alpha$ is still Spencer-scale. At the critical exponent $t_* = 2\alpha$, one has the endpoint bound

$$\operatorname{disc}(g_1, \dots, g_n) \le \begin{cases} C\sqrt{n}, & \Gamma_* \le 0, \\ C\sqrt{n \log\log(e^e + n)}, & \Gamma_* > 0. \end{cases}$$

Proof. The proof of Corollary 9.57 gives, with

$$r_\ell(\rho) = \left(\frac{\rho}{B_\ell}\right)^{1/\beta_\ell},$$

the estimate

$$N_{Z_\ell}(\rho) \le C_\ell \rho^{-t_\ell}\big(\log(e/r_\ell(\rho))\big)^{\Gamma_\ell}.$$

Since $B_\ell \ge 1$ and $0 < \beta_\ell \le 1$, the last logarithm is comparable to $\log(e/\rho)$, with constants depending only on B_ℓ, β_ℓ. Thus the same bound holds with $\log(e/\rho)^{\Gamma_\ell}$. Because there are only finitely many strata and polynomial gaps dominate all logarithmic powers, every term with $t_\ell < t_*$ is also bounded by a constant multiple of $\rho^{-t_*}(\log(e/\rho))^{\Gamma_*}$. For $t_\ell = t_*$, the definition of Γ_* gives the same domination. Hence

$$N_Y(\rho) \le C'\rho^{-t_*}\big(\log(e/\rho)\big)^{\Gamma_*} \qquad (0 < \rho \le 1).$$

Apply the direct compact-metric profile bound from Corollary 9.26 with

$$\rho = \min\left(\frac{1}{2}, (16L\sqrt{n})^{-1/\alpha}\right).$$

The approximation term is $O_{L,\alpha}(\sqrt{n})$, and

$$\frac{N_Y(\rho)}{n} \le C'' n^{t_*/(2\alpha)-1}\big(\log(e+n)\big)^{\Gamma_*}.$$

This is the displayed estimate. If $t_* < 2\alpha$, the factor after $e+$ is bounded uniformly in n. If $t_* = 2\alpha$, it is bounded for $\Gamma_* \le 0$, and for $\Gamma_* > 0$ contributes only $\log\log(e^e + n)$. □

Corollary 9.59 (Euclidean cubes via Cantor resolutions)**.** *Let $D \ge 1$, and give $[0,1]^D$ the max metric. If $g_1, \dots, g_n \in C([0,1]^D)$ are real-valued contractions with common (L,α)-Hölder modulus, then*

$$\operatorname{disc}(g_1, \dots, g_n) \le C_{D,L,\alpha}\sqrt{n \log\big(e + n^{D/(2\alpha)-1}\big)}.$$

In particular, $D \le 2\alpha$ gives

$$\operatorname{disc}(g_1, \dots, g_n) \le C_{D,L,\alpha}\sqrt{n}.$$

More generally, if $K \subset [0,1]^D$ is compact and $0 < \alpha \le 1$, the same estimates hold for common (L,α)-Hölder contractions on K.

Proof. Let

$$\Sigma_D = (\{0,1\}^D)^{\mathbb{N}}.$$

For distinct $\sigma, \tau \in \Sigma_D$, put

$$d_\Sigma(\sigma,\tau) = 2^{-\min\{k \ge 0 : \sigma_k \ne \tau_k\}},$$

and put $d_\Sigma(\sigma,\sigma) = 0$. The length-$k$ cylinder partition has 2^{Dk} atoms of diameter at most 2^{-k}. Given $0 < \rho \le 1$, choose k minimal with $2^{-k} < \rho$. These cylinders are contained in open ρ-balls and $2^k \le 2\rho^{-1}$, so

$$N_{\Sigma_D}(\rho) \le 2^D \rho^{-D} \qquad (0 < \rho \le 1).$$

Define the binary coding map $\pi : \Sigma_D \to [0,1]^D$ by

$$\pi(\sigma)_j = \sum_{k=0}^{\infty} \sigma_{k,j} 2^{-(k+1)}, \qquad 1 \le j \le D.$$

It is onto, and if the first differing symbol occurs at level k, then

$$\|\pi(\sigma) - \pi(\tau)\|_\infty \le 2^{-k} = d_\Sigma(\sigma,\tau).$$

Thus $[0,1]^D$ has a $(2^D, D, 1, 1)$-zero-dimensional Hölder resolution, and Corollary 9.24 proves the cube statement.

For a compact $K \subset [0,1]^D$ and $0 < \alpha \le 1$, each real-valued (L,α)-Hölder contraction g_i on K has the McShane extension

$$\widetilde{g}_i(x) = \inf_{y \in K}\big(g_i(y) + L\|x - y\|_\infty^\alpha\big), \qquad x \in [0,1]^D,$$

because $\|x-y\|_\infty^\alpha$ is a metric. Clipping $\widetilde{g}_i$ to the interval $[-1,1]$ preserves both the restriction to K and the L-Hölder constant. A signing for the clipped extensions on the cube restricts to the same signing on K, so the cube estimate gives the compact-subset estimate. □

Corollary 9.60 (Finite Lipschitz atlases)**.** *Let Y be a compact metric space with* $\operatorname{diam} Y \le 1$*. Suppose that*

$$Y = \bigcup_{\ell=1}^{m} \phi_\ell(K_\ell),$$

where $K_\ell \subset [0,1]^{D_\ell}$ is compact, the cube has the max metric, and $\phi_\ell : K_\ell \to Y$ is B_ℓ-Lipschitz. Put

$$D_* = \max_{1\le \ell \le m} D_\ell.$$

If $g_1, \dots, g_n \in C(Y)$ are real-valued contractions with common (L,α)-Hölder modulus, then

$$\operatorname{disc}(g_1, \dots, g_n) \le C_{\{D_\ell, B_\ell\}, m, L, \alpha} \sqrt{n \log\bigl(e + n^{D_*/(2\alpha)-1}\bigr)}.$$

In particular, every compact target admitting a finite D-dimensional Lipschitz atlas is Spencer-scale for common (L,α)-Hölder tuples whenever $D \le 2\alpha$:

$$\operatorname{disc}(g_1, \dots, g_n) \le C\sqrt{n}.$$

Thus compact Lipschitz D-manifolds, finite D-polyhedra, and finite unions of Lipschitz images of compact subsets of $\mathbb{R}^D$ obey the same Euclidean threshold $D \le 2\alpha$.

Proof. It is enough to bound the covering number of each chart image. If $B_\ell^+ = \max(1, B_\ell)$, then a max-metric grid cover of $[0,1]^{D_\ell}$ gives

$$N_{K_\ell}(r) \le C_{D_\ell} r^{-D_\ell} \qquad (0 < r \le 1).$$

Since ϕ_ℓ is B_ℓ-Lipschitz,

$$N_{\phi_\ell(K_\ell)}(\rho) \le N_{K_\ell}(\rho / B_\ell^+) \le C_{D_\ell} (B_\ell^+)^{D_\ell} \rho^{-D_\ell} \qquad (0 < \rho \le 1).$$

Corollary 9.52, applied to the finite union of the chart images, gives the displayed estimate with exponent D_*. The Spencer-scale consequence follows from $D_* \le 2\alpha$. The manifold and polyhedral examples have finite Lipschitz atlases of the stated dimension after choosing finitely many compact coordinate patches. □

Corollary 9.61 (Finite Hölder atlases)**.** *Let Y be a compact metric space with* $\operatorname{diam} Y \le 1$*. Suppose that*

$$Y = \bigcup_{\ell=1}^{m} \phi_\ell(K_\ell),$$

where $K_\ell \subset [0,1]^{D_\ell}$ is compact, the cube has the max metric, and $\phi_\ell : K_\ell \to Y$ is (B_ℓ, β_ℓ)-Hölder:

$$d_Y(\phi_\ell(x), \phi_\ell(y)) \le B_\ell \|x - y\|_\infty^{\beta_\ell}, \qquad 0 < \beta_\ell \le 1.$$

Put

$$s_* = \max_{1\le \ell \le m} \frac{D_\ell}{\beta_\ell}.$$

If $g_1, \dots, g_n \in C(Y)$ are real-valued contractions with common (L, α)-Hölder modulus, then

$$\operatorname{disc}(g_1, \dots, g_n) \le C_{\{D_\ell, B_\ell, \beta_\ell\}, m, L, \alpha} \sqrt{n \log\bigl(e + n^{s_*/(2\alpha)-1}\bigr)}.$$

In particular, $s_* \le 2\alpha$ gives the Spencer-scale estimate

$$\operatorname{disc}(g_1, \dots, g_n) \le C\sqrt{n}.$$

Equivalently, a finite D-dimensional β-Hölder atlas is Spencer-scale whenever $D \le 2\alpha\beta$. The Lipschitz-atlas corollary is the special case $\beta_\ell = 1$.

Proof. Let $B_\ell^+ = \max(1, B_\ell)$. The grid-cover estimate used in the proof of Corollary 9.60 gives

$$N_{K_\ell}(r) \le C_{D_\ell} r^{-D_\ell} \qquad (0 < r \le 1).$$

A set in K_ℓ of diameter at most $(\rho/B_\ell^+)^{1/\beta_\ell}$ has image of diameter at most ρ. Hence, for $0 < \rho \le 1$,

$$N_{\phi_\ell(K_\ell)}(\rho) \le N_{K_\ell}\Bigl((\rho/B_\ell^+)^{1/\beta_\ell}\Bigr) \le C_{D_\ell}(B_\ell^+)^{D_\ell/\beta_\ell} \rho^{-D_\ell/\beta_\ell}.$$

Apply Corollary 9.52 to the finite union of the chart images. The exponent is $\max_\ell D_\ell/\beta_\ell = s_*$, giving the displayed estimate. If $s_* \le 2\alpha$, Corollary 9.52 gives the Spencer-scale conclusion. □

Corollary 9.62 (Logarithmic Hölder atlases)**.** *Let Y be a compact metric space with* $\operatorname{diam} Y \le 1$*. Suppose that*

$$Y = \bigcup_{\ell=1}^{m} \phi_\ell(K_\ell),$$

where each K_ℓ is a compact metric space with $\operatorname{diam} K_\ell \le 1$*, and $\phi_\ell : K_\ell \to Y$ is (B_ℓ, β_ℓ)-Hölder:*

$$d_Y(\phi_\ell(x), \phi_\ell(y)) \le B_\ell d_{K_\ell}(x, y)^{\beta_\ell}, \qquad 0 < \beta_\ell \le 1.$$

Assume that, for each ℓ,

$$N_{K_\ell}(\rho) \le A_\ell \rho^{-d_\ell} \bigl(\log(e/\rho)\bigr)^{\gamma_\ell} \qquad (0 < \rho \le 1),$$

where $\gamma_\ell \in \mathbb{R}$. Put

$$t_\ell = \frac{d_\ell}{\beta_\ell}, \qquad t_* = \max_{1 \le \ell \le m} t_\ell, \qquad \Gamma_* = \max_{\ell:\, t_\ell = t_*} \gamma_\ell.$$

If $g_1, \dots, g_n \in C(Y)$ are real-valued contractions with common (L, α)-Hölder modulus, then

$$\operatorname{disc}(g_1, \dots, g_n) \le C\sqrt{n \log\Bigl(e + n^{t_*/(2\alpha)-1} \bigl(\log(e+n)\bigr)^{\Gamma_*}\Bigr)}.$$

Consequently $t_* < 2\alpha$ is Spencer-scale. At $t_* = 2\alpha$, the atlas is still Spencer-scale if $\Gamma_* \le 0$, and otherwise has only the critical-window loss

$$\operatorname{disc}(g_1, \dots, g_n) \le C\sqrt{n \log\log(e^e + n)}.$$

Proof. Let $B_\ell^+ = \max(1, B_\ell)$. A cover of K_ℓ by balls of radius $(\rho/B_\ell^+)^{1/\beta_\ell}$ maps to a cover of $\phi_\ell(K_\ell)$ by Y-balls of radius ρ. Hence

$$N_{\phi_\ell(K_\ell)}(\rho) \le A_\ell (B_\ell^+)^{d_\ell/\beta_\ell} \rho^{-d_\ell/\beta_\ell} \big(\log(e(B_\ell^+/\rho)^{1/\beta_\ell})\big)^{\gamma_\ell}.$$

For $0 < \rho \le 1$, the last logarithm is comparable to $\log(e/\rho)$, with constants depending on B_ℓ and β_ℓ. This remains true for signed powers γ_ℓ. Therefore

$$N_{\phi_\ell(K_\ell)}(\rho) \le C_\ell \rho^{-t_\ell} \big(\log(e/\rho)\big)^{\gamma_\ell}.$$

Now apply Corollary 9.53 to the finite union of chart images. The top polynomial exponent is t_*, and among charts with this exponent the top logarithmic power is Γ_*, giving the displayed estimate and the endpoint consequences. □

Corollary 9.63 (Exhaustive finite-scale Hölder atlases)**.** *Let Y be a compact metric space with* $\operatorname{diam} Y \le 1$. *Let $Y_r \subset Y$, $r \ge 1$, be compact sets whose one-sided Hausdorff errors*

$$\delta_r := \sup_{y \in Y} \operatorname{dist}(y, Y_r)$$

tend to 0. *Assume that there are parameters $s \ge 0$, $\gamma \in \mathbb{R}$, and $A_r \ge 1$ such that*

$$N_{Y_r}(u) \le A_r u^{-s} \big(\log(e/u)\big)^{\gamma} \qquad (0 < u \le 1,\ r \ge 1).$$

Let $g_1, \dots, g_n \in C(Y)$ be real-valued contractions with common (L, α)-Hölder modulus, and put

$$L_+ := \max(1, L), \qquad \rho_n := \min\left(\frac{1}{2}, \frac{1}{16 L_+^{1/\alpha} n^{1/(2\alpha)}}\right).$$

Choose any r_n with $\delta_{r_n} \le \rho_n/2$. Then

$$\operatorname{disc}(g_1, \dots, g_n) \le C_{s,\gamma,L,\alpha} \sqrt{n \log\Big(e + A_{r_n} n^{s/(2\alpha)-1} \big(\log(e+n)\big)^{\gamma}\Big)}.$$

Consequently, exhaustive stages may grow up to the unused entropy slack: if

$$A_{r_n} \le C_0 n^{1-s/(2\alpha)} \big(\log(e+n)\big)^{-\gamma} \qquad (n \ge 1),$$

then the tuple is Spencer-scale, $\operatorname{disc}(g_1, \dots, g_n) \le C\sqrt{n}$. *At the critical exponent $s = 2\alpha$, if*

$$A_{r_n} \le C_0 \big(\log(e+n)\big)^{\kappa},$$

then the same Spencer-scale conclusion holds when $\kappa + \gamma \le 0$, while $\kappa + \gamma > 0$ gives the critical-window bound

$$\operatorname{disc}(g_1, \dots, g_n) \le C\sqrt{n \log\log(e^e + n)}.$$

The stage covering hypothesis is automatic when the finite-atlas covering calculation on each Y_r gives the same A_r-budgeted profile.

Proof. Fix $0 < \rho \le 1/2$ and choose r with $\delta_r \le \rho/2$. If $\{B(y_j, \rho/2)\}_{j=1}^{M}$ covers Y_r, then every $y \in Y$ has $y' \in Y_r$ with $d_Y(y, y') \le \rho/2$, and y' lies in some $B(y_j, \rho/2)$. Hence $Y \subset \bigcup_j B(y_j, \rho)$, so

$$N_Y(\rho) \le N_{Y_r}(\rho/2) \le C_{s,\gamma} A_r \rho^{-s} \big(\log(e/\rho)\big)^{\gamma}.$$

The signed logarithmic power is harmless because $\log(2e/\rho)$ and $\log(e/\rho)$ are comparable on $0 < \rho \le 1/2$.

Apply the direct profile bound in Corollary 9.26 with $\rho = \rho_n$ and $r = r_n$. The Hölder approximation term satisfies

$$2nL(16\rho_n)^{\alpha} \le 2\sqrt{n},$$

and the entropy term is bounded by

$$\frac{N_Y(\rho_n)}{n} \le C_{s,\gamma,L,\alpha} A_{r_n} n^{s/(2\alpha)-1} \big(\log(e+n)\big)^{\gamma}.$$

This proves the displayed estimate. The Spencer-scale and critical-window consequences are immediate from the size of the quantity inside the logarithm. The final atlas assertion is the finite-stage covering calculation from Corollary 9.62, applied separately to each Y_r. □

Corollary 9.64 (Polynomial exhaustive atlas schedules)**.** *In the setting of Corollary 9.63, assume in addition that, for some $H, A, \eta > 0$, $\lambda \ge 0$, and $\kappa \in \mathbb{R}$,*

$$\delta_r \le H r^{-\eta}, \qquad A_r \le A r^{\lambda} \big(\log(e+r)\big)^{\kappa} \qquad (r \ge 1).$$

Put

$$s_{\mathrm{eff}} := s + \frac{\lambda}{\eta}, \qquad \Gamma_{\mathrm{eff}} := \gamma + \kappa.$$

Then every real-valued contraction tuple on Y with common (L, α)-Hölder modulus satisfies

$$\operatorname{disc}(g_1, \dots, g_n) \le C \sqrt{n \log\Big(e + n^{s_{\mathrm{eff}}/(2\alpha)-1} \big(\log(e+n)\big)^{\Gamma_{\mathrm{eff}}}\Big)}.$$

Consequently $s_{\mathrm{eff}} < 2\alpha$ is Spencer-scale. At $s_{\mathrm{eff}} = 2\alpha$, the same Spencer-scale conclusion holds when $\Gamma_{\mathrm{eff}} \le 0$, while $\Gamma_{\mathrm{eff}} > 0$ gives only the critical-window loss

$$\operatorname{disc}(g_1, \dots, g_n) \le C \sqrt{n \log\log(e^e + n)}.$$

Thus a polynomially improving exhaustion trades approximation rate against stage complexity by adding λ/η to the intrinsic stage exponent.

Proof. Use the signing scale ρ_n from Corollary 9.63, and choose

$$r_n = \left\lceil \max\left(1, \left(\frac{2H}{\rho_n}\right)^{1/\eta}\right) \right\rceil.$$

Then $\delta_{r_n} \le \rho_n/2$. Since $\rho_n \asymp_{L,\alpha} n^{-1/(2\alpha)}$, the choice of r_n gives

$$r_n^{\lambda} \le C n^{\lambda/(2\alpha\eta)} \quad \text{and} \quad \big(\log(e+r_n)\big)^{\kappa} \le C \big(\log(e+n)\big)^{\kappa},$$

with the usual comparison for signed logarithmic powers. Therefore

$$A_{r_n} n^{s/(2\alpha)-1} \big(\log(e+n)\big)^{\gamma} \le C n^{s_{\mathrm{eff}}/(2\alpha)-1} \big(\log(e+n)\big)^{\Gamma_{\mathrm{eff}}}.$$

Substituting this in Corollary 9.63 proves the displayed estimate. The Spencer-scale and critical-window conclusions follow by inspecting the term inside the outer logarithm. □

Corollary 9.65 (Stretched-exponential exhaustive atlas schedules)**.** *In the setting of Corollary 9.63, assume in addition that, for some $H, A, c, \theta > 0$, $\lambda \ge 0$, and $\kappa \in \mathbb{R}$,*

$$\delta_r \le H \exp(-c r^{\theta}), \qquad A_r \le A r^{\lambda} \big(\log(e+r)\big)^{\kappa} \qquad (r \ge 1).$$

Put

$$\Gamma_{\mathrm{fast}} := \gamma + \frac{\lambda}{\theta}.$$

Then every real-valued contraction tuple on Y with common (L,α)-Hölder modulus satisfies

$$\operatorname{disc}(g_1,\dots,g_n) \le C \sqrt{n \log\Big(e + n^{s/(2\alpha)-1} \big(\log(e+n)\big)^{\Gamma_{\mathrm{fast}}} \big(\log\log(e^e+n)\big)^{\kappa}\Big)}.$$

Consequently $s < 2\alpha$ is Spencer-scale. At $s = 2\alpha$, the same Spencer-scale conclusion holds if either $\Gamma_{\mathrm{fast}} < 0$, or $\Gamma_{\mathrm{fast}} = 0$ and $\kappa \le 0$. If $s = 2\alpha$ and $\Gamma_{\mathrm{fast}} > 0$, then

$$\operatorname{disc}(g_1,\dots,g_n) \le C \sqrt{n \log\log(e^e+n)},$$

while the borderline case $\Gamma_{\mathrm{fast}} = 0 < \kappa$ has the sharper triple-log loss

$$\operatorname{disc}(g_1,\dots,g_n) \le C \sqrt{n \log\log\log(e^{e^e}+n)}.$$

Proof. Use the signing scale ρ_n from Corollary 9.63. Choose

$$r_n = \Big\lceil C_0 \big(\log(e+n)\big)^{1/\theta} \Big\rceil,$$

with C_0 large enough, depending on H, c, L, α, θ, so that $\delta_{r_n} \le \rho_n/2$ for every $n \ge 1$. Then

$$r_n^{\lambda} \le C \big(\log(e+n)\big)^{\lambda/\theta}$$

and, for signed κ,

$$\big(\log(e+r_n)\big)^{\kappa} \le C \big(\log\log(e^e+n)\big)^{\kappa}.$$

Thus

$$A_{r_n} n^{s/(2\alpha)-1} \big(\log(e+n)\big)^{\gamma} \le C n^{s/(2\alpha)-1} \big(\log(e+n)\big)^{\Gamma_{\mathrm{fast}}} \big(\log\log(e^e+n)\big)^{\kappa}.$$

Substitution in Corollary 9.63 gives the displayed estimate. The endpoint cases follow by inspecting the logarithm: polynomial decay in $\log n$ absorbs any fixed power of $\log\log n$, positive powers of $\log n$ cost $\log\log n$, and a pure positive power of $\log\log n$ costs $\log\log\log n$. □

Definition 9.66 (Clopen Minkowski dimension)**.** For a compact zero-dimensional metric space X, define the upper clopen Minkowski dimension by

$$\overline{\dim}_{\mathrm{cl}}(X) := \limsup_{r\downarrow 0} \frac{\log M_X(r)}{\log(1/r)},$$

where $M_X(r)$ is the clopen partition growth from Corollary 9.14.

Definition 9.67 (Upper clopen Minkowski content)**.** For $s \geq 0$, define the upper clopen s-dimensional Minkowski content by

$$\overline{\mathcal{M}}^{s}_{\mathrm{cl}}(X) := \limsup_{r\downarrow 0} r^s M_X(r),$$

again with $M_X(r)$ as in Corollary 9.14.

Corollary 9.68 (Clopen-dimension threshold)**.** *Let X be a compact zero-dimensional metric space with $\overline{\dim}_{\mathrm{cl}}(X) < \infty$, and let $f_1, \dots, f_n \in C(X)$ be real-valued contractions with common (L, α)-Hölder modulus. Then*

$$\operatorname{disc}(f_1, \dots, f_n) \leq C_{X,L,\alpha} \sqrt{n \log(e+n)}.$$

If, in addition,

$$\overline{\dim}_{\mathrm{cl}}(X) < 2\alpha,$$

then the Spencer-scale bound holds:

$$\operatorname{disc}(f_1, \dots, f_n) \leq C_{X,L,\alpha} \sqrt{n}.$$

In particular, every compact zero-dimensional metric space of upper clopen Minkowski dimension 0 satisfies the Spencer-scale estimate for every positive Hölder exponent α.

Proof. If $D > \overline{\dim}_{\mathrm{cl}}(X)$, then by the definition of the limsup there are $C_D < \infty$ and $r_D > 0$ such that

$$M_X(r) \leq C_D r^{-D} \qquad (0 < r \leq r_D).$$

For the finitely many larger scales $r_D \leq r \leq \operatorname{diam} X$, the constant may be enlarged, so the same polynomial-growth hypothesis holds on $(0, 1]$ after rescaling the metric by an absolute factor. Applying Corollary 9.14 with this exponent D gives

$$\operatorname{disc}(f_1, \dots, f_n) \leq C_{D,L,\alpha,X} \sqrt{n \log\bigl(e + n^{D/(2\alpha)-1}\bigr)}.$$

If D is finite this is $O_{X,L,\alpha}(\sqrt{n \log(e+n)})$. If $\overline{\dim}_{\mathrm{cl}}(X) < 2\alpha$, choose D strictly between these two numbers; then $D/(2\alpha) - 1 < 0$, so the logarithm is bounded and the estimate is $O_{X,L,\alpha}(\sqrt{n})$. The dimension-zero case follows for every $\alpha > 0$. □

Corollary 9.69 (Critical clopen-content endpoint)**.** *Let X be a compact zero-dimensional metric space and let $f_1, \dots, f_n \in C(X)$ be real-valued contractions with common (L, α)-Hölder modulus. If*

$$\overline{\mathcal{M}}^{2\alpha}_{\mathrm{cl}}(X) < \infty,$$

then the Spencer-scale estimate holds:

$$\operatorname{disc}(f_1,\dots,f_n) \le C_{X,L,\alpha}\sqrt{n}.$$

Thus the strict inequality $\overline{\dim}_{\mathrm{cl}}(X) < 2\alpha$ *in Corollary 9.68 can be replaced at the critical dimension by finite upper clopen* 2α*-content.*

Proof. Finite upper content gives constants $C_0 < \infty$ and $r_0 > 0$ such that

$$M_X(r) \le C_0 r^{-2\alpha} \qquad (0 < r \le r_0).$$

Enlarging C_0 to cover the finitely many larger scales, after the same harmless metric rescaling used in Corollary 9.68, puts X under the polynomial clopen-growth hypothesis of Corollary 9.14 with $d = 2\alpha$. The logarithmic factor there is $\log(e + n^0) = O(1)$, giving the stated $O_{X,L,\alpha}(\sqrt{n})$ bound. □

Corollary 9.70 (Logarithmic critical window)**.** *Let X be a compact zero-dimensional metric space, and let $f_1,\dots,f_n \in C(X)$ be real-valued contractions with common (L,α)-Hölder modulus. Suppose that for some $A < \infty$ and $\beta \in \mathbb{R}$,*

$$M_X(r) \le A r^{-2\alpha}\big(\log(e/r)\big)^{\beta} \qquad (0 < r \le 1).$$

Then

$$\operatorname{disc}(f_1,\dots,f_n) \le \begin{cases} C_{A,L,\alpha,\beta}\sqrt{n}, & \beta \le 0, \\ C_{A,L,\alpha,\beta}\sqrt{n \log\log(e^e + n)}, & \beta > 0. \end{cases}$$

Thus a polylogarithmic excess over the critical growth $r^{-2\alpha}$ costs only a $\sqrt{\log\log n}$ factor, while finite or subcritical logarithmic content remains at Spencer scale.

Proof. Take $r = \min(1, (L\sqrt{n})^{-1/\alpha})$, as in Corollary 9.14. Then $2nLr^{\alpha} = O_{L,\alpha}(\sqrt{n})$ and

$$\frac{M_X(r)}{n} \le C_{A,L,\alpha}\big(\log(e+n)\big)^{\beta}$$

after increasing the constant to absorb the finitely many small n. Insert this into Corollary 9.14. If $\beta \le 0$, the logarithm there is $O_{A,L,\alpha,\beta}(1)$. If $\beta > 0$, it is

$$O_{A,L,\alpha,\beta}\big(\log\log(e^e + n)\big),$$

which gives the displayed estimates. □

Corollary 9.71 (Ultrametric and symbolic models)**.** *Let X be a compact ultrametric space, and let $N_X(r)$ be the least number of closed r-balls needed to cover X. Suppose*

$$N_X(r) \le A r^{-d} \qquad (0 < r \le 1).$$

If $f_1,\dots,f_n \in C(X)$ are real-valued contractions with common (L,α)-Hölder modulus, then

$$\operatorname{disc}(f_1,\dots,f_n) \le C_{A,d,L,\alpha}\sqrt{n \log\big(e + n^{d/(2\alpha)-1}\big)}.$$

In particular, $d \le 2\alpha$ gives $\operatorname{disc}(f_1,\dots,f_n) \le C_{A,d,L,\alpha}\sqrt{n}$.

For the one-sided q-symbol Cantor shift $\Sigma_q = \{1,\dots,q\}^{\mathbb{N}}$ with metric

$$d_\theta(x,y) = \theta^{\min\{k\geq 0 : x_k \neq y_k\}} \qquad (0<\theta<1),$$

with $d_\theta(x,x)=0$, put

$$s = \frac{\log q}{\log(1/\theta)}.$$

Then every common (L,α)-Hölder tuple on Σ_q satisfies

$$\operatorname{disc}(f_1,\dots,f_n) \leq C_{q,\theta,L,\alpha}\sqrt{n\log\bigl(e+n^{s/(2\alpha)-1}\bigr)},$$

and the Spencer-scale bound holds whenever $\alpha \geq s/2$.

Proof. For a fixed $r>0$, the relation $x \sim_r y$ iff $d(x,y)\leq r$ partitions an ultrametric space into closed r-balls. Each class has diameter at most r, and it is clopen: if y lies in the class, then the open ball $\{z : d(y,z)<r\}$ is contained in the same class. Thus $M_X(r)\leq N_X(r)$, and the first assertion is Corollary 9.14.

For Σ_q, the length-k cylinder partition has q^k clopen atoms of diameter at most θ^k. If $\theta^k \leq r < \theta^{k-1}$, then

$$M_{\Sigma_q}(r) \leq q^k \leq q\, r^{-s}.$$

The previous paragraph gives the displayed bound with $d=s$. The endpoint $\alpha = s/2$ is included because this is finite upper clopen s-content, equivalently by Corollary 9.69. □

Corollary 9.72 (Finite products add clopen dimension)**.** *Let X_j, $1\leq j\leq m$, be compact zero-dimensional metric spaces with clopen partition growth*

$$M_{X_j}(r) \leq A_j r^{-d_j} \qquad (0<r\leq 1),$$

and let $X=\prod_{j=1}^m X_j$ with the max product metric. Put

$$d_* = \sum_{j=1}^m d_j.$$

If $f_1,\dots,f_n \in C(X)$ are real-valued contractions with common (L,α)-Hölder modulus, then

$$\operatorname{disc}(f_1,\dots,f_n) \leq C_{\{A_j,d_j\},L,\alpha,m}\sqrt{n\log\bigl(e+n^{d_*/(2\alpha)-1}\bigr)}.$$

In particular, $d_*\leq 2\alpha$ gives the Spencer-scale bound

$$\operatorname{disc}(f_1,\dots,f_n) \leq C_{\{A_j,d_j\},L,\alpha,m}\sqrt{n}.$$

Consequently, for a finite product of symbolic Cantor shifts

$$\prod_{j=1}^m (\{1,\dots,q_j\}^{\mathbb{N}}, d_{\theta_j})$$

with the max metric, every common (L,α)-Hölder tuple has Spencer-scale discrepancy whenever

$$\alpha \geq \frac{1}{2}\sum_{j=1}^{m}\frac{\log q_j}{\log(1/\theta_j)}.$$

Proof. For each j, choose a clopen partition $\mathcal{P}_j$ of X_j with at most $M_{X_j}(r)$ atoms and atom diameters at most r. Then

$$\mathcal{P}_1 \times \cdots \times \mathcal{P}_m$$

is a clopen partition of X whose atoms have max-metric diameter at most r. Therefore

$$M_X(r) \leq \prod_{j=1}^{m} M_{X_j}(r) \leq \left(\prod_{j=1}^{m} A_j\right) r^{-d_*}.$$

Corollary 9.14 gives the first display and the Spencer-scale consequence. The symbolic product statement follows by applying Corollary 9.71 to each factor, where $d_j = \log q_j / \log(1/\theta_j)$. □

Remark 9.73 (Why there is no "$\mathsf{E}(A)$" for the limit). Proposition 9.10 explains the absence of any single entropy attached to the AF limit A itself. An infinite-dimensional C*-algebra has no finite hR; the only entropies in play are the stage entropies $\mathsf{E}(A_k)$, and Theorem 9.4 is forced to be a statement *relative to a chosen stage*. The supporting-stage entropy, not any intrinsic invariant of A, is what controls the discrepancy. A bounded-block tower has $\mathsf{E}(A_k) = O(R_k)$ and is the favourable case; a UHF tower has $\mathsf{E}(A_k) = R_k^2$ and is the boundary case where the entropy obstruction of Proposition 8.1 reappears at every depth.

9.6. **Summary statement.** Collecting the favourable reduction and the two obstructions gives the cleanest theorem the method supports in the limit. It is, honestly, "reduce to the supporting finite stage," made precise and made stable.

Theorem 9.74 (Matrix Spencer for AF algebras, summary form)**.** *Let $A = \overline{\bigcup_k A_k}$ be a separable unital AF C*-algebra with stage entropies $\mathsf{E}(A_k) = h_k R_k$, and let $A_1, \ldots, A_n \in A$ be contractions.*

(1) (Reduction.) *For every stage k and every joint contraction approximant of error $\delta_k = \delta_k(A_1, \ldots, A_n)$,*

$$\operatorname{disc}_{\mathrm{op}}(A_1, \ldots, A_n) \leq C\sqrt{n \log\left(e + \frac{\mathsf{E}(A_k)}{n}\right)} + 2n\delta_k,$$

and one may take the infimum over k.

(2) (Spencer regime.) *If the tuple is supported, exactly or to within $O(n^{-1/2})$, on a stage of entropy $\mathsf{E}(A_k) = O(n)$, then $\operatorname{disc}_{\mathrm{op}}(A_1, \ldots, A_n) = O(\sqrt{n})$. This holds for every tuple in a bounded-block tower whose support has reduced dimension $O(n)$, and in particular for every commutative AF algebra with $O(n)$ active minimal projections.*

(3) (Intrinsic-dimension upgrade.) *By the Akbas–Sra algebraic Matrix Spencer theorem, the same $O(\sqrt{n})$ conclusion holds whenever the tuple is supported, exactly or to within $O(n^{-1/2})$, on a stage with $\dim_{\mathbb{C}} A_k = O(n)$. This improves the preceding entropy criterion for heterogeneous direct-sum stages, but coincides with it on a single full matrix block.*

(4) (Low-Frobenius perturbative upgrade.) *In a finite-dimensional matrix representation, the preceding intrinsic-dimension AF-stage conclusion is stable under self-adjoint perturbations L_i with $\|L_i\|_{\mathrm{op}} \leq 1$ and $\|L_i\|_{\mathrm{F}}^2 \leq n/\log^3(en)$ (in particular, under rank-$n/\log^3(en)$ tails; see Corollaries 4.4 and 9.8).*

(5) (Full-span finite-lift ceiling.) *If the supporting stage contains a full $\mathrm{M}_D(\mathbb{C})$ block on which the compressed tuple spans the whole block, then the full-span faithful application of Theorem 1.3 cannot certify better than $\sqrt{n\log(e + D^2/n)}$ (Proposition 9.9); for full-span tuples in the M_{2^k} stage of a UHF tower this becomes the full-stage rate once $k > \frac{1}{2}\log_2 n$, and improving it would require leaving the full-span faithful finite-block route or avoiding the entropy dependence.*

(6) (Hard obstruction.) *If A is replaced by a non-AF von Neumann algebra, the finite-stage route can fail completely. For a generic AF element with infinite spectrum, the exact-stage statement is unavailable and only the approximate theorem applies, at the entropy of the approximating stages (Proposition 9.10). Quantitatively, those stage entropies must at least dominate the spectral packing numbers of the element being approximated (Proposition 9.11). In a II_1 factor a single diffuse self-adjoint contraction already escapes the exact finite-stage method.*

The self-adjoint versions hold for self-adjoint tuples. The constant C is that of Theorem 4.1 and is independent of k, of D, and of the ambient algebra.

Proof. Parts (1) and (2) are Theorem 9.4 with Corollaries 9.5 and 9.6, together with the bounded-block discussion of §9.4. The intrinsic-dimension upgrade is Corollary 9.7, and the low-Frobenius perturbative upgrade is Corollary 9.8. The full-span finite-lift ceiling is Proposition 9.9. The hard obstruction is Propositions 9.10 and 9.11. □

Remark 9.75 (The honest takeaway). The finite-lift theorem proved here is not a bound attached to an infinite-dimensional algebra as a single object. What survives the inductive limit is precisely the finite-stage theorem, made uniform in the ambient algebra and stable under norm approximation: the discrepancy of an AF tuple is controlled by the entropy of the finite stage that supports it, at the same rate $\sqrt{n\log(e + \mathsf{E}/n)}$ as in the finite case, with an approximation error that vanishes faster than $\sqrt{n}$ whenever the support is captured at scale $n^{-1/2}$. This stage-entropy bound is $O(\sqrt{n})$ for bounded-block towers and shallow UHF elements, where the supporting-stage entropy is $O(n)$. For exact finite-dimensional algebraic support, the stronger Akbas–Sra theorem replaces the sufficient entropy condition by the intrinsic-dimension condition $\dim_{\mathbb{C}} A_k = O(n)$. The Akbas–Sra perturbative theorem

also allows low-Frobenius self-adjoint tails on top of such a represented AF-stage core; low-rank tails with $\operatorname{rank} L_i \le n/\log^3(en)$ are an immediate special case. The finite-lift stage-entropy bound still degrades to the generic $\sqrt{n\log(e+D^2/n)}$ rate for full-span tuples/full-unit-ball models, by the same D^2 entropy obstruction (Proposition 8.1) that bars full-unit-ball faithful finite lifts from the square case of the conjecture in finite dimensions; and the exact finite-stage theorem disappears for diffuse elements of a von Neumann algebra, which have no finite supporting stage at all.

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